



Nova Lake Mobile Platform

Design Guide

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Revision History

Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
855310	0.4		NA	Initial draft release	May 2025
855310	0.5		NA	Initial release	November 2025
855310	0.5.1		NA	Revision Update- Only Power Map updated	November 2025
855310	0.7	NA	Introduction	• Added — Figure 1 on page 55 — Terminology Usage on page 55	January 2026
			PCB Stackup and Design Consideration	• Added — Description in PCB Stackup and Design Consideration on page 58 — Notes in Non Critical To Function (NCTF) Solder Balls and Testability Recommendation on page 98 — Adhesive Keep out Zone on page 98 • Updated — Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML), For DDR5 SODIMM/CSODIMM and LPDDR5x LPCAMM2 sub-section — Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML), For DDR5 SODIMM/CSODIMM and LPDDR5x Memory Down sub-section — Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML), For LPDDR5x Memory Down sub-section • Removed — Pad Definitions sub-section — NCTF Implementation - Continuity/Resistance Test Threshold sub-section — Land Pattern and Corner NCTF PCB Routing Guidance sub-section	

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Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
			General Design Considerations	• Added — 12 GT/s Transfer Speed Max RSS Length in mm (Refer Note1) in Table 66 on page 101 — Teardrop Structure Requirement at Junction of Via and Trace on page 113 • Removed 20-32GT/s signal details from Maximum Parallel Routing Restrictions on page 101	
			System Memory Interface Design Guidelines	• Added Point 3 in Via Placement and Pad Optimization on page 173 • Updated — Table 78 on page 129 — Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML), For LPDDR5x Memory Down sub-section — Processor Parameter Details in Table 124 on page 152 — BO3 in MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), CA/CS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines table — Processor Parameter Details in Table 153 on page 166 — BO length in MD x32 Type-4,0.8mm,10L sub-section • Removed — LPDDR0 and LPDDR1 LPCAMM2 Signal Name details from Table 77 on page 128 — Rules for Populating DIMM Slots sub-section	
			Display Interfaces	• Added — Max Number of vias table — eDP AUX MUX Topology sub-section — 355 mm cable assembly (micro coaxial) Note Detail in eDP HBR3 Main Link CTLE Topology Notes table	
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				— 355 mm cable assembly (micro coaxial) Note Detail in eDP HBR3 Main Link CTLE+DFE Topology Notes table — 355 mm cable assembly (micro coaxial) Note Detail in eDP HBR3 Main Link MUX CTLE Topology Notes table — 355 mm cable assembly (micro coaxial) Note Detail in eDP HBR3 Main Link MUX CTLE+DFE Topology Notes table • Updated — Table 173 on page 183 — Table 175 on page 184 • Removed — DisplayPort* Signal Descriptions sub-section — eDP AUX Topology Notes table — Main Link (Tx) Description Signal Name TCPx_TXx_P/N and TCPx_TXRXx_N/P from Table 201 on page 194 — TCP Port Signal Mapping for HDMI* table	
			High Speed I/Os Reference Documents	• Added New Document reference in Reference Documents on page 204	
			High Speed I/Os Platform Clock Design Guidelines	• Updated — Platform Clock Associated Signal Guidelines on page 205 — Table 222 on page 206 — RVP Final External Load Capacitor(Ce) Parameter Notes in Table 223 on page 208	
			High Speed I/Os MIPI CSI-2	• Added — Max Number of vias table in CSI CPHY Main Link Up To 1.0 Gbps Topology sub-section — Max Number of vias table in CSI CPHY Main Link Up To 2.0 Gbps Topology sub-section — Max Number of vias table in CSI CPHY Main Link Up To 2.5 Gbps Topology sub-section	

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			High Speed I/Os Discrete Thunderbolt™ Design Guidelines	• Updated — USB2.0 Group Signal and Description in Table 297 on page 246 — Table 300 on page 249 — Complete TCP DP Hot Plug Detect Retimer MUX With Barlow Ridge Implementation sub-section	
			High Speed I/Os USB-C* Sub-System	• Updated — TCP TBT4 sub-section — TCP TBT5 sub-section • Added BIOS/mFIT/VBT setting for TCPs on page 262	
			High Speed I/Os Processor - PCI Express* Design Guidelines	• Updated — PCIe Gen 1-3 sub-section — PCIe Gen 4 sub-section — PCIe Gen 5 sub-section — TBT Device (BR) PCIE Device Recommended Slot in Table 428 on page 327 • Removed PCI Express*Interface Configuration Details sub-section	
			High Speed I/Os CNVio Bus Design Guidelines	• Added Note in Probing on page 375 • Updated — CNVio3 sub-section — EMT to CITU in Tools for Debug on page 375 • Removed CNVio3 reuses CNVio2 clock traces (1 TX clock and 1 RX clock) in CNVio3 on page 367	
			Low Speed I/Os Asynchronous and Sideband Signals Design Guidelines	• Updated — Warning 3 in Topology Guidelines on page 383 — Table 519 on page 384	
			Low Speed I/Os Legacy Audio Interface Design Guidelines	• Updated — Signal name/ list Note Detail in DMIC 2-Load Branch Topology Notes table — Signal name/ list Note Detail in DMIC 2-Load Daisy Topology Notes table — HDA 1-Load (Device Down) Topology Diagram	
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Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
				• Removed Left and right DMIC from Intel® High Definition Audio (Intel HD Audio) and DMIC Topology Guidelines on page 412	
			Low Speed I/Os SoundWire* Interface Design Guidelines	• Added SNDW3_DATA3 Signal Name Description in Table 600 on page 433	
			Low Speed I/Os Serial Peripheral Interface (SPI0) Flash Design Guidelines	• Updated — M6 and M7 Max Length Segment Note, Max Length Total (mm), and Max Length Total Note in SPI0 Flash 76.8MHz and 100MHz 1 to 3-Load Branch (Device Down) MAF Topology Segment Lengths for Maximum 100MHz Routing Guidelines table — R1 Note Detail in SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology Notes table — M8 Max Length Segment Note in SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology Segment Lengths for Maximum 100MHz Routing Guidelines table — M3 and M4 Max Length Segment (mm) in SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology Segment Lengths for Maximum 76.8MHz Routing Guidelines table	
			Low Speed I/Os Touch Host Controller (THC) Design Guidelines	• Added Note Touch Guidelines on page 495 • Updated Touch Signal Descriptions on page 494	
			Low Speed I/Os Universal Asynchronous Receiver Transmitter (UART) and LSx Interface Design Guidelines	• Updated — LSx Interface on page 502 — LSX 1-Load to Retimer (Device Down) Topology Diagram	

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Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
			Platform Connectivity	• Updated — MrP to module in RF_RESET_B and ClkReq0 (Pin14 in M.2-2230, A43 in 1216) in RF Companion M.2 (2230) on page 603 — HrP side to CRF side in W_DISABLE1#, W_DISABLE2# in RF Companion M.2 (2230) on page 603 — Pulsar CNVio to CNVi IP in Table 723 on page 612	
			Electromagnetic Compatibility	• Updated — CNVio3 (STEP) Max Routing Length at Breakout Region (Microstrip) and Max Routing Length at Connector Region (Microstrip) in High Speed Differential Signal Routing Recommendations table — Third paragraph in ESD Diode Selection sub-section — ESD Diode Reverse Working Voltage in Discrete ESD Protection Diode Selection Guidelines table • Removed Type-C USB+DP (10 Gbps) from Common Mode Choke Selection sub-section	
			Power Delivery	• Added — Adaptive Thermal Monitor on page 696 — Extended Power Range (EPR) on page 702 — General Rail-to-Rail Sequencing details in Processor Rail Order Requirements on page 741 • Updated — 35-85W to 65W in Power Delivery Settings for Nova Lake on page 669 — VDDQ on page 673 — Tjmax to Maximum Operating Temperature in Terminology on page 678 — Table 743 on page 691 — Table 744 on page 692	
continued...					

Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
				— Timing Diagrams in Power Sequencing Timing Diagrams Legacy Signals on page 730	
			Power Integrity Recommendations	• Updated Complete Processor Power Rails subsection	
			Platform Debug and Test Hooks	• Updated Platform Debug and Test Hooks on page 900	
855310		0.4	• Added NVL 6C Content		March 2026
855310		1.0 0.7.1	NA	• Updated — Introduction on page 54 — Terminology Usage on page 55 — NVL 6C and NVL 8C/H16C Topology Differences on page 55 • Updated — H to Mobile in PCB Stackup and Design Consideration on page 58 — BO2 3,7 Routing Layers Trace K, CA/CS to CA/CS (%) value from 3.443 to 3.638 in Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC Type-3,0.8mm,8L CH-0, CH-1, CH-2, CH-3 Trace Spacing Specs table — BO2 3,7 Routing Layers Trace K, Byte to Byte (%) value from 3.443 to 3.638 in Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC Type-3,0.8mm,8L CH-0, CH-1, CH-2, — BO2 3,7 Routing Layers Trace K, DQS pair to DQ of same byte (%) value from 3.32 to 3.498 in Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC Type-3,0.8mm,8L CH-0, CH-1, CH-2, — BI Trace K, WCK pair to DQ of same byte (%) from 0.016 to 0.017 in Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-4,0.8mm,10L CH-0, CH-2, CH-4, CH-6 Trace Spacing Specs table	April 2026

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Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
			System Memory Interface Design Guidelines	• Updated — H to Mobile in System Memory Interface Design Guidelines on page 127 — POR Memory Configurations on page 127 — Table 109 on page 145 — DRAM Die Density (Gb) Parameter Details in Table 153 on page 166 — Points 3 and 4 in DDR Byte Swapping on page 172 — Table 168 on page 175	
			Display Interfaces	• Added White Paper reference in Table 174 on page 184 • Updated DDI_RCOMP Signal Name Details in Table 220 on page 202 • Removed TCP_RCOMP Signal Name Details from Table 220 on page 202	
			Embedded DisplayPort* (eDP*)	• Added — Description on NVL 8C/H16C in eDP AUX MUX Topology sub-section — Description on NVL 8C/H16C in eDP HBR3 Main Link MUX CTLE Topology sub-section — Description on NVL 8C/H16C in eDP HBR3 Main Link MUX CTLE+DFE Topology sub-section • Updated — Max Length Total from 200 to 220 after eDP HBR3 Main Link CTLE Topology Tx Routing Guidelines table — Max Length Total from 140 to 150 after eDP HBR3 Main Link CTLE Topology Tx Routing Guidelines table — Max Length Total Note after eDP HBR3 Main Link MUX CTLE+DFE Topology Tx Routing Guidelines table	
			HDMI*	• Updated — HDMI (Default nonreversed) to HDMI (Reversed) in Table 202 on page 194	
continued...					

Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
				- Max Length Segment (mm) in HDMI Retimer 12G Topology Tx, Post-Channel Other Signals table Removed - HDMI* TCP0-2 Port Type from Table 173 on page 183 - Cable Segment Details from HDMI DDC Topology Tx Routing Guidelines table	
			High Speed I/Os	Updated Reference Documents on page 204	
			Platform Clock Design Guidelines	Updated - CLK PCIe Gen 1-3 Add-in Card Clock Topology sub-section - CLK PCIe Gen1-3 Device Down Clock Topology Diagram title - CLK PCIe Gen 4 Add-in Card Clock Topology sub-section - CLK PCIe Gen4 Device Down Clock Topology Diagram title - CLK PCIe Gen 5 Add-in Card Clock Topology sub-section - CLK PCIe Gen5 Device Down Clock Topology Diagram title - Crystal Oscillator (38.4 MHz) Topology Diagram - Crystal Oscillator Topology Notes table - CLK PCIe Gen 1-3 Add-in Card Clock Topology Removed RF 200 kOhm from RVP Final External Load Capacitor (Ce) Parameter Notes Table 223 on page 208	
			MIPI CSI-2	Added - Document Reference in Table 253 on page 223 - Description on NVL 8C/H16C in CSI CPHY Main Link Up To 1.0 Gbps Topology sub-section - Description on NVL 8C/H16C in CSI CPHY Main Link Up To 2.0 Gbps Topology sub-section	

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Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
				— Description on NVL 8C/ H16C in CSI CPHY Main Link Up To 2.5 Gsps Topology sub-section • Updated — Point 4 in General Topology Guidelines: on page 231 — CSI CPHY Topology Guidelines on page 236	
			Discrete Thunderbolt™ Design Guidelines	• Updated — Description on NVL 8C/ H16C in TCP DP UHBR20 Retimer MUX With Barlow Ridge Topology sub-section — Description on NVL 8C/ H16C in TCP DP Hot Plug Detect Retimer MUX With Barlow Ridge Implementation sub-section — Description on NVL 8C/ H16C in TCP DP AUX Retimer MUX With Barlow Ridge Topology sub-section	
			USB-C* Sub-System	• Added Note after Table 314 on page 260 • Updated — Maximum via stub length Note Detail from 375um to 381um in TCP USB 3.2 Gen 1x1 Type-A Topology Notes table — Maximum via stub length Note Detail from 250um to 381um in TCP USB 3.2 Gen 2x1 Type-A Topology Notes table — TCP TBT5 sub-sections • Removed — TCP_RCOMP Signal Names Values from Table 316 on page 261 — Number of vias allowed Note Detail from TCP USB 3.2 Gen 1x1 Type-A Topology Notes table — Number of vias allowed Note Detail from TCP USB 3.2 Gen 2x1 Type-A Topology Notes table	
			Processor - PCI Express* Design Guidelines	• Updated — Tline Type in PCIe Gen 1-3 sub-sections	

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Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
				— Tline Type in PCIe Gen 4 sub-sections — Note Point in PCIe Gen 5 Root Port Power at Lower Gen Data Rates on page 326	
			Universal Serial Bus USB 3.2 Design Guidelines	• Updated — USB 3.2 Gen 1x1 Type-A External Topology sub-section — USB 3.2 Gen 2x1 Type-A External Topology sub-section	
			Processor Embedded USB2 (eUSB2) Design Guidelines	• Updated — Max Length Total Note after eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable Topology Rx, Tx Routing Guidelines table — Max Length Total Note after eUSB2 Repeater Post-Channel Type-C Flex or Internal Cable Topology Rx, Tx Routing Guidelines table	
			CNVio Bus Design Guidelines	• Updated — Maximum DC resistance for the trace on the motherboard Note Detail in CNVio3 General Guidelines table — M1 Segment Max Length Segment Note in CNVio3 M.2 Topology Rx,Tx Routing Guidelines table — M1 Segment Max Length Segment Note in CNVio3 Module Down Topology Rx,Tx Routing Guidelines table	
			Asynchronous and Sideband Signals Design Guidelines	• Updated — SVID 1-Load (Device Down) Topology sub-section • Removed — Package pins Note Detail from EPD_ON Topology Notes table — Package pins Note Detail from VDD2PWRGOOD Topology Notes table	
			Real Time Clock (RTC) Design Guidelines	• Updated — 32.768 kHz RTC Crystal Topology Diagram	

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Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
				— External capacitor values Note Detail in RTC (Real Time Clock) Topology Notes table — Component Selection Guidelines on page 399	
			SoundWire* Interface Design Guidelines	• Updated SoundWire* Interface Design Guidelines on page 432	
			Serial Peripheral Interface (SPI0) Flash Design Guidelines	• Updated — Serial Peripheral Interface (SPI0) Flash Design Guidelines on page 468 — SPI0 Flash 50MHz 2 to 3-Load Branch (Device Down) with G3 Wired-OR Topology sub-section — SPI0 Flash 50MHz 4-Load Branch (Device Down) with G3 Wired-OR Topology sub-section — SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology sub-sectionS	
			Electromagnetic Compatibility	• Updated — Crystal Ground Ring Requirements sub-sections — Minimize DLVR Input Power Planes' Exposure sub-section — PCIe4, PCIe5, and PCIe Refclk Interface Note details in High Speed Differential Signal Routing Recommendations table	
			Power Delivery	• Added Note in PWROK Signals Generation on page 728 • Updated — Table 741 on page 671 — 1.8V on page 672 — VDD2 on page 672 — Table 744 on page 692 — Document Reference in Important Notes on page 695 — Figure 450 on page 728	
continued...					

Document Number	Revision Number NVL 8C, 16C	Revision Number NVL 6C	Section	Description	Release Date
				— tPCH03 delay (12ms mins) to (2ms mins) in Processor Rail Order Requirements on page 741	
			Power Integrity Recommendations	• Added — Processor Decoupling and Layout Recommendations for NVL 8C/ H16C on page 748 • Updated — Note 2, 3, and 5 in Power Integrity Recommendations on page 747 — Image in VCCGT(H16C_12Xe/T3/0.8mm & 0.9mm PCB) sub-section — Image in VCCGT(H16C_12Xe/T4/0.8mm PCB) sub-section	

1.0 Introduction

This Design Guide provides motherboard implementation recommendations for the Nova Lake mobile platform with the following SKUs. Please refer [854352](#) for NVL HX Platform Design Guidelines.

- NVL 6C
- NVL 8C
- NVL H 16C 4Xe
- NVL H 16C 12Xe

NOTE

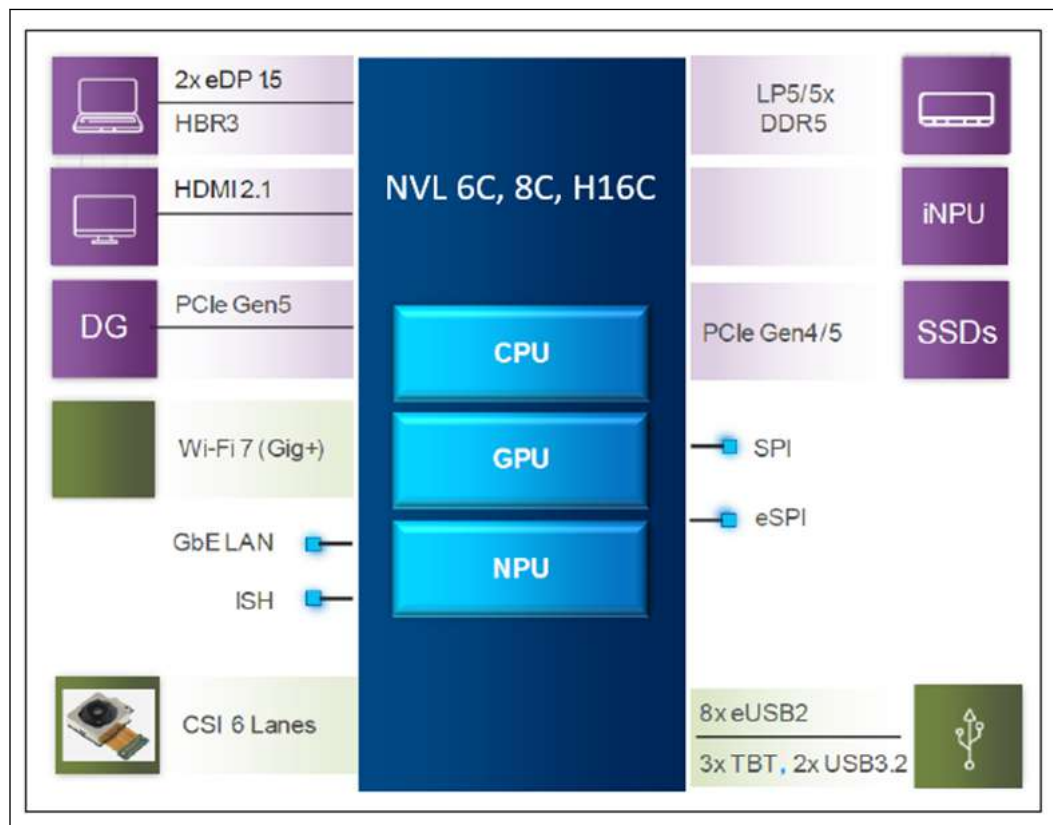
Refer to the [NVL 6C and NVL 8C/H16C Topology Differences](#) on page 55 for NVL 6C and 8C/16C Topology differences.

NOTE

For more details on the SKUs, refer to the *Nova Lake Mobile Processor External Design Specification, Volume 1 of 2* ([#844316](#))

The Platform Design Guide is developed to ensure maximum flexibility for board designers while reducing the risk of board related issues. Design recommendations are based on Intel's simulations and lab experience and are strongly recommended, if not necessary, to meet the timing and signal quality specifications. Design recommendations are based on the reference platforms designed by Intel. They should be used as an example but may not be applicable to particular designs.

The guidelines recommended in this document are based on experience, simulation, and preliminary validation work done at Intel while developing the Nova Lake processor-based platform. This work is ongoing, and these recommendations are subject to change. Metric units are used in some sections in addition to the standard use of U.S. customary system of units (USCS). If there is a discrepancy between the metric and USCS units, assume the USCS unit is most accurate. The conversion factor used is 1 inch (1000 mils) = 25.4mm.

Figure 1. NVL Mobile Processor Series Platform Diagram


1.1 Terminology Usage

This document uses the terms "initiator" and "target" for formerly known terms "master" and "slave".

1.2 NVL 6C and NVL 8C/H16C Topology Differences

NOTE

The table below captures only the differences between NVL 6C and 8C/H16C topologies. The other topologies not listed here are applicable for all the SKUs.

Chapter	Sub Chapter	Topology	Supported in NVL 8C/H16C	Supported in NVL 6C
PCB Stack-up	Generalized 8-Layer Motherboard Examples	Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML), For DDR5 SODIMM/CSODIMM and LPDDR5x LPCAMM2	Yes	Yes
	Generalized 10-Layer Motherboard Examples on page 70	Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML), For LPDDR5x Memory Down	Yes	No
		Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML), For DDR5 SODIMM/ CSODIMM and LPDDR5x Memory Down	Yes	Supported only for "MD x32 Type-3,0.9mm,10 L" Topology
		Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML), For LPDDR5x Memory Down	Yes	No
System Memory Interface Design Guidelines on page 127 ¹	DDR5 SODIMM/ CSODIMM 1DPC on page 129	SODIMM 1DPC Type-3,0.9mm,10L	Yes	No
	LPDDR5x Topologies on page 144	MD x32 Type-3,0.8mm,10L	Yes	No
		MD x32 Type-4,0.8mm,10L	Yes	No
High Speed I/Os	Display Interfaces on page 183	HDMI CR 1.65G Topology	No	Yes
		eDP HBR3 Main Link MUX CTLE Topology	Yes	No
		eDP HBR3 Main Link MUX CTLE+DFE Topology	Yes	No
		eDP Aux Mux Topology	Yes	No
	USB-C* Sub-System on page 254	TCP TBT5 Retimer Topology	Yes	No
		TCP TBT5 Retimer with Internal Cable Topology	Yes	No
		TCP TBT5 Cascaded Retimer Topology	Yes	No
		TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Redriver Topology	No	Yes
		TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Topology	No	Yes
		TCP AUX No Retimer Topology	No	Yes
TCP DP UHBR20 Retimer MUX With Barlow Ridge Topology		Yes	No	
continued...				

Chapter	Sub Chapter	Topology	Supported in NVL 8C/H16C	Supported in NVL 6C
		TCP DP Hot Plug Detect Retimer MUX With Barlow Ridge Implementation	Yes	No
		TCP DP AUX Retimer MUX With Barlow Ridge Topology	Yes	No
	CSI CPHY Topology Guidelines on page 236	CSI CPHY Main Link Up To 1.0 Gbps Topology	Yes	No
		CSI CPHY Main Link Up To 2.0 Gbps Topology	Yes	No
		CSI CPHY Main Link Up To 2.5 Gbps Topology	Yes	No
Electromagnetic Compatibility on page 627	EMC Component Selection -> Common Mode Choke Selection	Table: CMC Selection Guidelines-> Interface -> TCP USB 3.2+DP	No	Yes
	EMC Component Selection -> ESD Diode Selection	Table: Discrete ESD Protection Diode Selection Guidelines-> Interface -> TCP USB 3.2+DP	No	Yes
Power Integrity Recommendations on page 747		Processor Decoupling and Layout Recommendations for NVL 8C/ H16C on page 748	Yes	No
		Processor Decoupling and Layout Recommendations for NVL 6C on page 860	No	Yes
Note: 1. Please refer POR Memory Configurations on page 127 for 6C and 8C/H16C memory POR speeds				

2.0 PCB Stackup and Design Consideration

Different stackups can be implemented on the platform. It is important to note that variations in the PCB stackup, such as changes in the dielectric thickness, copper thickness, trace width, and spacing can impact the impedance, loss and jitter characteristics of all the interfaces. Such changes may either be intentional, or the results of variations encountered during the PCB manufacturing process. In either case, they must be properly considered when designing the interconnects.

Nova Lake mobile reference stack-ups are listed below:

1. Type-3, 0.8 mm, 8L, Mainstream, Mid Loss (ML), For DDR5 SODIMM/CSODIMM & LPDDR5x LPCAMM2.
2. Type-3, 0.9 mm, 10L, Mainstream, Premium Mid Loss (PML), For DDR5 SODIMM/CSODIMM & LPDDR5x Memory Down.
3. 3. Type-3, 0.8 mm, 10L, Thin PCB, Premium Mid Loss (PML), For LPDDR5x Memory Down.
4. Type-4 2-x-2+, 0.8 mm, 10L, Mainstream, Premium Mid Loss (PML), For LPDDR5x Memory Down.

All reference stackups in Nova Lake mobile PDG are based on Mid Loss PCB materials but with two different Df range. Type-3 8-Layer 0.8 mm stackup is based on Mainstream Mid Loss (MML) material with nominal Df 0.012. Type-3 10-Layer 0.8 mm/0.9 mm, and Type-4 2-x-2+ 10-layer 0.8 mm stackups are based on Premium Mid Loss (PML) material with nominal Df 0.009. These PCB materials are used on all Nova Lake mobile reference validation platforms (RVP).

All routing guidelines in this PDG are defined based on Nova Lake mobile reference stackups that meet the Tline Spec electrical targets for Impedance (Z), Insertion Loss (A) and Crosstalk (K), inclusive of $\pm 10\%$ impedance HVM variation. Customers should use Nova Lake mobile Intel® Platform Design Studio (# 869048) to run electrical analysis for PCB stackups that differ from the Intel reference stackups, ensure that electrical targets are met.

NOTES

1. Metric unit um and mm are used as the primary unit in this PDG.
2. All stackups assume processor is placed on Routing Layer 1.
3. Recommend, Cu foil of RTF type in all inner layers of Nova Lake mobile PCB to meet loss requirements. Outer layers (Top/Bottom) Cu foil type shall be of HTE type.

2.1 Reference Documents

Document Name	Document Number
Dual Stripline Crosstalk Behavior and Cross-Interface Coupling Technical White Paper	575518

2.2 Generalized 8-Layer Motherboard Examples

2.2.1 Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML), For DDR5 SODIMM/CSODIMM & LPDDR5x LPCAMM2

Pcb Type: Type-3

Pcb Thickness: 0.8mm

Pcb Layer Count: 8

Category: Mainstream

Material: Mid Loss (ML)

Variant: For DDR5 SODIMM/CSODIMM & LPDDR5x LPCAMM2

Table 1. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML), For DDR5 SODIMM/CSODIMM & LPDDR5x LPCAMM2

Routing Layer	Description	Tline Type	Description Memory	Tline Type Memory	Thickness (um)	Dielectric Constant	Loss Tangent
	S/M	None	S/M	None	13	4	0.031
1	BO, Main Route, Power	MS	BO, Power	MS	30		
	PP	None	PP	None	60	3.5	0.012
2	GND, Power	GND/Power	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.012
3	BO, Main Route, Power	DSL	BO, Main Route	SL	30		
	PP	None	PP	None	64	3.5	0.012
4	BO, Main Route, Power	DSL	GND	GND/Power	30		
	CORE	None	CORE	None	180	3.8	0.012
5	GND	GND/Power	BO, Main Route	SL	30		
	PP	None	PP	None	64	3.5	0.012
6	BO, Main Route, Power	SL	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.012
7	GND, Power	GND/Power	BO, Main Route	SL	30		
	PP	None	PP	None	60	3.5	0.012
8	BO, Main Route, Power	MS	Power	GND/Power	30		
	S/M	None	S/M	None	13	4	0.031

Table 2. Notes

Note
Dielectric thickness of 64um has been chosen specifically on SL layers to ensure POR memory speeds are met. Customers following thicker dielectrics will need to ensure cross talk co-efficient are equal to or less than those specified in memory Tline spec.

2.2.1.1 Tline Spec (Differential)

Table 3. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) Tline Spec (Differential)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
B*	MS	1, 8	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	94	-0.98	-0.124	0.304
B*	MS	1, 8	CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	250	375	94	-0.98	-0.175	0.171
B*	MS	1, 8	CNVio3	75	88	375	375	94	-0.98	-0.153	0.171
B*	MS	1, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	94	-0.98	-0.153	0.08
B*	MS	1, 8	CSI CPHY	81	88	88	375	91	-0.995	0.522	0.175
B*	MS	1, 8	TCP DP, TCP TBT4, TCP TBT5	81	88	500	500	91	-0.97	-0.108	0.084
B*	MS	1, 8	PCIe Gen 5	81	88	375	500	91	-0.97	-0.163	0.084
B*	MS	1, 8	PCIe Gen 4	81	88	375	375	91	-0.97	-0.163	0.175
B*	MS	1, 8	USB 3.2 Gen 1, USB 3.2 Gen 2	81	88	300	300	91	-0.97	-0.194	0.309
B*	SL	3, 6	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	78	-0.9	0.187	0.026
B*	SL	3, 6	CSI CPHY, CSI DPHY, eDP, HDMI, TCP AUX, TCP DP AUX	75	88	88	300	78	-0.9	1.713	0.026
B*	SL	3, 6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	78	-0.9	0.006	0.001
B*	SL	3, 6	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	75	88	375	375	78	-0.9	0.006	0.006
B*	SL	3, 6	PCIe Gen 4, PCIe Gen 5	75	88	230	500	78	-0.9	0.103	0.001
B*	SL	3, 6	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	78	-0.9	0.026	0.026
B*	DSL	3	CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	87	-0.83	2.915	0.306
B*	DSL	3	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	87	-0.83	0.807	0.306
B*	DSL	3	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	87	-0.83	0.155	0.052
B*	DSL	3	PCIe Gen 4, TCP DP, TCP TBT4, TCP TBT5	75	88	375	375	87	-0.83	0.155	0.155
B*	DSL	3	PCIe Gen 5	75	88	500	500	87	-0.83	0.052	0.052
B*	DSL	3	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	87	-0.83	0.306	0.306
B*	DSL	4	CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	99	-0.75	4.248	0.65
B*	DSL	4	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	99	-0.73	1.531	0.65
B*	DSL	4	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	99	-0.72	0.341	0.118
B*	DSL	4	PCIe Gen 4, TCP DP, TCP TBT4, TCP TBT5	75	88	375	375	99	-0.72	0.341	0.341
B*	DSL	4	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	99	-0.72	0.65	0.65
B*	DSL	4	PCIe Gen 5	88	88	500	500	95	-0.702	0.119	0.119
M*	MS	1, 8	CSI DPHY, eUSB2, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	115	95	300	300	80	-0.91	-0.278	0.346
M*	MS	1, 8	Differential Clock (Gen 3 and below support)	115	95	300	375	80	-0.91	-0.278	0.208
M*	MS	1, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	115	95	500	760	80	-0.91	-0.148	0.038
continued...											

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
M*	MS	1, 8	eDP, HDMI, PCIe Gen 4, TCP DP, TCP TBT4	115	95	500	500	80	-0.91	-0.148	0.107
M*	MS	1, 8	CNVio3, TCP TBT5	115	95	800	800	80	-0.91	-0.057	0.033
M*	MS	1, 8	UFS 4.0, USB 3.2 Gen 1, USB 3.2 Gen 2	115	95	750	500	80	-0.91	-0.068	0.107
M*	MS	1, 8	PCIe Gen 5, TCP TBT5 (Post-Channel)	130	125	800	800	80	-0.87	-0.073	0.042
M*	MS	1, 8	CSI CPHY	165	950	950	950	78	-0.826	-0.147	0.072
M*	MS	1, 8	HDMI (Post-Channel)	75	120	500	500	100	-0.936	-0.127	0.099
M*	SL	3, 6	CSI DPHY, eDP, eUSB2, HDMI, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	80	140	300	300	80	-0.88	0.027	0.027
M*	SL	3, 6	Differential Clock (Gen 3 and below support)	80	140	300	375	80	-0.88	0.027	0.006
M*	SL	3, 6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	80	140	500	760	80	-0.88	0.001	0
M*	SL	3, 6	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	80	140	375	375	80	-0.88	0.006	0.006
M*	SL	3, 6	PCIe Gen 4, PCIe Gen 5	80	140	230	500	80	-0.88	0.106	0.001
M*	SL	3, 6	USB 3.2 Gen 1, USB 3.2 Gen 2	80	140	450	500	80	-0.88	0.001	0.001
M*	SL	3, 6	CSI CPHY	85	170	170	500	78	-0.874	0.34	0.001
M*	DSL	3	CSI DPHY, PCIe Gen 1-3, TCP AUX, TCP DP AUX, eUSB2	105	115	300	300	80	-0.78	0.34	0.34
M*	DSL	3	Differential Clock (Gen 3 and below support)	105	115	300	375	80	-0.78	0.34	0.174
M*	DSL	3	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	105	115	500	760	80	-0.78	0.058	0.007
M*	DSL	3	eDP, HDMI, PCIe Gen 4, TCP DP, TCP USB 3.2	105	115	375	375	80	-0.78	0.174	0.174
M*	DSL	3	PCIe Gen 5	105	115	450	600	80	-0.78	0.09	0.025
M*	DSL	3	TCP TBT4, TCP TBT5, USB 3.2 Gen 1, USB 3.2 Gen 2	105	115	450	500	80	-0.78	0.09	0.058
M*	DSL	4	CSI DPHY, PCIe Gen 1-3, TCP AUX, TCP DP AUX, eUSB2	148	90	350	350	80	-0.65	0.402	0.402
M*	DSL	4	Differential Clock (Gen 3 and below support), PCIe Gen 4	148	90	375	375	80	-0.65	0.324	0.324
M*	DSL	4	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	148	90	540	760	80	-0.65	0.081	0.013
M*	DSL	4	eDP, HDMI, TCP DP, TCP USB 3.2	148	90	400	550	80	-0.65	0.262	0.073
M*	DSL	4	PCIe Gen 5, TCP TBT4, TCP TBT5	148	90	650	760	80	-0.65	0.032	0.013
M*	DSL	4	USB 3.2 Gen 1, USB 3.2 Gen 2	148	90	500	550	80	-0.65	0.112	0.073

Table 4. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) Tline Spec (Differential) Notes

Note
Routing Notes for CSI CPHY: (1) CPHY must be routed as a trio (2) If the port is supposed to support both CPHY and DPHY, it must be routed following CPHY guidelines (3) CPHY impedance target is 40 ohms single ended and 80 ohms differential (4) All traces in trio must follow same trace width (5) Intra-Pair guidelines must be used for trace to trace spacing in a trio (6) S-ES guidelines must be used for trio to trio in the same port (7) S-Non-ES must be used for port to port or CPHY to other interface (8) CSI CPHY in Microstrip (MS) routing is not recommended for long length, allowed only for short length (<5 mm) near the connector. For CNVio3 signal spacing: CNVio3 to non-CNVio3 pair spacing of 10H (H=Dielectric Thickness) is required. If this cannot be achieved, then GND shielding with proper stitching vias and 3H minimum spacing from GND shielding to CNVio3 is required. Please also refer to "CNVio3 Spacing Recommendation" diagram in CNVio3 section.

2.2.1.2 Tline Spec (Single Ended)

Table 5. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) Tline Spec (Single Ended)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K, Data - Any (%)	K, Clock - Any (%)
B*	MS	1, 8	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	57			-0.26	-1.665	-2.41
B*	MS	1, 8	Processor Sideband	75	88	88	57			-0.26	-1.665	-1.665
B*	MS	1, 8	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	57			-0.25	-2.41	-2.41
B*	MS	1, 8	XTAL, RTC	75	200	1270	57			-0.25	-2.394	-0.138
B*	SL	6	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	42			-0.27	3.69	0.669
B*	SL	6	Processor Sideband	75	88	88	42			-0.27	3.69	3.69
B*	SL	6	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	42			-0.27	0.669	0.669
B*	SL	6	XTAL, RTC	75	200	1270	42			-0.27	0.411	0.005
B*	DSL	3	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	52			-0.24	7.795	3.066
B*	DSL	3	Processor Sideband	75	88	88	52			-0.24	7.795	7.795
B*	DSL	3	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	52			-0.23	3.066	3.066
B*	DSL	3	XTAL, RTC	75	200	1270	52			-0.23	2.399	0.005
B*	DSL	4	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	67			-0.19	12.136	5.705
B*	DSL	4	Processor Sideband	75	88	88	67			-0.19	12.136	12.136
B*	DSL	4	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	67			-0.18	5.705	5.705
B*	DSL	4	XTAL, RTC	75	200	1270	67			-0.18	4.61	0.002
M*	MS	1, 8	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	160	125	375	40	36	44	-0.23	-2.882	-1.769
M*	MS	1, 8	CNVi BRI and RGI	160	250	250	40	36	44	-0.23	-2.494	-2.494
M*	MS	1, 8	Processor Sideband	160	125	125	40	36	44	-0.23	-2.882	-2.882
M*	MS	1, 8	Imaging Clock, SUSCLK, UFS Reference Clock	160	375	375	40	36	44	-0.23	-1.769	-1.769
M*	MS	1, 8	CLINK, I3C, SPI0 Flash	160	250	375	40	36	44	-0.23	-2.494	-1.769
M*	MS	1, 8	XTAL, RTC	160	200	1775	40	36	44	-0.23	-2.766	-0.074
M*	SL	6	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	85	125	375	40	36	44	-0.26	1.677	0.018
M*	SL	6	CNVi BRI and RGI	85	250	250	40	36	44	-0.26	0.146	0.146
M*	SL	6	Processor Sideband	85	125	125	40	36	44	-0.26	1.677	1.677
M*	SL	6	Imaging Clock, SUSCLK, UFS Reference Clock	85	375	375	40	36	44	-0.26	0.018	0.018
M*	SL	6	CLINK, I3C, SPI0 Flash	85	250	375	40	36	44	-0.26	0.146	0.018
M*	SL	6	XTAL, RTC	85	200	1775	40	36	44	-0.26	0.389	0.006
M*	DSL	3	DMIC, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	135	125	375	40	36	44	-0.22	4.326	0.423
continued...												

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K, Data - Any (%)	K, Clock - Any (%)
M*	DSL	3	CNVi BRI and RGI	135	250	250	40	36	44	-0.22	1.283	1.283
M*	DSL	3	Processor Sideband	135	125	125	40	36	44	-0.22	4.326	4.326
M*	DSL	3	Imaging Clock, SUSCLK, UFS Reference Clock	135	375	375	40	36	44	-0.22	0.423	0.423
M*	DSL	3	CLINK, I3C, SPI0 Flash, eSPI	135	250	375	40	36	44	-0.22	1.283	0.423
M*	DSL	3	XTAL, RTC	135	200	1775	40	36	44	-0.22	2.048	0.006
M*	DSL	4	DMIC, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	235	125	375	40	36	44	-0.15	5.837	0.683
M*	DSL	4	CNVi BRI and RGI	235	250	250	40	36	44	-0.15	1.979	1.979
M*	DSL	4	Processor Sideband	235	125	125	40	36	44	-0.15	5.837	5.837
M*	DSL	4	Imaging Clock, SUSCLK, UFS Reference Clock	235	375	375	40	36	44	-0.15	0.683	0.683
M*	DSL	4	CLINK, I3C, SPI0 Flash, eSPI	235	250	375	40	36	44	-0.15	1.979	0.683
M*	DSL	4	XTAL, RTC	235	200	1775	40	36	44	-0.15	3.038	0.002

Table 6. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) Tline Spec (Single Ended) Notes

Note
1) Based on Tline Spec (Single Ended) table recommendation, the CLK - Any (um) spacing requirement shall be applicable for: a) CLK to DATA signals on similar signaling group. b) CLK to DATA signals from other signaling group. c) CLK to CLK signals from other signaling group. 2) For CLK to pseudo static signals, the spacing between the CLK to others can be relaxed and used the DATA - Any (um) spacing requirement. Example of pseudo static signal: RESET, ALERT, Chip Select (CS), Request (REQ), INTERRUPT, INIT. 3) Processor Sideband IO refers to CATERR#, EPD_ON, FORCEPR#, THERMTRIP#, VDD2PWRGOOD signals.

2.2.1.3 Tline Spec (Memory)

Interface: DDR5

Notes

Table 7. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 Notes

Note
RCOMP to other nets spacing requirement is minimum 400 um.
DDR to Non-DDR nets spacing requirement is minimum 500 um.
DRAM RESET to be routed with 40 ohm impedance on Layer 3 and Layer 7 and distance to other nets (spacing requirement) is minimum 300 um.
Route RCOMP with 75 um TW (MR) on Layer 1, Layer 8. GND shield to any high speed IO with VSS trace/strip. GND reference required.
ALERT to other nets spacing requirement is minimum 400 um.

Trace Width

**Table 8. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC
Type-3,0.8mm,8L CH-0, CH-1, CH-2, CH-3 Trace Width Specs**

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	DQ	All	75		58		-0.45	
BO2	SL	3, 7	DQ	All	75		42		-0.47	
M	SL	3, 7	DQ	All	85		40		-0.458	
BI	MS	1	DQ	All	160		40		-0.398	
BO1	MS	1	DQS	All	75	88		95		-0.464
BO2	SL	3, 7	DQS	All	75	88		77		-0.469
M	SL	3, 7	DQS	All	80	88		75		-0.467
BI	MS	1	DQS	All	140	120		77		-0.407

**Table 9. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC
Type-3,0.8mm,8L CH-0, CH-3 Trace Width Specs**

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	All	75	88		95		-0.464
BO2	SL	5	CLK	All	75	88		85		-0.434
M	SL	5	CLK	All	105	88		75		-0.418
BI	MS	1	CLK	All	140	125		77		-0.407
BO1	MS	1	CA/CS	All	75		58		-0.45	
BO2	SL	5	CA/CS	All	75		50		-0.42	
M	SL	5	CA/CS	All	125		40		-0.404	
BI	MS	1	CA/CS	All	160		40		-0.398	

**Table 10. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC
Type-3,0.8mm,8L CH-1, CH-2 Trace Width Specs**

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	All	75	88		95		-0.464
BO2	SL	3, 7	CLK	All	75	88		77		-0.469
M	SL	3, 7	CLK	All	80	88		75		-0.467
BI	MS	1	CLK	All	140	125		77		-0.407
BO1	MS	1	CA/CS	All	75		58		-0.45	
BO2	SL	3, 7	CA/CS	All	75		42		-0.47	
M	SL	3, 7	CA/CS	All	85		40		-0.458	
BI	MS	1	CA/CS	All	160		40		-0.39844	

**Table 11. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC
Type-3,0.8mm,8L MISC Trace Width Specs**

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Z SE (Default) (ohm)	A SE (dB/inch @ 2GHz)
BO1	MS	1	RESET	All	75	58	
BO2	MS	1	RESET	All	160	40	
M	SL	3, 7	RESET	All	85	40	
BI	MS	1	RESET	All	160	40	
BO1	MS	1	ALERT	All	75	58	-0.42
BO2	MS	1	ALERT	All	160	40	-0.41
M	SL	3, 7	ALERT	All	85	40	-0.4
BI	MS	1	ALERT	All	160	40	-0.36

Trace Spacing

**Table 12. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC
Type-3,0.8mm,8L CH-0, CH-1, CH-2, CH-3 Trace Spacing Specs**

Trace	Routing Layers	Signal Groups	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	DQS pair to DQ of same byte (um)	DQ to CA/CS (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, DQS pair to DQ of same byte (%)
BO1	1		88	88	88	88	88	190	-1.902	2.241	-1.902	-1.91	2.246
BO2	3, 7		88	88	88	88	88	88	3.638	3.5	3.634	3.638	3.498
BO2	5	CA/CS, CLK	88	88					7.167	6.438			
M	3, 7		300	500	325	450	400	500	0.054	0.001	0.093	0.036	0.088
M	5	CA/CS, CLK	300	500		500			0.491	0.046		0.0448	
BI	1		300	500	325	400	400	500	-2.157	-0.619	-2.454	-2.451	-1.15

**Table 13. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) DDR5 SODIMM 1DPC
Type-3,0.8mm,8L CH-0, CH-1, CH-2, CH-3 Channel to Channel Trace Spacing Specs**

Trace	CA/CS to CA/CS (um)	DQ to DQ (um)
BO1	130	128
BO2	500	500
M	500	500
BI	500	500

LPDDR5x

Interface: LPDDR5x

Notes

Table 14. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) LPDDR5x Notes

Note
RCOMP to other nets spacing requirement is minimum 400 um.
DDR to Non-DDR nets spacing requirement is minimum 500 um.
DRAM RESET to be routed with 40 ohm impedance on Layer 1, Layer 3 and Layer 7 and distance to other nets (spacing requirement) is minimum 300 um.
Route RCOMP with 75 um TW (MR) on Layer 1, Layer 8. GND shield to any high speed IO with VSS trace/strip. GND reference required.

Trace Width

Table 15. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) LPDDR5x CAMM2 x128
Type-3,0.8mm,8L CH-0, CH-2, CH-4, CH-6 Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	Outer	75	88		94.3		-0.457
M	SL	3	CLK	Outer	80	88		75		-0.442
BI	MS	1	CLK	Outer	130	500		88.6		-0.391
BO1	MS	1	CLK	Outer	75	88		94.3		-0.457
M	SL	7	CLK	Outer	80	88		75		-0.442
BI	SL	1	CLK	Outer	130	500		88.6		-0.391
BO1	MS	1	CA/CS	Outer	75		57		-0.446	
BO2	SL	3	CA/CS	Outer	75		42		-0.441	
M	SL	3	CA/CS	Outer	85		40		-0.433	
BI	MS	1	CA/CS	Outer	130		44.8		-0.387	
BO1	MS	1	CA/CS	Outer	75		57		-0.446	
M	SL	7	CA/CS	Outer	85		40		-0.433	
BI	MS	1	CA/CS	Outer	130		44.8		-0.387	
BO1	MS	1	DQ	Outer	75		57		-0.439	
BO2	SL	3	DQ	Outer	88		38.8		-0.43	
M	SL	3	DQ	Outer	95		37		-0.429	
BI	MS	1	DQ	Outer	130		44.9		-0.39	
BO1	MS	1	RDQS	Outer	75	88		94		-0.462
BO2	SL	3	RDQS	Outer	80	88		75		-0.441
M	SL	3	RDQS	Outer	80	88		75		-0.44
BI1	SL	3	RDQS	Outer	80	88		75		-0.44
BI2	MS	1	RDQS	Outer	130	500		89		-0.391
BO1	MS	1	RDQS	Outer	75	88		94		-0.462
BO2	SL	5	RDQS	Outer	105	88		75		-0.383
M	SL	5	RDQS	Outer	105	88		75		-0.383
BI2	MS	1	RDQS	Outer	130	88		75		-0.383
BO1	MS	1	WCK	Outer	75	88		94		-0.462
BO2	SL	3	WCK	Outer	95	88		69.4		-0.434
M	SL	3	WCK	Outer	105	88		65		-0.43
BI1	SL	3	WCK	Outer	105	88		65		-0.431
BI2	MS	1	WCK	Outer	150	500		82.12		-0.386
BO1	MS	1	WCK	Outer	75	88		94		-0.462
M	SL	5	WCK	Outer	145	88		65		-0.43
BI2	MS	1	WCK	Outer	150	500		82.12		-0.386

**Table 16. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) LPDDR5x CAMM2 x128
Type-3,0.8mm,8L CH-1, CH-3, CH-5, CH-7 Trace Width Specs**

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	Inner	130	300		87		-0.39
BO2	SL	5	CLK	Inner	80	88		85.6		-0.396
M	SL	5	CLK	Inner	105	88		75		-0.384
BI	MS	1	CLK	Inner	130	500		88.6		-0.391
BO1	MS	1	CA/CS	Inner	130		44.6		-0.399	
BO2	SL	5	CA/CS	Inner	80		49.5		-0.387	
M	SL	5	CA/CS	Inner	125		40		-0.364	
BI1	SL	5	CA/CS	Inner	125		40		-0.359	
BI2	MS	1	CA/CS	Inner	130		44.8		-0.387	
BO1	MS	1	DQ	Inner	130		44.7		-0.389	
BO2	SL	7	DQ	Inner	75		41.4		-0.441	
BO3	SL	7	DQ	Inner	88		38.2		-0.442	
M	SL	7	DQ	Inner	95		37		-0.436	
BI	MS	1	DQ	Inner	130		44.9		-0.39	
BO1	MS	1	RDQS	Inner	130	300		86.8		-0.39
BO2	SL	7	RDQS	Inner	75	88		77		-0.449
M	SL	7	RDQS	Inner	80	88		75		-0.444
BI1	SL	7	RDQS	Inner	80	88		75		-0.445
BI2	MS	1	RDQS	Inner	130	500		88.6		-0.391
BO1	SL	1	RDQS	Inner	130	300		86.8		-0.39
BO2	SL	5	RDQS	Inner	80	88		85.7		-0.395
M	SL	5	RDQS	Inner	105	88		75		-0.383
BI2	MS	1	RDQS	Inner	130	500		88.6		-0.391
BO1	MS	1	WCK	Inner	150	300		80.5		-0.385
BO2	SL	7	WCK	Inner	75	88		77		-0.449
BO3	SL	7	WCK	Inner	95	88		68.7		-0.439
M	SL	7	WCK	Inner	105	88		65		-0.435
BI1	SL	7	WCK	Inner	105	88		65		-0.436
BI2	MS	1	WCK	Inner	150	500		82.1		-0.386

Table 17. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) LPDDR5x CAMM2 x128
Type-3,0.8mm,8L MISC Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Z SE (Default) (ohm)	A SE (dB/inch @ 2GHz)
BO1	MS	1	RESET	All	75	57.5	-0.423
BO2	MS	1	RESET	All	160	40	-0.398
M	SL	3, 7	RESET	All	85	40	-0.459
BI1	MS	1	RESET	All	160	40	-0.398

Trace Spacing

Table 18. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) LPDDR5x CAMM2 x128
Type-3,0.8mm,8L CH-0, CH-2, CH-4, CH-6 Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO1	88	88	88	100	88	300	88	88	-1.902	2.246	-1.861	-2.131	2.23	2.242	0.709
BO2	88	150	140	300	200	300	200	200	3.688	1.044	1.223	0.057	0.372	0.354	0.174
M	110	200	200	400	300	500	300	300	2.255	0.377	0.371	0.017	0.051	0.048	0.024
BI1					300		300	200					0.051	0.051	0.17
BI2					500		500	500					-0.987	-0.998	-0.435
BI	500	500	500	500		500			-1.183	-0.978	-1.128	-1.18			

Table 19. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) LPDDR5x CAMM2 x128
Type-3,0.8mm,8L CH-1, CH-3, CH-5, CH-7 Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO1	150	150	200	200	200	200	200	200	-2.776	-1.773	-2.67	-2.637	-1.796	-1.819	-0.744
BO2	88	150	88	140	88	300	88	88	7.023	6.339	3.455	1.795	3.314	3.314	1.604
BO3			180	300		300	200	200			0.493	0.048		0.31	0.152
M	235	250	200	400	300	500	300	300	4.511	1.436	0.322	0.012	0.043	0.04	0.02
BI1	160				300		300	200	3.134				0.318	0.307	0.15
BI2	500				500		500	500	-1.183				-0.987	-0.998	-0.435
BI		500	500	500		500				-0.978	-1.128	-1.18			

Table 20. Type-3, 0.8mm, 8L, Mainstream, Mid Loss (ML) LPDDR5x CAMM2 x128
Type-3,0.8mm,8L CH-0, CH-1, CH-2, CH-3, CH-4, CH-5, CH-6, CH-7 Channel to Channel Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	DQ to DQ (um)	DQS to DQS (um)	DQS to CA/CS (um)
BO1	88	100	300	300
BO2	200	300	300	300
BO3	200	500	300	300
M	500	500	500	500
BI1	300	500	500	500
BI2	500	500	500	500

2.3 Generalized 10-Layer Motherboard Examples

- The following stack-ups are supported for **NVL 8C/H16C** only.
 - Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML), For LPDDR5x Memory Down
 - Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML), For LPDDR5x Memory Down
- The Stack-up "Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML), For DDR5 SODIMM/CSODIMM and LPDDR5x Memory Down"

1. **NVL 8C/H16C**: Supported
2. **NVL 6C**: Supported only for "MD x32 Type-3,0.9mm,10L" Topology

2.3.1 Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML), For LPDDR5x Memory Down

Pcb Type: Type-3

Pcb Thickness: 0.8mm

Pcb Layer Count: 10

Category: Thin PCB

Material: Premium Mid Loss (PML)

Variant: For LPDDR5x Memory Down

Table 21. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML), For LPDDR5x Memory Down

Routing Layer	Description	Time Type	Description Memory	Time Type Memory	Thickness (um)	Dielectric Constant	Loss Tangent
	S/M	None	S/M	None	13	4	0.031
1	BO, Power	MS	BO, Power	MS	30		
	PP	None	PP	None	60	3.3	0.009
2	GND	GND/Power	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
3	BO, Main Route, Power	SL	BO, Main Route	SL	30		
	PP	None	PP	None	50	3.1	0.009
4	GND, Power	GND/Power	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
5	Main Route, Power	DSL	BO, Main Route	SL	30		
	PP	None	PP	None	50	3.1	0.009
6	Main Route, Power	DSL	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
7	GND	GND/Power	Main Route, Power	SL	30		
	PP	None	PP	None	50	3.1	0.009
8	BO, Main Route, Power	SL	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
9	GND	GND/Power	GND, Power	GND/Power	30		
	PP	None	PP	None	60	3.3	0.009
10	GND, Power, Main Route	MS	GND, Power	GND/Power	30		
	S/M	None	S/M	None	13	4	0.031

Table 22. Notes

Note
Dielectric thickness of 50um has been chosen specifically on SL layers to ensure POR memory speeds are met. Customers following thicker dielectrics will need to ensure cross talk co-efficient are equal to or less than those specified in memory Tline spec.

2.3.1.1 Tline Spec (Differential)

Table 23. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) Tline Spec (Differential)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
B*	MS	1, 10	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	75	88	200	300	95	-0.93	-0.063	0.312
B*	MS	1, 10	CSI DPHY, HDMI, eDP	75	88	250	375	95	-0.93	-0.134	0.175
B*	MS	1, 10	CNVio3	75	88	375	375	95	-0.93	-0.136	0.175
B*	MS	1, 10	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	95	-0.93	-0.136	0.082
B*	MS	1, 10	CSI CPHY	81	88	88	375	92	-0.95	0.649	0.18
B*	MS	1, 10	TCP DP, TCP TBT4, TCP TBT5	81	88	500	500	93	-0.92	-0.1	0.086
B*	MS	1, 10	PCIe Gen 5	81	88	375	500	93	-0.92	-0.146	0.086
B*	MS	1, 10	PCIe Gen 4	81	88	375	375	93	-0.92	-0.146	0.18
B*	MS	1, 10	USB 3.2 Gen 1, USB 3.2 Gen 2	81	88	300	300	93	-0.92	-0.165	0.318
B*	SL	3, 6, 8	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	76	-0.86	0.127	0.014
B*	SL	3, 6, 8	CSI CPHY, CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	76	-0.86	1.367	0.014
B*	SL	3, 6, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	76	-0.86	0.003	0
B*	SL	3, 6, 8	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	75	88	375	375	76	-0.86	0.003	0.003
B*	SL	3, 6, 8	PCIe Gen 4, PCIe Gen 5	75	88	230	500	76	-0.86	0.066	0
B*	SL	3, 6, 8	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	76	-0.86	0.014	0.014
B*	DSL	5, 6	CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	88	-0.75	2.558	0.133
B*	DSL	5, 6	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	88	-0.75	0.518	0.133
B*	DSL	5, 6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	88	-0.75	0.048	0.01
B*	DSL	5, 6	PCIe Gen 4, TCP DP, TCP TBT4, TCP TBT5	75	88	375	375	88	-0.75	0.048	0.048
B*	DSL	5, 6	PCIe Gen 5	75	88	500	500	88	-0.75	0.01	0.01
B*	DSL	5, 6	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	88	-0.75	0.133	0.133
M*	MS	1, 10	CSI DPHY, eUSB2, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	120	90	300	300	80	-0.86	-0.249	0.352
M*	MS	1, 10	Differential Clock (Gen 3 and below support)	120	90	300	375	80	-0.86	-0.249	0.212
M*	MS	1, 10	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	120	90	500	760	80	-0.86	-0.139	0.039
M*	MS	1, 10	eDP, HDMI, PCIe Gen 4, TCP DP, TCP TBT4	120	90	500	500	80	-0.86	-0.139	0.108
continued...											

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
M*	MS	1, 10	CNVio3, TCP TBT5	120	90	800	800	80	-0.86	-0.055	0.034
M*	MS	1, 10	UFS 4.0, USB 3.2 Gen 1, USB 3.2 Gen 2	120	90	750	500	80	-0.86	-0.065	0.108
M*	MS	1, 10	PCIe Gen 5, TCP TBT5 (Post-Channel)	135	125	800	800	80	-0.81	-0.071	0.044
M*	MS	1, 10	CSI CPHY	165	950	950	950	80	-0.75	-0.14	0.073
M*	MS	1, 10	HDMI (Post-Channel)	75	110	500	500	100	-0.89	-0.111	0.095
M*	SL	3, 6, 8	CSI DPHY, eDP, eUSB2, HDMI, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	75	140	300	300	80	-0.85	0.016	0.016
M*	SL	3, 6, 8	Differential Clock (Gen 3 and below support)	75	140	300	375	80	-0.85	0.016	0.003
M*	SL	3, 6, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	140	500	760	80	-0.85	0	0
M*	SL	3, 6, 8	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	75	140	375	375	80	-0.85	0.003	0.003
M*	SL	3, 6, 8	PCIe Gen 4, PCIe Gen 5	75	140	230	500	80	-0.85	0.069	0
M*	SL	3, 6, 8	USB 3.2 Gen 1, USB 3.2 Gen 2	75	140	450	500	80	-0.85	0.001	0
M*	SL	3, 6, 8	CSI CPHY	76	170	170	500	80	-0.85	0.254	0
M*	DSL	5, 6	CSI DPHY, eDP, eUSB2, HDMI, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	100	120	300	300	80	-0.7	0.129	0.129
M*	DSL	5, 6	Differential Clock (Gen 3 and below support)	100	120	300	375	80	-0.7	0.129	0.047
M*	DSL	5, 6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	100	120	500	760	80	-0.7	0.009	0
M*	DSL	5, 6	PCIe Gen 4, TCP DP	100	120	375	375	80	-0.7	0.047	0.047
M*	DSL	5, 6	PCIe Gen 5	100	120	500	500	80	-0.7	0.009	0.009
M*	DSL	5, 6	TCP TBT4, TCP TBT5, USB 3.2 Gen 1, USB 3.2 Gen 2	100	120	450	500	80	-0.7	0.018	0.009

Table 24. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) Tline Spec (Differential) Notes

Note
Routing Notes for CSI CPHY: (1) CPHY must be routed as a trio (2) If the port is supposed to support both CPHY and DPHY, it must be routed following CPHY guidelines (3) CPHY impedance target is 40 ohms single ended and 80 ohms differential (4) All traces in trio must follow same trace width (5) Intra-Pair guidelines must be used for trace to trace spacing in a trio (6) S-ES guidelines must be used for trio to trio in the same port (7) S-Non-ES must be used for port to port or CPHY to other interface (8) CSI CPHY in Microstrip (MS) routing is not recommended for long length, allowed only for short length (<5 mm) near the connector.
For CNVio3 signal spacing: CNVio3 to non-CNVio3 pair spacing of 10H (H=Dielectric Thickness) is required. If this cannot be achieved, then GND shielding with proper stitching vias and 3H minimum spacing from GND shielding to CNVio3 is required. Please also refer to "CNVio3 Spacing Recommendation" diagram in CNVio3 section.

2.3.1.2 Tline Spec (Single Ended)

Table 25. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) Tline Spec (Single Ended)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K, Data - Any (%)	K, Clock - Any (%)
B*	MS	1, 10	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	59			-0.25	-1.181	-2.1
B*	MS	1, 10	Processor Sideband	75	88	88	59			-0.25	-1.181	-1.181
B*	MS	1, 10	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	59			-0.23	-2.1	-2.1
B*	MS	1, 10	XTAL, RTC	75	200	1270	59			-0.23	-2.125	-0.133
B*	SL	3, 8	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	41			-0.26	2.894	0.469
B*	SL	3, 8	Processor Sideband	75	88	88	41			-0.26	2.894	2.894
B*	SL	3, 8	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	41			-0.26	0.469	0.469
B*	SL	6, 8	XTAL, RTC	75	200	1270	41			-0.26	0.276	0.007
B*	DSL	5, 6	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	51			-0.22	6.008	1.758
B*	DSL	5, 6	Processor Sideband	75	88	88	51			-0.22	6.008	6.008
B*	DSL	5, 6	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	51			-0.22	1.758	1.758
B*	DSL	5, 6	XTAL, RTC	75	200	1270	51			-0.22	1.246	0.005
M*	MS	1, 10	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	165	125	375	40	36	44	-0.21	-2.527	-1.667
M*	MS	1, 10	CNVi BRI and RGI	165	250	250	40	36	44	-0.21	-2.308	-2.308
M*	MS	1, 10	Processor Sideband	165	125	125	40	36	44	-0.21	-2.527	-2.527
M*	MS	1, 10	Imaging Clock, SUSCLK, UFS Reference Clock	165	375	375	40	36	44	-0.21	-1.667	-1.667
M*	MS	1, 10	CLINK, I3C, SPI0 Flash	165	250	375	40	36	44	-0.21	-2.308	-1.667
M*	MS	1, 10	XTAL, RTC	165	200	1775	40	36	44	-0.21	-2.524	-0.073
M*	SL	3, 8	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	80	125	375	40	36	44	-0.26	1.255	0.014
M*	SL	3, 8	CNVi BRI and RGI	80	250	250	40	36	44	-0.26	0.093	0.093
M*	SL	3, 8	Processor Sideband	80	125	125	40	36	44	-0.26	1.255	1.255
M*	SL	3, 8	Imaging Clock, SUSCLK, UFS Reference Clock	80	375	375	40	36	44	-0.26	0.014	0.014
M*	SL	3, 8	CLINK, I3C, SPI0 Flash	80	250	375	40	36	44	-0.26	0.093	0.014
M*	SL	6, 8	XTAL, RTC	80	200	1775	40	36	44	-0.26	0.269	0.007
M*	DSL	5, 6	DMIC, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	120	125	375	40	36	44	-0.21	2.916	0.1
M*	DSL	5, 6	CNVi BRI and RGI	120	250	250	40	36	44	-0.21	0.523	0.523
M*	DSL	5, 6	Processor Sideband	120	125	125	40	36	44	-0.21	2.916	2.916

continued...

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K, Data - Any (%)	K, Clock - Any (%)
M*	DSL	5, 6	Imaging Clock, SUSCLK, UFS Reference Clock	120	375	375	40	36	44	-0.21	0.1	0.1
M*	DSL	5, 6	CLINK, I3C, SPI0 Flash, eSPI	120	250	375	40	36	44	-0.21	0.523	0.1
M*	DSL	5, 6	XTAL, RTC	120	200	1775	40	36	44	-0.21	1.034	0.006

Table 26. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) Tline Spec (Single Ended) Notes

Note
1) Based on Tline Spec (Single Ended) table recommendation, the CLK - Any (um) spacing requirement shall be applicable for: a) CLK to DATA signals on similar signaling group. b) CLK to DATA signals from other signaling group. c) CLK to CLK signals from other signaling group. 2) For CLK to pseudo static signals, the spacing between the CLK to others can be relaxed and used the DATA - Any (um) spacing requirement. Example of pseudo static signal: RESET, ALERT, Chip Select (CS), Request (REQ), INTERRUPT, INIT. 3) Processor Sideband IO refers to CATERR#, EPD_ON, FORCEPR#, THERMTRIP#, VDD2PWRGOOD signals.

2.3.1.3 Tline Spec (Memory)

Interface: LPDDR5x

Notes

Table 27. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) LPDDR5x Notes

Note
RCOMP to other nets spacing requirement is minimum 400 um.
DDR to Non-DDR nets spacing requirement is minimum 500 um.
DRAM RESET to be routed with 40 ohm impedance on Layer 1, Layer 3 and Layer 5 and distance to other nets (spacing requirement) is minimum 250 um.
Route RCOMP with 75 um TW (MR) on Layer 1, Layer 10. GND shield to any high speed IO with VSS trace/strip. GND reference required.

Trace Width

**Table 28. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.8mm,10L CH-0, CH-2, CH-4, CH-6 Trace Width Specs**

Trace	Trace Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 26Hz)	A Diff (dB/inch @ 26Hz)
BO1	MS	1	CLK	Outer	75	88		95		-0.438
BO2	SL	7	CLK	Outer	78	88		75		-0.432
M	SL	7	CLK	Outer	78	88		75		-0.432
BI1	SL	7	CLK	Outer	78	88		75		-0.432
BI2	MS	1	CLK	Outer	130	400		89		-0.368
BO1	MS	1	CA/CS	Outer	75		57		-0.426	
BO2	SL	7	CA/CS	Outer	85		37		-0.427	
M	SL	7	CA/CS	Outer	85		37		-0.427	
BI1	SL	7	CA/CS	Outer	85		37		-0.427	
BI2	MS	1	CA/CS	Outer	130		45		-0.375	
BO1	MS	1	DQ	Outer	75		57		-0.426	
BO2	SL	3	DQ	Outer	88		37		-0.426	
M	SL	3	DQ	Outer	105		33		-0.423	
BI1	SL	3	DQ	Outer	105		33		-0.423	
BI2	MS	1	DQ	Outer	130		45		-0.375	
BO1	MS	1	RDQS	Outer	75	88		95		-0.444
BO2	SL	3	RDQS	Outer	78	88		75		-0.432
M	SL	3	RDQS	Outer	78	88		75		-0.432
BI1	SL	3	RDQS	Outer	78	88		75		-0.432
BI2	MS	1	RDQS	Outer	130	400		89		-0.368
BO1	MS	1	WCK	Outer	75	88		95		-0.444
BO2	SL	3	WCK	Outer	95	88		67		-0.426
M	SL	3	WCK	Outer	100	88		65		-0.425
BI1	SL	3	WCK	Outer	100	88		65		-0.425
BI2	MS	1	WCK	Outer	150	400		83		-0.363

**Table 29. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.8mm,10L CH-1, CH-3, CH-5, CH-7 Trace Width Specs**

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	Inner	130	300		88		-0.368
BO2	SL	7	CLK	Inner	78	88		75		-0.432
BO3	SL	7	CLK	Inner	78	88		75		-0.432
M	SL	7	CLK	Inner	78	88		75		-0.432
BI1	SL	7	CLK	Inner	78	88		75		-0.432
BI2	MS	1	CLK	Inner	130	400		89		-0.368
BO1	MS	1	CA/CS	Inner	130		45		-0.38	
BO2	SL	7	CA/CS	Inner	75		40		-0.433	
BO3	SL	7	CA/CS	Inner	85		37		-0.428	
M	SL	7	CA/CS	Inner	85		37		-0.427	
BI1	SL	7	CA/CS	Inner	85		37		-0.427	
BI2	MS	1	CA/CS	Inner	130		45		-0.375	
BO1	MS	1	DQ	Inner	130		45		-0.375	
BO2	SL	5	DQ	Inner	75		39		-0.433	
BO3	SL	5	DQ	Inner	88		37		-0.426	
M	SL	5	DQ	Inner	105		33		-0.423	
BI1	SL	5	DQ	Inner	105		33		-0.423	
BI2	MS	1	DQ	Inner	130		45		-0.375	
BO1	MS	1	RDQS	Inner	130	300		88		-0.368
BO2	SL	5	RDQS	Inner	78	88		75		-0.432
BO3	SL	5	RDQS	Inner	78	88		75		-0.432
M	SL	5	RDQS	Inner	78	88		75		-0.432
BI1	SL	5	RDQS	Inner	78	88		75		-0.432
BI2	MS	1	RDQS	Inner	130	400		89		-0.368
BO1	MS	1	WCK	Inner	150	300		82		-0.363
BO2	SL	5	WCK	Inner	75	88		76		-0.433
BO3	SL	5	WCK	Inner	95	88		67		-0.426
M	SL	5	WCK	Inner	100	88		65		-0.425
BI1	SL	5	WCK	Inner	100	88		65		-0.425
BI2	MS	1	WCK	Inner	150	400		83		-0.363

Table 30. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.8mm,10L MISC Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Z SE (Default) (ohm)	A SE (dB/inch @ 2GHz)
BO1	MS	1	RESET	All	75	58	-0.389
BO2	MS	1	RESET	All	165	40	-0.365
M	SL	3, 5	RESET	All	85	40	-0.427
BI1	MS	1	RESET	All	165	40	-0.365

Trace Spacing

Table 31. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.8mm,10L CH-0, CH-2, CH-4, CH-6 Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO1	88	88	88	100	88	200	88	88	-1.471	2.589	-1.483	-1.601	2.581	2.586	0.834
BO2	110	120	140	250	200	300	200	200	1.684	1.341	0.862	0.169	0.235	0.223	0.113
M	150	200	180	400	300	500	300	300	0.706	0.238	0.329	0.01	0.026	0.024	0.013
BI1	150	200	180	400	300	500	300	300	0.706	0.238	0.329	0.01	0.026	0.024	0.013
BI2	400	400	400	400	400	400	400	400	-1.506	-1.181	-1.505	-1.479	-1.18	-1.205	-0.498

**Table 32. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.8mm,10L CH-1, CH-3, CH-5, CH-7 Trace Spacing Specs**

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO1	150	150	200	200	200	200	200	200	-2.457	-1.473	-2.418	-2.416	-1.609	-1.671	-0.645
BO2	88	100	88	150	88	300	88	88	2.901	2.141	2.898	2.087	2.788	2.809	1.353
BO3	100	120	170	300	200	300	200	200	2.093	1.34	0.45	0.035	0.236	0.224	0.113
M	150	200	180	400	300	500	300	300	0.706	0.238	0.329	0.01	0.026	0.024	0.013
BI1	150	200	180	400	300	500	300	300	0.706	0.238	0.329	0.245	0.026	0.024	0.013
BI2	400	400	400	400	400	400	400	400	-1.506	-1.181	-1.505	-1.479	-1.18	-1.205	-0.498

**Table 33. Type-3, 0.8mm, 10L, Thin PCB, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.8mm,10L CH-0, CH-1, CH-2, CH-3, CH-4, CH-5, CH-6, CH-7 Channel
to Channel Trace Spacing Specs**

Trace	CA/CS to CA/CS (um)	DQ to DQ (um)	DQS to DQS (um)	DQS to CA/CS (um)
BO1	88	100	500	500
BO2	200	250	500	500
BO3	220	300	500	500
M	500	495	500	500
BI1	300	500	500	500
BI2	500	500	500	500

2.3.2 Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML), For DDR5 SODIMM/CSODIMM and LPDDR5x Memory Down

Pcb Type: Type-3

Pcb Thickness: 0.9mm

Pcb Layer Count: 10

Category: Mainstream

Material: Premium Mid Loss (PML)

Variant: For DDR5 SODIMM/CSODIMM and LPDDR5x Memory Down

Table 34. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML), For DDR5 SODIMM/CSODIMM and LPDDR5x Memory Down

Routing Layer	Description	Tline Type	Description Memory	Tline Type Memory	Thickness (um)	Dielectric Constant	Loss Tangent
	S/M	None	S/M	None	13	4	0.031
1	BO, Power	MS	BO, Power	MS	30		
	PP	None	PP	None	60	3.3	0.009
2	GND	GND/Power	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
3	BO, Main Route, Power	SL	BO, Main Route	SL	30		
	PP	None	PP	None	64	3.3	0.009
4	GND, Power	GND/Power	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
5	Main Route, Power	DSL	BO, Main Route	SL	30		
	PP	None	PP	None	64	3.3	0.009
6	Main Route, Power	DSL	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
7	GND	GND/Power	Main Route, Power	SL	30		
	PP	None	PP	None	64	3.3	0.009
8	BO, Main Route, Power	SL	GND	GND/Power	30		
	CORE	None	CORE	None	64	3.7	0.009
9	GND	GND/Power	GND, Power	GND/Power	30		
	PP	None	PP	None	60	3.3	0.009
10	GND, Power, Main Route	MS	GND, Power	GND/Power	30		
	S/M	None	S/M	None	13	4	0.031

Table 35. Notes

Note
Dielectric thickness of 64um has been chosen specifically on SL layers to ensure POR memory speeds are met. Customers following thicker dielectrics will need to ensure cross talk co-efficient are equal to or less than those specified in memory Tline spec.

2.3.2.1 Tline Spec (Differential)

Table 36. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Differential)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
B*	MS	1, 10	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	75	88	200	300	95	-0.93	-0.063	0.312
B*	MS	1, 10	CSI DPHY, HDMI, eDP	75	88	250	375	95	-0.93	-0.134	0.175
B*	MS	1, 10	CNVio3	75	88	375	375	95	-0.93	-0.136	0.175
B*	MS	1, 10	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	95	-0.93	-0.136	0.082
B*	MS	1, 10	CSI CPHY	81	88	88	375	92	-0.95	0.649	0.18
B*	MS	1, 10	TCP DP, TCP TBT4, TCP TBT5	81	88	500	500	93	-0.92	-0.1	0.086
B*	MS	1, 10	PCIe Gen 5	81	88	375	500	93	-0.92	-0.146	0.086
B*	MS	1, 10	PCIe Gen 4	81	88	375	375	93	-0.92	-0.146	0.18
B*	MS	1, 10	USB 3.2 Gen 1, USB 3.2 Gen 2	81	88	300	300	93	-0.92	-0.165	0.318
B*	SL	3, 6, 8	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	78	-0.83	0.189	0.027
B*	SL	3, 6, 8	CSI CPHY, CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	78	-0.83	1.725	0.027
B*	SL	3, 6, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	78	-0.83	0.006	0.001
B*	SL	3, 6, 8	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	75	88	375	375	78	-0.83	0.006	0.006
B*	SL	3, 6, 8	PCIe Gen 4, PCIe Gen 5	75	88	230	500	78	-0.83	0.105	0.001
B*	SL	3, 6, 8	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	78	-0.83	0.027	0.027
B*	DSL	5, 6	CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	87	-0.76	2.67	0.161
B*	DSL	5, 6	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	87	-0.75	0.583	0.161
B*	DSL	5, 6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	87	-0.75	0.062	0.014
B*	DSL	5, 6	PCIe Gen 4, TCP DP, TCP TBT4, TCP TBT5	75	88	375	375	87	-0.75	0.062	0.062
B*	DSL	5, 6	PCIe Gen 5	75	88	500	500	87	-0.75	0.014	0.014
B*	DSL	5, 6	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	87	-0.75	0.161	0.161
M*	MS	1, 10	CSI DPHY, eUSB2, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	120	90	300	300	80	-0.86	-0.249	0.352
M*	MS	1, 10	Differential Clock (Gen 3 and below support)	120	90	300	375	80	-0.86	-0.249	0.212
M*	MS	1, 10	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	120	90	500	760	80	-0.86	-0.139	0.039
M*	MS	1, 10	eDP, HDMI, PCIe Gen 4, TCP DP, TCP TBT4	120	90	500	500	80	-0.86	-0.139	0.108
M*	MS	1, 10	CNVio3, TCP TBT5	120	90	800	800	80	-0.86	-0.055	0.034
M*	MS	1, 10	UFS 4.0, USB 3.2 Gen 1, USB 3.2 Gen 2	120	90	750	500	80	-0.86	-0.065	0.108
M*	MS	1, 10	PCIe Gen 5, TCP TBT5 (Post-Channel)	135	125	800	800	80	-0.81	-0.071	0.044
M*	MS	1, 10	CSI CPHY	165	950	950	950	80	-0.75	-0.14	0.073
M*	MS	1, 10	HDMI (Post-Channel)	75	110	500	500	100	-0.895	-0.111	0.095

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Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
M*	SL	3, 6, 8	CSI DPHY, eDP, eUSB2, HDMI, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	80	130	300	300	80	-0.8	0.026	0.026
M*	SL	3, 6, 8	Differential Clock (Gen 3 and below support)	80	130	300	375	80	-0.8	0.026	0.006
M*	SL	3, 6, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	80	130	500	760	80	-0.8	0.001	0
M*	SL	3, 6, 8	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	80	130	375	375	80	-0.8	0.006	0.006
M*	SL	3, 6, 8	PCIe Gen 4, PCIe Gen 5	80	130	230	500	80	-0.8	0.107	0.001
M*	SL	3, 6, 8	USB 3.2 Gen 1, USB 3.2 Gen 2	80	130	450	500	80	-0.8	0.001	0.001
M*	SL	3, 6, 8	CSI CPHY	82	170	170	500	80	-0.8	0.351	0.001
M*	DSL	5, 6	CSI DPHY, eDP, eUSB2, HDMI, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	100	120	300	300	80	-0.71	0.162	0.162
M*	DSL	5, 6	Differential Clock (Gen 3 and below support)	100	120	300	375	80	-0.71	0.162	0.062
M*	DSL	5, 6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	100	120	500	760	80	-0.71	0.014	0.001
M*	DSL	5, 6	PCIe Gen 4, TCP DP	100	120	375	375	80	-0.71	0.062	0.062
M*	DSL	5, 6	PCIe Gen 5	100	120	500	500	80	-0.71	0.014	0.014
M*	DSL	5, 6	TCP TBT4, TCP TBT5, USB 3.2 Gen 1, USB 3.2 Gen 2	100	120	450	500	80	-0.71	0.024	0.014

Table 37. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Differential) Notes

Note
Routing Notes for CSI CPHY: (1) CPHY must be routed as a trio (2) If the port is supposed to support both CPHY and DPHY, it must be routed following CPHY guidelines (3) CPHY impedance target is 40 ohms single ended and 80 ohms differential (4) All traces in trio must follow same trace width (5) Intra-Pair guidelines must be used for trace to trace spacing in a trio (6) S-ES guidelines must be used for trio to trio in the same port (7) S-Non-ES must be used for port to port or CPHY to other interface (8) CSI CPHY in Microstrip (MS) routing is not recommended for long length, allowed only for short length (<5 mm) near the connector.
For CNVio3 signal spacing: CNVio3 to non-CNVio3 pair spacing of 10H (H=Dielectric Thickness) is required. If this cannot be achieved, then GND shielding with proper stitching vias and 3H minimum spacing from GND shielding to CNVio3 is required. Please also refer to "CNVio3 Spacing Recommendation" diagram in CNVio3 section.

2.3.2.2 Tline Spec (Single Ended)

Table 38. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Single Ended)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K _r Data - Any (%)	K _r Clock - Any (%)
B*	MS	1, 10	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	59			-0.25	-1.181	-2.1
B*	MS	1, 10	Processor Sideband	75	88	88	59			-0.25	-1.181	-1.181
B*	MS	1, 10	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	59			-0.23	-2.1	-2.1
B*	MS	1, 10	XTAL, RTC	75	200	1270	59			-0.23	-2.125	-0.133
B*	SL	3, 8	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	43			-0.25	3.717	0.677
B*	SL	3, 8	Processor Sideband	75	88	88	43			-0.25	3.717	3.717
B*	SL	3, 8	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	43			-0.25	0.677	0.677
B*	SL	6, 8	XTAL, RTC	75	200	1270	43			-0.25	0.416	0.005
B*	DSL	5, 6	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	88	175	51			-0.23	6.396	1.99
B*	DSL	5, 6	Processor Sideband	75	88	88	51			-0.23	6.396	6.396
B*	DSL	5, 6	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	51			-0.22	1.99	1.99
B*	DSL	5, 6	XTAL, RTC	75	200	1270	51			-0.22	1.437	0.005
M*	MS	1, 10	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	165	125	375	40	36	44	-0.21	-2.527	-1.667
M*	MS	1, 10	CNVi BRI and RGI	165	250	250	40	36	44	-0.21	-2.308	-2.308
M*	MS	1, 10	Processor Sideband	165	125	125	40	36	44	-0.21	-2.527	-2.527
M*	MS	1, 10	Imaging Clock, SUSCLK, UFS Reference Clock	165	375	375	40	36	44	-0.21	-1.667	-1.667
M*	MS	1, 10	CLINK, I3C, SPI0 Flash	165	250	375	40	36	44	-0.21	-2.308	-1.667
M*	MS	1, 10	XTAL, RTC	165	200	1775	40	36	44	-0.21	-2.524	-0.073
M*	SL	3, 8	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	85	125	375	40	36	44	-0.25	1.693	0.018
M*	SL	3, 8	CNVi BRI and RGI	85	250	250	40	36	44	-0.25	0.148	0.148
M*	SL	3, 8	Processor Sideband	85	125	125	40	36	44	-0.25	1.693	1.693
M*	SL	3, 8	Imaging Clock, SUSCLK, UFS Reference Clock	85	375	375	40	36	44	-0.25	0.018	0.018
M*	SL	3, 8	CLINK, I3C, SPI0 Flash	85	250	375	40	36	44	-0.25	0.148	0.018
M*	SL	6, 8	XTAL, RTC	85	200	1775	40	36	44	-0.25	0.394	0.006
M*	DSL	5, 6	DMIC, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	120	125	375	40	36	44	-0.21	3.213	0.134
M*	DSL	5, 6	CNVi BRI and RGI	120	250	250	40	36	44	-0.21	0.635	0.635
M*	DSL	5, 6	Processor Sideband	120	125	125	40	36	44	-0.21	3.213	3.213

continued...

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K, Data - Any (%)	K, Clock - Any (%)
M*	DSL	5, 6	Imaging Clock, SUSCLK, UFS Reference Clock	120	375	375	40	36	44	-0.21	0.134	0.134
M*	DSL	5, 6	CLINK, I3C, SPI0 Flash, eSPI	120	250	375	40	36	44	-0.21	0.635	0.134
M*	DSL	5, 6	XTAL, RTC	120	200	1775	40	36	44	-0.21	1.205	0.006

Table 39. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Single Ended) Notes

Note
1) Based on Tline Spec (Single Ended) table recommendation, the CLK - Any (um) spacing requirement shall be applicable for: a) CLK to DATA signals on similar signaling group. b) CLK to DATA signals from other signaling group. c) CLK to CLK signals from other signaling group. 2) For CLK to pseudo static signals, the spacing between the CLK to others can be relaxed and used the DATA - Any (um) spacing requirement. Example of pseudo static signal: RESET, ALERT, Chip Select (CS), Request (REQ), INTERRUPT, INIT. 3) Processor Sideband IO refers to CATERR#, EPD_ON, FORCEPR#, THERMTRIP#, VDD2PWRGOOD signals.

2.3.2.3 Tline Spec (Memory)

Interface: DDR5

Notes

Table 40. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) DDR5 Notes

Note
RCOMP to other nets spacing requirement is minimum 400 um.
DDR to Non-DDR nets spacing requirement is minimum 500 um.
DRAM RESET to be routed with 40 ohm impedance on Layer 1, Layer 3, Layer 5, Layer 7, Layer 10 and distance to other nets (spacing requirement) is minimum 300 um.
Route RCOMP with 75 um TW (MR) on Layer 1/10. GND shield to any high speed IO with VSS trace/ strip. GND reference required.
ALERT to other nets spacing requirement is minimum 400 um.

Trace Width

Table 41. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) DDR5 SODIMM 1DPC Type-3,0.9mm,10L CH-0, CH-1, CH-2, CH-3 Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	All	75	88		95		-0.44
BO2	SL	3, 5, 7	CLK	All	75	88		78		-0.41
BO3	SL	3, 5	CLK	All	80	88		75		-0.41
M	SL	3, 5, 7	CLK	All	80	88		75		-0.41
BI	MS	1	CLK	All	140	120		78		-0.38
BO1	MS	1	CA/CS	All	75		58		-0.42	
BO2	SL	3, 5, 7	CA/CS	All	75		42		-0.41	
M	SL	3, 5, 7	CA/CS	All	85		40		-0.4	
BI	MS	1	CA/CS	All	160		40		-0.36	
BO1	MS	1	DQ	All	75		58		-0.42	
BO2	SL	3, 5	DQ	All	75		42		-0.41	
M	SL	3, 5	DQ	All	85		40		-0.4	
BI	MS	1	DQ	All	160		40		-0.36	
BO1	MS	1	DQS	All	75	88		95		-0.44
BO2	SL	3, 5	DQS	All	75	88		78		-0.41
M	SL	3, 5	DQS	All	80	88		75		-0.41
BI	MS	1	DQS	All	140	120		78		-0.38

Table 42. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) DDR5 SODIMM 1DPC Type-3,0.9mm,10L MISC Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Z SE (Default) (ohm)	A SE (dB/inch @ 2GHz)
BO1	MS	1	RESET	All	75	58	
BO2	MS	1	RESET	All	160	40	
M	SL	3, 5, 7	RESET	All	85	40	
BI	MS	1	RESET	All	160	40	
BO1	MS	1	ALERT	All	75	58	-0.42
BO2	MS	1	ALERT	All	160	40	-0.41
M	SL	3, 5, 7	ALERT	All	85	40	-0.4
BI	MS	1	ALERT	All	160	40	-0.36

Trace Spacing

Table 43. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) DDR5 SODIMM 1DPC Type-3,0.9mm,10L CH-0, CH-1, CH-2, CH-3 Trace Spacing Specs

Trace	Routing Layers	Signal Groups	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	DQS pair to DQ of same byte (um)	DQ to CA/CS (um)	Reset to Anything (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, DQS pair to DQ of same byte (%)
BO1	1		88	88	88	88	88	190	100	-1.46	2.61	-1.46	-1.47	2.61
BO2	3, 5		88	88	88	88	88	88	100	3.72	3.59	3.72	3.73	3.59
BO2	7	CA, CLK, CS	88	88					100	3.72	3.59			
M	3, 5		300	500	325	450	400	450	150	0.06	0.01	0.04	0.01	0.01
M	7	CA, CLK, CS	300	500					150	0.06	0.01			
BI	1		300	500	325	400	400	500	150	-2.02	-0.57	-1.89	-1.52	-0.76

Table 44. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) DDR5 SODIMM 1DPC Type-3,0.9mm,10L CH-0, CH-1, CH-2, CH-3 Channel to Channel Trace Spacing Specs

Trace	Routing Layers	CA/CS to CA/CS (um)	DQ to DQ (um)	DQ to CA/CS (um)
BO1	1	185	174	185
BO2	3, 5	500	500	185
BO2	7	500	500	500
M	3, 5	500	500	500
M	7	500		
BI	1	500		

LPDDR5x

Interface: LPDDR5x

Notes

Table 45. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x Notes

Note
RCOMP to other nets spacing requirement is minimum 400 um.
DDR to Non-DDR nets spacing requirement is minimum 500 um.
DRAM RESET to be routed with 40 ohm impedance on Layer 1, Layer 3 and Layer 5 and distance to other nets (spacing requirement) is minimum 250 um.
Route RCOMP with 75 um TW (MR) on Layer 1, Layer 10. GND shield to any high speed IO with VSS trace/strip. GND reference required.

Trace Width

Table 46. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-3,0.9mm,10L CH-0, CH-2, CH-4, CH-6 Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	Outer	75	88		95		-0.438
BO2	SL	7	CLK	Outer	80	88		75		-0.41
M	SL	7	CLK	Outer	80	88		75		-0.41
BI1	SL	7	CLK	Outer	80	88		75		-0.41
BI2	MS	1	CLK	Outer	130	400		89		-0.368
BO1	MS	1	CA/CS	Outer	75		57		-0.426	
BO2	SL	7	CA/CS	Outer	85		40		-0.404	
M	SL	7	CA/CS	Outer	85		40		-0.402	
BI1	SL	7	CA/CS	Outer	85		40		-0.403	
BI2	MS	1	CA/CS	Outer	130		45		-0.375	
BO1	MS	1	DQ	Outer	75		57		-0.426	
BO2	SL	3	DQ	Outer	88		39		-0.404	
M	SL	3	DQ	Outer	95		37		-0.4	
BI1	SL	3	DQ	Outer	95		37		-0.4	
BI2	MS	1	DQ	Outer	130		45		-0.375	
BO1	MS	1	RDQS	Outer	75	88		95		-0.444
BO2	SL	3	RDQS	Outer	80	88		75		-0.411
M	SL	3	RDQS	Outer	80	88		75		-0.41
BI1	SL	3	RDQS	Outer	80	88		75		-0.41
BI2	MS	1	RDQS	Outer	130	400		89		-0.368
BO1	MS	1	WCK	Outer	75	88		95		-0.444
BO2	SL	3	WCK	Outer	95	88		70		-0.404
continued...										

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
M	SL	3	WCK	Outer	105	88		65		-0.402
BI1	SL	3	WCK	Outer	105	88		65		-0.402
BI2	MS	1	WCK	Outer	150	400		83		-0.363

Table 47. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-3,0.9mm,10L CH-1, CH-3, CH-5, CH-7 Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO1	MS	1	CLK	Inner	130	300		88		-0.368
BO2	SL	7	CLK	Inner	75	88		79		-0.413
BO3	SL	7	CLK	Inner	80	88		75		-0.41
M	SL	7	CLK	Inner	80	88		75		-0.41
BI1	SL	7	CLK	Inner	80	88		75		-0.41
BI2	MS	1	CLK	Inner	130	400		89		-0.368
BO1	MS	1	CA/CS	Inner	130		45		-0.38	
BO2	SL	7	CA/CS	Inner	75		42		-0.411	
BO3	SL	7	CA/CS	Inner	85		40		-0.404	
M	SL	7	CA/CS	Inner	85		40		-0.402	
BI1	SL	7	CA/CS	Inner	85		40		-0.403	
BI2	MS	1	CA/CS	Inner	130		45		-0.375	
BO1	MS	1	DQ	Inner	130		45		-0.375	
BO2	SL	5	DQ	Inner	75		42		-0.411	
BO3	SL	5	DQ	Inner	88		39		-0.401	
M	SL	5	DQ	Inner	95		37		-0.4	
BI1	SL	5	DQ	Inner	95		37		-0.4	
BI2	MS	1	DQ	Inner	130		45		-0.375	
BO1	MS	1	RDQS	Inner	130	300		88		-0.368
BO2	SL	5	RDQS	Inner	75	88		79		-0.414
BO3	SL	5	RDQS	Inner	80	88		75		-0.411
M	SL	5	RDQS	Inner	80	88		75		-0.41
BI1	SL	5	RDQS	Inner	80	88		75		-0.41
BI2	MS	1	RDQS	Inner	130	400		89		-0.368
BO1	MS	1	WCK	Inner	150	300		82		-0.363
continued...										

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO2	SL	5	WCK	Inner	75	88		79		-0.414
BO3	SL	5	WCK	Inner	95	88		70		-0.404
M	SL	5	WCK	Inner	105	88		65		-0.402
BI1	SL	5	WCK	Inner	105	88		65		-0.402
BI2	MS	1	WCK	Inner	150	400		83		-0.363

**Table 48. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.9mm,10L MISC Trace Width Specs**

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Z SE (Default) (ohm)	A SE (dB/inch @ 2GHz)
BO1	MS	1	RESET	All	75	58	-0.389
BO2	MS	1	RESET	All	165	40	-0.365
M	SL	3, 5	RESET	All	85	40	-0.427
BI1	MS	1	RESET	All	165	40	-0.365

Trace Spacing

**Table 49. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32
Type-3,0.9mm,10L CH-0, CH-2, CH-4, CH-6 Trace Spacing Specs**

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO1	88	88	88	100	88	200	88	88	-1.471	2.589	-1.483	-1.708	2.581	2.586	0.834
BO2	100	115	100	300	200	300	200	200	2.679	1.942	2.633	0.058	0.359	0.343	0.17
M	150	200	180	400	300	500	300	300	0.995	0.363	0.519	0.013	0.049	0.046	0.023
BI1	125	150	180	400	300	500	300	300	1.634	0.97	0.519	0.012	0.049	0.046	0.023
BI2	400	400	400	400	400	400	400	400	-1.506	-1.181	-1.505	-1.497	-1.18	-1.205	-0.498

Table 50. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-3,0.9mm,10L CH-1, CH-3, CH-5, CH-7 Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO1	150	150	200	200	200	200	200	200	-2.457	-1.473	-2.418	-2.387	-1.609	-1.671	-0.645
BO2	88	88	88	105	88	300	88	88	3.611	3.472	3.612	2.651	3.472	3.468	1.676
BO3	100	120	170	300	200	300	200	200	2.682	1.755	0.658	0.058	0.359	0.346	0.17
M	150	200	180	400	300	500	300	300	0.995	0.363	0.519	0.013	0.049	0.046	0.023
BI1	125	150	180	400	300	500	300	300	1.634	0.97	0.519	0.373	0.049	0.046	0.023
BI2	400	400	400	400	400	400	400	400	-1.506	-1.181	-1.505	-1.497	-1.18	-1.205	-0.498

Table 51. Type-3, 0.9mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-3,0.9mm,10L CH-0, CH-1, CH-2, CH-3, CH-4, CH-5, CH-6, CH-7 Channel to Channel Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	DQ to DQ (um)	DQS to DQS (um)	DQS to CA/CS (um)
BO1	88	100	500	500
BO2	220	300	500	500
BO3	220	300	500	500
M	500	500	500	500
BI1	300	500	500	500
BI2	500	500	500	500

2.3.3 Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML), For LPDDR5x Memory Down

Pcb Type: Type-4 2-x-2+

Pcb Thickness: 0.8mm

Pcb Layer Count: 10

Category: Mainstream

Material: Premium Mid Loss (PML)

Variant: For LPDDR5x Memory Down

Table 52. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML), For LPDDR5x Memory Down

Routing Layer	Description	Tline Type	Description Memory	Tline Type Memory	Thickness (um)	Dielectric Constant	Loss Tangent
	S/M	None	S/M	None	13	4	0.031
1	Power, Main Route	MS	Power	GND/Power	30		
	PP	None	PP	None	55	3.4	0.009
2	GND	GND/Power	BO, Main Route, Power	SL	20		
	PP	None	PP	None	60	3.4	0.009
3	BO, Main Route, Power	SL	GND	GND/Power	25		
	PP	None	PP	None	55	3.4	0.009
4	GND, Power	GND/Power	BO, Main Route	SL	15		
	CORE	None	CORE	None	64	3.7	0.008
5	GND	GND/Power	GND	GND/Power	30		
	PP	None	PP	None	75	3.7	0.009
6	BO, Main Route, Power	SL	BO, Main Route, Power	SL	30		
	CORE	None	CORE	None	64	3.7	0.008
7	GND	GND/Power	GND	GND/Power	15		
	PP	None	PP	None	55	3.4	0.009
8	BO, Main Route, Power	SL	BO, Main Route, Power	SL	25		
	PP	None	PP	None	60	3.4	0.009
9	GND	GND/Power	GND	GND/Power	20		
	PP	None	PP	None	55	3.4	0.009
10	Power, Main Route	MS	Power	GND/Power	30		
	S/M	None	S/M	None	13	4	0.031

Table 53. Notes

Note
Thinner dielectric thickness has been chosen specifically on SL layers to ensure POR memory speeds can be met for DQ/DQS/WCK signals. Customers following thicker dielectrics from POR stack-up will need to ensure cross talk co-efficient are equal to or less than those specified in memory Tline spec.

2.3.3.1 Tline Spec (Differential)

Table 54. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Differential)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
B*	SL	3, 8	PCIe Gen 4, PCIe Gen 5, UFS 4.0	75	88	200	375	78	-0.88	0.106	0.002
B*	SL	3, 8	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	78	-0.88	0.106	0.012
B*	SL	3, 8	CSI CPHY, CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	78	-0.88	1.301	0.012
B*	SL	3, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	78	-0.88	0.002	0
B*	SL	3, 8	CNVio3, TCP DP, TCP TBT4, TCP TBT5	75	88	375	375	78	-0.88	0.002	0.002
B*	SL	3, 8	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	78	-0.88	0.012	0.012
B*	SL	6	PCIe Gen 4, PCIe Gen 5, UFS 4.0	75	88	200	375	79	-0.8	0.242	0.01
B*	SL	6	CSI CPHY, CSI DPHY, HDMI, TCP AUX, TCP DP AUX, eDP	75	88	88	300	79	-0.8	1.925	0.037
B*	SL	6	Differential Clock (Gen 3 and below support), eUSB2, PCIe Gen 1-3, TCP USB 3.2	75	88	200	300	79	-0.8	0.242	0.037
B*	SL	6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	88	375	500	79	-0.8	0.01	0.001
B*	SL	6	CNVio3, TCP DP, TCP TBT4, TCP TBT5	75	88	375	375	79	-0.8	0.01	0.01
B*	SL	6	USB 3.2 Gen 1, USB 3.2 Gen 2	75	88	300	300	79	-0.8	0.037	0.037
M*	MS	1, 10	CSI DPHY, eUSB2, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	110	100	300	300	80	-0.89	-0.262	0.32
M*	MS	1, 10	Differential Clock (Gen 3 and below support)	110	100	300	375	80	-0.89	-0.262	0.193
M*	MS	1, 10	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	110	100	500	760	80	-0.89	-0.138	0.035
M*	MS	1, 10	eDP, HDMI, PCIe Gen 4, TCP DP, TCP TBT4	110	100	500	500	80	-0.89	-0.138	0.099
M*	MS	1, 10	CNVio3, TCP TBT5	110	100	800	800	80	-0.89	-0.05	0.03
M*	MS	1, 10	UFS 4.0, USB 3.2 Gen 1, USB 3.2 Gen 2	110	100	750	500	80	-0.89	-0.06	0.099
M*	MS	1, 10	PCIe Gen 5, TCP TBT5 (Post-Channel)	120	130	800	800	80	-0.86	-0.062	0.037
M*	MS	1, 10	CSI CPHY	150	875	875	875	79	-0.82	-0.151	0.076
M*	MS	1, 10	HDMI (Post-Channel)	75	140	500	500	100	-0.9	-0.133	0.1
M*	SL	3, 8	CSI DPHY, eDP, eUSB2, HDMI, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	75	115	300	300	80	-0.87	0.012	0.012
M*	SL	3, 8	Differential Clock (Gen 3 and below support)	75	115	300	375	80	-0.87	0.012	0.002
M*	SL	3, 8	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	75	115	500	760	80	-0.87	0	0
M*	SL	3, 8	PCIe Gen 4, PCIe Gen 5	75	115	230	500	80	-0.87	0.056	0
M*	SL	3, 8	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	75	115	375	375	80	-0.87	0.002	0.002
M*	SL	3, 8	USB 3.2 Gen 1, USB 3.2 Gen 2	75	115	450	500	80	-0.87	0	0
M*	SL	3, 8	CSI CPHY	80	170	170	500	80	-0.86	0.213	0
M*	SL	6	CSI DPHY, eDP, eUSB2, HDMI, PCIe Gen 1-3, TCP AUX, TCP DP AUX, TCP USB 3.2	80	120	300	300	80	-0.78	0.038	0.038
M*	SL	6	Differential Clock (Gen 3 and below support)	80	120	300	375	80	-0.78	0.038	0.01

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Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Intra-Pair (um)	S - ES (um)	S - Non-ES (um)	Z (Default) (ohm)	A (dB/inch @ 5GHz)	K, S - ES (%)	K, S - Non-ES (%)
M*	SL	6	Differential Clock (Gen 4 support), Differential Clock (Gen 5 support)	80	120	500	760	80	-0.78	0.001	0
M*	SL	6	CNVio3, TCP DP, TCP TBT4, TCP TBT5, UFS 4.0	80	120	375	375	80	-0.78	0.01	0.01
M*	SL	6	PCIe Gen 4, PCIe Gen 5	80	120	250	500	80	-0.78	0.098	0.001
M*	SL	6	USB 3.2 Gen 1, USB 3.2 Gen 2	80	120	450	500	80	-0.78	0.003	0.001
M*	SL	6	CSI CPHY	85	170	170	500	80	-0.77	0.429	0.001

Table 55. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Differential) Notes

Note
Routing Notes for CSI CPHY: (1) CPHY must be routed as a trio (2) If the port is supposed to support both CPHY and DPHY, it must be routed following CPHY guidelines (3) CPHY impedance target is 40 ohms single ended and 80 ohms differential (4) All traces in trio must follow same trace width (5) Intra-Pair guidelines must be used for trace to trace spacing in a trio (6) S-ES guidelines must be used for trio to trio in the same port (7) S-Non-ES must be used for port to port or CPHY to other interface (8) CSI CPHY in Microstrip (MS) routing is not recommended for long length, allowed only for short length (<5 mm) near the connector.
For CNVio3 signal spacing: CNVio3 to non-CNVio3 pair spacing of 10H (H=Dielectric Thickness) is required. If this cannot be achieved, then GND shielding with proper stitching vias and 3H minimum spacing from GND shielding to CNVio3 is required. Please also refer to "CNVio3 Spacing Recommendation" diagram in CNVio3 section.

2.3.3.2 Tline Spec (Single Ended)

Table 56. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Single Ended)

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K, Data - Any (%)	K, Clock - Any (%)
B*	SL	3, 8	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	67	75	175	44			-0.27	3.878	0.422
B*	SL	3, 8	Processor Sideband	67	75	75	44			-0.27	3.878	3.878
B*	SL	3, 8	Imaging Clock, SUSCLK, UFS Reference Clock	67	175	175	44			-0.27	0.422	0.422
B*	SL	3, 8	XTAL, RTC	67	200	1270	44			-0.27	0.243	0.006
B*	SL	6	CLINK, CNVi BRI and RGI, DMIC, eSPI, GSPI, HDA, I2C, I2S, I3C, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SPI0 Flash, SVID, THC-SPI, UART	75	75	175	44			-0.25	5.343	0.846
B*	SL	6	Processor Sideband	75	75	75	44			-0.25	5.343	5.343
B*	SL	6	Imaging Clock, SUSCLK, UFS Reference Clock	75	175	175	44			-0.24	0.846	0.846
B*	SL	6	XTAL, RTC	75	200	1270	44			-0.24	0.535	0.005

continued...

Trace	Tline Type	Routing Layers	I/O	Trace Width (um)	Data - Any (um)	Clock - Any (um)	Z (Default) (ohm)	Z (Min) (ohm)	Z (Max) (ohm)	A (dB/inch @ 1GHz)	K, Data - Any (%)	K, Clock - Any (%)
M*	MS	1, 10	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	125	125	375	44	39.6	48.4	-0.24	-2.493	-1.559
M*	MS	1, 10	CNVi BRI and RGI	125	250	250	44	39.6	48.4	-0.23	-2.227	-2.227
M*	MS	1, 10	Processor Sideband	125	125	125	44	39.6	48.4	-0.24	-2.493	-2.493
M*	MS	1, 10	Imaging Clock, SUSCLK, UFS Reference Clock	125	375	375	44	39.6	48.4	-0.23	-1.559	-1.559
M*	MS	1, 10	CLINK, I3C, SPI0 Flash	125	250	375	44	39.6	48.4	-0.23	-2.227	-1.559
M*	MS	1, 10	XTAL, RTC	125	200	1775	44	39.6	48.4	-0.23	-2.468	-0.066
M*	SL	3, 8	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	67	125	375	44	39.6	48.4	-0.27	1.273	0.011
M*	SL	3, 8	CNVi BRI and RGI	67	250	250	44	39.6	48.4	-0.27	0.083	0.083
M*	SL	3, 8	Processor Sideband	67	125	125	44	39.6	48.4	-0.27	1.273	1.273
M*	SL	3, 8	Imaging Clock, SUSCLK, UFS Reference Clock	67	375	375	44	39.6	48.4	-0.27	0.011	0.011
M*	SL	3, 8	CLINK, I3C, SPI0 Flash	67	250	375	44	39.6	48.4	-0.27	0.083	0.011
M*	SL	3, 8	XTAL, RTC	67	200	1775	44	39.6	48.4	-0.27	0.243	0.006
M*	SL	6	DMIC, eSPI, GSPI, HDA, I2C, I2S, ISH-SPI, LSX, Processor GPIO, SMLink, SoundWire, SVID, THC-SPI, UART	75	125	375	44	39.6	48.4	-0.24	2.119	0.026
M*	SL	6	CNVi BRI and RGI	75	250	250	44	39.6	48.4	-0.24	0.213	0.213
M*	SL	6	Processor Sideband	75	125	125	44	39.6	48.4	-0.24	2.119	2.119
M*	SL	6	Imaging Clock, SUSCLK, UFS Reference Clock	75	375	375	44	39.6	48.4	-0.24	0.026	0.026
M*	SL	6	CLINK, I3C, SPI0 Flash	75	250	375	44	39.6	48.4	-0.24	0.213	0.026
M*	SL	6	XTAL, RTC	75	200	1775	44	39.6	48.4	-0.24	0.535	0.005

Table 57. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) Tline Spec (Single Ended) Notes

Note
1) Based on Tline Spec (Single Ended) table recommendation, the CLK - Any (um) spacing requirement shall be applicable for: a) CLK to DATA signals on similar signaling group. b) CLK to DATA signals from other signaling group. c) CLK to CLK signals from other signaling group. 2) For CLK to pseudo static signals, the spacing between the CLK to others can be relaxed and used the DATA - Any (um) spacing requirement. Example of pseudo static signal: RESET, ALERT, Chip Select (CS), Request (REQ), INTERRUPT, INIT. 3) Processor Sideband IO refers to CATERR#, EPD_ON, FORCEPR#, THERMTRIP#, VDD2PWRGOOD signals.

2.3.3.3 Tline Spec (Memory)

Interface: LPDDR5x

Notes

Table 58. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x Notes

Note
RCOMP to other nets spacing requirement is minimum 400 um.
DDR to Non-DDR nets spacing requirement is minimum 500 um.
DRAM RESET to be routed with 40 ohm impedance on Layer 1, Layer 2 and Layer 4 and distance to other nets (spacing requirement) is minimum 250 um.
Route RCOMP with 75 um TW (MR) on Layer 1, Layer 10. GND shield to any high speed IO with VSS trace/strip. GND reference required.

Trace Width

Table 59. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-4,0.8mm,10L CH-0, CH-2, CH-4, CH-6 Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO	SL	2	DQ	Outer	80		40		-0.452	
M	SL	2	DQ	Outer	95		37		-0.445	
BI	SL	2	DQ	Outer	95		37		-0.444	
BO	SL	2	RDQS	Outer	80	75	75			-0.458
M	SL	2	RDQS	Outer	80	75	75			-0.458
BI	SL	2	RDQS	Outer	80	75	75			-0.458
BO	SL	2	WCK	Outer	90	60	69			-0.464
M	SL	2	WCK	Outer	105	75	65			-0.446
BI	SL	2	WCK	Outer	105	75	65			-0.446
BO	SL	8	CLK	Outer	75	65	75			-0.456
M	SL	8	CLK	Outer	75	65	75			-0.456
BI	SL	8	CLK	Outer	75	65	75			-0.456
BO	SL	8	CA/CS	Outer	80		40		-0.438	
M	SL	8	CA/CS	Outer	80		40		-0.46	
BI	SL	8	CA/CS	Outer	80		40		-0.46	

Table 60. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-4,0.8mm,10L CH-1, CH-3, CH-5, CH-7 Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Intra-pair (um)	Z SE (Default) (ohm)	Z Diff (Default) (ohm)	A SE (dB/inch @ 2GHz)	A Diff (dB/inch @ 2GHz)
BO	SL	4	DQ	Inner	95		37		-0.455	
M	SL	4	DQ	Inner	95		37		-0.456	
BI	SL	4	DQ	Inner	95		37		-0.455	
BO	SL	4	RDQS	Inner	85	75		75		-0.469
M	SL	4	RDQS	Inner	85	75		75		-0.469
BI	SL	4	RDQS	Inner	85	75		75		-0.469
BO	SL	4	WCK	Inner	90	60		70		-0.478
M	SL	4	WCK	Inner	110	75		65		-0.456
BI	SL	4	WCK	Inner	110	75		65		-0.456
BO	SL	6	CLK	Inner	80	75		75		-0.414
M	SL	6	CLK	Inner	80	75		75		-0.414
BI	SL	6	CLK	Inner	80	75		75		-0.414
BO	SL	6	CA/CS	Inner	85		40		-0.398	
M	SL	6	CA/CS	Inner	85		40		-0.396	
BI	SL	6	CA/CS	Inner	85		40		-0.396	

Table 61. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-4,0.8mm,10L MISC Trace Width Specs

Trace	Tline Type	Routing Layers	Signals	Pin group/Location	Trace Width (um)	Z SE (Default) (ohm)	A SE (dB/inch @ 2GHz)
BO1	MS	1	RESET	All	75	54.5	-0.41
BO2	MS	1	RESET	All	145	40	-0.392
M	SL	2, 4	RESET	All	85	40	-0.493
BI	SL	2, 4	RESET	All	85	40	-0.493

Trace Spacing

Table 62. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-4,0.8mm,10L CH-0, CH-2, CH-4, CH-6 Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO	100	110	88	88	150	88	150	150	2.038	1.574	2.385	2.4	0.551	0.528	0.257
M	200	300	180	200	200	500	300	300	0.222	0.023	0.262	0.17	0.168	0.016	0.008
BI	200	300	180	500	200	500	300	300	0.222	0.023	0.26	0.005	0.166	0.017	0.008

Table 63. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-4,0.8mm,10L CH-1, CH-3, CH-5, CH-7 Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	CLK to CA/CS (um)	DQ to DQ of same byte (um)	Byte to Byte (um)	RDQS pair to DQ of same byte (um)	DQ to CA/CS (um)	WCK pair to DQ of same byte (um)	WCK pair to DQS pair of same byte (um)	K, CA/CS to CA/CS (%)	K, CLK to CA/CS (%)	K, DQ to DQ of same byte (%)	K, Byte to Byte (%)	K, RDQS pair to DQ of same byte (%)	K, WCK pair to DQ of same byte (%)	K, WCK pair to DQS pair of same byte (%)
BO	100	180	88	150	150	88	150	150	3.152	0.688	2.07	0.6	0.487	0.472	0.238
M	180	300	180	200	200	500	300	300	0.717	0.076	0.245	0.2	0.154	0.015	0.007
BI	180	300	180	500	200	500	300	300	0.717	0.076	0.244	0.005	0.153	0.016	0.007

Table 64. Type-4 2-x-2+, 0.8mm, 10L, Mainstream, Premium Mid Loss (PML) LPDDR5x MD x32 Type-4,0.8mm,10L CH-0, CH-1, CH-2, CH-3, CH-4, CH-5, CH-6, CH-7 Channel to Channel Trace Spacing Specs

Trace	CA/CS to CA/CS (um)	DQ to DQ (um)
BO	100	88
M	500	500
BI	500	500

2.4 Non Critical To Function (NCTF) Solder Balls and Testability Recommendation

As we shift to thinner system designs, SMT is becoming more challenging with the corners being more susceptible to solder joint defects. Testing the corner NCTF Solder Joints (SJ's) is a good way to screen potential solder joint defects and check health of process. The Processor corner NCTF SJ's have some pins connected to the package POWER/VCC or GROUND/VSS planes which can be used to test connectivity.

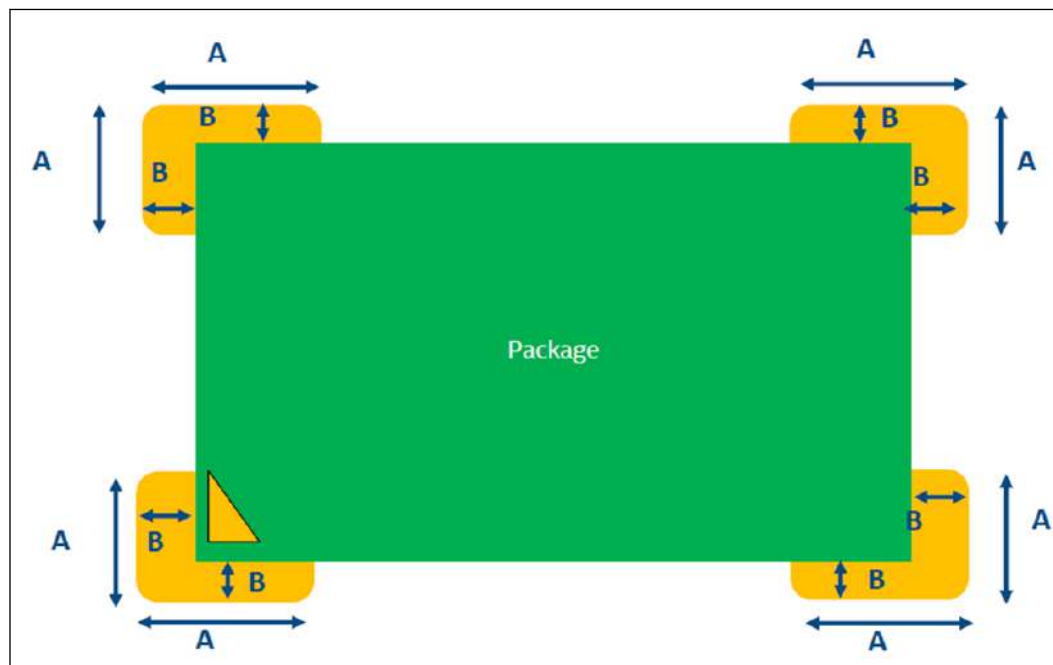
NOTES

- **For Corner Solder Joint Testability details, refer to the platform manufacturing guidelines.**
 - **For Non Critical To Function (NCTF) Solder Balls and Testability Recommendation, refer to the Intel Land Pattern Guidance: LPID 7672 (# 847894).**
-

2.5 Adhesive Keep out Zone

A combination of thinner system designs and higher thermally driven static compressive loads will have increased risk for board level solder joint reliability (SJR). Application of a board level adhesive will greatly mitigate the SJR risk. The specific combination of PCB thickness and static compressive load which requires adhesive can be found in the Nova Lake Mobile Platform Thermal Mechanical Design Guide (# 859348)

A KOZ is necessary in the board design to ensure adequate space for the adhesive dispense nozzle to not interfere with any board components during the HVM adhesive application process. KOZ may also be necessary to prevent the adhesive seeping to adjacent components. It is recommended to not have board components inside the KOZ.

Figure 2. Adhesive KOZ on Board Schematic**Table 65. Adhesive KOZ Dimensions**

Dimension	KOZ Value (mm)
A	5.7mm
B	2.2mm

NOTES

- Board components NOT allowed inside the KOZ
- PTHs or vias inside KOZ need to be capped with solder resist to prevent adhesive from flowing into them and creating voids.

3.0 General Design Considerations

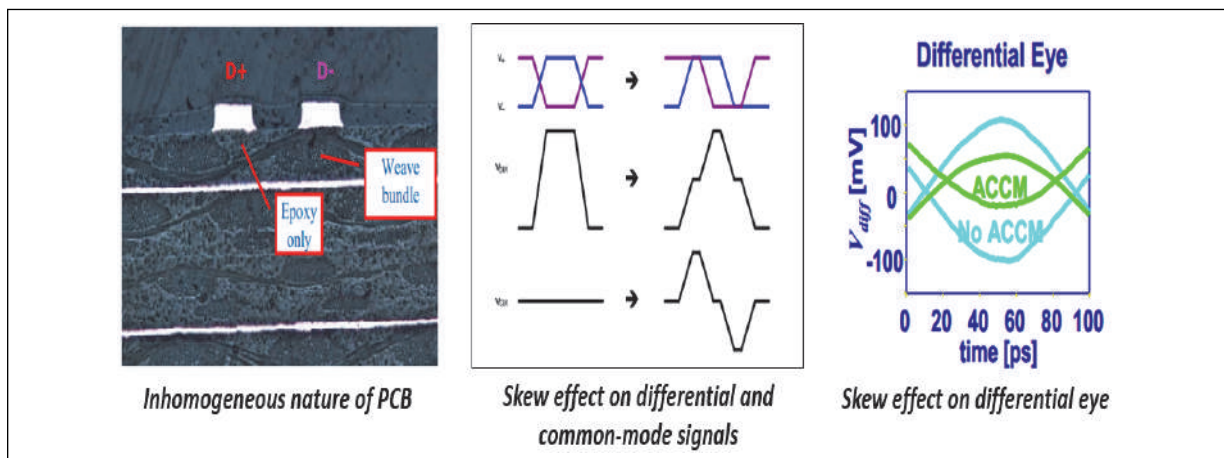
3.1 Fiberweave Impact for HSIOs Operating at ≥ 8 GT/s Speeds

Owing to negative signal integrity effects of fiberglass weave in the materials of which circuit boards are made, the Fiberweave impact to HSIO (High Speed I/O) running at ≥ 8 GT/s is significant, thus the need for mitigation to ensure robust system design.

- Fiber Weave Effect: Practical Impact Analysis and Mitigation Strategies White Paper (# 406926) discusses 17 options used to quantify the impact of Fiberweave and strategies studied to mitigate it.
- 10-degree PCB panel tilting is not the most cost-effective mitigation solution (among 17 options proposed), due to the low panel utilization and higher PCB cost.
- It is, therefore, Intel's advocacy for board routing efforts to address the cost concern. Examples of such routings, i.e. angled and zig-zag routing (minimum 10-degrees) are recommended.
- Customers are requested to prioritize floor planning on the board in such a way that the routing ends up being at an angle rather than orthogonal.

The Processor has Integrated Thunderbolt (TBT) controller, thus 20GT/s signal routing length can be as long as 205mm prior to connection to re-timer. Orthogonal routing with such length would result in adverse effect to channel margin, as described in Fiber Weave Effect: Practical Impact Analysis and Mitigation Strategies White Paper (# 406926)

Impact to channel margin can be shown below. This describes the difference in propagation velocities leading to skew between D+ and D- signals, therefore resulting in an increased of common mode noise.

Figure 3. Fiberweave Effects

3.1.1 Maximum Parallel Routing Restrictions

For 20-32GT/s signals, maximum allowable parallel RSS routing length is 12.7 mm. Due to such restrictions, zig-zag or angled routing (both need minimum 10 degrees) is recommended.

Below Figure specifies the maximum allowed routing parallel to the board edge for each transfer rate. It is given as a "Root Sum Square" (RSS, root of the sums squared) value. The given maximum RSS length is based on the assumption that weave type of PCB construction is not specified. In the cases that specific weaves with better skew control are used, the maximum RSS length can be relaxed, but quantification of the benefit is beyond the scope of this document.

Table 66. Maximum Root Square Sum (RSS) Length versus Transfer Speed

Transfer Speed	Max RSS Length in mm (Refer Note1)
2.5 GT/s	127
4 GT/s	101.6
5 GT/s	76.2
6.4 GT/s	63.5
8 GT/s	50.8
10 GT/s	50.8
12 GT/s	37
16 GT/s	25.4
20 GT/s	12.7
25.6 GT/s (40Gbps)	12.7
32 GT/s	12.7

NOTE

The length in the table represent total trace lengths that are aligned to the weave (that is, parallel to the manufacturing panel's edge). Actual routing may have a significant portion of the length at angles to the edge of the board those lengths should be considered in this analysis.

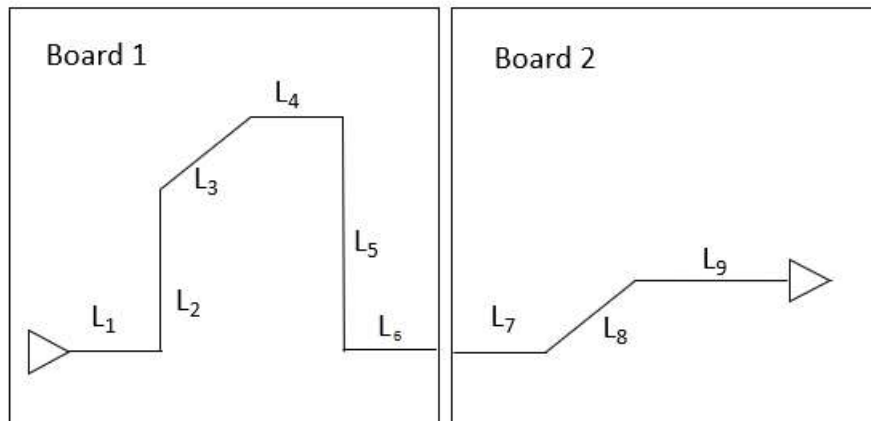
- The total length is the Root Square Sum of the total vertical and horizontal lengths that run parallel to the weave:

$$\text{Length} = \sqrt{(H_1)^2 + (H_2)^2 + (H_3)^2 + \dots + (V_1)^2 + (V_2)^2 + (V_3)^2 + \dots}$$

Where:

- H_x = length of each segment routed horizontally, and
- V_x = length of each segment routed vertically
- The routing length for angled routing are excluded in the calculation.
- Below is an example of how to calculate RSS length for a link that is routed across two boards.

$$\text{Length}_{\text{RSS}} = \sqrt{(L_1 + L_6)^2 + (L_2)^2 + (L_4)^2 + (L_5)^2 + (L_7)^2 + (L_9)^2}$$



- The length in the table represent total trace lengths that are aligned to the weave (i.e., parallel to the manufacturing panels's edge). Actual routing may have a significant portion of the length not parallel to the edge of the board - those lengths should not be considered in this analysis.
- For multiple board topologies, the lengths on the various boards are added equally.

3.1.2 Recommended Routing Guidelines

- Zig-zag or angled routing (both need minimum 10 degrees) is recommended after allowable RSS length.
- For angled routing, total length increases with angle of the segment.

- Customers are required to check their glass fabric material used related to weave pitch. Segment trace length needed to cross 2 weave bundles along its length.
- Each segment length can be independent segments of length.

Figure 4. Recommended Routing Illustrations

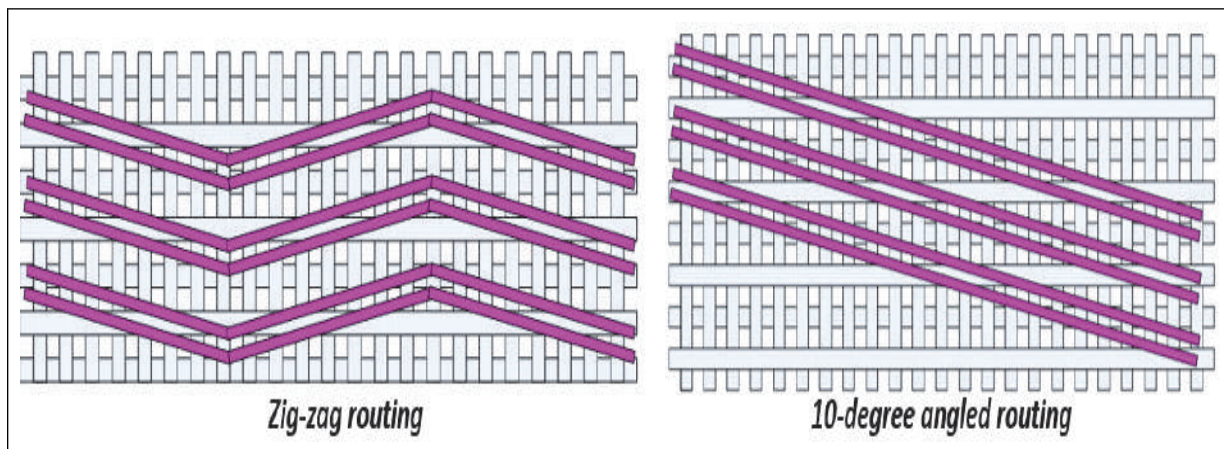
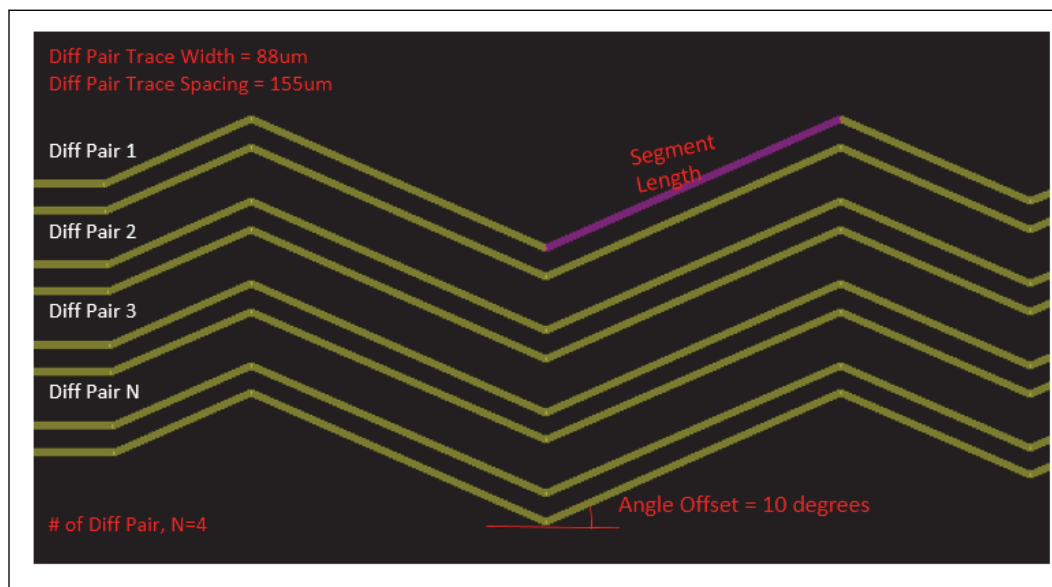


Figure 5. Zig-zag Routing Layout Example



NOTES

1. 10-degree routing angle: Length increases ~1-2%.
2. 20-degree routing angle: Length increases ~6%.

Figure 6. 1080 Glass Type Routing Layout Example

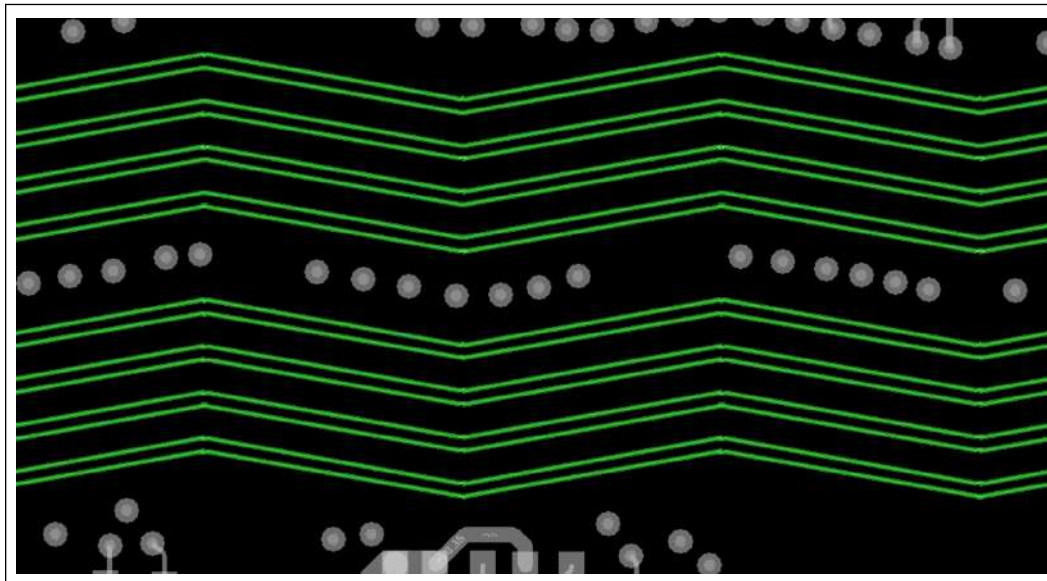
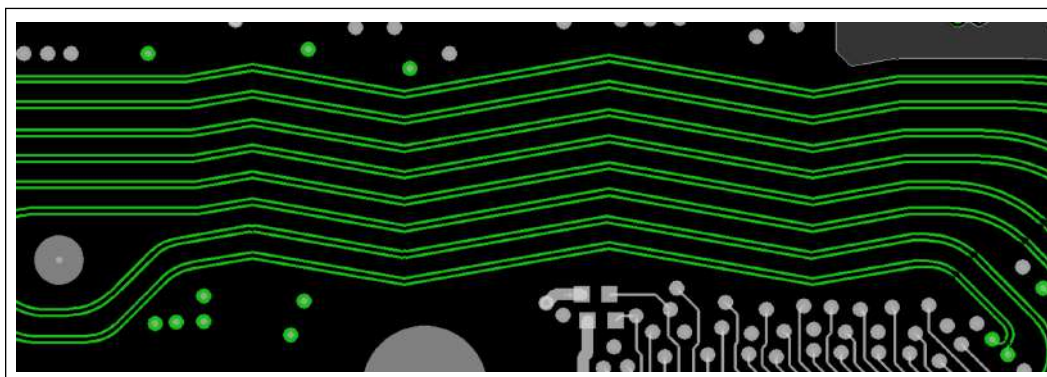
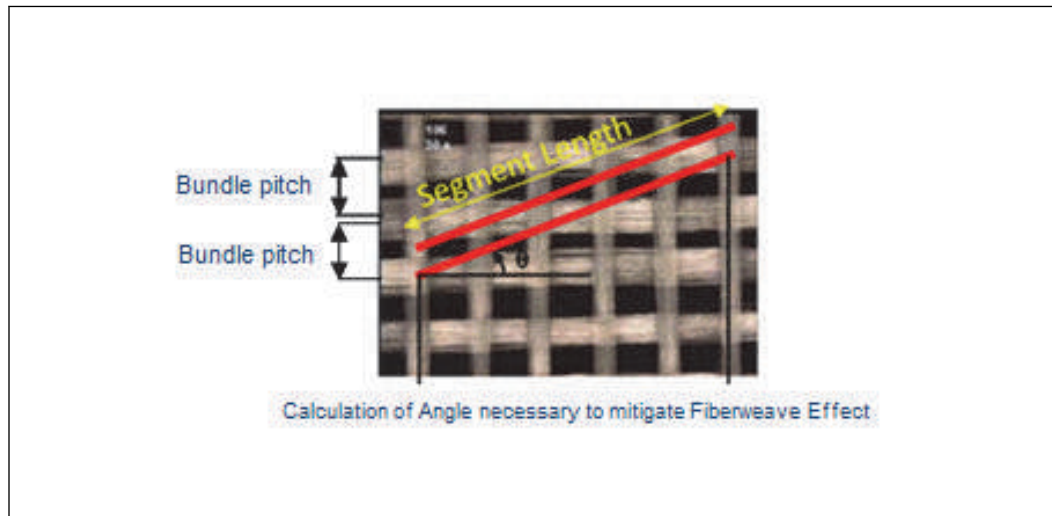


Figure 7. 1086 Glass Type Routing Layout Example



3.1.3 Zig-zag Segment Length Calculation Example For Improving Fabric

Customers can interchange the glass fabric with the same thickness for reduced Fiberweave effect. E.g., from 2113 to 2313, 1080 to 1078\1086, 106 to 1067. Work directly with your PCB vendor to evaluate options.

Figure 8. Calculation of Angle Necessary to Mitigate Fiberweave Effect**Typical case glass fabric material 106:**

Fiber bundle pitch: 469.9um

2X pitch: 939.8um

Zig-zag segment length to cross 2 pitches calculated as:

Segment Length= 2X pitch/Sin θ = 939.8/ Sin10 = 5412um

Replacement glass fabric material 1067:

Fiber bundle pitch : 363.2um

2X pitch: 726.4um

Zig-zag segment length to cross 2 pitches calculated as:

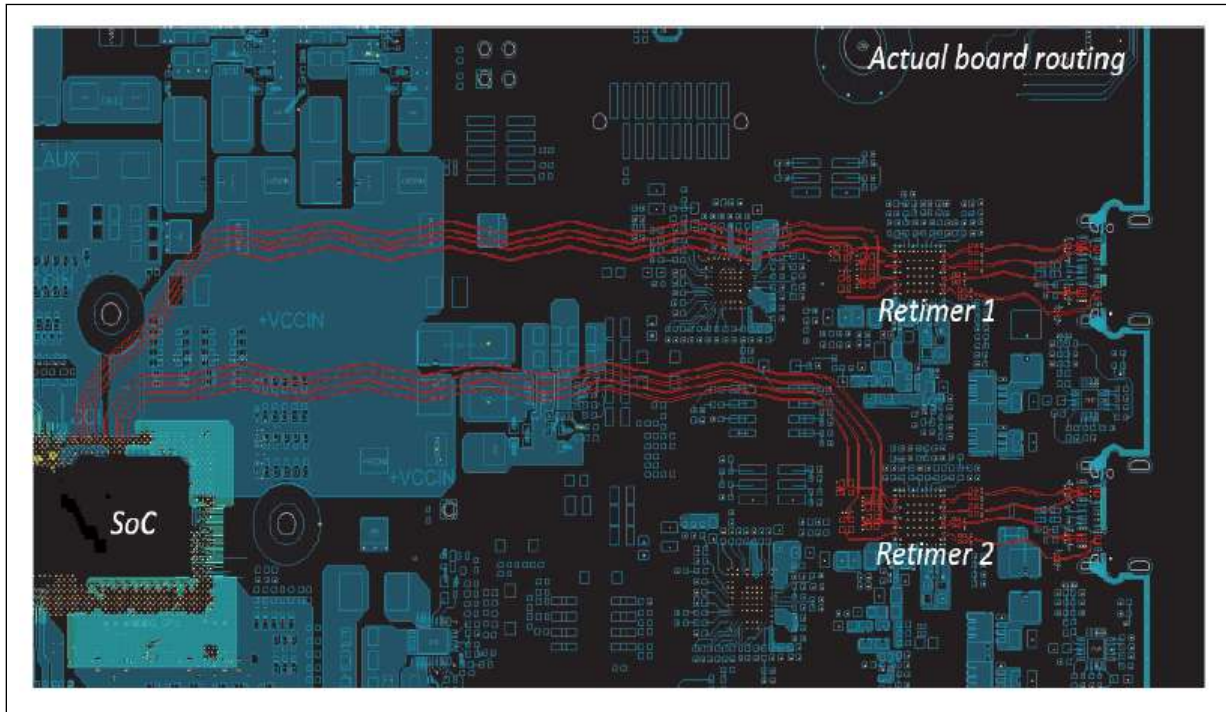
Segment Length= 2X pitch/Sin θ = 726.4/ Sin10 = 4183um

In this example calculation, 1067 glass fabric has a much finer fiber bundle pitch than 106. This helps reduce the overall trace length added from using zig-zag or angled routing at a minimum of 10 degrees.

Example Layout Implementation

Below is an example of which zig-zag routing is implemented in Intel reference design

Figure 9. Layout Implementation Example

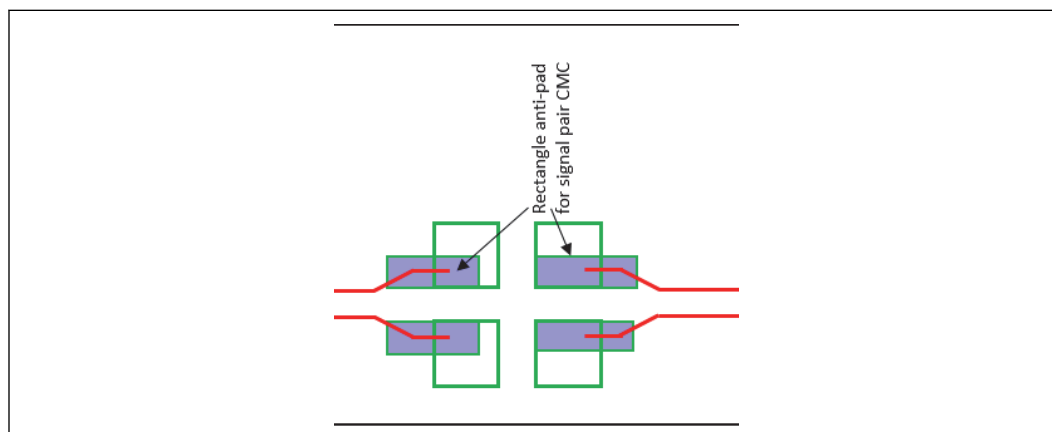


3.2 Component Footprint Plane Voiding Recommendation

Voiding the Ground/Power plane on the layer directly underneath SMT signal pads is necessary to minimize impedance mismatch caused by the SMT pads. This is recommended for large SMT pads of components such as ESD protection devices, Common Mode Chokes (CMC) and connectors. In general, the reference plane should be voided with the same size as the SMT pad. Voiding is only required on the layer directly underneath SMT signal pads and not required on any other internal layers. Voiding is not required for small component pads such as 0402 components and AC capacitors (unless explicitly mentioned for certain high-speed IOs such as Thunderbolt and PCIe*. For more details, refer to the IO chapters.)

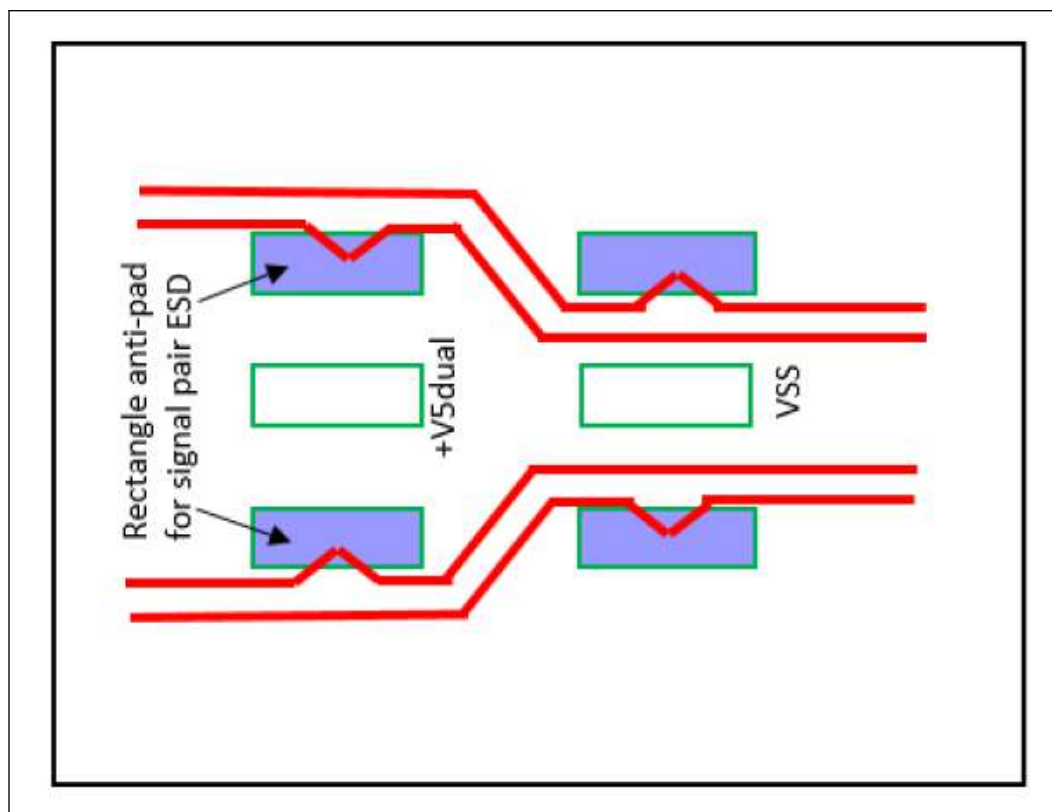
3.2.1 eUSB2/USB 3.2/USB 4 SMT Common-mode Choke Footprint and Ground Plane

Figure 10. eUSB2, USB 3.2 and USB 4 Common-mode Choke (CMC) Voiding Recommendation



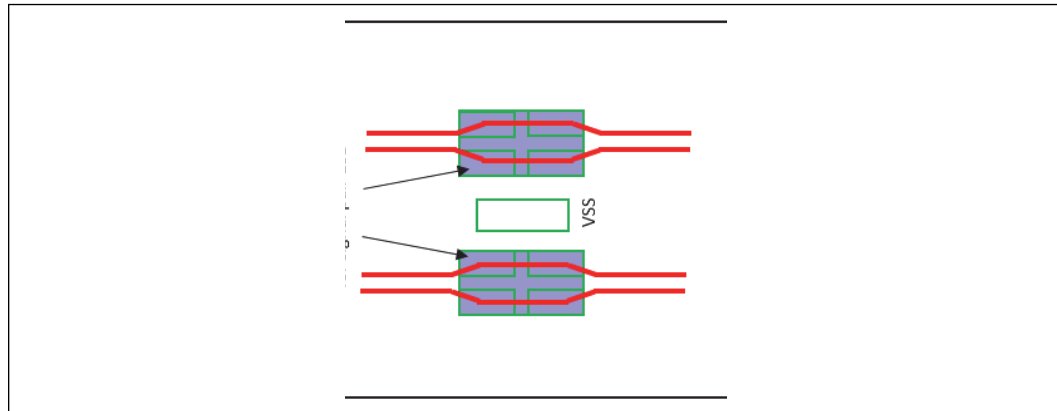
3.2.2 eUSB2 SMT Electrostatic Diode Footprint, Power and Ground Plane

Figure 11. eUSB2 Electrostatic Diode (ESD) Voiding Recommendation



3.2.3 USB3.2 SMT Electrostatic Diode Footprint, Power and Ground Plane

Figure 12. USB3.2 Electrostatic Diode (ESD) Voiding Recommendation



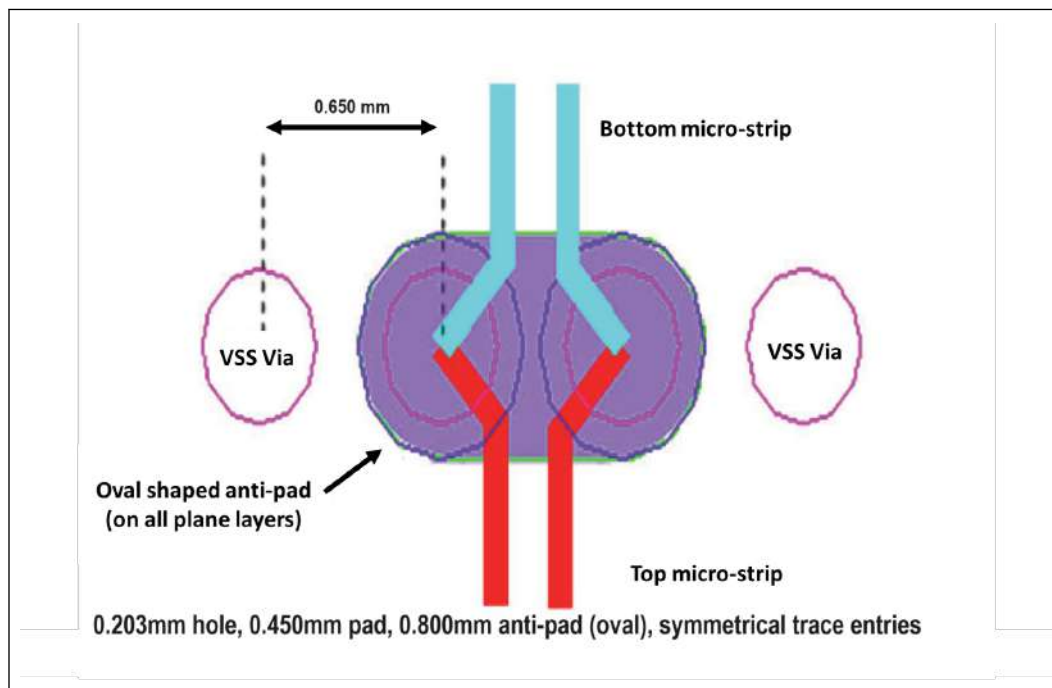
3.3 Differential Transitional Via Recommendations

Transitional vias will use oval-shapes anti-pads on all plane layers. This can be created using a rectangular-shaped void to overlap with the usual round-shaped via anti-pad. The vias must also have a symmetrical trace entry. Furthermore, Vss stitching vias must be placed at a symmetrical distance to the signal vias and must connect the reference planes above and below the incoming and outgoing traces.

Figure and Table below provides the transitional differential via pad stack details.

NOTE

This recommendation is not applicable to memory routing.

Figure 13. Differential Transitional Via Layout**Table 67. Differential Transitional Via Layout Recommendations**

Parameter	Recommendation
Via Diameter	203.3 μ m
Via Pad Size	450 μ m
Oval-Shaped Anti-Pad Size	800 μ m
Via-to-via Distance (centered)	650 μ m

3.4 General Dual-Stripline Support

In cases where high speed differential signals are routed in an 8- or 10- layer dual-stripline stack-up, follow the guidelines in table below. Figure below illustrates the meaning of Tx-over-Tx, Rx-over-Rx, and Tx-over-Rx.

For additional details and recommendations regarding dual-stripline routing, refer to Dual Stripline Crosstalk Behavior and Cross-Interface Coupling technical white paper (#575518).

Table 68. General Dual-Stripline Support

Tx -over-Rx Parallel Routing	Tx-over-Tx or Rx-over-Rx Parallel Routing	Tx-over-Rx or Tx-over-Tx or Rx-over-Rx Routing Orthogonal or with Angle
For CPU HSIOs, parallel routing is allowed if the offset is at least equal to the Signal-to-Non Equivalent-Signal spacing defined in the Tline specification sheet (attached with the PDG).	For CPU HSIOs, parallel routing is allowed if the offset is at least equal to the Signal-to-Equivalent-Signal spacing defined in the Tline specification sheet (attached with the PDG)	- Routing at an angle of at least 30 degrees significantly reduces layer-to-layer crosstalk. - Route differential pairs at a 30,45, 60, 90 etc degree angle.

Tx -over-Rx Parallel Routing	Tx-over-Tx or Rx-over-Rx Parallel Routing	Tx-over-Rx or Tx-over-Tx or Rx-over-Rx Routing Orthogonal or with Angle
For other IOs, route with an offset of 0.381mm or greater	For other IOs, a maximum of 50.8mm of parallel routing is allowed	- Allow maximum of 10 aggressor pairs to route over 1 DSL routing victim pair. - Figure 15 on page 111 shows orthogonal routing. - Figure 16 on page 111 shows angled routing at less than 90 degrees.

NOTE

Signal-to-Equivalent-Signal: Same interface AND direction of propagation AND swing.
Signal-to-Non Equivalent-Signal: Different interface OR direction of propagation OR swing.

NOTE

The guidelines under section 3.4 are not applicable for CNVio3. Refer to section: CNVio3 General Guidelines and Length Matching.

Figure 14. Tx-over-Rx, Tx-over-Tx and Rx-over-Rx

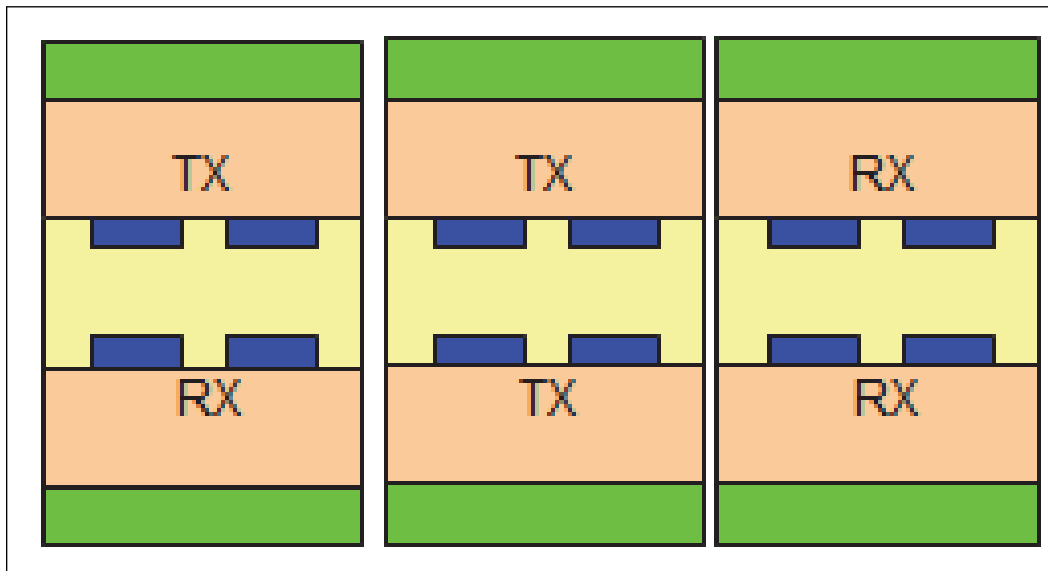


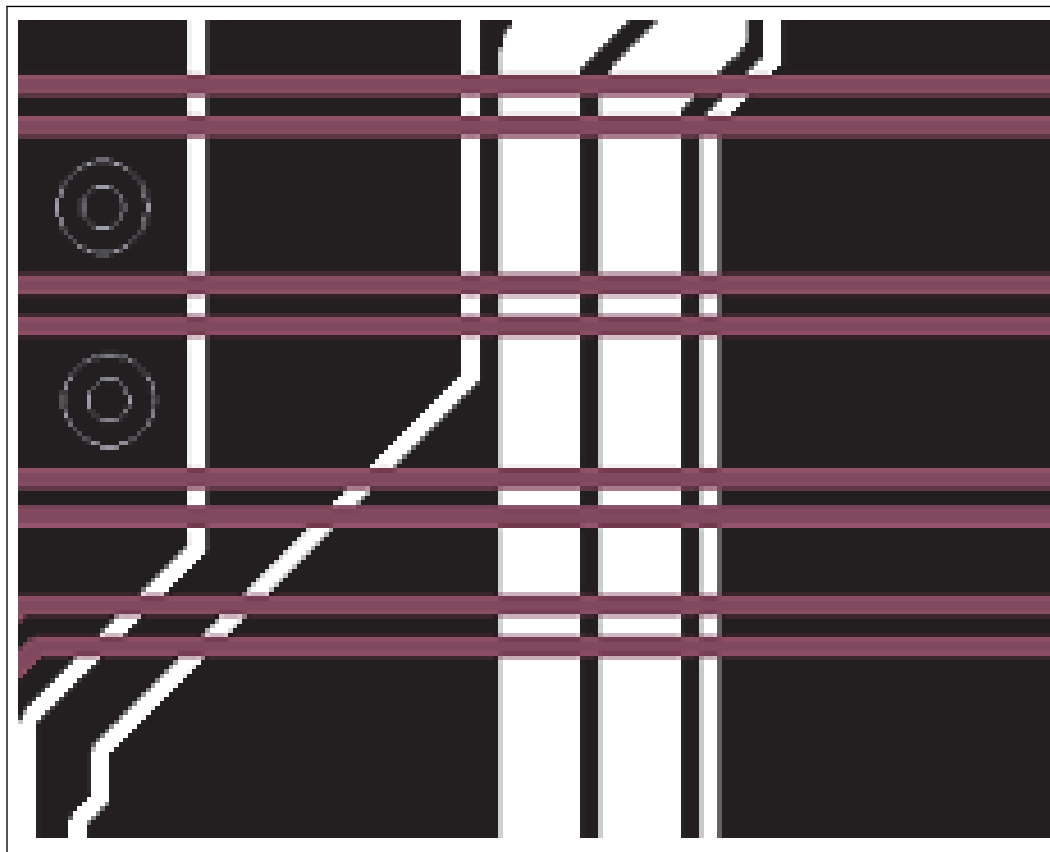
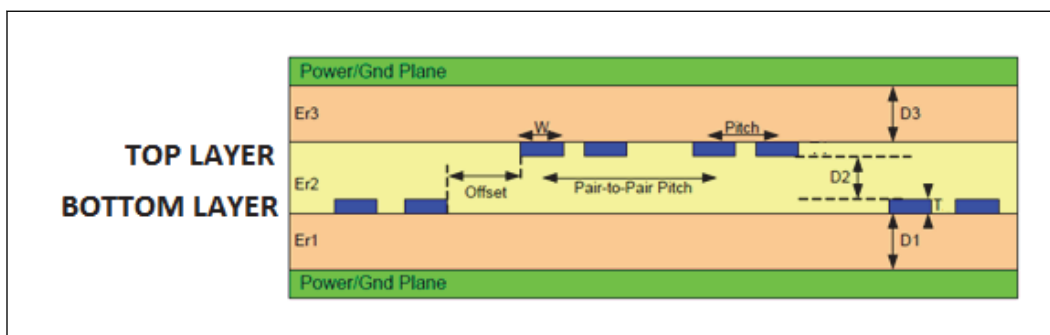
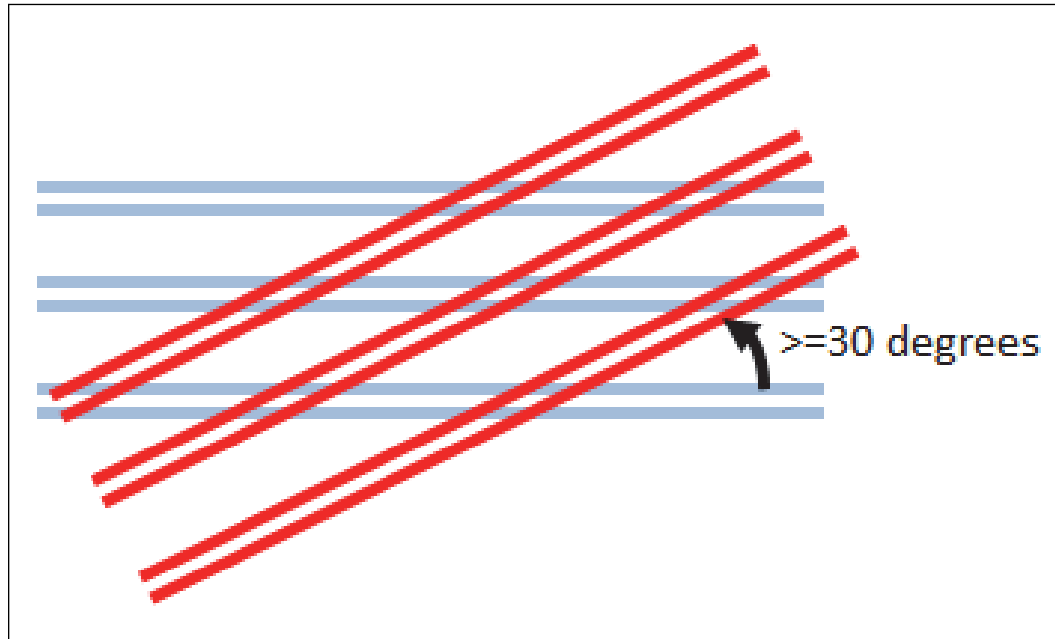
Figure 15. Dual-Stripline Orthogonal Routing Recommendation**Figure 16. Dual-Stripline High Speed Signal Dual Layer Routing Recommendation**

Figure 17. Recommended Routing Angle to Reduce Layer-to-Layer Crosstalk



3.5 Remove Non-functional Inner-layer Pads for Through Hole Connectors Vias

Remove non-functional inner-layer pads from through holes Via specially for through hole connectors. Not removing non-functional pads can cause an impedance to dip and impact may be limited to margin loss to compliance failures. Impact depends upon things like size of pads, pitch of connector pins, board thickness and reference impedance of routing (for 100 ohms dip to 60 ohms will have more impact than say 80 ohms dip to 60 ohms).

PCB vendors occasionally add inner layer pads, even when they are deliberately omitted in the Gerber file and some vendors may remove pads even when they are there in Gerber files.

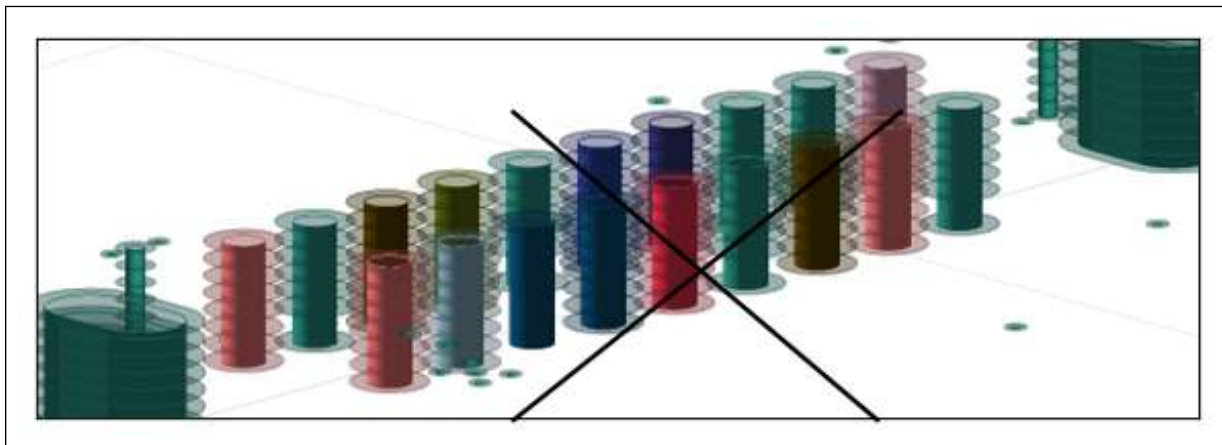
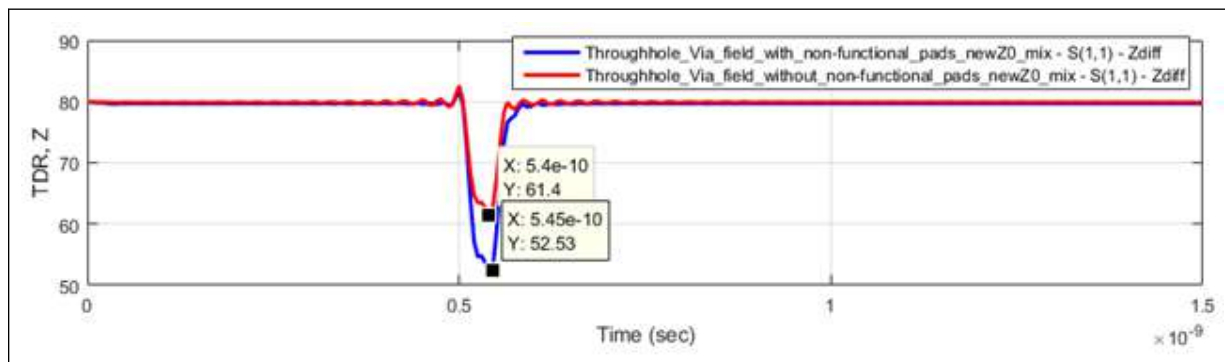
Some fabrication may remove non-functional pads for via, but not for those for via field of through hole connectors pins.

To be sure, remove nonfunctional pads from Gerber files and add an explicit Fab note to the tape-out database to not add nonfunctional pads.

Surface pads will always be present and pads in layer where there is connection should be present in each pin position. Connection should be done considering stub requirement of the interface, preferably without stub.

This should be done for all HSIO. Not required for ground, power pins and Low speed GPIOs connections.

This should be done for regular Vias or through holes via field of through-hole connectors and buried Vias for T4 boards.

Figure 18. Layout with Nonfunctional Pad Not Removed (Not recommended)**Figure 19. Layout with Nonfunctional Pad Removed (Recommended)****Figure 20. TDR Shows the Impact of Not Removing Nonfunctional Pads**

3.6 Teardrop Structure Requirement at Junction of Via and Trace

A Teardrop is a small copper fillet added at the junction where a narrow trace meets a via pad. Teardrops are used to:

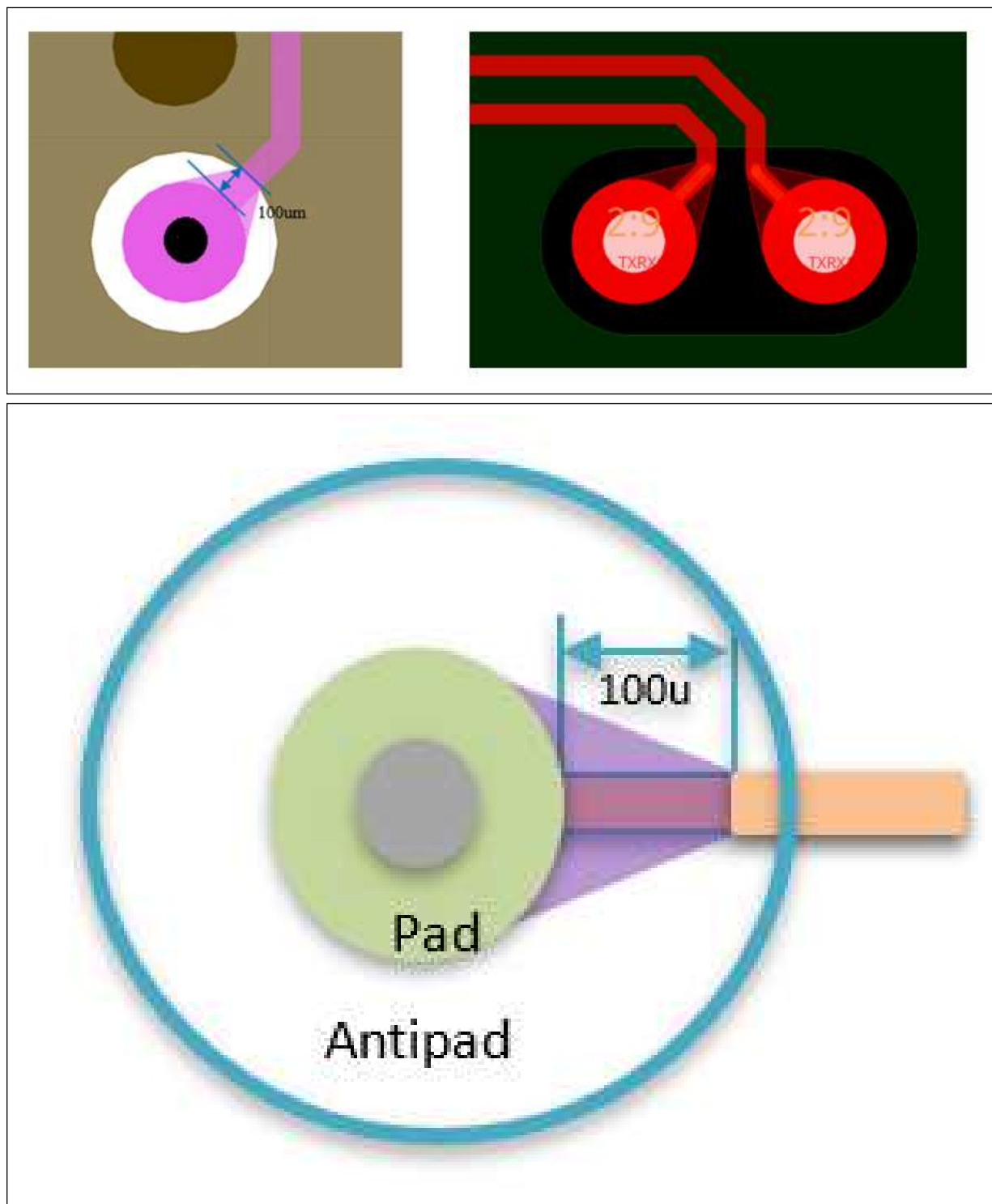
- Reduce stress concentration at the narrow-trace/large-pad junction, making the copper less likely to crack from drill wander, vibration, flexing, or thermal cycling.
- Increase manufacturing yield by giving more copper "capture" area if the drilled hole is slightly off-center, helping prevent open circuits or weak joints.
- In fine-pitch, HDI, or very thin trace designs, they add mechanical robustness with negligible extra area, which is especially useful on vias in BGA fan-out.

Electrically, teardrops can affect performance due to impedance mismatch, and this is critical for high-speed signals viz., memory in speeds > 8533 MT/s and HSIO > 10 Gbps.

To mitigate the impact of impedance mismatch, Intel recommends following guidelines for teardrop:

- Teardrop metal should be implemented fully within the antipad of via as shown in the figure below.
- Teardrop max length from via pad edge should be less than or equal to 100 μm (~ 4 mils). In case, teardrop shape is encroaching the metal outside the antipad area then voiding should be done in layers above and below the teardrop shape. However, make sure that teardrop void does not lead to Route over void (RoV) for other traces.
- If Teardrop is not implemented in layout, then this guidance should be provided to PCB supplier as part of Fab-notes in gerber release.

Figure 21. Recommended Teardrop Dimensions



3.7 Trace Length Matching Requirements

Follow the specific interface length matching guidelines if provided. If not provided, use the following as a general guideline.

Table 69. General Differential Pair Length Matching

Description	Routing Recommendation (mm)
Within Layer Max Mismatch	0.254
Total Length Max Mismatch	0.127
<i>Note:</i> Preferred to keep both 'within layer' mismatches and 'total length' mismatches within 0.127mm.	

The following are the length matching guidelines for differential-pairs:

- Each signal and its complement in a differential-pair should be length matched whenever possible on a layer-by-layer basis at the point of discontinuity. 'Within-layer' matching means that differential pairs should be matched within 0.254 mm before transitioning to a different layer. At any via transition the n/p mismatch for the entire route preceding the via cannot exceed 0.254 mm.
 - When trace length matching, the matching should be made as close as possible to the point where the length variation occurs, as shown in [Figure 22](#) on page 117, so the discontinuity will not propagate across the channel. For example, length matching in a chipset breakout area or connector pin field should occur within first 3.175 mm of the structure that causes mismatch.
- Serpentine layout introduces discontinuity to the channel and should be minimized so as to make it transparent to the signal. This is done by making its electrical length shorter than the signal rise time. In general, keeping serpentine routing length <2.54 mm is adequate. Trace spacing should not become greater than two times the original spacing. Refer [Figure 23](#) on page 117.
- In determining the overall length of a given signal in a differential-pair, use pad or pin edge-to-edge distances rather than the total etch present, unless the amount of trace routing inside each pad is identical. The amount of etch within a given pad is electrically part of the pad itself. In other words, only the etch outside of the pad edge is relevant to the overall length of a differential-pair. For example:
 - If the P signal of the differential-pair has an extra 0.127 mm of etch length compared to the N signal, and the extra 0.127 mm occurs due to extra etch extending into a component pad, the two signals should be considered identical length.
 - Similarly, trying to length match signals by adding etch inside a pad boundary (such as extra bends inside the pad itself) does not produce the intended length matching effect on the interconnection (refer [Figure 24](#) on page 118).

Figure 22. Example - Length Matching

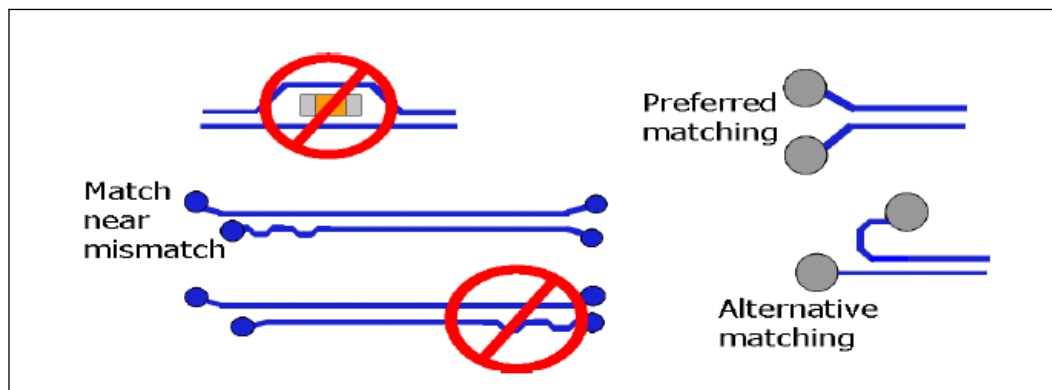


Figure 23. Example - Serpentine

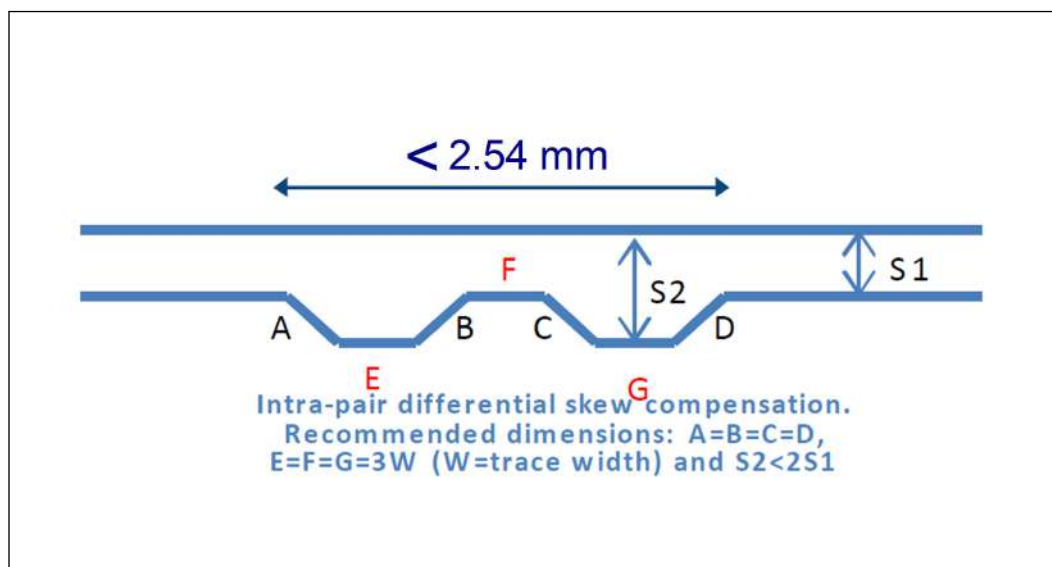


Figure 24. Example - Etch Located within a Pad

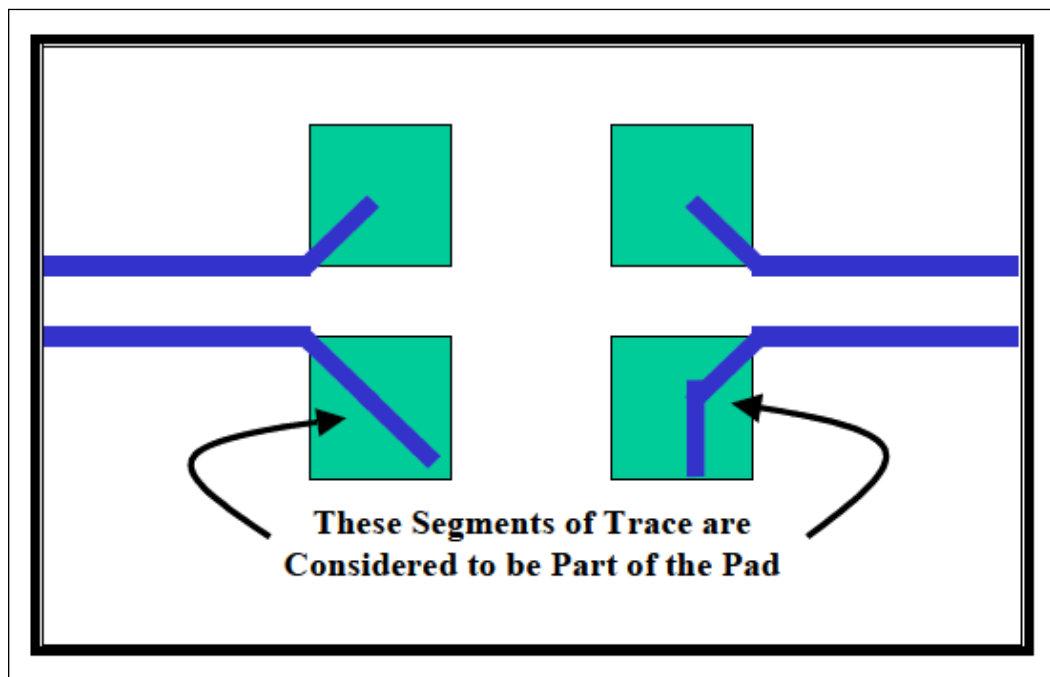


Figure 25. Example - Good Length Matching

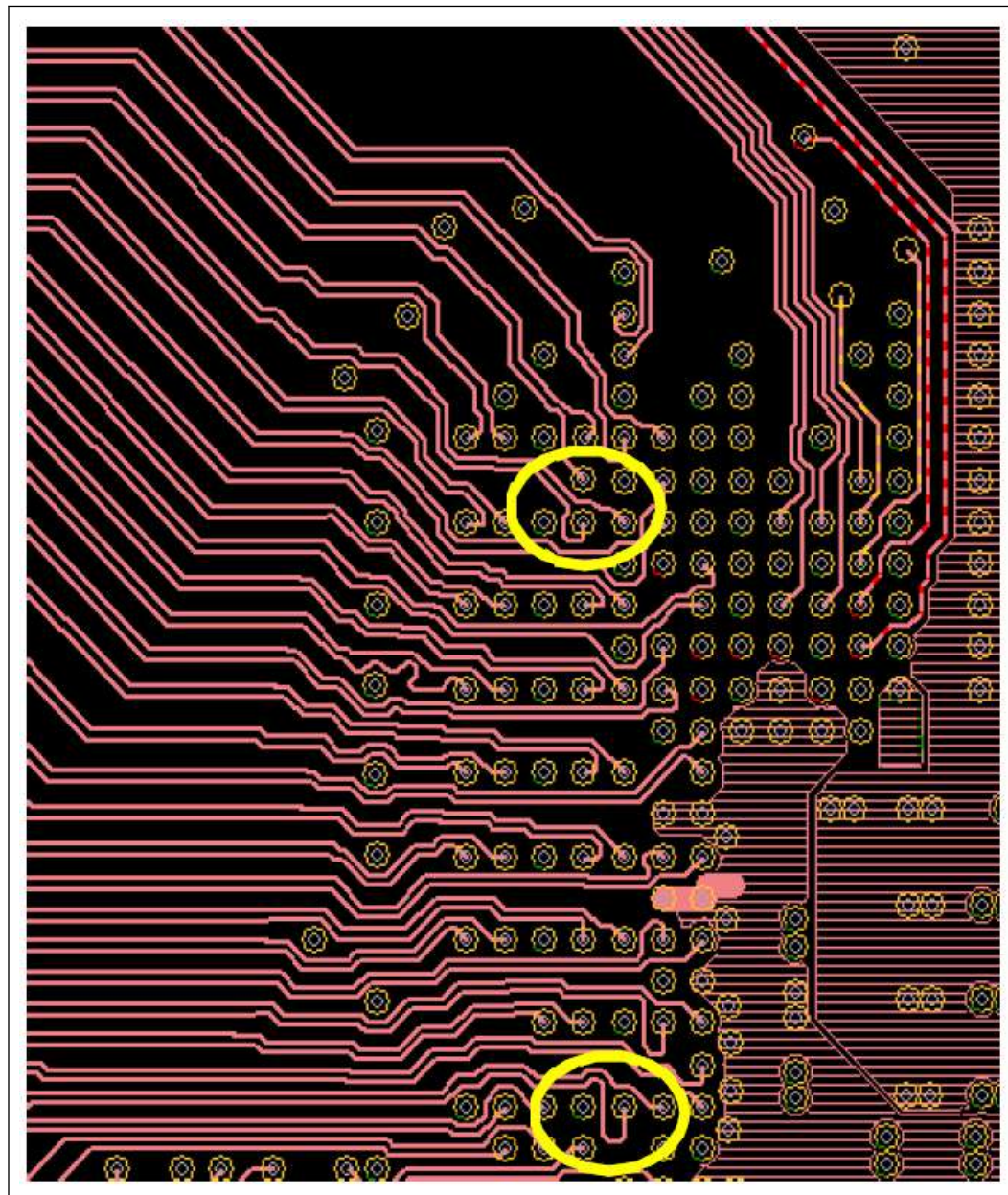
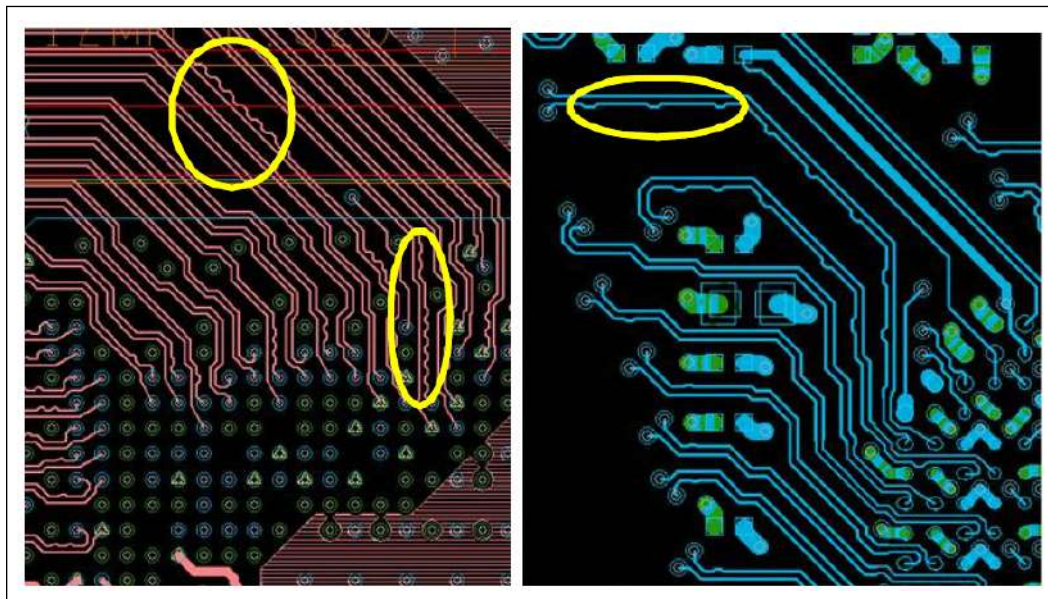


Figure 26. Example - Poor Length Matching



3.8 General Differential Routing Guidelines

Do not route traces under power connectors, other interface connectors, inductors, crystals, oscillators, clock synthesizers, magnetic devices or ICs that use and/or duplicate clocks. To minimize reflection, do not place stubs, test points, test vias on the route. Use vias and connector pads as test points instead. If a stub is unavoidable in the design, the total stub lengths on the particular trace should not be greater than 5.08 mm.

- It can be helpful for testability to route the TX and RX pairs for a given port on the same layer and close to each other to help ensure that the pairs share similar signaling characteristics. If the groups of traces are similar, a measure of RX pair layout quality can be approximated by using the results from actively testing the TX pair's signal quality.
- Keep signals clear of the core logic set. High current transients are produced during internal state transitions and can be very difficult to filter out.

3.9 Switching VR Circuit Guidelines

The following tables are applicable to Nova Lake platforms. It is strongly recommended to refer to the Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper (# 726825) for additional details as well as updates to the recommendations.

Table 70. Switching VR Circuit Clearance Requirements

Notes	Details
Switching VR Inductor footprint clearance to traces	The spacing required will scale according to VR switching currents. Min of 0.066mm per phase per amp of peak current (IccMax) for SE IOs. Min of 0.0304mm per phase per amp of peak current (IccMax) for Diff IOs.
<i>continued...</i>	

Notes	Details
	For example, if Icc max is 200A and there are 4 phases on the VR, then isolation needed for SE IOs is $0.066 \times 200 / 4 = 3.3\text{mm}$
Routing in layers under Inductor	No critical signal routing allowed upto 2 layers under inductor within 2.54mm distance from inductor foot prints. The immediate two layers under inductor needs to be GND/Power layers without any voids.
Inductor footprint to signal PTH via clearance	2.54mm for both Differential and single ended IO traces
Switching VR circuit power/ground Via to trace clearance	The spacing required will scale according to VR switching currents. Min of 0.066mm per phase per amp of peak current (IccMax) for SE IOs. Min of 0.0304mm per phase per amp of peak current (IccMax) for Diff IOs. For example, if Icc max is 200A and there are 4 phases on the VR, then isolation needed for SE IOs is $0.066 \times 200 / 4 = 3.3\text{mm}$
Switching VR circuit power /ground Via to signal via clearance	The spacing required will scale according to VR switching currents. Min of 0.2311mm per phase per amp of VR peak current (IccMax) for critical high speed SE IOs and Diff IOs. For example, if Icc max is 200A and there are 4 phases on the VR, then isolation needed for SE IOs is $0.2311 \times 200 / 4 = 11.557\text{mm}$ Slow speed critical single ended signals may need about 3.81mm of isolation. <i>Note:</i> Any via (both PWR and GND) part of the input or output side of VR needs to meet this clearance requirement.
Trace to VR power plane shape	0.635mm Min space between any critical IO trace to VR input or output power plane shape

Table 71. Summary of Component Placement and Layout Recommendations for Minimizing Phase Node Ringing

Element	Component/Placement/Layout Recommendation
DrMOS	Select components with low parasitic inductance and capacitance
DrMOS Input Power Flood	- Minimize path inductance with wide power shapes on each applicable layer. - Apply sufficient via stitching to minimize layer transition path inductance.
DrMOS Ground Flood	- Minimize path inductance with wide ground shapes on each applicable layer. - Apply sufficient via stitching to minimize layer transition path inductance.
DrMOS Input Decoupling	- Place input decoupling capacitors close to DrMOS VIN terminals. - Orient the ground pads of the input decoupling capacitors to minimize the current return path through the DrMOS. - Maximize the number of capacitors on the DrMOS layer to remove via inductance from the cap current path.
<i>continued...</i>	

Element	Component/Placement/Layout Recommendation
	Select low ESL capacitors.
Bootstrap Circuit	- Add a resistor to the bootstrap circuit. This establishes a RC time constant on the high-side FET gate and slows the turn-on (dV/dt), which reduces phase node ringing. - The designer must weigh the need to reduce ringing against the efficiency hit of adding the bootstrap resistor. - Incorporating a stuffing placeholder in the layout will provide the design with options during test/qualification. If a resistor is not needed, the location can be populated with a 0Ω resistor.
Snubber Circuit	- Add an RC snubber circuit from the phase node to ground. This establishes a RC time constant which slows the turn on/off of the high and low-side FETs (dV/dt), which reduces phase node ringing. - The designer must weigh the need to reduce ringing against the efficiency hit of adding the snubber RC circuit. - Incorporating RC stuffing placeholders in the layout will provide the design with options during test/qualification. If the snubber is not needed, the component locations can be depopulated.

Table 72. Summary of Victim Net and Phase Node Routing Recommendations

Element	Component/Placement/Layout Recommendation
Phase Node Shape	- Orient the inductor and DrMOS in a manner that enables the smallest phase node shape. - Minimize the size of the phase node shape to only encompass the inductor, DrMOS, bootstrap cap, and snubber cap component pads. - Do not route the phase node shape on inner layers of a Type-3 PCB.
Victim Net	- Where possible, avoid routing the net under phase nodes. - Route victim I/O under the minimal number of possible phase nodes. - Route victim I/O on the furthest available layer away from the phase node. - Maximize the number of power and ground planes/shapes between the phase node and victim net. - Minimize the length of victim net routing under the phase node. - Never length match (serpentine) I/O under the phase node (refer to the Figure 4-4 in) Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper (# 726825).
Vias	Do not place I/O vias in the boundary of the phase node shape. This will increase noise coupling amplitude. In addition, they will act as an antenna and spread noise contamination to internal routing layers.

Table 73. Summary of Inductor and Victim Net Routing Recommendations

Element	Component/Placement/Layout Recommendation
Inductor Selection	Use magnetically shielded inductors such as those embedded in ferrite or iron powder.
Inductor Footprint Plane Shape	- Extend inductor footprint plane shape in the X- and Y- directions to the pad edge, in order to minimize perforation/voiding thus preventing field leakage from the center of spiral coils. - Refer to the Figure 5-3 in Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper (# 726825) .
Victim Trace Separation from inductor	- Microstrip configuration: - High-current inductor ($\geq 10A$ per phase) - recommend 3 5mm spacing from inductor footprint to keep noise coupling to $< 5mV$. - Low-current inductor ($< 10A$ per phase) - recommend 3 3mm spacing from inductor footprint to keep noise coupling to $< 5mV$. - Refer to the Figure 5-4 (a) and Figure 5-5 in Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper (# 726825). - Stripline configuration (I/O routing at Layer 3; inductor at Layer 1): - High-current inductor ($\geq 10A$ per phase) - recommend 3 6.5mm spacing from inductor centerline to keep noise coupling to $< 5mV$. - Low-current inductor ($< 10A$ per phase) - recommend 3 1.6mm spacing from inductor centerline to keep noise coupling to $< 5mV$. - Assuming Layer 2 is full solid plane (PWR/GND) without plane perforation; total metal thickness (L2) $\geq 15\mu m$; dielectric thickness $\geq 105\mu m$. - Refer to the Figure 5-4 (a) and Figure 5-5 in Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper (# 726825). - Stripline configuration (I/O routing at Layer 4; inductor at Layer 1): - High-current inductor ($\geq 10A$ per phase) - recommend 3 2.2mm spacing from inductor centerline to keep noise coupling to $< 5mV$. - Low-current inductor ($< 10A$ per phase) - routing beneath inductor centerline is allowed with noise coupling $< 5mV$. - Assuming Layer 2 and Layer 3 are full solid planes (PWR/GND) without plane perforation; total metal thickness (L2+L3) $\geq 30\mu m$; dielectric thickness $\geq 160\mu m$.¹ - Refer to the Figure 5-4 (a) and Figure 5-6 in Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper (# 726825). - Stripline routing at Layer 5 and beyond is low risk. - Minimize I/O routing length under the inductor.
Victim Trace Separation from power /ground Via PTH	Allocate 3 5H spacing , where H is the dielectric thickness or height of the signal conductors above the nearest reference GND plane.
continued...	

Element	Component/Placement/Layout Recommendation
Victim Via Separation from inductor pad edge	Allocate 3 1mm spacing from inductor pad edge, in order to prevent inductor magnetic field leakage thru the via anti-pad voiding.
Victim Via Separation from power /ground Via PTH	Allocate 3 5mm spacing , in order to prevent via-to-via inductive coupling from power to power /ground Via victim net due to fast-switching current.
Victim Via	Avoid placing I/O micro vias and plated through hole vias under the power inductor.
Differential pair P-N skew	Keep differential P-N trace mismatch to <0.127mm (5mils) to minimize common-to-differential mode conversion.

NOTE

For systems using layer 3 for I/O routing, to minimize noise coupling, it may be necessary to route power or ground on layer 3 and move I/O to a deeper layer. .

3.10 RCOMP Recommendation

NOTE

RCOMP requirements are mandatory, specification defined and value should not be changed.

3.10.1 RCOMP

Figure 27. RCOMP Connection Diagram

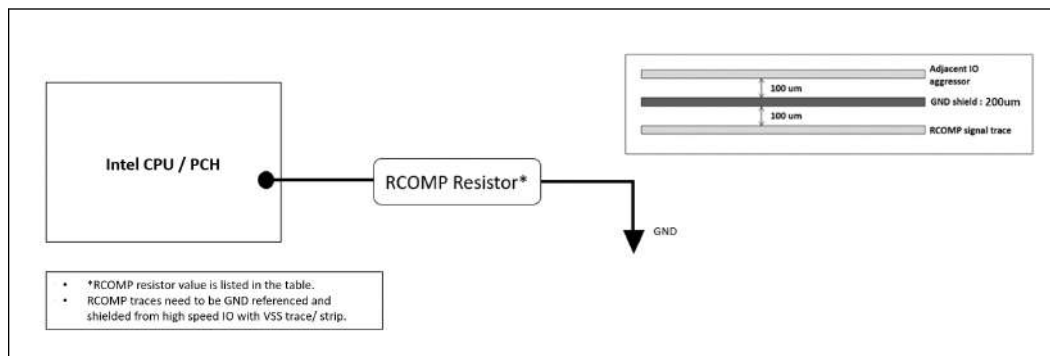
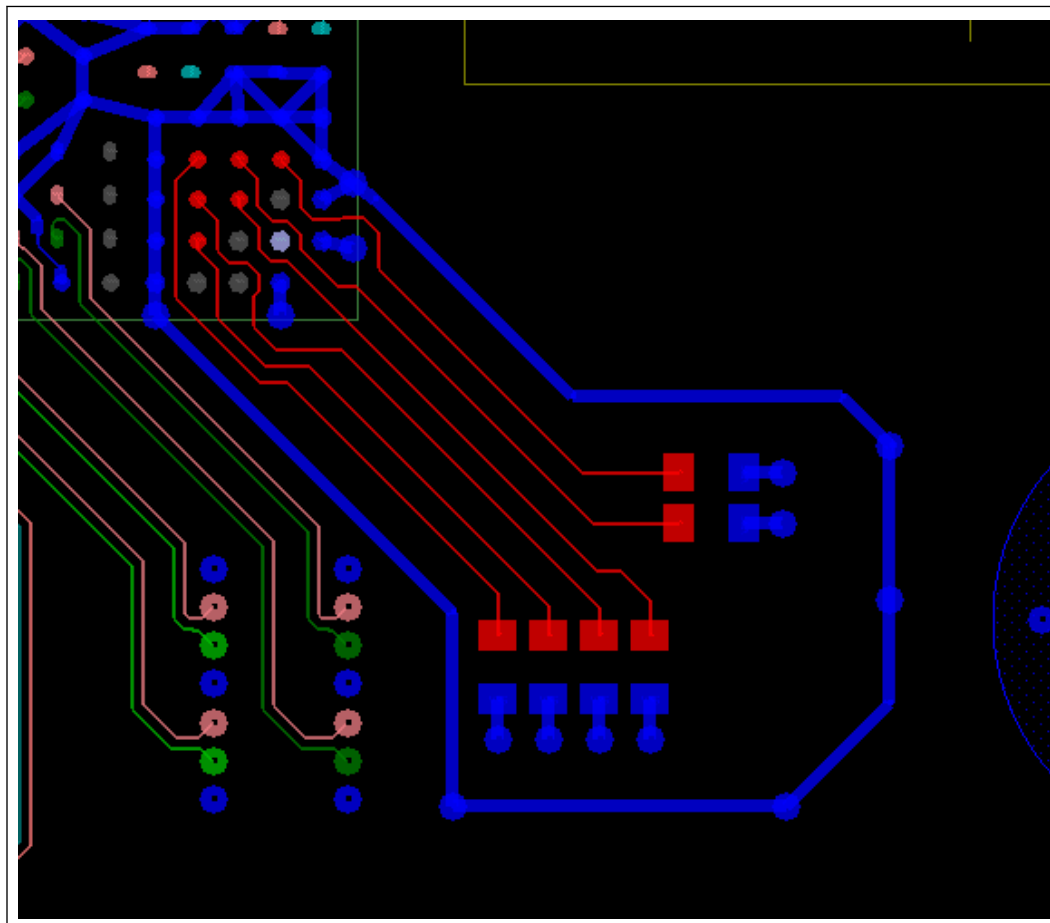


Figure 28. Example RCOMP with GND Shield**Table 74. RCOMP General Guidelines**

Detail
Trace width 75 μm with min trace spacing 100 μm .
Max total length on board = 9 mm. Max number of vias 1.
GND shield to any high speed IO with VSS trace/ strip. This is not required if the adjacent signal is another RCOMP signal. PTH vias are required to be punched from this GND shield to the referenced GND plane.
GND reference required. Noisy or switching references should be avoided.
Board external resistor size = 0201 (recommended).
Strongly recommended to follow reference layouts placement.

Table 75. RCOMP Configuration

Interface	Signal Name	Board RTerm (Ohm)	Max Board DC resistance (Ohm)	Tolerance (+ / -)%	Connection on other Side	Max Board Parasitic Capacitance (pF)	Resistor Stuffing	Routing Guide
Clock	CLK_S_RCOMP	180	0.1	1	GND	1.5	Mandatory	a) Unused RCOMP can be left as not connected. b) Refer to the RCOMP General Guidelines.
CNVi	CNV_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
Display	DDI_RCOMP	150	0.1	1	GND	1.5	Floating	a) Unused RCOMP can be left as not connected. b) Refer to the RCOMP General Guidelines.
CSI	CSI_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
PCIe	PCIe_A_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
PCIe	PCIe_B_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
eUSB2	eUSB2_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
USB3.2	USB32_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
UFS	UFS_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
DDR Interface	DDR_RCOMP	100	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
Soundwire	SNDW_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.

4.0 System Memory Interface Design Guidelines

Nova Lake mobile platform supports DDR5, LPCAMM2 and LPDDR5x memory technology. This chapter covers the guidelines to design a Nova Lake mobile platform with the following Plan of Record (POR) supported system memory interface configurations and features.

NOTE

For complete Nova Lake mobile processor system memory interface POR feature support details refer the latest release of the Nova Lake mobile Processor External Design Specification (EDS).

4.1 POR Memory Configurations

Table 76. POR Memory Configurations

Config#	Technology	Topology	Memory Device	Device Ballout	PCB Stack-up	Supported NVL SKU	Transfer Rate (MT/s)	IL/NIL
DDR5 SODIMM/CSODIMM 1DPC on page 129	DDR5	SODIMM/CSODIMM 1DPC B2B/SBS	Non-ECC SoDIMM/CSODIMM: A(1Rx8), B(2Rx8), C(1Rx16)	Non-ECC SODIMM/CSODIMM :262	T3/10L	8C, H16C 4Xe	8000	NIL
					T3/8L ⁸	6C	7200	
						8C, H16C 4Xe	8000	
LPDDR5x Memory Down Type 3	LPDDR5x	MD 4 x32	DDP QDP ODP	315	T3/10L	6C	1R/2R 7467 0.9mm T3	NIL
						8C, H16C 4Xe	1R/2R 8533 0.9mm T3	
							1R/2R 8533 0.8mm T3 ⁷	
LPCAMM2 Type 3 on page 145	LPDDR5x	LPCAMM2	DDP, QDP ODP	644	T3/8L	6C	1R 8533, 2R 7467	NIL

continued...

Config#	Technology	Topology	Memory Device	Device Ballout	PCB Stack-up	Supported NVL SKU	Transfer Rate (MT/s)	IL/NIL
						8C, H16C 4Xe	1R/2R 8533	
LPDDR5x Memory Down Type 4 on page 166	LPDDR5x	MD 4 x32	DDP QDP ODP	315	T4/10L	8C, H16C 4Xe	1R/2R 9600	NIL
Notes: 1. SBS - Side-by-side, DIMMs placed is side-by-side, in line placement. 2. B2B - Back-to-Back, DIMMs placed in parallel one to other. 3. SDP: Single Die Package, DDP: Dual Die Package , QDP: Quad Die Package, ODP: Octal Die Package 4. To avoid excessive latency due to ZQ topology, Intel specifies that each ZQ pin must have an unique ZQ resistor. 5. DDR5 SODIMM maximum speed is 5600. CSODIMM should be used above 5600. 6. DDR5 >6400 requires to define a main clock out of 2 clock differential pairs for each 32-bit channel. Refer to the Nova Lake DDR5 6400 CLK Setting Technical Advisory (# TBD) that provides guidance how to set the new BIOS policies for DDR5 >6400 CLK connectivity that is a different than RVP design. 7. Intel evaluate LPDDR5x 9600 with PCB Type 3 0.8mm, pending validation. 8. DDR5 CSODIMM 1DPC T3/8L 8000 pending validation.								

Table 77. POR Memory Signal List

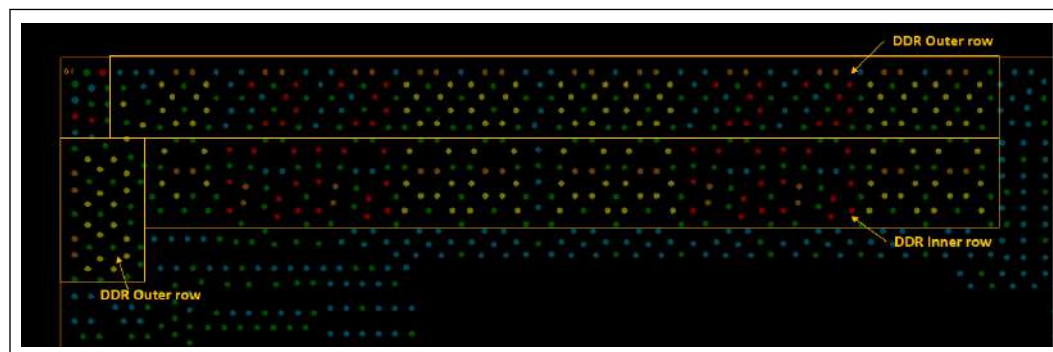
Processor and Memory Type	DDR5 CSODIMM Signal Name	LPDDR5/x Signal Name	LPCAMM2 Signal Name
Clock (CLK)	DDR[3:0]_CLK[3:0]_P DDR[3:0]_CLK[3:0]_N	DDR[7:0]_CLK_P DDR[7:0]_CLK_N	DDR[7:0]_CLK_P DDR[7:0]_CLK_N
WCK	NA	DDR[7:0]_WCK_[1:0]_P DDR[7:0]_WCK_[1:0]_N	DDR[7:0]_WCK_[1:0]_P DDR[7:0]_WCK_[1:0]_N
Control (CTRL)	DDR[0]_CS_[1:0] DDR[1]_CS_[1:0] DDR[2]_CS_[1:0] DDR[3]_CS_[1:0]	DDR_[7:0]_CS_[1:0]	DDR_[7:0]_CS_[1:0]
Command (CMD)	DDR[3:0]_CA_[12:0]	DDR_[7:0]_CA_[6:0]	DDR_[7:0]_CA_[6:0]
Strobe	DDR[3:0]_DQSP_[3:0] DDR[3:0]_DQSN_[3:0]	DDR[3:0]_DQSP_[1:0] DDR[3:0]_DQSN_[1:0] DDR[7:4]_DQSP_[1:0] DDR[7:4]_DQSN_[1:0]	DDR[3:0]_DQSP_[1:0] DDR[3:0]_DQSN_[1:0] DDR[7:4]_DQSP_[1:0] DDR[7:4]_DQSN_[1:0]
Data	DDR[1:0]_DQ_[3:0]_[7:0] DDR[3:2]_DQ_[3:0]_[7:0]	DDR[3:0]_DQ[1:0]_[7:0] DDR[7:4]_DQ[1:0]_[7:0]	DDR[3:0]_DQ[1:0]_[7:0] DDR[7:4]_DQ[1:0]_[7:0]
Reset	DRAM_RESET#	DRAM_RESET#	DRAM_RESET#
RCOMP	DDR_RCOMP	DDR_RCOMP	DDR_RCOMP

NOTE

DDR memory signals should be routed the same as the RVP design. For more details, refer to the 854352_NVL_UH_SCHCHK PDF in the zip file, from Rev 0.7 onwards.

4.2 DDR Outer and Inner Row Signals Package

The routing from the DDR to the package will be according to the following two rows: DDR outer signals and inner row signals.

Figure 29. DDR Outer and Inner Row Signals**NOTE**

This image is a reference, not the actual signals pin map.

4.3 DDR5 SODIMM/CSODIMM 1DPC

The topology "SODIMM 1DPC Type-3,0.9mm,10L" is supported for NVL 8C/H16C only.

Table 78. DDR5 SODIMM/CSODIMM Type-3/ Topology Guidelines

Parameter	Details
Processor	NVL 8C, H16C 4Xe
Memory Type	DDR5
Configuration	Channel A (x64) = One SODIMM/CSODIMM Channel B (x64) = One SODIMM/CSODIMM
Speed (MT/s)	8000
Channels	1 and 2
System Memory Voltage	DDR5 SODIMM/CSODIMM V _{bulk} is 5V SOC VDD2/VDDQ is 1.1V
Max Ranks Per Channel	2
DRAM Die Density (Gb)	16, 24 , 32
Max Capacity (GB)	128
Memory Types	SODIMM/CSODIMM 262 Balls RC-A, F(1RX8), B, G (2RX8), and C, H(1RX16)
PCB Layers/Type ¹	10L/Type-3 8L/Type-3 ⁷
DRAM Device Placement	Board can be single sided or dual sided, DIMMs are Back to Back Placement /Side-by-Side placement

continued...

Parameter	Details
Processor Memory Ball Map ²	Non-Interleaved
DRAM Device ODT Capability ³	Enabled

Notes:

1. Type-3 (T3)= PCB with zero build-up layers using Plated Through-hole (PTH) Vias with no Micro-vias.
2. Non-Interleave = The processor is optimized for a side-by-side ball map and placement of the DRAM memory down devices placement. Each memory channel's signals are grouped together on one side of the processor packages.
3. The DRAM Device ODT Capability can be enabled or disabled for reads and writes depending on the system memory interface margins. Enabling the DRAM device ODT increases the memory interface power and improves signal integrity/margins while disabling the DRAM device ODT decreases the memory interface power but degrades signal integrity/margins.
4. DDR5 SODIMM maximum speed is 5600. CSODIMM should be used above 5600.
5. Pending DRAM samples availability.
6. DDR5 strategy :
 - a. Intel NVL DDR5 validation plan includes 8000, 7200 and 6400 1bnm/1cnm memory or newer
 - b. Intel has no plans to validate DDR5 6400 1anm and 5600 1bnm/1cnm
 - c. Intel will support customer debug reactively. No plans to validate or support debug for 5600 1anm
7. DDR5 CSODIMM 1DPC T3/8L 8000 pending validation.
8. DDR POR speed refers to Processor top SKU. Other SKUs may use lower memory speed, refer to the ark.intel.com for top memory speed.

NOTE

DDR5 >6400 requires to define a main clock out of 2 clock Differential pairs for each 32-bit channel. Refer to the Nova Lake DDR5 >6400 CLK Setting Technical Advisory (#TBD) that provides guidance how to set the new BIOS policies for DDR5 >6400 CLK connectivity that is a different than RVP design.

4.3.1 SODIMM 1DPC Type-3,0.8mm,8L

Pcb Type & Thickness: Type-3, 0.8mm

Figure 30. DDR5 SODIMM Type-3 8L Signals Topologies

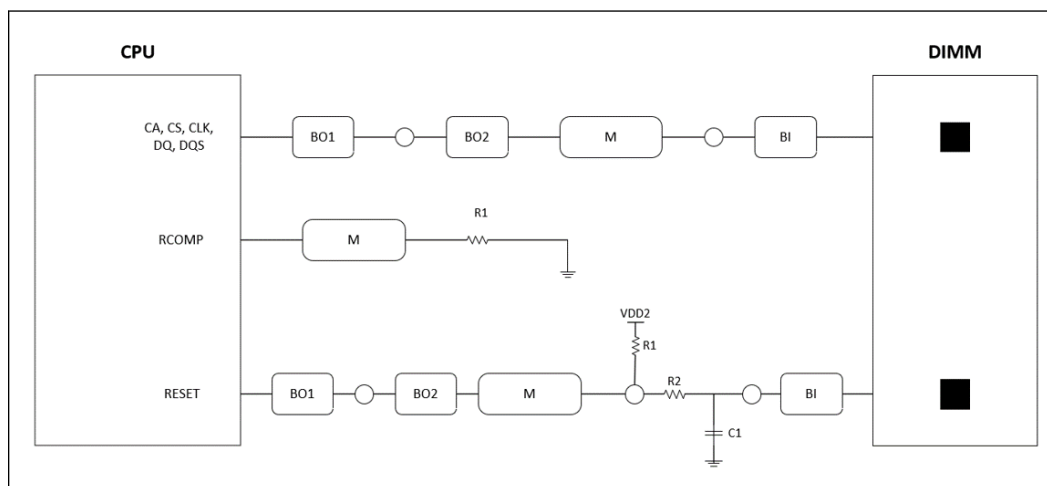


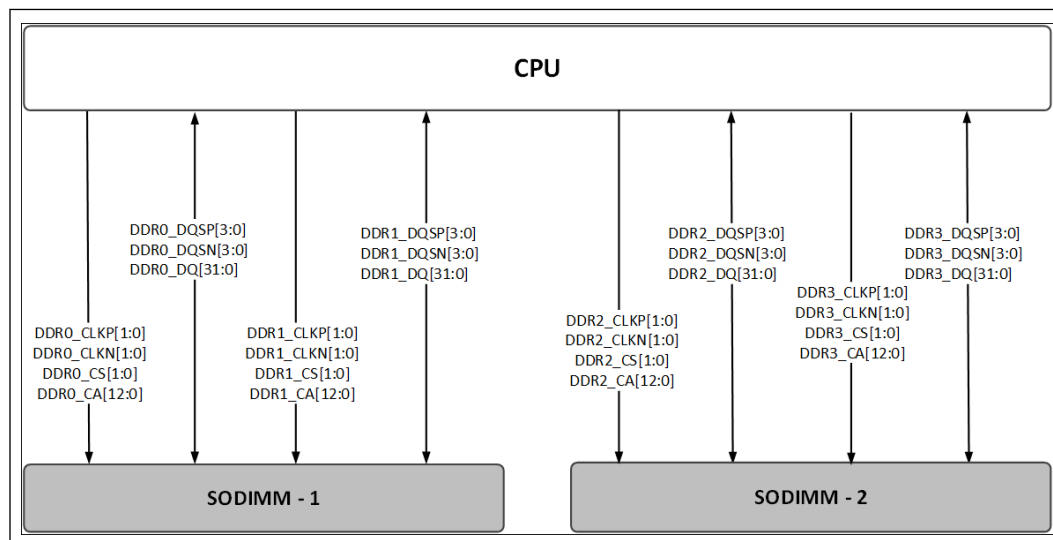
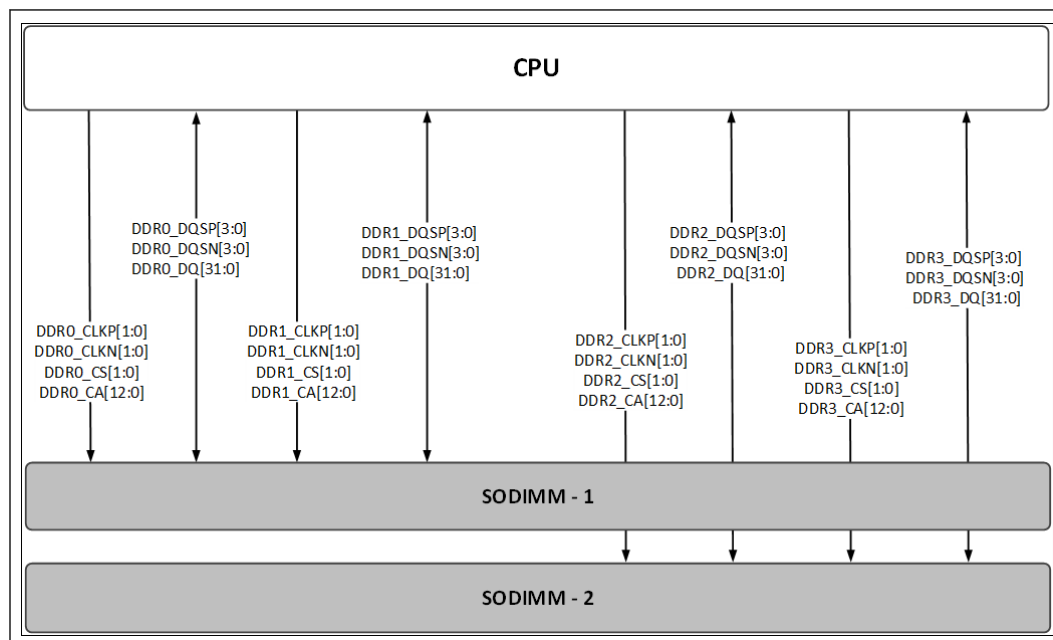
Figure 31. DDR5 SODIMM Side-by-Side Placement Block Diagram

Figure 32. DDR5 SODIMM Back-to-Back Placement Block Diagram


Figure 33. DDR5 SODIMM 1DPC Top-Pin Bottom-Pin Definition and Compensation

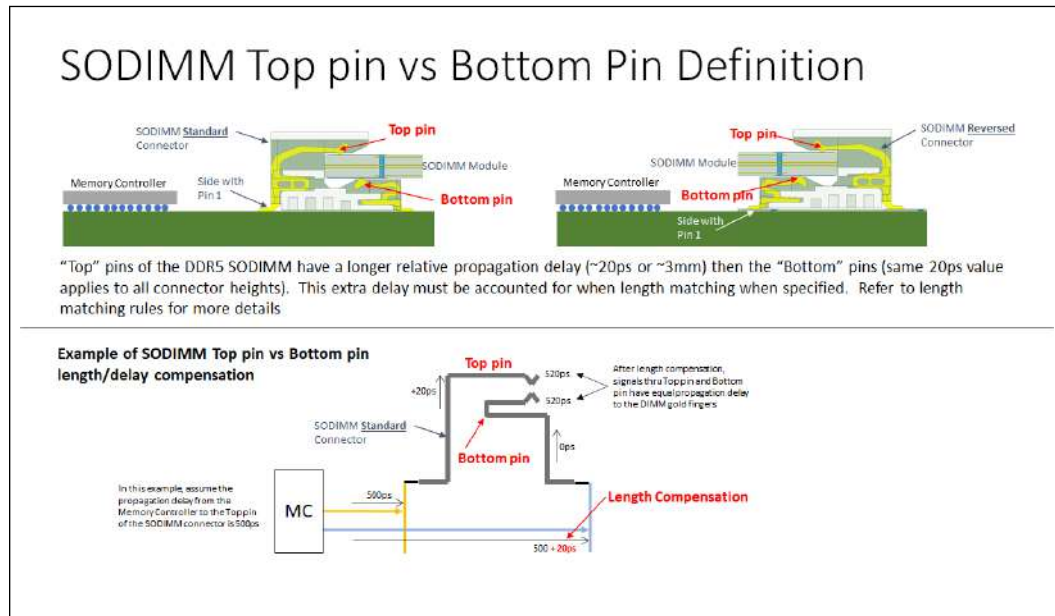


Table 79. SODIMM 1DPC Type-3,0.8mm,8L Notes

Note	Detail
PCB stackup	Type-3 stackup with minimum of 8 layer.
Layer assignment rules	Follow the reference design (RVP).
Reference plane	Continuous ground only.
Number of vias allowed	Max 2 transition vias.
Ground stitching	For VSS stitching using microvia, target a minimum S:G ratio of 1:1 at signal layer transition. For VSS stitching using PTH, target a minimum S:G ratio of 2:1 at signal layer transition. VSS via stitching pattern to be copied exactly from RVP / clip files.
Max MS BO/ BI length	Refer to RFI/ EMC guidelines for the max microstrip BO/ BI length requirement.
Serpentine routing	Use 3x of dielectric height for serpentine routing spacing.
Bit/ Byte swapping rule	Bit swapping is allowed within each Byte for all memory technologies. DDR5: Byte Swapping is allowed within each x32 channel.
Diff pair length matching	Match diff pair P-N within 0.2 mm at each layer.
MS BO and BI segment, PTH pattern	Keep DDR MS BO and BI as close as possible to reference board.
DDR to Non-DDR trace spacing	Minimum 500 um.
Unused signals	DDR signals on processor which are not applicable for DDR5 can be left unconnected
Rtt/ Ctt for RESET	Keep R1 empty; R2 = 0 Ohm; C1 = 0.1 uF (no stuff).
RCOMP value	100 Ohm

Table 80. Max number of vias

Signal Group	Max Number of Vias Allowed
CA/CS, CLK, DQ, DQS	2
RCOMP	1
ALERT	2

Length Matching**Table 81. SODIMM 1DPC Type-3,0.8mm,8L Length Matching**

Signal Group	Plus/Minus	Reference Group	Within/Group	Up to memory/connector	Length/Delay	Min/Max/Range	Min	Max
CLK_P	-	DQS	Channel (x32)	Same Connector	Length (mm)	Range	-38	63.5
DQ	-	DQS	Byte (x8)	Same Connector	Length (mm)	Range	-0.5	0.5
DQS	-	DQS	Pair (P-N)	Same Connector	Length (mm)	Max		0.125
CLK_P	-	CLK_N	Pair (P-N)	Same Connector	Length (mm)	Max		0.125
CLK_P	-	CLK_P	Channel (x32)	Same Connector	Length (mm)	Max		0.25
CLK_P	-	CS	Rank	Same Connector	Length (mm)	Range	-2.5	0
CS	-	CS	Channel (x32)	Same Connector	Length (mm)	Max		1.25
CLK_P	-	CA	Channel (x32)	Same Connector	Length (mm)	Range	-6	6
CA	-	CA	Channel (x32)	Same Connector	Length (mm)	Max		1.25
CLK	-	CLK	Across-channel	Same Connector	Length (mm)	Max		10

Mainstream, Mid Loss (ML), CA/CS, CH-1/CH-2, Inner**Table 82. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CA/CS, CH-1/CH-2, Inner Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.5
BO2	SL	20
M	SL	
BI	MS	4.5

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), CA/CS, CH-1/CH-2, Outer

Table 83. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CA/CS, CH-1/CH-2, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	4.5
BO2	SL	11
M	SL	
BI	MS	4.5

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), CA/CS, CH-0/CH-3, Inner

Table 84. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CA/CS, CH-0/CH-3, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.5
BO2	SL	7.5
M	SL	
BI	MS	4

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), CA/CS, CH-0/CH-3, Outer

Table 85. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CA/CS, CH-0/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	4.5
BO2	SL	5.6
M	SL	
BI	MS	3.5

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), CLK, CH-1/CH-2, Inner

Table 86. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CLK, CH-1/CH-2, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.5
BO2	SL	16
M	SL	
BI	MS	3

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), CLK, CH-1/CH-2, Outer

Table 87. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CLK, CH-1/CH-2, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	6.2	
BO2	SL	7	
M	SL		
BI	MS	3	

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), CLK, CH-0/CH-3, Inner

Table 88. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CLK, CH-0/CH-3, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.5
BO2	SL	7.7
M	SL	
BI	MS	1.5

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), CLK, CH-0/CH-3, Outer

Table 89. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), CLK, CH-0/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	6.2
BO2	SL	2
M	SL	
BI	MS	3

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), DQ, CH-0/CH-1/CH-2/CH-3, Inner

Table 90. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), DQ, CH-0/CH-1/CH-2/CH-3, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.6
BO2	SL	14
M	SL	
BI	MS	2

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), DQ, CH-0/CH-1/CH-2/CH-3, Outer

Table 91. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), DQ, CH-0/CH-1/CH-2/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	5.5
BO2	SL	11
M	SL	
BI	MS	2

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), DQS, CH-0/CH-1/CH-2/CH-3, Inner

Table 92. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), DQS, CH-0/CH-1/CH-2/CH-3, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.6
BO2	SL	14
M	SL	
BI	MS	2

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), DQS, CH-0/CH-1/CH-2/CH-3, Outer

Table 93. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), DQS, CH-0/CH-1/CH-2/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	8
BO2	SL	4
M	SL	
BI	MS	2

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), ALERT, MISC, All

Table 94. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), ALERT, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	7
BO2	MS	10.2
M	SL	
BI	MS	2.5

Max Length Total (mm): 115

Max Length Total Note: Max Total Length is inclusive of Package Length

Mainstream, Mid Loss (ML), RESET, MISC, All

Table 95. SODIMM 1DPC Type-3,0.8mm,8L Mainstream, Mid Loss (ML), RESET, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	13
BO2	SL	25
M	SL	
BI	MS	10

Max Length Total (mm): 200

Max Length Total Note: Max Total Length is inclusive of Package Length

4.3.2 SODIMM 1DPC Type-3,0.9mm,10L

Pcb Type & Thickness: Type-3, 0.9mm

Figure 34. DDR5 SODIMM Type-3 10L Signals Topologies

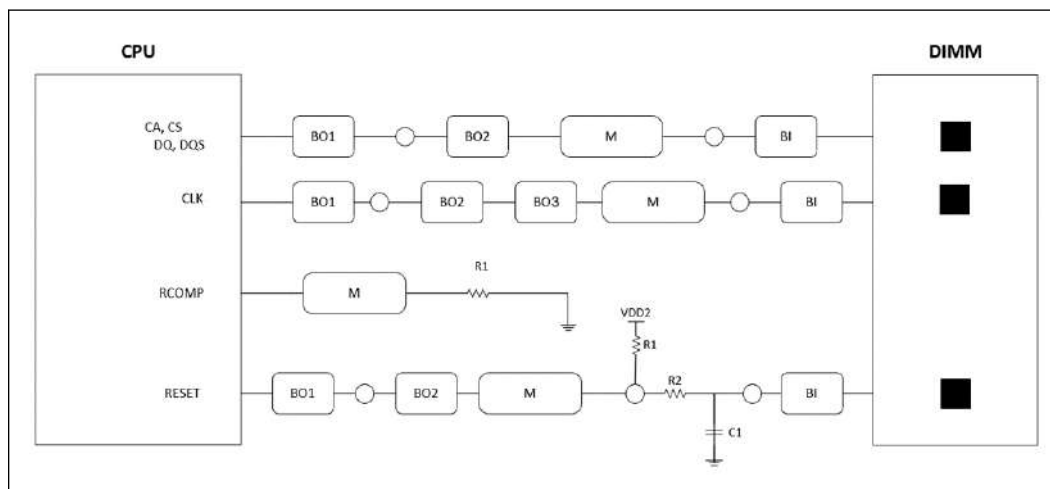


Figure 35. DDR5 SODIMM Back-to-Back Placement Block Diagram

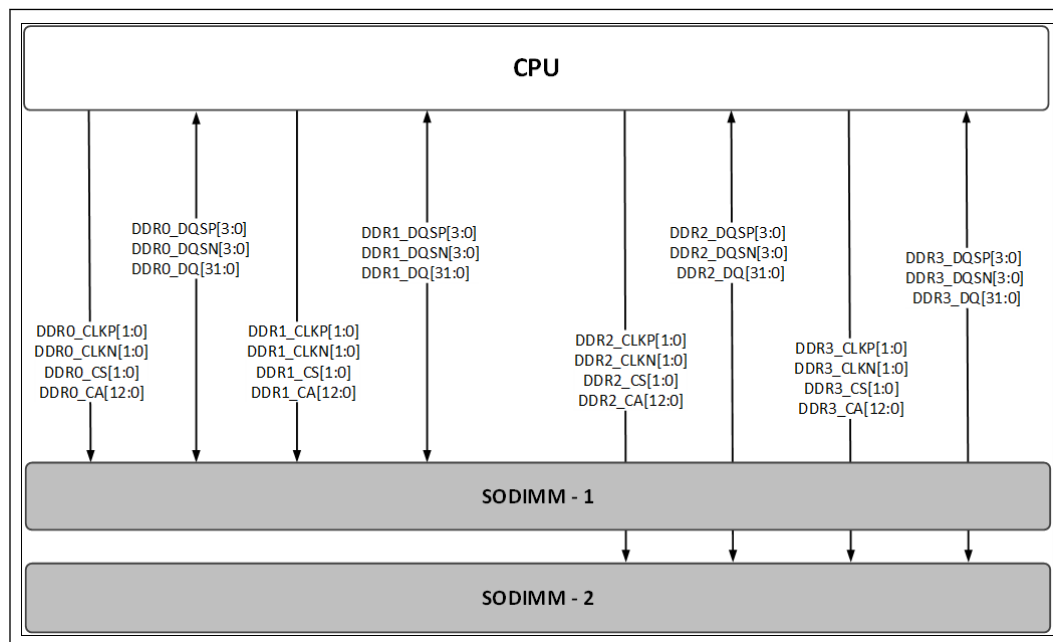


Figure 36. DDR5 SODIMM Side-by-Side Placement Block Diagram

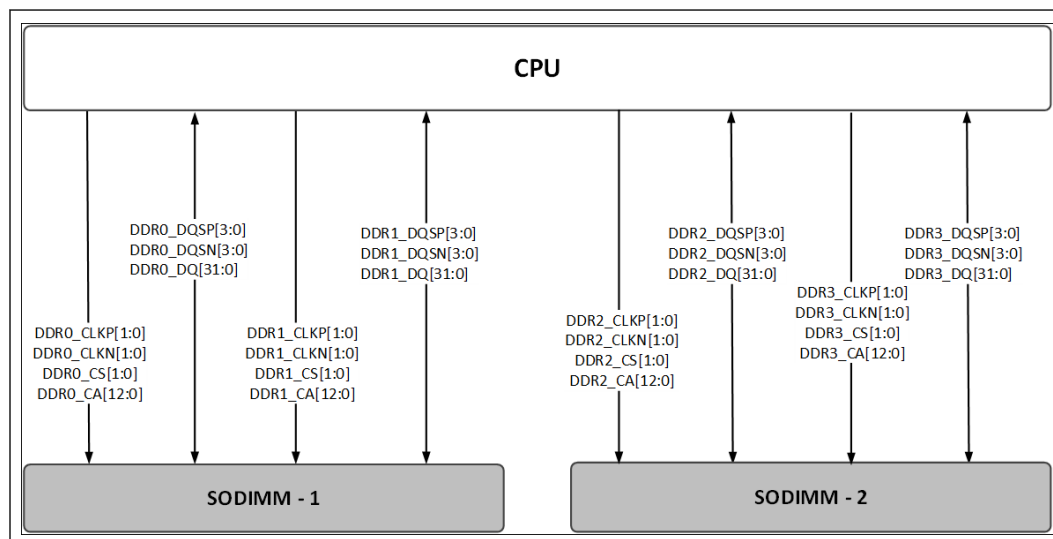


Figure 37. DDR5 SODIMM 1DPC Top-Pin Bottom-Pin Definition and Compensation

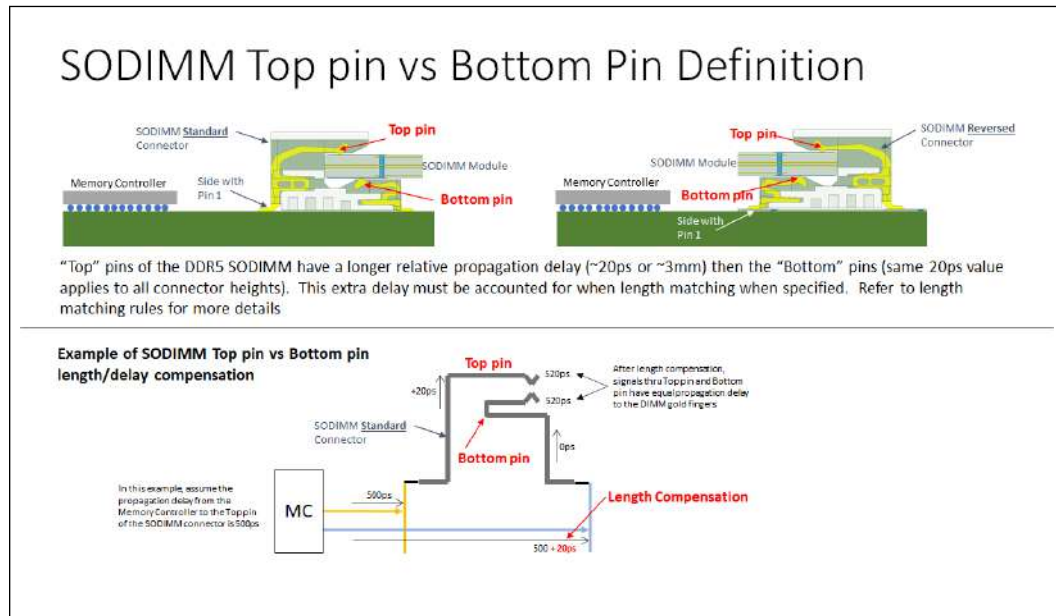


Table 96. SODIMM 1DPC Type-3,0.9mm,10L Notes

Note	Detail
PCB stackup	Type-3 stackup with minimum of 10 layer.
Layer assignment rules	Follow the reference design (RVP).
Reference plane	Continuous ground only.
Number of vias allowed	DQ/ DQS/ CA/ CS/ CLK: Max 2 vias. RESET signal may have max 5 via.
Ground stitching	For VSS stitching using microvia, target a minimum S:G ratio of 1:1 at signal layer transition. For VSS stitching using PTH, target a minimum S:G ratio of 2:1 at signal layer transition. VSS via stitching pattern to be copied exactly from RVP / clip files.
Max MS BO/ BI length	Refer to RFI/ EMC guidelines for the max microstrip BO/ BI length requirement.
Serpentine routing	Use 3x of dielectric height for serpentine routing spacing.
Bit/ Byte swapping rule	Bit swapping is allowed within each Byte for all memory technologies. DDR5: Byte Swapping is allowed within each x32 channel.
MS BO and BI segment, PTH pattern	Keep DDR MS BO and BI as close as possible to reference board.
DDR to Non-DDR trace spacing	Minimum 500 um.
Unused signals	DDR signals on CPU which are not applicable for DDR5 can be left unconnected

continued...

Note	Detail
Rtt/ Ctt for RESET	Keep R1 empty; R2 = 0 Ohm; C1 = 0.1 uF (no stuff).
RCOMP value	100 Ohm
CLK-to-CA/CS spacing	On L3 & L5, CLK & CA/CS may have a BO3 segment with CLK-to-CAC spacing of 375um after BO2

Table 97. Max number of vias

Signal Group	Max Number of Vias Allowed
CA, CLK, CS, DQ, DQS	2
RCOMP	1
ALERT	2
RESET	5

Length Matching**Table 98. SODIMM 1DPC Type-3,0.9mm,10L Length Matching**

Signal Group	Plus/Minus	Reference Group	Within/Group	Up to memory /connector	Length/Delay	Min/Max/Range	Min	Max
CLK_P	-	DQS	Channel (x32)	Same Connector	Length (mm)	Range	-38	63.5
DQ	-	DQS	Byte (x8)	Same Connector	Length (mm)	Range	-0.5	0.5
DQS	-	DQS	Pair (P-N)	Same Connector	Length (mm)	Max		0.125
CLK_P	-	CLK_N	Pair (P-N)	Same Connector	Length (mm)	Max		0.125
CLK_P	-	CLK_P	Channel (x32)	Same Connector	Length (mm)	Max		0.25
CLK_P	-	CS	Rank	Same Connector	Length (mm)	Range	-2.5	0
CS	-	CS	Channel (x32)	Same Connector	Length (mm)	Max		1.25
CLK_P	-	CA	Channel (x32)	Same Connector	Length (mm)	Range	-6	6
CA	-	CA	Channel (x32)	Same Connector	Length (mm)	Max		1.25
CLK	-	CLK	Across-channel	Same Connector	Length (mm)	Max		10

Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-1/CH-2/CH-3, Inner**Table 99. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-1/CH-2/CH-3, Inner Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.6	
BO2	SL	19.8	Total BO1 + BO2 should be less than 20.2mm
M	SL		
BI	MS	3.5	

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-1/CH-2/CH-3, Outer

Table 100. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-1/CH-2/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	6.2	
BO2	SL	10.4	Total BO1 + BO2 should be less than 13mm
M	SL		
BI	MS	3	

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-1/CH-2/CH-3, Inner

Table 101. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-1/CH-2/CH-3, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.5
BO2	SL	16
M	SL	
BI	MS	3

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-1/CH-2/CH-3, Outer

Table 102. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-1/CH-2/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	6.2	
BO2	SL	5.1	Total BO1 + BO2 should be less than 10.2mm
M	SL		
BI	MS	3	

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-1/CH-2/CH-3, Inner
Table 103. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-1/CH-2/CH-3, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.6	
BO2	SL	15.8	Total BO1 + BO2 should be less than 16.2mm
M	SL		
BI	MS	1.5	

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-1/CH-2/CH-3, Outer
Table 104. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-1/CH-2/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	5.5	
BO2	SL	11.4	Total BO1 + BO2 should be less than 15mm
M	SL		
BI	MS	1.5	

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), DQS, CH-0/CH-1/CH-2/CH-3, Inner
Table 105. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), DQS, CH-0/CH-1/CH-2/CH-3, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	0.6
BO2	SL	14.3
M	SL	
BI	MS	1.2

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), DQS, CH-0/CH-1/CH-2/CH-3, Outer

Table 106. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), DQS, CH-0/CH-1/CH-2/CH-3, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	8	
BO2	SL	4.9	Total BO1 + BO2 should be less than 11.8mm
M	SL		
BI	MS	1.5	

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), ALERT, MISC, All

Table 107. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), ALERT, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	7	
BO2	MS	10.2	
M	SL		
BI	MS	2.5	

Max Length Total (mm): 115

Max Length Total Note: Max total length is inclusive of package length.

Mainstream, Premium Mid Loss (PML), RESET, MISC, All

Table 108. SODIMM 1DPC Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), RESET, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	3
BO2	MS	35
M	SL	
BI	MS	3

Max Length Total (mm): 200

Max Length Total Note: Max total length is inclusive of package length.

4.4 LPDDR5x Topologies

4.4.1 LPCAMM2 Type 3

Table 109. System Memory Configuration Details Covered in this Section

Parameter	Details
Processor	NVL-H
Memory Type	LPCAMM2
Speed (MT/s)	1R/2R 8533 DDP, QDP, ODP (byte mode)
Channels	Channels: 0-7
System Memory Voltage	5V input to LPCAMM2 Module SoC VDDQ is 0.52V and VDD2 is 1.065V
Max Ranks Per Channel	2
DRAM Die Density (Gb)	16, 24
Max Capacity (GB)	96GB
PCB Layers/Type	10L/Type-3
DRAM Device Placement	Top Motherboard Layer, Placement
Processor Memory Ball Map ³	Non-Interleaved
DRAM Device ODT Capability	MRC trained
<i>Notes:</i> 1. DDP: Dual Die Package, QDP: Quad Die Package, ODP: Octal Die Package 2. No mixed vendor support within a channel or channel to channel and No mixed memory DRAM down type support (SDP, DDP, QDP) within a channel or channel to channel 3. Non-Interleave = The processor is optimized for a side-by-side ball map and placement of the DRAM memory down devices placement. Each memory channel's signals are grouped together on one side of the processor packages. 4. To avoid excessive latency due to ZQ topology, Intel specifies that each ZQ pin must have an unique ZQ resistor. 5. Intel recommends to follow the latest version LPCAMM2 connector CAD symbol from JEDEC Spec. 6. Pending LPCAMM2 availability modules and eco system readiness. 7. LPCAMM2 High Cap is not supported in NVL. 8. Refer to the LPCAMM validation strategy Nova Lake Platform Memory Enablement Guide (# 858687).	

4.4.1.1 CAMM2 x128 Type-3,0.8mm,8L

Pcb Type & Thickness: Type-3, 0.8mm

Figure 38. LPCAMM2 x128 Type-3 PCB DQ, DQS, CA, CS, CLK Topology Diagram

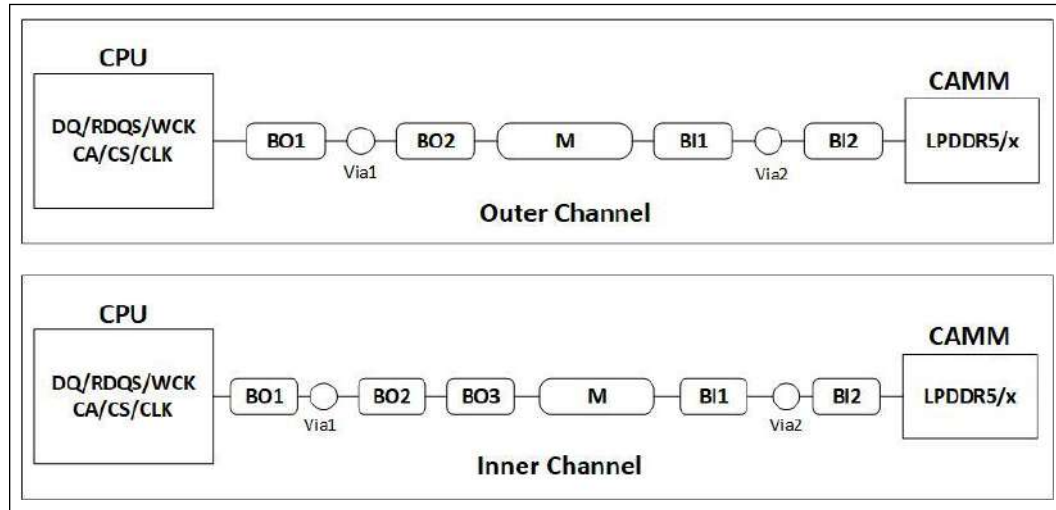


Figure 39. LPCAMM2 x128 Type-3 Reset and RCOMP Topology Diagram

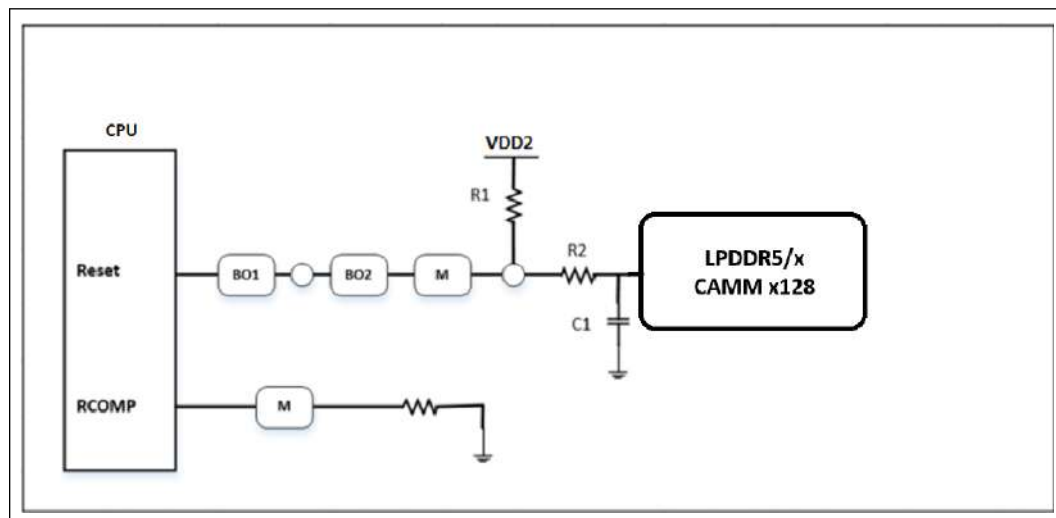
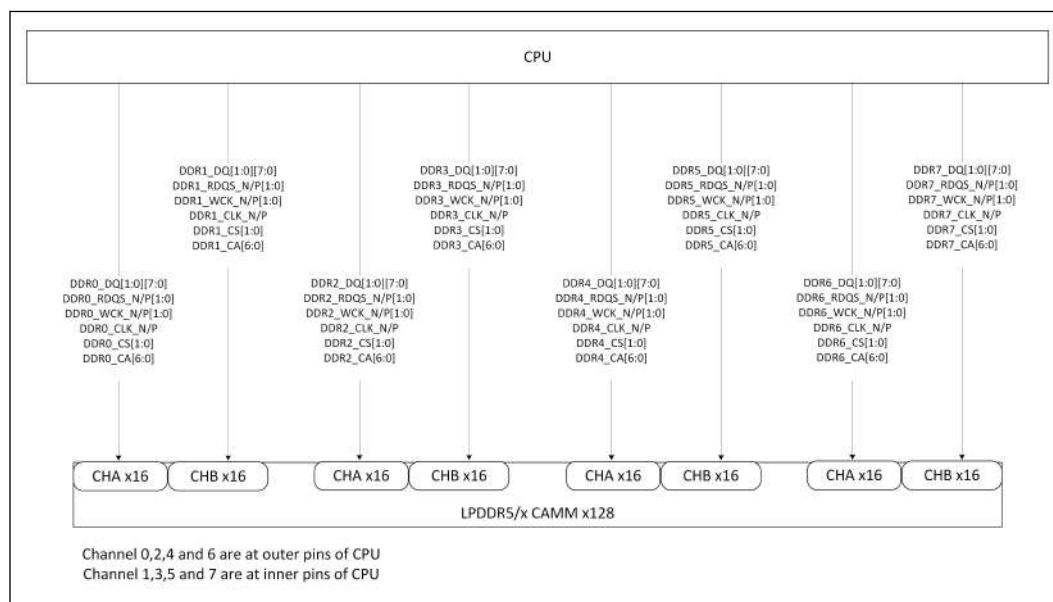


Figure 40. LPCAMM2 x128 Type-2 CAMM Placement and Block Diagram**Table 110. CAMM2 x128 Type-3,0.8mm,8L Notes**

Note	Detail
LPCAMM2 connector	Intel recommends to follow the latest version LPCAMM2 connector CAD symbol from JEDEC Spec.
PCB stackup	Type-3 stackup with minimum of 8 layers.
Layer assignment rules	Outer Row DQ signals on layer3 and Inner Row DQ signals on Layer 7. CAC Signals in layer 5.
Reference plane	Continuous ground only.
Number of vias allowed	DQ/ RDQS/WCK CA/ CS/ CLK: Max 2 vias,
GND stitching	For VSS stitching using microvia, target a minimum S:G ratio of 1:1 at signal layer transition. For VSS stitching using PTH, target a minimum S:G ratio of 2:1 at signal layer transition. VSS via stitching pattern to be copied exactly from RVP/ clip files.
Max MS BO/ BI length	Refer to RFI/ EMC guidelines for the maximum microstrip BO/ BI length requirement.
Serpentine routing	Use 3x of dielectric height for serpentine routing spacing.
Bit/ Byte swapping rule	Bit swapping is allowed within each Byte for all memory technologies. Byte swapping is allowed within a 16-bit channel. LPDDR5: x16 sub-channels can be swizzled within their x64 MC. When swapping channel DQ, the CA/CS signals must be swapped as well.
Unused signals	Leave unused DDR signals (PARITY, extra CA) floating (no connect). ALERT is also unused and can be left floating (no connect) at processor side. CS_2,3 signals to be terminated with pull up resistor 10Kohm to VDD2H_CPU. RFU P12 should be connected to GND. Pin A28 LPCAMM2 to be terminated with pull up resistor 22Kohm to 1.8V.
Rtt/ Ctt for RESET	Keep R1 empty; R2 = 0 Ohm; C1 = 0.1 uF (no stuff).
RCOMP value	100 Ohm

Table 111. Max number of vias

Signal Group	Max Number of Vias Allowed
CA, CLK, CS, DQ, RDQS, WCK	2
RCOMP	1

Length Matching
Table 112. CAMM2 x128 Type-3,0.8mm,8L Length Matching

Signal Group	Plus/Minus	Reference Group	Within/Group	Up to memory/connector	Length/Delay	Min/Max/Range	Min	Max
DQ	-	WCK	Byte (x8)	Same DRAM	Length (mm)	Range	-12	14
DQ	-	RDQS	Byte (x8)	Same DRAM	Length (mm)	Range	-3	3
WCK	-	CLK	Channel (x16)	Same DRAM	Length (mm)	Range	-5	10
DQ	-	DQ	Byte (x8)	Same DRAM	Length (mm)	Max		4
CLK	-	CA	Channel (x16)	Same DRAM	Length (mm)	Range	-5	5
CA	-	CA	Channel (x16)	Same DRAM	Length (mm)	Max		7.5
CLK	-	CS	Channel (x16)	Same DRAM	Length (mm)	Range	-12.5	12.5
CS	-	CS	Channel (x16)	Same DRAM	Length (mm)	Max		10
RDQS_P	-	RDQS_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
WCK_P	-	WCK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK_P	-	CLK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK	-	CLK	All Channels	Within DRAM	Length (mm)	Max		25

Mainstream, CA/CS, CH-0/CH-2/CH-4/CH-6, Outer
Table 113. CAMM2 x128 Type-3,0.8mm,8L Mainstream, CA/CS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	8.5	
BO2	SL		14.5 - BO1
M	SL		
BI	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, CA/CS, CH-1/CH-3/CH-5/CH-7, Inner

Table 114. CAMM2 x128 Type-3,0.8mm,8L Mainstream, CA/CS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL		9.5 - BO1
M	SL		
BI1	SL	3.2	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, CLK, CH-0/CH-2/CH-4/CH-6, Outer

Table 115. CAMM2 x128 Type-3,0.8mm,8L Mainstream, CLK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO1	MS	9
M	SL	
BI	MS	0.6

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, CLK, CH-1/CH-3/CH-5/CH-7, Inner

Table 116. CAMM2 x128 Type-3,0.8mm,8L Mainstream, CLK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL		5.5 - BO1
M	SL		
BI	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, DQ, CH-0/CH-2/CH-4/CH-6, Outer

Table 117. CAMM2 x128 Type-3,0.8mm,8L Mainstream, DQ, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	3.5	
BO2	SL		10 - BO1
M	SL		
BI	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, DQ, CH-1/CH-3/CH-5/CH-7, Inner

Table 118. CAMM2 x128 Type-3,0.8mm,8L Mainstream, DQ, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	6.7	
BO3	SL		13 - BO2
M	SL		
BI	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, RDQS, CH-0/CH-2/CH-4/CH-6, Outer

Table 119. CAMM2 x128 Type-3,0.8mm,8L Mainstream, RDQS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	8	
BO2	SL		10 - BO1
M	SL		
BI1	SL	11.5	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, RDQS, CH-1/CH-3/CH-5/CH-7, Inner

Table 120. CAMM2 x128 Type-3,0.8mm,8L Mainstream, RDQS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL		6.5 - BO1
M	SL		
BI1	SL	10	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, WCK, CH-0/CH-2/CH-4/CH-6, Outer

Table 121. CAMM2 x128 Type-3,0.8mm,8L Mainstream, WCK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	7	
BO2	SL		11 - BO1
M	SL		
BI1	SL	9	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, WCK, CH-1/CH-3/CH-5/CH-7, Inner

Table 122. CAMM2 x128 Type-3,0.8mm,8L Mainstream, WCK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	6	
BO3	SL		12 - BO2
M	SL		
BI1	SL	8.5	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Mid Loss (ML), RESET, MISC, All

Table 123. CAMM2 x128 Type-3,0.8mm,8L Mainstream, Mid Loss (ML), RESET, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	40	
BO2	SL		21 - BO1
M	SL		
BI1	MS		

Max Length Total (mm): 200

4.4.2 LPDDR5 Memory Down x32 Type 3

The topology "MD x32 Type-3,0.8mm,10L" is supported for **NVL 8C/H16C** only.

Table 124. System Memory Configuration Details Covered in this Section

Parameter	Details
Processor	NVL 8C, H16C 4Xe, H16C 12Xe
Memory Type	LPDDR5x x32
Speed (MT/s)	1R/2R 8533; 0.9mm T3 Stack-up 1R/2R 8533; 0.8mm T3 Stack-up ⁸
Channels	Channels: 0-7
Population Rules	2 DRAMs: - Use either channels 0 to 3 or 4 to 7 - From the above, connect any x16 channel to any DRAM 3 DRAMs: N/A 4 DRAMs: - Connect any x16 channel to any DRAM
System Memory Voltage	LPDDR5x DRAM VDDQ voltage is 0.52V, VDD2 is 1.065V
Max Ranks Per Channel	2
DRAM Die Density (Gb)	12, 16, 24 ⁹
Max Capacity (GB)	96GB
Memory Down Types ^{1,2} (Pkg Ranks - Die Bits - Pkg bits)	315 BGA DDP: 1-16-32 QDP: 2-16-32
PCB Layers / Type ³	10L/Type-3
DRAM Device Placement	Top Motherboard Layer, Single Sided for Processor and DRAM, Side by Side Placement
<i>continued...</i>	

Parameter	Details
Processor Memory Ball Map ⁴	Non Interleaved
DRAM Device ODT Capability ^{5,6}	MRC trained
<i>Notes:</i> 1. DDP: Dual Die Package, QDP: Quad Die Package, ODP: Octal Die Package 2. No mixed vendor support within a channel or channel to channel and No mixed memory DRAM down type support (SDP, DDP, QDP) within a channel or channel to channel 3. Non-Interleave = The processor is optimized for a side-by-side ball map and placement of the DRAM memory down devices placement. Each memory channel's signals are grouped together on one side of the processor packages. 4. The DRAM Device ODT will be trained by the MRC. 5. For LPDDR5x, to avoid excessive latency due to ZQ topology, Intel specifies that each ZQ pin must have a unique ZQ resistor. 6. LPDDR/x x16 sub-channels can be swizzle within their x64 Memory controller. When swapping channel DQ the CMD/CRTL signals must be swapper as well. 7. To avoid excessive latency due to ZQ topology, Intel specifies that each ZQ pin must have an unique ZQ resistor. 8. Intel evaluate LPDDR5x 9600 with PCB Type 3 0.8mm, pending validation 9. Pending DRAM samples availability and eco system readiness. 10. LPDDR5x Type 3 Thin PCB 0.8mm support in NVL-H16C 4Xe	

4.4.2.1 MD x32 Type-3,0.8mm,10L

Pcb Type & Thickness: Type-3, 0.8mm

Figure 41. LPDDR5/x x32 Type-3 PCB DQ, DQS, CA, CS, CLK Topology Diagram

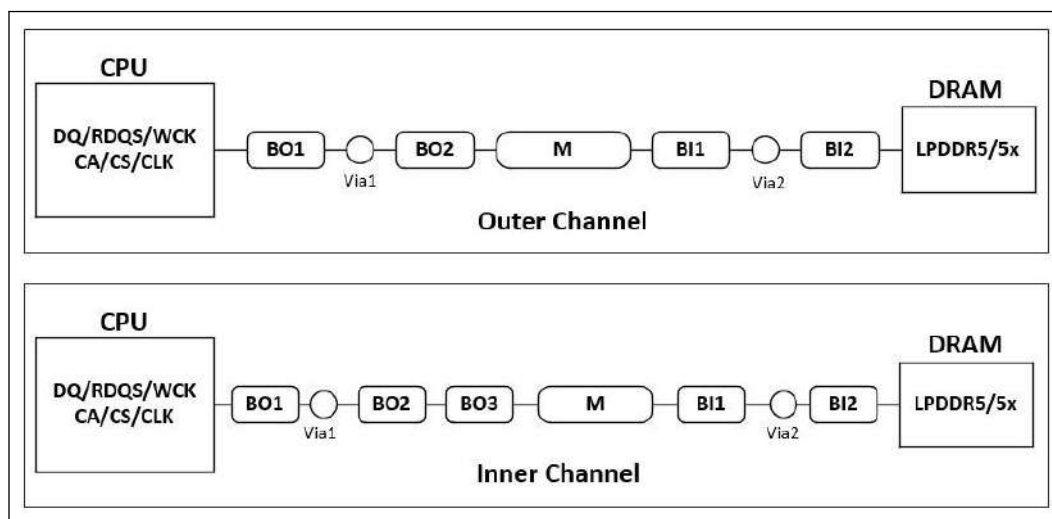


Figure 42. LPDDR5/x x32 Type-3 Reset and RCOMP Topology Diagram

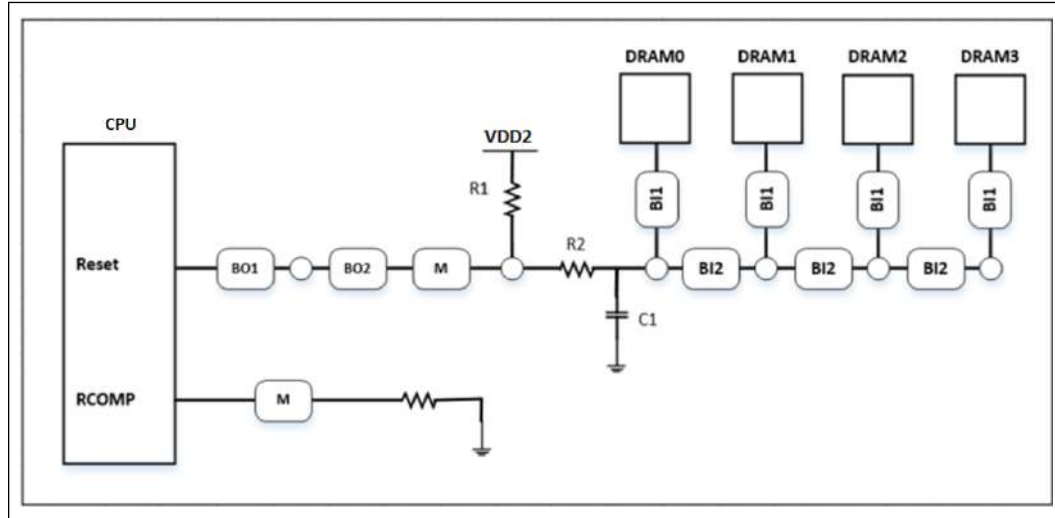


Figure 43. LPDDR5_x x32 Type-3 Memory Down Placement and Block Diagram

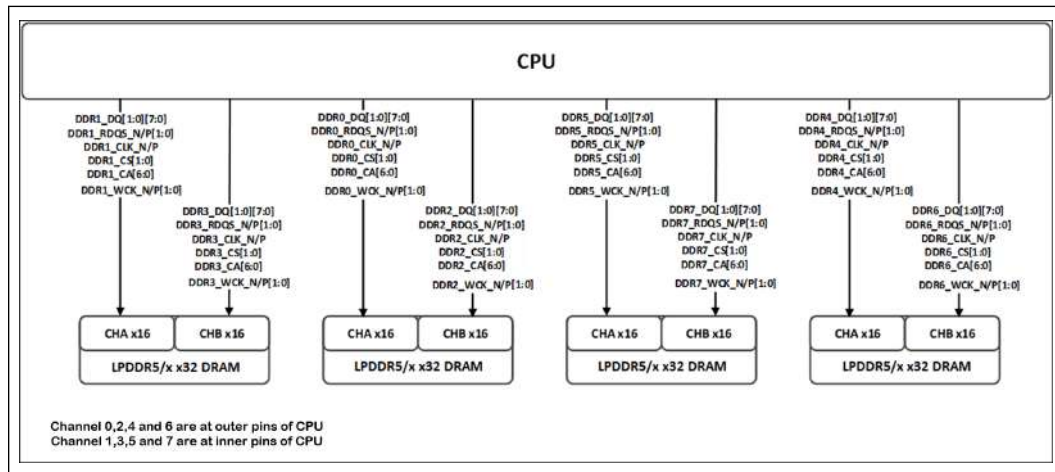


Table 125. MD x32 Type-3,0.8mm,10L Notes

Note	Detail
PCB stackup	Type-3 stackup with minimum of 10 layers.
Reference plane	Continuous ground only.
GND stitching	For VSS stitching using microvia, target a minimum S:G ratio of 1:1 at signal layer transition. For VSS stitching using PTH, target a minimum S:G ratio of 2:1 at signal layer transition. VSS via stitching pattern to be copied exactly from RVP/ clip files.
Max MS BO/ BI length	Refer to RFI/ EMC guidelines for the maximum microstrip BO/ BI length requirement.
Serpentine routing	Use 3x of dielectric height for serpentine routing spacing.
Bit/ Byte swapping rule	Bit swapping is allowed within each Byte for all memory technologies.
<i>continued...</i>	

Note	Detail
	Byte swapping is allowed within a 16-bit channel. LPDDR5: x16 sub-channels can be swizzled within their x64 MC. When swapping channel DQ, the CA/CS signals must be swapped as well.
Unused signals	Leave unused DDR signals (PARITY, extra CA) floating (no connect). ALERT is also unused and can be left floating (no connect) at processor side.
Rtt/ Ctt for RESET	Keep R1 empty; R2 = 0 Ohm; C1 = 0.1 uF (no stuff).
RCOMP value	100 Ohm
DDR to Non-DDR trace spacing	Minimum 500 um.

Table 126. Max number of vias

Signal Group	Max Number of Vias Allowed
CA, CLK, CS, DQ, RDQS, WCK	2
RCOMP	1

Length Matching**Table 127. MD x32 Type-3,0.8mm,10L Length Matching**

Signal Group	Plus/Minus	Reference Group	Within/Group	Up to memory /connector	Length/Delay	Min/Max/Range	Min	Max
DQ	-	WCK	Byte (x8)	Same DRAM	Length (mm)	Range	-12	14
DQ	-	RDQS	Byte (x8)	Same DRAM	Length (mm)	Range	-3	3
WCK	-	CLK	Channel (x16)	Same DRAM	Length (mm)	Range	-5	10
DQ	-	DQ	Byte (x8)	Same DRAM	Length (mm)	Max		4
CLK	-	CA	Channel (x16)	Same DRAM	Length (mm)	Range	-5	5
CA	-	CA	Channel (x16)	Same DRAM	Length (mm)	Max		7.5
CLK	-	CS	Channel (x16)	Same DRAM	Length (mm)	Range	-12.5	12.5
CS	-	CS	Channel (x16)	Same DRAM	Length (mm)	Max		10
RDQS_P	-	RDQS_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
WCK_P	-	WCK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK_P	-	CLK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK	-	CLK	All Channels	Within DRAM	Length (mm)	Max		25

Thin PCB, Premium Mid Loss (PML), CA/CS, CH-0/CH-2/CH-4/CH-6, Outer
Table 128. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), CA/CS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	7.5	
BO2	SL		10 - BO1
M	SL		
BI1	SL	2	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), CA/CS, CH-1/CH-3/CH-5/CH-7, Inner
Table 129. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), CA/CS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	10	
BO3	SL		10 - BO2
M	SL		
BI1	SL	14	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), CLK, CH-0/CH-2/CH-4/CH-6, Outer
Table 130. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), CLK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	7	
BO2	SL		10 - BO1
M	SL		
BI1	SL	9	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), CLK, CH-1/CH-3/CH-5/CH-7, Inner
Table 131. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), CLK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	8.5	
BO3	SL		10 - BO2
M	SL		
BI1	SL	8	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), DQ, CH-0/CH-2/CH-4/CH-6, Outer
Table 132. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), DQ, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	3.5	
BO2	SL		11 - BO1
M	SL		
BI1	SL	4	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), DQ, CH-1/CH-3/CH-5/CH-7, Inner
Table 133. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), DQ, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	5.5	
BO3	SL		10-BO2
M	SL		
BI1	SL	3	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), RDQS, CH-0/CH-2/CH-4/CH-6, Outer

Table 134. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), RDQS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	8	
BO2	SL		10 - BO1
M	SL		
BI1	SL	4	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), RDQS, CH-1/CH-3/CH-5/CH-7, Inner

Table 135. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), RDQS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	6	
BO3	SL		10-BO2
M	SL		
BI1	SL	2	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), WCK, CH-0/CH-2/CH-4/CH-6, Outer

Table 136. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), WCK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	9	
BO2	SL		10 - BO1
M	SL		
BI1	SL	4	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), WCK, CH-1/CH-3/CH-5/CH-7, Inner

Table 137. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), WCK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	7.5	
BO3	SL		11 - BO2
M	SL		
BI1	SL	2	
BI2	MS	0.6	

Max Length Total (mm): 60

Max Length Total Note: Max length is package + motherboard.

Thin PCB, Premium Mid Loss (PML), RESET, MISC, All

Table 138. MD x32 Type-3,0.8mm,10L Thin PCB, Premium Mid Loss (PML), RESET, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	40	
BO2	SL		21 - BO1
M	SL		
BI1	MS		

Max Length Total (mm): 200

4.4.2.2 MD x32 Type-3,0.9mm,10L

Pcb Type & Thickness: Type-3, 0.9mm

Figure 44. LPDDR5/x x32 Type-3 PCB DQ, DQS, CA, CS, CLK Topology Diagram

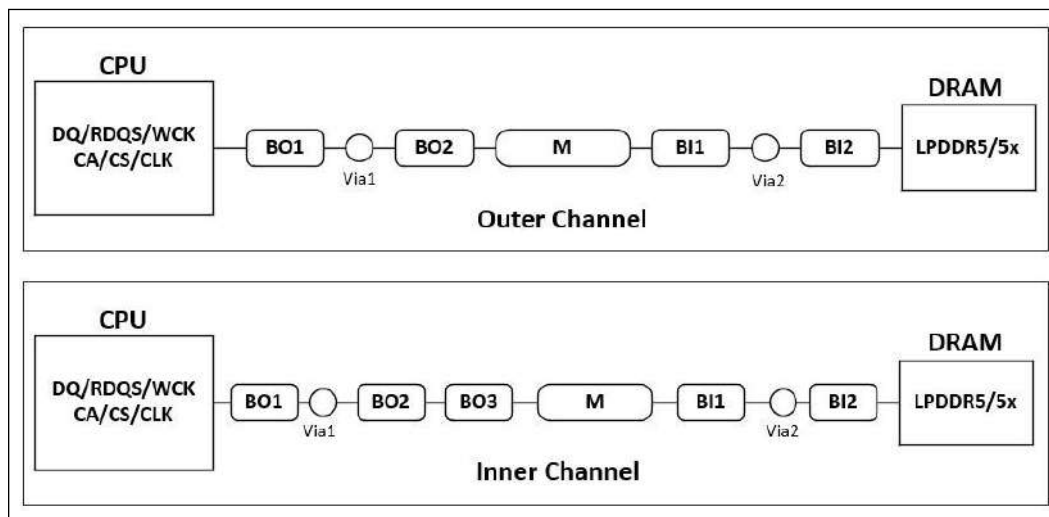


Figure 45. LPDDR5/x x32 Type-3 Reset and RCOMP Topology Diagram

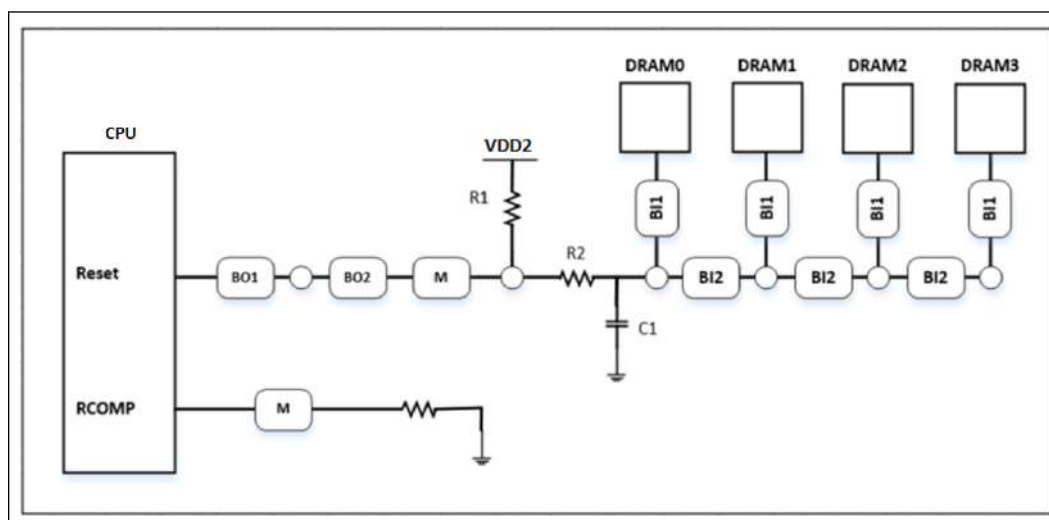


Figure 46. LPDDR5_x x32 Type-3 Memory Down Placement and Block Diagram

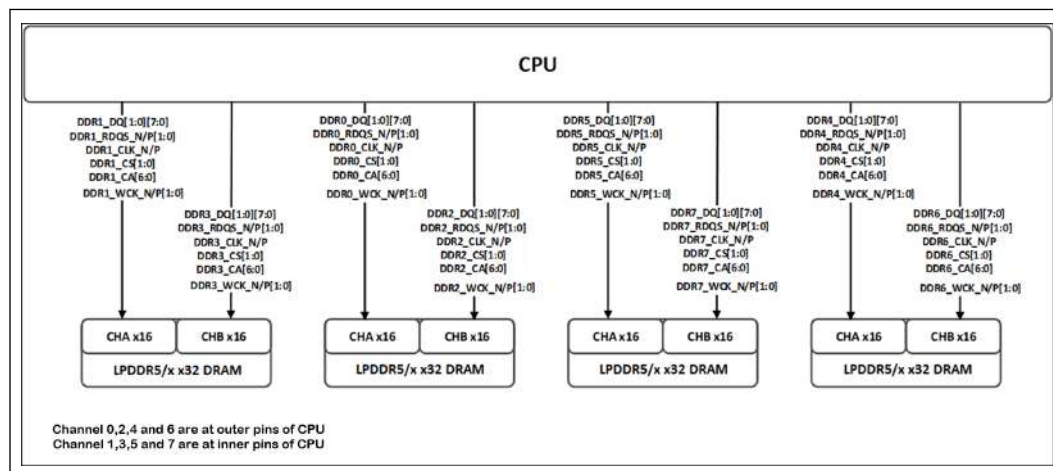


Table 139. MD x32 Type-3,0.9mm,10L Notes

Note	Detail
PCB stackup	Type-3 stackup with minimum of 10 layers.
Reference plane	Continuous ground only.
GND stitching	For VSS stitching using microvia, target a minimum S:G ratio of 1:1 at signal layer transition. For VSS stitching using PTH, target a minimum S:G ratio of 2:1 at signal layer transition. VSS via stitching pattern to be copied exactly from RVP/ clip files.
Max MS BO/ BI length	Refer to RFI/ EMC guidelines for the maximum microstrip BO/ BI length requirement.
Serpentine routing	Use 3x of dielectric height for serpentine routing spacing.
Bit/ Byte swapping rule	Bit swapping is allowed within each Byte for all memory technologies. Byte swapping is allowed within a 16-bit channel. LPDDR5: x16 sub-channels can be swizzled within their x64 MC. When swapping channel DQ, the CA/CS signals must be swapped as well.
Unused signals	Leave unused DDR signals (PARITY, extra CA) floating (no connect). ALERT is also unused and can be left floating (no connect) at processor side.
Rtt/ Ctt for RESET	Keep R1 empty; R2 = 0 Ohm; C1 = 0.1 uF (no stuff).
RCOMP value	100 Ohm
DDR to Non-DDR trace spacing	Minimum 500 um.

Table 140. Max number of vias

Signal Group	Max Number of Vias Allowed
CA, CLK, CS, DQ, RDQS, WCK	2
RCOMP	1

Length Matching

Table 141. MD x32 Type-3,0.9mm,10L Length Matching

Signal Group	Plus / Minus	Reference Group	Within / Group	Up to memory / connector	Length / Delay	Min / Max / Range	Min	Max
DQ	-	WCK	Byte (x8)	Same DRAM	Length (mm)	Range	-12	14
DQ	-	RDQS	Byte (x8)	Same DRAM	Length (mm)	Range	-3	3
WCK	-	CLK	Channel (x16)	Same DRAM	Length (mm)	Range	-5	10
DQ	-	DQ	Byte (x8)	Same DRAM	Length (mm)	Max		4
CLK	-	CA	Channel (x16)	Same DRAM	Length (mm)	Range	-5	5
CA	-	CA	Channel (x16)	Same DRAM	Length (mm)	Max		7.5
CLK	-	CS	Channel (x16)	Same DRAM	Length (mm)	Range	-12.5	12.5
CS	-	CS	Channel (x16)	Same DRAM	Length (mm)	Max		10
RDQS_P	-	RDQS_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
WCK_P	-	WCK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK_P	-	CLK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK	-	CLK	All Channels	Within DRAM	Length (mm)	Max		25

Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-2/CH-4/CH-6, Outer

Table 142. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	7.5	
BO2	SL		8 - BO1
M	SL		
BI1	SL	2	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), CA/CS, CH-1/CH-3/CH-5/CH-7, Inner

Table 143. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CA/CS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	7	
continued...			

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO3	SL		19 - BO2
M	SL		
BI1	SL	14	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-2/CH-4/CH-6, Outer

Table 144. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	7	
BO2	SL		9 - BO1
M	SL		
BI1	SL	9	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), CLK, CH-1/CH-3/CH-5/CH-7, Inner

Table 145. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), CLK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	8.5	
BO3	SL		18 - BO2
M	SL		
BI1	SL	8	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-2/CH-4/CH-6, Outer

Table 146. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	3.5	
BO2	SL		11 - BO1
M	SL		
BI1	SL	4	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), DQ, CH-1/CH-3/CH-5/CH-7, Inner

Table 147. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), DQ, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	7	
BO3	SL		13-BO2
M	SL		
BI1	SL	2	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), RDQS, CH-0/CH-2/CH-4/CH-6, Outer

Table 148. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), RDQS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	9	
BO2	SL		10 - BO1
M	SL		
BI1	SL	6	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), RDQS, CH-1/CH-3/CH-5/CH-7, Inner**Table 149. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), RDQS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	6	
BO3	SL		14-BO2
M	SL		
BI1	SL	2	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), WCK, CH-0/CH-2/CH-4/CH-6, Outer**Table 150. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), WCK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	8	
BO2	SL		10 - BO1
M	SL		
BI1	SL	4	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), WCK, CH-1/CH-3/CH-5/CH-7, Inner**Table 151. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), WCK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	0.5	
BO2	SL	6	
BO3	SL		15 - BO2
M	SL		
BI1	SL	2	
BI2	MS	0.6	

Max Length Total (mm): 70

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), RESET, MISC, All
Table 152. MD x32 Type-3,0.9mm,10L Mainstream, Premium Mid Loss (PML), RESET, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	40	
BO2	SL		21 - BO1
M	SL		
BI1	MS		

Max Length Total (mm): 200

4.4.3 LPDDR5x Memory Down Type 4

This topology is supported for NVL 8C/H16C only.

Table 153. System Memory Configuration Details Covered in this Section

Parameter	Details
Processor	NVL 8C, H16C 4Xe, H16C 12Xe
Memory Type	LPDDR5x
Speed (MT/s)	NVL-H: 1R/2R 9600 DDP, QDP, ODP
Channels	Channels: 0-7
Population Rules	2 DRAMs: - Use either channels 0 to 3 or 4 to 7 - From the above, connect any x16 channel to any DRAM 3 DRAMs: N/A 4 DRAMs: - Connect any x16 channel to any DRAM
System Memory Voltage	LPDDR5x DRAM VDDQ voltage is 0.52V, VDD2 is 1.065 V
Max Ranks Per Channel	2
DRAM Die Density (Gb)	12, 16, 24 ¹⁰
Max Capacity (GB)	96GB
Memory Down Types ^{1,2} (Pkg Ranks - Die Bits - Pkg bits)	315 BGA DDP:1-16-32 QDP:2-16-32
PCB Layers/Type ³	10L(2x2+)/Type-4
DRAM Device Placement	Top Motherboard Layer, Single Sided, Side by Side Placement
continued...	

Parameter	Details
Processor Memory Ball Map ⁴	Non Interleaved
DRAM Device ODT Capability ^{5,6}	MRC trained

Notes:

- DDP: Dual Die Package, QDP: Quad Die Package, ODP: Octal Die Package
- No mixed vendor support within a channel or channel to channel and No mixed memory DRAM down type support (SDP, DDP, QDP) within a channel or channel to channel
- Type-4 (T4) (2-x-2+) = PCB with 2 build-up layers Refer to **Stack-Up and PCB Considerations** chapter for more details in this document.
- Non-Interleave = The processor is optimized for a side-by-side ball map and placement of the DRAM memory down devices placement. Each memory channel's signals are grouped together on one side of the processor packages.
- The DRAM Device ODT will be trained by the MRC.
- For , to avoid excessive latency due to ZQ topology, Intel specifies that each ZQ pin must have a unique ZQ resistor.
- 16Gb die-64GB Pending DRAM samples availability.
- LPDDR5x x16 sub-channels can be swizzle within their x64 Memory controller. When swapping channel DQ the CMD/CRTL signals must be swapper as well.
- To avoid excessive latency due to ZQ topology, Intel specifies that each ZQ pin must have an unique ZQ resistor.
- Pending DRAM samples availability.

4.4.3.1 MD x32 Type-4,0.8mm,10L

Pcb Type & Thickness: Type-4 2-x-2+, 0.8mm

Figure 47. LPDDR5/x x32 Type4 DQ, RDQS, WCK, CA, CS, CLK Topology Diagram

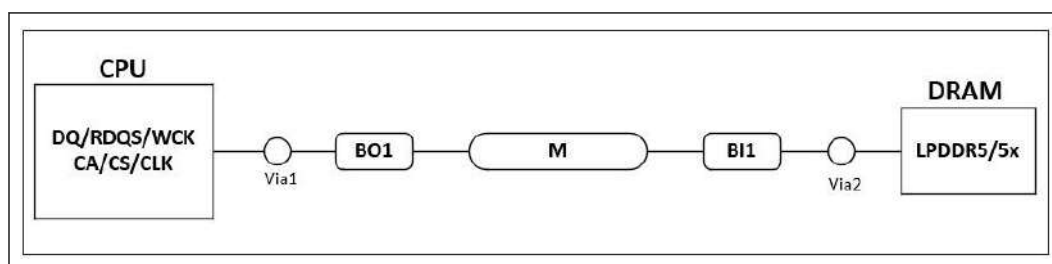


Figure 48. LPDDR5/x x32 Type-4 Reset and RCOMP Topology Diagram

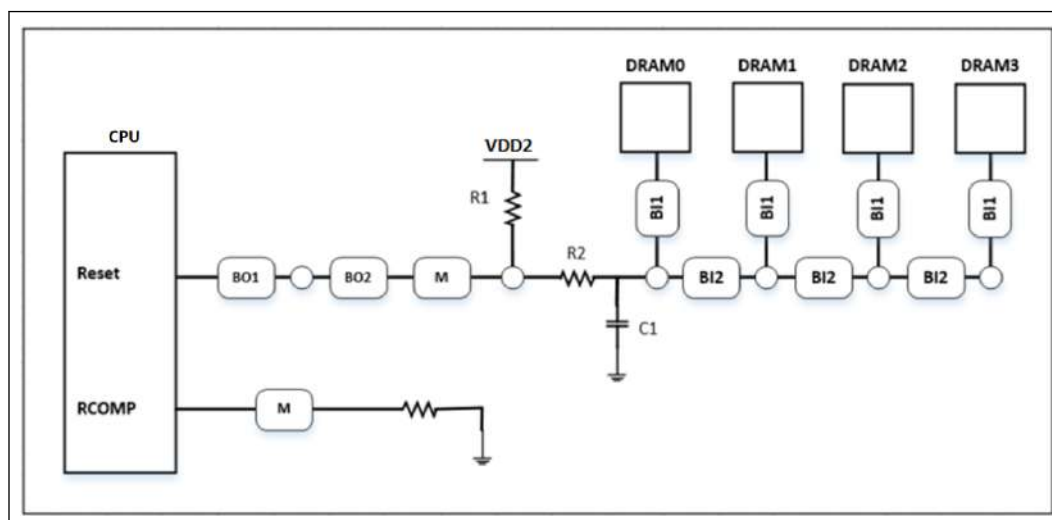
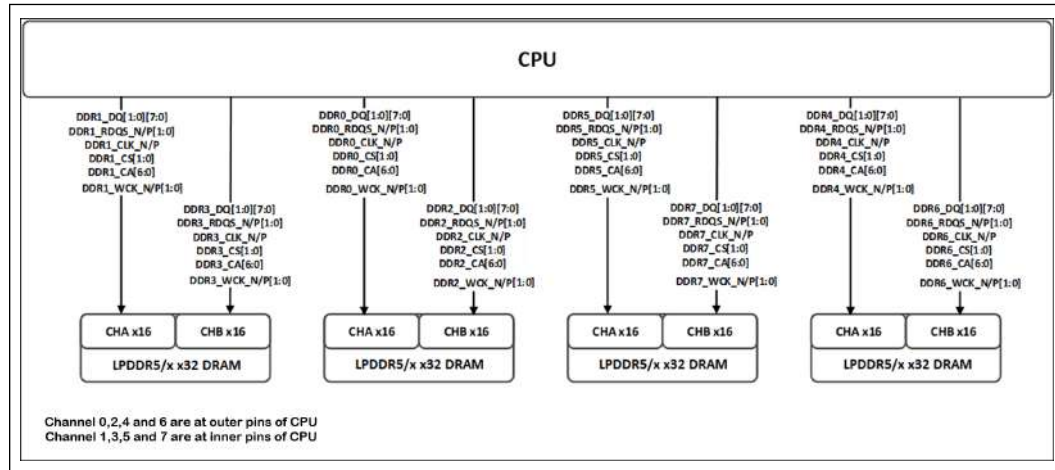


Figure 49. LPDDR5_x x32 Type-4 Memory Down Placement and Block Diagram

Table 154. MD x32 Type-4,0.8mm,10L Notes

Note	Detail
PCB stackup	Type-4 (2-x-2+) stackup with minimum of 10 layers.
Layer assignment rules	Outer row DQ on layer 2, Inner row DQ on layer 4, CAC can be on layer 6/8. Please follow RVP / Clip files for the layer assignment details.
Reference plane	Continuous ground only.
Number of vias allowed	DQ/ RDQS/ WCK: max 2 vias ; Only micro-vias allowed. CA/ CS/ CLK: max 2 vias; Both micro-vias & PTH allowed.
GND stitching	For VSS stitching using microvia, target a minimum S:G ratio of 1:1 at signal layer transition. For VSS stitching using PTH, target a minimum S:G ratio of 2:1 at signal layer transition. VSS via stitching pattern to be copied exactly from RVP / Clip files.
Serpentine routing	Use 3x of dielectric height for serpentine routing spacing. Do not apply serpentine routing in the break-out region.
Bit/ Byte swapping rule	Bit swapping is allowed within each Byte for all memory technologies. Byte swapping is allowed within a 16-bit channel. LPDDR5: x16 sub-channels can be swizzled within their x64 MC. When swapping channel DQ, the CA/CS signals must be swapped as well.
Unused signals	Leave unused DDR signals (PARITY, extra CA) floating (no connect). ALERT is also unused and can be left floating (no connect) at processor side.
Rtt/ Ctt for RESET	Keep R1 empty; R2 = 0 Ohm; C1 = 0.1 uF (no stuff).
RCOMP value	100 Ohm
DDR to Non-DDR trace spacing	Minimum 500 um.

Table 155. Max number of vias

Signal Group	Max Number of Vias Allowed	Notes
DQ, RDQS, WCK	2	Only Micro Vias allowed
CA, CLK, CS	4	2 micro via and 2 PTH
RCOMP	2	

Length Matching**Table 156. MD x32 Type-4,0.8mm,10L Length Matching**

Signal Group	Plus/Minus	Reference Group	Within/Group	Up to memory/connector	Length/Delay	Min/Max/Range	Min	Max
DQ	-	WCK	Byte (x8)	Same DRAM	Length (mm)	Range	-12	14
DQ	-	RDQS	Byte (x8)	Same DRAM	Length (mm)	Range	-3	3
WCK	-	CLK	Channel (x16)	Same DRAM	Length (mm)	Range	-5	10
DQ	-	DQ	Byte (x8)	Same DRAM	Length (mm)	Max		4
CLK	-	CA	Channel (x16)	Same DRAM	Length (mm)	Range	-5	5
CA	-	CA	Channel (x16)	Same DRAM	Length (mm)	Max		7.5
CLK	-	CS	Channel (x16)	Same DRAM	Length (mm)	Range	-12.5	12.5
CS	-	CS	Channel (x16)	Same DRAM	Length (mm)	Max		10
RDQS_P	-	RDQS_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
WCK_P	-	WCK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK_P	-	CLK_N	Pair (P-N)	Same DRAM	Length (mm)	Max		0.1
CLK	-	CLK	All Channels	Within DRAM	Length (mm)	Max		25

Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-2/CH-4/CH-6, Outer**Table 157. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), CA/CS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	SL	4	
M	SL		
BI	SL	4	

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), CA/CS, CH-1/CH-3/CH-5/CH-7, Inner

Table 158. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), CA/CS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	SL	5	
M	SL		
BI	SL	4	

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-2/CH-4/CH-6, Outer

Table 159. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), CLK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	3
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), CLK, CH-1/CH-3/CH-5/CH-7, Inner

Table 160. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), CLK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	4.5
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-2/CH-4/CH-6, Outer

Table 161. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), DQ, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	4
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), DQ, CH-1/CH-3/CH-5/CH-7, Inner

Table 162. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), DQ, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	4
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), RDQS, CH-0/CH-2/CH-4/CH-6, Outer

Table 163. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), RDQS, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	2
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), RDQS, CH-1/CH-3/CH-5/CH-7, Inner

Table 164. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), RDQS, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	5
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), WCK, CH-0/CH-2/CH-4/CH-6, Outer
Table 165. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), WCK, CH-0/CH-2/CH-4/CH-6, Outer Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	4.6
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), WCK, CH-1/CH-3/CH-5/CH-7, Inner
Table 166. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), WCK, CH-1/CH-3/CH-5/CH-7, Inner Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	SL	6
M	SL	
BI	SL	4

Max Length Total (mm): 55

Max Length Total Note: Max length is package + motherboard.

Mainstream, Premium Mid Loss (PML), RESET, MISC, All
Table 167. MD x32 Type-4,0.8mm,10L Mainstream, Premium Mid Loss (PML), RESET, MISC, All Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO1	MS	13	
BO2	MS		
M	SL		
BI	SL	10	

Max Length Total (mm): 200

4.4.4 DDR Byte Swapping

- Bit swapping is allowed within each Byte for all memory technologies.
- DDR5: Byte Swapping is allowed within each x32 channel.
- LPDDR5x: Byte swapping is allowed within a 16-bit channel.
- LPDDR5x: x16 sub-channels can be swizzled within their x64 MC.
- When swapping channel DQ, the CA/CS signals must be swapped as well.
- DDR5 x32 sub-channels can be swizzle within their x64 MC.

4.5 Via Placement and Pad Optimization

This section describes the following:

- Via Placement at CPU Side
- Via Placement at CSODIMM and DRAM Device Side
- Teardrop for via guidelines should be followed from [Teardrop Structure Requirement at Junction of Via and Trace](#) on page 113.

4.5.1 Via Placement at CPU Side

The following via stitching guidelines on the CPU-side are critical to follow to meet memory performance

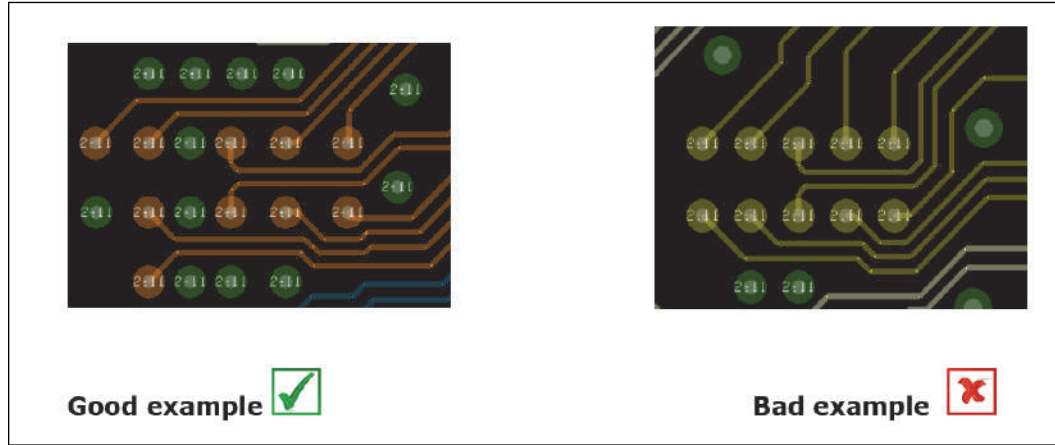
- Improper or insufficient ground vias for any given signal can cause a significant impact at higher data rates. For single-ended signaling, if a properly stitched return path via is not placed in close proximity to every signal via, a non-ideal return path is created.
- In a symmetrical stripline, the return current is shared equally between both reference planes. So, ensuring a ground stitching via which connects to both the reference planes of a signal layer is important.
- Follow the Reference Validation Platform (RVP) via placement at Processor side as much as possible for all Signal Groups.
- When a signal transitions to another routing layer, but the referencing plane is still kept the same, no stitching via is required.
- GND vias must be placed between signals from different signal groups and channels.
- Intel recommends to follow up GND stitching via and spacing rules to prevent return path and Cross-talk issues.
- Data and Strobe vias from different byte lanes must be separated by GND vias.
- A good example of CPU-side ground via stitching has sufficient ground coverage for every signal in a byte lane Figure below.
- Remove via pads from layers that are not connected to traces in [Figure 52](#) on page 175

NOTE

PTH via placement on the CPU side is primarily driven by pin pitch. Follow the RVP design for optimum placement of vias.

- A good example of CPU-side ground via stitching has sufficient ground coverage for every signal in a byte lane (Figure below).
- Remove via pads from layers that are not connected to traces ([Figure 52](#) on page 175).
- For Via separation Guidelines refer to [Via Placement at CSODIMM and DRAM Device Side](#) on page 174.

Figure 50. CPU-side GND via Stitching Examples (GND vias in Green)



4.5.2 Via Placement at CSODIMM and DRAM Device Side

Similar to the via placement guidelines on the CPU-side, the following via placement guidelines on the DRAM-side ensure design robustness at high memory frequencies.

- Follow the Reference Validation Platform (RVP) via placement at SODIMM and Memory DRAM Down side as much as possible for all Signal Groups.
- For HDI or Type-4 memory down system designs use GND Micro Vias from surface pin/pad (Top/Bottom) layer to connect to respective signal routing reference plane layers.
- GND vias must be placed between signals from different signal groups and channels.
- Data and Strobe vias from different byte lanes must be separated by GND vias.

NOTE

PTH(Plated Through Hole) via placement on the DRAM side is primarily driven by pin pitch. Follow the RVP design for optimum placement of vias.

- A good example of DRAM-side ground via stitching has sufficient ground coverage for every signal in a byte lane as shown in the figure below.
- Follow the via spacing guidelines covered in table below. At least one ground stitching via must be placed within every signal transition (DQ/DQS/DQS#/CLK/CTRL/CKE/CMD) whenever they change signal layers.
- Remove via pads from layers that are not connected to traces (Figure 52 on page 175).

Figure 51. DRAM-side GND via Stitching Examples (GND vias in Green)

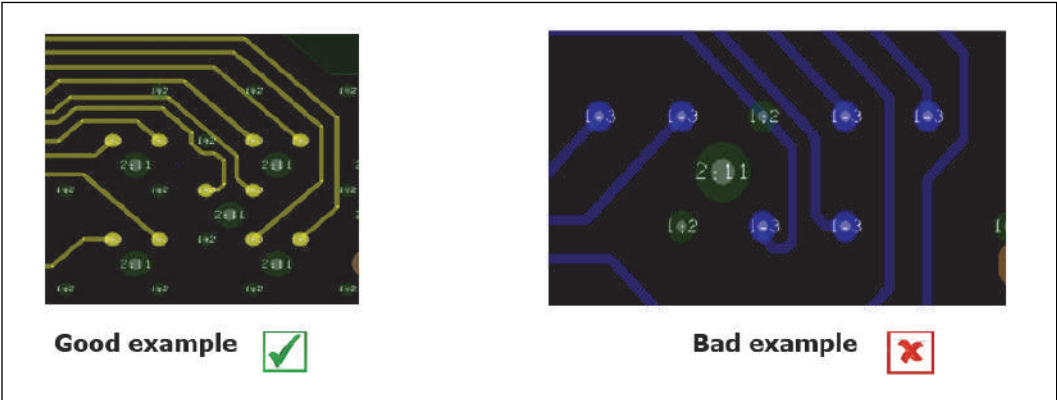


Figure 52. Remove PADs from Layers that Do Not Connect to Trace

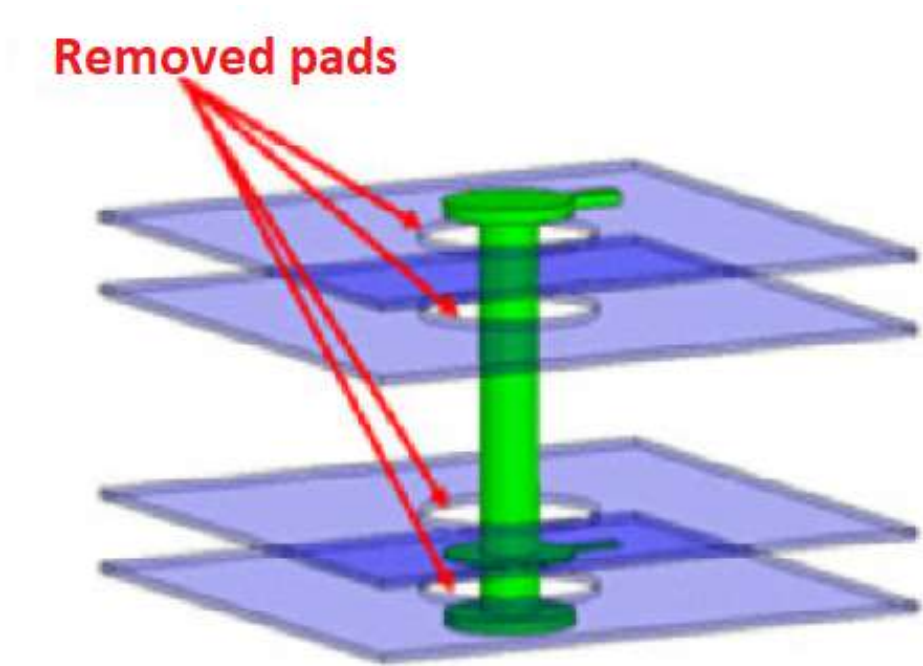


Table 168. Vias Separation Guidelines

CPU	Memory Type	Signal Group	Signal:GND Rule
NVL H	CSODIMM	DQ/DQS/DQS#	1:1 (microvia) 1:1-2:1 (PTH/Buried via)
		CLK/CTRL/CKE/CMD	1:1 (microvia) 1:1-2:1 (PTH/Buried via)
continued...			

CPU	Memory Type	Signal Group	Signal:GND Rule
		1 DIMM Per Channel	Pitch between DQ vias from different byte lanes
			Pitch between CMD to CTRL vias
Notes: 1. Via-to-via pitch is measures between via centers. 2. Signal GND via stitching pattern to be copy exact from RVP/Clip files.			

4.6 Memory Side Power Delivery

This section describes the following:

- DDR Power Delivery Memory Side

4.6.1 DDR Power Delivery Memory Side

The guidelines here are to support a **dual sided component** assembly.

The expectation is that the DRAM decoupling caps will be placed as close as possible to the DRAM devices to minimize the loop inductance of the caps.

The decoupling capacitor to DRAM device inductance loop can be improved by following common power integrity recommendations described in Power Integrity Recommendations.

4.6.1.1 Memory Device Decoupling

4.6.1.1.1 DDR5 Device Decoupling

Small Outline Dual In-line Memory Module (SODIMM)

DDR5 SODIMM comes with PMIC/ VR and decoupling on the module card itself, hence no additional decoupling guidelines are required. Nevertheless the source supply to the DDR5 SODIMM needs to follow the JEDEC/ vendor specification sheet.

4.6.1.1.2 LPDDR5x Device Decoupling

Memory Down Decoupling Config: 4X32

Description: LPDDR5x x32 memory down PI solution guidance with eDVFSC enabled on the platform.

eDVFSC requires an independent VDD2L rail on the platform. If the eDVFSC support is not desired on the platform, customers can short/merge DRAM VDD2L rail with the VDD2H rail. Customers should consult the DRAM vendors for design guidance on the merged VDD2H/VDD2L rail..

Table 169. Notes

Note
TYPE-4 (2-x-2+) DRAM power rail layer assignment is provided in LPDDR5x_TYPE4_1
TYPE-3 10L DRAM power rail layer assignment is provided in LPDDR5x_TYPE3_1
VSS referencing is recommended for all the Memory power rails
Information provided here is only for guidance. Customers are responsible for meeting all JEDEC SPECS on the Memory power rails

Table 170. Decoupling Solution And Filter Recommendation

Voltage Rail	Capacitor Form Factor and Value	Quantity	Notes	Reference Design
VDD2H	0201, 1 uF	16	4 caps per DRAM, 1 per side close to VDD2H BGAs. Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_2, LPDDR5x_TYPE3_2
VDD2H	0402, 10 uF	4	1 cap per DRAM. Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_2, LPDDR5x_TYPE3_2
VDDQ	0201, 1 uF	16	4 caps per DRAM, 1 per each corner close to VDDQ BGAs. Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_3, LPDDR5x_TYPE3_3
VDDQ	0402, 10 uF	4	1 cap per DRAM. Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_3, LPDDR5x_TYPE3_3
VDD1	0201, 1 uF	16	4 caps per DRAM, 1 per each VDD1 BGA. Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_4, LPDDR5x_TYPE3_4
VDD1	0402, 10 uF	4	1 cap per DRAM. Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_4, LPDDR5x_TYPE3_4
VDD2L	0201, 1uF	16	4 cap per DRAM, 1 per each corner close to VDD2L BGA's Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_5, LPDDR5x_TYPE3_5
VDD2L	0402, 10uF	4	1 cap per DRAM. Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPDDR5x_TYPE4_5, LPDDR5x_TYPE3_5

Figure 53. LPDDR5x_TYPE4_1

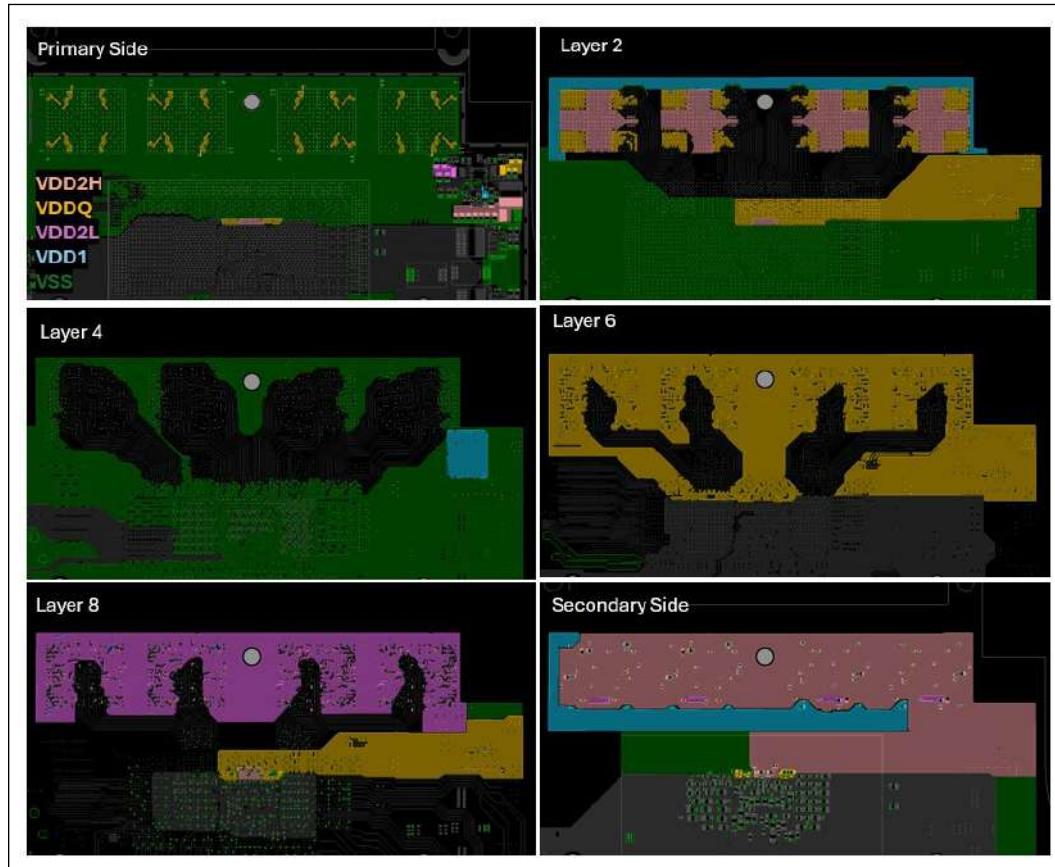


Figure 54. LPDDR5x_TYPE4_2

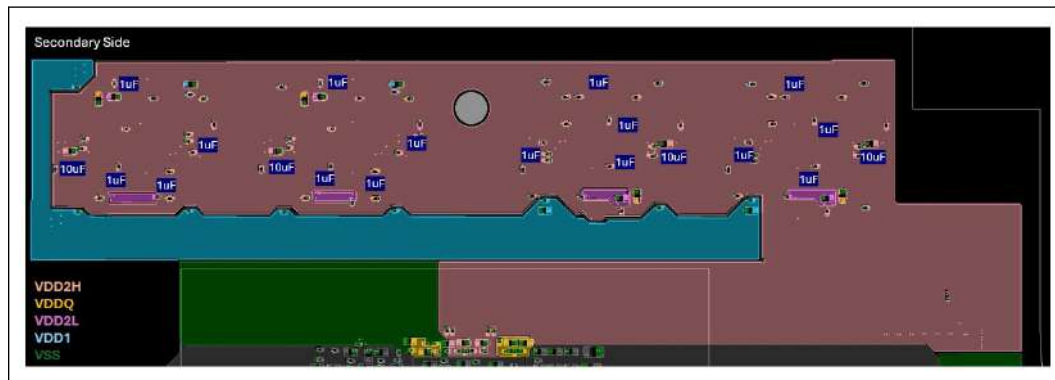


Figure 55. LPDDR5x_TYPE4_3

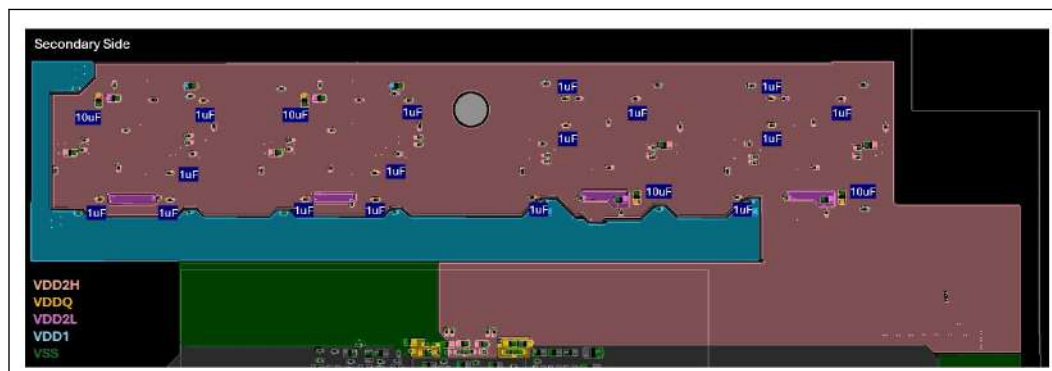


Figure 56. LPDDR5x_TYPE4_4

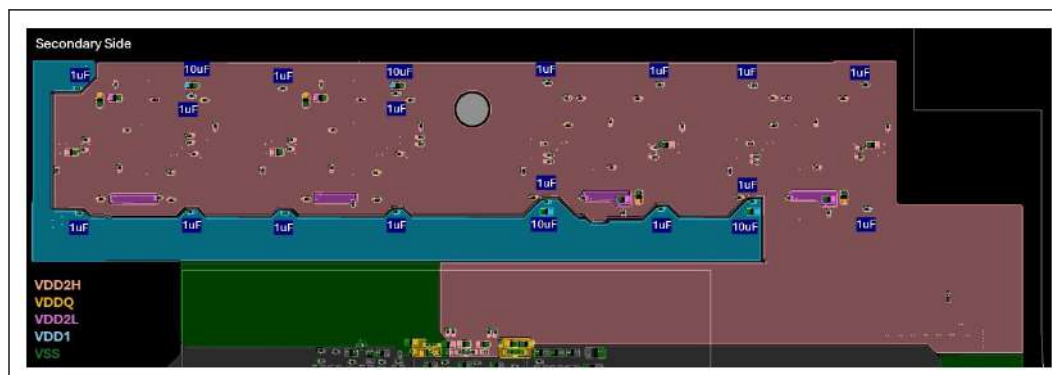


Figure 57. LPDDR5x_TYPE4_5

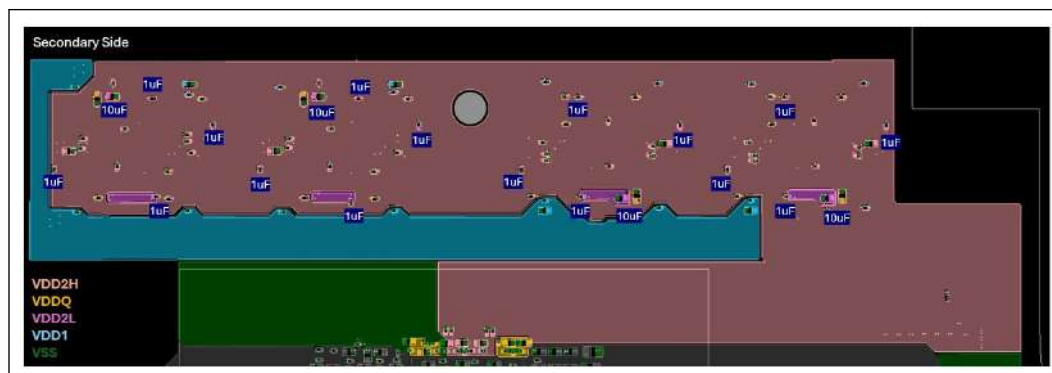


Figure 58. LPDDR5x_TYPE3_1

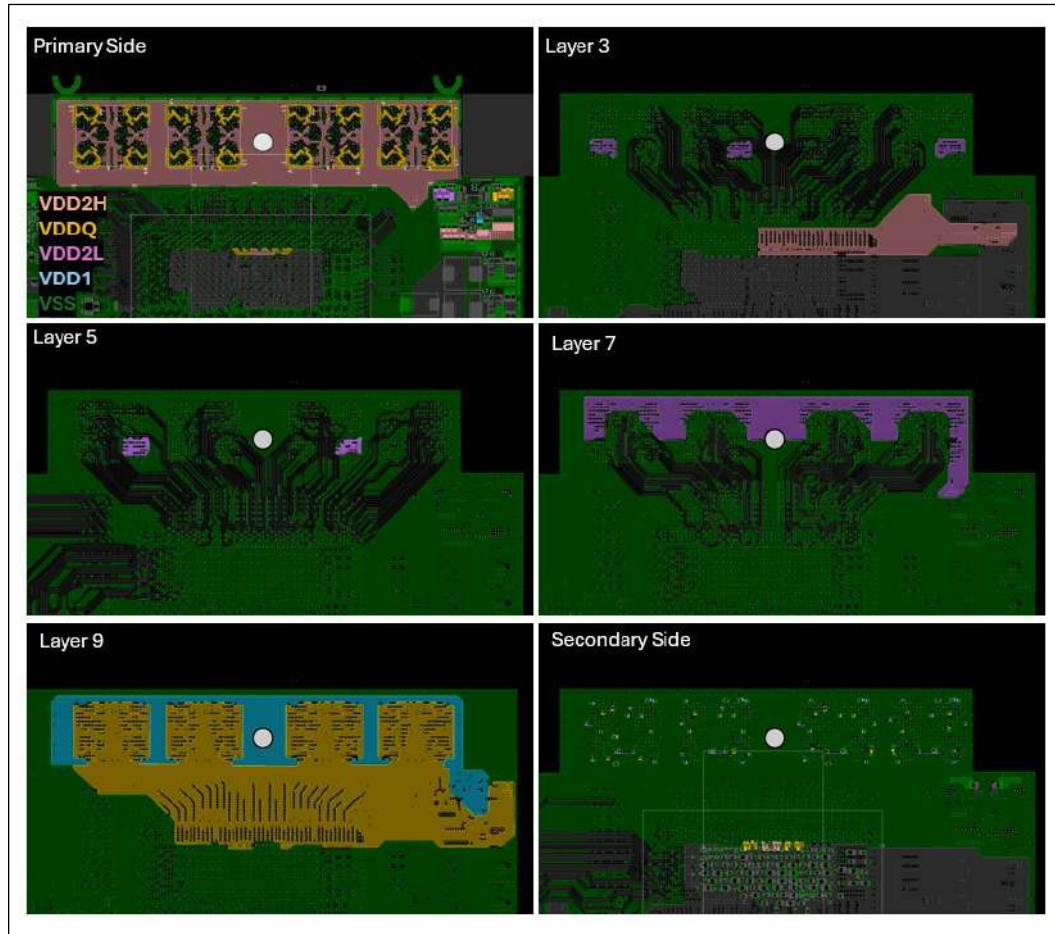


Figure 59. LPDDR5x_TYPE3_2



Figure 60. LPDDR5x_TYPE3_3

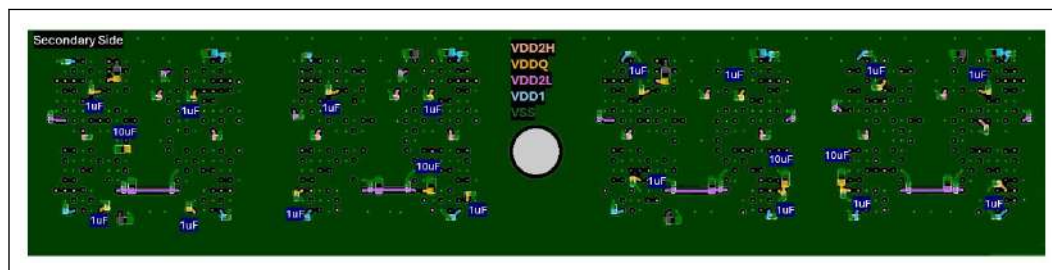


Figure 61. LPDDR5x_TYPE3_4

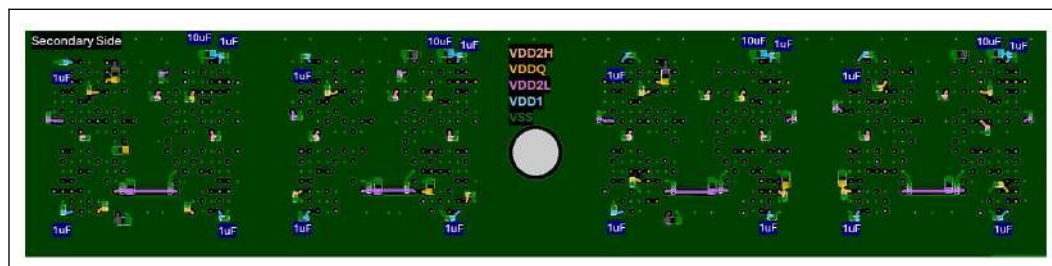
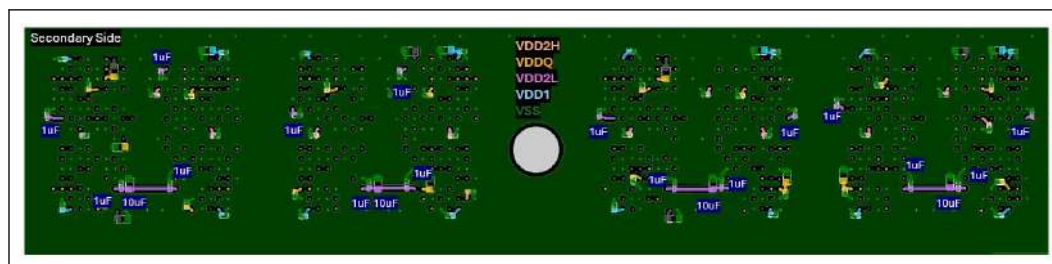


Figure 62. LPDDR5x_TYPE3_5



4.6.1.1.3 LPCAMM2 E1 Connector

LPCAMM2 E1 Module

Description: LPCAMM2 E1 module has a configurable VDDQ rail option i.e. VDDQ rail can be sourced from on module PMIC or from the Motherboard VRM. This section provides routing recommendations for using Motherboard VDDQ VRM shared with the SOC.

A shared VDDQ rail between the SOC and LPCAMM2 E1 module is required to support eDVFSQ feature. A shared VDDQ rail between the SOC and LPCAMM2 E1 module is required to support speeds above 7467 MT/s.

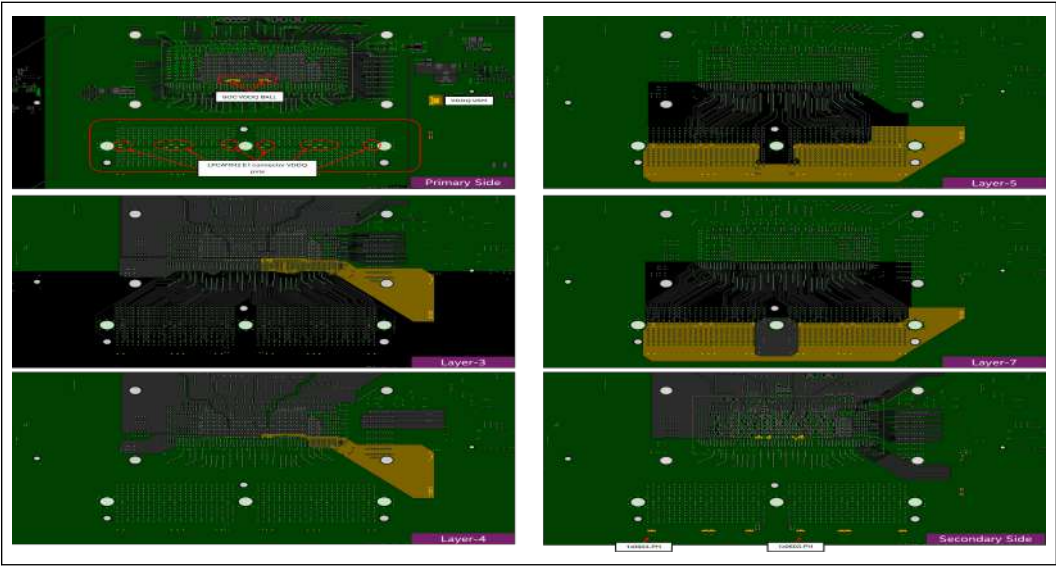
Table 171. Notes

Note
Information provided here is only for guidance. Customers are responsible for meeting all LPCAMM2 E1 connector JEDEC SPECS.
VSS referencing is recommended for the VDDQ power rail.

Table 172. Decoupling Solution And Filter Recommendation

Voltage Rail	Capacitor Form Factor and Value	Quantity	Notes	Reference Design
VDDQ	0603, PH	2	Customers may adjust the capacitor quantity/ form factor/ value to meet the JEDEC spec.	LPCAMM2_E1_1

Figure 63. LPCAMM2_E1_1



5.0 Display Interfaces

Table 173. Display Topology Summary

Interface	Port Type	Top supported technology	Topology type	Topology design guidance	Chapter
DisplayPort* (DP*)	TCP0-2	DP up to UHBR20	DP Retimer MUX With Barlow Ridge	TCP DP UHBR20 Retimer MUX With Barlow Ridge	6.1.6.8.1
			Aux	TCP DP AUX Retimer MUX With Barlow Ridge	6.1.6.9.1
	Common		HPDx#	TCP DP Hot Plug Detect Retimer MUX With Barlow Ridge	6.1.6.8.2
Embedded DisplayPort* (eDP*)	DDIA TCP0	eDP Up to HBR3	Direct	HBR3 Main Link CTLE	5.2.1.2
				HBR3 Main Link CTLE+DFE	5.2.1.3
			Muxed topology	HBR3 Main Link MUX CTLE	5.2.1.4
				HBR3 Main Link MUX CTLE+DFE	5.2.1.5
	Common		Aux	eDP Auxiliary	5.2.1.1
				eDP Auxiliary Mux	5.2.1.7
			HPDA#, HPD0#	eDP Hot Plug Detect	5.2.1.6
HDMI*	DDI4	HDMI Up to 12Gbps	Re-timer	HDMI Retimer 12G	5.3.1.1
			Re-driver	HDMI Limiting Redriver 12G	5.3.1.2
			DDC	HDMI DDC Topology	5.3.1.4
	Common		HPDx#	HDMI Hot Plug Detect	5.3.1.3

Reference Documents

Table 174. Digital Display Interface Reference Documents

Title	Document Number/Location
DisplayPort* Standard Version 2.1 Electrical Specification	VESA.org
VESA Embedded DisplayPort* Standard - Revision 1.5b	VESA.org
HDMI* Standard Version 2.1 Electrical Specification	HDMI.org
High Speed Internal Connector and Cable for Notebook Computers	602807
Alder Lake Platform Twisted Pair Cable Electrical Specification for Embedded DisplayPort Technical White Paper	644925
Nova Lake H Platform Non-POR Topology of Display, Storage, and Thunderbolt™ Technical White Paper	914060

5.1 DisplayPort* (DP*)

Table 175. DisplayPort* Bit Rates

Bit Rate	Lane Supports (Max # of Lanes)
1.62 Gb/s (RBR)	4
2.7 Gb/s (HBR)	4
5.4 Gb/s (HBR2)	4
8.1 Gb/s (HBR3)	4
10 Gb/s (UHBR10)	4
20 Gb/s (UHBR20)	4

5.2 Embedded DisplayPort* (eDP*)

eDP* Platform - Specific Important Information

Table 176. eDP* Bit Rates

Bit Rate	Lane Supports (Max # of Lanes)
1.62 Gb/s (RBR)	4
2.16 Gb/s	4
2.43 Gb/s	4
2.7 Gb/s (HBR)	4
3.24 Gb/s	4
4.32 Gb/s	4
<i>continued...</i>	

Bit Rate	Lane Supports (Max # of Lanes)
5.4 Gb/s (HBR2)	4
6.75 Gb/s	4
8.1 Gb/s (HBR3)	4

eDP* Signal Descriptions

Table 177. eDP* Signal Name

Description	DDI Signal Name	TCP Signal Name
Main Link (Tx)	DDIA_TX_P/N[3:0]	TCP0_TXRX[1:0]_P/N + TCP0_TX[1:0]_P/N
Aux	DDIA_AUX_P/N	TCP0_AUX_P/N
HPD	DDSP_HPDA#	DDSP_HPDA0#
Note: Rcomp resistor is required, for more information refer to RCOMP Recommendation on page 124		

Table 178. eDP* Main Link Signal Mapping Configuration

Signal Name		Default Configuration (nonreversed)	Reversed Configuration
DDI	TCP		
DDIA_TX_P/N[0]-[3]	TCP0_TXRX1_P/N	Main Link Lane 3	Main Link Lane 0
DDIA_TX_P/N[0]-[2]	TCP0_TX1_P/N	Main Link Lane 2	Main Link Lane 1
DDIA_TX_P/N[0]-[1]	TCP0_TXRX0_P/N	Main Link Lane 1	Main Link Lane 2
DDIA_TX_P/N[0]-[0]	TCP0_TX0_P/N	Main Link Lane 0	Main Link Lane 3
Notes: - For systems which implement reversed lane configuration it is required to set DDI Lane Reversal to enabled in VBT. DDI lane reversal option can be found under LFP1 configuration . - For systems which implement default signal configuration no additional configuration needed in VBT (default is disabled) .			

Table 179. eDP* Back light Control Signal Mapping

Description	Primary eDP* Signal Name	Secondary eDP* Signal Name
VDDEN - Main Power Enable	VDDEN	VDDEN2
BKLT_EN - Backlight Enable	BKLTEN	BKLTEN2
BKLT_CTRL - Backlight Control	BKLTCTL	BKLTCTL2

5.2.1 eDP

Description: Guidelines are same for eDP using DDI interface or TCP interface.

Table 180. eDP General Guidelines

Note	Detail
Voiding recommendation	It is recommended to void unused (non-functional) pads for differential vias, and pin pads for CMCs, ESDs and connectors to optimize the impedance matching in the channel. Voiding of 0402 or smaller AC capacitors pads is not necessary.

Table 181. eDP Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	Yes	25

5.2.1.1 eDP AUX Topology

Figure 64. eDP AUX Topology Diagram

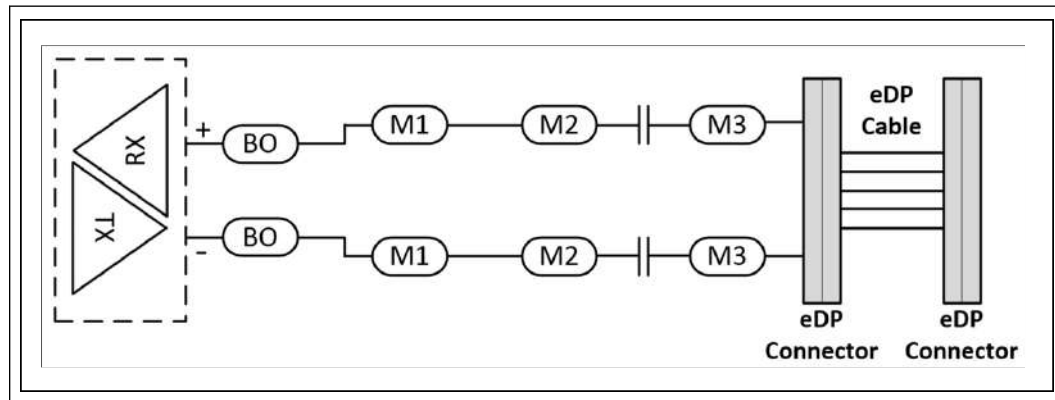


Table 182. eDP AUX Topology Notes

Note	Detail
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
AC cap value	75 nF to 200 nF including tolerance.
Cable assembly	70 - 110 Ω including tolerance. Insertion loss $\geq$ -1.0 dB @ 1 MHz.

Table 183. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Rx, Tx

Table 184. eDP AUX Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2+M3	DSL, MS, SL	450	$BO+M1+M2+M3 \leq 450$ mm
Cable	Cable	500	

Max Length Total (mm): 450

Max Length Total Note: Max length not including cable.

5.2.1.2 eDP AUX MUX Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 65. eDP AUX MUX Topology diagram

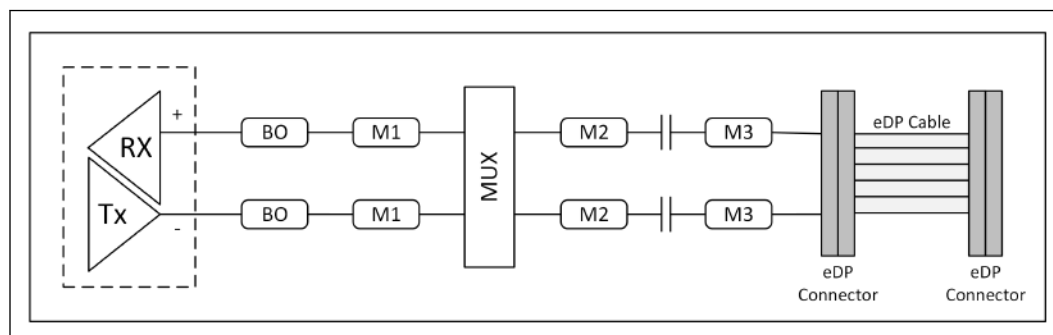


Table 185. eDP AUX MUX Topology Notes

Note	Detail
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
AC cap value	75 nF to 200 nF including tolerance.
Cable assembly	70 - 110 Ω including tolerance. Insertion loss $\geq$ -1.0 dB @ 1 MHz.

Table 186. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Rx, Tx

Table 187. eDP AUX MUX Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2+M3	DSL, MS, SL	450	$BO+M1+M2+M3 \leq 450$ mm
Cable	Cable	500	

Max Length Total (mm): 450

Max Length Total Note: Max length not including cable.

5.2.1.3 eDP HBR3 Main Link CTLE Topology

Figure 66. eDP HBR3 Main Link CTLE Topology Diagram

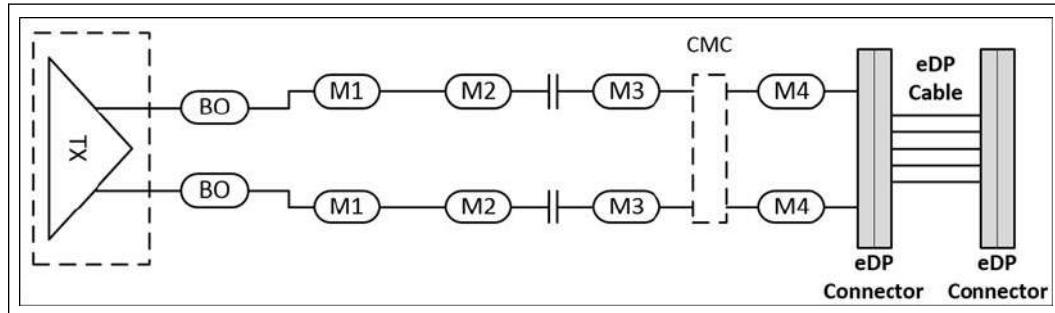


Table 188. eDP HBR3 Main Link CTLE Topology Notes

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
CMC	Refer Electromagnetic Compatibility Chapter (Common Mode Choke Selection).Optional. Populating will not reduce the supported length.
AC cap value	75 nF to 265 nF including tolerance.
355 mm cable assembly (micro coaxial)	Cable length: 355 mm for micro-coax with the following assumptions 25 mm assembly insertion loss ≥ -0.62 dB @ 4.05 GHz, including HVM tolerance 355 mm cable assembly insertion loss ≥ -5.1 dB @ 4.05 GHz, including HVM tolerance 80 - 100 ohms typical characteristic impedance with extra +/-10% tolerance
Twisted pair cable assembly	Refer to "Alder Lake Platform Twisted Pair Cable Electrical Specification for Embedded DisplayPort Technical White Paper" document #644925. This document describes the electrical parameter limits for the twisted pair cables assumed for this topology. It does not consider any EMI/ RFI risks.
TCON RX equalization	CTLE only.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
eDP HBR2 max length	Refer to eDP HBR3 Main Link CTLE+DFE Topology recommendation for HBR2 Main Link Topology max length.

Table 189. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	3	Total Channel

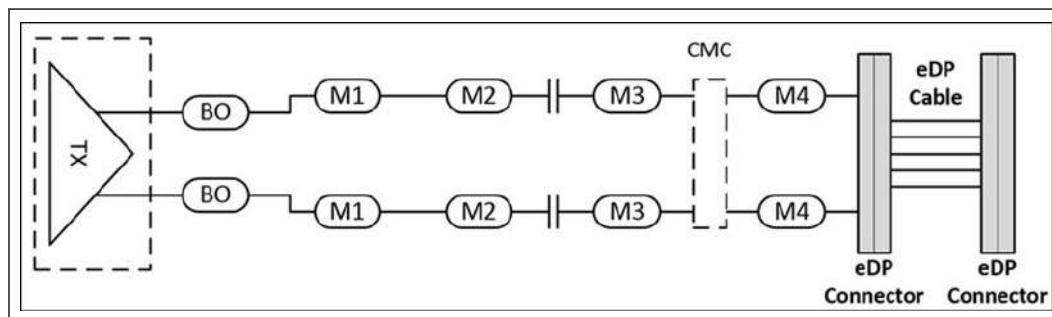
Tx

Table 190. eDP HBR3 Main Link CTLE Topology Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1	DSL, MS, SL	220	$BO + M1 + M2 + M3 + M4 \leq 220$ mm
M2+M3+M4	DSL, MS, SL	25	
Cable	Cable	355	

Max Length Total (mm): 220

Max Length Total Note: 1) Max length not including cable. 2) Refer to eDP HBR3 Main Link CTLE+DFE Topology recommendation for HBR2 max length. 3) Above mentioned table is for mid-loss, for T3 premium mid-loss Length of 10 mm can be increased for CTLE.

5.2.1.4 eDP HBR3 Main Link CTLE+DFE Topology**Figure 67. eDP HBR3 Main Link CTLE+DFE Topology Diagram****Table 191. eDP HBR3 Main Link CTLE+DFE Topology Notes**

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
CMC	Refer Electromagnetic Compatibility Chapter (Common Mode Choke Selection). Optional. Populating will not reduce the supported length.
AC cap value	75 nF to 265 nF including tolerance.
355 mm cable assembly (micro coaxial)	Cable length: 355 mm for micro-coax with the following assumptions 25 mm assembly insertion loss ≥ -0.62 dB @ 4.05 GHz, including HVM tolerance 355 mm cable assembly insertion loss ≥ -5.1 dB @ 4.05 GHz, including HVM tolerance 80 - 100 ohms typical characteristic impedance with extra +/-10% tolerance

continued...

Note	Detail
Twisted pair cable assembly	Refer to "Alder Lake Platform Twisted Pair Cable Electrical Specification for Embedded DisplayPort Technical White Paper" document #644925. This document describes the electrical parameter limits for the twisted pair cables assumed for this topology. It does not consider any EMI/ RFI risks.
TCON RX equalization	CTLE + DFE.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
eDP HBR2 max length	This recommendation is applicable to eDP HBR2 Main Link topology for panels comply to the VESA eDP HBR2 standard equalization.

Table 192. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	3	Total Channel

Tx

Table 193. eDP HBR3 Main Link CTLE+DFE Topology Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1	DSL, MS, SL	300	$BO+M1+M2+M3+M4 \leq 300$ mm
M2+M3+M4	DSL, MS, SL	25	
Cable	Cable	355	

Max Length Total (mm): 300

Max Length Total Note: 1) Max length not including cable. 2) This recommendation is applicable to eDP HBR2 data rate for panels comply to the VESA eDP HBR2 standard equalization. 3) Above table is for Mid-loss, for T3 premium mid-loss 10 mm length can be increased for CTLE+DFE.

5.2.1.5 eDP HBR3 Main Link MUX CTLE Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 68. eDP HBR3 Main Link MUX CTLE Topology Diagram

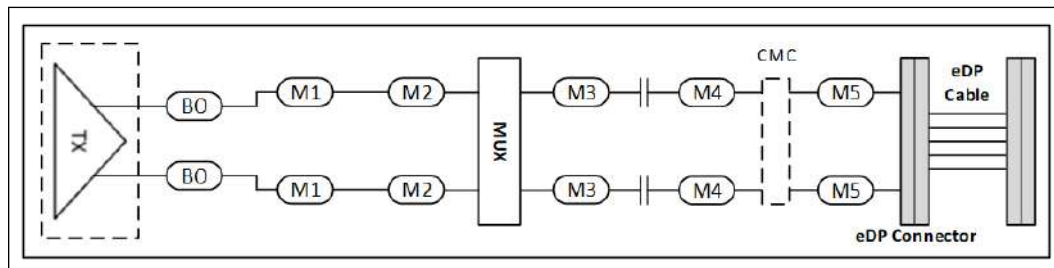


Table 194. eDP HBR3 Main Link MUX CTLE Topology Notes

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
CMC	Refer Electromagnetic Compatibility Chapter (Common Mode Choke Selection). Optional. Populating will not reduce the supported length.
AC cap value	75 nF to 265 nF including tolerance.
Mux component guidelines	Use a passive mux with the following parameters: Differential insertion loss @ 4.05 GHz > -2.2 dB Differential return loss @ 4.05 GHz < -8.0 dB Differential FEXT @ 4.05 GHz < -50 dB
355 mm cable assembly (micro coaxial)	Cable length: 355 mm for micro-coax with the following assumptions 25 mm assembly insertion loss $\geq$ -0.62 dB @ 4.05 GHz, including HVM tolerance 355 mm cable assembly insertion loss $\geq$ -5.1 dB @ 4.05 GHz, including HVM tolerance 80 - 100 ohms typical characteristic impedance with extra +/-10% tolerance
Twisted pair cable assembly	Refer to "Alder Lake Platform Twisted Pair Cable Electrical Specification for Embedded DisplayPort Technical White Paper" Document 644925. This document describes the electrical parameter limits for the twisted pair cables assumed for this topology. It does not consider any EMI/RFI risks.
TCON RX equalization	CTLE only.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
eDP HBR2 max length	Refer to eDP HBR3 MUX CTLE+DFE Topology recommendation for HBR2 max length.

Table 195. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	3	Total Channel

Tx**Table 196. eDP HBR3 Main Link MUX CTLE Topology Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1+M2	DSL, MS, SL	150	BO+M1+M2+M3+M4+M5 $\leq$ 150 mm
M3+M4+M5	DSL, MS, SL	25	
Cable	Cable	355	

Max Length Total (mm): 150

Max Length Total Note: 1) Max length not including cable. 2) Refer to eDP HBR3 MUX CTLE+DFE Topology recommendation for HBR2 max length, for T3 premium mid-loss Length of 10 mm can be increased for MUX CTLE.

5.2.1.6 eDP HBR3 Main Link MUX CTLE+DFE Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 69. eDP HBR3 Main Link MUX CTLE+DFE Topology Diagram

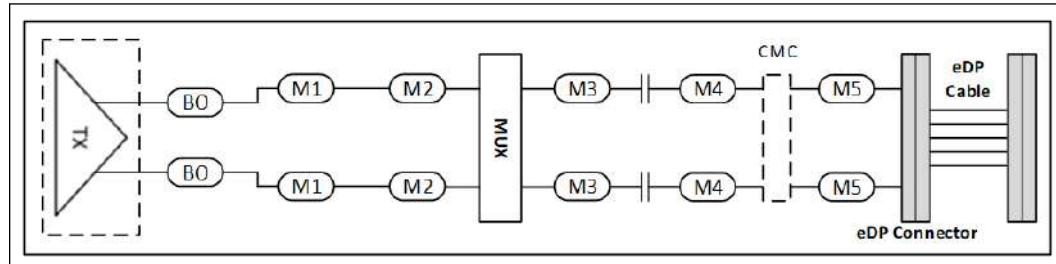


Table 197. eDP HBR3 Main Link MUX CTLE+DFE Topology Notes

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
CMC	Refer Electromagnetic Compatibility Chapter (Common Mode Choke Selection). Optional. Populating will not reduce the supported length.
AC cap value	75 nF to 265 nF including tolerance.
Mux component guidelines	Use a passive mux with the following parameters: Differential insertion loss @ 4.05 GHz > -2.2 dB Differential return loss @ 4.05 GHz < -8.0 dB Differential FEXT @ 4.05 GHz < -50 dB
355 mm cable assembly (micro coaxial)	Cable length: 355 mm for micro-coax with the following assumptions 25 mm assembly insertion loss >= -0.62 dB @ 4.05 GHz, including HVM tolerance 355 mm cable assembly insertion loss >= -5.1 dB @ 4.05 GHz, including HVM tolerance 80 - 100 ohms typical characteristic impedance with extra +/-10% tolerance
Twisted pair cable assembly	Refer to "Alder Lake Platform Twisted Pair Cable Electrical Specification for Embedded DisplayPort Technical White Paper" document #644925. This document describes the electrical parameter limits for the twisted pair cables assumed for this topology. It does not consider any EMI/ RFI risks.
TCON RX equalization	CTLE + DFE.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
eDP HBR2 max length	This recommendation is applicable to eDP HBR2 Mux topology for panels comply to the VESA eDP HBR2 standard equalization.

Table 198. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	3	Total Channel

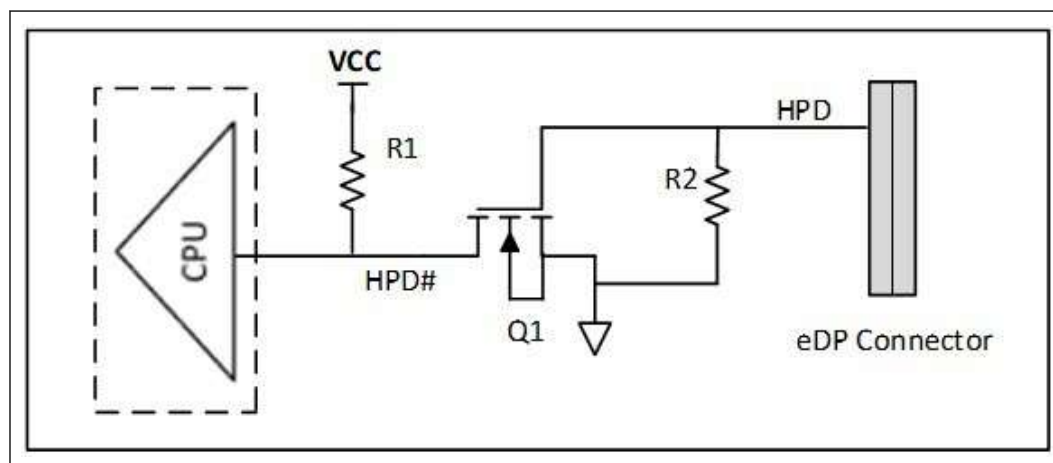
Tx

Table 199. eDP HBR3 Main Link MUX CTLE+DFE Topology Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1+M2	DSL, MS, SL	250	$BO + M1 + M2 + M3 + M4 + M5 \leq 250$ mm
M3+M4+M5	DSL, MS, SL	25	
Cable	Cable	355	

Max Length Total (mm): 250

Max Length Total Note: 1) Max length not including cable. 2) This recommendation is applicable to eDP HBR2 data rate for panels comply to the VESA eDP HBR2 standard equalization, for T3 premium mid-loss Length of 10 mm can be increased for MUX CTLE+DFE.

5.2.1.7 eDP Hot Plug Detect Implementation**Figure 70. eDP Hot Plug Detect Implementation Diagram****Table 200. eDP Hot Plug Detect Implementation Notes**

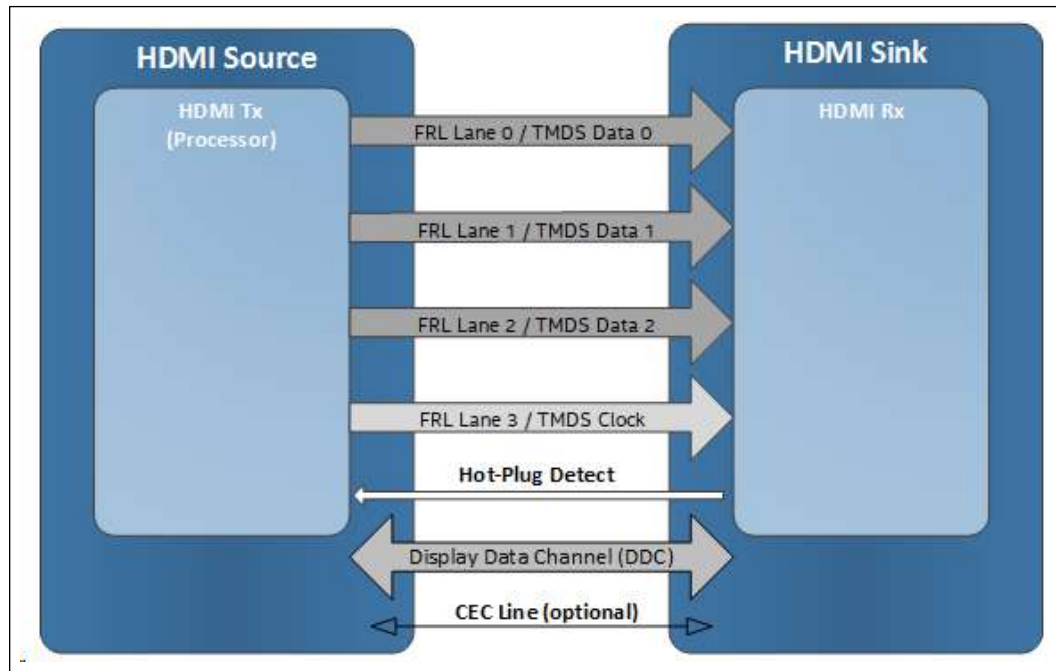
Note	Detail
Routing	Recommend 45 Ω nominal trace impedance with reasonable noise isolation.
VCC	1.25 V
R1	100 k Ω $\pm$ 5%
R2	100 k Ω $\pm$ 5%
Q1	For panel implementing eDP HPD: BSS138p or equivalent (0.85 V < VGS threshold < 1.95 V, input capacitance < 1 nF) For panel implementing eDP LVHPD: RU1J002YN or equivalent (0.35 V < VGS < 0.8 V, input capacitance < 1 nF)
Connectivity	Follow the vendor data sheet for any additional HPD connectivity requirements.

5.3 HDMI*

HDMI* Platform-specific Important Information

HDMI* includes three separate communications channels: TMDS or FRL, DDC, and the optional CEC (consumer electronics control) however CEC is not supported.

Figure 71. HDMI* Overview



HDMI* Signal Descriptions

Table 201. HDMI* Signals

Description	Signal Name	Note
Main Link (Tx)	DDI4_TX_P/N[3:0]	1
DDC	DDPx_CTRLCLK and DDPx_CTRLDATA	
Hot Plug Detect	DDSP_HPDx#	
<i>Note:</i>		

Table 202. DDI Port Signal Mapping for HDMI*

Signal Mapping			
Description	DDI	HDMI (Default non-reversed)	HDMI (Reversed)
Main Link (Tx)	DDI4_TXP/N[0]	HDMI Data_2	HDMI CLK
	DDI4_TXP/N[1]	HDMI Data_1	HDMI Data_0
<i>continued...</i>			

Signal Mapping			
Description	DDI	HDMI (Default non-reversed)	HDMI (Reversed)
	DDI4_TXP/N[2]	HDMI Data_0	HDMI Data_1
	DDI4_TXP/N[3]	HDMI CLK	HDMI Data_2
<i>Note:</i> For systems which implements reversed lane configuration it is required to set DDI Lane Reversal to enabled in VBT. DDI lane reversal option can be found under EFP1 configuration.			

5.3.1 HDMI

Table 203. HDMI General Guidelines

Note	Detail
Length matching between pair to pair (inter pair)	Within 13 mm (HDMI 2.1); 25 mm (HDMI 1.4).
Voiding recommendation	It is recommended to void unused (non-functional) pads for differential vias and pin pads for CMCs, ESDs and connectors to optimize the impedance matching in the channel. Voiding of 0402 or smaller AC capacitors pads is not necessary.
Unused signals	Unused data and AUX signals should be left no connected at the BGA ball.
Routing and return via requirement near vertical transition	For all vertical transitions, the differential transmission lines should be connected to the two vias symmetrically. Moreover, Vss return vias should be placed near the signal vias in an equidistant symmetrical manner and all signal vias should have identical Vss return vias around each of them.
Fiberweave notes	Must adhere to fiberweave related best routing practices (#406926).

Table 204. HDMI Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	Yes	13
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0

5.3.1.1 HDMI Retimer 12G Topology

Figure 72. HDMI Retimer 12G Topology Diagram

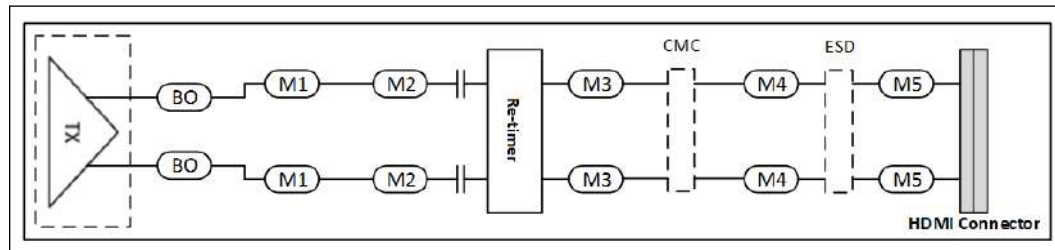


Table 205. HDMI Retimer 12G Topology Notes

Note	Detail
Number of vias allowed	Max 2 vias.
Maximum via stub length	250 um per via.
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
CMC	Refer Electromagnetic Compatibility Chapter (Common Mode Choke Selection). Optional. Populating will not reduce the supported length.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection). Optional if Retimer has integrated ESD that meets discharge requirements.
AC cap value	85 nF to 200 nF including tolerances.
Re-timer	Re-timer Rx must be spec compliant for HDMI 2.1a FRL source requirements at TP2_EQ
Max supported HDMI 1.4b TMDS data rate	3.4 Gbps
Max supported HDMI 2.1 TMDS data rate	6.0 Gbps
Max supported HDMI 2.1 FRL data rate	12 Gbps
Retimer sideband communication	Any port configured to support HDMI must implement sideband communications via the DDC (I2C) channel associated with that HDMI port as defined in the Video BIOS Table (VBT).
Connector	Connector shall meet all Category 3 requirements defined by the HDMI1.4b spec (Pin assignment, contact sequence, mechanical performance, electrical performance, environmental characteristics, connector drawings) in addition to the electrical performance requirements listed in HDMI2.1 spec which are defined by compliance parametric electrical tests HFR6-1 (Mated Connector Differential Impedance), HFR6-2 (Mated Connector Attenuation - Differential Insertion Loss) and HFR6-3 (Mated Connector Attenuation to Far-End Crosstalk Ratio) and must be included within the "Approved Category 3 Connector list" defined by the HDMI Licensing Administrator Inc.
Routing style	To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.

Table 206. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	2	Pre-Channel

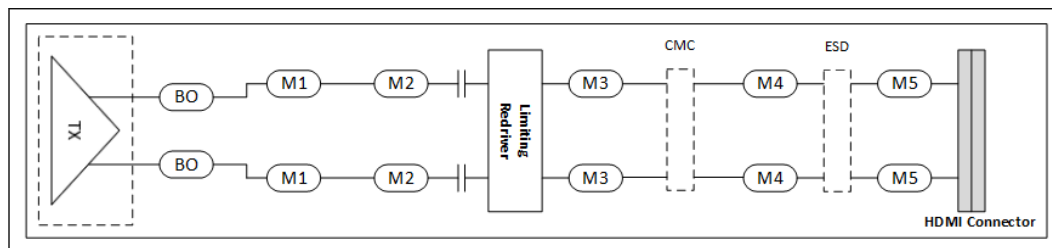
Tx, Pre-Channel**Table 207. HDMI Retimer 12G Topology Tx, Pre-Channel Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1+M2	DSL, MS, SL	220	$BO + M1 + M2 \leq 220$ mm

Max Length Total (mm): 220

Tx, Post-Channel**Table 208. HDMI Retimer 12G Topology Tx, Post-Channel Other Signals**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M3+M4+M5	MS	0	Refer Vendor Guidelines

5.3.1.2 HDMI Limiting Redriver 12G Topology**Figure 73. HDMI Limiting Redriver 12G Topology Diagram****Table 209. HDMI Limiting Redriver 12G Topology Notes**

Note	Detail
Maximum via stub length	250um per via
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
CMC	Refer Electromagnetic Compatibility Chapter (Common Mode Choke Selection) for more details. Populating will not reduce the supported length.
ESD	Consult with the redriver vendor to determine the appropriate ESD recommendation. Optional: If Redriver has integrated ESD that meets discharge requirements.
AC cap value	85 nF to 200 nF including tolerances.
Limiting Redriver	Redriver must be HDMI 2.1a FRL spec compliant.

continued...

Note	Detail
	Redriver part and equalization setting should be selected accordingly to compensate pre-channel losses. Refer to the redriver datasheet for detailed information on EQ settings and performance characteristics to achieve the best results. The exact pre-channel differential insertion loss for a given port with a specific routing and stackup can have a different maximum routing length limit. These recommendations are intended to ensure designs will have sufficient RX EQ to accommodate the expected worst-case loss.
Max supported HDMI 1.4b TMDS data rate	3.4 Gbps
Max supported HDMI 2.1 TMDS data rate	6.0 Gbps
Max supported HDMI 2.1 FRL data rate	12 Gbps
Retimer sideband communication	Any port configured to support HDMI must implement sideband communications via the DDC (I2C) channel associated with that HDMI port as defined in the Video BIOS Table (VBT).
Connector	Connector shall meet all Category 3 requirements defined by the HDMI1.4b spec (Pin assignment, contact sequence, mechanical performance, electrical performance, environmental characteristics, connector drawings) in addition to the electrical performance requirements listed in HDMI2.1 spec which are defined by compliance parametric electrical tests HFR6-1 (Mated Connector Differential Impedance), HFR6-2 (Mated Connector Attenuation - Differential Insertion Loss) and HFR6-3 (Mated Connector Attenuation to Far-End Crosstalk Ratio) and must be included within the "Approved Category 3 Connector list" defined by the HDMI Licensing Administrator Inc.

Table 210. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	2	Pre-Channel

Tx, Pre-Channel

Max Length Total Note: Note1*: Max/min routing length for HDMI Limiting Redriver 12G (12 Gbps) depends on redriver performance and its loss compensation capability when used in conjunction with Intel processor. Work with redriver vendor to ensure the selected part can provide adequate signal integrity for the end device.

Table 211. HDMI Limiting Redriver 12G Topology Tx, Pre-Channel Other Signals

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	0	Note1*
M1+M2	MS, DSL, SL	0	Note1*

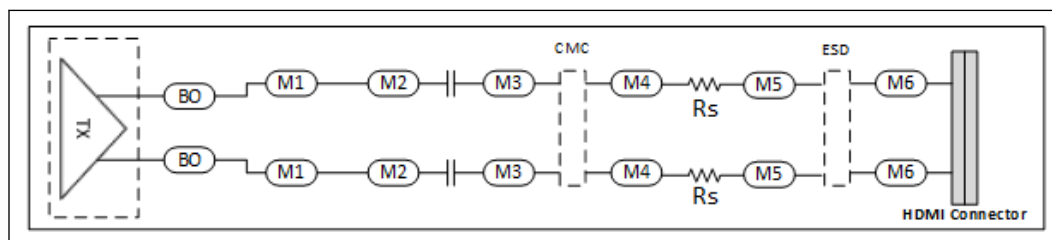
Tx, Post-Channel

Min Length Total Note: Refer to redriver vendor for minimum post-channel lengths.

Max Length Total Note: Note1*: Max/min routing length for HDMI Limiting Redriver 12G (12 Gbps) depends on redriver performance and its loss compensation capability when used in conjunction with Intel processor. Work with redriver vendor to ensure the selected part can provide adequate signal integrity for the end device.

Table 212. HDMI Limiting Redriver 12G Topology Tx, Post-Channel Other Signals

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M3+M4+M5	MS, DSL, SL	0	Note1*

5.3.1.3 HDMI CR 1.65G Topology**Figure 74. HDMI CR 1.65G Topology Diagram****Table 213. HDMI CR 1.65G Topology Notes**

Note	Detail
Maximum via stub length	250 um per via.
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
CMC	Refer Electromagnetic Compatibility Chapter (Common Mode Choke Selection) for more details. Populating will not reduce the supported length.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection).
AC cap value	Nominal 100 nF recommended (tolerance 75 nF to 200 nF).
Rs	0 Ω (Placeholder for potential through connection differential impedance tuning)
Max supported HDMI 1.4b TMDS data rate	1.65 Gbps

Table 214. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	3	Total Channel

Mainstream, Mid Loss (ML), Tx**Table 215. HDMI CR 1.65G Topology Mainstream, Mid Loss (ML), Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	13
M1+M2	MS, DSL, SL	180

Max Length Total (mm): 180

Figure 75. HDMI Hot Plug Detect Implementation Diagram



5.3.1.5 HDMI DDC Topology

Figure 76. HDMI DDC Direct Connect Topology Diagram

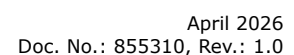
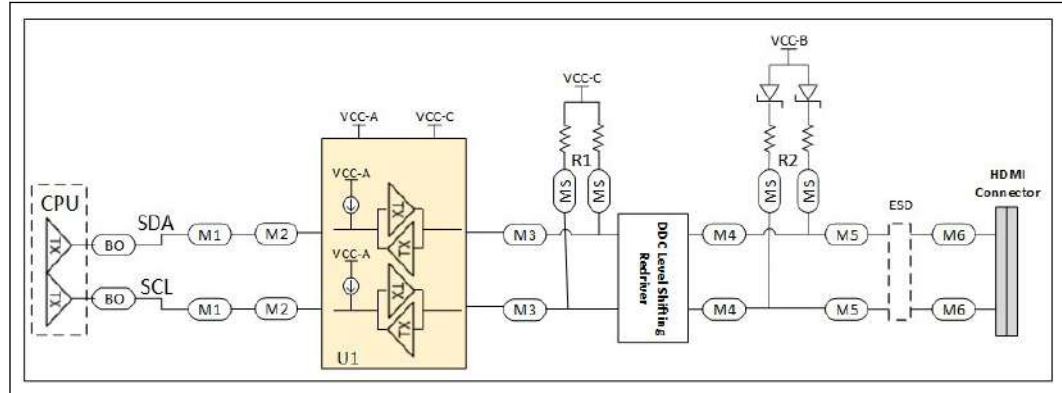


Figure 77. HDMI DDC Buffered Topology Diagram**Table 217. HDMI DDC Topology Notes**

Note	Detail
Routing	Total length from driver to HDMI connector is limited by spec capacitance limit of 50 pF including trace capacitance and discrete. Please refer to I2C section for calculation details.
Power rails	Power rails which will turn off in S3-S5 sleep states.
VCC-A	1.8 V
VCC-B	5 V
VCC-C	3.3 V
U1	I2C Voltage Level Translator, TCA9509 or equivalent.
R1 resistor	10 k Ω $\pm$ 5%
R2 resistor	1.5 k Ω $\pm$ 5%
Back drive current protection	Max capacitance for back drive current protection Schottky diode is 10 pF. Must use diode of Schottky type to minimize voltage drop.
I2C guidance	DDC signals follow I2C guidelines.
Alternative implementations	Depending on the Active Level Shifter implemented the DDC implementation may vary. Please refer to the vendors requirements for any changes in implementation.
Active level shifter part	Follow Active Level Shifter datasheet recommendation as SDA/SCL signals may be optional depending on the part requirements. Component may be removed from the topology diagram if SDA/SCL connections are not required.
ESD	For recommended part numbers, please refer to the EMC Component Selection section.
Cs cap value	47 pF nominal.
Edge rate control	An edge rate reduction mechanism such as a 47 pF capacitor (Cs) terminated to VSS is recommended to comply with DisplayPort maximum Aux slew rate specification .

Tx

Table 218. HDMI DDC Topology Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
B0	DSL, MS, SL	13
M1	DSL, MS, SL	254
M2	DSL, MS, SL	254
M3	DSL, MS, SL	254
M4+M5	DSL, MS, SL	38

Max Length Total (mm): 254

Table 219. HDMI DDC Topology Tx Other Signals

Segment	Tline Type	Max Length Segment (mm)
MS	DSL, MS, SL	25

5.4 Digital Display Interface Termination Guidelines

Table 220. Disabling and Termination Guidelines

Signal Name	Recommendation
DDIA_TX_N/P[3:0]	Left unconnected
DDIA_AUX_N/P	Left unconnected
DDSP_HPDA#	Pull high to 1.25v through 10Kohm resistor
DDSP_HPDAx# (TCP)	Pull high to 1.8v through 10Kohm resistor
TCPx_TX[1:0]_P/N TCPx_TXRX[1:0]_P/N	Left unconnected
DDI4_TX_N/P[3:0]	Left unconnected
DDI_RCOMP	Rcomp resistor is required as a placeholder (unstuffed), for more information refer to the RCOMP Recommendation on page 124.

5.5 Hybrid Graphics

Microsoft* Hybrid Graphics definition includes the following:

- The system contains a single integrated GPU and a single discrete GPU.
- It is a design assumption that the discrete GPU has a significantly higher performance than the integrated GPU.
- Both GPUs shall be physically enclosed as part of the system.
 - MS Hybrid DOES NOT support hot-plugging of GPUs.
 - OEMS should seek further guidance from MS before designing systems with the concept of hot-plugging.
- Starting with Th1 (WDDM 2.0), a previous restriction that the discrete GPU is a render-only device, with no displays connected to it, has been removed. A render-only configuration with NO outputs is still allowed, just NOT required.

It must be noted that systems that have outputs available off of the discrete GPU will NOT support previous versions of the OS (Windows* 8.1 and Older).

Microsoft* Windows 10 operating system enables the Windows 10 Hybrid graphics framework wherein the GPUs and their drivers can be simultaneously utilized to provide users with the benefits of both performance capability of discrete GPU (dGPU) and low-power display capability of the processor GPU (iGPU). For instance, when there is a high-end 3D gaming workload in progress, the dGPU will process and render the game frames using its graphics performance, while iGPU continues to perform the display operations by compositing the frames rendered by dGPU.

Each vendor [Intel as the vendor of integrated GPU / 3rd Party Graphics Vendor / Microsoft*] is responsible for their own component, and Microsoft owns the overall implementation including all Hybrid Graphics related documentation and guidance.

Hybrid Graphics has requirements which determine support on different platforms and these requirements may be different between versions of OS. Refer Microsoft documentation for detailed requirements per OS version.

The iGPU device is configured as the primary display adapter driving the embedded or local flat panel of the device. If dGPU is designated to drive the internal display, it gets configured as primary display adapter. In this scenario, and when there is no external display driven by iGPU, the system ceases to qualify as a hybrid graphics, and it rather behaves like any discrete graphics system where the capabilities of iGPU is not utilized.

Having a system configured where both iGPU and dGPU are driving displays needs to be carefully evaluated especially if there is difference in display capabilities of the two graphics adapters that would lead to user confusion or bad UX.

6.0 Processor I/Os

6.1 High Speed I/Os

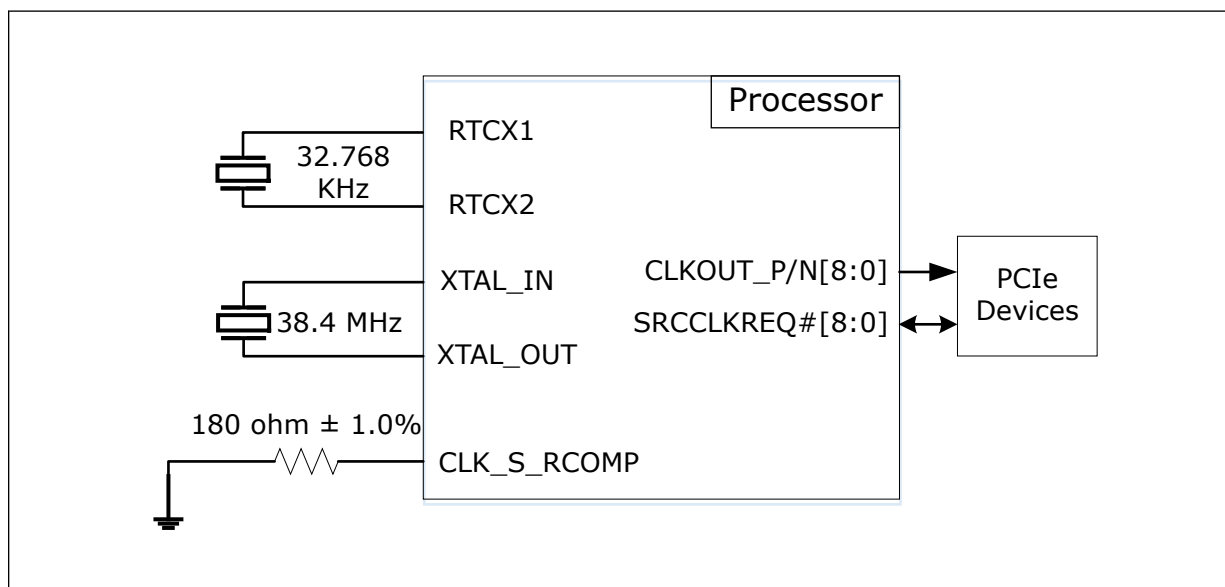
6.1.1 Reference Documents

Document Name	Document Number
Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper	726825
PCIe Gen4 Stub Length Requirement Relaxation Technical White Paper	723631
USB 3.1 Internal Cable and Connector Implementation Technical White Paper	571591
Fiber Weave Effect: Practical Impact Analysis and Mitigation Strategies - Technical Paper	406926
PCIe* 4.0 Re-Driver Vendor Status Update	575423
Thunderbolt - USB Type-C TBT Specification - Rev r1p5.D0 8	557111
Ice Lake Platform Laptop TBT Internal Cable and Connector Implementation Document Technical White Paper	575032
Thunderbolt JHL92x0R June Bridge Collateral	843990
Thunderbolt - Barlow Ridge HR Collateral	736855
Platform M.2 and Device Down-Based PCIe* Gen4 Cable Topology Technical White Paper	768992
Nova Lake-H and Nova Lake-HX Processor PCIe Gen5 CEM Add-In Card Topology Guidelines Technical Advisory	873804
Nova Lake H Platform Non-POR Topology of Display, Storage, and Thunderbolt™ Technical White Paper	914060
Mobile Platform Thunderbolt5™ Internal Cable and Connector Implementation Guide	873680

6.1.2 Platform Clock Design Guidelines

6.1.2.1 Interface Details

Figure 78. Clock Integration Distribution Diagram



6.1.2.2 Signal Descriptions

Table 221. Platform Clocks and Associated Signal Details and Descriptions

Note
For comprehensive information on platform clock details and signal descriptions, refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2 (844316).

6.1.2.3 Platform Clock Associated Signal Guidelines

- SRCCLKREQ#
- 38.4 MHz Input Clock
 - XTAL_IN
 - XTAL_OUT
- CLKOUT_P/N

NOTES

- All platforms are required to provide a 38.4 MHz input to the processor to enable the processor to generate all of its internal reference clocks and all of the single-ended and differential platform clock outputs.
- It is strongly recommended that customers designing a platform pay special attention and follow all of the XTAL_IN / XTAL_OUT routing guidelines. The crystal, components, and their XTAL_IN / XTAL_OUT signals are very sensitive to board noise and care must be taken when routing and placing these components and signals on the motherboard to provide shielding from any noisy signals or power planes near the components and traces. The details covered in this section provide the guidelines needed to ensure the best environment for these components and signals. Any deviation from these guidelines could result in some platform clock instabilities.

SRC Clock Request - SRCCLKREQ#[8:0]

Figure 79. SRC Request Signal Topology

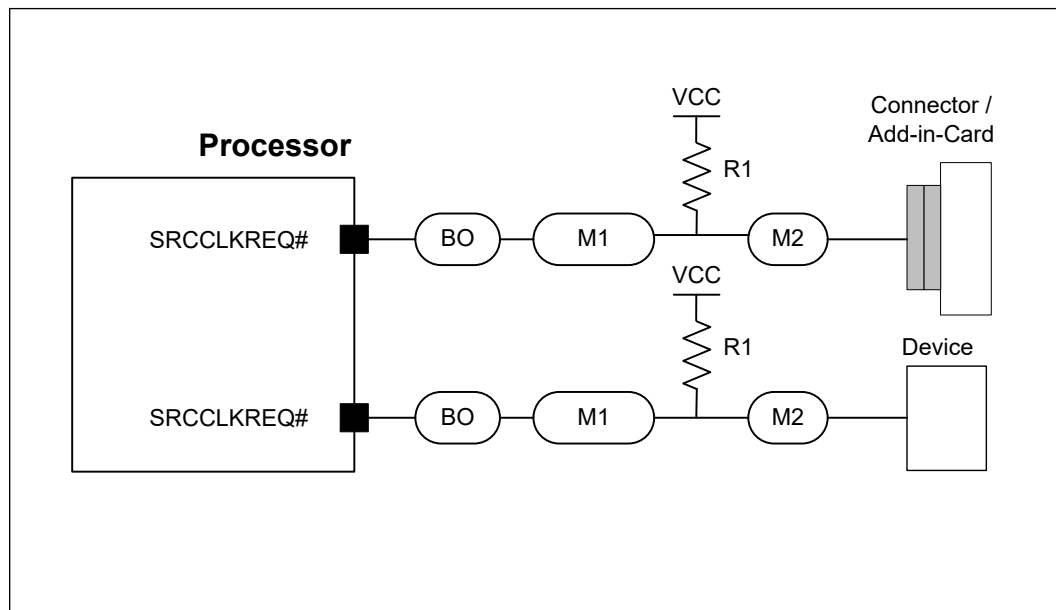


Table 222. SRC Request Signal Routing Guideline

Parameter	Segment	Stack - up	Rule
Reference Plane	BO , M1 , M2	MS/SL/DSL	Ground
Max Segment Length	BO	MS/SL/DSL	2.54 mm
	M2	MS/SL/DSL	2.54 mm
Total Motherboard Length	Processor-2-Connector (BO+M1+M2)	MS/SL/DSL	254 mm
	Processor-2-Device (BO +M1+M2)	MS/SL/DSL	304.8 mm
continued...			

Parameter	Segment	Stack - up	Rule
Resistor (R1)			10k Ohm $\pm 10\%$
Max Transition Via Count			Refer to the Note
<i>Notes:</i> 1. BO = Motherboard Break Out Segment, M = Motherboard Main Segments . 2. MS = Microstrip, SL = Stripline, DSL = Dual Stripline . 3. Ground referencing is preferred . However, single ended clocks can be routed referenced to other planes through the use of stitching capacitors to provide decoupling where the signal crosses planes. 4. At least one ground stitching via must be placed within 0.762 mm from every signal transition via whenever they change signal layers. 5. There are no Max Transition Via Count restrictions but it is recommended to minimize the number of via layer transitions on the SRCCLKREQ# signals. 6. PCIe* devices that support the clock request protocol and are powered on only in the SO state should use core well (SO) powered signals that can be mapped to output any PCI Express* clock as appropriate . 7. For SRCCLKREQ#[8:0] signals refer to Table 221 on page 205 8. Any un-used SRCCLKREQ# signal must be left as no connects. 9. Refer to the details covered in the "Stack-Up and PCB Considerations" for Trace Impedance, Differential Spacing, Group Spacing, and Isolation Spacing requirements			

38.4 MHz Crystal External Load Capacitor Requirements

The 38.4 MHz crystal is physically tuned to operate within the specified frequency range and ppm tolerance with a certain expected capacitive load present. The expected external capacitive load to be used (Ce) consists of the crystal capacitive load plus the pin and trace capacitances. The external load capacitors are important to minimize frequency variations from the crystal by compensating for variable PCB factors related to pin and trace capacitance. Care must be exercised in the selection of the external load capacitors to present the expected capacitive load specified for the crystal in use on the platform. To determine the External Load Capacitor Value, follow these steps:

Step 1: Ascertain the Initial External Load Capacitor Value (Ce) by employing one of the following approaches

- Manually compute the External Load Capacitor Value using the formula
 - $C_e = 2 * CL - (C_s + C_i)$, where:**
 - Ce = External Load Capacitor Value = Ce1 = Ce2
 - CL = Specified Crystal Capacitive Load
 - Found in the crystal vendor component data sheet
 - Cs = Mother Board Trace Capacitance
 - Including the crystal vendor defined pad capacitance
 - Ci = Processor Pin Capacitance = 8.5 to 10 pF
 - If utilizing a processor socket, it may be essential to consider and add its capacitance, which ranges from 0.5pF to 1.0pF.
- If incorporating a crystal vendor component that aligns with the Crystal Parameters specified in the "Reference Validation Platform (RVP) 38.4 MHz Crystal and External Load Capacitor Details" table, utilize the RVP Final External Load Capacitor (Ce) value provided as your initial Ce.

Step 2: Utilize the initial External Load Capacitor Value (Ce) on your motherboard to measure the Point in Time Frequency Tolerance to see if it meets the requirement called out in the "38.4 MHz Crystal Specifications" table. Should adjustments be required, utilize alternative Ce values iteratively until the criteria is fulfilled, thereby establishing the platform's Final External Load Capacitor (Ce) value. Ce must be calibrated to account for the combined effects of the motherboard PCB trace, XTAL component, and Processor Pin Capacitance. This calibration will ensure that the Point in Time Frequency Tolerance measured at XTAL_IN/XTAL_OUT falls within the specified range.

- The objective is to select a Ce that approaches zero PPM in measured Point in Time Frequency Tolerance.
- If necessary, collaborate with your crystal vendor to perform the Point in Time Frequency Tolerance measurements.

Table 223. Reference Validation Platform (RVP) 38.4 MHz Crystal and External Load Capacitor Details

Parameter	Values	Units	Notes
RVP Crystal Capacitive Load (CL)	10	pF	From crystal vendor data sheet.
RVP Board Trace Capacitance (Cs)	3.4	pF	Based off RVP stack-up and XTAL motherboard trace length = ~15mm.
RVP Processor Pin Capacitance (Ci) Processor Pin Capacitance Processor Socket Capacitance	9 to 11 8.5 to 10 0.5 to 1	pF	RVP Processor Pin Capacitance (Ci) value encompasses 0.5pF to 1pF for Processor Socket Capacitance.
RVP Crystal Equivalent Series Resistance (ESR)	≤ 30	Ω	From crystal vendor data sheet.
RVP Calculated Initial External Load Capacitor (Ce)	5.6 to 7.6	pF	External Load Capacitor Value using the formula $C_e = 2 * CL - (C_s + C_i)$
RVP Final External Load Capacitor (Ce)	8.0	pF	Final Ce value used on RVP for C1 and C2

Table 224. 38.4 MHz Crystal Specifications

Parameter	Values	Units	Notes
Nominal Frequency ^{1,2,3}	38.4	MHz	
Over Time Frequency Tolerance ⁴	± 100	PPM	Accounting for frequency deviation over the operating temperature range and crystal aging.
continued...			

Parameter	Values	Units	Notes
Equivalent Series Resistor	≤ 30	Ω	
Point in Time Frequency Tolerance ⁵	-5 to +15	PPM	Discrete snapshot of data at a single moment.
Notes: 1. Customers should verify that the vendor's published specification in the component data sheet meet the required conditions for frequency, frequency tolerance, temperature, oscillation mode and load capacitance as specified in the respective data sheet. 2. Perform conformance testing and EMC (FCC and EN) testing in real systems. 3. Independently measure the component's electrical parameters in real systems. Measure frequency at a test output to avoid test probe loading effects. Check the measured behavior is consistent from sample to sample and that the measurements meet the published specifications. For crystals, it is also important to examine the startup behavior while varying system voltage and temperature. 4. Customer should verify through measurement the crystal frequency error accounting for calibration, temperature, aging and physical platform implementation and ensure it is within ± 100 PPM with sufficient budgeting for aging. 5. This specification outlines the Point in Time Frequency Tolerance range associated with the Final External Load Capacitor (Ce). The Point in Time Frequency Tolerance, measured at XTAL_IN/XTAL_OUT, reflects the combined effects of the motherboard PCB trace, XTAL component, and Processor Pin Capacitance.			

6.1.2.3.1 Differential Clock (Gen 3 and below support)

Table 225. Differential Clock (Gen 3 and below support) Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
2	Pair (P-N)	Within	Length (mm)	Yes	0.254
3	Pair (P-N)	Total	Length (mm)	Yes	0.127
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

CLK PCIe Gen 1-3 Add-in Card Clock Topology

Figure 80. CLK PCIe Gen1-3 Add-in Card Clock Topology Diagram

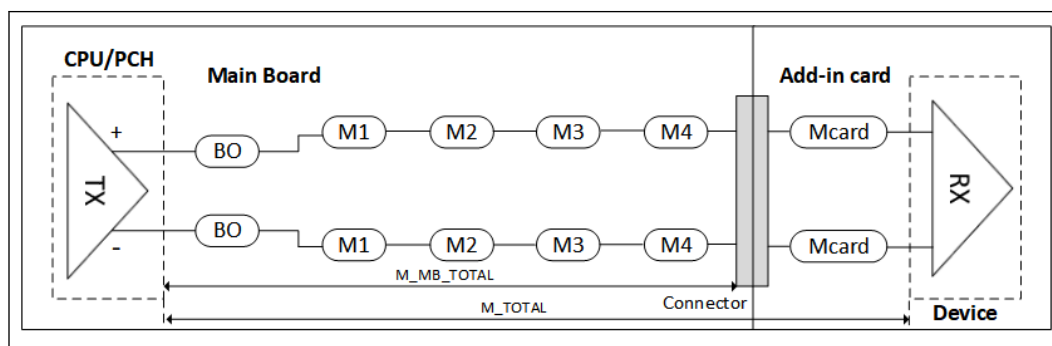


Table 226. CLK PCIe Gen 1-3 Add-in Card Clock Topology Notes

Note	Detail
Reference plane	Continuous ground only.
Add-in Card Clock Topology	This is valid for both M.2 and CEM Add-in Card.

Table 227. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	4	Total Channel

Tx

Table 228. CLK PCIe Gen 1-3 Add-in Card Clock Topology Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1+M2+M3	DSL, MS, SL	228.6	$BO + M1 + M2 + M3 + M4 \leq 228.6 \text{ mm}$
M4	MS	25.4	
Mcard	MS	76.2	

Min Length Total (mm): 76

Min Length Total Note: $M_MB_TOTAL = BO + M1 + M2 + M3 + M4$

Max Length Total (mm): 304.8

Max Length Total Note: $M_TOTAL = M_MB_TOTAL + Mcard$; If Mcard increases in length, M_MB_TOTAL must reduce in length accordingly.

CLK PCIe Gen 1-3 Device Down Clock Topology

Figure 81. CLK PCIe Gen1-3 Device Down Clock Topology Diagram

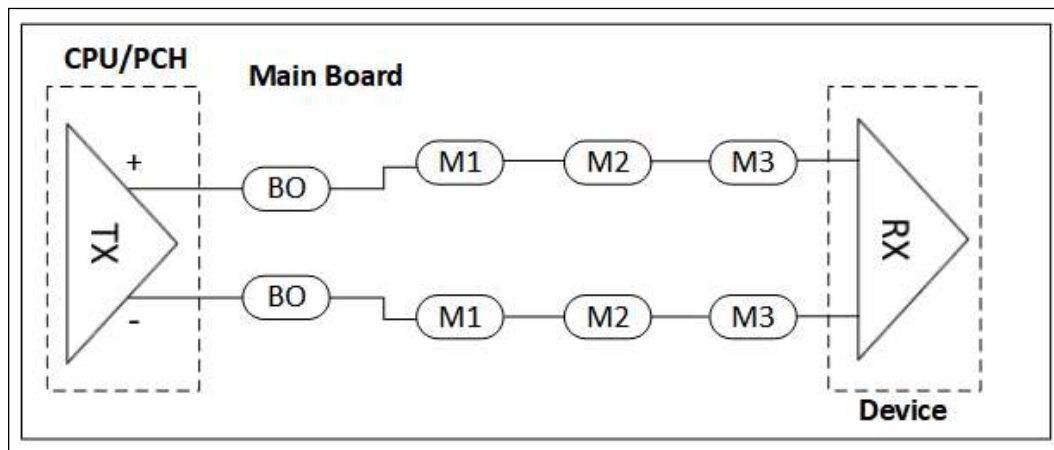


Table 229. CLK PCIe Gen 1-3 Device Down Clock Topology Notes

Note	Detail
Reference plane	Continuous ground only.

Table 230. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	4	Total Channel

Tx**Table 231. CLK PCIe Gen 1-3 Device Down Clock Topology Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1+M2	DSL, MS, SL	304.8	$BO + M1 + M2 + M3 \leq 304.8$ mm
M3	MS	25.4	

Min Length Total (mm): 89

Max Length Total (mm): 304.8

Max Length Total Note: $BO + M1 + M2 + M3 \leq 304.8$ mm**6.1.2.3.2 Differential Clock (Gen 4 support)****Table 232. Differential Clock (Gen 4 support) Length Matching**

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
2	Pair (P-N)	Within	Length (mm)	Yes	0.254
3	Pair (P-N)	Total	Length (mm)	Yes	0.127
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

CLK PCIe Gen 4 Add-in Card Clock Topology

Figure 82. CLK PCIe Gen4 Add-in Card Clock Topology Diagram

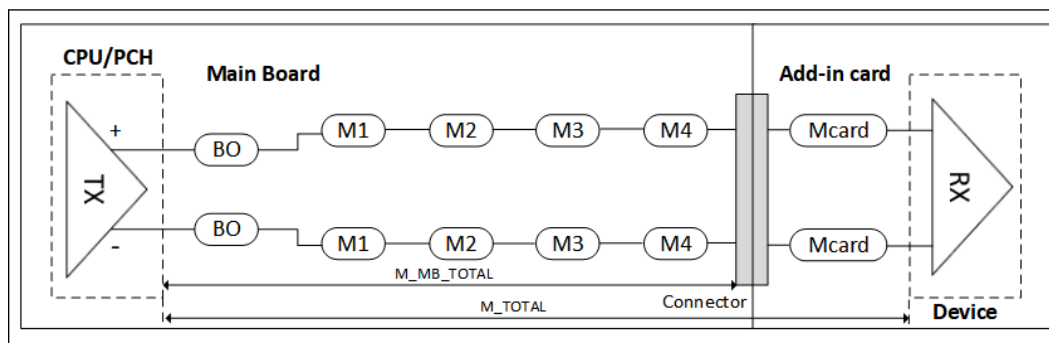


Figure 83. Differential Clock Shielding

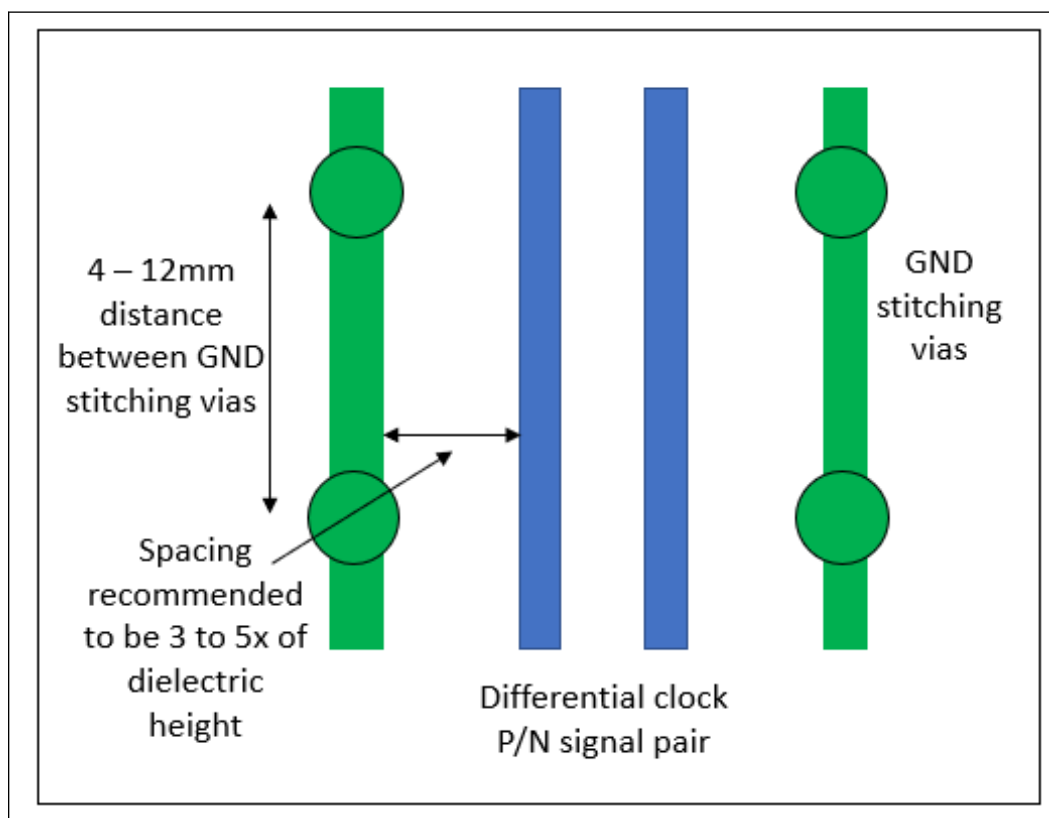


Table 233. CLK PCIe Gen 4 Add-in Card Clock Topology Notes

Note	Detail
Signal isolation	Differential SRC Clocks supporting PCIe Gen4 devices require 0.76 mm isolation to adjacent high speed IOs. Isolation to other Differential Clocks, eUSB2 and low-speed IOs can be maintained at 0.5 mm.
Reference plane	Continuous ground only.
EMC/ RF noise protection	[1] For signal sections routed as microstrip on surface layer, a GND ring/ shield is advised to be added with sufficient GND stitching vias for the length of the microstrip trace. [2] GND ring/ shield stitching vias should be placed at regular intervals of 4 - 12 mm. [3] It is advised to route in inner layer as stripline/ dual stripline routing. Limit microstrip routing and follow above shielding guidelines.
Add-in Card Clock Topology	This is valid for both M.2 and CEM Add-in Card.

Table 234. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	4	Total Channel

Tx**Table 235. CLK PCIe Gen 4 Add-in Card Clock Topology Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1+M2+M3	DSL, MS, SL	228.6	$BO+M1+M2+M3+M4 \leq 228.6$ mm
M4	MS	25.4	
Mcard	MS	76.2	

Min Length Total (mm): 76

Min Length Total Note: $M_MB_TOTAL = BO + M1 + M2 + M3 + M4$

Max Length Total (mm): 304.8

Max Length Total Note: $M_TOTAL = M_MB_TOTAL + Mcard$; If Mcard increases in length, M_MB_TOTAL must reduce in length accordingly.

CLK PCIe Gen 4 Device Down Clock Topology

Figure 84. CLK PCIe Gen4 Device Down Clock Topology Diagram

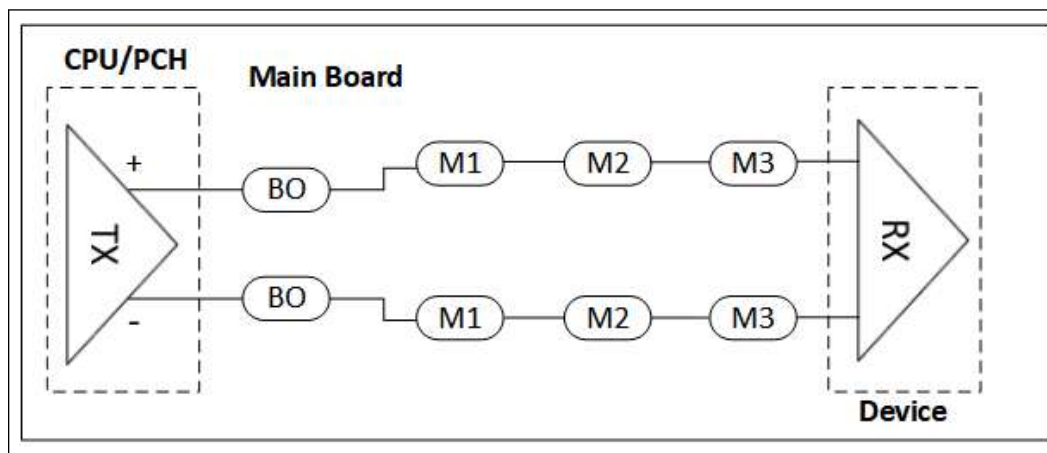


Figure 85. Differential Clock Shielding

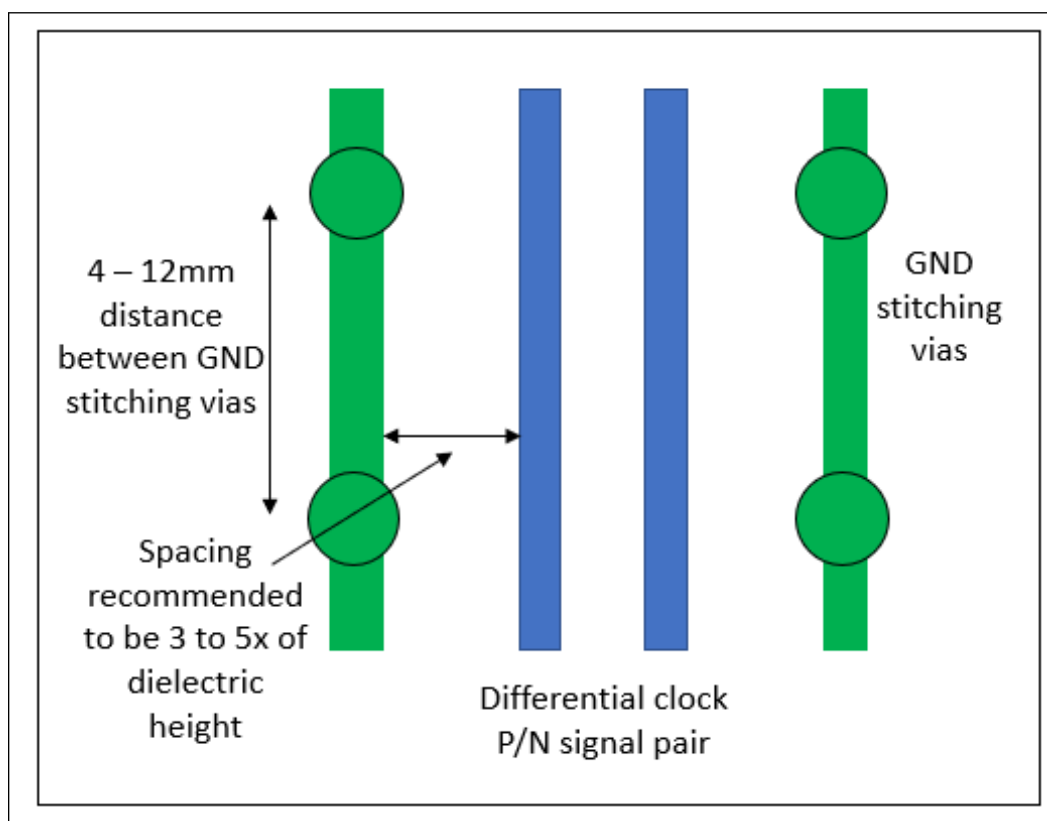


Table 236. CLK PCIe Gen 4 Device Down Clock Topology Notes

Note	Detail
Signal isolation	Differential SRC Clocks supporting PCIe Gen4 devices require 0.76 mm isolation to adjacent high speed I/Os. Isolation to other Differential Clocks, eUSB2 and low-speed I/Os can be maintained at 0.5 mm.
Reference plane	Continuous ground only.
EMC/ RF noise protection	[1] For signal sections routed as microstrip on surface layer, a GND ring/ shield is advised to be added with sufficient GND stitching vias for the length of the microstrip trace. [2] GND ring/ shield stitching vias should be placed at regular intervals of 4 - 12 mm. [3] It is advised to route in inner layer as stripline/ dual stripline routing. Limit microstrip routing and follow above shielding guidelines.

Table 237. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	4	Total Channel

Tx**Table 238. CLK PCIe Gen 4 Device Down Clock Topology Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1+M2	DSL, MS, SL	304.8	$BO+M1+M2+M3 \leq 304.8$ mm
M3	MS	25.4	

Min Length Total (mm): 89

Max Length Total (mm): 304.8

Max Length Total Note: $BO+M1+M2+M3 \leq 304.8$ mm**6.1.2.3.3 Differential Clock (Gen 5 support)****Table 239. Differential Clock (Gen 5 support) Length Matching**

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Within	Length (mm)	Yes	0.127
3	Pair (P-N)	Total	Length (mm)	Yes	0.127
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

CLK PCIe Gen 5 Add-in Card Clock Topology

Figure 86. CLK PCIe Gen5 Add-in Card Clock Topology Diagram

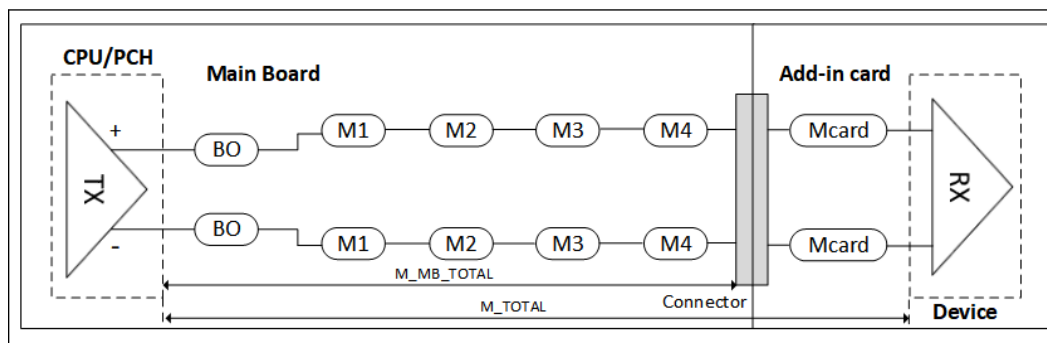


Figure 87. Differential Clock Shielding

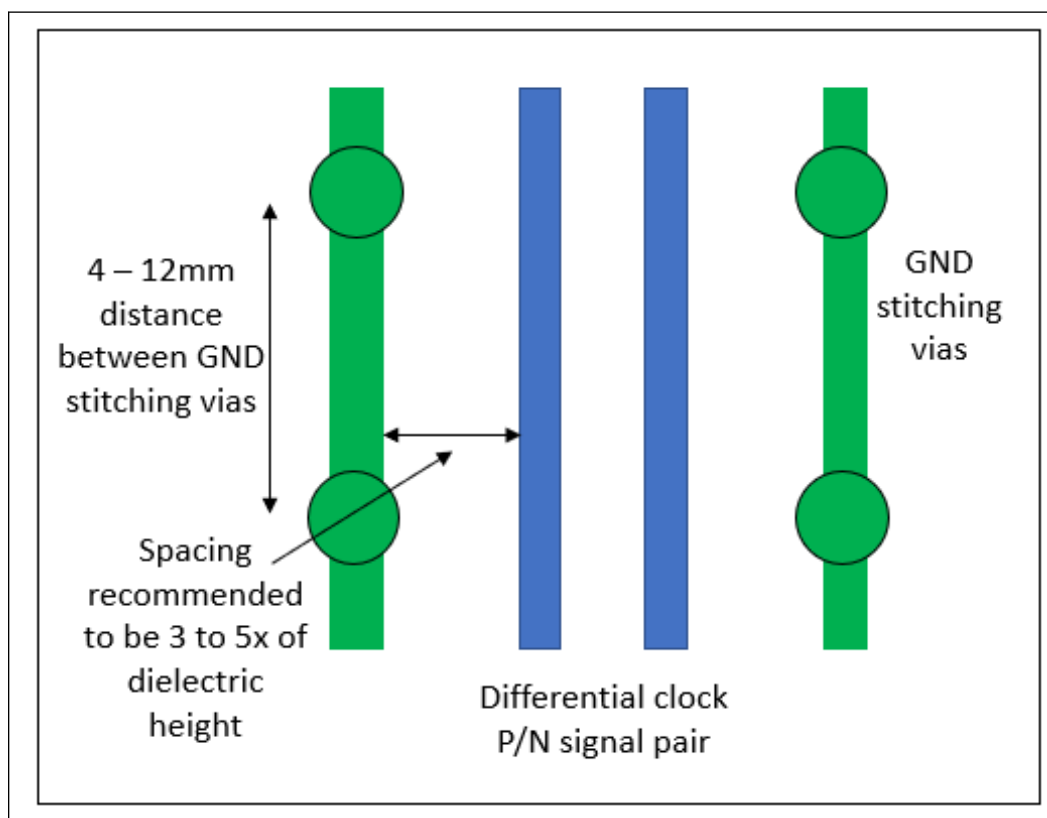


Table 240. CLK PCIe Gen 5 Add-in Card Clock Topology Notes

Note	Detail
Signal isolation	Differential SRC Clocks supporting PCIe Gen5 devices require 0.76 mm isolation to adjacent high speed IOs. Isolation to other Differential Clocks, eUSB2 and low-speed IOs can be maintained at 0.5 mm.
Reference plane	Continuous ground only
EMC/ RF noise protection	[1] For signal sections routed as microstrip on surface layer, a GND ring/ shield is advised to be added with sufficient GND stitching vias for the length of the microstrip trace. [2] GND ring/ shield stitching vias should be placed at regular intervals of 4 - 12 mm. [3] It is advised to route in inner layer as stripline/ dual stripline routing. Limit microstrip routing and follow above shielding guidelines. [4] GND ring/ shield spacing to signal should be 3x-5x the dielectric height of the stackup or approximately 300 um (whichever is larger). Clock signal spacing to other signals should follow the signal isolation guidelines . [5] GND ring/ shield width should be 2x the dielectric height of the stackup or approximately 140 um (whichever is larger).
Add-in Card Clock Topology	This is valid for both M.2 and CEM Add-in Card.

Table 241. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	4	Total Channel

Tx**Table 242. CLK PCIe Gen 5 Add-in Card Clock Topology Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1+M2+M3	DSL, MS, SL	177.8	$BO+M1+M2+M3+M4 \leq 177.8$ mm
M4	MS	25.4	
Mcard	MS	76.2	

Min Length Total (mm): 50.8

Min Length Total Note: $M_MB_TOTAL=BO+M1+M2+M3+M4$

Max Length Total (mm): 254

Max Length Total Note: $M_TOTAL=M_MB_TOTAL+Mcard$; If Mcard increases in length, M_MB_TOTAL must reduce in length accordingly.

CLK PCIe Gen 5 Device Down Clock Topology

Figure 88. CLK PCIe Gen5 Device Down Clock Topology Diagram

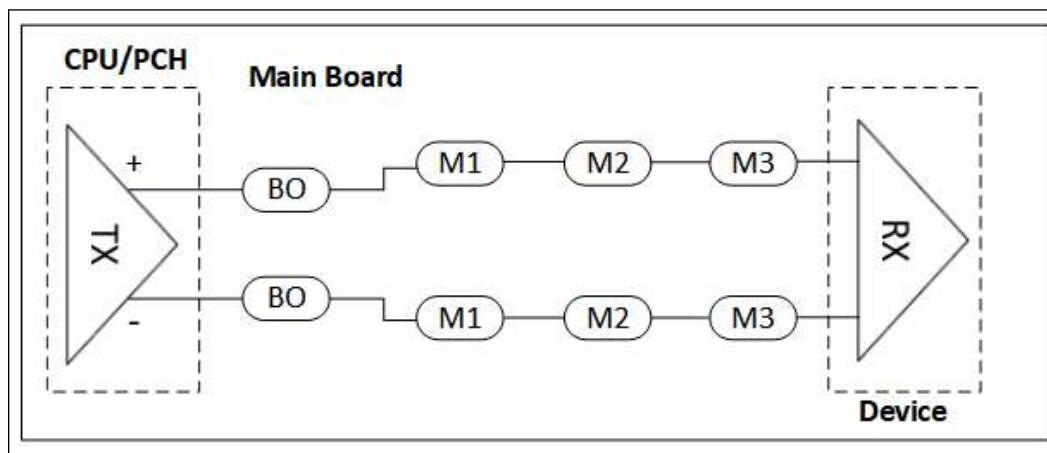


Figure 89. Differential Clock Shielding

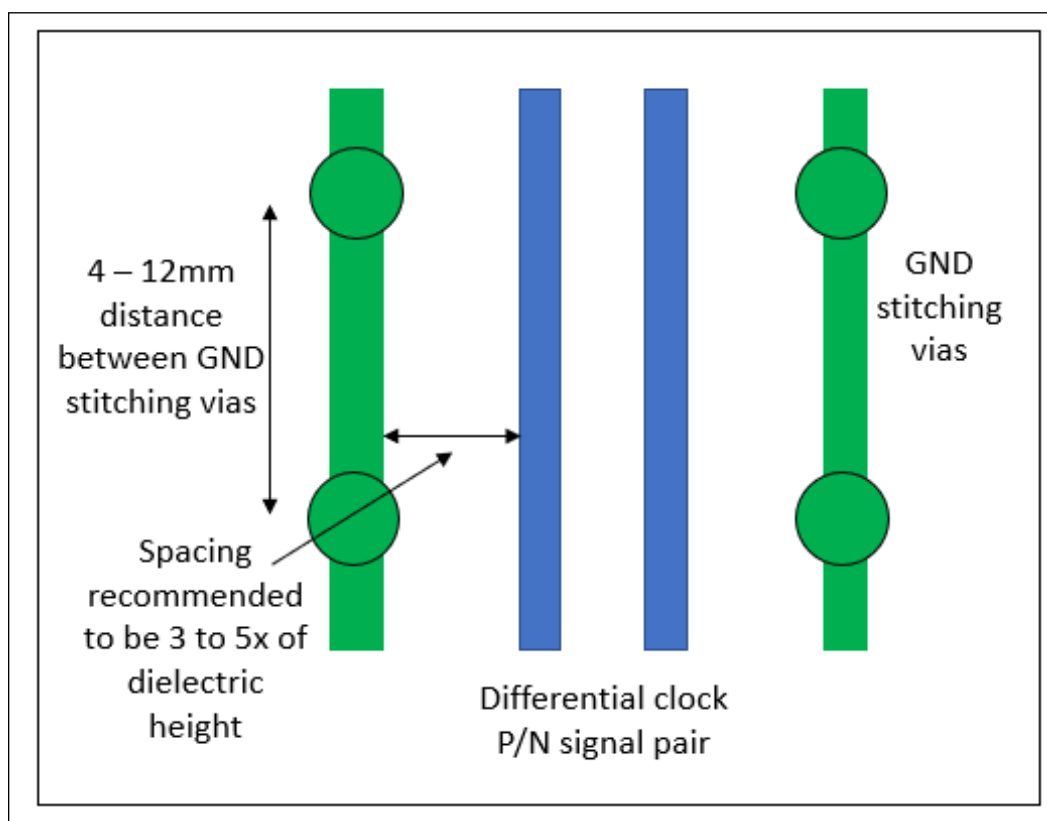


Table 243. CLK PCIe Gen 5 Device Down Clock Topology Notes

Note	Detail
Signal isolation	Differential SRC Clocks supporting PCIe Gen5 devices require 0.76 mm isolation to adjacent high speed IOs. Isolation to other Differential Clocks, eUSB2 and low-speed IOs can be maintained at 0.5 mm.
Reference plane	Continuous ground only
EMC/ RF noise protection	[1] For signal sections routed as microstrip on surface layer, a GND ring/ shield is advised to be added with sufficient GND stitching vias for the length of the microstrip trace. [2] GND ring/ shield stitching vias should be placed at regular intervals of 4 - 12 mm. [3] It is advised to route in inner layer as stripline/ dual stripline routing. Limit microstrip routing and follow above shielding guidelines. [4] GND ring/ shield spacing to signal should be 3x-5x the dielectric height of the stackup or approximately 300 um (whichever is larger). Clock signal spacing to other signals should follow the signal isolation guidelines . [5] GND ring/ shield width should be 2x the dielectric height of the stackup or approximately 140 um (whichever is larger).

Table 244. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	4	Total Channel

Tx**Table 245. CLK PCIe Gen 5 Device Down Clock Topology Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1+M2	DSL, MS, SL	254	BO+M1+M2+M3 ≤ 254 mm
M3	MS	25.4	

Min Length Total (mm): 76.2

Max Length Total (mm): 254

Max Length Total Note: $BO+M1+M2+M3 \leq 254 \text{ mm}$ **6.1.2.3.4 XTAL**

Segment Lengths

Table 248. Crystal Oscillator Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
MR	DSL, SL	

Max Length Total (mm): 25.4

Max Length Total Note: Maximum length is based on capacitor component sizes 0201

6.1.2.3.5 UFS Reference Clock

UFS Reference Clock 1-Load (Device Down) Topology

Figure 91. UFS Reference Clock Device Down Topology Diagram

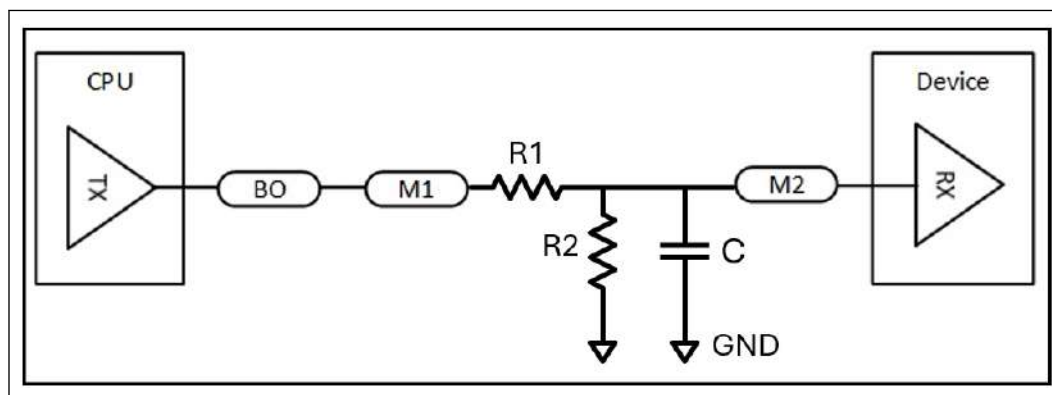


Table 249. UFS Reference Clock 1-Load (Device Down) Topology Notes

Note	Detail
Resistor values (for step down from 1.8 V to 1.2 V)	R1 = 47 Ω R2 = 150 Ω To be placed within 25.4 mm to UFS device
Capacitor, C	Placeholder shunt capacitor (unstuff by default). To be placed near R1 Refer to EMC chapter in the PDG for more details
Platform resistor tolerance	$\pm 5\%$
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Microstrip routing limitation	Due to potential RF interference concerns, it is recommended not to route UFS REFCLK signal on microstrip layer for the main route segment

Table 250. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	3	Total Channel

Segment Lengths

Table 251. UFS Reference Clock 1-Load (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
B0	MS, DSL, SL	12.7
M1	DSL, SL	165.1
M2	MS, DSL, SL	25.4

Min Length Total (mm): 38.1

Max Length Total (mm): 203.2

6.1.3 Flexible High Speed I/O

Flexible Input/Output (I/O) is a technology that allows the High Speed I/O (HSIO) lanes to be configured for connection to a Gigabit Ethernet (GbE) Controller, a PCIe* Controller, or an Extensible Host Controller Interface (xHCI) USB 3.2 Controller. Flexible I/O enables customers to optimize the allocation of the HSIO interfaces to better meet the I/O needs of their system.

NOTE

The Flexible I/O Lane support will vary depending on the SKU. Refer to the Flexible High Speed I/O section in the associated External Design Specification (EDS) Volume 1 for specific Flexible I/O implementation details.

Table 252. Acronyms

Acronyms	Description
USB	Universal Serial Bus
PCIe*	PCI Express* (Peripheral Component Interconnect Express*)
GbE	Gigabit Ethernet
HSIO	High Speed Input/Output

6.1.4 MIPI CSI-2

MIPI CSI-2 is the primary camera data interface to IPU8 (Imaging Processing Unit System). The platform supports up to 3 concurrently connected MIPI camera interfaces. Each of the 3 PHY ports supports x1/x2 in DPHY mode or T1/T2 in CPHY mode. Two of the PHYs can be aggregated to support a single camera x4 in DPHY mode or T3 in CPHY mode.

Table 253. Reference Specification

Title	Location
MIPI* CSI-2 Specifications	http://www.mipi.org/specifications/camera-interface
I2C Specifications	http://www.i2c-bus.org/
IPU8 System Design Guide	914612

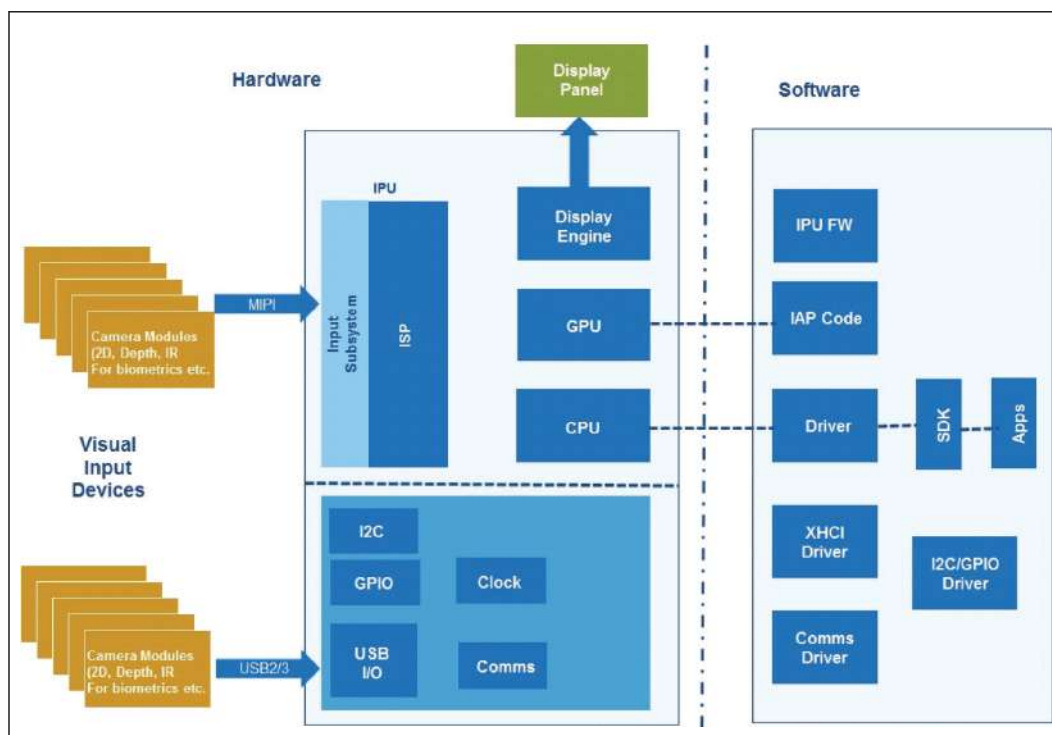
Feature Description

IPU8 includes the MIPI interfaces connected to the camera sensors. Refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2 (# [844316](#)) for supported PHY options.

6.1.4.1 CSI-2 Platform Specific Information

This platform has 18 DPHY (Data + Clocks) pins and 1 RCOMP/REXT pin.

The below diagram illustrates the different possible cameras and solutions to be supported on this platform along with the connection to Processor.

Figure 92. Visual Computing System Components Block Diagram

6.1.4.2 CSI PHY Support

Table 254. CSI PHY Support

Parameter/Port	NVL H POR
Total CDPHY Package Pins	24 + 1 Rcomp/REXT
Data and Clock Lane Count	In DPHY mode: 6 data lanes and 3 clock lanes In CPHY mode: 6 triodes
Camera Options	Up to 3 cameras: DPHY mode: x2, x2, x2 or x4, x2 CPHY mode: T2, T2, T2 or T3, T2
CSI Port A	If not being aggregated with port B: DPHY mode: x2 or x1 CPHY mode: T2 or T1 If being aggregated with port B: DPHY mode: x4 or x2 or x1 CPHY mode: T3 or T2 or T1
CSI Port B	If not being aggregated with port A: DPHY mode: x2 or x1 CPHY mode: T2 or T1 If being aggregated with port A: DPHY mode: Not supported CPHY mode: Not supported
CSI Port C	DPHY mode: x2 or x1 CPHY mode: T2 or T1

6.1.4.3 CSI-2 Lane Configurations

Table 255. CSI2 Configuration Options Non-aggregation Mode

PHY Instance	Config #1 - DPHY	Config #1 - CPHY
A 2T/X2	Port A Supports X1 (lane)/X2 (lane)	Port A Supports 2T/1T
B 2T/X2	Port B Supports X1/X2	Port B Supports 2T/1T
C 2T/X2	Port C Supports X1/X2	Port C Supports 2T/1T

Table 256. CSI2 Configuration Options Aggregation Mode

PHY Instance	Config #1 - DPHY	Config #1 - CPHY
A 3T/X4	Port A Supports X1 (lane)/X2 (lane)/x4 (lane)	Port A Supports 3T/2T/1T.
C 2T/X2	Port C Supports X1/X2	Port C Supports 2T/1T.

Table 257. NVL H LP CSI-2 DPHY Lane Allocation

Signal Name	CSI DPHY Connectivity Option #1	CSI DPHY Connectivity Option #2
CSI_A_CPHY_A0_DPHY_DP0	X2	X4
CSI_A_CPHY_B0_DPHY_DN0		
CSI_A_CPHY_C0_DPHY_CLK_P		
continued...		

Signal Name	CSI DPHY Connectivity Option #1	CSI DPHY Connectivity Option #2
CSI_A_CPHY_C1_DPHY_CLK_N		
CSI_A_CPHY_B1_DPHY_DN1		
CSI_A_CPHY_A1_DPHY_DP1		
CSI_B_CPHY_A0_DPHY_DP0 / CSI_A_CPHY_A2_DPHY_DP2	X2	
CSI_B_CPHY_B0_DPHY_DN0 / CSI_A_CPHY_B2_DPHY_DN2		
CSI_B_CPHY_C0_DPHY_CLK_P / CSI_A_CPHY_C2		
CSI_B_CPHY_C1_DPHY_CLK_N		
CSI_B_CPHY_B1_DPHY_DN1 / CSI_A_DPHY_DN3		
CSI_B_CPHY_A1_DPHY_DP1 / CSI_A_DPHY_DP3		
C SI_C_CPHY_A0_DPHY_DP0	X2	X2
CSI_C_CPHY_B0_DPHY_DN0		
CSI_C_CPHY_C0_DPHY_CLK_P		
CSI_C_CPHY_C1_DPHY_CLK_N		
CSI_C_CPHY_B1_DPHY_DN1		
CSI_C_CPHY_A1_DPHY_DP1		

Table 258. NVL HLP CSI-2 CPHY Lane Allocation

Pin Name	CSI CPHY Connectivity #1	CPHY Link	CSI CPHY Connectivity #2	CPHY Link
CSI_A_CPHY_A0_DPHY_DP0	CPHY A T2	T0	CPHY A T3	T0
CSI_A_CPHY_B0_DPHY_DN0				
CSI_A_CPHY_C0_DPHY_CLK_P		T1		T1
CSI_A_CPHY_C1_DPHY_CLK_N				
CSI_A_CPHY_B1_DPHY_DN1				
CSI_A_CPHY_A1_DPHY_DP1				
CSI_B_CPHY_A0_DPHY_DP0/ CSI_A_CPHY_A2_DPHY_DP2	CPHY B T2	T0	CPHY A T3	T2
CSI_B_CPHY_B0_DPHY_DN0/ CSI_A_CPHY_B2_DPHY_DN2				
CSI_B_CPHY_C0_DPHY_CLK_P/ CSI_A_CPHY_C2		T1		
CSI_B_CPHY_C1_DPHY_CLK_N				
CSI_B_CPHY_B1_DPHY_DN1/ CSI_A_DPHY_DN3				-
CSI_B_CPHY_A1_DPHY_DP1/ CSI_A_DPHY_DP3				
continued...				

Pin Name	CSI CPHY Connectivity #1	CPHY Link	CSI CPHY Connectivity #2	CPHY Link
CSI_C_CPHY_A0 _DPHY_DP0	CPHY C T2	T0	CPHY C T2	T0
CSI_C_CPHY_B0 _DPHY_DN0				
CSI_C_CPHY_C0 _DPHY_CLK _P				
CSI_C_CPHY_C1 _DPHY_CLK _N		T1		T1
CSI_C_CPHY_B1 _DPHY_DN1				
CSI_C_CPHY_A1 _DPHY_DP1				

6.1.4.4 CSI Signal Descriptions

CSI Signal Descriptions Non-aggregation Mode

Table 259. CSI DPHY Signal Descriptions

Design Pin Name	PHY Instance	DIR	DPHY Description	DPHY Link Type
CSI_A_CPHY_A0_DPHY_DP0	DPHY A X2	I	Data lane 0 positive terminal	Diff
CSI_A_CPHY_B0_DPHY_DN0		I	Data lane 0 negative terminal	Diff
CSI_A_CPHY_C0_DPHY_CLK_P		I	Clock lane positive terminal	Diff
CSI_A_CPHY_C1_DPHY_CLK_N		I	Clock lane negative terminal	Diff
CSI_A_CPHY_B1_DPHY_DN1		I	Data lane 1 negative terminal	Diff
CSI_A_CPHY_A1_DPHY_DP1		I	Data lane 1 positive terminal	Diff
CSI_B_CPHY_A0_DPHY_DP0/ CSI_A_CPHY_A2_DPHY_DP2	DPHY B X2	I	Data lane 0 positive terminal	Diff
CSI_B_CPHY_B0_DPHY_DN0/ CSI_A_CPHY_B2_DPHY_DN2		I	Data lane 0 negative terminal	Diff
CSI_B_CPHY_C0_DPHY_CLK_P/ CSI_A_CPHY_C2		I	Clock lane positive terminal	Diff
CSI_B_CPHY_C1_DPHY_CLK_N		I	Clock lane negative terminal	Diff
CSI_B_CPHY_B1_DPHY_DN1/ CSI_A_DPHY_DN3		I	Data lane 1 negative terminal	Diff
continued...				

Design Pin Name	PHY Instance	DIR	DPHY Description	DPHY Link Type
CSI_B_CPHY_A1_DPHY_DP1/ CSI_A_DPHY_DP3		I	Data lane 1 positive terminal	Diff
CSI_C_CPHY_A0_DPHY_DP0	DPHY C X2	I	Data lane 0 positive terminal	Diff
CSI_C_CPHY_B0_DPHY_DN0		I	Data lane 0 negative terminal	Diff
CSI_C_CPHY_C0_DPHY_CLK_P		I	Clock lane positive terminal	Diff
CSI_C_CPHY_C1_DPHY_CLK_N		I	Clock lane negative terminal	Diff
CSI_C_CPHY_B1_DPHY_DN1		I	Data lane 1 negative terminal	Diff
CSI_C_CPHY_A1_DPHY_DP1		I	Data lane 1 positive terminal	Diff
CSI_RCOMP		Analog	CSI Resistance Compensation	SE

Table 260. CSI CPHY Signal Descriptions

Design Pin Name	PHY Instance	DIR	CPHY Description	CPHY Link Type
CSI_A_CPHY_A0_DPHY_DP0	CPHY A	I	Trio 0 wire A	Trio
CSI_A_CPHY_B0_DPHY_DN0		I	Trio 0 wire B	Trio
CSI_A_CPHY_C0_DPHY_CLK_P		I	Trio 0 wire C	Trio
CSI_A_CPHY_C1_DPHY_CLK_N		I	Trio 1 wire C	Trio
CSI_A_CPHY_B1_DPHY_DN1		I	Trio 1 wire B	Trio
CSI_A_CPHY_A1_DPHY_DP1		I	Trio 1 wire A	Trio
CSI_B_CPHY_A0_DPHY_DP0/ CSI_A_CPHY_A2_DPHY_DP2	CPHY B T2	I	Trio 0 wire A	Trio
CSI_B_CPHY_B0_DPHY_DN0/ CSI_A_CPHY_B2_DPHY_DN2		I	Trio 0 wire B	Trio
CSI_B_CPHY_C0_DPHY_CLK_P/ CSI_A_CPHY_C2		I	Trio 0 wire C	Trio

continued...

Design Pin Name	PHY Instance	DIR	CPHY Description	CPHY Link Type
CSI_B_CPHY_C1 _DPHY_CLK_N		I	Trio 1 wire C	Trio
CSI_B_CPHY_B1 _DPHY_DN1/ CSI_A_DPHY_DN3		I	Trio 1 wire B	Trio
CSI_B_CPHY_A1 _DPHY_DP1/ CSI_A_DPHY_DP3		I	Trio 1 wire A	Trio
CSI_C_CPHY_A0 _DPHY_DP0	CPHY C T2	I	Trio 0 wire A	Trio
CSI_C_CPHY_B0 _DPHY_DN0		I	Trio 0 wire B	Trio
CSI_C_CPHY_C0 _DPHY_CLK_P		I	Trio 0 wire C	Trio
CSI_C_CPHY_C1 _DPHY_CLK_N		I	Trio 1 wire C	Trio
CSI_C_CPHY_B1 _DPHY_DN1		I	Trio 1 wire B	Trio
CSI_C_CPHY_A1 _DPHY_DP1		I	Trio 1 wire A	Trio
CSI_RCOMP		Analog	CSI Resistance Compensation	SE

CSI Signal Descriptions Aggregation Mode

Table 261. CSI DPHY Signal Descriptions

Design Pin Name	PHY Instance	DIR	DPHY Description	DPHY Link Type
CSI_A_CPHY_A0_DPHY_DP0	DPHY A X4	I	Data lane 0 positive terminal	Diff
CSI_A_CPHY_B0_DPHY_DN0		I	Data lane 0 negative terminal	Diff
CSI_A_CPHY_C0_DPHY_CLK_P		I	Clock lane positive terminal	Diff
CSI_A_CPHY_C1_DPHY_CLK_N		I	Clock lane negative terminal	Diff
CSI_A_CPHY_B1_DPHY_DN1		I	Data lane 1 negative terminal	Diff
CSI_A_CPHY_A1_DPHY_DP1		I	Data lane 1 positive terminal	Diff
continued...				

Design Pin Name	PHY Instance	DIR	DPHY Description	DPHY Link Type
CSI_B_CPHY_A0_DPHY_DP0/ CSI_A_CPHY_A2_DPHY_DP2		I	Data lane 0 positive terminal	Diff
CSI_B_CPHY_B0_DPHY_DN0/ CSI_A_CPHY_B2_DPHY_DN2		I	Data lane 0 negative terminal	Diff
CSI_B_CPHY_C0_DPHY_CLK_P/ CSI_A_CPHY_C2		I	Unused	Diff
CSI_B_CPHY_C1_DPHY_CLK_N		I	Unused	Diff
CSI_B_CPHY_B1_DPHY_DN1/ CSI_A_DPHY_DN3		I	Data lane 1 negative terminal	Diff
CSI_B_CPHY_A1_DPHY_DP1/ CSI_A_DPHY_DP3		I	Data lane 1 positive terminal	Diff
CSI_C_CPHY_A0_DPHY_DP0	DPHY C X2	I	Data lane 0 positive terminal	Diff
CSI_C_CPHY_B0_DPHY_DN0		I	Data lane 0 negative terminal	Diff
CSI_C_CPHY_C0_DPHY_CLK_P		I	Clock lane positive terminal	Diff
CSI_C_CPHY_C1_DPHY_CLK_N		I	Clock lane negative terminal	Diff
CSI_C_CPHY_B1_DPHY_DN1		I	Data lane 1 negative terminal	Diff
CSI_C_CPHY_A1_DPHY_DP1		I	Data lane 1 positive terminal	Diff
CSI_RCOMP		Analog	CSI Resistance Compensation	SE

Table 262. CSI CPHY Signal Descriptions

Design Pin Name	PHY Instance	DIR	CPHY Description	CPHY Link Type
CSI_A_CPHY_A0_DPHY_DP0	CPHY A T3	I	Trio 0 wire A	Trio
CSI_A_CPHY_B0_DPHY_DN0		I	Trio 0 wire B	Trio
CSI_A_CPHY_C0_DPHY_CLK_P		I	Trio 0 wire C	Trio
continued...				

Design Pin Name	PHY Instance	DIR	CPHY Description	CPHY Link Type
CSI_A_CPHY_C1 _DPHY_CLK_N		I	Trio 1 wire C	Trio
CSI_A_CPHY_B1 _DPHY_DN1		I	Trio 1 wire B	Trio
CSI_A_CPHY_A1 _DPHY_DP1		I	Trio 1 wire A	Trio
CSI_B_CPHY_A0_DPHY_DP0/ CSI_A_CPHY_A2 _DPHY_DP2		I	Trio 2 wire A	Trio
CSI_B_CPHY_B0_DPHY_DN0/ CSI_A_CPHY_B2 _DPHY_DN2		I	Trio 2 wire B	Trio
CSI_B_CPHY_C0_DPHY_CLK_P/ CSI_A_CPHY_C2		I	Trio 2 wire C	Trio
CSI_A_CPHY_C1_DPHY_CLK_N		I	Unused	Trio
CSI_B_CPHY_B1_DPHY_DN1/ CSI_A_DPHY_DN3		I	Unused	Trio
CSI_B_CPHY_A1_DPHY_DP1/ CSI_A_DPHY_DP3		I	Unused	Trio
CSI_C_CPHY_A0 _DPHY_DP0	CPHY C T2	I	Trio 0 wire A	Trio
CSI_C_CPHY_B0 _DPHY_DN0		I	Trio 0 wire B	Trio
CSI_C_CPHY_C0 _DPHY_CLK_P		I	Trio 0 wire C	Trio
CSI_C_CPHY_C1 _DPHY_CLK_N		I	Trio 1 wire C	Trio
CSI_C_CPHY_B1 _DPHY_DN1		I	Trio 1 wire B	Trio
CSI_C_CPHY_A1 _DPHY_DP1		I	Trio 1 wire A	Trio
CSI_RCOMP		Analog	CSI Resistance Compensation	SE

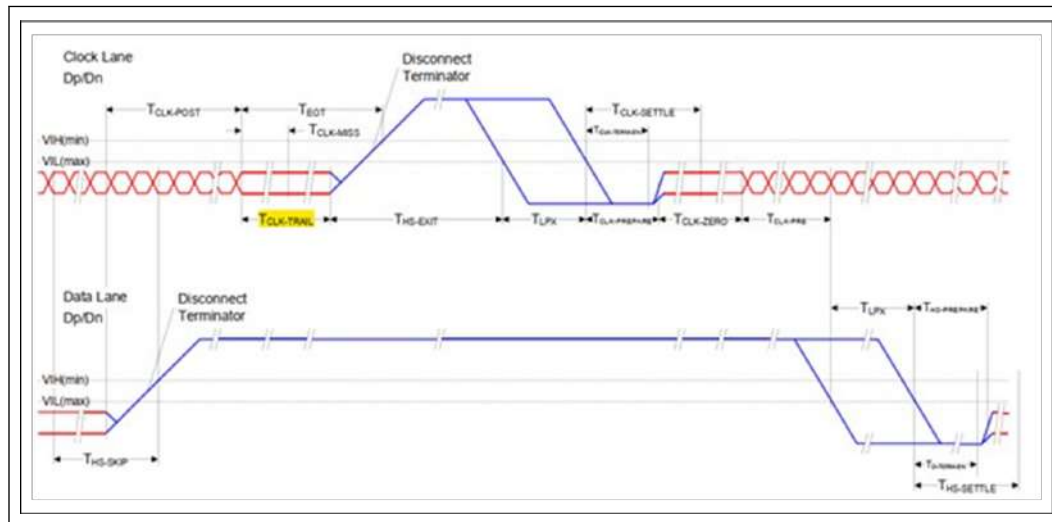
MIPI D-PHY High-Speed Operation in Aggregation Mode

The D-PHY Aggregation Mode feature in the IPU extends the PHY lanes by borrowing data lanes from a secondary PHY. The clock lane of the secondary PHY is disabled since the primary PHY provides the HS clock..

NOTE

In aggregation mode, you must extend the Tclk-trail from 60 ns to 75 ns, so the signals analog deserializer enable and deserializer clock dividers enable have time to de-assert before LP11.

Figure 93. Switching the Clock Lane Between Clock Transmission and Low-Power Mode



6.1.4.5 CSI DPHY Topology Guidelines

General Topology Guidelines:

1. Each DPHY configuration (ex: X4) should be routed on same layer.
2. Length matching between DPHY clock and data:
 - a. Board: within 1mm
 - For Cable + Connectors: within 1mm
3. Cable can be either coax or FPC cable (Flexible Printed Circuit Board)
4. For more information, refer to the IPU8 System Design Guide (# [914612](#)).

6.1.4.5.1 CSI DPHY

Table 263. CSI DPHY General Guidelines

Note	Detail
DPHY configuration routing recommendation	Recommend each DPHY configuration (example: X4) to be routed on the same layer.
CSI_RCOMP	200 Ω +/- 1% pull down to VSS. Provide good noise isolation. Platform Rdc < 0.1 Ω , board capacitance <1.5 pF.
Common mode choke (CMC)	Need is platform/ sensor specific and should be located near sensor.
Port connectivity	It is recommended that CSI A be one of the used ports if all ports are not used.
Loss numbers	1. PCB Loss number assumed: -0.36dB/inch (worst case) at 1.25 GHz 2. FPC cable loss number assumed: -0.316dB/inch (worst case) at 1.25 GHz 3. Cable length guideline is based on FPC cable. FPC cable loss should be within the loss number specified above. If micro-Coax is used, cable length can be longer as micro-Coax loss. Ex. 40 AWG cable has loss: -0.15dB/inch at 1.25 GHz. Please note that Cable loss numbers are general guidelines and actual numbers may be different due to the dimensions, type of material used.
continued...	

continued...

Note	Detail
	(Refer to the individual topologies for total loss numbers). These loss number take precedence over length mentioned in individual topologies below specially while using lower loss cable options like micro coaxial cables.

Table 264. CSI DPHY Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
6	DATA-CLK	Total	Length (mm)	Yes	1

CSI DPHY Main Link Up To 1.0 Gbps Topology

Figure 94. CSI DPHY Main Link Up To 1.0 Gbps Topology Diagram

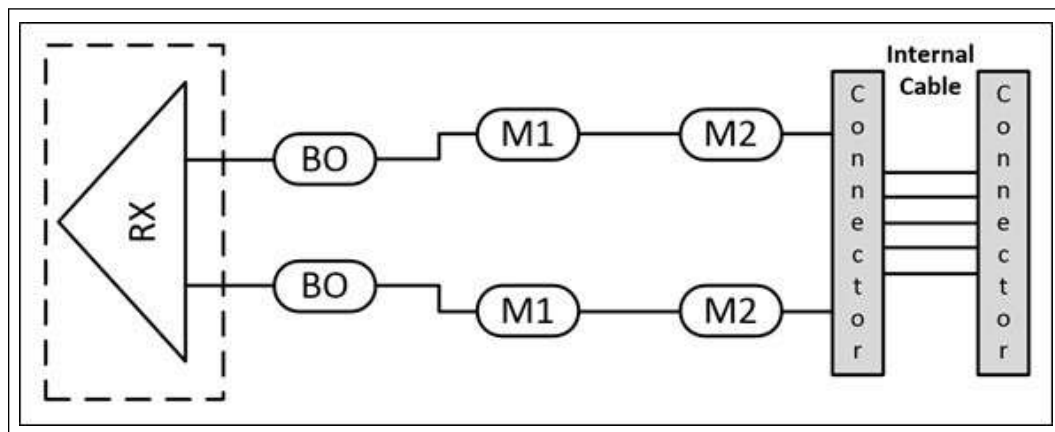


Table 265. CSI DPHY Main Link Up To 1.0 Gbps Topology Notes

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
510 mm cable assembly	Recommend 100 Ω +/- 10% including tolerances. Return loss < -15 dB @ $\leq$ 1.25 GHz. Crosstalk < -50 dB @ $\leq$ 1.25 GHz.
Total insertion loss (BO +M1+M2+Internal Cable)	For $\leq$ 1.0 Gbps: up to -10 dB @1.25 GHz.

Table 266. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	2	Total Channel

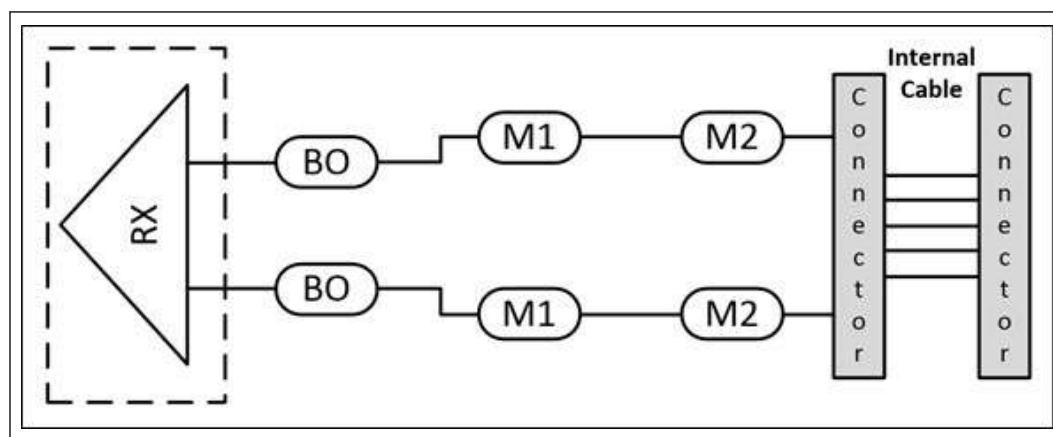
Rx

Table 267. CSI DPHY Main Link Up To 1.0 Gbps Topology Rx Routing Guidelines

Segment	Trline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1	DSL, MS, SL	305	$BO+M1+M2 \leq 305$ mm
M2	DSL, MS, SL	13	$BO+M1+M2+Cable \leq 635$ mm and Cable ≤ 510 mm
Cable	Cable	510	

Max Length Total (mm): 635

Max Length Total Note: Max length including PCB and cable.

CSI DPHY Main Link Up To 1.5 Gbps Topology**Figure 95. CSI DPHY Main Link Up To 1.5 Gbps Topology Diagram****Table 268. CSI DPHY Main Link Up To 1.5 Gbps Topology Notes**

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
305 mm cable assembly	Recommend $100 \Omega \pm 10\%$ including tolerances. Return loss < -15 dB @ ≤ 1.25 GHz. Crosstalk < -50 dB @ ≤ 1.25 GHz.
Total insertion loss (BO +M1+M2+Internal Cable)	For ≤ 1.5 Gbps: up to -7 dB @ 1.25 GHz.
Solution boundary	Length limited by DPHY v1.1 sensors that do not support skew calibration.

Table 269. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	2	Total Channel

Rx

Table 270. CSI DPHY Main Link Up To 1.5 Gbps Topology Rx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1	DSL, MS, SL	305	$BO + M1 + M2 \leq 305$ mm
M2	DSL, MS, SL	13	$BO + M1 + M2 + \text{Cable} \leq 380$ mm and Cable ≤ 305 mm
Cable	Cable	305	

Max Length Total (mm): 380

Max Length Total Note: Max length including PCB and cable.

CSI DPHY Main Link Up To 2.0 Gbps Topology

Figure 96. CSI DPHY Main Link Up To 2.0 Gbps Topology Diagram

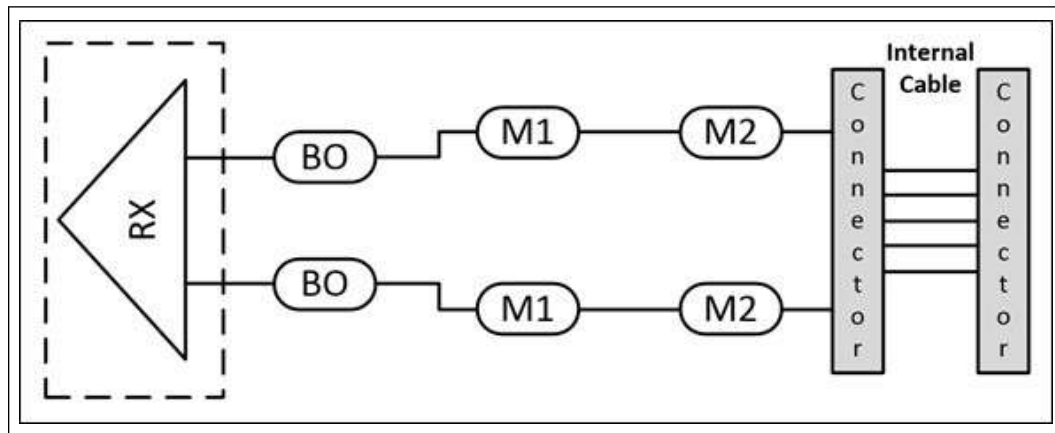


Table 271. CSI DPHY Main Link Up To 2.0 Gbps Topology Notes

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
305 mm cable assembly	Recommend $100 \Omega \pm 10\%$ including tolerances. Return loss < -15 dB @ ≤ 1.25 GHz. Crosstalk < -50 dB @ ≤ 1.25 GHz.
Total insertion loss (BO +M1+M2+Internal Cable)	For ≤ 2.0 Gbps: up to -7 dB @ 1.25 GHz.

Table 272. Max Number of vias

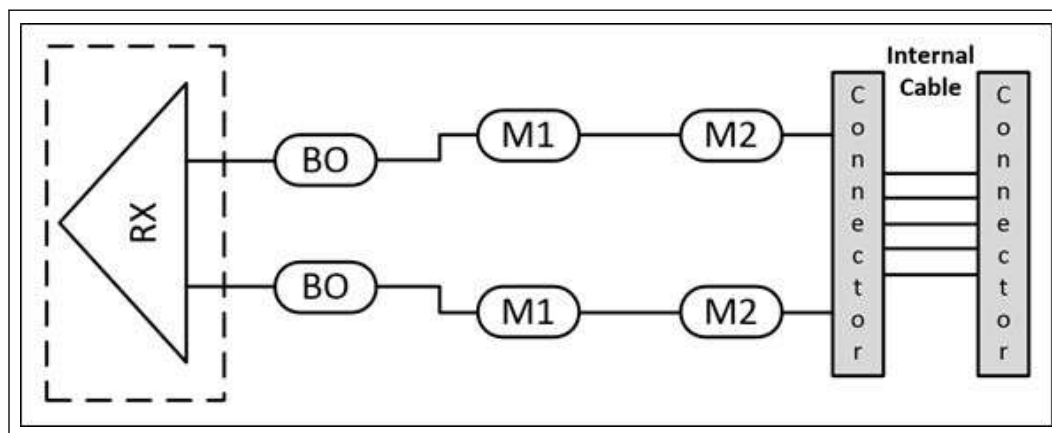
Signal Group	Max Number of Vias Allowed	Via Placement
Rx	2	Total Channel

Rx**Table 273. CSI DPHY Main Link Up To 2.0 Gbps Topology Rx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1	DSL, MS, SL	305	$BO + M1 + M2 \leq 305$ mm
M2	DSL, MS, SL	13	$BO + M1 + M2 + \text{Cable} \leq 380$ mm and $\text{Cable} \leq 305$ mm
Cable	Cable	305	

Max Length Total (mm): 380

Max Length Total Note: Max length including PCB and cable.

CSI DPHY Main Link Up To 2.5 Gbps Topology**Figure 97. CSI DPHY Main Link Up To 2.5 Gbps Topology Diagram****Table 274. CSI DPHY Main Link Up To 2.5 Gbps Topology Notes**

Note	Detail
Reference plane for microstrip route	Continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
Reference plane for stripline and dual stripline route	Both sides can either be continuous ground or continuous power that has low frequency peak to peak noise, ground preferred.
305 mm cable assembly	Recommend $100\ \Omega \pm 10\%$ including tolerances. Return loss < -15 dB @ ≤ 1.25 GHz. Crosstalk < -50 dB @ ≤ 1.25 GHz.
Total insertion loss (BO + M1 + M2 + Internal Cable)	For ≤ 2.5 Gbps: up to -7 dB @ 1.25 GHz.

Table 275. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	2	Total Channel

Rx
Table 276. CSI DPHY Main Link Up To 2.5 Gbps Topology Rx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	13	
M1	DSL, MS, SL	305	$BO + M1 + M2 \leq 305$ mm
M2	DSL, MS, SL	13	$BO + M1 + M2 + \text{Cable} \leq 380$ mm and Cable ≤ 305 mm
Cable	Cable	305	

Max Length Total (mm): 380

Max Length Total Note: Max length including PCB and cable.

6.1.4.6 CSI CPHY Topology Guidelines

For more information, refer to the IPU8 System Design Guide (# 914612).

6.1.4.6.1 CSI CPHY

Figure 98. CSI CPHY Cable Guidelines

Cable guidelines			
	Component		Notes
	Cable only	Cable assembly	
Impedance	50 ohm +/- 10%		Single ended
Insertion Loss	> -0.041 dB	> -0.72 dB	1", single ended, up to 1.25 Ghz
Return Loss	< -50 dB	< -28 dB	full length, single ended, up to 1.25 Ghz
Near end crosstalk	< -55 dB	< -25 dB	full length; single ended; up to 1.25 Ghz; From nearest aggressor
Far end crosstalk	< -63 dB	< -26 dB	full length; single ended; up to 1.25 Ghz; From nearest aggressor

Table 277. CSI CPHY General Guidelines

Note	Detail
CPHY configuration routing recommendation	Recommend each CPHY port to be routed on the same layer.
Common mode choke (CMC)	Need is platform/ sensor specific and should be located near sensor. CMC not accounted for in PDG.
Trio length matching	125 um for each trio; 1 mm between trios of a port.
Trio routing arrangement	All trios should be routed as A-B-C or C-B-A.
Referencing consideration	No routing over split planes allowed, or routing over noisy power planes. Only routing over solid planes are permissible. MIPI CSI CPHY signals should be treated as single ended nets.
continued...	

Note	Detail
Via placement	Via placement must be uniform within a trio to enable uniform routing and transition. During each via transition, it is imperative that return vias are placed symmetrically close to the data vias. Ensure minimal routing imbalances when traces are routed in/out from the via to reduce impedance imbalance i.e uniformity of spacing between wire A,B and C.
CSI port use preference	If some of the CSI ports are not used, it must have CSI_A as one of the used ports.
Loss numbers	1. PCB Loss number assumed: -0.42 dB/inch (worst case) at 1.25 GHz. 2. FPC cable loss number assumed: -0.316 dB/inch (worst case) at 1.25 GHz. 3. Cable length guideline is based on FPC cable. FPC cable loss should be within the loss number specified above. If micro-Coax is used, cable length can be longer as micro-Coax loss, for example 40 AWG cable has a loss of -0.15 dB/inch at 1.25 GHz. Please note that Cable loss numbers are general guidelines and actual numbers may be different due to the dimensions, type of material used. (Refer to the individual topologies for total loss numbers). These loss number take precedence over length mentioned in individual topologies below specially while using lower loss cable options like micro coaxial cables.
Reference plane	Microstrip: Full Ground reference. Stripline: Ground on both sides strongly preferred. If necessary, one side can be a IO rail power plane with $I_{ccmax} < 5$ A that has low amount of low frequency noise, but the other side must be continuous ground.
PCB and cable length restriction	For both PCB and cables, length can be 20%–80% of maximum length unless specified otherwise. This guideline doesn't apply in case cable is absent.

CSI CPHY Main Link Up To 1.0 Gbps Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 99. CSI CPHY Main Link Up To 1.0 Gbps Topology Diagram

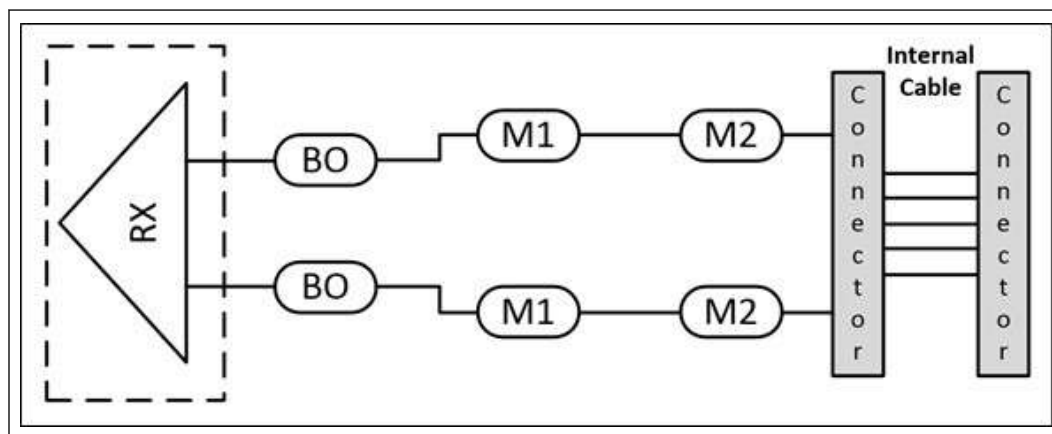


Table 278. CSI CPHY Main Link Up To 1.0 Gbps Topology Notes

Note	Detail
Number of vias allowed	Max 2 vias.
Total insertion loss (BO1+M1+M2+Internal Cable)	For ≤ 1.0 Gbps: up to -10 dB @1.25 GHz.

Table 279. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	2	Total Channel

Rx

Table 280. CSI CPHY Main Link Up To 1.0 Gsps Topology Rx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL	13	
M1	SL	495	$BO+M1+M2 \leq 495$ mm
M2	MS, SL	13	Microstrip routing should be limited to less than 1 mm near connector
Cable	Cable	495	$BO+M1+M2+Cable \leq 617$ mm

Max Length Total (mm): 617

Max Length Total Note: (1) Max length including cable (2) If the material used is Premium Mid Loss (PML), total length can be 635 mm and M1 length/cable length can be up to 510 mm

CSI CPHY Main Link Up To 2.0 Gsps Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 100. CSI CPHY Main Link Up To 2.0 Gsps Topology Diagram

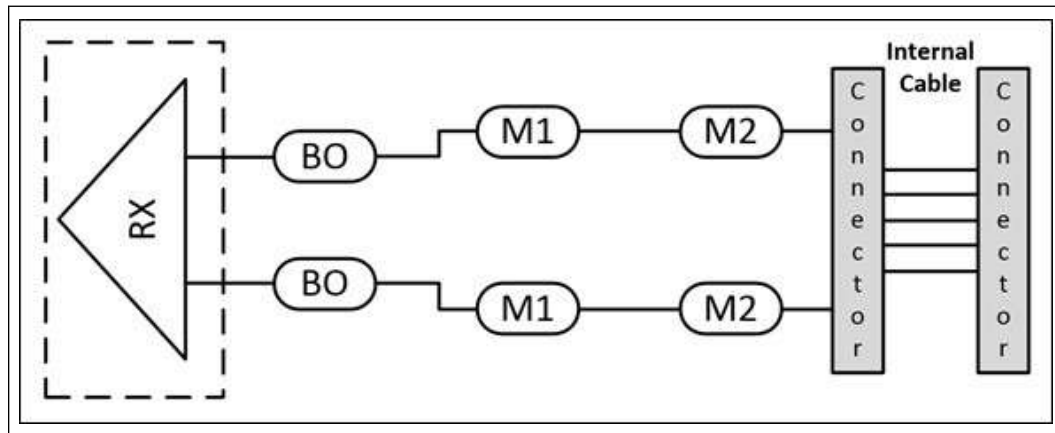


Table 281. CSI CPHY Main Link Up To 2.0 Gsps Topology Notes

Note	Detail
Number of vias allowed	Max 2 vias.
Total insertion loss (BO1+M1+M2+Internal Cable)	For 2.0 Gsps: up to -7 dB @ 1.25 GHz.

Table 282. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	2	Total Channel

Rx**Table 283. CSI CPHY Main Link Up To 2.0 Gbps Topology Rx Routing Guidelines**

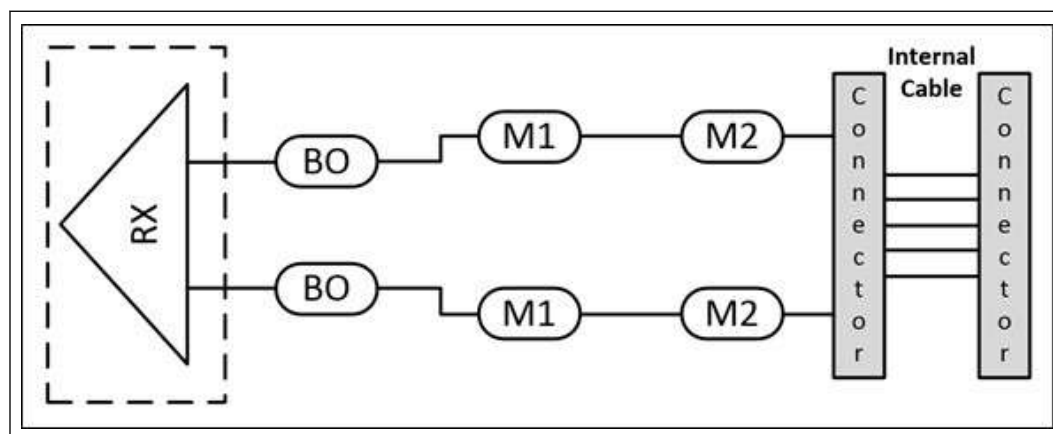
Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL	13	
M1	SL	295	$BO + M1 + M2 \leq 295$ mm
M2	MS, SL	13	Microstrip routing must be limited to less than 1 mm near connector
Cable	Cable	295	$BO + M1 + M2 + \text{Cable} \leq 370$ mm

Max Length Total (mm): 370

Max Length Total Note: (1) Max length including cable (2) For Premium Mid Loss (PML) material, maximum length can be 380 mm with M1 length and Cable length restricted to 305 mm

CSI CPHY Main Link Up To 2.5 Gbps Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 101. CSI CPHY Main Link Up To 2.5 Gbps Topology Diagram**Table 284. CSI CPHY Main Link Up To 2.5 Gbps Topology Notes**

Note	Detail
Number of vias allowed	Max 2 vias.
Total insertion loss (BO1+M1+M2+Internal Cable)	For 2.5 Gbps: up to -7 dB @ 1.25 GHz.

Table 285. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	2	Total Channel

Rx

Table 286. CSI CPHY Main Link Up To 2.5 Gbps Topology Rx Routing Guidelines

Segment	Trace Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL	13	
M1	SL	234	$BO + M1 + M2 \leq 234$ mm
M2	MS, SL	13	Microstrip routing must be limited to less than 1 mm near connector
Cable	Cable	234	$BO + M1 + M2 + \text{Cable} \leq 295$ mm

Max Length Total (mm): 295

Max Length Total Note: (1) Max length including cable (2) For Premium Mid Loss (PML) material, maximum length can be 305 mm with M1 length and Cable length restricted to 244 mm

6.1.5 Integrated Thunderbolt™ Design Guidelines

Thunderbolt™ technology (TBT) is a transformational high-speed, dual protocol I/O, and it provides flexibility and simplicity by encapsulating both data (PCIe*) and video (DisplayPort*) on a single cable connection that can daisy-chain up to six devices.

Thunderbolt controllers act as a point of entry or a point of exit in the Thunderbolt domain. The Thunderbolt domain is built as a daisy chain of Thunderbolt enabled products for the encapsulated protocols - USB, PCIe and DisplayPort. These protocols are encapsulated into the Thunderbolt fabric and can be tunneled across the Thunderbolt domain.

Thunderbolt controllers can be implemented in various systems such as PCs, laptops and tablets, or devices such as storage, docks, displays, home entertainment, cameras, computer peripherals, high end video editing systems, and any other PCIe based device that can be used to extend system capabilities outside of the system's box.

Thunderbolt™ 5 connection data rate is 40 Gbps per lane and is compatible with Thunderbolt™ 5 specification enabling a Thunderbolt link at up to 2x40Gbps, as well as backward compatible with Thunderbolt™ 4 (2x40Gbps) and Thunderbolt™ 3 (2x20Gbps) specifications.

NOTE

Thunderbolt™ 4/5 is a USB-C solution Intel brand which requires the following elements:

- eUSB2, USB3 (10G), USB3/DP implemented at the connector
- In addition, it requires USB4 implemented up to 40G, including TBT3 compatibility as defined by USB4/USB-PD specs.
- Thunderbolt™ 4/5 solutions use (and prioritize) the USB4 PD entry mode (while still supporting TBT3 alt mode).

6.1.5.1 Reference Documents

Table 287. Reference Documents

Title	Document Number
Thunderbolt JHL92x0R June Bridge Collateral	843990
Thunderbolt - Hayden Bridge Collateral	631030

6.1.5.2 Thunderbolt™ Port Power Requirements

Host Source Requirements

Single Port Hosts are required to minimally support 3A/5V on VBUS in Thunderbolt Mode.

Hosts with more than one port are required to provide power per the table below. All Thunderbolt ports shall be capable of providing Full VBUS power on a first-come, first-served basis up to the minimum number of required ports. Tablet PCs with screen sizes less than 14" and without keyboard shall provide a minimum of 7.5W on all Thunderbolt ports.

Thunderbolt Hosts shall provide 1.5W to the VCONN pin on all Thunderbolt ports. The VCONN voltage is required to be between 4.25V and 5.5V.

All Ports shall use the termination defined in USB Type-C r2.3 to broadcast their available Source current. The DFP/Source shall modify its termination to reflect any changes in Source current availability. Thunderbolt Ports shall use the 3.0A/5V pull-up.

Hosts must be DRPs to be able to support peer-to-peer communication.

Table 288. VBUS Power Provisioning with Number of Thunderbolt™ Ports

Power Standard	Voltage/Current	1 Port	2 Port	3 Port	4 Port
First Port VBUS Power	5V/3A ^{1,3}	1	1	1	1
Second Port VBUS Power	5V/1.5A ^{1,2,3,4}	NA	1	2	3
VCONN Power	1.5W TBT	1	2	3	4
Notes: 1. 5V/3A is required for one of every four ports. It is recommended that 5V/3A be provided per every two ports if possible. The power rules defined in USB PD shall be followed if power greater than 15W is provided. 2. 5V/1.5A minimum is required for the second, third, and fourth ports. 3. Power should be allocated on a first-come first-serve basis between the first, second, third and fourth ports. If the first port only consumes 5V/1.5A, then second port shall make 5V/3A available. If the first and second ports only consume 5V/1.5A each, the third port shall make 5V/3A available. If the first, second, and third ports only consume 5V/1.5A each, the fourth port shall make 5V/3A available. 4. Most Thunderbolt Bus Powered Devices have no functionality with Half of minimum power.					

Power Delivery

Refer to the USB PD r3.2 power compliance document for the compliance measurements. The document can be downloaded from <http://www.usb.org>.

Thunderbolt™ Power Provider VBUS Electrical Requirements

The VBUS Source Electrical Parameters USB PD r3.2 Specification. The VCONN Source requirements are specified in USB Type-C r2.3 Specification. The only specifications defined here are those which differ from or are in addition to the USB PD r3.2 and the USB Type-C r2.3 Specifications. This specification assumes compliance with USB Type-C r2.3 specification with additional conditions specifically required for Thunderbolt™.

Thunderbolt™ Power Providers must supply 5V and shall use the Fixed Voltage PDO. The requirement for Power Provides is shown in table below.

USB Type-C r2.3 and USB PD USB PD r3.2 specification can be downloaded from <http://www.usb.org>.

Table 289. Thunderbolt™ VBUS Source Electrical Parameters

Parameter	Description	Min	Typ	Max	Units	Note
VSrc_BPD5 ¹	Minimum voltage provided to port in Active when connected to a BPD	4.75	5	5.5	V	3A min current
Note: 1. Thunderbolt™ Specific Requirement						

Thunderbolt™ VCONN Source Electrical Requirements

The Thunderbolt™ Power Provider VCONN Electrical Parameters are specified in table below.

Table 290. Thunderbolt™ VCONN Source Electrical Requirements

Parameter	Description	Min	Typ	Max	Units	Note
VSrc_Susp	Minimum voltage provided to port in suspend	4.25	5	5.5	V	100mW min
VSrc_Active	Minimum power provided to port in Active	4.25	5	5.5	V	1.5W min

6.1.5.3 Thunderbolt™ Connector Guidelines

Refer to the Thunderbolt™ 4 and Thunderbolt™ 5 Cables and Connectors (# 621583) for the Thunderbolt™ 4 certified USB Type-C connectors.

6.1.5.4 Enable AMT Over TBT Dock Support

Requirements

PD and TBT controller should be connected to Aux power rail. SMLINK pins must be routed from the Processor to Thunderbolt™ (TBT) Controller, including relevant pull-up resistors.

For more details, refer to the Thunderbolt - Hayden Bridge Collateral (# 631030).

6.1.6 Discrete Thunderbolt™ Design Guidelines

Barlow Ridge are Thunderbolt™ controllers that acts as a point of entry or a point of exit in the Thunderbolt domain. The Thunderbolt domain is built as a daisy chain of Thunderbolt™ enabled products for the encapsulated protocols - PCIe* and DisplayPort. These protocols are encapsulated into the Thunderbolt™ fabric and can be tunneled across the Thunderbolt™ domain. Barlow Ridge Thunderbolt™ controller also acts as a flexible re-driver for DP protocol and a source for xHCI controller.

Thunderbolt™ technology (TBT) is a transformational high-speed, dual protocol I/O, and it provides flexibility and simplicity by encapsulating both data (PCIe*) and video (DisplayPort*) on a single cable connection that can daisy-chain up to six devices.

Barlow Ridge can be implemented in various systems such as PCs, laptops and tablets, or devices such as storage, docks, displays, home entertainment, cameras, computer peripherals, high end video editing systems, and any other PCIe* based device that can be used to extend system capabilities outside of the system's box.

Barlow Ridge USB4 connection data rate is 40Gbps per lane (25.6G symbols running PAM3) and is compatible with USB4 Ver2 specification enabling USB4 link at Gen4, as well as backward compatible with Gen3 and Gen2 lane speeds.

Barlow Ridge port connectivity through USB-C connector supports the following modes:

Connection Type	Mode	USB-C Port Details
USB	Native	Dual SSP/SS + Single HS/FS/LS USB connection
USB4	Native	USB4 lanes at 10Gbps/20Gbps/40Gbps
Thunderbolt3	Alternate	Dual TBT3 lanes at 20.265Gbps/10.1325Gbps
Display Port	Alternate	x1/x2/x4 Display Port lanes at 1.62/2.7/5.4/8.1/10/ 20Gbps
Multi-Function DP	Alternate	Single SSP/SS + USB2 connection + x2 Display Port lanes at 1.62/2.7/5.4/8.1/10/20Gbps
Note: UHBR10 is the minimum requirement for Barlow Ridge DP input to support DP2.1.		

Table 291. Acronyms

Acronyms	Description
TBT	Thunderbolt™ Technology
SSP	Super Speed Plus
SP	Super Speed
BR	Barlow Ridge

6.1.6.1 Supported Thunderbolt™ Configuration Options

Table 292. Supported Thunderbolt™ Configuration - Inputs

Thunderbolt™ Connector Number	PCIe* Configuration	DisplayPort* Sink
1 USB Type-C Connector	x4	3 x4
2 USB Type-C Connectors	x4	3 x4

Table 293. Supported Thunderbolt™ Configuration - Outputs

Thunderbolt™ Controller	Thunderbolt™ Link Connection
Barlow Ridge HR	2 high speed ports can be configured as: - Two USB Type-C connectors: - Thunderbolt™ (80G Symmetric mode/120G Asymmetric mode) - or Native DP2.1 ¹ - or Native USB3.2 Gen2x2 (20G)

6.1.6.2 Power Delivery

Thunderbolt™ Power Provider VBUS Electrical Requirements

Table 294. Thunderbolt™ VBUS Source Electrical Parameters

Parameter	Description	Min	Typ	Max	Units	Note
VSrc_BPD5 ¹	Minimum voltage provided to port in Active when connected to a BPD	4.75	5	5.5	V	3A min current

Note: 1. Thunderbolt™ Specific Requirement

Thunderbolt™ VCONN Source Electrical Requirements

The Thunderbolt™ Power Provider VCONN Electrical Parameters are specified in table below:

Table 295. Thunderbolt™ VCONN Source Electrical Requirements

Parameter	Description	Minimum	Type	Maximum	Units	Note
VSrc_Susp	Minimum voltage provided to port in suspend	4.25	5	5.5	V	100mW
VSrc_Active	Minimum power provided to port in Active	4.25	5	5.5	V	1.5W min

Thunderbolt™ BIOS

Thunderbolt™ introduces multilevel PCIe* trees with multiple bridges and endpoint devices. Windows* 10 RedStone2 and the versions before do not have fully built-in support for Thunderbolt™. Thunderbolt™ uses the native PCIe* driver in the OS, and supports hot plug/unplug via the BIOS.

For Windows* 10 RedStone3 and later versions, Thunderbolt™ hot plug/unplug are supported by native PCIe* driver.

Thunderbolt™ BIOS includes three main aspect:

1. Thunderbolt™ Hot Plug/unplug

Hot plug is a standard PCIe* feature and current Windows* OS only supports simple PCIe* topologies. In general case, Thunderbolt™ introduces multilevel PCIe* trees with multiples bridge and endpoint device. Current Windows* OS does not have built-in support for Thunderbolt™. In order to prevent PCIe* wrong resource allocation and misconfiguration, Thunderbolt™ BIOS uses ACPI issues software SMI and upon completion Notify (RP01,0) to trigger OS rescan of the underlying PCIe* sub-tree.

2. Security Levels

Barlow Ridge supports different levels of security for device connection and PCIe* tunneling.

3. Sx Handling

If the platform wants to enable wake support on some power state only, BIOS should know the host is at what power state, and then follow WakeEnabled flow or WakeDisabled flow. For System that provide Barlow Ridge power from two separate rails (VCC1v8_SX from a sustain rail while VCC1v8_S0 from S0 rail) the system must perform a pre-notice Sx entry procedure through the TBT2PCIe mailbox. For the case Barlow Ridge is not powering the CIO domain the BIOS needs to force Barlow Ridge power using POC_GPIO_6 to enable the mailbox functionality.

For detailed Thunderbolt™ BIOS specification and reference code , refer to Barlow Ridge BIOS Implementation Guide.

4. Software Connection Manager(SW CM)

The Connection Manager (CM) is the software/firmware entity that is responsible for enumerating, configuring, and managing a domain. The connection Manager is part of the TBT3/USB4 host system. As Barlow Ridge only supports SW CM, BIOS should set the SW CM Mode in default accordingly.

Reference Documents

Title	Document Number
Thunderbolt™ Diagnostic Tool (TDT) Customer Edition	604228
Thunderbolt Image Builder (TIB)	647686
Thunderbolt Debug Kit	Kit# 1019181
Thunderbolt - Barlow Ridge HR Collateral	736855

Compliance Specification

Table 296. Thunderbolt™ Compliance Specification

Title	Document Number
USB-IF USB4 Specification	http://www.usb.org
Thunderbolt Electrical - Host_Device Compliance Test Specification	524463
Thunderbolt Host Certification Collateral	618475

Thunderbolt™ Signal Descriptions

Thunderbolt™ Controller Signal Groups

Table 297. Barlow Ridge High Speed Signals

Group	Signal Name	Description
TBT Tx	- Barlow Ridge - TBTA_TX1_P, TBTA_TX1_N - TBTA_TX2_P, TBTA_TX2_N - TBTB_TX1_P, TBTB_TX1_N - TBTB_TX2_P, TBTB_TX2_N	TBT Transmitter Differential Pair
TBT Rx	- Barlow Ridge - TBTA_RX1_P, TBTA_RX1_N - TBTA_RX2_P, TBTA_RX2_N - TBTB_RX1_P, TBTB_RX1_N - TBTB_RX2_P, TBTB_RX2_N	TBT Receiver Differential Pair
PCIe* Tx	PCIE_TX[3:0]_P, PCIE_TX[3:0]_N	PCIe* Transmitter Differential Pair
PCIe* Rx	PCIE_RX[3:0]_P, PCIE_RX[3:0]_N	PCIe* Receiver Differential Pair
DP Sink	Barlow Ridge - DPSNK0_ML[3:0]_N - DPSNK0_ML[3:0]_P - DPSNK1_ML[3:0]_N - DPSNK1_ML[3:0]_P - DPSNK2_ML[3:0]_N - DPSNK2_ML[3:0]_P	DP Sink Main Link Receiver Differential Pair
USB 2.0	PA_USB2P_N PA_USB2N_N PB_USB2P_N PB_USB2N_N	USB2 Differential Pair

Note: 1. USB2.0: Recommended using split topology and not use USB2.0 from Barlow Ridge. The POR for Barlow Ridge down on board is using USB2.0 signals from PCH towards type-C DFPs.

6.1.6.3 Thunderbolt™ Topology Guidelines

Motherboard Down Topology

Thunderbolt™ I/O is a virtual wire from PC point of view that tunnels PCIe* and DisplayPort* stream. Considering PCIe*/DisplayPort* connection and few GPIOs for BIOS implementation, Thunderbolt™ controller onboard design is highly recommended. External DisplayPort* cable and GPIO cable may cause bad user experience, for example, no video or no Hot Plug support.

Thunderbolt™ Connector Guideline

Refer to the USB Type-C specification. The document can be downloaded from <http://www.usb.org>.

Thunderbolt designs must use a certified Thunderbolt connector. For more details, refer to the Thunderbolt™ 4 & Thunderbolt™ 5 Cables and Connectors (# 621583).

6.1.6.4 Thunderbolt™ Stack-up Guidelines

Stack-up and Layer Utilization Guidelines

Table 298. Thunderbolt™ Layer Utilization

Group	Max Number of Vias Allowed	Routing Layer	Pair-to-Pair Spacing
TBT Tx	2 (Refer Note1)	Top Layer	Tx to Tx >5h mil; Tx to Rx >13h mil
TBT Rx	2 (Refer Note1)	Bottom Layer	Rx to Rx >5h mil; Rx to Tx > 13h mil
<i>Note:</i> For mid-mount USB-C connector, PA_TX0, PB_TX0, PA_RX1, PB_RX1 connect to USB-C without Vias, PA_TX1, PB_TX1, PA_RX0, PB_RX0 connect to USB-C with 2 PTH Vias.			

Table 299. Thunderbolt™ Insertion Loss Specification

Thunderbolt™ Insertion Loss Specification	Insertion Loss Budget on PCB	Reserved Budget for TBT Controller Package and Receptacle	Total Insertion Loss Budget
@10Ghz (20Gbps) [Barlow Ridge]	4.3dB	3.2dB	7.5dB
@12.8Ghz (40Gbps) [Barlow Ridge]	5.6dB	4dB	9.6dB

6.1.6.5 Thunderbolt™ Optimization Guidelines

TBT Lane Routing and Component Placement

Via Placement and Via Pad Optimization

TBT lane differential routing makes layer transition, the “Top2Bottom Via” unused via pads and Thunderbolt™ connector DIP pin pads in the middle layers should be removed. Use the smallest possible Vias on board to optimize the impedance for the CIO interface.

6.1.6.6 Thunderbolt™ Debug Guidelines

PCIe* and DisplayPort* Configuration

For supported PCIe* configuration options, refer to the [Processor - PCI Express* Design Guidelines](#) on page 292. .

For DisplayPort* configuration, refer to the [DisplayPort* \(DP*\)](#) on page 184 VBIOS and three independent display configurations are also needed to be taken into consideration.

NVM Configuration

Support lane reversal mode such that the polarity traces (P/N) of a CIO/DP lane differential pair can be swapped (to N/P) for ease of routing. System developers must modify NVM setting (interface, offset, and value) to match the P/N configuration as used in actual lane trace routing. Refer to “Thunderbolt™ – NVM Image” for the latest release note about the offset/value setting for specific CIO/DP lanes. Each revision might have different instructions on how to control board-dependent channels lanes N/P swap configuration.

6.1.6.7 Thunderbolt™ Additional Guidelines

Thunderbolt™ Native PCIe* Mode

Thunderbolt™ native PCIe* mode is supported on Windows* 11 RedStone3 and later versions. Enabling native PCIe* mode on TBT provides benefits including the full support of Modern Standby, less platform BIOS development and validation efforts and shorter device enumeration time.

Modern Standby and Runtime D3 Support

Modern Standby and RTD3 support with Thunderbolt™ is supported on Windows* 11 RedStone3 and later versions with native PCIe* mode enabled.

Hardware Changes to Support TBT PCIe* RTD3

The following changes are needed to support TBT PCIe* RTD3 in hardware. These pins must be connected to dedicated PCH GPIO pins.

1. TBT_PERST_N: Signal must be isolated to allow a device D3 (cold) exit to occur without impacting other endpoints, with facilities to drive the required signal timing (example, via ACPI and/or Platform Driver).
 - a. TBT_PERST_N is a 1.8v logic level.
 - b. An AND gate is required between PCH GPIO that will function as BR PERST and Platform Reset signal .
 - c. A 100KOhm pull down on TBT_PERST_N pin is required.
2. TBT_PCIE_WAKE_N: Use of the WAKE pin for device-initiated wake during D3 (cold) without disrupting other devices.
3. RTD3_PWR_EN: Ability to coordinate/sequence power applied to a D3 (cold) domain. This pin must be connected to TBT controller directly instead of connecting to an external MOSFET that controls the power of TBT controller.

This pin must be connected to TBT controller directly instead of connecting to an external MOSFET that controls the power of TBT controller.

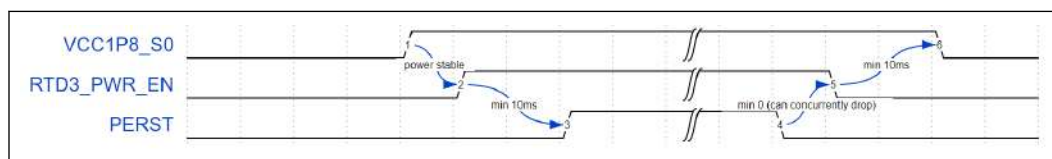
 - a. RTD3_PWR_EN (POC_GPIO_6) is a 1.8v logic level.
 - b. A 100Kohm pull down is required to keep the power state of Barlow Ridge OFF, till BIOS will enable it to meet power sequencing
4. TBT_CLKREQ_N: Connect to dedicated PCH PCIe* Clock Request pin.

Table 300. GPIO Assignment

Signal	PERST_N	WAKE_N	CLKREQ_N	RTD3_PWR_EN
NVL H GPIO	GPP_A11	GPP_E3	GPP_C13	GPP_E4

NOTE

These four signals are named as DG_PERST#, DG_PEWAKE#, DG_RTD3_PWR_EN and DG_CLKREQ# in Thunderbolt - Barlow Ridge HR Collateral (# 736855).

TBT PCIe* RTD3 Entry and Exit Sequence

- RTD3_PWR_EN GPIO (on NVL H) should be tri-state by default and shall not pull actively
- As long as VCC1P8_S0 is low, RTD3_PWR_EN and PERST shall be held low.
- VCC1P8_S0 goes active
 1. VCC1P8_S0 is enabled
 2. VCC1P8_S0 shall be high prior to RTD3_PWR_EN assertion
 - Power stable means crossing 90% of 1.8v and within $\pm 5\%$ sleeve of 1.8v
 3. PERST (active low) shall be released (go high) minimum 10mSec after RTD3_PWR_EN is high
- VCC1P8_S0 goes low due to PERST goes low
 1. PERST and RTD3_PWR_EN can concurrently drop
 2. RTD3_PWR_EN shall go low minimum 10mSec before VCC1P8_S0 goes low
 3. VCC1P8_S0 is disabled

6.1.6.8 TCP DP**Table 301. TCP DP General Guidelines**

Note	Detail
Voiding recommendation	It is recommended to void unused (non-functional) pads for differential vias and pin pads for CMCs, ESDs and connectors to optimize the impedance matching in the channel. Voiding of 0402 or smaller AC capacitors pads is not necessary.
Unused signals	Unused data and AUX signals should be left no connected at the BGA ball.

Table 302. TCP DP Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	Yes	25

6.1.6.8.1 TCP DP UHBR20 Retimer MUX With Barlow Ridge Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 102. TCP DP UHBR20 Retimer MUX With Barlow Ridge Topology Diagram

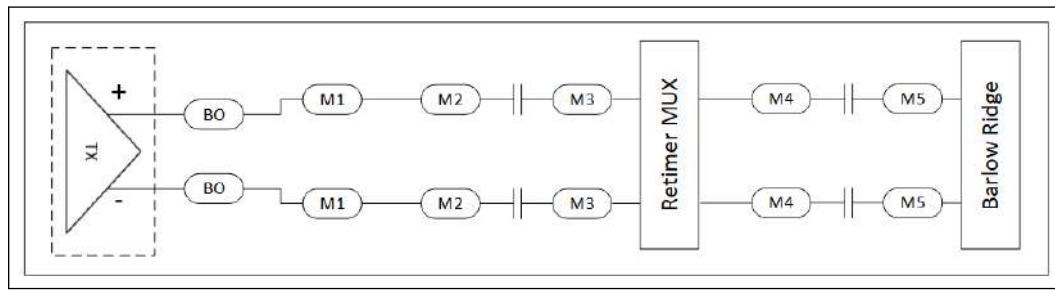


Table 303. TCP DP UHBR20 Retimer MUX With Barlow Ridge Topology Notes

Note	Detail
Maximum via stub length	250 um per via
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	Ground on both sides strongly preferred. If necessary, one side can be a power plane that has low amount of low frequency noise, but the other side must be continuous ground.
Reference plane for dual stripline route	Ground on both sides strongly preferred. If necessary, one side can be a power plane that has low amount of low frequency noise, but the other side must be continuous ground and it is the layer that is closer to the signal route.
AC cap value	Nominal 100 nF recommended (75 nF to 265 nF including tolerance).
Total insertion loss budget from processor pins to retimer/MUX pins	17 dB @ 10 GHz. Inclusive of HVM variation.
Supported interfaces	DP2.1 (UHBR20).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
RETIMER	Barlow ridge and Parade mux/retimer. Retimer Rx should meet VESA spec requirements.

Table 304. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	2	Total Channel

Tx, Pre-Channel**Table 305. TCP DP UHBR20 Retimer MUX With Barlow Ridge Topology Tx, Pre-Channel Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	13	
M1+M2	MS, DSL, SL	205	BO+M1+M2+M3 < Max Length Total
M3	MS, DSL, SL	10	

Max Length Total (mm): 205

Tx, Pre-Channel 2

Max Length Total Note: Note1*: Refer to Retimer/MUX vendor and Barlow Ridge documentation to determine min/max routing length.

Table 306. TCP DP UHBR20 Retimer MUX With Barlow Ridge Topology Tx, Pre-Channel 2 Other Signals

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4+M5	MS, DSL, SL	0	Note1*

6.1.6.8.2 TCP DP Hot Plug Detect Retimer MUX With Barlow Ridge Implementation

Description: This topology is supported for NVL 8C/H16C only.

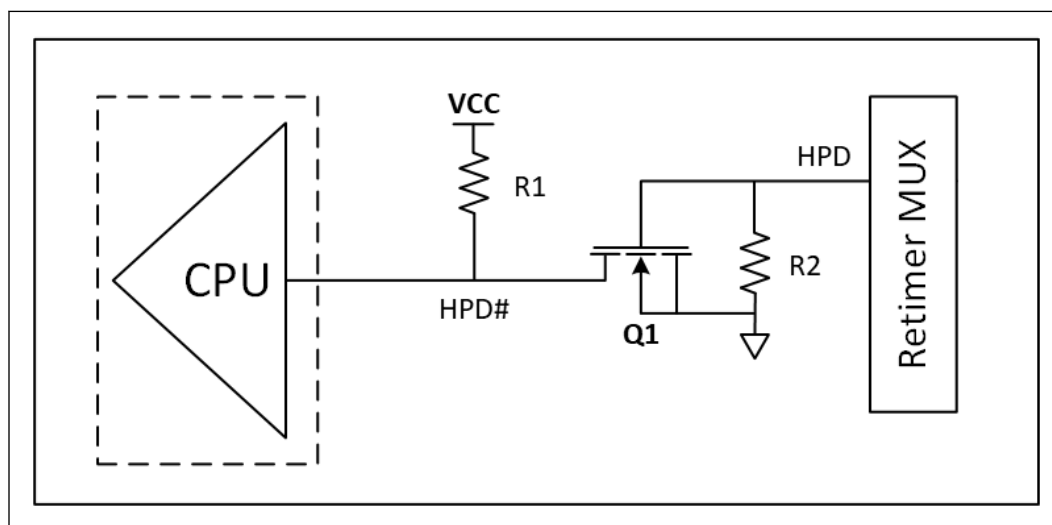
Figure 103. TCP DP Hot Plug Detect Retimer MUX With Barlow Ridge Implementation Topology Diagram

Table 307. TCP DP Hot Plug Detect Retimer MUX With Barlow Ridge Implementation Notes

Note	Detail
Routing	Recommend 40 Ω nominal trace impedance for T3 PCB stack-up and 44 Ω nominal trace impedance for T4 PCB stack-up, with reasonable noise isolation.
VCC	1.8 V
R1	100 k Ω $\pm$ 5%
R2	100 k Ω $\pm$ 5%
Q1	BSS138p or equivalent, 0.85 V < VGS threshold < 1.95 V, input capacitance < 1 nF.
Connectivity	Ensure to follow the Retimer MUX and Barlow Ridge data sheet for any additional HPD connectivity requirements.
ESD	Recommended to implement before the connector. Refer to RDC #736855 for Barlow Ridge guidelines.

6.1.6.9 TCP DP AUX

Table 308. TCP DP AUX Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

6.1.6.9.1 TCP DP AUX Retimer MUX With Barlow Ridge Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 104. TCP DP AUX Retimer MUX With Barlow Ridge Topology Diagram

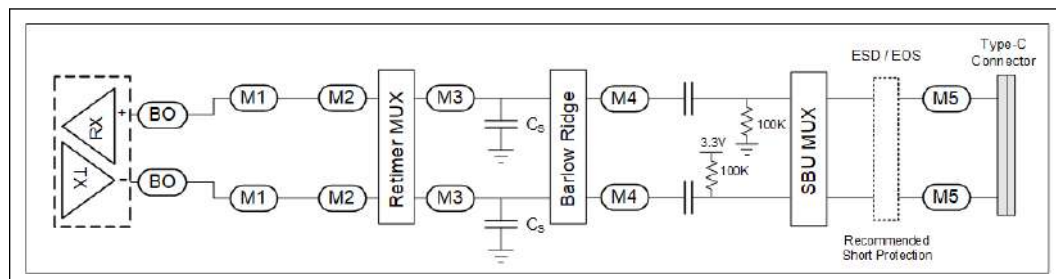


Table 309. TCP DP AUX Retimer MUX With Barlow Ridge Topology Notes

Note	Detail
Number of vias allowed	No explicit limit, use best known routing practices.
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
ESD/ EOS	Refer to USB4 ESD approved list in Barlow Ridge Host Router (BR HR) collateral.
Retimer	Barlow Ridge
MUX	MUX retimer should be vesa compliant.
Cs cap value	47 pF nominal.
Edge rate control	An edge rate reduction mechanism such as a 47 pF capacitor (Cs) terminated to VSS is recommended. Cs cap is optional.
Pull up voltage for DP_AUX_N	3.3V

Tx, Pre-Channel

Max Length Total (mm): 450

Max Length Total Note: Consult with the Retimer MUX vendor to confirm the maximum allowable routing length.

Table 310. TCP DP AUX Retimer MUX With Barlow Ridge Topology Tx, Pre-Channel Other Signals

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2	DSL, MS, SL	450	BO+M1+M2+M3+M4 < Max Length Total

Tx, Pre-Channel 2

Max Length Total Note: Refer to the Retimer MUX and Barlow Ridge documentation to establish the maximum allowable routing length.

Table 311. TCP DP AUX Retimer MUX With Barlow Ridge Topology Tx, Pre-Channel 2 Other Signals

Segment	Tline Type	Max Length Segment (mm)
M3	MS, DSL, SL	0

Tx, Post-Channel

Max Length Total Note: Refer to the Retimer MUX and Barlow Ridge documentation to establish the maximum allowable routing length.

Table 312. TCP DP AUX Retimer MUX With Barlow Ridge Topology Tx, Post-Channel Other Signals

Segment	Tline Type	Max Length Segment (mm)
M4+M5	MS, DSL, SL	0

6.1.7 USB-C* Sub-System

For more details, refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, "USB Type-C* Sub System" section, (# [844316](#)).

Below are functional block diagrams for USB-C for different SKUs. There are four main components needed to support USB-C* features POR for Intel platforms.

- **PD Controller:** PD Controller is a 3rd party discrete IC supporting various type-C* features such as detecting attach/orientation of cable, connecting VCONN/VBUS to connector, providing UFP/DFP roles and USB Power delivery protocol. It also has all the control logic needed to communicate with type-C* port partner.
- **EC:** EC can make decisions on USB PD policy along with help of OS and also helps support UCSI spec.
- **Re-timers:** These are discrete ICs needed on the motherboard to boost signal strength/condition the high speed signals. Requirements are based on different speeds the IC has to support.

NOTE

Only Intel re-timers are POR; 3rd party re-timers are not POR.

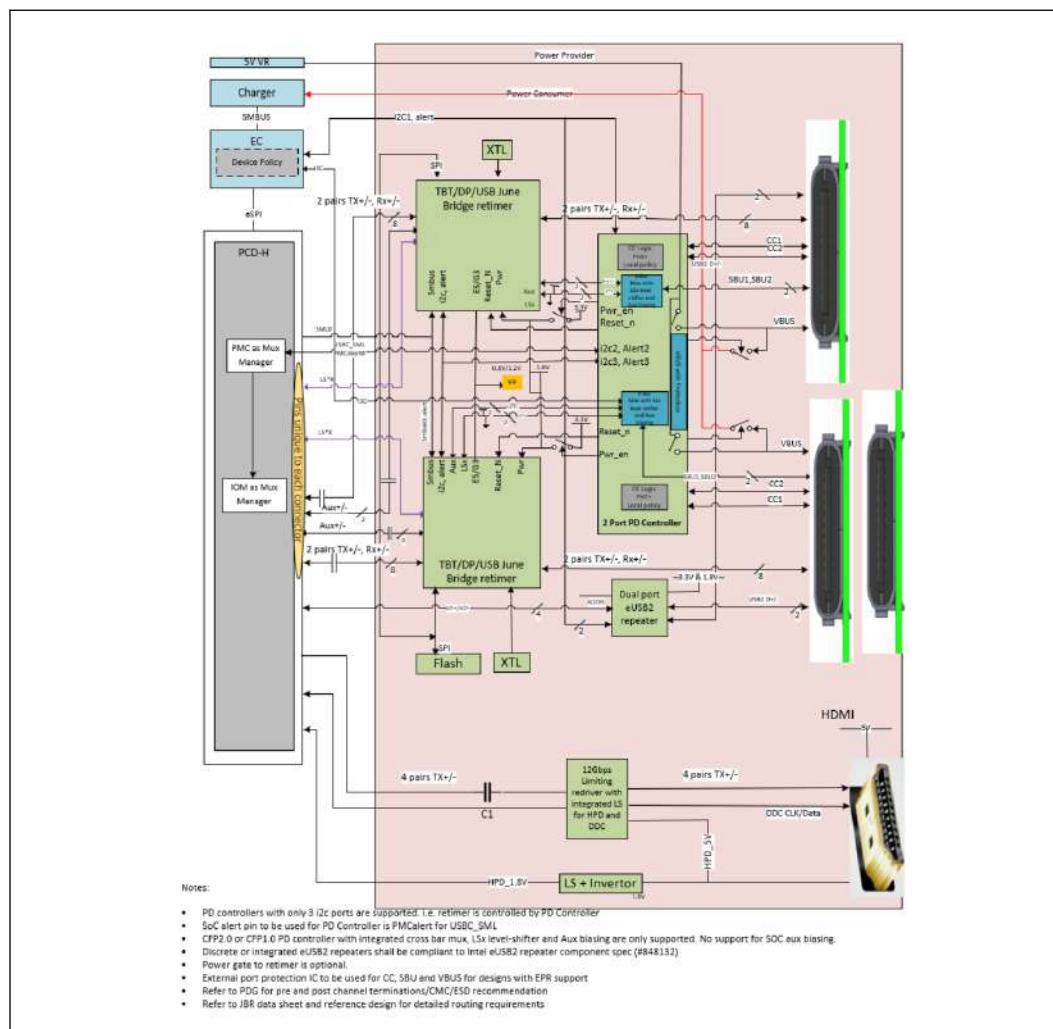
- **Miscellaneous Logic:** This is miscellaneous logic needed on the motherboard such as FETs for controlling power to/from VBUS, AC caps, discrete resistors/CMC/ESD etc. Depending on features integrated on PD Controller, the amount of misc. logic needed on the board will vary.

Details about these components is listed in next section.

PDG Documentation: All the discrete components (PD Controller, retimers, VBUS load switches) in the block diagrams below are suggested to be powered using "A" rail (i.e. rail active in Sx & S0) to support wakes in Sx and also to avoid any electrical leakage issues. EC could be on "A" rail.

Figures below are high level example implementation block diagram, actual implementation on schematic may vary.

Figure 105. TBT 40G Ports with Discrete eUSB2 Repeaters + HDMI 12G Limiting Redriver Topology



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Figure 107. TBT 80G Ports with Integrated eUSB2 Repeaters + HDMI 12G Limiting Redriver Topology + eDP (Secondary) Topology

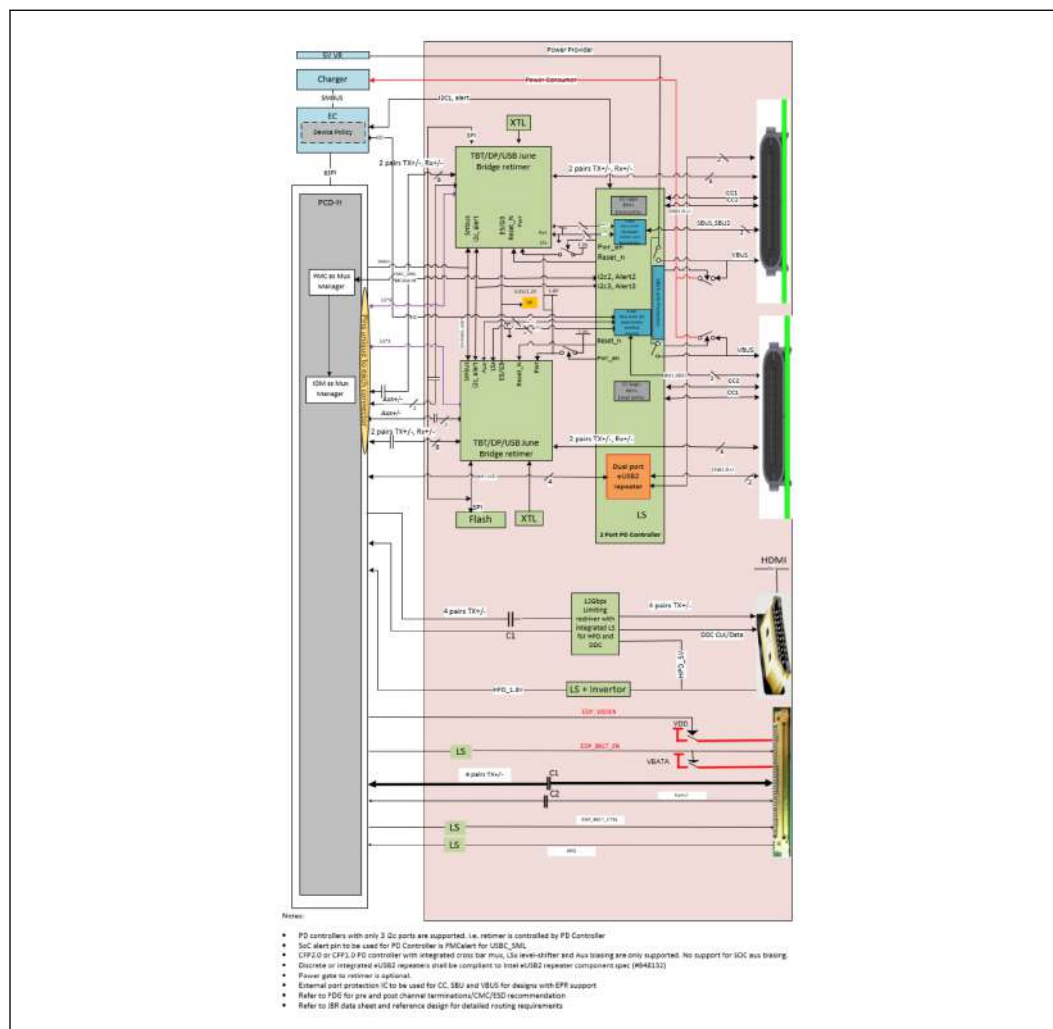
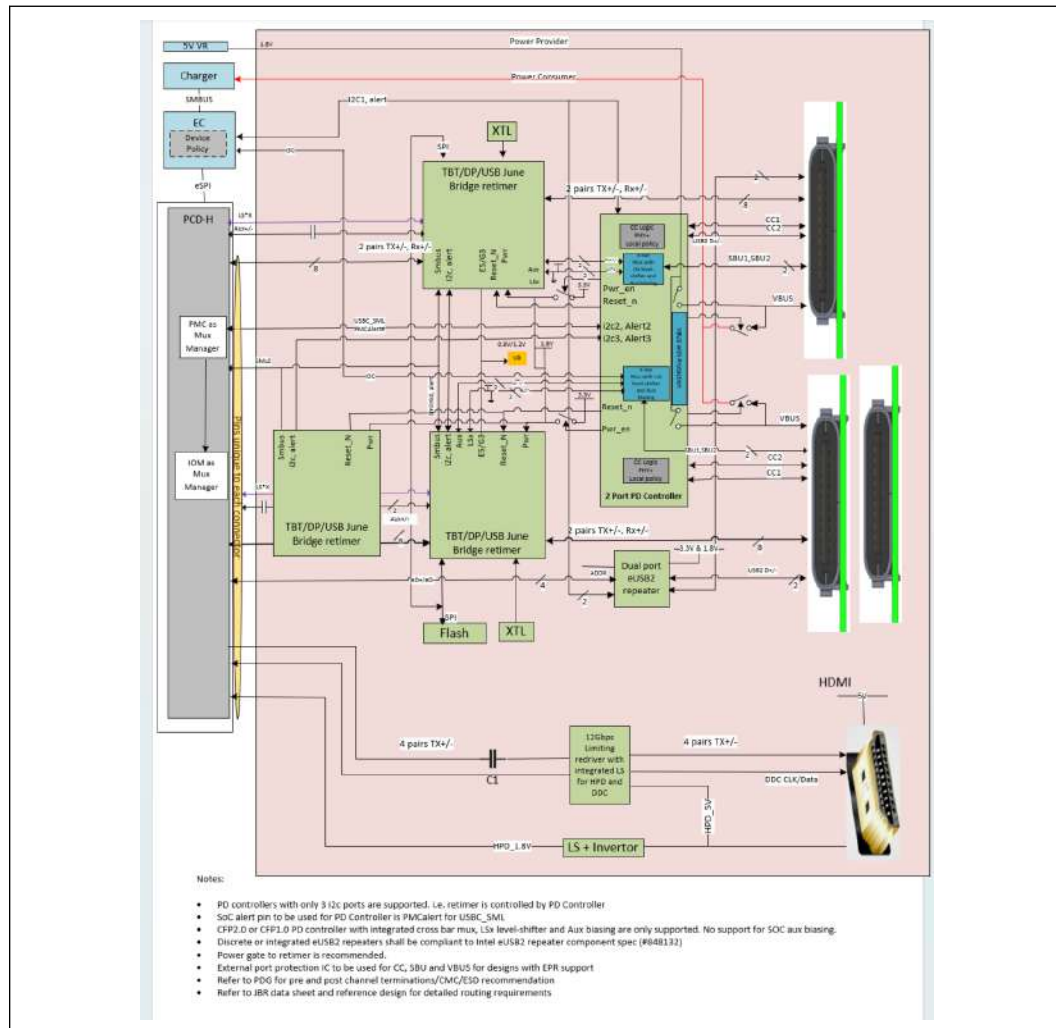


Figure 108. TBT 80G Ports with Short and Long Channel with Discrete eUSB2 Repeaters + HDMI 12G Retimer Topology



6.1.7.1 Aux/LSx Muxing

Aux and LSx are sideband signals used for DisplayPort and Thunderbolt Alternate modes, respectively. Aux is driven by the processor, operates at low voltage, and uses AC-coupled differential pair signaling. LSx is driven by the processor, operates at 1.8V, and uses DC-coupled UART signaling. Due to these differences, it is not possible to multiplex them internally within the processor; a platform-level multiplexer is required. Please refer the Reference Validation Platform (RVP) implementation. The Aux and LSx signals need to be routed to the multiplexer through the Intel re-timer/s because the re-timer monitors LSx lanes and configures the system for USB4 sleep or connect states. The DisplayPort re-timer operation mode is also auto-detected by Intel re-timer/s by monitoring Aux messaging during link establishment.

The LSx GPIOs in processor are mapped to specific USB-C* port in processor, so while routing LSx for a given USB-C* connector, this important mapping needs to be considered.

6.1.7.2 Aux Bias/Orientation/Isolation Control

For more details, refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, section "AUX BIAS Control (# 844316) .

6.1.7.3 USB-C* Connector

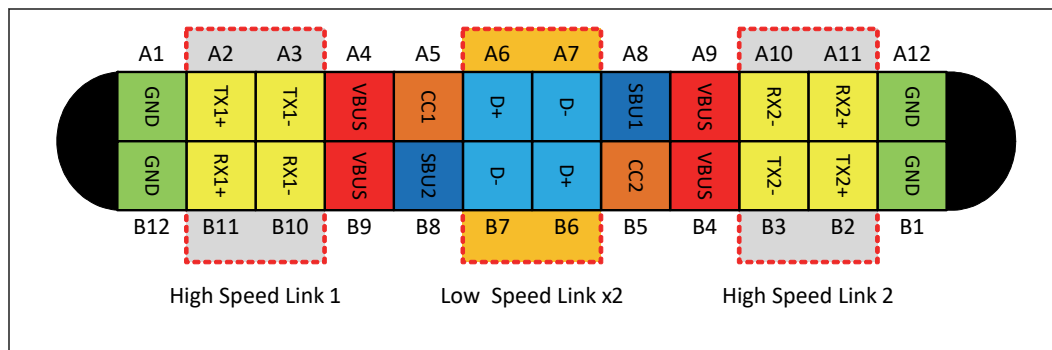
USB-C* connector consists of 24 signal pins designed in a symmetrical way. Refer to figure below.

There are up to six differential pairs available on a USB-C* connector, with 4 high speed differential pairs and two low speed pairs. There are two configuration channel signals (CC1/CC2) on the connector which is used for:

- Device detection
- Orientation detection
- Establish DFP and UFP roles
- Discover and configure power
- Discover and configure optional alternate/accessory modes

Unused CC pin will be reconfigured as VCONN, which is used to power the active cable.

Figure 109. USB-C* Receptacle Anatomy



With two configuration channel signals for communication, USB-C* connector can support USB 3.2/eUSB2 interfaces and any other vendor specific interfaces like DisplayPort and TBT in Alternate Mode.

In Alternate Mode, both the high speed and low speed link and both the SBU1/2 pins can be reconfigured to support the intended configuration. Refer the table below for a examples of the lane configuration.

Table 313. USB-C* Connector Signal Group and Description

Group	Pin Name	Description
Power	VBUS	VBUS, capable of up to 5A @ 36V with USB PD
	VCONN	VCONN is a power source to active cable which is capable of supplying up to 1.5W @ 5V. Unconnected CC pin will be reconfigured to VCONN when a device is plugged in.

continued...

Group	Pin Name	Description
Ground	GND	Ground
High Speed Differential	TX1+	First USB3.2 or USB 4 Diff TX pair or Display Alt Mode
	TX1-	
	RX1+	First USB3.2 or USB 4 Diff RX pair or Display Alt Mode Diff TX pair
	RX1-	
	TX2+	Second USB3.2 or USB 4 Diff TX pair or Display Alt Mode
	TX2-	
	RX2+	Second USB3.2 or USB 4 Diff RX pair or Display Alt Mode Diff TX pair
	RX2-	
Low Speed Differential	D+	Low Speed Differential Pair, primary for eUSB2 connection
	D-	
Control	CC1	Configuration Channel 1. Used for device detection, orientation detection and Alternate Mode configuration. Unconnected pins will be reconfigured to VCONN.
	CC2	Configuration Channel 2.Used for device detection, orientation detection and Alternate Mode configuration. Unconnected pins will be reconfigured to VCONN
Side band	SBU1/SBU2	Sideband signals for use by alternate: - DP mode -AUX signals - USB 4 mode - LSx signals

NOTE

Refer to the Processor-Low Speed Link (LSx) chapter for details on LSx layout guidelines.

Table 314. USB Type-C* Lanes Configuration

Lane1	Lane2	Comments
USB4v2 / TBT5	USB4v2 / TBT5	Both lanes operate at same speed, one of (40/20.6g/10.3g/20g/10g)
USB4v2 / TBT5	No connect	20.6g/10.3g/20g/10g
No connect	USB4v2 / TBT5	
USB 3.2	USB 3.2	Multi-Lane USB 3.2 (Host Only), 2x10G = 20G
USB 3.2	No connect	Any combination of: USB 3.2 Gen 1x1 (5Gb/s) USB 3.2 Gen 2x1 (10Gb/s)
No connect	USB 3.2	
USB 3.2	DPx2	Any of HBR3/HBR2/HBR1/HRBR for DP1.4a, DP2.1 (2x10/20Gbps) , and USB 3.2 (10 Gbps)
DPx2	USB 3.2	
DPx4	Both lanes at same DP rate - no support	Any of HBR3/HBR2/HBR1/HRBR for DP1.4a, DP2.1 (4x10/20Gbps)
continued...		

Lane1	Lane2	Comments
	for 2x DPx2 USB Type-C connector	

NOTE

NVL 6C is limited to only 40G/TBT4.

Table 315. Interchangeable DDI and USB-C* Connectors

Port	Connector Type
TCP0	USB-C
TCP1	USB-C
TCP2	USB-C
<i>Note:</i> - USB-C* connector may support USB4/DP/USB3. - Display port supported over Type-C connector over Intel retimer up to UHBR20 data rate. - HDMI supported only on Native connector (this product supports dedicated HDMI outside the Type-C subsystem).	

Table 316. TCSS Data, Sideband and Comp Signals

Signal Names	Values
TCP[2:0]_AUX_P TCP[2:0]_AUX_N	Common Lane: AUX N/P - pad
TCP[2:0]_TXRX[1:0]_P TCP[2:0]_TXRX[1:0]_N	Data Lane, RX, complementary signals N/P. Also serves as the secondary TX for display modes.
TCP[2:0]_TX[1:0]_P TCP[2:0]_TX[1:0]_N	Data Lane, TX, complementary signals P/N
<i>Note:</i> • Unused data and AUX signals should be left no connected at the BGA ball.	

Table 317. USB-C* Signals Mapping

Port ⁽¹⁾	Processor to Re-timer Signal Name	Re-timer to USB-C* Connector
Port 0	TCP0_TX0	TCP0_RT_TX0
	TCP0_TX1	TCP0_RT_TX1
	TCP0_TXRX0	TCP0_RT_TXRX0
	TCP0_TXRX1	TCP0_RT_TXRX1
Port 1	TCP1_TX0	TCP1_RT_TX0
	TCP1_TX1	TCP1_RT_TX1
	TCP1_TXRX0	TCP1_RT_TXRX0
	TCP1_TXRX1	TCP1_RT_TXRX1
Port 2	TCP2_TX0	TCP2_RT_TX0
<i>continued...</i>		

Port ⁽¹⁾	Processor to Re-timer Signal Name	Re-timer to USB-C* Connector
	TCP2_TX1	TCP2_RT_TX1
	TCP2_TXRX0	TCP2_RT_TXRX0
	TCP2_TXRX1	TCP2_RT_TXRX1
Note: For June Bridge re-timer pin information, refer to the Thunderbolt JHL92x0R June Bridge Collateral (# 843990 and for Thunderbolt - Hayden Bridge Collateral (# 631030).		

BIOS/mFIT/VBT setting for TCPs

Refer to the guidance 871426 outlining the required BIOS, mFIT, and VBT settings for TCPs per the requested protocol.

6.1.7.4 TCP AUX

Table 318. TCP AUX Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

6.1.7.4.1 TCP AUX June Bridge Retimer Topology

Figure 110. TCP AUX June Bridge Retimer Topology Diagram

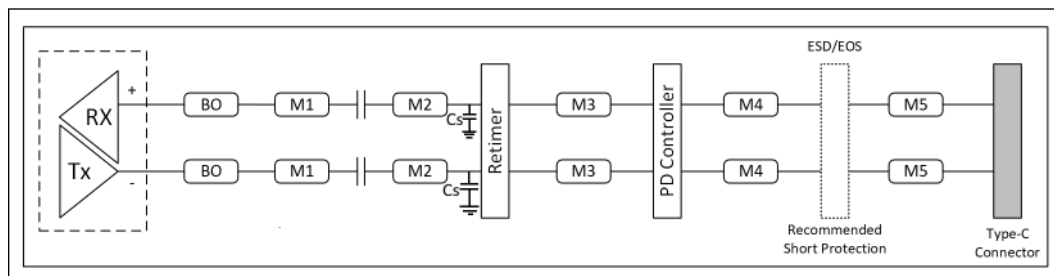


Table 319. TCP AUX June Bridge Retimer Topology Notes

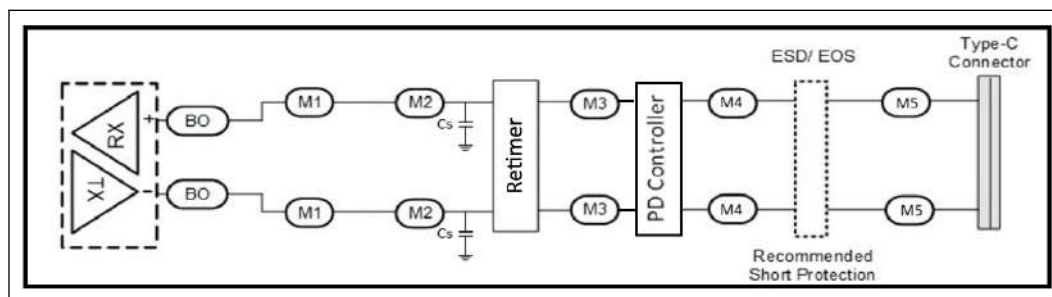
Note	Detail
Number of vias allowed	No explicit limit, use best known routing practices.
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
continued...	

Note	Detail
AC cap value	Nominal 100 nF recommended (75 nF to 200 nF including tolerance).
Retimer	Applies to June Bridge topology. Consult re-timer datasheet for exact Aux channel implementation requirements. Refer to RDC #843990
Optional edge rate control	An edge rate reduction mechanism may be needed to comply with DisplayPort maximum Aux slew rate specification dependent on the retimer's Aux implementation.
Cs cap value	47 pF nominal.
ESD	Refer to USB4 ESD approved list in June Bridge Re-timer collateral. refer to RDC #843990
PD Controller	Refer to RDC #843990 for June Bridge.

Rx,Tx**Table 320. TCP AUX June Bridge Retimer Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2	DSL, MS, SL	450	BO+M1+M2+M3+M4+M5 < Max Length Total
M3+M4+M5	DSL, MS, SL	127	

Max Length Total (mm): 450

6.1.7.4.2 TCP AUX Hayden Bridge Retimer Topology**Figure 111. TCP AUX Hayden Bridge Retimer Topology Diagram****Table 321. TCP AUX Hayden Bridge Retimer Topology Notes**

Note	Detail
Number of vias allowed	No explicit limit, use best known routing practices.
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
AC cap value	Nominal 100 nF recommended (75 nF to 200 nF including tolerance).
Retimer	Applies to Hayden Bridge topology. Consult re-timer collateral for exact Aux channel implementation requirements. Refer to RDC #631030
Optional edge rate control	An edge rate reduction mechanism may be needed to comply with DisplayPort maximum Aux slew rate specification dependent on the retimer's Aux implementation.

continued...

Note	Detail
Cs cap value	47 pF nominal.
ESD	Refer to USB4 ESD approved list in Hayden Re-timer collateral. refer to RDC #631030
PD Controller	Refer to RDC #631030 for Hayden Bridge.

Rx,Tx

Table 322. TCP AUX Hayden Bridge Retimer Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2	DSL, MS, SL	450	BO+M1+M2+M3+M4+M5 < Max Length Total
M3+M4+M5	DSL, MS, SL	127	

Max Length Total (mm): 450

6.1.7.4.3 TCP AUX Cascaded June Bridge Retimer Topology

Figure 112. TCP AUX Cascaded June Bridge Retimer Topology Diagram

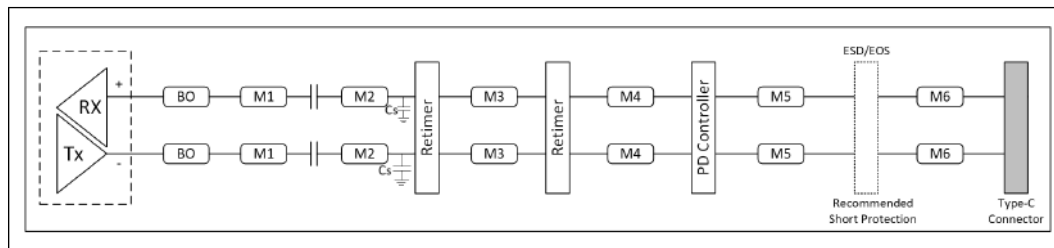


Table 323. TCP AUX Cascaded June Bridge Retimer Topology Notes

Note	Detail
Number of vias allowed	No explicit limit, use best known routing practices.
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
ESD/ EOS	Refer to USB4 ESD approved list in June Bridge Re-timer collateral. refer to RDC #843990
Retimer	Applies to June Bridge topology. Consult re-timer datasheet for exact Aux channel implementation requirements. Refer to RDC #843990
Cs cap value	47 pF nominal.
AC CAP value	Nominal 100 nF recommended (75 nF to 200 nF including tolerance).
Edge rate control	An edge rate reduction mechanism such as a 47 pF capacitor (Cs) terminated to VSS is recommended. Cs cap is optional.
PD Controller	Refer to RDC #843990 for June Bridge.

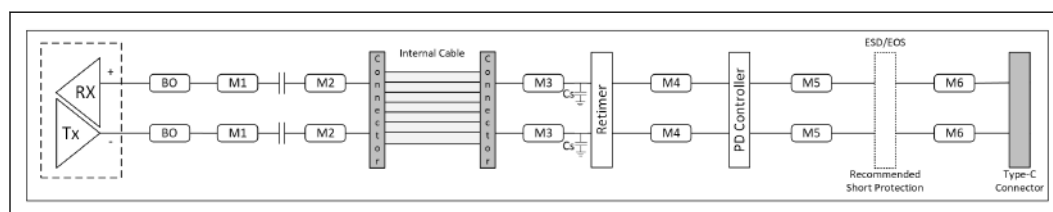
Rx,Tx

Table 324. TCP AUX Cascaded June Bridge Retimer Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2	DSL, MS, SL	450	BO+M1+M2+M3+M4+M5+M6 < Max Length Total
M3+M4+M5+M6	DSL, MS, SL	127	

Max Length Total (mm): 450

Max Length Total Note: M1+M2+M3+M4+M5+M6 <= 450 mm

6.1.7.4.4 TCP AUX June Bridge with Retimer Internal Cable Topology**Figure 113. TCP AUX June Bridge with Retimer Internal Cable Topology Diagram****Table 325. TCP AUX June Bridge with Retimer Internal Cable Topology Notes**

Note	Detail
Number of vias allowed	No explicit limit, use best known routing practices.
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
500 mm cable assembly	75 - 100 Ω including tolerance. Insertion loss $\geq$ -1.0 dB @ 1 MHz.
ESD / EOS	Refer to USB4 ESD approved list in June Bridge 40 Re-timer collateral. refer to RDC #843990
Retimer	Applies to June Bridge topology. Consult re-timer datasheet for exact Aux channel implementation requirements. Refer to RDC #843990
Cs cap value	47 pF nominal.
AC CAP Value	Nominal 100 nF recommended (75 nF to 200 nF including tolerance).
Edge rate control	An edge rate reduction mechanism such as a 47 pF capacitor (Cs) terminated to VSS is recommended. Cs cap is optional.
PD Controller	Refer to RDC #843990 for June Bridge.

Rx,Tx

Table 326. TCP AUX June Bridge with Retimer Internal Cable Topology Rx,Tx Routing Guidelines

Segment	Line Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2+M3	DSL, MS, SL	450	
M4+M5+M6	DSL, MS, SL	127	BO+M1+M2+M3+M4+M5+M6 < Max Length Total
Cable	Cable	500	

Max Length Total (mm): 450

Max Length Total Note: Cable length is not included in "Max length Total"

6.1.7.4.5 TCP AUX Hayden Bridge with Retimer Internal Cable Topology

Figure 114. TCP AUX Hayden Bridge with Retimer Internal Cable Topology Diagram

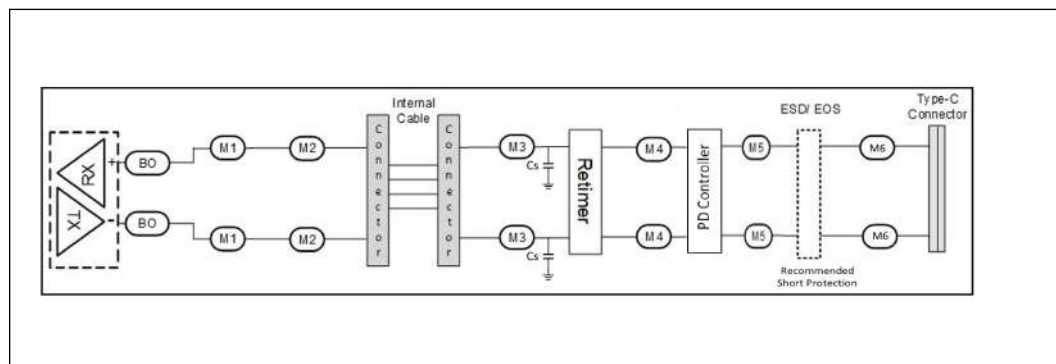


Table 327. TCP AUX Hayden Bridge with Retimer Internal Cable Topology Notes

Note	Detail
Number of vias allowed	No explicit limit, use best known routing practices.
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
500 mm cable assembly	75 - 100 Ω including tolerance. Insertion loss $\geq$ -1.0 dB @ 1 MHz.
Retimer	Applies to Hayden Bridge topology. Consult re-timer collateral for exact Aux channel implementation requirements. Refer to RDC #631030
Cs cap value	47 pF nominal.
AC CAP Value	Nominal 100 nF recommended (75 nF to 200 nF including tolerance).
Edge rate control	An edge rate reduction mechanism such as a 47 pF capacitor (Cs) terminated to VSS is recommended. Cs cap is optional.
ESD / EOS	Refer to USB4 ESD approved list in Hayden Re-timer collateral. refer to RDC #631030.
PD Controller	Refer to RDC #631030 for Hayden Bridge.

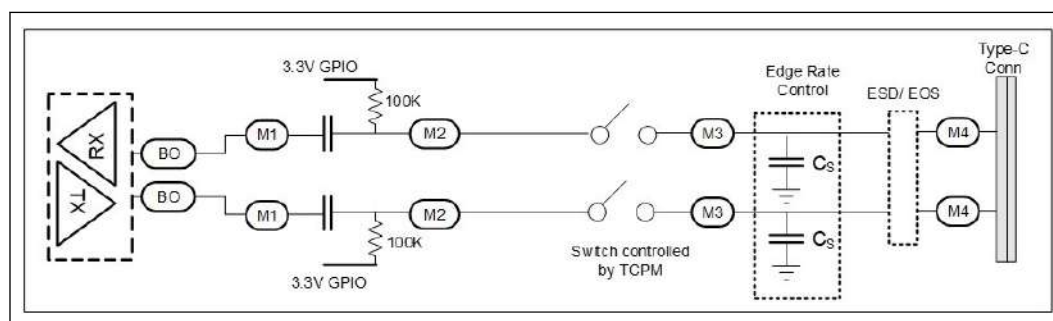
Rx,Tx

Table 328. TCP AUX Hayden Bridge with Retimer Internal Cable Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	25	
M1+M2+M3	DSL, MS, SL	450	
M4+M5+M6	DSL, MS, SL	127	BO+M1+M2+M3+M4+M5+M6 < Max Length Total
Cable	Cable	500	

Max Length Total (mm): 450

Max Length Total Note: Cable length is not included in "Max length Total"

6.1.7.4.6 TCP AUX No Retimer Topology**Figure 115. TCP AUX No Retimer Topology Diagram****Table 329. TCP AUX No Retimer Topology Notes**

Note	Detail
Number of vias allowed	No explicit limit, use best known routing practices.
Transmission line referencing	Reference planes can be continuous ground or continuous power that have low frequency peak to peak noise. Ground referencing is preferred.
ESD/ EOS	Refer to EMC chapter for parts recommendation. Recommend placeholder. Component must be stuffed if EMC tests fail
AC cap value	Nominal 100 nF recommended (75 nF to 200 nF including tolerance).
3.3 V GPIO	AUX termination set based on USB Type-C plug orientation requirements. Nominal 3.3 V or 0 V.
Cs cap value	47 pF nominal.
Edge rate control	An edge rate reduction mechanism such as a 47 pF capacitor (Cs) terminated to VSS is recommended to comply with DisplayPort maximum Aux slew rate specification.

Mainstream, Mid Loss (ML), Rx,Tx
Table 330. TCP AUX No Retimer Topology Mainstream, Mid Loss (ML), Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	25	
M1+M2+M3	MS, DSL, SL	450	BO+M1+M2+M3+M4 < Max Length Total
M4	MS, DSL, SL	25	

Max Length Total (mm): 450

6.1.7.5 TCP USB 3.2
Table 331. TCP USB 3.2 General Guidelines

Note	Detail
Discrete component part size for mainstream stackup	0402, if routing near maximum length it is recommended to use 0201 component size.
Voiding recommendation for mainstream stackup	It is recommended to void unused (non-functional) pads for differential vias and pin pads for CMCs, ESDs, AC cap and connectors to optimize the impedance matching in the channel.
Unused signals	Unused data and AUX signals should be left no connected at the BGA ball.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
Reference Plane	Microstrip: Must be continuous ground. Stripline: Ground on both sides strongly preferred. If necessary, one side can be a IO rail power plane with Iccmax < 5A that has low amount of low frequency noise, but the other side must be continuous ground. Dual Stripline: Ground on both sides strongly preferred. If necessary, one side can be a IO rail power plane with Iccmax < 5A that has low amount of low frequency noise, but the other side must be continuous ground and it is the layer that is closer to the signal route.

Table 332. TCP USB 3.2 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

6.1.7.5.1 TCP USB 3.2 Gen 1x1 Type-A Topology

Figure 116. TCP USB 3.2 Gen 1x1 Type-A Topology Diagram

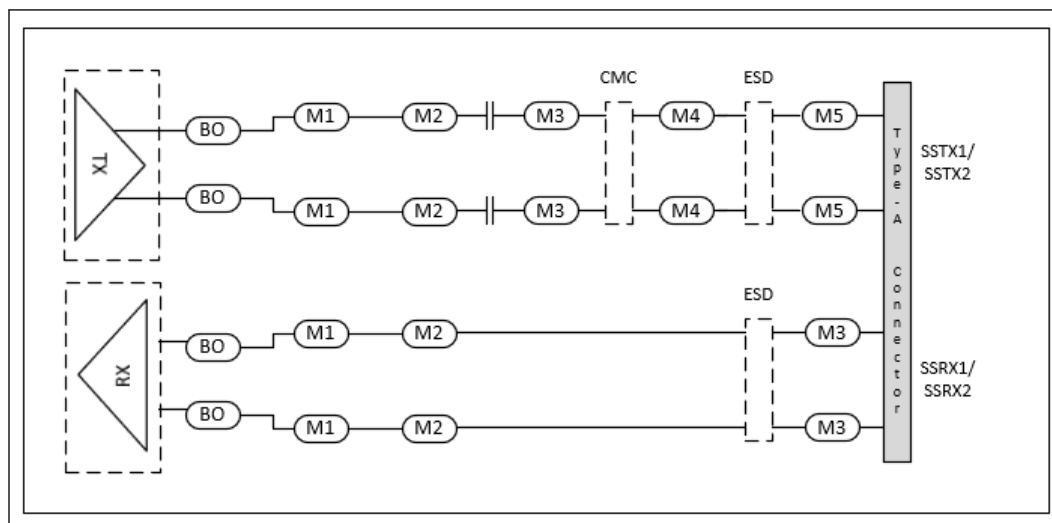


Table 333. TCP USB 3.2 Gen 1x1 Type-A Topology Notes

Note	Detail
Maximum via stub length	381 um
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground. Prefer ground on both sides.
Reference plane for dual stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground and it is the layer that is closer to the signal route. Prefer ground on both sides.
CMC	Refer to the EMC Component Selection section of the PDG for CMC recommendation(s).
ESD	Refer to the EMC Component Selection section of the PDG for ESD recommendation(s).
AC cap value	75 nF to 265 nF including tolerance, 25 V.
Supported interfaces	USB3.2 Gen1 (5 Gbps)
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.

Table 334. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	3	Total Channel	Route within 2 layers.

Rx, Tx

Table 335. TCP USB 3.2 Gen 1x1 Type-A Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	8	
M1+M2	MS, DSL, SL	229	$B0+M1+M2+M3+M4+M5 \leq 229$ mm
M3+M4+M5	MS, DSL, SL	25	

Max Length Total (mm): 229

6.1.7.5.2 TCP USB 3.2 Gen 2x1 Type-A Topology

Figure 117. TCP USB 3.2 Gen 2x1 Type-A Topology Diagram

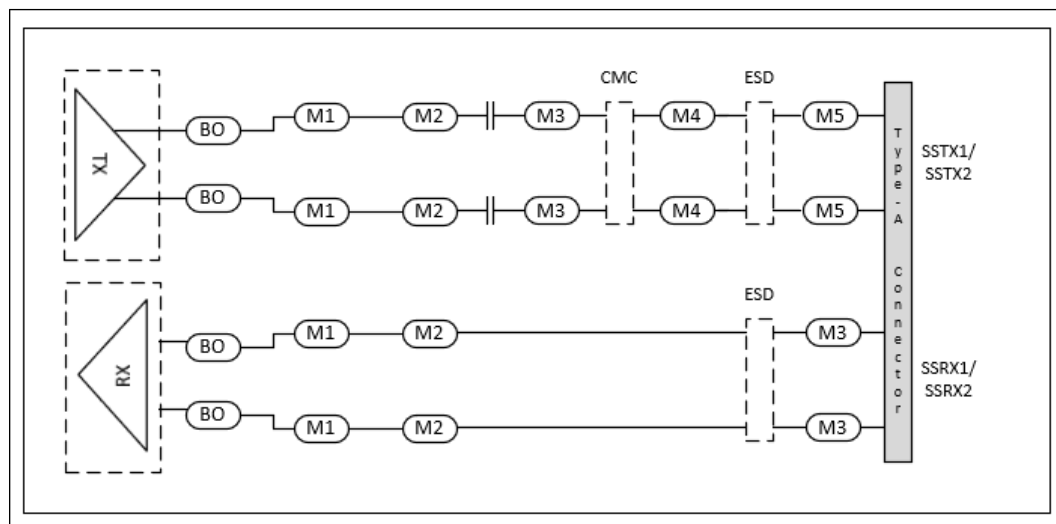


Table 336. TCP USB 3.2 Gen 2x1 Type-A Topology Notes

Note	Detail
Maximum via stub length	381 um
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground. Prefer ground on both sides.
Reference plane for dual stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground and it is the layer that is closer to the signal route. Prefer ground on both sides.
CMC	Refer to the EMC Component Selection section of the PDG for CMC recommendation(s).
ESD	Refer to the EMC Component Selection section of the PDG for ESD recommendation(s).

continued...

Note	Detail
AC cap value	75 nF to 265 nF including tolerance, 25 V.
Supported interfaces	USB3.2 Gen2 (10 Gbps)
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.

Table 337. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	3	Total Channel	Route within 2 layers.

Rx, Tx**Table 338. TCP USB 3.2 Gen 2x1 Type-A Topology Rx, Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	8	
M1+M2	MS, DSL, SL	118	BO+M1+M2+M3+M4+M5 ≤ 118 mm
M3+M4+M5	MS, DSL, SL	25	

Max Length Total (mm): 118

6.1.7.5.3 TCP USB 3.2 Gen 2x1 Type-A Redriver Topology**Figure 118. Requirement for Third Party Active Component**

Third Party Active MUX/Re-driver/Re-timer USB 3.2 VCM Requirements			
Electrical Parameter	Gen 1x1 (5 Gb/s)	Gen 2x1 (10 Gb/s)	Notes
$V_{RX-CM-DC-ACTIVE-IDLE-DELTA_P}$	200 mV peak (max)	200 mV peak (max)	Refer to USB 3.2 ECN VCM & TXDetect for full details on the measurement
$V_{RX-CM-DC_CONN}$	-0.5V (min1)	-0.5V (min1)	
	-0.3V (min2)	-0.3V (min2)	
	1.0V (max)	1.0V (max)	

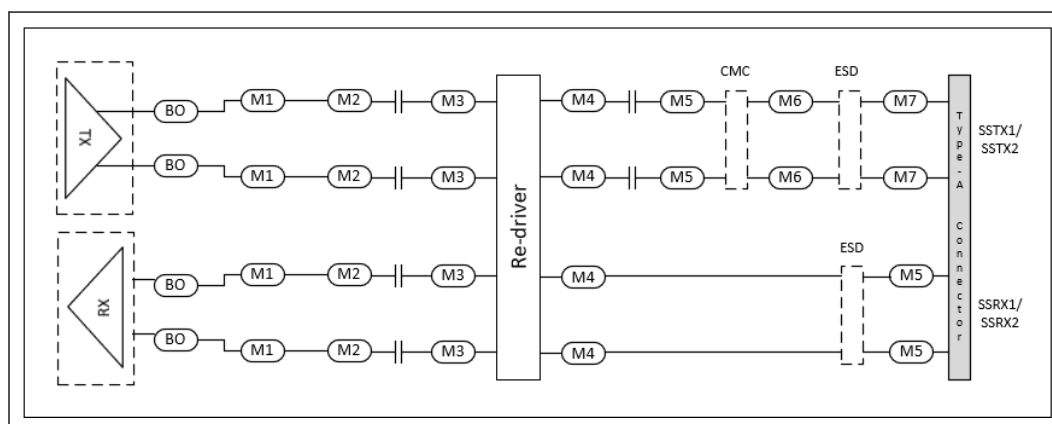
Figure 119. TCP USB 3.2 Gen 2x1 Type-A Redriver Topology Diagram

Table 339. TCP USB 3.2 Gen 2x1 Type-A Redriver Topology Notes

Note
Note1*: Max/ min routing length for USB3.2 Gen1 & 2 speeds (5/ 10 Gbps) depends on redriver performance and its loss compensation capability when used in conjunction with Intel processor. Work with repeater vendor to ensure the selected part can provide adequate signal integrity at the processor.

Rx, Tx
Table 340. TCP USB 3.2 Gen 2x1 Type-A Redriver Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	0	Note1*
M1	MS, DSL, SL	0	Note1*
M2+M3	MS	0	Note1*
M4+M5+M6+M7	MS	0	Note1*

Max Length Total Note: Refer to Note1

6.1.7.6 TCP USB 3.2 + DP

Table 341. TCP USB 3.2 + DP General Guidelines

Note	Detail
Discrete component part size for mainstream stackup	0402, if routing near maximum length it is recommended to use 0201 component size.
Recommended short protection circuit	It is recommended to include this circuit to comply with USB ECN \"RX AC Coupling Capacitor Option\" to provide ESD/ EOS protection and set RX bias at the Type-C connector. Intel's recommendation on DC rating of the Crx is 25 V (Extended Power Range not implemented)/50 V (Extended Power Range implemented). The customer should consider short event to VBUS that can be up to 20 V.
Voiding recommendation for mainstream stackup	It is recommended to void unused (non-functional) pads for differential vias and pin pads for CMCs, ESDs, AC cap and connectors to optimize the impedance matching in the channel.
Unused signals	Unused data and AUX signals should be left as no connect at the BGA ball.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).

Table 342. TCP USB 3.2 + DP Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	Yes	25
continued...					

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
4	RX-RX	Total	Length (mm)	Yes	25
5	TX-RX	Total	Length (mm)	Yes	25
6	DATA-CLK	Total	Length (mm)	No	0

6.1.7.6.1 TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Topology

Figure 120. TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Topology Diagram

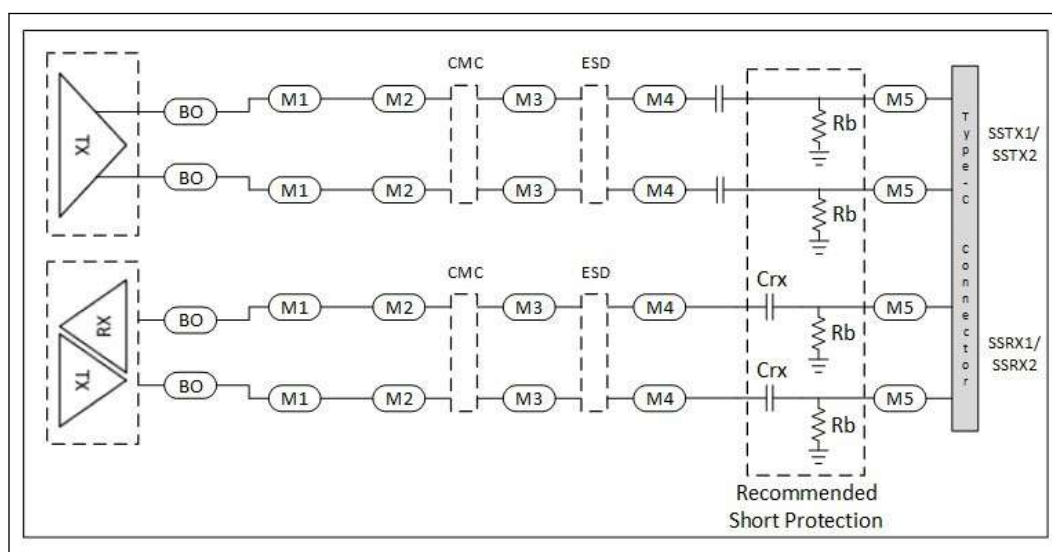


Table 343. TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Topology Notes

Note	Detail
Number of vias allowed	4 (route within 2 layers).
Maximum via stub length	250 um per via.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground. Prefer ground on both sides.
Reference plane for dual stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground and it is the layer that is closer to the signal route. Prefer ground on both sides.
CMC	Refer to the EMC Component Selection section of the PDG for CMC recommendation(s).
ESD	Refer to the EMC Component Selection section of the PDG for ESD recommendation(s).
AC cap value	75 nF to 265 nF including tolerance, 25 V (Extended Power Range not implemented)/ 50 V (Extended Power Range implemented).
continued...	

Note	Detail
Crx cap value	297 nF to 363 nF including tolerance, 25 V (Extended Power Range not implemented)/ 50 V (Extended Power Range implemented).
Rb resistor value	220 kΩ ± 5%, 25 V (Extended Power Range not implemented)/ 50 V (Extended Power Range implemented).
TX bleed resistor placeholder	Recommend having a placeholder for an 0201 Rb resistor between the AC cap and Type-C connector on the SSTX pairs.
Supported interfaces	USB 3.2 Gen 2x2 (10 Gbps per lane) and DP v1.4a (up to 8.1 Gbps per lane).
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.

Mainstream, Mid Loss (ML), Rx,Tx

Table 344. TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Topology Mainstream, Mid Loss (ML), Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS	8	
M1+M2	MS	118	BO+M1+M2+M3+M4+M5 ≤ 118 mm
M3+M4+M5	MS	25	

Max Length Total (mm): 118

6.1.7.6.2 TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Redriver Topology

Figure 121. TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Redriver Topology Diagram

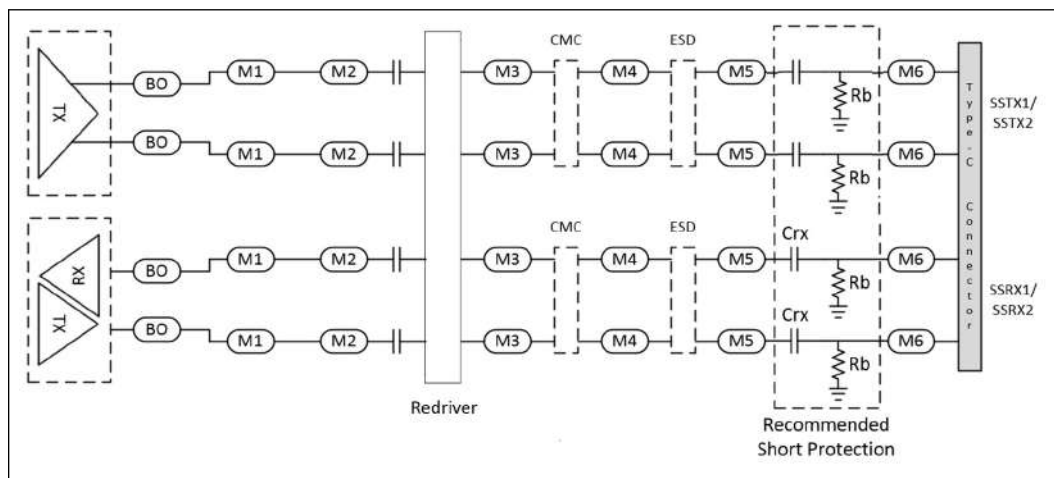


Table 345. TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Redriver Topology Notes

Note	Detail
Number of vias allowed	4 (route within 2 layers) before redriver, 2 after redriver.
Maximum via stub length	375 um per via between SOC and redriver. Redriver post-channel cannot have via stub.
continued...	

Note	Detail
Reference plane for micro-strip route	Must be continuous ground.
Reference plane for stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground. Prefer ground on both side.
Reference plane for dual stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground and it is the layer that is closer to the signal route. Prefer ground on both side.
CMC	For CMC refer to EMC section. Populating will not reduce the supported length.
ESD	For ESD refer to EMC section.
AC Cap Value	75 nF to 265 nF including tolerance, 25 V for cap closest to Type-C connector.
Crx Cap Value	297 nF to 363 nF including tolerance, 25 V.
Rb Resistor Value	220 k Ω $\pm$ 5%, 25 V.
TX Bleed Resistor Placeholder	Recommend having a placeholder for an 0201 Rb resistor between the AC Cap and Type-C connector on the SSTX pairs.
Redriver	Refer to Intel recommended parts section for redriver recommendation.
Supported Interfaces	USB 3.2 Gen 2, and DP v1.4a (8.1 Gbps).

Mainstream, Mid Loss (ML), Rx,Tx, Pre-Channel

Table 346. TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Redriver Topology Mainstream, Mid Loss (ML), Rx,Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL	0
M1+M2	DSL	0

Max Length Total Note: For max lengths follow the vendor Re-driver recommendation . Lengths will be varied based on Re-driver part

Mainstream, Mid Loss (ML), Rx,Tx, Post-Channel

Table 347. TCP USB 3.2 Gen 2x2 + DP HBR3 Type-C Redriver Topology Mainstream, Mid Loss (ML), Rx,Tx, Post-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
M3+M4+M5+M6	MS, DSL, SL	0

Max Length Total Note: Max lengths recommendation will depend on Vendor re-driver part number. Refer to vendor recommended re-driver part number

6.1.7.7 TCP TBT4

Table 348. TCP TBT4 General Guidelines

Note	Detail
Voiding recommendation for mainstream stackup	It is recommended to void unused (non-functional) pads for differential vias and pin pads for ESDs, AC caps and connectors to optimize the impedance matching in the channel.
Optimal routing	For optimal performance it is recommended to begin layout with TBT routing, use arc routing and maintain 3H spacing to vias/ void/ splits. It's also recommended that return loss specification of routing from retimer to Type-C connector is met. It is listed in section 5.5.1.1 of USB Type-C TBT Specification (Document #557111).
Discrete component part size for mainstream stackup	Recommend 0201 components. It is OK to use 0402 component between processor and retimer when NOT routing near maximum length. Strongly recommend 0201 components between retimer and Type-C connector, 0402 components may result in return loss spec violation.
Recommended short protection circuit	It is recommended to include this circuit to comply with USB ECN "RX AC Coupling Capacitor Option" to provide ESD/ EOS protection and set RX bias at the Type-C connector. See notes on each topology for Intel's recommendation on DC rating of the Crx. The customer should consider short event to VBUS that can be up to 25 V.
Unused signals	Unused data and AUX signals should be left unconnected at the BGA ball.
Routing and return via requirement near vertical transition	For all vertical transitions, the differential transmission lines should be connected to the two vias symmetrically. Moreover, Vss return vias should be placed near the signal vias in an equidistant symmetrical manner and all signal vias should have identical Vss return vias around each of them.
Recommendations and considerations for routing near switch mode power supply, routing under inductor, routing under phase node, and routing under DrMOS	Refer to "Switching VR Circuit Guidelines" in the "General Design Considerations" section as well as Technical White Paper Doc#: 726825
Dual-stripline routing recommendation	Dual-stripline routings refer to "General Dual-Stripline Support" section. When routing orthogonal or with angle on top of TBT5 DSL tlines, allow maximum of 3 aggressor pairs to route over 1 DSL routing victim pair.

Table 349. TCP TBT4 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.125
3	TX-TX	Total	Length (mm)	Yes	25
4	RX-RX	Total	Length (mm)	Yes	25
5	TX-RX	Total	Length (mm)	Yes	25
6	DATA-CLK	Total	Length (mm)	No	0

6.1.7.7.1 TCP TBT4 Retimer Topology

Figure 122. TCP TBT4 Retimer Topology Diagram

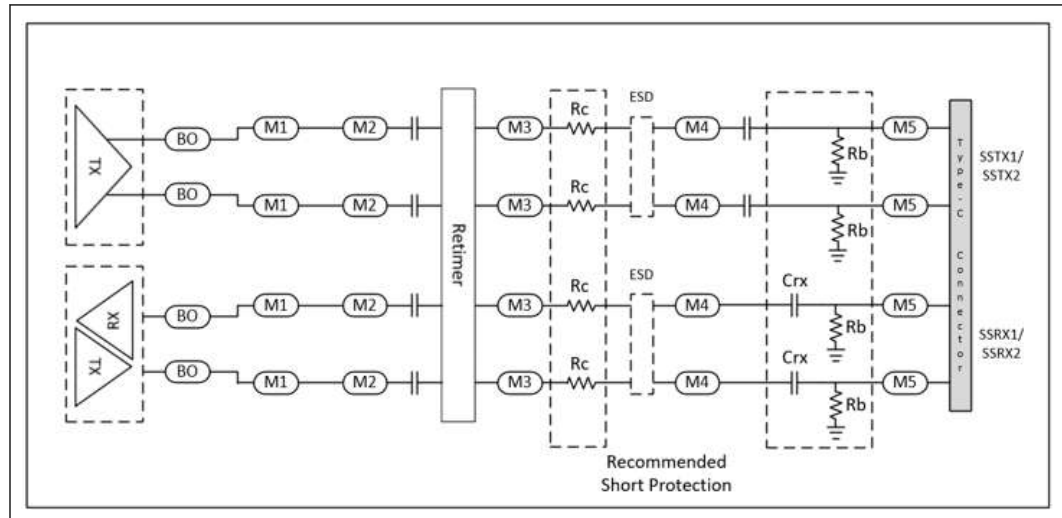


Table 350. TCP TBT4 Retimer Topology Notes

Note	Detail
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
Reference plane for dual stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
ESD	Please refer to Thunderbolt Retimer documentation for more details. Refer to RDC #843990 for June Bridge 40 Retimer guidelines, RDC #631030 for Hayden Bridge 40 Retimer guidelines.
AC Cap position	NA
AC cap value	135 nF to 265 nF including tolerance. For the post-channel cap - Form factor 0201 required - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported For the pre-channel cap - Form factor 0201 recommended - Rated voltage: 6.3 V. When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V.
Crx cap value	297 nF to 363 nF including tolerance. - Form factor: 0201 recommended. - Rated voltage: 25 V.

continued...

Note	Detail
	When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V.
Rb resistor value	220 k Ω $\pm$ 5% Form factor: 0201 recommended.
Rc resistor value	2.2 Ω $\pm$ 10% - Form factor: 0201 recommended. When the Extended Power Range Power Delivery (EPR) is implemented: i. If EPR is @ 28 V, then Rc should be 2.7 Ω $\pm$ 5%. ii. If EPR is @ 36 V, then Rc should be 3.24 Ω $\pm$ 5%. iii. If EPR is @ 48 V, then Rc should be 4.02 Ω $\pm$ 5%.
TX bleed resistor placeholder	Recommend having a placeholder for an 0201 Rb resistor between the AC cap and Type-C connector on the SSTX pairs.
Retimer	June Bridge 40, Hayden Bridge 40
Total insertion loss budget from retimer pins to Type-C connector pads	4.5 dB @ 10 GHz on PCB. Inclusive of HVM variation. Not inclusive of connector.
Supported interfaces	Thunderbolt 4.0, USB4.0 Gen3, USB3.2 Gen2x2 and DP2.1 (UHBR20).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).

Table 351. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	2	Pre-Channel	Maximum Via Stub Length is 250 μ m
Rx, Tx	2	Post-Channel	No Via Stub Allowed

Mainstream, Mid Loss (ML), Rx,Tx, Pre-Channel
Table 352. TCP TBT4 Retimer Topology Mainstream, Mid Loss (ML), Rx,Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	8	
M1+M2	MS, DSL, SL	205	Refer to Max length Total Note for details. DSL may cause via stub to be longer than the maximum required 250 μ m. Via stub mitigation routing technique may be employed.

Max Length Total (mm): 205

Max Length Total Note: BO+M1+M2 = 205 mm. If more than 90% of the routing is Stripline or Dual-stripline, the maximum total length can be extended to 225 mm (M1 + M2 = 217 mm).

Mainstream, Premium Mid Loss (PML), Rx,Tx, Pre-Channel

Table 353. TCP TBT4 Retimer Topology Mainstream, Premium Mid Loss (PML), Rx,Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	8	
M1+M2	MS, DSL, SL	205	Refer to Max length Total Note for details. DSL may cause via stub to be longer than the maximum required 250 um. Via stub mitigation routing technique may be employed.

Max Length Total (mm): 205

Max Length Total Note: BO+M1+M2 = 205 mm. T4 10L Premium Mid Loss (PML): If more than 90% routing is Stripline the maximum length can be extended to 225 mm (M1+M2 = 217 mm). T3 10L Premium Mid Loss (PML): If more than 90% is routing is Stripline or Dual stripline, the maximum total length can be extended up to 240 mm (M1+M2 = 232 mm).

Thin PCB, Premium Mid Loss (PML), Rx,Tx, Pre-Channel

Table 354. TCP TBT4 Retimer Topology Thin PCB, Premium Mid Loss (PML), Rx,Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	8	
M1+M2	MS, DSL, SL	205	Refer to Max length Total Note for details. DSL may cause via stub to be longer than the maximum required 250 um. Via stub mitigation routing technique may be employed.

Max Length Total (mm): 205

Max Length Total Note: BO+M1+M2 = 205 mm. If more than 90% of the routing is Stripline or Dual-stripline, the maximum total length can be extended to 225 mm (M1+M2 = 217 mm).

Rx,Tx, Post-Channel

Table 355. TCP TBT4 Retimer Topology Rx,Tx, Post-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
M3+M4+M5	MS	30

Max Length Total (mm): 30

6.1.7.7.2 TCP TBT4 Retimer with Internal Cable Topology

Figure 123. TCP TBT4 Retimer with Internal Cable Topology Diagram

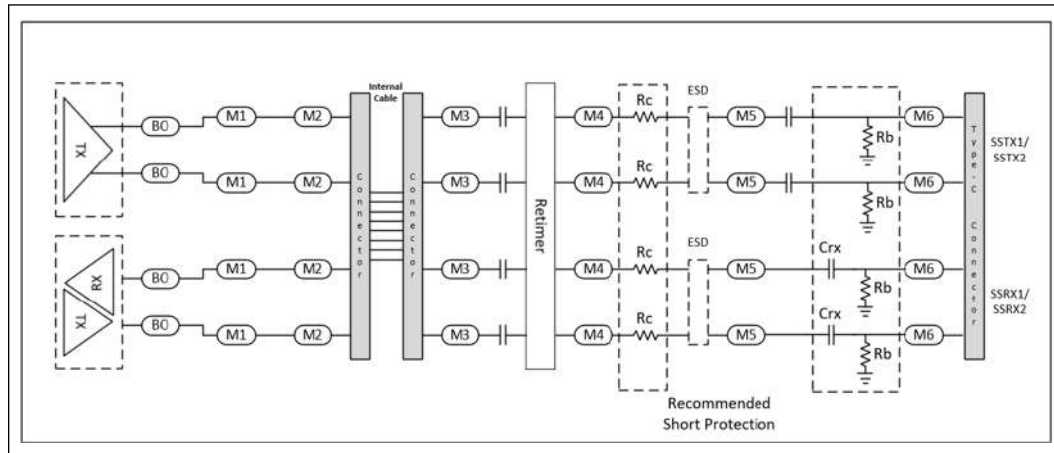


Table 356. TCP TBT4 Retimer with Internal Cable Topology Notes

Note	Detail
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
Reference plane for dual stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
ESD	Please refer to Thunderbolt Retimer documentation for more details. Refer to RDC #843990 for June Bridge 40 Retimer guidelines, RDC #631030 for Hayden Bridge 40 Retimer guidelines.
AC Cap position	NA
AC cap value	135 nF to 265 nF including tolerance. For the post-channel cap - Form factor 0201 required - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported For the pre-channel cap - Form factor 0201 recommended - Rated voltage: 6.3 V. When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V.
Crx cap value	297 nF to 363 nF including tolerance. - Form factor: 0201 recommended. - Rated voltage: 25 V. When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended.

continued...

Note	Detail
	- Rated voltage: 63 V.
Rb resistor value	220 k Ω $\pm$ 5% Form factor: 0201 recommended.
Rc resistor value	2.2 Ω $\pm$ 10% - Form factor: 0201 recommended. When the Extended Power Range Power Delivery (EPR) is implemented: i. If EPR is @ 28 V, then Rc should be 2.7 Ω $\pm$ 5%. ii. If EPR is @ 36 V, then Rc should be 3.24 Ω $\pm$ 5%. iii. If EPR is @ 48 V, then Rc should be 4.02 Ω $\pm$ 5%.
TX bleed resistor placeholder	Recommend having a placeholder for an 0201 Rb resistor between the AC cap and Type-C connector on the SSTX pairs.
Retimer	June Bridge 40, Hayden Bridge 40
Total insertion loss budget from retimer pins to Type-C connector pads	4.5 dB @ 10 GHz on PCB. Inclusive of HVM variation. Not inclusive of connector.
Supported interfaces	Thunderbolt 4.0, USB4.0 Gen3, USB3.2 Gen2x2 and DP2.1 (UHBR20).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).

Table 357. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	2	Pre-Channel	Maximum Via Stub Length is 250 μ m
Rx, Tx	2	Post-Channel	No Via Stub Allowed

Rx,Tx, Pre-Channel**Table 358. TCP TBT4 Retimer with Internal Cable Topology Rx,Tx, Pre-Channel Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	8	
M1+M2	MS, DSL, SL	110	DSL may cause via stub to be longer than the maximum required 250 μ m. Via stub mitigation routing technique may be employed.
M3	MS	20	DSL may cause via stub to be longer than the maximum required 250 μ m. Via stub mitigation routing technique may be employed.
Cable	Cable	255	This cable length assumes a micro-coax cable assembly that is electrically compliant with the specification given in document #575032.

Max Length Total (mm): 110

Max Length Total Note: BO+M1+M2+M3 = 110 mm. Max length not including cable.

Rx,Tx, Post-Channel

Table 359. TCP TBT4 Retimer with Internal Cable Topology Rx,Tx, Post-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
M4+M5+M6	MS, SL	30

Max Length Total (mm): 30

6.1.7.8 TCP TBT5

Figure 124. Void Above and Below Buried Vias PTH Pads in Type-4 Stackup

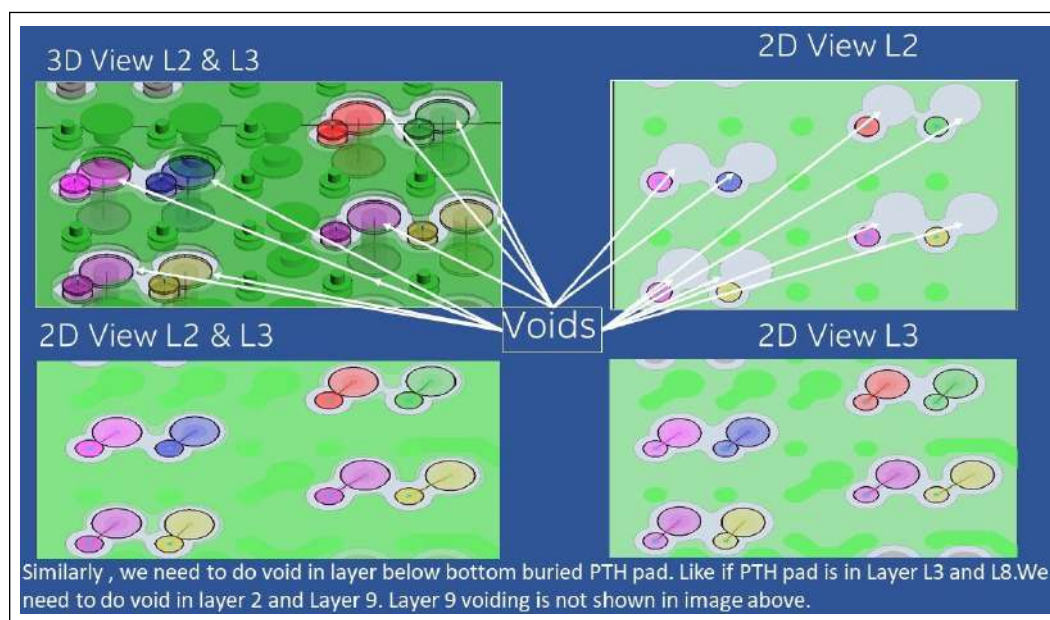


Figure 125. Voids Under SOC Ball Pads

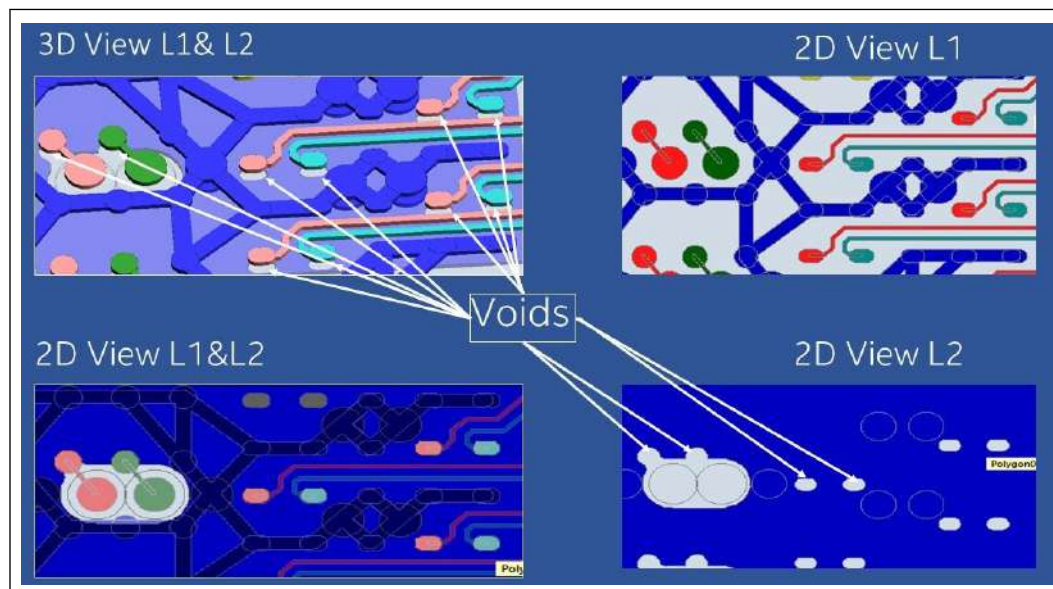


Figure 126. Component Pad Voiding Requirement

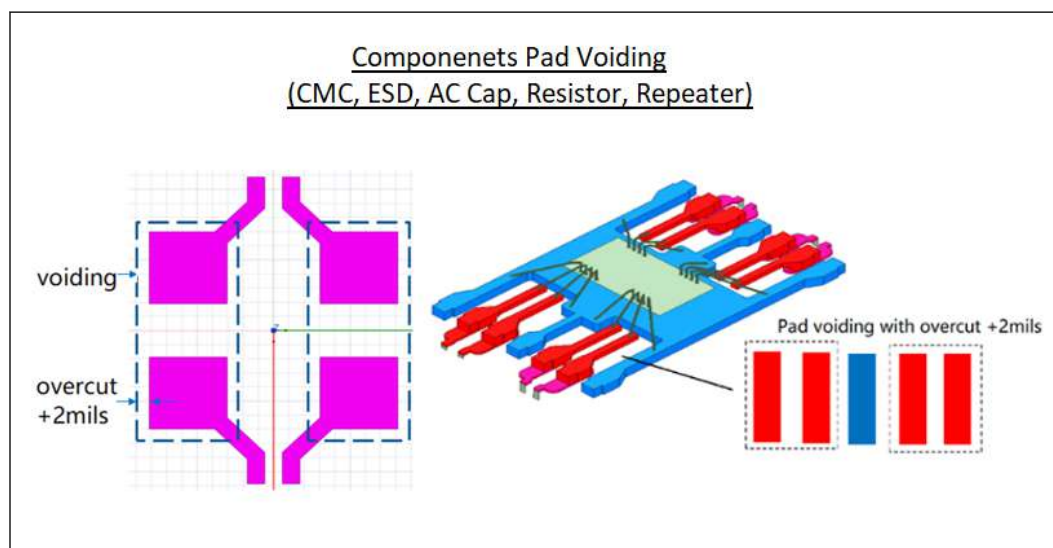


Table 360. TCP TBT5 General Guidelines

Note	Detail
Voiding recommendation for mainstream stackup	It is recommended to void unused (non-functional) pads for differential vias and pin pads for ESDs, AC caps and connectors to optimize the impedance matching in the channel.
Optimal routing	For optimal performance it is recommended to begin layout with TBT routing, use arc routing and maintain 3H spacing to vias/ void/ splits. It's also recommended that return loss specification of routing from retimer to Type-C connector is met. It is listed in section 5.5.1.1 of USB Type-C TBT Specification (Document #557111).

continued...

Note	Detail
Discrete component part size for mainstream stackup	Recommend 0201 components. It is OK to use 0402 component between processor and retimer when NOT routing near maximum length. Strongly recommend 0201 components between retimer and Type-C connector, 0402 components may result in return loss spec violation.
Recommended short protection circuit	It is recommended to include this circuit to comply with USB ECN "RX AC Coupling Capacitor Option" to provide ESD/ EOS protection and set RX bias at the Type-C connector. See notes on each topology for Intel's recommendation on DC rating of the Crx. The customer should consider short event to VBUS that can be up to 25 V.
Unused signals	Unused data and AUX signals should be left unconnected at the BGA ball.
Routing and return via requirement near vertical transition	For all vertical transitions, the differential transmission lines should be connected to the two vias symmetrically. Moreover, Vss return vias should be placed near the signal vias in an equidistant symmetrical manner and all signal vias should have identical Vss return vias around each of them.
Recommendations and considerations for routing near switch mode power supply, routing under inductor, routing under phase node, and routing under DrMOS	Refer to "Switching VR Circuit Guidelines" in the "General Design Considerations" section as well as Technical White Paper Doc#: 726825
Dual-stripline routing recommendation	Dual-stripline routings refer to "General Dual-Stripline Support" section. When routing orthogonal or with angle on top of TBT5 DSL tlines, allow maximum of 3 aggressor pairs to route over 1 DSL routing victim pair.

Table 361. TCP TBT5 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.125
3	TX-TX	Total	Length (mm)	Yes	25
4	RX-RX	Total	Length (mm)	Yes	25
5	TX-RX	Total	Length (mm)	Yes	25
6	DATA-CLK	Total	Length (mm)	No	0

6.1.7.8.1 TCP TBT5 Retimer Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 127. TCP TBT5 Retimer Topology Diagram

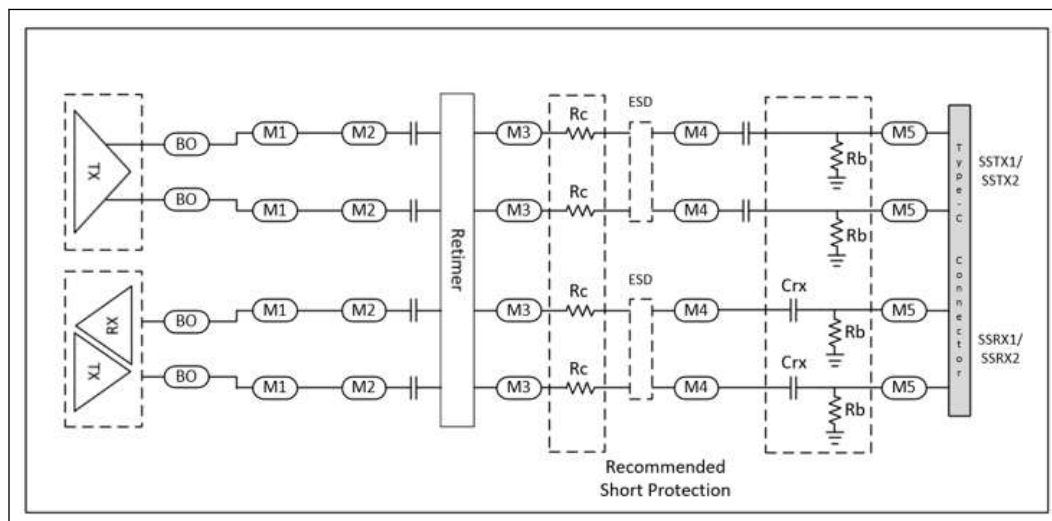


Table 362. TCP TBT5 Retimer Topology Notes

Note	Detail
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
Reference plane for dual stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
ESD	Please refer to Thunderbolt Retimer documentation for more details. Refer to RDC #843990 for June Bridge 80 Retimer guidelines.
AC Cap position	NA
AC cap value	135 nF to 265 nF including tolerance. For the post-channel cap - Form factor 0201 required - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V. For the pre-channel cap - Form factor 0201 recommended - Rated voltage: 6.3 V.
Component Placement	ESD Diode, AC cap, Rb Resistor and Rc Resistor should be placed as close as possible to USB-C connector up to 8mm radius
Crx cap value	297 nF to 363 nF including tolerance. - Form factor: 0201 required. - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported

continued...

Note	Detail
	When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V.
Rb resistor value	220 k Ω $\pm$ 5% Form factor: 0201 recommended.
Rc resistor value	2.2 Ω $\pm$ 10% - Form factor: 0201 recommended. When the Extended Power Range Power Delivery (EPR) is implemented: i. If EPR is @ 28 V, then Rc should be 2.7 Ω $\pm$ 5%. ii. If EPR is @ 36 V, then Rc should be 3.24 Ω $\pm$ 5%. iii. If EPR is @ 48 V, then Rc should be 4.02 Ω $\pm$ 5%.
TX bleed resistor placeholder	Recommend having a placeholder for an 0201 Rb resistor between the AC cap and Type-C connector on the SSTX pairs.
Retimer	June Bridge 80
Total insertion loss budget from retimer pins to Type-C connector	6 dB @ 12.8 GHz on PCB. Inclusive of HVM variation and including of type-C connector loss.
Supported interfaces	Thunderbolt 5.0, Thunderbolt 4.0, USB4 V2 Gen4, USB4.0 Gen3, USB3.2 Gen2x2 and DP2.1 (UHBR20).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).

Table 363. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	2	Pre-Channel	Maximum Via Stub Length is 250 μ m
Rx, Tx	2	Post-Channel	No Via Stub Allowed

Mainstream, Mid Loss (ML), Rx, Tx, Pre-Channel
Table 364. TCP TBT5 Retimer Topology Mainstream, Mid Loss (ML), Rx, Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	2	Minimize BO length to less than 2 mm in the escape area near ball for Microstrip lines on Top or Bottom layer. Don't need to consider BO in the inner layer.
M1+M2	MS, DSL, SL	200	Refer to Max length Total Note for details. DSL may cause via stub to be longer than the maximum required 250 μ m. Via stub mitigation routing technique may be employed.

Max Length Total (mm): 200

Max Length Total Note: BO+M1+M2 = 200 mm. If more than 90% of the routing is Stripline or Dual-stripline, the maximum total length can be extended to 225 mm (M1+M2 = 223 mm).

Mainstream, Premium Mid Loss (PML), Rx, Tx, Pre-Channel

Table 365. TCP TBT5 Retimer Topology Mainstream, Premium Mid Loss (PML), Rx, Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	2	Minimize BO length to less than 2 mm in the escape area near ball for Microstrip lines on Top or Bottom layer. Don't need to consider BO in the inner layer.
M1+M2	MS, DSL, SL	205	Refer to Max length Total Note for details. DSL may cause via stub to be longer than the maximum required 250 um. Via stub mitigation routing technique may be employed.

Max Length Total (mm): 205

Max Length Total Note: BO+M1+M2 = 205 mm. T4 10L Premium Mid Loss (PML): If more than 90% routing is Stripline the maximum length can be extended to 225 mm (M1+M2 = 223 mm). T3 10L Premium Mid Loss (PML): If more than 90% is routing is Stripline or Dual stripline, the maximum total length can be extended up to 240 mm (M1+M2 = 238 mm).

Thin PCB, Premium Mid Loss (PML), Rx, Tx, Pre-Channel

Table 366. TCP TBT5 Retimer Topology Thin PCB, Premium Mid Loss (PML), Rx, Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	2	Minimize BO length to less than 2 mm in the escape area near ball for Microstrip lines on Top or Bottom layer. Don't need to consider BO in the inner layer.
M1+M2	MS, DSL, SL	200	Refer to Max length Total Note for details. DSL may cause via stub to be longer than the maximum required 250 um. Via stub mitigation routing technique may be employed.

Max Length Total (mm): 200

Max Length Total Note: BO+M1+M2 = 200 mm. If more than 90% of the routing is Stripline or Dual-stripline, the maximum total length can be extended to 225 mm (M1+M2 = 223 mm).

Rx, Tx, Post-Channel

Table 367. TCP TBT5 Retimer Topology Rx, Tx, Post-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
M3+M4+M5	MS	30

Max Length Total (mm): 30

6.1.7.8.2 TCP TBT5 Retimer with Internal Cable Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 128. TCP TBT5 Retimer with Internal Cable Topology Diagram

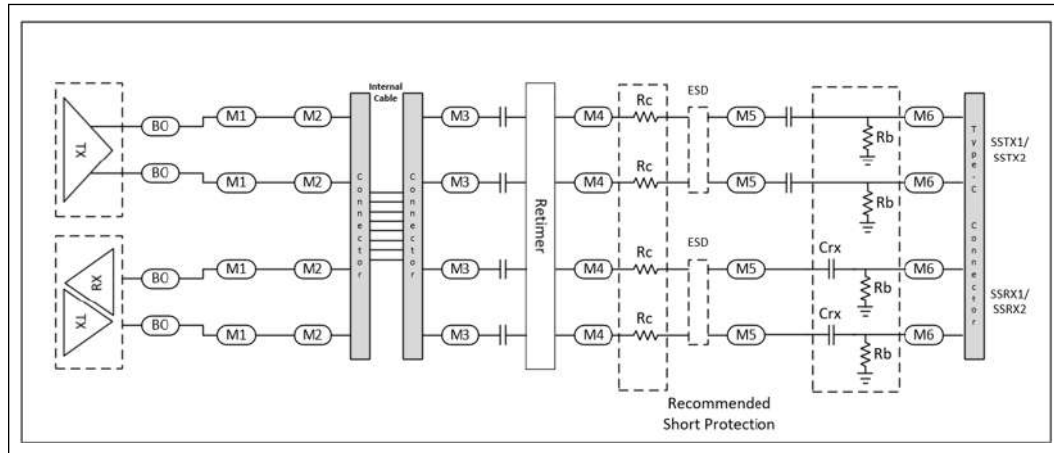


Table 368. TCP TBT5 Retimer with Internal Cable Topology Notes

Note	Detail
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
Reference plane for dual stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
ESD	Please refer to Thunderbolt Retimer documentation for more details. Refer to RDC #843990 for June Bridge 80 Retimer guidelines.
AC Cap position	NA
AC cap value	135 nF to 265 nF including tolerance. For the pre-channel cap - Form factor 0201 recommended - Rated voltage: 6.3 V. For the post-channel cap - Form factor 0201 required - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V.
Component Placement	ESD Diode, AC cap, Rb Resistor and Rc Resistor should be placed as close as possible to USB-C connector up to 8mm radius
Crx cap value	297 nF to 363 nF including tolerance. - Form factor: 0201 required. - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended.

continued...

Note	Detail
	- Rated voltage: 63 V.
Rb resistor value	220 k Ω $\pm$ 5% Form factor: 0201 recommended.
Rc resistor value	2.2 Ω $\pm$ 10% - Form factor: 0201 recommended. When the Extended Power Range Power Delivery (EPR) is implemented: i. If EPR is @ 28 V, then Rc should be 2.7 Ω $\pm$ 5%. ii. If EPR is @ 36 V, then Rc should be 3.24 Ω $\pm$ 5%. iii. If EPR is @ 48 V, then Rc should be 4.02 Ω $\pm$ 5%.
TX bleed resistor placeholder	Recommend having a placeholder for an 0201 Rb resistor between the AC cap and Type-C connector on the SSTX pairs.
Retimer	June Bridge 80
Total insertion loss budget from retimer pins to Type-C connector	6 dB @ 12.8 GHz on PCB. Inclusive of HVM variation and including of type-C connector loss.
Supported interfaces	Thunderbolt 5.0, Thunderbolt 4.0, USB4 V2 Gen4, USB4.0 Gen3, USB3.2 Gen2x2 and DP2.1 (UHBR20).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
Supported Cable	FFC and FPC cables, Refer to RDC #873680 document to learn more in detail including both FFC and FPC types of cable and reference electrical requirements.

Table 369. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	2	Pre-Channel	Maximum Via Stub Length is 250 μ m
Rx, Tx	2	Post-Channel	No Via Stub Allowed

Rx, Tx, Pre-Channel**Table 370. TCP TBT5 Retimer with Internal Cable Topology Rx, Tx, Pre-Channel Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	2	Minimize BO length to less than 2 mm in the escape area near ball for Microstrip lines on Top or Bottom layer. Don't need to consider BO in the inner layer.
M1+M2	MS, DSL, SL	90	DSL may cause via stub to be longer than the maximum required 250 μ m. Via stub mitigation routing technique may be employed.
M3	MS	20	
Cable	Cable	101.6	This cable length assumes a FPC cable with the insertion loss of 6.7 dB at 12.8 GHz Nyquist frequency. Refer to RDC #873680 document to learn more in detail including FFC type of cable and some reference electrical requirements.

Max Length Total (mm): 110

Max Length Total Note: For FPC cable maximum total length is 110 mm ($M1+M2 = 90$ mm & $BO+M1+M2+M3 = 110$ mm). If more than 90% is routing is Stripline or Dual stripline, the maximum total length can be extended up to 120 mm ($M1+M2 = 100$ mm & $BO+M1+M2+M3 = 120$ mm). If FFC cable used, maximum total length is 125 mm (Microstrip) or 145mm (90% Stripline). Max length not including cable.

Rx, Tx, Post-Channel

Table 371. TCP TBT5 Retimer with Internal Cable Topology Rx, Tx, Post-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
M4+M5+M6	MS, SL	30

Max Length Total (mm): 30

6.1.7.8.3 TCP TBT5 Cascaded Retimer Topology

Description: This topology is supported for NVL 8C/H16C only.

Figure 129. TCP TBT5 Cascaded Retimer Topology Diagram

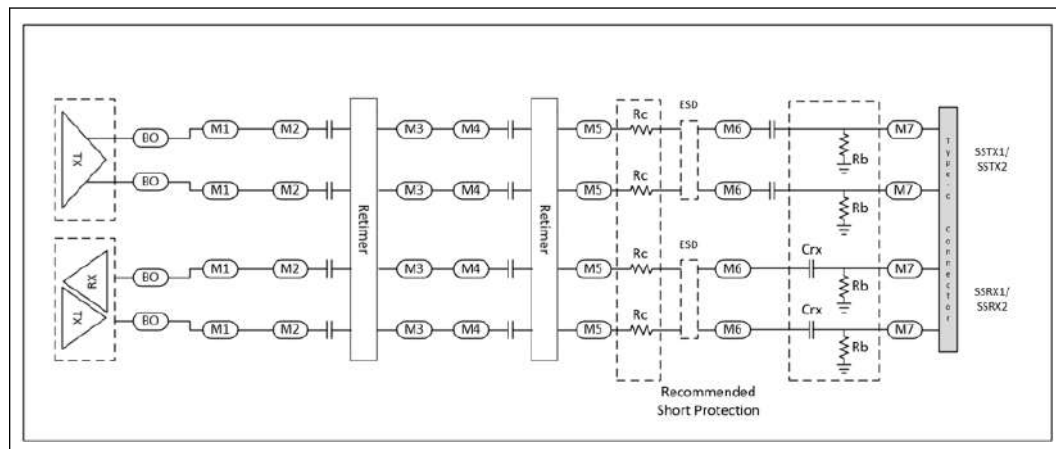


Table 372. TCP TBT5 Cascaded Retimer Topology Notes

Note	Detail
Routing style	Microstrip main route routing must be interleaved. To reduce system RFI/ EMI, it is recommended to use SL instead of MS. If MS routing is needed, refer to EMC PDG for RFI/ EMI mitigation.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
Reference plane for dual stripline route	Full ground reference is required. Dual reference/ Power reference not allowed.
ESD	Please refer to Thunderbolt Retimer documentation for more details. Refer to RDC #843990 for June Bridge 80 Retimer guidelines.
AC Cap position	NA

continued...

Note	Detail
AC cap value	135 nF to 265 nF including tolerance. For the post-channel cap - Form factor 0201 required - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported For the pre-channel cap - Form factor 0201 recommended - Rated voltage: 6.3 V. When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V.
Component Placement	ESD Diode, AC cap, Rb Resistor and Rc Resistor should be placed as close as possible to USB-C connector up to 8mm radius
Crx cap value	297 nF to 363 nF including tolerance. - Form factor: 0201 required. - Rated voltage: 6.3 V. - Rated voltage 25V if 60W supported When the Extended Power Range Power Delivery (EPR) is implemented: - Form factor: 0402 recommended. - Rated voltage: 63 V.
Rb resistor value	220 kΩ $\pm$ 5% Form factor: 0201 recommended.
Rc resistor value	2.2 Ω $\pm$ 10% - Form factor: 0201 recommended. When the Extended Power Range Power Delivery (EPR) is implemented: - If EPR is @ 28 V, then Rc should be 2.7 Ω $\pm$ 5%. - If EPR is @ 36 V, then Rc should be 3.24 Ω $\pm$ 5%. - If EPR is @ 48 V, then Rc should be 4.02 Ω $\pm$ 5%.
TX bleed resistor placeholder	Recommend having a placeholder for an 0201 Rb resistor between the AC cap and Type-C connector on the SSTX pairs.
Retimer	June Bridge 80
Total insertion loss budget from retimer pins to Type-C connector	6 dB @ 12.8 GHz on PCB. Inclusive of HVM variation and including of type-C connector loss.
Supported interfaces	Thunderbolt 5.0, Thunderbolt 4.0, USB4 V2 Gen4, USB4.0 Gen3, USB3.2 Gen2x2 and DP2.1 (UHBR20).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).

Table 373. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	2	Pre-Channel	Maximum Via Stub Length is 250 μ m
Rx, Tx	2	Post-Channel	No Via Stub Allowed

Rx, Tx, Pre-Channel

Table 374. TCP TBT5 Cascaded Retimer Topology Rx, Tx, Pre-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	2	Minimize BO length to less than 2 mm in the escape area near ball for Microstrip lines on Top or Bottom layer. Don't need to consider BO in the inner layer.
M1+M2	MS, DSL, SL	200	DSL may cause via stub to be longer than the maximum required 250 um. Via stub mitigation routing technique may be employed.

Max Length Total (mm): 200

Max Length Total Note: $BO + M1 + M2 \leq 200$ mm for Mainstream Mid Loss (ML), and $BO + M1 + M2 \leq 205$ mm for Premium Mid Loss (PML): If more than 90% of the routing is Stripline or Dual-stripline, the maximum total length can be extended to ≤ 225 mm ($M1 + M2 \leq 223$ mm). T3 Thin PCB need to meet the length in table above.

Rx, Tx, Pre-Channel 2

Table 375. TCP TBT5 Cascaded Retimer Topology Rx, Tx, Pre-Channel 2 Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M3+M4	MS, DSL, SL	200	$M3 + M4 \leq 200$ mm

Max Length Total (mm): 200

Max Length Total Note: $M3 + M4 \leq 200$ mm for Mainstream Mid Loss (ML), and $M3 + M4 \leq 205$ mm for Premium Mid Loss (PML): If more than 90% of the routing is Stripline or Dual-stripline, the maximum total length can be extended to ≤ 225 mm ($M3 + M4 \leq 223$ mm). T3 Thin PCB need to meet the length in table above.

Rx, Tx, Post-Channel

Table 376. TCP TBT5 Cascaded Retimer Topology Rx, Tx, Post-Channel Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
M5+M6+M7	MS, SL	30

Max Length Total (mm): 30

6.1.8 Processor - PCI Express* Design Guidelines

The guidelines in the chapter cover PCI Express* operation at 2.5GT/s, 5GT/s, 8GT/s, 16GT/s, and 32GT/s.

This processor supports 8 PCIe Gen4 lanes and 16 PCIe Gen5 lanes. There are 4 lanes in PCIE_C, 8 lanes in PCIE_D, and 4 lanes in PCIE_E. In that four of the PCIe Gen5 lanes (PCIE_C) are multiplexed with DMI.

NOTE

For the full PCI Express* Root Port Feature Details, refer to the PCI Express* Root Port Support Feature Details chapter in the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2 (# [844316](#)) .

6.1.8.1 Signal Description

Signal Name	Type	Description
PCIE_A[3:0]_TX_N PCIE_A[3:0]_TX_P	O	PCI Express* Controller A Differential Transmit Pairs These are the PCI Express* based outbound high-speed differential signals from PCIe Controller A
PCIE_A[3:0]_RX_N PCIE_A[3:0]_RX_P	I	PCI Express* Controller A Differential Receive Pairs These are the PCI Express* based inbound high-speed differential signals for PCIe Controller A
PCIE_B[3:0]_TX_N PCIE_B[3:0]_TX_P	O	PCI Express* Controller B Differential Transmit Pairs These are the PCI Express* based outbound high-speed differential signals from PCIe Controller B
PCIE_B[3:0]_RX_N PCIE_B[3:0]_RX_P	I	PCI Express* Controller B Differential Receive Pairs These are the PCI Express* based inbound high-speed differential signals for PCIe Controller B
PCIE_C[3:0]_TX_N / DMI_[3:0]_TX_N PCIE_C[3:0]_TX_P / DMI_[3:0]_TX_P	O	PCI Express* Controller C Differential Transmit Pairs These are the PCI Express* based outbound high-speed differential signals from PCIe Controller C In Processor only configuration PCIE_C[3:0]_TX_N/P are available In Processor + IOE configuration, the PCIE_C[3:0]_TX_N/P lanes function as the x4 DMI TX lanes.
PCIE_C[3:0]_RX_N / DMI_[3:0]_RX_N PCIE_C[3:0]_RX_P / DMI_[3:0]_RX_P	I	PCI Express* Controller C Differential Receive Pairs These are the PCI Express* based inbound high-speed differential signals for PCIe Controller C In Processor only configuration PCIE_C[3:0]_RX_N/P are available In Processor + IOE configuration, the PCIE_C[3:0]_RX_N/P lanes function as the x4 DMI RX lanes.
PCIE_D[7:0]_TX_N PCIE_D[7:0]_TX_P	O	PCI Express* Controller D Differential Transmit Pairs These are the PCI Express* based outbound high-speed differential signals from PCIe Controller D
PCIE_D[7:0]_RX_N PCIE_D[7:0]_RX_P	I	PCI Express* Controller D Differential Receive Pairs These are the PCI Express* based inbound high-speed differential signals for PCIe Controller D
PCIE_A_RCOMP PCIE_B_RCOMP	Analog	PCI Express* Controller A and B PHY Impedance Compensation Inputs ¹
PCIE_LINK_DOWN	O	PCI Express* Link Down Debug Signal PCIe link failure debug signal. PCIe Root Port(s) will assert this signal when a link down event occurs and is detected. For example when a link fails to train during an L1 sub-state exit event.
PCIE_E[3:0]_TX_N PCIE_E[3:0]_TX_P	O	PCI Express* Controller E Differential Transmit Pairs These are the PCI Express* based outbound high-speed differential signals from PCIe Controller E
PCIE_E[3:0]_RX_N PCIE_E[3:0]_RX_P	I	PCI Express* Controller E Differential Receive Pairs These are the PCI Express* based inbound high-speed differential signals for PCIe Controller E

6.1.8.2 Modern Standby - PCIe D3cold Enabling

Implementing PCIe end point device D3cold is required to enable its link state to enter L2 or deeper. PCIe Device Power Management D3cold state is where all core power is removed from end point device, and any auxiliary power is only supplied if the device is enabled to generate a wake event. Core power removal is controlled by the platform via external power controls, typically power FETs and may be required to limit the power consumption to PCIe Devices. Otherwise higher than expected device power could be observed with D3cold.

For complete D3cold hardware, software, and motherboard (PERST#, Wake#, device VDD_Core, and device VDD_Aux) recommendations reference the following material and consult your PCIe device vendor(s) for their D3cold required implementation and power consumption:

1. Nova Lake Platform Runtime D3 (RTD3) Hardware and Software Recommendations Design Guide (# TBD)
2. SLP_S0 Design and Debug Checklist (# 570300)

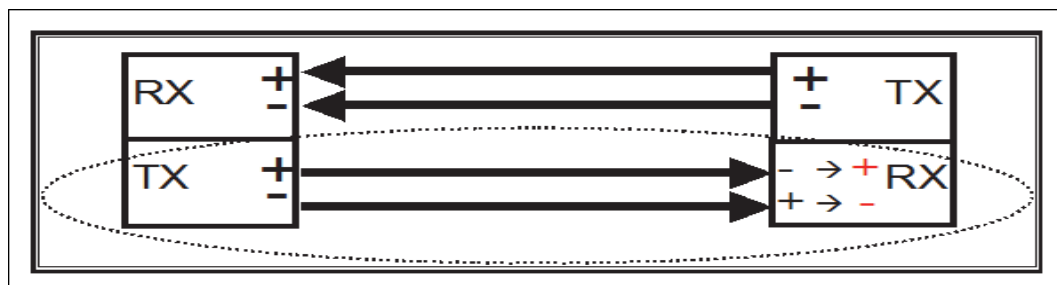
6.1.8.3 PCI Express* Lane Polarity Inversion

The PCI Express* Base Specification requires polarity inversion to be supported independently by all receivers across the Link where each differential pair within each Lane of the PCIe* Link handles its own polarity inversion. Polarity inversion does not imply direction inversion or direction reversal; that is, the Tx differential pair from one device must still connect to the Rx differential pair on the receiving device, per the PCIe* Base Specification.

NOTE

Polarity Inversion is not the same as "PCI Express* Controller Lane Reversal".

Figure 130. Polarity Inversion on a TX to RX Interconnect



6.1.8.4 PCIe Topologies

6.1.8.4.1 PCIe Gen 1-3

Figure 131. Type-4 PCB Via Stitching Recommendation

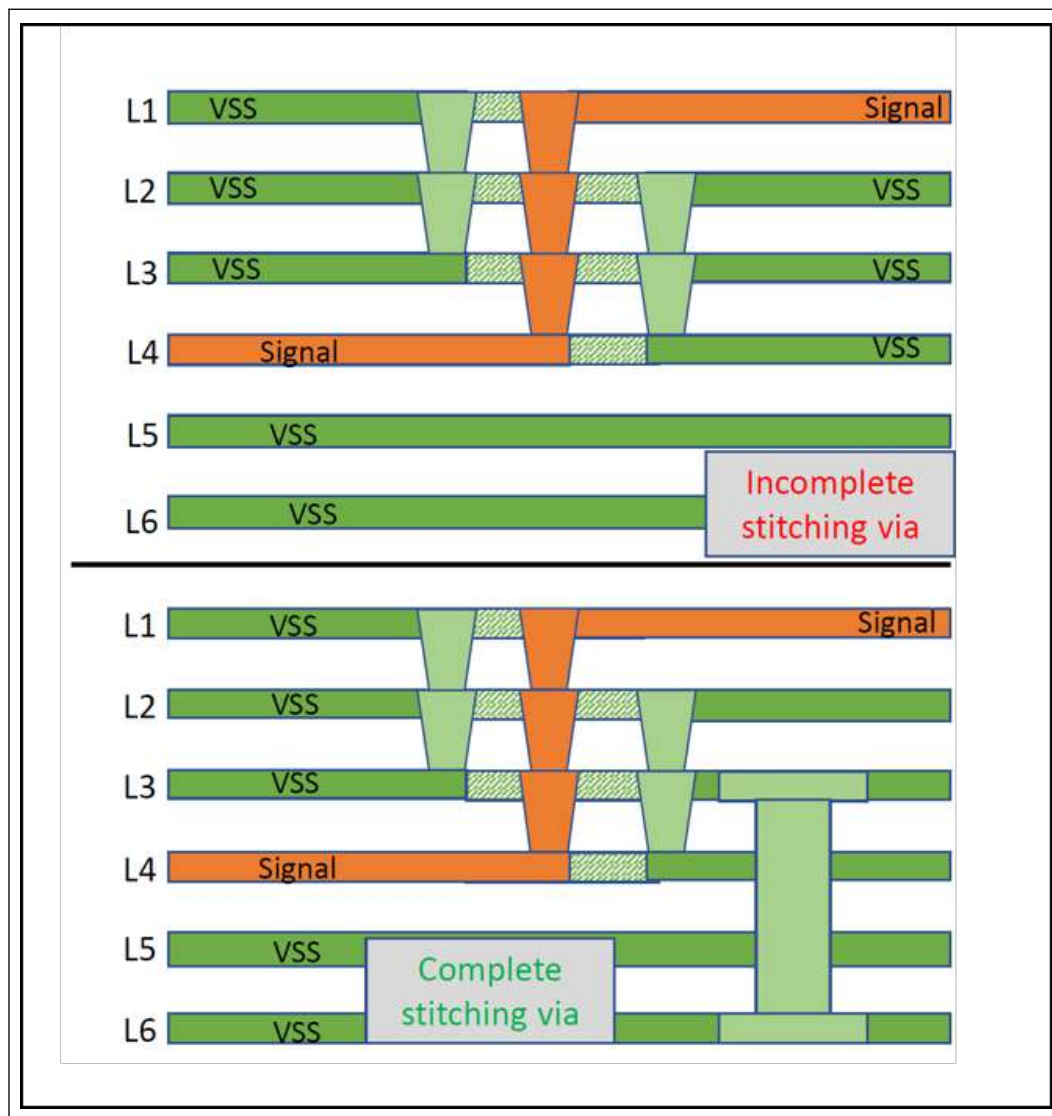


Figure 132. Recommended GND Stitching Via Placement

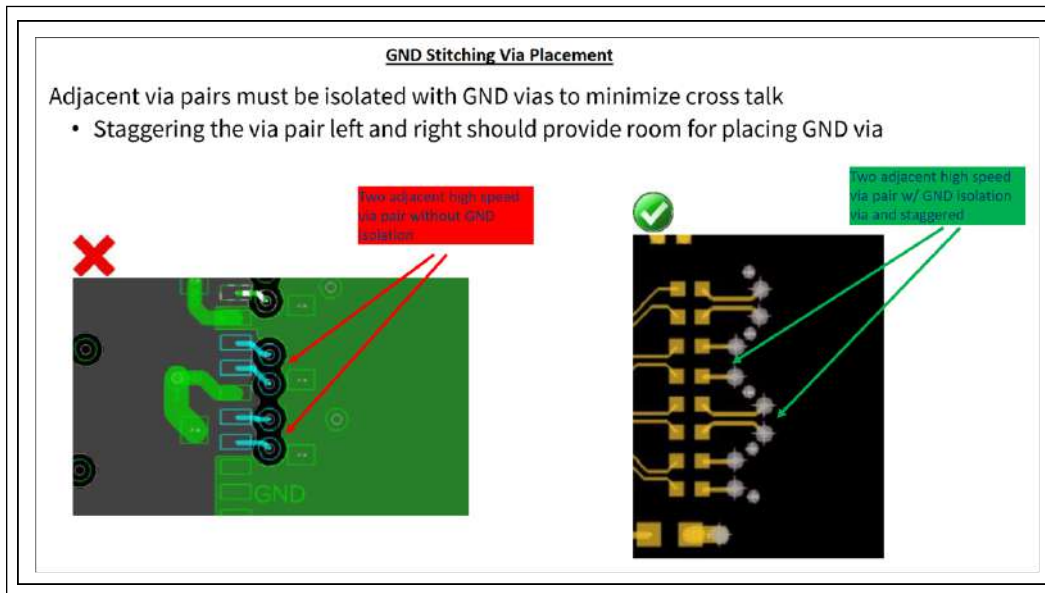


Figure 133. PCIE Board Layout Recommendation Toe Side HSIO Routing



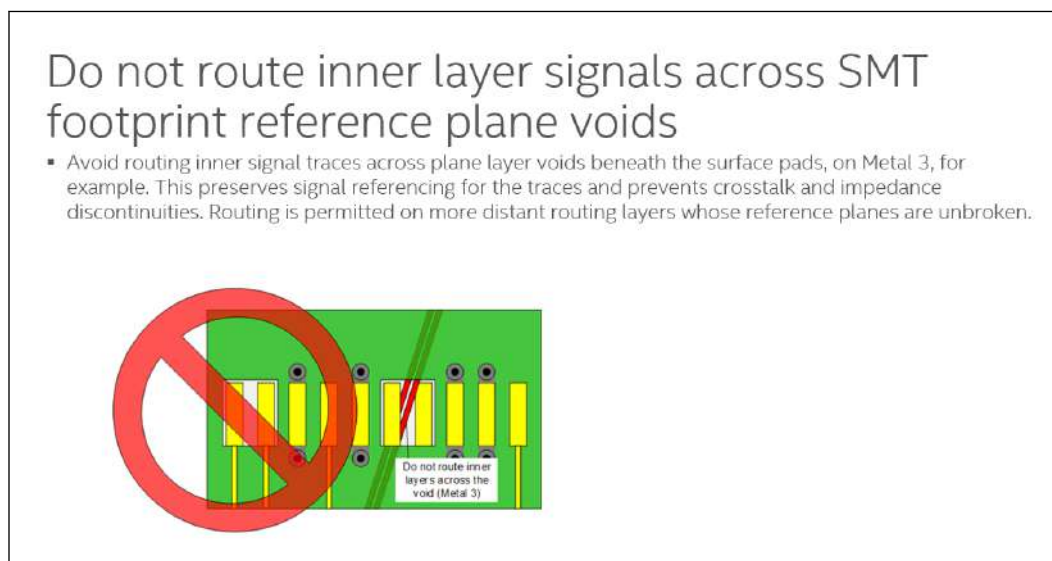
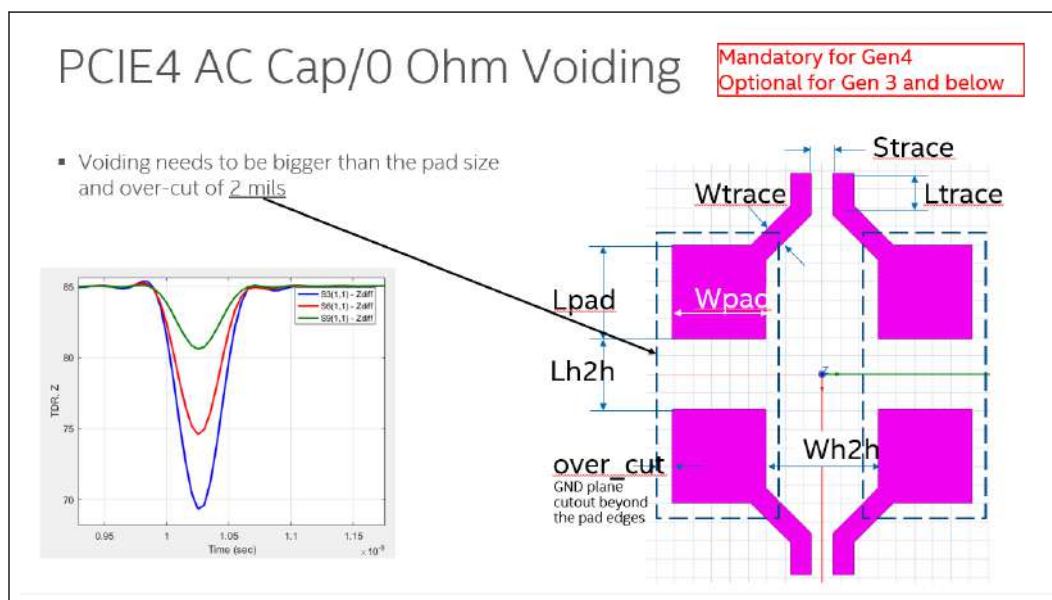
Figure 134. PCIe Board Layout Recommendation SMT footprint Voiding Reference**Figure 135. PCIe Board Layout Recommendation AC Cap Voiding**

Figure 136. PCIe4 M.2 Connector Voiding Recommendation

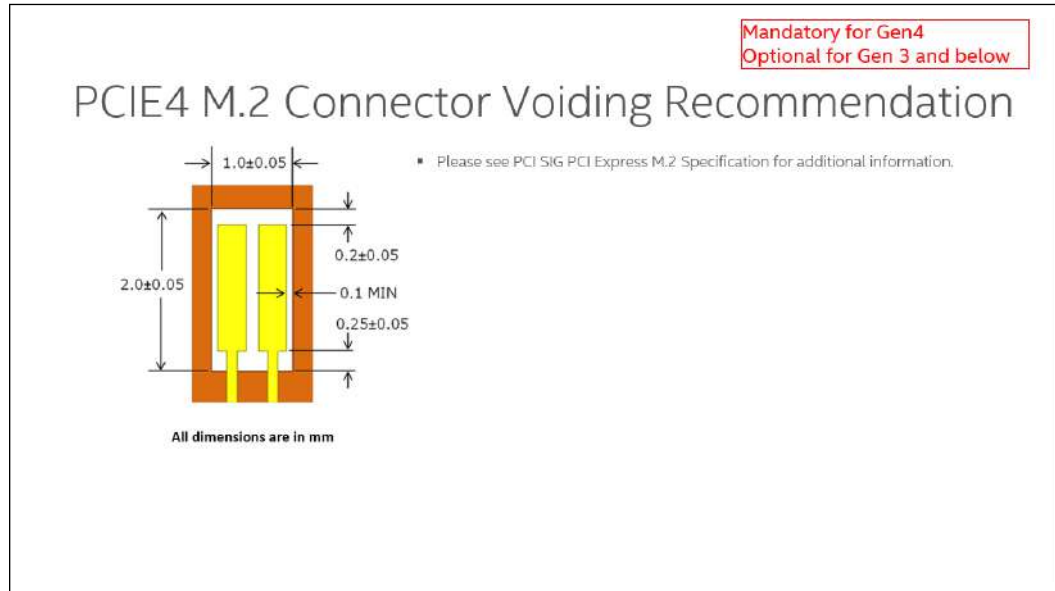


Figure 137. PCIe4 M.2 Connector Toe and Heel GND Via Recommendation

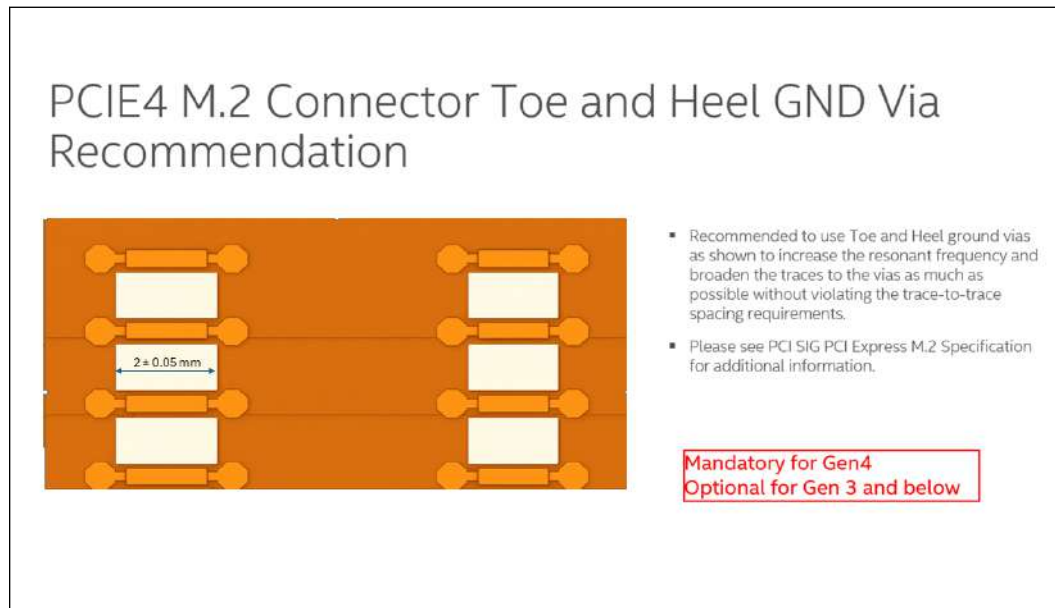


Table 377. PCIe Gen 1-3 General Guidelines

Note	Detail
Reference plane	(1) Continuous GND is recommended. (2) If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt).
<i>continued...</i>	

Note	Detail
	(3) If non-continuous power referencing is unavoidable on microstrip/surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt), but stitching caps are required over each plane split and need to be placed within 6 mm of a signal running across the plane split.
Voiding recommendation for mainstream stackup	It is recommended to void unused (non-functional) pads for differential vias and pin pads for ESDs, AC caps and connectors to optimize the impedance matching in the channel.
PCIe1, PCIe2, PCIe3 discrete component part size for mainstream stackup	It is recommended to use 0402 or smaller component sizes.
Unused signals	Unused data signals should be left as no connect at the ball.
Interleave requirement	Refer to topology section for details.
PCIe1, PCIe2, PCIe3 max via stub length	< 900 μm
Routing and return via requirement near vertical transition	For all vertical transitions, the differential transmission line should be connected to the two vias symmetrically. Moreover, Vss return vias should be placed near the signal vias in an equidistant symmetrical manner, and all signal vias should have identical Vss return vias around each of them.
PTH connector voiding requirement	Differential Oval Anti Pad required for all layers.
Voiding under CEM SMT connector pads	Required

Table 378. PCIe Gen 1-3 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

PCIe Gen 1 Device Down Topology

Figure 138. PCIe Gen1 Device Down Topology Diagram

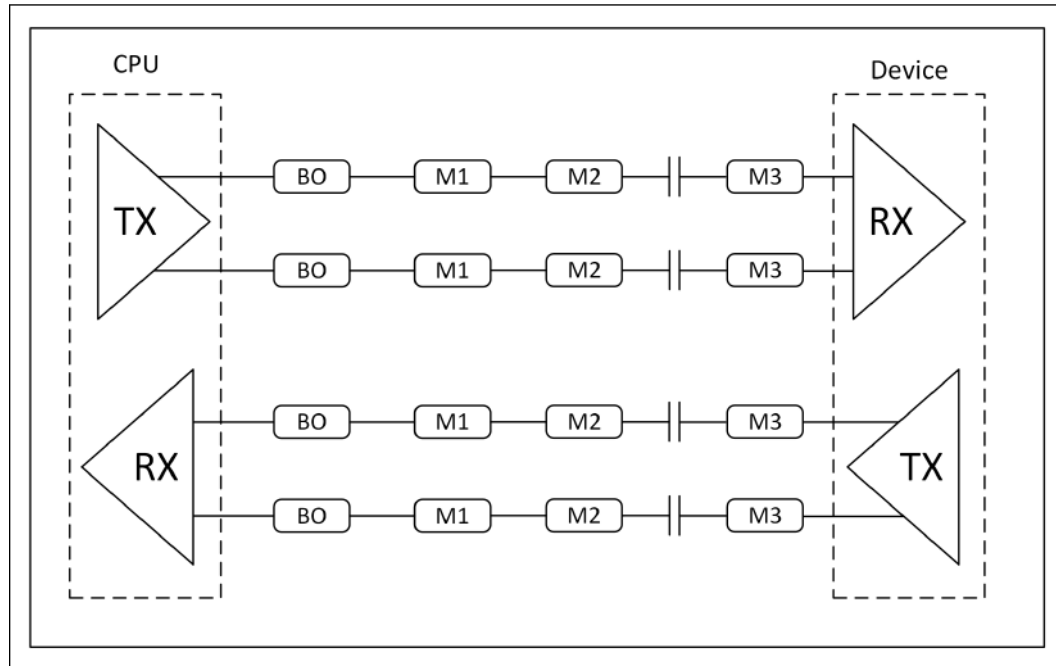


Table 379. PCIe Gen 1 Device Down Topology Notes

Note	Detail
AC capacitor value	Gen1: 75 nF - 265 nF Gen1 nominal = 100 nF
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The maximum total breakout length is still 15.2 mm.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.

Table 380. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	6	Total Channel	Gen1 (not counting microvia under package)

Rx,Tx

Table 381. PCIe Gen 1 Device Down Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1+M2	MS, DSL, SL	356	Max length : BO+M1+M2+M3 ≤ 356 mm
M3	MS, DSL, SL	8	

Max Length Total (mm): 356

Max Length Total Note: Max length is $BO + M1 + M2 + M3 \leq 356$ mm

PCIe Gen 1 Internal Cable Topology

Figure 139. PCIe Gen1 Internal Cable Topology Diagram

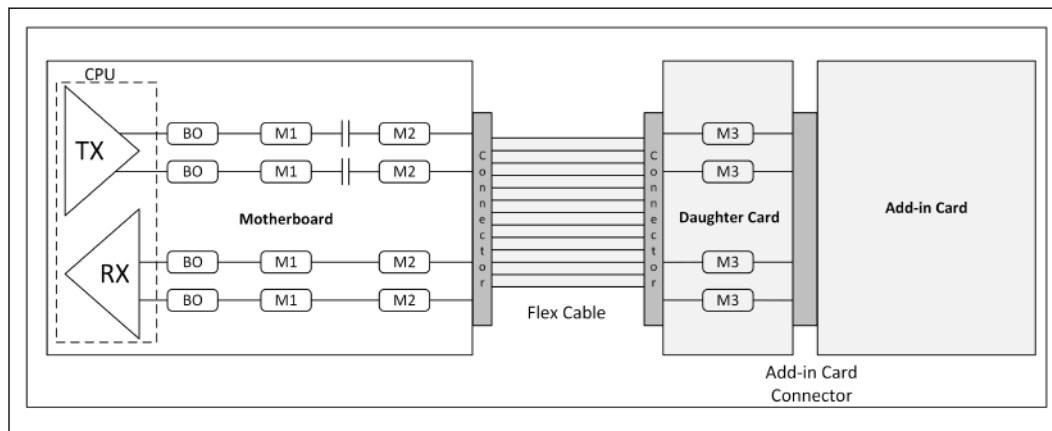


Table 382. PCIe Gen 1 Internal Cable Topology Notes

Note	Detail
Add-in-card	The total length of the add-in card is usually 38 mm to 50.4 mm. It is a combination of MS/ SL/ DSL (MS where required to connect to surface mounted components). It has a typical via count of 1. PCIe Gen1 M.2 Spec compliant add-in card
AC capacitor value	Gen1: 75 nF - 265 nF Gen1 nominal = 100 nF
Cable loss vs channel length trade-off table	See PCIe with internal cable routing length versus cable loss figure.
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Clarification on M2 segment for RX	The M2 segment for RX can also be routed on SL or DSL and not limited to MS.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.
Add-in Card Connector	Connector type M.2

Table 383. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	2	Total Channel	Gen1 (not counting microvia under package).
Rx	1	Total Channel	Gen1 (not counting microvia under package).
Rx, Tx	1	Post-Channel	Daughter Card/ Add-In card.

Rx,Tx

Table 384. PCIe Gen 1 Internal Cable Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1	MS, DSL, SL		See Cable Length vs. MB Length Table to determine segment max length.
M2	MS, DSL, SL	8	
M3	MS		See Cable Length vs. MB Length Table to determine segment max length.

Max Length Total (mm): 178

Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 178$ mm

Table 385. Cable Length vs. MB Length Table

Motherboard Length + I/O Card Length (inches)	Motherboard Length + I/O Card Length (mm)	Nominal Cable Loss up to 2.5 GHz (dB)
3	76.2	3.5
3.5	88.9	3
4.5	114.3	2.5
5.5	139.7	2
6.5	165.1	1.5
7	177.8	1

PCIe Gen 1 M.2 Topology

Figure 140. PCIe Gen1 M.2 Topology Diagram

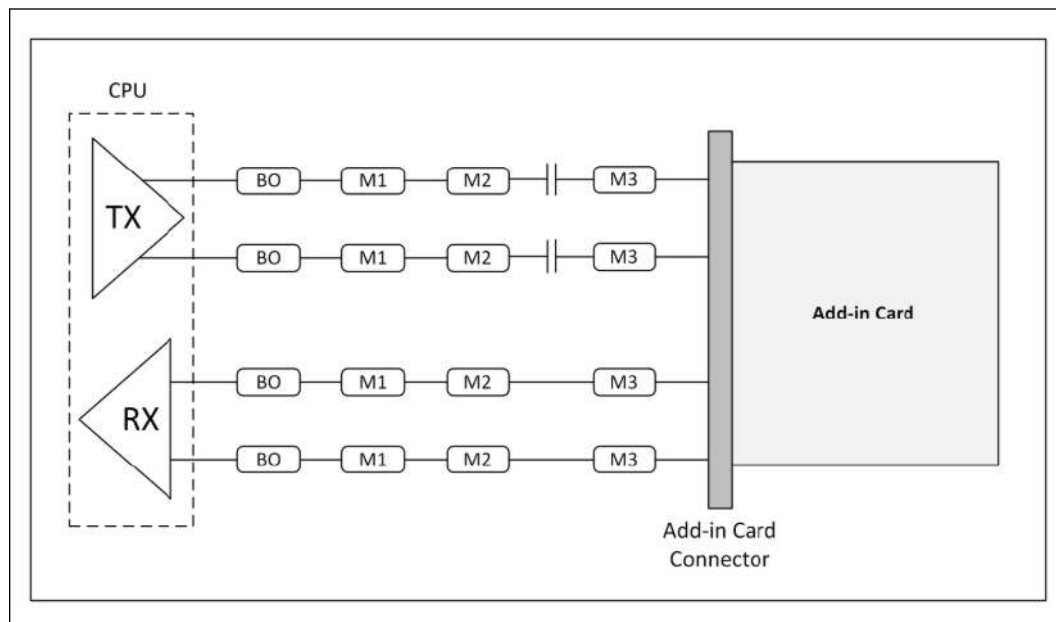


Table 386. PCIe Gen 1 M.2 Topology Notes

Note	Detail
Add-in-card	The total length of the add-in card is usually 38 mm to 50.4 mm. It is a combination of MS/ SL/ DSL (MS where required to connect to surface mounted components). It has a typical via count of 1. PCIe Gen1 M.2 Spec compliant add-in card
AC capacitor value	Gen1: 75 nF - 265 nF Gen1 nominal = 100 nF
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.
Add-in Card Connector	Connector type M.2

Table 387. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	4	Total Channel	Gen1 (not counting microvia under package).
Rx, Tx	1	Post-Channel	Express Card/ M.2/ Add-In Card.

Rx,Tx**Table 388. PCIe Gen 1 M.2 Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1 + M2	MS, DSL, SL	330	Max length : $BO+M1+M2+M3 \leq 330$ mm
M3	MS, DSL, SL	8	

Max Length Total (mm): 330

Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 330$ mm

PCIe Gen 2 Device Down Topology

Figure 141. PCIe Gen2 Device Down Topology Diagram

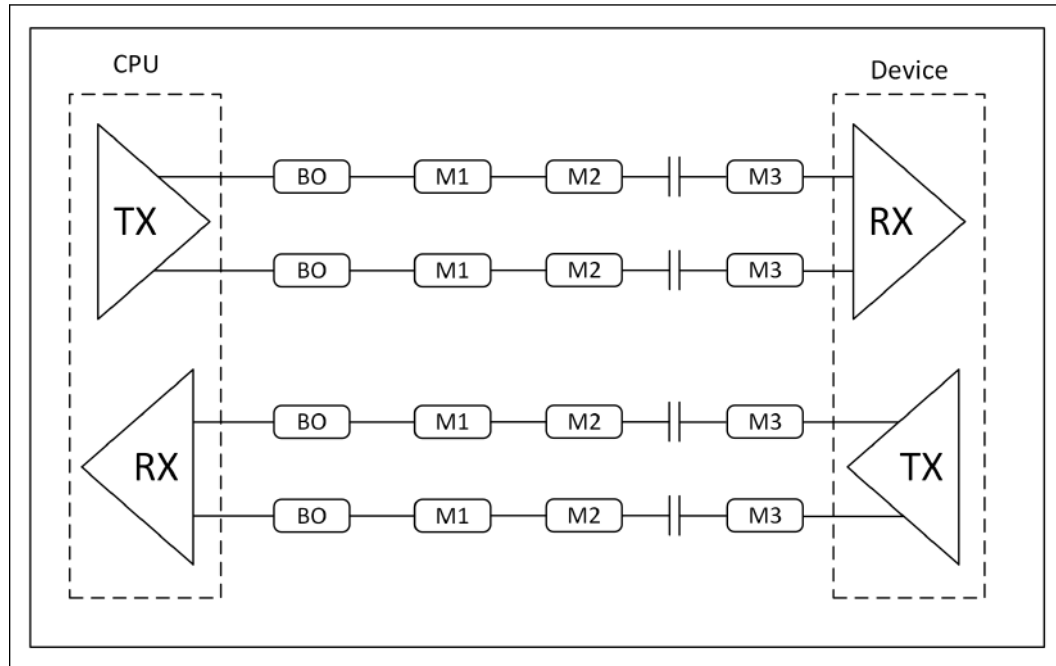


Table 389. PCIe Gen 2 Device Down Topology Notes

Note	Detail
AC capacitor value	Gen2: 75 nF - 265 nF Gen2 nominal = 100 nF
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.

Table 390. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	4	Total Channel	Gen2 (not counting microvia under package).

Rx,Tx

Table 391. PCIe Gen 2 Device Down Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1+M2	MS, DSL, SL	279	Max length is BO+M1+M2+M3 ≤ 279 mm
M3	MS, DSL, SL	8	

Max Length Total (mm): 279

Max Length Total Note: Max length is $BO + M1 + M2 + M3 \leq 279$ mm

PCIe Gen 2 Internal Cable Topology

Figure 142. PCIe Gen2 Internal Cable Topology Diagram

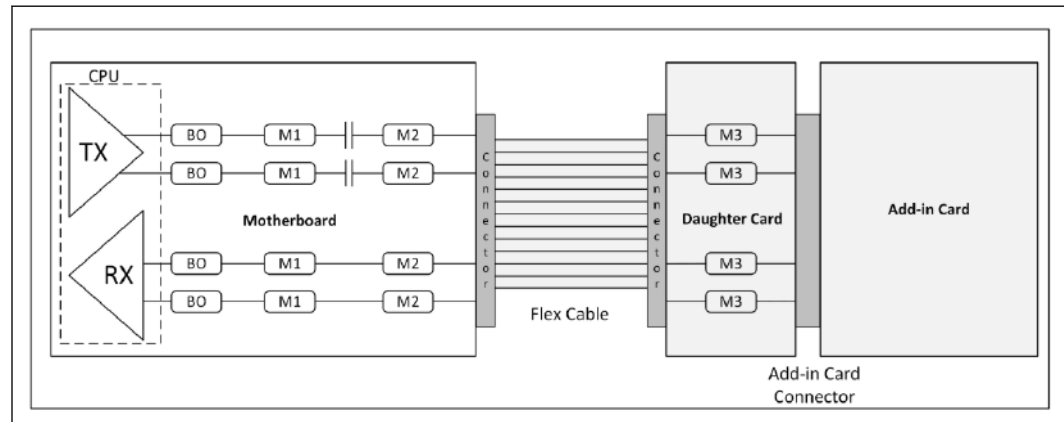


Table 392. PCIe Gen 2 Internal Cable Topology Notes

Note	Detail
Add-in-card	The total length of the add-in card is usually 38 mm to 50.4 mm. It is a combination of MS/ SL/ DSL (MS where required to connect to surface mounted components). It has a typical via count of 1. PCIe Gen2 M.2 Spec compliant add-in card
AC capacitor value	Gen2: 75 nF - 265 nF Gen2 nominal = 100 nF
Cable loss vs channel length trade-off table	See PCIe with internal cable routing length versus cable loss figure.
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Clarification on M2 segment for RX	The M2 segment for RX can also be routed on SL or DSL and not limited to MS.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.
Add-in Card Connector	Connector type M.2

Table 393. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	2	Total Channel	Gen2 (not counting microvia under package).
Rx	1	Total Channel	Gen2 (not counting microvia under package).
Rx, Tx	1	Post-Channel	Daughter Card/ Add-In Card.

Rx,Tx

Table 394. PCIe Gen 2 Internal Cable Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1	MS, DSL, SL		See Cable Length vs. MB Length Table to determine segment max length.
M2	MS, DSL, SL	8	
M3	MS		See Cable Length vs. MB Length Table to determine segment max length.

Max Length Total (mm): 178

Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 178$ mm

Table 395. Cable Length vs. MB Length Table

Motherboard Length + I/O Card Length (inches)	Motherboard Length + I/O Card Length (mm)	Nominal Cable Loss up to 2.5 GHz (dB)
3	76.2	3.5
3.5	88.9	3
4.5	114.3	2.5
5.5	139.7	2
6.5	165.1	1.5
7	177.8	1

PCIe Gen 2 M.2 Topology

Figure 143. PCIe Gen2 M.2 Topology Diagram

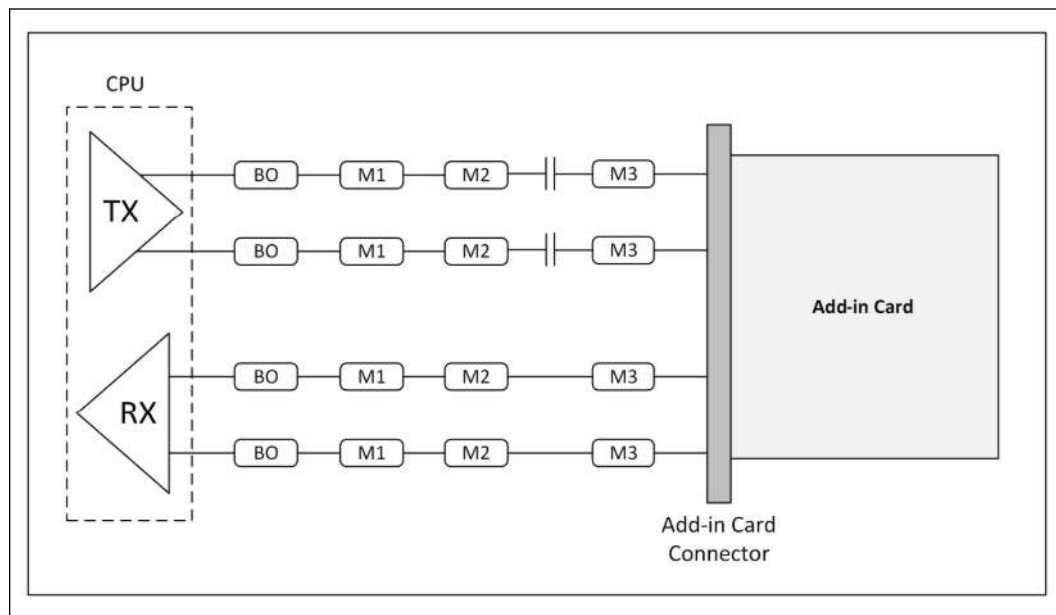


Table 396. PCIe Gen 2 M.2 Topology Notes

Note	Detail
Add-in-card	The total length of the add-in card is usually 38 mm to 50.4 mm. It is a combination of MS/ SL/ DSL (MS where required to connect to surface mounted components). It has a typical via count of 1. PCIe Gen2 M.2 Spec compliant add-in card
AC capacitor value	Gen2: 75 nF - 265 nF Gen2 nominal = 100 nF
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.
Add-in Card Connector	Connector type M.2

Table 397. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	4	Total Channel	Gen2 (not counting microvia under package).
Rx, Tx	1	Post-Channel	Express Card/ M.2/ Add-In Card.

Rx,Tx**Table 398. PCIe Gen 2 M.2 Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1 + M2	MS, DSL, SL	229	Max length is $BO+M1+M2+M3 \leq 229$ mm
M3	MS, DSL, SL	8	

Max Length Total (mm): 229

Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 229$ mm

PCIe Gen 3 Device Down Topology

Figure 144. PCIe Gen3 Device Down Topology Diagram

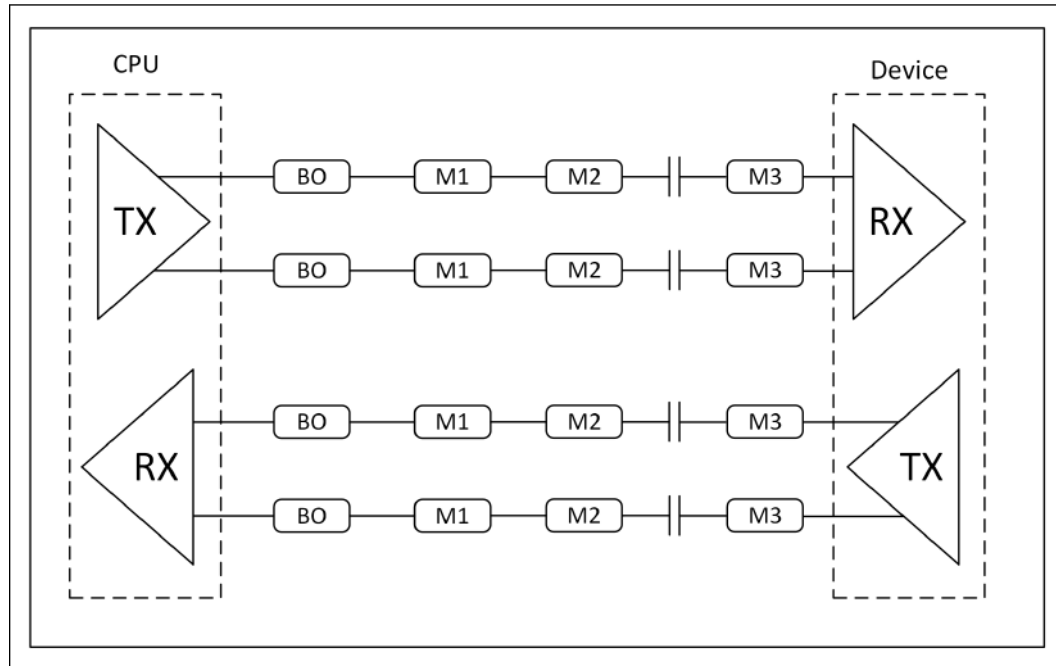


Table 399. PCIe Gen 3 Device Down Topology Notes

Note	Detail
AC capacitor value	Gen3: 176 nF - 265 nF Gen3 nominal = 220 nF
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.

Table 400. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	3	Total Channel	Gen3 (not counting microvia under package).

Rx,Tx

Table 401. PCIe Gen 3 Device Down Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1+M2	MS, DSL, SL	355	Max length is BO+M1+M2+M3 ≤ 355 mm
M3	MS, DSL, SL	8	

Max Length Total (mm): 355

Max Length Total Note: Max length is $BO + M1 + M2 + M3 \leq 355$ mm

PCIe Gen 3 Internal Cable Topology

Figure 145. PCIe Gen3 Internal Cable Topology Diagram

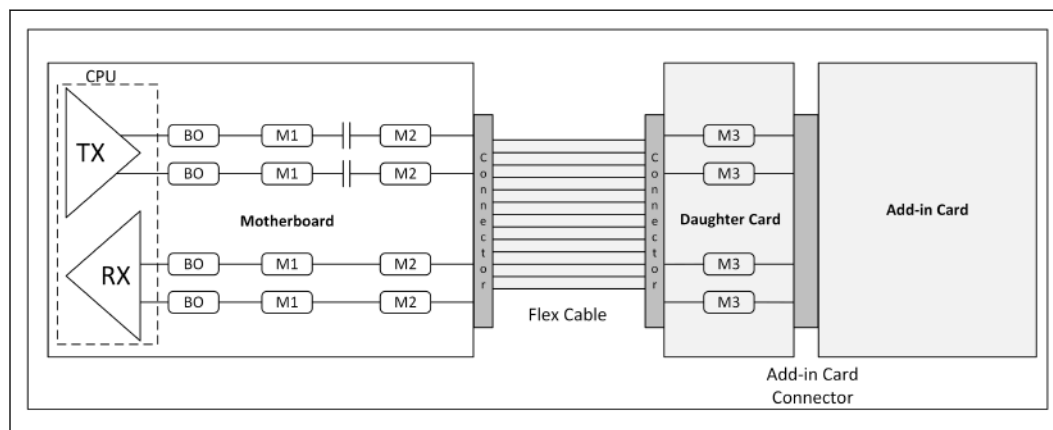


Table 402. PCIe Gen 3 Internal Cable Topology Notes

Note	Detail
Add-in-card	The total length of the add-in card is usually 38 mm to 50.4 mm. It is a combination of MS/ SL/ DSL (MS where required to connect to surface mounted components). It has a typical via count of 1. PCIe Gen3 M.2 Spec compliant add-in card
AC capacitor value	Gen3: 176 nF - 265 nF Gen3 nominal = 220 nF
Cable loss vs channel length trade-off table	See PCIe with internal cable routing length versus cable loss figure.
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Clarification on M2 segment for RX	The M2 segment for RX can also be routed on SL or DSL and not limited to MS.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.
Add-in Card Connector	Connector type M.2

Table 403. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	2	Total Channel	Gen3 (not counting microvia under package).
Rx	1	Total Channel	Gen3 (not counting microvia under package).
Rx, Tx	1	Post-Channel	Daughter Card/ Add-In Card.

Rx,Tx

Table 404. PCIe Gen 3 Internal Cable Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1	MS, DSL, SL		See Cable Length vs. MB Length Table to determine segment max length.
M2	MS, DSL, SL	8	
M3	MS		See Cable Length vs. MB Length Table to determine segment max length.

Max Length Total (mm): 127

Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 127$ mm

Table 405. Cable Length vs. MB Length Table

Motherboard Length + I/O Card Length (inches)	Motherboard Length + I/O Card Length (mm)	Nominal Cable Loss up to 4 GHz (dB)
3	76.2	3
4	101.6	2
5	127	1

PCIe Gen 3 M.2 Topology

Figure 146. PCIe Gen3 M.2 Topology Diagram

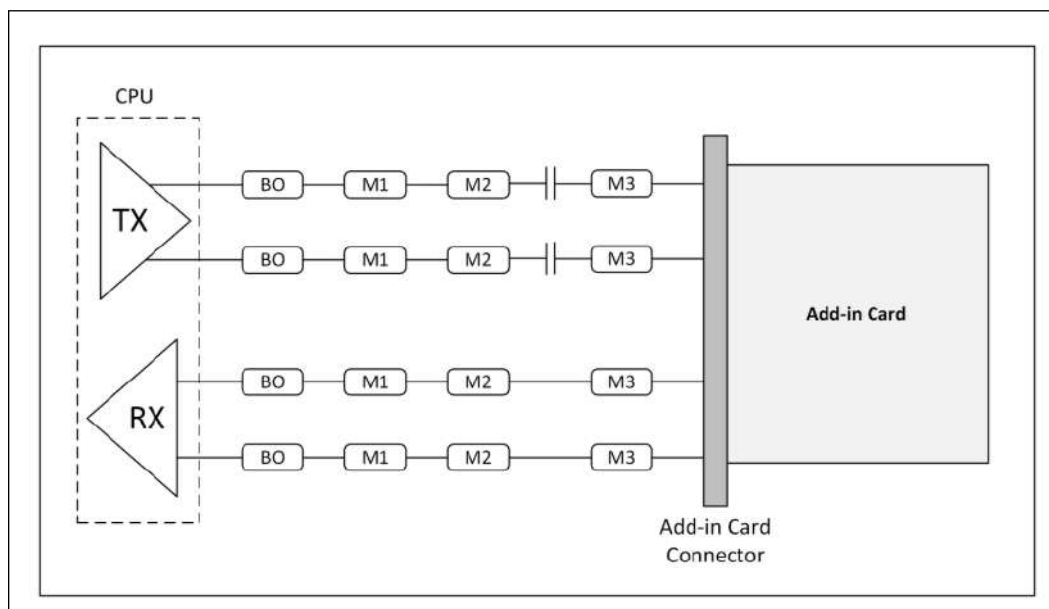


Table 406. PCIe Gen 3 M.2 Topology Notes

Note	Detail
Add-in-card	The total length of the add-in card is usually 38 mm to 50.4 mm. It is a combination of MS/ SL/ DSL (MS where required to connect to surface mounted components). It has a typical via count of 1. PCIe Gen3 M.2 Spec compliant add-in card
AC capacitor value	Gen3: 176 nF - 265 nF Gen3 nominal = 220 nF
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed. The total breakout length is still 15.2 mm.
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.
Add-in Card Connector	Connector type M.2

Table 407. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	3	Total Channel	Gen3 (not counting microvia under package).
Rx	2	Total Channel	Gen3 (not counting microvia under package).
Rx, Tx	1	Post-Channel	Express Card/ M.2/ Add-In Card.

Rx,Tx**Table 408. PCIe Gen 3 M.2 Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15.2	
M1 + M2	MS, DSL, SL		Max length is $BO+M1+M2+M3 \leq 254$ mm
M3	MS, DSL, SL	8	

Max Length Total (mm): 254

Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 254$ mm

6.1.8.4.2 PCIe Gen 4

Figure 147. Type-4 PCB Via Stitching Recommendation

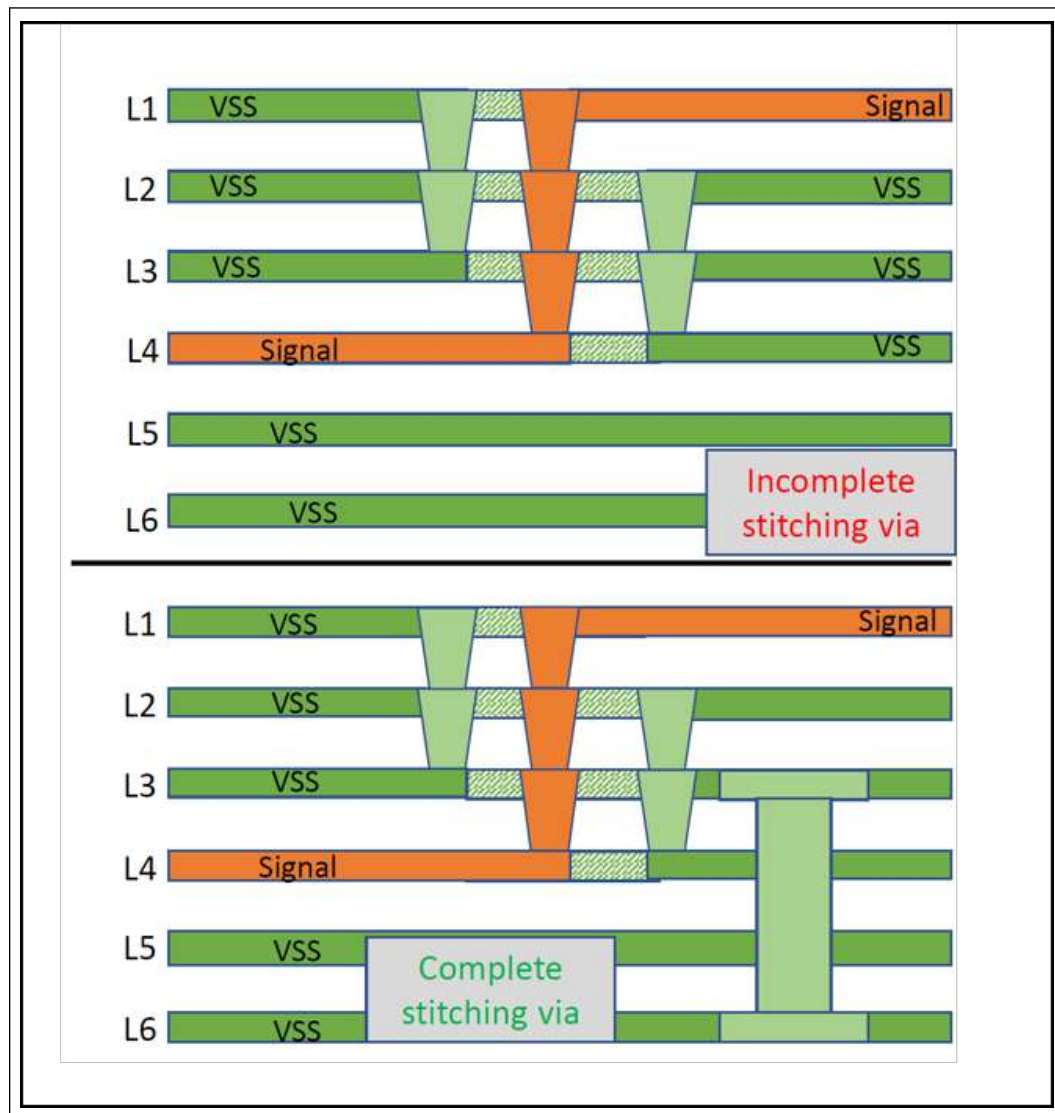


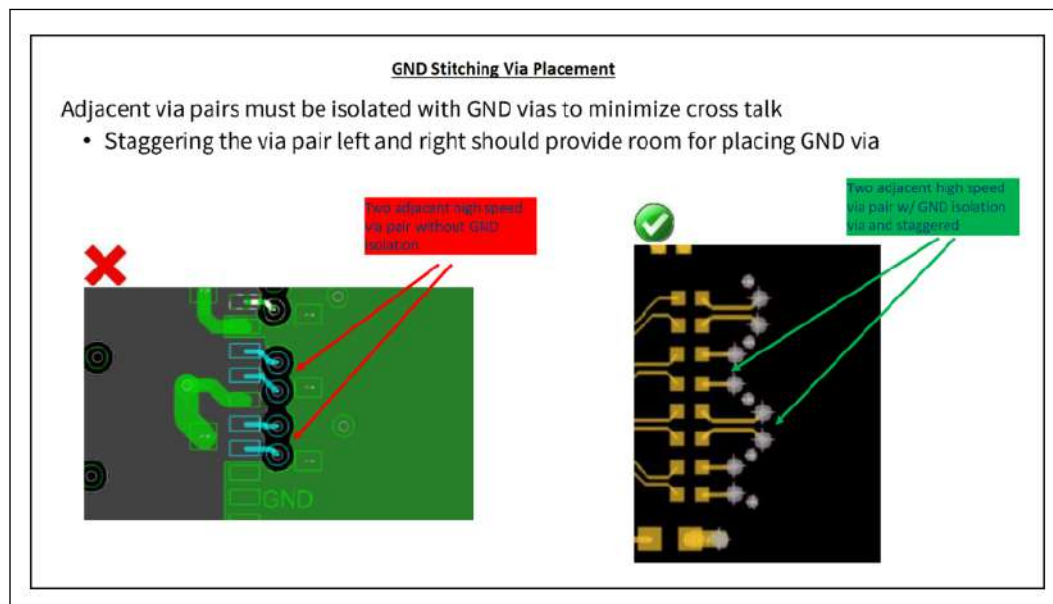
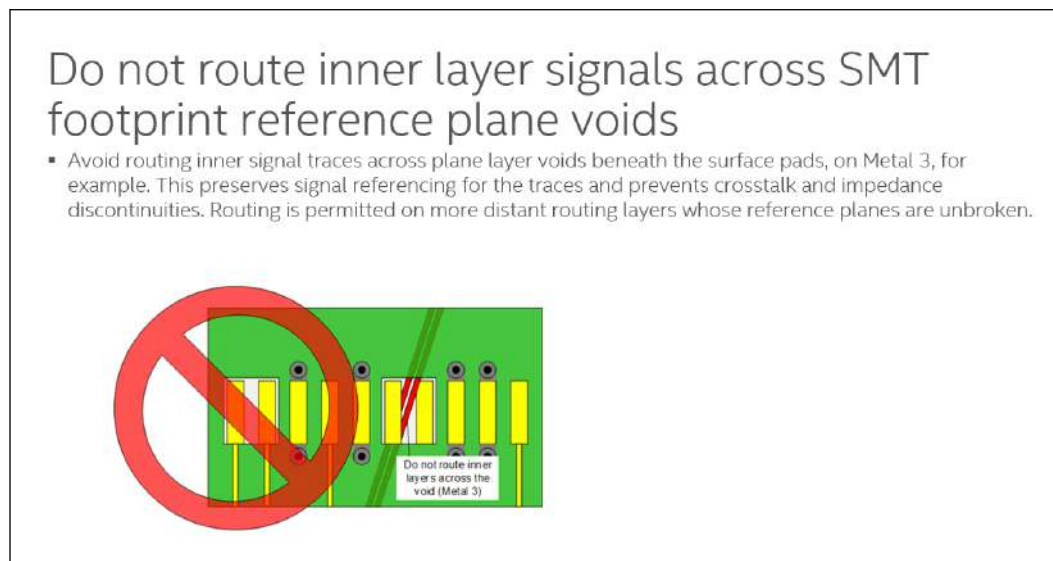
Figure 148. Recommended GND Stitching Via Placement**Figure 149. PCIe Board Layout Recommendation SMT footprint Voids Reference**

Figure 150. PCIe Board Layout Recommendation AC Cap Voiding

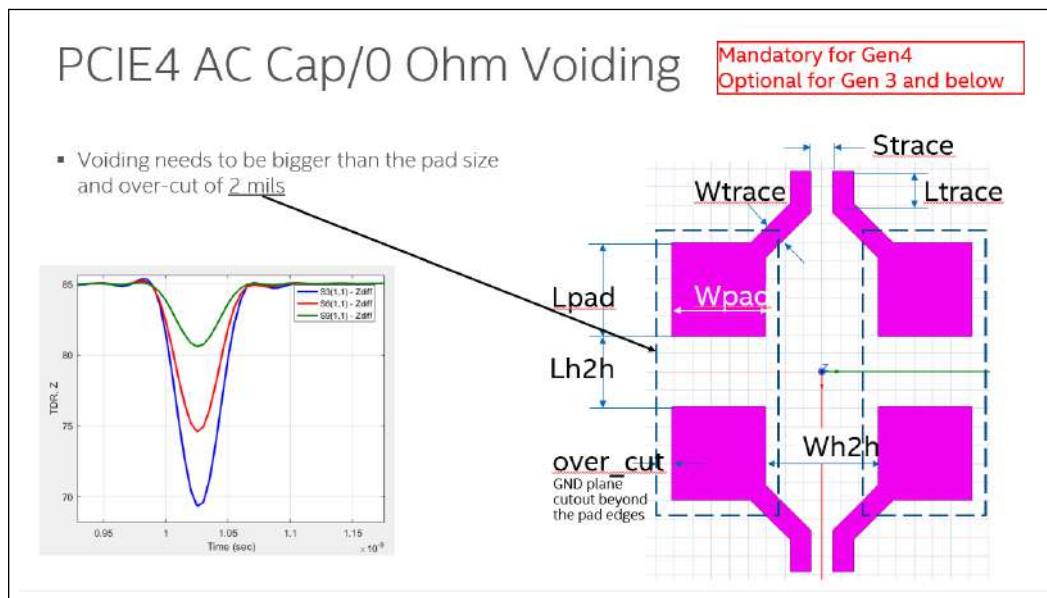


Figure 151. PCIe4 M.2 Connector Voiding Recommendation

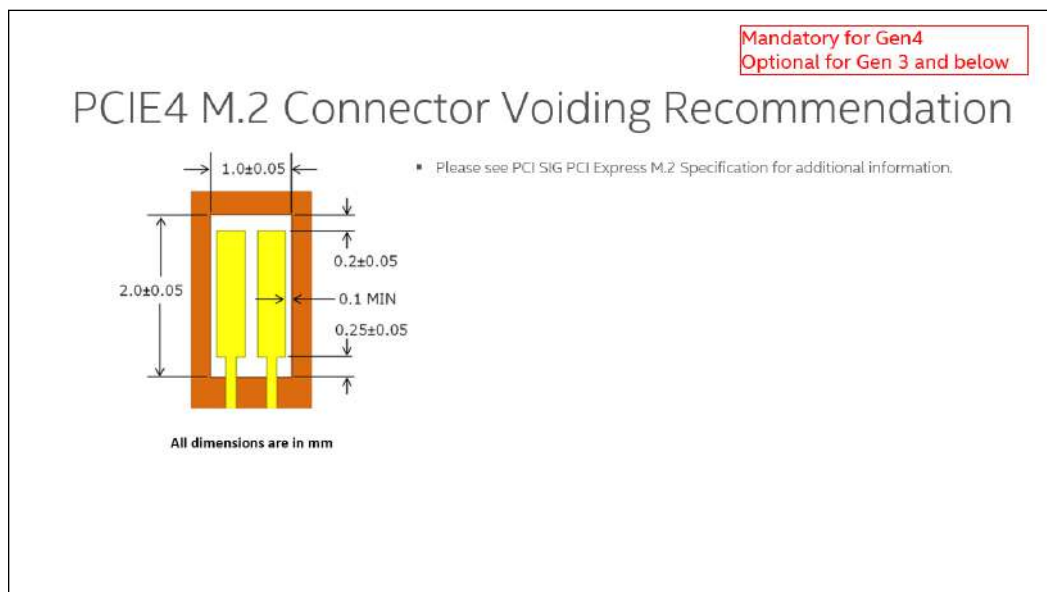


Figure 152. PCIe4 M.2 Connector Toe and Heel GND Via Recommendation

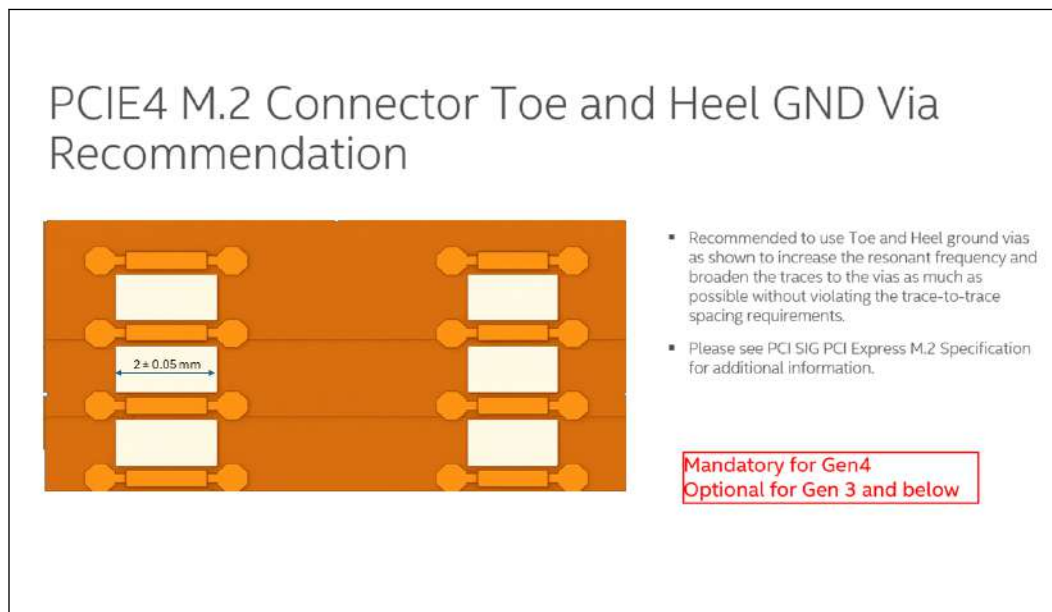


Figure 153. PCIe Board Layout Recommendation Toe Side HSIO Routing



Table 409. PCIe Gen 4 General Guidelines

Note	Detail
Reference plane	(1) Continuous GND is recommended. (2) If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt).

continued...

Note	Detail
	(3) If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt), but stitching caps are required over each plane split and need to be placed within 6 mm of a signal running across the plane split.
Voiding recommendation for mainstream stackup	It is recommended to void unused (non-functional) pads reference plane for differential vias and pin pads for ESDs, AC caps and connectors to optimize the impedance matching in the channel.
PCIe4 AC capacitor size for mainstream stackup	It is required to use 0402. If routing near maximum length it is recommended to use 0201 component size.
Unused signals	Unused data signals should be left as no connect at the ball.
Interleaved requirement	Refer to topology section for details.
PCIe4 max via stub length	< 250 um
Via stub length relaxation	PTH via stub length can be relaxed, refer to "PCIe Gen4 Stub Length Requirement Relaxation Technical White Paper" #723631 for details.
Routing and return via requirement near vertical transition	For all vertical transitions, the differential transmission line should be connected to the two vias symmetrically. Moreover, Vss return vias should be placed near the signal vias in an equidistant symmetrical manner, and all signal vias should have identical Vss return vias around each of them.
PTH connector voiding requirement	Differential Oval Anti Pad required for all layers. For PCIe4, see topology specific notes for additional guidance.
Voiding under CEM SMT connector pads	Required
Fiberweave for PCIe4	Must adhere to fiberweave related best routing practices (#406926).
PCIe4 redriver	Routing length can be extended beyond the PDG guidelines by using redriver. Refer to document #575423 for details.

Table 410. PCIe Gen 4 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.25
3	TX-TX	Total	Length (mm)	Yes	50
4	RX-RX	Total	Length (mm)	Yes	50
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

PCIe Gen 4 Device Down Topology

Figure 154. PCIe Gen4 Device Down Topology Diagram

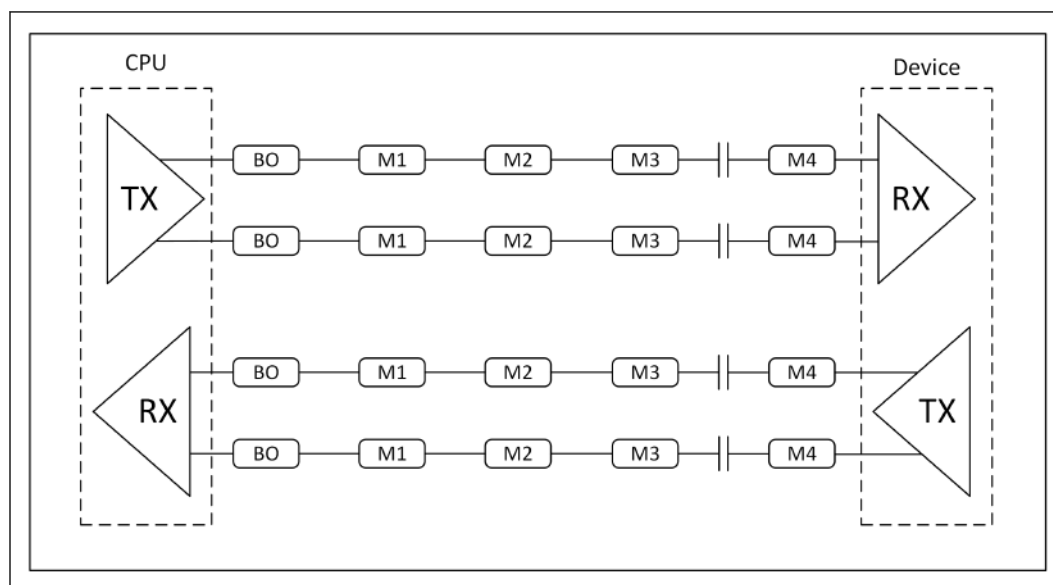


Table 411. PCIe Gen 4 Device Down Topology Notes

Note	Detail
Maximum via stub length for mainstream stackup	250 um per via.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground. Prefer ground on both sides.
Reference plane for dual stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground and it is the layer that is closer to the signal route. Prefer ground on both sides.
AC cap value	Nominal 220 nF recommended for Gen3 & Gen4 (spec range 75 nF to 265 nF including tolerance).
Routing type for mainstream stackup	Prefer non-interleaved routes in different layers.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
Device / End Point package insertion loss	Device / End Point needs to be within the 3 dB specification as per the PCIe Base Spec Rev 4.0 Version 1.0.
Device	PCIe Gen4 Spec Compliant device / Barlow Ridge

Table 412. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Rx,Tx

Table 413. PCIe Gen 4 Device Down Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1+M2+M3	MS, DSL, SL	292	Max length is $BO+M1+M2+M3+M4 \leq 292$ mm
M4	MS, DSL, SL	12.7	

Max Length Total (mm): 292

Max Length Total Note: Max length is $BO+M1+M2+M3+M4 \leq 292$ mm

PCIe Gen 4 M.2 Topology

Figure 155. PCIe Gen4 M.2 Topology Diagram

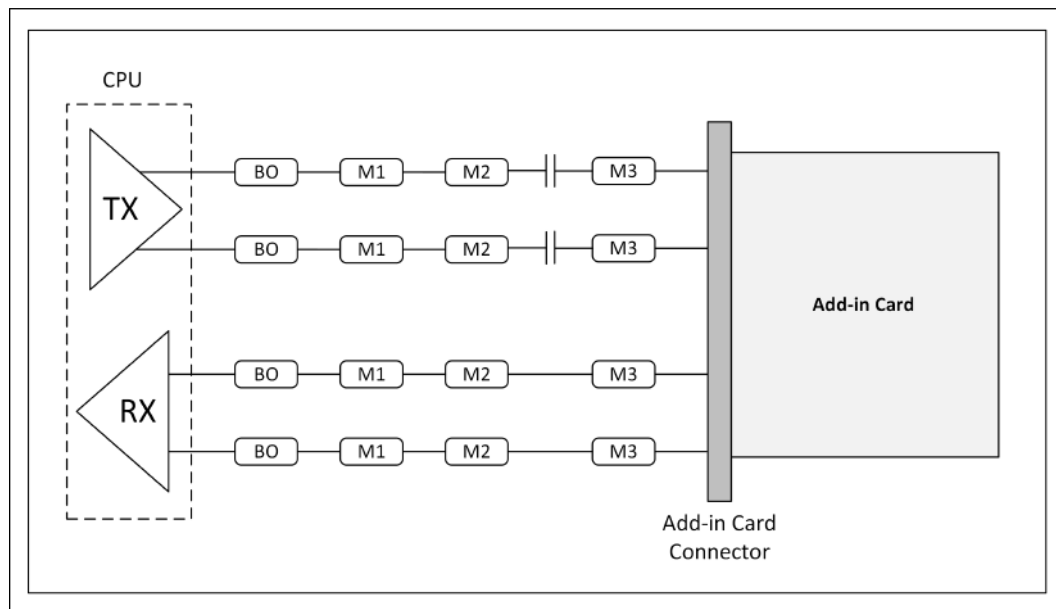


Table 414. PCIe Gen 4 M.2 Topology Notes

Note	Detail
Maximum via stub length for mainstream stackup	250 um per via.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground. Prefer ground on both sides.
Reference plane for dual stripline route	One side can be power plane that has low frequency peak-to-peak noise, the other side must be continuous ground and it is the layer that is closer to the signal route. Prefer ground on both sides.
AC cap value	Nominal 220 nF recommended for Gen3 and Gen4 (spec range 75 nF to 265 nF including tolerance).
continued...	

Note	Detail
Routing type for mainstream stackup	Prefer non-Interleaved routes in different layers.
M.2 connector Vss pad to via distance	Each M.2 connector Vss pad should be connected to its own Vss via. The distance between the edge of the pad and the edge of the via pad should be 0.5mm max.
Add-in card	Assuming ~50 mm of PCB routing and 2 vias with spec reference package. The total length matching RX-RX/TX-TX lane to lane include the budget within the module. PCIe Gen4 M.2 Spec compliant add-in card
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
Add-in Card Connector	Connector type M.2

Table 415. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	3	Total Channel

Rx,Tx**Table 416. PCIe Gen 4 M.2 Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1+M2	MS, DSL, SL	190	Max length is $BO+M1+M2+M3 \leq 190$ mm
M3	MS, DSL, SL	12.7	

Max Length Total (mm): 190

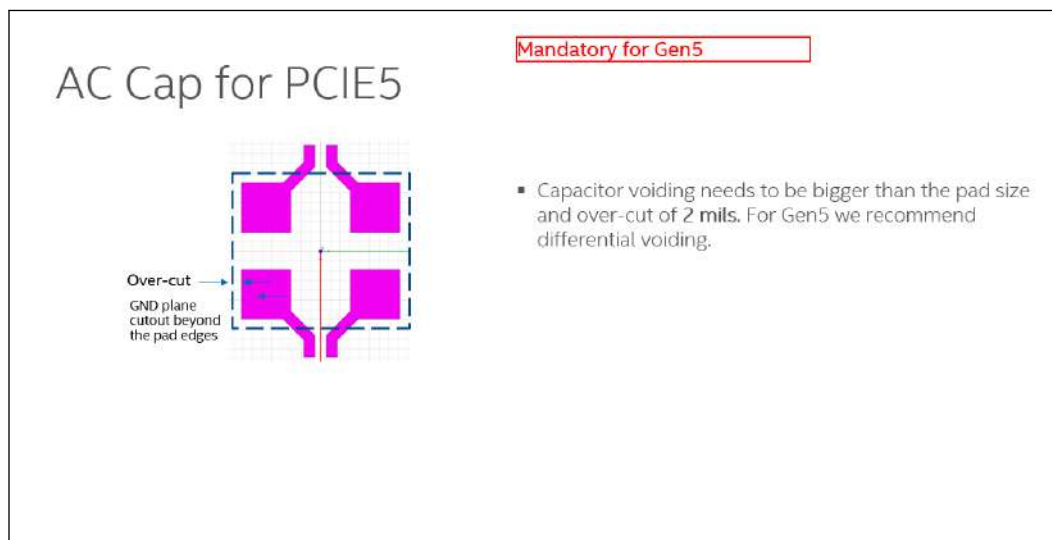
Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 190$ mm**6.1.8.4.3 PCIe Gen 5****Figure 156. Board Layout Recommendation: PCIe Gen5 AC Cap Voiding**

Figure 157. Void Above and Below Buried Vias PTH Pads in Type-4 Stackup

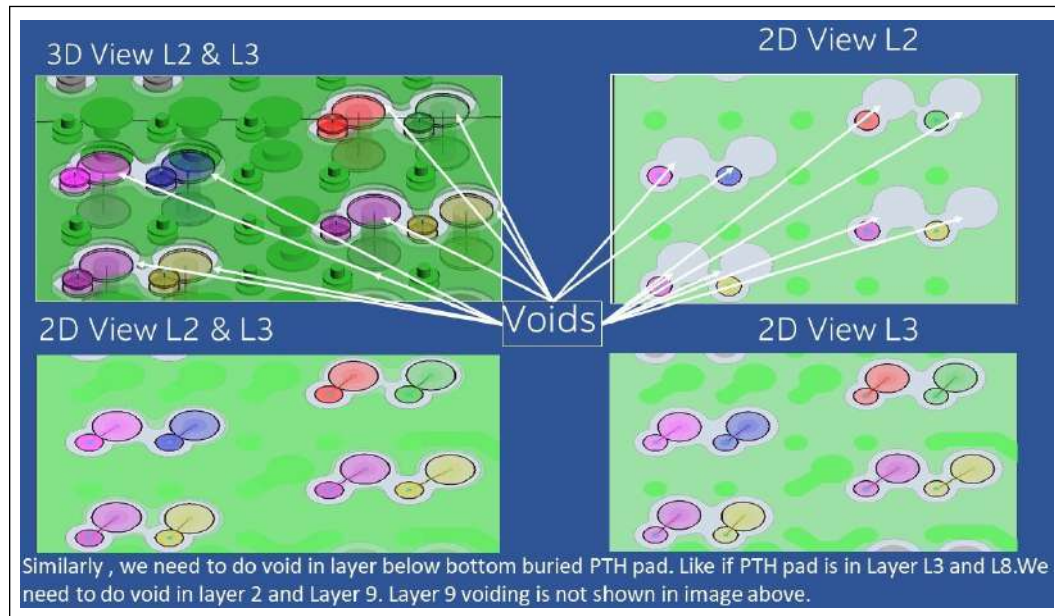


Figure 158. Voids Under SOC Ball Pads

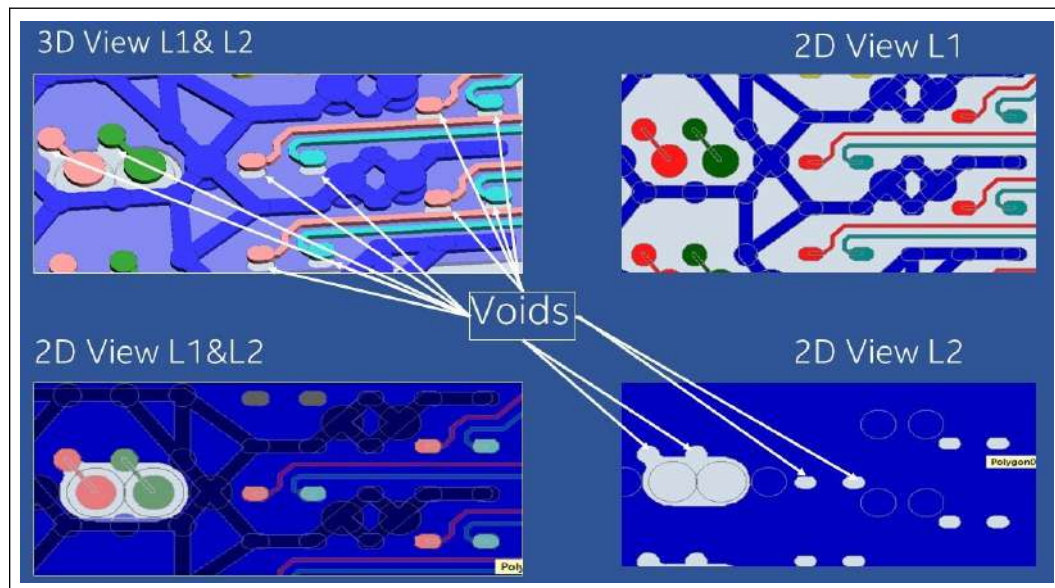


Table 417. PCIe Gen 5 General Guidelines

Note	Detail
Voiding recommendation	It is recommended to void unused (non-functional) pads for differential vias and pin pads for ESDs, AC caps and connectors to optimize the impedance matching in the channel. Ensure no routing in L3 below void in L2. If L3 is power, ensure power shape is symmetric to both p and n pads.
PCIe5 AC capacitor size	It is required to use 0201 .

continued...

Note	Detail
	Void reference layer under pad size of differential cap together. Extend void by 50 um on all sides of pads. Reference example provided per figure. Ensure spacing between P and N cap is about ~305 um and spacing between caps of different pair caps is more than 762 um.
Unused signals	Unused data signals should be left as no connect at the ball.
Interleaved requirement	Prefer non-interleaved routing. If Tx to Rx is routed same layer, need Signal-to-Non-Equivalent-Signal (S-Non-ES) spacing as given in Tlinespec between Tx to Rx.
Routing and return via requirement near vertical transition	For all vertical transitions, the differential transmission line should be connected to the two vias symmetrically. Moreover, Vss return vias should be placed near the signal vias in an equidistant symmetrical manner, and all signal vias should have identical Vss return vias around each of them.
Fiberweave for PCIe5	Must adhere to fiberweave related best routing practices (#406926).
Antipad for Via	800um diameter for anti-pad Via (T3 stack up). Please refer to chapter "General Design Considerations", subheading "Differential Transitional Via Recommendations" for image illustrating anti-pad diameter. Bigger Anti-pad should be provided for T4 buried PTH also, follow pad size radius+175um (diameter increased by 350um) say buried PTH pad radius size is 200um (400um diameter), anti-pad radius should be 375um (750um diameter).
Void under SOC ball pads	Second PCB layer should have void equal to pad size for T3 stack up PCB. And for T4 stack up with Via in pad, need to void around pad in top layer equal to 300 um diameter size. Ensure no routing in L3 below void in L2. If L3 is power, ensure power shape is symmetric to both p and n pads.
Void above and below buried via's PTH pads for T4 stackup	Need to provide void above and below buried via's PTH pads. Voids need to be 75 um more in radius than pad radius. Void is required as shapes over big pads cause a capacitive dip in impedance which results in lower margins. The same void should be provided in layers in which via is running through as well i.e., for PTH from L3 to L8, if there is no shape in L2 and L9 already (because of it being signal layer), nothing is required to be done for L2/L9. Void not required in L1/L10 for 2x2+ stack-up. for 1x1+ stack-up, all layers including top and bottom will need void. Ensure no routing above/below these voids for other signals in L1 & L10. In case surface layers (top and bottom) are power, ensure power shape is symmetric to both p and n pads. Refer to the image provided as a reference.
Reference plane for microstrip route	Must be continuous ground.
Reference plane for stripline route	Ground on both sides strongly preferred. If necessary, only a single continuous power reference plane is permitted on one side, but the other side must be a continuous ground. Simulation is needed to ensure electrical requirements are met.
Reference plane for dual stripline route	Ground on both sides strongly preferred. If necessary, only a single continuous power reference plane is permitted on one side, but the other side must be a continuous ground and it is the layer that is closer to the signal route. Simulation is needed to ensure electrical requirements are met.

Table 418. PCIe Gen 5 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.125
2	Pair (P-N)	Within	Length (mm)	Yes	0.125
3	TX-TX	Total	Length (mm)	Yes	50
4	RX-RX	Total	Length (mm)	Yes	50
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

PCIe Gen 5 Device Down Topology

Figure 159. PCIe Gen 5 Device Down Topology Diagram

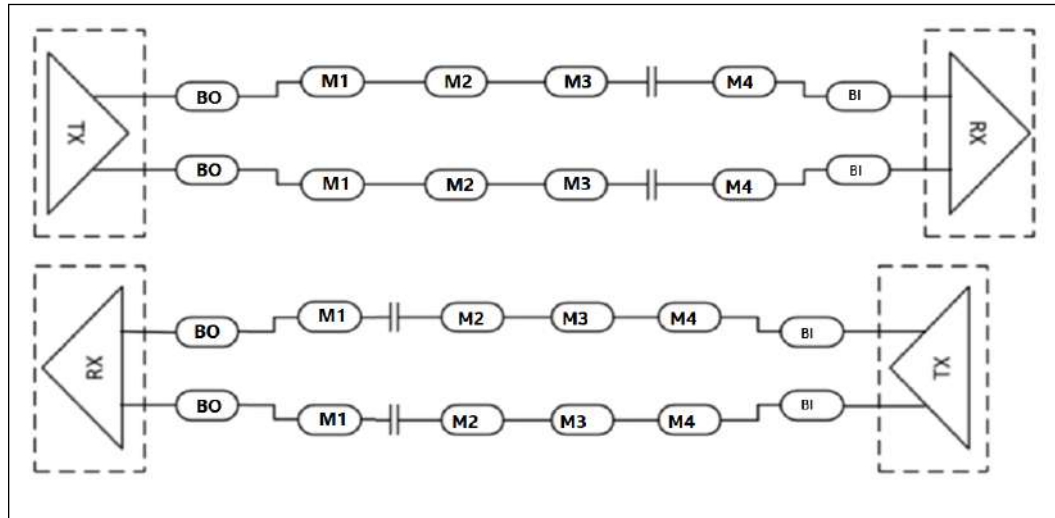


Table 419. PCIe Gen 5 Device Down Topology Notes

Note	Detail
Maximum via stub length	250 um
AC cap value	Nominal 220 nF recommended (spec range 176 nF to 265 nF including tolerance.).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
Device	PCIe Gen5 Spec compliant endpoint device.

Table 420. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Rx,Tx**Table 421. PCIe Gen 5 Device Down Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL	13	
M1+M2+M3+M4	MS, DSL, SL	200	$BO+M1+M2+M3+M4+BI \leq \text{MAX length Total}$
BI	MS, DSL, SL	13	

Max Length Total (mm): 200

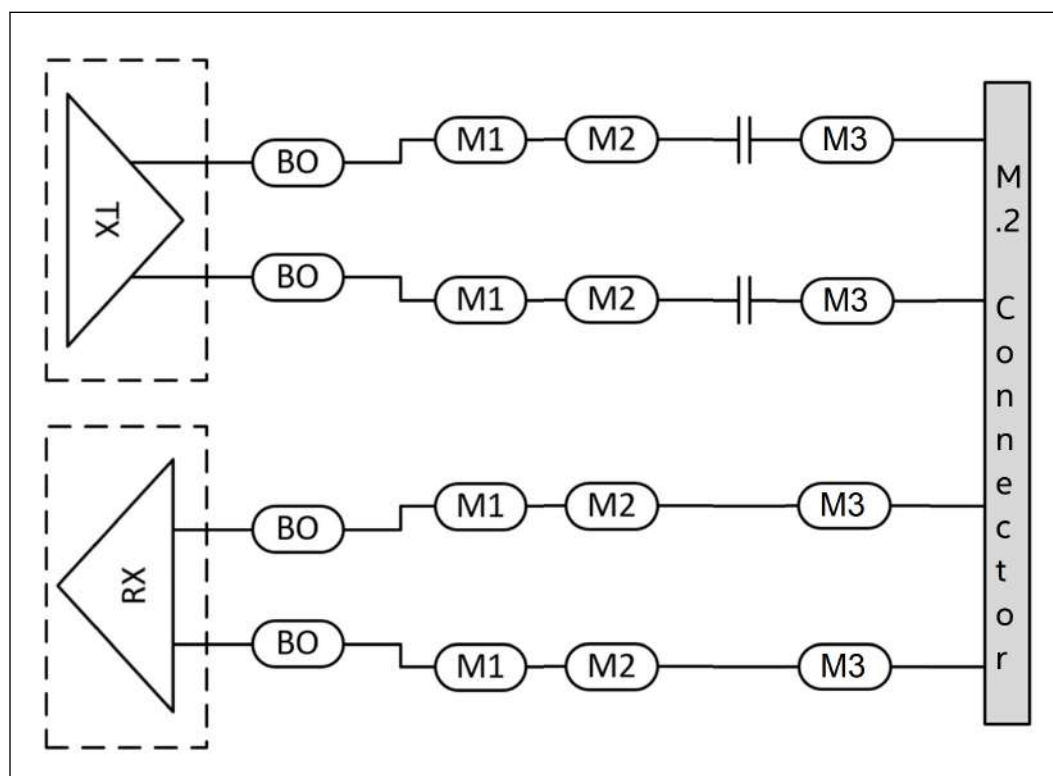
PCIe Gen 5 M.2 Topology**Figure 160. PCIe Gen 5 M.2 Topology Diagram**

Figure 161. Suggested Ground Void for Main Board M.2 Connector

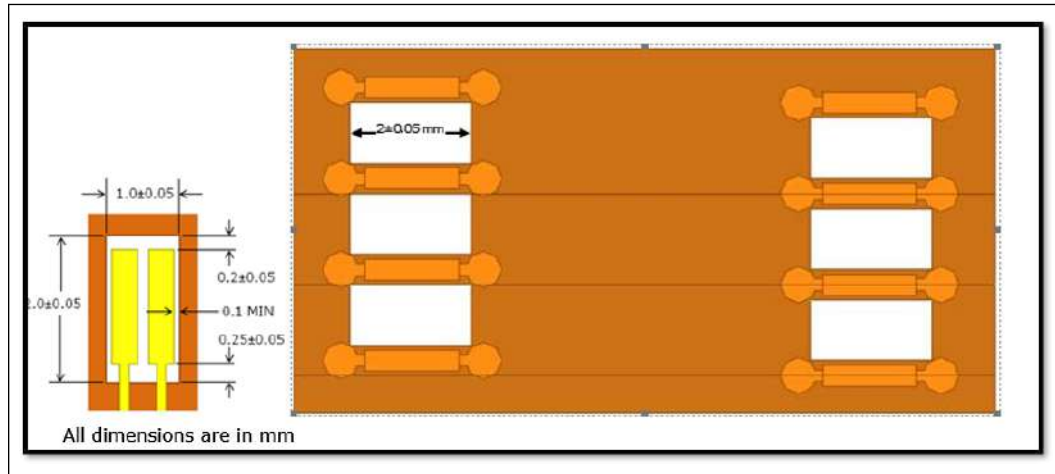


Figure 162. Suggested Toe and Heel Side Connection

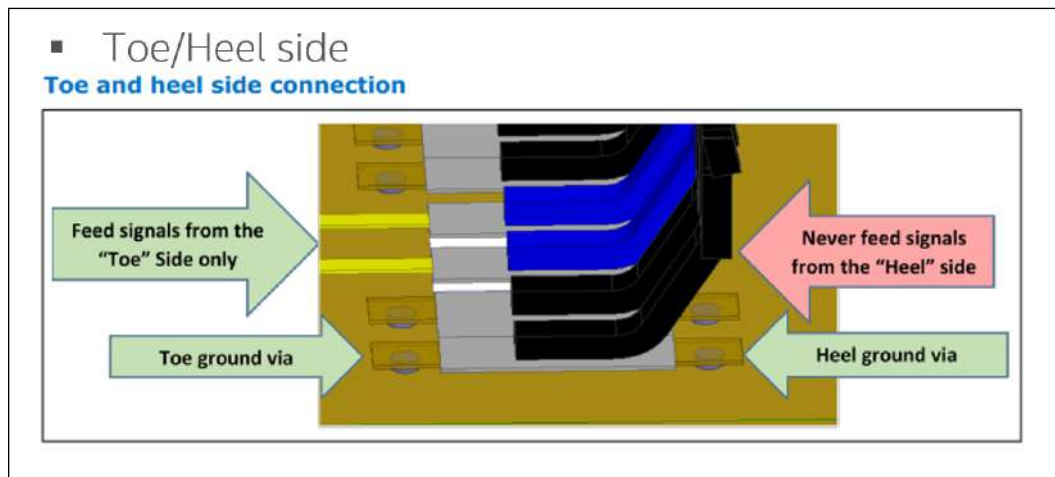


Table 422. PCIe Gen 5 M.2 Topology Notes

Note	Detail
Maximum via stub length	250 um per via.
AC cap value	Nominal 220 nF recommended (spec range 75 nF to 265 nF including tolerance).
Suggested Toe and Heel Side Connection	Each M.2 connector Vss pad should be connected to its own Vss via on both side Toe and edge. The distance between the edge of the pad and the edge of the via pad should be 255um max.
Suggested Ground Void for Main Board M.2 connector	Refer to image above. Top layer (or layer in which connector is placed) avoid ground shapes 200 um away from signal pads. Void in Top layer should be bigger than void in L2 layer.
continued...	

Note	Detail
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
Gen5 M.2 operating at Gen4	Please refer to PCIe Gen4 M.2 Topology chapter for lengths. those implementing gen4 topologies can follow trace geometry and spacing as given in Tlinespec for PCIe gen4 provided gen4 M.2 card meet spec requirements.
Add-in card	PCIe Gen5 M.2 Spec compliant add-in card.

Table 423. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	2	Total Channel

Rx,Tx**Table 424. PCIe Gen 5 M.2 Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1+M2	MS, DSL, SL	160	$BO+M1+M2+M3 \leq \text{MAX length Total}$
M3	MS, DSL, SL	12.7	

Max Length Total (mm): 160

Max Length Total Note: For Premium 10L T3 stack up, we can go up to 170 mm length instead of 160 mm. Premium Mid loss T4 and thin PCB need to meet original length in table above.

PCIe Gen 5 Port Operating at Gen 4 Data Rate Device Down Topology

Description: Platform should avoid using the PCIe Gen 5 capable ports for Gen 4 device. Because even in Gen 4 speed, Gen 5 IP will use more power as compared Gen 4 IP both working at Gen 4 data rate.

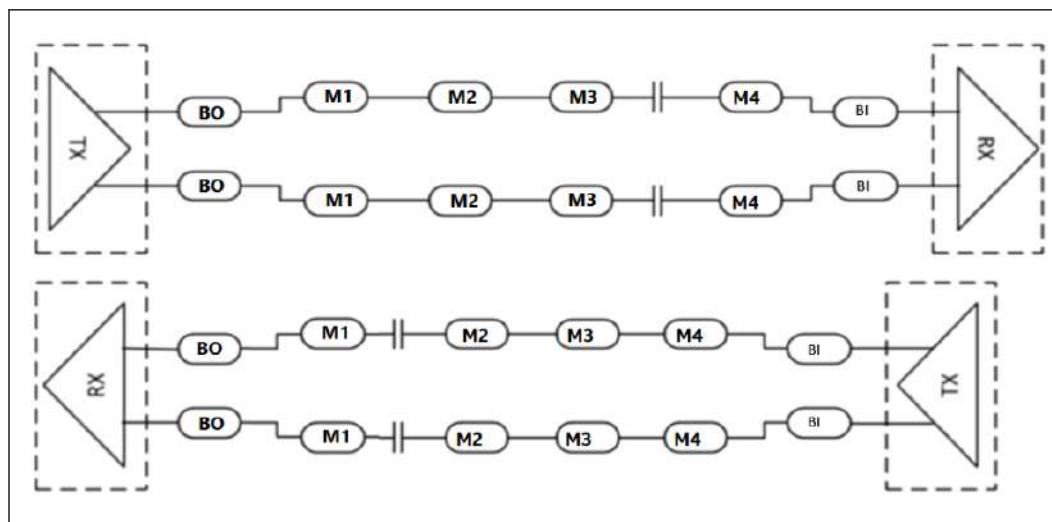
Figure 163. PCIe Gen 5 Port Operating at Gen 4 Data Rate Device Down Topology Diagram

Table 425. PCIe Gen 5 Port Operating at Gen 4 Data Rate Device Down Topology Notes

Note	Detail
Maximum via stub length	250 um
AC cap value	Nominal 220 nF recommended (spec range 176 nF to 265 nF including tolerance.).
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
Device	PCIe Gen4 Spec compliant endpoint device.

Table 426. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Rx,Tx

Table 427. PCIe Gen 5 Port Operating at Gen 4 Data Rate Device Down Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL	13	
M1+M2+M3+M4	MS, DSL, SL	270	BO+M1+M2+M3+M4+BI ≤ MAX length Total
BI	MS, DSL, SL	13	

Max Length Total (mm): 270

6.1.8.5 PCIe Gen 5 Root Port Power at Lower Gen Data Rates

- On Nova Lake platforms, PCIe Gen 5 root ports operating at lower Gen4 data rate may consume additional power.
- For semi-active workloads, operating a PCIe x4 Gen5-capable root port at a lower Gen4 data rate may experience an approximate power increase of 30 to 80mW over an equivalent PCIe x4 Gen4 root port operating at the same lower Gen4 data rate.
 - Power impact assumes that the root port width is x4 and within an approximate range of 5-25% utilization of the PCIe root port.
- Exact power difference will be dependent on actual PCIe root port utilization in any given workload and may differ than the estimated power impact listed above.

NOTES

- Semi-active workloads include Power and Performance Guide (PPG) KPIs, such as Streaming, Browsing, etc. Refer to the latest Nova Lake Mobile – Power And Performance Design Guide (#867669) for more details.
- Power estimates are based on a median unit, temperature at 40C.

6.1.8.6 General Routing Guidelines

Reference Planes

Referencing signals going over a board-to-board connector must be considered and stitching capacitors added at the connector as appropriate. For example, if a differential-pair is GND-referenced on the motherboard and the pair is V_{CC} referenced on the daughter card, then stitching capacitors must be added at the connector between V_{CC} and GND.

Lane to Controller Allocation

For the PCIe controller and lane configuration details, please refer EDS Vol1.

Different controllers are referred by referent function numbers.

Static Lane Reversal Support

The Processor has few soft strap options that will be driven by mFIT configuration. This will allow the platform to enable various controllers in the desired reversal mode. The root port supports static lane reversal. The target for the lane reversal support is to ease mother board layout. Card layout is not supported and should be handled by downstream device.

For the Gen5 PCIe x8 (8 lanes) PCIe_D only one reversal option is supported. This allows to design motherboard (designed Root Port 11) with straight or rotated processor relative to the PCI Express* slot. PCIe x8 interface is limited (no bifurcation support). Refer to the EDS for the configuration

Refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) . when the PCIe Gen5 PCIe_D is bifurcated into two Gen5 x4 PCIe interfaces, lane reversal configuration is not supported (assigned Root Ports 11 and 12).

6.1.8.7 Intel Validated Platform Processor PCIe Ports POR Configurations

Intel recommends customers to use Processor PCIe ports for various end points listed below. Refer to the below table for Intel validated configurations of NVL Processor CPU PCIe Ports.

Table 428. Recommended CPU PCIe Slot

PCIe Device	Intel Validated	Recommended Slot
Discrete Graphics	Yes	PCIe_D/E Validated PCIe_C approved
Storage Devices including Gen5 or lower NVMe SSD and SD7 or SD8 SDXC	Yes	Any for SSD PCIe_A/B for SDXC
TBT Device (BR)	Yes	PCIe_B
dLAN Connectivity	No	PCIe_A/B approved
WWAN Connectivity	No	Approved
WLAN	No	CNVi POR, PCIe_A/B approved
<i>continued...</i>		

PCIe Device	Intel Validated	Recommended Slot
Other (Audio, 3rd party connectivity)	No	PCIe_A/B Approved
dNPU	No	PCIe_C-E approved
Retimers/Redrivers	No	None

6.1.8.8 Port Reset Implementation

AND gate implementation (with inputs PLTRST# and GPP) should be used for resetting PCIe slots. This circuit provides an option to reset PCIe slots using BIOS-controlled GPP during debug, without resetting the entire platform.

A jumper should be added near the PCIe slots to select between the PLTRST# directly OR the ANDed output of PLTRST# and GPP. GPP should follow same routing guidelines as platform reset.

The BIOS needs to ensure that GPP is configured as output high for normal operation. This circuit will take effect and reset PCIe slots only when GPP is driven low.

An independent RST# circuit for PEG device typically being used for RTD3 use cases however, Intel recommends implementing a circuit that allows the BIOS to reset the PEG device independent of PLTRST#. Such circuitry may be utilized for RTD3 as well as for survivability cases where PEG devices may require additional functional resets

PCIe Gen 4/5 margining and functional link training can occasionally hang PCIe devices (slots or down). To ensure proper system functionality, Intel recommends implementing a circuit that allows the BIOS to reset the PCIe device, independent of PLTRST#. The reference circuit on the RVP uses GPP to control the auxiliary reset circuit, but the BIOS Reference Code allows a different GPIO to be used, if necessary for the system design flexibility. Unless necessary, customers are advised to use the default GPIO selection. Equivalent reset circuits are also acceptable. The auxiliary reset circuit is only required for PCIe Gen 4/5 implementations.

6.1.8.9 Debug Guidelines

The inclusion of test points and probing structures may impact the loss and jitter budgets of a PCI Express* interconnect. In general, test points and probe structures should not introduce stubs on the differential pairs or cause significant deviation from the recommendations given throughout this chapter. Existing vias, pads or pins should be used wherever possible to accommodate such structures. Careful consideration must be taken whenever additional probing structures are used.

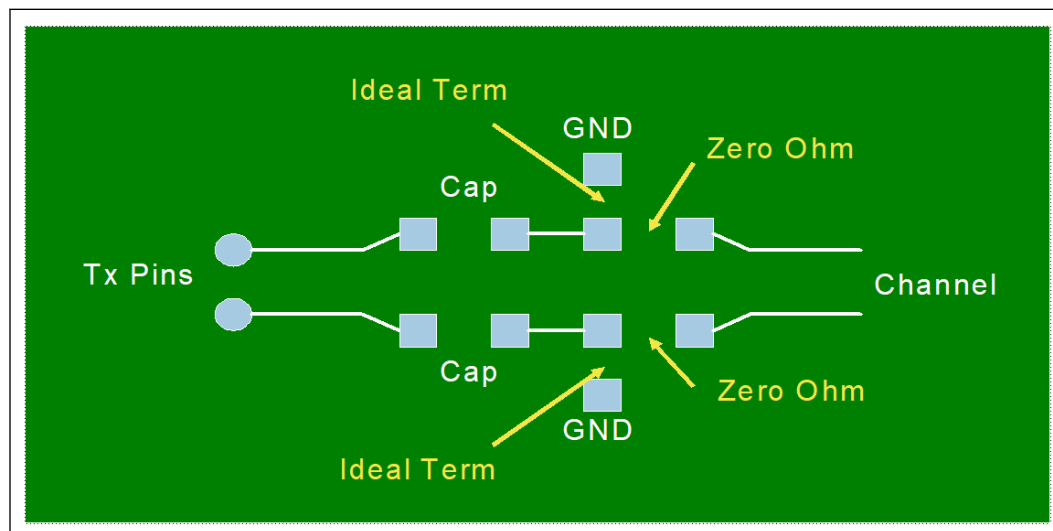
Probe Points for Testing Soldered Down PCI Express* Devices

In order to actively characterize the transmit path signal quality for a soldered down PCI Express* device, there must be a terminated probing point on the motherboard very near the soldered down device. This probing point may affect the overall signal quality of the system during normal operation due to the effects of stubs and impedance mismatches of the traces themselves.

The probing point requires populating a resistor stuffing option that can be used to break the path to the device very near the device pins and ideally terminate each line (TX+/TX-) to ground through a 50 Ω 1% resistor. Also, an AC coupling capacitor is required near the device pins. The resistor package/footprint must be as small as possible, preferably size 0201. The population of a 50- Ω resistor, as shown in the

figure below, will force the transmitting silicon (host or add-in card) to enter the compliance mode and begin transmitting the compliance packet. The population of two 0-Ω resistors will allow normal operation of the device. One replaces the AC coupling cap and the other connects the channel.

Figure 164. Board Preparation Example



6.1.9 Universal Serial Bus USB 3.2 Design Guidelines

Table 429. Processor USB 3.2 Compliance Documents

Title	Document Number / Location
USB 3.2 Specification	www.usb.org
USB 3.2 Front Panel Cable and Connector Implementation	http://www.usb.org/developers/docs/whitepapers/USB3p1_Front_Panel_CabCon_Implement_Doc_Rev1p0.pdf

6.1.9.1 Signal Descriptions

Signal Groups

The following table provides a list of the USB 3.2 interface signals and signal type.

Table 430. USB 3.2 Gen 2x1 (10 Gb/s) Interface Signals

Group	Signal Name
USB 3.2 Differential Transmit Pair: These are USB 3.2-based on high-speed differential signals. The signal should be mapped to a USB connector with one of the OC (overcurrent) signals.	USB32_1_TX_P USB32_1_TX_N USB32_2_TX_P USB32_2_TX_N
USB 3.2 Differential Receive Pair: These are USB 3.2-based high-speed differential signals. The signal should be mapped to a USB connector with one of the OC (overcurrent) signals.	USB32_1_RX_P USB32_1_RX_N USB32_2_RX_P

continued...

Group	Signal Name
	USB32_2_RX_N
Overcurrent Indicators: This signal set the corresponding bit in the xHCI controller to indicate that an overcurrent condition has occurred	GPP_E09/USB_OC0# GPP_B11/USB_OC1#/DDSP_HPD2#/ DISP_MISC3 GPP_B14/USB_OC2#/DDSP_HPD3#/ DISP_MISC4 GPP_B15/USB_OC3#
USB Resistor Bias, Analog connection points for an external resistor to ground. 200 Ohm +/- 1% to GND Board Rdc: < 0.1 Ohm Board capacitance: < 1.5 pF GND shield to any high speed IO with VSS trace/ strip GND reference required	USB32_RCOMP

6.1.9.2 Gen 1x1 (5 Gb/s) and Gen 2x1 (10 Gb/s) Topology Guidelines

The guidelines provided in this section applies to both Type-3 and Type-4 boards.

NOTE

It's not recommended to route the USB 3.2 signals to the USB connector for USB ports that is eUSB2 capable only.

USB 3.2 External Topology

This Topology is defined to be USB 3.2 signals routed directly to a standard Type-A connector. For additional information refer to the [PCB Stackup and Design Consideration](#) on page 58.

These guidelines were developed assuming that the USB 3.2 Device has capacitors on its Tx SS lines. A single 0 Ω resistor per signal trace should be available to be used to bypass the CMC content if it is proven that there is no EMI risk on USB 3.2 Tx/Rx signals. However, the 0 Ω resistor must be placed near the connector at the same recommended location as the CMC.

NOTE

USB-IF does not allow the usage of stacked USB Type-A connector for USB 3.2 Gen 2x1 capable port. <https://compliance.usb.org>

USB 3.2 Internal Cable Topology

This topology supports a low profile internal connector, internal cable solution.

For signal integrity considerations it is recommended that the USB 3.2 signals are routed to the A-connector via bottom-entry.

A single 0 Ω resistor per signal trace should be available to be used to bypass the CMC content if it is proven that there is no EMI risk on USB 3.2 Tx/Rx signals. However, this component must be placed near the connector at the same recommended location as the CMC.

NOTES

1. It is recommended that the internal connectors and cables adhere to the *High Speed Internal Connector and Cable Specification*.
 2. USB-IF does not allow the usage of stacked USB Type-A connector for USB 3.2 Gen 2x1 capable port <https://compliance.usb.org>.
-

USB 3.2 Re-driver Guidelines

Re-driver: An active component, which may have receiver equalization, transmitter de-emphasis, analog signal amplification, etc., to better "shape" the signals passing through it. Re-drivers do not perform retiming (i.e., the device does not implement any interface protocol and no re-sampling is performed). Because it changes the analog behavior of the incoming signals, the propagation of jitter, noise and other analog effects from one connected channel to the other must be considered.

NOTES

1. A re-driver's performance is implementation specific in terms of loss compensation.
 2. USB-IF does not allow the usage of stacked USB Type-A connector for USB 3.2 Gen 2x1 capable port <https://compliance.usb.org>.
-

Re-timer: An active component compliant to the updated Appendix E of the USB 3.2 specification that performs re-sampling of the incoming signal and re-transmitting the recovered signal per specification. A re-timer is protocol aware and compliant to link power management.

NOTES

1. The delay issue with a re-timer in USB 3.2 Gen 1x1 (5 Gb/s) operation is solved if the implementation follows the updated Appendix E of USB 3.2 Specification document.
 2. USB-IF does not allow the usage of stacked USB Type-A connector for USB 3.2 Gen 2x1 capable port <https://compliance.usb.org>.
-

For additional details and recommendations regarding the implementation of re-driver in USB 3.2 Gen 2x1 (10 Gb/s) channel, refer to the following documents.

Title	Document Number
Intel Channel Checker ICC for Client Re-driver CQC Analysis	572323

NOTE

Important Guidelines on Repeater/Active MUX topologies

1. Max pre-channel length depends on what the specific repeater selected can compensate for at both 5G and 10G speeds when used in conjunction with Intel Processor. Work with Repeater vendor for specific routing recommendations. The aforementioned DG is preliminary and superseded by vendor recommendations.
2. In addition, it is recommended to refer to the white paper pertaining to the repeater integration (USB 3.2 Repeater Integration Technical White Paper # 571574). The white paper illustrates in detail the fundamental differences between different repeater types (redriver vs retimer), and the efforts needed to enable such repeater in customer system designs.
3. It is strongly recommended that OEMs use active mux/retimer/redriver parts from 3rd party vendors which meet below requirement to avoid EOS on Rx lines for USB 3.2.

6.1.9.3 USB 3.2 Topology Guidelines

NOTE

It is strongly recommended to include the use of ESD protection devices on each USB data signals.

6.1.9.3.1 USB 3.2 Gen 1

Description: This PHY only supports configuration with standard Type-A connector.

Figure 165. USB 3.2 Stub Requirement

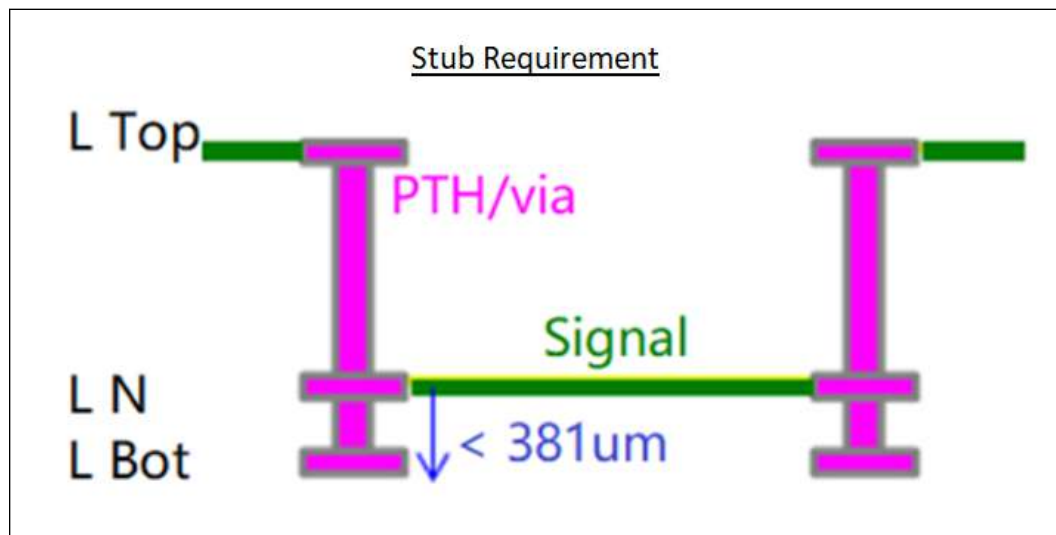


Figure 166. USB 3.2 Component Routing Requirement

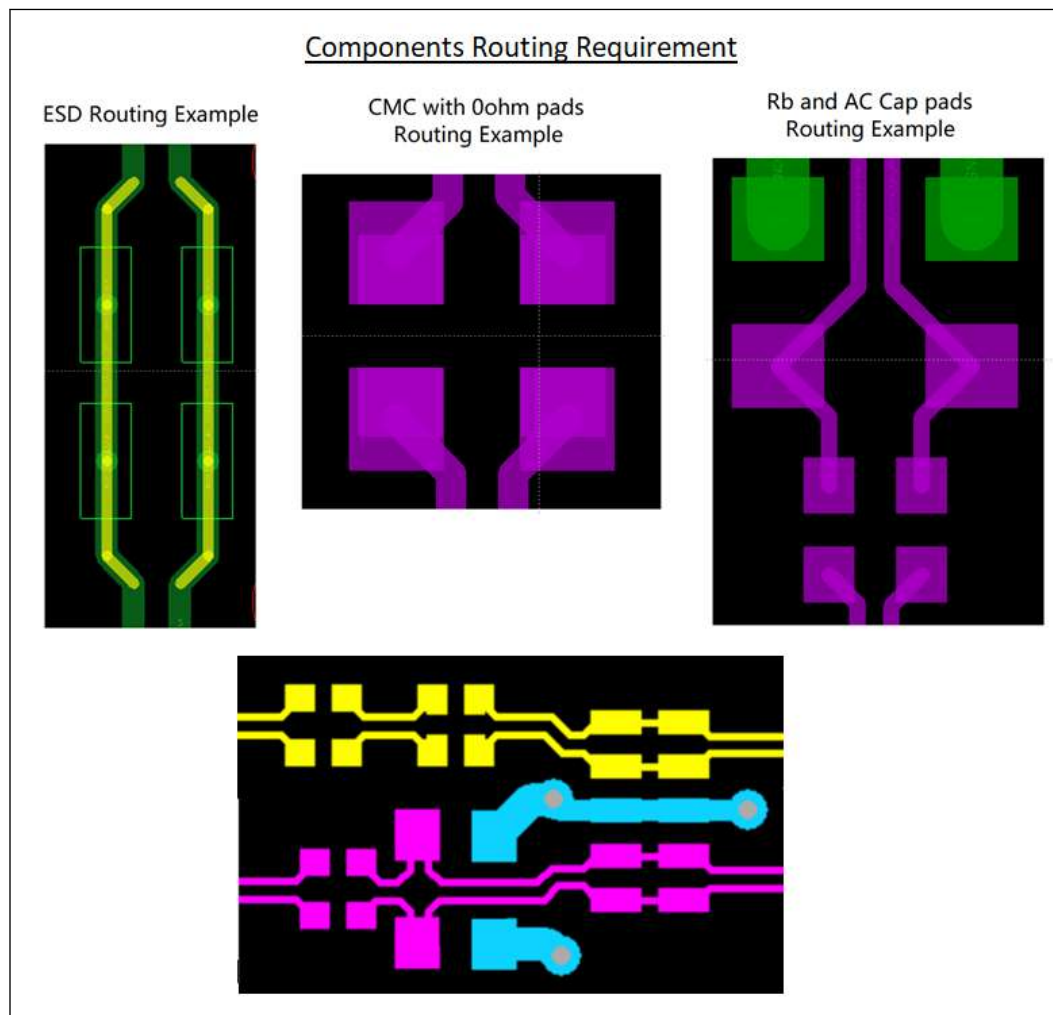


Figure 167. USB 3.2 Component Pad Voiding Requirement

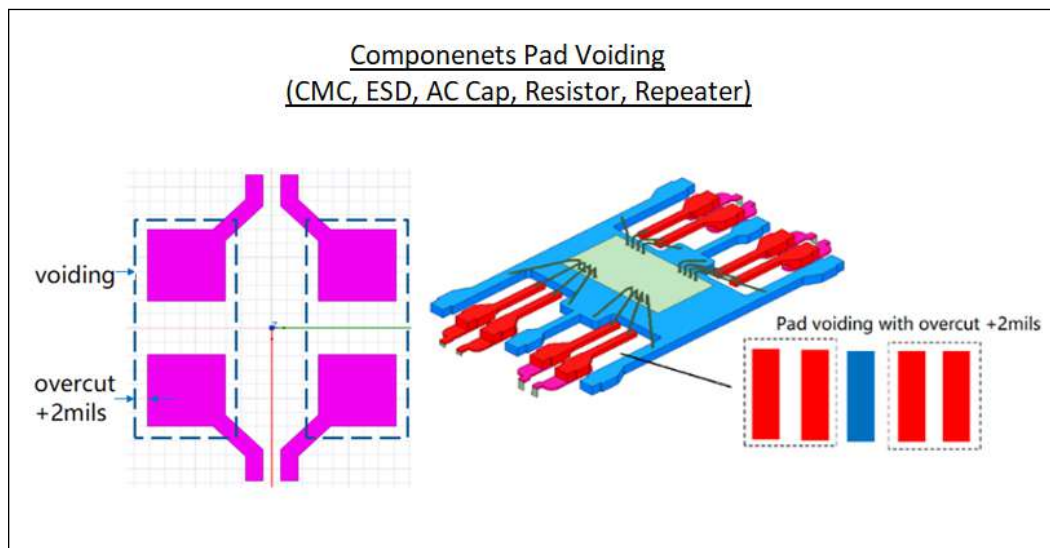


Figure 168. USB 3.2 Receptacle Ground Void Dimension Recommendations

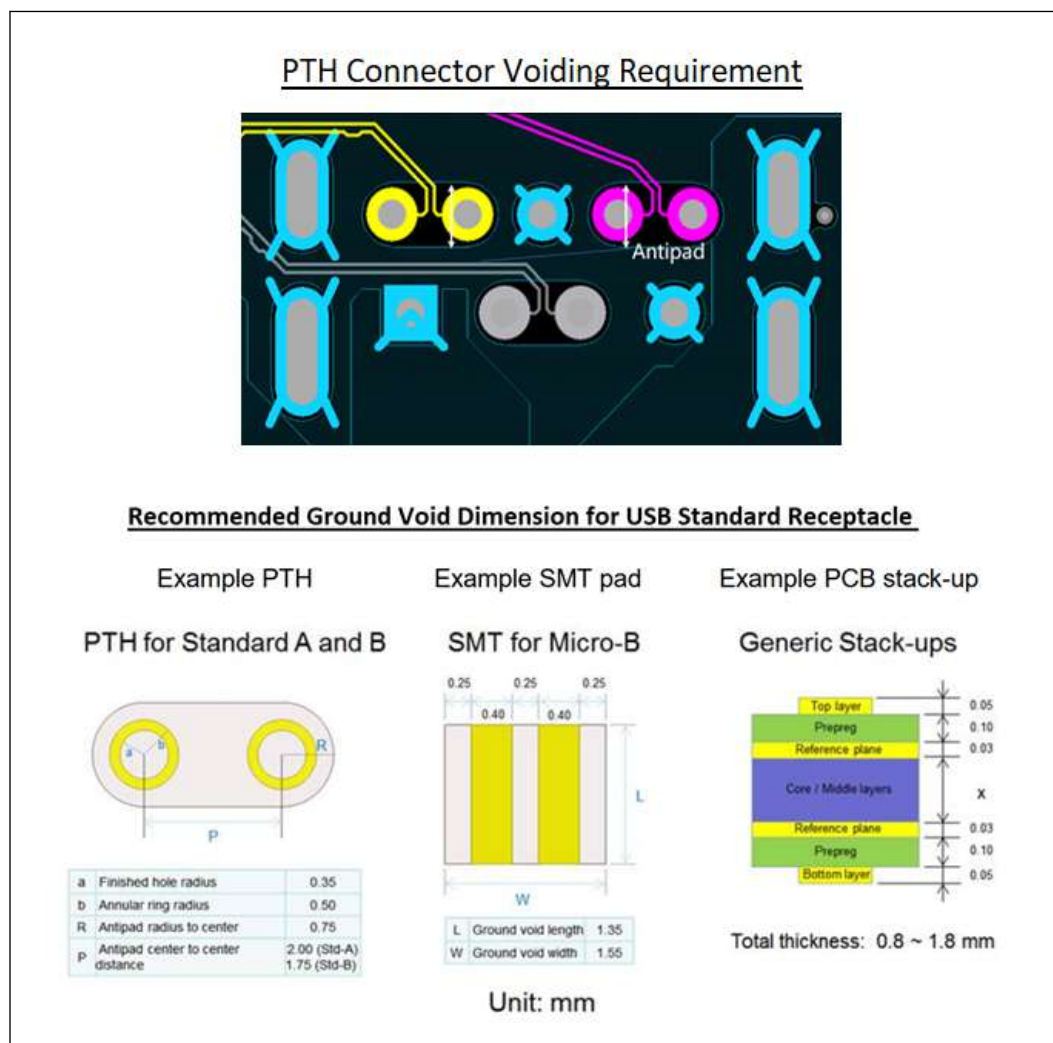


Figure 169. USB 3.2 Signals to PTH Transition Requirement

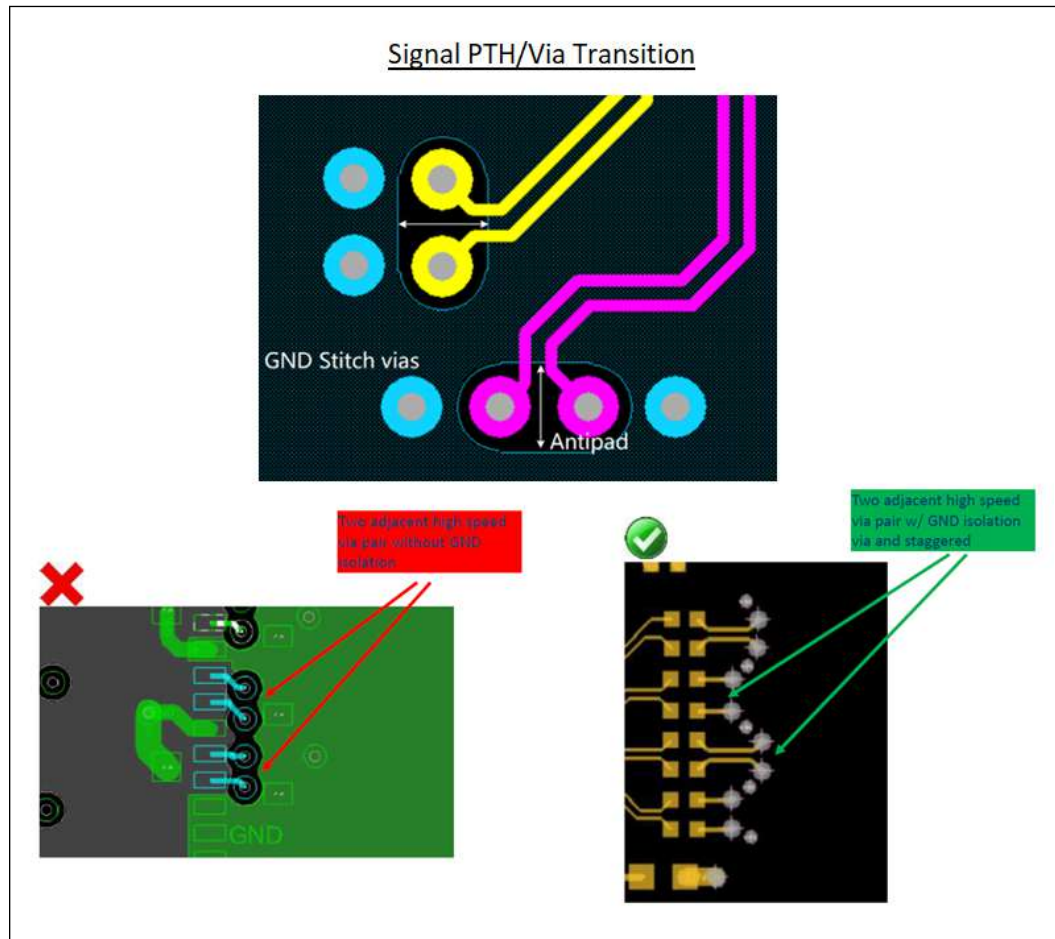


Table 431. USB 3.2 Gen 1 General Guidelines

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch < 0.254 mm. Total length mismatch < 0.127 mm.
Stub requirement	Channel and via stub requirement must meet < 381 um for both Tx and Rx signal pairs.
Tx AC cap value for USB 3.2 Gen 1 & Gen 2	AC capacitor value: 100 nF nominal (75 nF - 265 nF range).
Short circuit protection for Type-C topology (if applicable)	Crx cap value: 297 nF to 363 nF including tolerance, 25 V. Rb resistor value: 220 kΩ ± 5%.
Component size requirement (CMC, ESD, AC cap, Resistor)	Strongly recommended to use 0201 components for better impedance control.
Component voiding	All component pads must be voided on L2 and with GND reference on L3. Suggest having pad voiding size of 50.8 um overcut. Refer to Component Pad Voiding diagram.
<i>continued...</i>	

Note	Detail
Component routing	Signal pair to entry/ exist from components pad symmetrically. Maintain differential routing to meet trace target impedance whenever possible.
Signal PTH/ via transition	Differential pair voiding needed for each signal pair transition with recommended antipad size of 0.75 mm. Suggest having two symmetrical GND stitching vias for each signal pair transition.
Cable and connector recommendation	Strongly recommended to use SMT connectors. For PTH connectors (assuming connector is on the top), USB3.2 signals need to be routed from the bottom of the board such that there is no stub on for connector entry. Intel strongly recommends selecting cables and connectors which meet electrical requirements specified in both USB-IF Cable and Connector Specification dated June 2017 and Universal Serial Bus 3.1 Legacy Connectors and Cable Assemblies Compliance dated April 2018. The USB-IF does not certify legacy or USB Type-C(tm) Gen2 stacked connectors. The USB-IF does certify legacy or USB Type-C(tm) Gen1 stacked connectors. Refer USB-IF for more details.
PTH connector voiding requirement	Differential oval antipad required for all layers. Suggest using differential signal antipad with 2 mm size.
Repeater pad voiding	A differential void on L2 that is larger than the repeater with 50.8 um overcut is recommended. GND referencing is needed on L3. Refer to Component Pad Voiding diagram.
Routing scheme	Non-interleaved breakout (B0) is required to mitigate near-end crosstalk. The main route (MR) supports both interleaved routing and non-interleaved routing for maximum flexibility on stripline (SL). For microstrip (MS), interleaved routing scheme is recommended.
Reference plane	Continuous GND is recommended. If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). If non-continuous power referencing is unavoidable on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt), but stitching caps are required over each plane split and need to be placed within 6 mm of a signal running across the plane split.

Table 432. USB 3.2 Gen 1 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.127
2	Pair (P-N)	Within	Length (mm)	Yes	0.254
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

USB 3.2 Gen 1x1 Type-A External Topology

Figure 170. USB 3.2 Gen 1x1 Type-A External Topology Diagram

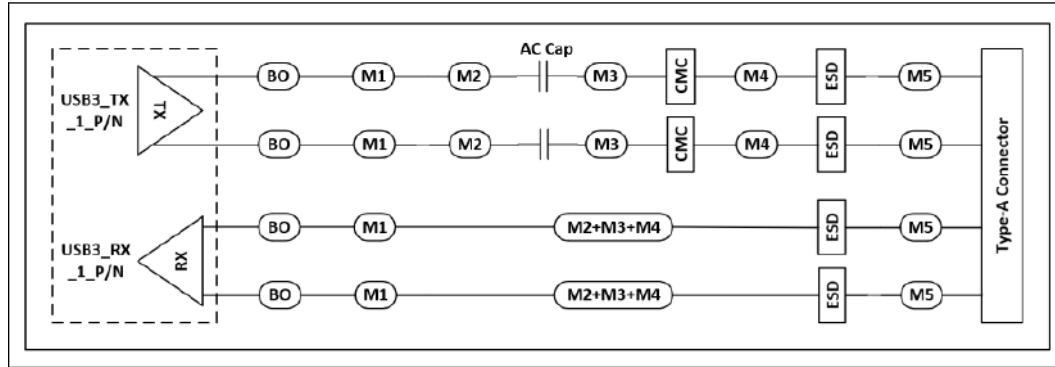


Table 433. Max Number of vias

Signal Group	Max Number of Vias Allowed	Notes
Rx, Tx	2	Max via count allow
Rx, Tx	1	Additional top-to-bottom via allow access to the connector

Rx, Tx

Table 434. USB 3.2 Gen 1x1 Type-A External Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.5	
M1	MS, DSL, SL	228	BO+M1+M2+M3+M4+M5 = 228 mm
M2+M3+M4+M5	MS	25.5	

Max Length Total (mm): 228

USB 3.2 Gen 1x1 Type-A Internal Cable Topology

Figure 171. USB 3.2 Gen 1x1 Type-A Internal Cable Topology Diagram

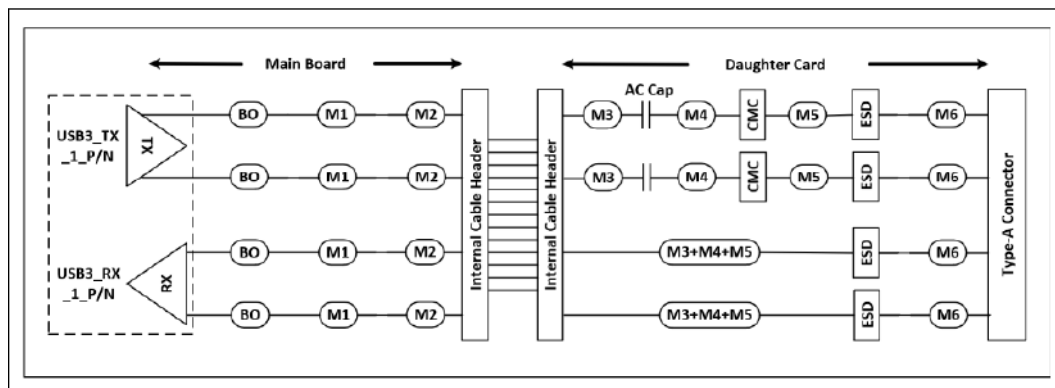


Table 435. USB 3.2 Gen 1x1 Type-A Internal Cable Topology Notes

Note
This assumes internal cable assembly insertion loss of ~2 dB @ 2.5 GHz, ~3 dB @ 5.0 GHz. Refer to USB3.2 "Front-Panel Internal Cable And Connector White Paper" for more details.

Table 436. Max Number of vias

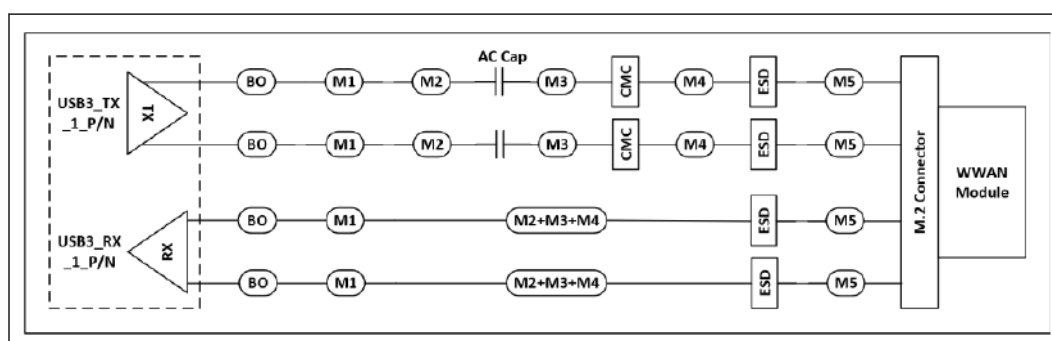
Signal Group	Max Number of Vias Allowed	Notes
Rx, Tx	2	Max via count allow
Rx, Tx	1	Additional top-to-bottom via allow access to the connector

Rx, Tx**Table 437. USB 3.2 Gen 1x1 Type-A Internal Cable Topology Rx, Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.5	
M1+M2	MS, DSL, SL	81.5	BO+M1+M2 ≤ 81.5 mm
M3+M4+M5+M6	MS	50.8	

Max Length Total (mm): 132.3

Max Length Total Note: For T4 Premium Mid-loss (PML), max length can extended to ≤135 mm (BO+M1+M2 ≤ 84.2 mm). For T3 Premium Mid-loss (PML), max length can extended to ≤139 mm (BO+M1+M2 ≤ 88.2 mm). For T3 Thin PCB, max length is ≤132.3 mm (BO+M1+M2 ≤ 81.5 mm).

USB 3.2 Gen 1x1 M.2 Topology**Figure 172. USB 3.2 Gen 1x1 M.2 Topology Diagram****Table 438. USB 3.2 Gen 1x1 M.2 Topology Notes**

Note	Detail
ESD Component	For M.2 topology, ESD is optional assuming ESD control is taken care during assembly process.

Table 439. Max Number of vias

Signal Group	Max Number of Vias Allowed	Notes
Rx, Tx	2	Max via count allow
Rx, Tx	1	Additional top-to-bottom via allow access to the connector

Rx, Tx

Table 440. USB 3.2 Gen 1x1 M.2 Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.5	
M1	MS, DSL, SL	228	$BO + M1 + M2 + M3 + M4 + M5 \leq 228 \text{ mm}$
M2+M3+M4+M5	MS	25.5	

Max Length Total (mm): 228

6.1.9.3.2 USB 3.2 Gen 2

Description: This PHY only supports configuration with standard Type-A connector.

Figure 173. USB 3.2 Stub Requirement

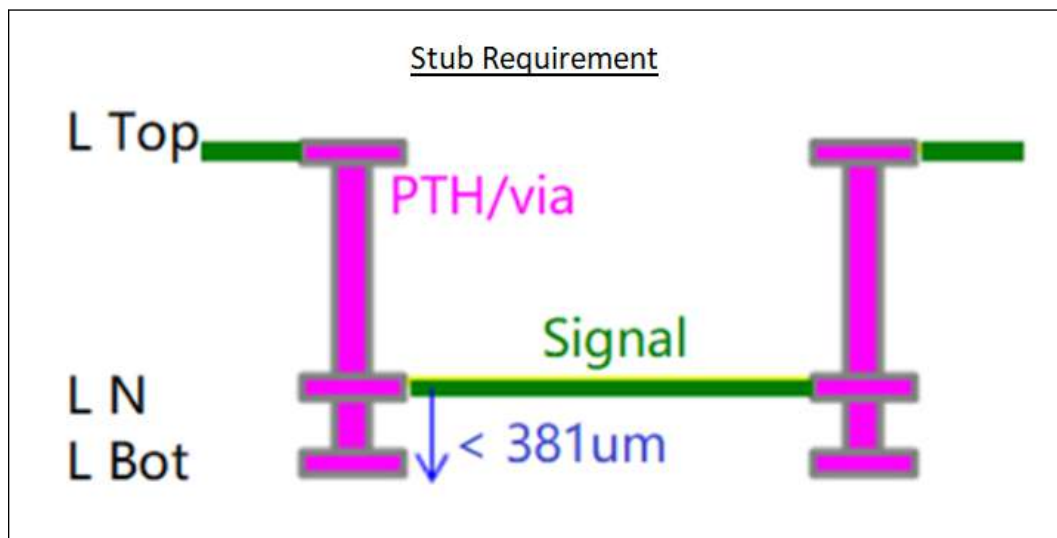


Figure 174. USB 3.2 Component Routing Requirement

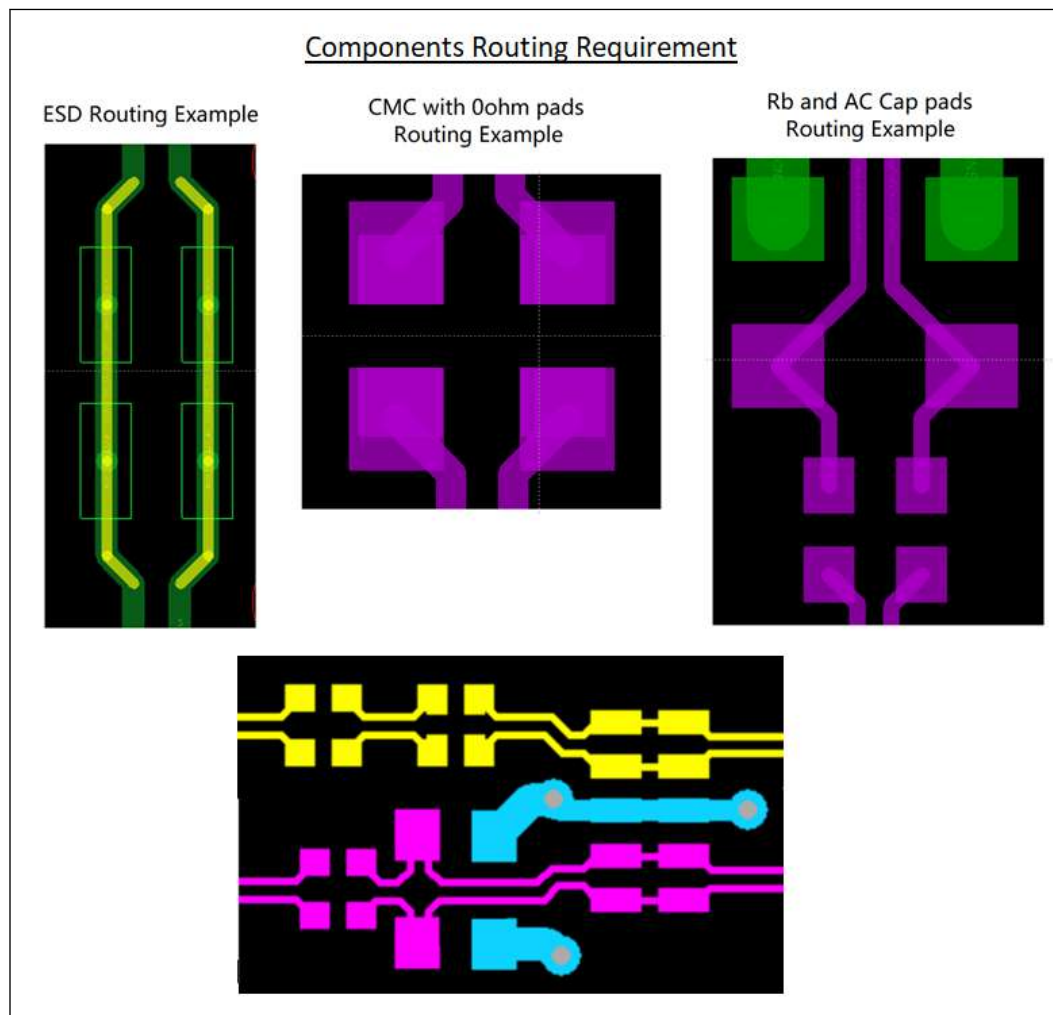


Figure 175. USB 3.2 Component Pad Voiding Requirement

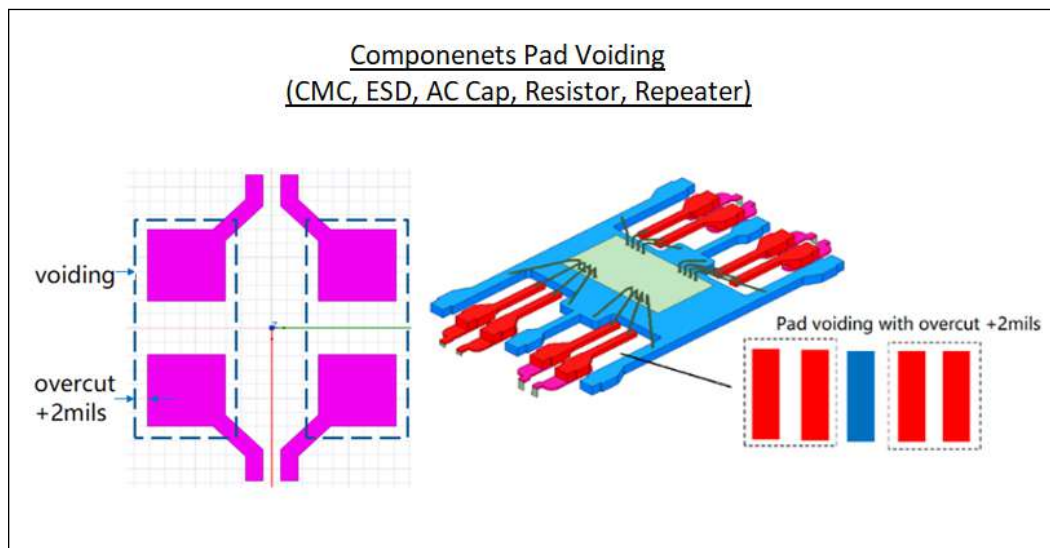


Figure 176. USB 3.2 Receptacle Ground Void Dimension Recommendations

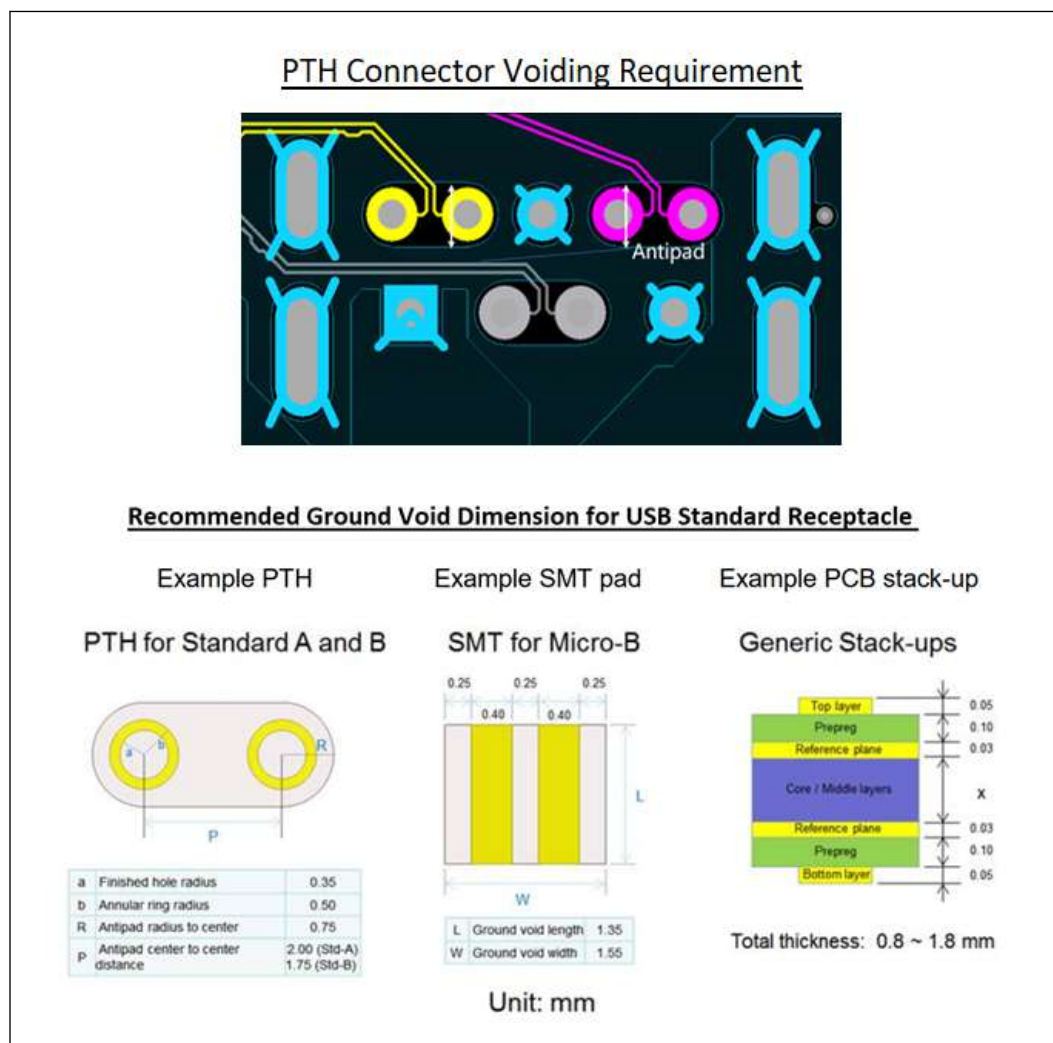


Figure 177. USB 3.2 Signals to PTH Transition Requirement

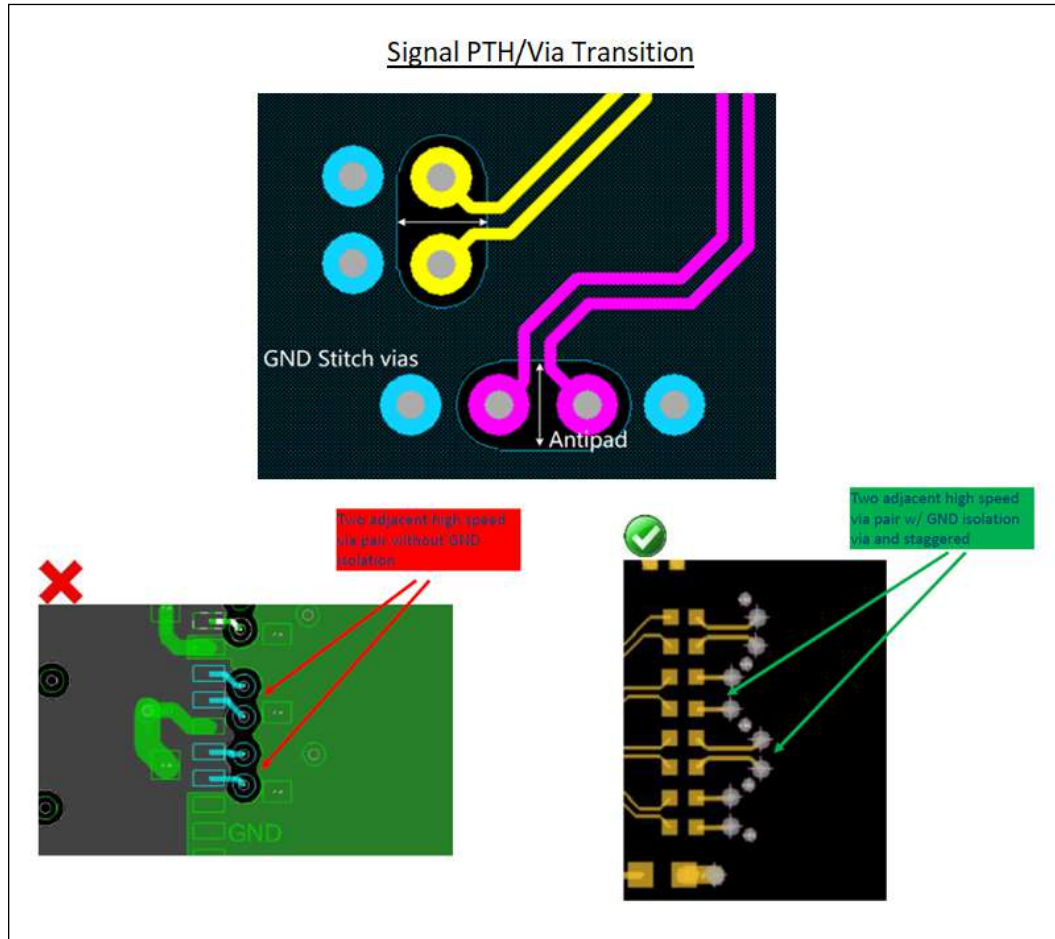


Table 441. USB 3.2 Gen 2 General Guidelines

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch < 0.254 mm. Total length mismatch < 0.127 mm.
Stub requirement	Channel and via stub requirement must meet < 381 um for both Tx and Rx signal pairs.
Tx AC cap value for USB 3.2 Gen 1 & Gen 2	AC capacitor value: 100 nF nominal (75 nF - 265 nF range).
Short circuit protection for Type-C topology (if applicable)	Crx cap value: 297 nF to 363 nF including tolerance, 25 V. Rb resistor value: 220 kΩ ± 5%.
Component size requirement (CMC, ESD, AC cap, Resistor)	Strongly recommended to use 0201 components for better impedance control.
Component voiding	All component pads must be voided on L2 and with GND reference on L3. Suggest having pad voiding size of 50.8 um overcut. Refer to Component Pad Voiding diagram.
<i>continued...</i>	

Note	Detail
Component routing	Signal pair to entry/ exist from components pad symmetrically. Maintain differential routing to meet trace target impedance whenever possible.
Signal PTH/ via transition	Differential pair voiding needed for each signal pair transition with recommended antipad size of 0.75 mm. Suggest having two symmetrical GND stitching vias for each signal pair transition.
Cable and connector recommendation	Strongly recommended to use SMT connectors. For PTH connectors (assuming connector is on the top), USB3.2 signals need to be routed from the bottom of the board such that there is no stub on for connector entry. Intel strongly recommends selecting cables and connectors which meet electrical requirements specified in both USB-IF Cable and Connector Specification dated June 2017 and Universal Serial Bus 3.1 Legacy Connectors and Cable Assemblies Compliance dated April 2018. The USB-IF does not certify legacy or USB Type-C(tm) Gen2 stacked connectors. The USB-IF does certify legacy or USB Type-C(tm) Gen1 stacked connectors. Refer USB-IF for more details.
PTH connector voiding requirement	Differential oval antipad required for all layers. Suggest using differential signal antipad with 2 mm size.
Repeater pad voiding	A differential void on L2 that is larger than the repeater with 50.8 um overcut is recommended. GND referencing is needed on L3. Refer to Component Pad Voiding diagram.
Routing scheme	Non-interleaved breakout (B0) is required to mitigate near-end crosstalk. The main route (MR) supports both interleaved routing and non-interleaved routing for maximum flexibility on stripline (SL). For microstrip (MS), interleaved routing scheme is recommended.
Reference plane	Continuous GND is recommended. If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). If non-continuous power referencing is unavoidable on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt), but stitching caps are required over each plane split and need to be placed within 6 mm of a signal running across the plane split.

Table 442. USB 3.2 Gen 2 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.127
2	Pair (P-N)	Within	Length (mm)	Yes	0.254
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

USB 3.2 Gen 2x1 Type-A External Topology

Figure 178. USB 3.2 Gen 2x1 Type-A External Topology Diagram

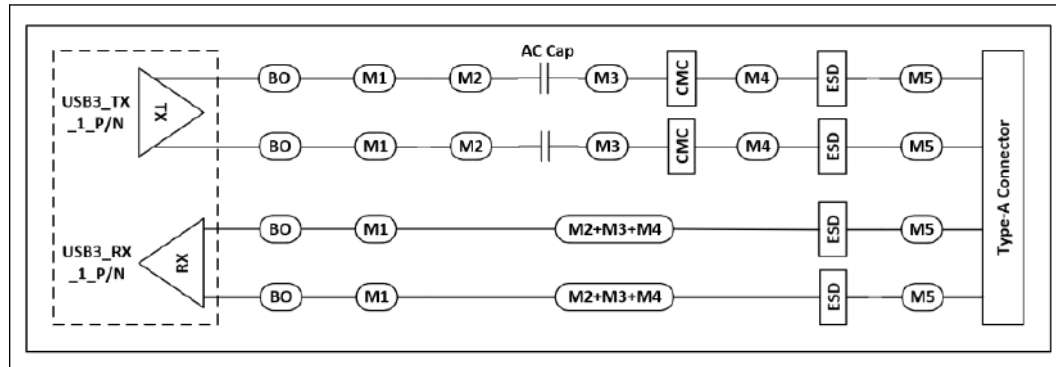


Table 443. Max Number of vias

Signal Group	Max Number of Vias Allowed	Notes
Rx, Tx	2	Max via count allow
Rx, Tx	1	Additional top-to-bottom via allow access to the connector

Rx, Tx

Table 444. USB 3.2 Gen 2x1 Type-A External Topology Rx, Tx Routing Guidelines

Segment	Line Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.5	
M1	MS, DSL, SL	101.6	$BO + M1 + M2 + M3 + M4 + M5 \leq 101.6$ mm
M2+M3+M4+M5	MS	25.5	

Max Length Total (mm): 101.6

Max Length Total Note: For T4 Premium Mid-loss (PML), length can extended to ≤ 104 mm. For T3 Premium Mid-loss (PML), length can extended to ≤ 105 mm. For T3 Thin PCB, max length is ≤ 101.6 mm.

USB 3.2 Gen 2x1 Type-A External With Retimer Topology

Figure 179. USB 3.2 Gen 2x1 Type-A External With Retimer Topology Diagram

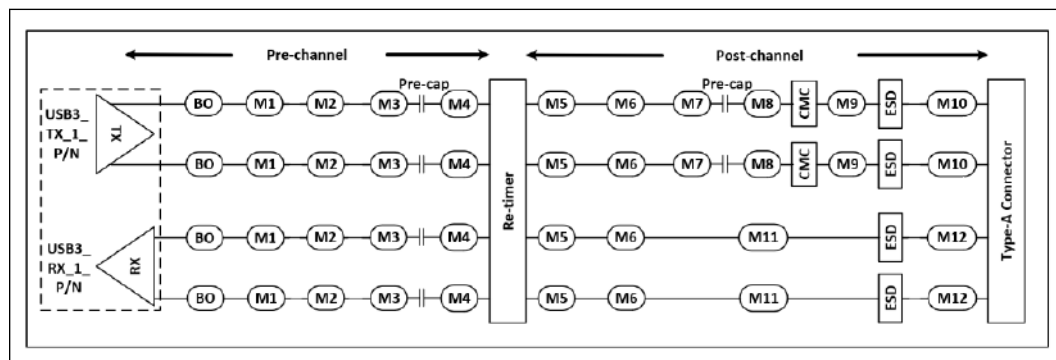


Figure 180. Requirement for Third Party Active Component

Third Party Active MUX/Re-driver/Re-timer USB 3.2 VCM Requirements			
Electrical Parameter	Gen 1x1 (5 Gb/s)	Gen 2x1 (10 Gb/s)	Notes
$V_{RX-CM-DC-ACTIVE-IDLE-DELTA_P}$	200 mV peak (max)	200 mV peak (max)	Refer to USB 3.2 ECN VCM & TXDetect for full details on the measurement
$V_{RX-CM-DC_CONN}$	-0.5V (min1)	-0.5V (min1)	
	-0.3V (min2)	-0.3V (min2)	
	1.0V (max)	1.0V (max)	

Table 445. USB 3.2 Gen 2x1 Type-A External With Retimer Topology Notes

Note
Note1*: Max/ min routing length for USB3.2 Gen1 & 2 speeds (5/ 10 Gbps) depends on redriver performance and its loss compensation capability when used in conjunction with Intel processor. Work with repeater vendor to ensure the selected part can provide adequate signal integrity at the processor.

Table 446. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	0	Pre-Channel	Work with the vendor to ensure that the maximum via count is met
Rx, Tx	0	Post-Channel	Work with the vendor to ensure that the maximum via count is met

Rx, Tx**Table 447. USB 3.2 Gen 2x1 Type-A External With Retimer Topology Rx, Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	0	Note1*
M1+M2	MS, DSL, SL	0	Note1*
M3+M4	MS	0	Note1*
M5+M6	MS, DSL, SL	0	Note1*
M7+M8+M9+M10 / M11+M12	MS	0	Note1*

Max Length Total Note: Note1*

USB 3.2 Gen 2x1, Gen 1x1 Type-A External With Redriver Topology

Figure 181. USB 3.2 Gen 2x1, Gen 1x1 Type-A External With Redriver Topology Diagram

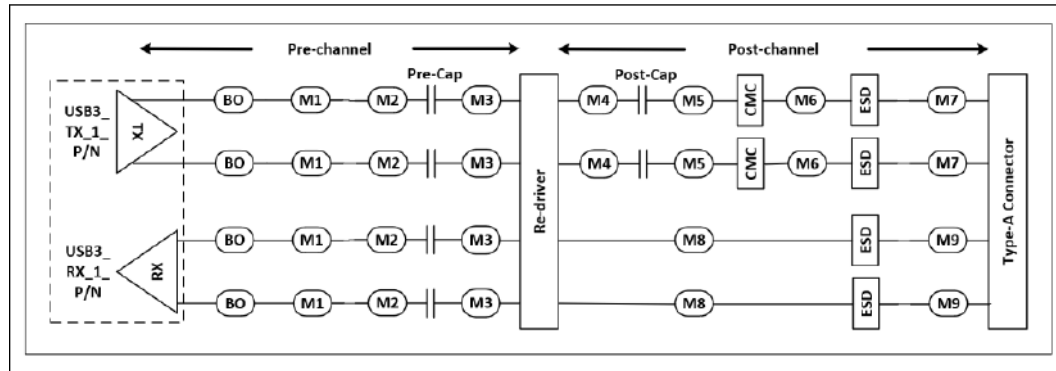


Figure 182. Requirement for Third Party Active Component

Third Party Active MUX/Re-driver/Re-timer USB 3.2 VCM Requirements			
Electrical Parameter	Gen 1x1 (5 Gb/s)	Gen 2x1 (10 Gb/s)	Notes
$V_{RX-CM-DC-ACTIVE-IDLE-DELTA_P}$	200 mV peak (max)	200 mV peak (max)	Refer to USB 3.2 ECN VCM & TXDetect for full details on the measurement
$V_{RX-CM-DC_CONN}$	-0.5V (min1)	-0.5V (min1)	
	-0.3V (min2)	-0.3V (min2)	
	1.0V (max)	1.0V (max)	

Table 448. USB 3.2 Gen 2x1, Gen 1x1 Type-A External With Redriver Topology Notes

Note
Note1*: Max/ min routing length for USB 3.2 Gen 1 & 2 speeds (5/ 10 Gbps) depends on redriver performance and its loss compensation capability when used in conjunction with Intel processor. Work with repeater vendor to ensure the selected part can provide adequate signal integrity at the processor.

Table 449. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	0	Pre-Channel	Work with the vendor to ensure that the maximum via count is met
Rx, Tx	0	Post-Channel	Work with the vendor to ensure that the maximum via count is met

Rx, Tx

Table 450. USB 3.2 Gen 2x1, Gen 1x1 Type-A External With Redriver Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	0	Note1*
M1	MS, DSL, SL	0	Note1*
M2+M3	MS	0	Note1*
M4+M5+M6+M7 / M8+M9	MS	0	Note1*

Max Length Total Note: Note1*

USB 3.2 Gen 2x1, Gen 1x1 Type-A Internal Cable With Redriver Topology

Figure 183. USB 3.2 Gen 2x1, Gen 1x1 Type-A Internal Cable With Redriver Topology Diagram

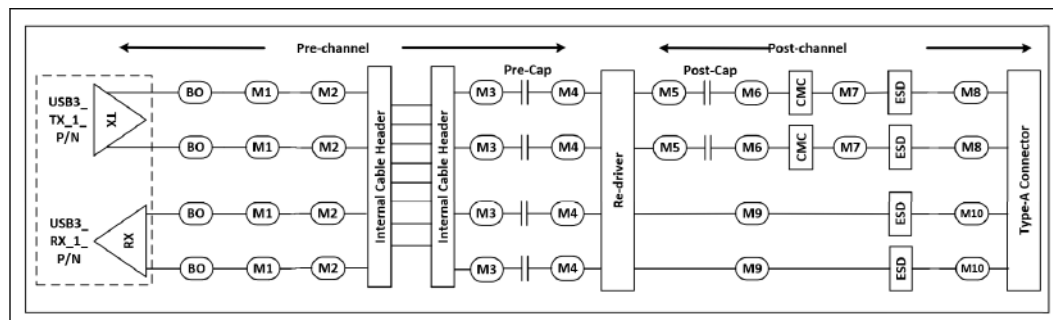


Figure 184. Requirement for Third Party Active Component

Third Party Active MUX/Re-driver/Re-timer USB 3.2 VCM Requirements			
Electrical Parameter	Gen 1x1 (5 Gb/s)	Gen 2x1 (10 Gb/s)	Notes
$V_{RX-CM-DC-ACTIVE-IDLE-DELTA_P}$	200 mV peak (max)	200 mV peak (max)	Refer to USB 3.2 ECN VCM & TXDetect for full details on the measurement
$V_{RX-CM-DC-CONN}$	-0.5V (min1)	-0.5V (min1)	
	-0.3V (min2)	-0.3V (min2)	
	1.0V (max)	1.0V (max)	

Table 451. USB 3.2 Gen 2x1, Gen 1x1 Type-A Internal Cable With Redriver Topology Notes

Note
This assumes internal cable insertion loss of ~3 dB @ 5 GHz. Refer to USB3.2 "Front-Panel Internal Cable And Connector White Paper" for more details.
Note1*: Max/ min routing length for USB 3.2 Gen 1 & 2 speeds (5/ 10 Gbps) depends on redriver performance and its loss compensation capability when used in conjunction with Intel processor. Work with repeater vendor to ensure the selected part can provide adequate signal integrity at the processor.

Table 452. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Rx, Tx	0	Pre-Channel	Work with the vendor to ensure that the maximum via count is met
Rx, Tx	0	Post-Channel	Work with the vendor to ensure that the maximum via count is met

Rx, Tx

Table 453. USB 3.2 Gen 2x1, Gen 1x1 Type-A Internal Cable With Redriver Topology Rx, Tx Routing Guidelines

Segment	Line Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	0	Note1*
M1	DSL, SL	0	Note1*
continued...			

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M2	MS	0	Note1*
M3+M4	MS	0	Note1*
M5+M6+M7+M8 / M9+M10	MS	0	Note1*

6.1.9.4 USB 3.2 Disabling and Termination Guidelines

If some of the USB 3.2 port(s) are not implemented on the platform, USB3Tp/n [x] and USB3Rp/n [x] signals may be left unconnected, where 'x' is the port number left no connect.

6.1.9.5 External End User Accessible Ports

Designers are recommended to provide a combination of USB 2.0 only ports and USB 3.2 ports for external use on their platforms to provide the greatest amount of flexibility for legacy device support.

6.1.9.6 USB Debug Port

The xHCI controller provides a debug port capability on all USB 3.2 ports

6.1.10 Processor Embedded USB2 (eUSB2) Design Guidelines

Table 454. Processor Embedded USB2 (eUSB2) Compliance Documents

Title	Doc #/Location
eUSB2 and USB 2.0 Specification and Compliance Requirements	www.usb.org

6.1.10.1 Signal Groups

Table 455. eUSB2 Signal Groups

Group	Signal Name	Description
Embedded USB2 (eUSB2) Transmit/Receive Differential Pair: These eUSB2 signal pair are routed to xHCI controller and should be mapped to a USB connector with one of the OC (overcurrent) eUSB2 signals.	eUSB2P_1	Universal Serial Bus Port Differential Pairs.
	eUSB2N_1	
	eUSB2P_2	
	eUSB2N_2	
	eUSB2P_3	
	eUSB2N_3	
	eUSB2P_4	
	eUSB2N_4	
	eUSB2P_5	
	eUSB2N_5	
	eUSB2P_6	
	eUSB2N_6	
	eUSB2P_7	
	eUSB2N_7	
	eUSB2P_8	
	eUSB2N_8	
Overcurrent Indicators:	GPP_E09/USB_OC0# GPP_B11/USB_OC1#/ DDSP_HP2#/DISP_MISC3	Overcurrent Indicators
continued...		

Group	Signal Name	Description
This signal set the corresponding bit in the xHCI controller to indicate that an overcurrent condition has occurred.	GPP_B14/USB_OC2#/ DDSP_HPDP3#/DISP_MISC4 GPP_B15/USB_OC3#	OC0#, OC1#, OC3#: when used, 10 kΩ pull-up to 1.8V (+1.8V based on the GPIO group) is required. OC2#: when used, 100 kΩ pull-up to 1.8V (+1.8V based on the GPIO group) is required. The OC2# pin is multiplexed on GPP_B14, which is a strap for Top Swap Override. A 100 kΩ pull-up ensures the strap functionality is not inadvertently asserted to enable Top Swap mode Override. Refer to the Pin Straps chapter in the EDS Vol1 for more info on the strap. When this pin is configured as GPIO, no pull-up resistor is required. Notes: • OC# pins are +1.8V tolerant. • OC# pins can be shared between USB ports. • Each USB connector should only have one OC# pin protection.
<i>Note:</i> The supported connectors are Type A Dual Stack and Type A Single Stack. The connectors must meet the USB2.0 Connector Specification.		

6.1.10.2 eUSB2 RCOMP Routing Guidelines

Table 456. eUSB2 RCOMP Routing Guidelines

Signal Names	Trace Width (W) and Isolation Spacing (S)	Resistor Value
eUSB2_RCOMP	Trace width 75μm with min trace spacing 100μm is recommended. Max total length on board 9mm. <i>Note:</i> Board Rdc: <0.10 Ohm Board capacitance: < 1.5 pF	200 Ω $\pm$ 1% connected to GND
Notes: 1. Although there is no hard value for max topology length as long as the DC resistance and Board capacitance routing spec is met, although it is recommended to route the eUSB2_RCOMP using 50 single-ended impedance and as short as possible. 2. Avoid routing next to clock pins or under stitching capacitors.		

6.1.10.3 eUSB2_RCOMP Routing Guidelines to Other Interfaces

If unable to implement a Vss shield for eUSB2_RCOMP, the minimum spacing between eUSB2_RCOMP and other traces should be at least , 0.400mm to provide proper isolation.

6.1.10.4 Overcurrent Protection

Nova Lake platform has implemented programmable USB Overcurrent signals. The 4 overcurrent pins are to be shared across the eUSB2 ports and USB 3.2 ports. This allows the platform designer flexibility in routing of the OC pins and allows for unused pins to be configured as GPIOs.

When configured for overcurrent protection, each of the OC pins needs to be connected to an external over current protection circuit. Each eUSB2 port may draw up to 500 mA. The overcurrent protection circuitry needs to be able to support at least 2 A to ensure proper functionality of eUSB2 compliant devices. However, if a fault device is plugged into the eUSB2 port, it should trigger an overcurrent when current is more than 500 mA is drawn. Designer must balance between optimum cost and protection level.

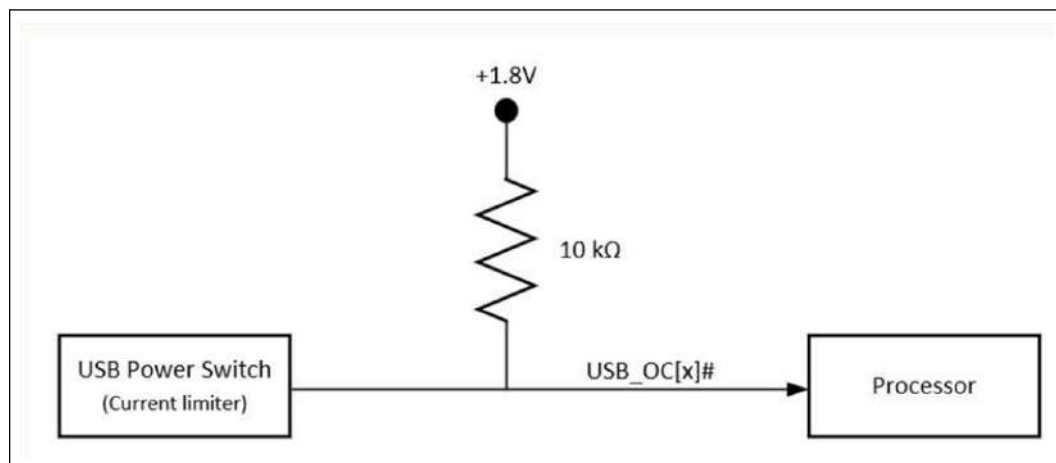
NOTES

1. When the current rating of the protection circuit has been decided, it is crucial to ensure that the overcurrent layout is designed to support the amount of current expected.
2. The overcurrent circuit should be designed to support up to 2 A, the overcurrent trace layout should be using a big fat trace or plane that can sustain at least 2 A.

Warning:

1. Failure to design the trace to support high current may cause the trace on motherboard to burn out and cause permanent damage to the protection circuit.
2. The overcurrent signals require 100 k Ω pull-up to voltage power-rail (+1.8V based on the GPIO group) when OC2# is used. Refer to the figure below.
3. The overcurrent signals require 10 k Ω pull-up to voltage power-rail (+1.8V based on the GPIO group) when OC0#, OC1#, and OC3# are used.

Figure 185. Sample Overcurrent Protection Circuit



Each overcurrent pin is configured to protect one or more USB ports by setting bits in the U2OCM1 register in the xHCI controller.

NOTE

It is the responsibility of system software (BIOS) to program the overcurrent registers of the given USB controller correctly and to make sure that each USB port is protected by only one overcurrent pin. Operation with more than one overcurrent pin mapped to a port is undefined.

Overcurrent Pin Mapping

It is not recommended that customers map more than four USB ports to a single OC pin. For more details on how to map OC pins to eUSB2 ports, refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2 (# 844316).

6.1.10.5 Integrated Bluetooth* and USB 2.0 Design Considerations

The integrated CNVi BT does not use a USB 2.0 port for connection. As a result, a USB 2.0 port is not required to be reserved for CNVi BT connection. For discreet BT card that uses the USB 2.0 interface, it can be connected to any available USB 2.0 port.

6.1.10.6 Supported Topologies

- **Back Panel/External Topology**

The external topology refers to the routing of USB signals to a standard USB-A/USB-C connector. It is required that each USB data line be routed with a common mode choke and ESD protection.

- **eUSB2 M.2 Topology**

- **Internal Cable Topology/Front Panel Topology**

This topology supports a low profile internal connector, internal cable solution and a daughter card.

NOTE

The Internal cable guidelines will be defined using insertion loss budget rather than cable length where a recommended mobile internal cable has a loss of 2 dB @ 2.5 GHz.

Design Constraint: Differential vias, CMC, ESD and connector voiding (strongly recommended) are required to optimize impedance mismatch in the channel.

It is required recommended to include the use of ESD protection devices on each USB data signals.

- **Device Down Topology**

- *eUSB2 External / Back Panel with MUX Topology:*

- Current electrical characteristic recommendation on Power switch/controller/MUX for charging is to have on-state resistance and on-state capacitance of no greater than 6 Ω and 6pF respectively.
- Current electrical characteristic limitation on Power switch/controller/MUX for charging is insertion loss of no greater than -0.6 dB and return loss of no greater than -13 dB.

6.1.10.7 eUSB2

Table 457. eUSB2 General Guidelines

Note	Detail
eUSB2 routing shall comply to USB 2.0 DCR ECN for D+ and D- maximum series DC resistance (DCR) specification.	
Recommendations and considerations for routing near switch mode power supply, routing under inductor, routing under phase node, and routing under DrMOS	Refer to Technical White Paper Doc#: 726825

Table 458. eUSB2 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.381
2	Pair (P-N)	Within	Length (mm)	Yes	0.254
3	TX-TX	Total	Length (mm)	No	0
4	RX-RX	Total	Length (mm)	No	0
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

6.1.10.7.1 eUSB2 Native Camera Topology

Figure 186. eUSB2 Native Camera Topology Diagram

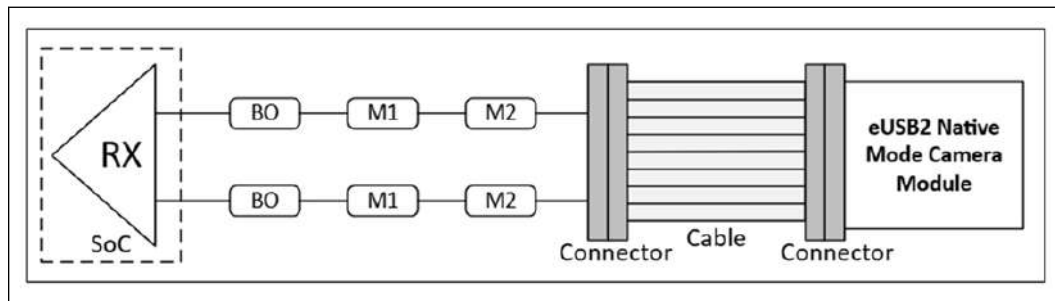


Table 459. eUSB2 Native Camera Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Camera module length	38mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.

Table 460. Max Number of vias

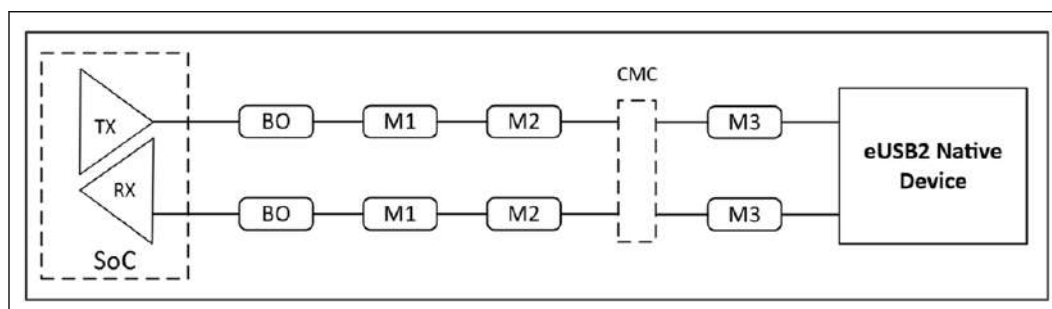
Signal Group	Max Number of Vias Allowed
Rx	3

Rx**Table 461. eUSB2 Native Camera Topology Rx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	13	
M1	MS, DSL, SL	326	$BO + M1 + M2 \leq 326$ mm
M2	MS, DSL, SL	13	0.1245 dB/in of differential loss at 240 MHz
Cable	Cable	400	0.111 dB/in of differential loss at 240 MHz. Cable differential impedance target should be 85 to 90 Ohms with +/- 10 Ohms variation.

Max Length Total (mm): 326

Max Length Total Note: This total length is excluding cable and camera module length. M1 and Cable lengths can be traded off according to their loss.

6.1.10.7.2 eUSB2 Native Device Down Topology**Figure 187. eUSB2 Native Device Down Topology Diagram****Table 462. eUSB2 Native Device Down Topology Notes**

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.
CMC	Optional. Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.

Table 463. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	3

Rx, Tx

Table 464. eUSB2 Native Device Down Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15	
M1	MS, DSL, SL	470	$BO + M1 + M2 + M3 + M4 \leq 470$ mm
M2	MS, DSL, SL	13	
M3	MS, DSL, SL	13	

Max Length Total (mm): 470

6.1.10.7.3 eUSB2 Repeater Pre-Channel Topology

Figure 188. eUSB2 Repeater Pre-Channel Topology Diagram

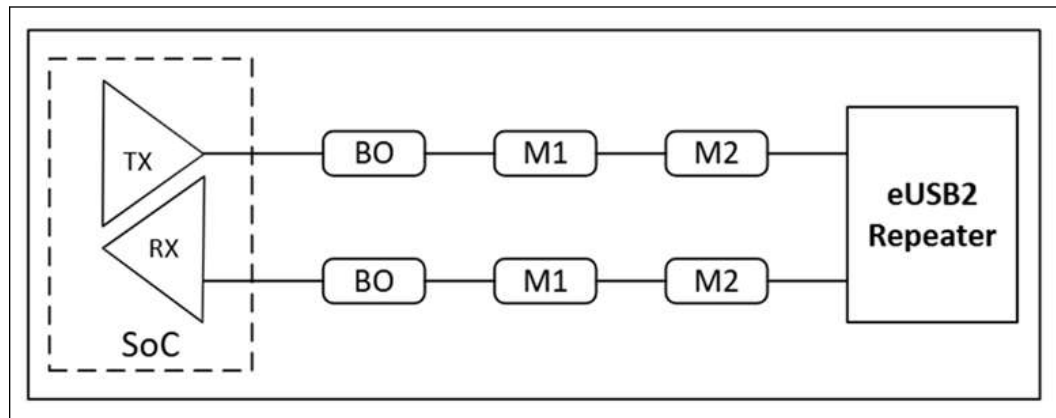


Table 465. eUSB2 Repeater Pre-Channel Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.

Table 466. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	3

Rx, Tx

Table 467. eUSB2 Repeater Pre-Channel Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	15	
M1	MS, DSL, SL	203	$BO + M1 + M2 \leq 203$ mm
M2	MS, DSL, SL	15	

Max Length Total (mm): 203

6.1.10.7.4 eUSB2 Repeater Post-Channel Device Down Topology

Figure 189. eUSB2 Repeater Post-Channel Device Down Topology Diagram

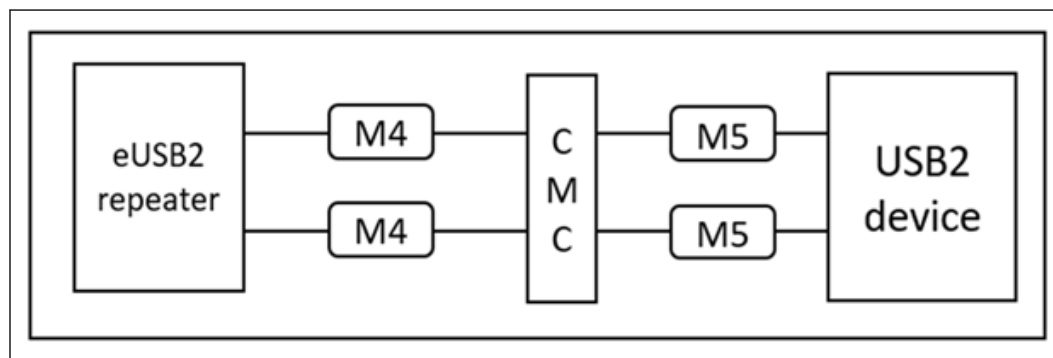


Table 468. eUSB2 Repeater Post-Channel Device Down Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.
CMC	Optional. Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.

Table 469. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	3

Rx, Tx

Table 470. eUSB2 Repeater Post-Channel Device Down Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	203	M4+M5 ≤ 203 mm
M5	MS, DSL, SL	25	

Min Length Total (mm): 13

Max Length Total (mm): 203

6.1.10.7.5 eUSB2 Repeater Post-Channel M.2 Topology

Figure 190. eUSB2 Repeater Post-Channel M.2 Topology Diagram

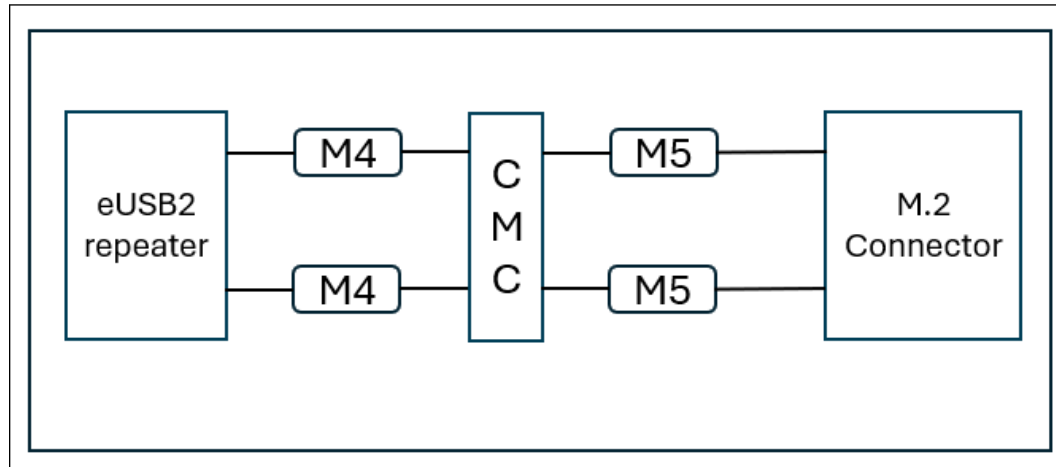


Table 471. eUSB2 Repeater Post-Channel M.2 Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.
CMC	Optional. Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.

Table 472. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	3

Rx, Tx

Table 473. eUSB2 Repeater Post-Channel M.2 Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	150	M4+M5 ≤ 150 mm
M5	MS, DSL, SL	25	

Min Length Total (mm): 13

Max Length Total (mm): 150

6.1.10.7.6 eUSB2 Repeater Post-Channel Type-A Topology

Figure 191. eUSB2 Repeater Post-Channel Type-A Topology Diagram

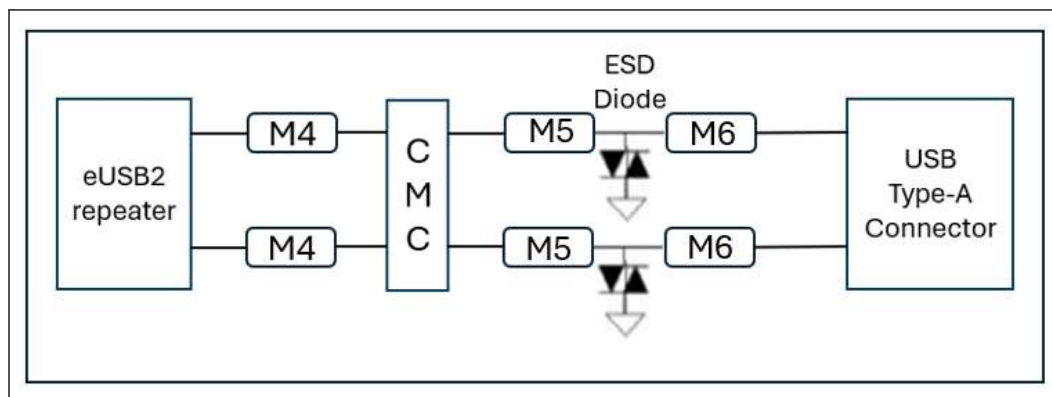


Table 474. eUSB2 Repeater Post-Channel Type-A Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.
CMC	Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection) for more details.

Table 475. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	3

Rx, Tx

Table 476. eUSB2 Repeater Post-Channel Type-A Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	203	$M4+M5+M6 \leq 203$ mm
M5+M6	MS, DSL, SL	25	

Min Length Total (mm): 13

Max Length Total (mm): 203

6.1.10.7.7 eUSB2 Repeater Post-Channel Type-A with MUX Topology

Figure 192. eUSB2 Repeater Post-Channel Type-A with MUX Topology Diagram

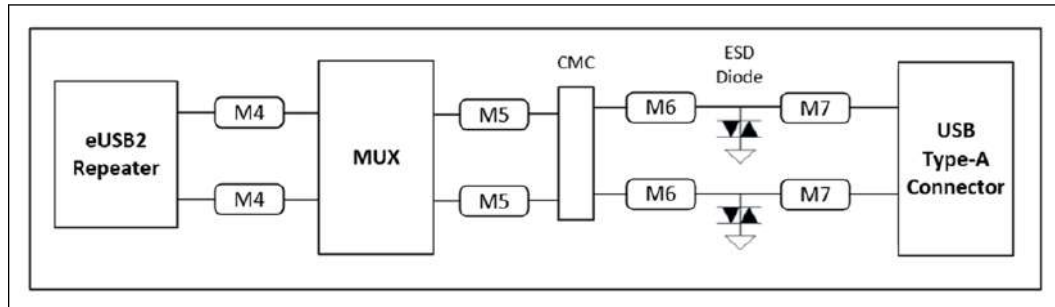


Table 477. eUSB2 Repeater Post-Channel Type-A with MUX Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.
Component R/C vs. channel length trade-off table	See Component Restrictions table.
Typical component R/C	On-state resistance and capacitance of 6 Ohm and 6 pF. Insertion loss ≥ -0.6 dB; Return loss ≤ -13 dB.
CMC	Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection) for more details.

Table 478. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	3

Rx, Tx

Table 479. eUSB2 Repeater Post-Channel Type-A with MUX Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	178	$M4+M5+M6+M7 \leq 178$ mm
M5	MS, DSL, SL	51	
M6+M7	MS, DSL, SL	25	

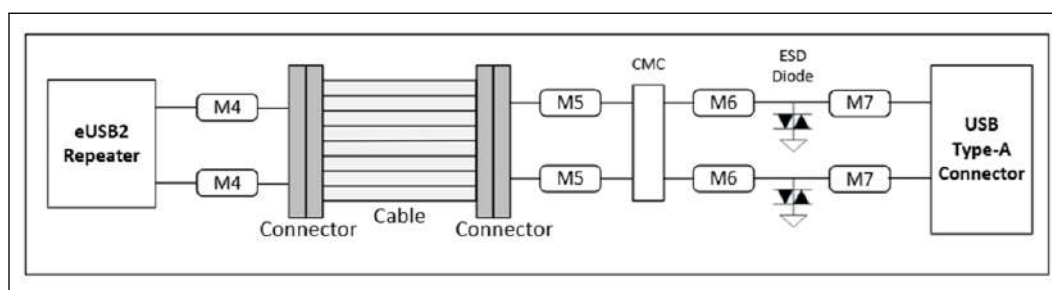
Min Length Total (mm): 13

Max Length Total (mm): 178

Max Length Total Note: See Component Restriction Table for component R/C vs. channel length trade-off recommendation.

Table 480. eUSB2.0 External_Back Panel with MUX Component Restriction Table

Motherboard total trace length (mm)	Max MUX on-state resistance, Ron	Max MUX on-state capacitance, Con	Equivalent MUX insertion loss at 240 MHz
178	4 Ω	4 pF	-0.45 to -0.46 dB
152	5 Ω	5 pF	-0.57 to -0.58 dB
127	6 Ω	6 pF	-0.74 to -0.75 dB
102	7 Ω	7 pF	-0.88 to -0.89 dB

6.1.10.7.8 eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable Topology**Figure 193. eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable Topology Diagram****Table 481. eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable Topology Notes**

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 4 vias.
Reference plane	Continuous GND recommended.
CMC	Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection) for more details.

Table 482. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	4

Rx, Tx

Table 483. eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	127	$M4+M5+M6+M7 \leq 127$ mm
Cable	Cable		Max cable loss: 2 dB at 2.5 GHz
M5	MS, DSL, SL	26	
M6+M7	MS, DSL, SL	25	

Min Length Total (mm): 13

Max Length Total (mm): 127

Max Length Total Note: Motherboard length (excluding cable) = 127 mm; Daughter card max length: 50.8 mm.

6.1.10.7.9 eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable without Daughter Card Topology

Figure 194. eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable without Daughter Card Topology Diagram

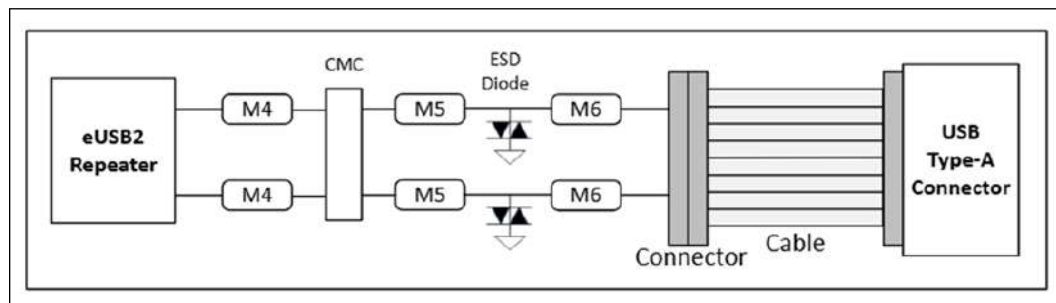


Table 484. eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable without Daughter Card Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 3 vias.
Reference plane	Continuous GND recommended.
CMC	Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection) for more details.

Table 485. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	3

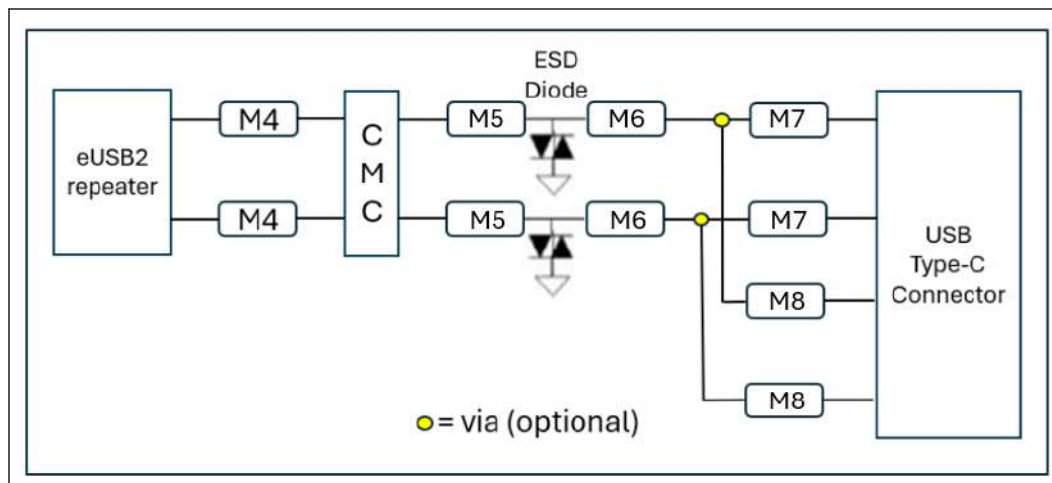
Rx, Tx**Table 486. eUSB2 Repeater Post-Channel Type-A Flex or Internal Cable without Daughter Card Topology Rx, Tx Routing Guidelines**

Segment	Trace Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	178	M4+M5+M6 ≤ 178 mm
M5+M6	MS, DSL, SL	25	
Cable	Cable		Max cable loss: 2 dB at 2.5 GHz

Min Length Total (mm): 13

Max Length Total (mm): 178

Max Length Total Note: Motherboard length (M4+M5+M6) = 178 mm

6.1.10.7.10 eUSB2 Repeater Post-Channel Type-C Topology**Figure 195. eUSB2 Repeater Post-Channel Type-C Topology Diagram****Table 487. eUSB2 Repeater Post-Channel Type-C Topology Notes**

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381mm .
Number of vias allowed	Max 4 vias (including one implemented to accommodate stub).
Reference plane	Continuous GND recommended.
Max stub length to Type-C connector (M8)	3.6 mm
CMC	Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection) for more details.

Table 488. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	4

Rx, Tx

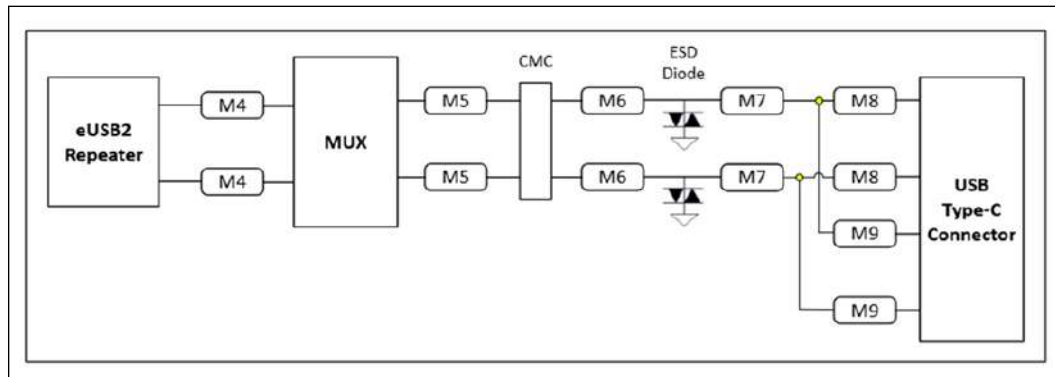
Table 489. eUSB2 Repeater Post-Channel Type-C Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	203	$M4+M5+M6+M7+M8 \leq 203$ mm
M5+M6+M7	MS, DSL, SL	25	
M8	MS, DSL, SL	3.6	

Min Length Total (mm): 13

Max Length Total (mm): 203

6.1.10.7.11 eUSB2 Repeater Post-Channel Type-C with MUX Topology

Figure 196. eUSB2 Repeater Post-Channel Type-C with MUX Topology Diagram

Table 490. eUSB2 Repeater Post-Channel Type-C with MUX Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm .
Number of vias allowed	Max 4 vias (including one implemented to accommodate stub).
Reference plane	Continuous GND recommended.
Max stub length for Type-C connector (M9)	3.6 mm
Component R/C vs. channel length trade-off table	See Module Component Restrictions figure.
Typical component R/C	On-state resistance and capacitance of 6 Ohm and 6 pF;

continued...

Note	Detail
	Insertion Loss ≥ -0.6 dB ; Return Loss ≤ -13 dB.
CMC	Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection) for more details.

Table 491. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	4

Rx, Tx**Table 492. eUSB2 Repeater Post-Channel Type-C with MUX Topology Rx, Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	178	$M4+M5+M6+M7+M8+M9 \leq 178$ mm
M5	MS, DSL, SL	51	
M6+M7+M8	MS, DSL, SL	25	
M9	MS, DSL, SL	3.6	

Min Length Total (mm): 13

Max Length Total (mm): 178

Max Length Total Note: See Component Restriction Table for component R/C vs. channel length trade-off recommendation.

Table 493. eUSB2.0 External_Back Panel with MUX Component Restriction Table

Motherboard total trace length (mm)	Max MUX on-state resistance, Ron	Max MUX on-state capacitance, Con	Equivalent MUX insertion loss at 240 MHz
178	4 Ω	4 pF	-0.45 to -0.46 dB
152	5 Ω	5 pF	-0.57 to -0.58 dB
127	6 Ω	6 pF	-0.74 to -0.75 dB
102	7 Ω	7 pF	-0.88 to -0.89 dB

6.1.10.7.12 eUSB2 Repeater Post-Channel Type-C Flex or Internal Cable Topology

Figure 197. eUSB2 Repeater Post-Channel Type-C Flex or Internal Cable Topology Diagram

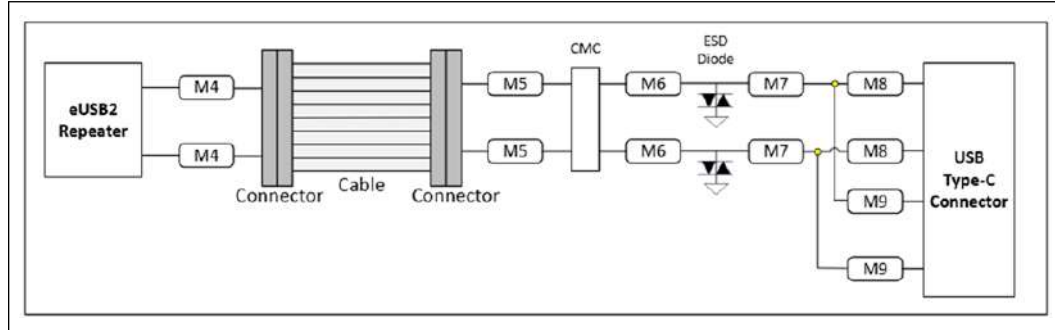


Table 494. eUSB2 Repeater Post-Channel Type-C Flex or Internal Cable Topology Notes

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.254 mm. Total length mismatch: 0.381 mm.
Number of vias allowed	Max 4 vias.
Reference plane	Continuous GND recommended.
Max stub length to Type-C connector (M9)	3.6 mm
CMC	Refer to Electromagnetic Compatibility chapter (Common Mode Choke Selection) for more details.
ESD	Refer to Electromagnetic Compatibility chapter (ESD Diode Selection) for more details.

Table 495. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	4

Rx, Tx

Table 496. eUSB2 Repeater Post-Channel Type-C Flex or Internal Cable Topology Rx, Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	127	
Cable	Cable		Max cable loss: 2 dB at 2.5 GHz
M5	MS, DSL, SL	26	
M6+M7+M8	MS, DSL, SL	25	
M9	MS, DSL, SL	3.6	

Min Length Total (mm): 13

Max Length Total (mm): 127

Max Length Total Note: Motherboard length (excluding cable) = 127 mm; Daughter card max length: 50.8 mm.

6.1.11 CNVio Bus Design Guidelines

The CNVio signals connects between the CRF RF companion module and the Processor. For details, refer to Wireless Connectivity Integration (CNVi) Design.

They are used as the main data bus for Wi-Fi* to transfer data between the Processor and the RF companion chip. The CNVio signals electrical characteristics are similar to the STEP (Serial Time Encoded Protocol) standard. The CNVio protocol used for this bus is proprietary.

CNVio3

CNVio3 is also known as STEP (Serial Time Encoded Protocol) interface, is a new proprietary differential interface to support Wi-Fi7 throughput requirements. CNVio3 is a serial digital interface and operates at 12Gbits/sec.

CNVio3 maps 2 bits per symbol into pulse widths. CNVio3 transmits output in a PWM signal. Each symbol has a positive pulse and a negative pulse. The transmission starts with a positive pulse and polarity is switched every symbol.

CNVio3 receiver decodes the pulse width back into the transmitted symbols.

Table 497. Signal Description

Signal Name	Description
CNV_RF_RESET#	RF companion (CRF) reset signal, active low.
CNV_BRI_DT	BRI bus TX.
CNV_BRI_RSP	BRI bus RX.
CNV_RGI_DT	RGI bus TX.
CNV_RGI_RSP	RGI bus RX.
CRF_CLKREQ	Processor to CRF wake indication
CNV_HSIF_TXP	CNVio bus Lane TX+
CNV_HSIF_TXN	CNVio bus Lane TX-
CNV_HSIF_RXP	CNVio bus Lane RX+
CNV_HSIF_RXN	CNVio bus Lane RX-
CNV_MFUART2_RXD	For CNVi and discrete connectivity: Optional WLAN/Bluetooth* WWAN co-existence signal (Input)
CNV_MFUART2_TXD	For CNVi and discrete connectivity : Optional WLAN/Bluetooth* WWAN co-existence signal (Output)
CNV_RCOMP	CNVi RCOMP is analog connection point.

6.1.11.1 CNVio Design Guidelines

6.1.11.1.1 CNVio3

Description: Serial Time Encoded Protocol (STEP) is a non-NRZ, non-PAM signaling architecture, which is referred to as CNVio3 in product design guidelines. CNVio3 (STEP) uses pulse width modulation and operates at a lower fundamental frequency than NRZ for similar data rates. It is a DC coupled asynchronous bus having high bandwidth, low power and small exit latency from low power states. The current version of CNVio3 has a data transfer rate of 12 Gbps to support the Intel WiFi7 integrated solution. Reflections and crosstalk are the primary performance limiters and hence CNVio3 requires a very clean channels.

Figure 198. CNVio3 Breakout and Break-in Guideline

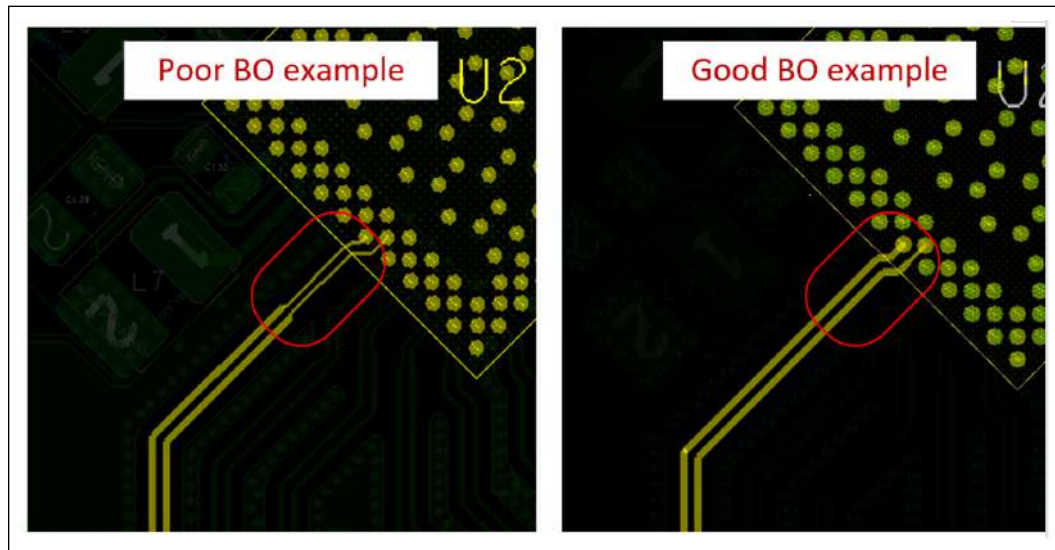


Figure 199. Module Down Pad Dimension and Voiding Requirements

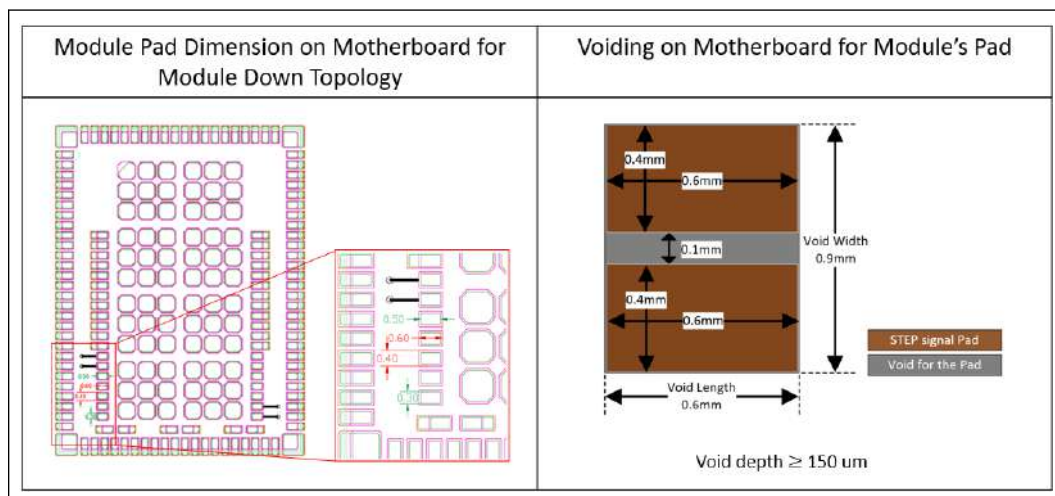


Figure 200. M.2 Connector Pad Voiding Recommendation

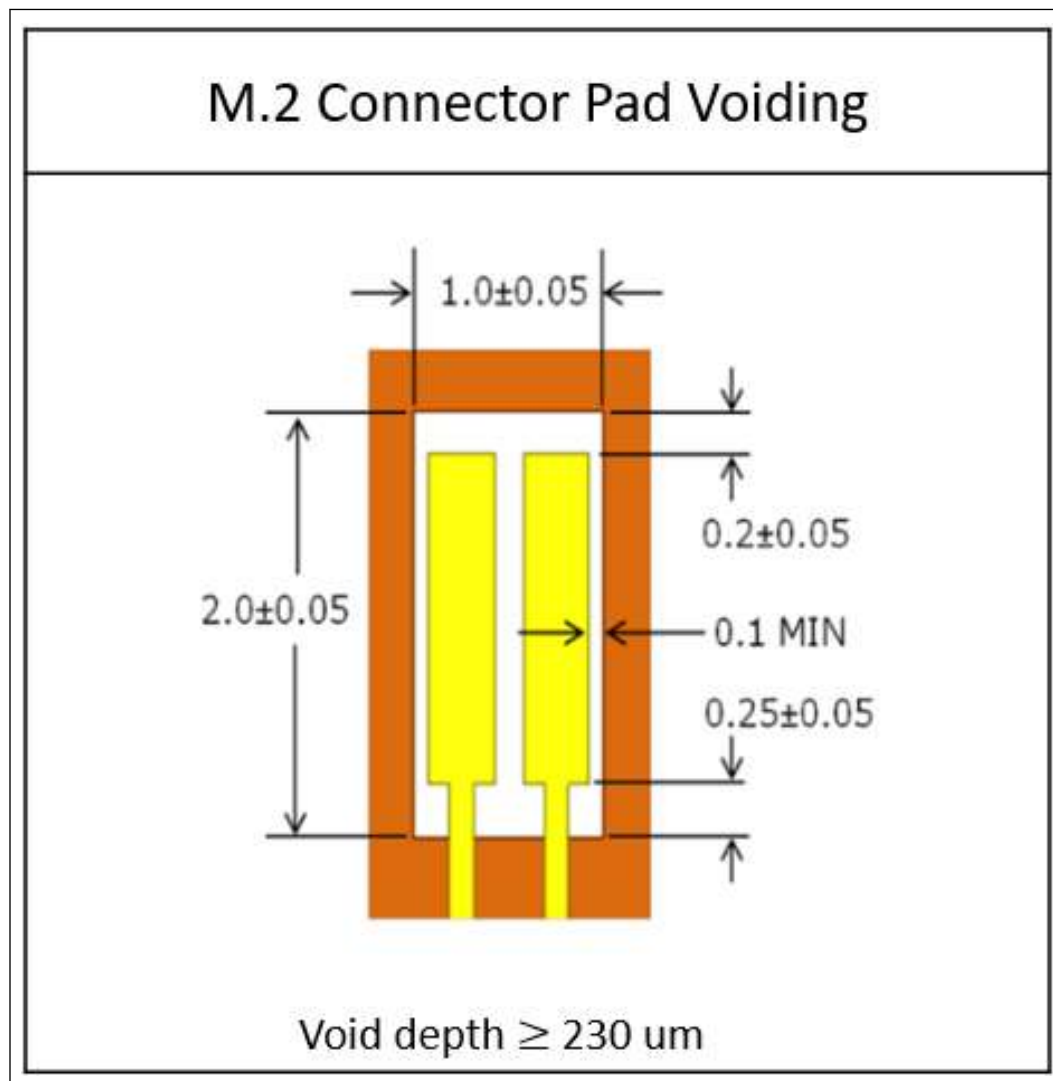


Figure 201. Void Above and Below Buried Vias PTH Pads in Type-4 Stackup

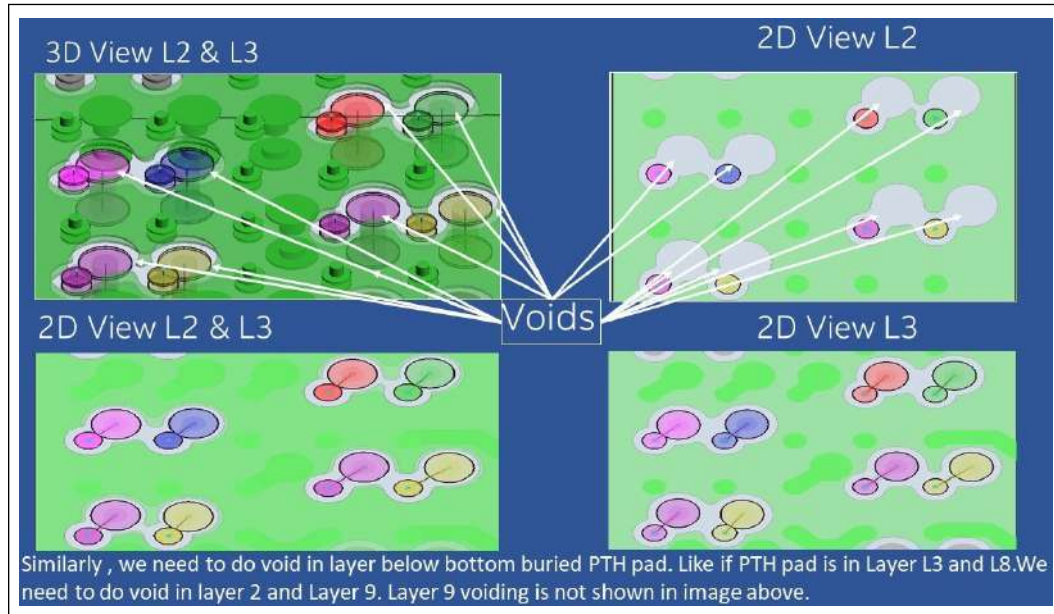


Figure 202. CNVIO3 Spacing Recommendation

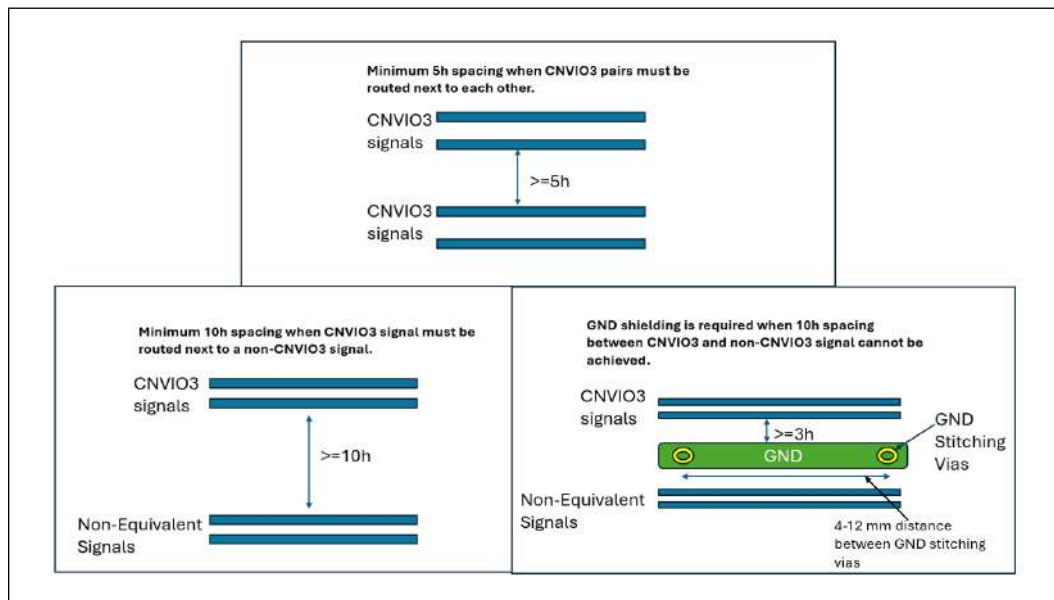


Table 498. CNVIO3 General Guidelines

Note	Detail
Length matching between P and N within a diff. pair	Within same layer mismatch: 0.178 mm
continued...	

Note	Detail
	Total length mismatch: 0.127 mm
Single ended equivalent impedance of differential channel	For a differential channel transmission line impedance of 80 Ω , the equivalent required single ended transmission line impedance should be 40 Ω .
Voiding recommendation for mainstream stackup	It is recommended to void reference plane for all component pads such as M.2 Connector pads/Module pads to optimize the impedance matching in the channel. Recommended voiding depth: - For M.2 connector pads: ≥ 230 μm to the closest ground reference layer. - For module pads: ≥ 150 μm to the closet ground reference layer.
Number of vias allowed	Max 2 vias.
Max via stub length for Type 3 PTH Vias	Max PTH via stub length is 0.215 mm.
Reference plane	Continuous GND is required on both sides.
Fiberweave	Must adhere to fiberweave related best routing practices (#406926).
CNVio3 BO/BI routing guide	CNVio3 BO/BI at the device requires the same impedance as main route. No neckdown into pin field. Refer to "CNVio3 BO/BI at Device Guideline" for more details.
Module pad dimension	Module pad dimension is 0.4 mm x 0.6 mm. Refer to the "Module Down Pad Dimension and Voiding Requirements" diagram.
Essential guidelines for routing near switch mode power supply, routing under inductor, routing under phase node, and routing under DrMOS	Do not route CNVio3 traces on any layer of the PCB under phase nodes, Inductor and DrMOS.
Differential Via optimization	Differential Impedance of a via should be centered around the characteristic impedance of the channel (± 10 Ω tolerance) for Optimum performance.
Maximum via stub for Type 4 Buried PTH Via	0.085 mm
Voiding for Type4 buried PTH vias	Need to provide void above and below buried via's PTH pads. Voids need to be 75 μm more in radius than pad radius. Void is required as shapes over big pads cause a capacitive dip in impedance which results in lower margins. Please refer to the image provided as a reference.
RCOMP for CNVio3	CNV_RCOMP need to connect to external resistor (200 Ω 1%)
Maximum DC resistance for the trace on the motherboard	$BO + M1 + M2 \leq 203$ mm, Max DC Resistance spec = 2 Ohms (typical) .Note: This DCR spec is calculated assuming Cu thickness as per PDG stackups only. Any reduction in Cu thickness w.r.t. PDG for the given max length can cause the DCR spec to be violated.Refer to #913498 for more details

Table 499. CNVio3 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.127
2	Pair (P-N)	Within	Length (mm)	Yes	0.178

CNVio3 M.2 Topology

Figure 203. CNVio3 M.2 Topology Diagram

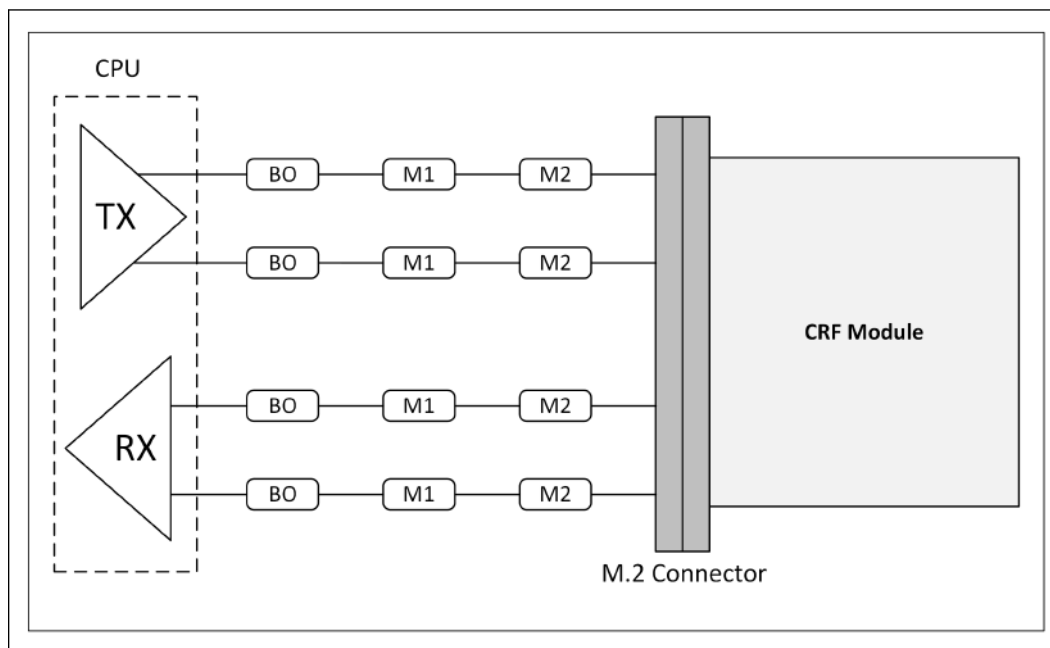


Figure 204. Ground Plane Voiding Under M.2 Connector Pads

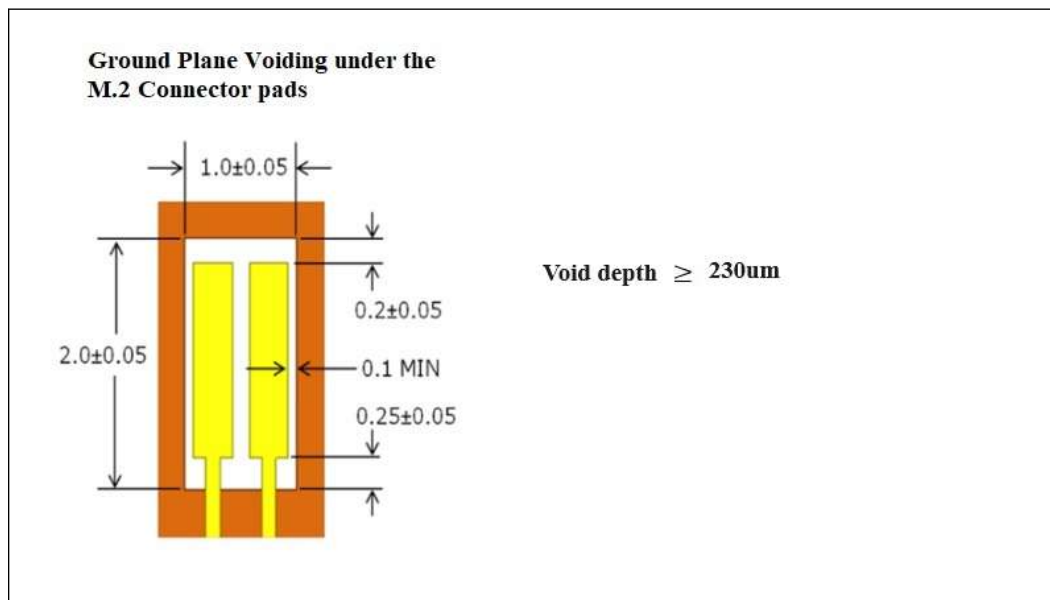


Figure 205. M.2 Connector Pin Assignment

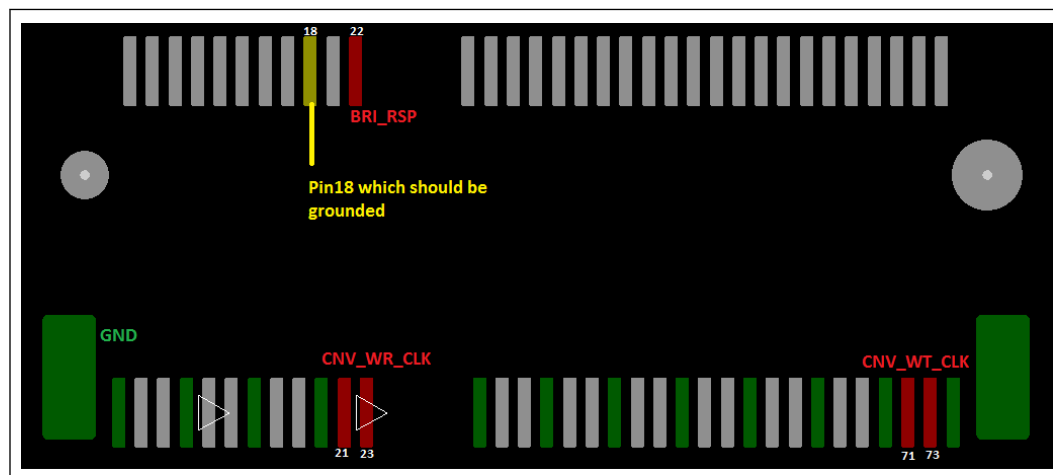


Table 500. CNVio3 M.2 Topology Notes

Note	Detail
M.2 Connector Pin 18	Pin 18 of the M.2 connector must be grounded on the motherboard side to prevent common mode coupling from the BRI signal onto CNVio3 signals, possibly resulting in functional failure of the CNVio3 bus.
Maximum Via stub allowed for Type-3 PTH Via	215 um
Maximum via stub allowed for Type-4 Buried PTH via	85 um. Via optimization is needed for higher via stub.

Table 501. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	2	Total Channel

Rx,Tx

Table 502. CNVio3 M.2 Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL	13	
M1	SL	203	BO + M1 + M2 ≤ 203 mm, Max DC Resistance spec = 2 Ohms (typical) .Note: This DCR spec is calculated assuming Cu thickness as per PDG stackups only. Any reduction in Cu thickness w.r.t. PDG for the given max length can cause the DCR spec to be violated.Refer to #913498 for more details
M2	MS	17	

Max Length Total (mm): 203

CNVio3 Module Down Topology

Figure 206. CNVio3 Module Down Topology Diagram

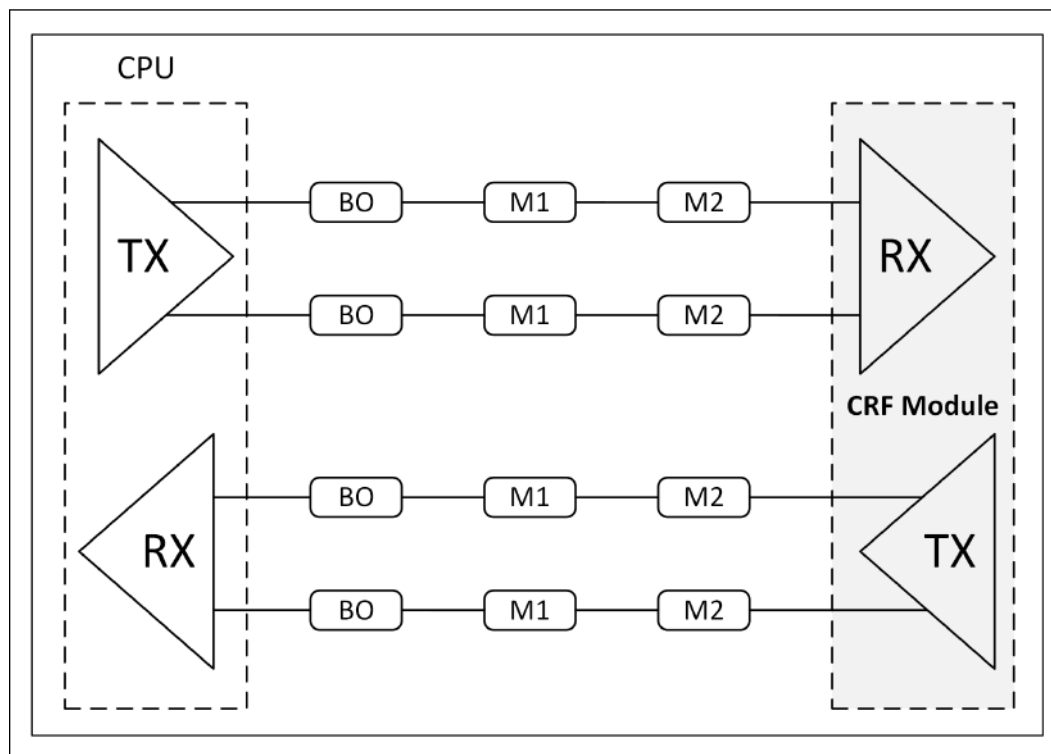


Figure 207. Module Down Pad Dimension and Voiding Requirements

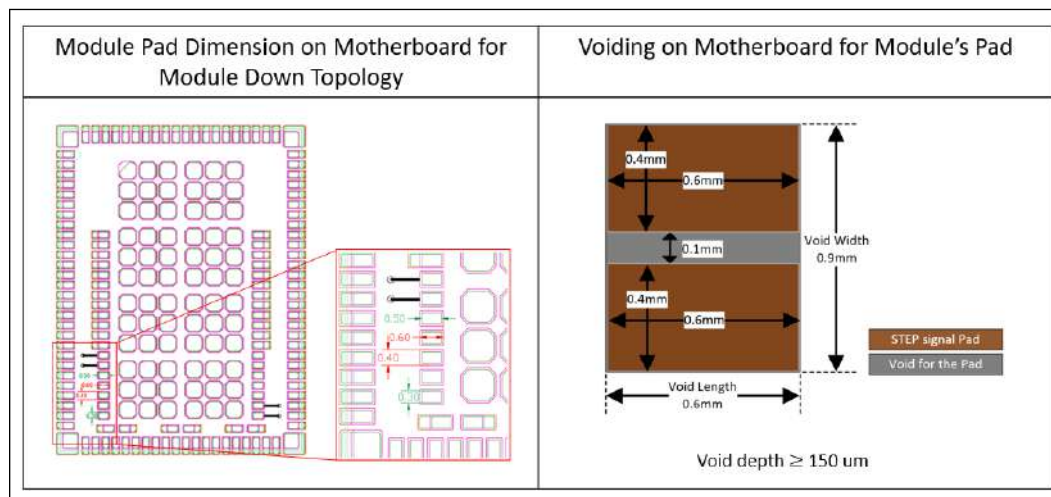


Table 503. CNVio3 Module Down Topology Notes

Note	Detail
Module pad dimension	Module pad dimension is 0.4 mm x 0.6 mm. Refer to the "Module Down Pad Dimension and Voiding Requirements" diagram.
Maximum Via stub allowed for Type-3 PTH vias	215 um
Maximum Via stub allowed for Type-4 Buried PTH Vias	85 um. Via Optimization is needed for higher via stub requirements.

Table 504. Max Number of vias

Signal Group	Max Number of Vias Allowed
Rx, Tx	2

Rx,Tx**Table 505. CNVio3 Module Down Topology Rx,Tx Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL	13	
M1	SL	203	BO + M1 + M2 ≤ 203 mm, Max DC Resistance spec = 2 Ohms (typical) .Note: This DCR spec is calculated assuming Cu thickness as per PDG stackups only. Any reduction in Cu thickness w.r.t. PDG for the given max length can cause the DCR spec to be violated.Refer to #913498 for more details
M2	MS	17	

Max Length Total (mm): 203

6.1.11.2 Tools for Analysis

Only CNVio2 can be simulated with 3rd party tools. For CNVio3, customers should use iPDS to compare the quality of the channels with the reference channel.

6.1.11.3 Tools for Debug

CNVio can be tested with the CNVio test tool such as the CITU tool. This tool will test the design marginality and can detect bus errors as well as very marginal signal integrity.

6.1.11.4 Probing

It is recommended to avoid adding any probing pads or test points to the CNVio traces. However in cases where probing the CNVio bus is needed it is recommended to use a special soft touch probe. A special CNVio analyzer U4421A from Agilent (Keysight) can be used with a special cable and retention module. When designing for using this probing set, the probe footprint is inserted within the CNVio traces while allowing the traces to maintain good impedance characteristics. An illustration of the probe and cable setup is shown in figure below.

CNVi BRI and RGI 1-Load Topology

Figure 209. CNVi BRI and RGI 1-Load M.2 Connector Topology Diagram

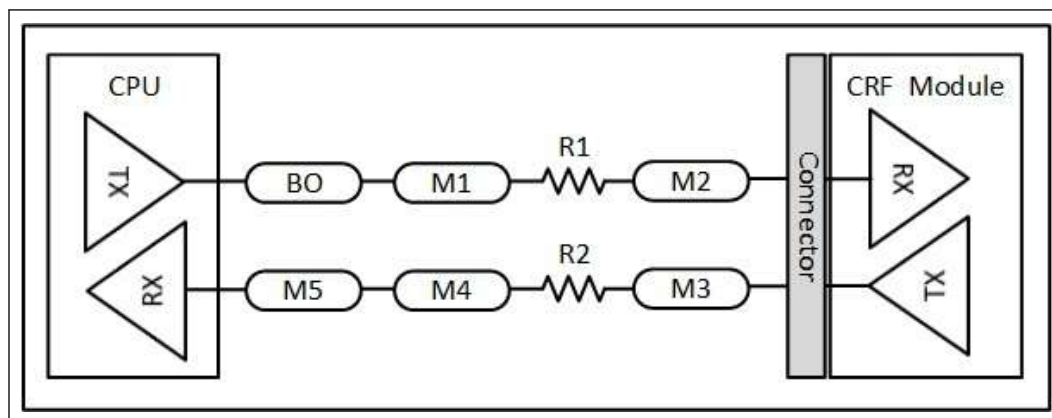


Figure 210. CNVi BRI and RGI 1-Load Module Down Topology Diagram

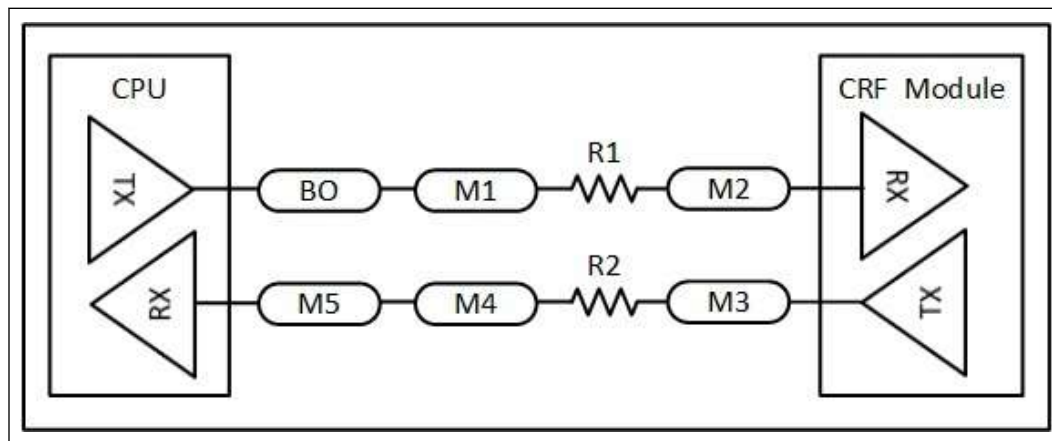


Table 506. CNVi BRI and RGI 1-Load Topology Notes

Note	Detail
Max frequency	40 MHz
M4 (BRI_RSP)	For microstrip and stripline routing only on platform, the total main board trace length can be extended up to 253.7 mm. For that, it is recommended to implement $M4 < 228.3$ mm (max).
Reference plane	[1] Continuous GND is recommended for microstrip line. [2] Dual continuous GND is recommended for strip line. [3] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [4] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
R1	$10\ \Omega \pm 5\%$ for max total length ≥ 25.4 mm.

continued...

Note	Detail
	To be placed 12.7 mm (can be extended to 25.4 mm) from the processor for CNV_BRI_DT and CNV_RGI_DT signal.
R2 (RGI)	For Wi-Fi 7 BE201, Wi-Fi 7 BE211 and Wi-Fi 7 BE213; 27 $\Omega \pm 5\%$ for max total length ≥ 25.4 mm. To be placed 12.7 mm from the connector for CNV_RGI_RSP signal.
R2 (BRI)	For Wi-Fi 7 BE211 and Wi-Fi 7 BE213; 27 $\Omega \pm 5\%$ for max total length ≥ 25.4 mm. For Wi-Fi 7 BE201; [1] 10 $\Omega \pm 5\%$ for max total length ≥ 101.6 mm [2] 27 $\Omega \pm 5\%$ for max total length < 101.6 mm To be placed 12.7 mm from the connector for CNV_BRI_RSP signal.
Length restriction for full Dual Stripline routing	M1 + M2 = 63.5 mm (max). M3 + M4 = 63.5 mm (max).
Trace spacing between DATA and DATA signals	0.250 mm
Signal name/ list	CNV_BRI_DT, CNV_RGI_DT, CNV_BRI_RSP, CNV_RGI_RSP.

Table 507. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	4	Total Channel
Tx	4	Total Channel

Segment Lengths for BRI_DT, RGI_RSP and RGI_DT signals
Table 508. CNVi BRI and RGI 1-Load Topology Segment Lengths for BRI_DT, RGI_RSP and RGI_DT signals Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
B0	MS, DSL, SL	12.7
M1	MS, DSL, SL	12.7
M2	MS, DSL, SL	228.6
M3	MS, DSL, SL	12.7
M4	MS, DSL, SL	228.6
M5	MS, DSL, SL	12.7

Max Length Total (mm): 254

Segment Lengths for BRI_RSP signal

Table 509. CNVi BRI and RGI 1-Load Topology Segment Lengths for BRI_RSP signal Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
M3	MS, DSL, SL	12.7
M4	MS, DSL, SL	152.4
M5	MS, DSL, SL	12.7

Max Length Total (mm): 177.8

Max Length Total Note: For routing involving DSL, it need to limit to 177.8 mm max total length. For routing with MS and SL, it can be extended up to 254 mm max total length.

6.1.11.6 Termination Requirements

Due to special power up sequencing of the CRF and I/O requirements, the Processor I/O shall be terminated according to the following considerations:

- The BRI_RSP and RGI_RSP signals from CRF to the Processor have I/O buffers which are not powered when the CRF is initially powered up. The supply to these buffers will only come up following the first clock request from the Processor. To avoid floating input, internal pull-ups of 20K are provided at the Processor I/O pins; ***no external pull-ups are required***.
- The **CNV_RF_RESET#** signal from the Processor to CRF is used as the main reset signal to the CRF during boot. This signal must have a stable value starting from initial power up of the platform and before the CRF power comes up. To ensure that the CNV_RF_RESET# signal is glitch free it is recommended that a **pull down** resistor be connected between the Processor pin and ground. A 75K ohm resistor to ground will ensure the stability of the signal. Since CNV_RF_RESET# is normally held at 1.8V it is expected to consume negligible amount of power.

6.1.11.7 RF Companion Specifications

For the RF Companion POR related to the platform, refer to the CCG 5Q Wireless Roadmap/Platform Matrix Customer Communication (# [601177](#)).

6.1.12 UFS 4.0

Description: UFS in NVL support Gear5 data rate.

Table 510. UFS 4.0 General Guidelines

Note	Detail
Stub requirement	Channel and via stub requirement must meet < 254 um for both Tx and Rx signal pairs.
Component size requirement (CMC)	Strongly recommended to use 0201 components for better impedance control.
Component voiding	All component pads must be voided on L2 and with GND reference on L3. Suggest having pad voiding size of 50.8 um overcut. Refer to Component Pad Voiding diagram.
continued...	

Note	Detail
Component routing	Signal pair to entry/ exist from components pad symmetrically. Maintain differential routing to meet trace target impedance whenever possible.
Signal PTH/ via transition	Differential pair voiding needed for each signal pair transition with recommended antipad size of 0.75 mm. Suggest having two symmetrical GND stitching vias for each signal pair transition.
Routing scheme	Non-interleaved breakout (B0) is required to mitigate near-end crosstalk. The main route (MR) supports both interleaved routing and non-interleaved routing for maximum flexibility on stripline (SL). For microstrip (MS), interleaved routing scheme is recommended.
Reference plane	Continuous GND is recommended. If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). If non-continuous power referencing is unavoidable on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt), but stitching caps are required over each plane split and need to be placed within 6 mm of a signal running across the plane split.

Table 511. UFS 4.0 Length Matching

Matching Group Id	Matching Groups	Total/Within Layer	Length/Delay	Required	Maximum Mismatch
1	Pair (P-N)	Total	Length (mm)	Yes	0.127
2	Pair (P-N)	Within	Length (mm)	Yes	0.127
3	TX-TX	Total	Length (mm)	Yes	12.7
4	RX-RX	Total	Length (mm)	Yes	12.7
5	TX-RX	Total	Length (mm)	No	0
6	DATA-CLK	Total	Length (mm)	No	0

6.1.12.1 UFS 4.0 Gear 5 Device Down Topology

Figure 211. UFS 4.0 Gear 5 Device Down Topology Diagram

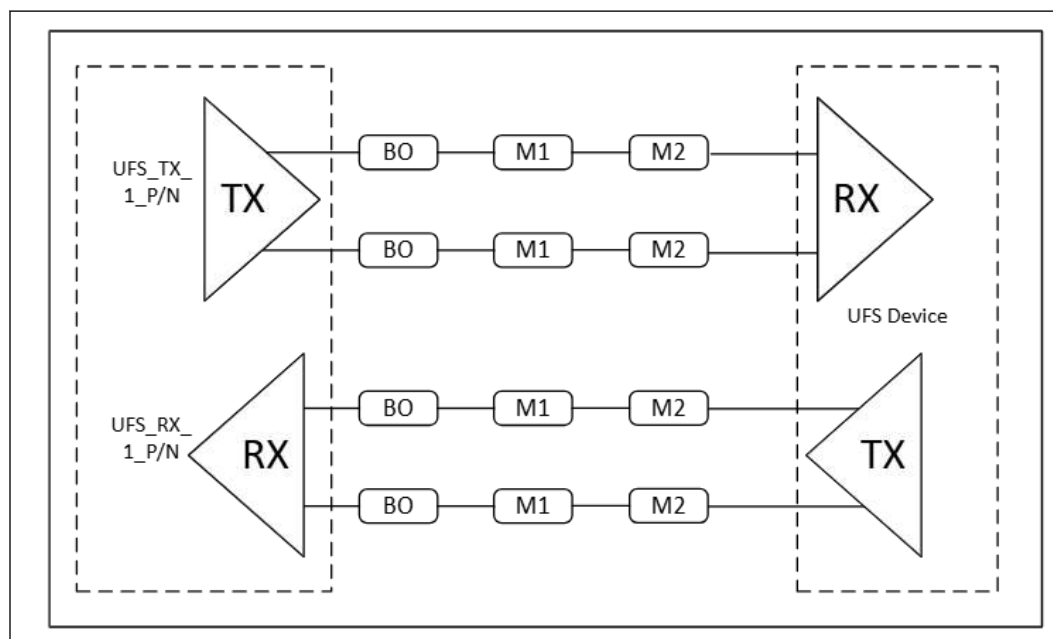


Table 512. UFS 4.0 Gear 5 Device Down Topology Notes

Note	Detail
Reference plane	Continuous ground recommended.

Table 513. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	2	Total Channel

Rx,Tx

Table 514. UFS 4.0 Gear 5 Device Down Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	SL	12.7	
M1	SL	127	Max length is $BO + M1 + M2 \leq 127$ mm
M2	MS, SL	12.7	

Min Length Total (mm): 63.5

Min Length Total Note: $BO + M1 + M2 \geq 63.5$

Max Length Total (mm): 127

Max Length Total Note: Max length is $BO + M1 + M2 \leq 127$ mm

6.1.12.2 UFS 4.0 Gear 4 Device Down Topology

Figure 212. UFS 4.0 Gear 4 Device Down Topology Diagram

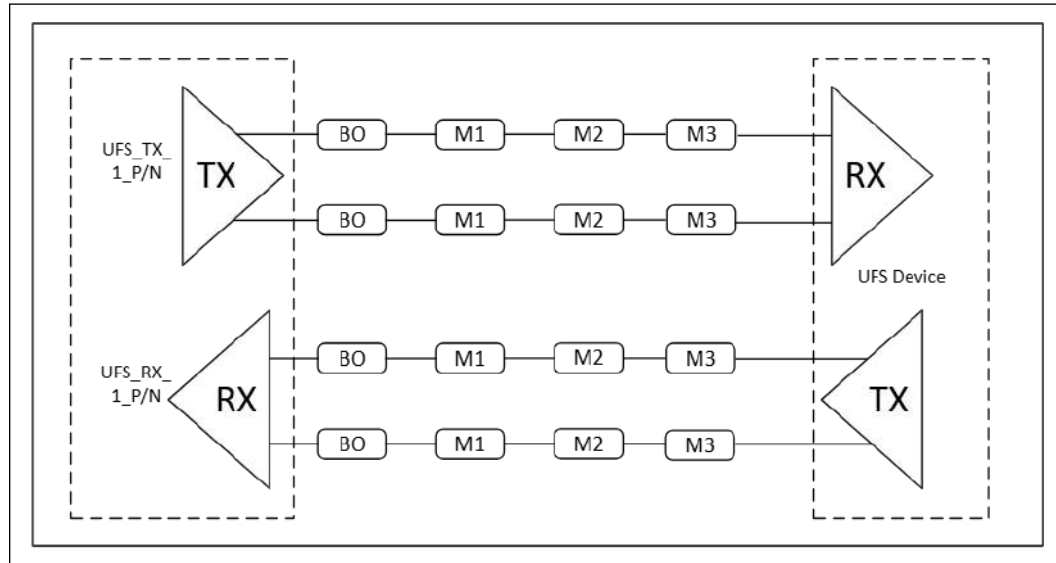


Table 515. UFS 4.0 Gear 4 Device Down Topology Notes

Note	Detail
Reference plane	Continuous ground recommended.
Breakout length and spacing	An initial breakout segment of 4 mm in length with a tighter pair-to-pair spacing of 0.11 mm - 0.16 mm is allowed.

Table 516. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	3	Total Channel

Rx,Tx

Table 517. UFS 4.0 Gear 4 Device Down Topology Rx,Tx Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	SL	12.7	
M1	SL	150	Max length is BO+M1+M2+M3 ≤ 150 mm
M2+M3	MS, SL	12.7	

Min Length Total (mm): 63.5

Min Length Total Note: $BO+M1+M2+M3 \geq 63.5\text{mm}$

Max Length Total (mm): 150

Max Length Total Note: Max length is $BO+M1+M2+M3 \leq 150\text{ mm}$

6.2 Low Speed I/Os

6.2.1 Reference Documents

Document Name	Document Number
Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper	726825
Intel® Smart Sound Technology ACPI Configuration BIOS Specification	609917
Intel® Touch Host Controller Integration Guide	778513

6.2.2 Pin Straps

Refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for details on signal description and strapping options.

6.2.3 Asynchronous and Sideband Signals Design Guidelines

This section describes the topologies and layout recommendations for the asynchronous signals. Refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.

6.2.3.1 Signal Descriptions

Signal Groups

Table 518. Asynchronous and Sideband Legacy Signal Group

Signal Name	Description
FORCEPR#	FORCEPR# goes active when the processor temperature monitoring sensor(s) detects that the processor has reached its maximum safe operating temperature. This indicates that the processor Thermal Control Circuit (TCC) has been activated, if enabled, activate the TCC when the temperature limit of the VR is reached.
CATERR#	This signal indicates that the system has experienced a catastrophic error and cannot continue to operate. The processor will set this for non-recoverable machine check errors or other unrecoverable internal errors. On the processor, CATERR# is used for signaling the following types of errors: Legacy MCERRs - CATERR# is asserted for 16 BCLKs. Legacy IERRs - CATERR# remains asserted until warm or cold reset.
THERMTRIP#	Processor output signal: Processor Asserted by the processor to indicate a thermal trip event which may cause processor sever damage, causing the processor to transition to an S5 state.
BPM[1:0]	Processor bi-directional signal that indicates the status of breakpoints and programmable counters used for monitoring processor performance
SVID VIDSOUT VIDSCK VIDALERT#	VIDALERT#, VIDSCLK, and VIDSCLK comprise a three signal serial synchronous interface (SVID) used to transfer power management information between the processor and the voltage regulator controllers. Alert signal must be routed between Clk and Data signals to minimize Cross-Talk.

6.2.3.2 Topology Guidelines

This section describes topologies and layout recommendations for the legacy signals. Although these signals toggle with relatively low frequency, most of them have very high-edge rates.

WARNING

Inappropriate routing or lack of termination can seriously decrease signal quality and lead to electrical specification violations and even logical system failures. The following guidelines apply to all legacy signals described in this section.

- Watch for termination recommendations. If any of the signals that require motherboard termination are pulled-up to a voltage higher than VCCPRIM_1P8, the reliability and power consumption of the processor may be affected. Therefore, it is very important to follow the recommended pull-up voltage for these signals.
- Changing reference plane is not recommended and may cause signal integrity degradation. In any case, if such routing cannot be avoided, use stitching vias and bypass capacitors between the reference planes near the layer transition.
- General trace spacing requirements (for 40 Ω for Type3, 45 Ω for Type4 characteristic impedance traces) specified in the following table.

Table 519. Asynchronous and Sideband Signal General Routing Guideline

Trace Type	Stack-up (MS/SL/DSL)	Units	Minimum Spacing (S)
Microstrip	MS	mm	0.125
Stripline	SL	mm	0.125
Dual-Stripline	DSL	mm	0.125

6.2.3.3 FORCEPR# Topology

The signal can be driven either by Embedded Controller (EC) or by VR. If connected from FORCEPR# to particular manufacturer temp sensor and not VR as described in the topology, such FORCEPR# connection must have the drive capabilities as defined in the future available VR specification.

Consider the following:

1. FORCEPR# is capable of toggling hundreds of times per second to signal a hot temperature condition.
2. If FORCEPR# is not used, then it must be terminated with a 1K Ω $\pm$ 5% pull-up resistor to VCCPRIM_1P8.

The Embedded Controller (EC) output DC levels are typically 0 V and 3.3 V, while processor input DC levels are 0 V and VCCPRIM_1P8. Processor's high voltage level is lower than Embedded Controller (EC)'s high voltage level, therefore an onboard voltage level shifter is required. An inverting logic level shifter must be used. If Embedded Controller (EC) asserts FORCEPR# signal by driving high, the level shifter must invert it and drive the processor side FORCEPR# low.

NOTE

The Topology demonstrated is designed to be unidirectional, the processor is the receiver.

- A device is any (OD) driver that is connected to the FORCEPR# signal, for example VR.

- R1-n series resistors are needed to keep the FORCEPR# signal at safe level (reduce Under-shoot) if a device and/or Inv OD pull-down resistor is too low (driver is too strong). The total pull-down resistance of each device should be between 75 and 200 Ω . Total pull-down resistance is defined as the sum of internal pull-down resistance of the device and the series resistor R_i
- Examples:
 - If internal pull-down resistor of device1 is 75 Ω , no additional (on board.) series resistor is needed, R_1 can be omitted.
 - If internal pull-down resistor of device2 is 15 Ω , additional (on board) series resistor is needed to increase the total pull-down resistance of device2, $60 < R_2 < 185$ (R_2 can be 70 Ω).
 - If Embedded Controller (EC) requires an inverter - an open drain inverter that meets Embedded Controller (EC) logical LOW/HIGH voltage levels needs to be selected. The OD PU resistor at the inverter output needs to be selected to meet the invert R_{ON} and satisfy the requirements of the next stage.

6.2.3.3.1 FORCEPR# Topology

Figure 213. FORCEPR# Topology Diagram

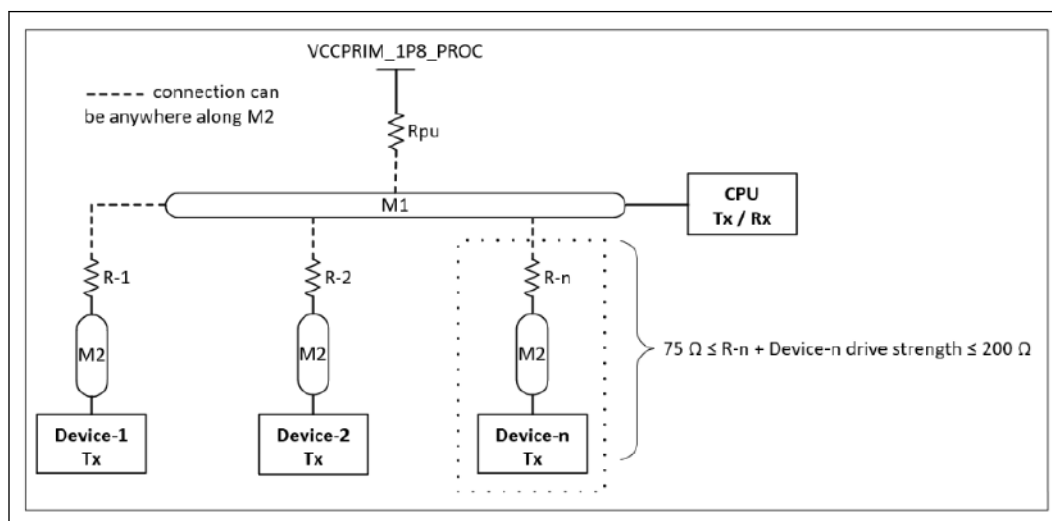


Table 520. FORCEPR# Topology Notes

Note	Detail
R-1, R-2, R-n	$75 \Omega \leq R-n + \text{Dev-n drive strength} \leq 200 \Omega$. [1] Drive strength of each device should be controlled within 75 Ω to 200 Ω , use an additional resistor if the device drive strength is too strong. [2] Optionally, if the device drive strength is $\geq 75 \Omega$ (weak), R-n is not needed. [3] Recommended to be placed < 127 mm from respective device.
Rpu	2.2 k Ω $\pm$ 20%. Total length can be extended up to 1524 mm. Refer to "FORCEPR# Topology Routing Length vs Rpu Guideline Table" for details.
Signal name/ list	FORCEPR#

Segment Lengths

Table 521. FORCEPR# Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
B0 + M1 + M2	MS, DSL, SL	889	Max length for B0 < 127 mm and M2 < 127 mm. Total length can be extended up to 1524 mm. Refer to "FORCEPR# Topology Routing Length vs Rpu Guideline Table" for details.

Max Length Total (mm): 889

Table 522. FORCEPR# Topology Routing Length vs Rpu Guideline Table

Max Total Board Routing Length (mm)	Max # of Connected Devices	Max Rpu
254	4	5.6 kΩ ± 10%
254	5	4.7 kΩ ± 10%
381	5	3.9 kΩ ± 10%
508	5	3.3 kΩ ± 10%
762	5	2.7 kΩ ± 10%
889	5	2.2 kΩ ± 10%
1143	7	1.8 kΩ ± 10%
1524	7	1.5 kΩ ± 10%

6.2.3.4 CATERR# Topology Guidelines

CATERR# indicates system has experienced a catastrophic error and cannot continue to operate. The processor will set this for non-recoverable machine check errors and other internal unrecoverable errors. CATERR# is an asynchronous open drain driver signal with no on-die termination (ODT). This signal can be connected to a LED if desired. Also, this signal should have an exposed test point for easy debug access.

6.2.3.4.1 CATERR# Topology

Figure 214. CATERR# Topology Diagram

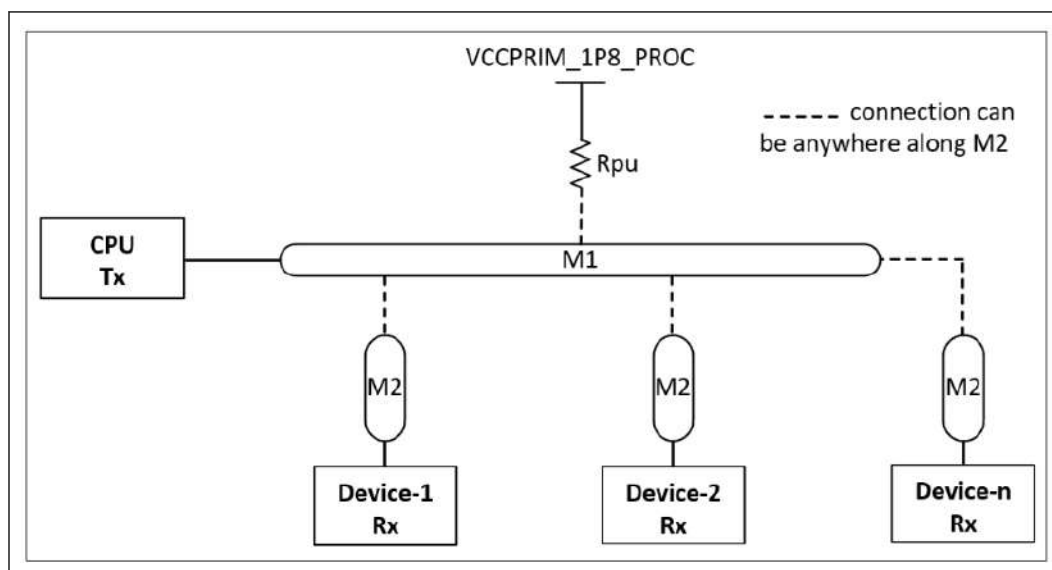


Table 523. CATERR# Topology Notes

Note	Detail
Rpu	2.2 kΩ ± 20%. Total length can be extended up to 1524 mm. Refer to "CATERR# Topology Routing Length vs Rpu Guideline Table" for details.
Signal name/ list	CATERR#

Segment Lengths

Table 524. CATERR# Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
B0 + M1 + M2	MS, DSL, SL	1016	Max length for B0 < 127 mm. Total length can be extended up to 1524 mm. Refer to "CATERR# Topology Routing Length vs Rpu Guideline Table" for details.

Max Length Total (mm): 1016

Table 525. CATERR# Topology Routing Length vs Rpu Guideline Table

Max Total Board Routing Length (mm)	Max # of Connected Devices	Max Rpu
254	3	5.6 kΩ ± 10%
254	5	4.7 kΩ ± 10%
381	5	3.9 kΩ ± 10%
508	5	3.3 kΩ ± 10%
762	5	2.7 kΩ ± 10%
continued...		

Max Total Board Routing Length (mm)	Max # of Connected Devices	Max Rpu
1016	5	2.2 k Ω $\pm$ 10%
1270	5	1.8 k Ω $\pm$ 10%
1524	5	1.5 k Ω $\pm$ 10%

6.2.3.5 THERMTRIP# Topology Guidelines

RLS series resistor is needed to keep the THERMTRIP# signal at safe level (reduce Under-shoot) if OD LS pull-down resistor is too low (driver is too strong). The total pull-down resistance of OD LS should be between 55 and 200 Ω . Total pull-down resistance is defined as the sum of internal pull-down resistance of the OD LS and the series resistor RLS. Examples:

- If internal pull-down resistor of OD LS is 55 Ω , no additional (on board) series resistor is needed => RLS can be omitted.
- If internal pull-down resistor of OD LS is 15 Ω , additional (on board) series resistor is needed to increase the total pull-down resistance of OD LS in range: $40 < R_{LS} < 185$, in order to meet the guideline: $55 < R_{LS} + R_{ODLS} < 200$.

6.2.3.5.1 THERMTRIP# Topology

Figure 215. THERMTRIP# Topology Diagram

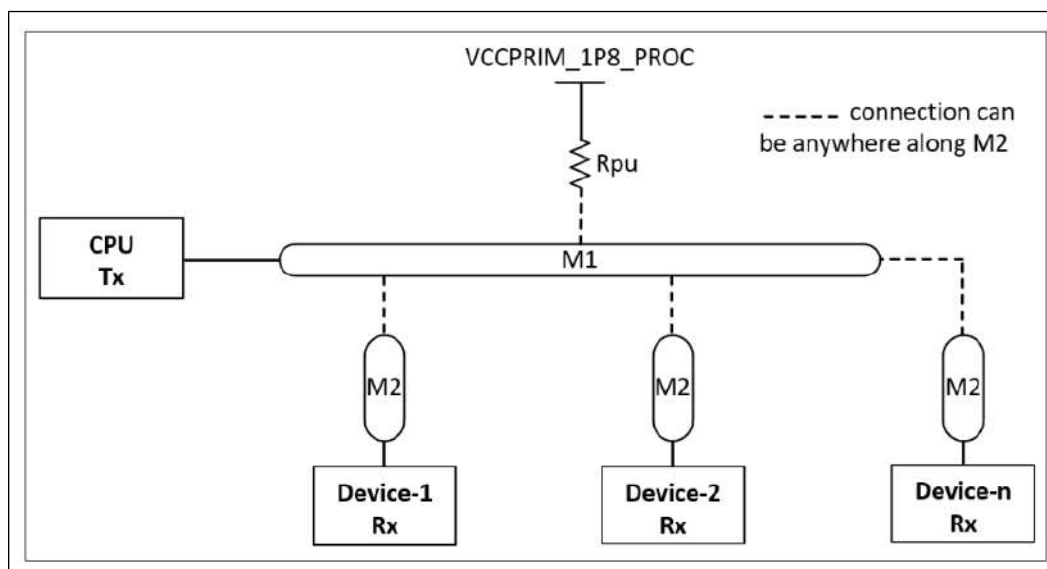


Table 526. THERMTRIP# Topology Notes

Note	Detail
Rpu	2.2 k Ω $\pm$ 20%. Total length can be extended up to 1524 mm. Refer to "THERMTRIP# Topology Routing Length vs Rpu Guideline Table" for details.
Signal name/ list	THERMTRIP#

Segment Lengths

Table 527. THERMTRIP# Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
B0 + M1 + M2	MS, DSL, SL	1016	Max length for B0 < 127 mm. Total length can be extended up to 1524 mm. Refer to "THERMTRIP# Topology Routing Length vs Rpu Guideline Table" for details.

Max Length Total (mm): 1016

Table 528. THERMTRIP# Topology Routing Length vs Rpu Guideline Table

Max Total Board Routing Length (mm)	Max # of Connected Devices	Max Rpu
254	3	5.6 k Ω $\pm$ 10%
254	5	4.7 k Ω $\pm$ 10%
381	5	3.9 k Ω $\pm$ 10%
508	5	3.3 k Ω $\pm$ 10%
762	5	2.7 k Ω $\pm$ 10%
1016	5	2.2 k Ω $\pm$ 10%
1270	5	1.8 k Ω $\pm$ 10%
1524	5	1.5 k Ω $\pm$ 10%

6.2.3.6 SVID

6.2.3.6.1 SVID 1-Load (Device Down) Topology

Figure 216. SVID 1-Load (Device Down) Topology Diagram

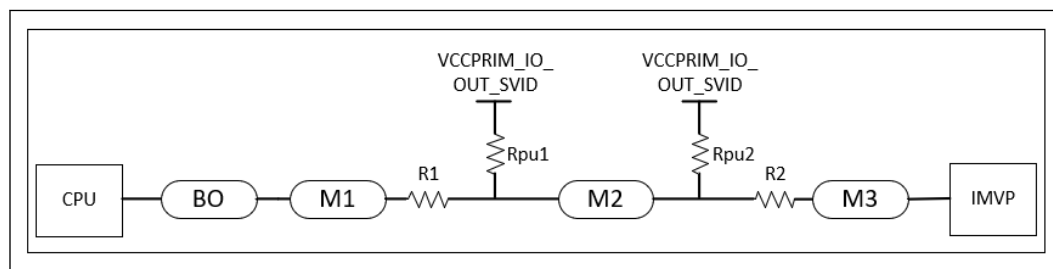


Table 529. SVID 1-Load (Device Down) Topology Notes

Note	Detail
VIDSOUT platform resistors	Rpu1 = 160 Ω , Rpu2 = 160 Ω , R1 = 10 Ω , R2 = 15 Ω .
VIDSCK platform resistors	Rpu1 = Empty, Rpu2 = 82 Ω , R1 = 0 Ω / Short pad, R2 = 15 Ω .
VIDSALERT# platform resistors	Rpu1 = 100 Ω , Rpu2 = Empty, R1 = 0 Ω / Short pad, R2 = 0 Ω / Short pad.
Platform resistor tolerance	$\pm$ 5%
continued...	

Note	Detail
Resistor Placement Recommendation (Rpu1, R1, Rpu2, R2)	The Rpu1 & R1 resistor should be place close to processor and Rpu2 & R2 resistor should be place close to IMVP.
Clarification about Rpu1/ Rpu2/ R1/ R2 = Empty/ 0 Ω / Short Pad	[1] Rpu1 or Rpu2 = Empty; The specific side of the trace related to Rpu1 = Empty or Rpu2 = Empty does not need a pull-up resistor. [2] R1 or R2 = 0 Ω/ Short Pad; The signal does not need a series resistor for the specific part on the trace which is being represented by R1 or R2. Be aware, the trace still needs to remain connected end to end. * It is strongly recommended to refer for the implementation on RVP.
Gen5 VRTT tool validation	Stuff VIDSOUT R1 with 0 Ω if VIDSOUT signal swing fails VIL with Gen5 VRTT tool.
Maximum trace DC resistance	7 Ω
Signal PTH/ via transition	Suggest having a GND stitching vias for each signal transition. The GND stitching vias to the signal vias need to be within 5.08 mm from one another.
Reference plane	Continuous ground only.
Essential guidelines for routing near switching voltage regulator circuit	It is strongly recommended to meet routing near switching voltage regulator guidelines for SVID signals. Refer to "Switching VR Circuit Guideline" chapter in PDG for the details. (Technical White Paper Doc#: 726825)
Route ordering	When routing at minimum spacing route ALERT between DATA and CLK.
Length matching between VIDSOUT and VIDSK	$\pm$ 2.54 mm
Signal name/ list	VIDSOUT, VIDSK, VIDALERT#.

Table 530. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	6	Total Channel
Rx	6	Total Channel

Segment Lengths

Table 531. SVID 1-Load (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	24	
M1	MS, DSL, SL	52	
M2	MS, DSL, SL	153.8	
M3	MS, DSL, SL	75	

Max Length Total (mm): 304.8

6.2.3.7 EPD_ON

6.2.3.7.1 EPD_ON Topology

Figure 217. EPD_ON Topology Diagram

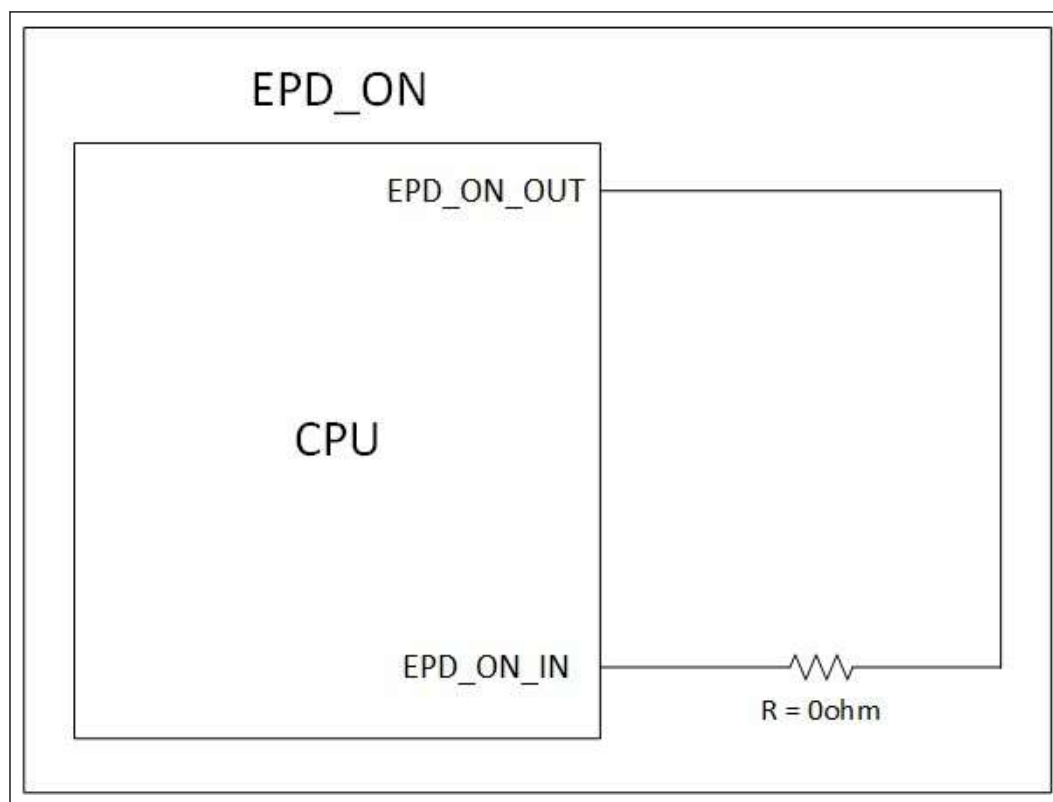


Table 532. EPD_ON Topology Notes

Note	Detail
Signal name/ list	EPD_ON_OUT, EPD_ON_IN
Routing restriction	Route the signal far away from the noisy source/signal. Refer to "Switching VR Circuit Guideline" chapter in PDG for the details. (Technical White Paper Doc#: 726825). Follow the small routing length as possible.

6.2.3.8 VDD2PWRGOOD

6.2.3.8.1 VDD2PWRGOOD Topology

Figure 218. VDD2PWRGOOD Topology Diagram

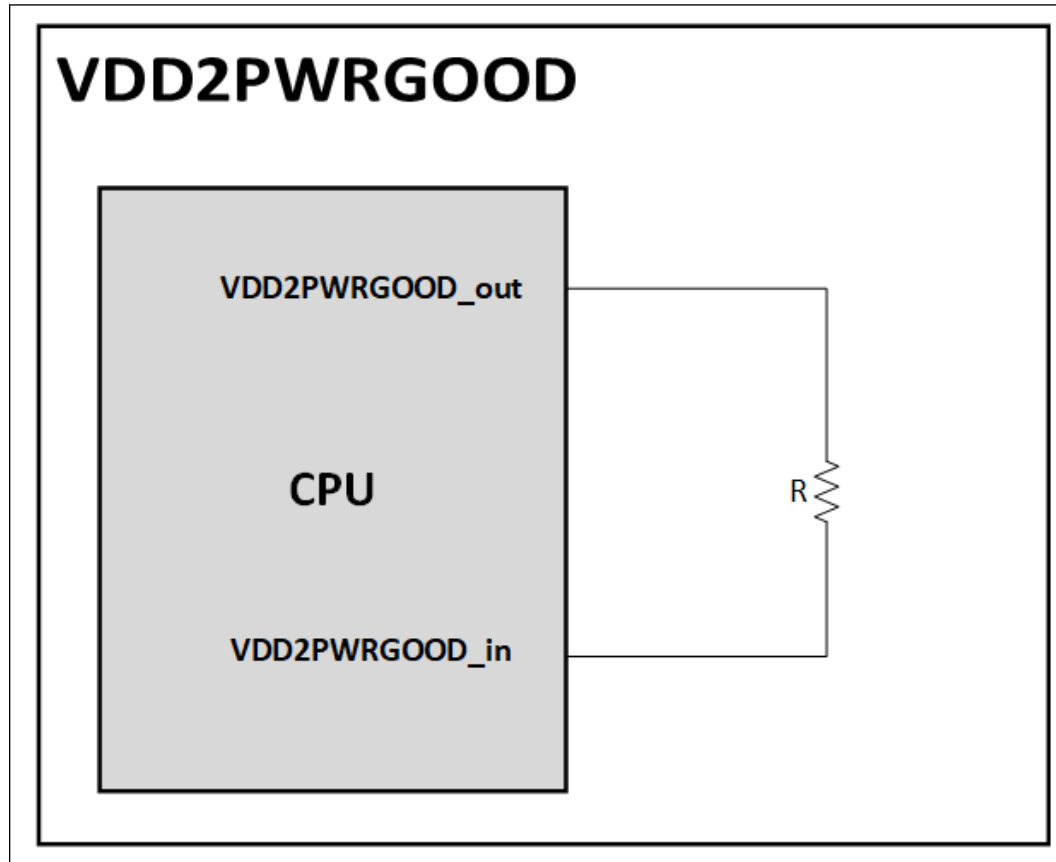


Table 533. VDD2PWRGOOD Topology Notes

Note	Detail
Routing restriction	Route the signal far away from the noisy source/signal. Refer to "Switching VR Circuit Guideline" chapter in PDG for the details. (Technical White Paper Doc#: 726825). Follow the small routing length as possible.
Signal name/ list	VDD2PWRGOOD_IN, VDD2PWRGOOD_OUT
R	0 Ω

6.2.3.9 ESD Protection for Asynchronous Signals

Asynchronous signals are not subject to signal integrity issues but could be victims of ESD events resulting in catastrophic system failures. Therefore, some of them deserve extra care during board routing: good referencing from top or bottom layer, placement of noise filters, no exposure to the edges of a board.

Refer to the Electromagnetic Compatibility chapter for the detailed info on ESD sensitive nets and their treatments.

6.2.4 Imaging Clock

Signal

There are three different Imaging Clock out signals from the processor.

Table 534. Imaging Clock Signal Descriptions

Signal Names	Type	Description
GPP_D04/IMGCLKOUT0	O	Imaging Clock
GPP_D00/IMGCLKOUT1	O	Imaging Clock
GPP_F07/RSVD/IMGCLKOUT2	O	Imaging Clock

6.2.4.1 Imaging Clock 1-Load (Add-In Card) Topology

Figure 219. Imaging Clock 1-Load (Add-In Card) Topology Diagram

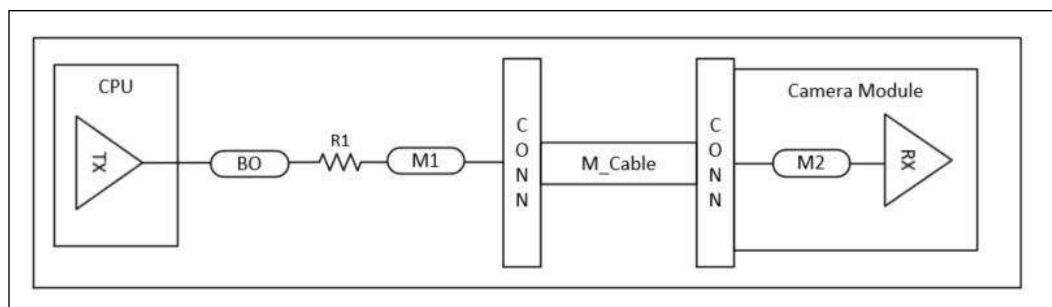


Table 535. Imaging Clock 1-Load (Add-In Card) Topology Notes

Note	Detail
Device duty cycle spec	Input clock duty cycle spec of $\pm 10\%$ or above (example: $\pm 15\%$). Refer to vendor's datasheet for the device electrical spec. If the information is not published in the device datasheet, do contact the device vendor for more details.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
R1	$10\ \Omega \pm 5\%$. Recommended to be placed 25.4 mm from the processor.
Max Length Segment	Max segment length for M1, M2 and M_cable are interchangeable as long as total max length M1 + M2 + M_cable is $< 393.7\text{ mm}$.
Trace spacing between CLK and other signals	0.375 mm
Processor buffer driver strength	33 Ω .
Signal name/ list	IMGCLKOUT[0:2].

Table 536. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	3	Total Channel

Segment Lengths

Table 537. Imaging Clock 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1	DSL, MS, SL	241.3	
M2	DSL, MS, SL	25.4	
M_Cable	Cable	127	

Max Length Total (mm): 406.4

6.2.5 Real Time Clock (RTC) Design Guidelines

The Processor contains a real time clock (RTC) with 256 bytes of battery -backed SRAM. The internal RTC module provides two key functions: keeping date and time and storing system data in its RAM when the system is powered down.

The RTC module requires an external oscillating source 32.768 kHz connected on the RTCX1 and RTCX2 balls. Figure below shows the external circuitry that comprises the oscillator of Processor RTC.

The Processor uses a crystal circuit to generate a low-swing 32 kHz input sine wave. This input is amplified and driven back to the crystal circuit via the RTCX2 signal. Internal to Processor, the RTCX1 signal is amplified to drive internal logic.

Table 538. Reference Specification

Title	Document Number
Design Considerations for Platforms Without a Coin Cell Battery - White Paper	549657
Real Time Clock Reset Timing Technical Advisory	610459

6.2.5.1 Signal Description

Signal Names	Description
RTCX1	Crystal Input 1
RTCX2	Crystal Input 2
SUSCLK	RTC Clock
RTCST#	RTC Reset
SRTCST#	Secured RTC Reset

6.2.5.2 Topology Guidelines

VCCRTC_PROC External Circuit

In this platform, the maximum voltage for VCCRTC_PROC is being reduced to minimize leakage on the ESD diodes and prevent RTC oscillator problems. The designer must ensure that the voltage at VCCRTC_PROC (output of the Low IQ LDO) doesn't exceed the voltage specified in the 'Power Delivery' section.

General RTC Layout Considerations

Since the RTC circuit is very sensitive and requires highly accurate oscillation, reasonable care must be taken during layout and routing of the RTC circuit. Tables below provide some general length guidelines for the RTC crystal circuit.

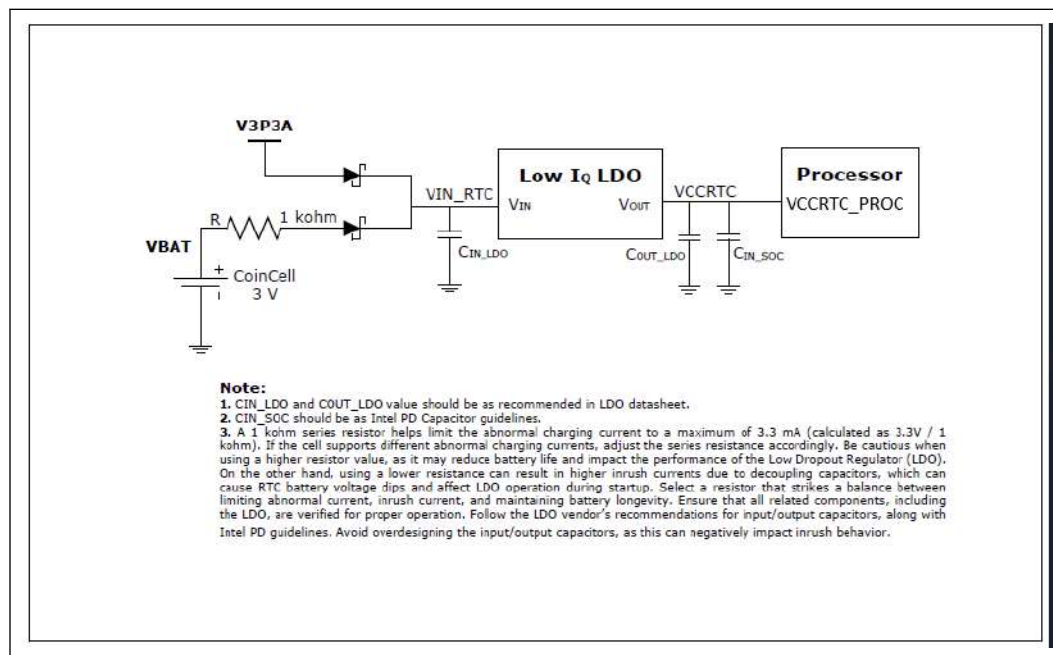
RTC External Battery Connection

The RTC requires an external battery connection to maintain its functionality and its RAM while Processor is not powered by the system.

The voltage of the battery can affect the RTC accuracy. In general, when the battery voltage decays, the RTC accuracy also decreases.

The battery must be connected to Processor using an isolation Schottky diode circuit. The Schottky diode circuit allows RTC well to be powered by the battery when the system power is not available, but by the system power when it is available. To do this, the diodes are set to be reverse biased when the system power is not available. The following figure is an example of a diode circuit that is used.

Figure 220. Schottky Diode Circuit to Connect RTC External Battery



A standby power supply should be used in a mobile system to provide continuous power to the RTC when available, which will significantly increase the RTC battery life and thereby the RTC accuracy.

Using a rechargeable coin battery or a capacitor with a short projected discharged time can increase the risk of:

- an un-provisioned Intel® Active Management Technology (Intel® AMT)
- unusable anti-replay blobs
- Intel® Management Engine (Intel® ME) file system corruption

The first two cases are recoverable: Intel® Active Management Technology (Intel® AMT) can be re-provisioned, and applications using the anti-replay-blobs can be re-enrolled. The third case, the corruption of the Intel® Management Engine (Intel® ME) file system, is irretrievable, but has a very low probability of occurring even when a full discharge occurs.

RTC Coin-cell Less Implementation

Modern implementation of the RTC can also be designed without a coin-cell battery. Details on this implementation is available in the *Design Considerations for Platforms Without a Coin Cell Battery - White Paper*.

6.2.5.2.1 RTC (Real Time Clock) Topology

Figure 221. 32.768 kHz RTC Crystal Topology Diagram

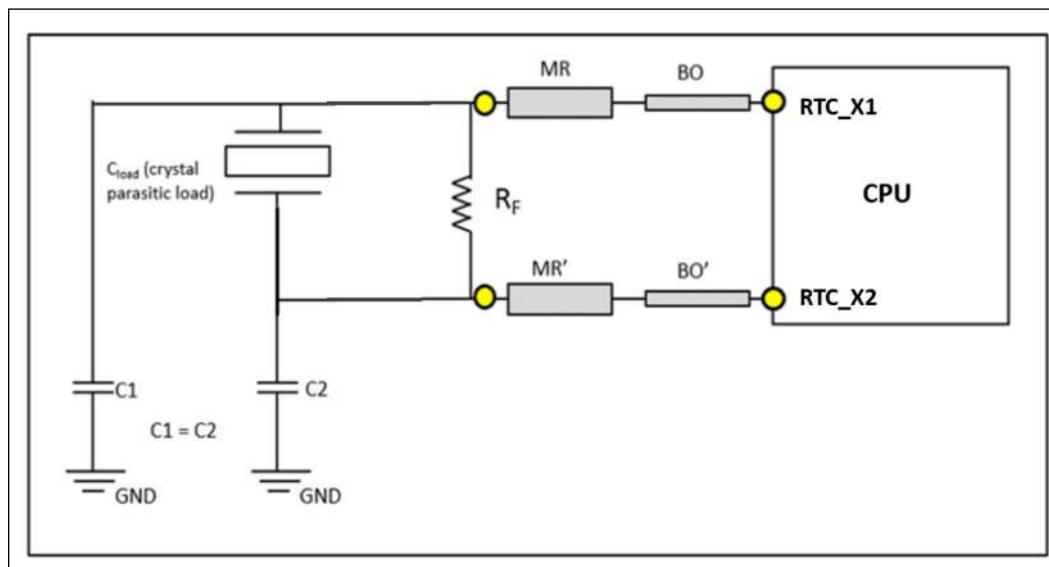


Table 539. RTC (Real Time Clock) Topology Notes

Note	Detail
Length matching between RTC_X1 and RTC_X2	Total length mismatch: 1.27 mm
GND shielding	GND shielding to adjacent signals (especially high speed IO) is recommended.
Reference plane	Continuous GND only; do not reference to power planes. If routed as stripline or dual stripline, both top and bottom reference/ adjacent planes MUST be solid continuous ground.
continued...	

Note	Detail
External capacitor values	External capacitors C1 and C2 must be matched ($C1 = C2$). Capacitor values should be tuned accordingly with respect to load capacitance for RTC crystal Due to vendor variation in crystal characteristic, layout variation and PCB trace, calculation is recommended to determine the C1/C2 value.
Main route (MR)	Main route (MR) is recommended to be limited to SL/ DSL to limit coupling and impact from EMI/ RFI sources.
Rf	10 MOhms

Table 540. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	2	Total Channel	recommended to minimize number of vias

Segment Lengths**Table 541. RTC (Real Time Clock) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
MR	DSL, SL	

Max Length Total (mm): 20.3

Max Length Total Note: Max length of 20.3 mm is based on 0402 resistor and capacitor sizes assumption. Max length can be 25.4 mm if component sizes are reduced from 0402 to 0201.

There must not be a jumper for SRTCRST# pin. The SRTCRST# does not impact the implementation of CMOS clearing. Refer to [Figure 222](#) on page 398 for external circuit for RTC.

NOTES

- RC time delay for SRTCRST# should be the same as RC delay for RTCRST#.
- RTCRST# and SRTCRST# cannot be shorted together.

6.2.5.5 Well Input Strap Requirements

All RTC-well inputs (RSMRST#, RTCRST#, SRTCRST#, INTRUDER#, PLT_PWROK) must be either pulled up to VCCRTC_PROC or pulled down to ground while in the G3 state. RTCRST#, when configured as shown in [External RTCRST# Circuit](#) on page 398 meets this requirement. RSMRST# should have a weak external pull-down to ground, and INTRUDER# should have a weak external pull-up to VCCRTC_PROC. This will prevent these nodes from floating in G3, and correspondingly will prevent ICCRTC leakage that can cause excessive coin-cell drain. The PLT_PWROK input signal should also be configured with an external weak pull-down.

6.2.5.6 Component Selection Guidelines

• RTC Clock Crystal Capacitors (C1 and C2)

To maintain the RTC accuracy, the external capacitor values C1 and C2 should be chosen to provide the manufacturer's specified load capacitance (Cload) for the crystal when combined with the parasitic capacitance of the trace, socket (if used), and package.

$$C_E = 2 * C_L - (C_S + C_I)$$

- C_E = External Load Capacitor Value = C1 = C2
- C_L = Specified Crystal Capacitive Load
 - Found in the crystal vendor component data sheet
- C_S = Mother Board Trace Capacitance = 4.8 pF
- C_I = Processor Pin Capacitance = 5.0 to 6.1 pF

RVP RTC Clock Crystal Capacitor (C1 and C2) Example

- $C_E = 2 * C_L - (C_S + C_I)$
- — RVP $C_L = 12.5$ pF
 - $C_S = 4.8$ pF
 - $C_I = 5.0$ to 6.1 pF
- RTC C_E (Min) = $2 * 12.5 - (4.8 + 6.1) = 14.1$ pF
- RTC C_E (Max) = $2 * 12.5 - (4.8 + 5.0) = 15.2$ pF
- **Capacitor Selection Considerations (C3 and C4)**

DC voltage characteristics vary among different types and sizes of capacitor. Capacitance variation with voltage, temperature and age should be taken into account when selecting components for the RTC external circuit to ensure the actual capacitance matches expectations, especially for C3 and C4 shunt

capacitors. Over time, as systems have decreased in size, ceramic capacitors have also become smaller for equivalent nominal values of capacitance. Large amounts of nominal capacitance can now be obtained in very small form factors. For instance, 0201 capacitors with values at or above 1uF are now common (e.g. X5R or X7R 0201, 1uF capacitors). Ceramic capacitors tend to exhibit a loss of capacitance with applied DC voltage. High capacitance, small form factor (0402, 0201, and lower) parts, tend to exhibit more significant capacitance loss due to DC bias than their physically larger counterparts, for the same applied voltage. This phenomenon can result in lower than expected time constants for circuits utilizing small form factor capacitors. The impact may also vary by manufacturer for "equivalent" parts. The data sheet for each capacitor considered should be checked to make sure that, under DC conditions, the effective capacitance works as intended in a given circuit application.

6.2.5.7 SUSCLK

6.2.5.7.1 SUSCLK 1-Load (Add-In Card) Topology

Figure 223. SUSCLK 1-Load (Add-In Card) Topology Diagram

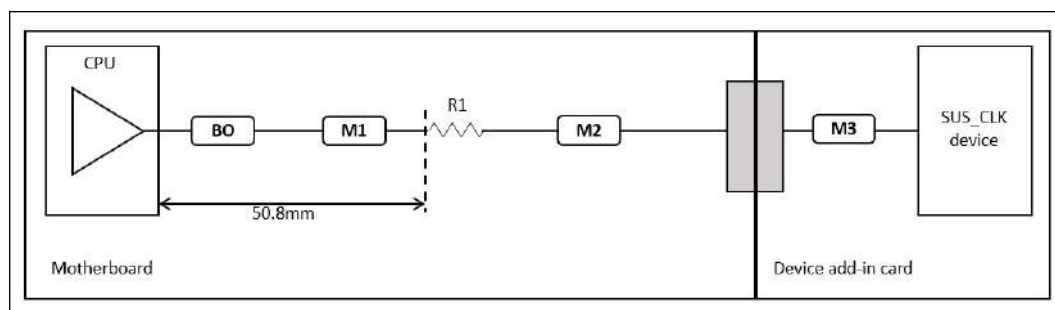


Table 542. SUSCLK 1-Load (Add-In Card) Topology Notes

Note	Detail
R1	0 Ω placeholder is recommended, but not mandatory. If failing device overshoot/undershoot, a 33 Ω resistor can be stuffed. If possible work with device vendor to determine necessity of R1. Must be placed within 50.8 mm from processor.
Signal name/ list	SUSCLK

Table 543. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	6	Total Channel	It is recommended to minimize number of vias.

Segment Lengths

Table 544. SUSCLK 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	38.1
M2	DSL, MS, SL	177.8
M3	DSL, MS, SL	50.8

Max Length Total (mm): 279.4

6.2.5.7.2 SUSCLK 1-Load (Device Down) Topology

Figure 224. SUSCLK 1-Load (Device Down) Topology Diagram

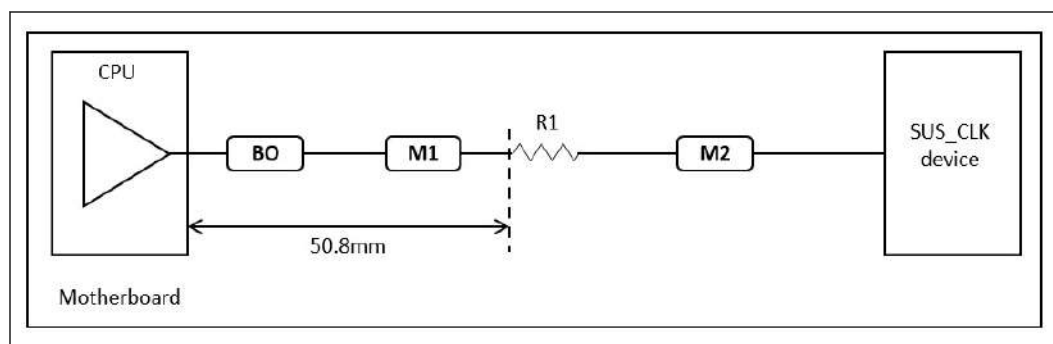


Table 545. SUSCLK 1-Load (Device Down) Topology Notes

Note	Detail
R1	0 Ω placeholder is recommended, but not mandatory. If failing device overshoot/undershoot, a 33 Ω resistor can be stuffed. If possible work with device vendor to determine necessity of R1. Must be placed within 50.8 mm from processor.
Signal name/ list	SUSCLK

Table 546. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	6	Total Channel	It is recommended to minimize number of vias.

Segment Lengths

Table 547. SUSCLK 1-Load (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	38.1
M2	DSL, MS, SL	228.6

Max Length Total (mm): 279.4

6.2.5.7.3 SUSCLK 2-Load (Device Down and Add-In Card) Topology

Figure 225. SUSCLK 2-Load (Device Down and Add-In Card) Topology Diagram

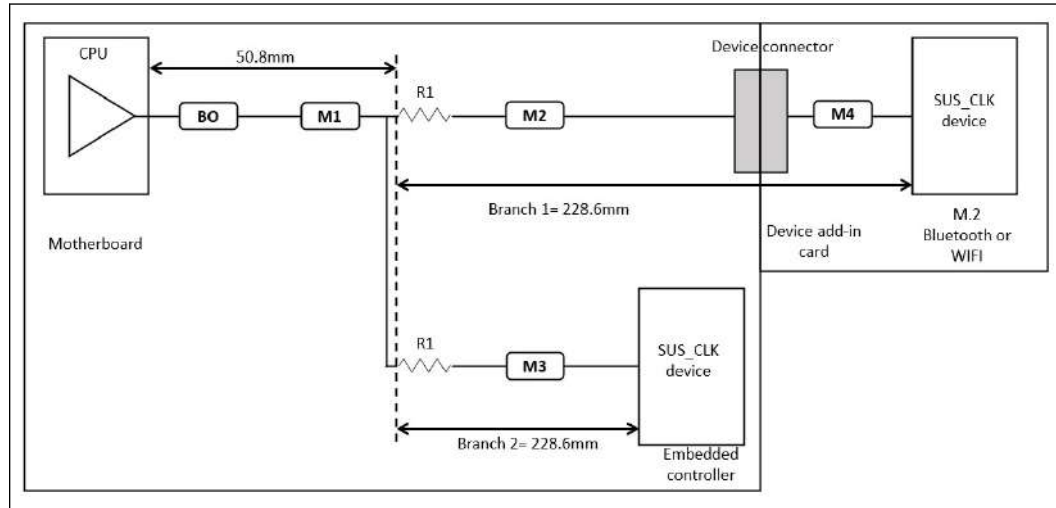


Table 548. SUSCLK 2-Load (Device Down and Add-In Card) Topology Notes

Note	Detail
Branch length	Branch 1 = M2 + M4 = 228.6 mm. Branch 2 = M3 = 228.6 mm.
Length matching between branches	Length matching is not required between branches.
R1	33 Ω . Must be placed within 50.8 mm from processor.
Signal name/ list	SUSCLK

Table 549. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement	Notes
Tx	6	Total Channel	It is recommended to minimize number of vias.

Segment Lengths

Table 550. SUSCLK 2-Load (Device Down and Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	38.1
M2	DSL, MS, SL	177.8
M3	MS, DSL, SL	228.6
M4	DSL, MS, SL	50.8

Max Length Total (mm): 279.4

Max Length Total Note: 1) Max length between processor to the furthest end device = 279.4 mm; 2) Total topology length = 508 mm.

6.2.6 eSPI Interface Guidelines

The Enhanced Serial Peripheral Interface (eSPI) is intended for connecting an EC to the platform and for connecting to the IOE.

eSPI operates at 1.8V only. This interface is not shared and distinct from the SPI interface used for flash device and TPM. The eSPI interface supports up to 50MHz and up to Quad Mode.

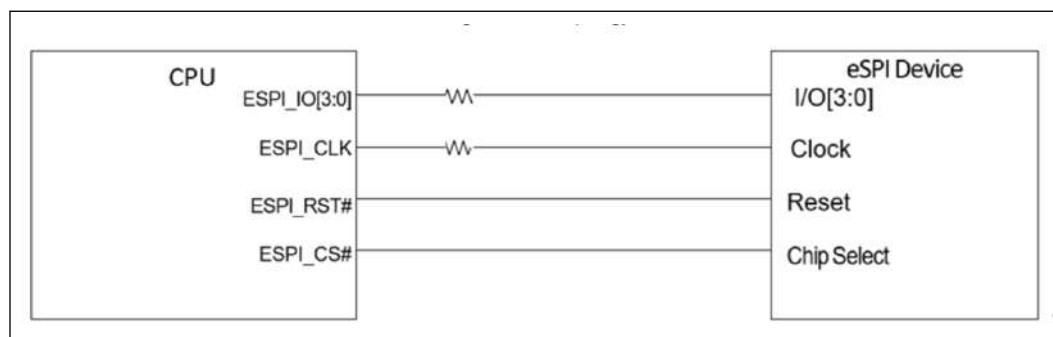
The interface supports both MAF (Host Attached Flash) and SAF (Device Attached Flash).

6.2.6.1 Signal Description

Table 551. eSPI Signals

Signal Name	Group	Description
ESPI_IO[3:0]	Data	Bi-directional data signals used to transfer data between Processor and eSPI target device
ESPI_CLK	Clock	eSPI Clock output from Processor
ESPI_CS[0:3]#	Control	eSPI chip selects
ESPI_RESET#	Control	eSPI reset signal
ESPI_ALERT[0:3]#	Control	eSPI Asynchronous Alert signal input

Figure 226. eSPI Signal Load Topology



6.2.6.1.1 eSPI

eSPI 1-Load (Device Down) Topology

Figure 227. eSPI 1-Load (Device Down) Topology Diagram

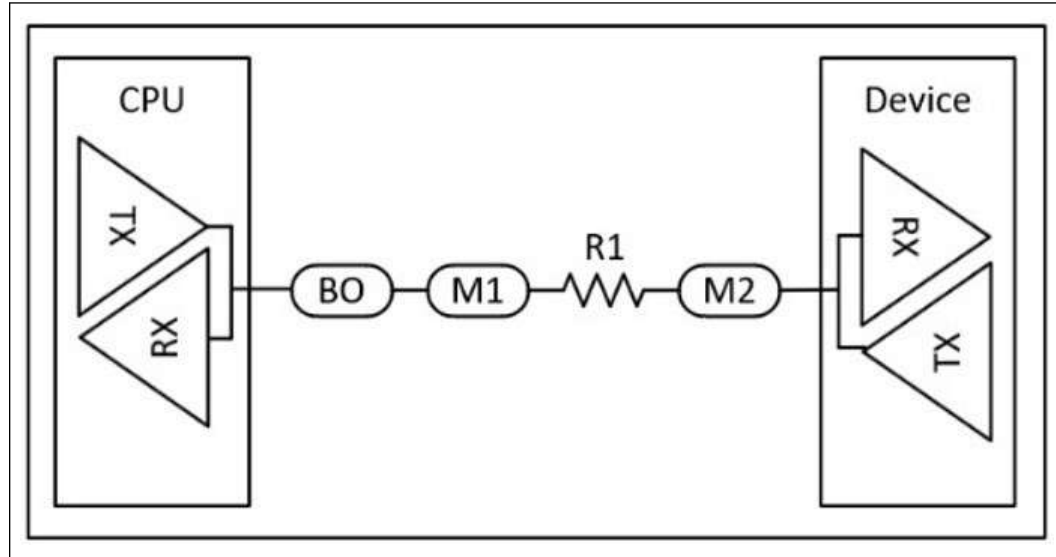


Table 552. eSPI 1-Load (Device Down) Topology Notes

Note	Detail
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	50 MHz
Device assumption	Compliant to Intel ESPI base specification 1.0 (66 MHz and above).
R1	30 Ω $\pm$ 5% on ESPI_CLK and ESPI_IO[0:3].
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	[1] 0.125 mm for MS and SL Tline Type [2] 0.250 mm for DSL Tline Type
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer drive strength	33 Ω
Signal name/ list	ESPI_CLK, ESPI_IO[0:3], ESPI_CS0#, ESPI_RESET#, ESPI_ALERT0#.

Table 553. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths**Table 554. eSPI 1-Load (Device Down) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
M1	MS, DSL, SL	101.6
M2	MS, DSL, SL	101.6

Min Length Total (mm): 63.5

Min Length Total Note: BO + M1 + M2 > 63.5 mm

Max Length Total (mm): 215.9

6.2.7 Generic Serial Peripheral Interface (GSPI)

6.2.7.1 Platform- Specific Important Information

The generic SPI (GSPI) interfaces support devices which use SPI serial protocols for transferring data.

Each interface consists of 4 wires: a clock (CLK), one chip select (CS) and 2 data lines (MOSI and MISO).

The CPU generic SPI is full-duplex synchronous serial interface. The interface operates in master mode only. Serial data formats may range from 4 to 32 bits in length. Maximum frequency depends on the topology

NOTE

The GSPI is not the same as the CPU SPI interface for flash devices. This GSPI is used mainly for finger print sensor and touch panel support on the platform.

6.2.7.2 Signal Descriptions

Signal Groups**Table 555. GSPI Signals**

Group	Signal Name	Description
Clock	GSPI0_CLK	Clock signals
	GSPI1_CLK	
	GSPI2_CLK	
continued...		

Group	Signal Name	Description
	GSPI0A_CLK GSPI1A_CLK	
Data	GSPI0_MISO GSPI1_MISO GSPI2_MISO GSPI0A_MISO GSPI1A_MISO	Initiator In Target Out signals
	GSPI0_MOSI GSPI1_MOSI GSPI2_MOSI GSPI0A_MOSI GSPI1A_MOSI	Initiator Out Target In signals
Chip Select	GSPI0_CS0# GSPI1_CS0# GSPI2_CS0# GSPI0A_CS0# GSPI1A_CS0#	Chip select signals

6.2.7.2.1 GSPI

GSPI 1-Load (Add-In Card) Topology

Figure 228. GSPI 1-Load (Add-In-Card) Topology Diagram

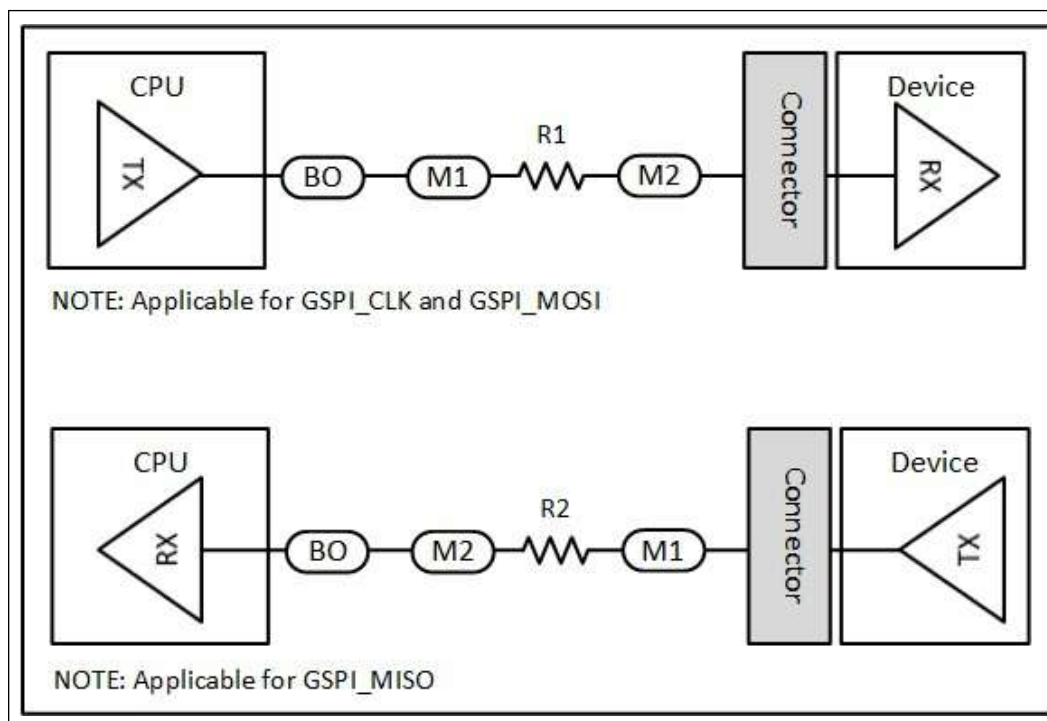


Table 556. GSPI 1-Load (Add-In Card) Topology Notes

Note	Detail
Reference plane	(1) Continuous GND is recommended. (2) If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). (3) If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	25 MHz
R1	0 Ω placeholder. To be placed on GSPI[0:2]_CLK, GSPI[0:1]A_CLK, GSPI[0:2]_MOSI and GSPI[0:1]A_MOSI.
R2	22 $\Omega \pm 5\%$. To be placed on GSPI[0:2]_MISO and GSPI[0:1]A_MISO.
Device AC electrical and timing characteristics	[1] Impedance: 50 Ω to 100 Ω. [2] 0.85 ns < rise/ fall time measured at 20%-80% < 1.9 ns based on 10 pF test load. [3] 1.6 pF $\leq$ input capacitance, $C_{in} \leq 1.75$ pF. [4] Data input setup time < 9 ns. [5] Data input hold time < 5 ns. [6] 0 ns < CLK to DATA output valid time < 14 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer driver strength	50 Ω .
Signal name/ list	GSPI[0:2]_CLK, GSPI[0:2]_MISO, GSPI[0:2]_MOSI, GSPI[0:2]_CS0#, GSPI[0:1]A_CLK, GSPI[0:1]A_MOSI, GSPI[0:1]A_MISO, GSPI[0:1]A_CS0#.

Table 557. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths

Table 558. GSPI 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1	MS, DSL, SL	12.7	
M2	MS, DSL, SL	114.3	M2 can support up to 190.5mm but the frequency needs to be reduced to 20MHz only. M2 length should comprise of M2+AIC lengths.

Min Length Total (mm): 50.8

Max Length Total (mm): 139.7

Max Length Total Note: Total length can be further increased to 215.9 mm but the bus frequency needs to reduce to 20 MHz.

GSPI 1-Load (for Touch Panel) Topology

Figure 229. GSPI 1-Load (for Touch Panel) Topology Diagram

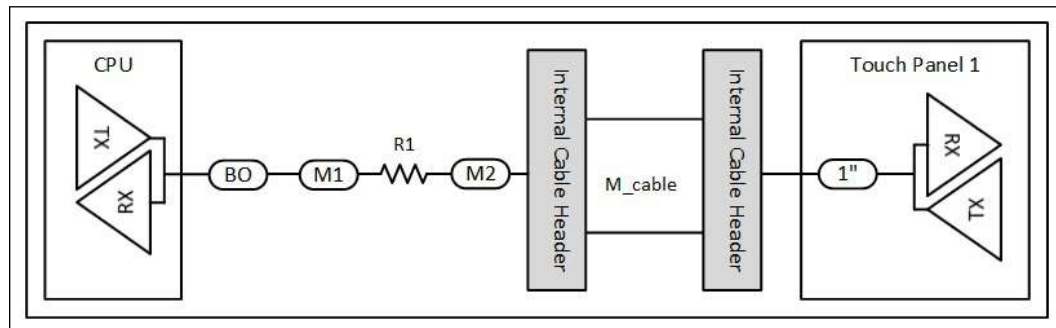


Table 559. GSPI 1-Load (for Touch Panel) Topology Notes

Note	Detail
Reference plane	(1) Continuous GND is recommended. (2) If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). (3) If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	12 MHz
R1	18 $\Omega \pm 5\%$. To be placed 12.7 mm from the connector on GSPI[0:1]_CLK, GSPI[0:1]_MOSI, GSPI[0:1]_MISO, GSPI0A_CLK, GSPI0A_MOSI, and GSPI0A_MISO.
M_cable	Recommended to use shielded cable.
Length matching between CLK and DATA/ CS# signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer driver strength	40 Ω .
Signal name/ list	GSPI[0:1]_CLK, GSPI[0:1]_MISO, GSPI[0:1]_MOSI, GSPI[0:1]_CS0#, GSPI0A_CLK, GSPI0A_MOSI, GSPI0A_MISO, GSPI0A_CS0#.

Table 560. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	7	Total Channel
Tx	7	Total Channel

Segment Lengths**Table 561. GSPI 1-Load (for Touch Panel) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
M1	MS, DSL, SL	228.6
M2	MS, DSL, SL	12.7
M_cable	Cable	457.2

Min Length Total (mm): 78.74

Max Length Total (mm): 711.2

6.2.7.3 Debug Guidelines/Recommendations

GSPI signals are multiplexed with GPIOs and default to GPIO functionality. If GSPI interface is not used, the signals can be used as GPIOs instead. If the GPIO functionality is also not used, the signals can be left as no-connect.

6.2.7.4 Tools

Intel does not promote any specific tool for this interface.

6.2.8 ISH-SPI

6.2.8.1 ISH-SPI 1-Load (Add-In Card) Topology

Figure 230. ISH-SPI 1-Load (Add-In Card) Topology Diagram

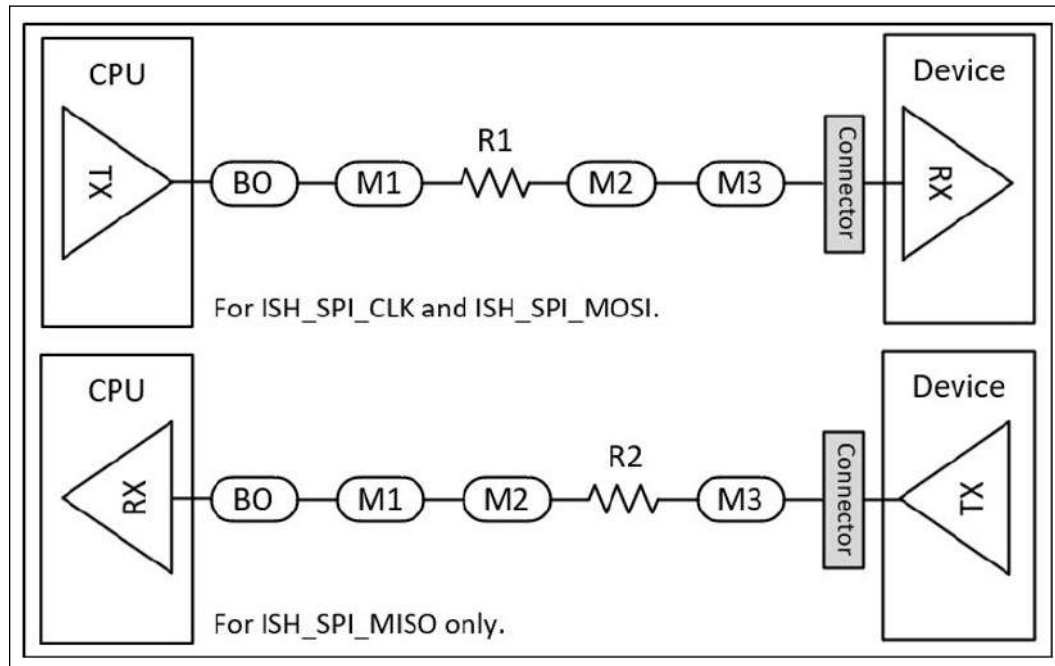


Table 562. ISH-SPI 1-Load (Add-In Card) Topology Notes

Note	Detail
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	25 MHz
R1	0 Ω placeholder. To be placed on ISH_SPI_CLK, ISH_SPIA_CLK, ISH_SPI_MOSI and ISH_SPIA_MOSI.
R2	0 Ω placeholder. To be placed on ISH_SPI_MISO and ISH_SPIA_MISO.
Device AC timing characteristics	Data input setup time < 5 ns. Data input hold time < 5 ns. 1 ns < CLK to DATA output valid time < 15 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
continued...	

Note	Detail
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer drive strength	50 Ω
Signal name/ list	ISH_SPI_CLK, ISH_SPI_MISO, ISH_SPI_MOSI, ISH_SPI_CS#, ISH_SPIA_CLK, ISH_SPIA_MISO, ISH_SPIA_MOSI, ISH_SPIA_CS#.

Table 563. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths**Table 564. ISH-SPI 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
M1	MS, DSL, SL	12.7
M2	MS, DSL, SL	215.9
M3	MS, DSL, SL	12.7

Min Length Total (mm): 76.2

Max Length Total (mm): 254

6.2.9 Legacy Audio Interface Design Guidelines

This platform is a next-generation architecture for audio that allows audio output and input streams to be processed by the host IA processor or for processing to be offloaded from IA processing via the next generation Intel® Smart Sound Technology DSP.

This design guideline chapter includes the routing details for different legacy audio interfaces that connect to the Processor. These are as follows:

- Intel® High Definition Audio (Intel® HD Audio) Interface
- Digital Microphone (DMIC) Interface
- Integrated Interchip Sound (I²S) Interface

6.2.9.1 Platform Specific Important Information

In this section, Intel HD Audio interface is compliant with Microsoft* Universal Audio Architecture (UAA). Contact your IHV (Independent Hardware Vendor) for information on Intel HD Audio, I²S, and DMIC based products.

On the Nova Lake platform the Intel HD Audio, I²S, and DMIC interfaces are multiplexed with each other as well as other interfaces to allow flexibility in pin usage. Refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# [844316](#)) for specific SKU pins assignments and multiplexing details.

6.2.9.2 Signal Description

Table 565. Legacy Audio Signals

Signal Name	Description
Intel® High Definition Audio Interface	
HDA_RST#	Initiator hardware reset to external codec(s)
HDA_SYNC	48 KHz fixed rate sample sync to the codec(s)
HDA_BCLK	24.000 MHz serial data clock generated by the Intel® HD Audio controller.
HDA_SDO	Serial TDM data output to the codec(s)
HDA_SDI[1:0]	Serial TDM data inputs from the codec(s)
I²S Interface	
I2S_MCLK1_OUT	I ² S* Initiator Clock Output
I2S[2:0]_SCLK	I ² S Serial Bit Clocks for connections to I ² S devices.
I2S[2:0]_TXD	I ² S Transmit Data (Serial Data Out) for connection to I ² S devices.
I2S[2:0]_RXD	I ² S Receive Data (Serial Data In) for connection to I ² S devices.
I2S[2:0]_SFRM	I ² S Serial Frame for connection to I ² S devices.
DMIC Interface	
DMIC_CLK_A[1:0]	Serial data clock to digital microphone module.
DMIC_DATA[1:0]	Serial data input from the digital microphone module

6.2.9.3 Intel® High Definition Audio (Intel HD Audio) and DMIC Topology Guidelines

The CPU supports a single Intel HD Audio Interface that supports up to one device.

The DMIC interface has been changed from previous chipset generations allowing up to 1 clock for each DMIC interfaces.

- Intel HD Audio 1-Load Topology (Device Down)
- DMIC 2-Load Branch Topology (Add-In Card) - Shared clock and data signal to each DMIC module branched on the add-in card.
- DMIC 2-Load Daisy Topology - Shared clock and data signal to each DMIC module in a daisy configuration.

Disabling and Termination Guidelines for High Definition Audio Interface

When HDA_SDIN[1:0] interface is not implemented on the platform the signal pin(s) may be left unconnected.

6.2.9.3.1 DMIC

DMIC 2-Load Branch Topology

Figure 231. DMIC 2-Load Branch CLK Topology Diagram

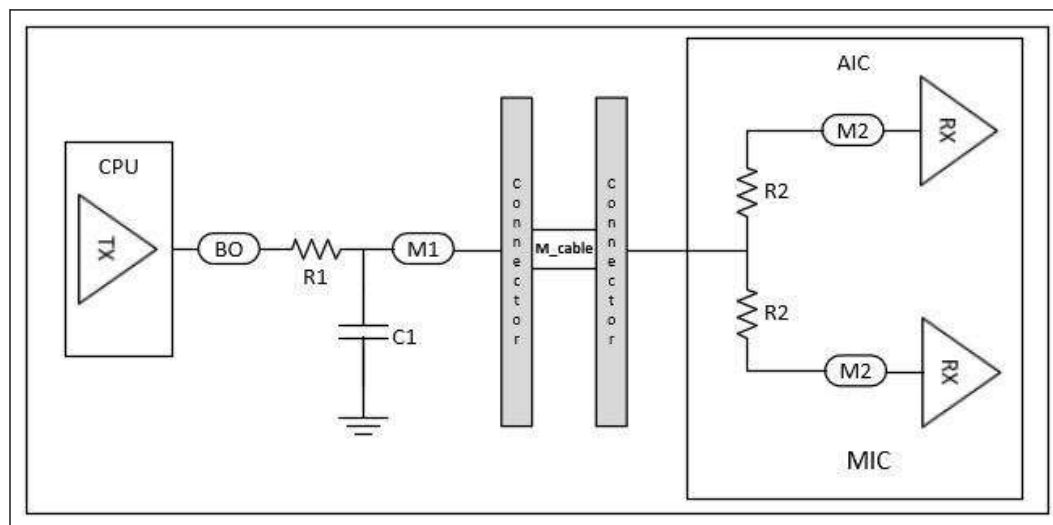


Figure 232. DMIC 2-Load Branch DATA Topology Diagram

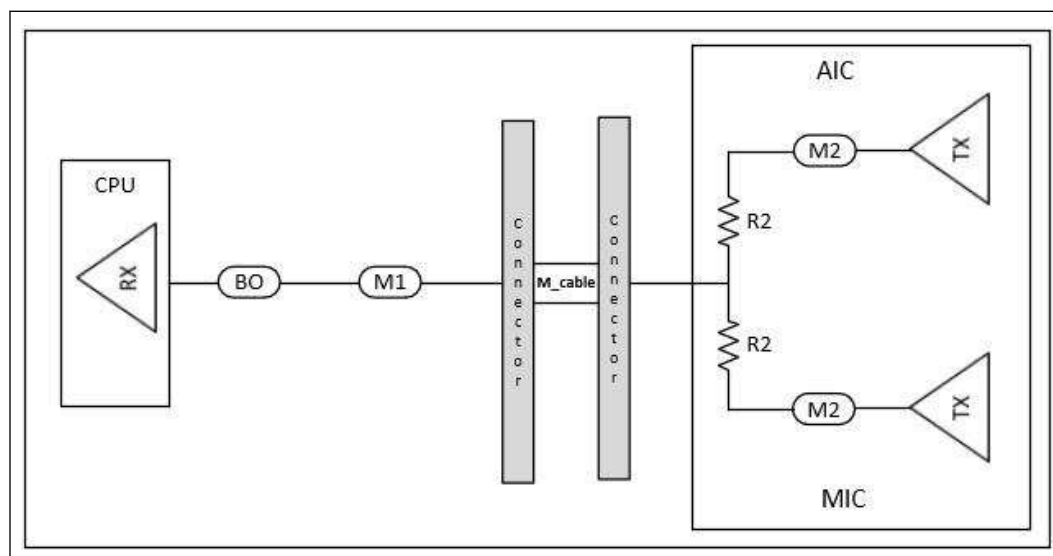


Table 566. DMIC 2-Load Branch Topology Notes

Note	Detail
R1 & C1	[1] EMI RC filter placeholder: Stuff R1 with 0 Ω and unstuff C1 by default. Refer to EMC section for details. [2] To be placed only for DMIC_CLK. [3] C1 is recommended to be placed close to R1. C1 is recommended to be placed 12.7 mm from the processor, but can be extended to 25.4 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum. If M1 is routed on MS, EMI RC filter should be placed at the beginning of the MS routing.
R2	50 Ω .

continued...

Note	Detail
	[1] R2 is recommended to be placed immediate after the branches. [2] If the branches not immediately split after connector, please follow rule [1].
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Length matching between branch	Branch 1 and branch 2 need to be length matched within 2.54 mm.
Length matching between CLK and DATA signals	N/A
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer driver strength	[1] GPP_D: 40 Ω . [2] GPP_S: 50 Ω , slew 11.
Signal name/ list	DMIC_CLK_A[0:1], DMIC_DATA[0:1].

Table 567. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths

Table 568. DMIC 2-Load Branch Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	254
M2	DSL, MS, SL	127
M_cable	Cable	508

Max Length Total (mm): 901.7

Max Length Total Note: 1) Max length between processor to the furthest end device = 901.7 mm; 2) Total topology length = 1028.7 mm.

DMIC 2-Load Daisy Topology

Figure 233. DMIC 2-Load Daisy CLK Topology Diagram

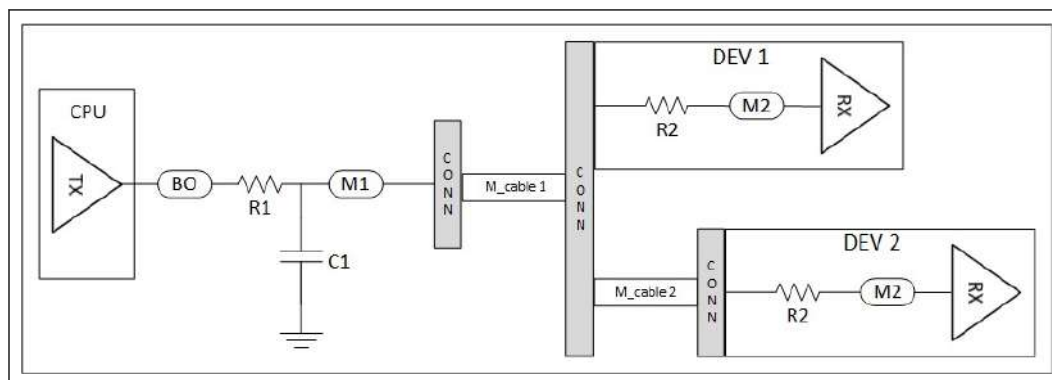


Figure 234. DMIC 2-Load Daisy DATA Topology Diagram

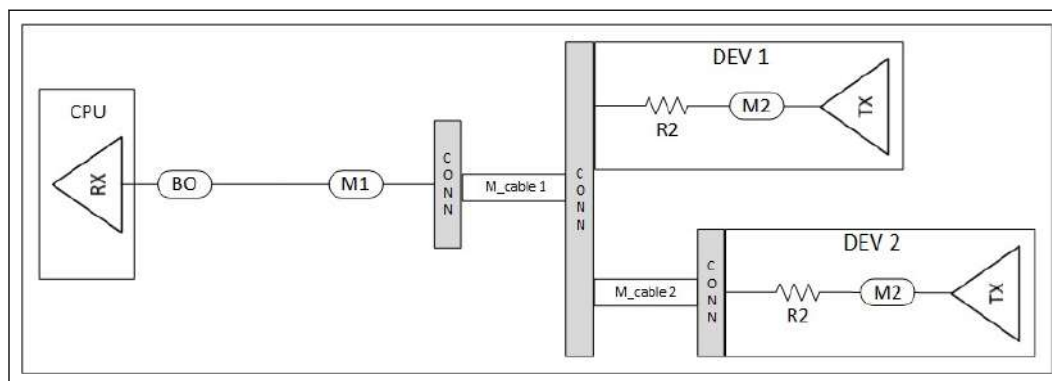


Table 569. DMIC 2-Load Daisy Topology Notes

Note	Detail
R1 & C1	[1] EMI RC filter placeholder: Stuff R1 with 0 Ω and unstuff C1 by default. Refer to EMC section for details. [2] To be placed only for DMIC_CLK. [3] C1 is recommended to be placed close to R1. C1 is recommended to be placed 12.7 mm from the processor, but can be extended to 25.4 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum. If M1 is routed on MS, EMI RC filter should be placed at the beginning of the MS routing.
R2	50 Ω. R2 is recommended to be placed immediate after the branches from connector.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Length matching between CLK and DATA signals	N/A

continued...

Note	Detail
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer driver strength	[1] GPP_D: 40 Ω. [2] GPP_S: 50 Ω, slew 11.
Signal name/ list	DMIC_CLK_A[0:1], DMIC_DATA[0:1].

Table 570. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths

Table 571. DMIC 2-Load Daisy Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	254
M2	DSL, MS, SL	10
M_cable_1	Cable	524
M_cable_2	Cable	101

Max Length Total (mm): 901.7

Max Length Total Note: 1) Max length between processor to the furthest end device = 901.7 mm; 2) Total topology length = 911.7 mm.

6.2.9.3.2 HDA

HDA 1-Load (Device Down) Topology

Figure 235. HDA 1-Load (Device Down) Topology Diagram

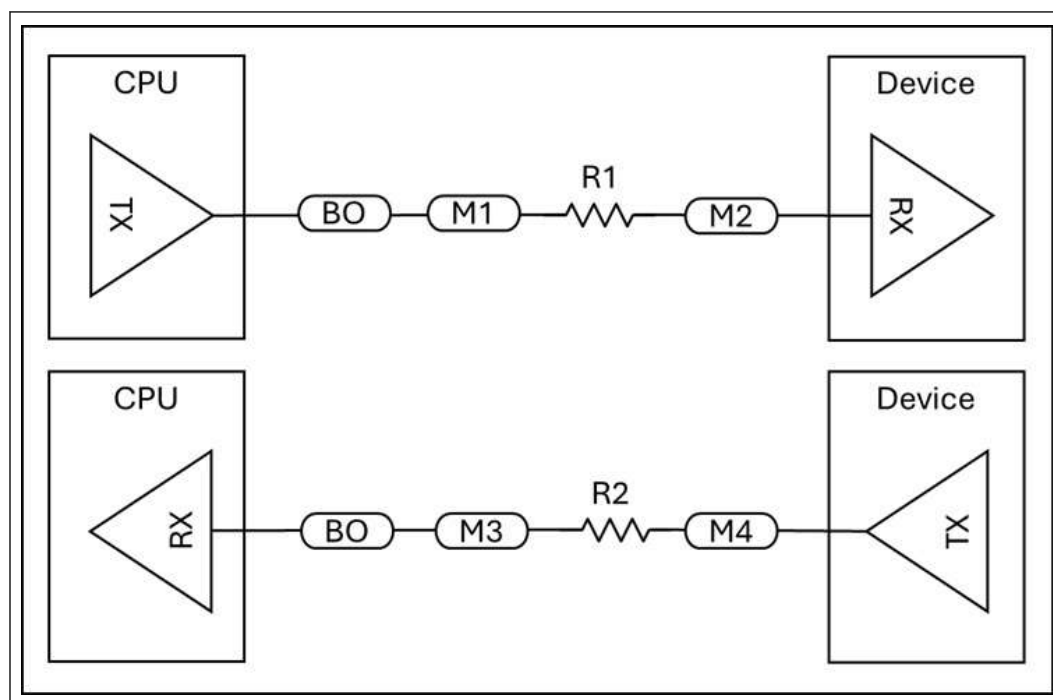


Table 572. HDA 1-Load (Device Down) Topology Notes

Note	Detail
R1	0 Ω placeholder. Recommended to be placed 12.7 mm from the processor, but can be extended to 76.2 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
R2	33 $\Omega \pm 10\%$. Recommended to be placed 12.7 mm from device.
Reference plane	(1) Continuous GND is recommended. (2) If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). (3) If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Dual stripline (DSL)	No parallelism between CLK and DATA signals, as well as DATA and DATA signals.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
continued...	

Note	Detail
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer driver strength	40 Ω .
Signal name/ list	HDA_SDO, HDA_SDI0, HDA_BCLK, HDA_SYNC, HDA_RST#.

Table 573. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel

Segment Lengths

Table 574. HDA 1-Load (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
B0	MS, DSL, SL	12.7
M1	MS, DSL, SL	25.4
M2	MS, DSL, SL	329.6
M3	MS, DSL, SL	342.3
M4	MS, DSL, SL	12.7

Max Length Total (mm): 367.7

6.2.9.4 Intel® I2S* Topology Guidelines

When the I2S interface is used, the following guidelines should be used for routing to end devices. The following topologies are supported:

- I2S Host Mode: 1-Load (Device Down) Topology
- I2S Host Mode: 1-Load (Add-In Card) Topology
- I2S Host Mode: 1-Load (Cable and Add-In Card) Topology
- I2S Host Mode: 2-Load Branch (Cable and Add-In Card) Topology
- I2S Host Mode: 4-Load Daisy Chain (Cable and Add-In Card) Topology
- I2S Device Mode: 1-Load (Device Down) Topology
- I2S Device Mode: 1-Load (Add-In Card) Topology
- I2S MCLK: 1-Load (Add-In Card) Topology

6.2.9.4.1 I2S

Table 575. I2S General Guidelines

Note	Detail
Host/ device mode terminology	Host Mode refers to the mode where the clock is driven by Intel processor. Device Mode refers to the mode where the clock is driven by external device.

I2S Host Mode: 1-Load (Device Down) Topology

Figure 236. I2S Host Mode: 1-Load (Device Down) Topology Diagram

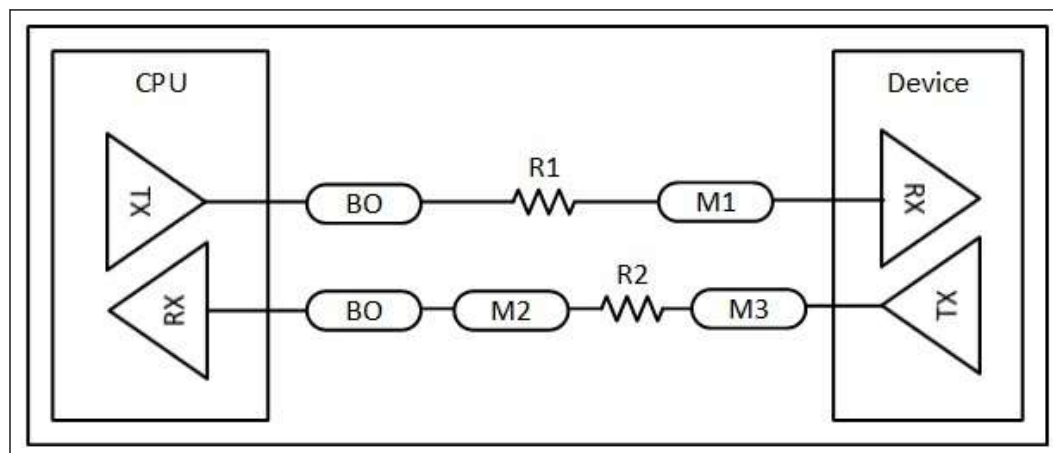


Table 576. I2S Host Mode: 1-Load (Device Down) Topology Notes

Note	Detail
R1	15 $\Omega \pm 5\%$ Recommended to be placed 12.7 mm from processor, but can be extended to 50.8 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
R2	27 $\Omega \pm 5\%$. Recommended to be placed 12.7 mm from device.
Max frequency	12.288 MHz
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Device AC timing characteristics	Data input setup time < 10 ns. Data input hold time < 10 ns. 0 ns < CLK to DATA output valid time < 30 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Trace spacing between port to port	0.375 mm
Processor buffer driver strength	[1] GPP_D (I2S0): 50 Ω . [2] GPP_S (I2S1/ I2S2): 50 Ω , slew 11.
Signal name/ list	I2S[0:2]_SCLK, I2S[0:2]_SFRM, I2S[0:2]_TXD, I2S[0:2]_RXD.

Table 577. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths

Table 578. I2S Host Mode: 1-Load (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	368.3
M2	DSL, MS, SL	355.6
M3	MS, DSL, SL	12.7

Min Length Total (mm): 50.8

Min Length Total Note: Tx: BO + M1= 50.8 mm, Rx: BO + M2 + M3 = 50.8 mm.

Max Length Total (mm): 381

I2S Host Mode: 1-Load (Add-In Card) Topology

Figure 237. I2S Host Mode: 1-Load (Add-In Card) Topology Diagram

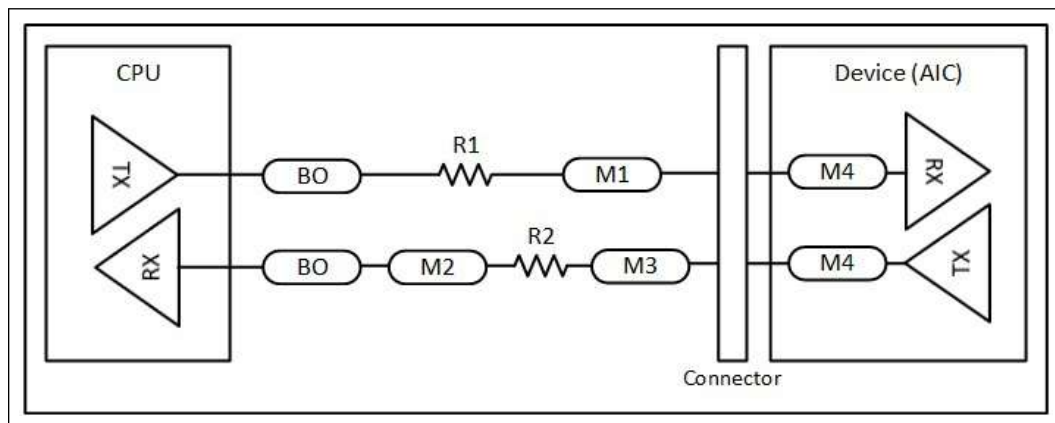


Figure 238. I2S Signal Ganging with CNVi GPIO Topology Diagram

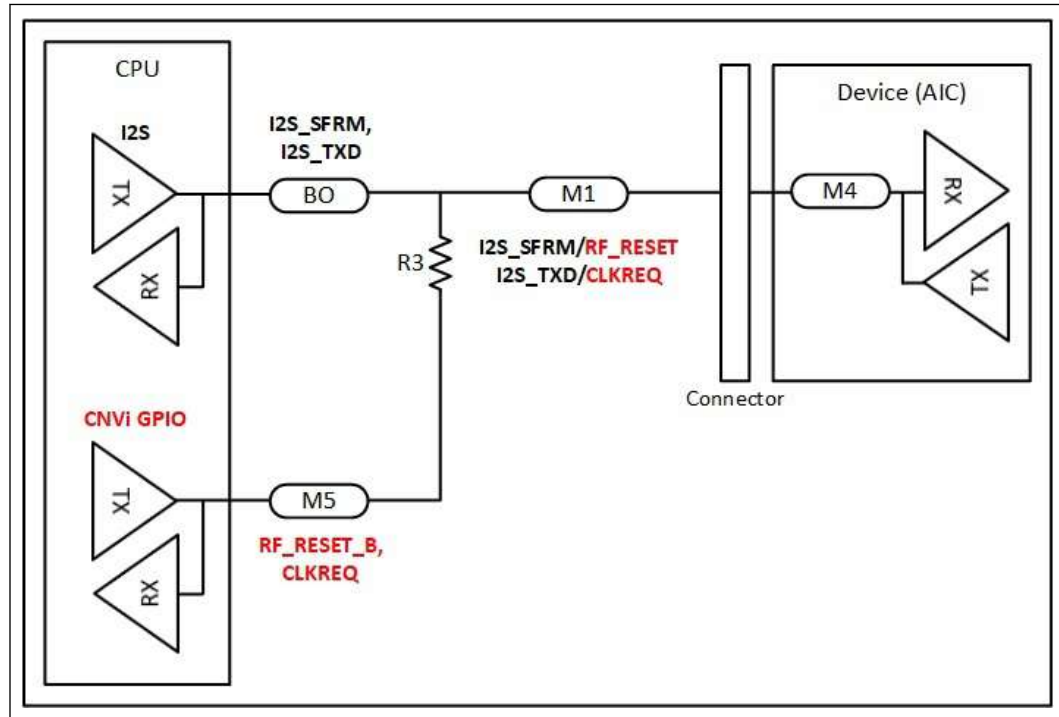


Table 579. I2S Host Mode: 1-Load (Add-In Card) Topology Notes

Note	Detail
R1	15 $\Omega \pm 5\%$. Recommended to be placed 12.7 mm from processor, but can be extended to 50.8 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
R2	27 $\Omega \pm 5\%$. Recommended to be placed 12.7 mm from connector/ header.
R3 (applicable to I2S to discrete CNV module topology only)	1k Ω . Used for ganging I2S signals with CNVi GPIO signals. Required on the CNVi GPIO signals, to be placed <2" from the ganging point.
Additional guidelines for I2S to discrete CNV module topology	[1] For I2S topology connecting to discrete connectivity module, there is a requirement to share/ gang I2S signals with CNVi GPIO signals (RF_RESET_B and CLKREQ) onto the same physical line: - I2S_SFRM with RF_RESET_B - I2S_TXD with CLKREQ [2] A 1k Ω resistor (R3) is required on the CNVi GPIO signals, to be placed <2" from the ganging point. [3] Location of the ganging is not restricted and can be implemented anywhere along the bus. [4] Refer to "I2S signal ganging with CNVi GPIO" diagram for guidance on the implementation. Note that the I2S series resistors (R1 and R2) are still needed and considered in a separate topology diagram.
Max frequency	12.288 MHz
Reference plane	[1] Continuous GND is recommended.

continued...

Note	Detail
	[2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Device AC timing characteristics	Data input setup time < 10 ns. Data input hold time < 10 ns. 0 ns < CLK to DATA output valid time < 30 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Trace spacing between port to port	0.375 mm
Processor buffer driver strength	[1] GPP_D (I2S0): 50 Ω. [2] GPP_S (I2S1/ I2S2): 50 Ω, slew 11.
Signal name/ list	I2S[0:2]_SCLK, I2S[0:2]_SFRM, I2S[0:2]_TXD, I2S[0:2]_RXD.

Table 580. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths

Table 581. I2S Host Mode: 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
B0	DSL, MS, SL	12.7	
M1	DSL, MS, SL	355.6	
M2	DSL, MS, SL	330.2	
M3	DSL, MS, SL	25.4	
M4	MS, DSL, SL	12.7	
M5	MS, DSL, SL		Denotes length of the ganging CNVI GPIO signal for I2S discrete connectivity topology. There is no max length restriction for this segment.

Min Length Total (mm): 50.8

Min Length Total Note: Tx: B0 + M1 = 50.8 mm, Rx: B0 + M2 + M3 = 50.8 mm.

Max Length Total (mm): 381

I2S Host Mode: 1-Load (Cable and Add-In Card) Topology

Figure 239. I2S Host Mode: 1-Load (Cable and Add-In Card) Topology Diagram

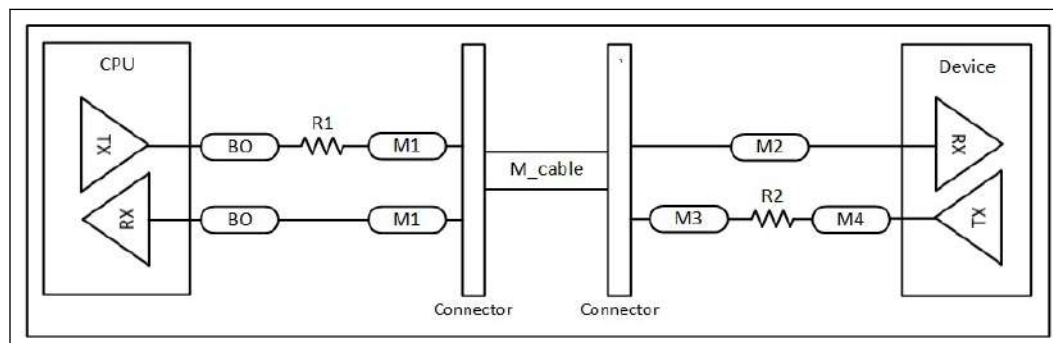


Table 582. I2S Host Mode: 1-Load (Cable and Add-In Card) Topology Notes

Note	Detail
R1	15 $\Omega \pm 5\%$. Recommended to be placed 12.7mm from processor, but can be extended to 50.8mm maximum if required. The length of Breakout (BO) section is still 12.7mm.
R2	33 $\Omega \pm 5\%$. Recommended to be placed 12.7 mm from the connector/ header.
Max frequency	12.288 MHz
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Device AC timing characteristics	Data input setup time < 10 ns. Data input hold time < 10 ns. 0 ns < CLK to DATA output valid time < 30 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Trace spacing between port to port	0.375 mm
Processor buffer driver strength	[1] GPP_D (I2S0): 50 Ω . [2] GPP_S (I2S1/ I2S2): 50 Ω , slew 11.
Signal name/ list	I2S[0:2]_SCLK, I2S[0:2]_SFRM, I2S[0:2]_TXD, I2S[0:2]_RXD.

Table 583. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths

Table 584. I2S Host Mode: 1-Load (Cable and Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Line Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	342.9
M2	DSL, MS, SL	25.4
M3	DSL, MS, SL	12.7
M4	MS, DSL, SL	12.7
M_Cable	Cable	152.4

Min Length Total (mm): 50.8

Min Length Total Note: Tx: $BO + M1 + M_Cable + M2 = 50.8$ mm, Rx: $BO + M1 + M_Cable + M3 + M4 = 50.8$ mm.

Max Length Total (mm): 533.4

I2S Host Mode: 2-Load Branch (Cable and Add-In Card) Topology

Figure 240. I2S Host Mode: 2-Load Branch (Cable and Add-In Card) Topology Diagram

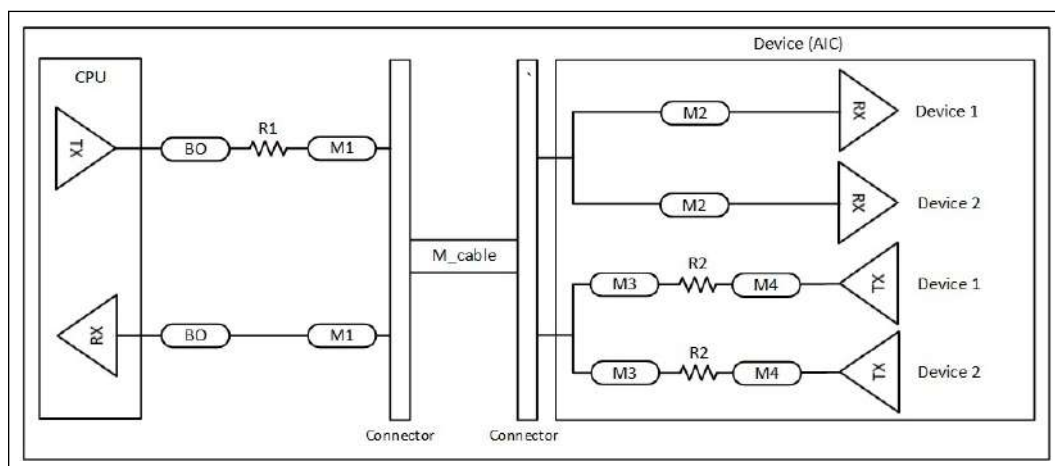


Table 585. I2S Host Mode: 2-Load Branch (Cable and Add-In Card) Topology Notes

Note	Detail
R1	[1] GPP_D (I2S0): $47\ \Omega \pm 10\%$.
<i>continued...</i>	

Note	Detail
	[2] GPP_S (I2S1/ I2S2): 0 Ω placeholder. Recommended to be placed 25.4 mm from processor, but can be extended to 50.8 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
R2	22 $\Omega \pm 5\%$. Recommended to be placed 12.7 mm from device.
Max frequency	12.288 MHz
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Device AC timing characteristics	Data input setup time < 10 ns. Data input hold time < 10 ns. 0 ns < CLK to DATA output valid time < 30 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Trace spacing between port to port	0.375 mm
Processor buffer driver strength	[1] GPP_D (I2S0): 50 Ω . [2] GPP_S (I2S1/ I2S2): 50 Ω , slew 11.
Signal name/ list	I2S[0:2]_SCLK, I2S[0:2]_SFRM, I2S[0:2]_TXD, I2S[0:2]_RXD.

Table 586. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths**Table 587. I2S Host Mode: 2-Load Branch (Cable and Add-In Card) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	25.4
M1	DSL, MS, SL	279.4
M2	DSL, MS, SL	38.1
<i>continued...</i>		

Segment	Line Type	Max Length Segment (mm)
M3	DSL, MS, SL	25.4
M4	MS, DSL, SL	12.7
M_Cable	Cable	25.4

Min Length Total (mm): 50.8

Min Length Total Note: Tx: BO + M1 + M_Cable + M2 = 50.8 mm, Rx: BO + M1 + M_Cable + M3 + M4 = 50.8 mm.

Max Length Total (mm): 368.3

Max Length Total Note: 1) Max length between processor to the furthest end device = 368.3 mm; 2) Total topology length = 406.4 mm.

I2S Host Mode: 4-Load Daisy Chain (Cable and Add-In Card) Topology

Figure 241. I2S Host Mode: 4-load Daisy Chain (Cable and Add-In Card) Topology Diagram

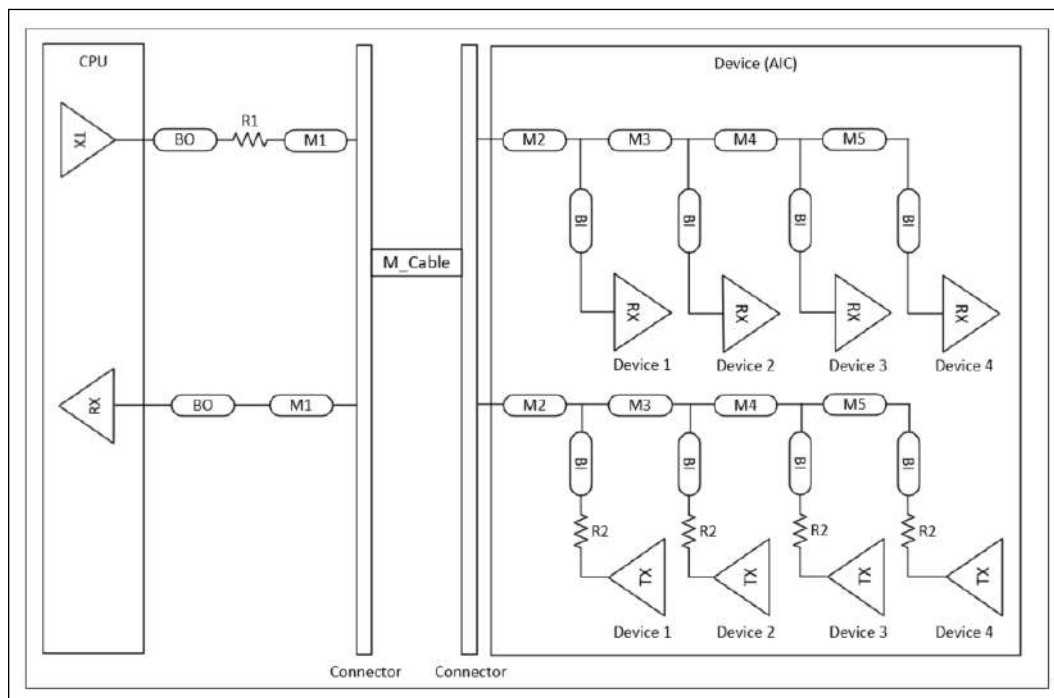


Table 588. I2S Host Mode: 4-Load Daisy Chain (Cable and Add-In Card) Topology Notes

Note	Detail
R1	[1] GPP_D (I2S0): $56 \Omega \pm 10\%$. [2] GPP_S (I2S1/ I2S2): 0Ω placeholder. Recommended to be placed 25.4 mm from processor, but can be extended to 50.8 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
R2	$27 \Omega \pm 10\%$.

continued...

Note	Detail
	Recommended to be placed 12.7 mm from the device.
Max frequency	6.144 MHz
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Device AC timing characteristics	Data input setup time < 10 ns. Data input hold time < 10 ns. 0 ns < CLK to DATA output valid time < 30 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Trace spacing between port to port	0.375 mm
Processor buffer driver strength	[1] GPP_D (I2S0): 50 Ω. [2] GPP_S (I2S1/ I2S2): 50 Ω, slew 11.
Signal name/ list	I2S[0:2]_SCLK, I2S[0:2]_SFRM, I2S[0:2]_TXD, I2S[0:2]_RXD.

Table 589. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths**Table 590. I2S Host Mode: 4-Load Daisy Chain (Cable and Add-In Card) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	25.4
M1	DSL, MS, SL	279.4
M2	DSL, MS, SL	38.1
M3	DSL, MS, SL	25.4
M4	MS, DSL, SL	50.8
M5	MS, DSL, SL	25.4
BI	MS, DSL, SL	25.4
M_Cable	Cable	25.4

Min Length Total (mm): 50.8

Min Length Total Note: Tx: $BO + M1 + M_Cable + M2 = 50.8 \text{ mm}$, Rx: $BO + M1 + M_Cable + M3 + M4 = 50.8 \text{ mm}$.

Max Length Total (mm): 495.3

Max Length Total Note: 1) Max length between processor to the furthest end device = 495.3 mm; 2) Total topology length = 571.5 mm.

I2S Device Mode: 1-Load (Device Down) Topology

Figure 242. I2S Device Mode: 1-Load (Device Down) Topology Diagram

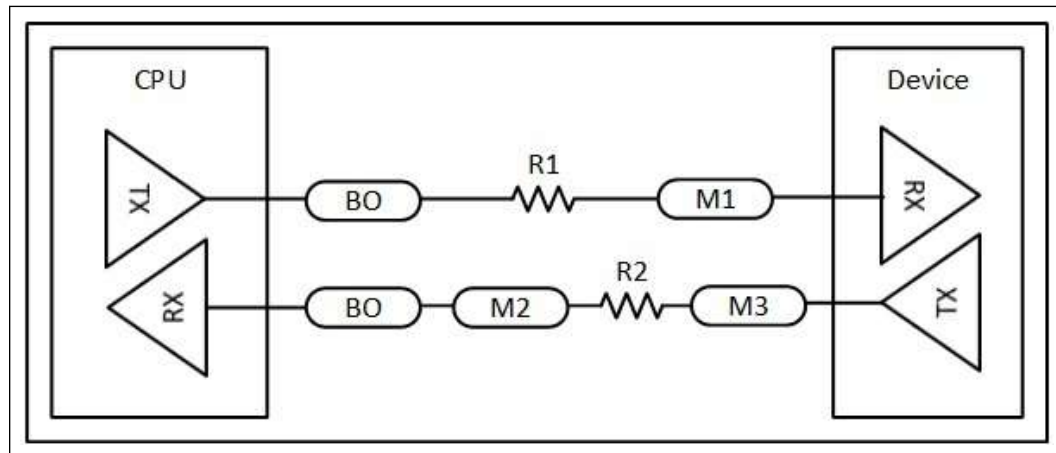


Table 591. I2S Device Mode: 1-Load (Device Down) Topology Notes

Note	Detail
R1	0 Ω placeholder. Recommended to be placed 12.7 mm from processor, but can be extended to 50.8 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
R2	51 $\Omega \pm 5\%$. Recommended to be placed 12.7 mm from device.
Max frequency	12.288 MHz
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Device AC timing characteristics	Data input setup time < 10 ns. Data input hold time < 10 ns. 0 ns < CLK to DATA output valid time < 30 ns.
Length matching between CLK and DATA signals	12.7 mm
continued...	

Note	Detail
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Trace spacing between port to port	0.375 mm
Processor buffer driver strength	[1] GPP_D (I2S0): 50 Ω. [2] GPP_S (I2S1/ I2S2): 50 Ω, slew 11.
Signal name/ list	I2S[0:2]_SCLK, I2S[0:2]_SFRM, I2S[0:2]_TXD, I2S[0:2]_RXD.

Table 592. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

Segment Lengths**Table 593. I2S Device Mode: 1-Load (Device Down) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	215.9
M2	DSL, MS, SL	203.2
M3	MS, DSL, SL	12.7

Min Length Total (mm): 50.8

Min Length Total Note: Tx: BO + M1 = 50.8 mm, Rx: BO + M2 + M3 = 50.8 mm.

Max Length Total (mm): 228.6

I2S Device Mode: 1-Load (Add-In Card) Topology

Figure 243. I2S Device Mode: 1-Load (Add-In Card) Topology Diagram

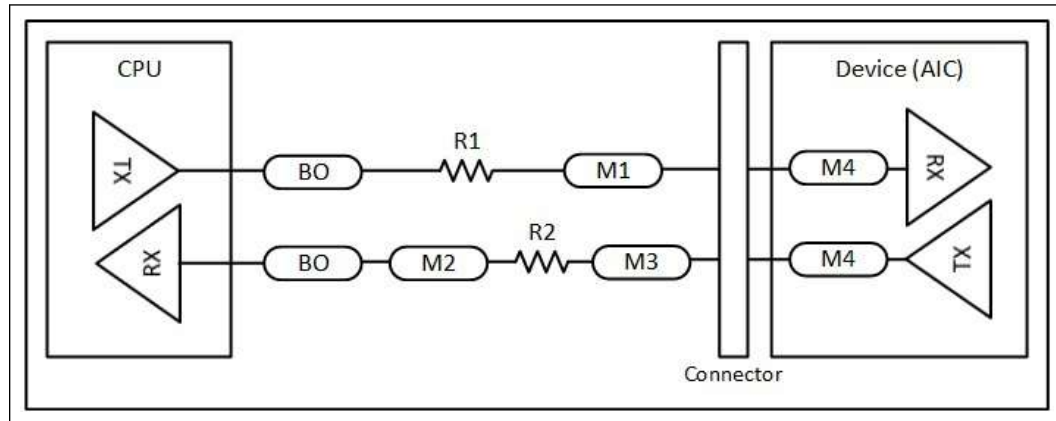


Table 594. I2S Device Mode: 1-Load (Add-In Card) Topology Notes

Note	Detail
R1	0 Ω placeholder. Recommended to be placed 12.7 mm from processor, but can be extended to 50.8 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
R2	51 $\Omega \pm 5\%$. Recommended to be placed 12.7 mm from connector/ header.
Max frequency	12.288 MHz
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Device AC timing characteristics	Data input setup time < 10 ns. Data input hold time < 10 ns. 0 ns < CLK to DATA output valid time < 30 ns.
Length matching between CLK and DATA signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Trace spacing between port to port	0.375 mm
Processor buffer driver strength	[1] GPP_D (I2S0): 50 Ω . [2] GPP_S (I2S1/ I2S2): 50 Ω , slew 11.
Signal name/ list	I2S[0:2]_SCLK, I2S[0:2]_SFRM, I2S[0:2]_TXD, I2S[0:2]_RXD.

Table 595. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel
Tx	7	Total Channel
Rx	7	Total Channel

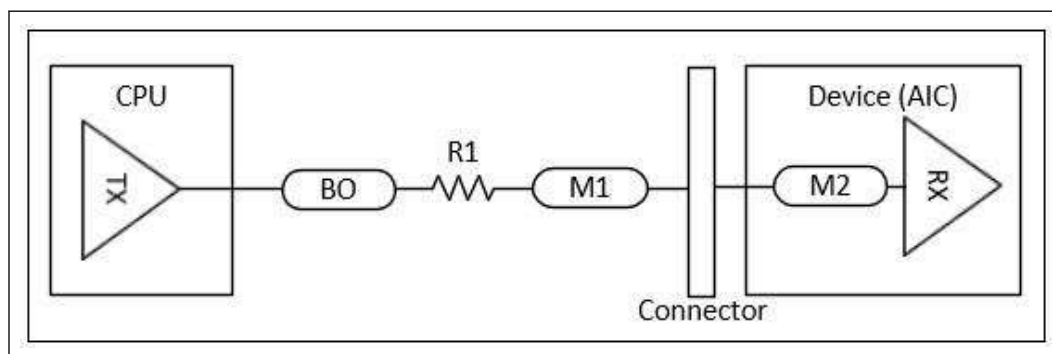
Segment Lengths**Table 596. I2S Device Mode: 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	MS, DSL, SL	190.5
M2	DSL, MS, SL	177.8
M3	DSL, MS, SL	12.7
M4	DSL, MS, SL	25.4

Min Length Total (mm): 50.8

Min Length Total Note: Tx: BO + M1 = 50.8 mm, Rx: BO + M2 + M3 = 50.8 mm.

Max Length Total (mm): 228.6

I2S MCLK: 1-Load (Add-In Card) Topology**Figure 244. I2S MCLK: 1-Load (Add-In Card) Topology Diagram****Table 597. I2S MCLK: 1-Load (Add-In Card) Topology Notes**

Note	Detail
R1	33 Ω $\pm$ 5%. Recommended to be placed 12.7 mm from processor, but can be extended to 50.8 mm maximum if required. The length of breakout section (BO) is still 12.7 mm maximum.
Reference plane	[1] Continuous GND is recommended.

continued...

Note	Detail
	[2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Minimum length	38.1 mm
Trace spacing between CLK and other signals	0.375 mm
Processor buffer driver strength	50 Ω .
Signal name/ list	I2S_MCLK1_OUT

Table 598. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Tx	7	Total Channel

Segment Lengths

Table 599. I2S MCLK: 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
B0	DSL, MS, SL	12.7	
M1	DSL, MS, SL		$M1 + M2 < 368.3$
M2	DSL, MS, SL		$M1 + M2 < 368.3$

Max Length Total (mm): 381

6.2.10 SoundWire* Interface Design Guidelines

For the Lake platform, SoundWire* is still the newest audio interface targeting (but not limited to) the phone, mobile, and desktop market as an alternative to Slimbus. It's main advantage is the connection simplicity with a 2 wire multi-drop topology. For the Lake processor there are 4 separate SoundWire interfaces that can be used.

For the Nova Lake platform, SoundWire* is an audio interface targeting, but not limited to, the phone, mobile, and desktop markets. Its main advantage is simplified connectivity for audio devices. The Nova Lake processor supports four separate SoundWire interfaces that can be used.

6.2.10.1 Platform Specific Important Information

On the Nova Lake platform the SoundWire* interface and other audio interfaces can be configured by itself or at the same time with other interfaces for connection to end devices. Refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# [844316](#)) assignments.

6.2.10.2 Signal Description

Table 600. SoundWire* Signals

Signal Name	Description
SNDW0_DATA0	SoundWire 0 Data 0: Serialized data line containing frame and data being transmitted /Received
SNDW0_CLK	SoundWire 0 Clock : Serial bit clock used to control the timing of a transfer.
SNDW1_DATA0	SoundWire1 Data 0: Serialized data line containing frame and data being transmitted /Received
SNDW1_CLK	SoundWire 1 Clock : Serial bit clock used to control the timing of a transfer.
SNDW2_DATA0	SoundWire 2 Multilane Data 0: Supports multilane topology for high fidelity codecs and multimedia container formats
SNDW2_DATA1	SoundWire 2 Multilane Data 1: Supports multilane topology for high fidelity codecs and multimedia container formats
SNDW2_DATA2	SoundWire 2 Multilane Data 2: Supports multilane topology for high fidelity codecs and multimedia container formats
SNDW2_CLK	SoundWire 2 Clock : Serial bit clock used to control the timing of a transfer.
SNDW3_DATA0	SoundWire 3 Multilane Data 0: Supports multilane topology for high fidelity codecs and multimedia container formats
SNDW3_DATA1	SoundWire 3 Multilane Data 1: Supports multilane topology for high fidelity codecs and multimedia container formats
SNDW3_DATA2	SoundWire 3 Multilane Data 2: Supports multilane topology for high fidelity codecs and multimedia container formats
SNDW3_DATA3	SoundWire 3 Multilane Data 3: Supports multilane topology for high fidelity codecs and multimedia container formats
SNDW3_CLK	SoundWire Clock 3: Serial bit clock used to control the timing of a transfer.

6.2.10.3 Topology Guidelines

There are several topologies supported in Nova Lake for SoundWire*.

6.2.10.3.1 SoundWire

SoundWire Large System: 1-Load (Add-In Card) Topology

Figure 245. SoundWire Large System: 1-Load (Add-In Card) Topology Diagram

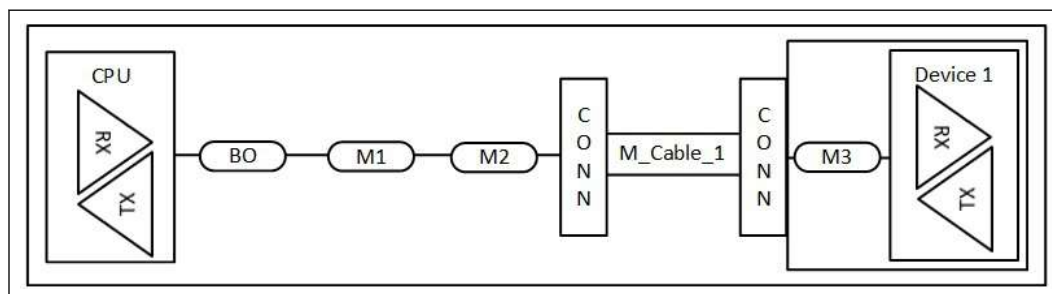


Table 601. SoundWire Large System: 1-Load (Add-In Card) Topology Notes

Note	Detail
M_Cable characteristic impedance target	50 Ω $\pm$ 10%.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	12.288 MHz
SoundWire ports supported	Single-lane: Port #0, Port #1, Port #2, Port #3. Multi-lane: Port #2 (up to 3 data lane), Port #3 (up to 4 data lane).
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	For total length < 600 mm: [1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 7 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 9 Clock Period For total length $\geq$ 600 mm: [1] CLDS = 6 Clock Period [2] DODS/DODSE/DODSE2 = 7 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 9 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 11.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of 20 Ω $\pm$ 30% 2) Device series resistor of 56 Ω $\pm$ 5% Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	Single-lane: SNDW[0:3]_CLK, SNDW[0:3]_DATA0. Multi-lane: SNDW[2:3]_CLK, SNDW2_DATA[0:2], SNDW3_DATA[0:3], SNDW2A_DATA1.

Table 602. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths

Table 603. SoundWire Large System: 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1+M2	DSL, MS, SL	300
M3	DSL, MS, SL	150
M_Cable	Cable	350

Min Length Total (mm): 98.1

Max Length Total (mm): 812.7

SoundWire Large System: 2-Load Star Topology

Figure 246. SoundWire Large System: 2-Load Star (Add-In Card) Topology Diagram

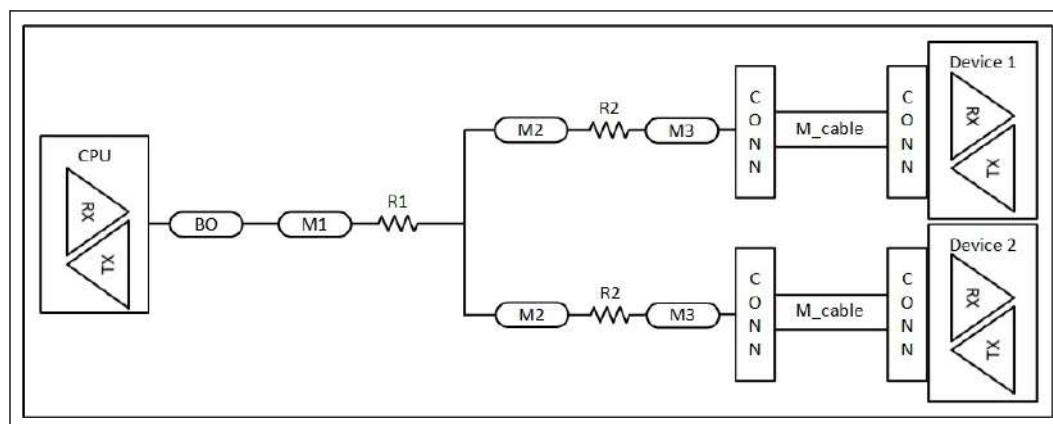


Figure 247. SoundWire Large System: 2-Load Star (Device Down) Topology Diagram

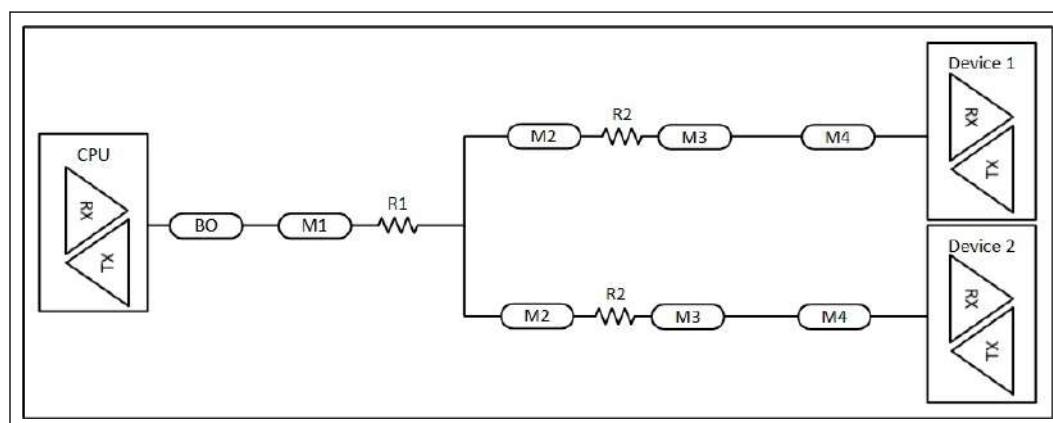


Table 604. SoundWire Large System: 2-Load Star Topology Notes

Note	Detail
M_Cable characteristic impedance target	50 Ω $\pm$ 10%.
R1	0 Ω placeholder. To be placed 10 mm to 15 mm from the branch.
R2	0 Ω placeholder. To be placed 5 mm to 10 mm after the branch.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Length matching between branch for Device 1 & 2 (Option 1: M_cable or Option 2: M4)	10 mm
Max frequency	9.6 MHz
SoundWire ports supported	Port #0, Port #1, Port #2, Port #3.
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	For total length < 850.9 mm: [1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period For total length $\geq$ 850.9 mm: [1] CLDS = 9 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 11.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of 20 Ω $\pm$ 30% 2) Device series resistor of 56 Ω $\pm$ 5% Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	SNDW[0:3]_CLK, SNDW[0:3]_DATA0.

Table 605. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths

Table 606. SoundWire Large System: 2-Load Star Topology Segment Lengths Routing Guidelines

Segment	Trline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	170
M2	MS	5
M3	MS	10
M4	DSL, MS, SL	550
M_Cable	Cable	550

Min Length Total (mm): 165.1

Max Length Total (mm): 747.7

Max Length Total Note: 1) Max length between processor to the furthest end device = 747.7 mm; 2) Total topology length = 1312.7 mm.

SoundWire Large System: 4-Load Star (Device Down) Topology

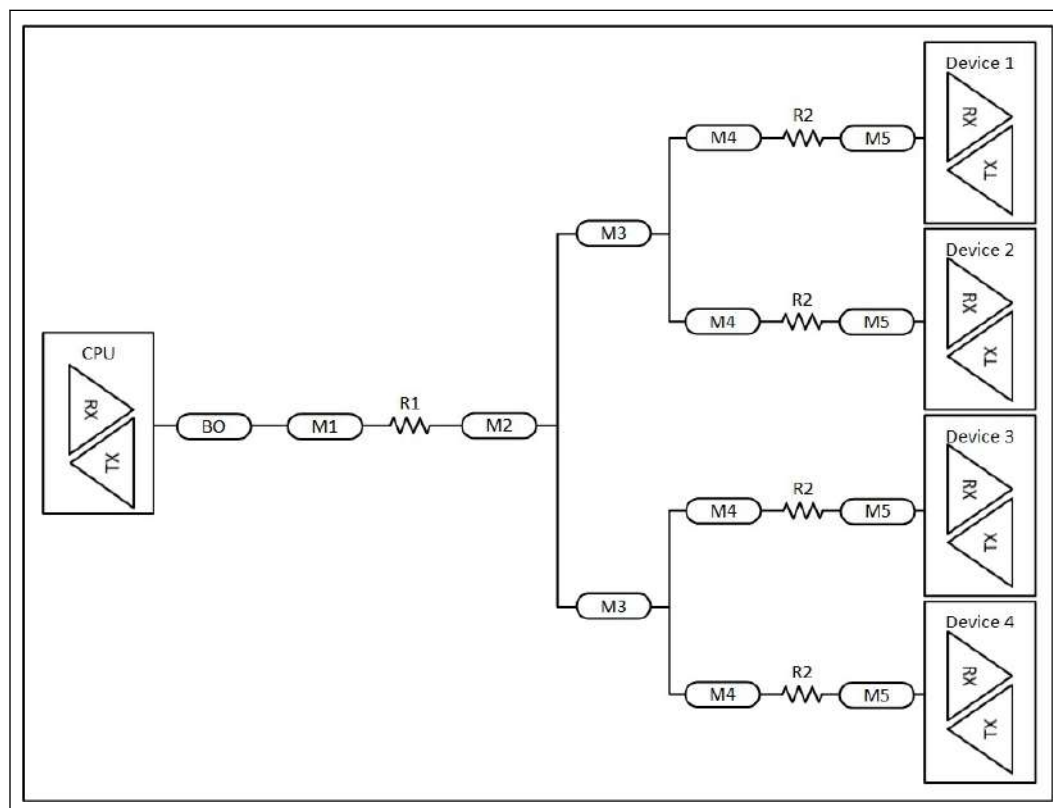
Figure 248. SoundWire Large System: 4-Load Star (Device Down) Topology Diagram

Table 607. SoundWire Large System: 4-Load Star (Device Down) Topology Notes

Note	Detail
R1	0 Ω placeholder. To be placed 10 mm to 15 mm from the first branch from processor.
R2	0 Ω placeholder. To be placed 5 mm to 10 mm from the device.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Length matching between branch (M3_1 & M3_2)	10 mm
Max frequency	9.6 MHz
SoundWire ports supported	Port #0, Port #1, Port #2, Port #3.
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	For total length < 571.5 mm: [1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period For total length $\geq$ 571.5 mm: [1] CLDS = 9 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 11.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of 20 $\Omega \pm 30\%$ 2) Device series resistor of 91 $\Omega \pm 5\%$ Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	SNDW[0:3]_CLK, SNDW[0:3]_DATA0.

Table 608. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths

Table 609. SoundWire Large System: 4-Load Star (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	80
M2	DSL, MS, SL	10
M3	DSL, MS, SL	400
M4	DSL, MS, SL	20
M5	DSL, MS, SL	10

Min Length Total (mm): 127

Max Length Total (mm): 532.7

Max Length Total Note: 1) Max length between processor to the furthest end device = 532.7 mm; 2) Total topology length = 1022.7 mm.

SoundWire Large System: 2-Load Dumbbell Topology

Figure 249. SoundWire Large System: 2-Load Dumbbell (Device Down) Topology Diagram

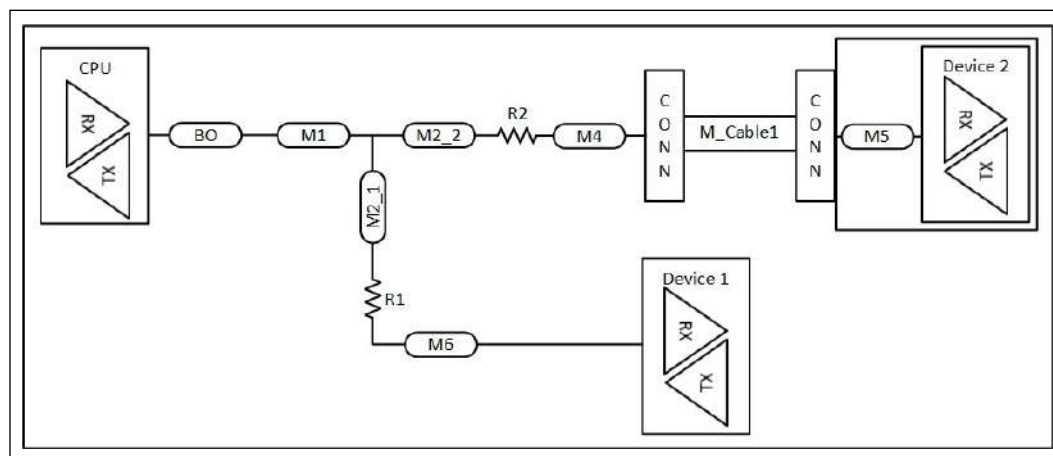


Figure 250. SoundWire Large System: 2-Load Dumbbell (Add-In Card) Topology Diagram

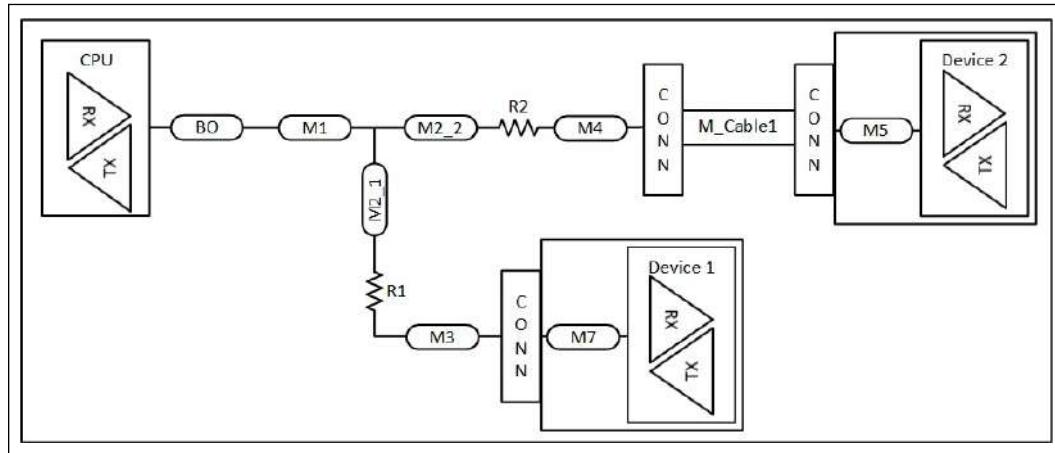


Table 610. SoundWire Large System: 2-Load Dumbbell Topology Notes

Note	Detail
M_Cable1 characteristic impedance target	50 $\Omega \pm 10\%$.
R1	Device Down topology: 10 $\Omega \pm 10\%$. To be placed on SNDW_CLK only, and 10 mm from the branching. Add-In Card topology: 22 $\Omega \pm 10\%$. To be placed on SNDW_CLK only, and 10 mm from the branching.
R2	Device Down topology: 10 $\Omega \pm 10\%$. To be placed on SNDW_CLK only, and 10 mm from the branching. Add-In Card topology: 22 $\Omega \pm 10\%$. To be placed on SNDW_CLK only, and 10 mm from the branching.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	9.6 MHz
SoundWire ports supported	Port #0, Port #1, Port #2, Port #3.
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	For total length < 1003.3 mm: [1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period For total length $\geq$ 1003.3 mm: [1] CLDS = 9 Clock Period

continued...

Note	Detail
	[2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 11.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of 20 $\Omega \pm 30\%$ 2) Device series resistor of 68 $\Omega \pm 5\%$ Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	SNDW[0:3]_CLK, SNDW[0:3]_DATA0.

Table 611. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths**Table 612. SoundWire Large System: 2-Load Dumbbell Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
B0	DSL, MS, SL	12.7
M1	DSL, MS, SL	187.3
M2_1	DSL, MS, SL	10
M2_2	DSL, MS, SL	10
M3	DSL, MS, SL	200
M4	DSL, MS, SL	40
M5	DSL, MS, SL	100
M6	DSL, MS, SL	300
M7	DSL, MS, SL	100
M_Cable1	Cable	500

Min Length Total (mm): 152.4

Max Length Total (mm): 850

Max Length Total Note: 1) Max length between processor to the furthest end device = 850 mm; 2) Total topology length = 1160 mm.

SoundWire Large System: 3 to 4-Load Dumbbell Topology

Figure 251. SoundWire Large System: 3 to 4-Load Dumbbell Option 1 Topology Diagram

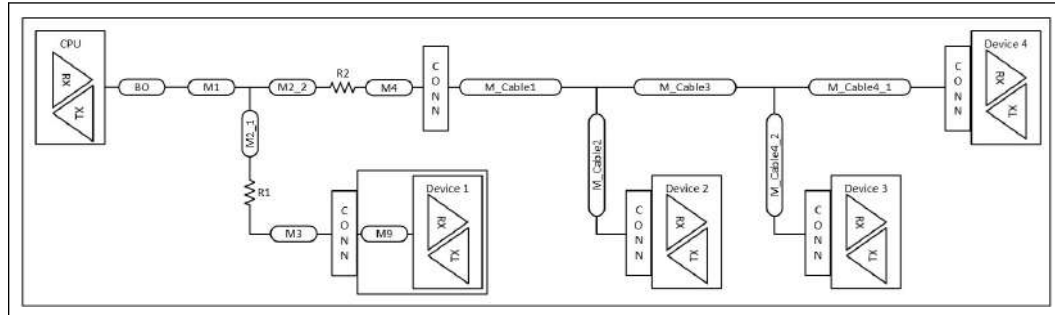


Figure 252. SoundWire Large System: 3 to 4-Load Dumbbell Option 2 Topology Diagram

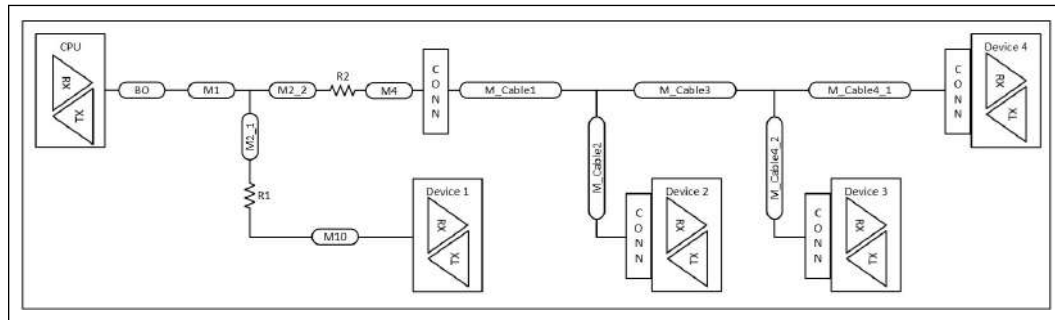


Figure 253. SoundWire Large System: 3 to 4-Load Dumbbell Option 3 Topology Diagram

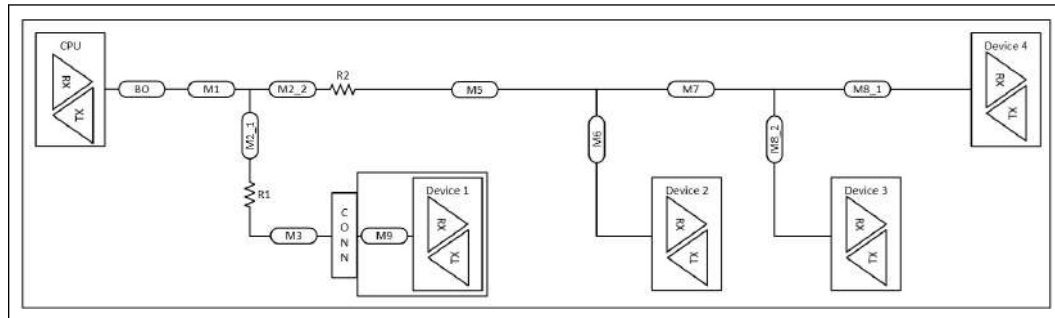


Table 613. SoundWire Large System: 3 to 4-Load Dumbbell Topology Notes

Note	Detail
M_Cable_[1:2] characteristic impedance target	50 Ω $\pm$ 10%.
R1	15 Ω $\pm$ 10%. To be placed on SNDW_CLK only, and 10 mm from the branching.
R2	15 Ω $\pm$ 10%. To be placed on SNDW_CLK only, and 10 mm from the branching.
Multiload support	This topology supports up to 4 devices. For 3-Load topology, only device 2/ 3/ 4 can be removed.

continued...

Note	Detail
Length Extension	Total length for 3-Load topology can be extended up to 520 mm by increase segment length: [1] $M4 + M_{\text{cable1}} = 300 \text{ mm}$ [2] $M5 = 300 \text{ mm}$
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	9.6 MHz
SoundWire ports supported	Port #0, Port #1, Port #2, Port #3.
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	For total length < 508 mm: [1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period For total length $\geq 508 \text{ mm}$: [1] CLDS = 9 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 11.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of $20 \Omega \pm 30\%$ 2) Device series resistor of $91 \Omega \pm 5\%$ Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	SNDW[0:3]_CLK, SNDW[0:3]_DATA0.

Table 614. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths

Table 615. SoundWire Large System: 3 to 4-Load Dumbbell Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
M1	MS, DSL, SL	137.3
<i>continued...</i>		

Segment	Tline Type	Max Length Segment (mm)
M2_1	MS, DSL, SL	10
M2_2	MS, DSL, SL	10
M3	MS, DSL, SL	150
M5	MS, DSL, SL	200
M6	MS, DSL, SL	50
M7	MS, DSL, SL	30
M8_1	MS, DSL, SL	30
M8_2	MS, DSL, SL	30
M9	MS, DSL, SL	100
M10	MS, DSL, SL	250
M4 + M_Cable1	MS, DSL, SL, Cable	200
M_Cable2	Cable	50
M_Cable3	Cable	30
M_Cable4_1	Cable	30
M_Cable4_2	Cable	30

Min Length Total (mm): 127

Max Length Total (mm): 420

Max Length Total Note: 1) Max length between processor to the furthest end device = 420 mm; 2) Total topology length = 760 mm.

SoundWire Large System: 1 to 4-Load Daisy Chain Topology

Figure 254. SoundWire Large System: 1 to 4-Load Daisy Chain Topology Diagram

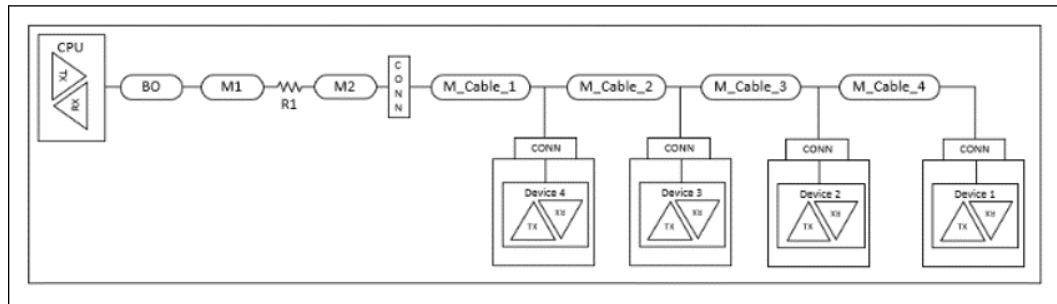


Table 616. SoundWire Large System: 1 to 4-Load Daisy Chain Topology Notes

Note	Detail
M_Cable_[1:4] characteristic impedance target	50 Ω $\pm$ 10%.
R1	0 Ω placeholder. To be placed on SNDW_CLK only, and maximum 10 mm from the connector.
continued...	

Note	Detail
Multiload support	This topology supports up to 4 devices where device placements are interchangeable.
M_Cable_1 maximum length	250 mm
Max allowable cable length between nearest device and furthest device from processor	300 mm
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	9.6 MHz
SoundWire ports supported	Port #0, Port #1, Port #2, Port #3.
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	[1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 01.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of 20 $\Omega \pm 30\%$ 2) Device series resistor of 56 $\Omega \pm 5\%$ Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	SNDW[0:3]_CLK, SNDW[0:3]_DATA0.

Table 617. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths

Table 618. SoundWire Large System: 1 to 4-Load Daisy Chain Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	170
M2	MS	10
M_Cable_1 + M_Cable_2 + M_Cable_3 + M_Cable_4	Cable	400

Min Length Total (mm): 127

Max Length Total (mm): 592.7

Max Length Total Note: 1) Max length between processor to the furthest end device = 592.7 mm; 2) Total topology length = 592.7 mm.

SoundWire Large System: 3-Load Cable Stick (Add-In Card) Topology

Figure 255. SoundWire Large System: 3-Load Cable Stick (Add-In Card) Topology Diagram

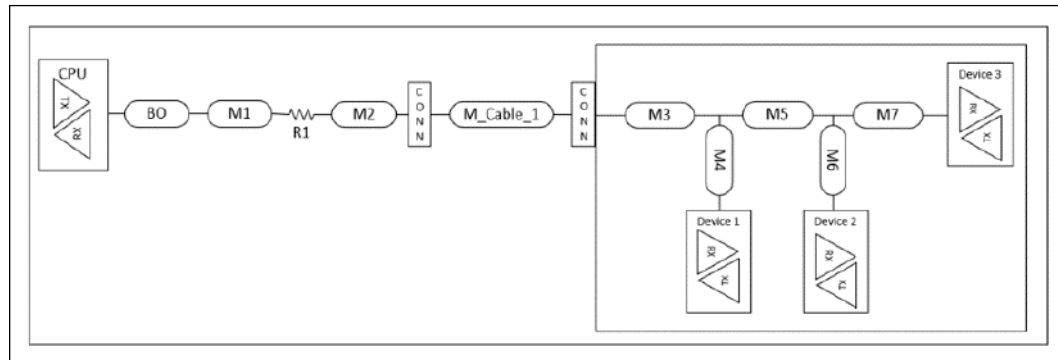


Table 619. SoundWire Large System: 3-Load Cable Stick (Add-In Card) Topology Notes

Note	Detail
M_Cable characteristic impedance target	50 Ω $\pm$ 10%.
R1	0 Ω placeholder. To be placed on SNDW_CLK only, and maximum 10 mm from the connector.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	9.6 MHz
SoundWire ports supported	Port #0, Port #1, Port #2, Port #3.
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	For total length < 850.9 mm: [1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period For total length $\geq$ 850.9 mm: [1] CLDS = 9 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period

continued...

Note	Detail
	[3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 01.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of 20 $\Omega \pm 30\%$ 2) Device series resistor of 56 $\Omega \pm 5\%$ Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	SNDW[0:3]_CLK, SNDW[0:3]_DATA0.

Table 620. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths**Table 621. SoundWire Large System: 3-Load Cable Stick (Add-In Card) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	170
M2	DSL, MS, SL	10
M3	DSL, MS, SL	200
M4	DSL, MS, SL	50
M5	DSL, MS, SL	30
M6	DSL, MS, SL	30
M7	DSL, MS, SL	30
M_Cable_1	Cable	500

Min Length Total (mm): 127

Max Length Total (mm): 952.7

Max Length Total Note: 1) Max length between processor to the furthest end device = 952.7 mm; 2) Total topology length = 1032.7 mm.

SoundWire Small System: 1 to 4-Load Daisy Chain (Device Down) Topology

Figure 256. SoundWire Small System: 1 to 4-Load Daisy Chain (Device Down) Topology Diagram

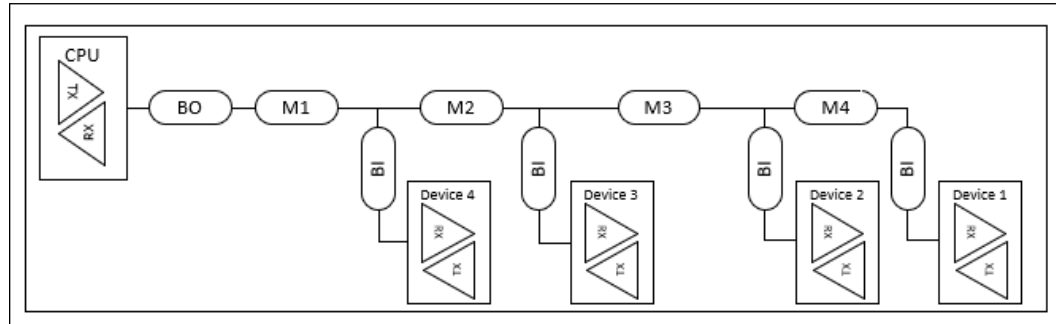


Table 622. SoundWire Small System: 1 to 4-Load Daisy Chain (Device Down) Topology Notes

Note	Detail
Multiload support	This topology supports up to 4 devices where device placements are interchangeable.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	12.288 MHz
SoundWire ports supported	1 to 4-Load topology: Single-lane: Port #0, Port #1, Port #2, Port #3. 1-Load topology: Multi-lane: Port #2 (up to 3 data lane), Port #3 (up to 4 data lane).
Length matching between CLK and DATA signals	5.08 mm
Processor TX & RX timing register configuration	[1] CLDS = 5 Clock Period [2] DODS/DODSE/DODSE2 = 10 Clock Period [3] DOAIS/DOAISE/DOAISE2 = 10 Clock Period Please refer to "Intel Smart Sound Technology ACPI Configuration" for further details on how to configure timing register. (BIOS Specification Doc#: 609917)
Processor buffer driver strength	50 Ω , slew 11.
Device buffer driver strength & resistor combination	1) Device buffer driver strength of 20 $\Omega \pm 30\%$ 2) Device series resistor of 56 $\Omega \pm 5\%$ Refer to section on "Additional Guidelines - SoundWire* Interface Design Guidelines" for resistor placement to achieve the recommended device drive impedance target.
Signal name/ list	Single-lane: SNDW[0:3]_CLK, SNDW[0:3]_DATA0.
continued...	

Note	Detail
	Multi-lane: SNDW[2:3]_CLK, SNDW2_DATA[0:2], SNDW3_DATA[0:3], SNDW2A_DATA1.

Table 623. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel

Segment Lengths**Table 624. SoundWire Small System: 1 to 4-Load Daisy Chain (Device Down) Topology Segment Lengths Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	40
M2	DSL, MS, SL	60
M3	DSL, MS, SL	130
M4	DSL, MS, SL	40
BI	DSL, MS, SL	10

Min Length Total (mm): 62.7

Max Length Total (mm): 292.7

Max Length Total Note: 1) Max length between processor to the furthest end device = 292.7 mm; 2) Total topology length = 322.7 mm.

6.2.10.4 Additional Guidelines - SoundWire* Interface Design Guidelines**Adjusting Drive Impedance for Down Devices**

In the topology section the device buffer strength target recommendations are different depending on the Topology and its configuration. If devices used do not have buffer strengths that match the target recommendations and do not have the ability to adjust the buffer strengths through software you can adjust the impedance via an external resistor (R1) in series near the device, between the device and processor. To determine R1 the formula is: $R1 = \text{Device Buffer Impedance Target} - \text{Actual Device Buffer Impedance}$.

Figure 257. SoundWire* Device Drive Impedance Adjustment

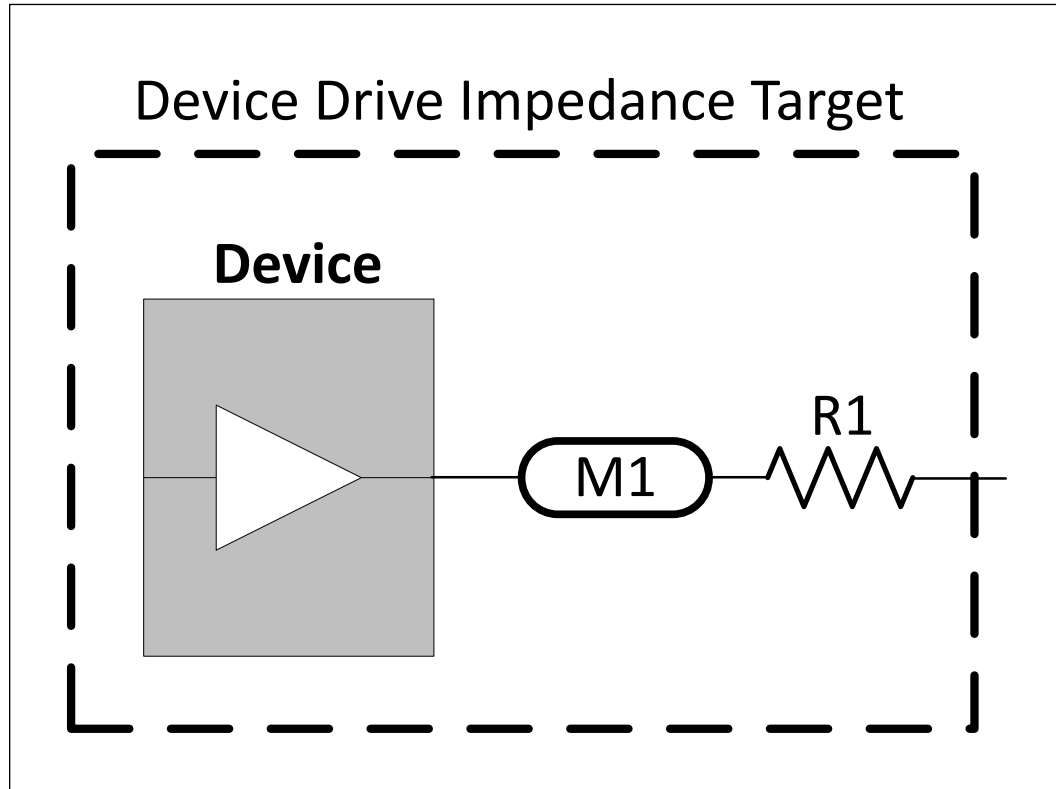


Table 625. Additional Guidelines - SoundWire* Interface Design Guidelines Notes

Note	Detail
M1	10 mm
R1	$R1 = \text{Device Drive Impedance Target} - \text{Device Buffer Strength}.$ For example: Device Drive Impedance Target = 80 Ω . Device Buffer Strength = 45 Ω . Then $R1 = 80 \Omega - 45 \Omega = 35 \Omega$.

6.2.11 I²C Interface Design Guidelines

CPU includes six I²C controllers, providing six independent I²C interfaces. Each interface is a two-wire I²C serial interface consisting of a serial data line and a serial clock.

The I²C interface support 1.8 V.

6.2.11.1 Platform Specific Important Information

Table 626. Reference Specifications

Title
I2C Specification Version 5

6.2.11.2 Signal Descriptions

Signal Groups

Table 627. I²C* Signals

Group	Signal Name	Description
Clock	I2C[5:0]_SCL	I ² C clock signals
Data	I2C[5:0]_SDA	I ² C data signals
Clock	I2C[3:5]A_SCL	I ² C clock signals(Alternate)
Data	I2C[3:5]A_SDA	I ² C data signals (Alternate)
Clock	ISH_I2C[2:0]_SCL	ISH I ² C clock signals
Data	ISH_I2C[2:0]_SDA	ISH I ² C data signals
Clock	ISH_I2C2A_SCL	ISH I ² C data signals (Alternate)
Data	ISH_I2C2A_SDA	ISH I ² C data signals (Alternate)
Clock	THC_I2C[1:0]_SCL	THC I ² C clock signals
Data	THC_I2C[1:0]_SDA	THC I ² C data signals

6.2.11.3 Topology Guidelines

General Design Considerations

I²C bus is a multi-device bus (i.e. more than one device can be connected to an I²C bus). The total number of devices connected to the bus is determined by the total capacitive load on the bus.

I²C clock and data signals require pull-up resistors.

Do not mix and match devices supporting different speeds on the same I²C bus.

I²C signals are multiplexed with GPIOs and default to GPIO functionality (as input). If I²C interfaces are not used, the signals can be used as GPIOs instead. If the GPIO functionality is also not used, the signals can be left as no-connect.

Detailed Routing Requirements

System designers must consider the total bus capacitance which includes both package and device pin capacitance and board trace length capacitance when designing I²C bus. The total bus capacitance must not exceed 100pF for High Speed Mode and 400pF for other modes (Standard/Fast/Fast mode plus/High speed Mode).

6.2.11.3.1 I2C

Table 628. I2C General Guidelines

Note	Detail
Note 1	Multiple options for pull-up/ pull-down and current assist strengths are provided for a particular total bus capacitance. Choose the optimal value for system based on design considerations like power, BOM and cost.
Note 2	Trace capacitance per inch is approximately $CL = 85\sqrt{\epsilon_r / Z_0}$ pF, ϵ_r - dielectric constant of material, Z_0 - trace impedance.
Note 3	Bus capacitance calculation assumptions: [1] Device capacitance = 5 pF to 10 pF per I2C device. [2] Motherboard board trace capacitance = 2 pF to 5 pF per inch of trace length. [3] Processor pin capacitance = 8 pF to 10 pF.

I2C Topology

Table 629. I2C Topology Notes

Note	Detail
Stackup layer (microstrip/ stripline/ dual stripline)	MS/ SL/ DSL.
Characteristic impedance	30 Ω to 70 Ω .
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Trace spacing between SCL and other signals	[1] 0.375 mm: Frequency of other signals > 1 GHz. [2] 0.250 mm: Frequency of other signals > 100 MHz. [3] 0.125 mm: Frequency of other signals $\leq$ 100 MHz.
Trace spacing between SDA and other signals	0.125 mm
Trace segment length	Refer to bus capacitance table.
Total trace length (SDA, SCL)	Individual trace lengths should meet total bus capacitance (Cb) requirements for the mode of operation. Refer to the pull-up/ pull-down tables for the supported bus capacitance.
Length matching between SDA and SCL signals	25.4 mm
Pull-up resistor placement	No restriction.
Pull-up resistor value tolerance	$\pm$ 5%
Max stub length to pull-up resistor	25.4 mm
continued...	

Note	Detail
Board trace characteristic assumptions for default pull-up recommendation	Board trace resistance per length = 20 Ω per meter. Board trace capacitance per length = 208 pF per meter.
Extended length support	For extended length support, refer to "855310_NVL_H_I2C_SMLINK_LSIO_Flexi_PDG". Potential enablers: [1] Reduced trace resistance. [2] Reduced trace capacitance. [3] Optimized pull-up resistor placement.
Signal name/ list	I2C[0:5]_SDA, I2C[0:5]_SCL, I2C[3A:5A]_SDA, I2C[3A:5A]_SCL, ISH_I2C[0:2]_SDA, ISH_I2C[0:2]_SCL, ISH_I2C2A_SDA, ISH_I2C2A_SCL, THC_I2C[0:1]_SCL, THC_I2C[0:1]_SDA.

I2C Standard Mode (100 kHz)

Frequency: 100 kHz

Table 630. I2C Standard Mode (100 kHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	8.2 k Ω	100 Ω
Up to 100 pF	4.7 k Ω	100 Ω
Up to 200 pF	2.2 k Ω	100 Ω
Up to 300 pF	1.5 k Ω	100 Ω
Up to 400 pF	1.2 k Ω	100 Ω

I2C Fast Mode (400 kHz)

Frequency: 400 kHz

Table 631. I2C Fast Mode (400 kHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	3.3 k Ω	100 Ω
Up to 100 pF	1.5 k Ω	100 Ω
70 pF to 200 pF	820 Ω	100 Ω
70 pF to 400 pF	620 Ω	100 Ω

I2C Fast Mode Plus (1 MHz)

Frequency: 1 MHz

Table 632. I2C Fast Mode Plus (1 MHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	1.2 k Ω	100 Ω
Up to 100 pF	680 Ω	100 Ω
60 pF to 300 pF	300 Ω	50 Ω

I2C High Speed Mode (1.7 MHz) - SCL

Frequency: 1.7 MHz

Table 633. I2C High Speed Mode (1.7 MHz) - SCL Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength	Current Assist
Up to 50 pF	1.5 k Ω	100 Ω	None
45 pF to 80 pF	1 k Ω	100 Ω	None
75 pF to 165 pF	470 Ω	100 Ω	None
130 pF to 210 pF	360 Ω	50 Ω	None
200 pF to 250 pF	300 Ω	50 Ω	None
235 pF to 265 pF	470 Ω	50 Ω	010
263 pF to 310 pF	470 Ω	50 Ω	011
300 pF to 350 pF	470 Ω	50 Ω	100
335 pF to 400 pF	470 Ω	50 Ω	110

I2C High Speed Mode (1.7 MHz) - SDA

Frequency: 1.7 MHz

Table 634. I2C High Speed Mode (1.7 MHz) - SDA Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength	Current Assist
Up to 50 pF	1.8 k Ω	100 Ω	None
45 pF to 100 pF	1.2 k Ω	100 Ω	None
50 pF to 165 pF	910 Ω	100 Ω	None
130 pF to 350 pF	430 Ω	50 Ω	None
95 pF to 400 pF	360 Ω	50 Ω	None

I2C High Speed Mode (3.4 MHz) - SCL

Frequency: 3.4 MHz

Table 635. I2C High Speed Mode (3.4 MHz) - SCL Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength	Current Assist
Up to 50 pF	750 Ω	100 Ω	None
40 pF to 70 pF	560 Ω	100 Ω	None
55 pF to 100 pF	390 Ω	50 Ω	None

I2C High Speed Mode (3.4 MHz) - SDA

Frequency: 3.4 MHz

Table 636. I2C High Speed Mode (3.4 MHz) - SDA Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength	Current Assist
Up to 50 pF	1.5 k Ω	100 Ω	None
45 pF to 70 pF	1 k Ω	100 Ω	None
55 pF to 100 pF	750 Ω	50 Ω	None

I2C for DDR5 SPD Topology**Table 637. I2C for DDR5 SPD Topology Notes**

Note	Detail
Stackup layer (microstrip/ stripline/ dual stripline)	MS/ SL/ DSL.
Characteristic impedance	30 Ω to 70 Ω .
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Trace spacing between SCL and other signals	[1] 0.375 mm: Frequency of other signals > 1 GHz. [2] 0.250 mm: Frequency of other signals > 100 MHz. [3] 0.125 mm: Frequency of other signals $\leq$ 100 MHz.
Trace spacing between SDA and other signals	0.125 mm
Trace segment length	Refer to bus capacitance table.
Total trace length (SDA, SCL)	Individual trace lengths should meet total bus capacitance (Cb) requirements for the mode of operation. Refer to the pull-up/ pull-down tables for the supported bus capacitance.
Length matching between SDA and SCL signals	25.4 mm
Pull-up resistor placement	No restriction.
Pull-up resistor value tolerance	$\pm$ 5%
Max stub length to pull-up resistor	25.4 mm
Max stub length to device	127 mm
Board trace characteristic assumptions for default pull-up recommendation	Board trace resistance per length = 20 Ω per meter. Board trace capacitance per length = 208 pF per meter.
Extended length support	For extended length support, refer to "855310_NVL_H_I2C_SMLINK_LSIO_Flexi_PDG". Potential enablers: [1] Reduced trace resistance. [2] Reduced trace capacitance. [3] Optimized pull-up resistor placement.
continued...	

Note	Detail
Max SPD device input capacitance	5 pF
General recommendations for DDR5 SPD	[1] It is recommended to have a dedicated bus for DDR5 SPD usage. [2] If dedicated bus is not possible, it is recommended to use a voltage translator to shift the signal down to 1V to match the SPD device.
Signal name/ list	I2C[0:5]_SDA, I2C[0:5]_SCL, I2C[3A:5A]_SDA, I2C[3A:5A]_SCL.

I2C Standard Mode (100 kHz)

Frequency: 100 kHz

Table 638. I2C Standard Mode (100 kHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	8.2 kΩ	100 Ω
Up to 100 pF	4.7 kΩ	100 Ω
Up to 200 pF	2.7 kΩ	100 Ω
Up to 300 pF	1.8 kΩ	100 Ω
Up to 400 pF	1.5 kΩ	50 Ω

I2C Fast Mode (400 kHz)

Frequency: 400 kHz

Table 639. I2C Fast Mode (400 kHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	3.9 kΩ	100 Ω
45 pF to 100 pF	2.7 kΩ	100 Ω
85 pF to 250 pF	1.2 kΩ	50 Ω

I2C Fast Mode Plus (1 MHz)

Frequency: 1 MHz

Table 640. I2C Fast Mode Plus (1 MHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	2.2 kΩ	100 Ω
Up to 115 pF	1 kΩ	50 Ω

6.2.11.4 Clock Frequency Guidelines

I2C clock frequency is determined by several factors, including, but not limited to:

- Platform load
- PVT variations and its effect on clocks and interconnects within Processor
- VIL/VIH sensitivity of Processor pads

The aim is to provide a reference value of HCNT and LCNT for various values of permissible SCL rise times for typical conditions. Values for intermediate SCL rise times may be interpolated from the below values in the table. Fine tuning of the recommended values may also be undertaken to change clock frequency to the desired value. A unit change in HCNT or LCNT results in 10 ns change in high and low clock period respectively. Spike filter length can be changed in units of 10 ns. Increment in spike filter length should be accompanied with equal decrement in HCNT and vice versa.

NOTE

Spike filter length assumed to be 10 ns (min value) for all the recommended values.

For HCNT/LCNT register details, refer to Nova Lake H/Hx External Design Specification (EDS) Volume 2 of 2, (# 855520).

Table 641. LPSS I2C0

Mode	Frequency																			
	SS (100 kHz)										FM (400 kHz)									
Min LCNT	473										133									
Min HCNT	399										59									
Rise time (ns)	50	150	250	350	450	550	650	750	850	950	34	62	90	118	146	174	202	230	258	286
LCNT	593	583	573	563	553	543	533	523	513	503	166	166	166	166	166	166	166	165	162	159
HCNT	399	399	399	399	399	399	399	399	399	399	77	75	72	69	66	63	61	59	59	59
	FM+ (1 MHz)										HS1 (1.7 MHz)									
Min LCNT	53										35									
Min HCNT	25										11									
Rise time (ns)	6	18	30	42	54	66	78	90	102	114	23	29	35	41	47	53	59	65	71	77
LCNT	66	66	66	66	66	65	64	63	62	60	39	39	39	39	39	39	39	38	38	37
HCNT	30	29	28	27	25	25	25	25	25	25	15	14	13	13	12	12	11	11	11	11

Table 642. LPSS I2C1/2/3

Mode	Frequency																			
	SS (100 kHz)										FM (400 kHz)									
Min LCNT	473										133									
Min HCNT	399										59									
Rise time (ns)	50	150	250	350	450	550	650	750	850	950	34	62	90	118	146	174	202	230	258	286
LCNT	593	583	573	563	553	543	533	523	513	503	166	166	166	166	166	166	166	165	162	159
HCNT	399	399	399	399	399	399	399	399	399	399	77	75	72	69	66	63	61	59	59	59
	FM+ (1 MHz)										HS1 (1.7 MHz)									
Min LCNT	53										35									

continued...

Mode	Frequency																			
Min HCNT	25										11									
Rise time (ns)	6	18	30	42	54	66	78	90	102	114	23	29	35	41	47	53	59	65	71	77
LCNT	66	66	66	66	66	65	64	63	62	60	39	39	39	39	39	39	39	38	38	37
HCNT	30	29	28	27	25	25	25	25	25	25	15	14	13	13	12	11	11	11	11	11

Table 643. LPSS I2C4/5

Mode	Frequency																			
	SS (100 kHz)										FM (400 kHz)									
Min LCNT	473										133									
Min HCNT	399										59									
Rise time (ns)	50	150	250	350	450	550	650	750	850	950	34	62	90	118	146	174	202	230	258	286
LCNT	59 2	582	572	562	552	542	532	522	512	502	166	166	166	166	166	166	166	164	162	159
HCNT	39 9	399	399	399	399	399	399	399	399	399	77	74	71	69	66	63	60	59	59	59
	FM+ (1 MHz)										HS1 (1.7 MHz)									
Min LCNT	53										35									
Min HCNT	25										11									
Rise time (ns)	6	18	30	42	54	66	78	90	102	114	23	29	35	41	47	53	59	65	71	77
LCNT	66	66	66	66	66	65	64	62	61	60	39	39	39	39	39	39	39	38	38	37
HCNT	30	29	27	26	25	25	25	25	25	25	14	14	13	13	12	11	11	11	11	11

6.2.11.5 Tools

Intel does not promote any specific tool for this interface.

6.2.12 I3C

6.2.12.1 I3C 1-Island (Device Down) Topology

Figure 258. I3C 1-Island (Device Down) Topology Diagram

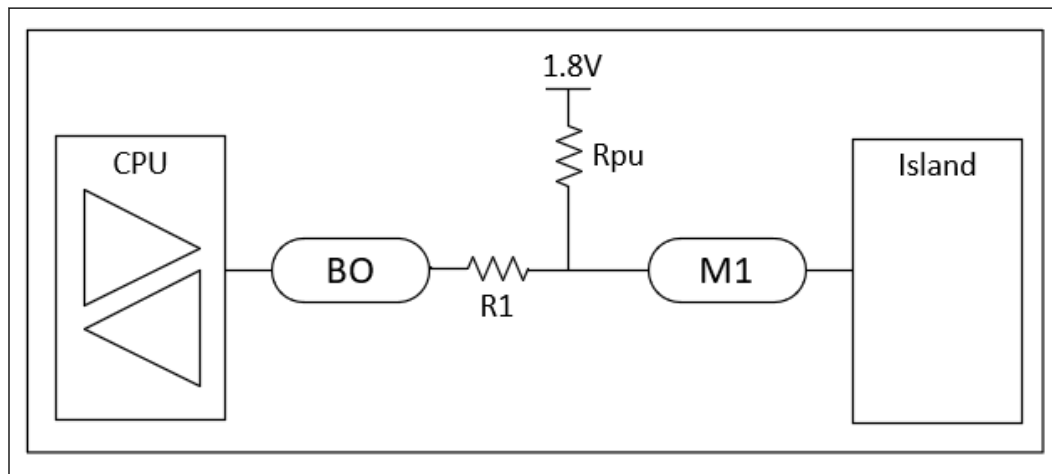


Table 644. I3C 1-Island (Device Down) Topology Notes

Note	Detail
Mixed bus topology	This topology supports usages with a mix of I2C and I3C devices. For mixed I2C and I3C bus, routing should follow I3C PDG.
Reduced max length	Actual max length for I3C can go up to 1016 mm. Reduced max length is due to legacy I2C Fast Mode Plus (FM+) requirement.
Island terminology	'Island' refers to 1 to 3 devices connected and placed near to each other, where the max capacitance per island is 20 pF. For multiple devices within an island, the max stub length to each device is 25.4 mm.
Max frequency	2 MHz
Rpu	1 kΩ.
R1	20 Ω. Placed within 25.4 mm distance from processor.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Processor Write Hold Time Programming	If total length > 450 mm, it is recommended to tune the SDA_TX_HOLD timing register to improve hold time during processor write mode.
Length matching between SDA and SCL signals	25.4 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
continued...	

Note	Detail
Processor buffer driver strength	40 Ω.
Max device buffer driver strength	90 Ω
Signal name/ list	I3C[0:3]_SDA, I3C[0:3]_SCL, I3C[0:1]A_SDA, I3C[0:1]A_SCL, I3C3A_SDA, I3C3A_SCL, ISH_I3C[0:1]_SDA, ISH_I3C[0:1]_SCL, I3C1B_SDA, I3C1B_SCL.

Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode

Table 645. I3C 1-Island (Device Down) Topology Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	25.4
M1	MS, DSL, SL	990.6

Max Length Total (mm): 1016

Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode Plus

Table 646. I3C 1-Island (Device Down) Topology Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode Plus Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	25.4
M1	MS, DSL, SL	533.4

Max Length Total (mm): 558.8

6.2.12.2 I3C 2-Island Daisy Chain (Device Down) Topology

Figure 259. I3C 2-Island Daisy Chain (Device Down) Topology Diagram

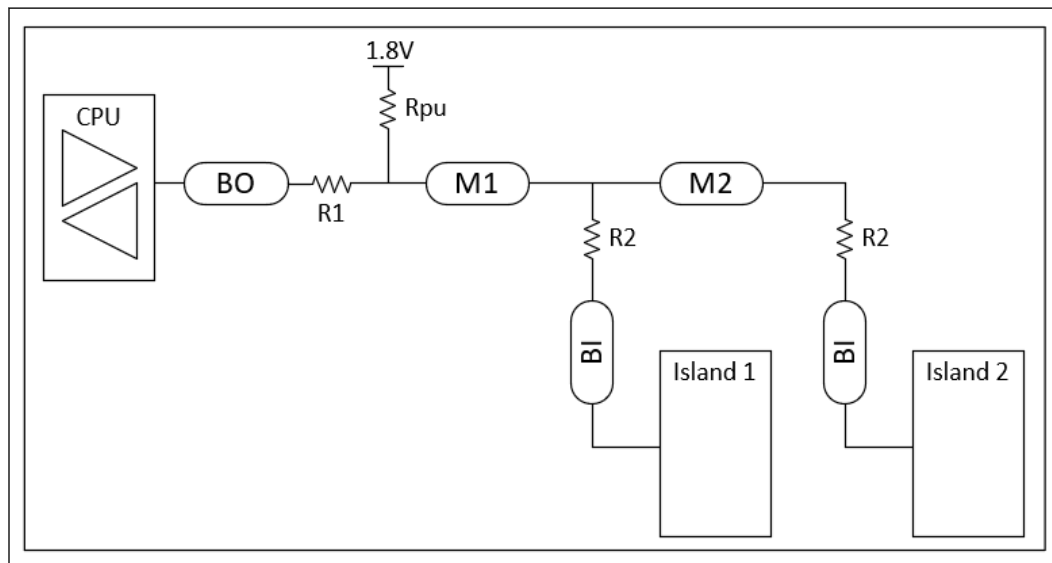


Table 647. I3C 2-Island Daisy Chain (Device Down) Topology Notes

Note	Detail
Mixed bus topology	This topology supports usages with a mix of I2C and I3C devices. For mixed I2C and I3C bus, the routing needs to follow the I3C PDG.
Reduced max length	Actual max length for I3C can go up to 762 mm. Reduced max length is due to legacy I2C Fast Mode Plus (FM+) requirement.
Island terminology	'Island' refers to 1 to 3 devices connected and placed near to each other, where the max capacitance per island is 20 pF. For multiple devices within an island, the max stub length to each device is 25.4 mm.
Max frequency	2 MHz
Rpu	1 kΩ.
R1	20 Ω. Placed within 25.4 mm distance from processor.
R2	20 Ω. Placed within 25.4 mm distance to device.
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Processor Write Hold Time Programming	If total length > 450 mm, it is recommended to tune the SDA_TX_HOLD timing register to improve hold time during processor write mode.
Length matching between SDA and SCL signals	25.4 mm
continued...	

Note	Detail
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Max device buffer driver strength	90 Ω
Processor buffer driver strength	40 Ω .
Signal name/ list	I3C[0:3]_SDA, I3C[0:3]_SCL, I3C[0:1]A_SDA, I3C[0:1]A_SCL, I3C3A_SDA, I3C3A_SCL, ISH_I3C[0:1]_SDA, ISH_I3C[0:1]_SCL, I3C1B_SDA, I3C1B_SCL.

Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode

Table 648. I3C 2-Island Daisy Chain (Device Down) Topology Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	25.4
M1	MS, DSL, SL	381
M2	MS, DSL, SL	381
BI	MS, DSL, SL	25.4

Max Length Total (mm): 762

Max Length Total Note: 1) Max length between processor to the furthest end device = 762 mm; 2) Total topology length = 787.4 mm.

Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode Plus

Table 649. I3C 2-Island Daisy Chain (Device Down) Topology Segment Lengths For Mixed I3C/I2C Bus, Up To I2C Fast Mode Plus Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	25.4
M1	MS, DSL, SL	254
M2	MS, DSL, SL	254
BI	MS, DSL, SL	25.4

Max Length Total (mm): 558.8

Max Length Total Note: 1) Max length between processor to the furthest end device = 558.8 mm; 2) Total topology length = 584.2 mm.

6.2.12.3 I3C 1-Load 10MHz (Device Down) Topology

Figure 260. I3C 1-Load 10MHz (Device Down) Topology Diagram

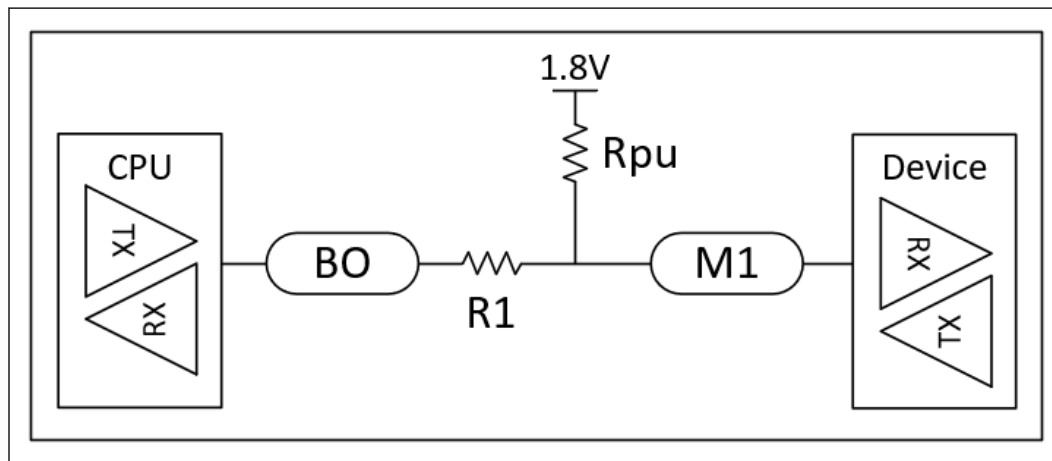


Table 650. I3C 1-Load 10MHz (Device Down) Topology Notes

Note	Detail
R1	[1] $33\ \Omega \pm 5\%$ for max total length < 127 mm. [2] $20\ \Omega \pm 5\%$ for max total length ≥ 127 mm. Placed within 25.4 mm distance from processor.
Rpu	1 k Ω .
Max frequency	10 MHz
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Length matching between SDA and SCL signals	25.4 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Processor buffer driver strength	40 Ω .
Max device buffer driver strength	90 Ω
Signal name/ list	I3C[0:3]_SDA, I3C[0:3]_SCL, I3C[0:1]A_SDA, I3C[0:1]A_SCL, I3C3A_SDA, I3C3A_SCL, ISH_I3C[0:1]_SDA, ISH_I3C[0:1]_SCL, I3C1B_SDA, I3C1B_SCL.

Table 651. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	7	Total Channel

Segment Lengths

Table 652. I3C 1-Load 10MHz (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	25.4
M1	MS, DSL, SL	355.6

Max Length Total (mm): 381

6.2.13 SMLink Interface Design Guidelines

6.2.13.1 Platform Specific Important Information

SMLink Description

SMLink interface is based on SMBus protocol. The processor incorporates four SMLink interfaces, SMLink0, SMLink1, USB_C_SML and SML0B. SMLink0 is used for Re-timer configuration for USB Type-C interface and between CPU's LAN controller and the USB_C_SML is used for PD to PMC communication.

6.2.13.2 Signal Descriptions

Signal Groups

Table 653. SMLink Signals

Group	Signal Names	Description
Clock	SML[1:0]CLK SML0BCLK USB-C_SMLCLK PSE_SMLCLK	SMLink Clock signals
Data	SML[1:0]DATA SML0BDATA USB-C_SMLDATA PSE_SMLDATA	SMLink Data lines
ALERT	SML[1:0]ALERT# SML0BALERT# PMCALERT# PSE_SMLALERT#	SMLink ALERT signals USB Type-C PD controller / Re-timer Alert.

6.2.13.3 Topology Guidelines

General Design Considerations

- The maximum bus capacitance is 400 pF.

- SMLink devices that can operate in S3 must be powered by the suspend power supply.

6.2.13.3.1 SMLink

Table 654. SMLink General Guidelines

Note	Detail
Note 1	Multiple options for pull-up/ pull-down and current assist strengths are provided for a particular total bus capacitance. Choose the optimal value for system based on design considerations like power, BOM and cost.
Note 2	Trace capacitance per inch is approximately $CL = 85\sqrt{\epsilon_r} / Z_0$ pF, ϵ_r - dielectric constant of material, Z_0 - trace impedance.
Note 3	Bus capacitance calculation assumptions: [1] Device capacitance = 5 pF to 10 pF per SMLink device. [2] Motherboard board trace capacitance = 2 pF to 5 pF per inch of trace length. [3] Processor pin capacitance = 8 pF to 10 pF.

SMLink Topology

Table 655. SMLink Topology Notes

Note	Detail
Stackup layer (microstrip/ stripline/ dual stripline)	MS/ SL/ DSL.
Characteristic impedance	30 Ω to 70 Ω .
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Trace spacing between CLK and other signals	[1] 0.375 mm: Frequency of other signals > 1 GHz. [2] 0.250 mm: Frequency of other signals > 100 MHz. [3] 0.125 mm: Frequency of other signals $\leq$ 100 MHz.
Trace spacing between DATA and other signals	0.125 mm
Trace segment length	Refer to bus capacitance table.
Total trace length (DATA, CLK)	Individual trace lengths should meet total bus capacitance (Cb) requirements for the mode of operation. Refer to the pull-up/ pull-down tables for the supported bus capacitance.
Length matching between DATA and CLK signals	25.4 mm
Pull-up resistor placement	No restriction.
Pull-up resistor value tolerance	$\pm$ 5%
Max stub length to pull-up resistor	25.4 mm
<i>continued...</i>	

Note	Detail
Board trace characteristic assumptions for default pull-up recommendation	Board trace resistance per length = 20 Ω per meter. Board trace capacitance per length = 208 pF per meter.
Extended length support	For extended length support, refer to "855310_NVL_H_I2C_SMLINK_LSIO_Flexi_PDG". Potential enablers: [1] Reduced trace resistance. [2] Reduced trace capacitance. [3] Optimized pull-up resistor placement.
Signal name/ list	SML[0:1]CLK, SML[0:1]DATA, SML[0:1]ALERT#, SML0BCLK, SML0BDATA, SML0BALERT#, PSE_SMLCLK, PSE_SMLDATA, PSE_SMLALERT#, USBC-SMLCLK, USBC-SMLDATA.

SMLink Standard Mode (100 kHz)

Frequency: 100 kHz

Table 656. SMLink Standard Mode (100 kHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	5.6 k Ω	100 Ω
Up to 100 pF	2.7 k Ω	100 Ω
Up to 200 pF	1.5 k Ω	100 Ω
Up to 300 pF	1 k Ω	100 Ω
Up to 400 pF	820 Ω	100 Ω

SMLink Fast Mode (400 kHz)

Frequency: 400 kHz

Table 657. SMLink Fast Mode (400 kHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	1.8 k Ω	100 Ω
Up to 100 pF	1.5 k Ω	100 Ω
75pF to 200 pF	560 Ω	100 Ω
85 pF to 300 pF	430 Ω	50 Ω

SMLink Fast Mode Plus (1 MHz)

Frequency: 1 MHz

Table 658. SMLink Fast Mode Plus (1 MHz) Settings

Total Bus Capacitance (Cb)	External Pull-up Strength	Intel Silicon Pull-down Strength
Up to 50 pF	820 Ω	100 Ω
Up to 100 pF	470 Ω	100 Ω
Up to 200 pF	270 Ω	50 Ω

6.2.13.4 Detailed Routing Requirements

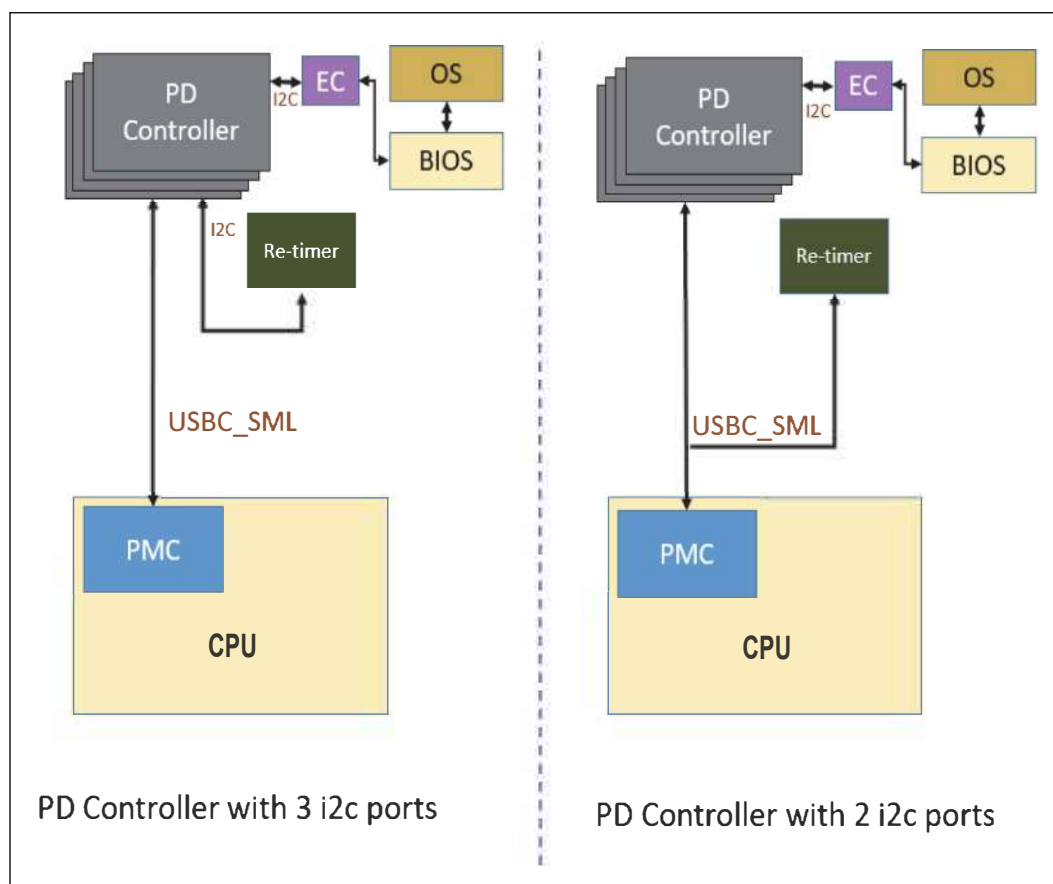
Since SMLink trace length is dependent on total capacitance of the system bus, designer needs to take into consideration the number and types of devices/slots on the motherboards.

External pull up is required which is also based on the load capacitance and bus speed.

Designer needs to calculate the total capacitance for trace routing and the pins of each device connected on the bus.

6.2.13.5 SMLink Connectivity Recommendation

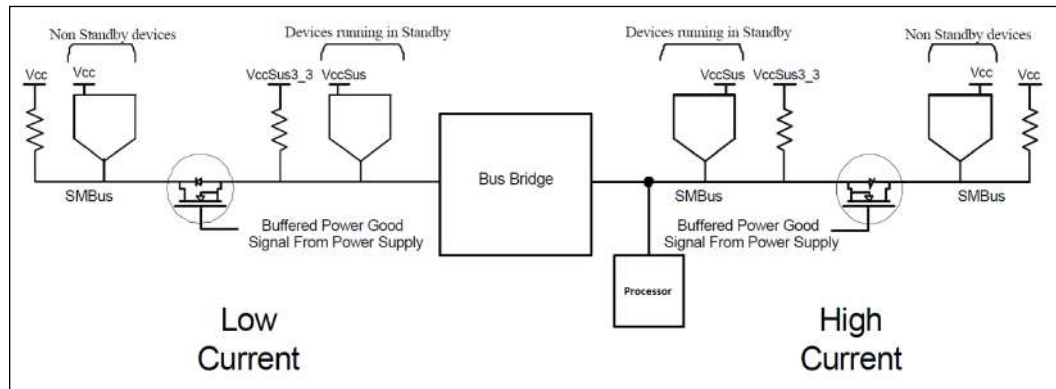
Figure 261. SMLink Connectivity for USB Type-C PD Controller



High Power/Low Power Mixed Architecture

This design allows for current isolation of high and low current devices while also allowing SMLink devices to communicate while in S3. Suspend power well leakage is minimized by keeping non-essential devices on the core supply. This is accomplished by the use of a "FET" to isolate the devices powered by the core and suspend supplies. Refer Figure below:

Figure 262. High Power/Low Power Mixed Vcc_SUSPEND / Vcc_CORE_ Architecture



Added Considerations for Mixed Architecture:

- The bus bridge must be powered by suspend power well.
- Devices that are powered by the suspend power well must not drive into other devices that are powered off. This is accomplished with the “bus switch”.
- The bus bridge can be a device like the Philips* PCA9515.

6.2.13.6 Tools

Intel does not promote any specific tool for this interface.

6.2.14 Serial Peripheral Interface (SPI0) Flash Design Guidelines

The following provides general guidelines for compatibility and design recommendations for supporting flash devices.

The Serial Peripheral Interface (SPI0) supports 2 SPI0 flash devices via 2 chip select (SPI0_CS0# and SPI0_CS1#). The maximum size of flash supported is determined by the SFDP-discovered addressing capability of each device. Each component can be up to 16 MB (32 MB total addressable) using 3-byte addressing. Each component can be up to 64 MB (128 MB total addressable) using 4-byte addressing. Another chip selects (SPI0_CS2#) is also available and only used for TPM on SPI0 support.

The SPI0 interface supports 1.8V only. Descriptor mode is required for all platforms with Processor.

The SPI0 controller supports a maximum operating frequency of 80 MHz.

A SPI0 flash device supporting SFDP (Serial Flash Discovery Parameter) is required for all Processor design. A SPI0 flash device on SPI0_CS0# with a valid descriptor **MUST** be attached directly to the Processor.

NOTE

Refer to the platform SPI Programming guide for the supported SPI0 flash operating frequency.

Table 659. SPI0 Signals

Signal Name	Group	Description
SPI0_MOSI	Data	SPI0 serial output data from Processor to the SPI0 flash device. This Pin will also function as Input during Dual and Quad I/O operation
SPI0_MISO	Data	SPI0 serial input data from the SPI0 flash device to Processor. This Pin will also function as Output during Dual and Quad I/O operation
SPI0_IO2	Data	SPI0 I/O to comprehend the support for the Quad I/O operation
SPI0_IO3	Data	SPI0 I/O to comprehend the support for the Quad I/O operation
SPI0_CLK	Clock	SPI0 Clock output from Processor
SPI0_CS0#	Chip Select0	SPI0 chip select 0
SPI0_CS1#	Chip Select1	SPI0 chip select 1 signal is used as the second chip select when 2 flash devices are used. Do not use when only one SPI0 flash is used
SPI0_CS2#	Chip Select2	Chip Select is dedicated to support TPM on SPI0.

6.2.14.1 Guidelines

SPI0 flash must be directly connected to the Processor SPI0 bus. Also, refer to the Serial Flash vendor documentation for additional Serial Flash specific design considerations.

NOTE

For SPI0 topology with Processor and EC G3 Flash sharing in Wire-OR Configuration, it is recommended that when VCC_SPI is on, all primary wells must be ON before accessing the SPI bus.

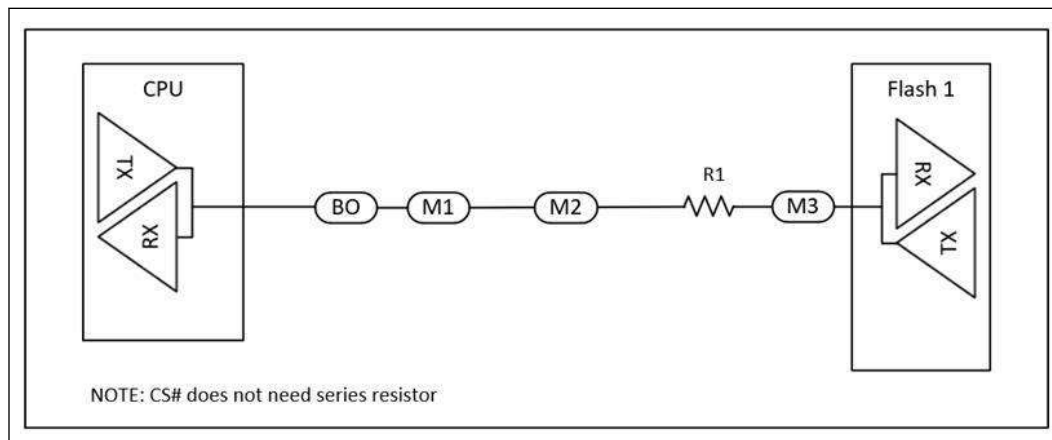
6.2.14.1.1 SPI0 Flash**SPI0 Flash 50MHz 1 to 4-Load Branch (Device Down) MAF Topology****Figure 263. SPI0 50MHz 1-Load MAF (Device Down) Topology Diagram**

Figure 264. SPI0 50MHz 2-Load Branch MAF (Device Down) with 2 Flash Topology Diagram

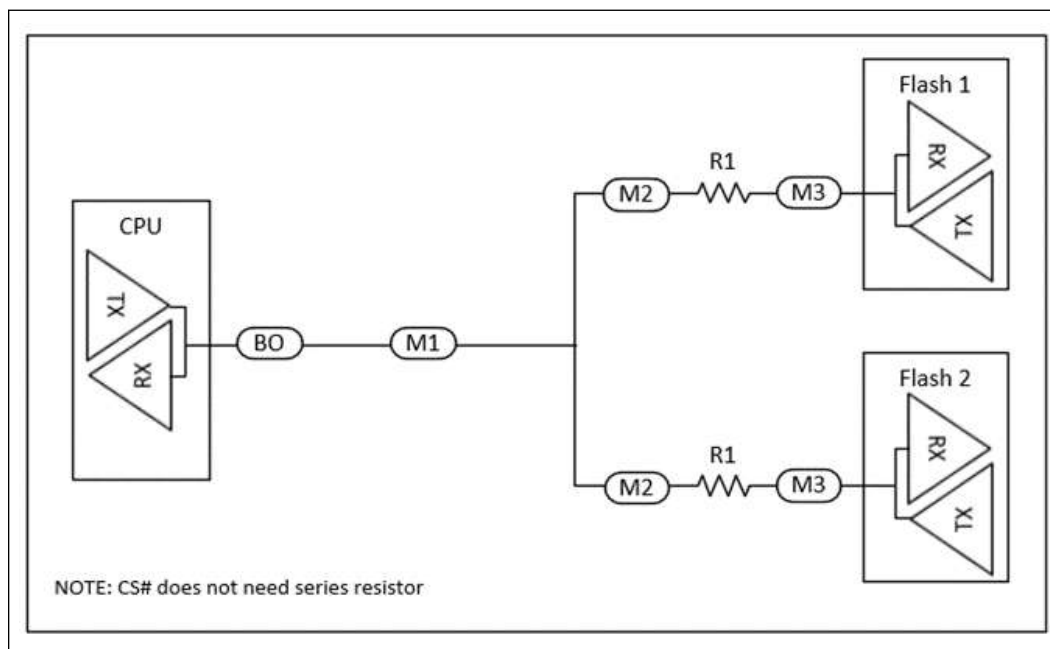


Figure 265. SPI0 50MHz 2-Load Branch MAF (Device Down) with Flash and TPM Topology Diagram

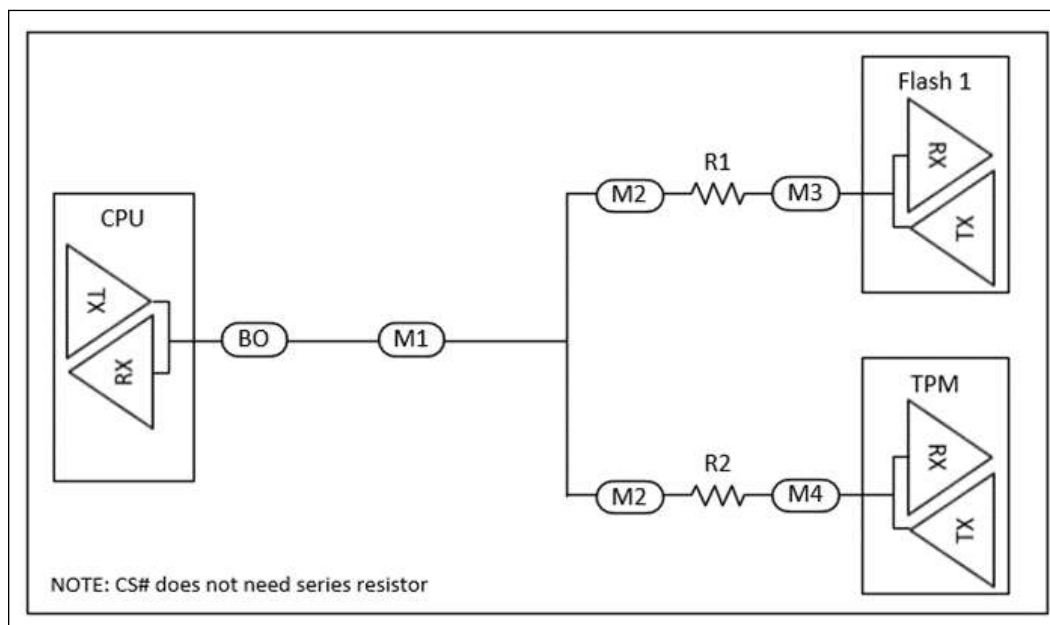


Figure 266. SPI0 50MHz 3-Load Branch MAF (Device Down) Topology Diagram

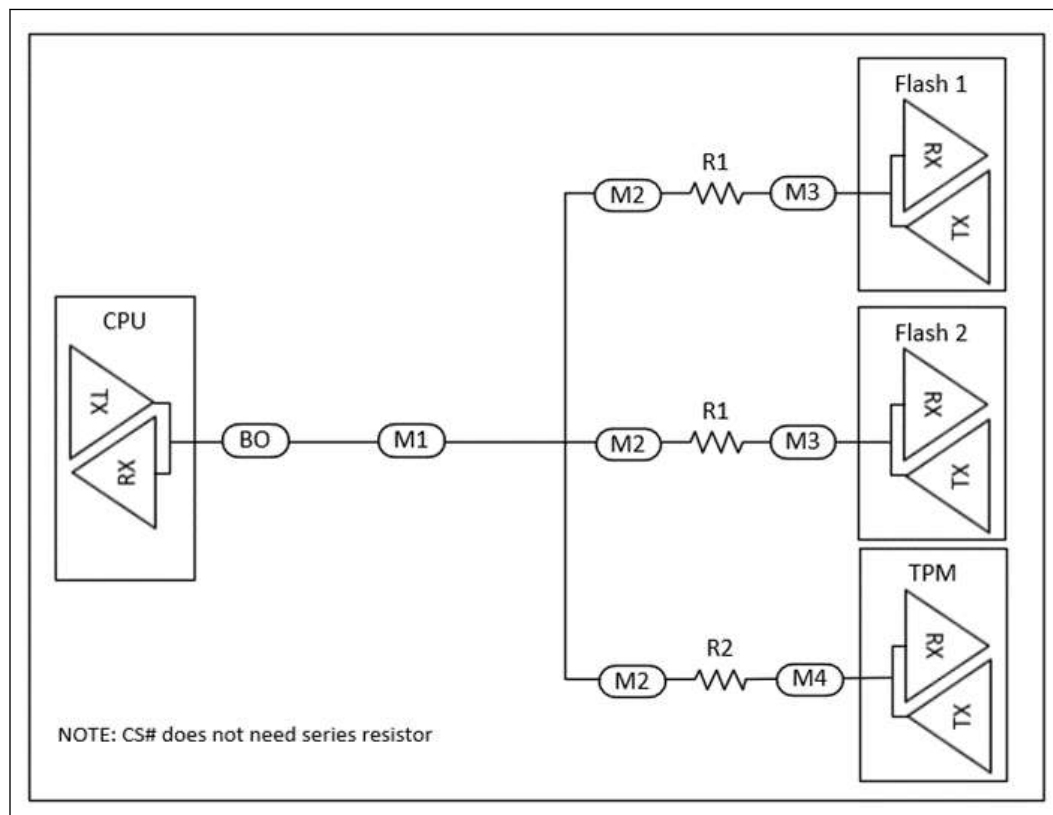


Figure 267. SPI0 50MHz 4-Load Branch MAF (Device Down) Topology Diagram

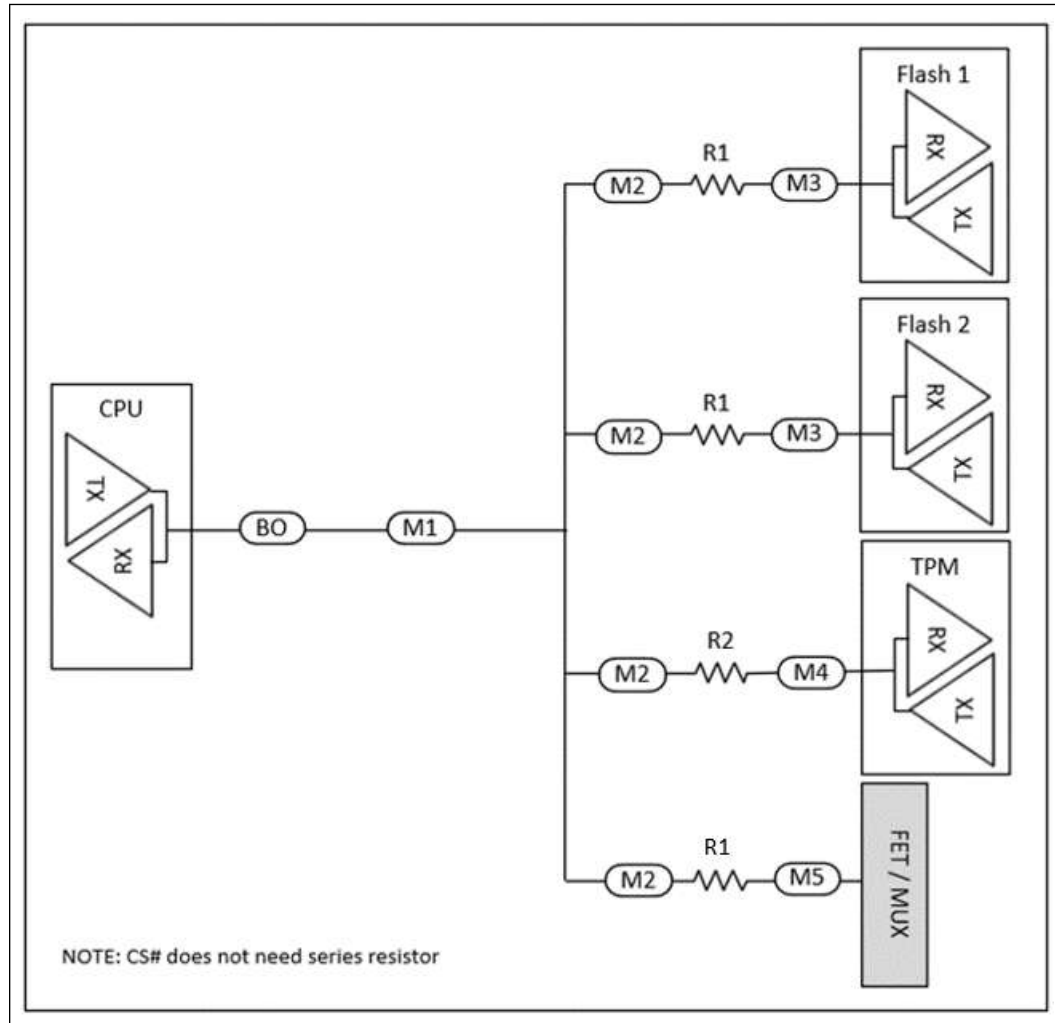


Table 660. SPI0 Flash 50MHz 1 to 4-Load Branch (Device Down) MAF Topology Notes

Note	Detail
Reference plane	Continuous ground only.
Max frequency	Flash 1 & Flash 2: 50 MHz. TPM: 33 MHz.
Flash Programmer	Require additional isolation circuitry to protect processor from back-feed voltage when connect to flash programmer.
R1	24 $\Omega \pm 5\%$. To be placed within 5.08 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
R2	51 $\Omega \pm 5\%$. To be placed within 5.08 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.

continued...

Note	Detail
Length from processor to flash 1 or flash 2	Length from processor to flash 1 or flash 2: = BO + M1 + M2 + M3
Flash device electrical characteristics recommendation for 50MHz support: Option 1	[1] Driver strength spec: 50 Ω or stronger across worse case corners. [2] 0.4 ns $\leq$ rise/ fall time measured at 30% to 70% $\leq$ 1.45 ns based on 30 pF test load. [3] Output capacitance, Cout $\leq$ 16 pF (including RPMC die). [4] Input capacitance, Cin $\leq$ 12 pF (including RPMC die). [5] Data input setup time $\leq$ 3 ns. [6] Data input hold time $\leq$ 3 ns. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 6.5 ns (50% to 50%), based on 30 pF test load.
Flash device electrical characteristics recommendation for 50MHz support: Option 2	[1] Driver strength spec: 40 Ω or stronger across worse case corners. [2] 0.4 ns $\leq$ rise/ fall time measured at 30% to 70% $\leq$ 1.25 ns based on 30 pF test load. [3] Output capacitance, Cout $\leq$ 14 pF (including RPMC die). [4] Input capacitance, Cin $\leq$ 12 pF (including RPMC die). [5] Data input setup time $\leq$ 3 ns. [6] Data input hold time $\leq$ 3 ns. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 9 ns (50% to 50%), based on 30 pF test load.
Stuffing resistor	If one of devices is unstuffed (rework), treat it as (n -1)-Load topology thus place stuffing resistor as close as possible to the branch to avoid long stub length.
Length matching between CLK and DATA/ CS# signals	12.7 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Signal name/ list	SPI0_CLK, SPI0_MOSI, SPI0_MISO, SPI0_IO2, SPI0_IO3, SPI0_CS[0:1]# (for flash), SPI0_CS2# (for TPM)

Table 661. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	6	Total Channel

Segment Lengths for Maximum 50MHz**Table 662. SPI0 Flash 50MHz 1 to 4-Load Branch (Device Down) MAF Topology Segment Lengths for Maximum 50MHz Routing Guidelines**

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1	MS, DSL, SL	124.46	
M2	MS, DSL, SL	5.08	
<i>continued...</i>			

Segment	Trace Type	Max Length Segment (mm)	Max Length Segment Note
M3	MS, DSL, SL	35.56	
M4	MS, DSL, SL	50.8	This length only applicable for TPM branch
M5	MS, DSL, SL	38.1	This length only applicable for Debug Tool branch

Min Length Total (mm): 30.48

Min Length Total Note: Min length between processor to the closest end device = 30.48 mm.

Max Length Total (mm): 177.8

Max Length Total Note: 1) Max length between processor to Flash = 177.8 mm; 2) Total topology length for: 2-load topology with flash and TPM = 233.68 mm; 2-load topology with 2 flash = 218.44 mm; 3-load topology = 274.32 mm; 4-load topology = 317.5 mm

SPI0 Flash 50MHz 2 to 5-Load Branch (Device Down) with G3 Flash Sharing Topology

Figure 268. SPI0 Flash 50MHz 2-Load Branch (Device Down) with G3 Flash Sharing Topology Diagram

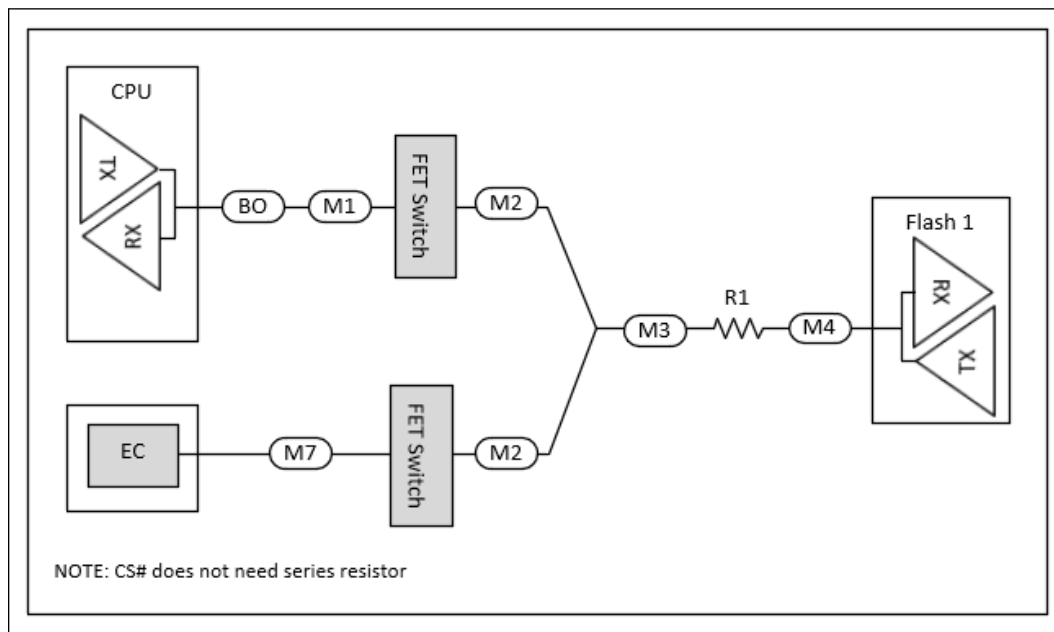


Figure 269. SPI0 Flash 50MHz 3-Load Branch (Device Down) with G3 Flash Sharing of 2 Flash Topology Diagram

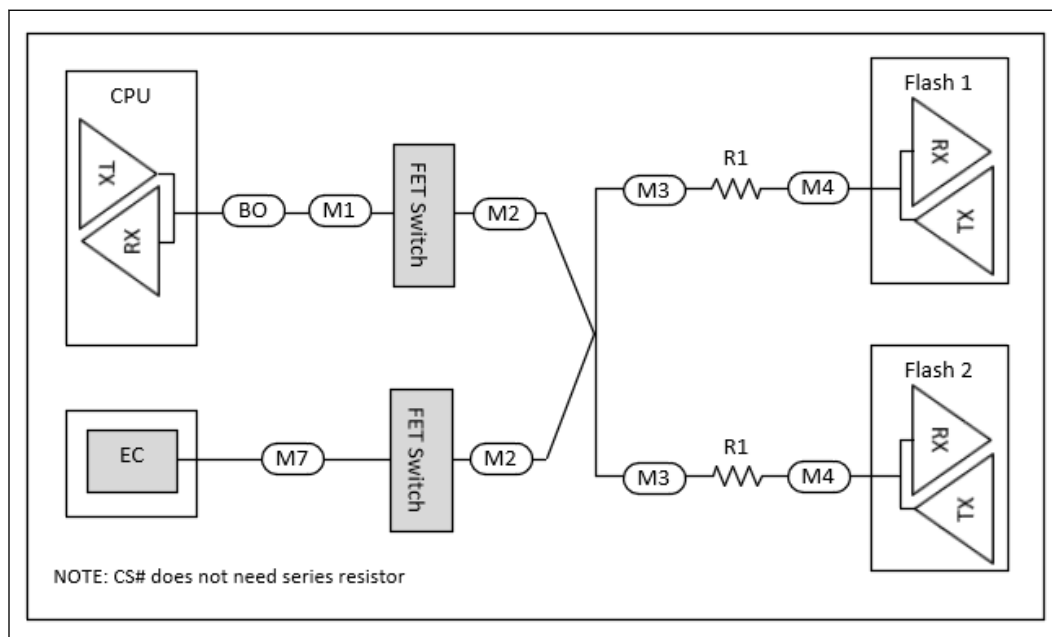


Figure 270. SPI0 Flash 50MHz 3-Load Branch (Device Down) with G3 Flash Sharing of Flash and TPM Topology Diagram

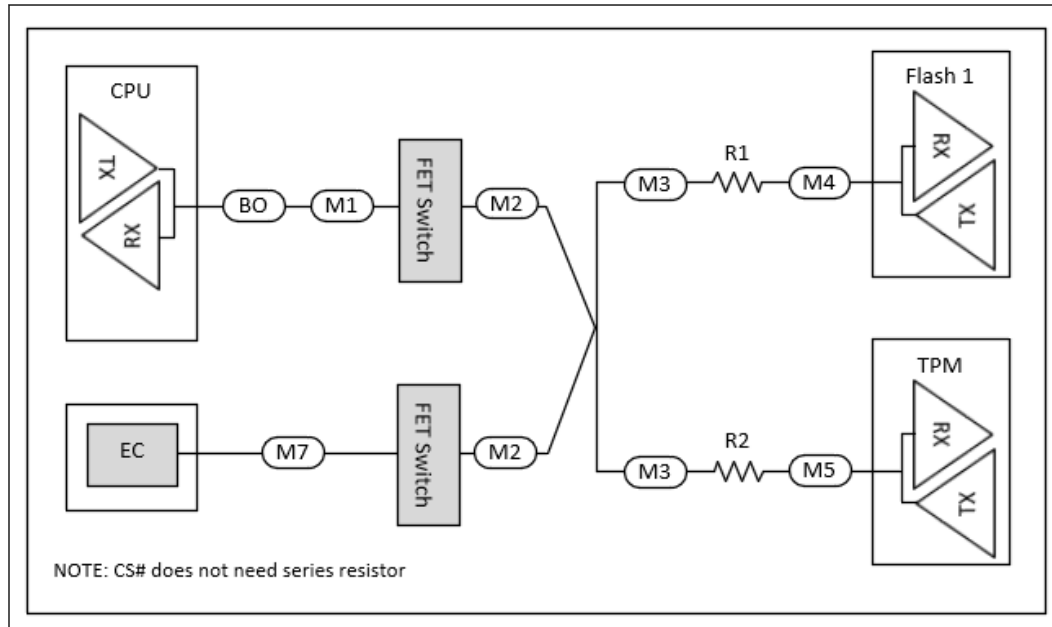


Figure 271. SPI0 Flash 50MHz 4-Load Branch (Device Down) with G3 Flash Sharing Topology Diagram

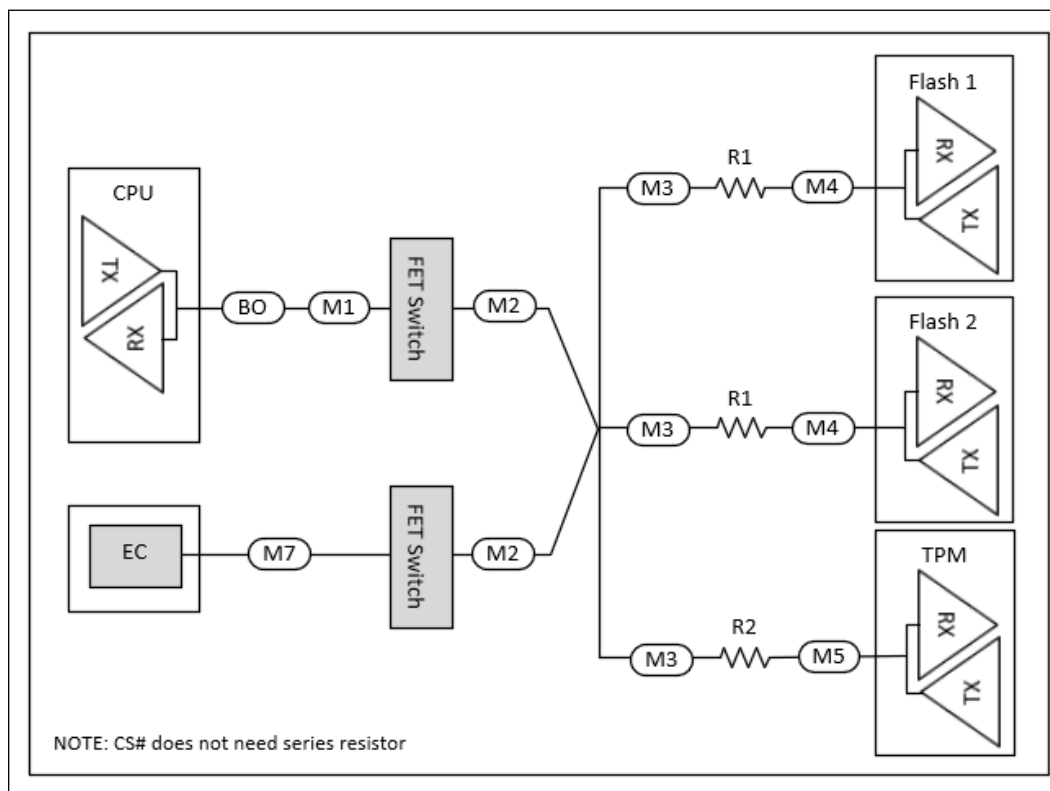


Figure 272. SPI0 Flash 50MHz 5 Load Branch (Device Down) with G3 Flash Sharing Topology Diagram

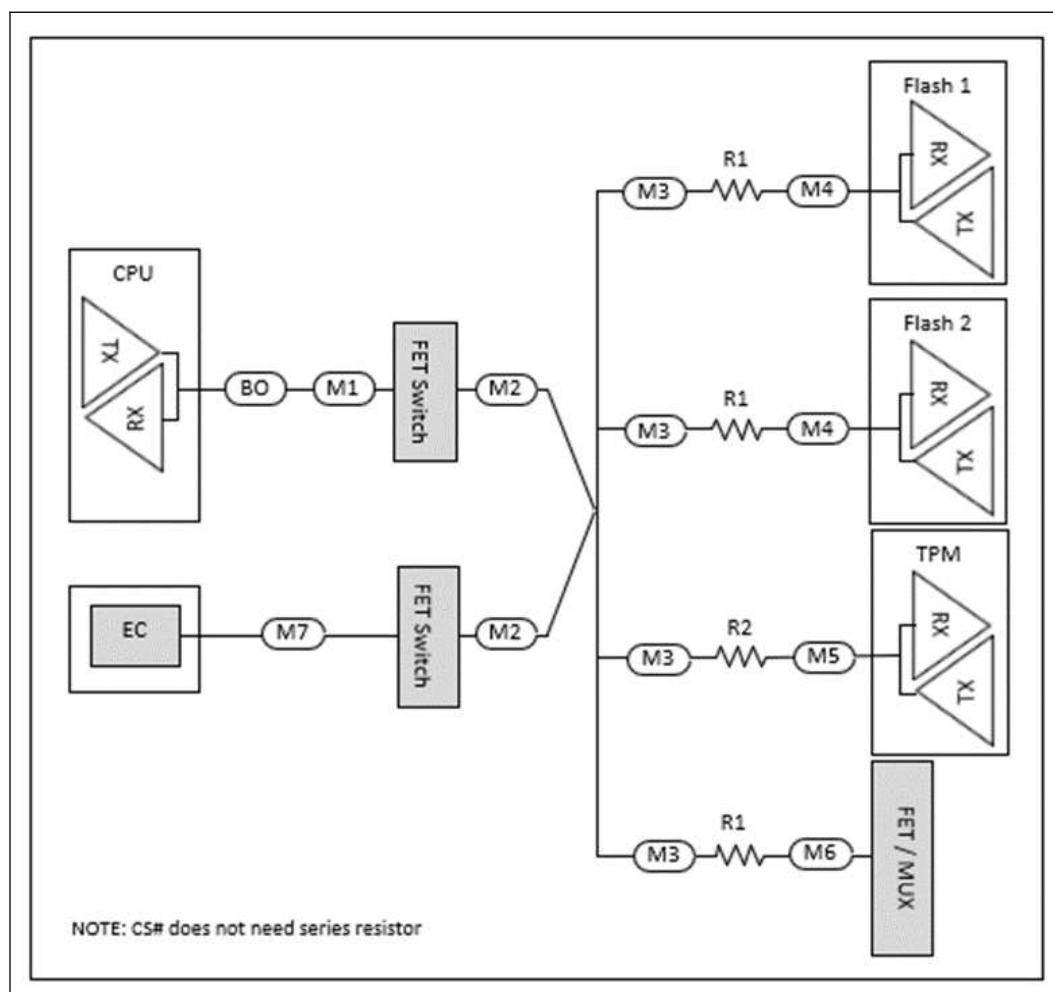


Table 663. SPI0 Flash 50MHz 2 to 5-Load Branch (Device Down) with G3 Flash Sharing Topology Notes

Note	Detail
Reference plane	Continuous ground only.
Max frequency	Flash 1 & Flash 2: 50 MHz. EC: 50 MHz. TPM: 33 MHz.
R1	24 $\Omega \pm 5\%$. To be placed within 5.08 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
R2	51 $\Omega \pm 5\%$. To be placed within 5.08 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
Length from processor to flash 1 or flash 2	Length from processor to flash 1 or flash 2:

continued...

Note	Detail
	= BO + M1 + M2 + M3 + M4
Flash device electrical characteristics recommendation for 50MHz support: Option 1	[1] Driver strength spec: 50 Ω or stronger across worse case corners. [2] 0.4 ns $\leq$ rise/ fall time measured at 30% to 70% $\leq$ 1.45 ns based on 30 pF test load. [3] Output capacitance, Cout $\leq$ 16 pF (including RPMC die). [4] Input capacitance, Cin $\leq$ 12 pF (including RPMC die). [5] Data input setup time $\leq$ 3 ns. [6] Data input hold time $\leq$ 3 ns. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 6.5 ns (50% to 50%), based on 30 pF test load.
Flash device electrical characteristics recommendation for 50MHz support: Option 2	[1] Driver strength spec: 40 Ω or stronger across worse case corners. [2] 0.4 ns $\leq$ rise/ fall time measured at 30% to 70% $\leq$ 1.25 ns based on 30 pF test load. [3] Output capacitance, Cout $\leq$ 14 pF (including RPMC die). [4] Input capacitance, Cin $\leq$ 12 pF (including RPMC die). [5] Data input setup time $\leq$ 3 ns. [6] Data input hold time $\leq$ 3 ns. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 9 ns (50% to 50%), based on 30 pF test load.
Stuffing resistor	If one of devices is unstuffed (rework), treat it as (n - 1)-Load topology thus place stuffing resistor as close as possible to the branch to avoid long stub length.
FET component criteria	Ron < 4.5 Ω . Cin < 6 pF.
Length matching between CLK and DATA/ CS# signals	12.7 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Signal name/ list	SPI0_CLK, SPI0_MOSI, SPI0_MISO, SPI0_IO2, SPI0_IO3, SPI0_CS[0:1]# (for flash), SPI0_CS2# (for TPM)

Table 664. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	6	Total Channel

Segment Lengths for Maximum 50MHz
Table 665. SPI0 Flash 50MHz 2 to 5-Load Branch (Device Down) with G3 Flash Sharing Topology Segment Lengths for Maximum 50MHz Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1	MS, DSL, SL	119.38	
M2	MS, DSL, SL	5.08	
M3	MS, DSL, SL	5.08	
<i>continued...</i>			

Segment	Trace Type	Max Length Segment (mm)	Max Length Segment Note
M4	MS, DSL, SL	35.56	
M5	MS, DSL, SL	50.8	This length only applicable for TPM branch
M6	MS, DSL, SL	38.1	This length only applicable for Debug Tool branch
M7	MS, DSL, SL	127	

Min Length Total (mm): 30.48

Min Length Total Note: Min length between processor to the closest end device = 30.48 mm.

Max Length Total (mm): 177.8

Max Length Total Note: 1) Max length between processor to Flash = 177.8 mm; 2) Total topology length for: 2-load topology = 309.88 mm; 3-load topology with flash and TPM = 365.76 mm; 3-load topology with 2 flash = 350.52 mm; 4-load topology = 406.4 mm; 5-load topology = 449.58 mm

SPI0 Flash 50MHz 2 to 3-Load Branch (Device Down) with G3 Wired-OR Topology

Figure 273. SPI0 2-Load Branch (Device Down) with G3 Wired-OR Topology Diagram

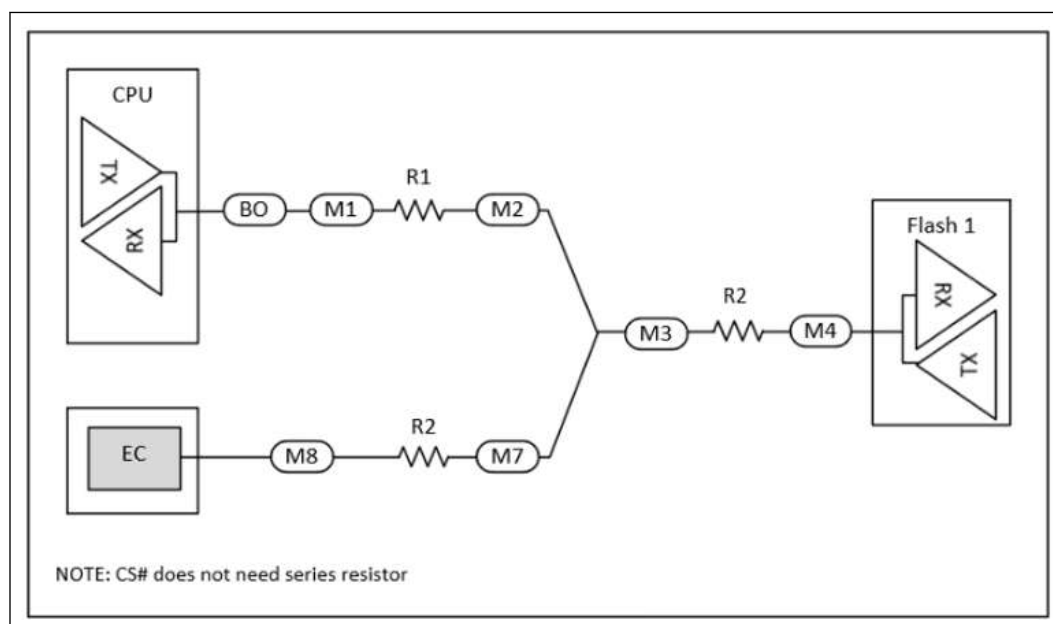


Figure 274. SPI0 3-Load Branch (Device Down) with G3 Wired-OR of Flash and TPM Topology Diagram

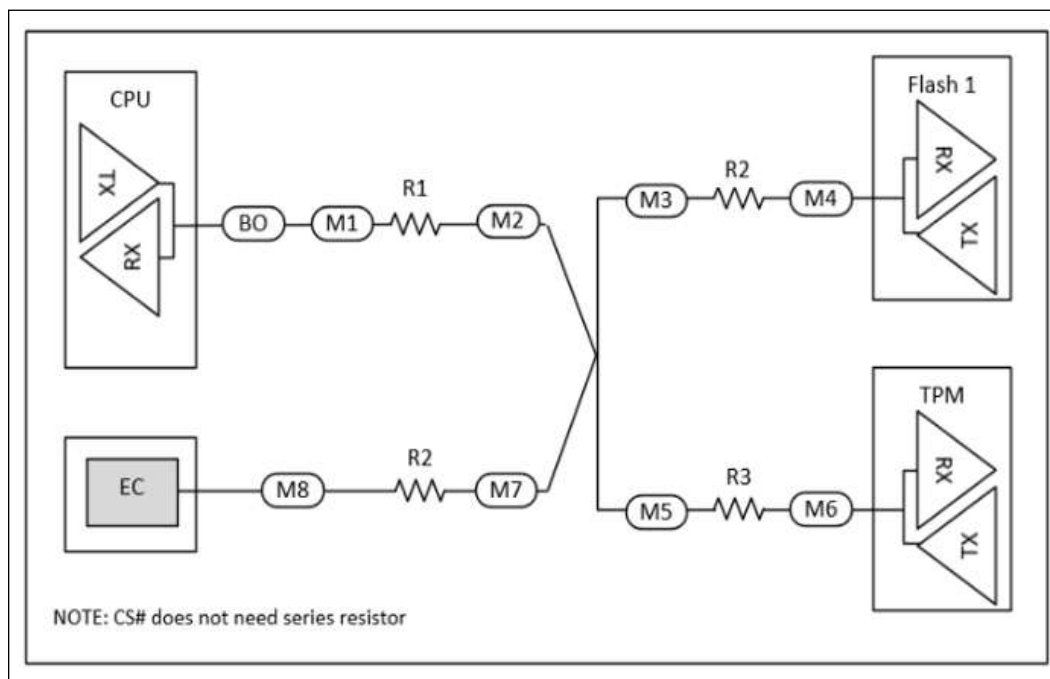


Figure 275. SPI0 3-Load Branch (Device Down) with G3 Wired-OR of Flash Topology Diagram

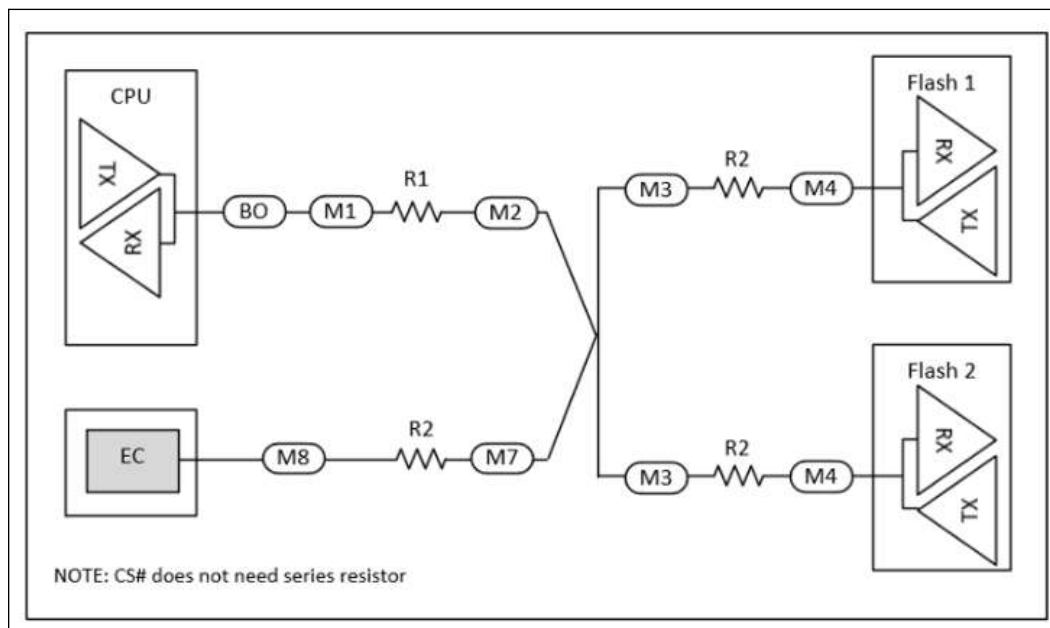


Table 666. SPI0 Flash 50MHz 2 to 3-Load Branch (Device Down) with G3 Wired-OR Topology Notes

Note	Detail
Reference plane	Continuous ground only.
Max frequency	Flash 1 & Flash 2: 50 MHz. EC: 50MHz. TPM: 33 MHz.
Flash Programmer	Require additional isolation circuitry to protect processor from back-feed voltage when connect to flash programmer.
EC and processor branch requirement	Delta between BO + M1 + M2 and M7 + M8 shall not exceed 25.4 mm.
Flash branch requirement, M3 + M4	SPI branches of segment M3 + M4 need length matching within 2.54mm.
R1	[1] DATA: $18 \Omega \pm 5\%$. To be placed within 15.24 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] CLK: $13 \Omega \pm 5\%$. To be placed within 15.24 mm from the branch on SPI0_CLK.
R2	[1] DATA: $18 \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] CLK: $24 \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK.
R3	$51 \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
Flash device electrical characteristics recommendation for 50MHz support: Option 1	[1] Driver strength spec: 50Ω or stronger across worse case corners. [2] $0.4 \text{ ns} \leq \text{rise/ fall time}$ measured at 30% to 70% $\leq 1.45 \text{ ns}$ based on 30 pF test load. [3] Output capacitance, $C_{out} \leq 16 \text{ pF}$ (including RPMC die). [4] Input capacitance, $C_{in} \leq 12 \text{ pF}$ (including RPMC die). [5] Data input setup time $\leq 3 \text{ ns}$. [6] Data input hold time $\leq 3 \text{ ns}$. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 6.5 ns (50% to 50%), based on 30 pF test load.
Flash device electrical characteristics recommendation for 50MHz support: Option 2	[1] Driver strength spec: 40Ω or stronger across worse case corners. [2] $0.4 \text{ ns} \leq \text{rise/ fall time}$ measured at 30% to 70% $\leq 1.25 \text{ ns}$ based on 30 pF test load. [3] Output capacitance, $C_{out} \leq 14 \text{ pF}$ (including RPMC die). [4] Input capacitance, $C_{in} \leq 12 \text{ pF}$ (including RPMC die). [5] Data input setup time $\leq 3 \text{ ns}$. [6] Data input hold time $\leq 3 \text{ ns}$. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 9 ns (50% to 50%), based on 30 pF test load.
Length matching between CLK and DATA/ CS# signals	12.7 mm
continued...	

Note	Detail
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Signal name/ list	SPI0_CLK, SPI0_MOSI, SPI0_MISO, SPI0_IO2, SPI0_IO3, SPI0_CS[0:1]# (for flash), SPI0_CS2# (for TPM).

Table 667. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	6	Total Channel

Segment Lengths for Maximum 50MHz
Table 668. SPI0 Flash 50MHz 2 to 3-Load Branch (Device Down) with G3 Wired-OR Topology Segment Lengths for Maximum 50MHz Routing Guidelines

Segment	Trace Type	Max Length Segment (mm)	Max Length Segment Note
B0	MS, DSL, SL	12.7	
M1	MS, DSL, SL	73.66	M1 can support up to 86.36mm for 2-load topology.
M2	MS, DSL, SL	15.24	
M3	MS, DSL, SL	12.7	
M4	MS, DSL, SL	50.8	
M5	MS, DSL, SL	12.7	This length only applicable for TPM branch only
M6	MS, DSL, SL	76.2	This length only applicable for TPM branch only
M7	MS, DSL, SL	12.7	
M8	MS, DSL, SL	114.3	

Min Length Total (mm): 58.42

Min Length Total Note: Min length between processor to the closest end device = 58.42 mm.

Max Length Total (mm): 165.1

Max Length Total Note: 1) Max length between processor to Flash for: 2-load topology = 177.8 mm; 3-load topology = 165.1 mm; 2) Total topology length for: 2-load topology = 304.8 mm; 3-load with flash only topology = 355.6 mm; 3-load with flash and TPM topology = 381 mm.

SPI0 Flash 50MHz 4-Load Branch (Device Down) with G3 Wired-OR Topology

Figure 276. SPI0 4-Load Branch (Device Down) with G3 Wired-OR Topology Diagram

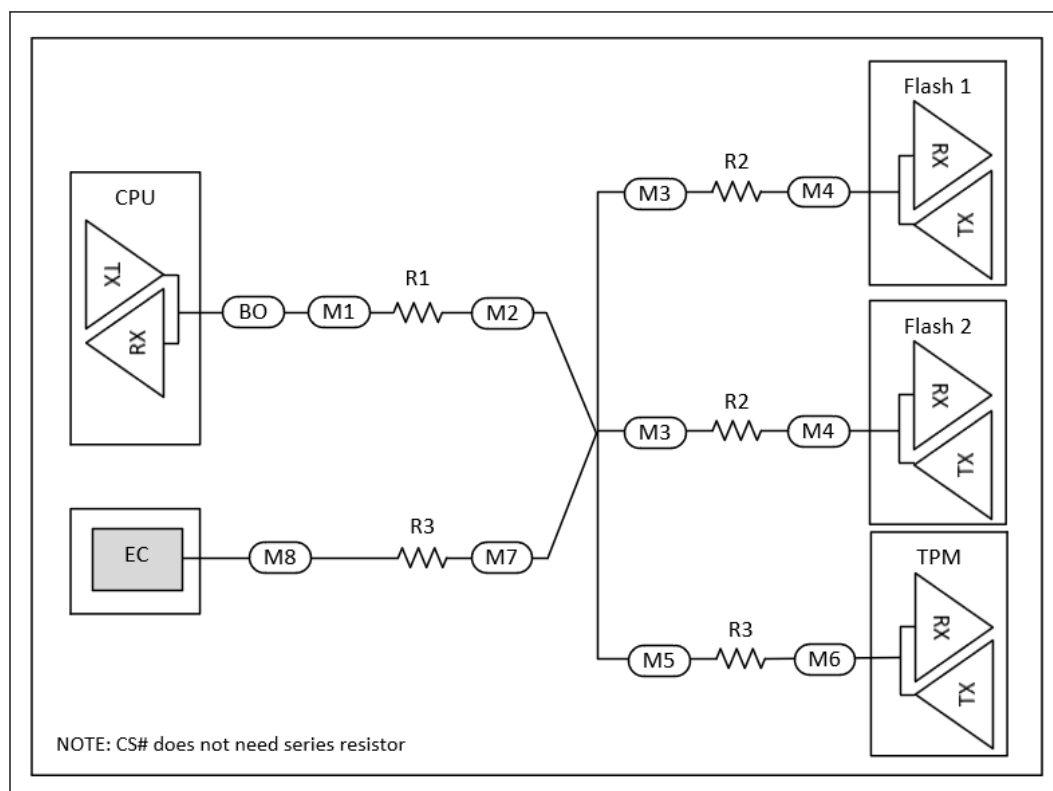


Table 669. SPI0 Flash 50MHz 4-Load Branch (Device Down) with G3 Wired-OR Topology Notes

Note	Detail
Reference plane	Continuous ground only.
Max frequency	Flash 1 & Flash 2: 50 MHz. TPM & EC: 33 MHz.
Flash Programmer	Require additional isolation circuitry to protect processor from back-feed voltage when connect to flash programmer.
EC and processor branch requirement	Delta between BO + M1 + M2 and M7 + M8 shall not exceed 12.7 mm.
Flash branch requirement, M3 + M4	SPI branches of segment M3 + M4 need length matching within 2.54mm.
R1	[1] DATA: $13\ \Omega \pm 5\%$. To be placed within 10.16 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] CLK: $13\ \Omega \pm 5\%$. To be placed within 10.16 mm from the branch on SPI0_CLK.
R2	[1] DATA: $13\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.

continued...

Note	Detail
	[2] CLK: $24\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK.
R3	$68\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
Flash device electrical characteristics recommendation for 50MHz support: Option 1	[1] Driver strength spec: $50\ \Omega$ or stronger across worse case corners. [2] $0.4\ \text{ns} \leq \text{rise/fall time measured at 30\% to 70\%} \leq 1.45\ \text{ns}$ based on 30 pF test load. [3] Output capacitance, $C_{out} \leq 16\ \text{pF}$ (including RPMC die). [4] Input capacitance, $C_{in} \leq 12\ \text{pF}$ (including RPMC die). [5] Data input setup time $\leq 3\ \text{ns}$. [6] Data input hold time $\leq 3\ \text{ns}$. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 6.5 ns (50% to 50%), based on 30 pF test load.
Flash device electrical characteristics recommendation for 50MHz support: Option 2	[1] Driver strength spec: $40\ \Omega$ or stronger across worse case corners. [2] $0.4\ \text{ns} \leq \text{rise/fall time measured at 30\% to 70\%} \leq 1.25\ \text{ns}$ based on 30 pF test load. [3] Output capacitance, $C_{out} \leq 14\ \text{pF}$ (including RPMC die). [4] Input capacitance, $C_{in} \leq 12\ \text{pF}$ (including RPMC die). [5] Data input setup time $\leq 3\ \text{ns}$. [6] Data input hold time $\leq 3\ \text{ns}$. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 9 ns (50% to 50%), based on 30 pF test load.
Length matching between CLK and DATA/ CS# signals	12.7 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Signal name/ list	SPI0_CLK, SPI0_MOSI, SPI0_MISO, SPI0_IO2, SPI0_IO3, SPI0_CS[0:1]# (for flash), SPI0_CS2# (for TPM).

Table 670. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	6	Total Channel

Segment Lengths for Maximum 50MHz
Table 671. SPI0 Flash 50MHz 4-Load Branch (Device Down) with G3 Wired-OR Topology Segment Lengths for Maximum 50MHz Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
B0	MS, DSL, SL	12.7
M1	MS, DSL, SL	66.04
M2	MS, DSL, SL	10.16
M3	MS, DSL, SL	12.7
M4	MS, DSL, SL	50.8
<i>continued...</i>		

Segment	Tline Type	Max Length Segment (mm)
M5	MS, DSL, SL	12.7
M6	MS, DSL, SL	50.8
M7	MS, DSL, SL	12.7
M8	MS, DSL, SL	101.6

Min Length Total (mm): 45.72

Min Length Total Note: Min length between processor to the closest end device = 45.72 mm.

Max Length Total (mm): 152.4

Max Length Total Note: 1) Max length between processor to Flash = 152.4 mm; 2) Total topology length = 393.7mm.

SPI0 Flash 76.8MHz and 100MHz 1 to 3-Load Branch (Device Down) MAF Topology

Figure 277. SPI0 Flash 100MHz and 76.8MHz 1 Load (Device Down) MAF Topology Diagram

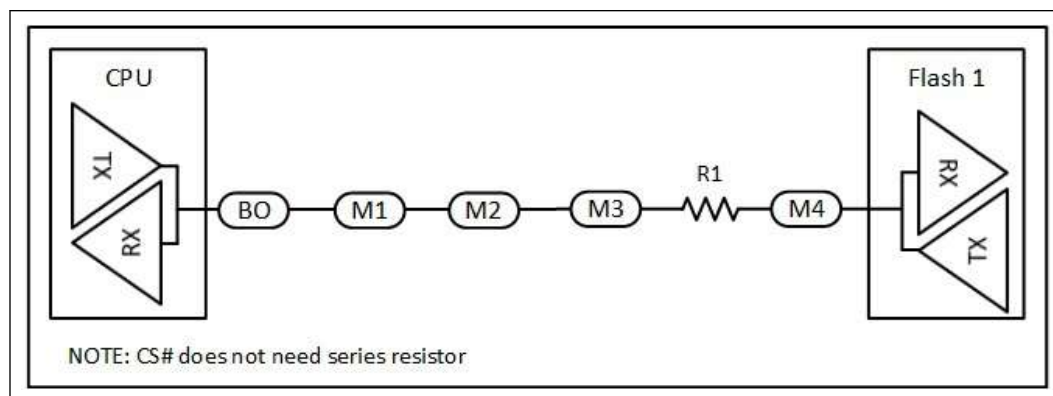


Figure 278. SPI0 Flash 100MHz and 76.8MHz 2 Load Branch (Device Down) MAF with Flash and TPM Topology Diagram

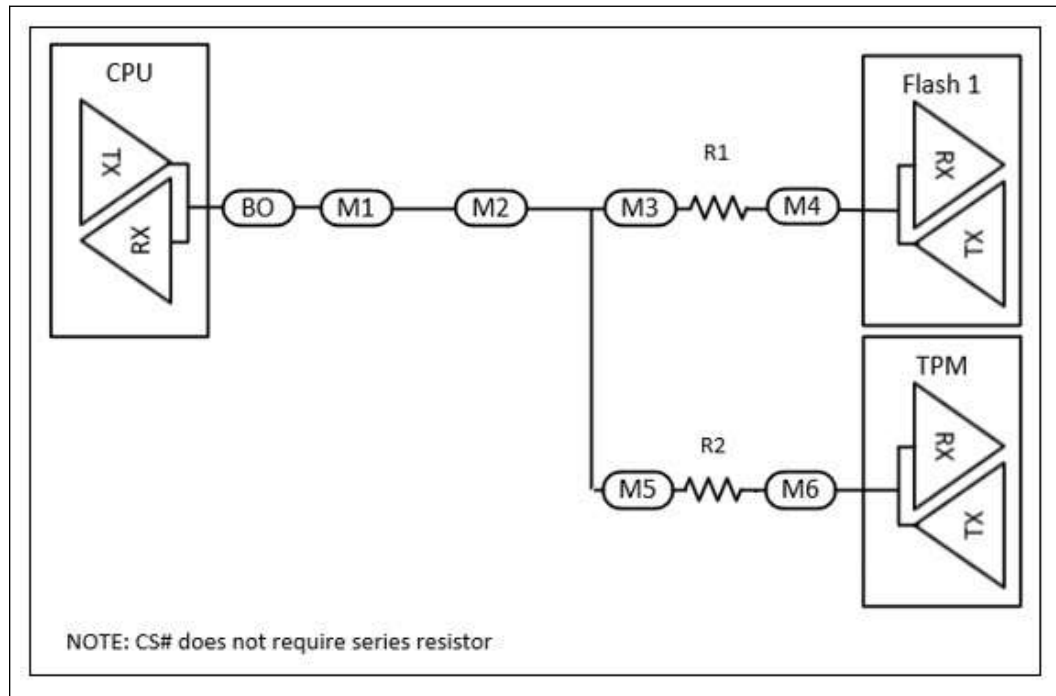


Figure 279. SPI0 Flash 100MHz and 76.8MHz 3 Load Branch (Device Down) MAF Topology Diagram

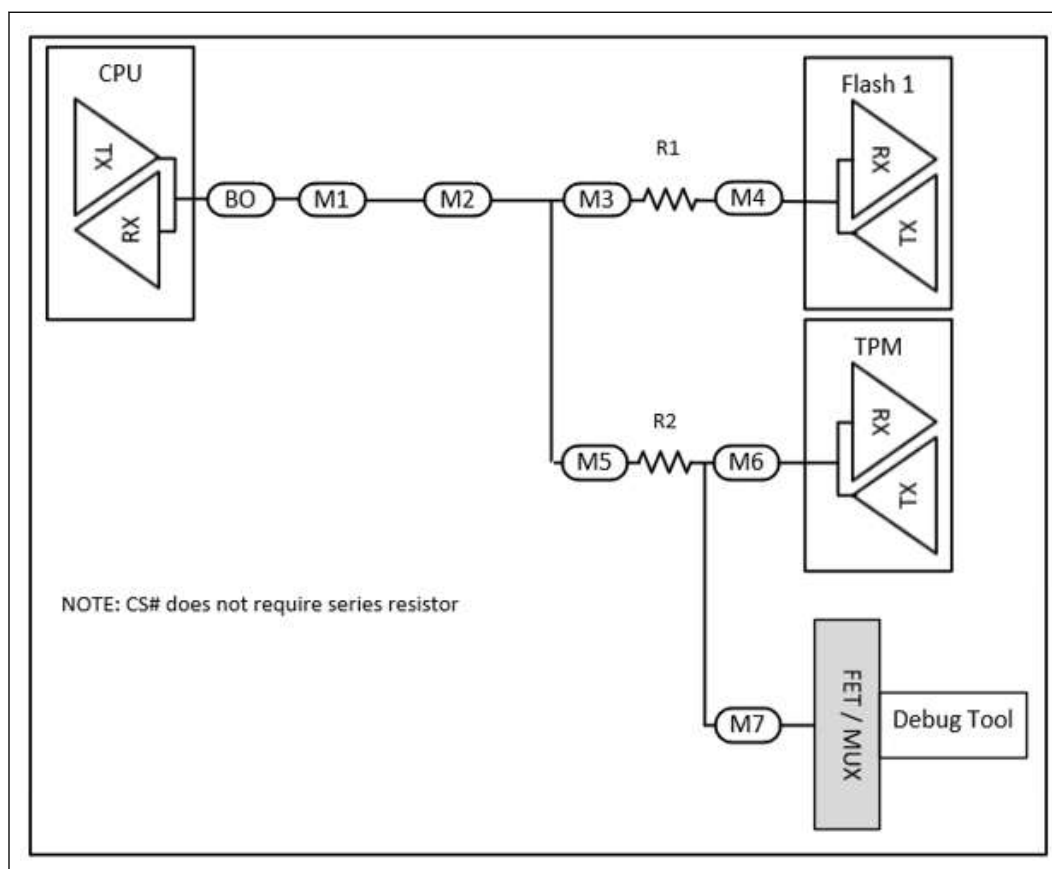


Table 672. SPI0 Flash 76.8MHz and 100MHz 1 to 3-Load Branch (Device Down) MAF Topology Notes

Note	Detail
Reference plane	Continuous ground only.
Max frequency	Flash 1: 100 MHz. TPM and Debug Tool: < 14.3 MHz.
Flash Programmer	Require additional isolation circuitry to protect processor from back-feed voltage when connect to flash programmer.
R1	[1] 22 Ω $\pm$ 5%. To be placed within 12.7 mm from flash device on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] 10 Ω $\pm$ 5%. To be placed within 12.7 mm from flash device on SPI0_CLK.
R2	430 Ω $\pm$ 5%. To be placed within 2.54 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
Length from processor to Flash 1	Length from processor to flash 1: = BO + M1 + M2 + M3 + M4

continued...

Note	Detail
Routing scheme	Interleave 1 CS with every 2 DATA lanes is recommended. Recommend to route DATA and CLK on same layer.
Flash device electrical characteristics recommendation for 100MHz and 76.8MHz support	[1] Driver strength spec: 50 Ω or stronger across worse case corners. [2] 0.4 ns $\leq$ rise/ fall time measured at 30% to 70% $\leq$ 2 ns based on 30 pF test load. [3] Output capacitance, $C_{out} \leq 16$ pF (including RPMC die). [4] Input capacitance, $C_{in} \leq 12$ pF (including RPMC die). [5] Data input setup time ≤ 1.75 ns. [6] Data input hold time ≤ 2 ns. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 6 ns (50% to 50%), based on 30 pF test load.
Length matching between CLK and DATA/ CS# signals	12.7 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Signal name/ list	SPI0_CLK, SPI0_MOSI, SPI0_MISO, SPI0_IO2, SPI0_IO3, SPI0_CS0# (for flash), SPI0_CS2# (for TPM)

Table 673. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	3	Total Channel

Segment Lengths for Maximum 100MHz
Table 674. SPI0 Flash 76.8MHz and 100MHz 1 to 3-Load Branch (Device Down) MAF Topology Segment Lengths for Maximum 100MHz Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1	MS, DSL, SL	40.64	M1 can support up to 78.74 mm for 1-Load topology
M2	MS, DSL, SL	10.16	
M3	MS, DSL, SL	25.4	
M4	MS, DSL, SL	12.7	
M5	MS, DSL, SL	2.54	This length only applicable for TPM and Debug Tool branch
M6	MS, DSL, SL	76.2	This length only applicable for TPM branch
M7	MS, DSL, SL	25.4	This length only applicable for Debug Tool branch

Min Length Total (mm): 38.1

Min Length Total Note: BO + M1 + M2 + M3 + M4 = 38.1 mm

Max Length Total (mm): 101.6

Max Length Total Note: 1) Max length between processor to Flash for ; 1-load topology = 139.7 mm; 2 & 3 load topology = 101.6mm 2) Total topology length for: 1-load topology = 139.7 mm; 2-load topology = 180.34 mm; 3-load topology = 205.74 mm

Segment Lengths for Maximum 76.8MHz

Table 675. SPI0 Flash 76.8MHz and 100MHz 1 to 3-Load Branch (Device Down) MAF Topology Segment Lengths for Maximum 76.8MHz Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1	MS, DSL, SL	40.64	M1 can support up to 66.04 mm for 1-Load topology
M2	MS, DSL, SL	25.4	
M3	MS, DSL, SL	35.56	
M4	MS, DSL, SL	12.7	
M5	MS, DSL, SL	2.54	This length only applicable for TPM and Debug Tool branch
M6	MS, DSL, SL	76.2	This length only applicable for TPM branch
M7	MS, DSL, SL	25.4	This length only applicable for Debug Tool branch

Min Length Total (mm): 38.1

Min Length Total Note: $BO + M1 + M2 + M3 + M4 = 38.1$ mm

Max Length Total (mm): 127

Max Length Total Note: 1) Max length between processor to Flash for; 1load = 152.4 mm; 2 & 3 load = 127 mm; 2) Total topology length for: 1-load topology = 152.4 mm; 2-load topology = 205.74 mm; 3-load topology = 231.14 mm

SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology

Figure 280. SPI0 Flash 100MHz and 76.8MHz 2 Load Branch (Device Down) with G3 Flash Sharing Topology Diagram

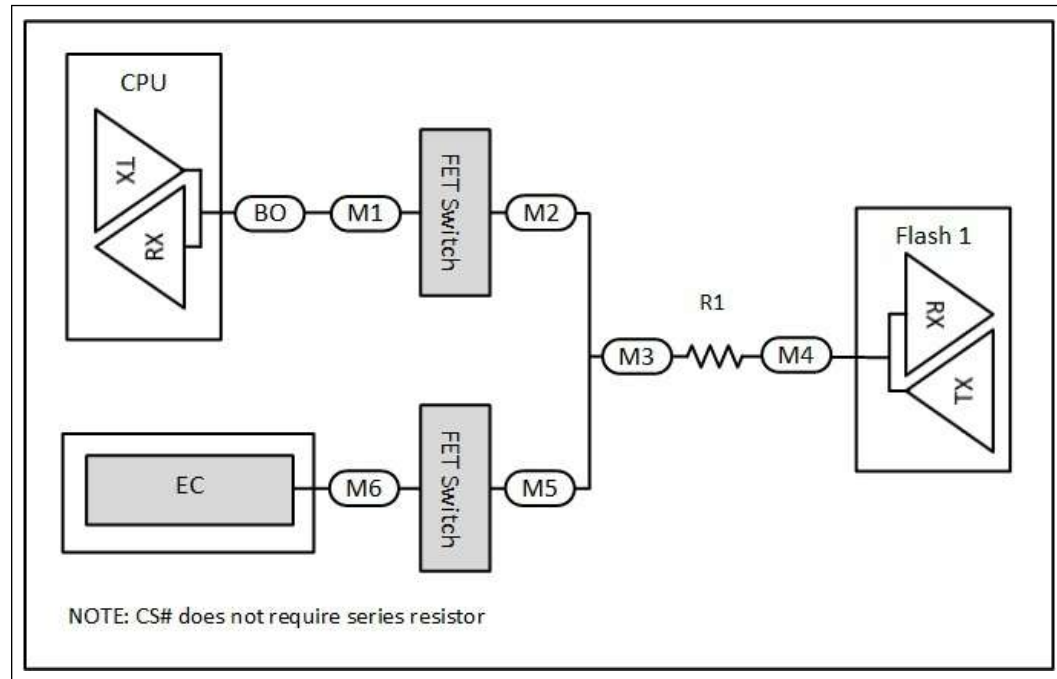


Figure 281. SPI0 Flash 100MHz and 76.8MHz 3 Load Branch (Device Down) with G3 Flash Sharing of Flash and TPM Topology Diagram

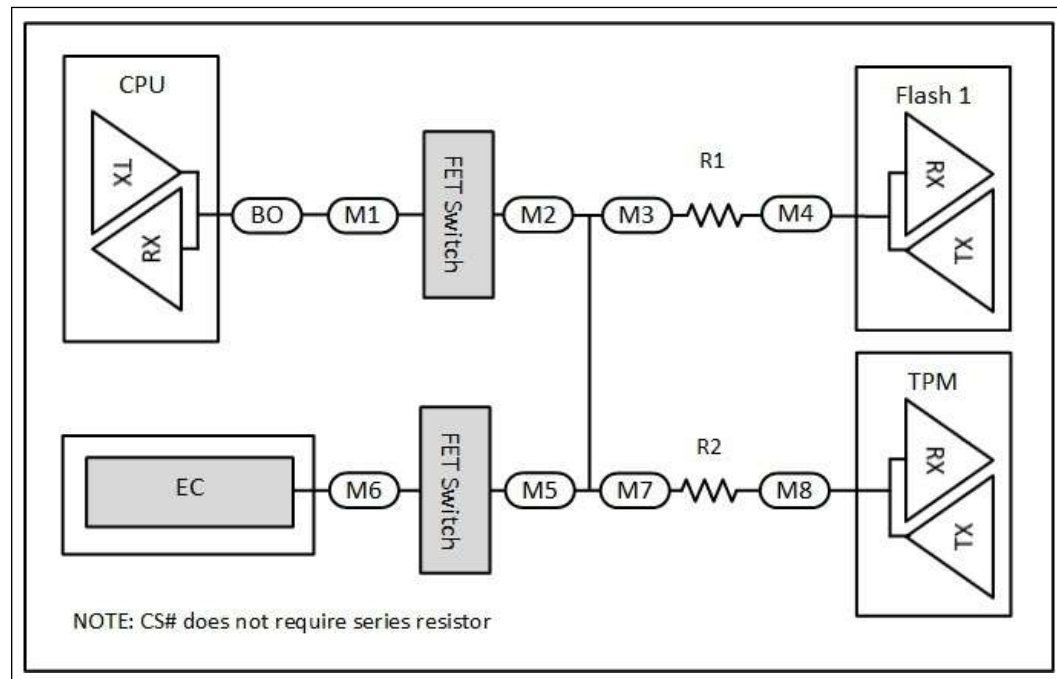


Figure 282. SPI0 Flash 100MHz and 76.8MHz 4 Load Branch (Device Down) with G3 Flash Sharing Topology Diagram

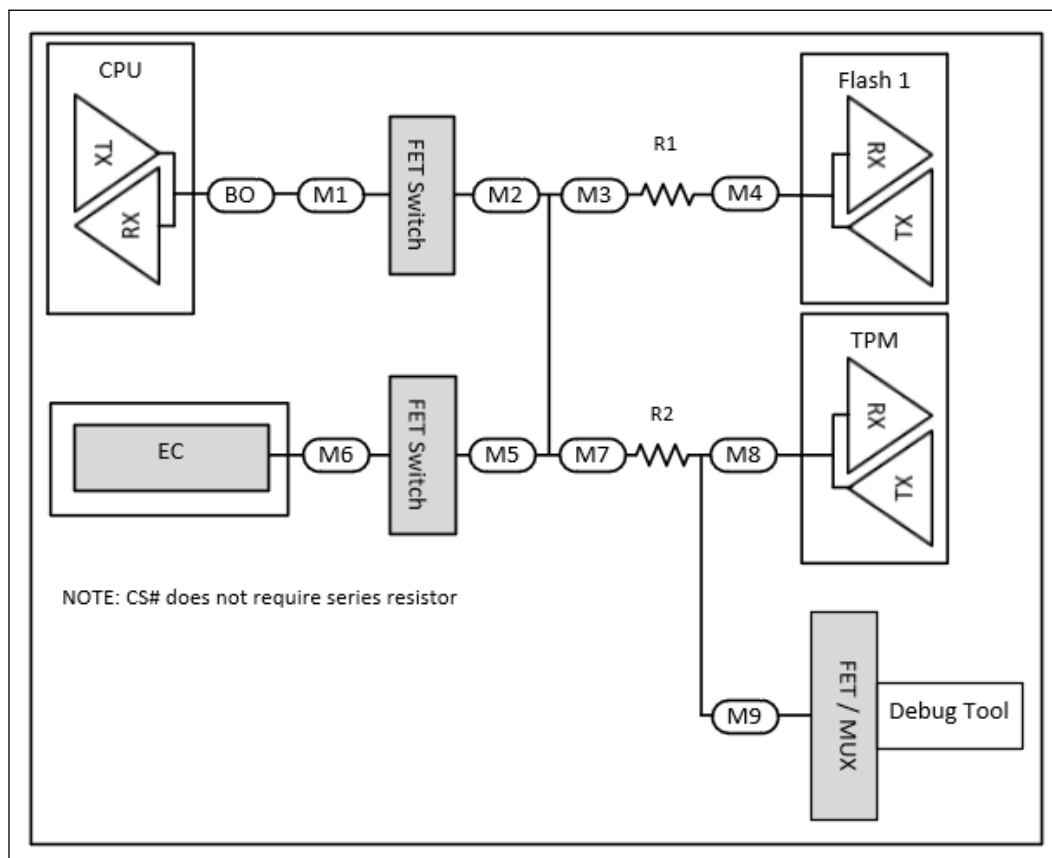


Table 676. SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology Notes

Note	Detail
Reference plane	Continuous ground only.
Max frequency	Flash 1: 100 MHz. TPM and Debug Tool: < 14.3 MHz. EC: 50MHz.
R1	[1] $20\ \Omega \pm 5\%$. To be placed within 12.7 mm from flash device on SPI0_CLK. [2] $16\ \Omega \pm 5\%$. To be placed within 12.7 mm from flash device on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
R2	$430\ \Omega \pm 5\%$. To be placed within 2.54 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.
Length from processor to Flash 1	Length from processor to flash 1: = BO + M1 + M2 + M3 + M4
Routing scheme	Interleave 1 CS with every 2 DATA lanes is recommended. Recommend to route DATA and CLK on same layer.

continued...

Note	Detail
Flash device electrical characteristics recommendation for 100MHz support	[1] Driver strength spec: 40 Ω or stronger across worse case corners. [2] 0.4 ns $\leq$ rise/ fall time measured at 30% to 70% $\leq$ 2 ns based on 30 pF test load. [3] Output capacitance, Cout $\leq$ 16 pF (including RPMC die). [4] Input capacitance, Cin $\leq$ 12 pF (including RPMC die). [5] Data input setup time $\leq$ 1.75 ns. [6] Data input hold time $\leq$ 2 ns. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 6 ns (50% to 50%), based on 30 pF test load.
Flash device electrical characteristics recommendation for 76.8MHz support	[1] Driver strength spec: 50 Ω or stronger across worse case corners. [2] 0.4 ns $\leq$ rise/ fall time measured at 30% to 70% $\leq$ 2 ns based on 30 pF test load. [3] Output capacitance, Cout $\leq$ 16 pF (including RPMC die). [4] Input capacitance, Cin $\leq$ 12 pF (including RPMC die). [5] Data input setup time $\leq$ 1.75 ns. [6] Data input hold time $\leq$ 2 ns. [7] CLK to DATA output valid time, min: 1 ns, based on 30 pF test load. [8] CLK to DATA output valid time, max: 6 ns (50% to 50%), based on 30 pF test load.
FET component criteria	Ron < 4.5 Ω . Cin < 6 pF.
Length matching between CLK and DATA/ CS# signals	12.7 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Signal name/ list	SPI0_CLK, SPI0_MOSI, SPI0_MISO, SPI0_IO2, SPI0_IO3, SPI0_CS0# (for flash), SPI0_CS2# (for TPM)

Table 677. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	3	Total Channel

Segment Lengths for Maximum 100MHz
Table 678. SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology Segment Lengths for Maximum 100MHz Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
B0	MS, DSL, SL	12.7	
M1	MS, DSL, SL	30.48	
M2	MS, DSL, SL	5.08	
M3	MS, DSL, SL	25.4	
M4	MS, DSL, SL	12.7	
M5	MS, DSL, SL	2.54	
<i>continued...</i>			

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
M6	MS, DSL, SL	127	
M7	MS, DSL, SL	2.54	This length only applicable for TPM and Debug Tool branch
M8	MS, DSL, SL	76.2	This length only applicable for TPM branch
M9	MS, DSL, SL	25.4	This length only applicable for Debug Tool branch

Min Length Total (mm): 30.48

Min Length Total Note: BO + M1 + M2 + M3 + M4 = 30.48mm

Max Length Total (mm): 86.36

Max Length Total Note: 1) Max length between processor to Flash = 86.36 mm; 2) Total topology length for: 2-load topology = 215.9 mm; 3-load topology = 294.64 mm; 4-load topology = 320.04 mm

Segment Lengths for Maximum 76.8MHz

Table 679. SPI0 Flash 76.8MHz and 100MHz 2 to 4-Load Branch (Device Down) with G3 Flash Sharing Topology Segment Lengths for Maximum 76.8MHz Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, DSL, SL	12.7	
M1	MS, DSL, SL	55.88	M1 can support up to 68.58 mm for 2-Load topology
M2	MS, DSL, SL	5.08	
M3	MS, DSL, SL	15.24	
M4	MS, DSL, SL	12.7	
M5	MS, DSL, SL	2.54	
M6	MS, DSL, SL	127	
M7	MS, DSL, SL	2.54	This length only applicable for TPM and Debug Tool branch
M8	MS, DSL, SL	76.2	This length only applicable for TPM branch
M9	MS, DSL, SL	25.4	This length only applicable for Debug Tool branch

Min Length Total (mm): 30.48

Min Length Total Note: BO + M1 + M2 + M3 + M4 = 30.48mm

Max Length Total (mm): 101.6

Max Length Total Note: 1) Max length between processor to Flash: 2 load topology = 114.30 mm; 3 & 4 load topology = 101.60 mm; 2) Total topology length for: 2-load topology = 243.84 mm; 3-load topology = 309.88 mm; 4-load topology = 335.28 mm

6.2.15 Touch Host Controller (THC) Design Guidelines

NOTE

THC-SPI interface not supported in Windows OS.

The Touch Host Controller utilizes standard SPI and I2C interface to connect external touch devices. The THC supports two ports for the interfacing with devices. THC also supports the GPIO based interrupt from touch IC's.

6.2.15.1 Touch Signal Descriptions

Signal Names	Group	Description
THC0_SPI1_CLK	Clock	THC0 SPI1 Clock
THC0_SPI1_IO0	Data	THC0 SPI1 serial output data to the Touch Screen device. This Pin will also function as Input during Dual and Quad I/O operation
THC0_SPI1_IO1	Data	THC0 SPI1 serial input data from the Touch Screen device. This Pin will also function as Output during Dual and Quad I/O operation
THC0_SPI1_IO2	Data	THC0 SPI1 I/O to comprehend the support for the Quad I/O operation
THC0_SPI1_IO3	Data	THC0 SPI1 I/O to comprehend the support for the Quad I/O operation
THC0_SPI1_CS#	Chip Select	THC0 SPI1 chip select
THC0_SPI1_RST#	Reset	THC0 SPI1 reset
THC0THC0_SPI1_RST_INT#	Interrupt	THC0 SPI1 interrupt
THC1_SPI2_CLK	Clock	THC1 SPI2 Clock
THC1_SPI2_IO0	Data	THC1 SPI2 serial output data to the Touch Screen device. This Pin will also function as Input during Dual and Quad I/O operation
THC1_SPI2_IO1	Data	THC1 SPI2 serial input data from the Touch Screen device. This Pin will also function as Output during Dual and Quad I/O operation
THC1_SPI2_IO2	Data	THC1 SPI2 I/O to comprehend the support for the Quad I/O operation
THC1_SPI2_IO3	Data	THC1 SPI2 I/O to comprehend the support for the Quad I/O operation
THC1_SPI2_CS#	Chip Select	THC1 SPI2 chip select
THC1_SPI2_RST#	Reset	THC1 SPI2 reset
THC1_SPI2_INT#	Interrupt	THC1 SPI2 interrupt
THC0_SPI1_DSINC	Display Frame Sync	THC1-SPI Port1 External Display Frame Sync
continued...		

Signal Names	Group	Description
THC1_SPI2_DSSYNC	Display Frame Sync	THC1-SPI Port2 External Display Frame Sync
THC_I2C[0:1]_SCL	Clock	THC I2C clock signals
THC_I2C[0:1]_SDA	Data	THC I2C data signals

6.2.15.2 Touch Guidelines

Touch device must be directly connected to the THC[0:1]_I2C/THC[0:1]_SPI bus and interrupt pins need to be connected to THC[0:1]_INT# . Also, refer to the Serial Touch device vendor documentation for additional Serial Touch specific design considerations.

NOTE

THC-SPI and THC-I2C interface are not supported in Windows OS.

6.2.15.2.1 THC-SPI 1-Load (for Touch Panel) Topology

Figure 283. THC-SPI 1-Load (for Touch Panel) Topology Diagram

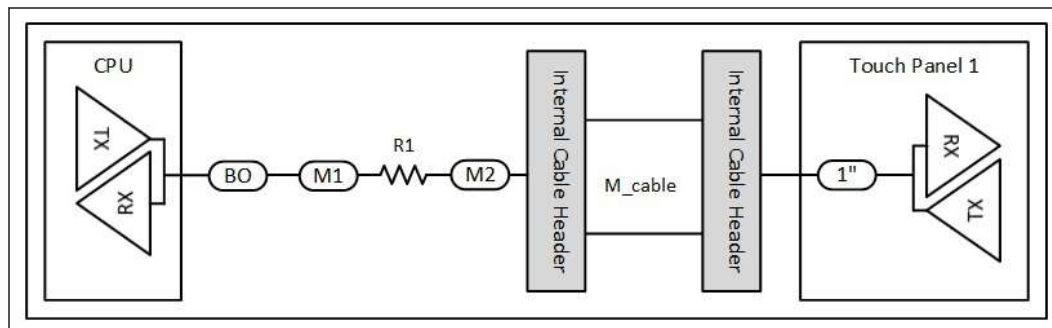


Table 680. THC-SPI 1-Load (for Touch Panel) Topology Notes

Note	Detail
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Max frequency	42.67MHz (Please contact Intel for higher frequency enabling)
R1	18 $\Omega \pm 5\%$ To be placed 12.7 mm from the connector on THC[0:1]_SPI_CLK and THC[0:1]_SPI_IO[0:3].
M_cable	Recommended to use shielded cable.
M_cable electrical characteristics at 100 MHz	-0.075 dB/inch < Insertion loss < -0.052 dB/inch -32.8 dB < Return loss < -46.2 dB

continued...

Note	Detail
	-45.4 dB < Crosstalk < -47.2 dB 42 Ω < Impedance < 62 Ω
Minimum length, total	78.74 mm
Length matching between CLK and DATA/ CS# signals	12.7 mm
Trace spacing between DATA and DATA signals	0.125 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Devices Electrical Assumptions	Please refer to "SPI Interface and Electrical Assumptions for the Touch Device" chapter in Intel Touch Host Controller Integration Guide for further details. (Touch Integration Guide Doc#: 778513)
Processor buffer driver strength	40 Ω .
Signal name/ list	THC0_SPI_CLK, THC0_SPI_IO[0:3], THC0_SPI_CS#, THC0_RST#, THC0_INT#, THC1_SPI_CLK, THC1_SPI_IO[0:3], THC1_SPI_CS#, THC1_RST#, THC1_INT#.

Table 681. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx, Tx	4	Total Channel
Tx	4	Total Channel
Rx	4	Total Channel

Segment Lengths

Table 682. THC-SPI 1-Load (for Touch Panel) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	MS, SL, DSL	12.7	
M1	MS, SL, DSL	228.6	
M2	MS, SL, DSL	12.7	
M_cable	Cable	304.8	For different frequency option, refer to "THC 1-Load (for Touch Panel) Topology M_cable and Processor Duty Cycle Guideline Table".

Min Length Total (mm): 78.74

Max Length Total (mm): 558.8

Table 683. THC 1-Load (for Touch Panel) Topology M_cable and Processor Duty Cycle Guideline Table

Max Frequency (MHz)	M_cable Max Length (mm)	Low time ratio (%)
42.67	304.8	50
36.57	508	57
<i>continued...</i>		

Max Frequency (MHz)	M_cable Max Length (mm)	Low time ratio (%)
32	457.2	50
25.6	762	50
16	1219.2	50

6.2.15.2.2 THC-I2C

For THC-I2C design guidelines, refer to the [I²C Interface Design Guidelines](#) on page 450.

6.2.16 Controller Link

Controller Link interface is the management communication link between the Processor and Intel Wireless cards. Correct operation of this bus is required for wireless network communication and wake up events. It is designed for either an internal Controller Link connection for CNVi or an external bus, as described in the rest of this section (also called “discrete connectivity”).

For implementing the discrete connectivity, refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# [844316](#)) for full Controller Link details.

On platform, the CLINK data and clock interface is supplied from nominal 1.25V VCCIO rail, which is a change from previous generations. No change in the CL_RST# signal, which continues to be supplied with the 1.8V rail. CLINK interface is driven with bi-directional clock, which is supplied with the data. When data driven from SOC to WNIC (Wireless Network Interface Card), the interface clock rate is 30MHz.

In the opposite direction, the interface clock rate is 20MHz. The frequency of the transmitted clock is half the bit rate as the data is sampled on both rising and falling edges

NOTE

Refer to the WNIC product datasheet for supported data rate and clock.

Table 684. Intel® CSME Controller Link Signals and Signal-Integrity Design Guidelines

Group	Signal Name	Description
Clock	CL_CLK	Controller Link Clock: Bi-directional clock that connects to a Wireless LAN Device supporting Intel® Active Management Technology.
Data	CL_DATA	Controller Link Data: Bi-directional data that connects to a Wireless LAN Device supporting Intel® Active Management Technology.
Reset	CL_RST#	Controller Link Reset: Controller Link reset that connects to a Wireless LAN Device supporting Intel® Active Management Technology.

6.2.16.1 CLINK 1-Load (Add-In Card) Topology

Figure 284. CLINK 1-Load (Add-In Card) Topology Diagram

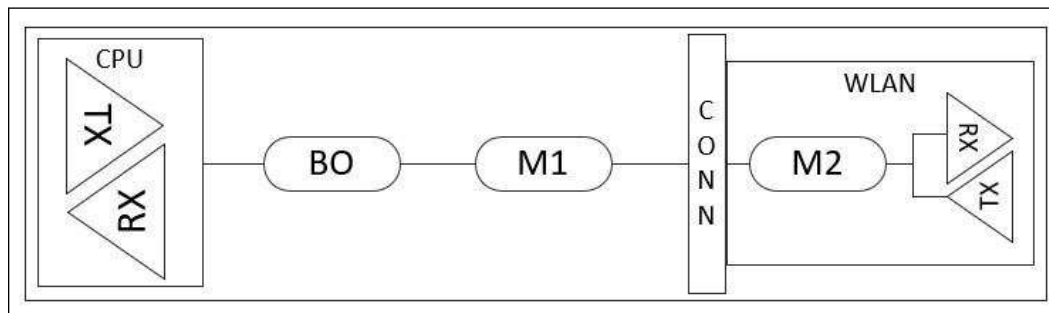


Table 685. CLINK 1-Load (Add-In Card) Topology Notes

Note	Detail
Maximum trace DC resistance	14 Ω
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Length matching between CLK and DATA signals	25.4 mm
Trace spacing between DATA and DATA signals	0.250 mm
Trace spacing between CLK and DATA/ other signals	0.375 mm
Signal name/ list	CL_CLK, CL_DATA, CL_RST#.

Table 686. Max Number of vias

Signal Group	Max Number of Vias Allowed	Via Placement
Rx	5	Total Channel
Tx	5	Total Channel

Segment Lengths

Table 687. CLINK 1-Load (Add-In Card) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)	Max Length Segment Note
BO	DSL, MS, SL	12.7	
M1	DSL, MS, SL	241.3	
M2	DSL, MS, SL	38.1	

Max Length Total (mm): 292.1

6.2.17 Universal Asynchronous Receiver Transmitter (UART) and LSx Interface Design Guidelines

6.2.17.1 UART Interface

The platform integrates three Universal Asynchronous Receiver Transmitter (UART) controllers. The controllers can be used in the low-speed, full-speed, and high-speed modes. The controllers are based on the 16550 industry standard and communicates with serial data ports that conform to the RS-232 interface protocol.

Pull-up resistor is required on all Data and Control signals to ensure the correct default state.

NOTE

Bluetooth* devices are not supported on the Processor UART interfaces.

6.2.17.2 UART Signal Descriptions

Table 688. UART Signals

Group	Signal Name	Description
Data	UART[2:0]_RXD	Receive Data Input signals
	UART[2:0]_TXD	Transmit Data Output signals
	ISH_UART0_RXD	Receive Data Input signals
	ISH_UART0_TXD	Transmit Data Output signals
	ISH_UART1A_RXD	Receive Data Input signals
	ISH_UART1A_TXD	Transmit Data Output signals
	ISH_UART1_RXD	Receive Data Input signals
	ISH_UART1_TXD	Transmit Data Output signals
Control	UART0_RTS#, UART2_RTS#	Request to Send signals
	UART0_CTS#, UART2_CTS#	Clear to Send signals
	ISH_UART0_RTS#	Request to Send signals
	ISH_UART0_CTS#	Clear to Send signals

6.2.17.2.1 UART0/ UART1 1-Load (Device Down) Topology

Figure 285. UART0/ UART1 1-Load (Device Down) Topology Diagram

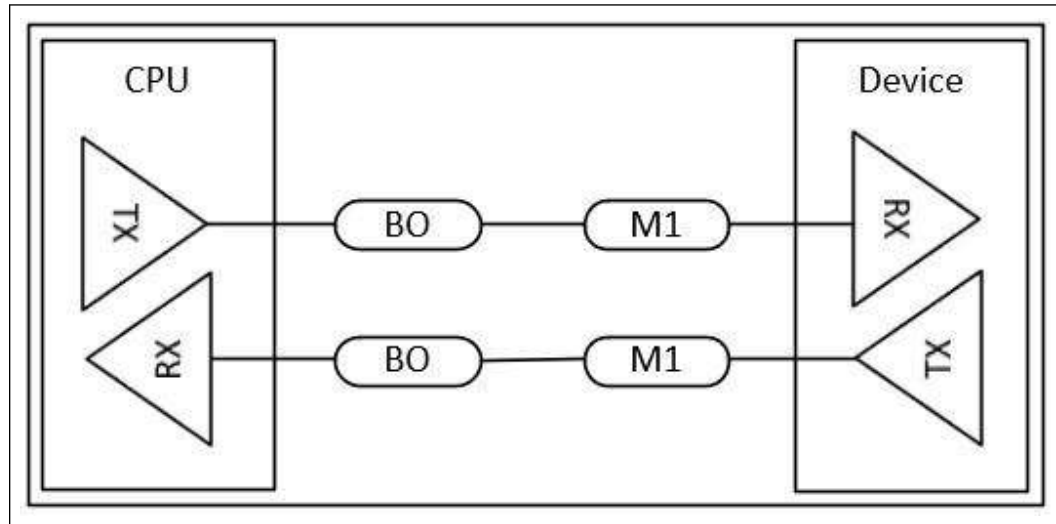


Table 689. UART0/ UART1 1-Load (Device Down) Topology Notes

Note	Detail
Trace spacing between DATA and DATA/ others signals	0.125 mm
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Signal name/ list	UART[0:1]_RXD, UART[0:1]_TXD, UART0_RTS#, UART0_CTS#, UART1A_RXD, UART1A_TXD, ISH_UART[0:1]_RXD, ISH_UART[0:1]_TXD, ISH_UART0_RTS#, ISH_UART0_CTS#, ISH_UART1A_RXD, ISH_UART1A_TXD.
Processor buffer driver strength	50 Ω.

Segment Lengths

Table 690. UART0/ UART1 1-Load (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	12.7
M1	DSL, MS, SL	622.3

Max Length Total (mm): 635

6.2.17.2.2 UART2 1-Load (Device Down) Topology

Figure 286. UART2 1-Load (Device Down) Topology Diagram

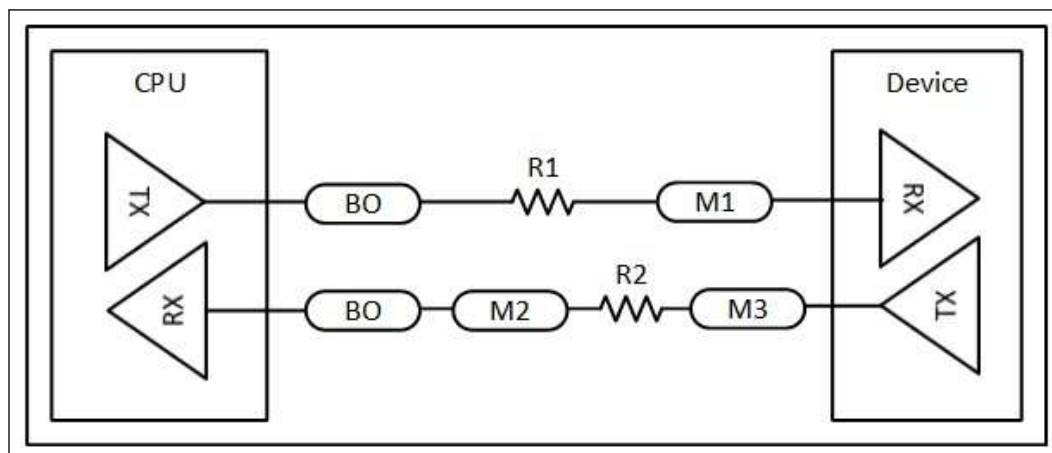


Table 691. UART2 1-Load (Device Down) Topology Notes

Note	Detail
R1	13 Ω . Recommended to be placed 12.7 mm from processor, but can be extended to 50.8 mm maximum if required.
R2	0 Ω placeholder. If failing device overshoot/ undershoot, a 13 Ω resistor can be stuffed. If possible work with device vendor to determine necessity of R2.
Trace spacing between DATA and DATA/ others signals	0.125 mm
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
Processor buffer driver strength	40 Ω .
Signal name/ list	UART2_RXD, UART2_TXD, UART2_RTS#, UART2_CTS#.

Segment Lengths

Table 692. UART2 1-Load (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
M1	MS, DSL, SL	622.3
M2	MS, DSL, SL	609.6
M3	MS, DSL, SL	12.7

Max Length Total (mm): 635

6.2.17.3 LSx Interface

The processor LSx interface enables communication between one Thunderbolt controller to another. NVL-H supports 3 pairs of LSx signals. The LSx signals use a UART interface. It operates at a 1 MHz baud rate and shares identical electrical characteristics/routing guidelines as UART.

6.2.17.4 LSx Signal Description

Table 693. LSx Signals

Group	Signal Name	Description
Data	TBT_LSx[0:2]_RXD	Receive Data Input signals
	TBT_LSx[0:2]_TXD	Transmit Data Output signals

6.2.17.4.1 LSX 1-Load to Retimer (Device Down) Topology

Figure 287. LSX 1-Load to Retimer (Device Down) Topology Diagram

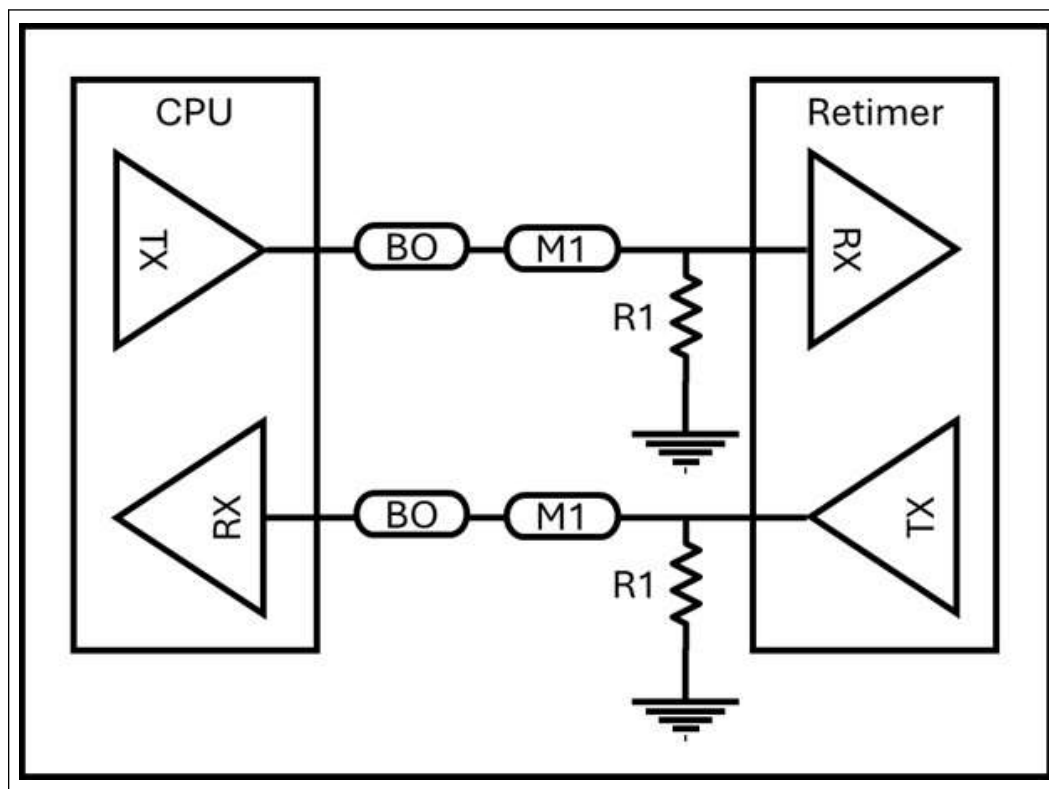


Table 694. LSX 1-Load to Retimer (Device Down) Topology Notes

Note	Detail
Trace spacing between DATA and DATA/ others signals	0.125 mm
Reference plane	[1] Continuous GND is recommended. [2] If continuous GND cannot be implemented, a combination of GND on one side and continuous power plane on the other side with the condition that the GND plane be the closer reference (dual-reference) can be implemented. The continuous power plane requires low peak-to-peak noise and low current switching speed (di/dt). [3] If non-continuous power referencing is required on microstrip/ surface layer, signal can reference over power planes with a low peak-to-peak noise and low current switching speed (di/dt).
R1	A pull-down resistor is required when using the June Bridge Retimer. For detailed specifications and implementation guidance, please refer to 843990 Thunderbolt JHL92x0R June Bridge Collateral. This pull-down resistor is not required when using the Hayden Bridge Retimer.
Processor buffer driver strength	50 Ω .
Signal name/ list	TBT_LSX[0:2]_TXD, TBT_LSX[0:2]_RXD.

Segment Lengths

Table 695. LSX 1-Load to Retimer (Device Down) Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	DSL, MS, SL	25.4
M1	DSL, MS, SL	622.3

Max Length Total (mm): 647.7

6.2.17.5 Tools

Intel does not promote any specific tool for this interface.

6.2.18 General Purpose I/Os Design Guidelines

6.2.18.1 GPIO Checklist

Care should be taken on GPIOs that default to a Native Function when they are used as GPIO function. These GPIOs should not be used if control/termination is required during Sx/Reset. These GPIO may be reset to the Native function on Sx/Reset (depending on GPIO PadRstCfg bit setting) with specific pin state, and may cause issue.

Refer to the Nova Lake Mobile GPIO Implementation Summary, (# [856577](#)) for details on GPIO multiplex and straps.

All unused GPIO pins that default to GPIO functionality do not need termination. For unused GPIO pins that default to a native function, termination on the pins depend on the termination requirement of the native function.

6.2.18.2 Processor GPIO CMOS Buffer Type Topology

Figure 288. Processor GPIO 1-Load Topology Diagram

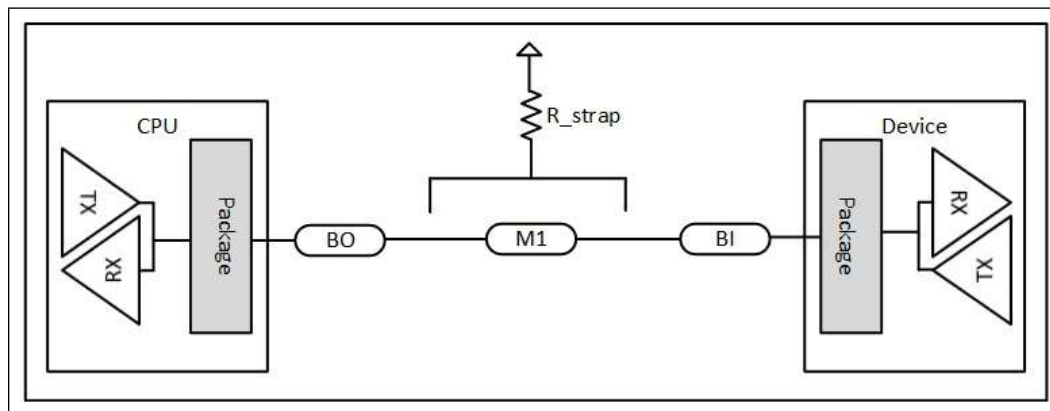


Figure 289. Processor GPIO Multi-Load Branch Topology Diagram

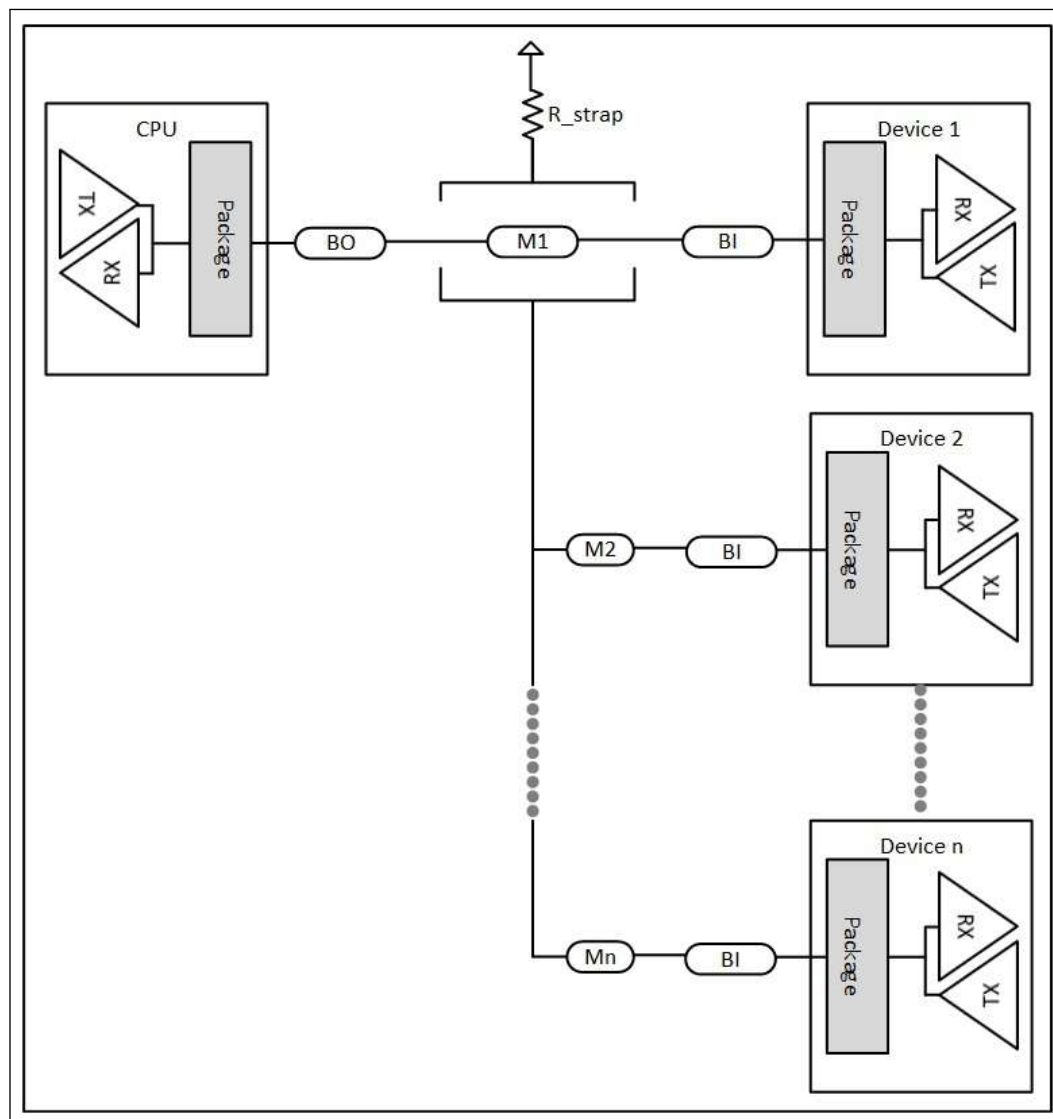


Figure 290. Formula for Strapping or Pull-up Resistor Calculation

$$R_{strap} \text{ or } R_{pu} = \frac{t_r}{0.8473 \times C_b}$$

Figure 291. Example of Strapping or Pull-up Resistor Calculation

$$R_{strap \text{ or } R_{pu}} = \frac{0.000001s}{0.8473 \times (33 \text{ pF} + 20 \text{ pF})} = 22.3 \text{ k}\Omega$$

Table 696. Processor GPIO CMOS Buffer Type Topology Notes

Note	Detail
Total trace length	Total trace length options as below: [1] Total trace length BO + M1 + BI =< 177.8 mm, or [2] Total trace length BO + M2 + BI =< 177.8 mm, or [3] Total trace length BO + Mn + BI =< 177.8 mm.
Strapping resistor, R_strap	20 kΩ on need basis, for strapping purposes. Resistor can be either pull-up or pull-down depending on strapping design.
Number of vias allowed	No restriction.
Reference plane	No restriction.
Note 1	R_strap resistor is optional depending on signal needs and can be anywhere along M1, M2, Mn lane. Refer to EDS or device datasheet for the requirement.
Note 2	Main route of Mn can start anywhere along M1 as long as the total length of each branch from processor to device is less than 177.8 mm.
Note 3	R_strap resistor value depends on power sequence/ strapping's timing and DC voltage level requirement (refer below for calculation details).
Note 4	Level shifter may be needed depending on actual device implementation. Generally no length restriction after level shifter.
Note 5	For processor input-only signal, if device output is open drain buffer type, then proper pull-up resistor, Rpu is needed and refer to "Rise Time and R_strap or Rpu Calculation" section for details.
Note 6	Noise sensitive GPIO pin should be properly shielded/ with sufficient trace spacing (e.g.: 3x of trace width). This is to prevent undesirable random system reset.
Rise time and R_strap or Rpu calculation	
R_strap or Rpu formula	Refer image [Formula for Strapping or Pull-up Resistor].
Bus capacitance rule of thumb, Cb	Device = 10 pF per device. Board trace = 3.3 pF per 25.4 mm of trace length.
Example	[1] Total bus length = 10 inches --> 10 x 3.3 pF = 33 pF. [2] Branch topology with two devices --> 2 x 10 pF = 20 pF. [3] With a rise time requirement (tr) = 1 us. Refer image [Example for Strapping or Pull-up Resistor Calculation].

Segment Lengths

Table 697. Processor GPIO CMOS Buffer Type Topology Segment Lengths Routing Guidelines

Segment	Tline Type	Max Length Segment (mm)
BO	MS, DSL, SL	12.7
M1	MS, DSL, SL	177.6
BI	MS, DSL, SL	12.7

Max Length Total (mm): 203

6.2.19 Miscellaneous Signals

RSVD_PULLDOWN

For GPP pin details, refer to the GPIO implementation summary in Nova Lake Mobile GPIO Implementation Summary (# [856577](#)).

For the following signals, refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# [844316](#)):

RSVD_PULLDOWN must be connected to VSS(GND) through a 50-ohm resistor unless otherwise specified in the notes below:

NOTES

1. ER4 needs 200-400 Ohm resistor to VSS(GND).
 2. EE39 needs 50 Ohm resistor to VSS(GND).
-

7.0 Platform Ingredients

The Platform Ingredients chapter covers the following information:

- Intel Security Firmware Engines
- Intel® Integrated Sensor Solution
- Discrete Trusted Platform Module

7.1 Intel® Security Firmware Engines

Intel® Security Firmware Engines are internal processors running Intel® CSE, Intel® ESE FW and Intel® GSC FW. The security firmware engines are connected to the flash, loading Intel® CSE FW that is running Intel® AMT section (where applicable), controlling EC for power states transition, and more. [Figure 292](#) on page 509 illustrates an abstract overview of the NVL-H mobile architecture.

7.1.1 Acronyms

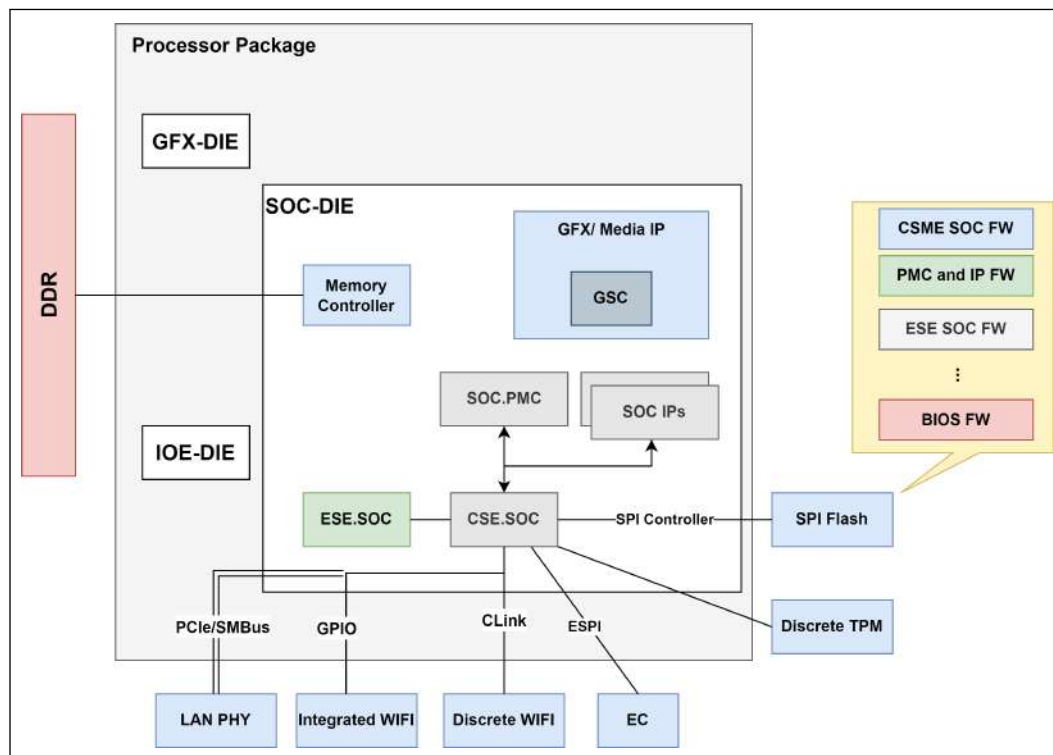
Acronyms	Description
NVL	Nova Lake
EC	Embedded Controller
CM0	Intel® CSE power state. The host CPU is in S0 and Intel® CSE is operational.
CM3	Intel® CSE power state. The host CPU is in Sx and Intel® CSE is operational.
CMoff	Intel® CSE power state. The host CPU is in Sx and Intel® CSE is not operational.
Sx	System state S3, S4 or S5
TBD	To Be Defined

7.1.2 Reference Documents

7.1.3 Intel® Security Firmware Engine Overview

Intel® Security Firmware Engine interacts with the on-board EC with proper policy decisions during system power state transitions. The below figure illustrates the Sx power need to remain on for indicating power to LAN, WLAN and more.

Figure 292. Intel® NVL-H Package Architecture



7.1.4 Intel® CSE to EC Interface Signals

Intel® CSE subsystem in SoC die must communicate with the EC to indicate when Sx power wells needs to remain on, to indicate when LAN and/or WLAN power are/is needed and to know the system power source (AC or battery) to make proper policy decisions. To allow this communication, several sideband signals are required between Processor and EC.

7.1.5 Optimization Guidelines

Intel® CSE Guidelines on Corporate SKUs

There are two additional areas of board support that should be considered when designing Intel® CSE corporate requirements. One is the ability to access Intel® CSE in an out-of-band manner (via LAN or Wireless LAN), second is the to obtain out-of-band connectivity for Intel® CSE in CM3 power state.

Out-of-band accessibility of Intel® CSE:

- Intel® LAN (including SLP_LAN# signal support) is required for LAN connectivity to Intel® CSE and supported applications, such Intel® Active Management Technology (Intel® AMT).
- Intel® Wireless LAN products is required for wireless network connectivity to Intel® CSE and supported applications, such Intel® Active Management Technology (Intel® AMT).

- In terms of Intel Connectivity integration (CNVi) solution, the internal controller link is used to manage the CNVi IP which is a hard macro embedded in Intel SoC. The external RF companion module (CRF) does not allow switching off its power in any system state where the processor is powered on, in order to maintain synchronization at all times. Refer to Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.

Out-of-band accessibility of Intel® CSE in CM3 power state:

- Refer to the [Platform Power Sequencing Specification](#) on page 722.

Given different considerations, it is possible to design three different types of platforms:

- Platforms that support CM3 power states **and** use Intel® LAN PHY or Intel® Wireless LAN solutions.
- Platforms that do not support CM3 power states **and** use Intel® LAN PHY or Intel® Wireless LAN PHY solutions.
- Platforms that support CM3 power states **and** use a third-party LAN PHY solution.

Table 698. Corporate SKU Firmware Behavior

Power Policy	Power Flow Changes
PP1 Intel® CSE is on in S0 Only	Intel® CSE will be ON after G3-exit, Deep Sx exit, Type 8 GRST (which is Power Button Override) and will be OFF after reaching its intended Sx state.
PP2 Intel® CSE is on in Sx State	None

Table below summarizes the power rails, power states and signals.

Table 699. Power Delivery Summary for Intel® CSE Subsystem

What It Powers	Rail	Sx / CM3	Sx / CMoff	Enabled By	Power OK Indicator
Common					
Platform 5V rail	V5.0A	On	On	NA	NA
DRAM VDD	V1.35 / V1.2	On in S3	On in S3	SLP_S4#	NA
DRAM VTT	VDDQ/2	Off	Off	SLP_S3#	NA
Processor	V1.0 VCC	Off	Off	SLP_S3#	PLT_PWROK
M3 Support + Intel® LAN PHY / Intel® Wireless LAN Solution					
SPI interface	V1.8 or V3.3 VCCSPI	On	On	SLP_SUS#	RSMRST#
Wireless LAN Device	Power to the Wireless LAN Device	On	Off	3.3v PRIM	NA
LAN PHY	Power to the LAN PHY	On	Off	SLP_LAN#	NA
No M3 Support + Intel® LAN PHY / Intel® Wireless LAN Solution					
SPI Interface	V1.8 or V3.3 VCCSPI	On	On	SLP_SUS#	RSMRST#
Wireless LAN Device	Power to the Wireless LAN Device	NA	Off	3.3v PRIM	NA
LAN PHY	Power to the LAN PHY	NA	Off	SLP_LAN#	NA
continued...					

What It Powers	Rail	Sx / CM3	Sx / CMoff	Enabled By	Power OK Indicator
M3 Support + A third-party LAN PHY Solution					
SPI Interface	V1.8 or V3.3 VCCSPI	On RSMRST#	On	SLP_SUS#	
<i>Notes:</i> 1. Intel® ME Local RAM is powered by VCCSRAM_1P0 2. SLP_WLAN# power control shall be skipped in CNVi flexible design (same as Intel RVP).					

7.1.6 SPI Flash Descriptor Security Override

To support Intel® CSE, a SPI flash is required on the platform with sufficient space to hold system BIOS, Intel® CSE firmware, Intel® Integrated Sensor Solution firmware and any additional third party data storage. The SPI flash needs to be powered in states S0 through S5, independent of Intel® CSE power state.

GPP_B04/BK0/SBK0/ISH_GP0/USB-C_GPP_B04 signal is used for Flash Descriptor Security Override (which is also known as Intel® CSE Debug Mode). This signal input should be tied to a jumper which connects to power through an external pull-up (1 K Ω $\pm$ 5%) only. When this signal is asserted, the Flash Security gets overridden for ease of image programming.

Customers are encouraged to implement this jumper - only on pre-productions designs, to be able to re-flash the SPI Flash device by customer technicians and/or by Intel Debug teams when asking help from them. This jumper may be removed from the board design before final production.

7.1.7 Intel® APS

Intel® APS is a tool for Intel® Platform Enablement Test Suite (Intel® PETS). Intel® PETS tool is used to test compliancy to Intel® CSE requirements and is a validation vehicle to assist with the reproduction of power management and power sequencing issues. The Intel® APS signals are required to be present at an accessible location on the board. Routing these signals to a connector is strongly recommended and will contribute significantly for accelerating debug and system validation.

An Intel® APS header allows for easy connection to a platform without extensive rework and/or intrusion by grouping signals required by Intel® APS to a common connector. It is strongly recommended to use one of the two proposed connector options described below.

The first option is to use an 18-pin Flat Flex Cable (FFC)/Flexible Printed Circuitry (FPC) connector, such as the 10051922-1810elf. It's a Lower Side Contact, Side Entry, Surface Mount ZIF Connector with 0.5 mm pitch. The pinout of the ZIF connector is shown in the figure below. This ZIF connector should be placed in an accessible location for accessing it without opening the chassis. It could be somewhere under the mobile system keyboard or inside the SSD/DDR module compartment. The keyboard or the SSD/DDR back cover could be lifted and the FFC/FPC could be plugged into the proposed ZIF connector, as shown in the below three figures. After well placing the Intel® APS adapter, all the necessary Intel® APS signal lines can be connected to the 18-pin 0.1" (2.54 mm) header of Intel® APS Adapter, as shown in the Figure "Intel® Automatic Power Switcher (Intel® APS) Adapter Connected". Refer Intel representative for information of the Intel® APS Adapter and the FFC/FPC.

Figure 293. Location of ZIF Connector, FFC/FPC and Intel® APS Adapter

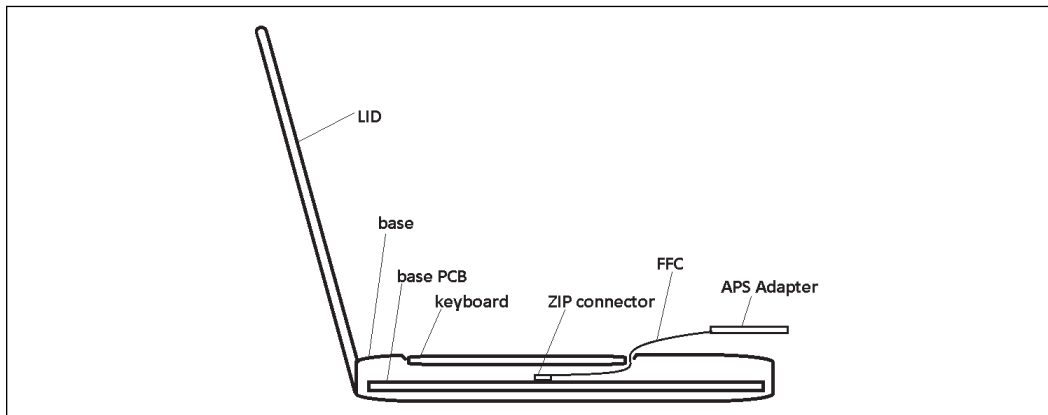


Figure 294. Intel® Automatic Power Switcher (Intel® APS) Adapter with an FFC

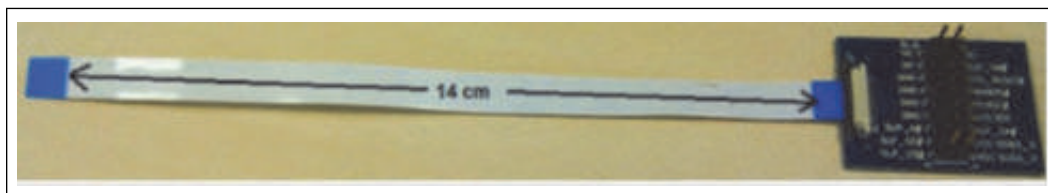


Figure 295. Intel® Automatic Power Switcher (Intel® APS) Adapter

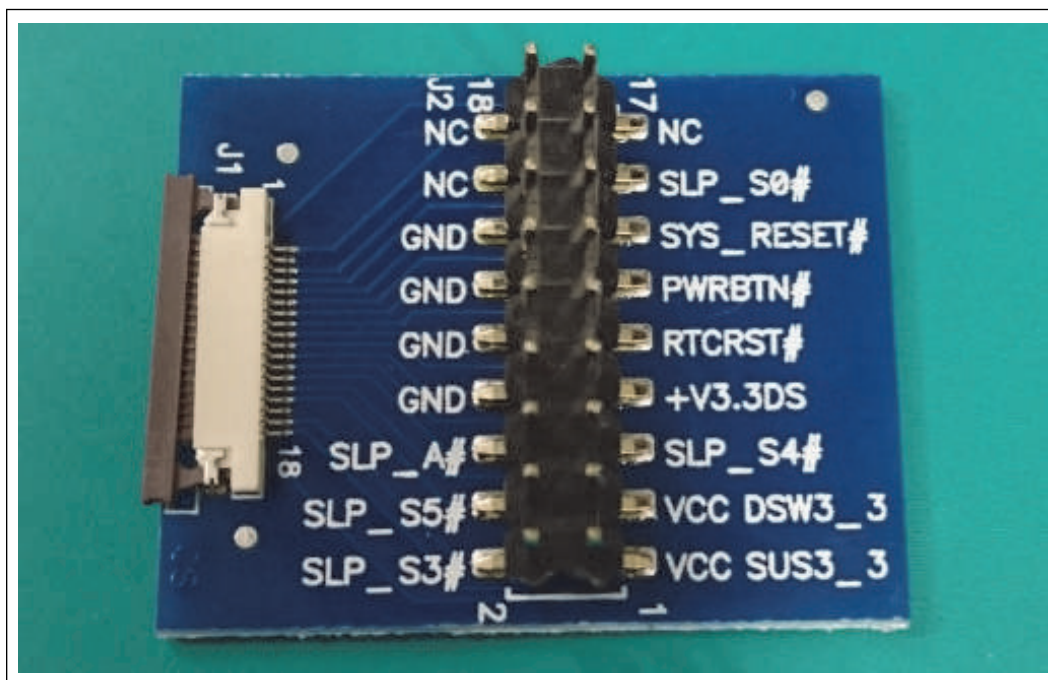
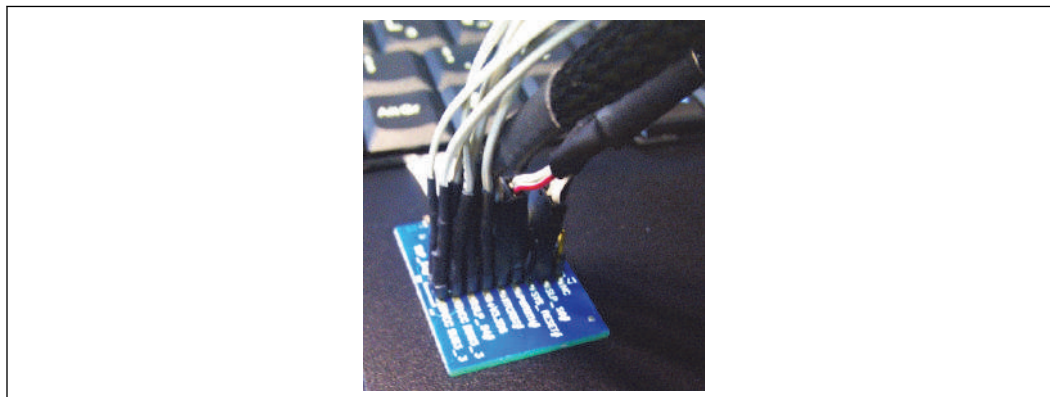
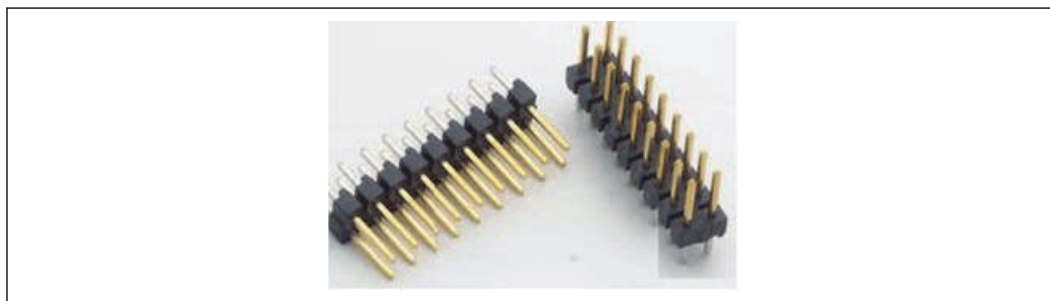


Figure 296. Intel® Automatic Power Switcher (Intel® APS) Adapter Connected

The second connector option is to use a single 18-pin (2 rows of 9 pins) male header with pin width of 0.64 mm, pin height of 5.84 mm and a 2.54 mm distance (pitch) between pins, as shown in the figure below. The pin-out of the 18-pin male header is shown in the Table "Intel® Automatic Power Switcher (Intel® APS) Connector". These pins can be connected directly to Intel® APS. The Intel® APS Adapter is not necessary.

When system gets to the production stage, and if the Intel® CSE subsystem is fully functional and is fully validated on pre-production systems, the Intel® APS connector may become "UN_STUFF" (meaning, not being placed on the board). It is recommended to leave the footprint of Intel® APS connector with 'APS' label on silk screen next to the footprint on the system board. Solder down the Intel® APS connector to the foot print if the system is going to be sent to Intel for Intel® CSE subsystem debugging

Figure 297. Intel® Automatic Power Switcher (Intel® APS) Header**Table 700. Intel® Automatic Power Switcher (Intel® APS) Connector**

Pin #	Pin Names	Description	Intel® APS Pin Names
1	VCCPRIM_3P3	3.3V Primary Power Well	SUS
2	SLP_S3#	When asserted (0) system is in S3	S3
3	NC	No Connection	
4	SLP_S5#	When asserted (0) system is in S5	S5
5	SLP_S4#	When asserted (0) system is in S4	S4
6	SLP_A#	When asserted (0) Intel® CSE is in Moff	M1/OFF
<i>continued...</i>			

Pin #	Pin Names	Description	Intel® APS Pin Names
7	NC	No Connection	
8	GND	Ground	GND
9	RTCRST#	When asserted (0) CMOS is cleared	
10	GND	Ground for RTCRST#	
11	PWRBTN#	When asserted (0) Power Button is pushed	PWR B
12	GND	Ground for PWRBTN#	PWR B GND
13	SYS_RESET#	When asserted (0) Reset Button is pushed	RST B
14	GND	Ground for SYS_RESET#	RST B GND
15	SLP_S0#	When asserted (0) system is in deterministic idle state	
16	NC	No Connection	
17	NC	No Connection	
18	NC	No Connection	

Table 701. Pin Location for Dual-in-Line Connector

Pin Names	Pin #	Header	Pin #	Pin Names
VCCPRIM_3P3	1		2	SLP_S3#
NC	3		4	SLP_S5#
SLP_S4#	5		6	SLP_A#
NC	7		8	GND
RTCRST#	9		10	GND for RTCRST#
PWRBTN#	11		12	GND for PWRBTN#
SYS_RESET#	13		14	GND for SYS_RESET#
SLP_S0#	15		16	NC
NC	17		18	NC

NOTE

The unused or unconnected signal sampling lines which are INEX3 to INEX5 from the Intel® APS should be connected to GND to prevent unexpected measurement results or other potential problems. If unused signal lines are not connected to GND this could result in possible damage to the Intel® APS hardware. Because of this, it is strongly recommended that additional GND pins are provided for all unused INEX sampling lines. All other NVL-H platforms should provide GND pins for the five Deep Sx related signals.

Firmware Debug Hooks

The Firmware debug and its debug hooks are fully defined in Chapter “Platform Debug and Test Hooks”.

- The Firmware Debug Hook is used to support multiple components on the system, such as the CPU, Intel® ME and etc.

7.1.8 Firmware Debug Hooks

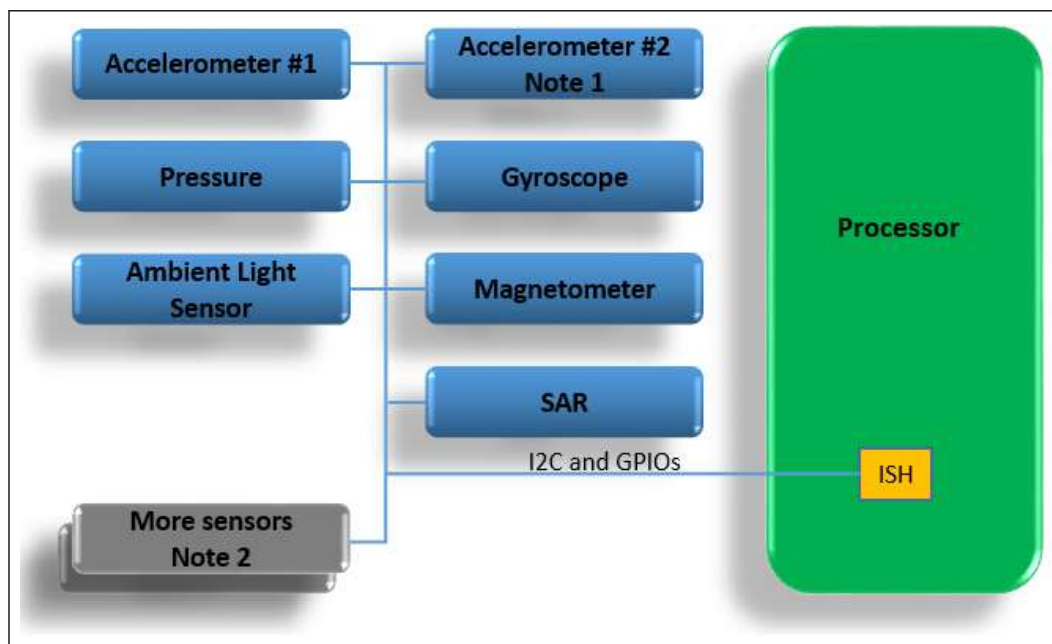
The Firmware Debug Hook is used to support multiple components on the system, such as the CPU, Intel® CSE and etc. [Platform Debug and Test Hooks](#) on page 900.

7.2 Intel® Integrated Sensor Solution (IISS)

This section provides the implementation recommendations of IISS for 2-in-1 (Detachable, Convertible/360-Clamshell), pAIO (personal All In One) and AIO (All In One) system designs. The intent of this document assists the usage of this IISS and sensor devices in hardware design to meet the Microsoft Windows* 11 Operating Systems sensor recommendations. It is highly recommended to follow all of the design guidelines and perform all checklists. Do not deviate from any of the recommendations outlined in this document to prevent or minimize any functionality or responsiveness risks.

Figure below shows the block diagram of the Processor and the sensors.

Figure 298. IISS Block Diagram



NOTES

1. It is for Convertible/360-Clamshell design
2. More sensors are optional, as described in "BOM Innovation" section below

Sensors

Intel provides a list of sensors as IISS POR BOM. Intel provide support and perform full validation cycle on these sensors.

IISS firmware algorithms support only unprocessed and unfiltered raw data coming directly from the sensors. Any pre-process of the sensor data with internal FW which is inside a sensor is not support by IISS Firmware.

Sensors Extensibility Options

The designer has flexibility to choose different vendor for the sensor types that are not included in the proposed IISS POR BOM. Intel have developed the IISS Firmware Development Kit (FDK) to enable the extensibility. Please contact local ISH CE for a FDK release. Following are the extensibility of BOM:

- **BOM Baseline**
 - Feature Set: Intel baseline (which is Windows* HCK Support).
 - BOM: Intel Integrated Sensor Solution (IISS) POR BOM
 - Intel differentiated: pressure sensor, activity context, device context and gesture.
 - uDriver: Intel own FW integration, calibration and validation.
 - Support: Intel provide full support which includes validation on Intel systems.
- **BOM Flexibility**
 - Feature Set: Intel baseline.
 - BOM: OxM can replace sensor which is one of IISS supported sensor types from sensor vendor(s).
 - uDriver: OxM owns FW integration, calibration and validation.
 - Support: Intel provide FW and FDK. OxM integrate custom uDriver.
- **BOM Innovation**
 - Feature Set: Custom set with Intel baseline functionality supported.
 - BOM: OxMs can choose to add new types of sensors and/or algorithms.
 - uDriver: OxM owns FW integration, calibration and validation.
 - Support: Intel provide FW and FDK. OxM integrate custom uDriver and algorithms.

7.2.1 Acronyms

Acronyms	Description
NVL	Nova Lake
BOM	Bill Of Material
EC	Embedded Controller
FDK	Firmware Development Kit
IISS	Intel® Integrated Sensor Solution (as-known-as Intel® ISH)
Intel® ME	Intel® Management Engine
Intel® ISH	Intel® Integrated Sensors Hub
OxM	ODM or OEM
<i>continued...</i>	

Acronyms	Description
POR	Plan Of Record
uDriver	Micro Driver
ZIF	Zero Insertion Force (connector)

7.2.2 Reference Documents

Document	Document Number
Mobile Computer Platform Magnetometer Design Integration Technical White Paper	640580
Intel® Integrated Sensor Solution POR BOM Reference Material	570092
Integrating Motion and Orientation Sensors (by Microsoft*)	https://msdn.microsoft.com/en-us/library/windows/hardware/dn642102(v=vs.85).aspx
ISH EC Communication for Platform State Update Interface Specification	603384

7.2.3 Schematics Design

Intel highly recommends thoroughly understanding and implementing all of the design requirements outlined in the device datasheet avoid from sensor malfunctioning and support problems. The physical locations of the sensors on the main board or the daughter board depend on the board placement and the form factor.

Intel is not a sensor vendor. Follow manufacturers' datasheet and application notes for sensor design. It is recommended to include signal description, function description, trace length and air gap information in schematics where ever possible. Refer to the datasheet for the pin definitions and the reference schematics for the detail interconnections.

Sensors - I2C Bus Considerations

- Sensors can be connected to ISH_I2C0, ISH_I2C1 or ISH_I2C2. It is required to ensure the total capacitance loading does not exceed the maximum bus load. There is no specified location for ISH_I2C pull-up resistors, but it is recommended to place them on the main board especially when the sensors are on the daughter board.

Sensors - GPIO Considerations

- For sensor interrupt output, each of Intel® ISH managed GPIO can be configured as an input pin and connects to the output pin of a sensor with proper PU resistor to Vio if the sensor supports Low-Active. Be noticed, Intel® ISH FW does not support shared interrupt.
- For a Convertible/360-Clamshell design, it needs a primary accelerometer placed in the lid and a 2nd accelerometer in the base. It is strongly recommended to add additional route(s) from the output of the hall sensor(s) to Intel® ISH managed GPI(s). By doing this, it can help Intel® ISH FW to precisely distinguish between 0° (which is Lid-close mode) and 360° (which is Tablet-mode). The status of Platform-Lid-Mode can be delivered by Intel® ISH managed GPO to a discrete companion chip like an EC. One pin can represent two modes, two pins can represent four modes and so on. For example, if the angle between the lid and the

base has been divided into 4 regions, 0~9 degrees represent Lid-close mode, 351~360 degrees represent Tablet mode, 10~200 represents Clamshell mode, 201~350 represents Stand/Tent mode. The companion chip reads these Intel® ISH managed GPO(s) for knowing which mode it is now. Based on this information, powering On/Off Keyboard, Touchpad, LED and more are feasible.

- For an Embedded-Controller Interface (ECI) enabled design, the pin count of Intel® ISH managed GPIOs can be saved for other sensors. Currently the status of Platform-Lid-Mode, On-Table Detection and Dynamic Keyboard Lighting can be delivered through ECI. Be noticed, the Intel® ISH managed I2C is the host and the EC managed I2C is the device on the bus. Refer to # 603384 and Section 3.3.5.1 of # 570093 for more detail.
- For an On-Table Detection design, it is preferred to place the primary accelerometer in the lid as the source of virtual sensor of On-Table Detection in a Convertible/360-Clamshell design. Contact local ISH CE for the limitation and more detail if On-Table Detection is enabled on a Low-cost Clamshell design which only supports screen rotation on Microsoft Windows 11 if the primary accelerometer is placed in the base.

The output of the virtual sensor could be from one of below:

- An ISH managed GPO to the companion chip.
- An ISH managed I2C to the companion chip with ECI enabled.
- A customized Windows API which is compatible with Microsoft Sensor API.
- Intel® Dynamic Tuning Technology (Intel® DTT) framework.
- For an In-Bag Detection design, it is preferred to place the 2nd accelerometer in the base as the source of virtual sensor of In-Bag Detection in a Convertible/360-Clamshell design.

The output of the virtual sensor could be from one of below:

- An ISH managed GPO to the companion chip.
- A customized Windows API which is compatible with Microsoft Sensor API.
- For alternate pin-mux, the primary pin and alternate pin are exclusive to each other and cannot be enabled at the same time. Intel® ISH does not support it. Each internal of Intel® ISH managed pins maps to one pin at Processor package level. Choose one of the primary and the alternate pins and get the necessary support from BIOS team to fit the design.

Sensors - SPI Bus Considerations

Avoid from duplication, refer to ISH-SPI in this document.

Sensors - UART Considerations

Avoid from duplication, refer to generic UART in this document.

Sensors - I3C Considerations

Avoid from duplication, refer to generic I3C in this document.

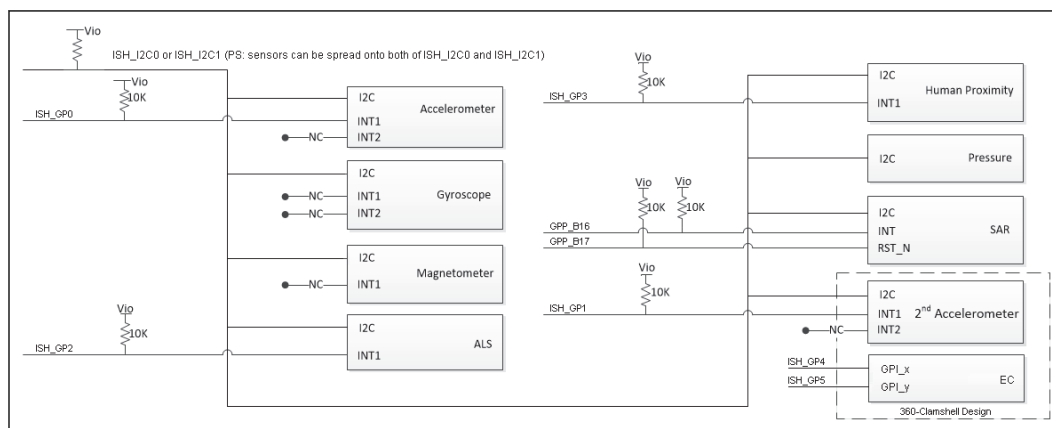
Sensors - Voltage Considerations

Refer to Nova Lake GPIO document for more detail. Place the level-shifter(s) for voltage translating between ISH interface and sensor component(s) when needed.

Sensors - Example of I2C Bus and Interrupt Connection

- Figure below shows an example of interrupt connections to Accelerometer, Gyroscope, Magnetometer, Barometer and etc. All Intel® ISH managed I2C buses and interrupt lines should have their own pull-up resistors for supporting Open-Drain configuration with Intel® ISH FW. Refer to the datasheet from sensor vendor for more detail.

Figure 300. I2C Bus and Interrupt Connection Example



7.2.4 Layout and Assembly

General Rules

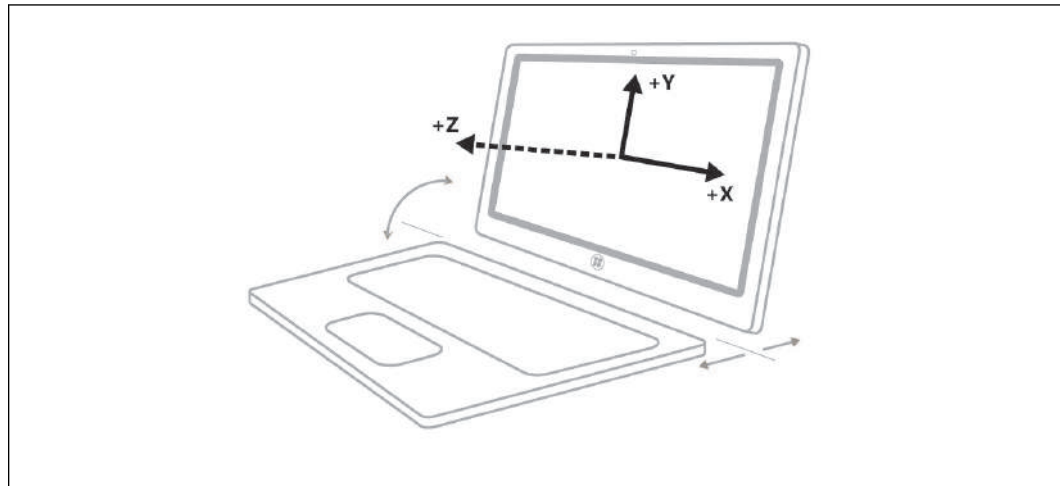
- Make the sensor placement to be the first step when outlining a new PCB layout. Plan accordingly accommodating to both of board and system level.**
- Understand and follow all the layout and design requirements from the sensor vendors and manufacturers.
- Place the sensor 2 or 3 cm away from the hot devices to avoid large temperature gradient.
- Place the sensor at interference-free locations on the board in the system for all user scenarios.
 - Place the accelerometer in a stable position and isolated from any vibration.
 - Place the magnetometer away from magnetic interferences.
 - Place the ambient light sensor away from any unwanted lighting sources (for example: a LED indicator) in the system. Also make sure that the ambient light is not blocked.
- Do NOT place the sensor in a region of PCB that can warp as it can cause package stress which causes shift.
- Do NOT place the sensor at the position which has unsymmetrical stress coming from PCB deformation as it will cause drift.
- Design for manufacturing: Design a system that the board is easy to be assembled and ensure the board is fixed firmly without flex after the assembly.

- It is strongly recommended not to place any metal structure such as via(s), trace(s), plane(s) and others in the area just below the sensor component on the top/bottom metal layer.

Accelerometer

- The X, Y and Z axes of the sensor chip orientation is recommended to match the diagram below.

Figure 301. Accelerometer X, Y and Z Orientation

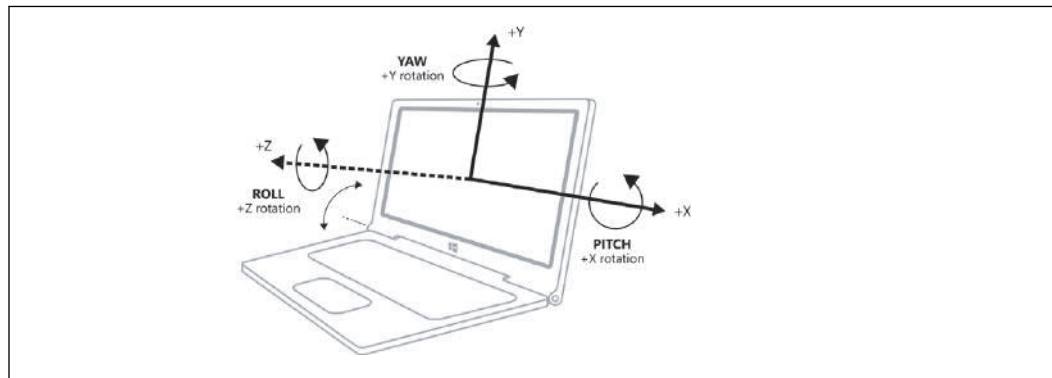


- Accelerometer must be mounted securely, near a screwed standoff point and cannot be moved independently.
- Accelerometer must be placed in a stable position and isolated from any vibration including transmitted vibration, vibration of the air pressure change and etc.
- It is recommended that accelerometer is kept a distance of 30 mm away from audio speakers, fans and haptic vibration motors.
- Many modern electronic accelerometers have the ability to perform on-board filtering (e.g., removing DC gravity components in the signal). Intel® ISH specifically disables all on-board accelerometer filtering. All signal conditioning and signal thresholding of accelerometer are performed within Intel® ISH FW.

Gyroscope

- The X, Y and Z axes of the sensor chip orientation is recommended to match the diagram below.

Figure 302. Gyroscope X, Y and Z Orientation



- Gyroscope must be mounted securely, near a screwed standoff point and cannot be moved independently.
- Gyroscope must be placed in a stable position and isolated from any vibration including transmitted vibration, air pressure change vibration and etc.
- It is recommended that gyroscope is kept a distance of 30 mm away from audio speakers, fans and haptic vibration motors.

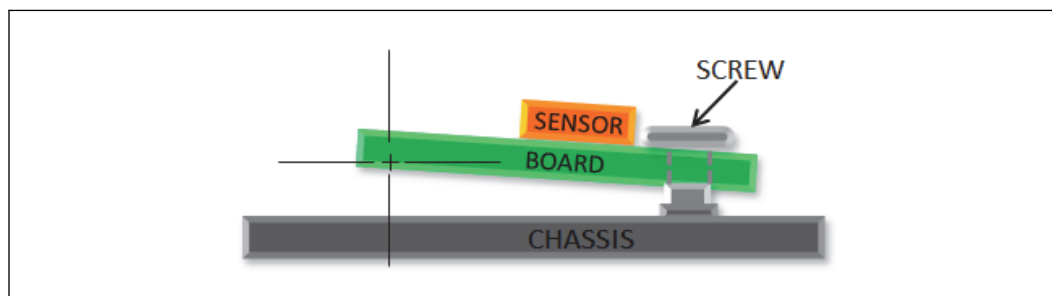
Inconsistent Assembling of Accelerometer and Gyroscope

- **Issue Replication Steps:**

- System laid flat on leveled surface and Microsoft Sensor Diagnostic Tool showed system appeared to be moving (For example, XYZ accelerometer values are non-zeroes).
- Output from Gyroscope device showed system appeared to be rotating.

- **Possible root cause 1:**

Sensor device was placed near screw holes. Board bends when screwed to chassis causes the sensor to skew, not leveled in referenced to the chassis.

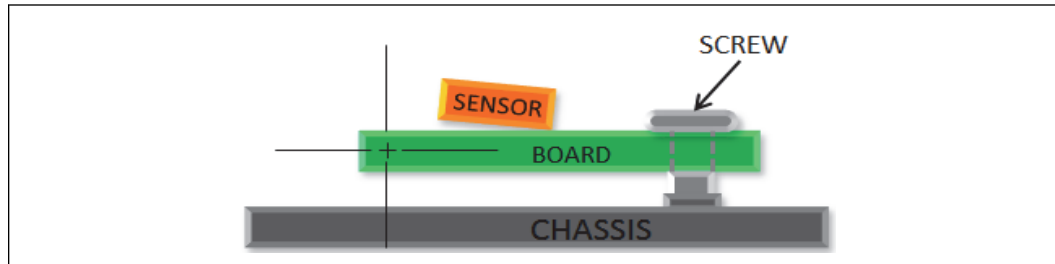


WORKAROUND:

1. Relocate sensor device away from screw hole.
2. Perform per-System Calibration.

- **Possible root cause 2:**

Assembly house SMT machines inconsistent placements. Sensors soldered on the board are skewed in referenced to the main board or the daughter board.



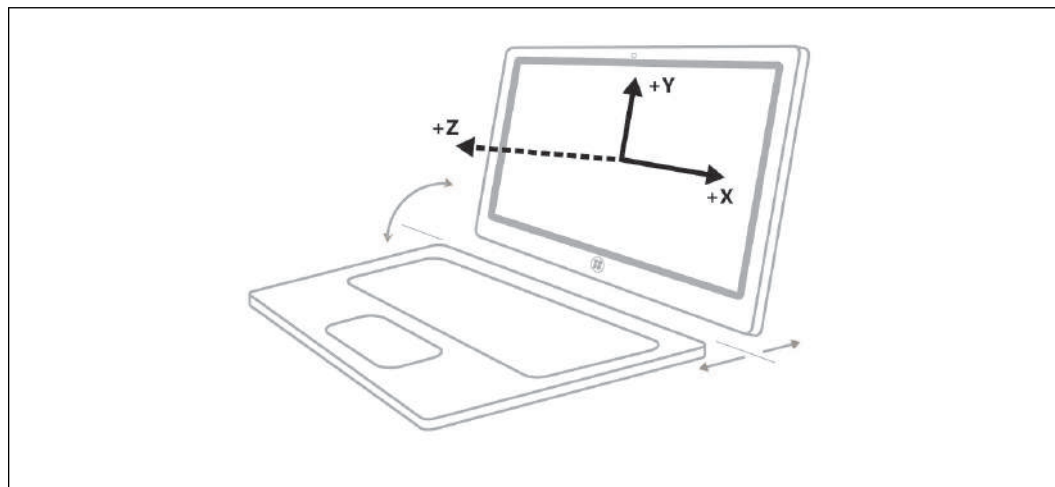
WORKAROUND:

1. Provide tighter SMT machine tolerance level.
2. Perform per-System Calibration.

Magnetometer

- The X, Y and Z axes of the sensor chip orientation is recommended to match the diagram below.

Figure 303. Magnetometer X Y and Z Orientation



- **Make the magnetometer placement be the first step when outlining a new PCB layout.** The ideal location for the magnetometer is to be as far as possible away from magnetized components, high current components and traces. The best locations are often to be found on edges and/or at corners of the PCB.
- Must place it at least 40 mm away from any rare earth magnets (speaker magnets, LID magnets, etc.) and inductors.
- Must place it at least 30 mm away from any large/thick (structural) ferrous metals, the components which includes the magnet of a hard disk, digitizer and the components with the magnetism of 100 Gauss.
- Keep a minimum distance of 10 mm away from the components which includes microphone, vibration motor, magnetic switch, transformer, camera module, beeper, battery, SAW filter, antenna, power amplifier, choke, LCD rear case and the components with the magnetism of 5 Gauss.

- Keep a minimum distance of 5 mm away from the components which includes memory socket, SIM card socket, USB connector, other connectors, screw, nut, spring, EMI shielding part and the components with the magnet-conductive material (which includes **Iron, Cobalt and Nickel**).
- Keep a minimum distance of 2 mm away from the surrounding resistor and capacitor since **Nickel plating** is usually present on solder end caps of surface mount components.
- **Keep it away from the power traces (which include power or ground plane) on all the PCB layers, component on both PCB sides and out of the board with the variable current higher than 10 mA. For instance: keep a minimum distance of 20 mm away from the trace that carries the variable current of 100 mA or keep a minimum distance of 25 mm away from the trace that carries the variable current of 200 mA.**
- **Constant current has the interference as magnet. Make sure the interference does not cause the output of the magnetometer over the measurement range of the magnetometer.**
- Do NOT cover the magnetometer with any material that can block the static magnetic field from magnet and earth.
- As far as possible away from mechanical based HDDs, vibrators, hearing aid coils and battery connector which contain ferromagnetism, high frequency circuits and power circuits.
- Make sure the LCD back shell, shield (for RF, BT...) and mechanical part (key, screw, connector) do not contain ferromagnetism material (Fe, Co, Ni).
- Apply the same rules to the opposite side of the PCB.
- Apply the same rules to any of daughter boards or other components sandwiched in proximity to the magnetometer.
- Make sure the rules are followed on all the usage modes. (Example: The magnetometer is in the lid. Consider components on the base when system is on tablet mode in a Convertible/360-Clamshell design.)

All digitizer solutions that employ a ferrous-based shield are unsuited for deployment with the magnetometer. Digitizers that require a ferrous-based shield are not supported by Intel® ISH FW due to the excessive soft iron distortion induced on the magnetometer. Other platforms possessing digitizer solutions that do not require a ferrous-based shield must conform to the general platform guidance for deploying magnetometer.

Ambient Light Sensor (ALS)

- The ALS must be placed in the lid as close as possible to the camera module (if it is present). In order to allow effective control of screen brightness based on ambient light. The performance of the display brightening is ultimately defined by the performance of ambient light sensor and dimming to specified LCD luminance levels in response to ambient light changes. While the firmware has the ability to control the ALS signal gain and the LCD luminance response and proper mechanical design is still required to ensure the right user experience. Several hardware features must be considered when integrating ambient light sensors into system.
- **ALS Characteristics:**

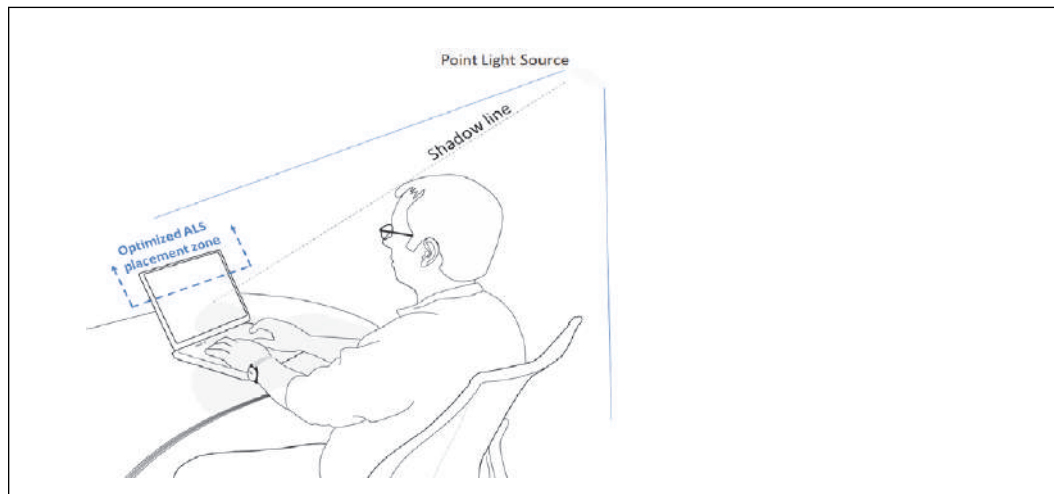
Characteristic	Specification	Notes
Illuminance Range	1-100k Lux	As output through sensor module which includes optics, surfaces, calibration data and etc. It's not only related to the bare sensor.
Accuracy	+/- 10%	As output through sensor module which includes optics, surfaces, calibration data and etc. It's not only related to the bare sensor.
Type	Digital	16-bit resolution.
ALS IR/UV rejection	Supported	Important for consistent measurement.

- **Aperture Placement:**

The light sensor aperture is required to be located on the same plane as the primary display which is facing to the user. This is necessary because the sensor measures the ambient light conditions facing the user for supporting adaptive brightness and light-aware applications that display their content on the screen.

- Avoid placing the light sensor in areas of the computer that are likely to be obscured from the light source or by shadows.
- Figure below illustrates an example of user scenario in which a direct light source is behind the user. A shadow is cast over the lower half of the screen and the base of the computer. The optimal light sensor placement is near the top of the screen and facing the user in this scenario.

Figure 304. Ambient Light Sensor Cross Section



- **Aperture Size:**

The aperture must be wide enough to detect light without significant attenuation when the light source is at an angle to the device or from a diffused source. A small hole or deep hole could cause a "Shadow Effect" which reduces the usability of the ALS sensor.

OxM should work with their sensor vendor or manufacturer to determine the minimal and optimal aperture size.

- **Optical filter:**

In some instances, an optical film or ink coating may be applied to hide the ALS optical cavity. The filter must be designed to allow at least 80% of visible wavelength of light to pass. It means a light meter placed behind the film must register at least 80% of the LUX value from a direct light source that it registers with the film removed.

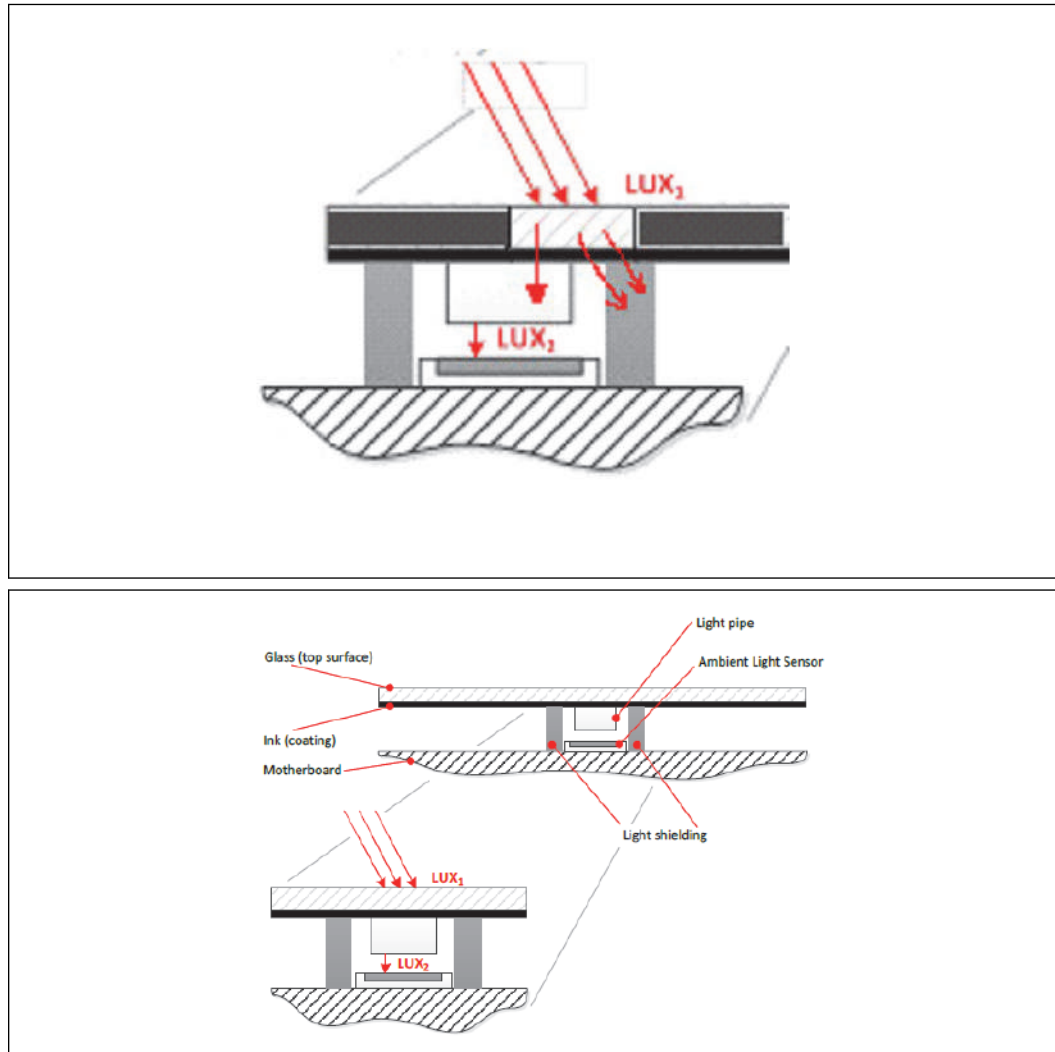
- **Light pipe:**

A light pipe constructed from a light-conducting material may optionally be deployed to increase the light-gathering ability of the optical cavity. This may be required in cases where the ALS optical cavity is excessively deep, inhibiting light gathering from off-axis (i.e., angled) and diffused light sources.

- **Cover Consistency of Glass/Cover**

- The glass or plastic covering material must be consistent across the units of the same model. Failure to follow this recommendation will result in inaccurate readings from the ALS based on the per model calibration parameters.
- Care must be taken when placing the Glass/Cover. A misalignment would mask out some of the light and produce unexpected readings from the ALS.

Figure 305. Ambient Light Sensor Cross Section



Pressure Sensor

- In order to achieve the specified performance for your design, the following recommendations when mounting a pressure sensor on PCB:
 - The clearance above the sensor package shall be 0.1mm at minimum.
 - For the device housing appropriate venting needs to be provided in case the ambient pressure shall be measured.
 - Liquids shall not come into direct contact with the device.
 - During operation pressure sensor is sensitive to air pressure variation, fast heating and light heating which can influence the accuracy of the measurement.
 - The pressure sensor should not be placed at the position where has unwanted fast air pressure variation caused by fans, speaker and etc.

- The pressure sensor shall not be placed close the fast heating parts in case of gradients $> 3^{\circ}\text{C}/\text{sec}$. It is recommended to follow the vendor's application note and contact the vendor representative for details.
- The area below the sensor (on the same side of the board) must be defined as keep-out area. It is strongly recommended not to place any structure in top metal layer underneath the sensor.

Specific Absorption Rate (SAR) Sensor

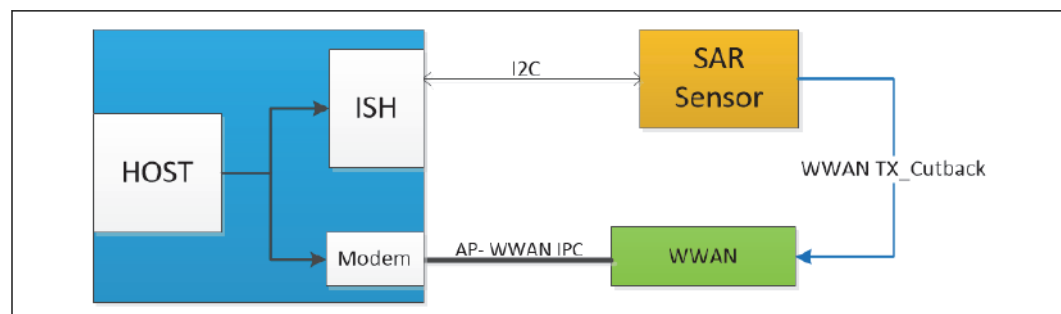
- **SAR**

SAR sensor is used as a proximity sensor to identify conductive objects (human body parts such as ear, head, lap and torso) presence near the transmitting system antenna. SAR requirements limit the levels of transmitted RF power to ensure that the power absorbed by a human body is below certain limits established by regulatory bodies around the world. This may require that a device reduce its RF transmitting power when it is in close proximity to the human body. In these systems, the SAR sensor activation is used to allow the device to operate at a higher power (and increased operating range) when the antennas are not close to a body but then reduce the transmission power of the relevant antenna to ensure compliance with the RF exposure requirements when close to the human body. Many regulatory bodies globally are aligning to International RF exposure guidelines or the FCC limits.

SAR needs to be evaluated for transmitters operating anywhere in the RF spectrum (9KHz - 300GHz) and so it is relevant to cellular technologies, Wireless LAN technologies and many other wireless technologies supporting data and/or voice.

A simple SAR sensor control over Antenna power is described in the figure below. In this way, the SAR sensor directly controls the WWAN transmitter power. This is the most efficient way remain compliant with the RF exposure requirements without any additional logic.

Figure 306. Simple SAR Sensor Control over WWAN Antenna Power



- **Regulation Sources**

The following are the 2 RF exposure requirements for devices that are used in close proximity to the body (within 20 cm). As this can be a changing field, the data published in this section is as it was known in the time of publishing of this document:

- **United States:** The relevant FCC Requirements for the FCC requires that devices used in close proximity to the body have a SAR level below 1.6 watts per kilogram (W/kg) measured over the volume containing a mass of 1 gram of tissue that is absorbing the RF energy.

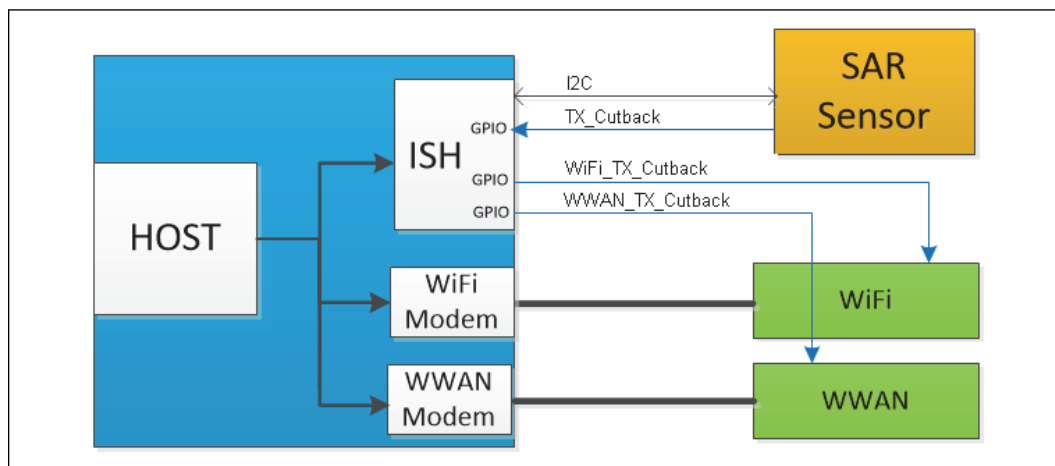
- European Union: CENELEC specify SAR limits within the EU, following IEC standards (IEC 62209-2:2010). For devices used in close proximity to the body, the SAR limit is 2.0 W/kg measured over the volume containing a mass of 10 grams of tissue that is absorbing the RF energy.

- **Co-SAR**

In more sophisticated systems where the system support both WWAN phone and IP phone (e.g., such as WWAN and Wi-Fi).

Figure below shows a recommended way to control antenna power using an algorithm that runs on the Operating system (OS). In this way, the OS has control over each antenna for controlling the transmitted power to ensure compliance with RF exposure requirements.

Figure 307. Co-SAR Sensor Control over WWAN Antenna and Wi-Fi Antenna Power



- **Design Guidelines**

1. There are two main relevant usage models that should be covered by SAR sensors used to reduce transmission power of the antenna:
 - When bringing the system to the users' ear, so phone speaker (and RF antennas) is closed to the users' ear and head. This is mainly relevant to "Mobile Phone" systems, such as handsets and phablets.
 - When placing the system on or close to other human body parts, such as the lap or torso. This is mainly relevant to laptop/clamshell, tablets, phablets and similar types of devices.
2. The sensor electrode may be a simple copper area on a PCB or FPC (Flexible PCB) for example with an adjacent reference ground area. Sensor electrode capacitance (to ground) will vary when a conductive object is moving in its proximity.
3. The sensor electrode needs to be located close to the antenna or antennas that it is controlling. Having the sensor be away from the antenna makes trigger distances unreliable. The size, shape and position of the sensor electrode are largely dependent on the system usage model. Typically, the sensing area for on ear detection is around the speaker at the top of the mobile phone and near the camera.
4. Conductive tape (copper tape) works well for prototypes, while FPC is suggested for mass production.

5. Keep the SAR sensor area away from area that human finger touches, such as a touch screen, systems' buttons, keyboard and etc. There is no need to throttle antenna power in such a case as the extremity SAR requirements are 2.0 W/kg over 10g of tissue mass.
 6. Robust electrode design is the one that minimizes environmental (parasitic) capacitance noise value and variation of it.
 7. Power reduction values need to consider both the stand-alone use of a transmitting antenna and simultaneous transmissions from multiple antennas, where supported. The combined SAR value from these antennas needs to meet the relevant limits.
 8. System design considerations need to ensure that the default power state is always the reduced power state such that if the SAR proximity sensor circuitry is failed.
 9. The software controls need to be robust and must not be user accessible (i.e. the user must not be able to modify the power reduction settings or disable the sensor). Firmware and software updates need to be carefully tested to ensure that the proximity sensor functions are not bypassed.
 10. Where the proximity sensor enables large power reduction regulatory bodies such as the FCC may expect to refer a description of how the sensor performance is verified across production units. This may necessitate the use of test software, test fixtures and sampling at the production floor. Large power reduction values can trigger device surveillance testing by regulatory agencies so quality control at the production level is important.
 11. In general, wherever possible, place the transmitting antennas in areas not accessible to human body parts, so eliminate the need for SAR sensor circuits and the need to lower transmission power. This is difficult in tablets, 2-in-1 and phones which are small and difficult to have an antenna that would not be in close proximity. For tablets it makes sense to keep the antennas toward the front (screen side) of the device to maximize separation from person. Other items to suggest here might be the ability to keep antennas that might operate simultaneous (Wi-Fi and WWAN in hot spot mode, for example) as far apart as possible to reduce the need for power reduction in those modes.
- **Device Requirements**

The electrode and circuit design need to meet the following minimum requirements:

 1. The sensor electrode needs to be placed on the inner side of the back cover, close to the transmission antenna, and properly shielded with ground to minimize the interference from the surrounding electronics. The shield needs to be tied to the platform ground to remove the ground bouncing noise and false proximity detection from the platform ground traces.
 2. Sensor should be placed very close to the electrode to reduce noise. Routing from sensor electrode to sensor needs to be coaxial cable with ground shielding.
 3. Sensor proximity range requirement needs to be determined after SAR testing, but can start with 10 mm from the back and side close to where the transmission antennae are placed in the form factor system.
 4. Sensor detect range should be 0-60pF with distance of 20 mm from a human body part.
 5. SAR sensor circuit power needs to be less than 0.1 mW.

6. SAR sensor proximity range for human body and other non-conductive material (such as tablet top) should have big enough delta so that it can be programmed/tuned to detect only the human body proximity and no other non-conductive material.

7.2.5 Form Factor Considerations

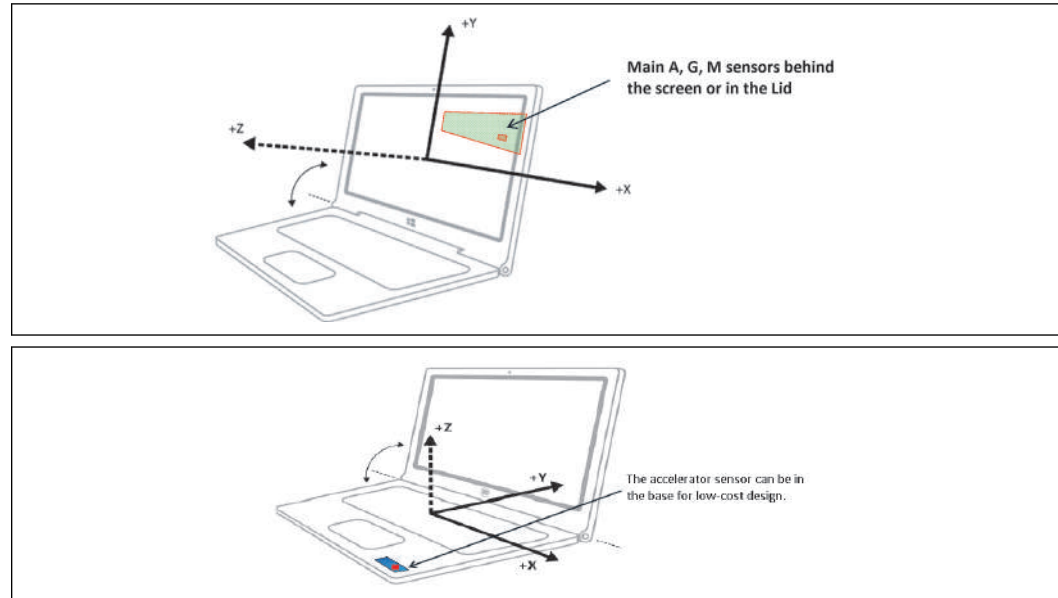
This section illustrates the sensor placement for form factors such as Clamshell, 2-in-1 (Detachable, Convertible/360-Clamshell, Twist, Slider, Ferris Wheel, etc.), and pAIO (portable All In One).

- For all the form factors, it is must to place the primary accelerometer, gyro and magnetometer sensors in the lid or behind the screen.
- For Convertible/360-Clamshell designs, please must place the 2nd accelerometer in the base.
- For Low-cost Clamshell designs which only support screen rotation, the primary accelerometer is the only available sensor which can be placed in the base. On-Table Detection is supported by the primary accelerometer placed in the lid. Please contact local ISH CE for more detail if On-Table Detection is enabled on Low-cost Clamshell designs.

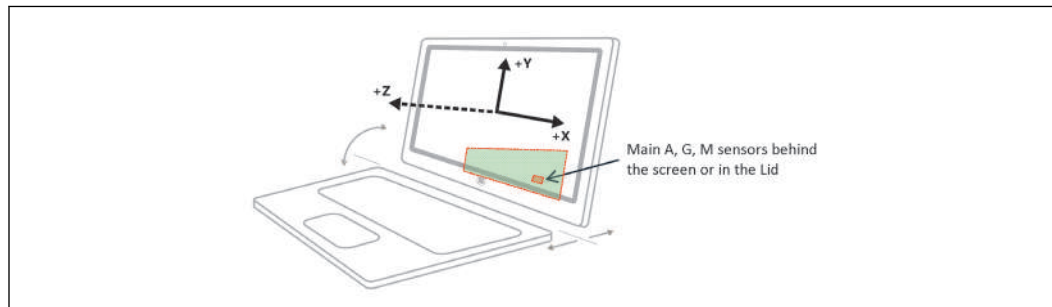
System/Chassis Integration Options

- Clamshell: Main sensors behind the screen or in the lid. For low-cost design, the accelerometer sensor can be placed in the base.

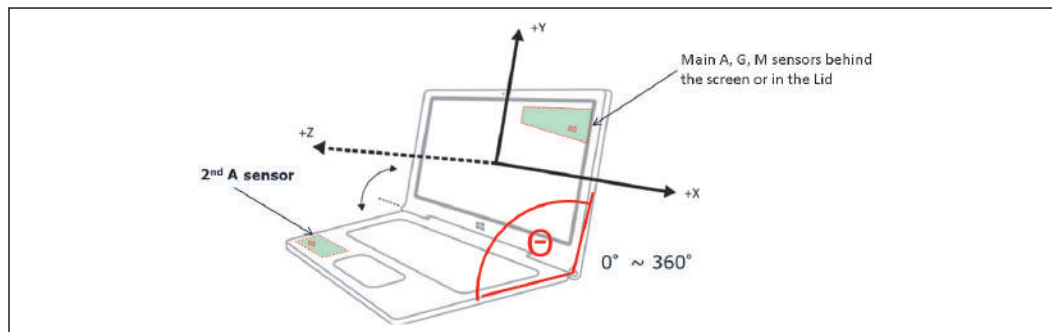
Figure 308. System/Chassis Integration Options: Clamshell



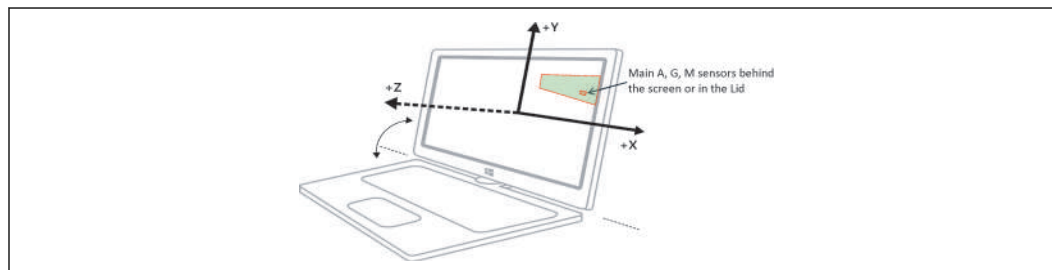
- 2-in-1 Detachable: Main sensors behind the screen or in the lid.

Figure 309. System/Chassis Integration Options: 2-in-1 Detachable

- 2-in-1 Convertible/360-Clamshell: Main sensors behind the screen or in the Lid. The 2nd accelerometer is in the base.

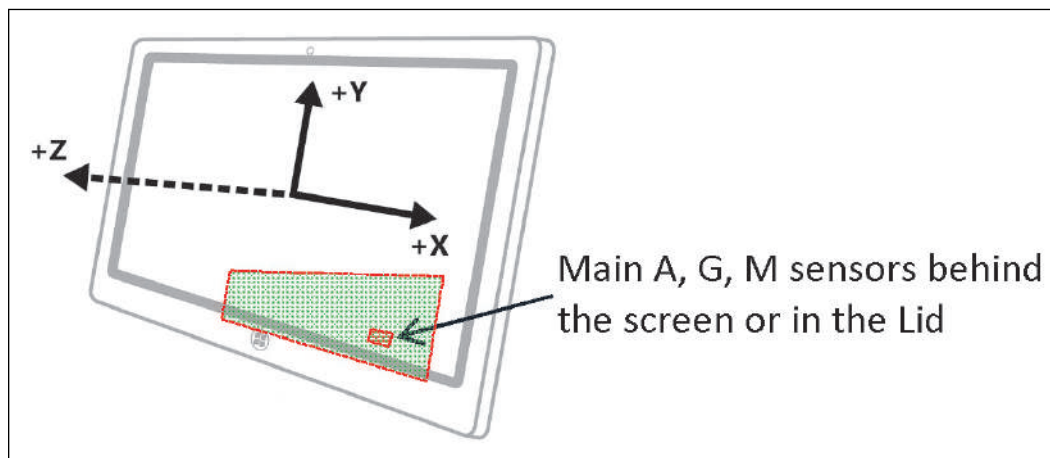
Figure 310. System/Chassis Integration Options: 2-in-1 Convertible/360-Clamshell

- Other 2-in-1: Main sensors behind the screen or in the lid.

Figure 311. Other 2-in-1: Twist

- pAIO: Main sensors behind the screen (Lid).

Figure 312. System/Chassis Integration Options: pAIO



7.2.6 System Debug

System debug hook is an important part of the HW design. It provides the ability to access the system without harming the chassis too much. Otherwise, it will be difficult for Intel to assist within the stipulated time frame.

The debug hook may be eliminated from production systems. For high dense designs, customer is proposed to replace it with test pads. Intel expects these debug hooks to be implemented on customer's system.

Debugging Nova Lake IISS may be divided into two main areas:

- Firmware debug
- Sensor debug

NOTE

Connecting the I2C debug device that has built-in I2C pull-up resistors will change the final value of pull-up resistors. The wrong pull-up resistors may cause Intel® ISH I2C bus not to work. Refer to the Chapter "I2C Interface Design Guidelines".

Firmware Debug Hooks

The Firmware debug and its debug hooks are fully defined in Chapter "Platform Debug and Test Hooks".

NOTE

The Firmware Debug Hook is used to support multiple components on the system, such as the CPU, Intel® ME and etc. It is not specific to Intel® ISH.

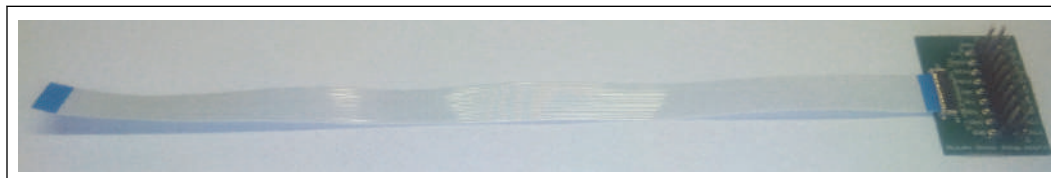
Sensors Debug Hooks

IISS requires part of all below signals to be routed to a connector which should be mounted on the system board in an easily accessible location.

Pin Name	Function
ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 and ISH_I2C2	® ISH managed ISH_I3C/ISH_I2C buses
ISH_GP[0:11] and ISH_GP[5A,6A,7A,8A,9A,10A,11A]	Intel® ISH managed GPIO pins
ISH_SPI	Intel® ISH managed SPI bus
ISH_UART[0:1]	Intel® ISH managed UART ports
Vio	Sensor Reference Voltage
GND	Ground

In order to access this connector with external test/debug equipment, Intel developed an adaptor in the figure below. To obtain such adaptor, contact local ISH CE.

Figure 313. Intel® ISH Debug Adapter and Ribbon Cable



Intel recommend customer placing a small connector on the PCB. It provides all the sensor debug traces located in an easily accessible location such as under the keyboard, inside the DDR compartment, the WLAN module compartment or the SSD compartment. Please refer to the figures below.

Figure 314. Proposed Location for ZIF Connector - Under the Keyboard

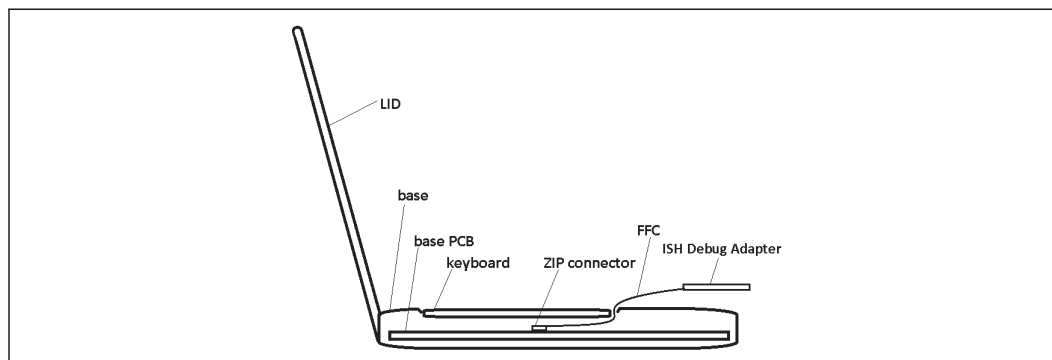
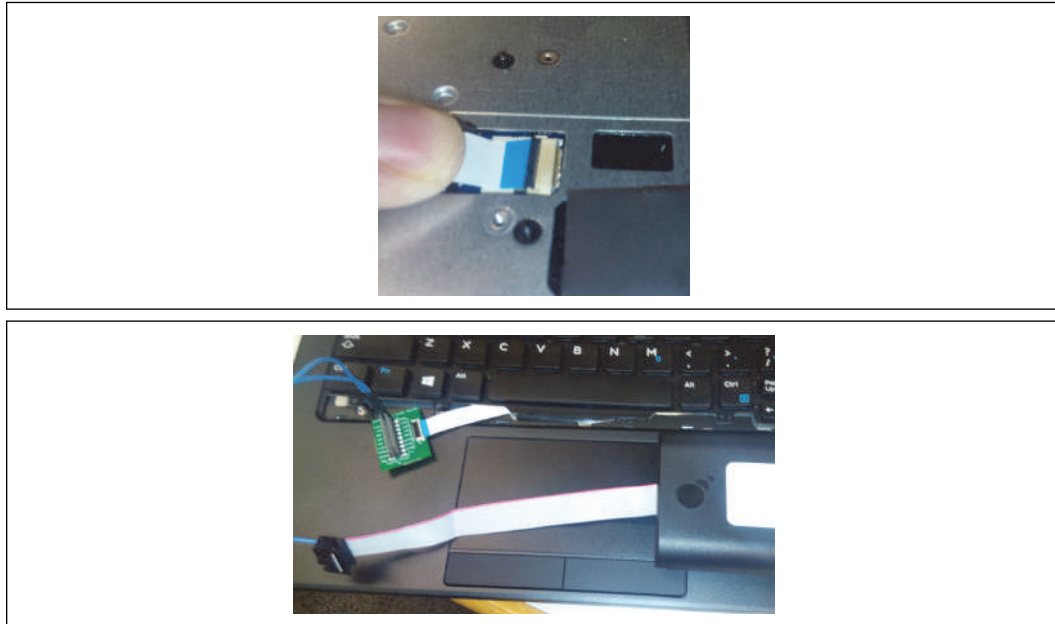


Figure 315. Proposed Location for ZIF Connector - Under the keyboard



The ZIF connector is an 18-pin FPC/FFC Lower Side Contact, Side Entry, Surface Mount connector with 0.5 mm pitch, such as the 10051922-1810elf. The pinout of the ZIF connector is shown in Table below.

Table 703. 18-pin ZIF Connector Pinout

Pin	Pin Name	Pin	Pin Name
1	ISH_I3C0_SCL/ISH_I2C0_SCL	10	GND
2	ISH_I3C0_SDA/ISH_I2C0_SDA	11	ISH_I2C2_SCL
3	ISH_I3C1/ISH_I2C1_SCL	12	ISH_I2C2_SDA
4	ISH_I3C1/ISH_I2C1_SDA	13	Reserved
5	GND	14	Reserved
6	GPIOa ¹	15	Reserved
7	GPIOb ¹	16	Reserved
8	GPIOc ¹	17	GND
9	GPIOd ¹	18	Vio
<i>Notes:</i> 1. GPIOa, GPIOb, GPIOc and GPIOd are place holders for customer designed GPIOs. 2. It is strongly recommended that the name 'Nova Lake IISS' be labeled on silk screen next to the ZIF connector/pads. The pin names are also recommended. 3. Reserved pins could be used for ISH_SPI or ISH_UART or any of other purposes.			

There is another option of the debug hook in high dense design. Test pads of all necessary IISS signals can be reserved on the PCB. Solder them with wires to an external 2x9 pin header for debugging.

Table below defines the pin configuration of the external 2x9 pin, 0.1" spacing header.

Table 704. Pin Location for Dual-in-Line Connector

Pin Name	Pin	Header	Pin	Pin Name
ISH_I3C0_SCL/ISH_I2C0_SCL	1		2	ISH_I3C0_SDA/ ISH_I2C0_SDA
ISH_I3C1/ISH_I2C1_SCL	3		4	ISH_I3C1/ISH_I2C1_SDA
GND	5		6	GPIOa
GPIOb	7		8	GPIOc
GPIOd	9		10	GND
ISH_I2C2_SCL	11		12	ISH_I2C2_SDA
Reserved	13		14	Reserved
Reserved	15		16	Reserved
GND	17		18	Vio

After the Nova Lake IISS is fully functional and fully validated on pre-production systems, the debug connector may be un-stuffed on production stage of systems. It is recommended to leave connector footprint on the PCB without being populated for future debugging.

NOTE

Solder the debug connector to the footprint before the system is going to be sent to Intel for IISS debug support.

NOTE

Reserved pins could be used for ISH_SPI or ISH_UART or any of other purposes.

7.3 Discrete Trusted Platform Module

Trusted Platform Module (TPM) is a Trusted Computing Group (TCG) low cost security solution to increase confidence on system security. The TCG Website with all current specifications and revisions can be found at this link,

<https://trustedcomputinggroup.org/>.

The Trusted Platform Module (TPM) is a device that resides on the motherboard and is connected to Processor using the Serial Peripheral Interface (SPI) bus to communicate with the rest of the platform.

Today most protection against computer viruses and unauthorized intrusions consists of adding and updating software that installs outer barriers and surveillance tools. The goal of Safer Computing is to go much deeper, integrating a level of trust into the actual hardware and pre-operating system environments. Applications intended for e-business are based on trust in the communication partner and the reliability of the connection.

The objective of the TPM is to establish a baseline of platform integrity and enhance system security. TPM's are available from several integrated circuit vendors in the form of a silicon component and accompanying software. When integrated into the PC,

a Trusted Platform Module provides protected storage of platform data allowing for platform-level authentication toward the goal of making data files, transactions and communication more trustworthy.

7.3.1 Platform-Specific Important Information

For more information on trusted platform modules and Safer Computing, refer to the trusted computing group web site: <http://www.trustedcomputinggroup.org>.

7.3.2 Signal Descriptions

Table 705. TSSOP-28 Pin Assignments

Pin Names	Pin(s)	Type	Description	SPI Pin-Assignments
PIRQ	20	BI/O	PIRQ#: SPI Interrupt, active low, open collector	M-(Mandatory)
MOSI	23	BI	MOSI: As defined in Section SPI Hardware Protocol in TPM Spec	M
MISO	26	BI		M
LPCPD#	28	I	Implementation of this pin SHALL allow for the pin to be strapped HIGH.	O-(Optional)
SPI_CLK	21	I	SPI_CLK: As defined in Section Clocking	M
SPI_CS#	22	I	SPI_CS#: As defined in SPI Interface Specification	M
LRESET#/SPI_RST#	16	I	SPI_RST#: Active Low	M
SERIRQ	27	BI	As defined in the LPC Interface Specification	O
CLKRUN#/GPIO	15	BI	Same as PCI CLKRUN#. Active Low, internal pull-down. Only needed by peripherals that need DMA or bus mastering in a system that can stop the PCI bus (generally mobile devices). Implementation of CLKRUN# is TPM and chipset vendor specific GPIO will default to low.	O
PP/GPIO	7	I, BI	Physical Presence, active high, internal pull-down. Used to indicate Physical Presence to the TPM. GPIO will default to low	O
XTALOI/32k in	13	I	32 kHz crystal input or 32 kHz clock input	O
XTALO	14	O	32 kHz crystal output	O
GPIO/SM_CLK	2	BI	Defaults as a GPIO. GPIO will default high. Also used as System Management Bus (SMB) Clock signal	O
GPIO/SM_DAT	1	BI	Defaults as a GPIO. GPIO will default high Also used as System Management Bus (SMB)Data signal.	O
continued...				

Pin Names	Pin(s)	Type	Description	SPI Pin-Assignments
GPIO-Express-00	6	BI	GPIO assigned to TPM_NV_INDEX_GPIO_00, internal pull-up Open-Collector output (when configured as output).	O
VNC	3		Vendor-controlled No Connect. This pin will be defined by the TPM vendor or can be a GPIO. There is no defined default state for this signal.	O
TESTI	8	I	This pin will be pulled low on the motherboard. Pull high to enable Test mode. Pull low to disable Test mode and enable GPIO/BADD on pin 9(TESTBI).	O
TESTBI/BADD/GPIO	9	8	TESTBI: Test port. Internal pull-up If TESTI is pulled low, TESTBI acts as a GPIO and (optionally) BADD. GPIO will default high. BADD (optional, defaults high, use external pull-down to signal "low") can be used to select the legacy I/O base address. This logic is manufacturer specific, as well as the selected addresses. Setting is read at Startup.	O
POWER				
VDD			This is either a 3.3 volt or 1.8V DC power rail supplied by the motherboard to the module. The maximum power for this interface is 250 mA. Available from S0-S2.	M
GND	4, 11, 18, 25	I	Zero volts. Expected to be connected to main motherboard ground.	M
VBAT	12	I	Battery input, may be 3.3V. Available from S0-S5 and in G3 state.	O
VSB	5	I	Standby DC power rail, may be 3.3V or 1.8V. Available from S0-S5.	O

Table 706. QFN-32 Pin Assignments

Pin Names	Pin(s)	Type	Description	SPI Pin-Assignments
SPI_PIRQ#	18	BI/O	PIRQ#: SPI Interrupt, active low, open collector	M
SPI_CLK	19	I	SPI_CLK:	M
SPI_CS#	20	I	SPI_CS#:	M
SPI_RST#	17	I	SPI_RST#: Active Low	M
GPIO	3, 4	BI	GPIO defaults to low.	O
VNC/GPIO	6, 13, 29, 30	I,BI	Vendor defined no-connect.	
GPIO will default to low	O			
continued...				

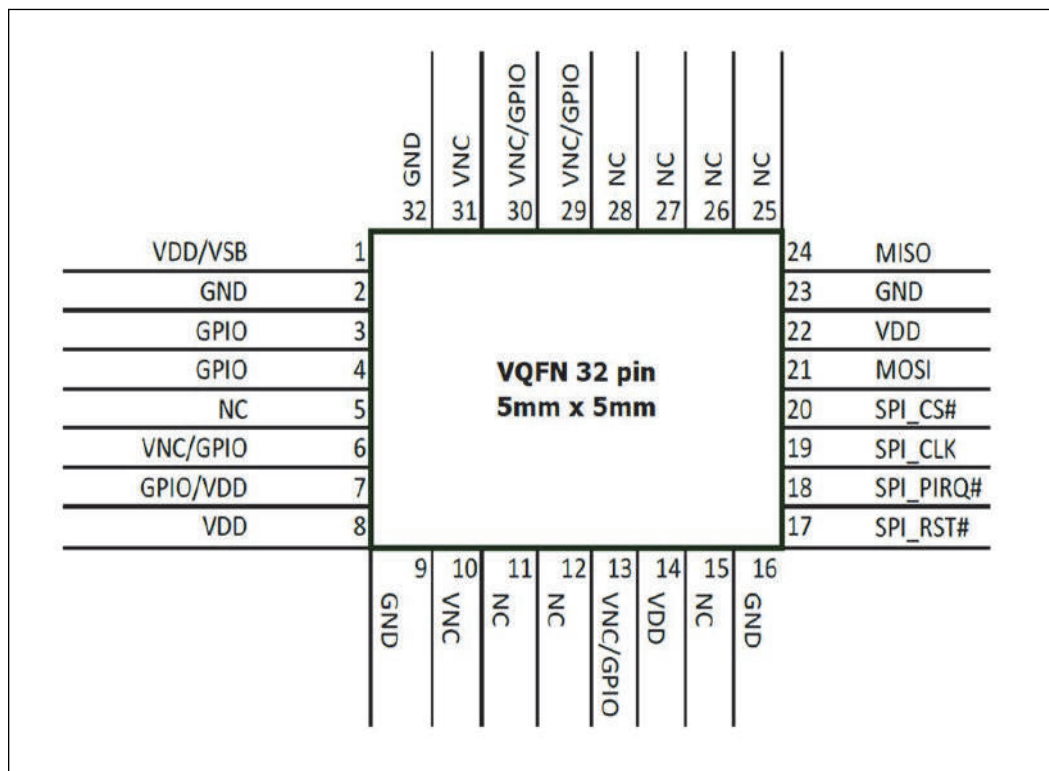
Pin Names	Pin(s)	Type	Description	SPI Pin-Assignments
MOSI	21	BI	MOSI - As defined in Section 6.4 SPI Hardware Protocol	M
MISO	24	BI	MISO -	M
VNC	10, 31		Vendor-controlled No Connect. This pin will be defined by the TPM vendor or can be a GPIO. There is no defined default state for this signal.	O
Power				
VDD	14 22	I	This is a either a 3.3 volt or 1.8 volt DC power rail supplied by the motherboard to the module. The maximum power for this interface is 250 mA. Available from S0-S2.	M
VDD/VSB	1	I	This is a either a 3.3 volt or 1.8 volt DC power rail supplied by the motherboard to the module. The maximum power for this interface is 250 mA. Available from S0-S2. If defined as VSB, this is a 3.3V supply.	M
GND	2, 9, 16, 23, 32	I	Zero volts. Expected to be connected to main motherboard ground.	M
NC	5, 11, 12, 15, 25, 26, 27, 28		No connect	O
GPIO/Power	7	I	GPIO defaults to low, e.g. Physical Presence	
Power is vendor defined	O			

7.3.3 Topology Guidelines

Figure 316. TPM Combo TSSOP-28 Pin out

GPIO/SM_DAT	1	28	LPCPD#
GPIO/SM_CLK	2	27	SIRQ
VNC	3	26	LAD0/MISO
GND	4	25	GND
VSB	5	24	VDD
GPIO-Express-00	6	23	LAD1/MOSI
PP/GPIO	7	22	LFRAME#/SPI_CS#
TestI	8	21	LCLK/SPI_CLK
TestBI/BADD/GPIO	9	20	LAD2/PIRQ#
VDD	10	19	VDD
GND	11	18	GND
VBAT	12	17	LAD3
xtalI/32k in	13	16	LRESET#/SPI_RST#
xtalO	14	15	CLKRUN#/GPIO

Figure 317. TPM SPI VQFN-32 Pin out



7.3.4 Component Selection Guidelines

The TPM in the PC Client platform serves as the Root of Trust. As such, the hardware implementation of the TPM on the motherboard has to account for how the TPM is connected to the other components of the platform which form the trust chain, such as the CPU. It is important that the TPM reset, clock and power signals support the TPM's function as the RTM and RTR and cannot be easily circumvented. Motherboard manufacturers should take care to ensure that the physical connections and routing

- The _TPM_INIT (LRESET#/SPI_RST#) signal SHALL be connected to the platform CPU Reset signal such that it complies with the requirements, specified in Section 1.2.7 of the TPM Profile (PTP) Specification available on TCG website, [HOST Platform Reset](#) in the PC Client Implementation Specification for Conventional BIOS.
- The TPM's main power pins (VDD) SHALL be connected such that the TPM is powered during ACPI states S0-S2 and MAY be powered in S3-S5.
- If a TPM implements the optional VBAT and/or VSB pins, the pins MAY be connected to a battery or auxiliary power source. The motherboard manufacturer SHOULD consult their TPM documentation.

Software Interface to SPI-TPM

This section provides guidelines for platform OEM's and ISV's to aid in design of platforms and software using an SPI TPM. The following sections are informative only, as they describe recommended behavior.

The SPI interface has been architected to be transparent to the driver and application layers in a TPM-enabled software stack. There are some SPI properties which will produce different results than LPC in cases where software does not follow good design practice. In these cases, this specification addresses the TPM requirements so that none of the TPM's security is impacted by bad software design, at the risk of a potentially poor user experience. As such, software and drivers should follow these recommendations to ensure a robust implementation.

- SW should continue to use the memory mapped 0xFED4_xxxx address range to access the SPI-TPM

7.3.5 Debug Guidelines/Recommendations

Discrete Trusted Platform Module (TPM) Design Considerations

This design guide defines a new software interface to the TPM for TPM 2.0, in addition to the FIFO interface. This interface, the Command Response Buffer Interface, has been defined so that it may be implemented in a TPM which also contains a FIFO interface. The Command Response Buffer (CRB) Interface is intended to be physical-bus agnostic, so that it could be implemented on a SPI interface, as specified in this specification or on another physical interface not specified. In order for a TPM to be compliant with this specification, however, it is required to implement at least one of the interfaces is defined.

Refer to the TPM vendor documentation for additional TPM specific design considerations.

8.0 Platform Connectivity

8.1 Intel® Ethernet 1G LAN Design Considerations and Guidelines

The Processor incorporates an integrated 10/100/1000 Mb/s Media Access Controller (MAC) that can be used with an external Intel® Ethernet Connection I219 Physical Layer Transceiver (PHY). Its bus master capabilities enable the component to process high-level commands and perform multiple operations. This can decrease processor use by off-loading communication tasks from the processor.

For specific design and implementation information about the I219 Physical Layer Transceiver (PHY), refer the following reference documents.

Table 707. LAN Design Reference Documents

Document	Document Number
Intel® Ethernet Connection I219 Series (Jacksonville)- All Collateral	https://www.intel.com/content/www/us/en/support/products/82187/ethernet-products/gigabit-ethernet-controllers-up-to-2-5gbe/intel-ethernet-connection-i219-series.html
Intel® Client Ethernet Message of the Week	574707
Intel® Ethernet Connection I219: Product Brief	333229
Intel® Ethernet Connection I219 Datasheet	544486
Intel® Ethernet Connection I219 Reference Schematic	544767
Intel® Ethernet Connection I219 Discrete Inductor Selection Guide	605890
Intel® Ethernet Connection I219 Design Checklist	544506
Intel® Ethernet Connection I218/ I219 Specification Update	571189
Intel® Ethernet Connection I219 PXE Software (UEFI and Legacy)	574961
Intel® Ethernet Connection I219 Windows NDIS Driver Production Version	638909
Intel® Ethernet Connection I219 MDDS	626590
Intel® Ethernet Connection I219 LAN Tools	626873

8.2 Intel® Ethernet 2.5G LAN Design Considerations and Guidelines

This section provides guidelines for connecting interfaces, using special pins, selecting components, and layout guidance. Unused interfaces should be terminated with pull-up or pull-down resistors. There is one reserved pin (pin #8), which should pull-down with 0 Ohm resistor. Please check the I226 collateral package or ask your Intel representative if there is question about schematic or layout designs.

Table 708. LAN Design Reference Documents

Document	Document Number
Intel® Ethernet Controller I226 (Foxville) - All Collateral	https://www.intel.com/content/www/us/en/products/sku/210599/intel-ethernet-controller-i226v/specifications.html
Intel® Client Ethernet Message of the Week	574707
Intel® Ethernet Controller I226 Product Brief	621753
Intel® Ethernet Controller I226 Datasheet	596659
Intel® Ethernet Controller I226 Frequently Asked Questions	574703
Ethernet Product Group (EPG) Product Lifecycle Guidance (includes I219, and I225/226)	499897
Intel® Ethernet Controller I226 Specification Update (External)	615084
Intel® Ethernet Controller I226 Reference Schematic	576602
Intel® Ethernet Controller I225/ I226 Cadence Symbol Files	576601
Intel® Ethernet Controller I226 Design Checklist	576704
Intel® Ethernet Controller I226 Design Considerations	607353
Intel® Ethernet Controller I225/ I226 Boundary Scan Description Language (BSDL)	613949
Intel® Ethernet Controller I225 NVM	615202
Intel® Ethernet Controller I225 NVM for vPro Docking	638915
Intel® Ethernet Controller I226 PXE Software (UEFI and Legacy)	615203
Intel® Ethernet Controller I226 LAN Tools	615722
Intel® Ethernet Controller I225/ I226 IEEE Tool User Guide	618457
Intel® Ethernet Controller I225/ I226 Windows NDIS Driver Production Version	638910
Intel® Ethernet Controller I225 MDDS	626344
Intel® Ethernet Controller I226 Manufacturing Instruction Manual	633982
Intel® Ethernet Controller I225/ I226 Input/Output Buffer Information Specification (IBIS) model	630782
<i>continued...</i>	

Document	Document Number
Intel vPro® for Discrete LAN Solution Bring Up Guide	633269
Intel® Ethernet Controller I225/ I226 Time-Sensitive Networking Software Manual	682924
Intel® Ethernet Controller I226 NVM	686540
Intel® Ethernet Controller I226 NVM for vPro Docking	727433
Intel® Ethernet Controller I225/ I226 Windows 11 NetAdapterCx Production Driver	686541
Intel® Ethernet Controller I226 MDDS	727054
Intel® Ethernet Controller I225/ I226 Windows Capsule Update SDK	730256

8.2.1 Connecting PCIe Interface

The I226 connects to the host system using a PCIe interface. The lane has to be AC-coupled between its corresponding transmitter and receiver, with the AC-coupling capacitor located close to the transmitter side (within 1 inch). Each end of the link is terminated on the die into nominal 100 differential DC impedance. Board termination is not required.

Link Width Configuration

The I226 is a MAC/PHY PCIe device that supports link width x1. During link configuration, the platform and the I226 negotiates on a common link width. The link width result must be x1.

Polarity Inversion and Lane Reversal

To ease routing, designers have the flexibility to the lane reversal modes supported by the I226. During the link training sequence, the polarity inversion of differential pair is detected and automatically corrected.

PCIe Reference Clock

The I226 requires a 100 MHz differential reference clock called PE_CLK_p and PE_CLK_n. This signal is typically generated from a Peripheral Controller Hub and provides it to the I226. For Network Interface Cards (NICs), the clock is furnished at the PCIe connector and the signal must be routed to the PCIe connector.

NOTE

The frequency tolerance for the PCIe reference clock is ± 300 ppm.

PCIe Bias Resistor

For proper biasing of the PCIe analog interface, a 200 Ω 1% pull-down resistor must be connected between the RBIAS (pin #20) pin and ground in order to calibrate the PCIe analog modules. To avoid noise coupled onto this reference signal, place the bias resistor close to the I226 and keep traces as short as possible. Do not change this resistor value.

Miscellaneous PCIe Signals

The I226 signals power management events to the system by pulling low the PE_WAKE_N signal. This signal operates like the LAN WAKE signal. Note that somewhere in the system, this signal has to be pulled high to the auxiliary 3.3 V supply rail.

The PE_RST# signal, which serves as the familiar reset function for the I226, needs to be connected to the host system's corresponding signal.

PCIe Layout Recommendations

For information regarding the PCIe signal routing, refer to the PCIe Design Guidelines.

8.2.2 Connecting 2.5GBASE-T MDI Interfaces

The MDI line interface on the I226 is capable of driving up to +100 meters of CAT-5E or CAT-6 unshielded twisted pair (100 Ω differential impedance) on 10 Mb/s, 100 Mb/s, 1 GbE, or 2.5 GbE. It is designed to drive this via a quad, 50 Ω center tapped 1:1 transformer connected to an RJ-45 PCBmount jack. Solutions that combine the transformer and RJ-45 jack into a single device are also supported.

The MDI line interface on the I226 supports automatic A/B and C/D pair swaps and inversions (MDI-X). It also supports provisioned ABCD to DCBA pair reversal (called the MDI Lane Swap feature) for ease of routing via a hardware strap. Please see the MDI Lane Swap Configuration Feature section for details.

PHY Calibration Resistor

For proper biasing of the MDI analog interface, a 22 K Ω 1% pull-down resistor must be connected between the RCAL pin (pin#42) and ground in order to calibrate the I226 MDI analog front-end. To avoid noise coupled onto this reference signal, place the calibration resistor close to the I226 and keep traces as short as possible. Do not change this resistor value.

MDI Circuit Guidelines

The MDI discrete design and integrated magnetic components were chosen for inclusion in the reference design and Bill of Material (BOM). These components are capable of delivering the performance required for this demanding application.

For systems with enhanced ESD performance requirements, the additional enhanced ESD parts might be needed. Consult with the ESD part vendors for their latest technology and technique. Also consult with Intel representative for the schematic and layout implementation

MDI Layout Guidance

The magnetics can be implemented either as a discrete module or an integrated solution combining the magnetics and the RJ-45 jack.

For the discrete magnetics, the I226 designs consist of an RJ-45 jack, Bob Smith termination and discrete magnetics all separated parts (refer Figure below).

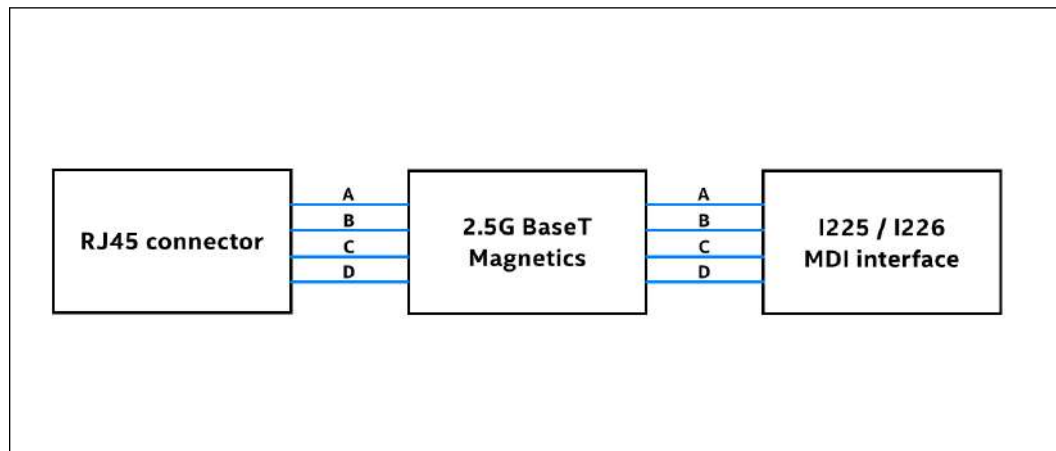
For the integrated magnetics, the I226 designs is simple. The RJ-45 connector has Bob Smith termination and discrete magnetics integrated inside (refer [Figure 322](#) on page 549).

Minimizing the amount of space needed for the PHY is important because other interfaces compete for physical space on the LAN on Motherboard (LOM) near the connector. The PHY circuits need to be as close as possible to the connector.

Figure below illustrates some basic placement distance guidelines. It shows four differential pairs and the layout can be generalized for a 2.5 GbE system with four analog pairs. The ideal placement for the I226 is approximately two inches behind the magnetics module for both the discrete and integrated solutions.

- **Discrete Magnetics**

Figure 318. Discrete Magnetics



The I226 impedance of the Bob Smith termination is $75\ \Omega$ Channel A, B, C and D, whereas the PHY input impedance is $100\ \Omega$.

The I226, referred as a LAN silicon in Figure 1-2 and Figure 1-3, must be at least 1.5 inches from the I/ O back panel. To help reduce EMI, the following recommendations should be followed:

- Minimize the length of the MDI interface
- Place the MDI traces no closer than 0.5 inch (1.3 cm) from the board edge.

Figure 319. Device Placed Less Than One Inch from the RJ-45 Connector

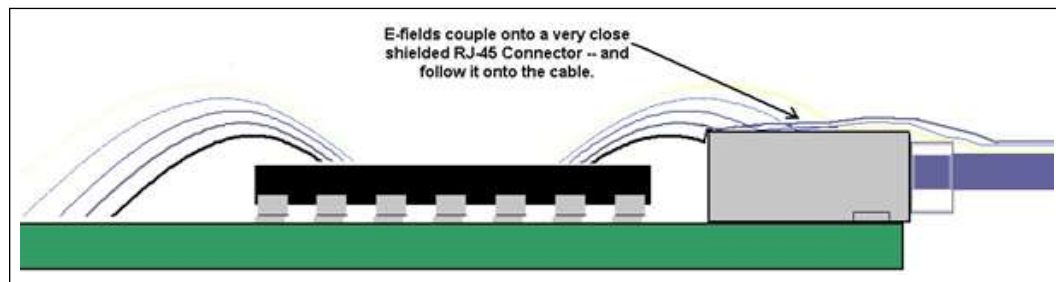


Figure 320. Discrete Magnetic Connector Layout Topology

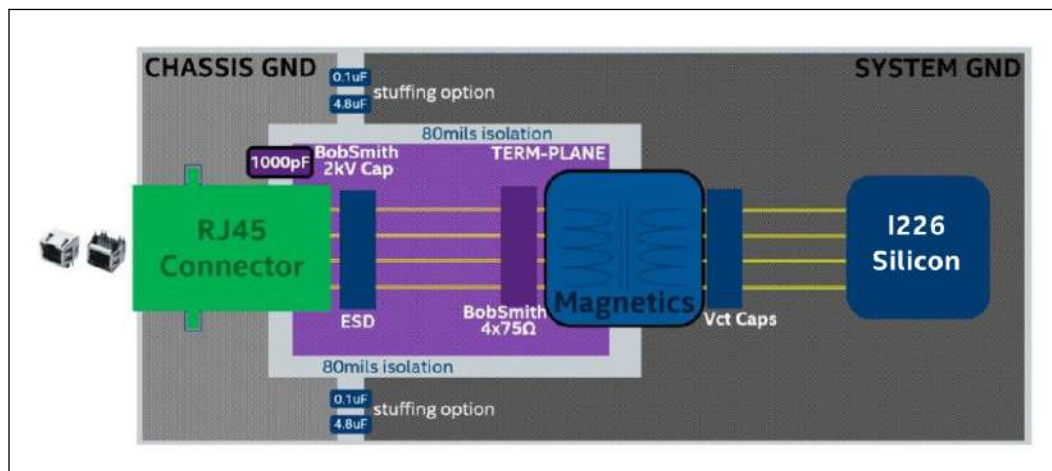
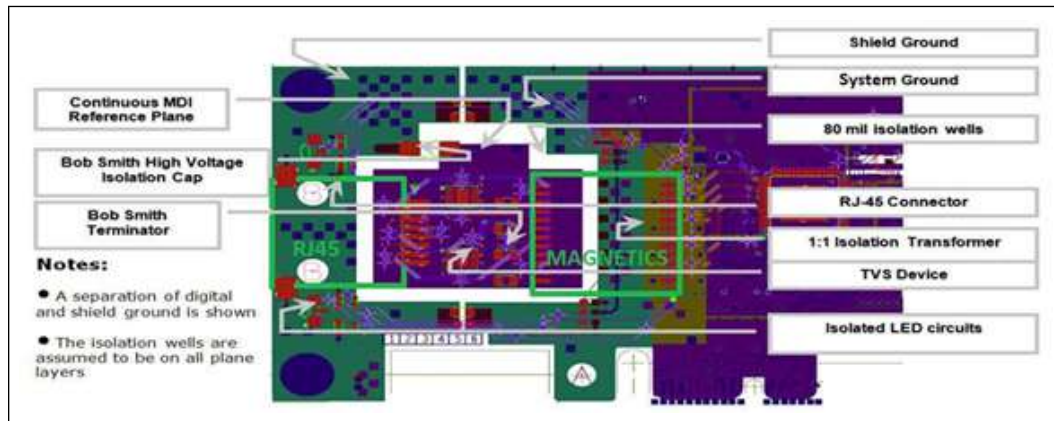


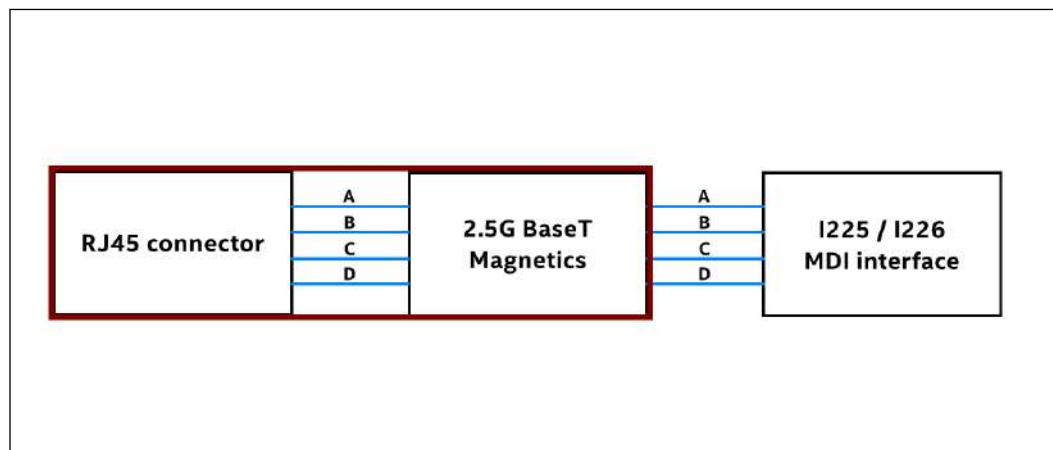
Figure 321. Discrete Magnetic Layout Example



- The ground is split underneath the magnetics, with the MDI Reference Plane GND (the purple island) being present under the front-end (Figure above), and the circuit system ground under the I226 side of the magnetics. The MDI traces on the line side are referenced to the MDI reference plane. Splitting the ground planes beneath the transformer minimizes noise coupling between the primary and secondary sides of the transformer and between the adjacent coils in the magnetics. This arrangement also improves the common mode choke functionality of magnetics module.
- Between the RJ45 connector and the magnetics device, the MDI Reference Plane GND (the purple island) can be voided. When using this voiding technique, this MDI segment (cable side of the magnetics) will have no reference plane. Please make sure using the trace width/separation to target 100- Ohm differential impedance. The MDI segment on the silicon side of the magnetics still references to the system ground.
- The Bob Smith termination ($4 \times 75 \Omega$) to be implemented from the line side transformer center taps to the reference plane.
- Bob Smith termination and the shield of the RJ-45 jack are both connected to chassis ground, which are the two large circular metallization in the MDI (refer [Figure 322](#) on page 549).
- The resistors in the Bob Smith termination are required to be 0805's to handle the cable discharge voltages.
- Similarly, the magnetics should be placed as close as possible to the RJ-45 jack.
- In [Figure 321](#) on page 548, the Hi-POT clearance for cable discharge is shown and designed with ≥ 2.032 mm of clearance from the ground. Therefore, the MDI reference plan has 2.032 mm clearance away from any other signals (including shield ground and system ground). Note that the breakdown voltage in FR4 is lowest in the x-y axes planar to the PCB and highest in the z-axis between layers.
- The MDI reference plane is coupling to the shield ground via the Bob Smith high voltage isolation capacitor. This helps ESD performance.
- The shield ground can be coupled to the system ground via 2-4 small value capacitors. Spread and distribute these capacitors out. The capacitor value can be tuned for improving the EMI performance.

- **Integrated Magnetics**

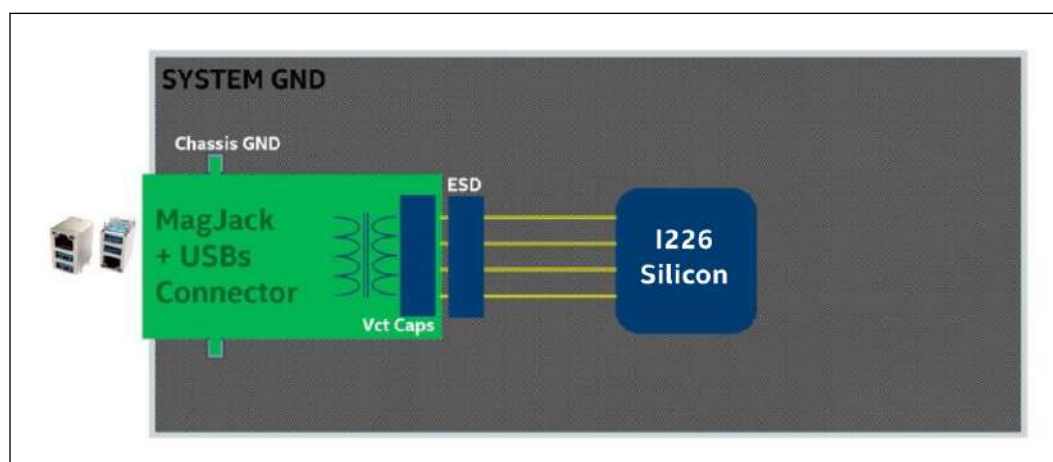
Figure 322. Integrated Magnetic



At high level, the guidance on MDI trace implementation between the I226 and the integrated module remains the same as with discrete magnetics, including length requirements and target impedance.

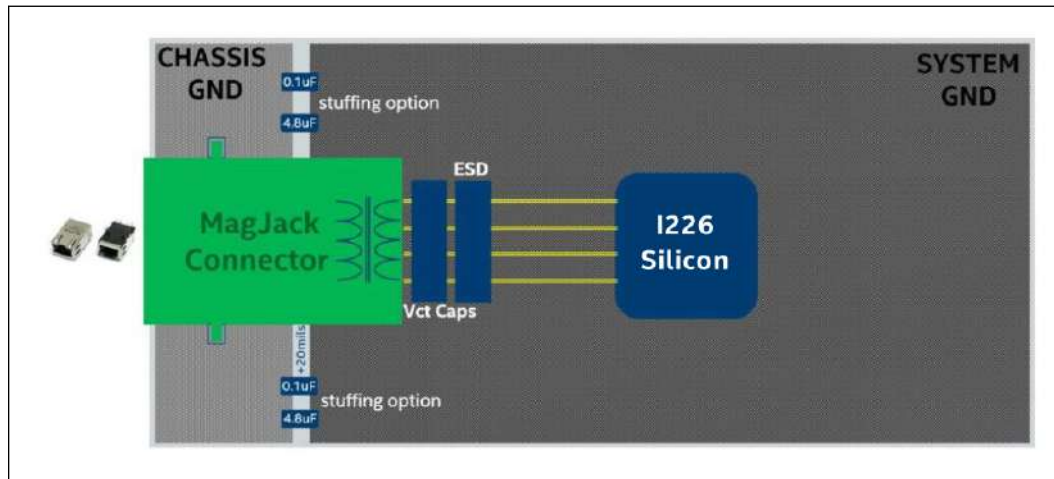
However, the RJ45 connector has the magnetics and the Bob Smith termination network integrated inside. Therefore, no need to take care the high-voltage isolation routing. Do not split the connector shield ground with the system ground. Just connect the RJ45 shield ground directly to the system ground.

Figure 323. Integrated Magnetic/USB Connector Layout Topology



The MagJack/ICM integrated magnetics connector can have the slit Ground implementation for the EMI improvement. The stuffing cap values can be adjusted to gain 1-2dBs

Figure 324. Integrated Magnetic Connector Layout Topology with Ground Slit for EMI Improvement



- **MDI Differential Pair Trace Routing for LAN Design**

Trace routing considerations are important to minimize the effects of crosstalk and propagation delays on sections of the board where high-speed signals exist. Signal traces should be kept as short as possible to decrease interference from other signals, including those propagated through power and ground planes.

- **Signal Trace Geometry**

One of the key factors in controlling trace EMI radiation are the trace length and the ratio of trace-width to trace-height above the reference plane. To minimize trace inductance, high-speed signals and signal layers that are close to a reference or power plane should be as short and wide as practical. Ideally, the trace-width to trace-height above the ground plane ratio is between 1:1 and 3:1. To maintain trace impedance, the width of the trace should be modified when changing from one board layer to another if the two layers are not equidistant from the neighboring planes.

Each pair of signals should target a differential impedance of $100 \Omega \pm 15\%$.

A set of trace length calculation tools are made available from Intel to aid with MDI topology design. Contact your Intel representative for tool availability.

When designing a board layout, the automatic router feature of the CAD tool must not route the differential pairs without intervention. In most cases, the differential pairs require manual routing.

NOTE

Measuring trace impedance for layout designs targeting 100Ω often results in lower actual impedance due to over-etching. Designers should verify actual trace impedance and adjust the layout accordingly. If the actual impedance is consistently low, a target of 105Ω to 110Ω should compensate for over-etching.

- **Matching Traces within a Pair (P and N)**

P and N for each MDI pair should be matched to within 0.127 mm on the PCB to prevent common-to-differential and differential-to-common conversion due to the length mismatch.

If in-pair length matching is not possible using bends or small loops, serpentine routing (zig-zag of a shorter trace) is acceptable if only one to three meanders are routed within 5.08 mm of the source of the skew or the end(s) of any otherwise unmatched lengths of a differential trace segment. For example, near device pins and/or at or near the connector pins and possibly at differential signal vias. Refer to the Intel® Ethernet Controller I226 Checklists for more details.

Table 709. MDI Routing Summary

Parameter	Main Route Guidelines	Breakout Guidelines ¹	Notes
Signal group	MDI_P[3:0] MDI_N[3:0]		
Microstrip*/Stripline* uncoupled single-ended impedance specification	50 ±10%		
Microstrip/Stripline uncoupled differential impedance specification	100 ±15%		2 , 3
Microstrip nominal trace width	Design dependent	Design dependent	
Microstrip nominal trace space	Design dependent	Design dependent	3
Microstrip/Stripline trace length	< 4 inches		
Microstrip/Stripline pair-to-pair space (edge-to-edge)	7 times the dielectric thickness		
Microstrip/Stripline bus-to-bus spacing	7 times the dielectric thickness		
Matching traces within a pair (P and N)	< 0.127 mm		
Keep pair-to-pair length differences	< 2 inches		
Notes: 1. Pair-to-pair spacing 7 times the dielectric thickness for a maximum distance of 12.7 mm from the pin. The phase tolerance between MDI_P and MDI_N is <0.127 mm. 2. Board designers should ideally target 100 Ohm ±10%. If it's not feasible (due to board stackup) it is recommended that board designers use a 95 Ohm ±10% target differential impedance for MDI with the expectation that the center of the impedance is always targeted at 95 Ohm The ±10% tolerance is provided to allow for board manufacturing process variations and not lower target impedances. The minimum value of impedance cannot be lower than 90 Ohm. 3. Simulation shows 80 differential trace impedances degrade MDI return loss measurements by approximately 1 dB from that of 90 Ohm.			

Table 710. Maximum Trace Lengths Based on Trace Geometry and Board Stackup

Dielectric Thickness (um)	Dielectric Constant (DK) at 1 MHz	Width / Space/ Width (um)	Pair-to-Pair Space (um)	Nominal Impedance (Ohm)	Impedance Tolerance (+/-%)	Maximum Trace Length (inches) ¹
68.58	4.05	101.6/254/101.6	482.6	95 ²	17 ²	3.5
68.58	4.05	101.6/254/101.6	482.6	95 ²	15 ²	4
continued...						

Dielectric Thickness (um)	Dielectric Constant (DK) at 1 MHz	Width / Space/ Width (um)	Pair-to-Pair Space (um)	Nominal Impedance (Ohm)	Impedance Tolerance (+ / -%)	Maximum Trace Length (inches) ¹
68.58	4.05	101.6/254/101.6	482.6	95	10	5
83.82	4.1	106.68 / 228.6 / 106.68	584.2	100 ²	17 ²	4
<i>Notes:</i> 1. Longer MDI trace lengths can be achievable, but might make it more difficult to achieve IEEE conformance. Simulations have shown deviations are possible if traces are kept short. Longer traces are possible; use cost considerations and stack-up tolerance for differential pairs to determine length requirements. 2. Deviations from 100 Ohm nominal and/or tolerances greater than 15% decrease the maximum length for IEEE conformance.						

NOTE

Use the MDI differential trace calculator to determine the maximum MDI trace length for your trace geometry and board stackup. Contact your Intel representative for access.

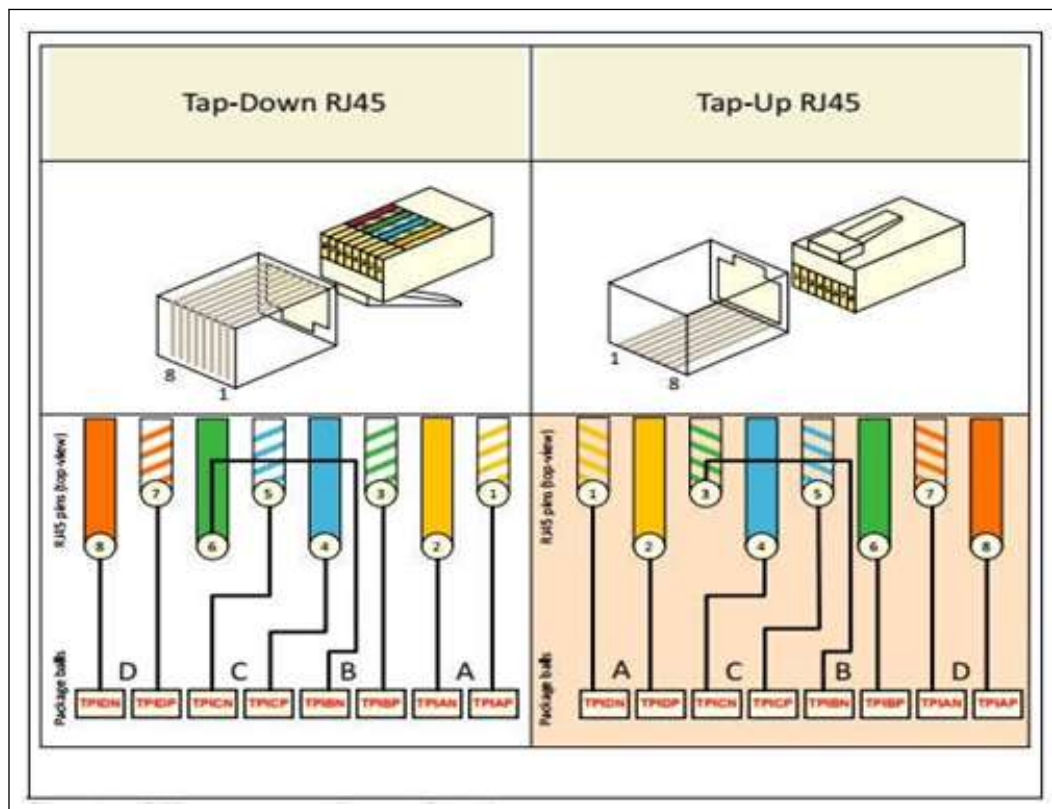
The following factors can limit the maximum MDI differential trace lengths for IEEE conformance:

- Dielectric thickness
- Dielectric constant
- Nominal differential trace impedance
- Trace impedance tolerance
- Copper trace losses
- Additional devices, such as switches, in the MDI path might impact IEEE conformance. Board geometry should also be factored in when setting trace length.

PHY MDI Lane Swap Configuration

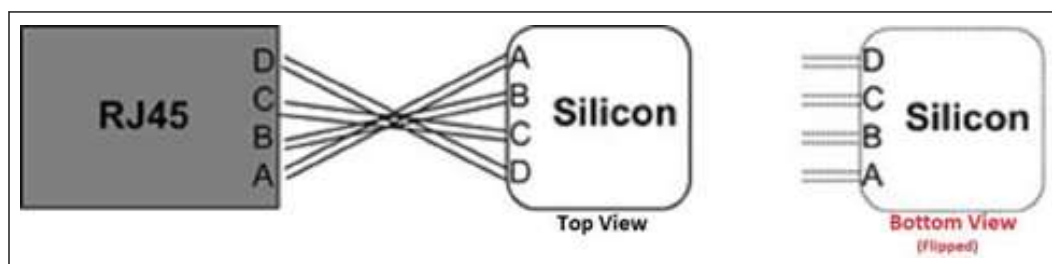
In general, the RJ45 connector on the system PCB can have two different structures with the tap up or the tap down as shown in Figure below. The difference between tap-up and tap-down is a swap of positioning ABCD and DCBA orders. The I226 pin#18 strapping allows system designers to perform the MDI lane swap configuration. As a result, a PCB layout can avoid the crossing routing problem.

Figure 325. MDI Lane Swap Configuration



MDI Crossing Issue: When selecting RJ45 connectors, the RJ45 MDI lanes order might not line up with the Silicon MDI lanes order. This causes the MDI crossing routing problem in layout.

Figure 326. Crossing Routing Issue Between RJ45 and Silicon

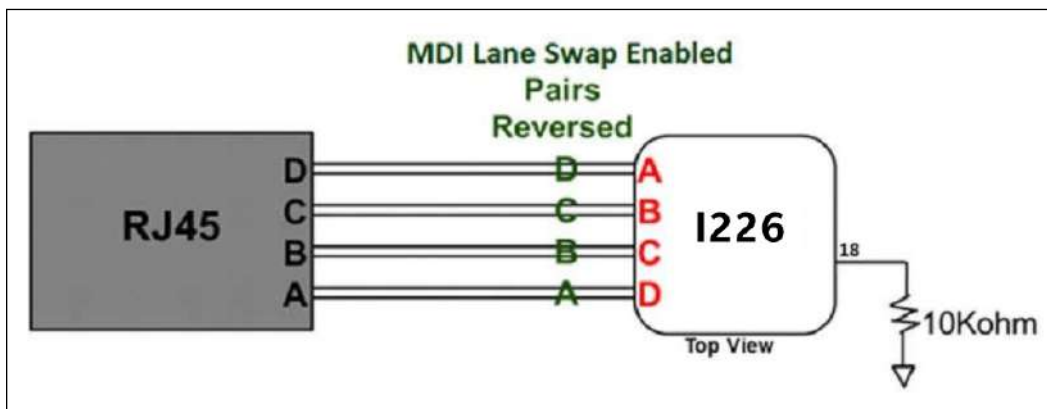


Silicon: Pin#18 allows the MDI lane swap configuration. This is a hardware strap.

Default Configuration (Unstrap): Pin#18 has its internal pull-up resistor. The pin can leave unconnected.

MDI Swap Configuration (Strapped): Pin#18 needs a 10 K ohms pull-down. This hardware strap activates the MDI lane swap feature. ABCD order are reversed.

Figure 327. MDI Lane Swap Enabled by Strapping



Center Tap Connection Via Capacitors to Ground

The I226 has a voltage-mode driver. Therefore, it is required that a center tap be decoupled to ground via capacitors.

When using an integrated magnetic, a 0.1 μF capacitor should be connected from each center tap pin to ground. Most of integrated magnetic connector will have 4x 0.1 μF capacitor inside. Therefore, the I226 does not need any external capacitor. Please add 1x 0.1 μF capacitor (empty) as a place holder.

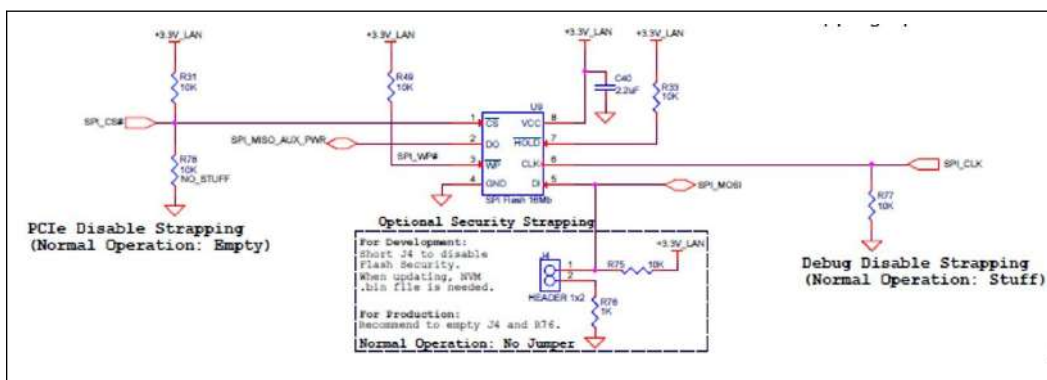
When using a discrete magnetic, add a 0.1 μF capacitor to each center tap pin of the magnetic. The I226 should have a total of 4x 0.1 μF capacitors at the magnetic center tap. Please see I226 Reference Schematic for details.

8.2.3 Connecting Flash Interface

Connecting Flash

Intel recommends having a Flash socket for validation, but it is not required for production releases. Following is a typical SPI connection for the I226. Please refer Datasheet and I226 Reference Schematic for details of the strapping options.

Figure 328. Typical SPI Flash Connection



The I226 provides support for a SPI Flash device that is made accessible to the system through the following:

- Flash Base Address register. The PCIe Control register at an offset that will be identified in a later release of this document
- An address range of the IOADDR register, defined by the I/O Base Address register (PCIe Control register at an offset that will be identified in a later release of this document).
- Expansion ROM Base Address register. The PCIe Control register is at an offset that will be identified in a later release of this document.

Supported Flash Devices

The I226 uses a SPI Flash. Several words of the Flash are accessed automatically by the I226 after reset to provide pre-boot configuration data before it is accessed by host software. The remainder of the Flash space is available to software for storing the MAC address, serial numbers, and additional information.

8.2.3.1 Connecting Miscellaneous Signals

LAN Disable

The I226 has the LAN_DISABLE_N pin (pin #2) that can be used for disabling Ethernet functions (disable the silicon). For normal operation, pull this pin up with a 10 K Ω resistor to 3.3 Vdc.

8.2.4 Connecting Light Emitting Diodes (LEDs)

The I226 provides three programmable outputs to directly drive the LEDs for link activity and speed indications. Each of the three LED outputs can be individually configured to select the particular event, state, or activity, which is indicated on that output. In addition, each LED can be individually configured for output polarity, as well as for blinking versus non-blinking (steady-state) indication.

The LED ports are fully programmable through the Flash interface (LEDCTL register). In addition, the hardware-default configuration for all LED outputs can be specified via a Flash field, thus supporting LED displays configurable to a particular OEM preference.

System designers need to provide separate current limiting resistors for each LED connected.

Since the LEDs are likely to be placed close to the board edge and to external interconnect, take care to route the LED traces away from potential sources of EMI noise. In some cases, it might be desirable to attach filter capacitors.

LED Configuration #1

This configuration has been used widely in Sever/Workstation/Desktop/AIC/etc.

Expectation:

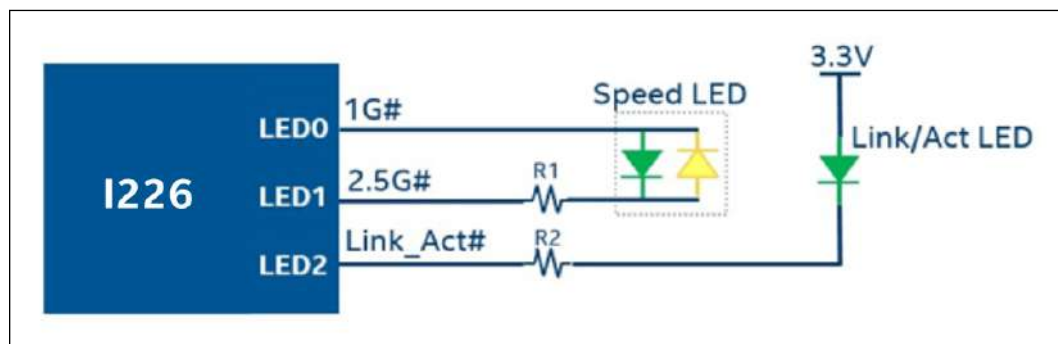
Speed LED is a dual-color. Link-Activity LED is a single-color.

Highest speed LED is Green. Second highest speed LED is Yellow. All other-speed LEDs are OFF.

Figure 329. LED Configuration #1 Expected Behavior

Speed LED		
10 Mbps	OFF	
100 Mbps	OFF	
1000 Mbps	Yellow	←2 nd Highest Speed
2500 Mbps	Green	←Highest Speed
Link-Activity LED		
Link-up	Green	
Tx/Rx Activity	Blinking Green	

Figure 330. Typical Schematic Setup for LED Configuration #1Design



NOTES

- All LED pins are active-low signals.
- R1/R2 are current-limit resistors.

Figure 331. I226 NVM Setting for Configuration #1: Offset 0x1C = 0x0508 and Offset 0x1F = 0x8427

NVM word offset 0x1C = 0x0508 (for LED1 Setting)															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	0
Reserved				TXAMP				Blink1	Invert1	Reserved			LED1=0x8 (for 2.5G#)		

NVM word offset 0x1F = 0x8427 (for LED0 and LED2 Setting)															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	0	1	0	0	0	0	1	0	0	1	1	1
Blink2	Invert2	Reserved			LED2=0x4 (for Link_Act#)				Blink0	Invert0	Global Blink	Reserved		LED0=0x7 (for 1G#)	

Design Notes:

Invert = 0 (LED is an active low signal)

Blink2 = 1 (LED2 is solid = link-up and blinking = Tx/Rx)

Global Blink = 1 (fast blinking mode for all LEDs)

LED Configuration #2

This configuration is for the I226 design, which needs individual color for each Ethernet speed.

Expectation:

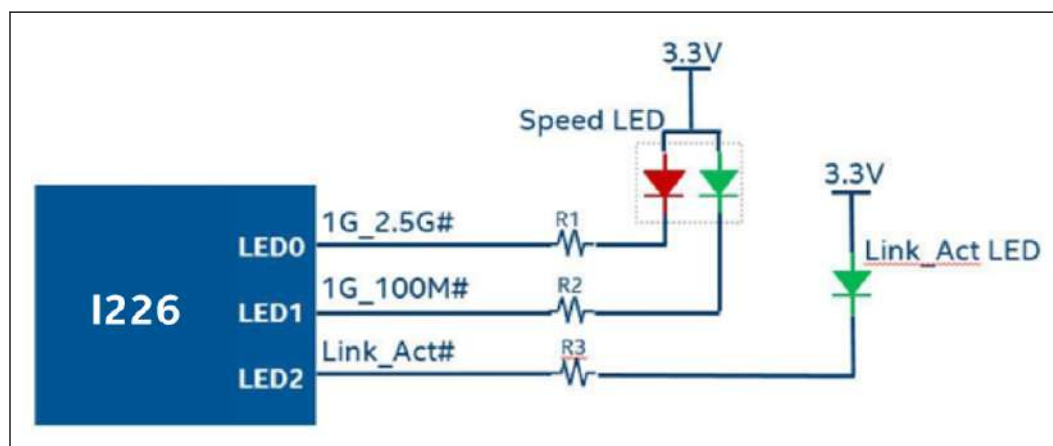
The speed LED can display three different colors. The Link-Activity LED is a single-color.

Each speed LED has a different color. This has an advantage for distinct displays of all speeds at assigned colors. The RJ45 cost expected to be higher.

Figure 332. LED Configuration #2 Expected Behavior

Speed LED		
10 Mbps	OFF	
100 Mbps	Green	← A
1000 Mbps	Yellow	← Mixed A+B color, both LEDs turned on at a same time.
2500 Mbps	Red	← B
Link-Activity LED		
Link-up	Green	
Tx/Rx Activity	Blinking Green	

Figure 333. Typical Schematic Setup for LED Configuration #2



Design Notes:

- All LED pins are active-low signals.
- R1/R2/R3 are current-limit resistors.

- Each speed can have its own color by this 3-pin LED. Traditionally, 1 GbE is yellow and 100 Mb/s is green. This setup creates a new red indication for 2.5 GbE.

Figure 334. Mixed-color Examples for the LED Configuration #2

Example#	Color A	Color B	Mixed A+B
1	Red	Green	= Yellow
2	Red	Yellow	= Orange
3	Orange	Green	= Yellow

Figure 335. I225 NVM Setting for Configuration #2: Offset 0x1C = 0x050D and Offset 0x1F = 0x842E

NVM word offset 0x1C = 0x050D (for LED1 Setting)															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	1	0	1	0	0	0	0	1	1	0	1
Reserved				TXAMP				Blink1	Invert1	Reserved		LED1=0xD (for 1G_100M#)			
NVM word offset 0x1F = 0x842E (for LED0 and LED2 Setting)															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	0	1	0	0	0	0	1	0	1	1	1	0
Blink2	Invert2	Reserved			LED2=0x4 (for Link_Act#)				Blink0	Invert0	Global Blink	Reserved		LED0=0xE (for 1G_2.5G#)	

Design Notes:

- Invert = 0 (LED is an active low signal)
- Blink2 = 1 (LED2 is solid = Link-up and blinking = Tx/Rx)
- Global Blink = 1 (fast blinking mode for all LEDs)

8.2.5 Connecting JTAG Port

The I226 contains a test access port (3.3 V only) conforming to the IEEE 1149.1-2001 Edition (JTAG) specification. To use the test access port, connect these balls to pads accessible by specific test equipment.

For normal operation, pull-up JTCK, JTDO, JTMS and JTDI signals with 10 K Ω resistors to 3.3 Vdc. J_RST_N signals should also be pulled up as well.

8.2.6 Crystal Design Considerations

All designs require an external clock. The only option for this clock source is a 25 MHz external crystal. The I226 uses the crystal to generate clocks for the high speed interfaces.

The chosen crystal vendor should be consulted early in the design cycle. Crystal manufacturers familiar with networking equipment clock requirements might provide assistance in selecting an optimum, low- cost solution.

The following parameters are required when selecting the 25 MHz external crystal component for I226.

Table 711. External Crystal Specifications

Parameter Name	Symbol	Recommended Value	Max/Min Range	Conditions
Frequency	f_o	25 [MHz]		@25 [°C]
Vibration Mode		Fundamental		
Frequency Tolerance @25 °C	Df/f_o @25°C	±30 [ppm]		@25 [°C]
Temperature Tolerance	Df/f_o	±30 [ppm]		0 to +70 [°C]
Series Resistance (ESR)	R_s		60 [Ohm] max	@25 [MHz]
Crystal Load Capacitance	C load	18 [pF]		
Shunt Capacitance	C_o		5 [pF] max	
Drive Level	D_L		100 [milliW] max	
Aging	Df/f_o	±5 ppm per year	±5 ppm per year max	
Calibration Mode		Parallel		
Insulation Resistance			500 [Mohms] min	@ 100 Vdc
<i>Note:</i> The selected Crystal Component must meet or exceed the specified drive Level (D_L) The crystal Component datasheet should show 100uW or more.				

Quartz Crystal

Quartz crystals are the mainstay of frequency components due to low cost and ease of implementation. They are available from numerous vendors in many package types with various specification options.

Vibrational Mode

Crystals are available in both fundamental and third overtone. Unless there is a special need for third overtone, use fundamental mode crystals. At any operating frequency, third overtone crystals are thicker and more rugged than fundamental mode crystals. Third overtone crystals are more suitable for use in military or harsh industrial environments. Third overtone crystals require a trap circuit (extra capacitor and inductor) in the load circuitry to suppress fundamental mode oscillation as the circuit powers up. Selecting values for these components is beyond the scope of this document.

Frequency Tolerance

The frequency tolerance for an Ethernet Platform LAN Connect is dictated by the IEEE 802.3 specification as ±50 parts per million (ppm). This measurement is referenced to a standard temperature of 25 °C. It is recommended that a frequency tolerance of ±10 ppm be used.

Temperature Stability and Environmental Requirements

Temperature stability is a standard measure of how the oscillation frequency varies over the full operational temperature range (and beyond). Several optional temperature ranges are currently available, including -10 °C to +70 °C for industrial environments. Some vendors separate operating temperatures from temperature stability. Manufacturers can also list temperature stability as 50 ppm in their datasheets.

NOTE

Crystals also carry other specifications for storage temperature, shock resistance, and reflow solder conditions. Crystal vendors should be consulted early in the design cycle to discuss the application and its environmental requirements.

Calibration Mode

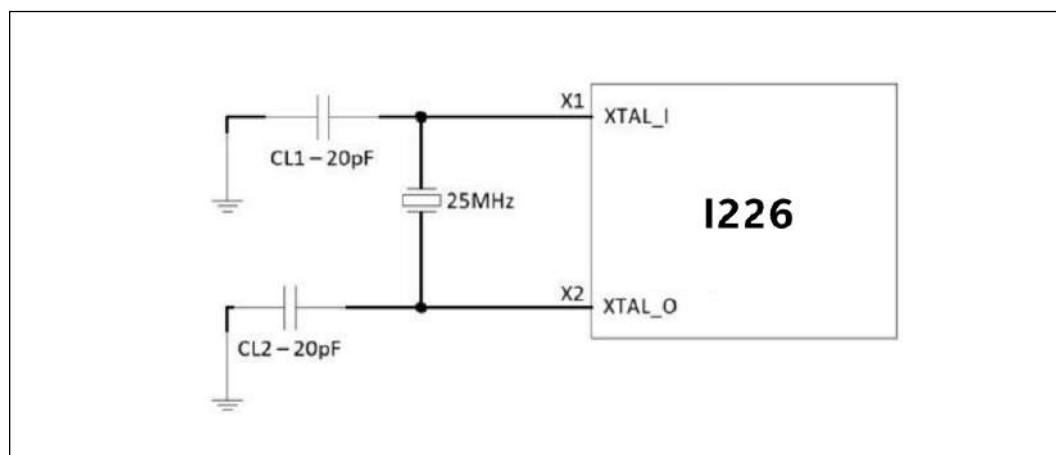
The terms series-resonant and parallel-resonant are used to describe crystal oscillator circuits.

Specifying parallel mode is critical to determining how the crystal frequency is calibrated at the factory. A crystal specified and tested as series resonant oscillates without problem in a parallel-resonant circuit, but the frequency is higher than nominal by several hundred parts per million. The purpose of adding load capacitors to a crystal oscillator circuit is to establish resonance at a frequency higher than the crystal's inherent series resonant frequency.

Reference Crystal Circuit

Figure below shows a typical schematic of the I226 oscillating circuit. Pin X1 and X2 refers to XTAL1 and XTAL2 in the I226, respectively. The crystal and the capacitors form a feedback element for the internal inverting amplifier. This combination is called parallel-resonant, because it has positive reactance at the selected frequency. In other words, the crystal behaves like an inductor in a parallel LC circuit.

Figure 336. Crystal Circuit



Crystal Load Capacitance

The formula for crystal load capacitance is as follows:

$$C_L = \frac{(C1 \cdot C2)}{(C1 + C2)} + C_{stray}$$

where $C1 = C2 = 20$ pF and C_{stray} = allowance for additional capacitance in pads, traces and the chip carrier within the Ethernet device package. An allowance of 3 pF to 10 pF accounts for lumped stray capacitance.

The calculated crystal load capacitance is 18 pF with $C1 = C2 = 20$ pF and an estimated stray capacitance of about 8 pF. Individual stray capacitance components can be estimated and added. For example, surface mount pads for the load capacitors add approximately 2.5 pF in parallel to each capacitor. This technique is especially useful if Y1, C1 and C2 must be placed farther than approximately one-half (0.5) inch from the device. Thin circuit boards generally have higher stray capacitance than thick circuit boards.

The oscillating frequency should be measured with a precision frequency counter where possible. The load specification or values of C1 and C2 should be fine-tuned for the design. As the actual capacitance load increases, the oscillating frequency decreases.

NOTE

C1 and C2 can vary by as much as 5% (approximately 1 pF) from their nominal values.

Shunt Capacitance

The shunt capacitance parameter is relatively unimportant compared to load capacitance. Shunt capacitance represents the effect of the crystal's mechanical holder and contacts. The shunt capacitance should equal a maximum of 5 pF.

Equivalent Series Resistance (ESR)

ESR is the actual component of the crystal's impedance at the calibration frequency, which the inverting amplifier's loop gain must overcome. ESR varies inversely with frequency for a given crystal family. The lower the ESR, the faster the crystal starts up. Use crystals with an ESR value of 60 Ω or better.

Driver Level

Drive level refers to power dissipation in use. The allowable drive level for a Surface Mounted Technology (SMT) crystal is less than its through-hole counterpart, because surface mount crystals are typically made from narrow, rectangular AT strips, rather than circular AT quartz blanks.

I226 Ethernet controllers can drive crystals up to 100 mW value. Therefore, the crystal datasheets should support at least of 100 mW. Otherwise, the silicon will over drive the crystal.

Aging

Aging is a permanent change in frequency (and resistance) occurring over time. This parameter is most important in its first year because new crystals age faster than old crystals. Use crystals with a maximum of ± 5 ppm per year aging.

Reference Crystal

The normal tolerances of the discrete crystal components can contribute to small frequency offsets with respect to the target center frequency. To minimize the risk of tolerance-caused frequency offsets causing a small percentage of production line units to be outside of the acceptable frequency range, it is important to account for those shifts while empirically determining the proper values for the discrete loading capacitors, C1 and C2. Even with a perfect support circuit, most crystals oscillate slightly higher or slightly lower than the exact center of the target frequency. Therefore, frequency measurements (which determine the correct value for C1 and C2) should be performed with an ideal reference crystal. When the capacitive load is exactly equal to the crystal's load rating, an ideal reference crystal is perfectly centered at the desired target frequency.

Reference Crystal Selection

There are several methods available for choosing the appropriate reference crystal. These are listed below.

- If a Saunders and Associates (S&A) crystal network analyzer is available, discrete crystal components can be tested until one is found with zero or nearly zero ppm deviation (with the appropriate capacitive load). A crystal with zero or near zero ppm deviation is a good reference crystal to use in subsequent frequency tests to determine the best values for C1 and C2.
- If a crystal analyzer is not available, the selection of a reference crystal can be done by measuring a statistically valid sample population of crystals, which has units from multiple lots and approved vendors. The crystal, which has an oscillation frequency closest to the center of the distribution, should be the reference crystal used during testing to determine the best values for C1 and C2.
- It might also be possible to ask the approved crystal vendors or manufacturers to provide a reference crystal with zero or nearly zero deviation from the specified frequency when it has the specified CLoad capacitance.

When choosing a crystal, keep in mind that to comply with IEEE specifications for 100/1000/2.5GBASE-T Ethernet LAN, the transmitter reference frequency must be precise within ± 50 ppm. Intel recommends using a transmitter reference frequency that is accurate to within ± 10 ppm to account for variations in crystal accuracy due to crystal manufacturing tolerance

Circuit Board

Since dielectric layers of the circuit board are allowed some reasonable variation in thickness, stray capacitance from the printed board (to the crystal circuit) also vary. If the thickness tolerance for the outer layers of dielectric is controlled within ± 17 percent of nominal, the circuit board should not cause more than ± 2 pF variation to the stray capacitance at the crystal. When tuning crystal frequency, it is recommended that at least three circuit boards are tested for frequency. These boards should be from different production lots of bare circuit boards. Alternatively, a larger sample population of circuit boards can be used. A larger population increases the probability of obtaining the full range of possible variations in dielectric thickness and the full range of variation in stray

capacitance. Next, the exact same crystal and discrete load capacitors (C1 and C2) must be soldered onto each board and the LAN reference frequency should be measured on each circuit board. The circuit board, which has a LAN reference frequency closest to the center of the frequency distribution, should be used while performing the frequency measurements to select the appropriate value for C1 and C2.

- **Temperature Changes**

Temperature changes can cause crystal frequency to shift. Frequency measurements should be done in the final system chassis across the system's rated operating temperature range.

8.2.7 PCB Guidelines

This section describes the general PCB design guidance targeted as supplementary information in addition to the specific requirements and recommendations for individual interfaces. For items not directly covered in the specific interface sections, these guidance recommendations apply to the design.

Via Usage

Use vias to optimize signal integrity. Figure below shows correct via usage. Figure below shows the type of topology that should be avoided.

Figure 337. Correct Via Usage

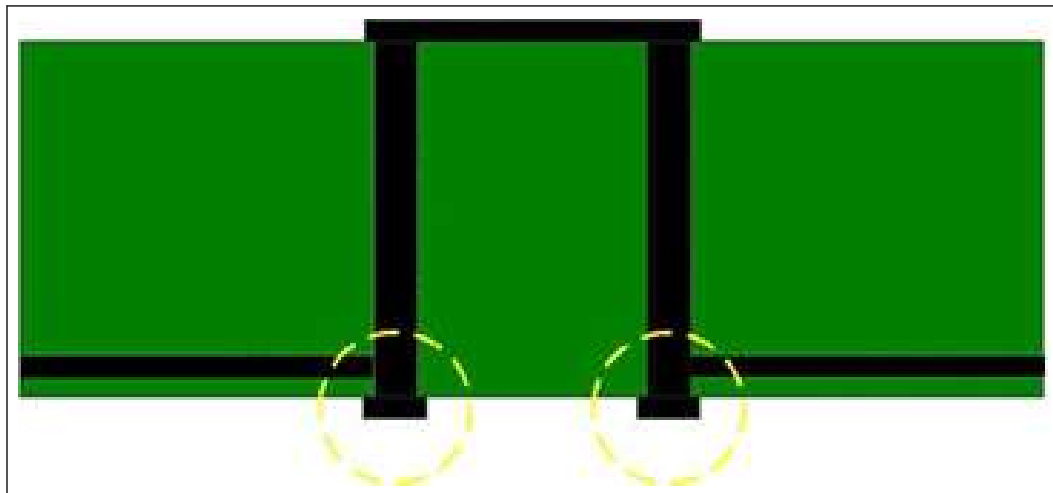
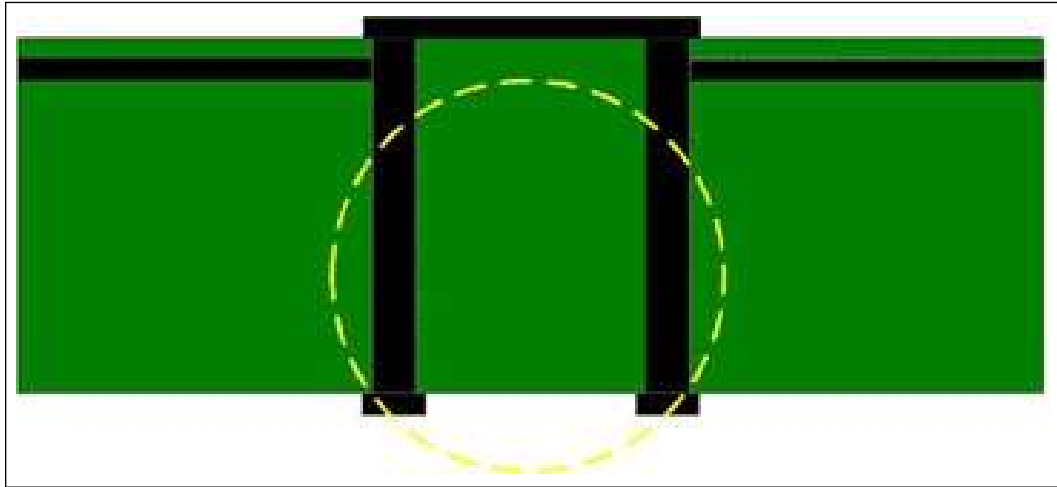
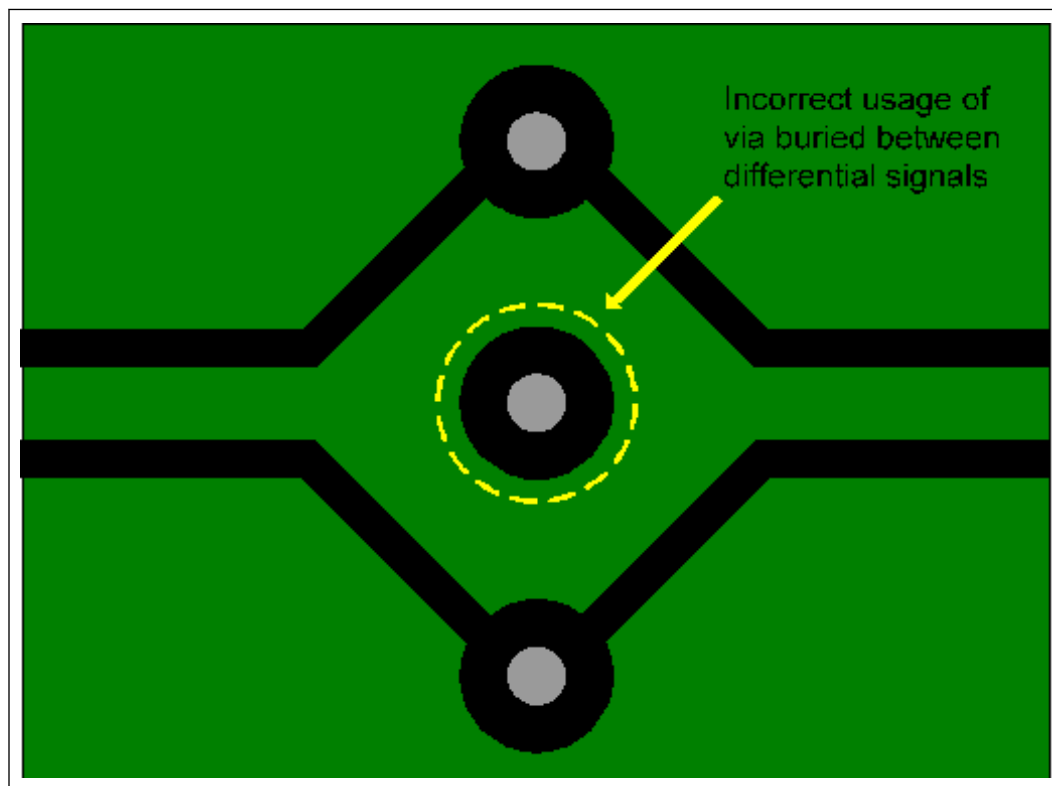


Figure 338. Incorrect Via Usage

Any via stubs on the MDI differential signal traces must be less than 0.889 mm in length. Keeping MDI signal via stubs less than or equal to 0.508 mm is preferable.

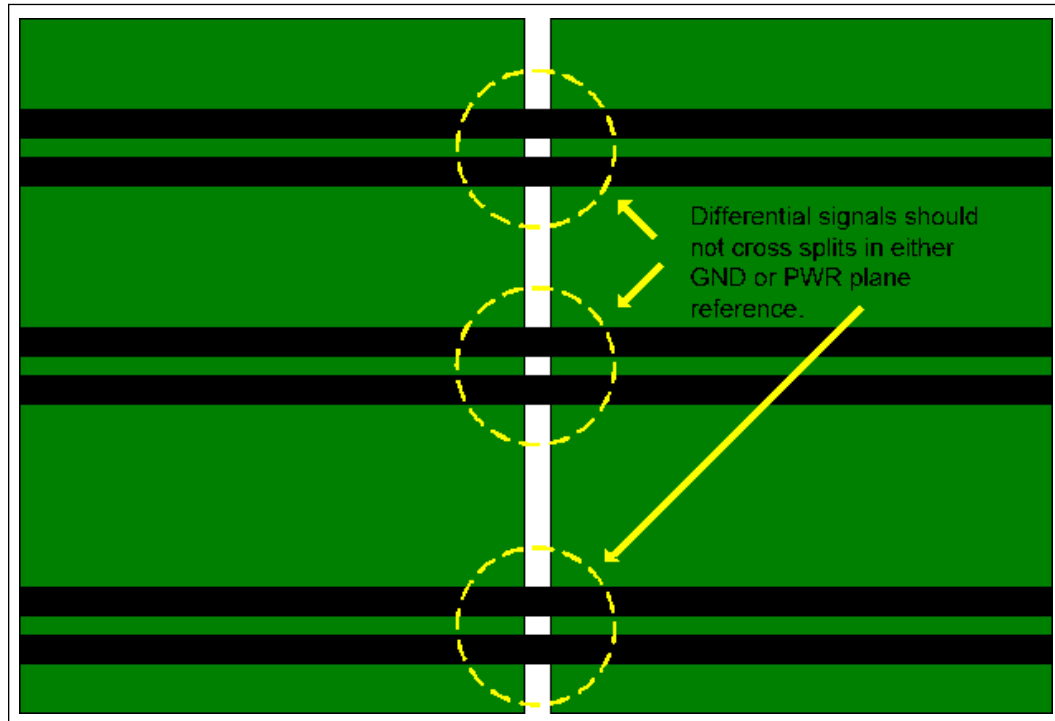
Place ground vias adjacent to signal vias used for the MDI interface. DO NOT embed vias between the high-speed signals, but place them adjacent to the signal vias. This helps to create a better ground path for the return current of the AC signals, which also helps address impedance mismatches and EMC performance.

It is recommended that, in the breakout region between the via and the capacitor pad, target a Z0 for the via to capacitor trace equal to 50 Ohm. This minimizes impedance imbalance.

Figure 339. Incorrect Via Usage for Differential Pair**Reference Planes**

Do not cross plane splits with the MDI high-speed differential signals. This causes impedance mismatches and negatively affects the return current paths for the board design and layout. Refer Figure below.

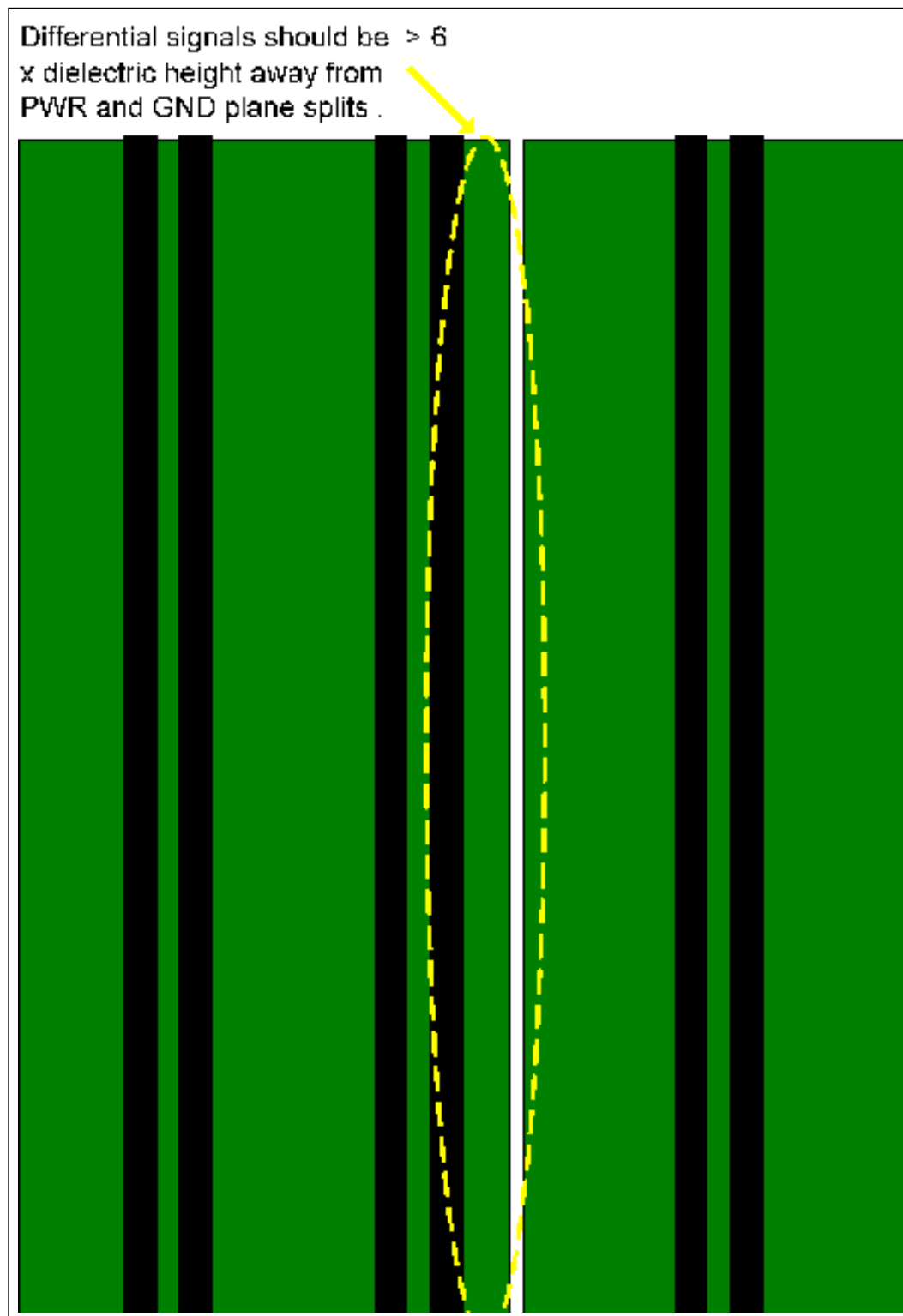
Figure 340. Improper Differential Signal Routing - Plane Split



Traces should not cross power or ground plane splits if at all possible. Traces should stay seven times the dielectric height away from plane splits or voids. If traces must cross splits, capacitive coupling should be added to stitch the two planes together to provide a better AC return path for the high-speed signals. To be effective, the capacitors should have low ESR and low equivalent series inductance

NOTE

Even with plane split stitching capacitors, crossing plane splits is extremely high risk for 5GBASE-T designs

Figure 341. Differential Signal Routing - Plane Split and Void Proximity

It is recommended that the MDI signals stay at least seven times the dielectric height away from any power or ground plane split. This improves impedance balance and return current paths.

If a high-speed signal needs to reference a power plane, ensure that the height of the secondary (power) reference plane is at least 3 x the height of the primary (ground) reference plane.

Reducing Circuit Inductance

Traces should be routed over a continuous reference plane with no interruptions. If there are vacant areas on a reference or power plane, the signal conductors should not cross the vacant area. Routing over a void in the reference plane causes impedance mismatches and usually increases radiated noise levels. Noisy logic grounds should NOT be located near or under high-speed signals or near sensitive analog pin regions of the LAN silicon. If a noisy ground area must be near these sensitive signals or IC pins, ensure sufficient decoupling and bulk capacitance in these areas. Noisy logic and switching power supply grounds can sometimes affect sensitive DC subsystems such as analog to digital conversion, operational amplifiers, etc.

All ground vias should be connected to every ground plane; and similarly, every power via should be equally potential power planes. This helps reduce circuit inductance. Another recommendation is to physically locate grounds to minimize the loop area between a signal path and its return path. Rise and fall times should be as slow as possible while still meeting the relevant electrical requirements because signals with fast rise and fall times contain many high frequency harmonics, which can radiate significantly. The most sensitive signal returns closest to the chassis ground should be connected together. This results in a smaller loop area and reduces the likelihood of crosstalk. The effect of different configurations on the amount of crosstalk can be studied using electronics modeling and simulation software.

Signal Isolation

To maintain the best signal integrity, keep digital signals far away from the analog traces. A good rule to follow is no digital signal should be within 7x to 10x dielectric height of the differential pairs. If digital signals on other board layers cannot be separated by a ground plane, they should be routed at a right angle (90 degrees) to the differential signal traces. If there is another Ethernet controller on the board, take care to keep the differential pairs away from that circuit. The same thing applies to switching regulator traces.

Rules to follow for signal isolation:

- Separate and group signals by function on separate board layers if possible. Maintain a separation that is at least seven times the thinnest adjacent dielectric height between all differential pairs (Ethernet) and other nets, but group associated differential pairs together.
- Over the length of the trace run, each differential pair should be at least seven times the thinnest adjacent dielectric height away from any parallel signal traces.
- Physically group together all components associated with one clock trace to reduce trace length and radiation.
- Isolate other I/O signals from high-speed signals to minimize crosstalk. Crosstalk can increase radiated EMI and can also increase susceptibility to EMI from other signals.

- Avoid routing high-speed LAN traces near other high-frequency signals associated with a video controller, cache controller, processor, or other similar devices.

Traces for Decoupling Capacitors

Traces between decoupling and I/O filter capacitors should be as short and wide as practical. Long and thin traces are more inductive and reduce the intended effect of decoupling capacitors. Also, for similar reasons, traces to I/O signals and signal terminations should be as short as possible. Vias to the decoupling capacitors should be sufficiently large in diameter to decrease series inductance.

Power and Ground Planes

Good grounding requires minimizing inductance levels in the interconnections and keeping ground returns short, signal loop areas small, and locating decoupling capacitors at or near power inputs to bypass to the signal return. This significantly reduces EMI radiation.

These general following guidelines help reducing circuit inductance:

- Route traces over a continuous plane with no interruptions. Do not route over a split power or ground plane. If there are vacant areas on a ground or power plane, avoid routing signals over the vacant area. Routing signals over power or ground voids increases inductance and increases radiated EMI levels.
- Use distance and/or extra decoupling capacitors to separate noisy digital grounds from analog grounds to reduce coupling. Noisy digital grounds can affect sensitive DC subsystems.
- All ground vias should be connected to every ground plane, and every power via should be connected to all power planes at equal potential. This helps reduce circuit inductance.
- Physically locate grounds between a signal path and its return. This minimizes the loop area.
- Avoid fast rise/fall times as much as possible. Signals with fast rise and fall times contain many high frequency harmonics, which can radiate EMI.
- Do not route high-speed signals near switching regulator circuits.
- It's acceptable to put ground fill or thieving on the trace layers, but preferably not closer than 1.27 mm to the differential traces and the connector pins.
- If differential traces must be routed on another layer, the signal vias should carry the signal to the opposite side of the PCB (to be near the top of the PCB), AND if the high-speed signals are being routed between two connectors on the same board, before the signal traces reach the second connector, they must return to the original signal layer (before reaching the connector pin). This strategy keeps via stubs short without requiring back drilling.
- Each time differential traces make a layer transition (pass through a pair of signal vias), there must be at least one ground via located near each signal via. Two ground vias near each signal via is better. Refer Figures below.

Figure 342. Return Path Vias for Differential Signals - Acceptable Example

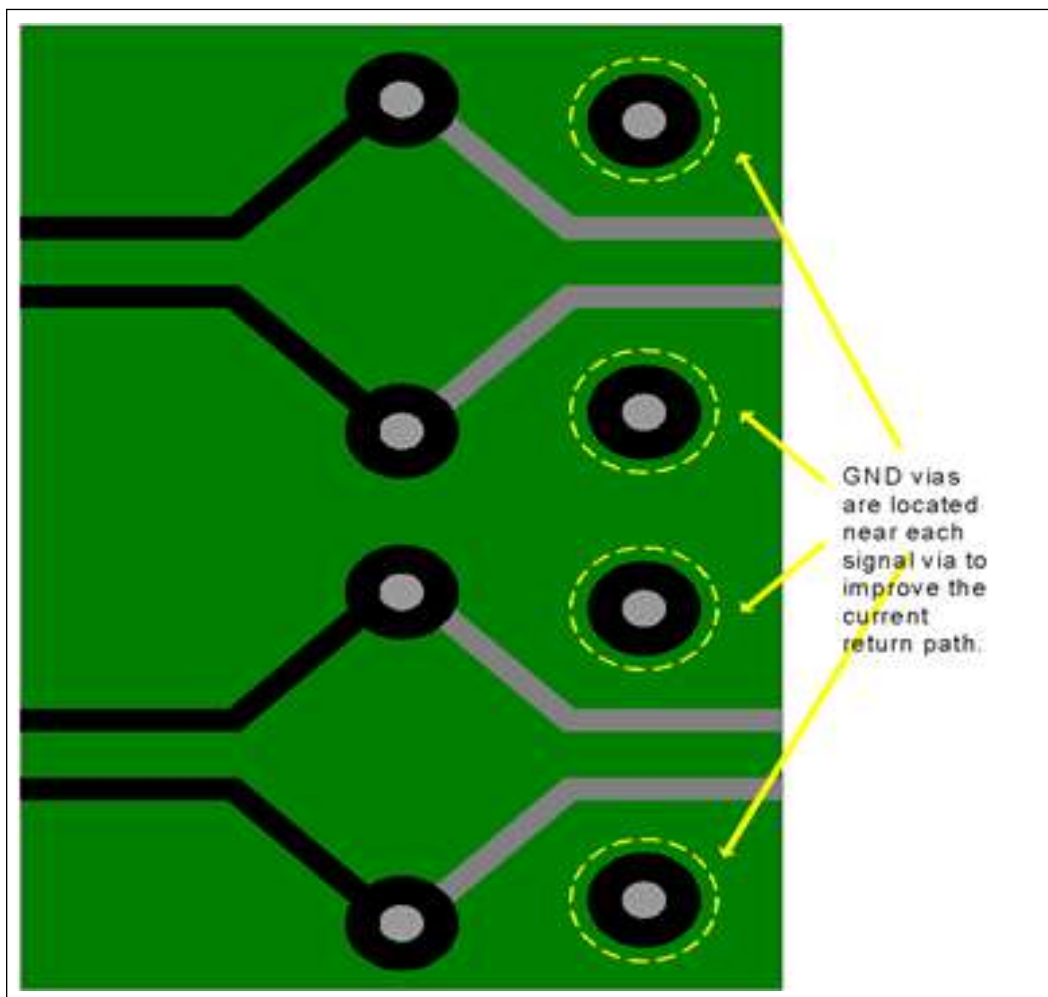
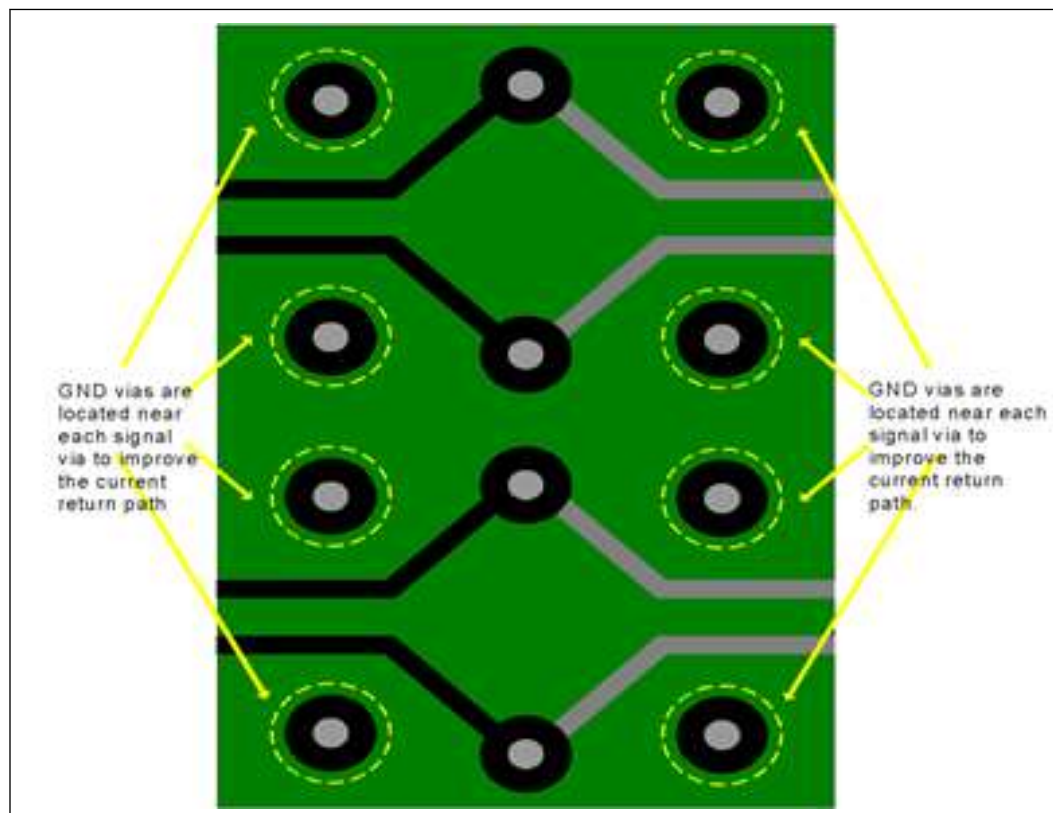
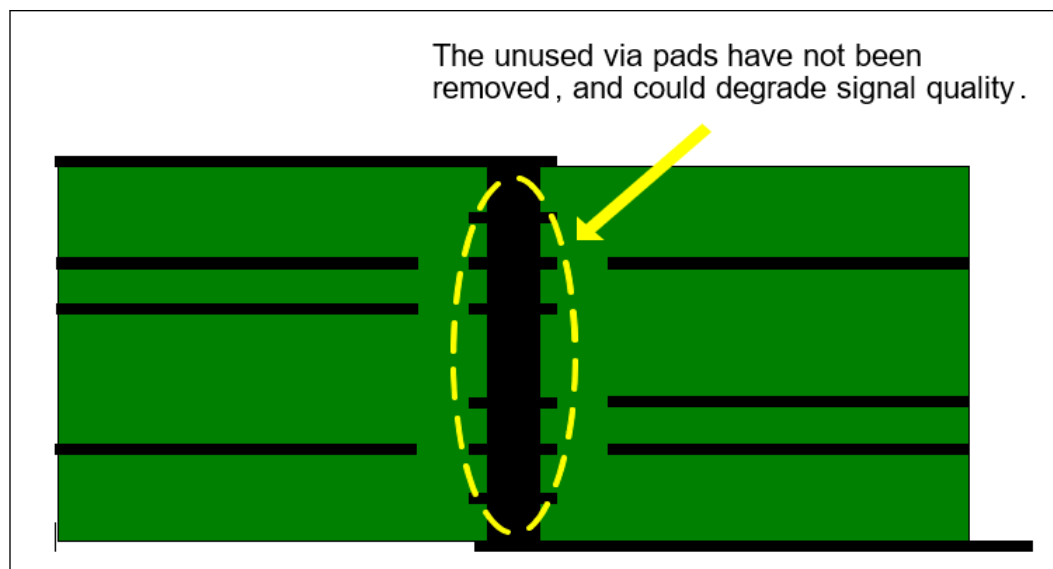


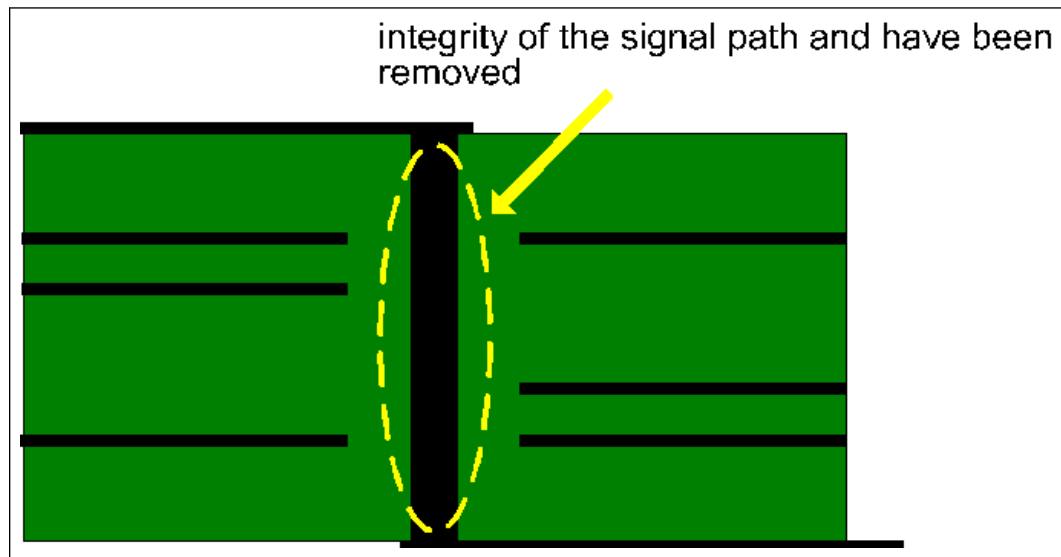
Figure 343. Return Path Vias for Differential Signals - Optimal Example

If the PCB fabrication process permits it, it's best to remove signal via pads on unconnected metal layers. Refer Figure below.

Figure 344. Signal Vias: Improper Padstack Example

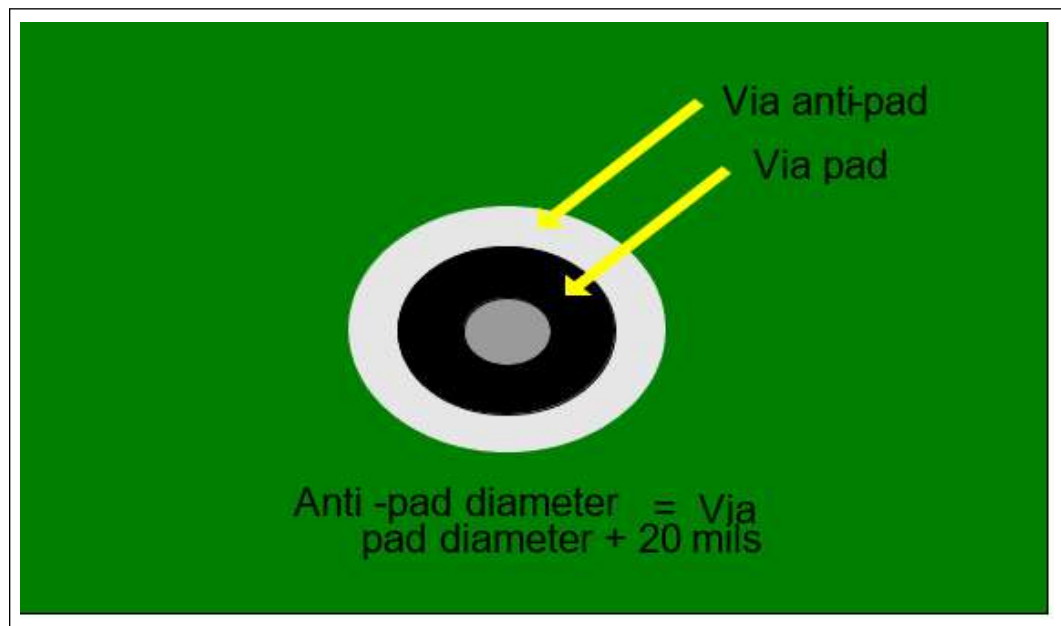
The unused via pads degrade the signal

Figure 345. Signal Vias: Optimal Padstack



On metal layers where signal vias need to have via pads, it is desirable to reduce capacitance between the signal vias and ground-plane layers. The anti-pad diameters should be up to 0.508 mm larger than the via pad diameters. Refer Figure below. Clearance between the pad and the surrounding metal should be ≥ 0.254 mm

Figure 346. Anti-Pad Geometry



Each time differential signal vias pass through a plane layer, within each differential pair, the anti-pads should overlap. Refer Figure below.

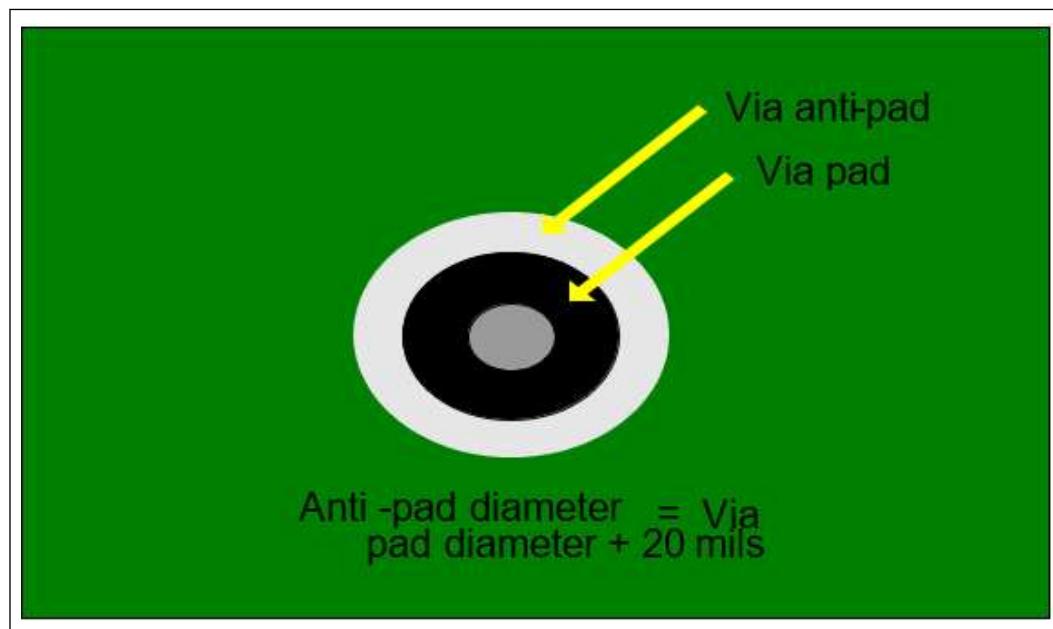
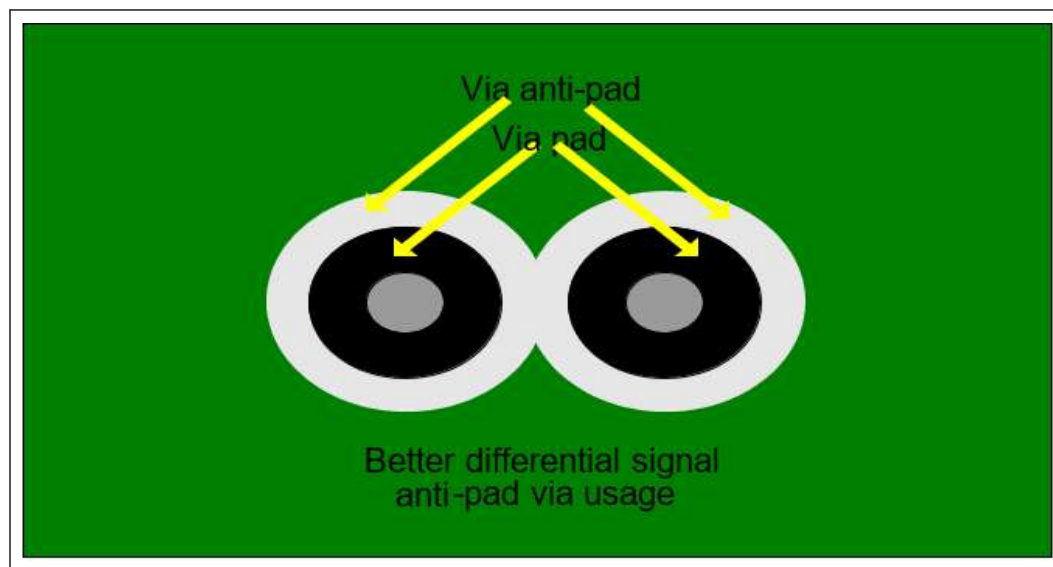
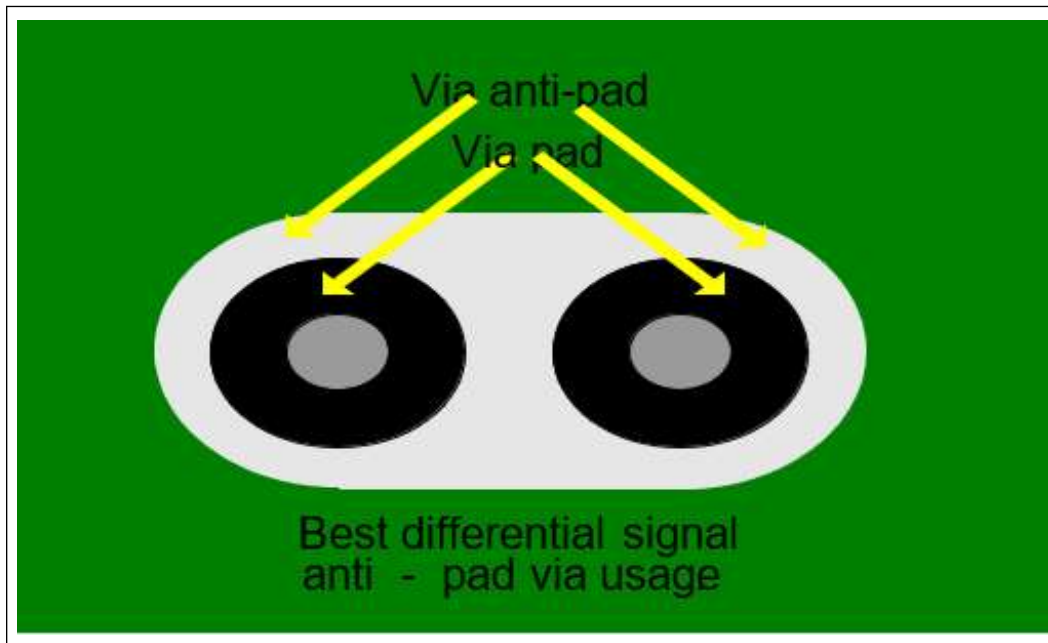
Figure 347. Differential Signal Vias: Improper Anti-pad Geometry**Figure 348. Differential Signal Vias: Acceptable Example**

Figure 349. Differential Signal Vias: Optimal Example



Differential High Speed Layout Guideline

The following shows the Good vs Bad Practices for typical differential high-speed signals. This guideline should be applied when routing the 4x MDI pairs and PCIe TX/RX pairs of I226.

Figure 350. Differential High Speed Layout Guidelines

Fan-out/ Break-out			
Balanced routing			
Direction change			
Length matching			
Keep-outs/ Clearances			
Stubs			
Layer change			

Good
 Fair
 Bad

8.2.8 Component Selection (Bill Of Material)

Listed parts have been used successfully in the past designs.

NOTE

No particular part is required to use with the I226 silicon. It is recommended to start with the known quality parts, which Intel knows that they work. At a high-level, any part within the I226 specifications should work with the silicon.

Any selected parts must go through the validations for production design quality.

NOTE

Refer to the recommended parts list in the I226 Datasheet.

8.3 Antenna Design Guidelines

A broad range of wireless modules are available suited to different regulatory, performance, and user requirements. This chapter shall discuss the considerations for antenna integration in Nova Lake processor designs including, but not limited to, antenna performance, placement options, and RF system-level integration. There is no additional Antenna design requirement to support Dual Chain Bluetooth starting available in Whale Peak 2 (Intel® Wi-Fi 7 BE211) and following Gen Intel Wi-Fi modules.

The processor may support radios for:

- Wi-Fi (WLAN)
- Bluetooth (BT)
- GNSS
- 5G sub 6GHz NR

8.3.1 Antenna Integration

The successful antenna integration on the system is determined through three prime objectives:

1. Ensure good individual antenna efficiency in the frequency bands supported by the respective modem, for all antennas and all modes of operation.
2. Ensure adequate isolation between antennas to:
 - a. Minimize the spatial correlation and mutual coupling between antennas for radios supporting MIMO or diversity, and maximize the benefit that can be derived from the additional transceiver chain.
 - b. Enable the simultaneous operation use-case scenario of multiple radios (co-existence) to be supported without any potential performance degradation. Co-existence of multiple wireless modules and their antennas can become a major problem if not considered in the initial stages of design.
3. Minimize the platform noise coupling into each antenna (in-band noise) which may potentially impact the radio's receiver performance.

8.3.2 Antenna Performance

The individual antenna performance can be characterized by the frequency coverage, efficiency, and peak gain. The isolation between antennas is defined by the S_{21} as measured from the antenna ports on a network analyzer. For radios supporting MIMO or diversity, the MIMO antennas will need to meet the recommended gain imbalance and envelope correlation coefficient limits. These antenna metrics are commonly referred to as passive antenna metrics since the antennas are not connected to their respective modems during the measurement. For A-GNSS, most major carriers have defined radiated performance requirements for wireless devices which include the Total Radiated Power (TRP) and Total Isotropic Sensitivity (TIS). Refer to the target carrier(s) radiated performance requirements for details.

The following tables summarize the passive antenna performance recommendations for each wireless system, the isolation recommendations, and the recommendations for different modes of operation.

The envelope correlation coefficient can be expressed in the following methods depending on the available figures of merit:

Table 712. Antenna Performance Recommendations

Antenna	Frequency GHz	Type	Max size of Antenna Element mm	Efficiency ¹ dB	Peak Gain ^{3,5,6} dBi	VSWR ⁴	MIMO Metrics
WLAN	2.40 - 2.49 5.15 - 5.85 5.925 - 7.125	PIFA/ Slot ⁷	35x8x1	-3.9 (2.4G) -4.4 (5G) -4.4 (6G)	<3 (2.4G) <5.0 (5G) <5.0 (6G)	2:1 (2.4G) 2:1 (5G) 2:1 (6G)	Correlation Coefficient < 0.3 Gain imbalance < 1 dB
Bluetooth	2.40 - 2.49	PIFA/ Chip ⁷	35x8x1	-3.9	<3	2:1	N/A
GNSS ²	1.56 - 1.61	PIFA	75x13x2	-3.5	<2	2:1	N/A
M.2 HSPA (2G, 3G)	0.82 - 0.96 1.71 - 2.17	PIFA	75x13x2	-4 (f<1GHz) -3 (f>1GHz)	<2	3:1 (<1 GHz) 2:1 (>1 GHz)	Correlation Coefficient : -> ecc < 0.5 (f<1 GHz) -> ecc < 0.4 (f>1 GHz) Gain imbalance < 1 dB
M.2 HSPA (2G, 3G, LTE)	0.70 - 0.96 1.42 - 1.51 1.71 - 2.17 2.30 - 2.40 2.50 - 2.70 GNSS ²	PIFA	75x13x2	-3.5 (f<1.6 GHz) -3 (f>1.6 GHz)	<2	3:1 (<1 GHz) 2:1 (>1 GHz)	Correlation Coefficient : -> ecc < 0.5 (f<1.6 GHz) -> ecc < 0.4 (f>1.6 GHz) Gain imbalance < 1 dB
M.2 HSPA (2G, 3G, LTE, LTE-LAA)	0.70 - 0.96 1.42 - 1.51 1.71 - 2.17 2.30 - 2.40 2.50 - 2.70 3.3 - 3.70 5.0 - 6.0 GNSS ²	PIFA	75x13x2	-3.5 (f<1.6 GHz) -3 (f>1.6 GHz)	<2	3:1 (<1 GHz) 2:1 (>1 GHz)	Correlation Coefficient : -> ecc < 0.5 (f<1.6 GHz) -> ecc < 0.4 (f>1.6 GHz) Gain imbalance < 1 dB

continued...

Antenna	Frequency GHz	Type	Max size of Antenna Element mm	Efficiency ¹ dB	Peak Gain ^{3,5,6} dBi	VSWR ⁴	MIMO Metrics
4.5G/LTE Sub 6GHz Main & Aux	0.60 - 0.96 1.42 - 1.51 1.71 - 2.17 2.30 - 2.40 2.50 - 2.70 3.3 - 4.20 4.4 - 5.0 GNSS ²	PIFA	75x13x2	-3.5 (f<1.6 GHz) -3 (f>1.6GHz)	<2	3:1 (<1 GHz) 2:1 (>1 GHz)	Correlation Coefficient : -> ecc < 0.5 (f<1.6 GHz) -> ecc < 0.4 (f>1.6 GHz) Gain imbalance < 1 dB
4.5G/LTE Sub 6GHz MIMO 3&4	1.71 - 2.17 2.30 - 2.40 2.50 - 2.70 3.3 - 4.20 4.4 - 5.0 5.0 - 6.0	PIFA	45x10x2	-6 (MIMO RX) -3.5 (GNSS)	<2	3:1 (<1 GHz) 2:1 (>1 GHz)	Correlation Coefficient : -> ecc < 0.4 Gain imbalance < 1 dB
5G/LTE sub 6GHz Main ⁸ & Aux	0.61-0.96 1.45-1.51 1.71-2.20 2.30-2.70 3.30-4.20 4.40-5.00 5.125-5.925 GNSS ²	PIFA	75x13x3	-3.5 (f<1.6GHz) -3 (f>1.6GHz) -3 (GNSS)	<2	3:1 (<1 GHz) 2:1 (>1 GHz)	Correlation Coefficient : -> ecc < 0.5 (f<1.6 GHz) -> ecc < 0.4 (f>1.6 GHz) Gain imbalance < 1 dB
5G/LTE Sub 6GHz MIMO ⁸ 3 ⁹ &4 ¹⁰	1.452-2.20 ⁹ 1.805-2.20 ¹⁰ 2.30-2.695 3.30-4.20 4.40-5.0	PIFA	45x10x2	-3 (MIMO3) -6 (MIMO4)	<2	3:1 (<1 GHz) 2:1 (>1 GHz)	Correlation Coefficient : -> ecc < 0.5 (f<1.6 GHz) -> ecc < 0.4 (f>1.6 GHz) Gain imbalance < 1 dB
Notes: 1. Efficiency targets include the cable loss. Actual cable loss will vary based upon the chassis integration. 2. Peak gain targets include the cable loss. Refer to the module's regulatory modular approval for details. 3. VSWR 2:1 is equivalent to -10 dB S ₁₁ (Reflection Coefficient), and VSWR 3:1 is equivalent to -6 dB S ₁₁ . 4. For TRP evaluation refer to module EPS document for actual conducted Tx Power depending on Geo. For TIS evaluation refer to conducted sensitivity. 5. During the Wi-Fi modular certification, reference antenna uses less than 5 dBi Peak gain requirement at 5GHz and 5.6 dBi for UNII7~8 band. Table shows the recommendation mainly targeting to best UX from 3.5 dBi. 6. For Regulatory purpose Intel handles modular certification with PIFA antenna only. 7. 5G support 2X2 TX MIMO and dedicated TX antennas are 5G Main and MIMO3.							

Table 713. Isolation Recommendations

Antenna Ports	Wi-Fi	Bluetooth*	GNSS	5G sub6 NR/ LTE sub 6GHz Main & Aux	5G sub6 NR/ LTE sub 6GHz MIMO 3&4
Wi-Fi	25 dB nominal	25 dB nominal ¹ (15 dB min)	25 dB	25 dB ² (15 dB LTE Band7, 40, 65, 66 ³)	25 dB nominal ² (15 dB min)
Bluetooth*	25 dB nominal ¹ (15 dB min)	N/A	25 dB	25 dB (20 dB LTE Band7, 40, 65, 66 ³)	25 dB nominal ¹ (15 dB min)
continued...					

Antenna Ports	Wi-Fi	Bluetooth*	GNSS	5G sub6 NR/ LTE sub 6GHz Main & Aux	5G sub6 NR/ LTE sub 6GHz MIMO 3&4
GNSS	25 dB	25 dB	N/A	25 dB (shared in Aux)	25 dB
5G sub6 NR/ LTE sub 6GHz Main & Aux	25 dB (15 dB LTE Band7, 40, 65, 66 ³)	25 dB (20 dB LTE Band7, 40, 65, 66 ³)	25 dB (shared in Aux)	25 dB	25 dB
5G sub6 NR/ LTE sub 6GHz MIMO 3&4	25 dB nominal ¹ (15 dB min)	25 dB nominal ¹ (15 dB min)	25 dB	25 dB	25 dB
Notes: 1. BT-Wi-Fi isolation translates to Wi-Fi main to Wi-Fi Aux isolation for the case where BT antenna is shared with Wi-Fi. For best performance, recommended isolation > 35 dB. Nominal performance with limited degradation can be seen with isolation > 25dB. Co-existence solution allows functionality down to 15 dB isolation with reduced performance. 2. Without LTE band 7 and without a co-existence mechanism between Wi-Fi modules, 30 dB isolation is the best, but 25 dB is acceptable as indicated shown above 3. Without a coexistence scheme between WLAN , additional filtering at WLAN modules is required to operate at 15 dB isolation. Careful consideration of the total system solution is required for this case					

Table 714. Recommendations for Different Modes of Operation

Description of Lid Mode	Clamshell (open Lid)	Tablet	Closed Lid
Antenna performance recommendations	As in Antenna Performance Recommendations	As in Antenna Performance Recommendations	3 dB lower than values in Antenna gain Performance Recommendations

1. With the far-field complex antenna radiation patterns, which can be measured in antenna test chambers, the envelope correlation coefficient can be evaluated with the following formula:

$$\rho_e = \frac{\left| \int_0^{2\pi} \int_0^\pi \left(XPR \cdot F_{\theta 1}(\theta, \phi) F_{\theta 2}^*(\theta, \phi) P_\theta(\theta, \phi) + F_{\phi 1}(\theta, \phi) F_{\phi 2}^*(\theta, \phi) P_\phi(\theta, \phi) \right) \sin \theta d\theta d\phi \right|^2}{\left| \int_0^{2\pi} \int_0^\pi \left(XPR \cdot |F_{\theta 1}(\theta, \phi)|^2 P_\theta(\theta, \phi) + |F_{\phi 1}(\theta, \phi)|^2 P_\phi(\theta, \phi) \right) \sin \theta d\theta d\phi \right| \cdot \left| \int_0^{2\pi} \int_0^\pi \left(XPR \cdot |F_{\theta 2}(\theta, \phi)|^2 P_\theta(\theta, \phi) + |F_{\phi 2}(\theta, \phi)|^2 P_\phi(\theta, \phi) \right) \sin \theta d\theta d\phi \right|}$$

where ρ_e is the envelope correlation coefficient, F_1 and F_2 are the complex vector field strengths of antenna 1 and 2 respectively, XPR is the cross-polarization ratio, and P is the incident field distribution function.

2. In the case where the complex antenna radiation pattern cannot be obtained, an alternative method, which requires the S-parameters and the individual radiation efficiencies, can be utilized to determine the correlation coefficient in the following formula:

$$\rho = \frac{-S_{11}S_{12}^* - S_{21}S_{22}^*}{\sqrt{(1 - |S_{11}|^2 - |S_{21}|^2)(1 - |S_{22}|^2 - |S_{12}|^2)}\eta_1\eta_2}$$

where ρ is the envelope correlation coefficient, S_{11} , S_{12} , S_{21} , S_{22} are the S-parameters of the two port antenna system measured on a network analyzer, S_{12}^* is the complex conjugate of S_{12} , and η_1 and η_2 are the antenna radiation

efficiencies measured with the unused antenna port terminated with a 50 Ohm load. However, this equation may provide the accurate prediction of correlation coefficient in certain ideal situation only.

3. In the case where only the S-parameters are available, the envelope correlation coefficient can be estimated with the following formula:

$$\rho_e = \frac{|S_{11}^* S_{12} + S_{21}^* S_{22}|^2}{(1 - |S_{11}|^2 - |S_{21}|^2)(1 - |S_{22}|^2 - |S_{12}|^2)}$$

The envelope correlation coefficient calculation methods shown above are described in the following papers:

- Ying et al.; "Characterization of multi-channel antenna performance for mobile terminal by using near field and far field parameters," COST 273 TD(04)(095), Goteborg, Sweden, 2004.
- Hallbjorner, P.; "The significance of radiation efficiencies when using S-parameters to calculate the received signal correlation from two antennas," Antennas and Wireless Propagation Letters, IEEE, vol.4, no., pp. 97- 99, 2005
- S.Blanch, J. Romeu and I. Corbella, "Exact expression of antenna system diversity performance from input parameter description", Electronic Lett., vol 39, 705-707, 2003

8.3.3 Antenna Placement

The following must be considered when deciding the placement of antennas:

- **Modes of Operation:** For devices which enable different modes of operation (i.e. Clamshell, Tablet, and Closed Lid), careful consideration of the environment around the antenna in each configuration while deciding the antenna placement is necessary to deliver the desired user experience in all modes of operation. Furthermore, it should be verified that the antennas are placed in such a manner that the user does not accidentally cover all antennas while in an intended mode of operation. This would result in loss of connectivity or deteriorating of the signal for the affected radios.
- **Keep-Out-Zone (KOZ):** A metal-free KOZ is typically required to integrate the antenna in traditional designs. The physical size of the antenna will depend on the target frequencies, bandwidth, and performance requirements of the antenna. Antennas with larger geometric footprints tend to provide better performance. The non-metallic materials (i.e. plastics) also need to be assessed since different materials properties may have different effects to the antenna characteristics.
- **Lossy Structures:** Separation of the antenna radiation elements from lossy structures, such as the LCD enclosure, will provide better bandwidth. Also, larger bandwidth considerations require larger separation from nearby metallic structures.
- **Grounding:** Unbalanced antennas require a ground reference, which is typically the chassis or the back of the display. The size of the ground reference needs to be considered as part of the antenna design.
- **Cable Routing:** Each antenna will require a coaxial cable to connect to the radio module's RF port. Cable routing that avoids regions with noise and minimizes cable lengths is recommended to minimize the platform noise impact and cable loss. In the case where the module is in the base and the antenna is in the

display, the coaxial cable will need to be fed through the hinges. The size of the mechanical hinge places a restriction on the number of cables that can be fed through and the diameter of the cables. Typical RF coax cables can range in outer diameter (OD) from 0.81 mm to 1.37 mm; however, the RF connector receptacle defined by the M.2 module specification will only accept mated plugs with either 0.81 mm OD cables or 1.13 mm OD cables. Refer the respective radio module's product description for the RF connector type.

- **Cable Loss:** Coaxial cables will introduce an insertion loss depending on the diameter, length, and frequency of interest. Thicker coaxial cables tend to be less lossy while being more expensive compared to thinner cables. Lower loss cables are also available from vendors which will reduce the total cable loss especially for long cables. Double shielded cables are useful in ensuring that radio module sensitivity degradation is limited due to the cable picking up platform noise when routed through the platform.
- **RF Exposure (SAR):** Regulatory bodies, such as FCC and CE, define detailed requirements for the allowable RF exposure for devices which generate RF electromagnetic fields. Specific Absorption Rate (SAR) is a measure of the rate in which energy is absorbed in body tissue due to EM energy from an antenna, and is a prime consideration for placement of antennas, especially for higher power radios. In general, the antennas should be located in areas which will not be in close proximity to the human body in any mode of operation. Refer to the published guidance from FCC for more details on the RF exposure requirements.
- **Isolation between antenna ports:** To improve the isolation between antennas, the following methods can be employed which include, but is not limited to:
 1. Spatial diversity - larger spatial separation between antennas is one way of increasing the isolation between ports of radio modules that operate in bands that are close to one another.
 2. Polarization diversity - in some instances where the antennas have polarization discrimination, the relative orientation of the antennas can be used to improve isolation and enhance performance.
 3. Pattern diversity - utilizing antennas with different radiation patterns and orienting the peak gain direction of the antenna in different directions may also improve isolation.
- **Platform Noise:** Various components in the system, such as display/touch circuits, memory, PLLs, cameras, clocks, USB/HDMI ports, SSD, DDR, and other high speed IO, may generate noise which can couple to the antennas and/or coax cables and thus degrade the wireless radio's receiver performance. Careful consideration needs to be employed with respect to antenna placement and cable routing to avoid these noise sources and minimize the platform noise impact to the receivers. With detachable tablets and convertible notebooks, the proximity of the antenna to the base components needs to be especially considered during the initial floor planning. In addition to antenna placement, the particular component which was highly suspected to be radiation source should require good RF shielding for significant isolation improvement.
 - Internal studies have found that 3D cameras, which utilize the USB3 interface to communicate with the motherboard, is a source of broadband RF noise which may impact the antennas in the vicinity. The camera noise may impact the radio's receiver performance through the antenna in two ways:

1. Conducted noise which flows from the camera through the shared ground plane (chassis) and in to the antenna through the antenna ground connection. To reduce the noise coupled in to antenna, the ground plane should be manipulated to re-direct the current flow from the camera away from the antenna.
2. Radiated noise from the 3D camera interface (especially the USB3 connector), may couple to the antenna and hence desensitize the radio receiver. In general, the antennas should be separated from the 3D camera and USB3 bus to avoid any degradation to the radio's receiver performance. Studies have shown that the antenna should be at least 15 mm away from the USB3 connector for minimal impact to Wi-Fi radios. Shielding of the camera and USB3 connector is also recommended.

Antenna Placement Options for Multi-Mode Devices

Four potential antenna placement options are shown in the figures below for clamshells or multi-mode devices such as convertibles and detachable Ultrabook™. The target form factor for the placement options is around 13 to 14 inch notebook, but it is not limited to this size. Recent platform ID trend introduces Bezel-less display platforms and there is no space available in LCD bezel area where could be the prime location for antennas. The module selection includes support for 2x2 MIMO Wi-Fi/BT, and 4X4 5G. GNSS is assumed to be combined with 5G Aux antenna. GNSS is combined with 5G AUX antenna (AGPS/GPS) as 5G NR will support 2X2 TX MIMO from 5G Main and MIMO3 antennas.

For multi-mode devices with Tablet and/or Closed-Lid modes, corresponding metal keep-out-zones will need to be reserved on the base to minimize any potential impact to the antennas when configured in these modes.

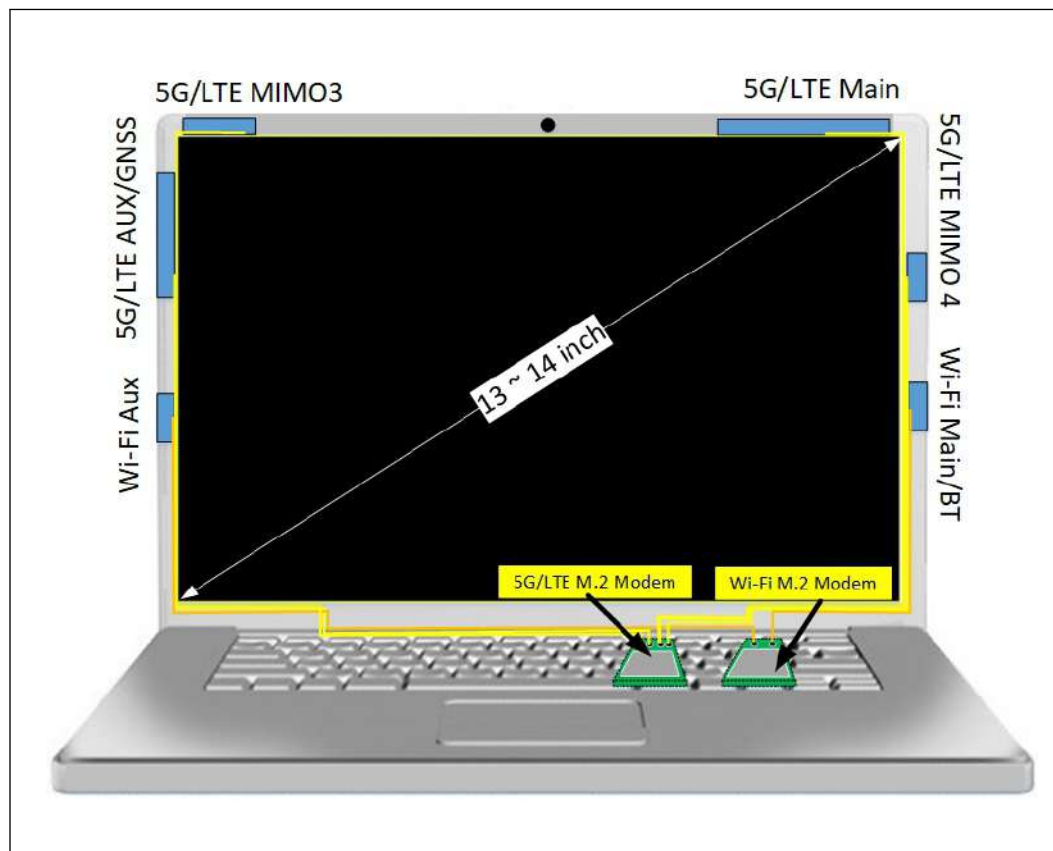
Figure 351. Antenna Placement Option 1: Clamshell Mode Bezel Integration

Figure 352. Antenna Placement Option 2: Clamshell/2in1 Base Integration 1

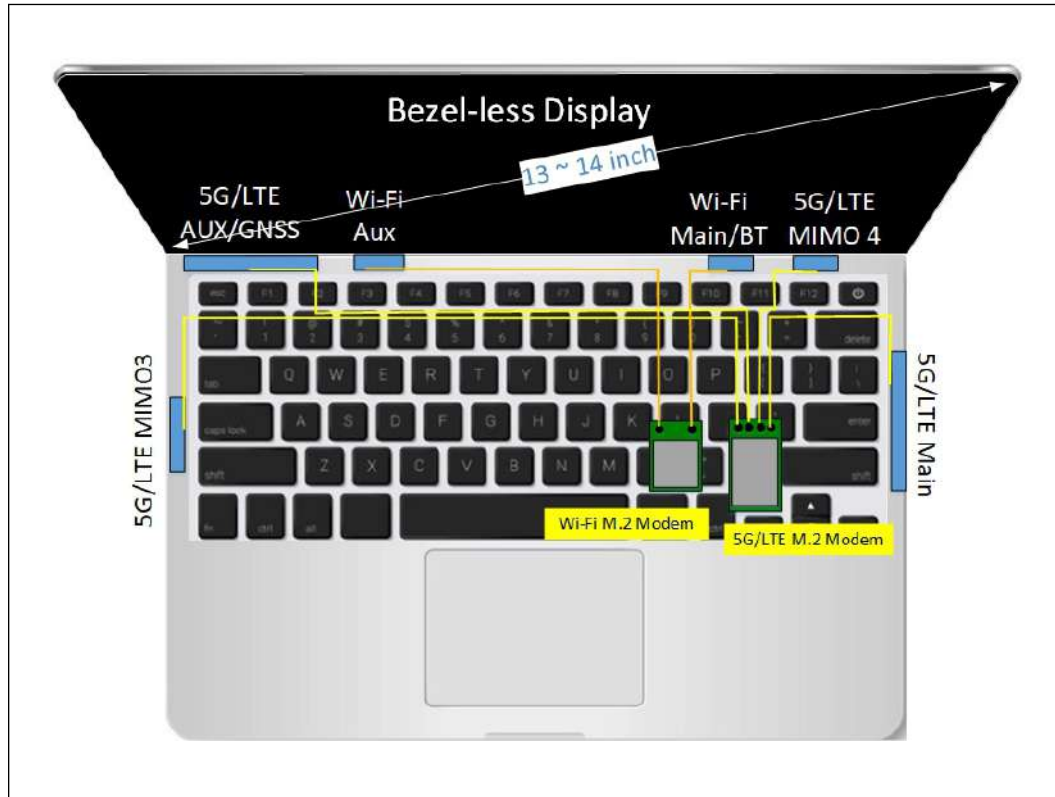


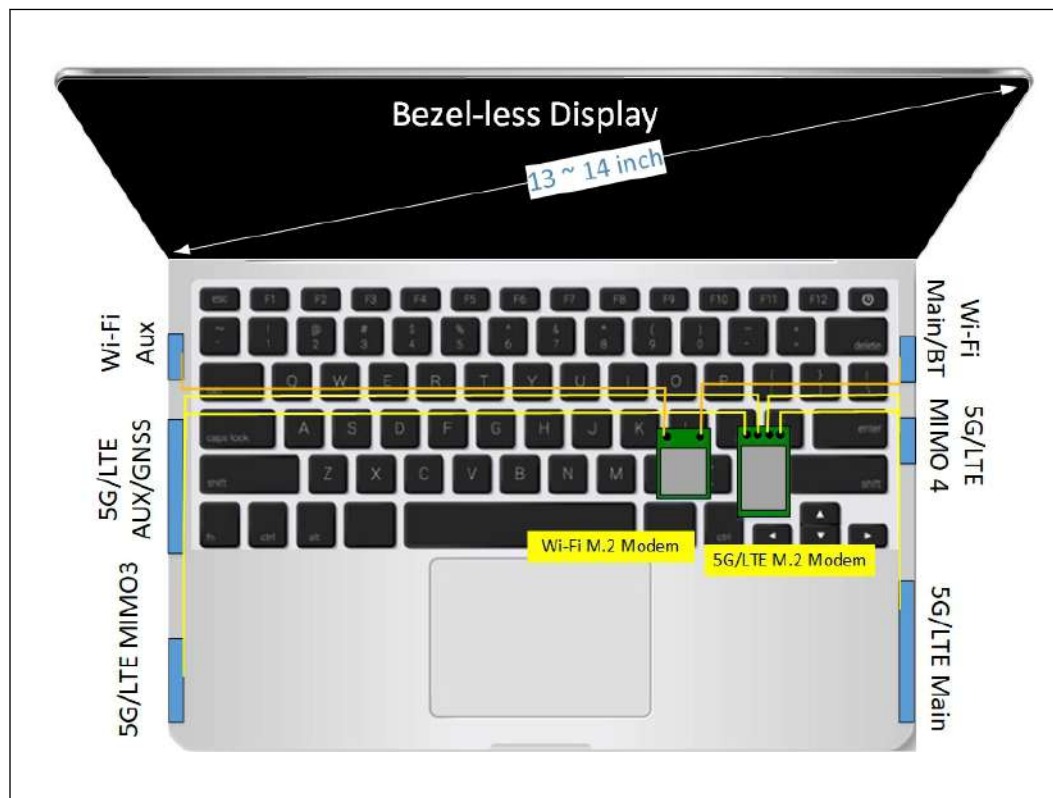
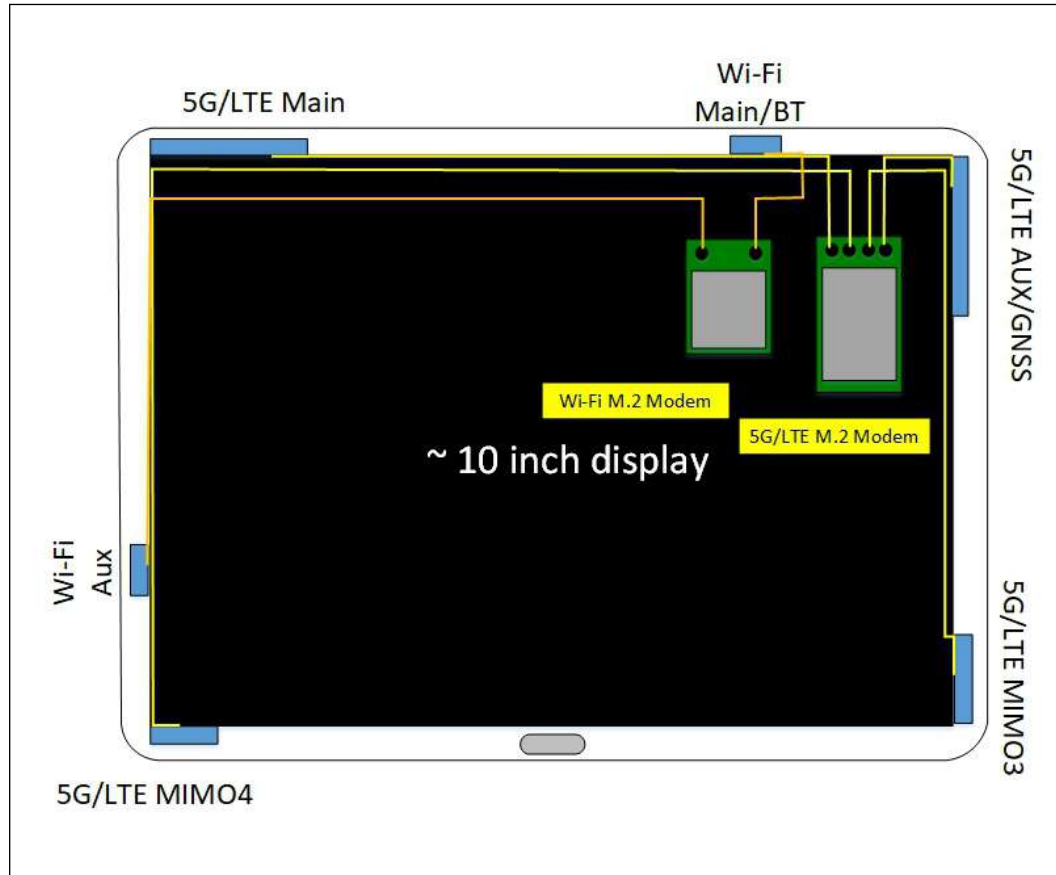
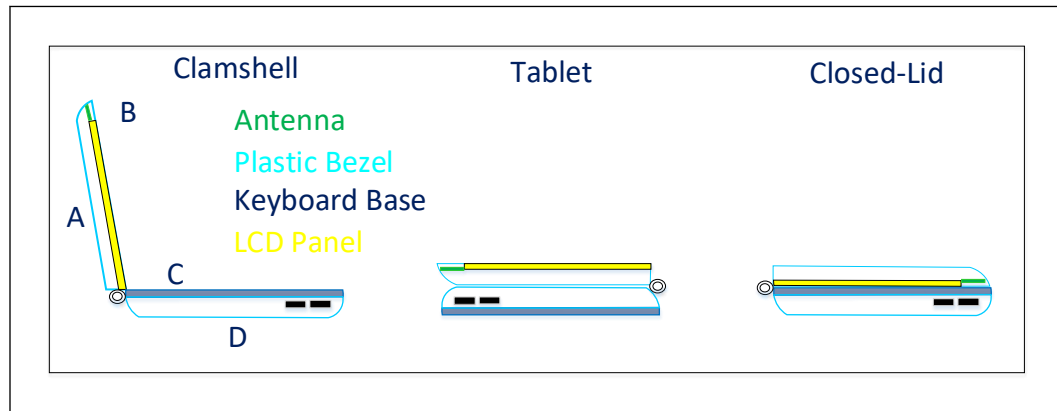
Figure 353. Antenna Placement Option 3: Clamshell/2in1 Base Integration 2

Figure 354. Antenna Placement Option 4: Tablet/Detachable Integration



Previously, most Clamshell-only notebooks have satisfied the SAR exclusion thresholds defined by FCC in which one of the requirements being the antennas are mounted around the display with the minimum antenna-to-user separation distance. In those cases, the SAR testing and transmit power reduction could be potentially avoided. With multi-mode devices, the various modes of operations will require special attention to the impact to SAR due to the antenna placement and may require transmit power reduction to meet regulatory requirements. In general, the antennas in the display should be mounted closer to the B-surface to increase the separation distance from the body and reduce the SAR. The figure below provides recommendations for antenna placement in a non-metallic convertible chassis, with considerations for SAR.

Figure 355. Placement Recommendation with Regulatory SAR Considerations

- In the case of a detachable tablet, or the Tent Mode test configuration for convertible notebooks defined by Intel Regulatory, the antenna on the A-surface is the worst-case location for body-SAR and will result in significant transmit power reduction for Wi-Fi.

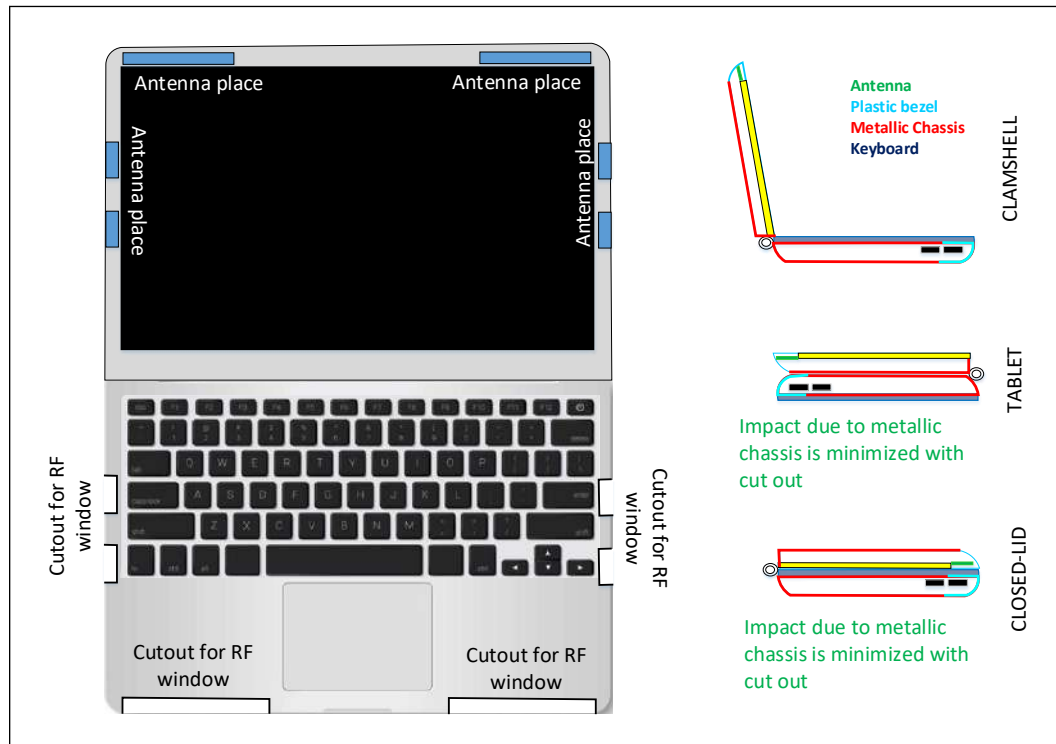
Antenna Placement in Metallic Chassis

Many platforms with the U-processor design utilize metal materials for the surface covers, commonly referred to as metallic chassis. The antenna must be placed such that there is sufficient clearance from metal to enable good wireless performance. When there are metal surfaces surrounding the antenna, it is recommended to have plastic sections or insert molded plastic in the metal covers to provide an RF transparent window which will not degrade the antenna performance and thus enable good wireless connectivity in all modes of operation. Other potential options include integrating the antenna on the base (keyboard side) of the platform or utilizing the hinge area. Care should be taken to ensure potential EMI from adjacent I/O connectors do not interfere with the antenna reception. Multiple antenna SKUs and/or revisions will imply multiple chassis SKUs and/or revisions, so this option will need to be considered very early in the design cycle. Utilization of the A-surface for the antennas may result in higher SAR and will need to be assessed before committing to such a design.

The following placement options may be considered:

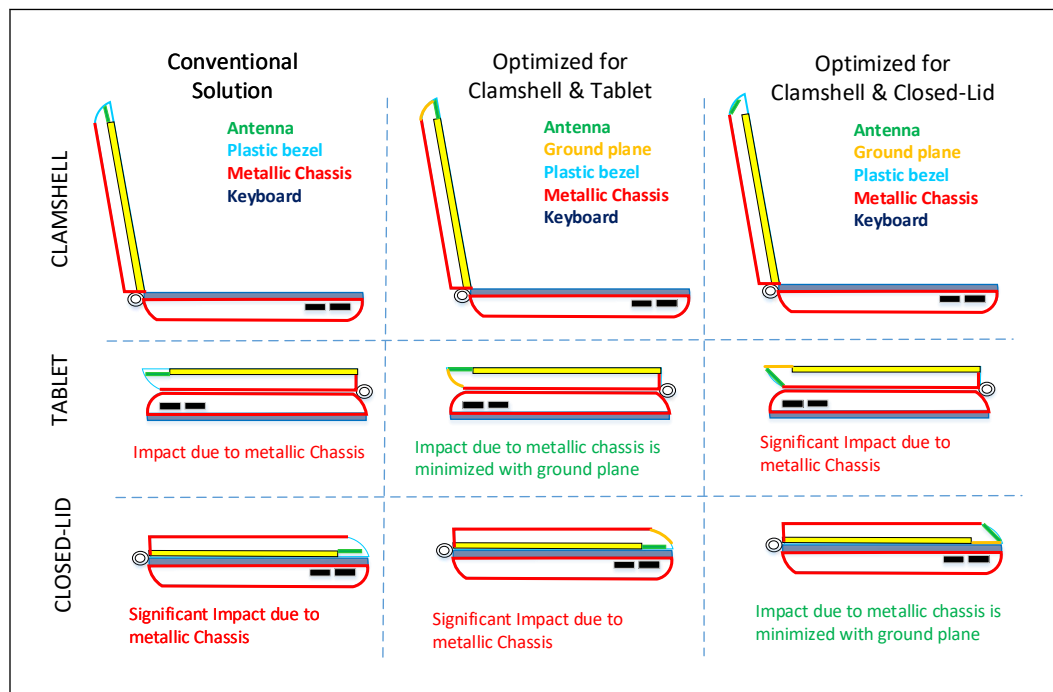
1. **Metal Chassis with cutouts in display and base:** In the case of a metallic chassis, cutouts may be employed to provide required clearance for the antenna on the A and B-surfaces. When in Tablet or Closed-Lid modes, the antenna must not reside closer than 5 mm to metal (or EMI coating) on the D-surface or C-surface, respectively. A suitable plastic cutout, with a recessed EMI shield maybe required to achieve both ESD and EMI protection and good Tablet and Closed-Lid performance. The figure below depicts this solution.

Figure 356. Placement Options for Metallic Chassis with Cutouts in Display and Base



2. **Metallic chassis with cutout in the display:** A dielectric loaded antenna maybe placed for Wi-Fi operation on the B-surface, with a cutout on the A-surface. Such a dielectric loaded antenna can operate with a metallic B-surface, which also shields the antenna from the changes introduced by closing the lid. The figure below depicts this solution.

Figure 357. Placement Options for Metallic Chassis with Cutouts in Display



- Due to the Tent Mode test configuration for convertible notebooks defined by Intel Regulatory, or in the case of a detachable tablet, the antenna on the A-cover is the worst-case for the body SAR requirement in touch condition and will result in significant transmit power reduction for 5G and possibly even Wi-Fi. (Same applies for detachable tablet notebooks)
3. **Base mounted Wi-Fi antennas:** Antennas can be placed in the hinge area on the base of a metallic chassis notebook, with a plastic cutout near the hinge. The hinge must be such that when the lid closes, the area in the base where the antennas are housed is not shielded completely by the lid. Special consideration must be given to maximizing the separation between the antenna and the human body below the D-surface to pass SAR requirements with low-risk. Platform noise pickup from base mounted antennas needs to be measured before committing to a particular location as there is a higher risk of impact from platform noise due to the closer proximity of the antennas to the motherboard in clamshell or convertible designs.
 4. **Slot antennas cut into the chassis:** Slots can be cut into the A-surface of the antenna, near the apex of the lid and the cutout can be covered with plastic inserts. Such an antenna would be integrated into the chassis and special care must be given to the feeding mechanism, especially when materials not amenable to soldering such as aluminum are used.

8.3.4 Platform Quiet Zone

Design for Wireless (DFW) is the design principle for wireless engineers to engage platform integration considering a wireless architecture in highest priority. Among multiple rules for DFW, quiet zone is one of the key factors in early design stage to prevent unnecessarily frequent post debugging process or expensive design iterations due to the simple ignorance of wireless integration in early stage.

Mobile Platform consists of multiple active components such as main board, battery, LCD panel and various cables and so on. Especially, main board integrates majority of active components including radio modules and they generate electromagnetic interference signals when they are active and being trapped inside the chassis or radiates out of the chassis. Well known sources are HSIO, USB, TBT, DDR, CPU and so on. Antenna integration without this consideration, radio sensitivity will significantly be degraded and eventually drop the desired wireless signals and provide bad wireless user experience. As an example, figure depicted below shows common RF noise sources near the integrated antennas inside the mobile platform without considering quite zone concept applied in early design stage

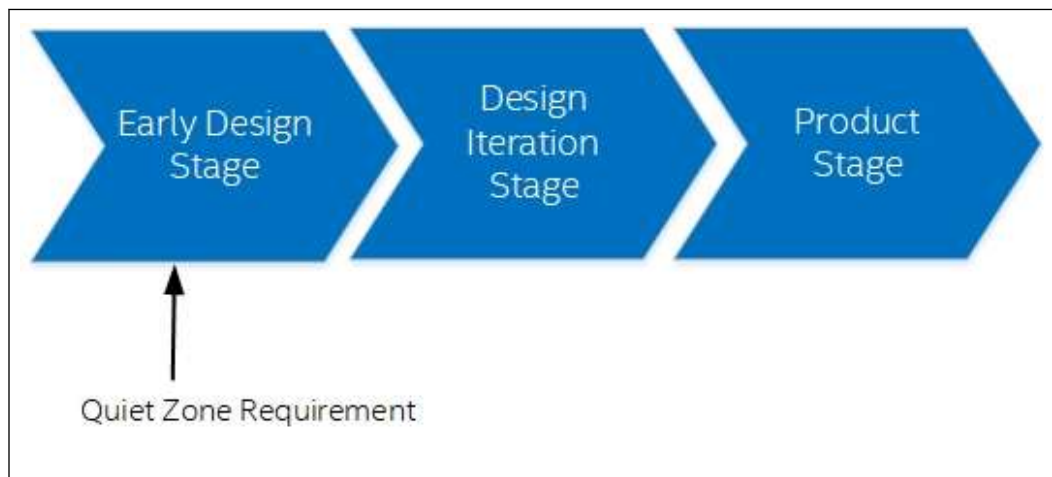
Figure 358. Well Known Noise Sources Inside Platform Near Integrated Antenna



In order to achieve the best OTA performance from the platform, antenna must be placed the location where there is no or minimum RF noise or proper EMI/EMC shielding needs to be applied to prevent the noise pick up from the integrated antenna

Quiet Zone Definition and Design Procedure

In definition, Quiet Zone (QZ) is the location/area inside the platform where the measured/simulated RF noise level meets or exceeds the NOISE Specifications within the desired frequency band and can be defined as preferred antenna and RF coax cable routing location.

Figure 359. Design Procedure and QZ Requirement Applying Stage

Quiet Zone Measurement Characterization

Quiet zone could be identified from the platform noise level measured at the integrated antenna. Thus, it would be simply measured by spectrum analyzer at the controlled environment as shown in the figure below. It is noted that DUT should be powered on and categorize in two states - idle mode and work-load mode. And these two states are defined in Table below.

Table 715. Platform Idle and Workload Mode Configuration

Idle mode DUT configuration	DUT is powered on and no CPU/Memory access - steady state of the device
CPU	~ 5%
Disk (HDD)	~ 2%
GPU	~ 1%
Memory	2 to 3 GB
Workload mode DUT configuration	DUT is powered on and additional workload with given stress
CPU	60 ~ 80 %
Disk	~ 80%
GPU	~ 15%
Memory	6 ~7 GB

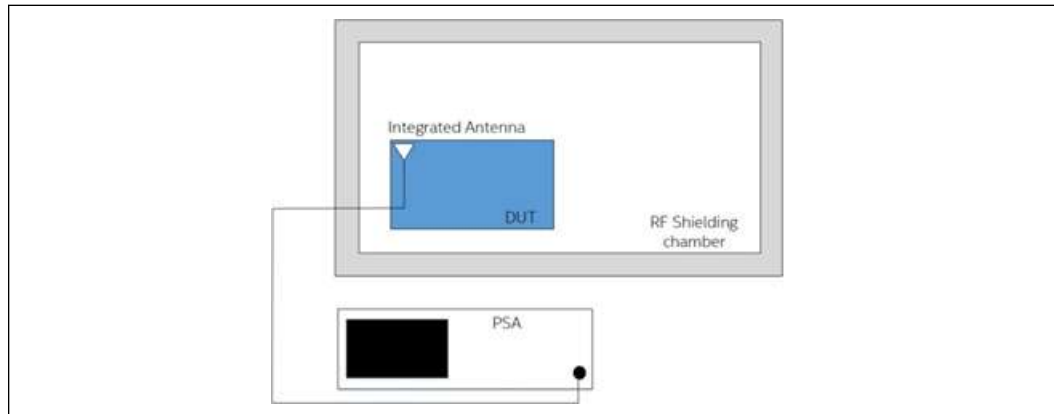
It is noted that this workload configuration represents the actual usage such as web-browsing and zoom video conference etc. And this table would be updated in the future per other applications or usage requirements.

In order to characterize the quiet zone, table below will be a reference to characterize the QZ status check at the antenna location. For the measurement consistency, RBW is set to 100KHz and three states are defined to understand the risk of radio RX desense

Table 716. Quiet Zone Characterization from PSA Measurement

QZ Characterization	Good	Better	Best
Measured reading in idle mode	-111 dBm/100KHz	-113 dBm/100KHz	-115 dBm/100KHz ¹
Measured reading in workload mode	-109 dBm/100KHz	-111 dBm/100KHz	-113 dBm/100KHz
1. Result in best is based on 8 dB system loss from antenna loss (-4dB) and desense loss (-4dB)			

Figure 360. Simple Quiet Zone Measurement Method



Quiet Zone Evaluations Tools

There are multiple ways of quiet zone measurement and evaluation inside the platforms. Well known conventional method to measure the quiet zone requires physical radio and antenna cable access through assembled chassis. This introduces undesired measurement uncertainty and longer lab preparation time to collect the data.

Therefore, non-invasive method would be preferable for quicker and accurate evaluation at the antenna port. Example tool could be:

- NPT - Noise Profiling Tool for 5G/LTE (naming convention is subject to change)
- NDT - Noise Diagnostic Tool for Wi-Fi

NTP tool is internally available and currently working further optimization process for test automation and will be available for next gen 5G quiet zone evaluation.

NDT tool is available and distributed for OEM customer use for Wi-Fi noise profiling.

Another useful tool for early design stage would be full 3D Electromagnetic simulation tool. This will require full 3D CAD information of chassis and main board details (layout file). Current existing tool such as CST, HFSS, Feko, XFDTD etc. require expensive computational resources to include all detail features for full 3D analysis. However, simplified 3D EM simulation method under development would provide efficient way to determine the quiet zone inside the platform and could be transferred to OEMs. This tool development is targeting to deploy in Lake platform time frame.

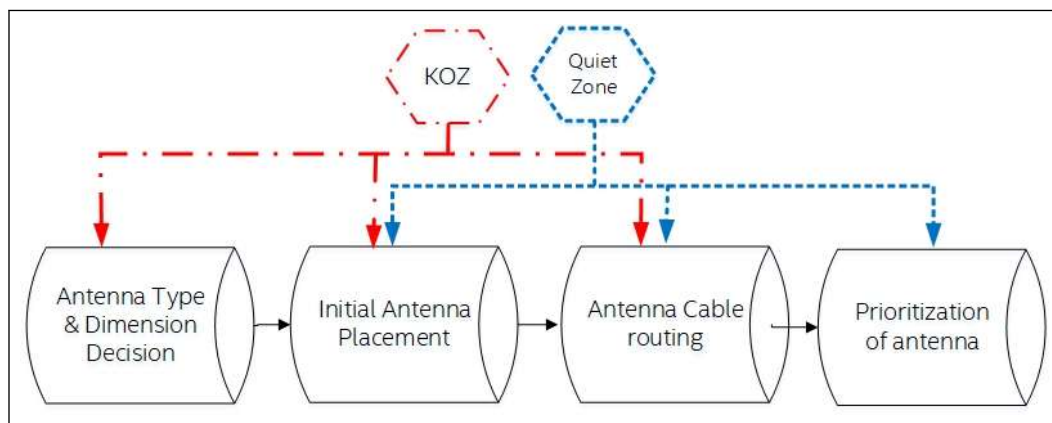
Design Principle Using Quiet Zone Concept

In early stage of platform design, there are multiple decision point for antenna integration process to consider. Quiet zone and Keep out zone concept play the key role to determine the acceptability in each stage as following:

- Antenna type and possible dimension
- Initial antenna placement inside the platform
- Antenna cable routing inside the chassis
- Antenna priority decision based on UX and certification requirement

Figure below indicate the applying stage of indicated rules for correct floor planning in early design stage before the assembly.

Figure 361. Early Design Stage Quiet Zone and Keep out Zone Requirement Application



8.3.5 RF System-Level Integration Recommendations

Wi-Fi

Platforms typically support 2x2 (and some 3x3) MIMO configurations for Wi-Fi. In addition to meeting the efficiency requirements set forth in [Antenna Performance](#) on page 577, some additional system level recommendations are listed in table below.

Table 717. Additional Recommendations for Wi-Fi MIMO Operation

SN	Additional Recommendations for Wi-Fi MIMO Operation	
1	Difference in Noise coupling into Main and Aux antennas	< 1 dB
2	Absolute value of noise coupling into Main and Aux antennas	< 3dB de-sense from noise floor

In addition the antenna specifications, the following considerations are brought to attention:

- Modern Standby Support (previously Connected Standby or InstantGo)
 - The platform may need to maintain the Wi-Fi connection while in standby and in any mode of operation such as clamshell, tablet, closed lid, detached, and/or convertible modes. SAR requirements will need to be met and active states must be achievable in each of the above operation modes.
- Specific Absorption Rate (SAR)

- In situations where the platform cannot meet regulatory RF exposure requirements, the radio module's RF transmit power may need to be reduced.
- In-Device Co-Existence Interface

Bluetooth

A majority of platforms are expected to use the combo Wi-Fi-BT modules. The combo modules implement co-existence schemes which allow for simultaneous operation of Wi-Fi and Bluetooth with isolation down to 15 dB; however, higher isolation offers better performance and robustness and is recommended as shown in [Table 713](#) on page 578.

If a discrete Bluetooth* module is implemented, the isolation between the BT antenna and each Wi-Fi antenna will need to meet the isolation recommendations in [Table 713](#) on page 578. In platforms where Wi-Fi Aux antenna is also shared as the BT antenna, the Wi-Fi to BT isolation becomes the same as Wi-Fi Main to Wi-Fi Aux isolation.

GNSS

Global Navigation Satellite System (GNSS) includes GPS, GLONASS, Galileo, Beidou, and other regional systems. Two potential integration options (and corresponding antenna choices) are supported for GNSS:

1. Discrete GNSS solution: A standalone GNSS antenna is required, which will be connected to the RF front end of the discrete GNSS modem implementation.

8.4 M.2-1A

A new M.2-1A connector will introduce into 2230 module type 1A (B). It will introduce for FmP4 to support higher current consumption. Therefore, in order to support FmP4, platform shall introduce new M.2-1A connector and deliver current consumption up to 4A.

NOTE

For the RF Companion POR related to the platform, refer to the CCG 5Q Wireless Roadmap/Platform Matrix Customer Communication, (# [601177](#)).

8.5 Standard M.2 Connectivity Design Considerations

This section describes the following:

- Standard (Discrete) Connectivity Guidelines
- PCIe* Host Interface Errata

8.5.1 Standard (Discrete) Connectivity Guidelines

The platform supports both Integrated (CNVi) and Discrete (Standard M.2) connectivity modules on the M.2 socket. These are the guidelines for using the standard M.2 Wi-Fi/BT combo modules.

This section includes several important implementation aspects the OEM should take into consideration when implementing a platform that would accommodate this product.

Signal Connection Pitfalls

- The OEM should make sure to follow the M.2 definitions of signal names and directions (I/O TX/Rx etc.) and avoid confusion between platform side and device side.

NOTE

Some lines are bidirectional, such as PCIe CLKREQ, PEWAKE.

Pullups and Pulldowns

The OEM should apply pullups and pulldowns in the platform side according to below table.

Table 718. Socket 1 Pullups and Pulldowns

I/F	Signals	PU/PD Guidelines	Rationale
W_Disable#	Power valid to PERST# Input active	PU	May be required by M.2 cards; in Intel Wireless products there is already a PU at the silicon level.
PCIe*	PEWAKE#	PU	Open drain, required by M.2
	CLKREQ#	PU	Open drain, required by M.2
	PERST#	PD	Required by Intel platform design guidelines
	Other PCIe* signals	None	PCIe* specification

The OEM must avoid using PU/PD when not needed or when required not to be used. Unless this rule is followed, it would result in a back-bias condition, in which the IO is getting voltage before the device side is ready for it.

IO Connection Scenarios and Best Practices

The motherboard designer should address the following requirements for the sake of avoiding failsafe problems, reducing unneeded leakage and for following best practice design rules:

- Level-shifter back-bias prevention
 - Level shifter shall not set value in A side when not getting voltage in B side.
 - Rationale:** Prevent back-bias and wrong logic condition.
 - Level shifters shall be back-bias protected.
 - Rationale:** The level shifter is often supplied with a different supply than the IO connected to it. During ramp up/down states there might be back-bias scenario.

PCIe*

- PCIe* is the standard M.2 interface for Wi-Fi.
- PERST# is used for device error recovery flow management by OS. For this purpose, it is mandatory to use a dedicated GPIO for PERST#.

- PERST# should also be connected to a platform global reset signal.
- Refer to the RVP for implementation details.
- PERST# and PEWAKE# are used for D3 flows it is possible to share these signals with other system components.
 - PERST# signal must be shared. As example Wi-Fi PERST# can be connected to a platform global PCIe reset signal with AND gate implementation.
 - Connected standby flows are using RTD3hot. RTD3hot is not using PERST#, and not using PEWAKE#
 - Non-connected standby flows are using D3cold. D3cold is using PERST# and PEWAKE#.
- If, in S3-S5 the platform puts all the devices to D3cold, and will assert PERST# to all of them, then, as a result PERST# can be shared.
- Wi-Fi core can still be kept active in D3, but in that case, since it is D3cold, Wi-Fi will ignore PERST# even if it puts Wi-Fi into PCIe reset.
- Wi-Fi core can wake the system using PEWAKE# and there is added value in non-CS platforms to have dedicated PEWAKE# signals, but it is not mandatory (depend on Platform decision/needs).
- To conclude, even in non-connected standby flows the signals can be shared.

Connectivity Module Power Control

The platform 3.3V rail shall be supplied to the WLAN NIC in all relevant platform states, per the table below:

Platform ATE	WNIC 3.3V POWER
S0	ON
S0ix	ON
PS-ON	ON
S3/S4/S5	ON
DEEPSX	OFF
PSEUDO G3	OFF

When a load switch (i.e., power switch) is present and WLAN NIC gets a 3.3V rail through this switch.

Power Feed

It is required to have decoupling caps on the power feeds in each end of the connector.

- 10uF+0.1uF+0.01uF at one end of socket in support of 3.3 V3V pins 2 and 4.
- 10uF+0.1uF+0.01uF at the other end of the socket in support of 3.3 V3V pins 72 and 74.
- Prevent antenna cable routing across the decoupling capacitors.

BIOS

The customer will need to apply the relevant Connectivity BIOS objects and methods, along with their appropriate settings, according to the device in use. Wrong values in the BIOS will result in unexpected behavior and degraded performance.

8.5.2 PCIe* Host Interface Errata

NOTE

Below errata topics are not applicable for PCIe Gen4.

This section includes several important implementation aspects about PCIe* host interface. The platform designer should take these items into account as part of the platform design.

Squelch Detect Mechanism

The Squelch (SQ) detect mechanism may not consistently identify Processor wake signaling (TS1 symbols) as valid above SQ Maximum threshold of 175 mV, as defined in the PCIe* specification.

It is important to follow proper platform design and layout guidelines as defined in the PCIe* CEM specification to ensure Processor wake signaling (Electrical Idle Detect Threshold - *Vrx-idle-det-diff-p-p* parameter) in the range of:

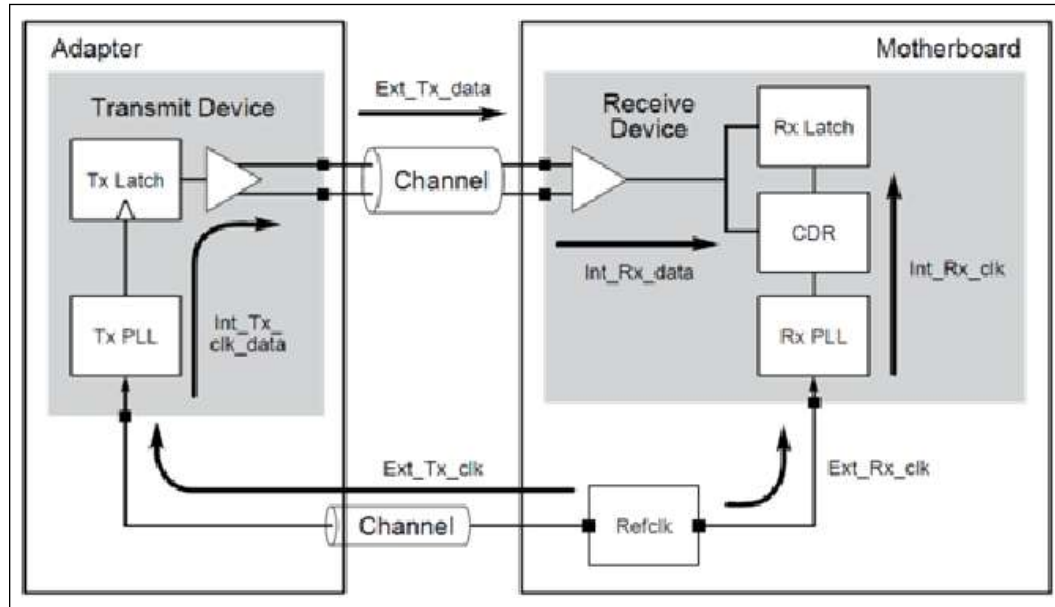
- 65 mV to 280 mV

Customers should design/plan appropriately for all adapters that may be used in a given platform.

PCIe* Common Clock Configuration

PCIe* on both platform side (Processor) and WLAN side must operate using the PCIe* common clock configuration. Both Processor and WLAN are configured for this by default (bit 6 is set in registers 0x50). From hardware perspective, the same reference clock (Refclk) must be used for both Processor and Wi-Fi card, as shown in Figure below.

Figure 362. PCIe* Common Clock Configuration



Source: PCI Express Rev 2.0 spec (PCI_Express_Base_Rev_2.0_20Dec06a, figure 4-50)

Enabling PCIe* Controllers with ASPM

ASPM defines the L states of the PCIe* connections, L0, L0s, L1 and L2.

The device does not support L0s. This is in order to benefit from the power saving that is achieved with L1 state, while avoiding platform integration complexity which is involved with using L0s state.

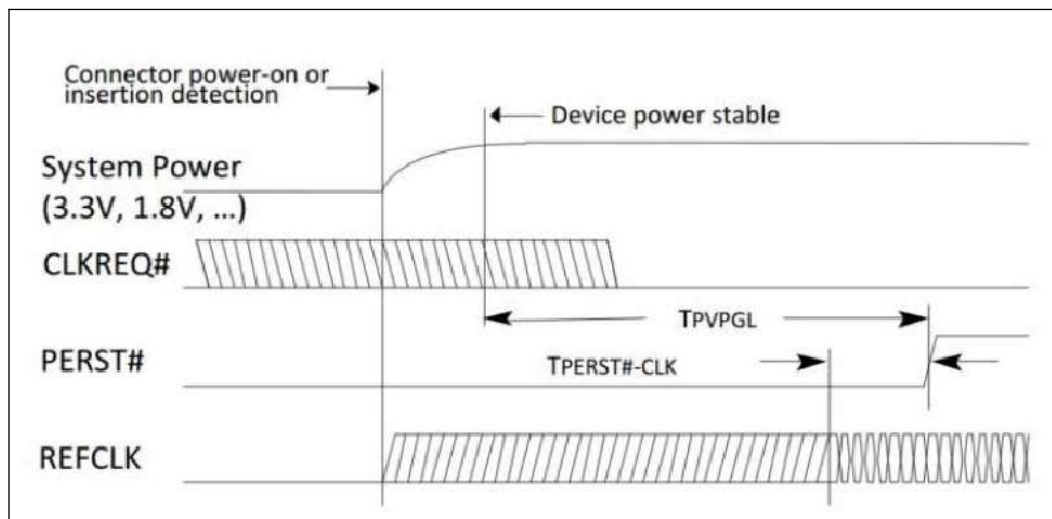
The device supports ASPM optionality ECN, allowing support of L1 without L0s. Therefore, there is no need for special BIOS actions as with previous Intel's wireless products.

CLKREQ# Timing

Latest PCIe* M.2 Specification does not specify a constraint on when PCIe* device should assert CLKREQ# after Power Valid (which is a point where the V3.3 rail reached nominal level).

Refer to the below timing diagram and timing tables taken from PCIe* M.2 specification and as can be seen, CLKREQ# assertion timing is not defined.

Figure 363. PCIe* CLKREQ# Timing



Source: M.2 specification

Table 719. Power-up CLKREQ# Timing

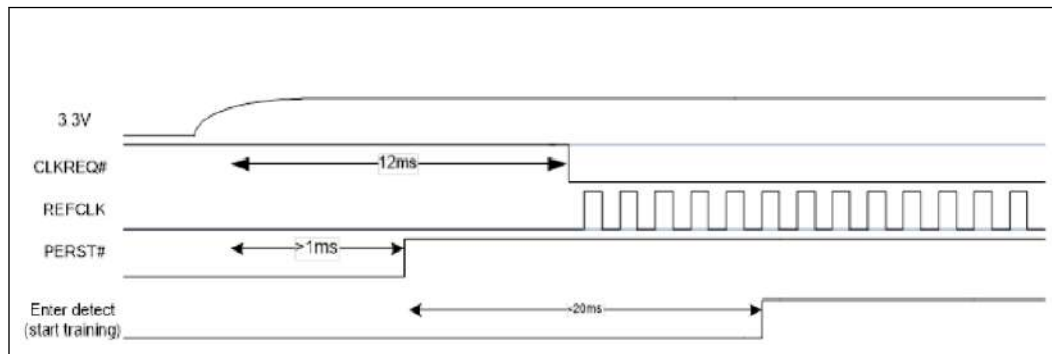
Symbol	Parameter	Min	Max	Units
T_{PVPL}	Power valid to PERST# Input active	Implementation specific		ms
$T_{PERST\#-CLK}$	REFCLK stable before PERST# inactive	100		μ s

WLAN asserts CLKREQ# within 12ms from Power valid as shown in below figure, these devices comply with the requirement to enter DETECT within 20ms of PERST# de-assertion.

NOTE

Refer to M.2 device specification to meet the POWER rail to PERST# timing requirements.

Figure 364. Cold Reset



The PERST# signal rise time needs to meet the limitations listed in below table. This guarantees robust out-of-reset flow of the device and better immunity to noise that can be carried by this signal. In addition, it is required that this signal rise in a monotonic way and avoid a step-like rise.

Power-up PERST# Timing

Table 720. Power-up PERST# Timing

PERST# Rise Time	
Parameter	Intel® Wireless Devices
	Note: Refer the CRF module EPS for details.
PERST# Rise Time	< 150nSec Monotonic rise
PERST# Ripple/Glitch During Rise Time	< 100mV glitch during the rise time phase

PCI Setting for Gen 2x1 (10 Gb/s) and De-emphasis

Some platforms experienced difficulties enumerating the Wi-Fi device after warm boot. The cause of this issue is the ability to choose the right PCI interface speed. A workaround for this issue is to disable the PCI De-emphasis option in the BIOS and set speed to Gen 2x1 (10 Gb/s).

- De-emphasis disabled.
- Speed – AUTO (to allow Gen 2x1 (10 Gb/s) speed).

8.6 Wireless Connectivity Integration (CNVi) Design Considerations

8.6.1 Connectivity Integration (CNVi)

Connectivity integration (CNVi) is a general term referring to a family of connectivity solutions. The common component solution is the CNVi IP , which is a hard macro embedded in various Intel SOC chips.

The CNVi solution also contains an external RF companion module (CRF) and RF antennas. This module can be implemented in the following variations:

1. M.2 (2230)
2. Solder Down M.2 (1216)
3. Solder Down Module

Refer to the following kits:

- Intel® Wi-Fi Companion RF (CRF) Symbol Kit (ZIP) ([787089](#))
- Intel® Wi-Fi M.2 1216 Discrete Symbol Kit (ZIP) ([627347](#))

8.6.2 Integrated Connectivity Concept

Integrated Connectivity (CNVi) is a architecture for wireless connectivity devices. The concept of CNVi is to move a large part of the functional content of the connectivity chip from the radio chip into the Processor. As a result, a large portion of the chip logic

and memory resources is moved out of the radio chip, reducing platform bill of material (BOM) size and cost, and improving accessibility to Processor re-sources (audio, memory, etc.).

In the CNVi architecture, the MACs of the Wi-Fi and Bluetooth®, including processors, logic and memory are relocated from the radio chip into the Processor chip. Signal processing, Analog and RF functions stay in the radio chip, which is called a Companion RF chip (CRF) in CNVi terminology.

The part of the connectivity IP which is ported into the Processor is called CNVi IP. The CNVi IP interfaces with the rest of the Processor functions through Processor-internal interfaces and buses, and does not require any host interfaces at the platform level. On the other hand, inter-facing the CNVi IP and the Companion RF (CRF) chip does require platform signals to be routed between the Processor and the Companion RF chip (Intel-proprietary interfaces, based on existing M.2 signals).

The Integrated Connectivity architecture has the MAC part of the Wi-Fi and BT cores located inside the Processor. As a result, the Host interfaces of Wi-Fi and BT are no longer part of the M.2 module, which is an RF companion module. These Host Interfaces reside in the Processor and are not exposed to the platform.

The figure below shows the Platform-level view of CNVi, including all related system components.

Figure 365. CNVi Platform Block Diagram

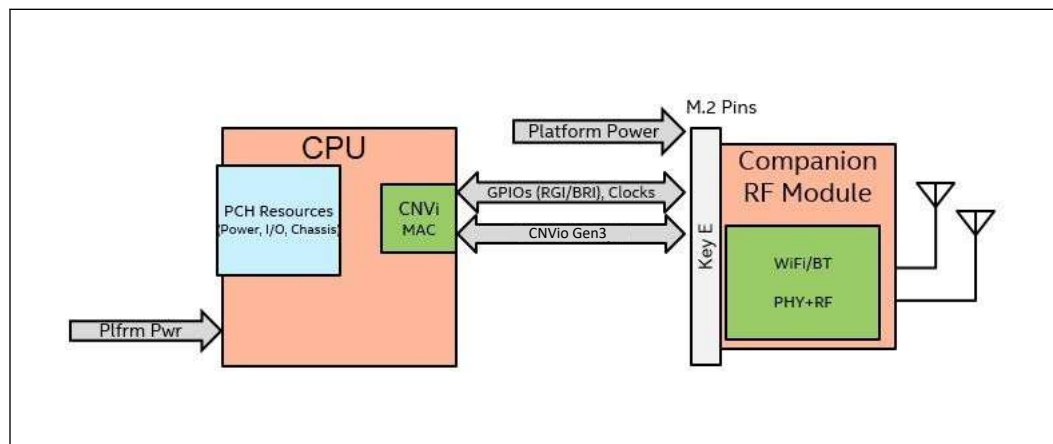
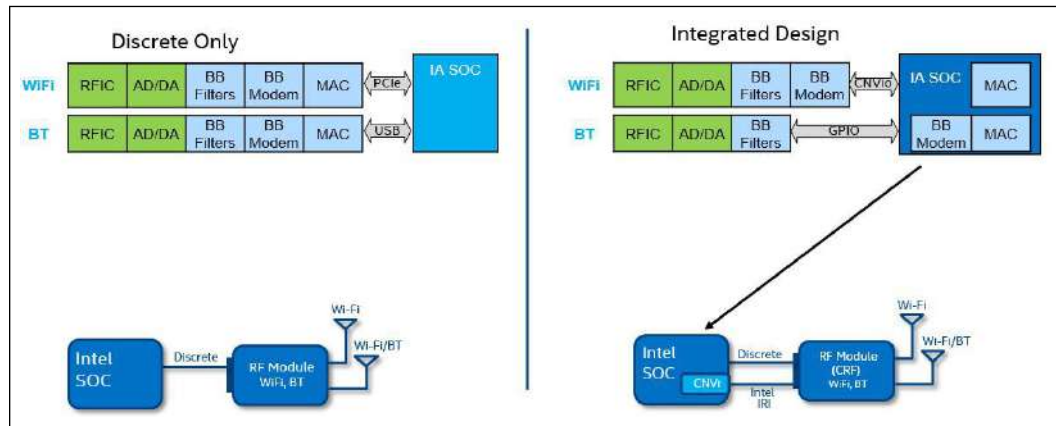


Figure 366. CNVi Discrete vs Integrated Architecture



The CNVi platform system includes the following blocks:

System-on-Chip (SoC)

CNVi IP is integrated with Processor part of Multi-chip module (MCM).

Platform VRs

Provides a single 3.3V rail to the external connectivity circuitry.

CNVi Related Straps

These are pin-straps used on Processor pins. These straps control Processor initialization functions related to CNVi implementation.

Refer "Power-up sequence" Sections for the signals and power-up flow.

Connectivity CRF/Discrete

The connectivity block can have different mechanical implementations depending on the form factor and functionality. Connectivity modules are characterized by the following properties:

- Integration level: Intel RF companion chip (CRF) or Discrete (Intel or TPV)
- Form factor: M.2 2230 (Connectorized module) or M.2 1216 or (Solder down module) Chip-on-board (COB)
- Includes both Wi-Fi and BT Cores
- Wi-Fi streams (depending on the used CRF)
- BT Stream is similar for both Previous generation and New Generation.

Connectivity Antennas

2x2 Configuration include:

- Wi-Fi main antenna
- Shared Wi-Fi/Bluetooth Antenna (combining Wi-Fi chain-1 and Bluetooth signals together)

1x1 Configuration include:

- Wi-Fi/Bluetooth Antenna with Diversity option (can select one of the 2 available antennas).

Table 721. Antenna Connector Functionality

Connector	Functionality
Wi-Fi + BT	ANT1
Wi-Fi	ANT2 (M.2-2230) / ANT3 (M.2-1216)

8.6.3 CNVi Form Factors

The modules described in this section are:

- RF Companion M.2 (2230)
- RF Companion Solder-down M.2 (1216)

RF Companion M.2 (2230)

The RF Companion 2230 module has the same mechanical outline as the standard M.2 connectivity Type 2230-S3-E card.

The standard M.2 Key E connector pin out is modified to accommodate the proprietary RF companion signals, hence called “Hybrid Key-E”. This unique design allows inserting the RF Companion module into a standard M.2 Key E socket.

This feature is only possible when the Motherboard design follows specific guidelines described in this document.

The M.2 socket with the Hybrid Key-E scheme is intended to be used with a proprietary pinout. This scheme is called “Hybrid Key-E” due to the mechanical similarity to a Key-E connector and the ability to support both Companion RF and Discrete modules. When designing a motherboard with this scheme, it is possible to have the same M.2 socket supporting two different connectivity cards:

1. CNVi Companion RF module (CRF)
2. Standard M.2 discrete module (Discrete)

The ability to swap between these cards on the same M.2 socket, while using the same motherboard design, is an important feature desired by PC platform OEMs.

When designing the motherboard M.2 socket and routing according to the Hybrid Key-E scheme, and subject to certain assumptions that will be defined later, the following basic properties are guaranteed:

- Inserting either of the optional cards (CRF, Discrete) to the M.2 socket will be safe (meaning no damage to the motherboard or card will occur).
- Both options can be used and will function as desired, subject to the following:
 1. For CNVi: CNVio/RGI/BRI and few of the CNVi-CRF signals are in place.
 2. For Discrete: all external interfaces (PCIe/USB/Controller Link) are available and connected.

The pinout for the Hybrid Key-E socket on the motherboard is shown in next Figure .The inner columns show the Companion RF proprietary signals at their assigned pins. The Companion RF signals, listed with the prefix “/”, signify that they

are electrically MUX'd inside the Processor, and are shared. Due to this internal Processor sharing, these signals do not require any jumpers to select between the two functions.

NOTE

There are six (6) such signals.

- **Module Pinout**

The RF Companion 2230 module Pinout is shown in Table below.

Table 722. RF Companion Module 2230 Pin List

Pin #	Pin Name Platform Pinout	Pin Name Spec (Product side) Pinout	Pin Name Module Pinout	Direction w/respect to CRF Module	CRF Voltage on Module Side	Connection on Platform/Usage
1	GND		GND			GND
2	3.3 V	3.3 V	3.3 V		3.3 V	3.3 V Supply
3	USB_D+	USB_D+	NC	IO	3.3 V	Not used by CRF Shall be connected to USB for Discrete support
4	3.3 V	3.3 V	3.3 V		3.3 V	3.3 V Supply
5	USB_D-	USB_D-	NC	IO	3.3 V	Not used by CRF Shall be connected to USB for Discrete support
6	LED1#	LED1#	LED1#	O	OD	LED 1 (Main, Wi-Fi)
7	GND		GND			GND
8	PCM_CLK/I2S SCK	PCM_CLK/I2S SCK	NC	IO	1.8 V	Not used by CRF Optional UART+PCM interface when used with Discrete
9	WGR_D1N	SDIO_CLK/ SYSCLK	WGR_D1N	O	CNVio phy	CNVio bus RX lane 1
10	PCM_SYNC/I2S WS/RF_RESET_B	PCM_SYNC/I2S WS	RF_RESET_B	I	1.8 V	The PCM functionality is not used by CRF, but using this signal as RF reset indication (active low, PD) Optional UART+PCM interface when used with Discrete
11	WGR_D1P	SDIO_CMD	WGR_D1P	O	CNVio phy	CNVio bus RX lane 1
12	PCM_IN/I2S SD_IN	PCM_OUT/I2S SD_OUT	NC	O	1.8 V	Not used by CRF Optional UART+PCM interface when used with Discrete
13	GND	SDIO_DATA0	GND			
14	PCM_OUT/I2S SD_OUT	PCM_IN/I2S SD_IN	CLKREQ0	I	1.8 V	CRF: Wake/activity request from Processor side
continued...						

Pin #	Pin Name Platform Pinout	Pin Name Spec (Product side) Pinout	Pin Name Module Pinout	Direction w/respect to CRF Module	CRF Voltage on Module Side	Connection on Platform/Usage
						Optional PCM interface when used with Discrete
15	WGR_D0N	SDIO_DATA1	WGR_D0N	O	CNVio phy	CNVio bus RX lane 0
16	LED2#	LED2#	LED2#	O	OD	LED secondary (BT)
17	WGR_D0P	SDIO_DATA2	WGR_D0P	O	CNVio phy	CNVio bus RX lane 0
18	GND		LNA_EN			This a special purpose test pin of the CRF module. Should be connected to Ground on the platform.
19	GND	GND/SDIO DATA3	GND			
20	UART-Wake	UART WAKE#	NC	O	3.3 V	Not used by CRF Optional UART+PCM interface when used with Discrete
21	WGR_CLKN	SDIO_WAKE#	WGR_CLKN	O	CNVio phy	CNVio bus RX clock
22	UART RXD/ BRI_RSP	UART TXD	BRI_RSP	O	1.8 V	BRI bus RX, PD in CRF Optional UART+PCM interface when used with Discrete
23	WGR_CLKP	SDIO_RESET#	WGR_CLKP	O	CNVio phy	CNVio bus RX clock
24	Connector Key	Connector Key	Module Key			
25	Connector Key	Connector Key	Module Key			
26	Connector Key	Connector Key	Module Key			
27	Connector Key	Connector Key	Module Key			
28	Connector Key	Connector Key	Module Key			
29	Connector Key	Connector Key	Module Key			
30	Connector Key	Connector Key	Module Key			
31	Connector Key	Connector Key	Module Key			
32	UART TXD/ RGI_DT	UART RXD	RGI_DT	I	1.8 V	BRI bus TX, PU in CRF Optional UART+PCM interface when used with Discrete
33	GND		GND			GND
34	UART CTS/ RGI_RSP	UART RTS	RGI_RSP	O	1.8 V	RGI bus RX, PD in CRF Optional UART+PCM interface when used with Discrete
35	PETp0	PERp0	NC		PCIe phy	Not used by Harrison Peak Shall be connected to PCIe for Discrete support
continued...						

Pin #	Pin Name Platform Pinout	Pin Name Spec (Product side) Pinout	Pin Name Module Pinout	Direction w/respect to CRF Module	CRF Voltage on Module Side	Connection on Platform/Usage
						PCIe phy signals.
36	UART RTS/ BRI_DT	UART CTS	BRI_DT	I	1.8 V	BRI bus TX, PU in CRF
37	PETn0	PERn0	NC		PCIe phy	Not used by CRF Shall be connected to PCIe for Discrete support PCIe phy signals.
38	Controller Link RESET	<Vendor Defined> Controller Link RESET	NC			Not used by CRF Optional Controller Link interface when used with Discrete
39	GND		GND			GND
40	Controller Link DATA	<Vendor Defined> Controller Link DATA	NC			Not used by CRF Optional Controller Link interface when used with Discrete
41	PERp0	PETp0	NC		PCIe phy	Not used by CRF Shall be connected to PCIe for Discrete support PCIe phy signals.
42	Controller Link CLK	<Vendor Defined> Controller Link CLK	NC			Not used by CRF Optional Controller Link interface when used with Discrete
43	PERn0	PETn0	NC		PCIe phy	Not used by CRF Shall be connected to PCIe for Discrete support PCIe phy signals.
44	COEX3 (I/O)	COEX3 (I/O)	NC			Not used by CRF Optional Coex interface with LTE modem when used with Discrete
45	GND		GND			GND
46	COEX2	COEX2				Not used by CRF Optional Coex interface with LTE modem when used with Discrete
47	REFCLKP0	REFCLKP0	NC		PCIe phy	Not used by CRF Shall be connected to PCIe for Discrete support PCIe phy signals.
48	COEX1	COEX1	NC			Not used by CRF Optional Coex interface with LTE modem when used with Discrete
continued...						

Pin #	Pin Name Platform Pinout	Pin Name Spec (Product side) Pinout	Pin Name Module Pinout	Direction w/respect to CRF Module	CRF Voltage on Module Side	Connection on Platform/Usage
49	REFCLKN0	REFCLKN0	NC		PCIe phy	Not used by CRF Shall be connected to PCIe for Discrete support PCIe phy signals.
50	SUSCLK (32 kHz)	SUSCLK (32 kHz)	C_P32K (32 kHz)	I	3.3 V	CRF also supports 1.8 V electrical levels on this signal PD in CRF
51	GND		GND			GND
52	PERST0#	PERST0#	NC		3.3 V	Not used by CRF Shall be connected to PCIe for Discrete support
53	CLKREQ0#	CLKREQ0#	NC		3.3 V	PCIe clock request. Not used by CRF Shall be connected to PCIe for Discrete support
54	W_DISABLE2#	W_DISABLE2#	W_DISABLE2#	I	3.3 V	CRF also supports 1.8 V electrical levels on this signal PU in CRF
55	PEWAKE0#	PEWAKE0#	NC		3.3 V	Not used by CRF Shall be connected to PCIe for Discrete support
56	W_DISABLE1#	W_DISABLE1#	W_DISABLE1#	I	3.3 V	CRF also supports 1.8 V electrical levels on this signal PU in CRF
57	GND		GND			GND
58	PER1p0	I2C DATA/ UART_TX	NC		1.8 V	Not used by CRF
59	WT_D1N	WT_D1N <PERp1>	WT_D1N		CNVio phy	CNVio bus TX lane 1
60	PER1n0	I2C CLK/ UART_RX	NC		1.8 V	Not used by CRF
61	WT_D1P	WT_D1P <PER1n1>	WT_D1P	I	CNVio phy	CNVio bus TX lane 1
62	GND	ALERT#/ UART_RTS	NC		1.8 V	Not used by CRF
63	GND		GND			GND
64	PET1p0	REFCLK0/ UART_CTS <Reserve>	REFCLK0	O	1 V	Not used by CRF
65	WT_D0N	WT_D0N	WT_D0N	I	CNVio phy	CNVio bus TX lane 0
continued...						

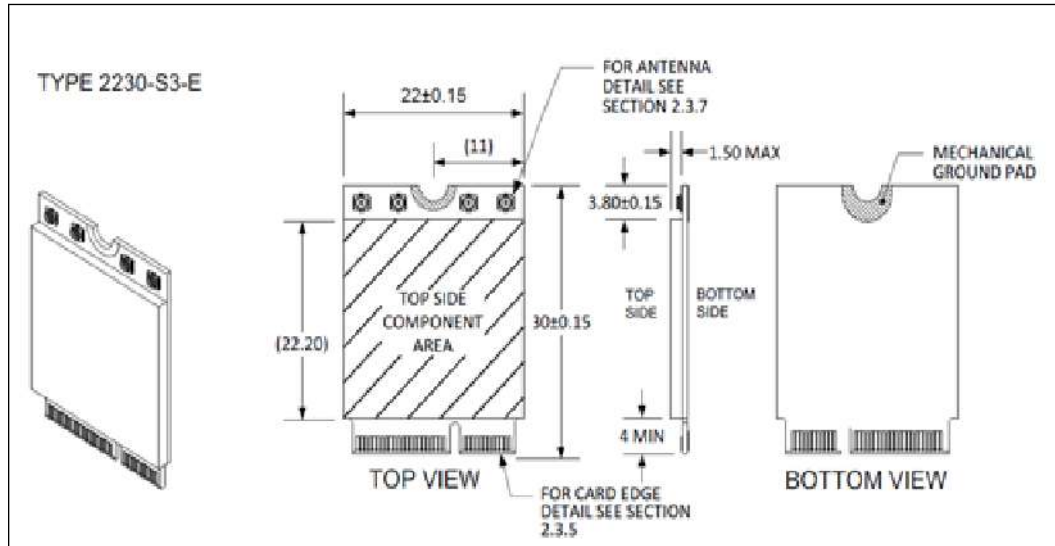
Pin #	Pin Name Platform Pinout	Pin Name Spec (Product side) Pinout	Pin Name Module Pinout	Direction w/ respect to CRF Module	CRF Voltage on Module Side	Connection on Platform/Usage
		<PET1p1>				
66	PET1n0	PERST1#	NC			Not used by CRF
67	WT_D0P	WT_D0P <PET1n1>	WT_D0P	I	CNVio phy	CNVio bus TX lane 0
68	CLKREQ1#	CLKREQ1#	NC			Not used by CRF
69	GND		GND			GND
70	UIM_POWER_ SRC/GPIO1/ PEWAKE1#	UIM_POWER_ SRC/GPIO1/ PEWAKE1#	NA			Not used by CRF
71	WT_CLKN	WT_CLKN <REFCLKP1>	WT_CLKN	I	CNVio phy	CNVio bus TX clock
72	3.3 V	3.3 V	3.3 V			3.3 V Supply
73	WT_CLKP	WT_CLKP <REFCLKN1>	WT_CLKP	I	CNVio phy	CNVio bus TX clock
74	3.3 V	3.3 V	3.3 V			3.3 V Supply
75	GND		GND			GND

Note: PU is Pull up and PD is pull down.

- **Module Mechanical Dimensions**

The M.2 2230 CRF Module mechanical diagram is shown in figure below.

Figure 367. Module Mechanical Diagram



NOTE

Source is PCI-SIG M.2 Specification.

- **Hybrid Key-E (2230) Pin-Out**

The RF Companion M.2 2230 module is intended to be used with a proprietary pin out “Hybrid Key E” scheme due to the mechanical similarity to a Key-E connector.

The pinout for Hybrid Key-E socket on MB is shown in figure below. The CRF signals listed with the prefix “/” signify that they are electrically MUX’d inside the Processor and are shared.

Figure 368. Hybrid Key E Pinout- Platform Side View

Single Key Platform Slot Pinout				
	PS - Bottom			CS - Top
	Standard M.2 Key E	Next Gen ICNV Signals	Next Gen ICNV Signals	Standard M.2 Key E
74	W3P3A		GND	75
72	W3P3A	WT_CLKP	REFCLKN1	73
70	PEWake1# (IO)(0/3.3V)	WT_CLKN	REFCLKP1	71
68	CLKREQ1# (IO)(0/3.3V)		GND	69
66	PERST1# (IO)(0/3.3V)	WT_D0P	PERn1	67
64	RESERVED	WT_D0N	PERp1	65
62	ALERTW (I)(0/1.8V)	REFCLK0 (I)(0/3.3V)	GND	63
60	I2C_CLK (IO)(0/1.8V)	AAWRP_IRQM Not Used	WT_D1P	61
58	I2C_DATA (IO)(0/1.8V)	AAWRP_I2C_CLK Not Used	WT_D1N	59
56	W_DISABLE1# (O)(0/3.3V)		GND	57
54	W_DISABLE2# (O)(0/3.3V)		PEWake0# (IO)(0/3.3V)	55
52	PERSTOP# (O)(0/3.3V)		CLKREQ0# (IO)(0/3.3V)	53
50	SUSCLK (32kHz) (IO)(0/3.3V)	C_P32K (3.3V Tolerant)	GND	51
48	CODEX_TXD (IO)(0/1.8V)		REFCLKN0	49
46	CODEX_RXD (IO)(0/1.8V)		REFCLKP0	47
44	CODEX3 (IO)(0/1.8V)		GND	45
42	CLink CLK		PERn0	43
40	CLink DATA		PERp0	41
38	CLink RESET (O)(0/3.3V)		GND	39
36	LPSS UART RTS (O)(0/1.8V) / BRI_OT (MUX'd in PCH/SoC)		PETn0	37
34	LPSS UART CTS (I)(0/1.8V) / RGI_RSP (MUX'd in PCH/SoC)		PETp0	35
32	LPSS UART Tx (O)(0/1.8V) / RGI_OT (MUX'd in PCH/SoC)		GND	33
E	Connector Key		Connector Key	E
	Connector Key		Connector Key	
	Connector Key		Connector Key	
	Connector Key		Connector Key	
	Connector Key		Connector Key	
22	LPSS UART Rx (I)(0/1.8V) / BRI_RSP (MUX'd in PCH/SoC)	WGR_CLKP	SDIO Reset# (O)(0/1.8V)	23
20	UART Wake# (I)(0/3.3V)	WGR_CLKN	SDIO Wake# (I)(0/1.8V)	21
18	GND	GND	SDIO DAT3 (IO)(0/1.8V)	19
16	LED2# (I)(0/0)	WGR_D0P	SDIO DAT2 (IO)(0/1.8V)	17
14	PCM_OUT (O)(0/1.8V) / CLKREQ0 (MUX'd in PCH/SoC)	WGR_D0N	SDIO DAT1 (IO)(0/1.8V)	15
12	PCM_IN (I)(0/1.8V)	GND	SDIO DAT0 (IO)(0/1.8V)	13
10	PCM_SYNC (IO)(0/1.8V) / RF_RESET_B (MUX'd in PCH/SoC)	WGR_D1P	SDIO CMD# (O)(0/1.8V)	11
8	PCM_CLK (O)(0/1.8V)	WGR_D1N	SDIO CLK# (O)(0/1.8V)	9
6	LED1# (I)(0/0)	GND	GND	7
4				5
2	W3P3A	GND	GND	1

- **Special Considerations for Hybrid Key-E Scheme**

The “Hybrid Key-E” scheme relies on assigning multiple functions to M.2 connector pins and to pins. This causes a significant reduction in the amount of signals that needs to be routed between the Processor and the M.2 module, at the expense of additional dependency between modes, and the loss of some functionality. The dependencies between multiple function pins and M.2 card functionality is described in the next Sections.

- **Shared M.2 Socket Pins**

The following M.2 pins are shared between different functions:

— **V3P3A, GND**

These are the M.2 card power supply (3.3V) and Ground pins, respectively. Both have multiple pins on the connector. These pins have the same purpose in either Discrete or CNVi implementations, and therefore are not affected by the Hybrid Key-E scheme

— **PCIe-1/CNVio**

These are 6 pins which are assigned to the PCIe-1 bus in the M.2 standard pinout. This bus has 3 differential pairs, two for the PCIe data lanes (one per direction) and one for the PCIe clock. In the "Hybrid Key-E" scheme these signals are used for CNVi interface from CNVi to the RF companion chip. Due to this sharing, the "Hybrid Key -E" scheme does not support PCIe-1.

— **SDIO/CNVio**

These are 8 pins which are assigned to the SDIO bus in the M.2 standard pinout. This bus has 8 signals, 4 bi-directional for the SDIO data, one be-directional command signal, one clock (Processor to M.2) and 2 control (Reset Processor to M.2, Wake M.2 to Processor). In the "Hybrid Key-E" scheme these signals are used for CNVi interface the RF companion chip to MAC (6 for CNVi and 2 for ground). Due to this sharing, the "Hybrid Key-E" scheme does not support SDIO.

— **PCM/ClockReq and Reset**

The standard M.2 defines 4 pins for a dedicated PCM audio serial bus for BT. In the "Hybrid Key-E" scheme, two of these signals are used for CRF Clock Request - a wake indication from Processor to CRF and Reset (For Processor to RF companion). Since the PCM serial bus is connected to GPIO pins, and the CNVi clock request and reset pins are also connected to GPIO pins, it is possible to have support for both PCM bus and CNVi signals by changing the GPIO multiplexing function select. Due to this sharing, the "Hybrid Key-E" scheme can still support PCM (for discrete BT connectivity with PCM support).

— **UART (BT) / BRI and RGI**

The standard M.2 defines 4 pins for a dedicated UART serial bus for Bluetooth. In the "Hybrid Key-E" scheme, all these signals are used for CNVi BRI (Bluetooth* Radio Interface) and RGI (Radio Generic interface), each comprising of 2 signals (one per direction). Since the UART serial bus is connected to GPIO pins, and the CNVi BRI, RGI are also connected to GPIO pins, it is possible to have support for both UART bus and CNVi signals by changing the GPIO multiplexing function select. Due to this sharing, the "Hybrid Key-E" scheme can still support BT UART (for discrete connectivity with BT-UART support).

In the CRF, we have Pull-Up on the RGI/BRI _DT and Pull Down on the RGI/BRI_RST signal.

— **SUSCLK/P_32K**

These signals are both functionally similar. In the M.2 standard, this pin is optionally connected to a 32kHz RTC clock.

In the Hybrid Key-E scheme for CNVi, it should be connected to a 32kHz clock. In the CRF, there is an option to use either an internal 32kHz clock or this external clock. The decision can be made by an auto-detect mechanism that checks whether the external 32kHz clock is indeed active (configurable option), and selects it if so, else it will choose the internal clock.

The “price” is additional power consumption in low-power states. If an external clock is used, it MUST be driven by a valid 32kHz clock any time the RF companion module is powered on (no glitches are allowed), and a BIOS/ACPI based indication is needed as well (refer “*Intel Connectivity Platforms BIOS Guidelines*”).

The 32kHz clock can come from a pin or from a different source on the platform, depending on the platform used. Note that the External 32kHz accuracy is assumed to be 20ppm.

This signal is 1.8V based and is tolerant to 3.3V.

— **RF_RESET_B**

This is an Intel proprietary signal between the Processor and the CRF, used as internal RESET indication from the Processor to the CRF used during the init flow of the CNVi based modules. This signal has an internal Pull-Down in module side.

— **ClkReq0 (Pin14 in M.2-2230, A43 in 1216)**

This is an Intel proprietary signal between the Processor and the CRF , used as Wake indication from the Processor to the CRF

This signal has an internal Pull-Down in module side. It is also shared with Optional PCM interface that can be used with Discrete Module solution (Non CRF)

• **Non-shared M.2 Socket Pins**

The functions of these pins impacted by the “Hybrid Key-E” scheme as will be described below:

— **V3P3A, GND**

These are the M.2 card power supply (3.3V) and Ground pins. These pins have the same purpose in either discrete or CNVi implementations and therefore are not affected by the “Hybrid Key-E” scheme.

— **PEWake1#, CLKREQ1#, PERST1#**

These are 3 control signals used by the PCIe-1 bus in standard M.2 cards. Although these signals are not shared with any other function, they have no usage in a “Hybrid Key-E” scheme design. This is because the PCIe-1 bus by itself is not usable (refer PCIe-1/CNVio sharing above).

— **PCIe-0 Bus**

This consists of 6 signals (3 differential pairs) used for PCIe data to and from the Processor, and a single pair used for the PCIe bus clock. In a CNVi RF companion these signals are left unused. In standard discrete these signals are being used as the Wi-Fi bus interface.

— **W_DISABLE1#, W_DISABLE2#**

These are used for Wi-Fi and BT RF-Kill control respectively. Asserting these signals effectively shuts off the RF transmission or the relevant core.

The M.2 modules support receiving a wireless disable (RF-KILL) command through the two RF-KILL pins for turning off Wi-Fi and BT, respectively. These pins can be connected to a platform switch or to Processor GPIOs (recommendation- if possible do not use GPIOs that have Platform impact as “bootstraps” during platform init).

The RF-Kill signals (Wireless disable*) have an internal Pull-Up in HrP side (they are “Active Low” signals).

— **PEWake0#, CLKREQ0#, PERST0#**

These are 3 control signals used by the PCIe-0 bus in standard M.2 cards. They should be routed to Processor pins which are assigned to GPIOs in Discrete or Combo modes.

— Coex UART Interface

This consist of 3 signals (two UART bus signals and one GPIO) used for Wi-Fi - LTE coexistence signaling in the M.2 standard definition.

Since these pins are left unused in the RF companion, they can still be used with a discrete M.2 card even when designing with the "Hybrid Key-E" scheme.

In platforms that are designed to support a "Hybrid Key-E" there should be 3 pins connected to each signal, to allow the modem to connect to the M.2 pin (in the discrete connectivity case).

— CLink Interface

This I/F is relevant, and is a MUST for vPRO based platforms. This consist of 3 signals (clock, data and reset). This bus in an Intel proprietary bus. Since these pins are left unused in the RF companion, but they will still be used with a discrete M.2 card even when designing with the "Hybrid Key-E" scheme on vPRO based platforms.

— LED1, LED2

These are optional pins that are assigned to drive LEDs on the platform in the standard M.2 cards. They are used for the same function when using CNVi and therefore are not affected by the "Hybrid Key-E" scheme.

— USB Bus

The standard M.2 defines 2 pins for a differential USB bus. This has no usage in the RF companion.

When using a standard M.2 discrete card, the pins will have the standard functionality and are not affected by the "Hybrid Key-E" scheme. The different connectivity interfaces as used by a discrete connectivity (Standard M.2, Intel connectivity or TPV), CNVi RF companion card are summarized in table below.

Table 723. Connectivity Interfaces for Different M.2 2230 Cards

M.2 interface	CNVi	Discrete
PCIe-1	M.2 pins are not connected to the CRF. Wi-Fi uses internal IOSF to interface the host.	Used for Wi-Fi host interface
PCIe-2	Not functional Pins are connected to CRF and CNVi IP and can't be used as PCIe.	Not functional Pins are connected to CNVi IP and can't be used as PCIe.
Wi-Fi SDIO	Not functional Pins are connected to CRF and CNVi IP and can't be used as SDIO.	Not functional Pins are connected to CNVi IP and can't be used as SDIO.
Wi-Fi Controller Link	M.2 pins are not connected to the CRF. Wi-Fi uses internal Controller Link to interface the ME.	Used for Wi-Fi CSME interface
Wi-Fi RF-Kill	Used (optional)	Used (optional)
BT USB	M.2 pins are not connected to the CRF. BT uses internal U2U to interface the host.	Used for BT USB interface
continued...		

M.2 interface	CNVi	Discrete
BT UART	M.2 pins are not connected to the CRF. BT uses internal UART to interface the host.	Used for BT UART interface
BT I2S (Audio)	M.2 pins are not connected to the CRF. BT uses internal I2S to interface the host.	Used for BT I2S interface
BT wake	M.2 pin is not connected to the CRF. BT uses internal vGPIO.	Used for BT wake signal
BT RF-Kill	Used (optional)	Used (optional)
UART bus to ISH	Pins are connected to Refclock (38.4M system clock). An internal UART is used to connect Pulsar to ISH	Pins are connected to Refclock (38.4M system clock). Can be used for ISH UART connection.
I2C bus to A4WP	Unused.	Unused.
Power supply	Used per M.2 standard	Used per M.2 standard
GND	Used per M.2 standard	Used per M.2 standard

- **Other General Recommendations**

The Following Pins are not in use and, at least on the Intel RVPs it is recommended to Not Connect or connect to test points:

- Pins 58,60,62,64, 68, 70 will be TP (or disconnected).
- Pin 66 can be connected to the “platform reset” BUT notice that Intel Wireless Products will not respond to this specific signal and will ignore it.

Soldered Down M.2 (SD-1216)

The RF Companion SD-1216 module has the same mechanical outline as the standard M.2 connectivity type 1216-S3. The standard M.2 land pattern is modified to accommodate the proprietary RF Companion Module signals. As opposed to the Connectorized module, the SD-1216 module is not sharing the standard M.2 pads for the RF companion functions. Instead, it has an additional set of solder pads which don't overlap with the standard solder pads. This allows having a single motherboard design that can accommodate either a standard M.2 1216 card or an RF companion 1216 card. However unlike the connectorized case, swapping cards requires removing a soldered down module and can't be done by a simple socket card exchange. Additionally the assembly tooling and BOM should change between a discrete and CNVi motherboards assembly.

- **Soldered Down 1216 Module Pinout**

As explained above, the RF companion module M.2 1216 card has a proprietary set of pads and does no reuse the standard M.2 pads. The special pad-out required for supporting CNVi and discrete on the same motherboard real-estate is shown in figure below. The corresponding pin-description and CRF signals are shown in figure and table below, respectively.

Figure 369. SD-1216 Module Pad-out for Supporting CNVi and Discrete 1216 Modules

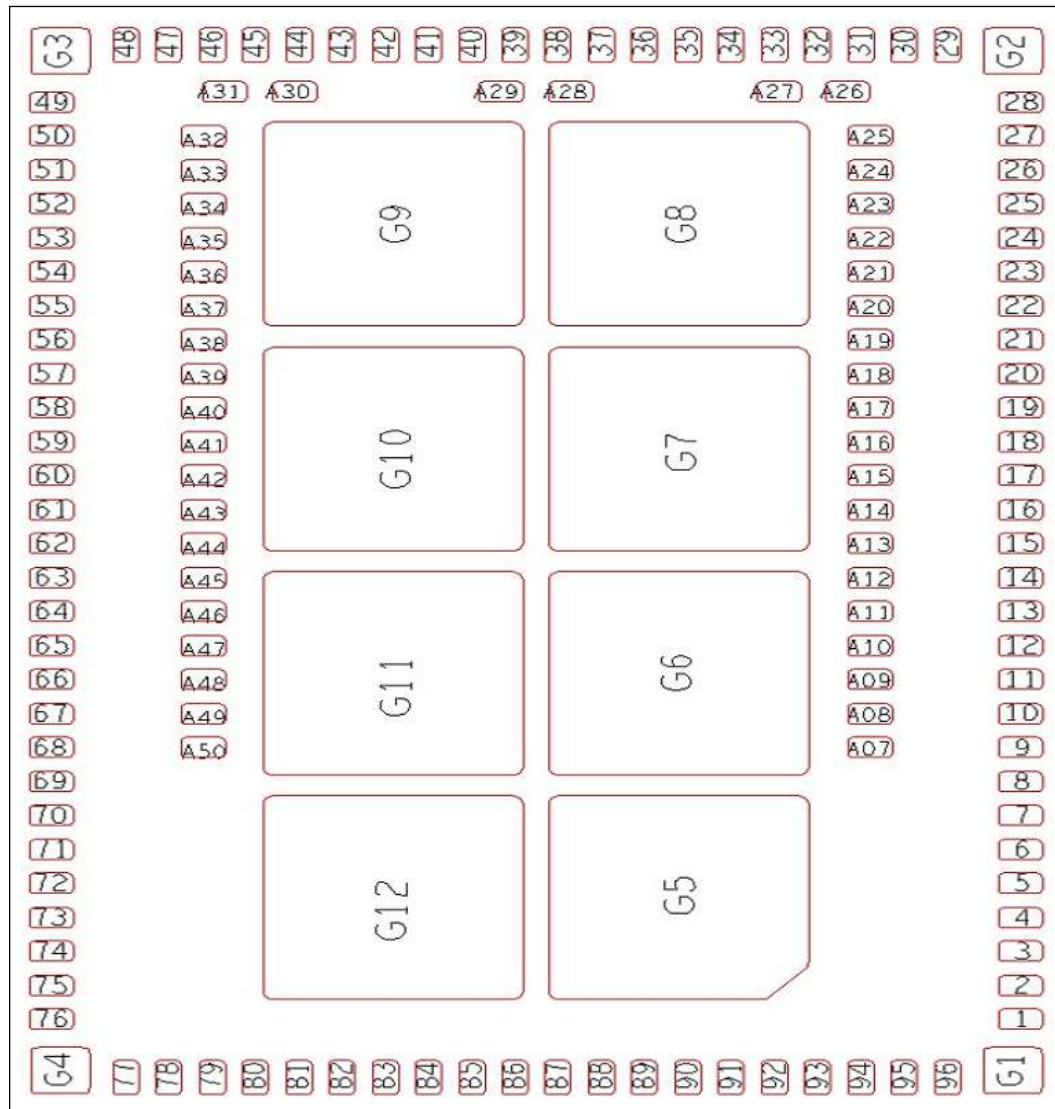
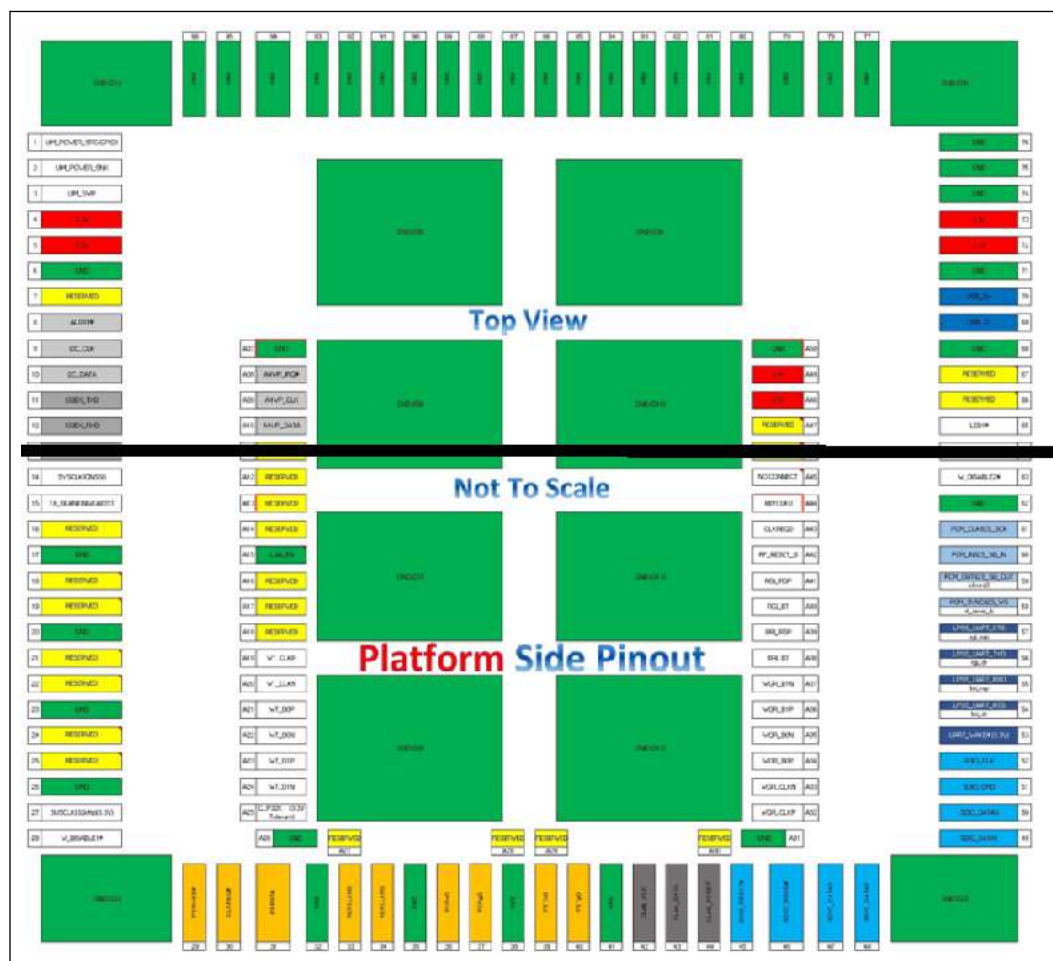


Figure 370. Pin Out Scheme for Dual CNVi/Discrete 1216 Footprint



Refer document #627347 and 630287 for LP pattern guidelines.

Table 724. RF Companion Module 1216 Pin List (CNVi Only)

Pin #	Pin Name	Function When CNVi is Used	Function when Standard M.2 is Used
1	UIM_POWER_SRC/GPIO1	Not used	UIM_POWER_SRC/GPIO1
2	UIM_POWER_SNK	Not used	UIM_POWER_SNK
3	UIM_SWP	Not used	UIM_SWP
4	3.3V	3.3V	3.3V
5	3.3V	3.3V	3.3V
6	GND	GND	GND
7	RESERVED	Not used	RESERVED
8	ALERT#	Not used	ALERT#
9	I2C_CLK	Not used	I2C_CLK

continued...

Pin #	Pin Name	Function When CNVi is Used	Function when Standard M.2 is Used
10	I2C_DATA	Not used	I2C_DATA
11	COEX_TXD	Not used	COEX_TXD
12	COEX_RXD	Not used	COEX_RXD
13	COEX3	Not used	COEX3
14	SYSCLK/GNSS0	Not used	SYSCLK/GNSS0
15	TX_BLANKING/GNSS1	Not used	TX_BLANKING/GNSS1
16	RESERVED	Not used	RESERVED
17	GND	GND	GND
18	RESERVED	Not used	Not used
19	RESERVED	Not used	Not used
20	GND	GND	GND
21	RESERVED	Not used	Not used
22	RESERVED	Not used	Not used
23	GND	GND	GND
24	RESERVED	Not used	Not used
25	RESERVED	Not used	Not used
26	GND	GND	GND
27	SUSCLK(32kHz)(3.3V)	Not used	SUSCLK(32kHz)(3.3V)
28	W_DISABLE1#	W_DISABLE1#	W_DISABLE1#
29	PEWAKE#	Not used	PEWAKE#
30	CLKREQ#	Not used	CLKREQ#
31	PERST#	Not used	PERST#
32	GND	GND	GND
33	REFCLKN0	Not used	REFCLKN0
34	REFCLKP0	Not used	REFCLKP0
35	GND	GND	GND
36	PERn0	Not used	PERn0
37	PERp0	Not used	PERp0
38	GND	GND	GND
39	PETn0	Not used	PETn0
40	PETp0	Not used	PETp0
41	GND	GND	GND
42	CLink CLK	Not used	CLink CLK
43	CLink DATA	Not used	CLink DATA
44	CLink RESET	Not used	CLink RESET
continued...			

Pin #	Pin Name	Function When CNVi is Used	Function when Standard M.2 is Used
45	SDIO_RESET#	Not used	Not used
46	SDIO_WAKE#	Not used	Not used
47	SDIO_DATA3	Not used	Not used
48	SDIO_DATA2	Not used	Not used
49	SDIO_DATA1	Not used	Not used
50	SDIO_DATA0	Not used	Not used
51	SDIO_CMD	Not used	Not used
52	SDIO_CLK	Not used	Not used
53	UART_WAKE# (3.3V)	Not used	UART_WAKE# (3.3V)
54	LPSS_UART_RTS/bri_dt	Not used	LPSS_UART_RTS/bri_dt
55	LPSS_UART_RXD/bri_rsp	Not used	LPSS_UART_RXD/bri_rsp
56	LPSS_UART_TXD/rgi_dt	Not used	LPSS_UART_TXD/rgi_dt
57	LPSS_UART_CTS/rgi_rsp	Not used	LPSS_UART_CTS/rgi_rsp
58	PCM_SYNC/I2S_WS	Not used	PCM_SYNC/I2S_WS
59	PCM_OUT/I2S_SD_OUT	Not used	PCM_OUT/I2S_SD_OUT
60	PCM_IN/I2S_SD_IN	Not used	PCM_IN/I2S_SD_IN
61	PCM_CLK/I2S_SCK	Not used	PCM_CLK/I2S_SCK
62	GND	GND	GND
63	W_DISABLE2#	W_DISABLE2#	W_DISABLE2#
64	LED2#	LED2#	LED2#
65	LED1#	LED1#	LED1#
66	RESERVED	Not used	RESERVED
67	RESERVED	Not used	RESERVED
68	GND	GND	GND
69	USB_D-	Not used	USB_D-
70	USB_D+	Not used	USB_D+
71	GND	GND	GND
72	3.3V	3.3V	3.3V
73	3.3V	3.3V	3.3V
74	GND	GND	GND
75	GND	GND	GND
76	GND	GND	GND
77	GND	GND	GND
78	GND	GND	GND
79	GND	GND	GND
<i>continued...</i>			

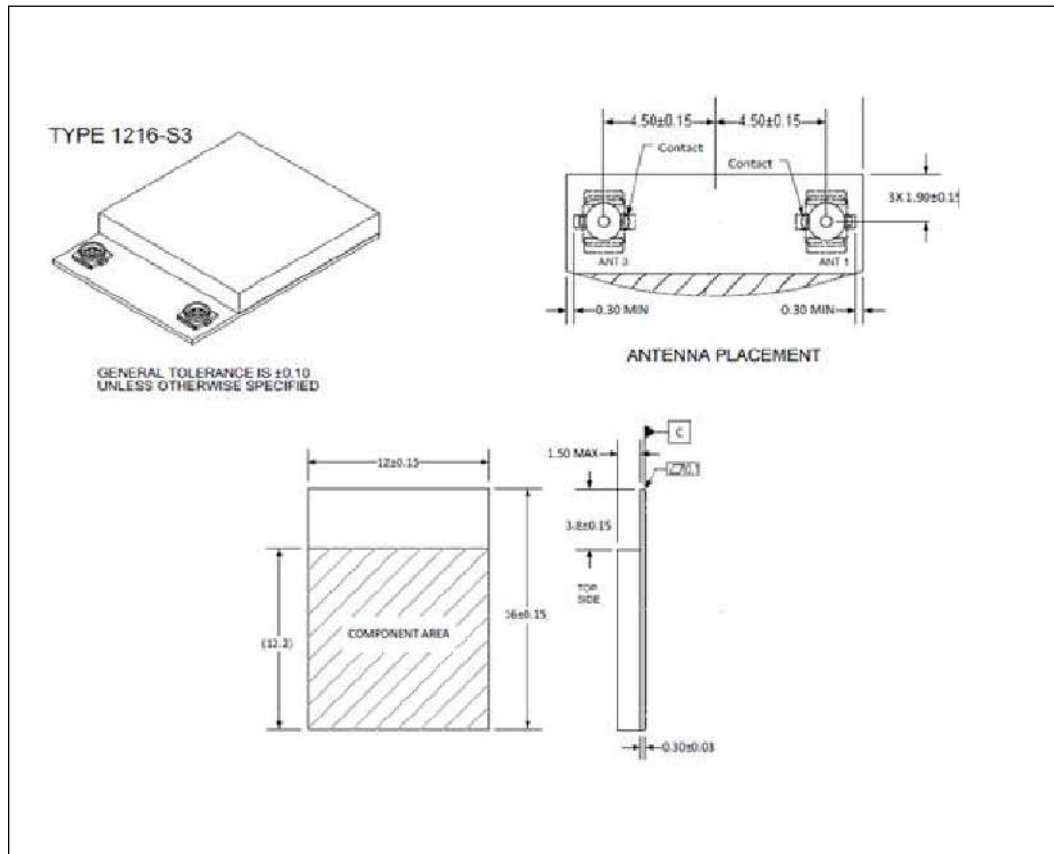
Pin #	Pin Name	Function When CNVi is Used	Function when Standard M.2 is Used
80	GND	GND	GND
81	GND	GND	GND
82	GND	GND	GND
83	GND	GND	GND
84	GND	GND	GND
85	GND	GND	GND
86	GND	GND	GND
87	GND	GND	GND
88	GND	GND	GND
89	GND	GND	GND
90	GND	GND	GND
91	GND	GND	GND
92	GND	GND	GND
93	GND	GND	GND
94	GND	GND	GND
95	GND	GND	GND
96	GND	GND	GND
G1	GND	GND	GND
G2	GND	GND	GND
G3	GND	GND	GND
G4	GND	GND	GND
G5	GND	GND	GND
G6	GND	GND	GND
G7	GND	GND	GND
G8	GND	GND	GND
G9	GND	GND	GND
G10	GND	GND	GND
G11	GND	GND	GND
G12	GND	GND	GND
A07	GND	GND	Not Applicable
A08	A4WP_IRQ#	PU – Pull Up PD – Pull Down Not Applicable	Not Applicable
A09	A4WP_CLK	PU – Pull Up PD – Pull Down Not Applicable	Not Applicable
A10	A4WP_DATA	PU – Pull Up	Not Applicable
continued...			

Pin #	Pin Name	Function When CNVi is Used	Function when Standard M.2 is Used
		PD – Pull Down Not Applicable	
A11	RESERVED	RESERVED	Not Applicable
A12	RESERVED	RESERVED	Not Applicable
A13	RESERVED	RESERVED	Not Applicable
A14	RESERVED	RESERVED	Not Applicable
A15	LNA_EN	GND	Not Applicable
A16	RESERVED	RESERVED	Not Applicable
A17	RESERVED	RESERVED	Not Applicable
A18	RESERVED	RESERVED	Not Applicable
A19	WT_CLKP	WT_CLKP	Not Applicable
A20	WT_CLKN	WT_CLKN	Not Applicable
A21	WT_D0P	WT_D0P	Not Applicable
A22	WT_D0N	WT_D0N	Not Applicable
A23	WT_D1P	WT_D1P	Not Applicable
A24	WT_D1N	WT_D1N	Not Applicable
A25	C_P32K	C_P32K	Not Applicable
A26	GND	GND	Not Applicable
A27	RESERVED	RESERVED	Not Applicable
A28	RESERVED	RESERVED	Not Applicable
A29	RESERVED	RESERVED	Not Applicable
A30	RESERVED	RESERVED	Not Applicable
A31	GND	GND	Not Applicable
A32	WGR_CLKP	WGR_CLKP	Not Applicable
A33	WGR_CLKN	WGR_CLKN	Not Applicable
A34	WGR_D0P	WGR_D0P	Not Applicable
A35	WGR_D0N	WGR_D0N	Not Applicable
A36	WGR_D1P	WGR_D1P	Not Applicable
A37	WGR_D1N	WGR_D1N	Not Applicable
A38	BRI_DT	BRI_DT	Not Applicable
A39	BRI_RSP	BRI_RSP	Not Applicable
A40	RGI_DT	RGI_DT	Not Applicable
A41	RGI_RSP	RGI_RSP	Not Applicable
A42	RF_RESET_B	RF_RESET_B	Not Applicable
A43	CLKREQ0	Disconnected/floating on M.2 side	Not Applicable
continued...			

Pin #	Pin Name	Function When CNVi is Used	Function when Standard M.2 is Used
A44	REFCLK0	REFCLK0	Not Applicable
A45	NO CONNECT	NO CONNECT	Not Applicable
A46	RESERVED	RESERVED	Not Applicable
A47	RESERVED	RESERVED	Not Applicable
A48	3.3V	No connect	Not Applicable
A49	3.3V	No connect	Not Applicable
A50	GND	GND	Not Applicable

- Soldered Down 1216 Module Mechanical Dimensions**

Figure 371. 1216 Module Mechanical Diagram



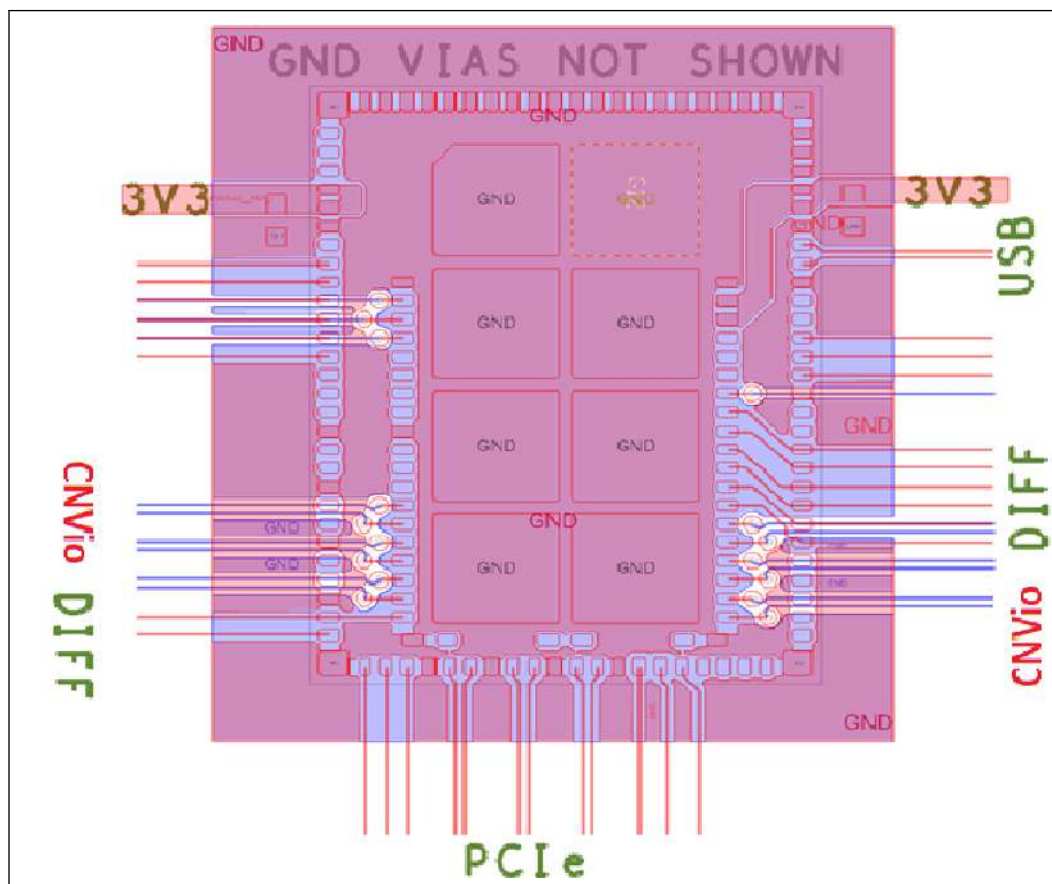
- Breakout Example for Soldered Down Module**

The soldered down modules have a special pad shape which combines the standard M.2 pad ring on the outside with the new inner ring of the CNVi pads. It is recommended to consider all signal properties when designing the motherboard for dual discrete/CNVi design. An example for the breakout layout for a type-3 board (with no micro-via) is shown in the figure below.

NOTE

In order to keep the picture clear, this example shows only two signal layers (top layer and 3rd layer) while the other layers are not shown. Also there is an assumption that the second layer shall be ground.

Figure 372. Board Layout Example Showing Breakout from 1216 Pads



8.6.4 Platform Considerations

Selecting Connectivity Solution

The platform motherboard can be designed to support discrete connectivity (either Intel or TPV), integrated connectivity (CNVi).

NOTE

All CNVi 2230 SKUs can support a “Hybrid Key-E” routing with no jumpers. When considering a TPV module one should only use a Key-E module if the motherboard design is “Hybrid Key-E”. A Key-A module will not fit into this scheme.

NOTE

Refer to document titled "601177-CCG 5Q Wireless Roadmap/Platform Matrix Customer Communication" for the RF Companion POR related to the platform.

NOTE

Refer to CRF/Discrete module EPS (External Product Specification) for information on the CRF/Discrete module Generation SKU supported on the platform.

8.6.5 Signal Connection Pitfalls

The OEM should make sure to follow the M.2 definitions of signal names and directions (I/O, Tx/Rx, etc.) and avoid confusion between the platform side and device side.

NOTE

Few lines are bidirectional, such as PCIe CLKREQ, PEWAKE.

8.6.6 Internal USB Port

Specifically for NVL-H, the USB port assigned to discrete solution is Port number 4.

NOTE

Bluetooth* transfer is exclusively supported over internal PCIe only.

8.6.7 Pull-ups and Pull-downs

Internal/Integrated PU/PD Settings

The OEM should consider the pull-ups and pull-downs on the CRF card, as described in below table.

Table 725. Socket 1 Pull-ups and Pull-downs

I/F	Signals	PU/PD in CNVr (CRF)	PU/PU in CNVi (Processor)	Comments
BRI/RGI (BT UART)	RGI_DT/ BRI_DT	PU (~120-150Kohm), RGI_DT shall be applied with 1K pull down during power-on Init	-	Shared with BT UART Expected power wasted while active with 100K <32uW at each pin RGI_DT is used by the platform to strap the presence of the CRF, as such it is a strong pull-down by the CRF (1K) as long as RF_RESET_B = 0 BRI_RSP is used by the CNVi to strap the CRF type; i.e, when to apply PLL speed: when BRI_RSP = 0 – PLL at
	RGI_RSP/ BRI_RSP	none	PU = 20Kohm This can be set by the BIOS after boot-up; needs to be enabled ONLY if CNVi is not used in the platform	

continued...

I/F	Signals	PU/PD in CNVr (CRF)	PU/PU in CNVi (Processor)	Comments
				1280M (JfP), when BRI_RSP = 1 – PLL at 1320 (HrP and other future parts)
W_Disable* #	Wi-Fi/BT RF Kill	PU (~110–130Kohm)	-	
Slow CLK	SUSCLK (32kHz)	PD (~100Kohm)	-	If available on platform
Init Signals	RF_RESET_B	PD (~100Kohm)	-	Shared with PCM_SYNC
	CLKREQ0	PD (~200Kohm)	-	Shared with PCM_OUT

Platform PU/PD Requirements

The OEM should apply pull-ups and pull-downs on the platform side according to below table.

Table 726. Platform Pull-up and Pull-down Requirements

I/F	Signals	PU/PD in Platform	Comments
BRI/RGI (BT UART)	RGI_DT	PU (20kohm)	This pull is required so that the Processor will be able to reliably detect that the CRF is present at power-up. However, it is possible to increase the resistor to 50K or even to 100K instead of 20K.
Init Signals Init Signals	RF_RESET_B	PD (75Kohm)	It is highly encouraged to increase this resistor (or allow to switch it off when CNVi is active; not sure this is possible at the platform level). This resistor consumes power (43uW) all the time.

The OEM must avoid using a specific PU/PD when not needed or when required to not be used. Unless this rule is followed, a back-bias condition will result, where the IO is getting voltage before the device side is ready for it.

8.6.8 IO Connection Scenarios and Best Practices

The motherboard designer should address the following requirements for the sake of avoiding failsafe problems, reducing unneeded leakage and for following best practice design rules:

- Level-shifter back-bias prevention
 - Level shifter shall not set value in A side when not getting voltage in B side.
 - **Rationale:** Prevent back bias and wrong logic condition.
 - Level shifters shall be back-bias protected.
 - **Rationale:** The level shifter is often supplied with a different supply than the IO connected to it. During ramp up/down states there might be back-bias scenario.

8.6.9 Connectivity Module Power Control

When designing the platform for CNVi, it is recommended to either not have this switch in the design, or to hard-wire it to be always-on by following the Processor (ON when the Processor is ON). This is because the operation of the RF companion module does not allow switching off its main power in any system state where the Processor is powered on, in order to maintain synchronization with the at all times.

Customer Implementation for WLAN Module	CRF 3.3V Power Rail on WLAN Module
CNVi only	Always ON
Discrete PCIe* only	Always ON
CNVi/Discrete PCIe*	Always ON

NOTE

1. CRF module was designed with low-leakage power, and so therefore does not require any external power control to be used during normal operation.

8.6.10 Wi-Fi Wireless Disable and RF-Kill

W_DISABLE1# (pin 56 in M.2 2230 pinout, pin 28 in M.2 1216 SD pinout) serves as HW RF-Kill for the Wi-Fi radio. The pin is recommended to be left unconnected if, and only if, the HW RF-Kill signal is not required.

Asserting the W_DISABLE#_1 signal will result in a complete shutdown of the RF of the Wi-Fi part. The result from the user perspective is similar to that of disabling the Wi-Fi device from the laptop.

NOTE

HW RF-Kill is part of the spec of the M.2 modules.

On some platforms this signal is also mapped as a GPIO; if so, it is recommended to not use the GPIO that is mapped to this signal for any Boot-Stepping function without considering the electrical behavior of this signal.

8.6.11 M.2 Bluetooth® HW RF-Kill

W_DISABLE2# (pin 54 in M.2 2230 pinout, pin 63 in M.2 1216 SD pinout) serves as HW RF-Kill for the Bluetooth® radio.

Asserting W_DISABLE#_2 signal will result in a complete shutdown of the Bluetooth® part. The result from the user perspective is similar to that of removing the Bluetooth® device from the platform.

It is recommended to not use the GPIO that is mapped to this signal for any Boot-Stepping function without considering the electrical behavior of this signal.

8.6.12 Bluetooth* Over USB or PCIe Option

NOTE

Bluetooth* transfer is exclusively supported over internal PCIe only. If PCIe interface to be selected, BIOS sets the bt_pcie register bit and BT_EN bit should remain in default value. The selection of PCIe interface will bring the interface out of reset and perform all relevant PCIe interface readiness preparations.

8.6.13 Power Supply De-coupling

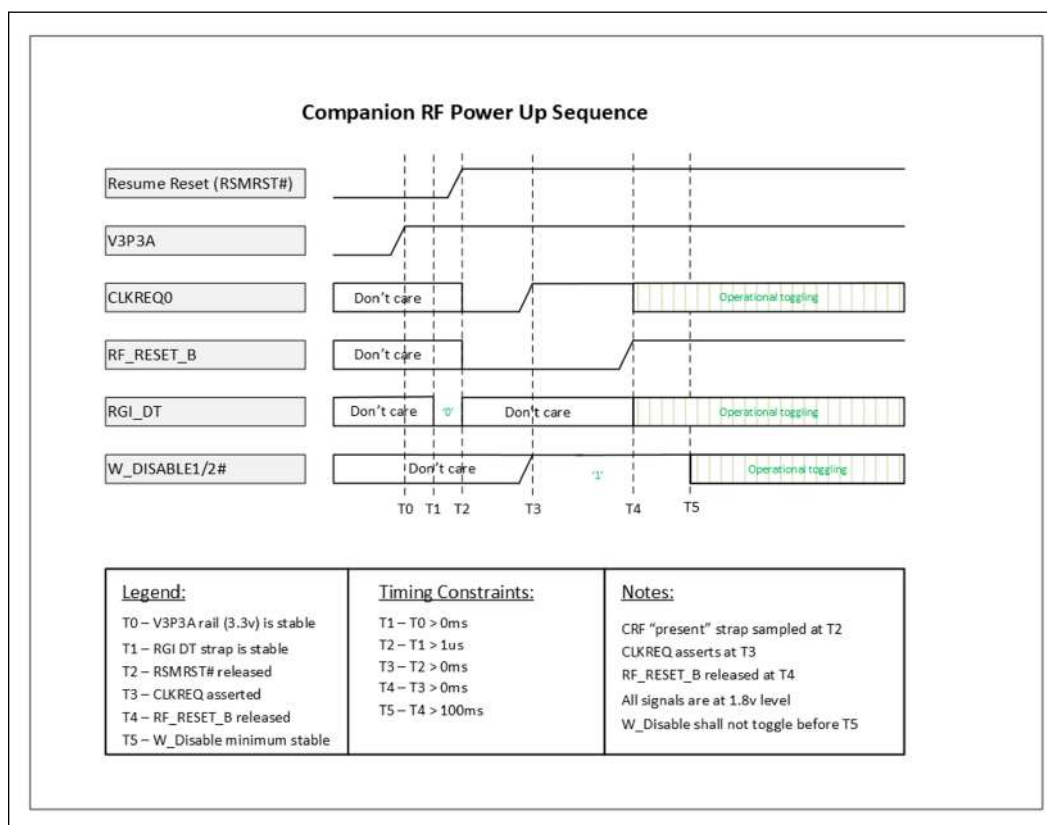
It is required to have decoupling caps on the power feeds in each end of the connector.

- 10uF+0.1uF+0.01uF at one end of socket in support of 3.3 V pins 2 and 4 (in 2230 modules) or pins 4 and 5 (in 1216 modules)
- 10uF+0.1uF+0.01uF at the other end of the socket in support of 3.3 V pins 72 and 74 (in 2230 modules) or pin 72 and 73 (in 1216 modules)
- Prevent antenna cable routing across the decoupling capacitors.

8.6.14 BIOS

The CNVi modules require specific BIOS support. A full description of BIOS support needed for platform can be found in the Connectivity BIOS Guide documents.

8.6.15 CNVi Power Up Sequence



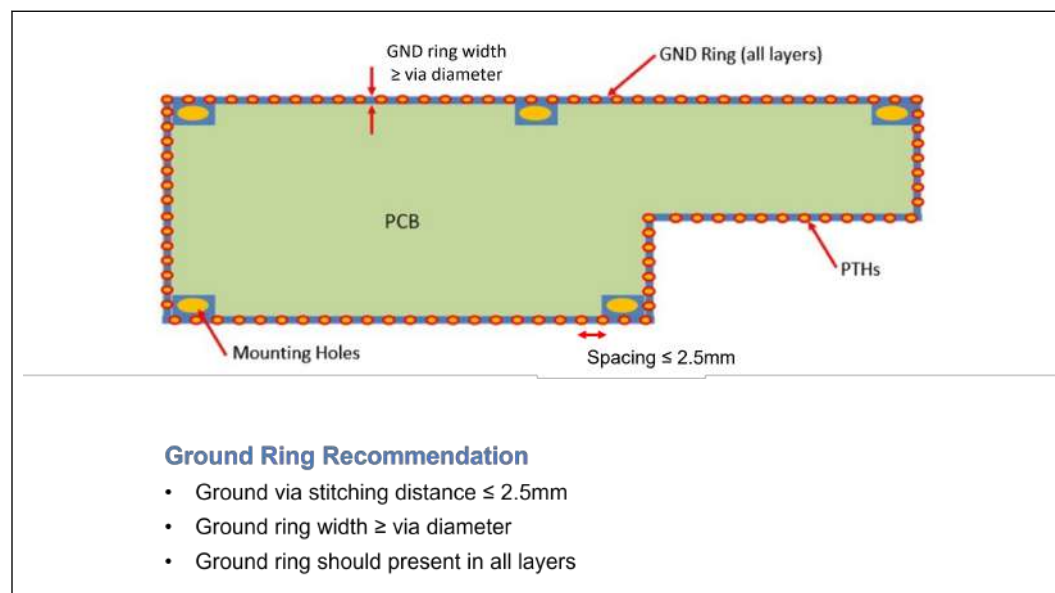
9.0 Electromagnetic Compatibility

9.1 EMC General Considerations

9.1.1 Ground Ring

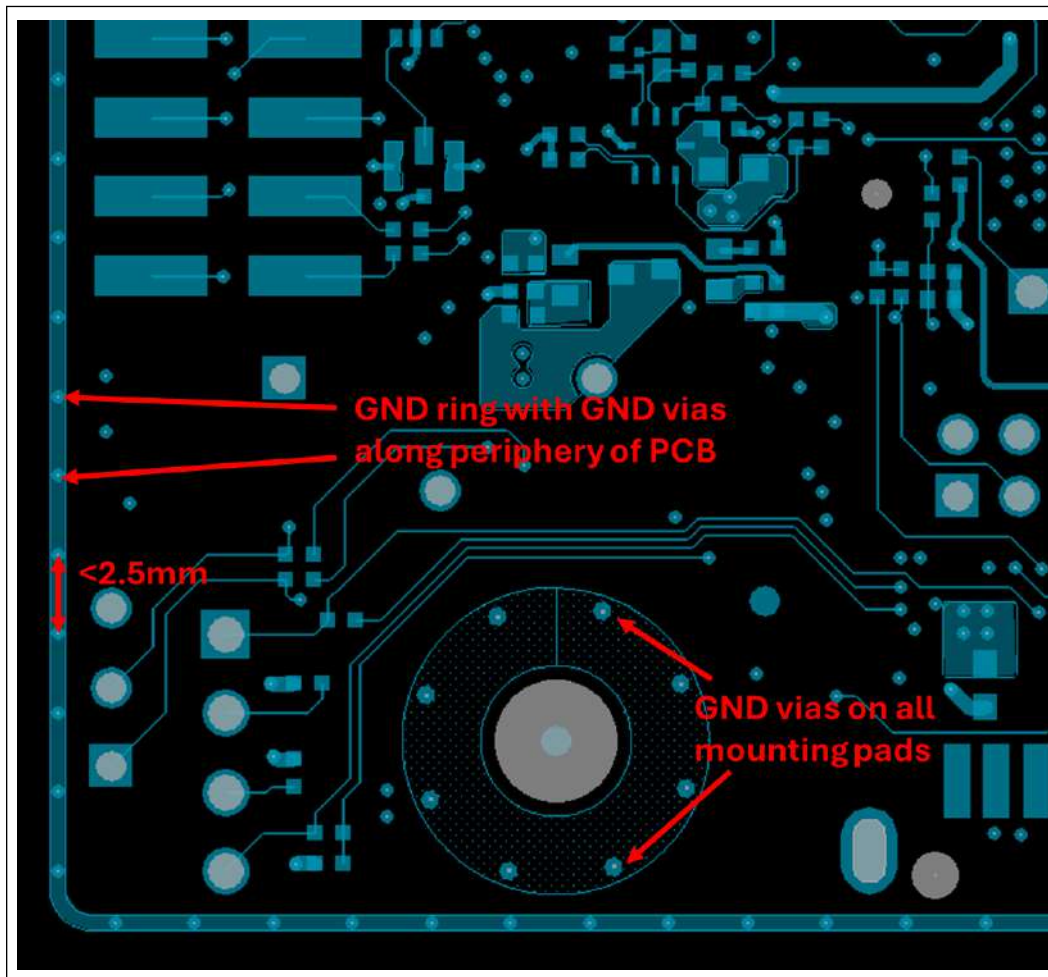
PCB ground (GND) rings on all layers with GND PTH vias are recommended along the edge of the board. Multiple mounting holes should be used connecting the PCB GND to chassis ground. It can significantly reduce the PCB edge coupled noise.

Figure 373. Guidelines for PCB GND Ring and GND Stitching Vias



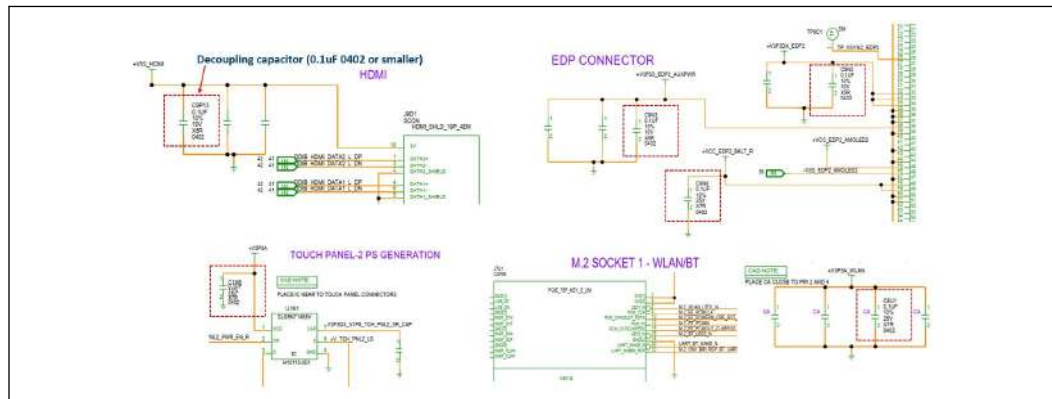
Multiple GND stitching vias are required on all mechanical mounting pads including for internal device modules (e.g. SSD, WiFi module) and I/O connectors (e.g. USB, DP) throughout the platform.

Figure 374. Ground Ring with Ground Vias



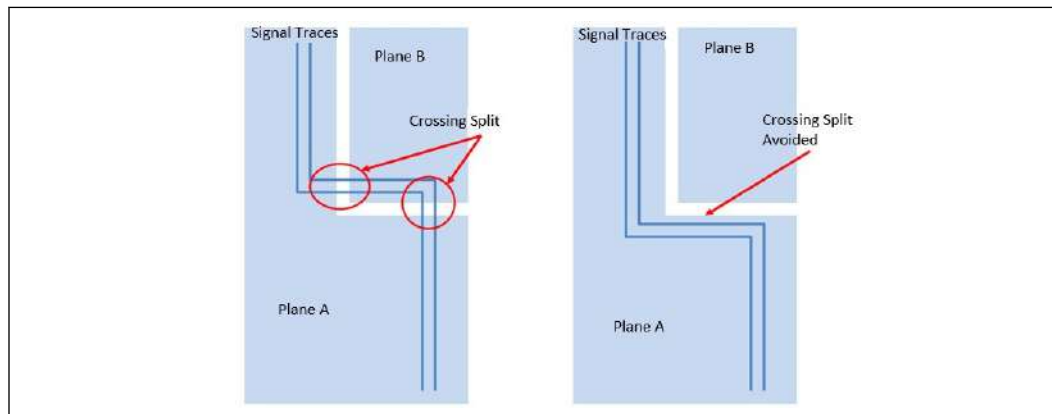
9.1.2 Connector Decoupling

Place a decoupling capacitor (0.1 uF, 0402 or smaller) on the power pins of all internal and external connectors to reduce power noise. These connectors include HDMI, DP, eDP, USB, Type C, FPC connectors, M.2, PCIe edge connector, etc.

Figure 375. Decoupling Capacitors Placement on Connector Power Pins

9.1.3 Crossing Splits in Reference Plane

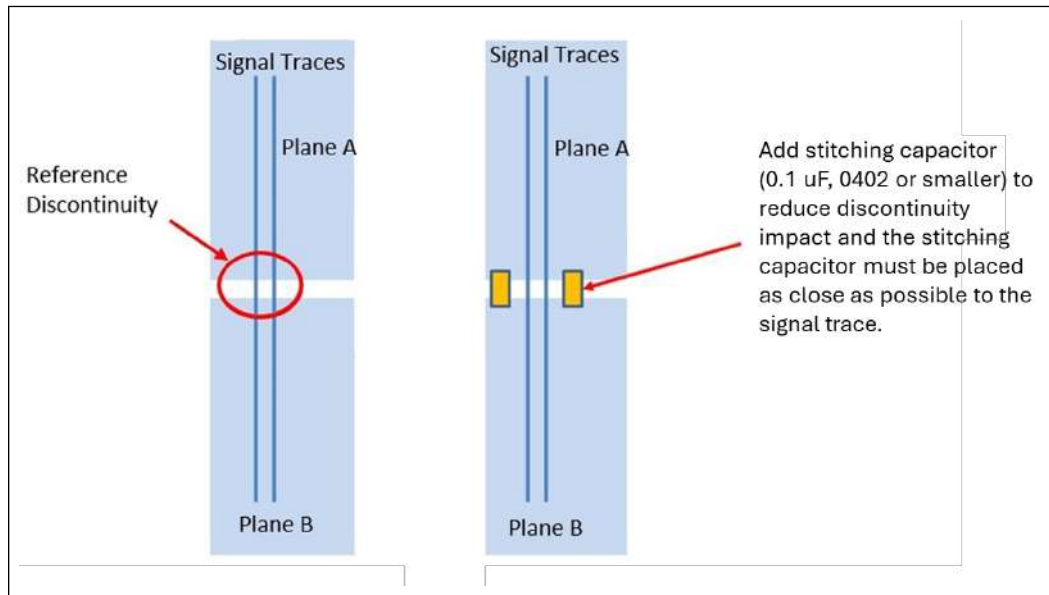
Crossing a split in the reference plane increases emissions from signal traces. Avoid crossing plane splits by moving the traces or reshape the reference plane.

Figure 376. Avoid Signal Traces Routings Across Split Planes

9.1.4 Reference Discontinuity

Changing reference plane from GND to power plane, or from power plane A to power plane B, may significantly increase noise emission. It is recommended to have stitching capacitors (0.1 uF, 0402 or smaller) to bridge the two reference planes and provide current return path and the stitching capacitor must be placed as close as possible to the signal trace.

Figure 377. Reduce Signal Referencing Discontinuities Impact



9.2 Crystal RF Immunity

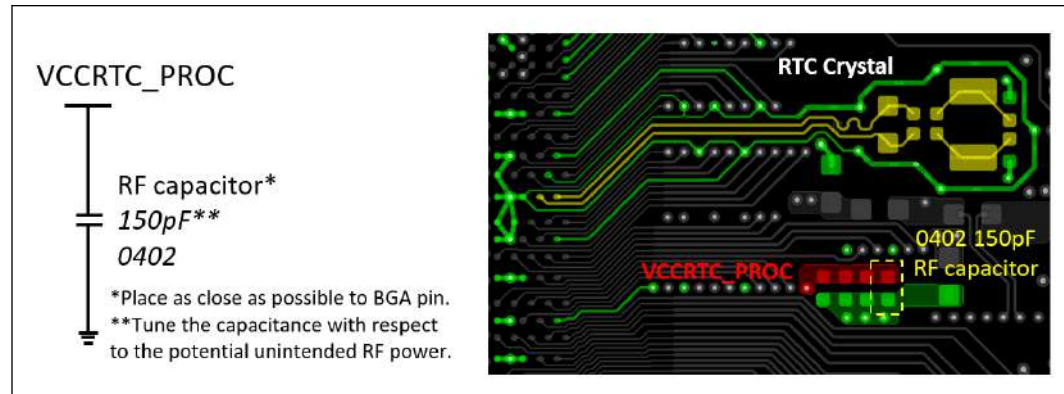
The crystal oscillator that provides the input clock to the processor has been identified to be very sensitive to RF power from nearby radios. Radiated power from internal WWAN and WLAN antennas as well as from external WWAN/ WLAN devices placed near the platform can couple to the crystal and corrupt the clock signal causing issues such as screen flickering and system hang. Similarly, the crystal used with the Thunderbolt retimer is very sensitive to RF power, and noise from nearby antennas can result in connection loss with the device plugged into the Thunderbolt port. Refer to retimer collaterals in the "EMC/RF Reference Documents" chapter for XTAL specifications. The following RF Immunity Guidelines apply to both the 38.4 MHz crystal, RTC (real-time clock) crystal as well as the Thunderbolt retimer crystal.

- Follow crystal routing guidelines very carefully. (Refer to "XTAL" section under "High Speed I/Os" chapter).
- Implement a ground ring surrounding crystal and related components. Refer to the figure below. Do not connect external load caps to GND ring, but rather directly to GND plane on adjacent layer.
- Use a shielded crystal component.
- Ensure that clock and high-speed I/O traces are not routed in proximity to the crystal components and traces (XTAL_IN/OUT, RTC_IN/OUT).
- Place crystal far away from antennas and potential RF power coupling paths such as unshielded/ungrounded cables. The exact distance needed depends on the platform design. Platforms without board shield may have higher risk.
- Chassis shielding and design plays a critical role in RF immunity. Refer to the "EMC Mechanical Considerations" chapter and collateral #602554.
- Do not place the crystal component on the PCB layout that is facing the keyboard, touchpad or other input devices.
- 0 Ohm resistors R1 and R2 are required as potential stuffing options.

The following RF guidelines apply to RTC (real-time) crystal:

- Reserve a placeholder for the RF cap on the power plane for RTC. The capacitance of the stuffed RF cap can be tuned with respect to the potential unintended RF power. The footprint should be as close as possible to processor side.

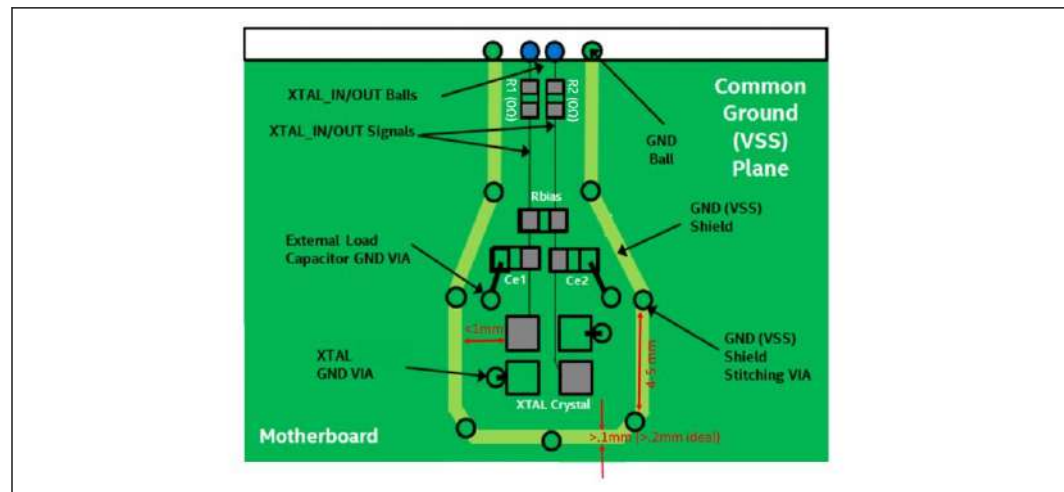
Figure 378. RF Capacitor Placement Recommendation for RTC Crystal



9.2.1 Crystal Ground Ring Requirements

Adding a ground ring surrounding the crystal component, related components and routing can help reduce RF immunity risk.

Figure 379. Guidelines for Ground Ring Implementation for RF Immunity



9.3 Memory RFI Mitigation

Memory interface and DRAM devices generate narrowband and broadband RFI noise. The noise may interfere with radio bands. The following guidelines help to reduce the RFI risk due to memory.

9.3.1 Memory RFI Mitigation Consideration

Table 727. Summary of Memory RFI Mitigation Consideration

Design Area	Recommendations
Memory signals on Type-4 board	Full stripline routing required
Memory signals main routing on Type-3 boards	Main routing full stripline required
Memory signals breakout routing on Type-3 boards	DQ microstrip breakout signal = 6.5 mm CLK, WCK, DQS, CA microstrip breakout signal = 9 mm
Memory device (SODIMM, solder down memory, LPCAMM2) shielding	Shield requirement depends on antenna placement and system design. Refer to the PDG chapter "On-Board Shielding".
Embed memory power plane in inner layers of the board	Follow "Power Integrity Recommendations" for memory power plane closely to embed these power planes. If embedding memory power plane is not possible, recommend either to shield exposed power plane or to add placeholders for RF capacitors.
DDR Radio Frequency Interference Mitigation (RFIM) Feature	Recommended. Refer to RFIM white paper (Document # 640438)
Antenna barricade technology	Recommended. Refer to "Antenna Barricade technology" section under "EMC Mechanical Considerations" PDG chapter or contact Intel for more information.

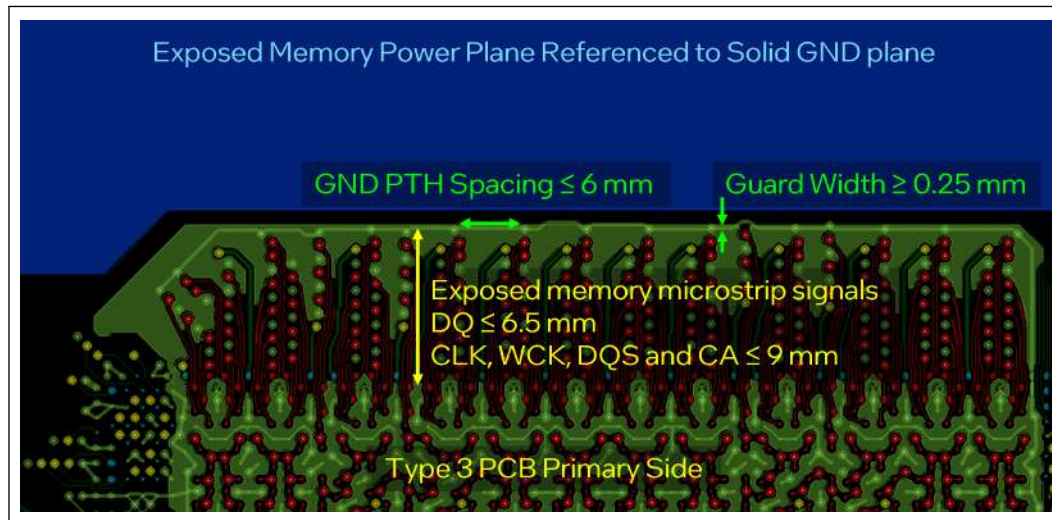
9.3.2 Memory Signals Design Guidelines

For Type-4 board, it is recommended that memory signals (DQ/ CLK/ DQS/ WCK/ CA) are fully routed in inner layers. For Type-3 board, it is recommended that main route of memory signals is routed as stripline; with minimum microstrip breakout and break-in length allowed, as shown in table below. An isolation ground guard ring is recommended on the surface layer between memory power plane and memory breakout signal routing.

Table 728. Memory Signals Microstrip Trace Breakout Length Requirement for Type-3 board

Signal Types	Total Microstrip Breakout Length (mm)
DQ	<= 6.5
CLK, WCK, DQS, CA	<= 9

* Break-in microstrip length should be ≤ 1.5 mm based on system dependency.

Figure 380. RFI Suppression Guidelines on Memory Signal and Power Plane Exposure

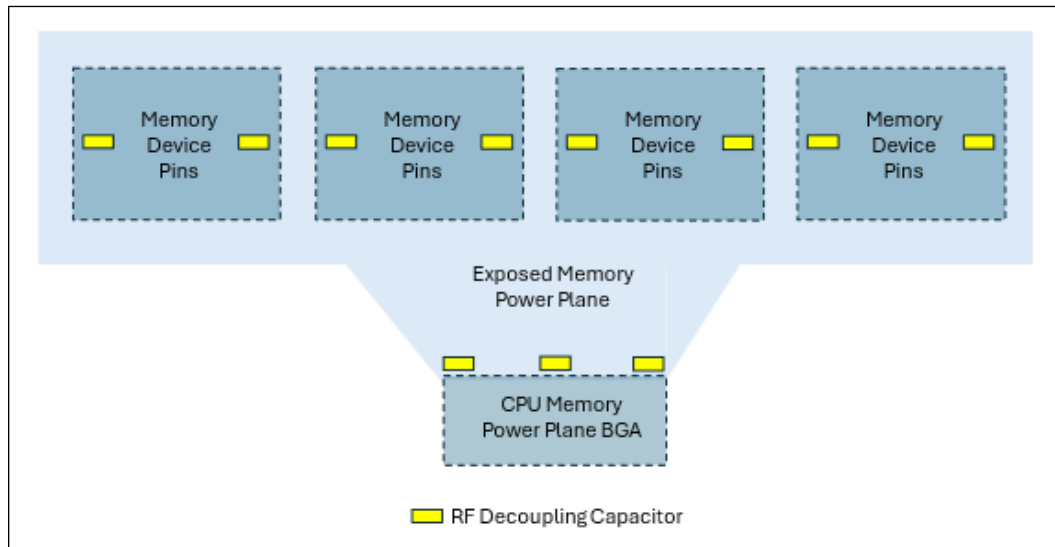
9.3.3 Memory Power Plane Routing Guidelines

It is strongly recommended to embed memory power planes in inner PCB layers with reference to solid ground on adjacent layers. Prioritizing embedding VDDQ over VDD2. For more details, refer to “Power Integrity Recommendations” to embed these power planes. Minimize exposure of memory power plane size on top and bottom layer of the board and keeping minimum routing for BGA and capacitors only.

If embedding memory power plane is not possible, please consider either one of the following recommendations:

- Option 1: Consider covering exposed memory power planes with shield or absorber.
- Option 2: Consider adding placeholders for platform RF capacitors to mitigate power plane noise. Refer to the figures and guidelines below.

Figure 381. Exposed Memory Power Plane RF Decoupling Capacitors



RF decoupling capacitors placement in processor area:

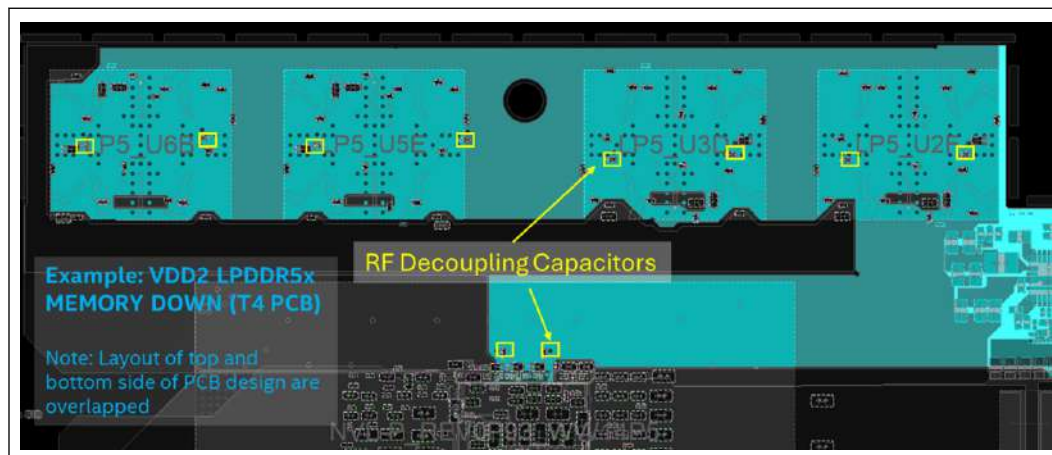
- 12pF (0402) or 15pF (0201)
- < 4 mm proximity from BGA of processor memory power plane
- 2 to 3 capacitors are recommended
- Reserve as placeholder (capacitors unstuffed)

RF decoupling capacitors placement in memory module area:

- 12pF (0402) or 15pF (0201)
- < 4 mm proximity from pins of memory devices
- At least 2 capacitors per memory module
- Reserve as placeholder (capacitors unstuffed)

2.4 GHz band effective RF capacitor values are to be found by RF interference measurements as their effectiveness depends on RF capacitor size, locations, number, and pad and via parasitic. Begin with RF capacitor values tested on Intel reference board and refine them further as needed.

The figures below show an example of RF decoupling capacitor placement on VDD2 LPDDR5x memory down in Type4 PCB:

Figure 382. RF Decoupling Capacitor Placement on VDD2 LPDDR5x Memory Down in Type 4 PCB

9.3.4 Memory Devices

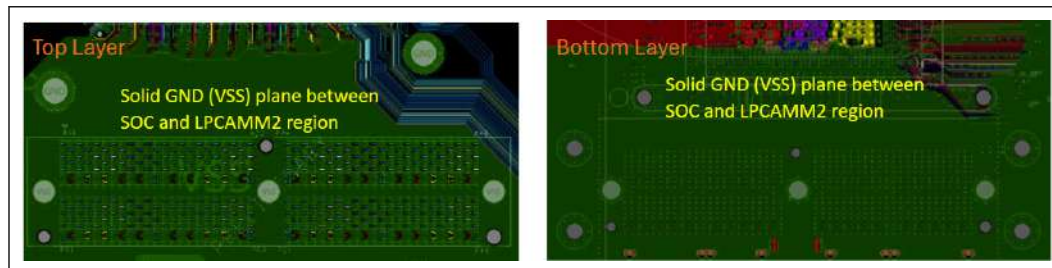
Memory devices (ex. DRAM, SODIMM, LPCAMM2) can radiate broadband and narrowband RFI noise and may cause radio frequency interference. A combination of EMI shield and antenna placement is one of the solutions to mitigate noise from memory devices.

SODIMM memory devices can cause higher RFI noise than the soldered down DRAM devices and it is recommended to use an EMI shield for the SODIMM memory devices and/or use the DDR RFI Mitigation (RFIM) feature.

Refer to the sections “On-Board Shielding” and “Intel DDR RFI Mitigation (RFIM) Software Feature” for more information.

LPCAMM2 PCB Design Guidelines

1. Avoid exposed power planes
 - Avoid exposed memory power plane on top and bottom side of motherboard. Distribute GND stitching vias across the GND plane.

Figure 383. Avoid Exposed Memory Power Plane on Top and Bottom Side of Motherboard

2. LPCAMM2 back plate grounding and module retention screws
 - For improved RFI performance, it is recommended to ensure a ground connection between the LPCAMM2 module and the PCB.

- Expose ground pads with multiple ground stitching vias around mechanical mounting/screw holes on the bottom side of the motherboard PCB (to allow ground contact of the back plate).
- A back plate with a custom nut (or EMI washer, alternatively) is recommended to ensure ground connection between back plate and PCB bottom layer ground.
- The recommended back plate and threaded insert/custom nut dimensions are the following figure (in mm).
- Conductive screws without any coating/paint are recommended to secure the module to the motherboard. The screw's dimensions guidelines (in mm) are based on a recommended LPCAMM2 module-level shield.

Figure 384. Provide solder mask opening for GND (VSS) around mounting holes

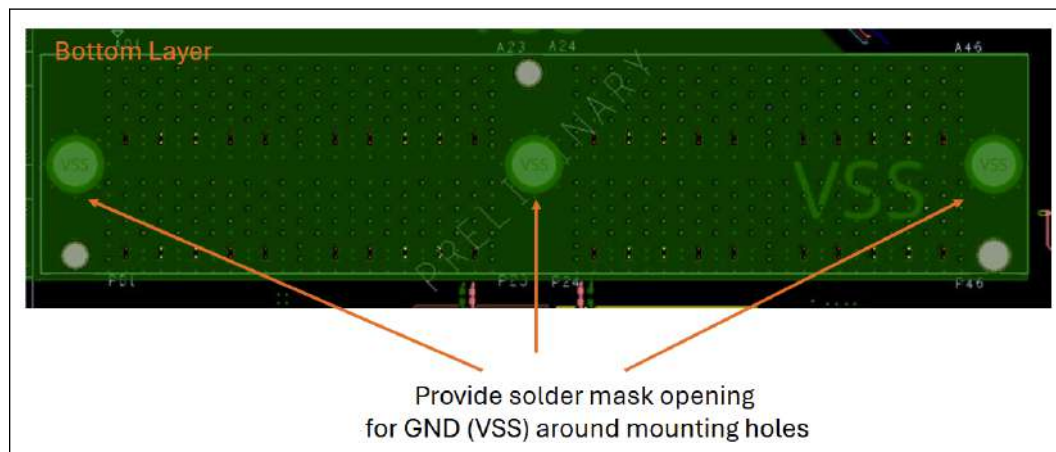


Figure 385. Back plate with threaded insert or nut for ground contact to PCB

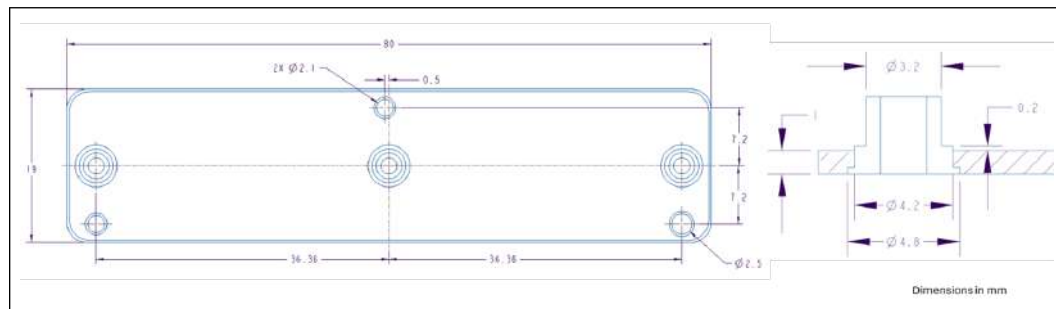
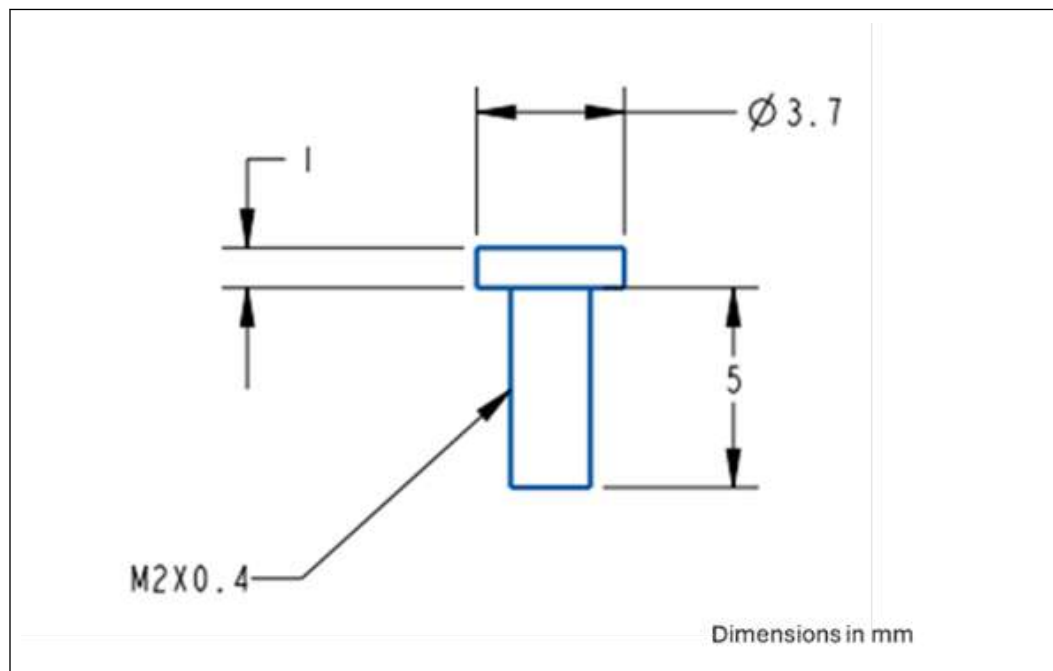


Figure 386. Recommended LPCAMM2 module screw dimensions

3. LPCAMM2 Module-level Shield

- A module-level EMI shield can be installed on the LPCAMM2 module as illustrated in the pictures below. The shield must make ground contact to the ground ring on the periphery of the module. A top plate/EMI shield with conductive foam gaskets can be used to achieve the ground contact of the shield with the module PCB. The figures below show the recommended dimensions for the shield and conductive gaskets.

Figure 387. Optional on-module shield for LPCAMM2

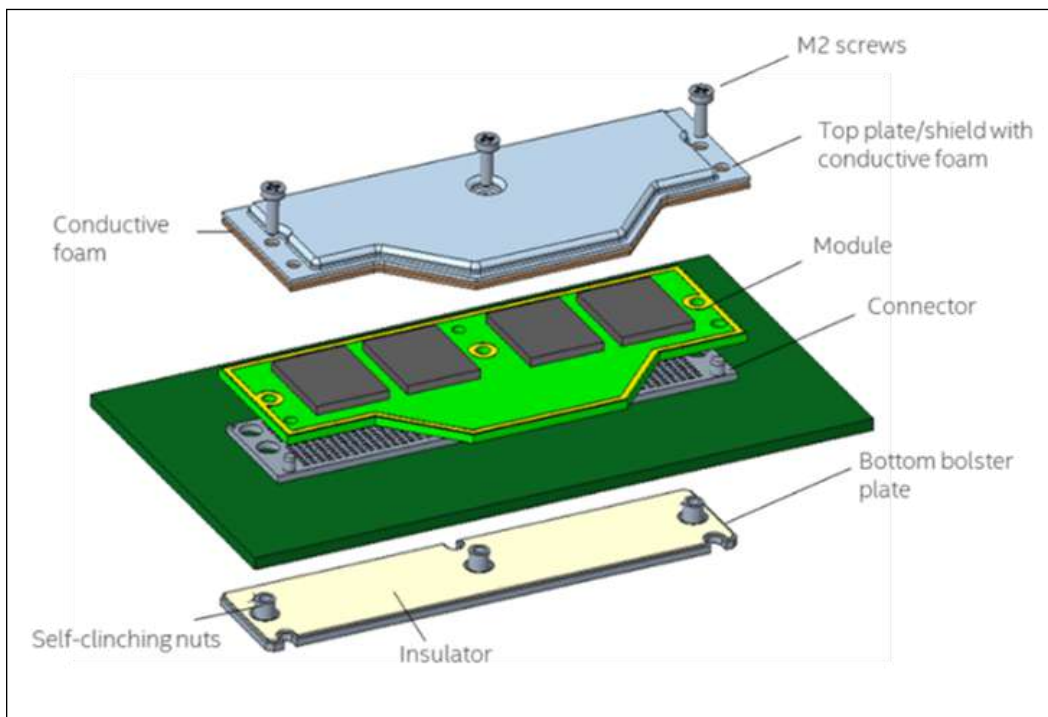


Figure 388. Recommended dimensions for LPCAMM2 on-module shield

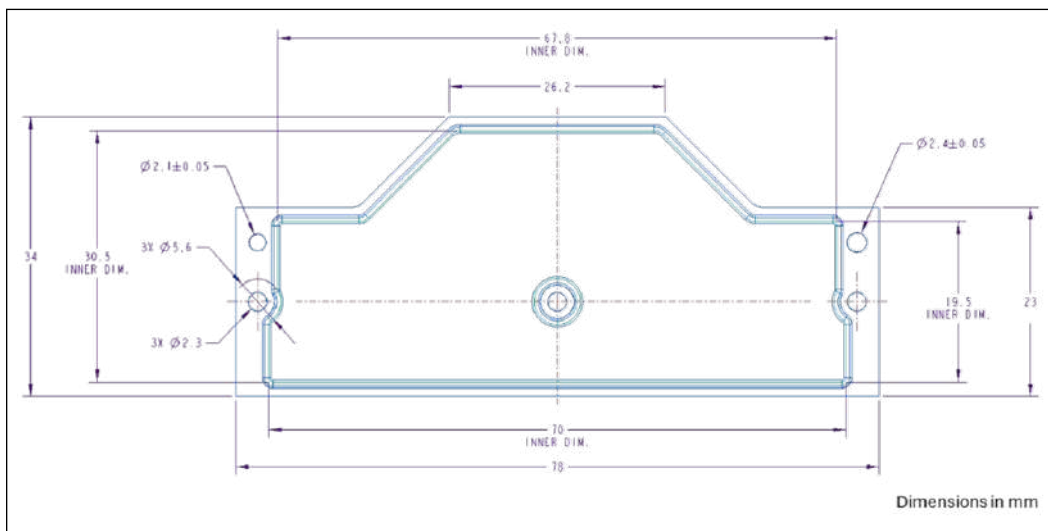
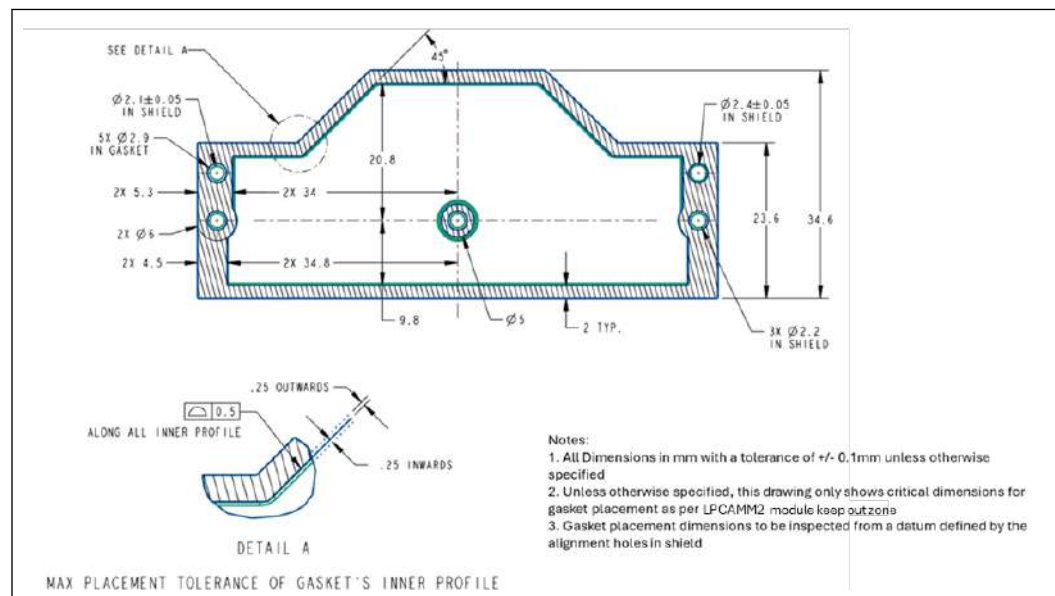


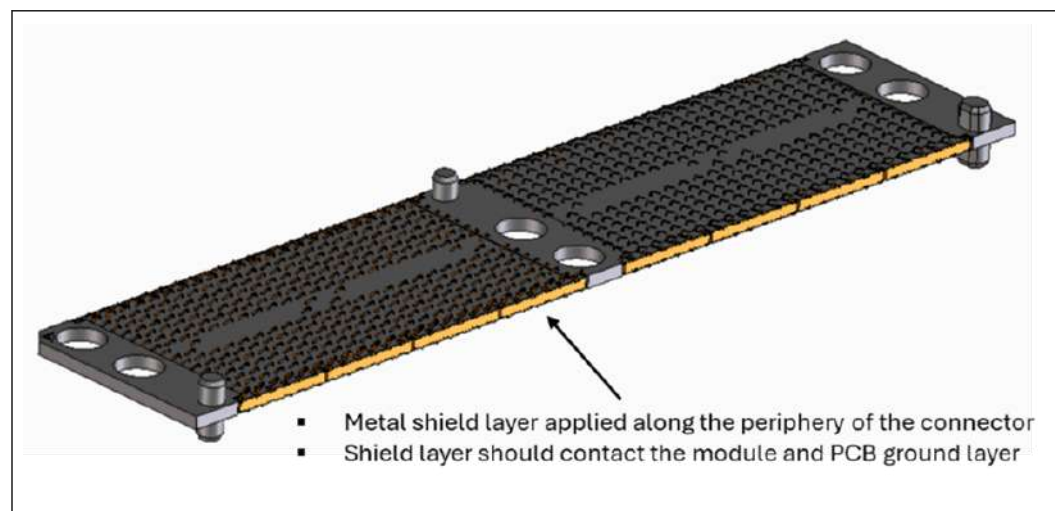
Figure 389. Critical dimensions for conductive gasket for LPCAMM2 on-module shield



4. RFI-friendly LPCAMM2 connector option

- To mitigate noise radiation from the LPCAMM2 connector, an optional RFI-friendly connector has been developed with connector vendors. The connector design is comprised of a thin metal shield layer along the periphery of the connector. The shield layer may wrap around the entire connector or only along the south-edge of the connector near the differential pins. The shield layer should contact the module ground and PCB ground using compression contacts. An illustration of the connector is shown below. Customers should contact their LPCAMM2 connector vendor for part numbers and details.

Figure 390. RFI-friendly LPCAMM2 Connector



9.3.5 Intel DDR RFI Mitigation (RFIM) Software Feature

DDR RFIM is a frequency-shifting RFI-mitigation software feature. The DDR RFIM feature is primarily aimed at resolving narrowband RFI from memory (DDR and LPDDR technologies and both soldered-down and modules) for the Wi-Fi in high frequency bands (5 to 7GHz). By changing the DDR data rate, using Intel dynamic voltage and frequency scaling (DVFS) technology, the narrowband memory noise at the clock harmonic frequency can be shifted out of a radio band of interest, thus mitigating RFI to the radio. The change of the DDR data rate is dynamic based on the radio channel in use.

Dynamic DDR data rate selection and control can be done at the request of the Intel integrated Wi-Fi firmware (CNVi) or at the request of Intel's DTT DDR RFIM policy. In systems with CNVi installed, the feature is enabled by default. In systems with discrete Intel or 3rd party Wi-Fi, the DDR RFIM policy must be enabled in DTT. For both cases, the OEM has the option to customize the DDR RFI table (list of Wi-Fi channels affected by each DDR frequency of the system) based on the measured platform noise.

Intel also offers a semiautomatic tool to characterize the platform DDR noise and customize the DDR RFI table that is an input to RFIM. It is highly recommended to use the tool to characterize the noise from the different memory DVFS points. Please refer to white papers #640438 and #816192 for detailed information.

9.4 DLVR RFI Design Guidelines

The Digital Linear Voltage Regulator (DLVR) may induce RF noise on input power planes or power rails that couple to radio antennas, resulting in RF interference within radio bands.

The following guidelines help to reduce the RFI risk due to DLVR.

9.4.1 Minimize DLVR Input Power Planes' Exposure

It is recommended that the DLVR input power plane (VCCCORE) size be minimized on surface layers while considering PI requirements. Refer to power integrity requirement for resistance and VR sense feedback recommendations.

9.4.2 RF Decoupling Capacitors on DLVR Input Power Planes

High-frequency decoupling capacitors on the DLVR input power plane(s) can help in DLVR noise-reduction. Since there are system-level dependencies for the RFI, in certain scenarios (like a system with aggressive antenna placement to DLVR input power plane or WWAN radio), it is recommended to have placeholders for decoupling capacitors. The following table and figures show the recommended values and placement for the decoupling capacitors and an example of layout implementation.

Figure 391. RF Decoupling Capacitor Guideline for DLVR

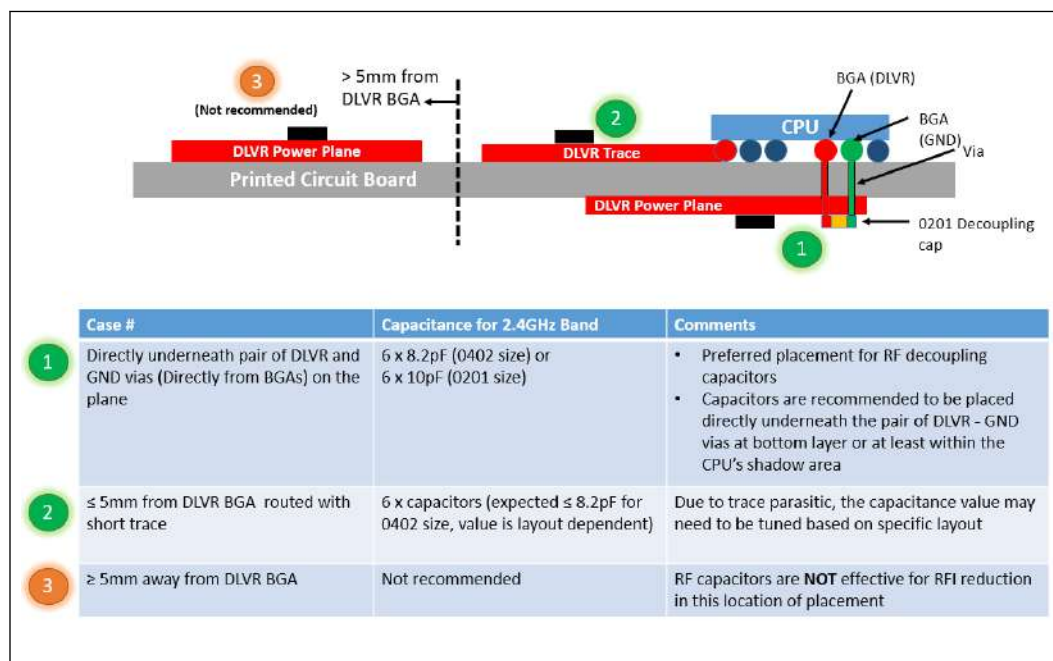
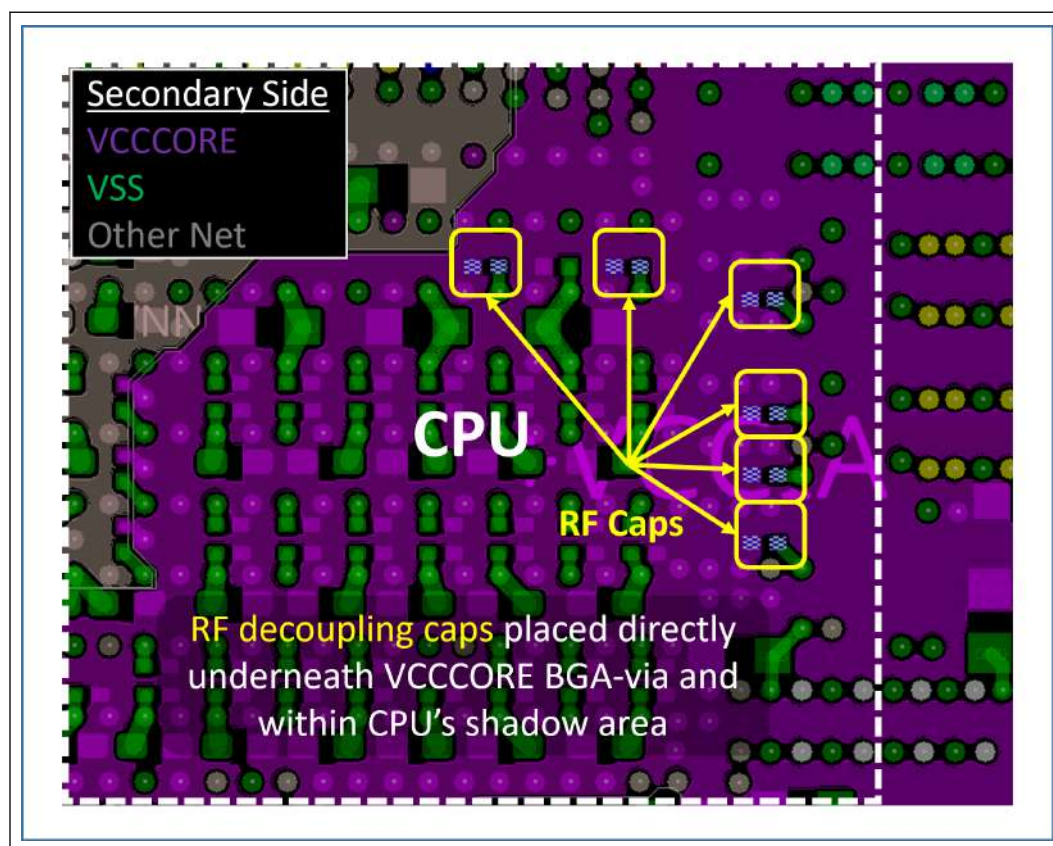


Figure 392. Example of RF Cap Placement for DLVR power plane



9.5 NPU RFI Mitigation

Neural Processing Unit (NPU) operates in a wide range of frequencies and its maximum operation frequency is getting higher gen-over-gen. NPU maximum frequency may operate at Wi-Fi band and interfere with Wi-Fi channels depending on the workload. Intel may update this content based on internal validation. Please contact Intel representative for details.

9.5.1 NPU RFI Mitigation Considerations

The following recommendations apply to the system that supports NPU maximum frequency of Wi-Fi frequency band:

Table 729. NPU RFI Mitigation Considerations

Design Recommendation	Details
Proper thermal grounding	Refer to "EMC Mechanical Considerations" PDG chapter for example.
Reserve on-board shielding especially SOC area or core area	Ensure proper shielding ground contact for shielding effectiveness. Refer to "Shielding Ground Contact" PDG section under "On-Board Shielding" chapter for details.
Antenna Barricade Technology	Refer to "Antenna Barricade Technology" PDG section under "EMC Mechanical Considerations" chapter for more details.
Implement NPU power plane (VCCSA) RF decoupling capacitor placeholder	Refer to "RF Decoupling Capacitors on NPU Power Plane" PDG section for guidelines.
Minimize NPU power plane size (VCCSA) on top and bottom of the motherboard	

9.5.2 RF Decoupling Capacitors on NPU Power Plane

It is recommended to have placeholders for high-frequency decoupling capacitors on the VCCSA (NPU power plane). These decoupling capacitors may help in reducing potential NPU noise in certain scenarios such as a system with aggressive antenna placement to NPU power plane. The following table and figure show the recommended values and placement for the decoupling capacitors:

Figure 393. RF Decoupling Capacitor Guideline for NPU Power Plane

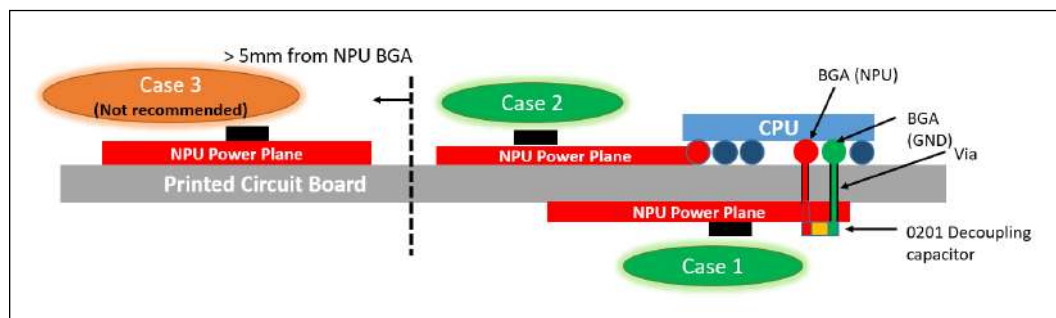


Table 730. NPU Power Plane RF Capacitor Placement Guidelines

Case #	Location of Decoupling Capacitor	NPU RF Capacitors for 2.4GHz Band	Comments
Case 1	Directly underneath pair of NPU power via and GND vias (Directly from BGAs) on the plane	0201 size: 6 x capacitors (Suggested value: between 22pF to 33pF)	Preferred placement for RF decoupling capacitors. Capacitors are recommended to be placed directly underneath the pair of NPU power via - GND vias at bottom layer or at least within the processor's shadow area. The value of the RF capacitor may need to be tuned with respect to the potential unintended RF power within Wi-Fi bands.
Case 2	≤ 5mm from NPU power via BGA	0201 size: 6 x capacitors (Suggested value: between 22pF to 33pF)	The value of the RF capacitor may need to be tuned with respect to the potential unintended RF power within Wi-Fi bands.
Case 3 (Not Recommended)	> 5mm away from NPU power via BGA	Not recommended	RF capacitors are NOT effective for RFI reduction in this location of placement.

9.6 Graphic (GFX) Power Planes RFI Design Guidelines

Graphics (GFX) may induce RF noise on power planes or rails that couple to radio antennas, resulting in RF interference within radio bands.

The following guidelines help to reduce the RFI risk due to GFX power planes.

9.6.1 RF Decoupling Capacitors on GFX Power Planes

It is recommended to have placeholders for high-frequency decoupling capacitors on the VCCGT (GFX power plane). These decoupling capacitors may help in reducing potential GFX noise in certain scenarios such as a system with aggressive antenna placement to GFX power planes. The following table and figure show the recommended values and placement for the decoupling capacitors:

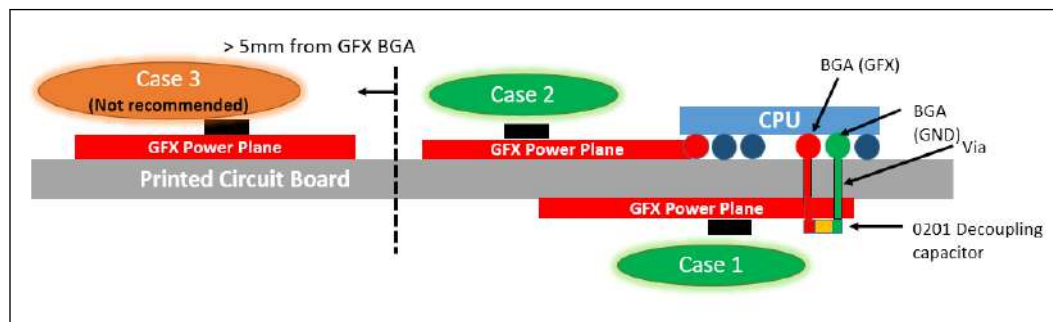
Figure 394. RF Decoupling Capacitor Guideline for GFX Power Planes

Table 731. Graphic Power Plane RF Capacitor Placement Guideline

Case #	Location of Decoupling Capacitor	GFX RF Capacitors for 2.4GHz Band	Comments
Case 1	Directly underneath pair of GFX power via and GND vias (Directly from BGAs) on the plane	0201 size: 6 x capacitors (Suggested value: between 22pF to 33pF)	Preferred placement for RF decoupling capacitors. Capacitors are recommended to be placed directly underneath the pair of GFX power via - GND vias at bottom layer or at least within the processor's shadow area. The value of the RF capacitor may need to be tuned with respect to the potential unintended RF power within Wi-Fi bands.
Case 2	≤ 5mm from GFX power via BGA	0201 size: 6 x capacitors (Suggested value: between 22pF to 33pF)	The value of the RF capacitor may need to be tuned with respect to the potential unintended RF power within Wi-Fi bands.
Case 3 (Not recommended)	> 5mm away from GFX power via BGA	Not recommended	RF capacitors are NOT effective for RFI reduction in this location of placement.

9.7 HSIO RFI Mitigation

Exposed high speed I/O routing traces are considered as RFI source. The RFI level depends on interface signal characteristics such as data rate and distance between antenna and signal trace.

9.7.1 Minimize Exposed High Speed I/O Channel Routing

It is recommended that the routing of I/O interfaces in the table below be embedded in inner layers and covered by solid ground. Due to routing limitations, full stripline routing may not be possible at breakout region and the routing at the connector region. Please refer to the table below for maximum microstrip length.

Table 732. High Speed Differential Signal Routing Recommendations

Interface	Data Rate	Max Routing Length at Breakout Region (Microstrip)	Max Routing Length at Connector Region (Microstrip)	Cable Shielding	Note
eUSB2	480 Mbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
UFS reference clock	38.4MHz	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	Add a placeholder shunt capacitor (0402 or 0201 size) near to UFS device. Refer to "UFS Reference Clock" (High Speed I/Os chapter) for more information about capacitor placement.
UFS 4.0 Gear 5	20 Gbps (Rate A), 23.32 Gbps (Rate B)	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
CNVio3 (STEP)	12 Gbps	Refer to "CNVio Bus Design Guidelines" section under "High Speed I/Os" chapter	Refer to "CNVio Bus Design Guidelines" section under "High Speed I/Os" chapter	Not applicable	Keep out distance of CNVio3 signals (microstrip) to Wi-Fi antenna > 15 mm. Refer to Intel document #610694 for the latest Intel Wi-Fi module RFI/EMC recommendations.
DMIC Clk		< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
eDP 1.5	8.1 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
HDMI 2.0	5.94 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
HDMI 2.1	12 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
PCIe4	16 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	Refer to CLK PCIe Gen4 Add-in Card Clock Topology or CLK PCIe Gen4 Device Down Clock Topology (under High Speed I/Os) for more EMC/ RF noise protection recommendations.
PCIe5	32 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	Refer to CLK PCIe Gen5 Add-in Card Clock Topology or CLK PCIe Gen5 Device Down Clock Topology (under High Speed I/Os) for more EMC/ RF noise protection recommendations.
PCIe Refclk	100 MHz	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	Refer to CLK PCIe Gen4/Gen5 Add-in Card Clock Topology or CLK PCIe Gen4/Gen5 Device Down Clock Topology (under High Speed I/Os) for more EMC/ RF noise protection recommendations.
SPI Clk		< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
continued...					

Interface	Data Rate	Max Routing Length at Breakout Region (Microstrip)	Max Routing Length at Connector Region (Microstrip)	Cable Shielding	Note
TBT5	80 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
USB3.2 Gen1Type A	5 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	
USB3.2 Gen2Type A	10 Gbps	< 38 mm	< 25 mm	Shielded FPC/ Shielded FFC/ Coax	

For mechanical information, please refer to "EMC Mechanical Considerations" chapter.

9.7.2 eDP RFI Mitigation

RF Interference from Embedded Display Port (eDP) Interface and TCON

In mobile designs, RF interference from eDP interface has become more problematic due to the increase of thinner and lighter devices, which limit options for effective cable shielding and antenna isolation. While the data rates increase, the high-frequency data and clocks generate RF noise that falls into Wi-Fi radio bands. Besides RF noise from eDP, timing controller (TCON), used to convert the eDP signal and drive the individual pixels of the display, could be another source of interference.

Narrowband Spur Noise from eDP Data and TCON Clock

To prevent interference with radio bands, select the eDP data rates that ensure the fundamental frequencies and harmonics of the eDP signals do not fall into Wi-Fi radio bands. The eDP data rates 2.43 Gbps and 6.75 Gbps should be avoided to mitigate the RF noise impact on Wi-Fi radios. Other mitigation solutions such as using high-quality shielding cable for eDP and TCON connection/board or enabling spread spectrum clocking may provide effective but limited noise reduction as an alternative solution.

Figure 395. Mitigate RF interference by preventing using data rate falling into Wi-Fi radios

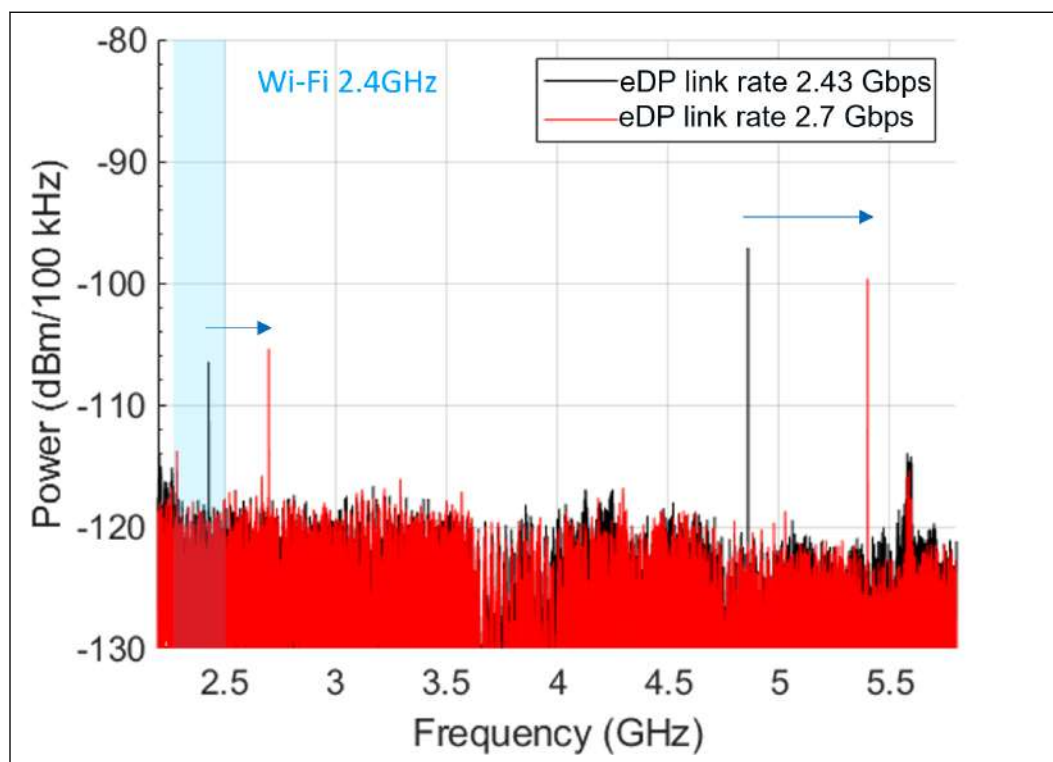
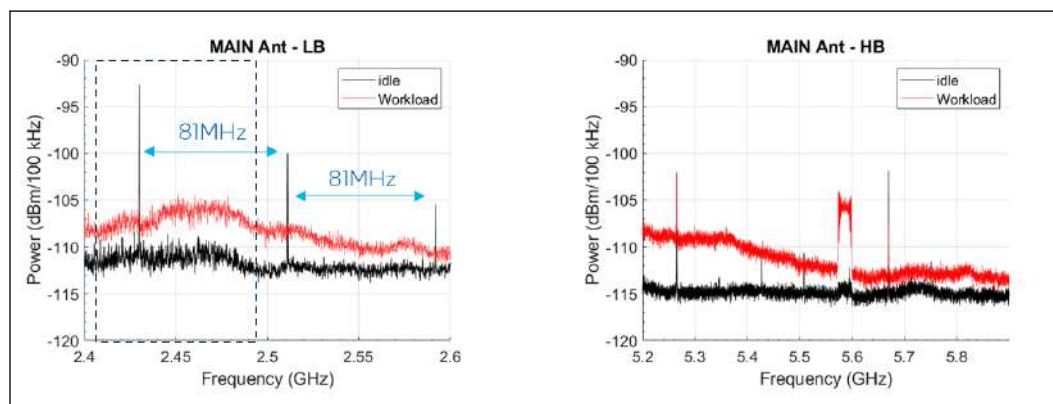


Figure 396. TCON clock noise falling into Wi-Fi bands



9.8 EMC Component Selection

9.8.1 Common Mode Choke Selection

The common mode choke (CMC) selection is shown below.

Alternatively, PCB CMC can be also considered for eDP 1.4, eDP 1.5, Type-C USB+DP (10 Gbps), USB3.2 Gen 1 and USB3.2 Gen2 signal. Refer to #546122 for Intel PCB Common Mode Choke License Technology Technical White Paper.

Table 733. CMC Selection Guidelines

Interface	Data Rate	CMC Placement	CMC Cutoff Frequency (3dB Bandwidth)	CMC Insertion Loss	CM Rejection (10-dB Band)
eDP 1.5	8.1 Gb/s	< 25 mm routing length from connector	> 10 GHz	< 1.2 dB @ 5 GHz	1.8 - 3.8 GHz
HDMI 2.0 #	≤ 6 Gb/s	< 25 mm routing length from connector	> 6 GHz	< 0.35 dB @ 1.5 GHz, < 0.5 dB @ 3.0 GHz	0.3 - 3.8 GHz
HDMI 2.1	12 Gb/s	< 25 mm routing length from connector	> 10 GHz	< 1.0 dB @ 6 GHz	0.3 - 3.8 GHz
USB3.2 Gen 1/2	≤ 10 Gb/s	< 25 mm routing length from connector	> 7.5 GHz	< 1.5 dB @ 2.5 GHz, < 1.2 dB @ 5 GHz	1.8 - 3.8 GHz
TCP USB 3.2 + DP	10 Gb/s	<25 mm routing length from connector	>10 GHz	<1.2 dB @ 5 GHz	1.8-3.8 GHz
eUSB2.0*	480 Mb/s	< 25 mm routing length from connector		< 0.2 dB @ 240 MHz	0.5 - 2.0 GHz

To provide longer channel length the insertion loss criteria for HDMI interface is set to low loss value. The EMI/ RF Interference issues caused by HDMI common mode noise are mitigated using CMC and chassis level mitigations such as connector shell grounding etc.

* Common mode chokes with a target common mode impedance of 80 to 90 Ohm at 100 MHz generally provide adequate noise attenuation for eUSB2.0. Higher impedance CMC generally have a greater damaging effect on signal quality. Signal quality must be checked for low-speed, full-speed and high-speed USB operations in case if high impedance component is used.

9.8.2 CMC Placement Guideline for USB3 Interface

Table 734. CMC Placement Guidelines for USB3 Interface

Port Usage Scenario	CMC Guideline
USB3 only	Place Tx CMCs; Rx CMCs are optional
USB3 & DP	Place both Tx and Rx CMCs due to emission from DP
USB3 & TBT	Do not place either Tx or Rx CMCs

9.8.3 ESD Diode Selection

ESD diode selection recommendations below only apply to interfaces with:

- Hotplug input/output port application (external ports that exposed to end user)

- Direct connection between SOC and connector (Eg, without redriver / retimer in between)

For interfaces that require repeater or bridge chip:

- Intel retimer: Please refer to Thunderbolt retimer or controller documentation for more details, document number can be found in PDG chapter "EMC/RF Reference Documents".
- Third party repeater or bridge chip: Customer may contact vendor for necessary ESD protection.

The selection of the most suitable ESD diode depends on the customer's design targets. It is recommended to conduct a SEED analysis for further ESD design optimization. Customers may refer to the chapter "System Efficient ESD Design (SEED)" for more details.

Table 735. Discrete ESD Protection Diode Selection Guidelines

Interface	Data Rate	ESD Diode Placement	ESD Diode Shunt Capacitance	ESD Diode R _{dyn}	ESD Diode Reverse Working Voltage
HDMI 2.1 Direct Connect Rate ≤ 6 Gbps (Data)	≤ 6 Gbps	< 25 mm routing length from connector	< 0.4 pF	< 0.5 Ohm	>4.0 V
HDMI 2.1 Direct Connect Rate > 6Gbps (Data)	> 6 Gbps	< 25 mm routing length from connector	< 0.4 pF	< 0.5 Ohm	>4.0 V
HDMI connector HPD/DDC pins		< 25 mm routing length from connector			>5.0 V
USB3.2 Gen1/2	≤10 Gbps	< 25 mm routing length from connector	< 0.35 pF	< 0.5 Ohm	>3.3 V
TCP USB 3.2 + DP	10 Gbps	< 25 mm routing length from connector	< 0.35 pF	< 0.5 Ohm	> 1.0 V

9.8.4 Other Component Selection

Table 736. Other Component Selection Guidelines

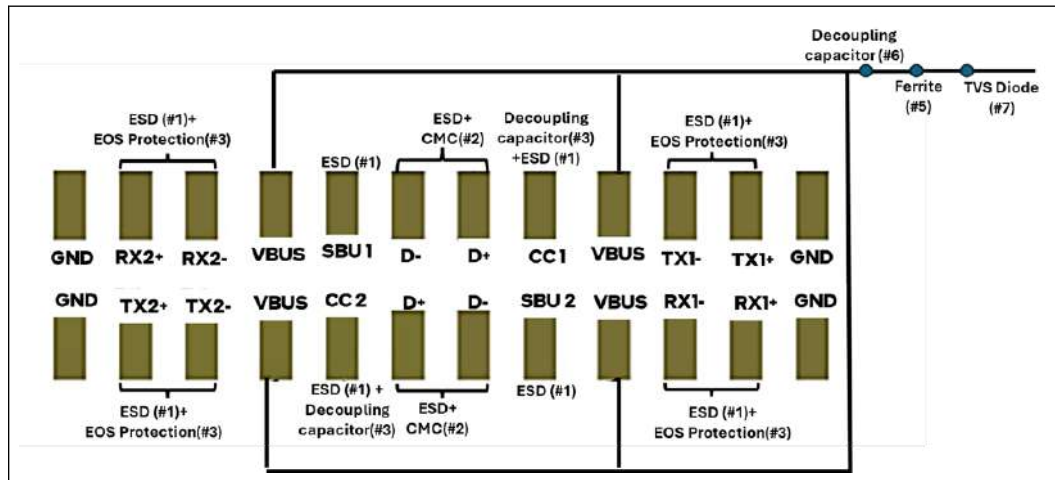
Interface/ IO Connector	Component Requirement	Recommended Details
Type C Connector ESD Protection (#1)	ESD protection for Tx/Rx, CC, SBU lines	Refer "ESD Diode Selection" chapter for more details.
Type C Connector D +/- (#2)	ESD and CMC protection for eUSB2 data lines	Refer "ESD Diode Selection" and "Common Mode Choke" chapter for more details respectively.
Type-C connector CC pins (#3)	Decoupling capacitors on CC1 and CC2 pins	Refer to USB PD controller Specification, if not 200 pF is recommended.
Type-C connector Tx/ Rx AC caps (#4)	AC caps for Tx, Rx lanes	In order to avoid EOS damage due to shorting between VBUS and adjacent pins, AC cap with voltage rating ≥ 50 V VBUS maximum voltage rating is required for both Tx and Rx lanes; consider EPR requirement if
continued...		

Interface/ IO Connector	Component Requirement	Recommended Details
		applicable. The AC cap needs to be next to the connector and followed by ESD diode. Refer to SI PDG for values/placement.
Type-C connector VBUS pins	Ferrite bead on VBUS (#5)	Optional. Selection based on current rating requirement of the platform.
Type-C connector VBUS pins	Decoupling capacitors on VBUS signal pins (x4) (#6)	Refer to Type-C Specifications for more details. (If no specific value is given then it is recommended to use 0.1 uF, 0402 or smaller)
Type-C connector VBUS pins	VBUS ESD Protection (#7)	Select the TVS diode as per the VBUS voltage ratings.
AC JACK (#8)	Ferrite bead on power and ground pins of AC JACK	Ferrite bead rated current >= Maximum current rating for the power line. Selection criteria: >80 Ohm Impedance between 100-300 MHz
Touch SPI Interface	Placeholder for shunt capacitor	Touch SPI interface nets are single ended nets operating at >=20MHz frequency, so recommended to have shunt capacitor placeholder for clock and IO nets.
Voltage regulator	EMI filter at the voltage regulator input pin (ferrite bead)	Optional. Switching Voltage regulators can show broadband noise in the LTE low band, adding ferrite bead can help in reducing the noise. Ferrite bead selection is based on the current rating of the voltage regulator input pin.

Components for Type-C Interface

The image below indicates various components added on the Type C connector for ESD, EMI and EOS protection. For more details about component selection refer "Other Component Selection (#1 to #7)" table.

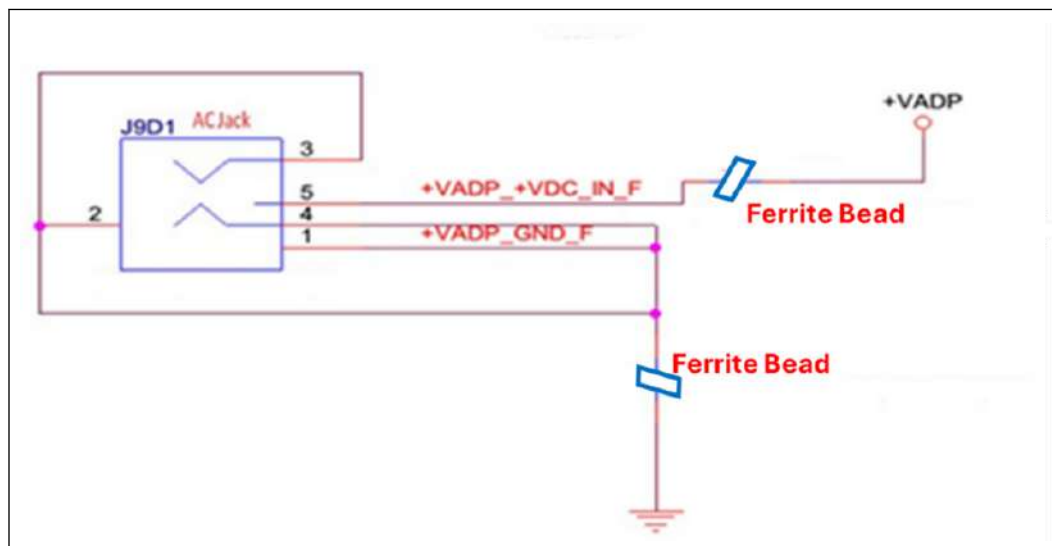
Figure 397. Components For Type C Interface



Ferrite Bead for Power and Ground Pins of AC Jack

To prevent EMI noise passing through adapter cable adding ferrite bead at the connector pin can help. For more details refer to AC Jack (#8) details from "Other Component Selection" table.

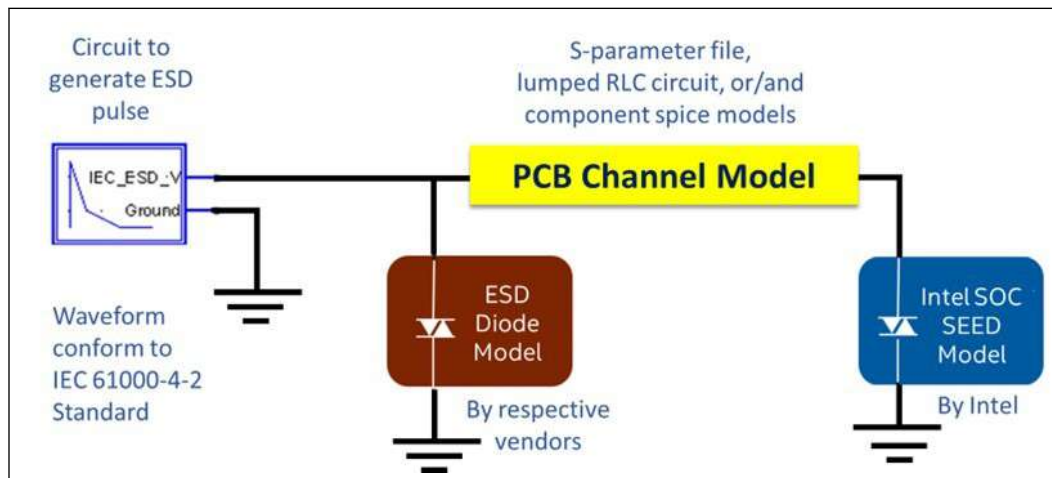
Figure 398. Ferrite Beads for AC Jack



9.8.5 System Efficient ESD Design (SEED)

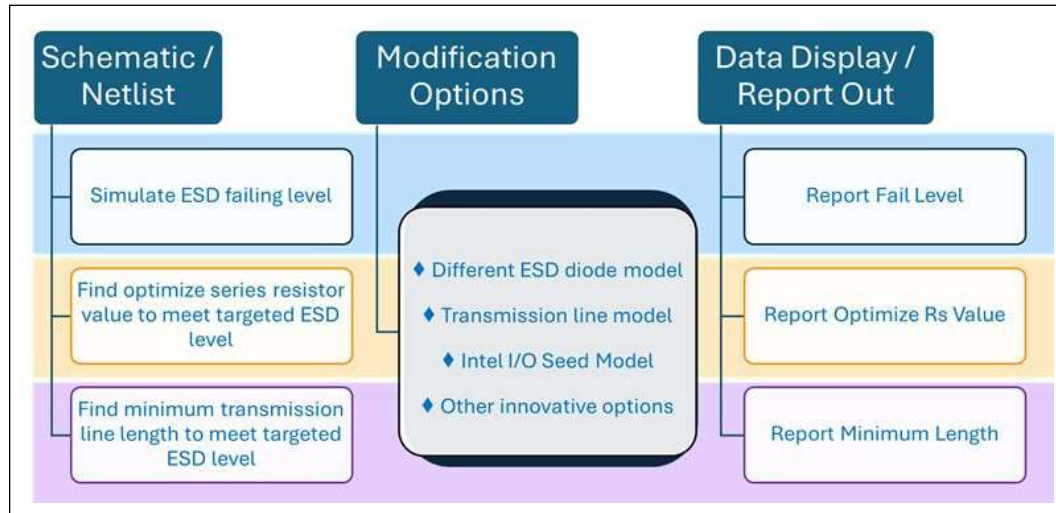
PDG chapter “ESD Diode Selection” serves as a general recommendation for system level ESD protection. To further optimize selection of ESD diode, customers may consider using SEED simulation concept during system design phase. SEED defines an ESD robustness design concept that assures matching between PCB protection components and IC IO behavior during system ESD discharges event. It allows selection of ESD diode which is optimized based on IO characteristics and PCB design. It also promotes early assessment using a simulation to achieve first-time-right PCB design. This design concept applies to hot plug input/output port applications, that includes USB3, Type-C, Display Port, and HDMI.

Figure 399. SEED Block Diagram



SEED simulation deck is available in Advanced Design System (ADS) and HSPICE. Refer to Intel System Efficient ESD Design (SEED) Models User Guide (#702999) for step-by-step simulation guidance. Customers may contact the Intel representative for platform specific SEED Models.

Figure 400. Intel® SEED Analysis Template

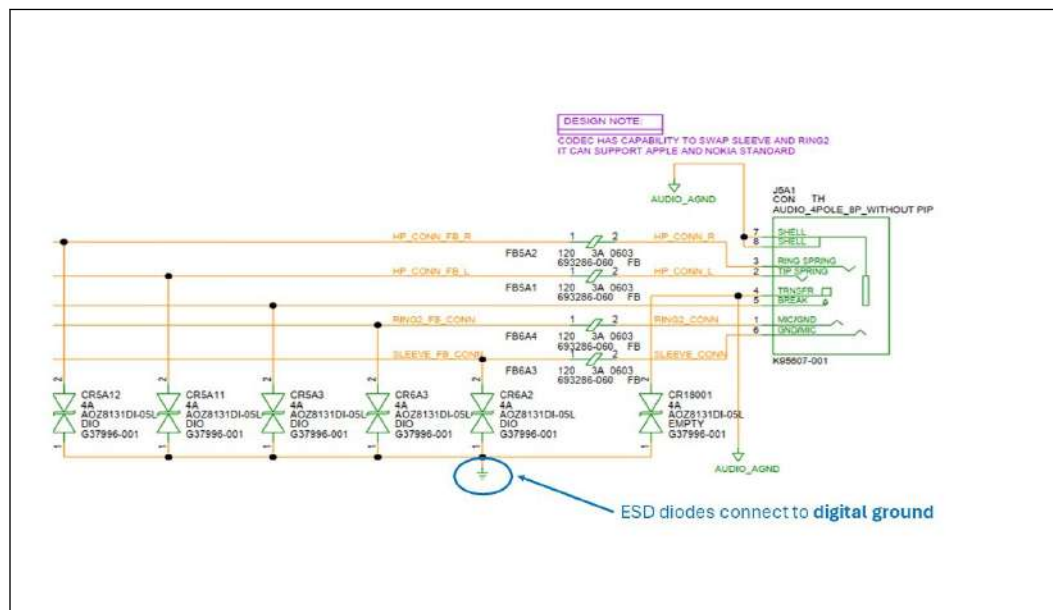


9.9 EMC Audio

9.9.1 Audio ESD Protection

Audio jack needs to have ESD protection. The ESD diodes need to connect to digital / chassis ground.

Figure 401. Audio Jack ESD Protection



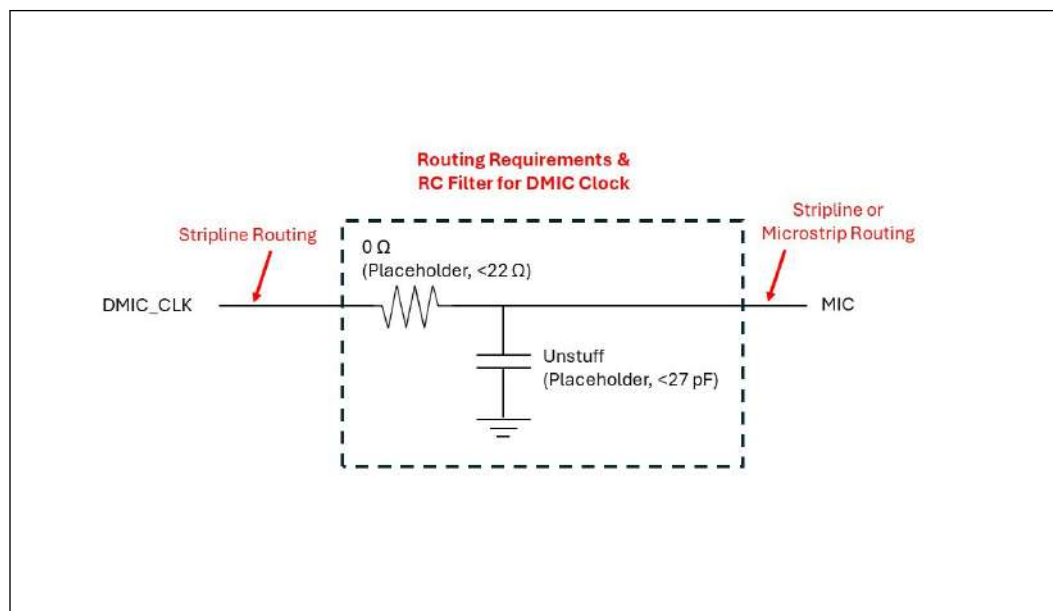
9.9.2 DMIC EMI/ RFI Risk

DMIC clock may have EMI/ RFI risk due to its high-order harmonics.

It is recommended to have stripline routing and a placeholder for an RC filter before being routed in microstrip/ cable to reduce high-frequency harmonics.

The RC value may vary according to the board design to meet the electrical requirements.

Figure 402. Routing Requirements and RC Filter for DMIC Clock



9.10 Wi-Fi/WWAN Radios Performance vs. Platform Noise

Noise from multiple aggressors, such as switching power supplies, high-speed digital clocks or data can generate harmonics or broadband noise that interferes with the radio bands on the platform. Noise could couple to the platform antenna hence reducing the radios performance like throughput, latency, and sensitivity by reducing the signal-to-noise ratio (SNR).

This section provides the RF platform noise threshold that can set the baseline for determining if the noise level in the radio bands is tolerable to maintain wireless performance, therefore ensuring the best user experience. To meet the noise requirement, customers could implement the mitigation solutions illustrated in the EMC General Considerations guidelines.

9.10.1 Platform Noise Threshold Recommendation

Each type of noise source has its own characteristic noise profile therefore has its own frequency spectrum that could have a different impact on radio performance. Scrambled data signals for high-speed interfaces (e.g. USB 3.0, DDR5, PCIE) are broadband, covering a wide frequency range and radio band in the GHz range. On the other hand, clock signals are highly periodic and have a relatively narrowband spectrum, which leads to the generation of harmonics. Unlike scrambled data, these harmonics are periodic and predictable.

Table 737. Recommended Noise Threshold for Wi-Fi and WWAN Radios

Noise Profile	Wi-Fi In-band Noise Threshold (dBm/100 kHz)	WWAN In-band Noise Threshold (dBm/100 kHz)
Narrowband noise (<= 600 kHz Bandwidth)	-106	-98
Broadband Noise	-117	-117

The table above shows the recommended noise threshold based on experimental results. By artificially changing the noise profile and level, the figures below represent the throughput and sensitivity impact on Wi-Fi and WWAN. The noise threshold is defined according to the noise level that can ensure radio performance.

Figure 403. Wi-Fi Throughput Degradation by RF Noise Level

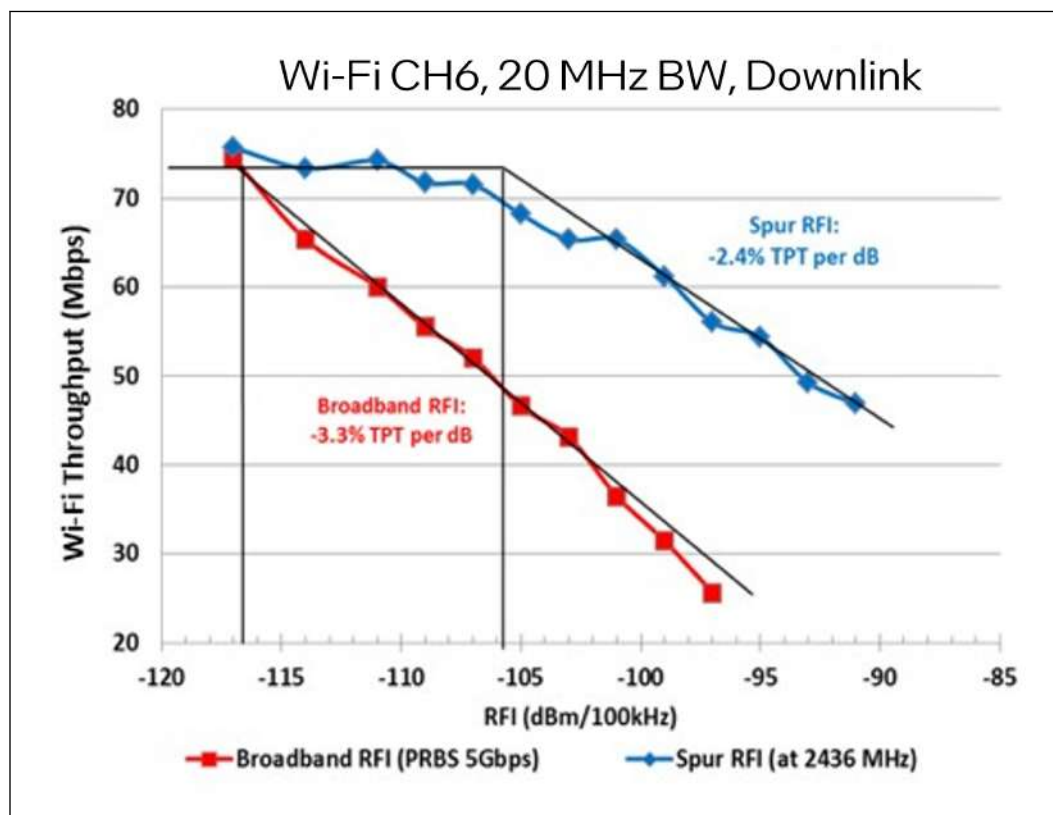
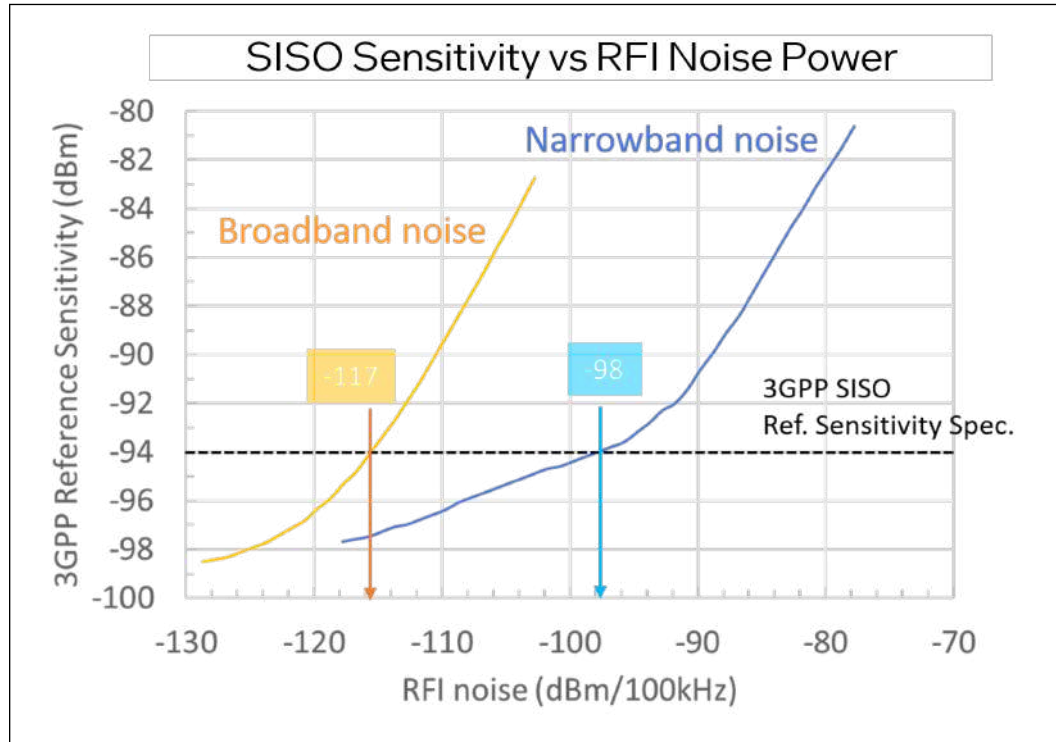


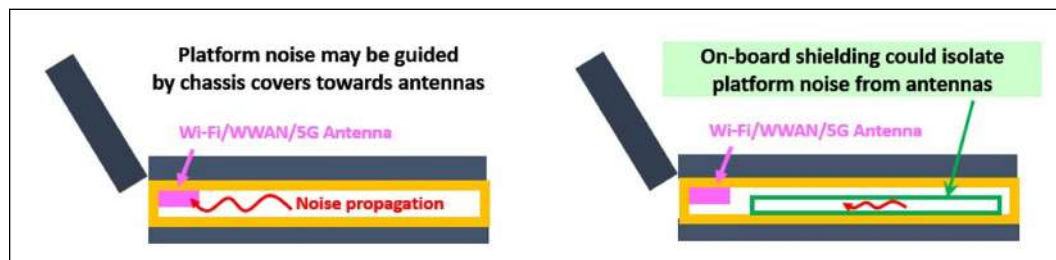
Figure 404. WWAN Sensitivity Degradation by RF Noise Level



9.11 On-Board Shielding

Platform on-board shielding can be an effective mitigation option against RF interference with Wi-Fi/WWAN/5G antennas. It can help to reduce the platform noise radiation and reduce susceptibility to external RF noise, particularly the memory and SSD.

Figure 405. General On-Board Shielding Consideration



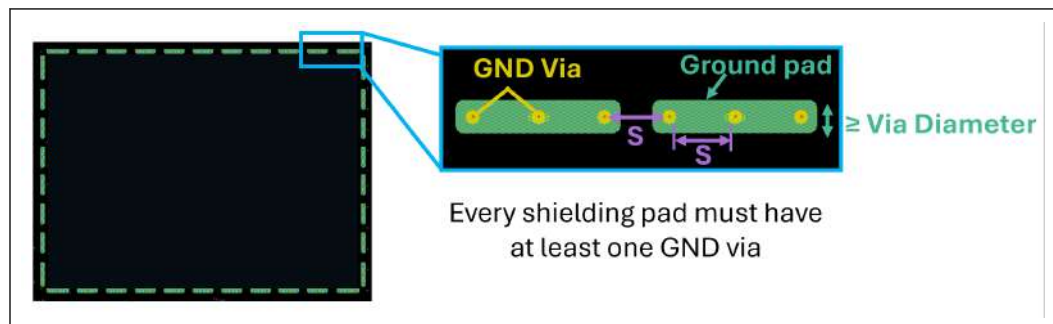
9.11.1 Shielding Ground Contact

To ensure good shielding effectiveness and minimum noise leakage it is critical to have proper electrical contact of the PCB ground layer with the shielding mechanical enclosure, ideally leaving no electrical openings. Poor grounding contact or gaps/seams in the shield can reduce the shielding effectiveness. Also, the size of any

aperture and opening in a mechanical shield enclosure needs to be minimized, keeping it smaller than 2.5 mm. A conductive gasket can also be used to ground the shield to the PCB.

The following figure shows a general recommendation for spacing between grounding vias (S). The general recommendation is to use via spacing $\leq \lambda/6$ (e.g., approximately 4 mm for 7.5 GHz). To improve shielding effectiveness, use closer spacing.

Figure 406. On-Board Shield Grounding General Guideline



9.11.2 Thermal and EMI Co-Design Considerations

An integrated thermal and EMI shield co-design solution can be implemented in a system. The thermal/EMI shield solution needs to have proper electrical contact to the PCB ground layer to guarantee adequate shielding effectiveness.

An example of an EMI shield with co-optimized EMC-thermal solution consists of a large metal frame interposer that covers GPU, GDDR devices, GPU voltage regulator, processor, DDR devices and voltage regulator. The metal frame has an electrically conductive interface and connects to the on-board GND pad opening.

Figure 407. Example Implementation of EMC-Thermal Solution

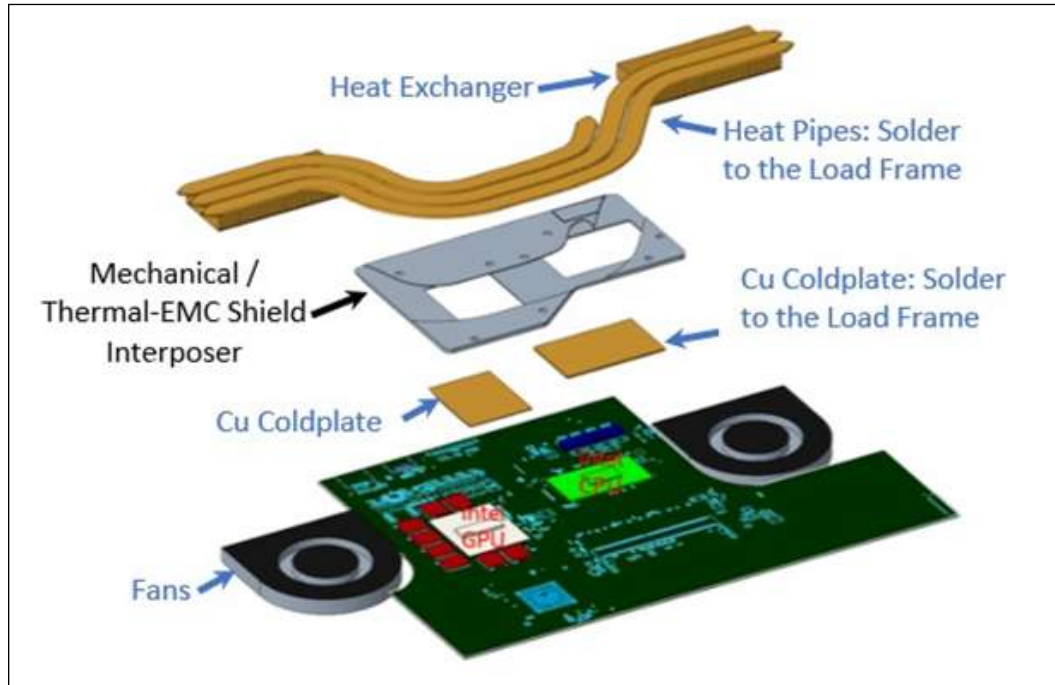
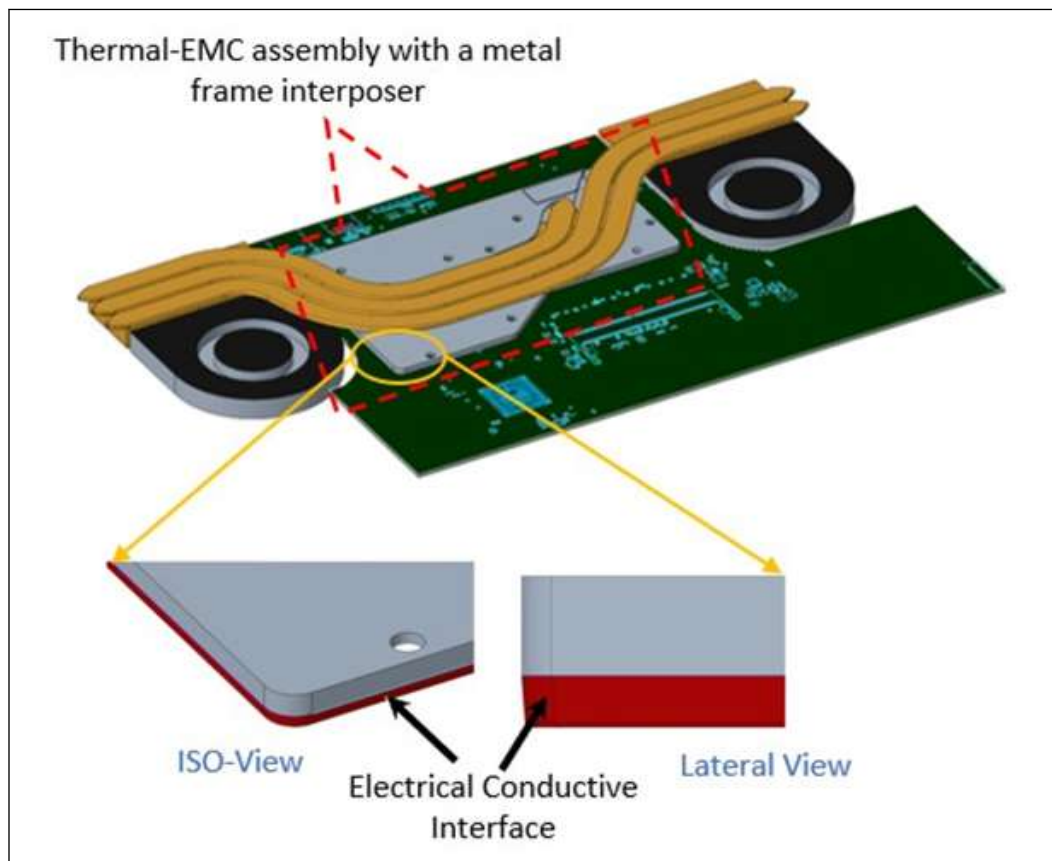


Figure 408. Metal Frame Interposer with Electrically Conductive Interface to Board

9.12 EMC Mechanical Considerations

Mechanical design of the system plays a critical role in meeting requirements for EMI, ESD, RFI, RF Immunity and EFT. The broad categories of consideration in mechanical design are chassis design, connector selection and grounding, high speed cable design and grounding.

- All faces of the chassis should have a conductive EMI coating (in cases where metallic chassis is not used) to provide RF shielding. The sidewalls of the chassis should also have a continuous EMI coating.
- Conductive contacts between the faces of the chassis are required to reduce the size of the openings along the chassis seams (the faces of the chassis should make continuous electrical contact).
- Avoid non-conductive coating/paint on screws that are used to secure the motherboard to the chassis or used between chassis faces and expose solder mask on the PCB for the screws to make contact to the PCB ground layers.
- Large metal parts of the system should be connected to chassis ground at more than one point. Examples of large metal parts include the PCB ground layers, chassis, metal frame of the display, metal casing of the camera modules, FPC cables, etc. Conductive gaskets or screws may be used to tie large metal structures together at multiple contact points.

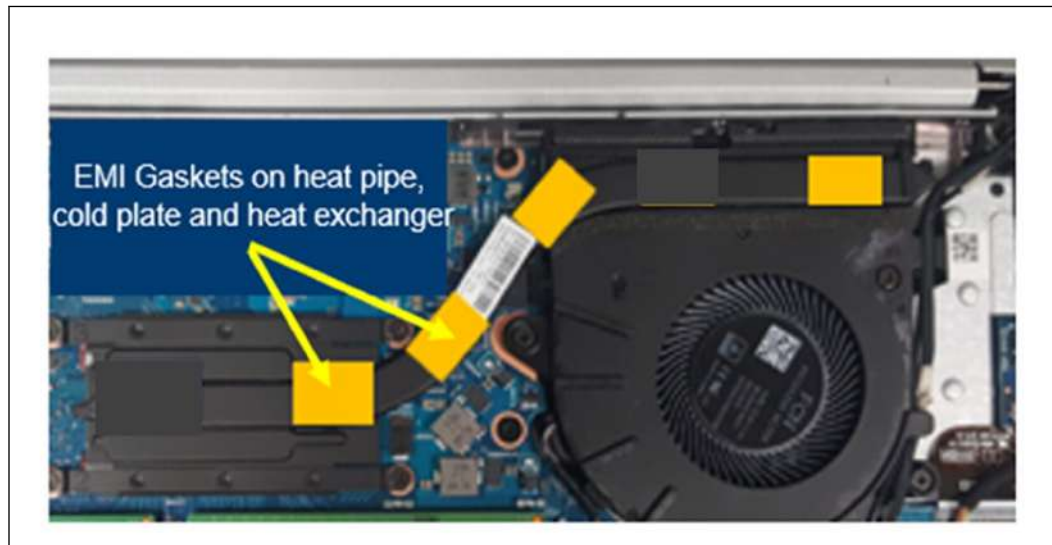
- The thermal solution (heat pipe, cold plate) should be grounded to eliminate or reduce any resonance effects that could couple noise in the vicinity of the thermal solution.
- Refer to Intel White Paper #602554 for further guidelines on the EMI coating thickness and selection criteria, conductive contact between chassis faces and thermal solution grounding.
- The figures below illustrate the guidance for chassis design considerations:

Figure 409. EMI Coating on Chassis Covers and Sidewalls



Figure 410. Faces of the Chassis should make Continuous Electrical Contact



Figure 411. Thermal Solution Grounding

9.12.1 Type A Connector Selection

- To minimize RFI, select a type A receptacle shell that has a backside shield with multiple ground contact points to the PCB ground layers.
- Ensure that the mechanical latches form a ground contact with the plug shell when inserted.
- Apertures/openings in the receptacle shell should be minimized.
- The receptacle connectors should be connected to metal chassis or enclosures through grounding fingers, screws, or conductive gaskets

9.12.2 Type C Connector Selection

- The receptacle shell should have multiple connection points to the PCB ground layers.
- Apertures in the receptacle should be minimized. If apertures are unavoidable, a maximum aperture size of 1.5 mm is recommended.
- The receptacle connectors should be connected to metal chassis or enclosures through grounding fingers, screws, or conductive gaskets.
- The mid-plate in the connector receptacle should be directly connected to system PCB ground layers.
- Select a connector where the RFI pad in the receptacle has multiple grounding contacts to the receptacle shell.

The following figures illustrate the recommendations for USB Type C connector selection and for further details, refer to the USB Type C Cable and Connector Specification (https://www.usb.org/cable_connector).

Figure 412. Type-C Connector Selection

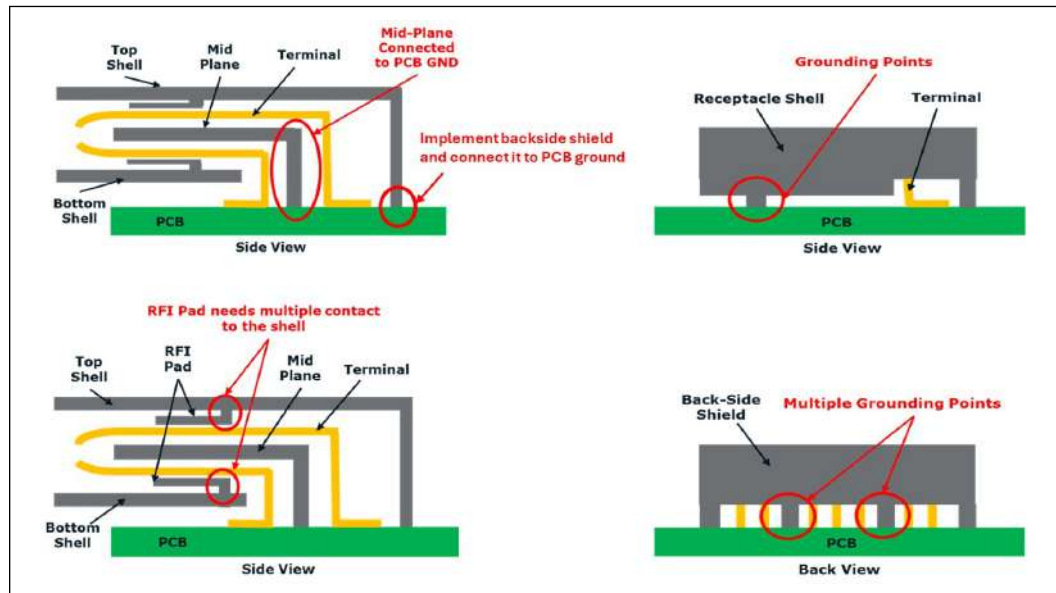
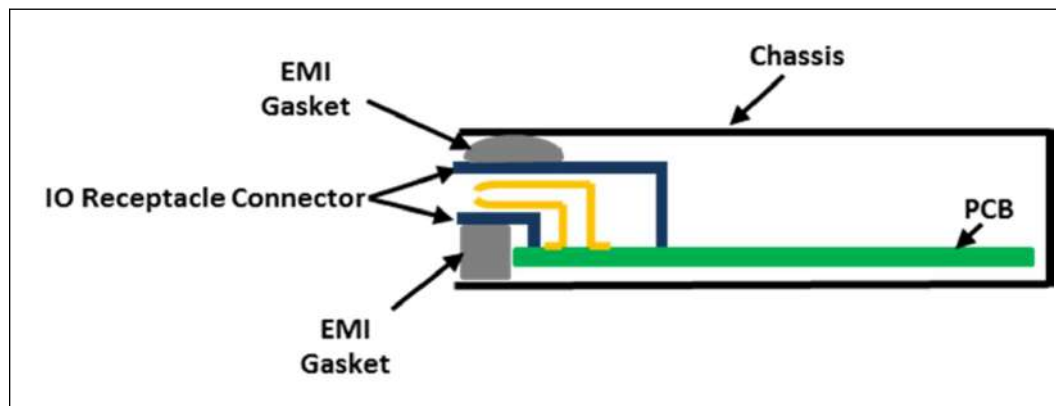


Figure 413. Conductive Gaskets to Ground IO Connectors to Chassis



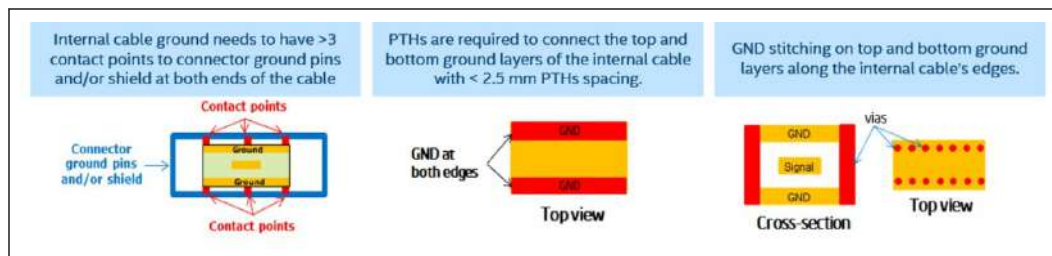
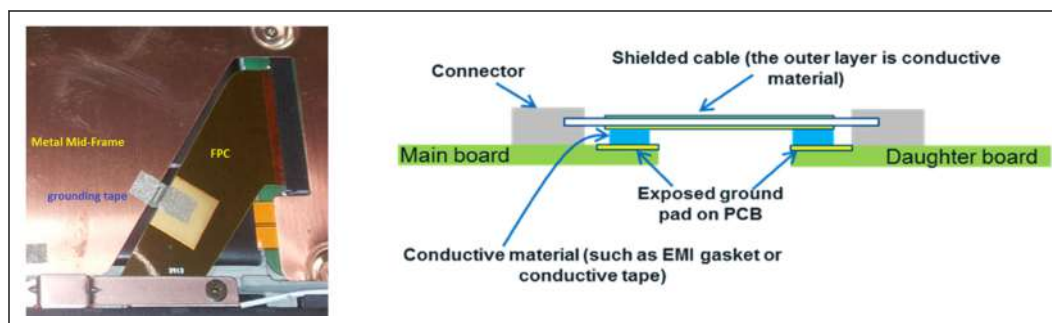
9.12.3 Cable Routing and Shielding, Ground Cables to Chassis

Electromagnetic noise can radiate from internal cables carrying high-speed signals causing RF interference or cables can couple noise to sensitive circuits or nets on the PCB and cause RF immunity issues. Internal cables should be well shielded to reduce the noise radiation from it.

The following are key considerations while selecting and designing with internal cables (FPC/FFC):

- The internal cable should be shielded with either full ground layers on top and bottom, or by using an EMI shielding film (silk screening with silver/copper ink, aluminum/copper film lamination).

- For the shield to be effective, the shield layer should not be floating as a floating shield does not provide a return current path and may even make the noise radiation worse. To improve the ground contact of the shield, a direct PCB ground or chassis ground connection of the shield is recommended. Multiple grounding points may be needed for long FPCs (typically every 50mm)
- The routing of the internal cables is also critical to reducing the amount of noise that can couple to the sensitive circuits. For further guidelines on cable routing, refer to Intel White Paper #602554.
- The figures below illustrate the guidance for internal cable shielding and grounding

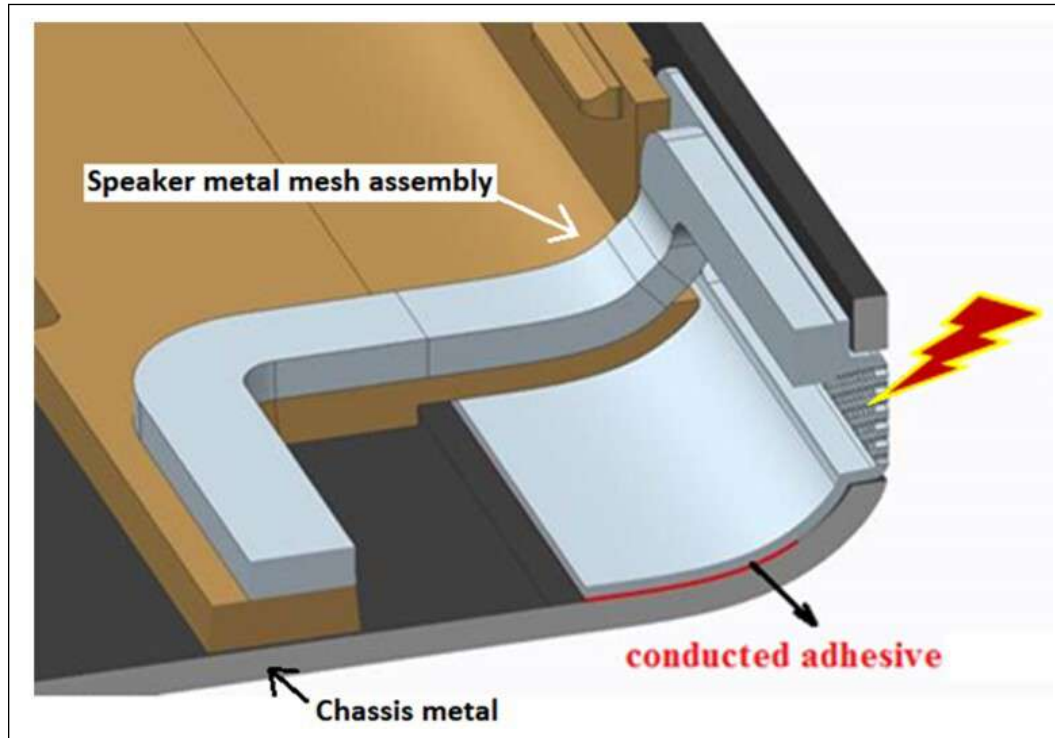
Figure 414. Internal Cable Design Criteria for EMI**Figure 415. Direct PCB or Chassis Ground Connection of the Cable Shield**

9.12.4 Minimize ESD Chassis Entry Points

It is recommended to design the chassis/system industrial design to minimize ESD entry and thereby improve the system level ESD protection. Following are two examples of ESD entry points and mitigation methods:

- Example 1: Speaker openings - In systems with metal mesh for the speaker openings, the ESD energy can be diverted to the system ground by connecting the metal mesh to system ground. This will minimize the ESD entry through the speaker lines into the PCB.
- Example 2: SIM/SD trays - SIM/SD cards are very sensitive to ESD. Avoid any air gaps around the tray that otherwise can lead to ESD entry. Divert ESD into system ground by connecting the metal tray of the SIM/SD to the chassis ground or PCB ground layers.

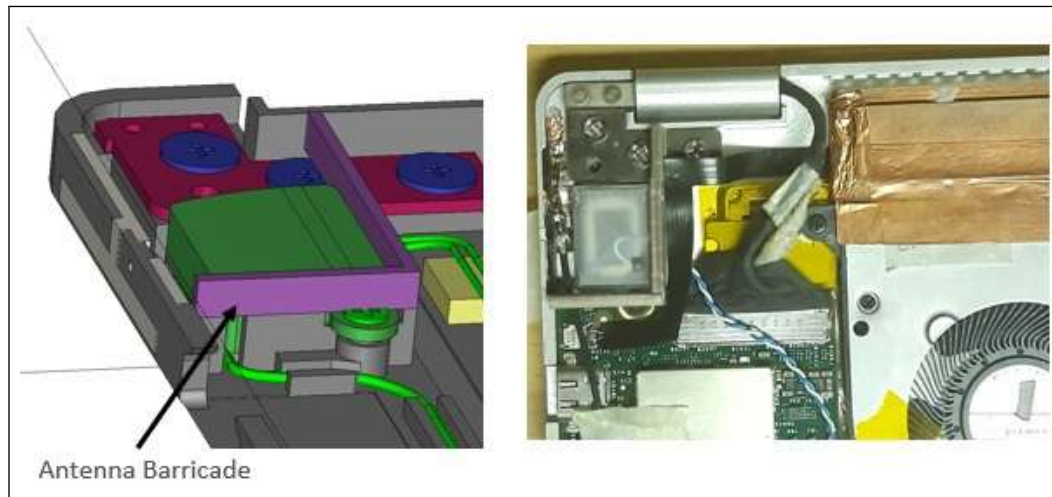
Figure 416. Minimize ESD Entry Points in the Chassis



9.12.5 Antenna Barricade Technology

The chassis design may not be effective enough at higher frequency bands (6 GHz bands). Noise leakage observed in the seams, slots and between metal-to-metal bonding. Antenna barricade, metal enclosure around antenna, is an effective solution to mitigate RFI noise at platform level by physically and electromagnetically isolating the antenna from potential sources of interference. Experimental data shows ~10 dB noise reduction with antenna barricade. To avoid any opening/leakage EMI gasket material between barricade and chassis can be applied at multiple locations. However, antenna barricade needs to be planned upfront as it is part of chassis design.

Intel has developed a license technology on antenna barricade technology and design. Contact Intel for more information.

Figure 417. Example of Antenna Barricade in Mobile Design

9.13 EMC ESD Considerations

9.13.1 ESD Sensitive Net Mitigation

The following net categories may be sensitive to ESD:

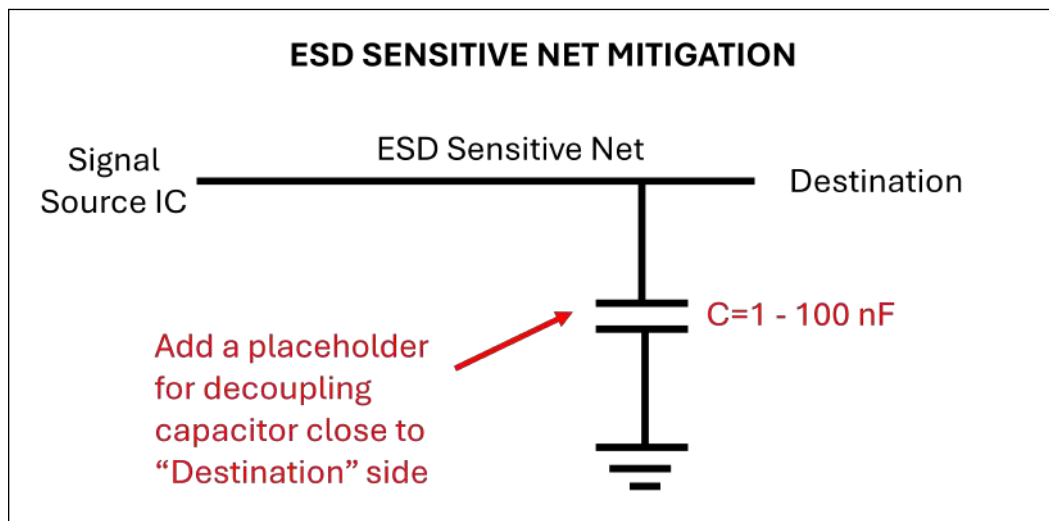
- Processor-Hot or PCH-Hot: *PROCHOT*, *FORCEPR#*, *PCHHOT*
- Reset: *RESET*, *RST*, *RSTB*, *DRAM_RESET*
- Power-Ok, Power-Good: *PWROK*, *PWRGD*
- Crystal: XTAL_IN and XTAL_OUT, RTC

System-level ESD issues may be mitigated by one of the following:

- Routing sensitive nets as stripline.
- Adding placeholder for decoupling capacitors on the sensitive nets to reduce the ESD level injected into the victim ICs (excluding crystal oscillator). The value of decoupling capacitor varies on different nets and systems. The recommended value is 1-100 nF.

For crystal oscillator, follow the guidelines from "XTAL" (High Speed I/Os chapter), "Real Time Clock (RTC) Design Guidelines" (Low speed I/Os chapter) and "XTAL Ground Ring Requirement" (Electromagnetic Compatibility chapter).

Figure 418. ESD Sensitive Net Mitigation



9.14 EMC/RF Reference Documents

Table 738. Reference Documents for Platform EMC/RF Considerations

PDG Sub-section(s)	Document Number	Document Name
(1) Crystal RF Immunity (2) ESD Diode Selection	631030	Thunderbolt - Hayden Bridge Collateral
(1) Crystal RF Immunity (2) ESD Diode Selection	843990	Thunderbolt JHL92x0RP June Bridge Collateral
(1) Crystal RF Immunity (2) EMC Mechanical Consideration	602554	Chassis Design Guidelines for Improving RF Immunity and Electromagnetic Compatibility White Paper Revision 1.0
(1) Memory RFI Mitigation Consideration (2) Intel DDR RFI Mitigation (RFIM) Software Feature	640438	Intel® DDR Memory Radio-Frequency Interference Mitigation (RFIM) Policy Enabling and Validation Technical White Paper
Intel DDR RFI Mitigation (RFIM) Software Feature	816192	Intel® DDR Radio-Frequency Interference Mitigation (RFIM) Feature Semi-Automation of Table Generation Technical White Paper
Minimize Exposed High Speed I/O Channel Routing	610694	Intel® WLAN Module Layout Checklist
System Efficient ESD Design	702999	Intel System Efficient ESD Design (SEED) Models User Guide
Common Mode Choke Selection	546122	Intel PCB Common Mode Choke License Technology Technical White Paper
Antenna Barricade Technology	636597	Antenna Barricade Technology
<i>continued...</i>		

PDG Sub-section(s)	Document Number	Document Name
-	608288	System Level Procedure for Identifying Possible RF Immunity Issues Technical Advisory WW01 2019
-	621488	New DRAM Technology and Compelling EMI Shield Mitigation Solutions Radio Frequency Interference (RFI) Analysis Technical White Paper
-	611571	Flexible Flat Cable for High-Speed Signaling Design Technical White Paper

10.0 Power Delivery

This chapter provides guidelines for the platform power delivery solution. The platform power delivery is comprised of IMVP 9.3 voltage regulators (VRs), Rest of Platform (ROP) VRs, load switches, AC/DC adapter, power and current monitors, USB Type C power delivery and Power Sequencing.

The platform design guide will cover the hardware aspects of power delivery design. For system integration related details such as Platform Monitoring and Control (P_{sys}), Intel® Dynamic Tuning Power Boss, Windows* PMAx Driver, and USB PD Considerations with PL4, refer to Power Delivery Power Management Design Guide (# 634413)

- IMVP 9.3 VRs regulate power to the CPU, graphics and system agent. Details are provided in [Processor Power Delivery Guidelines](#) on page 686.
- Rest of Platform (ROP) VRs regulate power to the rest of the SoC and rest of the system components.
- AC Adapter considerations are mentioned in [AC Adapter Considerations](#) on page 715
- USB Type C Power Delivery is covered in [USB-C* Sub-System](#) on page 254
- Acoustic Noise Mitigation is covered in [Acoustic Noise Mitigation](#) on page 714
- Power Sequencing is covered in [Platform Power Sequencing Specification](#) on page 722

Table 739. Reference Documents

Document Title	Document Number
Power Delivery Power Management (PDPM) Design Guide	634413
Reading and Overriding Power Delivery via BIOS Technical Advisory	626935
Nova Lake- H16C 12Xe VCCGT Fast V-Mode Validation Technical advisory	871402
New BIOS Debug Logs for VR Power/Thermal Technical Advisory	614453
Buck Boost Charger Specification-Power Supply Design Guide	571381
Acoustic Noise Mitigation Options and Validation Directions Technical Advisory	575216
Serial VID (SVID) Protocol Specification	456098
Modern Standby Battery Charger Power Optimization Technical Advisory	621813
Battery Voltage Droop and Brick Over Current Protection Circuit Design White Paper	556361
2020 Intel Dynamic Tuning Technology (Intel DTT) Configuration Guide	618762
Intel Battery Power Optimizer (Intel BPO) (Formerly Intel Dynamic Battery Power Technology v1, v2, v3) Technical Advisory	626679
Intel Dynamic Tuning 8.x BIOS Specification	541817
DC Performance Tuning Guidelines Technical Advisory	618459
<i>continued...</i>	

Document Title	Document Number
Acoustic Noise Mitigation Options and Validation Directions Technical Advisory	575216
Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2	844316
Reactive PL4 with Fast PROCHOT#/FORCEPR# Technical Advisory	635170
IMVP9.3 Voltage Regulation Validation Plan	790157
Nova Lake Platform BIOS Firmware Architecture Specification (FAS)	850519
Nova Lake Platform Turbo and Thermal Power Management Guide for Intel® Core™-Based Processors Implementation Guide	TBD
Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification	861123

Table 740. NVL- H Tool Links

Document Title	Document Number
Gen5 VR Test Tool	https://designintools.intel.com/gen5-vr-test-tool-base-kit.html
416-pin MTM Header	https://designintools.intel.com/416-pin-male-to-male-header.html
Software License	https://designintools.intel.com/gen5-vr-test-tool-host-software-1-year-feature-activation-subscription.html

10.1 Power Maps and Rest of Platform (ROP)

10.1.1 Power Map Tool

The Platform Power Map diagrams key specifications of the voltage regulators (IMVP9.3 and ROP VRs), load switches and the loads. To provide more flexibility in power design, power maps are provided as a separate configurable tool. This tool is also where Baseline and Performance power limits (PL2, PL4), VCC_{CORE} , VCC_{GT} , and VCC_{SA} ICC_{MAX} , $ICC_{MAX,APP}$, I_{PL2} , VR thermal design current (TDC), VR TDC Time Window (Min) and T_{RISE} (Min) can be found for VR design and validation. Refer to Nova Lake Platform Power Map (*this is part of this release package*) to generate a power map that accompanies this section.

NOTE

The processor EDS specification documents take precedence and will override any conflicts with what is being stated herein.

10.1.2 Power Delivery Settings for Nova Lake

NVL-H SKU supports PL1 range to 65W. Recommended power settings (PL1, PL2, PL4, and Icc specs) are provided in the Power Map.

- Intel BIOS setting will present the recommendation as described above.
- Intel recommends customers configure the BIOS based on system power and thermal capability.
- Minimum and Maximum Assured Power (PL1) supported for NVL H is provided in the Nova Lake Mobile Platform Thermal and Mechanical Design Guide, (# 859348).

10.1.3 Rails/Non-IMVP Voltage Regulators

Rest of Processor rails consist of VCCPRIM_VNNAON, VCCPRIM_IO, as well as system shared rails 1.8V, VCCRTC_PROC, VDD2 and VDDQ.

Rest of platform rails should be designed for ramp times between 100µs to 2ms unless otherwise specified or required for a specific load. Where a conflict between this document and an official specification occurs, the official specification is the defining entity.

VCCPRIM_VNNAON

It has a nominal fixed voltage of 0.77V and is supplied in system states S0 through S5.

Differential or "pseudo-differential" package sensing is highly recommended for this rail to maintain accurate tolerance and account for ground bounce.

Differential or "pseudo-differential" package sensing is highly recommended for this rail to maintain accurate tolerance and account for ground bounce. The Nova Lake package has a dedicated VCCPRIM_VNNAON_SENSE pin used to connect the VR's positive remote sense to. The negative remote sense should be routed to a VSS/GND via near the VCCPRIM_VNNAON_SENSE pin. For a VR that supports "pseudo-differential" sensing, please refer to the vendor's application notes on what VR node to connect the negative remote sense to.

Due to noise concerns, it is highly recommended to limit VR switching frequency to 1MHz on this rail to avoid 1st harmonic (fundamental frequency) power supply ripple above 1MHz. Higher-order harmonics are generally not a concern.

NOTE

Refer to "VCCPRIM_VNNAON (0.6mm)" and "VCCPRIM_VNNAON (0.7mm)" in the [Power Integrity Recommendations](#) on page 747 chapter for filter implementation.

VCCPRIM_VNNAON has tight voltage tolerance, high Iccmax, and high current slew rate ($\Delta i/\Delta t$) coupled with high inductor to package sense point path resistance (Rpath) leading to a potential overshoot issue. It is recommended to follow the following guidelines to reduce overshoot:

- Design the VR and its required output capacitor to make sure that the tolerance (DC setting error, ripple, droop/overshoot) is met. Use the VRTT to validate as described in Rest of Platform (ROP) Voltage Regulator Test Plan (# [729382](#)).
 - Implement the output decoupling capacitor placement as per the recommendation in [Power Integrity Recommendations](#) on page 747.

NOTE

The capacitor requirements given in the Power Integrity section only address high frequency (>1MHz) requirements and do not include capacitors needed for voltage regulator response (<1MHz). VR capacitors should be placed as close to the load as possible.

- Work with the VR vendor for further optimization.

- Minimize Rpath from inductor to package pins as much as feasible. Rpath of <4 mOhm is highly recommended, with <2 mOhm preferred to avoid a large number of backside capacitors under the package.
- The Power Integrity capacitor recommendations are for high frequency (>1MHz) decoupling requirements and do not include the Power Delivery capacitors related to voltage regulator response (<1MHz). For best transient response, PD capacitors are recommended to be placed as close to the load as possible.
- The Power Map contains an HSIO calculator for the user to calculate the required Iccmax
 - Δi is proportional to Iccmax
 - Lower HSIO usage leads to lower VNNAON Iccmax which will help in lowering overshoot risk
- In the case that the design has undershoot (Vmin) margin but does not have overshoot(Vmax) margin, it is recommended to lower the VNNAON nominal voltage
 - For example, the nominal voltage could be set to 0.76V instead of 0.77V to provide +60mV and -40mV tolerance budgeting, provided the VCCPRIM_VNNAON rail is still meeting 0.72V Vmin and 0.82V Vmax at the package sense point

Table 741. VCCPRIM_VNNAON Parameters

Symbol	Description	Value	Notes
TDC	Thermal Design Current	$I_{CC\ MAX} / 2$	
Icc MAX	Icc MAX current 10ms max	Refer to Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (# 861123)	
Icc step	Current step for transient measurement	11A/us for NVL 16C and 8C. 10A/us for NVL 6C)	
di/dt	Maximum slew rate for transient measurement	$I_{CC\ step} / 1\mu s$	
Tolerance	Voltage tolerance at package sense pin	Refer to Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (# 861123)	

For VCCPRIM_VNNAON_PI Model, refer to the Nova Lake H Power Integrity Model (# 866209).

VCCPRIM_IO

VCCPRIM_IO is a required rest of Processor power rail for Nova Lake. It has a nominal voltage of 1.25V and is supplied in system states S0 through S5.

Differential or "pseudo-differential" package sensing is highly recommended for this rail to maintain accurate tolerance and account for ground bounce. The Nova Lake package has a dedicated VCCPRIM_IO_SENSE pin used to connect the VR's positive remote sense to. The negative remote sense should be routed to a VSS/GND via near the VCCPRIM_IO_SENSE pin.

For a VR that supports "pseudo-differential" sensing, refer to the vendor's application notes on what VR node to connect the negative remote sense to positive.

Due to noise concerns, it is highly recommended to limit VR switching frequency to 1MHz on this rail to avoid 1st harmonic (fundamental frequency) power supply ripple above 1MHz. Higher-order harmonics are generally not a concern.

Table 742. VCCPRIM_IO Parameters

Symbol	Description	Value	Notes
TDC	Thermal Design Current	Not specified	
ICC _{MAX}	ICC _{MAX} current 10ms max	Refer to Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (# 861123)	
ICC _{step}	Current step for transient measurement	0 to 70% IccMAX and 30 to 100% IccMAX	
di/dt	Maximum slew rate for transient measurement	2.5A/μs	
Tolerance	Voltage tolerance at package sense pin	Refer to Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (# 861123)	

1.8V

The 1.8V rail is required rest of Processor power rails for Nova Lake with nominal voltage of 1.8V . The associated SoC pin names are VCCPRIM_1P8 . These rails are supplied in system states S0 through S5.

For rail specifications, refer to . Nova Lake Platform Power Map and Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (# [861123](#))

Refer to the VCCPRIM_1P8 section in the Power Integrity chapter for filter implementation.

VDD2

The VDD2 rail is a required rest of Processor power rail for Nova Lake. It has a nominal voltage dependent on the DRAM technology used: 1.065V for LPDDR5 and 1.1V for DDR5 and is supplied in system states S0 through S3.

For rail specifications, refer to Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (# [861123](#)).

NOTE

The LPDDR5 JEDEC spec defines VDD2 as 1.05V -2.85%/ + 6.66%. Intel specifies nominal VDD2 voltage in the middle of the range at 1.065V +/-5%.

Symbol	Description	Value	Notes
TDC	Thermal design current	Not Specified	
ICC _{MAX}	ICC _{MAX} current 10ms max	Refer to the Nova Lake Platform Power Map and Nova Lake Processor Electrical Specification (# TBD) for SoC ICCmax.	
ICC _{step}	Current step for transient measurement	0 to 70% ICCmax and 30 to 100% ICCmax	Test VDD2 based on combined ICCmax (SoC +Memory current)
di/dt	Maximum slew rate for transient measurement	2.5A/μs	
Tolerance	Voltage tolerance at BGA	Refer to the Nova Lake Processor Electrical Specification (# TBD).	

VCCRTC_PROC

VCCRTC_PROC is a required rest of Processor power rail for Nova Lake that powers the Real Time Clock circuitry. It has a nominal voltage of 1.2V. with a tolerance of ±5% and is supplied at all times.

Refer to the [Real Time Clock \(RTC\) Design Guidelines](#) on page 394 for recommendations on powering this rail.

For rail specifications, refer to Refer to the Nova Lake Processor Electrical Specification (# TBD)..

VDDQ

The VDDQ rail is a required rest of Processor power rail for Nova Lake that powers the memory subsystem. It has a nominal voltage of 0.52V and only used when LPDDR5 memory is connected. When DDR5 is used, the VDDQ can be tied to VDD2 voltage.

TDC Thermal Design Current	Not specified
ICCMAX	Refer to the Nova Lake H Platform Power Map and Nova Lake Processor Electrical Specification (# TBD)
Icc step current step for transient measurement	Current Step: dI/dt:- 2.5A/us
Tolerance	Refer to the Nova Lake H Processor Electrical Specification (# TBD)

In order to support LPCAMM2 E1 modules, Intel recommends using the "VDDQ common Method" as defined in the JEDEC standard. System boards designed with the common method will supply VDDQ to the CAMM2 connector and must also tie the "VDDQ_DISABLE" CAMM connector pin to "VIN_BULK" CAMM connector pin as recommended in JEDEC standard. To ensure the proper power-up sequence for LPCAMM2 memory rails, the "PWR_GOOD" output from the CAMM2 connector should be part of the mother board VDDQ regulator enable logic. The mother board VDDQ regulator must complete its soft-start within the memory power-up sequence specified in the memory JEDEC standard. During the power-down sequence VDDQ must quickly discharge the output voltage to meet the memory power down sequence specified in the memory JEDEC standard.

For systems (LPDDR5 Memory down / LPCAMM2 E1 modules) using a common VDDQ rail for the processor and DRAM, the mother board VDDQ regulator shall consider supporting memory DVFSQ mode. DVFSQ requires supporting inrush current capability calculated with the maximum slew rate chosen for the DVFSQ mode (for example, 15 mV/usec $\pm$ 10%), total capacitance (including memory CAMM2 module/DRAM de-coupling and Processor VDDQ rail de-coupling plus mother board de-coupling capacitance if applicable). Refer to the latest JEDEC standards.

Following table provides the process to calculate VDDQ VR's DVFSQ slew rate and check VDDQ rail slew rate against the minimum. 10mV/us and Maximum 20mV/us, applicable in the case when VDDQ Voltage Regulator operates in "Current limit" mode (operates at constant inrush Current limit value).

In case of "fixed slew rate" control mode (example 15 mV/usec $\pm$ 10%), VDDQ Voltage Rail is expected to ramp with the fixed slew rate, for example with CAMM2 module VDDQ rail capacitance of 200uF, VDDQ regulator is expected to provide the slew rate of 15mV/us as the VR inrush current doesn't reach VDDQ VR inrush Current limit.

Minimum Slew Rate	Maximum Slew Rate
$ILimit.min = ILimit - \text{current tolerance(A)}$ minimum value of VR inrush current limit	$ILimit.Max = ILimit + \text{current tolerance(A)}$ Maximum value of VR inrush current limit
$Cout.Total.Max = Cout.Total + \text{cap. Tolerance}$ Maximum value of Total decoupling capacitance in uF	$Cout.Total.min = Cout.Total - \text{cap. Tolerance}$ minimum value of Total decoupling capacitance in uF
$Slew\ rate = (ILimit.min \div Cout.Total.Max) \geq 10$ mV/us To meet latency requirements	$Slew\ rate = (ILimit.Max \div Cout.Total.min) \leq 20$ mV/us To meet JEDEC Max. slew rate requirement

Example: Calculation with decoupling cap and inrush current assumptions :

CAMM2 module VDDQ rail capacitance = 356uF

SoC VDDQ rail de-coupling = 88uF

VDDQ Voltage Regulator decoupling cap = 94uF

No additional VDDQ rail capacitance on mother board is accounted in the calculation.

Capacitor manufacturing tolerance = +/- 20% (worst case analysis)

Total capacitance = 356uF (module) + 88uF (SoC) + 2x47uF (VR) = 538 uF.

VDDQ Voltage Regulator inrush current limit = 7A +/- 0.5A.

VDDQ Rail load current assumed to be 0A for the calculation.

DC bias capacitance de-rating is also ignored in the calculation.

Minimum Slew Rate	Maximum Slew Rate
$ILimit.min = 7 - 0.5 = 6.5\ A$	$ILimit.Max = 7 + 0.5 = 7.5\ A$
$Cout.Total.Max = 538 * 1.2 = 645.6\ uF$	$Cout.Total.min = 538 * 0.8 = 430.4\ uF$
$Slew\ rate = 6.5 * 1000 / 645.6 = 10.0682\ mV/us$	$Slew\ rate = 7.5 * 1000 / 430.4 = 17.4257\ mV/us$
Slew rate of 10.0682 mV/us $\geq$ 10 mV/us - OK.	Slew rate of 17.4 mV/us $\leq$ 20 mV/us - OK.

Maximum Slew rate value might over take by fixed slew rate set value (example: 15mV/usec) for the above calculation.

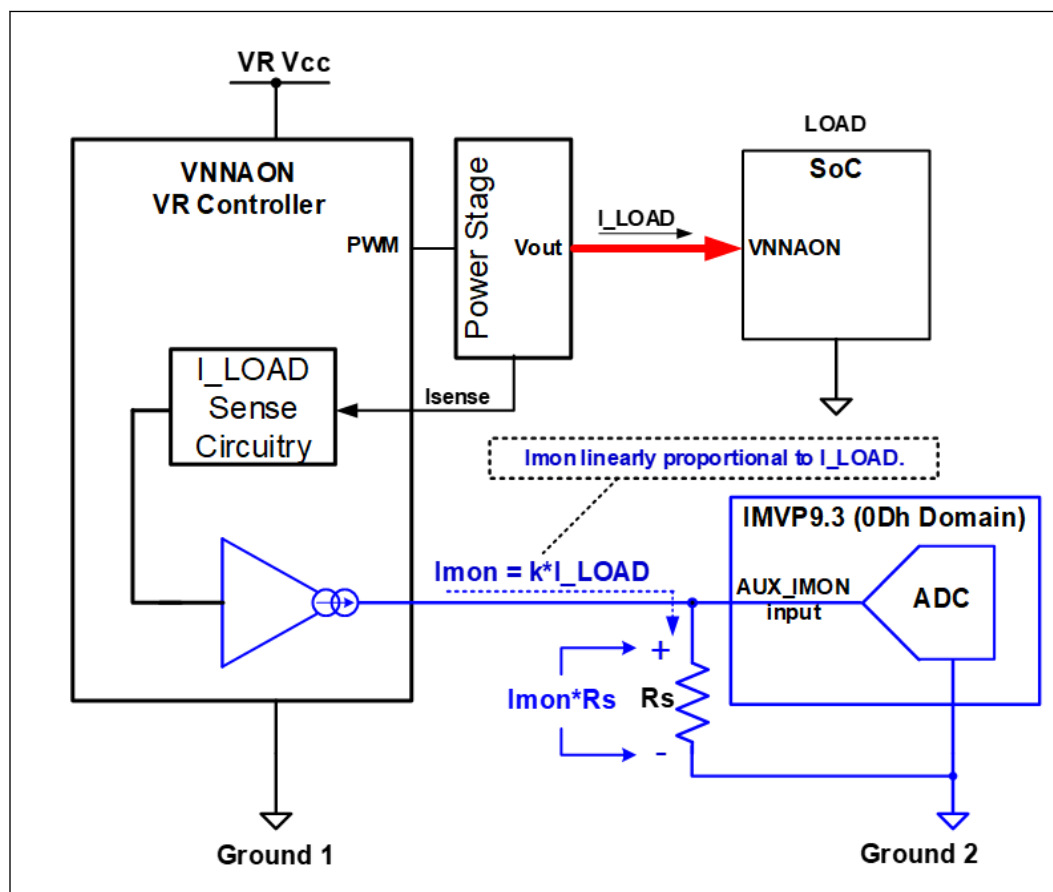
10.1.3.1 VNNAON IMON

This section specifies the current monitoring (Imon) requirements of the VNNAON voltage regulator.

10.1.3.1.1 Background

Imon output pin is an analog current mode output implemented in the VNNAON voltage regulator. Its implementation is depicted in Figure 419 on page 675.

Figure 419. VNNAON Imon Application



This output sends a current mode signal to the platform that represents the current the VR is sourcing to the load (linearly proportional to the VNNAON load current). This signal is digitized by the platform ADC and is used to calculate the power being sourced from the VNNAON voltage regulator output.

10.1.3.1.2 Active States

If the Low Power Mode (LPM) pin is implemented, when the signal goes low the voltage regulator may disable IMON functionality to save power.

Imon must be stable $\leq 200\mu\text{s}$ after exit from LPM.

10.1.3.1.3 Electrical Requirements

Here are the electrical requirements for Imon:

- **Output Type:** Current mode output proportional to I_LOAD.
 - Converted to voltage at destination ADC by shunt resistor Rs.
- **Full Scale Voltage (FSV):** settable from 500mV to 3.3V.
 - Full scale voltage set by value of shunt resistor Rs for I_LOAD=ICCMAX.
 - Amplifier should be designed to output current capability that results in a full-scale voltage range of 500mV to 3.3V for Rs range of 4.7k to 47k respectively.
- Imon Rise Time for (0A to ICCMAX step): $t_r \leq 10\mu s$ (10% to 90%, BW=35 kHz min).

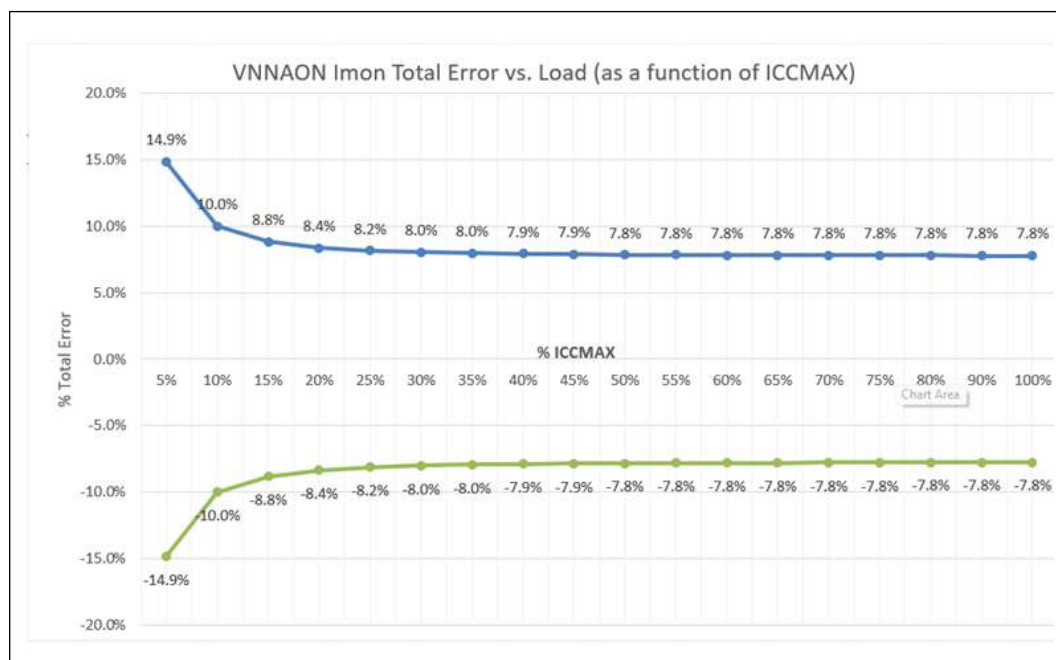
10.1.3.1.4 Imon Output Total Error Requirements

Imon output of the VNNAON VR must comply with the error requirements depicted in below figure.

Assumptions:

- Assumes DCR sensing is worst case tolerance.
- Inductor DCR 4mOhm +/-7%.
- NTC tol $\leq \pm 3\%$:
 - Assumes DCR thermal compensation.
 - May not be applicable if DCR sense is not used.
- Isense signal path offset <0.5mV.
- Other methods of sensing may be tighter tolerance.

Figure 420. VNNAON Imon Output Total Error Requirements



10.1.4 Rest of Platform (ROP) VR Enabling

Intel has worked together with various vendors to enable both discrete and point of load (POL) voltage regulators that meet the new power delivery requirements in Nova Lake. For a partial list of VR controllers that can be used for the rest of the Processor rails.

10.1.5 Processor Load Switches

Load switches are not required on any Nova Lake Processor rails.

10.2 Reactive Peak Power Management

Nova Lake introduces the ability to manage peak power events.

Designers use *proactive* reductions of the Processor PL4 level to prevent catastrophic system crashes, however, the drawback of lowering PL4 is that the reductions in performance happened proactively, guaranteeing the power levels by only allowing lower performance frequency states. From a performance standpoint, this is undesirable behavior. Project Athena testing has shown responsiveness key experience indicators (KEIs) are closely tied to the PL4 setting.

To preclude the over design of platforms, the approach taken here is to allow for *reactive* peak power management. Two reactive features have been introduced:

1. **Fast V-mode:** This feature insulates VR FETs / Inductors, from observing full PL4 current as well as upstream input power devices from observing full PL4 power. $ICC_{MAX,APP}$, which is less than ICC_{MAX} , represents the real workload maximum expected current. Fast V-mode allows for platform power subsystems to be designed to $ICC_{MAX,APP}$, not ICC_{MAX} , while providing a performance improvement over proactive PL4 / ICC_{MAX} reduction(s).
2. **Reactive PL4:** 2S (two cells in series) battery systems, while being efficient in power conversion, are at risk of brownout during peak power events, hence they tend to request lower PL4 levels. This PL4 level fluctuates depending on the remaining state of charge (RSOC) of the battery. The system can implement the Reactive PL4 mechanism called "PL4 Boost" given the:
 - a. Effective capacitance on V_{SYS} , and
 - b. Power removal reaction speed due to a system rail undervoltage event.

The Processor uses PL4 Boost to calculate a higher performance frequency with a potentially higher Pmax than the programmed PL4 value. Upon FORCEPR# assertion, the programmed PL4 level is respected. Oscillatory assertions are addressed when identified.

10.2.1 Fast V-Mode

An SoC integrated droop detector monitors when the SoC load current exceeds a preset threshold by monitoring for SoC voltage domain droop. The SoC compares the IMVP VR output voltage with a preset threshold voltage (V_{trip}) and when the IMVP VR output voltage reaches this V_{trip} , the SoC throttles itself to reduce load current.

IMVP9.1 VRs enhance this detector by adding a cycle-by-cycle current limiting feature where the IMVP VR quickly enters current limit (becomes a current source) with the VR output limited to a preset value (I_{TRIP}) as set in the VR ICC_limit register.

Refer to the IMVP9.3 Pulse Width Modulation VR Vendor Enabling Specification (# 740011) for more information.

10.2.1.1 FVM Benefits

- The SoC detector and IMVP cycle by cycle limiting work together to allow the voltage regulator inductors and FETs to be sized for realistic workload current (ICCMAX.APP) instead of virus currents (ICCMAX).
- Upstream power sources to be sized for realistic workload currents since Fast V-mode shields them from SoC large dynamic loading events that occur when the SoC current exceeds the worst case realistic maximum current (ICCMAX.APP).

10.2.1.2 Terminology

1. **IccMax** is the maximum current the processor can draw, typically seen running a virus application (stress applications specifically designed to push the SoC to maximum power). This is the current used to design the voltage regulator output decoupling.
2. **PMax (PL4)** is the maximum instantaneous electrical power drawn by the SoC (per power rail), typically seen running a virus app (stress applications specifically designed to reach maximum consumed power) with an all core turbo and Maximum Operating Temperature scenario.
3. **IccMax.app** is less than IccMax and is the electrical current drawn by the SoC (per power rail) while running a typical user realistic application(s) scenario at P0nmax and Maximum Operating Temperature. It corresponds to PMax.App.
4. **PMax.App (PL4.App)** is less than Pmax and is the electrical power drawn by the SoC (per power rail) representing a typical user realistic application(s) scenario at P0nmax and Maximum Operating Temperature. Pmax.App values may differ from SKU to SKU or generation to generation. PMax.app represents worst case power with real world workloads. Pmax.App occurs at IccMax.App
5. **ITRIP** is VR output current limit threshold above which the VR goes into current limit. This current is greater than IccMax.App and less than IccMax. It is determined by customer-specific VR current capability and the VR IMON measurement tolerance (TOL_{IMON}). Itrip should be set as follows.

$$I_{trip} = I_{ccMax.App} / (1 - TOL_{IMON})$$

Setting Itrip this way ensures the VR never enters current limit below IccMax.APP, which avoids the risk of throttling on realistic workloads.
6. **ITRIP Max** is the peak current seen by the VR FETs and inductors, accounting for the VR controller current measurement tolerance (TOL_{IMON}). The SoC VR and system input power sources must be able to sustain this current for at least 10ms.

NOTE

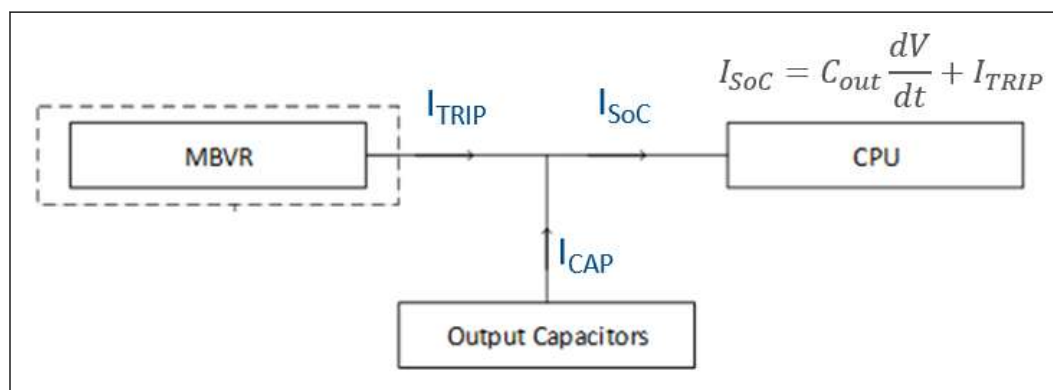
TOL_IMON can be sometimes called as ITRIP_TOL or ITRIP_ACCURACY

7. **VTRIP** is the output voltage level below which the SoC will self-throttle. This voltage is internally set in the SoC and is not visible external to the SoC. For the purposes of VRTT testing, a Vtrip value will be provided for use with the VRTT FVM test scripts for FVM compliance testing.

10.2.1.3 Operation

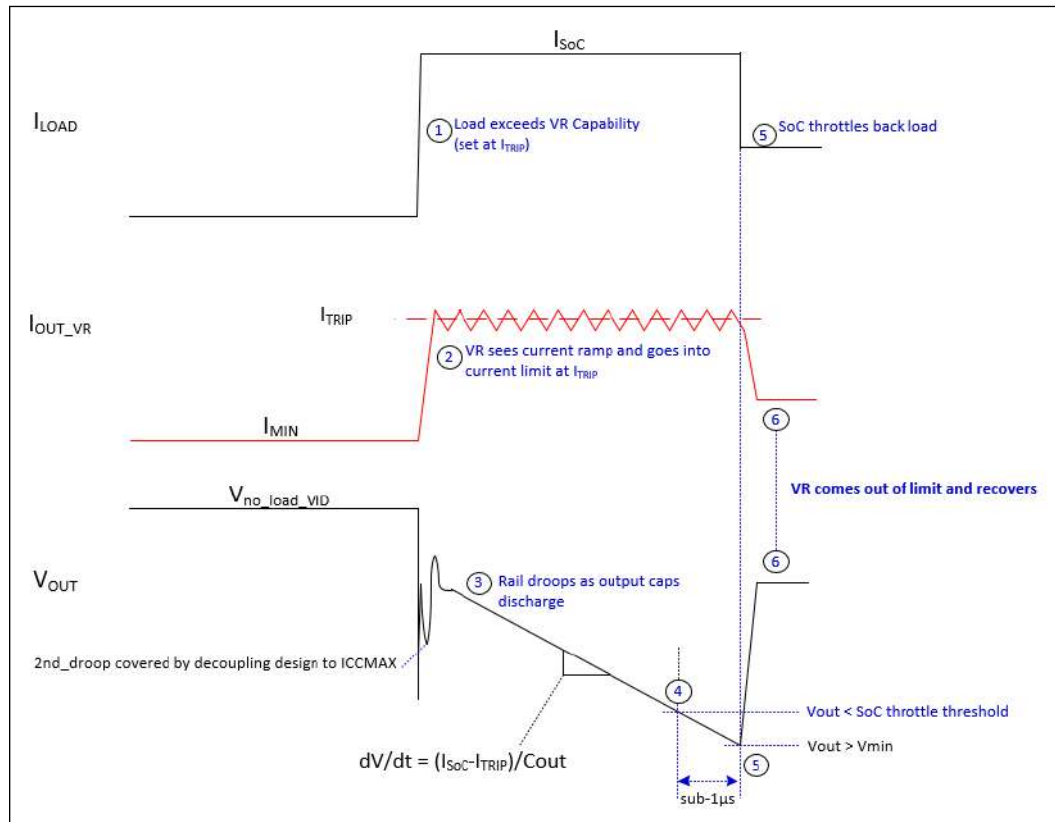
Fast V-Mode works to protect the voltage regulator and upstream input power sources from transient currents that exceed the worst-case realistic current (ICCMAX.APP) generated by realistic workloads. Referring to the figure below, when the load current (I_{SoC}) is below the regulator current limit threshold (I_{TRIP}), the VR regulates the output voltage normally as required by the IMVP specification.

Figure 421. Power Delivery Currents During Fast V-mode Limit



When the load current (I_{SoC}) exceeds I_{TRIP} the VR goes into current limit and the following events occurs, as shown in the figure below:

Figure 422. Fast V-mode Limit Example



1. The SoC applies a load that exceeds the VR cycle-by-cycle limit (I_{TRIP}).
2. The VR regulates/limits the average VR output current to I_{TRIP} .
3. The output decoupling makes up the difference between the current demanded by the load (I_{SOC}) and the current provided by the regulator (I_{TRIP}).

$$I_{CAP} = I_{SOC} - I_{TRIP}$$

The output droops at a rate that set by the total output decoupling and I_{CAP} :

$$dV/dt = I_{CAP} / C_{OUT_TOTAL}$$
4. When V_{out} falls below the SoC throttle threshold (V_{TRIP}), SoC throttling is triggered.
5. The SoC throttles back its load current below I_{TRIP} .
6. The VR exit current limit and recovers to normal operation.

10.2.1.4 Designing for Fast V-mode

The parameters for designing to Fast V-mode requirements can be found in the power map tool for the appropriate SoC.

The power map introduces a new option to support the Fast V-mode feature. Fast V-mode acts in the following manner.

- Limits the VR output current so that the VR inductors and FETs do not see currents in excess of I_{TRIP_MAX} .
- Limits the power seen by upstream power sources to $P_{max.APP}/(VR_efficiency)$ for each SoC rail that support Fast V-mode.
- Does not prevent VR output decoupling from seeing the SoC $ICCMAX$.

When designing for Fast V-mode:

- **VR FETs and Inductors:** design to sustain I_{TRIP_MAX} for 10ms peaks. They do not need to be designed for $ICCMAX$.
- **VR output decoupling:** design for $ICCMAX$ transient since the decoupling will see full SoC $ICCMAX$.
- **IMVP 0x21h:** designer should set reg0x21h to I_{TRIP_MAX} value to maximize the I_{MON} ADC dynamic range.

NOTE

For Fast V-mode systems, the SoC $ICCMAX$ must be set to I_{trip} or less when Fast V-mode is disabled to ensure SoC currents do not exceed FET and inductor design limits.

When Fast V-mode is disabled with system designed for full $ICCMAX$:

- **VR FETs and Inductors:** design to sustain $ICCMAX$ for 10ms peaks.
- **VR Output Decoupling:** design for $ICCMAX$ transient since the decoupling will see full SoC $ICCMAX$.
- **IMVP 0x21h:** designer should set reg0x21h to $ICCMAX$ value to maximize the I_{MON} ADC dynamic range.

10.2.1.5 Setting Fast V-mode Current Limit Threshold (I_{TRIP})

Platforms should never throttle below $ICCMAX.APP$ as this increases the probability of throttling with realistic workloads. If we assume $ICCMAX.APP$ is the lowest allowable current limit point, and we assume that the VR has a non-zero current measurement error (TOL_{IMON}), then the I_{TRIP} setting that guarantees the VR never trips below $ICCMAX.APP$ can be written as:

$$I_{TRIP} = ICCMAX.APP / (1 - TOL_{IMON})$$

where:

- I_{TRIP} is the nominal VR controller current limit threshold set as a function of $ICCMAX.APP$ and VR current measurement error (TOL_{IMON}).
- $ICCMAX.APP$ is a design parameter specified by Intel
- TOL_{IMON} is the peak VR controller current measurement error for the selected VR controller

Example: A VR with +/-6.5% I_{MON} error means the one sided $TOL_{IMON} = 0.065$

10.2.1.6 Output FET and Inductor Selection

As stated above, because the VR will have a non-zero IMON tolerance, the nominal VR current limit threshold (I_{TRIP}) must be offset higher than ICCMAX.APP to guarantee that the VR never goes into current limit below ICCMAX.APP. Because I_{TRIP} is the nominal current limit threshold, the worst case current seen by the FETs and inductors is:

$$I_{TRIP_MAX} = I_{TRIP} \times (1 + TOL_{IMON})$$

where:

- TOL_{IMON} is the peak VR controller current measurement error.
- I_{TRIP_MAX} is the worst case current the FETs and inductors would see as a function of nominal current limit threshold (I_{TRIP}) and VR current measurement error (TOL_{IMON}).

Example: A VR with +/-6.5% IMON error means the one sided $TOL_{IMON} = 0.065$

FETs and inductors must be able to support I_{TRIP_MAX} for 10ms peaks.

10.2.1.7 Output Capacitance

The output capacitance must be designed to support ICCMAX transients and must meet all the usual electrical requirements associated with IMVP voltage rails (load line, transient droop, overshoot, and other IMVP rail specs) as called out in this PDG, and in the IMVP Validation Test Plan.

Additionally, Fast V-mode requires the decoupling solution must be large enough to ensure that, while the VR is in current limit, the rail does not decay at a rate greater than FVM Max ΔV over time $t_{THROTTLE}$ as specified in the power map tool.

Where:

- FVM Max ΔV is the maximum allowable voltage delta during VR current limit .
- $t_{THROTTLE}$ is SoC throttle time.

10.2.1.8 Testing for Fast V-mode Compliance

The following description references the Figure below.

For Fast V-mode to work properly the SoC requires that the maximum dV/dt during limit not exceed a specified amount.

NOTE

Refer to the power map for the Fast V-Mode requirements for FVM Max ΔV and $t_{THROTTLE}$.

T Testing for dV/dt compliance is performed using the following tools:

- VRTT Tool
- Client Fast V-mode Test script
- Updated IMVP test plan with added Fast V-mode Compliance tab

Platforms that support Fast V-mode must pass compliance testing as called out in the IMVP Validation Test Plan using updated VRTT Test Scripts. Testing includes the following:

1. All standard IMVP Validation Test plan testing for cases where the VR does NOT go into current limit (legacy testing).
2. New testing for large transient to ICCMAX with the VR going into limit. The VR must pass with no Vmin violation.
3. Steady state rail decay test while the VR is in current limit. The VR must meet a requirement that the decay dV/dt that does not exceed a maximum allowable dV (FVM Max ΔV) over a specified time ($t_{THROTTLE}$) as specified by Intel requirements.

These tests are included in the latest release of the IMVP Validation Test Plan and VRTT test scripts. While the first two tests are similar to legacy testing for loadline and transient Vmin testing, the third test is a new test that checks to make sure that the rate of decay during VR current limit does not exceed the allowable dV (FVM Max ΔV) over time $t_{THROTTLE}$. The VRTT tool tests for this in the following manner.

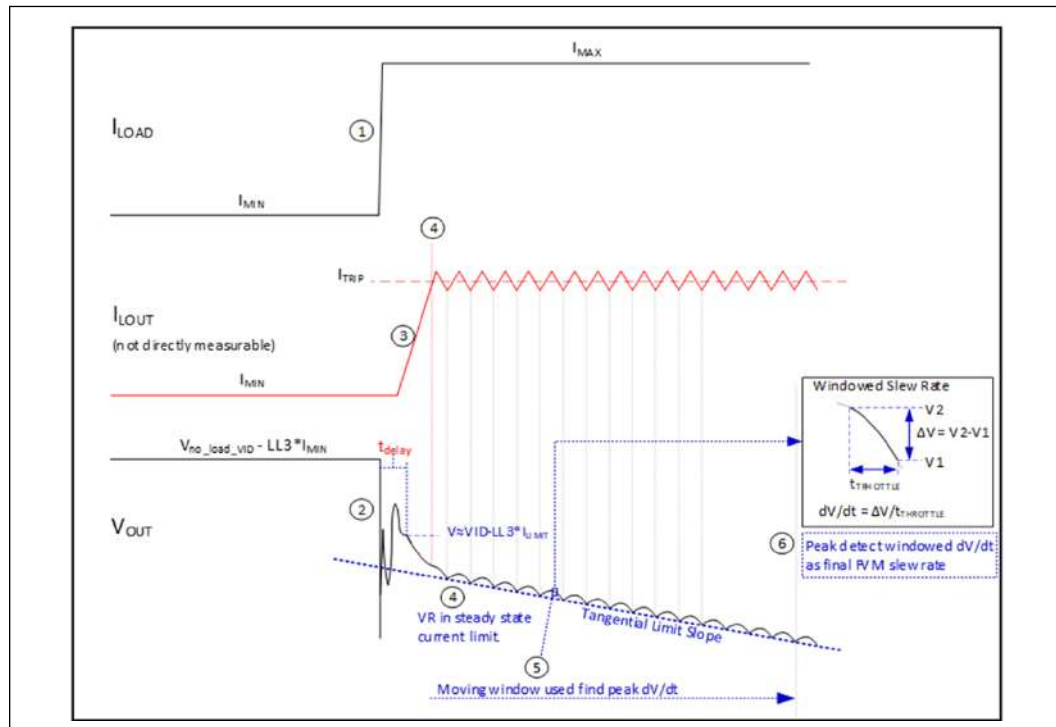
1. VRTT applies a load of ICCMAX and captures the waveforms as shown in the Figure below.

NOTE

The inductor current (I_{LOUT}) is shown for clarity; the tool cannot see this current.

2. Decoupling covers the initial I_{CCMAX} transient until the VR starts to respond.
3. The waveform data is post processed using a sliding time window of duration $t_{THROTTLE}$ to find the measured dV over time $t_{THROTTLE}$, the time it takes the SoC to throttle back once a throttle is triggered.
4. The test reports the DV, which must be less than the required DVFVM specified in the power map tool for a pass result.

Figure 423. Fast V-mode Compliance Testing Methodology



During a current limit condition where the VR is acting as a current source the load current is.

$$I_{SOC} = (C_{OUT} * dV/dt) + I_{TRIP}$$

During current limit, the cap current is set by the SoC load current (I_{SOC}) and current limit of the VR (I_{TRIP}). In the event that a design does not meet the maximum allowable Fast V-mode dV (FVM Max ΔV) over the dt ($t_{THROTTLE}$) designers may choose one of the following actions to reduce the dV/dt:

- Increase VR I_{TRIP} to reduce capacitor current (I_{CAP})
- Reduce the SoC I_{CCMAX} setting to lower I_{CAP} for fixed I_{TRIP}
- Add C_{OUT} to reduce the dV/dt for fixed I_{SOC} and I_{TRIP}

While option three seems like an obvious and simple solution that will allow a pass for Fast V-mode testing, it will come at the expense of elevated light load power loss when entering and exiting low power states where the IMVP rails go to 0V condition. This will directly impact battery life for realistic workloads where the IMVP VR goes 0V often. As an example, a 1000uF decoupling solution that ramps to 1.0V at a 1kHz rate will mean 1W of added loss.

Given the infrequent nature of load currents that exceed I_{TRIP} relative to realistic workload currents, and the large impact of increased C_{OUT} on battery life, it is better to first attempt a reasonable increase VR I_{TRIP} capability, or reasonable reduction in I_{CCMAX} , as a correction for Fast V-mode dV/dt failures. It is recommended that increased C_{OUT} should only be used as a last resort for correction of Fast V-mode dV/dt failures.

10.2.1.9 Enabling Fast-Vmode In System

In order to enable Fast V-mode the following must occur on transition to S0:

1. The ICCMAX supported by the decoupling (same as the previous ICCMAX parameter in previous gen platforms)
2. BIOS must write the VR current limit setting (I_{TRIP}) to the VR and SoC via the BIOS mailbox. Refer to the Nova Lake Processor Family BIOS Specifications (# 861572).

10.2.2 Reactive PL4 with PL4 Boost

The PL4 Boost feature enables higher peak performance and/or responsiveness for 2S battery systems in low remaining state of charge (RSOC) conditions. Responsiveness gains are a result of the Processor using higher frequency states while having a reactive mechanism in place to quickly reduce loading.

Using 2S batteries allows for the most efficient power conversion and battery density per volume versus 3S batteries, however, in low RSOC conditions there is risk of brownouts due to system rail voltage droop when using high PL4 settings.

The IMVP controller can be used to monitor V_{SYS} voltage and assert PSYS_CRIT# (if the pin exists) or VR_HOT# in the event of an undervoltage event. This function is referred to as the *system input power critical path*, or PSYS_CRIT#, in the Serial VID (SVID) Protocol Specification (#456098). Some configuration of the controller is needed:

1. Set the PSYS_CRIT# mode to voltage droop monitoring mode by setting the VsysModePeak bit in the IMVP controller to V_{SYS} (trigger below threshold)
2. Set the threshold using the PsysCrLvl bits to the appropriate minimum system voltage supported
3. Program the debounce time in the PsysCritAssertBebounce registers
4. The PSYS_CRIT# (if available) or VR_HOT# pin should be connected to the Processor's FORCEPR# signal.

PL4_BOOST is programmed by the BIOS as an increase to PL4 that the battery and/or power supply can support in the presence of a fast throttling response to the system rail drooping below the threshold set.

Fast Throttling: The Processor power consumption would drop below the programmed PL4 value within 1 μ sec after FORCEPR# is asserted by the platform.

In the absence of a programmed PL4_BOOST value, the Processor operating point is guaranteed not to exceed PL4.

When PL4_BOOST is valid, the maximum Processor power is the lower of:

- PL4 + PL4_BOOST
- The highest operating point at which the fast FORCEPR# response is guaranteed to render Processor power below PL4 within 1 μ sec after FORCEPR# assertion

Along with Processor FORCEPR# reaction time of 1 μ sec, the total response time must also account for platform level delays:

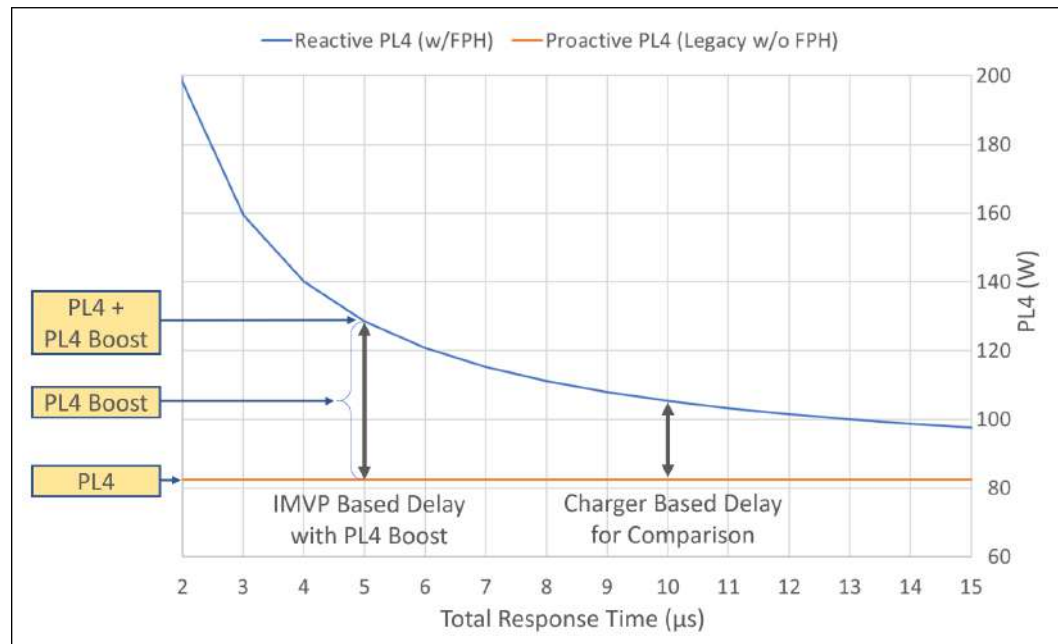
- VR propagation delay: consult the IMVP VR data sheet

- PSYS_CRIT# debounce time as programmed - for example 2 μ sec using IMVP and 7 μ sec using a charger. Consult your supplier data sheet.

The potential PL4 Boost available is a function of the total response time, V_{SYS} input capacitors, and system/battery resistance. Refer Figure 496 for an example of the PL4 gain possible depending on the total response time.

The difference between the possible reactive PL4 and the proactive PL4 is known as PL4_Boost.

Figure 424. PL4 Boost Given Total Reaction Delay



The above figure illustrates the effective PL4 capability of the system when using the IMVP based method versus the Charger based method. Using the IMVP controller to monitor V_{SYS} voltage provides additional PL4 when used with PL4 Boost.

PL4 Boost design and validation aspects are provided in Reactive PL4 with Fast FORCEPR# Technical Advisory (# 635170).

10.3 Processor Power Delivery Guidelines

This section provides the general processor voltage rail design guidelines.

10.3.1 General Processor Power Delivery Considerations

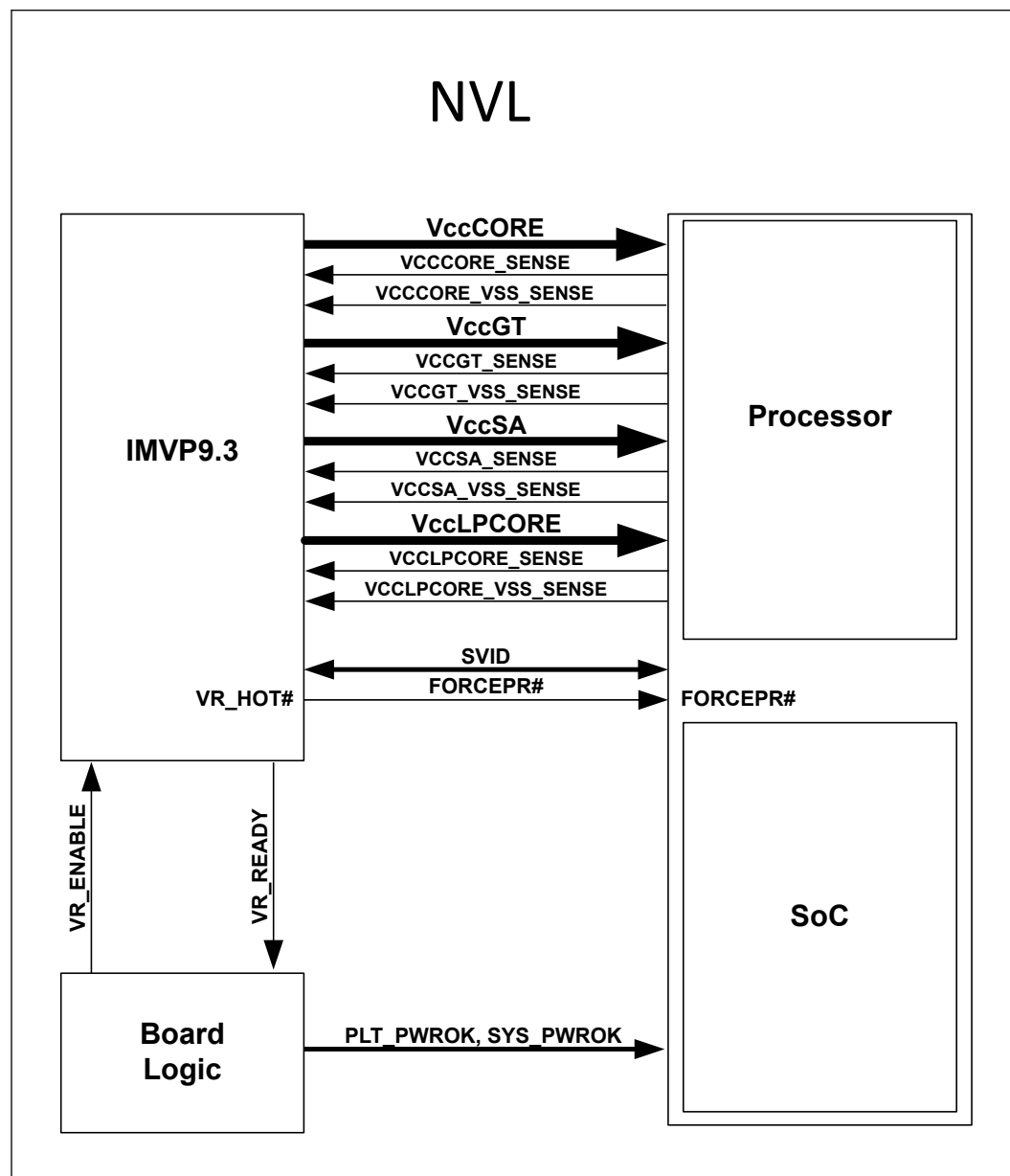
Nova Lake requires an IMVP9.3 VR controller to power the processor core, graphics, and system agent. IMVP9.3 includes several required features for Nova Lake:

1. Cycle-by-cycle current limiting for the "Fast V-mode" feature
2. No early SVID ALERT signaling when performing a SETVID down

Refer to the IMVP9.3 Pulse Width Modulation VR Vendor Enabling Specification (# [740011](#)) for further details.

The following figure is a generic representation of the IMVP9.3 VR. Other features and implementations are also possible.

Figure 425. IMVP9.3 VR Block Diagram



Important points to note in this block diagrams are:

- IMVP9.3 requires differential remote sensing. Differential sense inputs for a given rail must be the **ONLY** feedback point back to controller. Refer to the [Table 743](#) on page 691.

- VR_READY is a required controller output signal. It indicates that VR is operational. VR_READY signal is also used for appropriate power up sequencing during a platform boot.
- The IMVP9.3 compliant controller is required to provide VR_HOT# (VR Thermal Throttling) output signal to indicate operation close to regulator maximum thermal limit. It is used by the system to prevent catastrophic thermal damage.
- VR_ENABLE (IMVP9.3) is a required signal provided by platform logic. It enables the outputs of the IMVP9.3 regulators.

10.3.2 Testing and Validation

The nominal processor voltage is determined by an SVID code provided to the VR. The VID nominal value indicates the reference point that the VID provides for the static and ripple voltage tolerance. Due to VR tolerances, PCB parasitic and current draw variations, the average voltage seen at the processor may be slightly higher or lower than the reference value according to the loadline spec or tolerances.

The VR must meet the static, ripple, and loadline tolerance limits for the entire range of processor load. Loadline, static and ripple tolerance window specifications must be met as measured differentially at Vcc_SENSE/Vss_SENSE pins.

Frequency Domain Impedance Measurement (FDIM) testing is also expected to be completed up to 1MHz on every IMVP9.3 power rail.

Refer to the IMVP9.3 Validation Plan for guidance on validating power delivery and FDIM test requirements.

10.3.3 Audible Noise Reduction

Acoustic Noise is caused by different factors in the system, like mechanical noise that comes from the fan or electrical.

In case of electrical noise issues, the appropriate Audible Noise Reduction option depends on the noise type. Solving acoustic noise is not.

- To avoid acoustic noise issues, during board design the BOM selection needs to be considered. Using smaller capacitors such as 0402 can help reduce the intensity of vibration and noise resulting from voltage switching.
- Noise mitigation is best done at the source.
- Symmetrical mounting on Top and Bottom of the PCB (using TH via), using capacitors with the same size, can help mitigate vibration intensity. By doing so, a large part of the vibrations generated by the capacitors is canceled therefore reducing the acoustic noise.
- Minimize the length and impedance of traces connecting capacitors to power and ground planes to reduce parasitic effects.
- Use short and wide traces for power distribution to minimize voltage drops and impedance.
- Implement a solid ground plane to provide a low-impedance return path for high-frequency currents and reduce ground loops.
- Implement a solid ground plane to provide a low-impedance return path for high-frequency currents and reduce ground loops.

- Connect all ground points directly to the ground plane without loops or shared paths to minimize noise pickup.

At least three conditions need to be simultaneously present for acoustic noise. They are periodic dv/dt in the audible range, sufficient dv magnitude, and insufficient damping, e.g. typical of a heavily-reliant or all-MLCC decoupling solution. The noise amplitude increases with current in $I = C * dv / dt$.

Refer to Acoustic Noise Mitigation Options and Validation Directions Technical Advisory (# 575216) for ways to mitigate acoustic noise.

NOTE

For further details on mitigating acoustic noise, refer to the Acoustic Noise Mitigation (# 575216). By using one of the software solutions in the technical advisory, may impact power and performance, interrupt response time, and/or device latency.

10.3.4 Auto Power Management

IMVP Power State (PSx) settings are based on *potential* current, not actual current, so at times the PSx selection is not going to be the most efficient place for the VR to be. To improve VR efficiency, Auto Power Management (APM) has been introduced in the IMVP specification. APM allows the VR to automatically phase shed and enter discontinuous mode (DCM) as needed, as long as the VR can remain within all specs.

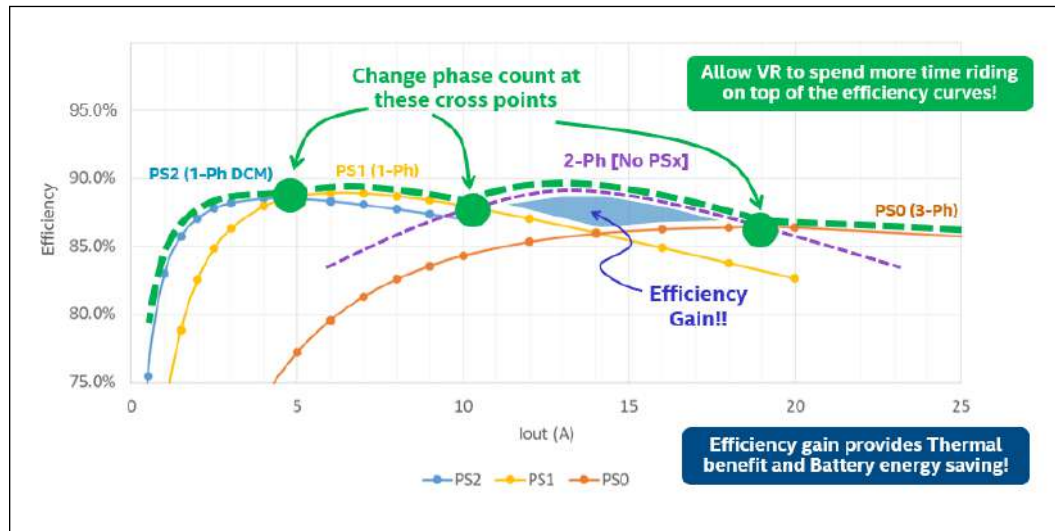
Some of the benefits of APM include:

1. Increases opportunity for VR to be in most efficient place by shedding/adding phases and enabling DCM as needed.
2. Allows VR to operate with optimal phase count (between one phase and all phases) in PS1 and PS0 as shown in Figure below.
3. Effectively reduces VR loss in Battery Life and Performance workloads, which may improve VR thermals.

Nova Lake has combined and relaxed the voltage tolerance band + ripple to provide more opportunities for APM to be used. Refer General Processor VR Parameters for tolerance.

It is recommended to choose a VR Controller that supports Auto Power Management with a short response time and with good light load efficiency as evidenced by the results of the IMVP9.3 Light Load Efficiency Test Plan (# 830609). Programming the phase transition thresholds at the efficiency cross points shown in Figure below accurately can also improve the effectiveness of APM.

Figure 426. Auto Power Management Efficiency Gain Example



10.3.5 Vcc_SENSE/Vss_SENSE Package Sensing

Motherboards are subject to layout area constraints that could significantly compromise a VR's placement on the motherboard and its connection to the processor pins. To ensure that the power delivery network's (PDN) impedance profile stays at/below the maximum loadline target for frequencies above the VR's Gain-Bandwidth (G-BW), the placement and connection of the decoupling capacitors as well the shapes of the power plane(s) is critical. At frequencies within the VR's G-BW control, the VR layout placement may be compensated using remote sensing of the voltage at the die through Vcc_SENSE/Vss_SENSE signals.

The processor implements the concept of "package sensing" for IMVP9.3 voltage rails; the following examples use Vcc/Vss as general labels.

NOTE

For optimal performance and noise rejection, the feedback signals for the IMVP9.3 controller should be routed as if they are a differential pair connecting the Vcc and Vss bumps of the processor to the IMVP9.3 controller Vcc_SENSE and Vss_SENSE pins respectively.

NOTE

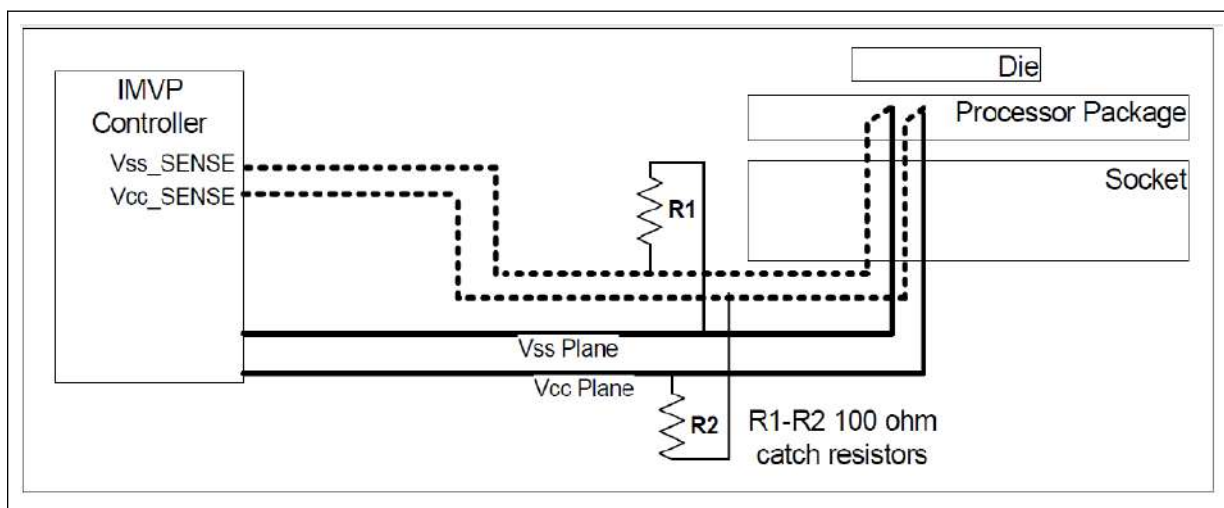
This is true for non-IMVP9.3 rails too. This implementation extends the VR controller's regulation range, enclosing the motherboard parasitic within its feedback loop minimizing the impact on voltage regulation. The conceptual implementation of Vcc_SENSE/Vss_SENSE package sensing is illustrated in Figure below with recommendations in Table below.

Table 743. Package Sensing Recommendations

Signal Names	R1,R2	Trace Length Match
VccCORE_SENSE /VccCORE_VSS_SENSE	100 Ohm	<0.635 mm
VccGT_SENSE / VccGT_VSS_SENSE		
VccSA_SENSE / VccSA_VSS_SENSE		
VCC_LP_ECORE_SENSE / VCC_LP_ECORE_VSS_SENSE		

To minimize any stray noise pickup to the Vcc*_SENSE/ Vss*_SENSE lines

- Sense traces should be referenced to a solid ground plane
- Avoid crossing over plane splits
- Maintain 25-mil separation distance away from any other dynamic signals

Figure 427. Processor Vcc_SENSE/Vss_SENSE Package Sensing Example

- Figure above demonstrates the purpose/function of the feedback “catch” resistors. As discussed above package sensing effectively encloses the board parasitic within the feedback loop of the VR; however, a problem arises when the DUT is not present; the feedback loop is open and the VR does not have any feedback. Aberrant or unstable VR behavior may result, including over voltage, errant/ sporadic VR operation, etc. This may potentially damage motherboard components like the bulk decoupling capacitors. The “catch” resistors serve to close this feedback path, “bypassing” the Vcc_SENSE/Vss_SENSE connections through the processor and providing the necessary closed-loop feedback to the VR.
- Operation: The processor Vcc_SENSE/Vss_SENSE package traces are shunted to the motherboard Vccx and Vssx planes via the catch resistors R1 and R2. R1 and R2 are sized large in relation to the sense lines impedance so not to cause error when the DUT is present.
- Some additional guidelines:
 - R1, R2 should be placed within 50mm (2 inches) of the processor socket, minimizing any potential error due to Vcc_SENSE/Vss_SENSE line resistance.

- Preserve the transmission line integrity of Vcc_SENSE/Vss_SENSE lines by avoiding stub routing to R1, R2 catch resistors.

NOTE

Catch resistors are optional for systems where the processor (or VRTT interposer) is soldered down.

10.4 IMVP 9.3 Voltage and Current Requirements

An IMVP 9.3 compliant VR supplies the required voltage and current. The VID nominal value indicates the reference point for the static and ripple voltage tolerance.

Each IMVP 9.3 VR output must meet the specifications in the processor EDS at remote sense processor pins.

Due to VR tolerances, PCB parasitic and current draw variations, the average voltage at the processor may be slightly higher or lower than the reference value. However, the VR must meet processor static and ripple tolerance limits over the entire range of processor load. The following conditions apply to the specifications:

- Applies to all frequencies unless specific frequencies are listed
- Applies to full temperature range
- Icc is measured at the nominal VID Vcc value under maximum signal loading conditions such as running a power virus program.
- Applies to full input power rail range
- The nominal VID voltage, voltage tolerance, and maximum current values are for reference only. Refer to the appropriate component documentation for the most current specifications.

An IMVP 9.3 compliant VR must be designed to:

- Temporarily support “I_{PL2}” currents during a Max Turbo Power (PL2) event for the respective tau duration their VR can thermally support.
- Momentarily support the processor’s maximum current (VR’s electrical design constraint)
- Operate within spec under the worst-case dynamic/load transient events

Refer to the processor’s EDS for detailed processor specifications. Values shown in this document represent voltage regulator design guidelines, not processor specifications.

Table 744. General Processor VR Parameters

Symbol	Description	Parameter Values	Notes
VCCCORE, VID	VID Range	0 V - 1.52 V	
VCCCORE, VID	Offset Voltage	250mV	Refer to the Note 6
VCC _{GT} VID	VID Range	0 V - 1.52 V	
VCCSA VID	VID Range	0 V - 1.52 V	
<i>continued...</i>			

Symbol	Description	Parameter Values	Notes
V _{CCCore} , V _{CCGT} , V _{CCSA} V _{CCLECORE} V _{Boot}	VBoot Setting	0 V	
VID Range	Voltage	0-1.52V	
VID1	Test VID	0.9	
VID2	Test VID in Low Freq Mode	0.6	
VID3	Test VID in PS3	0.3	
I _{cc_PS1}	Max Current in PS1 state	15A	
I _{cc_PS2}	Max Current in PS2 state	5A	Transient loads will be from 1A to 5A
I _{cc_PS3}	Max Current in PS3 state	1A	Transient loads are not expected
V _{_TOB}	DC tolerance band	Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (861123)	
V _{_TOB} + ripple	Tolerance band plus ripple in PS0, PS1, PS2, PS3	Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (861123)	Combined Tolerance band plus ripple helps to enable VR Auto Power Management (APM)
Max Slew rate	Refer to the Power Map		
For VID changes > 50mV	90 mV/us		
R _{AC_LL} (mΩ) MAX	Loadline slope in response to dynamic load increase events 1kHz-1MHz	Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (861123)	
Notes: 1. Refer to applicable processor Electrical Design Specification (EDS) for latest electrical parameters. 2. VR should be designed to efficiently deliver power at TDC and beyond. 3. These are Intel default values and are used in the calculation of Min and Max LL slope tolerances in this table. Values obtained from the VR vendor based on the VR design may be used here as well, but they will change the calculation results. 4. I_{cc_PS1}, I_{cc_PS2} and I_{cc_PS3} cutoff thresholds should not be used as OCP values. Refer to the Over-Current Protection (OCP) section in the IMVP specification for determining OCP threshold. 5. Customers are strongly recommended to program the platform BIOS VR power state (PS) cutoff threshold settings to align with system IMVP VR power state and efficiency measurement results. For VR Power State Cutoff threshold tuning refer IMVP9.3 Light Load Efficiency Test Plan 6. IMVP9.3 controller to support offset voltage (33h) capability as required in IMVP9.3 specification. The offset voltage of 33H register is an 8-bit wide, two's complement number read write register. • Default 00h = no offset • Bit 7 = sign bit (as part of two's complement) • 00000001 = offset + 1 VID step • 00000010 = offset + 2 VID step • 11111111 = offset - 1 VID step			

Table 745. Processor VR Overshoot and Design Values

For the VR Overshoot and VR Design Values, refer to the Power Map and Nova Lake Electrical Specification (# TBD).

NOTE

In addition to the processor VR design values mentioned above, designers must also ensure that the auxiliary rails like 1.8V, 3.3V, 5V (whichever applicable) for POR (Power-On Reset) timing requirements are met during both power-up and power-down sequences. This includes handling scenarios such as surprise power-downs or abnormal shutdowns. Furthermore, no over-voltage should be applied to the processor power rails during these transitions..

10.5 Power Delivery Guidance for Systems with Discrete Graphics

A VccGT voltage regulator is required for all SKUs that feature SoC integrated graphics, even if integrated graphics is not being used. For designs with discrete graphics, refer to the Power Map for the minimum size requirements of the VccGT VR.

10.6 BIOS Settings

"Performance" configuration power/current numbers aim for the highest CPU and Graphics performance. For some SKUs, "Baseline" configuration power /current numbers are provided in the power map to provide a balance between performance and platform constraints.

NOTE

Initial BIOS is always configured to the Performance settings. The ODM/OEM needs to override package power PL2, PL4, Vcc_{CORE}/ Vccl_{PEcore}/Vcc_{GT}/Vcc_{SA} Iccmax, I_{PL2}, load line and Vcc_{CORE} VR TDC Current Limit and TDC Time Window (Min) numbers in the BIOS according to their platform or system capability

For the details of the BIOS settings, refer to the Nova Lake Processor Family Core and Uncore BIOS Specification.

10.6.1 AC Load Line (LL) Design Recommendations

The following are important design recommendations for ACLL:

- ACLL is dictated by decoupling and motherboard layout and package.
- BIOS ACLL impacts the nominal CPU VID voltage.
- VR HW and BIOS ACLL must match and meet the EDS specification.
- ACLL should be tuned using VRTT measurements to match the Intel ACLL specification in the External Design Specification, Volume 1 of 2
- In the event the design can support a shallower (lower value) load line, for example due to a superior board layout, customers can take this advantage of this by configuring the BIOS accordingly. This may result in reduced power consumption, increased average frequency, and/or higher turbo residency
- Failing to program the ACLL settings via BIOS will result in out of spec operation such as: higher voltage than needed, higher power and thermals, and hastened aging of the processor which will reduce reliability.

10.6.2 DC Load Line (LL) Design Recommendations

The following are important design recommendations for DC LL:

- DCLL is programmed into VR HW.
- BIOS DCLL impacts the power calculation done by CPU.
- VR HW and BIOS DCLL should be programmed to match ACLL measured on the customer system, not the EDS specification value.
- BIOS DC LL must be programmed the same as VR HW DCLL measured on the customer system, not the EDS spec value.
- If VR HW DCLL is shallower than ACLL spec, the processor will operate at an increased voltage resulting in higher power and thermals and hastened aging of the processor which will reduce reliability .

10.6.3 Trusted Load Line (LL)

Background

Programmable loadline was introduced to allow flexibility in OEM design around Intel loadline targets, being able to trade off cost versus power in VR-down designs. A more expensive board with less resistance can have a shallower loadline and thus run at lower power on average. Intel guides OEMs to design boards to meet EDS specification. Intel guides OEMs to test boards with VRTT methodology, measuring board resistance. Intel guides OEMs to program characterized loadline data through BIOS mailbox, later to be consumed by PCODE during boot, to ensure correct power delivery and stability. Intel design guidance recommends operating at correct loadline resistance, as operating at an incorrect loadline resistance will result in minimum voltage violations which may result in system lock-up, blue screening, or data corruption. Threat actors may use undervolting to break security and cryptography (Plundervolt, VOLTpwn, VoltJockey, PMFault). Since PCODE cannot verify loadline values, untrusted BIOS may misrepresent loadline to PCODE, producing undervolting. Intel is introducing a new trusted loadline mechanism which may provide additional security after end-of-manufacturing. Intel recommends OEMs to adopt the trusted loadline mechanism.

Guidance

A system designer must provide VRTT-characterized loadline data in CPU straps, in order to ensure that a CPU will only request safe VIDs. A system designer must configure minimum loadline straps through mFIT configuration tool. Minimum loadline straps enforce a lower bound on both BIOS-provided ACLL and BIOS-provided DCLL, thus ensuring adequate LL compensation. This configuration must be done in addition to, and not as a substitute for, BIOS loadline settings, which must also be provided, as in prior generations. Strap values can be reconfigured as often as needed before end-of-manufacturing. Once a system goes through end-of-manufacturing flow, strap values are locked and cannot be further altered, enhancing the security of the system. Since loadline values represent resistance, values do not change over the lifetime of the platform, and can thus be fixed.

Important Notes

1. For supported rails, PCODE computes $DCLL = \max(\text{BIOS_DCLL}, \text{MIN_LL_STRAP})$ and $ACLL = \max(\text{BIOS_ACLL}, \text{MIN_LL_STRAP})$.

2. Minimum loadline straps default to 0mΩ, thus ensuring legacy behavior by default, whereby BIOS LL settings take precedence.
3. The requirement applies to core domains only (VCC_CORE, VCC_LP_ECORE), since only these domains do run secure workloads.
4. Overclocking SKUs are exempt and may keep the default minimum of 0mΩ, implying legacy behavior.
5. For precise strap information and format, refer to Nova Lake Mobile 6C, 8C, H16C, and HX Processor Electrical Specification (# 861123).
6. Programming minimum loadline straps is recommended as a security best practice.
7. Intel recommends OEMs follow the end-of-manufacturing (EoM) process.

10.6.4 Adaptive Thermal Monitor

- Designing VR for continuous PL2 (legacy) is non-optimal as PL2 is time limited and leads to larger than required VR size.
- VR_TDC (IRMS) allows better current-thermal correlation compared to linear averaging, but proper tuning is required and more guard bands as current is used as proxy for temperature.
- Intel is improving performance gen over gen and, hence increasing the sustained current numbers. Closed loop temperature monitoring is required under these scenarios.
- All SVID rails IMVP9.3 controllers do support Tmon (0x17h) and CPU will monitor their Tmon periodically. All SVID rails VR temperature thresholds for TOPD can be set through mailbox below (SET_TARGET_TEMPERATURE). When Tmon temperature exceeds TOPD temperature threshold set, thermal control algorithm will kick in to manage VR thermal in such a way that VR temperature is kept equal or lower than temperature threshold set. Frequency will be reduced slowly and not as step with LFM. For optimized design, TOPD temperature threshold is recommended to be set 2C or 3C lower than Thermal Alert# threshold.

Disclaimers:

- This feature is under validation.
- Performance Improvement in the score vary from system to system and Power Delivery to Power Delivery designs
- Helps when PL1max = PL2 scenarios
- It is recommended to use X7R caps at the VR input

Figure 428. Adaptive Thermal Monitor Block Diagram

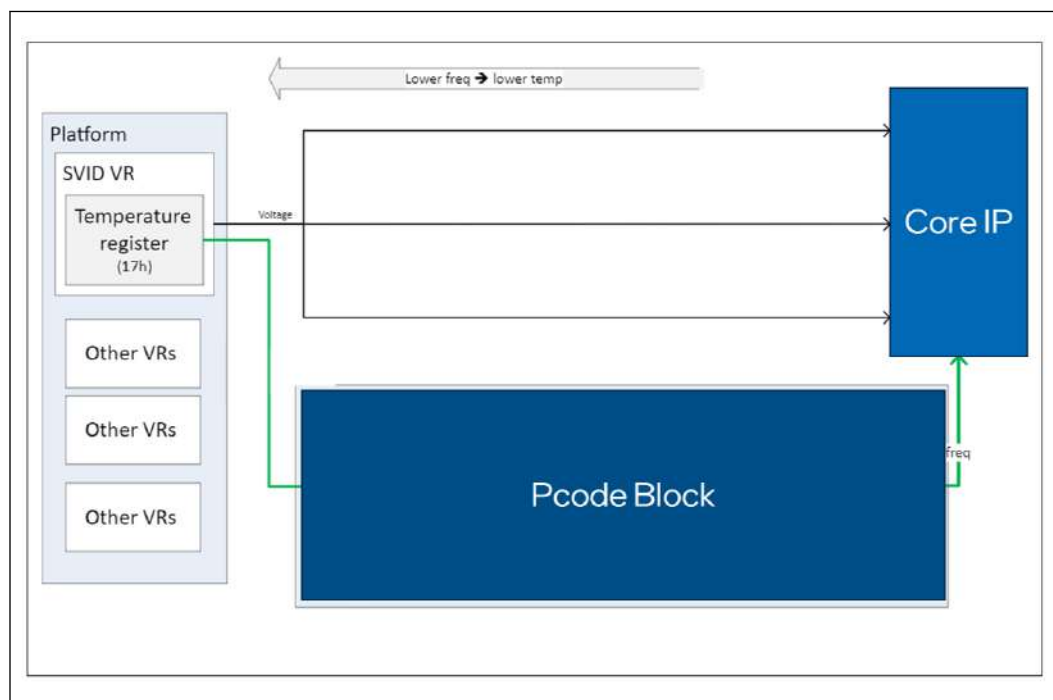


Figure 429. BIOS Settings

SET_TARGET_TEMPERATURE

Sub-Opcode: 0x16

Lock: BIOS_CPL

Set target temperature for the selected VR address.

Note: VR Address parameter is needed for this command. BIOS is expected to be the sole user of this interface.

Name				PARAM2
Type				IN
Bits	Access	Default	Description	
12:0	WO	0b	VR_ADDRESS	

Name				DATA
Type				INPUT
Bits	Access	Default	Description	
7:0	WO	0b	TEMP_LIMIT Temperature limit, in US temperature (in Celsius) format	
8:8	WO	0b	ENABLE When set, ensures that VR average current does not exceed the limit value over an exponential weighted moving average of the time window.	

NOTE

It is recommended to use X7R capacitor at the VR input and near the FET which handles a wider range (-55°C to +125°C) with slightly better stability. VR can be optimized based on the TDC calculator.

10.7 Battery Charging System

This section reviews battery charger topologies and requirements in mobile system designs.

During CPU Turbo Boost periods, the system may overload the input power source.

It is required that the battery charger has protection mechanisms such as FAST FORCEPR# to mitigate overloading of the power source. There are two types of charger topologies capable of doing this:

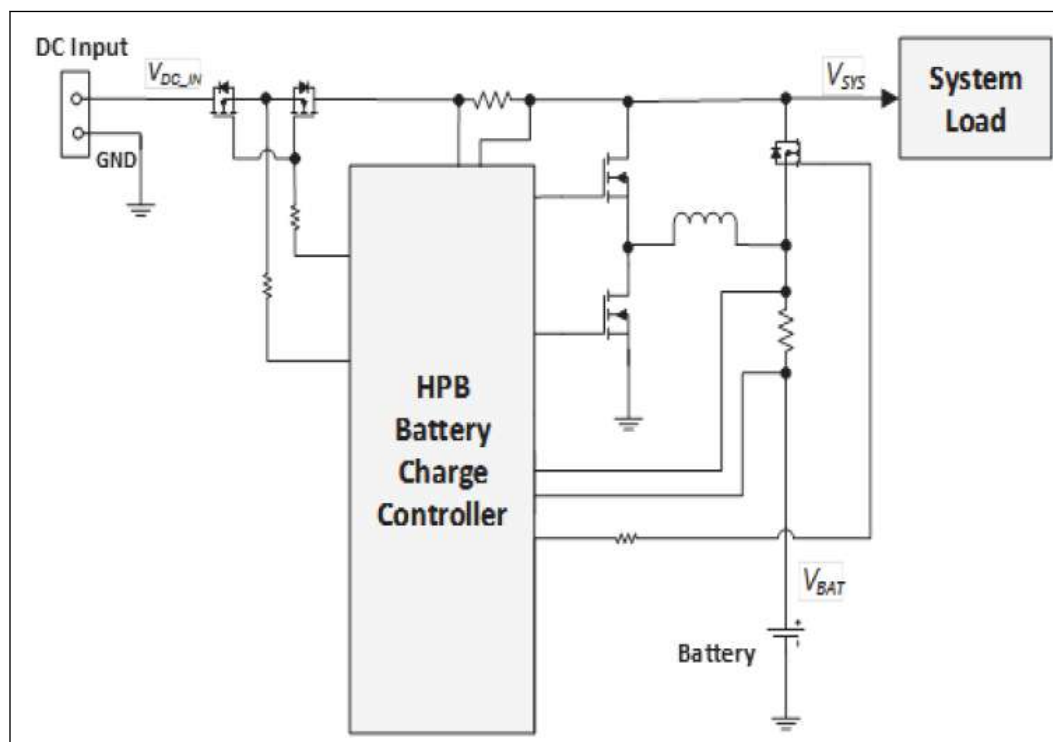
1. Hybrid Power Boost (HPB) battery charger
2. Narrow VDC (NVDC) battery charger

NVDC battery chargers are often capable of operating in both NVDC and HPB modes.

10.7.1 Hybrid Power Boost (HPB) Battery Charger

In the Hybrid Power Boost configuration, the synchronous buck converter runs as a normal buck converter when the adapter provides power to the system and is charging the battery. When the adapter power is not sufficient, the synchronous buck converter runs in reverse to boost the battery voltage to around 20V. Thus the battery supplements the adapter whenever the adapter power is not sufficient. This requires no circuit changes from a traditional adapter. The change is required in the control circuit (the Battery charger controller).

The advantage of this system over a traditional charger is that the battery is able to assist the adapter during turbo workloads. This system has the disadvantage that the light load efficiency is pretty low as it is difficult to achieve high light load efficiency at high input voltage.

Figure 430. Hybrid Power Boost (HPB) Battery Charger

10.7.2 Narrow VDC (NVDC) Battery Charger

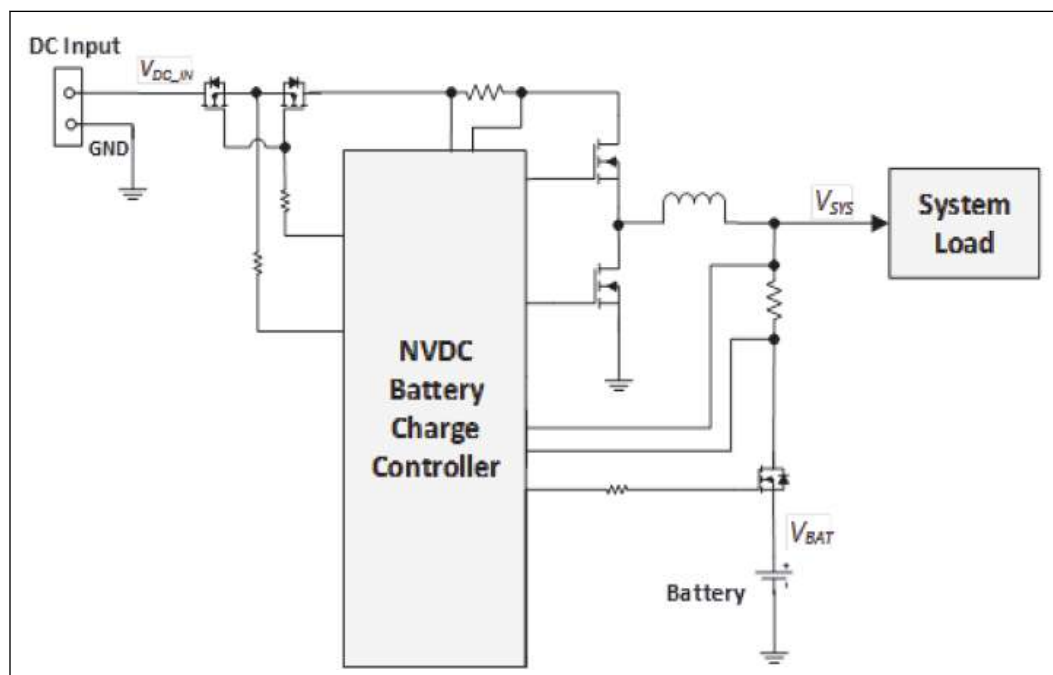
Figure 431. Narrow VDC (NVDC) Battery Charger

Figure above shows the Narrow VDC (NVDC) topology. Here, the system bus (V_{sys}) is not connected directly to the adapter. It is connected to the output of the buck converter. Hence, NVDC operates only as a buck converter, both when NVDC charges the battery and when the battery supplements the adapter and provides power to the system. NVDC implementation reduces the switch-over period between the charging mode and the hybrid power mode. NVDC implementation allows the system to minimize the period of overloading the input power source when CPU is in Turbo Boost mode.

The advantage of using the NVDC system is that the overall system efficiency is better compared to the Hybrid Power Boost charger. The system can be designed for a smaller voltage rating since the system has a lower V_{in} . The disadvantage is that the charger components' size and power dissipation increases.

10.7.3 Benefits and Trade-Offs

Table 746. Benefits and Trade-offs of the Different Chargers

PD Architecture	Turbo support	Advantages	Limits
Traditional charger	No. Turbo is limited to the capability of the adapter or Battery (varies widely by state of charge)	A very common for system with less dynamic load.	VRs with large range of input voltage are less efficient at light load. The AC adapter has to be sized to supply peak system power for ~10ms.
Hybrid power boost	Yes	Higher input voltage is more suitable to larger system with large power consumption.	VRs with large range of input voltage are less efficient at light load. The AC Adapter has to supply peak system power for 1ms.
Narrow VDC (NVDC)	Yes	Design optimized with lower input voltage has better light load efficiency. FETs can be optimized for lower input voltage and VRs can be smaller and faster.	Charger's components (FET/Inductor) size are often larger and power dissipation is often greater.

For more information on P_{SYS} , refer to Power Delivery Power Management Design Guide (#634413)

10.7.4 Supporting 5V-20V USB Charging

USB-C PD specification suggests the following operating modes for the USB-C PD supported ports:

1. System battery sources power to a sink device connected to a USB- PD port (5V, up to 3A OUT)
2. 5V USB AC Adapter sources power to the system(5V, up to 3A IN)
3. USB PD AC Adapter sources power to the system and battery (9V, 15V, 20V up to 3A or 5A IN)

The Buck Boost charger provides a simple, cost effective and space saving solution that support all of these charging modes.

Figure 432. USB-PD Subsystem Detailed View

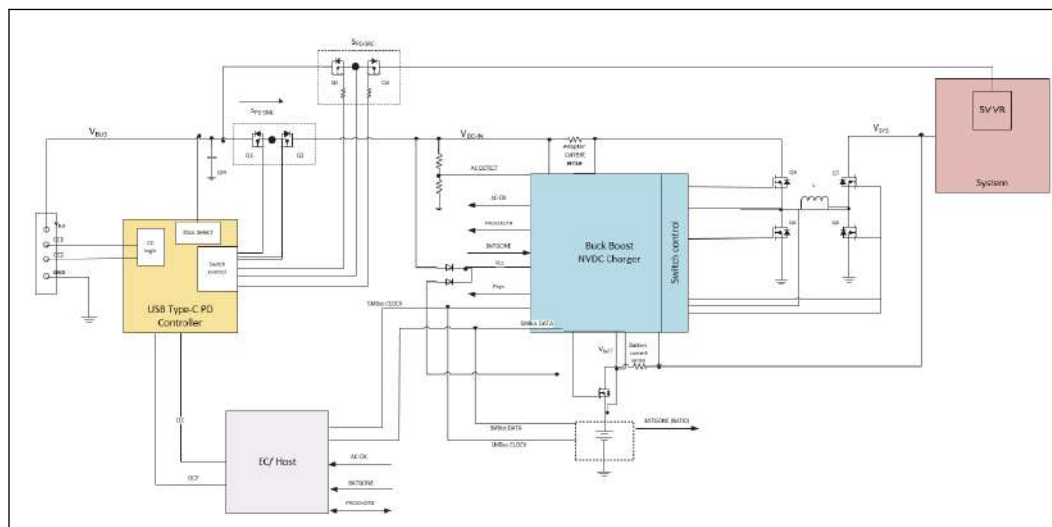


Figure 433. Modes of Operation

OPERATION							
	Mode	Power Flow	Topology	Q1	Q2	Q3	Q4
1	Battery Powering USB (e.g. mouse, KB, HD, charging phone/tablet; $V_{batt} > V_{usb}$)	Batt→USB	Buck CV	Control FET	Sync Diode	ON	OFF
2	Generic USB Charging Battery ($V_{usb} < V_{batt}$, i.e. $V_{usb} = 5V$)	Batt←USB	Boost CC/CV	Sync Diode	Control FET	ON	OFF
3	USB PD Charging Battery + Sourcing System ($V_{usb} > V_{batt}$, i.e. $V_{usb} = 12V/20V$)	Batt←USB	Buck CC/CV	ON	OFF	Control FET	Sync Diode
4	Battery Operation w/ No USB (No adapter or USB plugged in)	None	Short Protection	OFF	OFF	OFF	OFF

Features Required in Buck Boost Charger

1. Adapter voltages supported: 3.2V - 21.5V Sustained (recommended)
2. Battery configurations supported: 2S - 4S (5.6V - 18V)
3. Two level programmable input current limit (sustained and peak capability power sources)
4. Default input current limit programmable by hardware pin
5. Platform power monitoring (P_{sys})
6. Fast FORCEPR# - System voltage, Input current (and voltage), Battery discharge current
7. Low power mode during battery only Sx/ S0ix mode (consume <1.5mW)

For detailed Buck Boost charger requirements, refer to the Buck Boost Battery Charger Specification Architecture Guide (# [571381](#))

Design Considerations for 5V VBUS Operation

It is important to consider the following factors when using a buck boost charger on the platform:

1. With a 5V USB input, the inrush current on the system bus should be limited when the charger and system rails are enabled. This is especially important under a dead battery power on scenario.
2. The maximum resistance from the USB VBUS to the input of the charger should be <50-60mOhm in order to operate at 5V.
3. Dead battery power on sequence with a port controller needs to take into account a way to power on the TCPM/ EC and meeting the pSnkStdby requirement. If the TCPM is powered using the V3.3A rail, care should be taken to ensure that inrush current does not exceed the default current limit set on the charger.

10.7.5 Extended Power Range (EPR)

USB Power Delivery (PD) 3.1 and Power Conversion Considerations

USB PD 3.2 introduces Extended Power Range (EPR), increasing the maximum supported power output to 240W (48V @ 5A), compared to the 100W limit (20V @ 5A) of the earlier Standard Power Range (SPR).

Table 747. SPR and EPR Voltage and Power Levels

PDO Normative Voltage	PDO Type	Current	Power
5V	SPR	3A	15W
9V	SPR	3A	27W
15V	SPR	3A	45W
20V	SPR	5A	100W
28V	EPR	5A	140W
36V	EPR	5A	180W
48V	EPR	5A	240W

To utilize the full potential of EPR, a power conversion stage is required on the platform to step down the 48V VBUS to the system or battery voltage. The primary challenge in this conversion is thermal management, especially at high voltage and power levels.

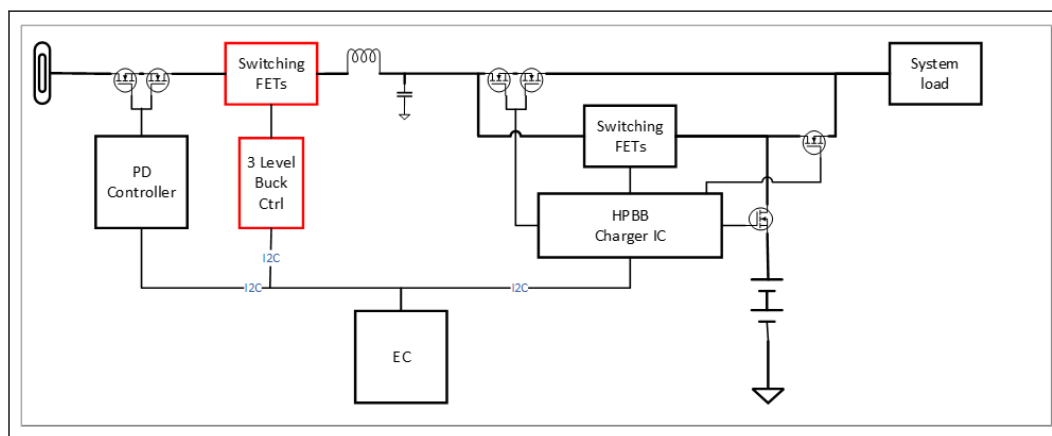
A single stage power conversion is preferable but due to thermal challenges a more expensive two-stage approach is acceptable in NVL-Hx platform. The two-stage approach consists of:

1. A buck converter with max input of 48V stepping down to an intermediate voltage (e.g., 18V).
2. A hybrid power buck (HPB) charger accepting the intermediate 18V and then supplying to system and battery charger.

Power Supply Implementation

The below figure shows the hardware connection block diagram with the 3-level buck converter. The Buck converter comes with a Pass-through-mode feature (PTM) to support SPR Power data objects (PDOs) when a non-EPR Source is connected. It needs to be configured by EC to enter/exit PTM mode.

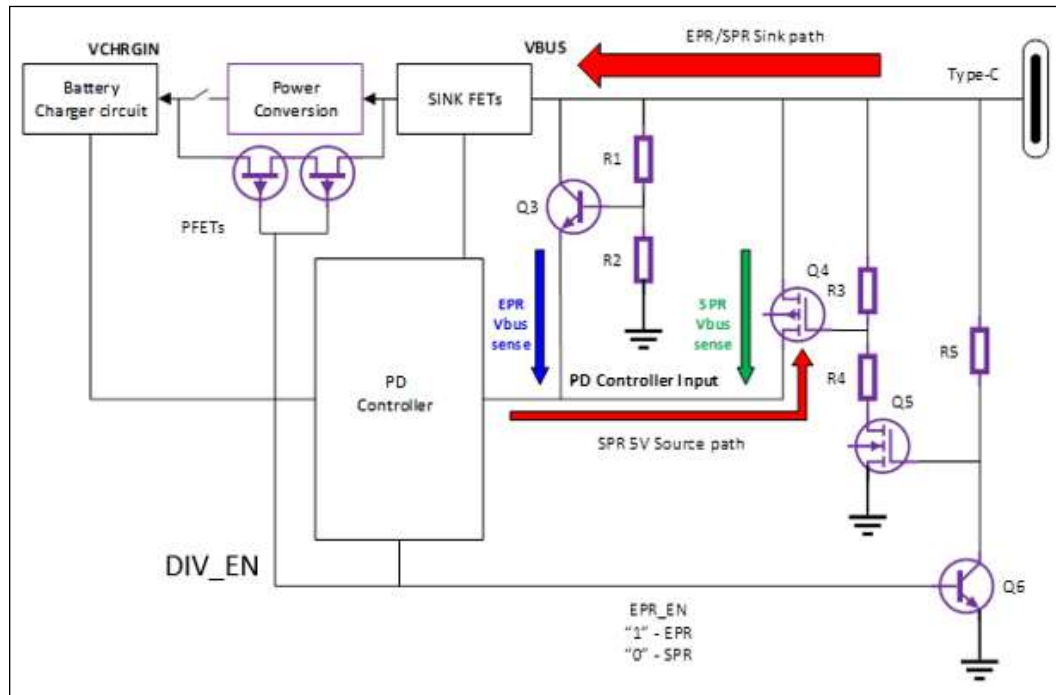
Figure 434. Hardware Connection Block Diagram



PD Controllers and Charger ICs available today do not have the capability to withstand up to 48V at its VBUS pins. Most operate at 28V typ max. When EPR voltages above 28V is negotiated, additional hardware needs to be added to block against VBUS >24V and scaling down the VBUS voltage which is connected to and monitored by PD Controller. A GPIO pin generated from PD Controller, named DIV_EN is asserted as shown in the waveform below to engage the hardware as well as to inform the PD Controller firmware that incoming voltage to VBUS is scaled down. DIV_EN is asserted during the EPR contract accept message from platform PD Controller to the Source adapter. DIV_EN is de-asserted after receiving PS_RDY message from Source adapter after VBUS transitions from EPR to SPR voltage.

The Power conversion block and the PFETs shown in the diagram below are part of the 3-level buck block. The PFETs provides the Pass-through-mode (PTM) path.

Figure 435. Power Conversion Block Diagram



Sink Path Control

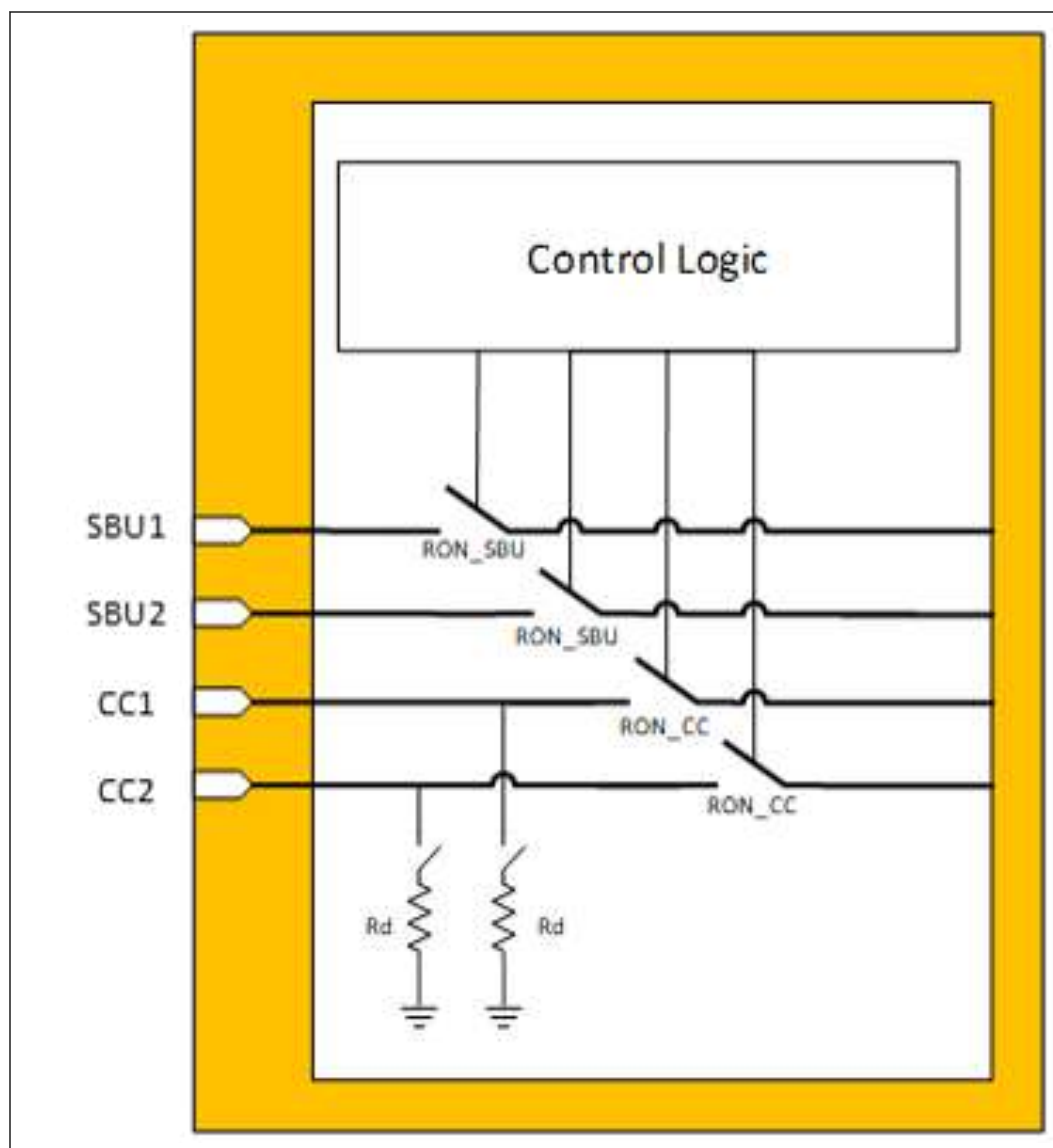
The Sink path will be controlled by EC except in these cases:

1. In no battery or dead battery conditions
2. In good battery condition, PD Controller auto disables prior to contract change and auto enables after contract change

Safety Precautions

Due to high voltage of 48V, the following should be observed.

1. There is a possibility of shorting CC/SBU lines against VBUS up to 48V. To prevent damage to downstream circuitry if a short happens, an over voltage protection IC can be used.



For high-speed signals that are AC coupled the capacitor used must be rated up to 50V or above. HSIO topology integrates VBUS-short /EOS protection up to 20V (SPR). To support higher VBUS up to EPR, two components must be adjusted.

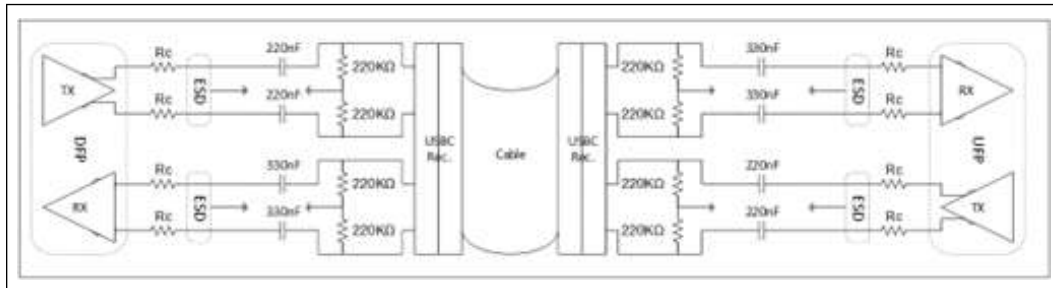
- CTX (220nF) and CRX (330nF) package size shall change to 0402 with proper DC rating based on maximum VBUS voltage.
- R_C (Current Limit Resistor) value shall change based on maximum V_{BUS} voltage.

Table 748. RC and CTX Required with Different EPR Voltages

	SPR	EPR @ 28v	EPR @ 36v	EPR @ 48v
220nF/ 330nF	25v 0201	35v 0402 ¹	50v 0402 ¹	63v 0402
R_C	$2.2\Omega \pm 5\%$	$2.7\Omega \pm 5\%$	$3.24\Omega \pm 5\%$	$4.02\Omega \pm 5\%$

NOTE

¹ 63V 0402 recommended by safety standard.

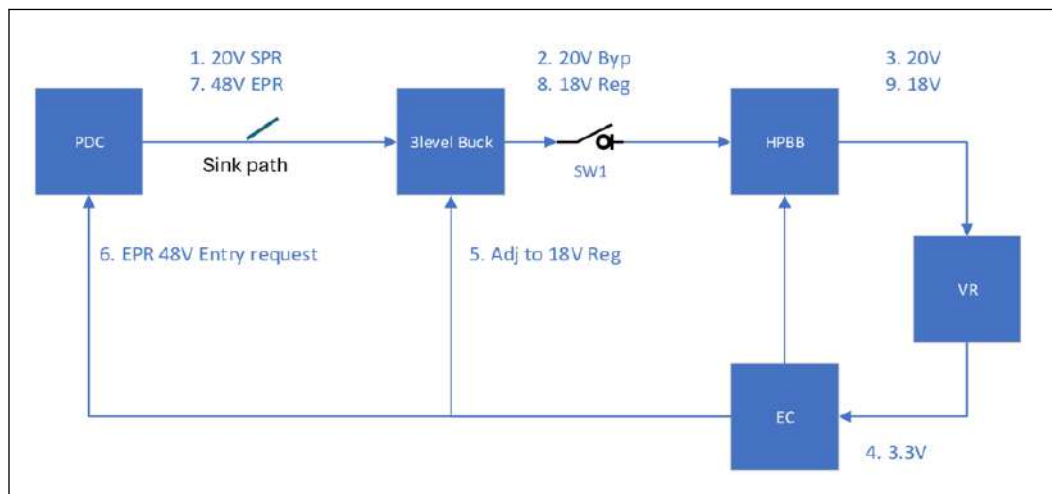


2. When adapter is removed
 - a. 48V may still be present at the output of the Sink path due to presence of capacitors. The Buck converter with PTM mode, should ensure that voltage is properly discharged before allowing PTM mode entry to prevent a direct 48V presence at the HPBB charger IC. One way to determine if voltage is safely discharged is to check the BCACOK signal that is coming out of Charger IC. A deassertion of this signal implies that input voltage has gone below the Charger IC input voltage setting and is safe to engage PTM mode.
 - b. The possibility of arcing at the connector may occur if 48V drops too quickly due to heavy loading. Further explanation is provided in the Appendix section of the USBPD3.1 specification.
3. Never allow PD Controller to negotiate to EPR voltages directly without getting notification from EC. EC needs to configure the first stage converter from Pass-through mode to Regulated mode prior to EPR voltage negotiation.
4. Safety regulatory requires platform to fail safely in the event of an unexpected fault voltage at VBUS (up to 55V). It is acceptable for platform to be non-functional in the event of such unexpected fault scenario. However, there must be no hazardous conditions such as thermal events.

Connection Flow

Upon connection, power to the platform does not come up immediately but through a series of negotiations. The steps are outlined below.

Figure 436. Connection Flow



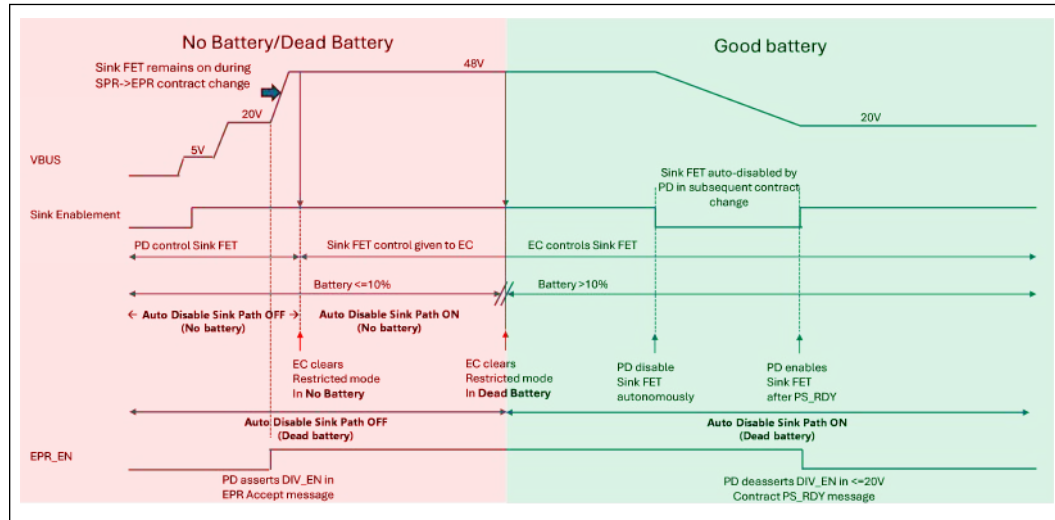
1. USBPD on platform negotiates for power up to 20V (SPR) with USB Source adapter. EC closes sink path by notifying PD.
2. Buck converter is in pass-through mode (default upon power up).
3. Charger IC (HPBB) receives 20V input power.
4. EC receives 3.3V power in case of no battery/dead battery conditions.
5. EC configures Buck converter to regulation mode for Vout of 18V via I2C. EC reduces load as it prepares transition from SPR to EPR (to meet Isnkstdby requirement). The table below shows load reduction mechanisms in different battery conditions until EPR voltage is completely negotiated.

Table 749. Load Reduction Mechanisms

Good Battery	Dead Battery	No Battery
EC halts charging	EC halts charging	EC holds boot
	EC holds boot	

6. EC notifies PD Controller to allow negotiation to EPR voltages while Sink path is still closed. Since load is reduced, EPR voltage can be negotiated without exceeding Isnkstdby (pre requirement of USBPD specifications).

Figure 437. Battery Conditions



In the initial contract while in no battery or dead battery, the Sink path will be controlled by PD controller as EC is unpowered.

When there is a contract change while battery is in good battery mode, the Sink path will be auto-disabled and re-enabled by PD controller to meet $I_{snkstdby}$. Auto-disabling by PD controller occurs during contract change and occurs only when Restricted mode is set in PD controller.

Limitations

There are several items to take note with respect to this implementation.

1. The Psys is sensed by the Charger IC at the output of the first stage converter and hence does not account for the power loss of the first stage converter.
2. Since the first stage converter is a buck-only converter this implementation only supports PDOs above battery voltage, i.e. 5V, 9V, 15V PDOs are not supported in a 4S battery system.

Layout Considerations

The important layout consideration with EPR is the creepage and clearance distance with respect to a max voltage of 55V. According to IPC standards, the requirement for different conductors is stated as below along with the voltage range. B1 refers to internal layers of the PCB requiring a minimum spacing of 0.1mm for 55V. B2 refers to the external conductors (top and bottom of PCB) with minimum spacing of 0.6mm. Creepage distance is the same as clearance distance if the PCB has a flat surface.

Figure 438. Electrical Conductor Spacing

Voltage Between Conductors (DC or AC Peaks)	Minimum Spacing						
	Bare Board				Assembly		
	B1	B2	B3	B4	A5	A6	A7
0-15	0.05 mm [0.00197 in]	0.1 mm [0.0039 in]	0.1 mm [0.0039 in]	0.05 mm [0.00197 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]
16-30	0.05 mm [0.00197 in]	0.1 mm [0.0039 in]	0.1 mm [0.0039 in]	0.05 mm [0.00197 in]	0.13 mm [0.00512 in]	0.25 mm [0.00984 in]	0.13 mm [0.00512 in]
31-50	0.1 mm [0.0039 in]	0.6 mm [0.024 in]	0.6 mm [0.024 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.4 mm [0.016 in]	0.13 mm [0.00512 in]
51-100	0.1 mm [0.0039 in]	0.6 mm [0.024 in]	1.5 mm [0.0591 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.5 mm [0.020 in]	0.13 mm [0.00512 in]
101-150	0.2 mm [0.0079 in]	0.6 mm [0.024 in]	3.2 mm [0.126 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
151-170	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	3.2 mm [0.126 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
171-250	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	6.4 mm [0.252 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
251-300	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	12.5 mm [0.4921 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.8 mm [0.031 in]
301-500	0.25 mm [0.00984 in]	2.5 mm [0.0984 in]	12.5 mm [0.4921 in]	0.8 mm [0.031 in]	0.8 mm [0.031 in]	1.5 mm [0.0591 in]	0.8 mm [0.031 in]
> 500 See para. 6.3 for calc.	0.0025 mm /volt	0.005 mm /volt	0.025 mm /volt	0.00305 mm /volt	0.00305 mm /volt	0.00305 mm /volt	0.00305 mm /volt

B1 - Internal Conductors
 B2 - External Conductors, uncoated, sea level to 3050 m [10,007 feet]
 B3 - External Conductors, uncoated, over 3050 m [10,007 feet]
 B4 - External Conductors, with permanent polymer coating (any elevation)
 A5 - External Conductors, with conformal coating over assembly (any elevation)
 A6 - External Component lead/termination, uncoated, sea level to 3050 m [10,007 feet]
 A7 - External Component lead termination, with conformal coating (any elevation)

10.8 Battery and Fuel Gauging System

This section suggests system architecture for the battery and fuel gauging systems provides information on how Turbo can be performed during discharge, given a two series Li pack configuration, providing for maximum performance.

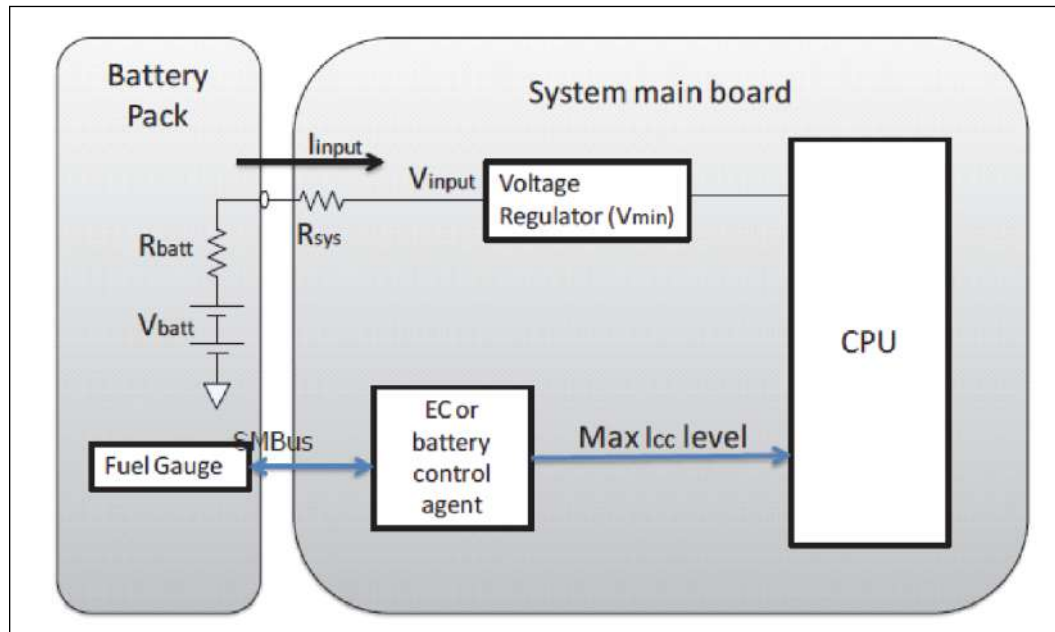
We have adopted a cost / functionality strategy for this platform. The market is divided into three segments (shown below) which can be identified by differences in their battery and fuel gauging system.

1. **Volume Segment:** Battery and fuel gauging system designed to meet nominal battery runtime at nominal cost targets. Fuel gauging system provides nominal accuracy. Approaching full Processor performance.
2. **Premium Segment:** Battery and fuel gauging system designed to meet maximum battery life. Fuel gauging system provides best accuracy, and supports Dynamic Battery Power Technology. Full Processor performance.
3. **Value Segment:** Reduced battery life to meet cost target. Fuel gauging system may sacrifice accuracy to save cost. Nominal Processor performance.

10.8.1 Intel® Battery Performance Optimizer (BPO)

Intel® Battery Performance Optimizer (Intel® BPO), previously known as Dynamic Battery Power Technology (DBPT). Figure below shows the high level architecture of Intel® BPO. The input voltage of system regulator, V_{input} , is lower than the open circuit voltage of battery cells, V_{batt} . The voltage drop depends on the load current, I_{input} , and the total parasitic resistance from cells to regulator, R_{batt} and R_{sys} .

Figure 439. Intel® BPO Top Level Block Diagram

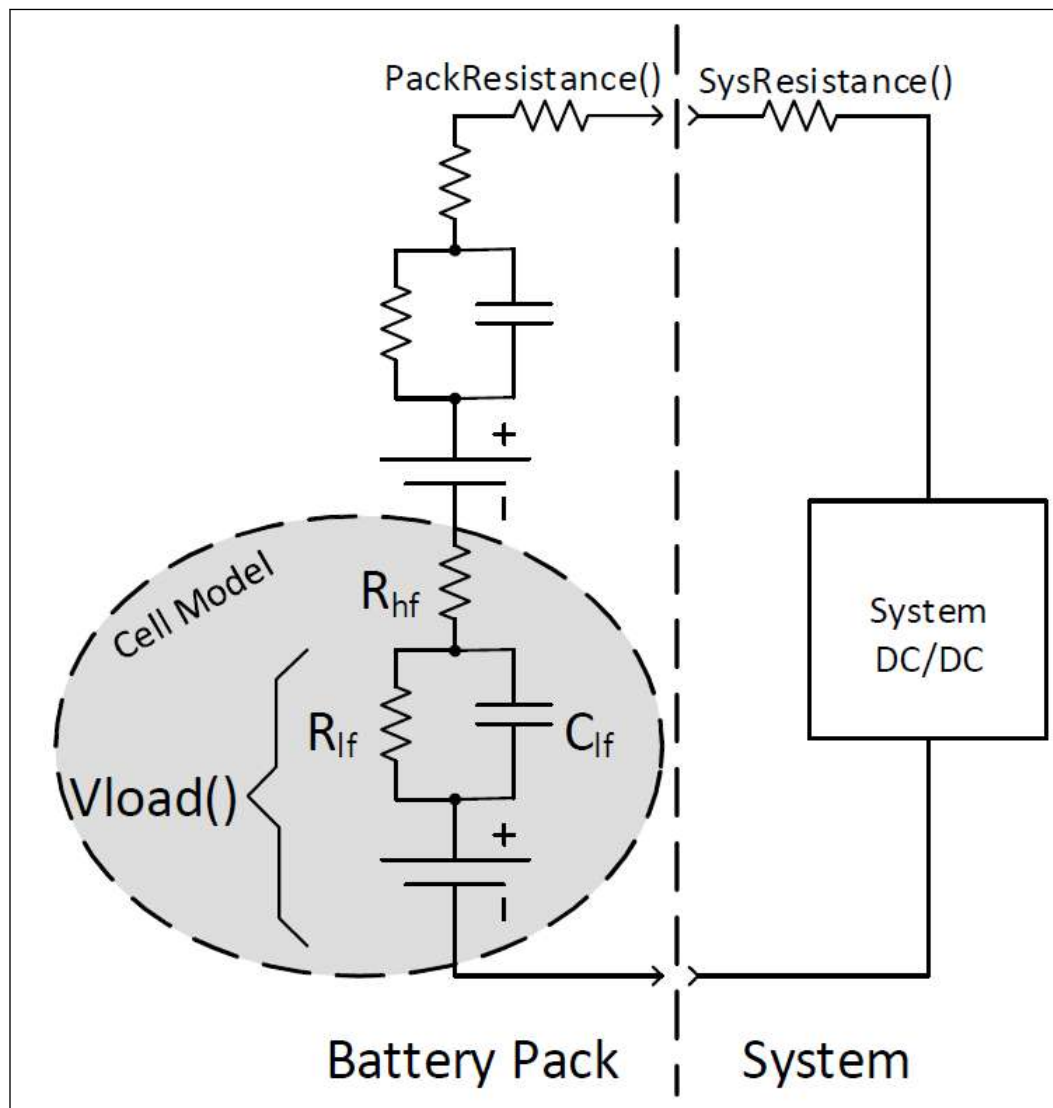


Where R_{batt} includes both cell resistance and parasitic resistance due to the cell interconnect, sense resistor, FET, fuse, connector of battery pack. R_{sys} includes the resistance of power/ground metal, sense resistor, FET, and other parasitic resistance on the system main board.

In order to prevent V_{input} from violating the required minimum system input voltage, V_{min} , I_{input} has to be adjusted according to the level of V_{batt} as the battery is discharged, as the impedance of the battery changes with state of charge, temperature, and age.

As processors Turbo, there is a need to support higher power levels for short duration power events. To assist in supporting these power spikes for a short duration, the following model allows for determining the packs behavior during these events, so that the system can determine what additional power it can get from the motherboard capacitance, AC adapters, etc. The model we have chosen is commonly used to model the behavior of the cells, and is shown in Figure below .

Figure 440. Cell Model and Battery System Diagram



The model of a single cell is shown within the greyed circle. The model consists of an internal cell voltage, which is closely tied to the state of charge of the cell. The low frequency resistance and capacitance, R_{lf} and C_{lf} respectively, have a very long time constant and model the behavior the cells shows to take 10's of minutes to settle to a final terminal voltage. The final component of the model is the high frequency resistance of the cell R_{hf} .

Without Intel® BPO, the pack voltage must be chosen so that under load the pack voltage cannot fall below V_{min} . Ways to achieve this are to choose cell counts of 3 or more series Li cells. An alternative is to allow Turbo events on discharge, only to a predetermined critical threshold based on state of charge of the pack. The issue with this approach is that the impedance of the battery changes based on age and temperature, and the solution involves reducing Processor performance in an overly conservative way. Intel® BPO allows for the Processor to perform optimally, at any given battery impedance.

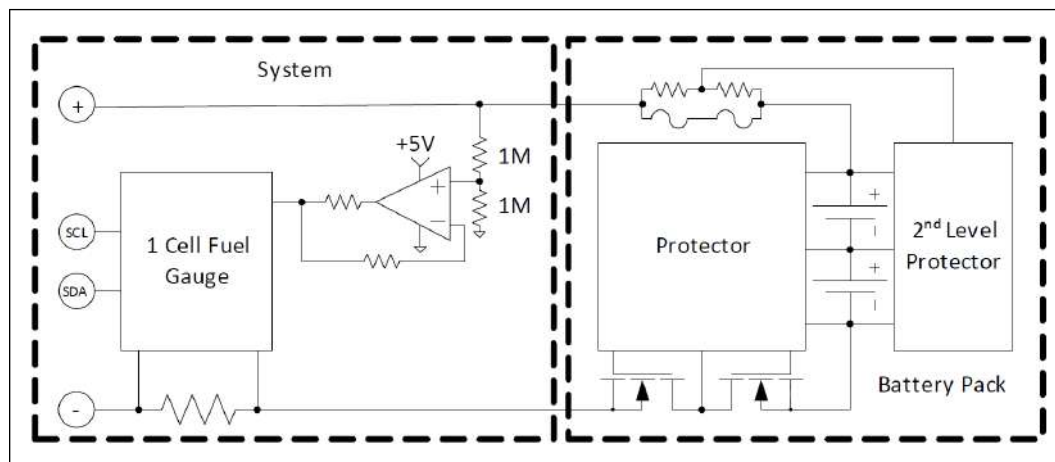
10.8.2 Fuel Gauging

By choosing between the following fuel gauging types, it is possible to tailor the fuel gauging solution cost, by trading off functionality. The following choices of fuel gauging solution are ordered below, starting with the most accurate, highest featured. As the list continues the features are reduced, along with the cost.

In-system Fuel Gauge without BPO

This solution for In-system fuel gauging provides for adequate fuel gauging accuracy. This solution also foregoes support for BPO. The solution utilizes the selection of fuel gauging parts aimed at the cost competitive one cell fuel gauging market. As the Processor solutions require 5V rails, this solution uses an Op-Amp to divide the pack voltage of a 2 series Li pack configuration in half, and supply this precision divided voltage to the fuel gauge solution designed to handle one series cell. A typical block diagram of such a system is shown below.

Figure 441. 1S Fuel Gauge with 2S Pack Block Diagram



NOTE

In such a system the Op-Amp and the divider resistors should be chosen to provide a very accurate division ratio, as inaccuracy in this division will directly affect fuel gauging accuracy. This means selection of an Op-Amp with a very low input offset voltage. In addition a matched set of divider resistors, or high precision resistors should be evaluated for their effect upon accuracy. Note that the Op-Amp will not be required to swing to the power or ground rails if powered by a 5V rail, so cost can be saved by choosing a low power, low input offset amplifier, which does not provide rail to rail output swing. The bandwidth of the amplifier is of some concern, but a bandwidth of 200kHz should be adequate to give good results. It is suggested to consult the fuel gauge manufacturer to determine what bandwidth their gauging solution requires.

In Windows* solutions, a reporting of current is required by the operating system, so be sure to choose a single cell gauging solution which reports current for such systems. Other operating systems may not require reporting of current. Single cell gauging solutions which monitor only voltage, can provide a bit lower price point while typically giving up a couple percent of accuracy.

10.8.3 Battery Pack Size and Configuration

The battery pack is typically size to meet goals for runtime and peak power. In order to achieve the highest performance for the processor, please specify the battery based on guidelines for power capability of the pack from the Table below.

Table 750. Battery Discharge Capability Recommendations

Recommended Peak Duration	10ms	100 sec	Constant Discharge
Recommended Max Discharging Power Level if P_{SYS} is not used	$PL4 + P_{ROP_MAX}$	Max Turbo Power (PL2) + P_{ROP_MAX}	$PL1 + P_{ROP_MAX}$
Recommended Max Discharging Power Level if P_{SYS} is used	$PL4 + P_{ROP_MAX}$	$P_{SYS}PL2$	$P_{SYS}PL1$
Peak Duration with PL3/ $P_{SYS}PL3$	configurable down to 2ms ¹	NA	NA
Duty Cycle with PL3/ $P_{SYS}PL3$	configurable down to 4%	NA	NA

NOTE

Refer to .

Once the power capabilities to support a performance level are decided, a choice of pack configuration remains.

The choice is often between two series and three series configurations of Li based cells. Two series configurations result in higher currents, when compared to a three series configuration, for equivalent system power consumption. This would normally tend toward choosing the higher voltage solution, as power losses increase with the square of the current. The other factor is that with a lower voltage range, the DC/DC converter can typically be optimized for a better overall efficiency, which tends to favor the two series solution. Another factor to consider, is that a three series configuration will likely not have issues with reaching low voltages during Turbo events, where a two series configuration may reach low voltages under load, and require support of BPO to manage this. For the lower power consumption systems in this line, an NVDC charger configured for two series cells seems to achieve the greatest system efficiency. For higher power consumption systems in this line, a three series cell configuration will achieve the greatest system efficiency. The tradeoff between the two approaches can be gauged, by looking at the I^2R losses throughout the power system in the two approaches.

A trade-off that is often overlooked is that of fuel gauging accuracy, and pack cost. If we think of battery pack cost in terms of \$/Wh (dollars per watt-hour), then any inaccuracy in the fuel gauge can be expressed as a cost in dollars, for a given pack size. For example, given a 50Wh pack, and a cell cost of \$0.40/Wh, the cost of cells would be \$20. When choosing between a 6% accurate fuel gauge, and a 1% accurate fuel gauge, it makes sense to look at the cost of that potentially lost capacity, when looking at the price difference between the gauges. With a difference of 5% in accuracy, and a cell cost of \$20, the 1% accurate gauge could be \$1 less expensive ($\$20 \times 5\% = \1), and still offer better performance. The 6% accurate gauge would need to increase the cell cost by \$1 to allow for the potentially decreased runtime due to

gauge performance. In addition the more accurate gauge will offer a solution that is smaller and lighter, as the pack capacity that is purchased is fully used, rather than just sized up to allow for fuel gauge inaccuracy.

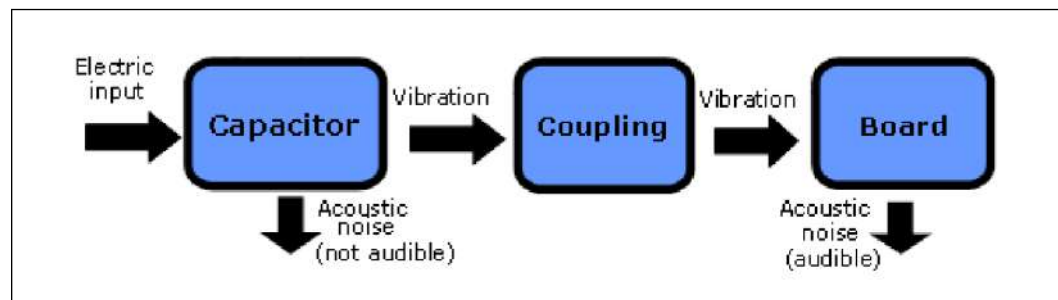
10.9 Acoustic Noise Mitigation

Audible acoustic noise can be generated by electronic components, such as capacitors or inductors, when they vibrate due to large, fast and periodic voltage or current changes. Since platform power management techniques (example: processor C-state management) can cause such voltage transitions, it is important to address and minimize the acoustic risk.

These vibrations are transferred through the rigid mounting of the electronic components to the board. Multi-layered ceramic capacitors (MLCC) are most prone to this, because they are of rigid construction and rigidly mounted to the board. The circuit board acts as a sounding board, and audible acoustic noise is radiated.

This acoustic noise generation scenario is demonstrated in below figure. In such a scenario, there are a few architectural design guidelines that should be followed in order to minimize the risk of generating audible acoustic noise.

Figure 442. Acoustic Noise Generation Flow Chart



To illustrate some of the complexity involved, the vibration pattern generated by a single capacitor when mounted in the center of an otherwise unpopulated board is shown in below figure.

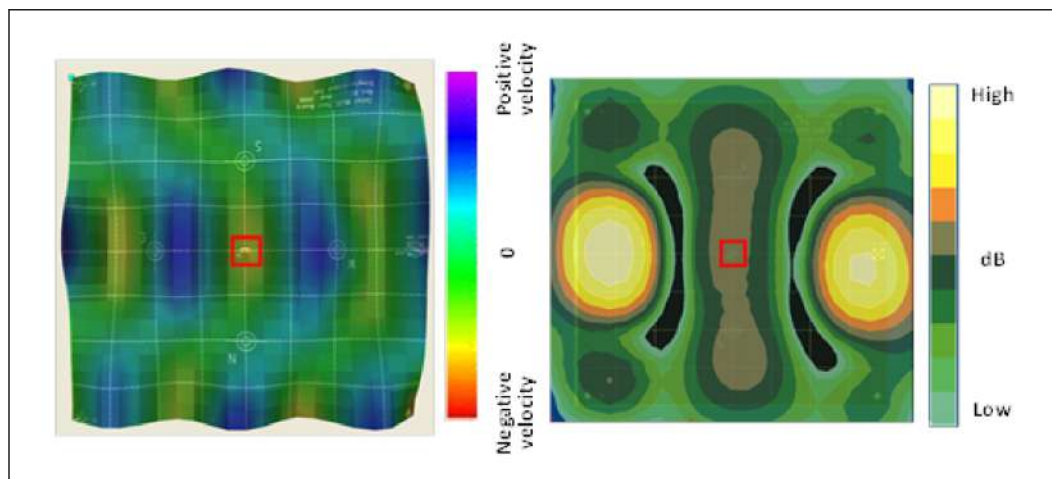
The left-hand pattern shows “peak” and “valley” deflections throughout the board. The right-hand pattern shows the acoustic response generated by the board vibrations due to the capacitor vibration.

In this example, the acoustic maxima are symmetrically located on the left side and the right side of the board, because that is the orientation of the mounting pads that transmit the vibration. That the acoustic maxima locations do not correspond to the vibration source (i.e. capacitor) creates an added complexity for troubleshooting the acoustic issue. Therefore, it is not advised to use microphone scanning to identify the vibrating source component.

Figure 443. Example Vibration (Left) and Acoustic Radiation (Right) Pattern Generated by One Capacitor Mounted in Center of Test Board

NOTE

Capacitor Location Denoted by Red Square.



NOTE

For more acoustic noise solution kindly refer to the **Acoustic Noise Mitigation Guidance Technical Advisory (# 575216)**. Using Software solution to mitigate acoustic noise provided by the Technical advisory option may impact power and performance, interrupt response time, and device latency.

10.10 AC Adapter Considerations

To support Turbo performance as well as to provide more flexibility on the size and cost of the AC adapter, this section will cover the important aspects of adapter requirements and trade-offs including thermal power considerations.

10.10.1 Power Budgeting for AC Adapter

The adapter power rating is divided into the sustained power level and transient power level. The sustained power level is the maximum constant power that the adapter can supply to the system. The transient power level is the maximum peak power that the adapter can supply within a given duration of time, usually in milliseconds.

The adapter power rating has to meet the power demand of system, which can be divided in to Processor power and rest-of-platform (ROP) power. For the system with Turbo, the sustained Processor power is Max Turbo Power (PL2) and the peak Processor power is PL4.

Default and Extreme Power Limits (Max Turbo Power (PL2), PL4) are now provided in the Power Map.

The ROP power is determined by the components other than Processor. The number of USB ports, the display module, the memory module, etc., significantly impact on the ROP power level. The P_{SYS} feature can monitor the ROP power in real-time. If P_{SYS} is not available, the designer has to estimate the typical ROP power level (P_{ROP_typ}) and the worst case maximum ROP power level (P_{ROP_max}) in order to size the adapter.

10.10.2 Transient Power Requirement for Turbo

The selection of charger PD architecture is going to affect the transient power requirements on the AC adapter in order to adequately support Turbo operation. The duration of transient power level depends on how long the battery takes to start providing power.

10.10.3 Sustained Power Requirement at Dead Battery Level

For the system with either Hybrid Power Boost (HPB) or NVDC charger, the sustained power requirement of the adapter can be estimated based on the scenario of a dead battery when the battery does not have enough charge to power the system and the AC adapter has to provide all the power. Intel recommends Battery Performance Optimizer (BPO) to better determine the battery power capability in real-time.

P_{SYS} , the platform power monitoring and control feature on this platform can be used to maintain a desired P_{SYSPL2} level where the overly conservative estimation of ROP power can be reduced. Instead of using the worst case maximum ROP power level to estimate the sustained power level for AC adapter, the pre-defined P_{SYSPL2} can be used.

Table 751. Sustained and Transient Power Levels with Different PD Architectures

Charger Type	Adapter Supplemented by Battery	Sustained Power Level with Dead Battery	Transient Power Level	Transient Power Level with Dead Battery Level
Traditional	No	Without P_{SYS} : Max Turbo Power (PL2) + P_{ROP_max} With P_{SYS} : P_{SYSPL2}	Up to 10ms: $PL4 + P_{ROP_max}$	Up to 10ms: $PL4 + P_{ROP_max}$
Hybrid Power Boost (HPB)	Yes		Up to 1.5ms: $PL4 + P_{ROP_max}$	
NVDC	Yes		Same as sustained power level	

NOTE

The above power loading levels have to be divided by the conversion efficiency to get the final adapter output power levels. The P_{SYSPL} and PL values need to be updated regularly as the power capability of adapter and battery changes. In Nova Lake Platform Turbo and Thermal Power Management Guide for Intel® Core™ based Processors User Guide (# TBD), the flow chart and equations are provided to calculate P_{SYSPL} and PL values based on adapter power capability and battery power capability (BPO) for 4 design options; NVDC with P_{SYS} , NVDC without P_{SYS} , HPB with P_{SYS} , and HPB without P_{SYS} .

10.10.4 TCSS Power Adders Guidance

This section provides Power Adders Guidance for Integrated TCSS not reflected in the processor baseline workload. Power Adders are sustained thermal power for non-IA IP on the device. Power Adders may be negligible or significant and impact the overall thermal budget for the processor. The recommended Processor PL1 power targets for the scenarios provided below reflect adjustments to the baseline power required to preserve base frequency as Process oriented to the sustained long-term thermal capability. The system designer should consider the power impact of different I/O connections on the processor and platform when determining the thermal budget and as Process oriented cooling solution to preserve user experiences or performance. The scenarios shown in table below represents sustained workload situations. Minimum Assured Power (cTDP Down Power) [W] down will still maintain Processor performance without undesired behavior from these Power Adders. In the case of bursty workloads, the impact on processor performance may be negligible and the adders provided may not be needed to be factored into the design.

Table 752. TCSS Scenarios with Integrated Thunderbolt

TCSS Scenario	Description	TCSS Power Adder (W)
Baseline	TC-Cold NDA, (Internal Display (eDP), 4K resolution (3840x2160)	0
DP Alternate Mode	1x4K at 60Hz external display, extended mode	1.3
Thunderbolt™ DP Mode	1x4K at 60Hz external display, extended mode	1.8
Thunderbolt™ dual DP Mode	2x4K at 60Hz external display, extended mode	2.8
Thunderbolt™ DP Mode, Max IO	1x4K at 60Hz external display, IO power scaled to max to simulate full BW	3.0

NOTES

- All values are pre-Si estimates on scaled measured data and are subject to change
- Test were performed with top bin with fixed Processor Base Frequency (P1) [MHz] for median parts at Tj Max at 100°C
- Workload used Cinebench* R15 MCT on external display in extended mode with other displays

Figure 444. Scenario 1: Alternate DP Mode

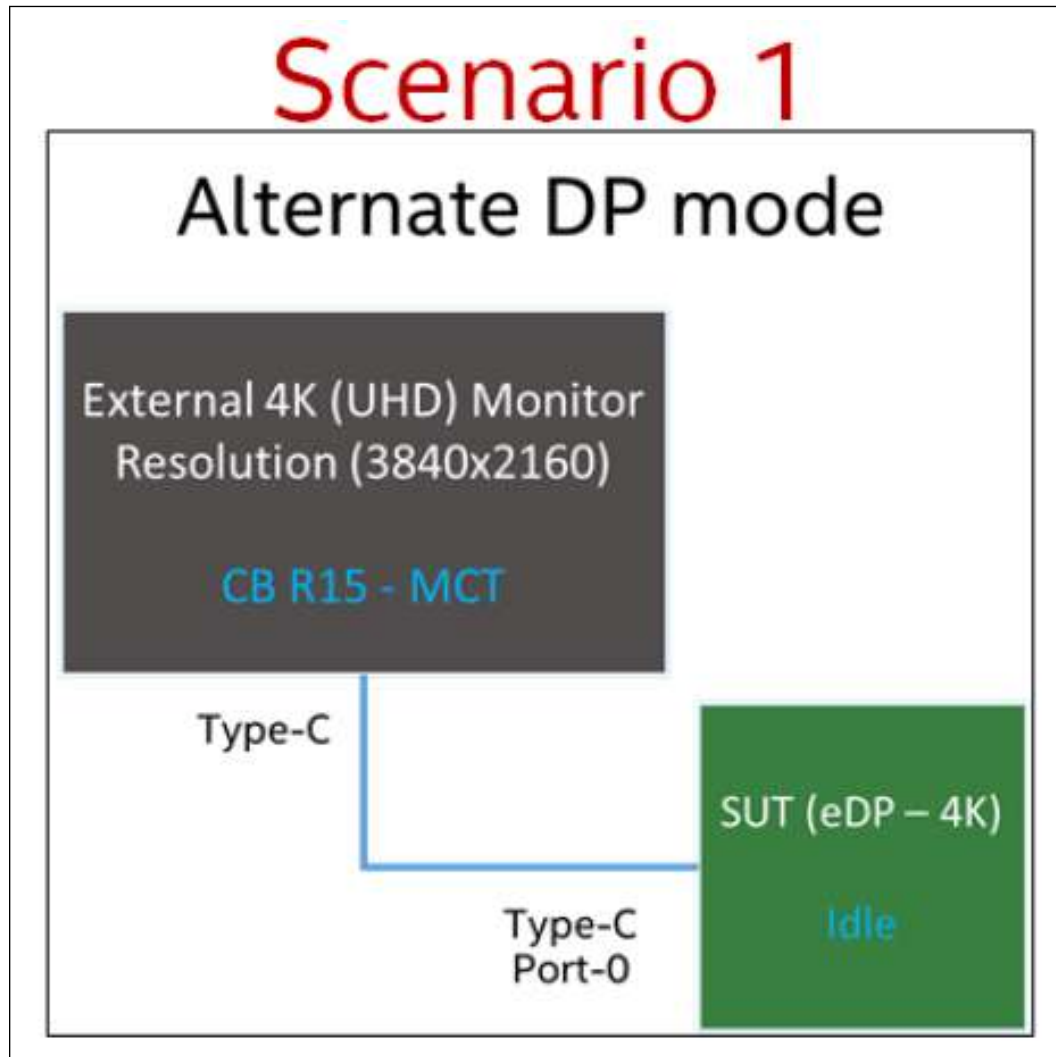


Figure 445. Scenario 2: TBT DP Mode

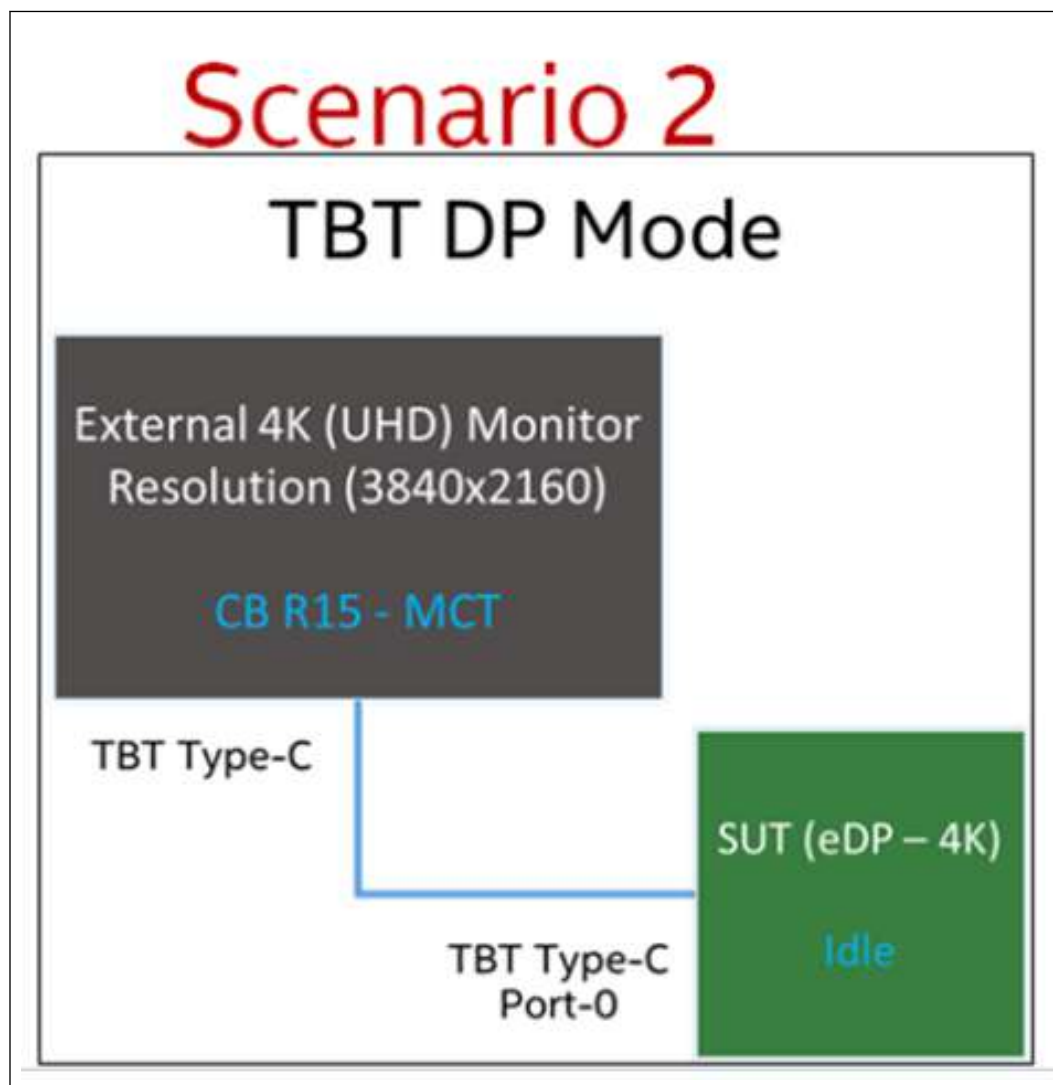


Figure 446. Scenario 3: TBT Dual DP Mode

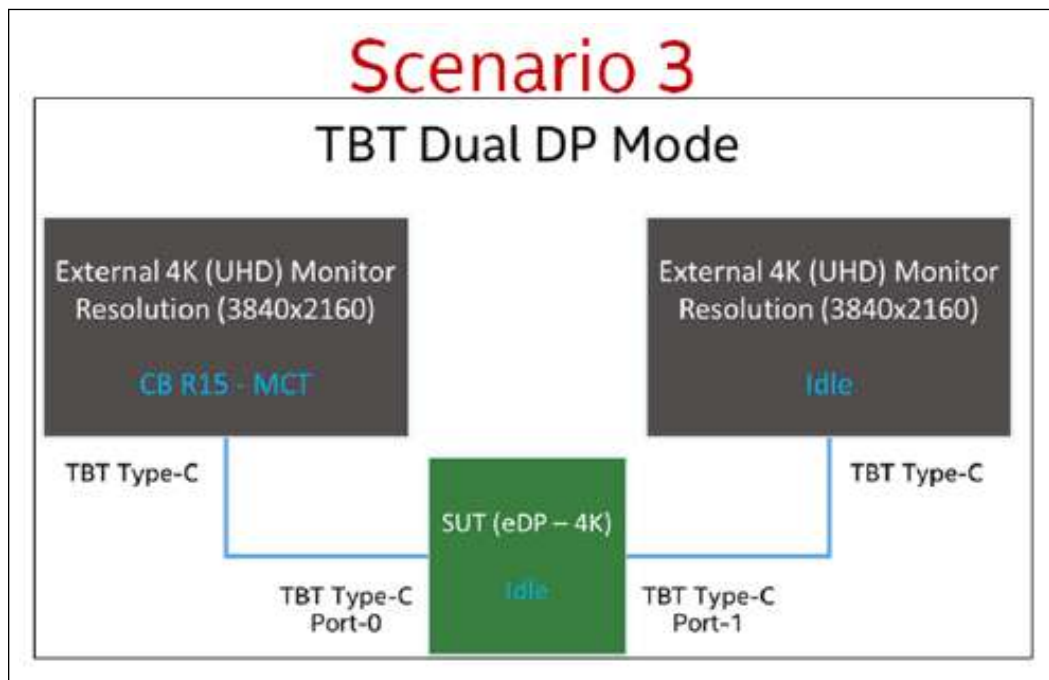
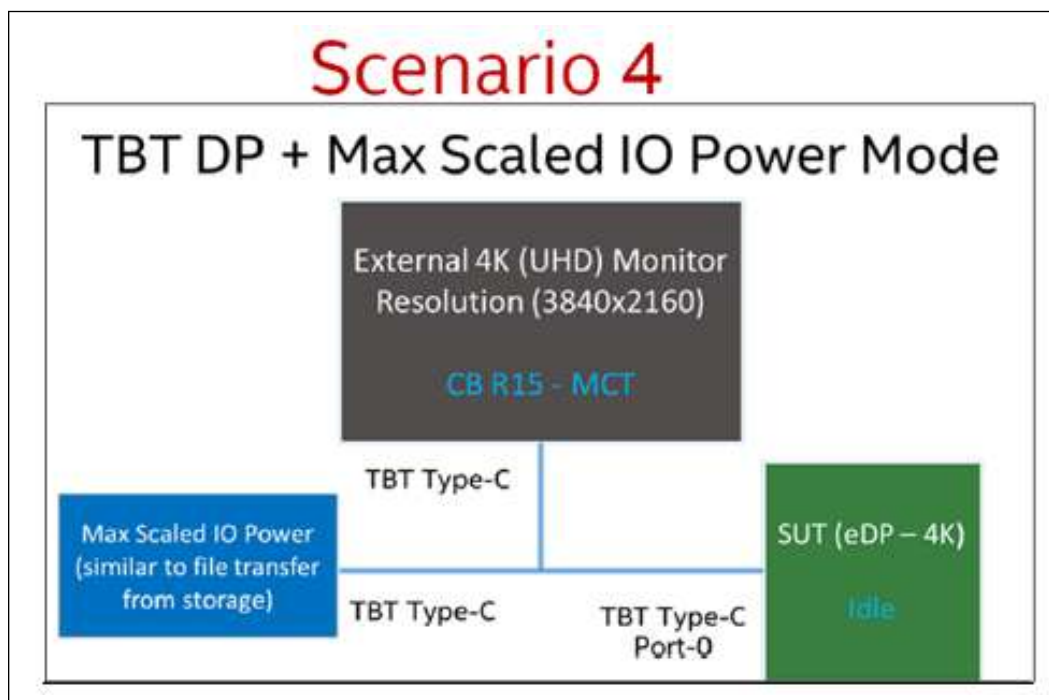


Figure 447. Scenario 4: TBT DP + Max Scaled IO Power Mode



10.10.5 P_{sys} and Other Considerations for Using Smaller AC Adapter

There are some cases where a smaller adapter with lower sustained power level can be used for better portability or lower adapter cost. However, there will be impacts on the dead battery operation, Turbo performance, and charging time.

If an AC adapter with sustained power level less than Max Turbo Power (PL2) + P_{ROP_max} , it is recommended to have $P_{sys}PL2$ enabled and set $P_{sys}PL2$ to the sustained power rating of AC adapter. However, there could be performance impact if the sustained power of adapter is much less than the Max Turbo Power (PL2) + P_{ROP_max} and user loaded the ROP power close to the worst case maximum. Here is an example.

For example, in a system with Max Turbo Power (PL2) of 25W, PL4 of 46W, P_{ROP_max} of 15W, and conversion efficiency of 90%, the sustained power level of adapter can be calculated with the following equation. An adapter with a sustained power rating of 45W is recommended for this system.

$$P_{sustained} = (\text{Max Turbo Power (PL2)} + P_{ROP_max}) / \text{Efficiency}$$

However, if a smaller adapter with 30W sustained power rating is used with this system, without P_{sys} , the Max Turbo Power (PL2) will have to be reduced down to 12W, which is even less than PL1 of 15W. In this scenario, both Max Turbo Power (PL2) and PL1 have to be reduced to 12W when the battery is dead. If the system has a traditional charger, the Max Turbo Power (PL2) and PL1 have to be reduced to 12W no matter if the battery is dead or not as long as the adapter is connected.

On the other hand, if the system has P_{sys} enabled, the performance will be managed to keep the $P_{sys}PL2$ at 30W. If the rest of platform power is lowered to 10W, the Processor will have 17W of power budget and the performance should be better than the case without P_{sys} .

With a smaller adapter, the charging time from 0% to 80% state of charge will likely be impacted. However, the actual charging time impacts depends on many other factors besides the sustained power rating of adapter, such as the power state of system, battery size, constant current charging rate. As the sustained power rating is reduced, the power available for charging is going to be reduced first.

10.10.6 Protection Mechanisms

The electrical protection mechanisms cover the following two scenarios: the magnitude of system peak power exceeds the adapter transient power limit, and the duration of system peak power event exceeds the duration of adapter transient capability.

In the event when system peak power might exceed the adapter transient power limit, the over-current-protection (OCP) is designed to sense the adapter current and trigger FORCEPR# if the over-current condition is met. For the discrete implementation, refer to the Battery Voltage Droop and Brick Over Current Protection Circuit Design (# 556361). There are chargers with integrated FORCEPR# circuit available as well.

However, after FORCEPR# is triggered, the Processor frequency is going to be throttled to the absolute minimum causing significant performance loss. Because running at minimum frequency builds up thermal headroom for turbo, it is possible

that the processor can cause repeated triggering of FORCEPR#. To prevent this scenario from happening, the system EC may reduce PL4 as a preemptive measure if an under-sized adapter is connected.

10.11 Platform Power Sequencing Specification

This section provides the following information:

- Platform Power Sequence requirements for the platform
- Platform Voltage Rails Status in the different S-States/M-States
- Voltage Rail Sequencing Requirements

The block diagrams and timing diagrams in this document are references to help demonstrate the timing requirements for the platform, and the actual timing specifications that need to be met by the system are defined in section [Power Sequencing Timing Diagrams Legacy Signals](#) on page 730 and . Depending on the system architecture, all applicable timings need to be met to ensure proper functionality of the Processor.

10.11.1 Key Changes to Platform Sequence Architecture

Removal of few power rails and Signals

A few power rails and signals were removed/added from this platform with respect to its predecessor.

NOTES

- Power rails which are not available in this platform are VCCPRIM_3P3, VCCST. VCCST_EN signal is not available.
 -
 - Newly added power rails are VCC_LP_ECORE, VDDQ
-

NOTE

VCCRTC nominal voltage is 1.2V. Hence Ultra low bias current LDO be used to convert coinell voltage or 3.3V from battery (coinless) to 1.2V. There is a major change in bringing up the PRIM rails in Nova Lake Platform - the lower power rails should be enabled first followed by higher power rails. i.e., The power-up sequence should be in the order - VCCPRIM_VNNAON -> VCCPRIM_IO -> VCCPRIM_1P8_PROC for Processor. DSx power state is NOT supported. Hence there will not be any classification of DSx and Non-DSx mentioned in the below sections. All other power states which were supported in the predecessor platform are supported.

10.11.2 Sequencing Interface Signals List and Power Rails

Table 753. Platform Sequencing Signals List

Signal Name	Source	Destination	Description
RTCRST#	Platform	Processor	When asserted, this signal resets the register bits in the RTC well.
SRTCRST#	Platform	Processor	This signal resets the manageability register bits in the RTC well when the RTC battery is removed.
RSMRST#	Platform	Processor	This signal is used for resetting the Primary power plane logic. This signal must be asserted for at least 10 ms after the Primary power wells are valid. When de-asserted, this signal is an indication that the power wells are stable. RSMRST# is 1.25V IO as it belongs to RTC power plane.
PRIMPWRDNACK	Processor	Platform	Asserted high by Processor when it does not require Primary well to be powered. This signal is required to be connected to EC for platforms with or without M3 support This signal gives an added flexibility for the EC to turn OFF the Prim Rails when not needed by the Processor. PRIMPWRDNACK can be High/Low in Sx/Moff based on the ME power policy selected. On Platforms that don't support M3, EC must keep Prim rails powered ON If: 1. PRIMPWRDNACK is de-asserted low Else, EC has the option to turn-off the Prim rails On Platforms that supports M3, EC must keep Prim rails powered ON If: 1. PRIMPWRDNACK is de-asserted low OR 2. It's the first 200 ms after Prim rails power has been applied. Else, EC has the option to turn-off the Prim rails. <i>Note:</i> 1. In eSPI mode this signal is a virtual wire on the eSPI interface and the hard wire signal from Processor is disabled. Refer to the Enhanced Serial Peripheral Interface [eSPI] Compatibility Specification For 2015 / 2016 Client, Desktop and Server Platforms (# 508740) and Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.
SUSCLK	Processor	Platform	This clock is an output of the RTC generator circuit for use by other chips for refresh clock.
ACPRESENT	Platform	Processor	Used on mobile systems to determine presence of AC power or battery power.
PRIMACK#	Platform	Processor	Platform can leave this signal unconnected since it is not used, and this signal has an internal pull-up that ensures it is high, as required for the boot sequence to proceed. Note, there is no assertion order required between PRIMACK# and PRIMPWRDNACK assertion on G3 exit. <i>Note:</i> In eSPI mode this signal is a virtual wire on the eSPI interface and the hard wire signal from Processor is disabled. Refer to the Enhanced Serial Peripheral Interface [eSPI] Compatibility Specification For 2015 /2016 Client, Desktop and Server Platforms (# 508740) and for Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) more details .
PWRBTN#	Platform	Processor	Signal driven from EC to Processor indicating a system request to go into Sleep State OR if the system is already in the Sleep State then it will cause a wake event.
continued...			

Signal Name	Source	Destination	Description
SLP_A#	Processor	Platform	This signal is used to control power to devices on the platform in conjunction with the Intel® ME sub-system. This signal will be asserted in M-off state. If M3 is not supported then SLP_A# will have the same timings as SLP_S3#. <i>Note:</i> In eSPI mode this signal is also virtual wire on the eSPI interface, in addition to the hard signal from Processor. Refer to the Enhanced Serial Peripheral Interface [eSPI] Compatibility Specification For 2015 /2016 Client, Desktop and Server Platforms (# 508740) and Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details
SLP_LAN#	Processor	Platform	LAN Sub-System Sleep Control. Controls power to the LAN PHY. When "low", indicates that power can be shut off to the external wired LAN (GbE) PHY. <i>Note:</i> In eSPI mode this signal is also virtual wire on the eSPI interface, in addition to the hard signal from Processor. Refer to the Enhanced Serial Peripheral Interface [eSPI] Compatibility Specification For 2015 /2016 Client, Desktop and Server Platforms (# 508740) and Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.
SLP_S5#	Processor	Platform	This signal is for power plane control. When asserted (low), it will shut-off power to all non-critical systems in S5 (Soft Off) states. <i>Note:</i> In eSPI mode this signal is also virtual wire on the eSPI interface, in addition to the hard signal from Processor. Refer to the Enhanced Serial Peripheral Interface [eSPI] Compatibility Specification For 2015 /2016 Client, Desktop and Server Platforms (# 508740) and Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.
SLP_S4#	Processor	Platform	S4 Sleep Control. This signal is for power plane control. When asserted (low), it will shut-off power to all non-critical systems in S4 (Suspend to Disk) and lower (S5). <i>Note:</i> In eSPI mode this signal is also virtual wire on the eSPI interface, in addition to the hard signal from Processor. Refer to the Enhanced Serial Peripheral Interface [eSPI] Compatibility Specification For 2015 /2016 Client, Desktop and Server Platforms (# 508740) and Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.
SLP_S3#	Processor	Platform	S3 Sleep Control. This signal is for power plane control. When asserted (low), it will shut-off power to all non-critical systems in S3 (Suspend to RAM) and lower (S4, S5). <i>Note:</i> In eSPI mode this signal is also virtual wire on the eSPI interface, in addition to the hard signal from Processor. Refer to the Enhanced Serial Peripheral Interface [eSPI] Compatibility Specification For 2015 /2016 Client, Desktop and Server Platforms (# 508740) and Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.
VDD2_VDDQ_VR_PG	Platform VR	Platform Logic	Indicates that the DRAM power supplies are stable and within specification. Note: This is a separate Platform Logic signal and not to be confused with VDD2PWRGOOD_IN, VDD2PWRGOOD_OUT pins of SoC.
Platform S0 Rails	Platform	Processor/ Processor	These are all the non-core platform rails.
RSMRST_PWRGD	Platform	Platform	The signal represents power good for VCCPRIM rails and other S5 rails. This is platform generated signal
continued...			

Signal Name	Source	Destination	Description
ALL_SYS_PWRGD	Platform	Platform	This signal represents the power good for all the rest of platform voltage rails.
IMVP VR_READY	SoC	Platform Logic/SoC	IMVP VR_READY is an active-high output that indicates that the IMVP9.3 start-up sequence is complete. Assertion of VR_READY indicates that the IMVP9.3 VR is ready to accept an SVID commands and is operating properly. Refer to the VR_READY definition in the IMVP9.3 spec for additional details.
PLT_PWROK	Platform	Processor	When asserted, PLT_PWROK indicates that all the main Processor Primary rails and all the other Processor rails are up. PLT_PWROK is 1.25V IO and belongs to RTC power plane.
DRAM_RESET#	Processor	Processor	Controls reset to the memory subsystems (DDR5/LPDDR5)
PLTRST#	Processor	Processor	The Processor asserts PLTRST# to reset devices on the platform (e.g., SIO, LAN etc.). Asserted during power-up and when S/W initiates a hard reset sequence through the Reset Control register.
SPI0	Processor	Flash device	S erial P eripheral I nterface between Processor and BIOS Flash Devices.
LAN_WAKE#	PHY	Processor	Can be used by the LAN PHY as a wake signal. <i>Note:</i> In eSPI mode this signal is also virtual wire on the eSPI interface, in addition to the hard signal from Processor. Refer eSPI Compatibility Specification (508740) and Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.
ESPI_RESET#	Processor	Platform	Controls reset to eSPI
SLP_S0#	Processor	Platform	S0 Sleep Control. When Processor is idle and processor is in C10 state, this pin will assert indicate VR controller can go into a light load mode. This signal can also be connected to EC for other power management related optimizations. <i>Note:</i> In eSPI mode this signal is a hard wire only and not a virtual wire.
PROC_C10_GATE#	Processor	Platform	C10 Indication Signal. When Processor is idle and processor is in C10 state, this pin will assert to indicate that Processor is in C10 state. This signal can also be connected to EC for other power management related optimizations. Recommended to have a test point for debug purpose even when not used.
SKTOCC#	Processor	Platform	Socket Occupied: Pulled down directly in the processor package to the ground. System board designers may use this signal to determine if the processor is present for safety purposes, it helps to avoid accidentally applying power to the socket while nothing is installed into the socket. If the customers do not want to use or do not need to use the pin (PKG without socket), they can leave it floating.

Table 754. Power Sequence Related Power Rails

Name	Source	Destination	Description
VCCRTC_PROC	Platform	Processor	1.2V Processor RTC Well. This power is not expected to be shut off in any of the sleep states unless the RTC battery is removed or completely drained. Platform should convert RTC coinell voltage (nominal - 3.3V) to 1.2V through Ultra low bias current LDO
VCCPRIM_1P8/ VCCPRIM_IO/ VCCPRIM_VNNAON	Platform	Processor	Processor Primary (I/O) Rails
continued...			

Name	Source	Destination	Description
VPP/VDD1/VDDQ	Platform	Memory / Processor	Memory power rail, voltage dependent on memory technology
VDD2	Platform	Processor	Processor Memory power rail, voltage dependent on memory technology
VCC_CORE	Platform	Processor	Processor Core Rail
VCCGT	Platform	Processor	Graphics Power Rail
VCCSA	Platform	Processor	System Agent Power Rail

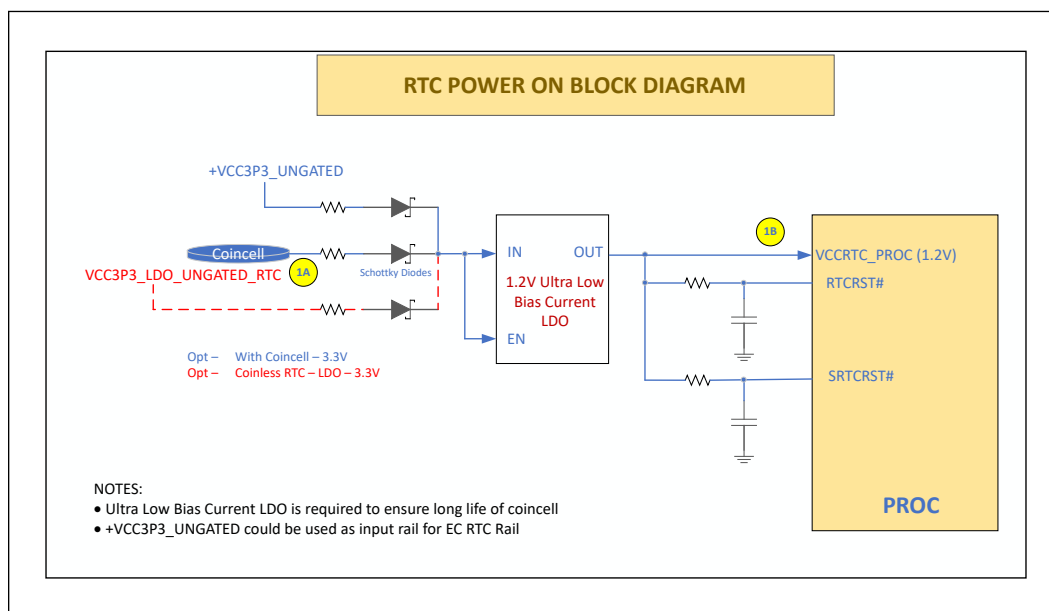
10.11.3 RTC and PRIM Rail Architecture

VCCRTC_PROC Circuitry

One of the key changes of this platform from its predecessor is VCCRTC_PROC Circuitry.

VCCRTC_PROC rail nominal voltage is 1.2V. Hence an ultra low bias current LDO is required to convert coincell voltage to 1.2V as required by processor. This ultra low bias current LDO is to ensure long life of coincell. RTC Power On Block Diagram is provided below as reference.

Figure 448. RTC Power On Block Diagram



Pseudo-G3 Overview

In Pseudo-G3 only VCCRTC_PROC power rail is powered ON while all other rails are OFF

Pseudo-G3 State Definition

- Entry to pG3 is permitted when PMC drives PRIMWRDNACK High. PRIMWRDNACK can be High/Low in Sx/CMOff based on the ME Power Policy selected (via EXT_PRIM_PD_EN bit)

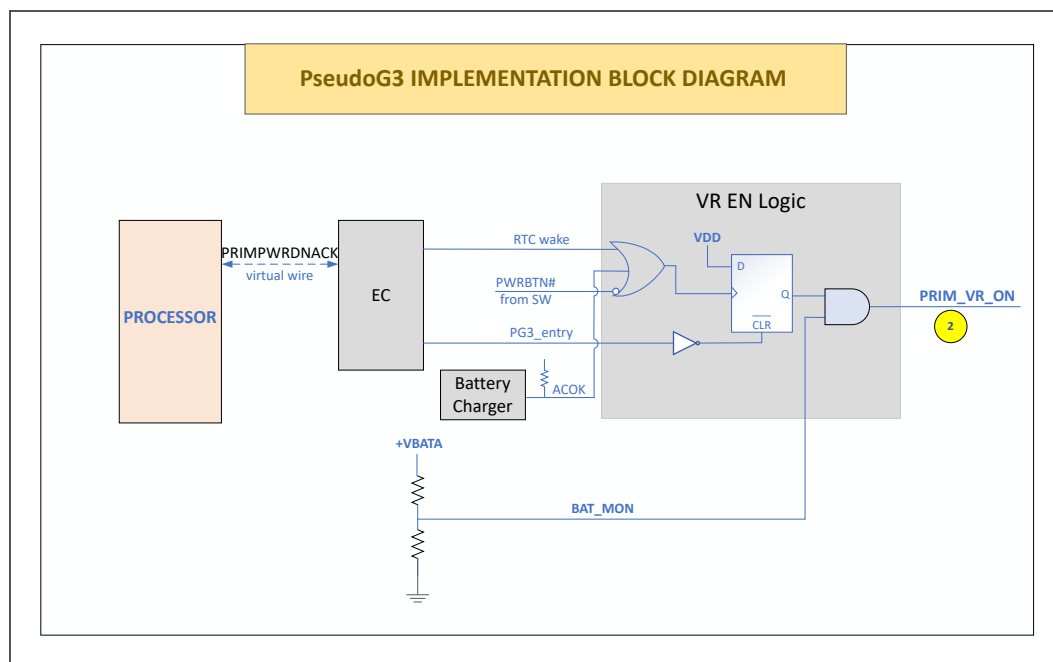
2. EC/platform wake logic kept ON. Platform should target EC power consumption of 5mW in Pseudo-G3 when PRIMPWRDNACK is high
3. Platform Logic / EC powers down Primary/DSW power rails while on battery. EC shall not power down Primary/DSW power rails when on AC. Network wakes will continue to work on AC.
4. pG3 to S0 wake events are power button toggle, ACPI TAD/RTC Timer, OEM specific wake events.

Advantages of Pseudo-G3

1. Low Power Consumption - approximately 100uW of Processor Power
2. Power Gating cost is saved in Pseudo-G3
3. It meets all Regulatory Requirements

There are many ways to implement Pseudo-G3. The below figure is one of such ways but pG3 entry won't happen with AC mode. ACOK is considered as a wake event and would exit pG3. Refer to the RVP for circuit reference.

Figure 449. Pseudo-G3 Implementation Reference Diagram



PRIM Rail Architecture

Power Up

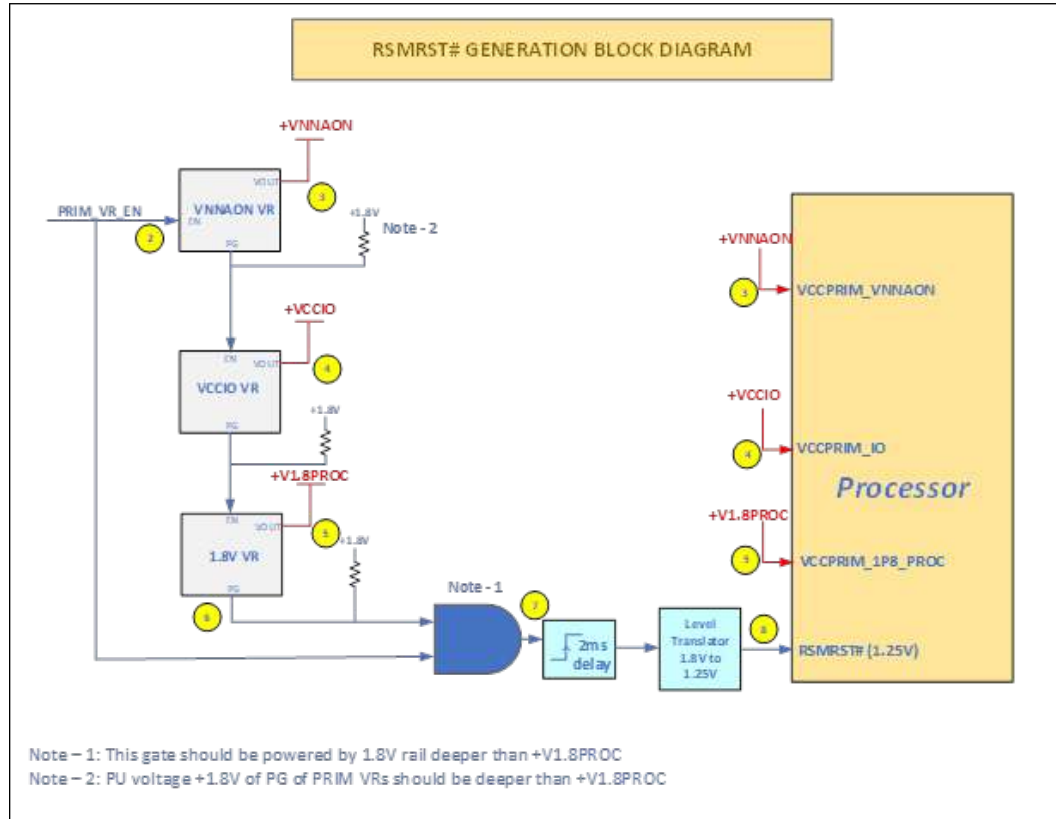
- PRIM rails should power ON:
Low voltage power rail should ramp first and higher voltage power rail last.

Power Down

No required order, but all PRIM rails should ramp down within a few us from PRIM VR EN low

Refer the Figure below for details. Also refer to Rail to Rail Power Sequencing Requirements

Figure 450. PRIM Rail Architecture Block Diagram



10.11.4 PWROK Signals Generation

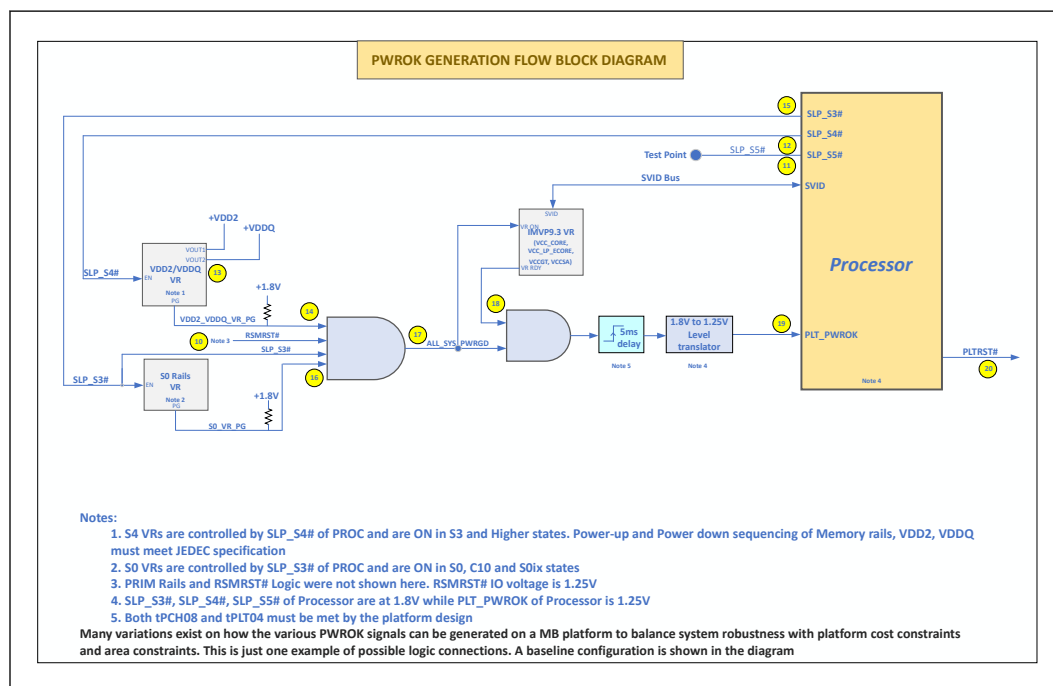
The power sequencing consists of multiple stages of power state transitions. During each stage, different VRs will be turned on through various control signals from the Processor such as the SLP_S4# and SLP_S3#. In response, the platform will ramp the required voltage rails in the required order and then asserted the various power ok signals required by the processor, and other platform components, after the necessarily timing requirements have been met. The figure below shows a high level representative control signal and power good logic diagram for rail control and PLT_PWROK generation. This figure shows the high-level recommended power-on sequencing, and power ok generation, from SLP_S4# de-assertion to PLTRST# de-assertion, for the platform.

During G3, S5 to S0 transitions, the platform will need to generate the RSMRST# signals. In this phase of the power up sequence, the Primary power well voltage rails are ramped to the processor.

During pG3 to S0 and S5 to S0 transitions, the platform needs to generate the PLT_PWROK signals to the Processor. In this phase of the power up sequence, the platform and processor S0 rails are ramped up.

The PLT_PWROK signal is expected to be asserted by the platform to indicate to the processor that all required Processor voltage rails are up and stable, and that the Processor may continue the final boot sequence leading up to PLTRST# de-assertion. This includes, but is not limited to starting to turn on clocks and executing other internal pre-reset activities required to boot the system.

Figure 451. PWROK Generation Flow Diagram



NOTE

In the above diagram, VDD2/VDDQ VR assumed to be satisfying JEDEC specification of Memory technology of interest for both power up and power down sequencing. Otherwise, discrete platform logic for each memory rails would be required.

10.11.5 Power States

NOTE:

On G3 exit, prior to the first transition to S0, S5 power may be higher than S5 power after the first S0 to S5 transition. Some SoC settings required to achieve minimum S5 power are loaded during first boot to S0 after a G3 exit. Consequently, entry into S5 from S0 will result in a more power-optimized S5 state than entry into S5 from G3 without an S5-S0-S5 transition.

Table 755. System with M3 State Supported

Rails	S0/M0 ³	C10	S0ix/M-off ⁴	S3/M ₃	S3/M-off	S4 and S5/M3	S4 and S5/M-off	PseudoG3 ¹	G3 ¹
VCCRTC_PROC	ON	ON	ON	ON	ON	ON	ON	ON	ON
<i>continued...</i>									

Rails	S0/M0 ³	C10	S0ix/M-off ⁴	S3/M ₃	S3/M-off	S4 and S5/M3	S4 and S5/M-off	PseudoG3 ¹	G3 ¹
VBATA (DC)	ON	ON	ON	ON	ON	ON	ON	ON	No Power
VCCPRIM_1P8_PROC	ON	ON	ON	ON	ON	ON	ON	No Power	No Power
VCCPRIM_IO	ON	ON	ON	ON	ON	ON	ON	No Power	No Power
VCCPRIM_VNNAON	ON	ON	ON	ON	ON	ON	ON	No Power	No Power
VDD2	ON	ON	ON	ON	ON	OFF	OFF	No Power	No Power
VDDQ	ON	ON	ON	ON	ON	OFF	OFF	No Power	No Power
VCCSA	ON	0V ²	0V ²	OFF	OFF	OFF	OFF	No Power	No Power
VCC_CORE	ON	0V ²	0V ²	OFF	OFF	OFF	OFF	No Power	No Power
VCC_LP_ECORE	ON	0V ²	0V ²	OFF	OFF	OFF	OFF	No Power	No Power
VCCGT	ON	0V ²	0V ²	OFF	OFF	OFF	OFF	No Power	No Power

Notes: 1. G3 is a state where all rails are off at the platform level, but RTC well is still powered. Pseudo-G3 a term that describes a platform specific state where all main SoC rails are off, SoC RTC rail is powered, and additional platform level wake logic is powered to allow for exiting this low power state, back to normal operation. Pseudo-G3 is not an official ACPI power state; it is a platform centric term used to differentiate between G3 and G3 with wake capability at the platform level.

2. This supply is expected to be ON with 0V setting in PS4 state. Specific power up latencies apply when exiting this state.

3. S0/M0 state includes all Package C-states from C0-C10.

4. Assume SLP_S0# and PROC_C10_GATE# have asserted from the Processor.

5. NA

6. V1.8M, V3.3M are not shown in the table. For no M3 support on external devices, V3.3M/V1.8M will be OFF in Sx/M3.

10.11.6 Power Sequencing Timing Diagrams Legacy Signals

Table 756. Legend for Signals in Transition Waveforms

Color/Legend	Comments
Signal Names	Timing of these signals is set by Processor
Signal Names	Timing of these signals should be met by the platform
Signal Names	Voltage rails or chip-to-chip buses
Grey Highlight	Indicates unstable state

Figure 452. Timing Diagram for G3 to S5

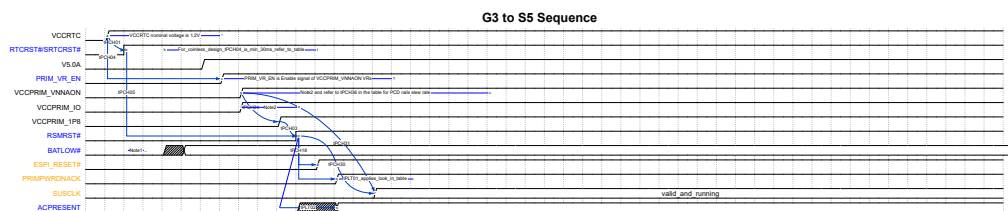
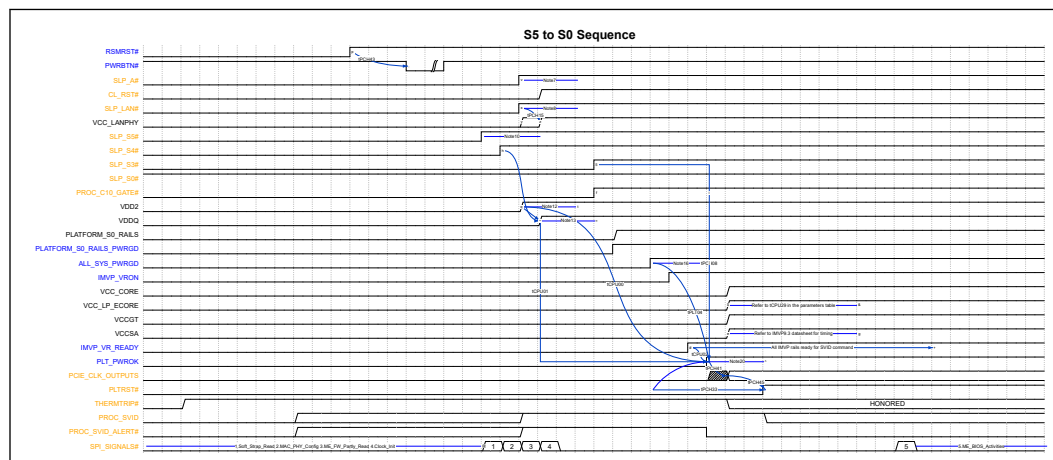


Figure 453. Timing Diagram for S5 to S0



NOTE

General Note: Some of legacy signals shown in these diagrams are not available as hard signals when eSPI is used.

1. Processor will sample BATLOW# on the rising edge of RSMRST#
 2. Refer [Rail-to-Rail Power Sequencing Requirements](#) on page 741 on Prim rail-to-rail power and power down dependencies
 3. NA
 4. NA
 5. NA
 6. NA
 7. SLP_A# always goes high with or before SLP_S3#. Depending on Processor settings SLP_A# may go high, then low, then high again all before SLP_S3#, but will go high no later than SLP_S3# on an Sx to S0 transition, or the Sx to S0 portion of G3 exit, Global Reset, and Deep Sx exit.

In the event of a global reset after SLP_A# is de-asserted, during the power up sequence, SLP_A# will assert and the controller monitoring the sleep signals should reset its timeouts.
 8. High for WoL=1, Low for WoL=0. SLP_LAN# may rise before, but no later than SLP_A#.
 9. On first exit from G3, SLP_WLAN# de-asserts with SLP_S3# de-assertion
 10. Delay between SLP_S5#, SLP_S4#, and SLP_S3# exaggerated for drawing purposes. If the system EC is driving these signals in eSPI mode the minimum delay between SLP_S3#, SLP_S4#, and SLP_S5# is not guaranteed
 11. NA
 12. Required with DDR5 and LPDDR5 memory configurations
 13. VDD2 must ramp after VPP on DDR5 based systems, thus VDD2, enabled by SLP_S4#, may ramp up after SLP_S3# de-assertion due to VR ramp timing and configuration. The sequence is set by the JEDEC specification for the memory technology of interests. See the JEDEC specification for the DDR technology used.
 14. NA
 15. NA
 16. ALL_SYS_PWRGD is assumed to logically AND together the pwrgood signals for the major system power rails
 17. NA
 18. NA
 19. NA
 20. The Platform should ensure that PLT_PWROK does not glitch when RSMRST# is de-asserted
-

Additional Notes:

Some of legacy signals shown in these diagrams, like PRIMPWRDNACK, are not available as hard signals when eSPI is used, they are Virtual Wires.

The state of the PRIMPWRDNACK signal is used by the EC to determine if Processor requires the suspend-well to stay powered.

- PRIMPWRDNACK
 - Platform not supporting M3 -
 - EC must keep Prim rails powered ON If, PRIMPWRDNACK is de-asserted low
 - Else, EC has the option to turn-off the Primary rails
 - Platform supporting pG3/M3 -
 - EC must keep Prim rails powered ON If,
 1. PRIMPWRDNACK is de-asserted low **OR**
 2. It's the first 200 ms after Primary rails power has been applied.
 - Else, EC has the option to turn-off the Prim rails.
- Primary rails should **never** be active while VccRTC rail is inactive.

Figure 454. Timing Diagram for S0 to S0ix to S0

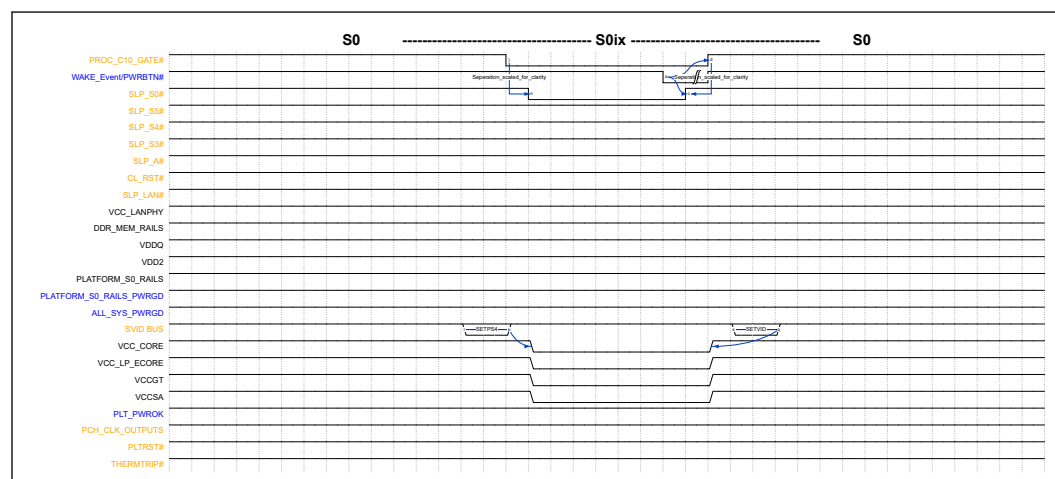


Figure 455. Timing Diagram for S0 to G3 - 1 of 2 - transition from S0/M0 to S5

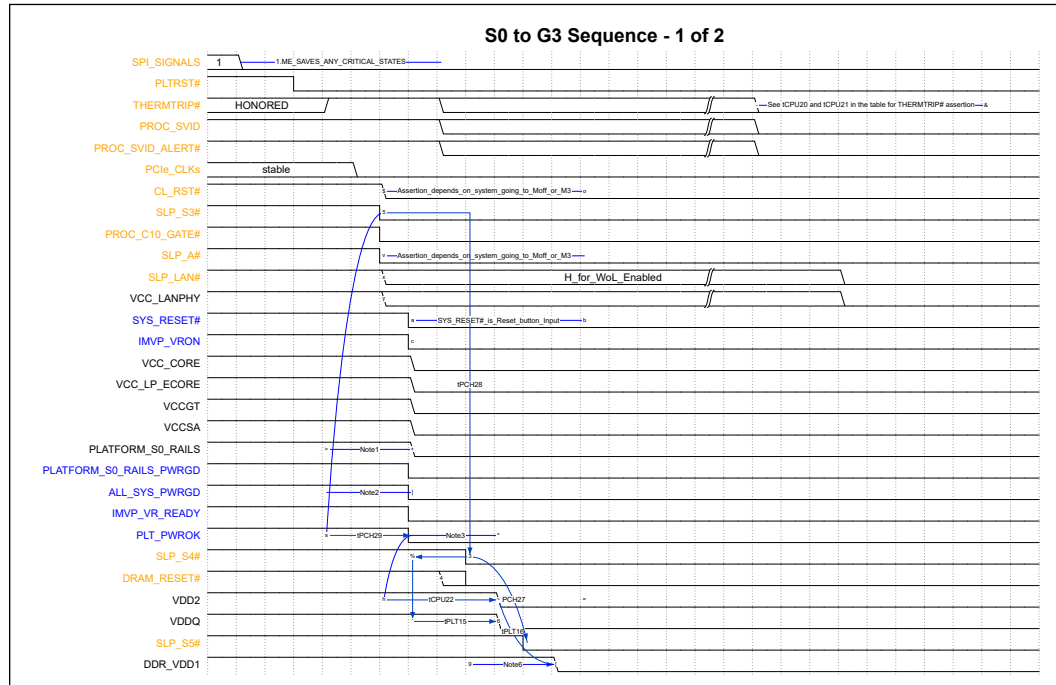


Figure 456. Timing Diagram for S0 to G3 - 2 of 2 - transition from S0/M0 to S5

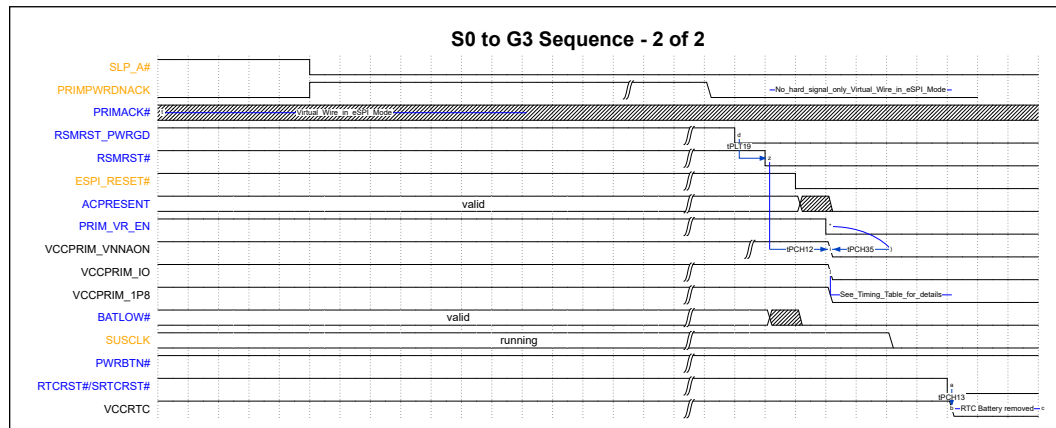


Figure 457. Timing Diagram for Cold Reset (1 of 2)

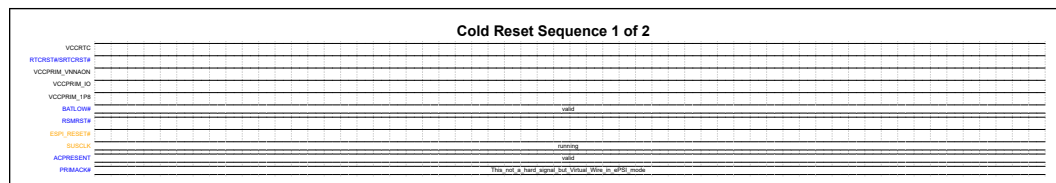


Figure 458. Timing Diagram for Cold Reset (2 of 2)

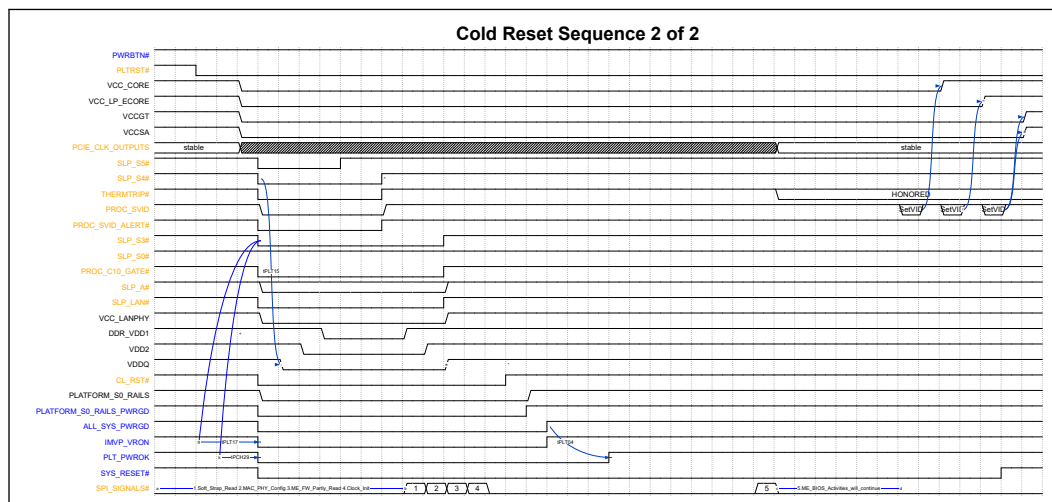
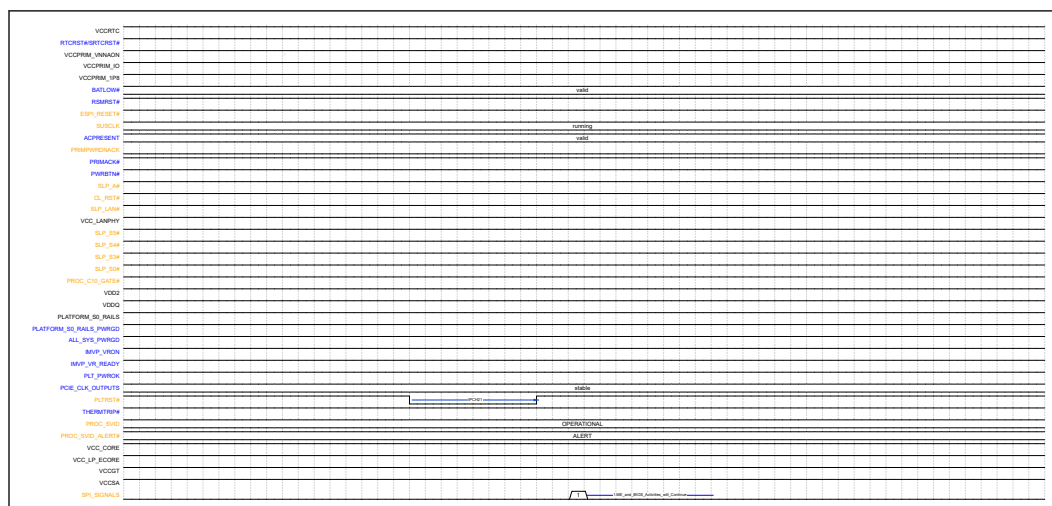


Figure 459. Timing Diagram for Warm Reset



Power Sequencing Timing Requirements

The following table defines the timing parameters used in the timing diagrams in previous sections. The timing parameters are put into three categories: tCPU, tPCH, and tPLT. tCPU parameters are mostly required by the processor, tPCH parameters are required or controlled by the processor, and tPLT parameters are mostly controlled by the platform.

The timing parameters are defined by Min, Max and Typical specifications. The Min and Max timings refer to the minimum or maximum timings allowed between the first and second signals in the Description column, as are the timing boundaries that must be followed. The Typical column refer to the typical timing values measured on Intel boards during validation, which do not imply a requirement but can be used as a reference.

The timing parameter or label or value or description were written in bold font to highlight the changes with respect to the previous generation platform.

Table 757. Platform Sequencing Timing Parameters

Label	Applicable SKU	Required By	Controlled By	Min.	Max.	Typical	Units	Note #	Description
tCPU00		PROC	PLT	2			ms	6, 19	VDD2 and VDDQ stable to PLT_PWROK assertion
tCPU01	All	PROC	PLT	2			ms	6, 19, 30	All DDR rails stable to PLT_PWROK assertion
tCPU02	All	PROC	PLT	0			ms	6, 19	VCCSA, VCC_CORE, VCC_LP_CORE and VCCGT ready for SVID command to PLT_PWROK assertion. Note: Cursor reference should be from IMVP VR READY High to PLT_PWROK High for measurement purpose.
tCPU20	All	PROC	PLT + PCH		500		ms		THERMTRIP# assertion until all IMVP VRs are disabled and not sourcing power
tCPU21	All	PROC	PCHALL		10		ms		THERMTRIP# assertion until Processor issues global reset and assertion of SLP_Sx signals. This is set by Processor. This timing is provided for informational purposes only. The platform is not required to do anything to meet it.
tCPU22	All	PROC	PLT	1			us	37	PLT_PWROK de-assertion to any of VDD2 and VDDQ below specification for normal S0 to Sx transitions. It is recommended that the platform take PLT_PWROK low within 1us of less of SLP_S3# assertion (low).
tCPU29	All	PROC	PLT		100		mV/us	13	Processor power rail instantaneous slew rate.
tPCH01	Mobile	PROC	PLT	9			ms	1, 26	VCCRTC_PROC = 1.08V to the point in time where voltage on the RTC resets equals 0.65 times the voltage present on the VCCRTC_PROC rail during ramp. This measurement should be made from VCCRTC_PROC = 1.08V to the first of RTCRST# or SRTCST# reaching 0.65 * VCCRTC_PROC. Refer Real Time Clock Reset Timing Technical Advisory # 610459.
tPCH03	All	PROC	PLT	2			ms		VCCPRIM rails stable (@95% of full value) to RSMRST# high.
tPCH04	All		PROC	9			ms	29, 43	VCCRTC_PROC stable (@90% of full value) to start of VCCPRIM rails ramp for systems with coined RTC battery
	All		PROC	30			ms	29, 43	VCCRTC_PROC stable (@90% of full value) to start of VCCPRIM rails ramp for systems with coinless RTC. Refer to # 549657 for Design considerations without coincell RTC Battery.
tPCH05	All	PROC	PLT	1			us	43	RTCRST# high (voltage above ViH_min) to RSMRST# high (when voltage crosses ViL_max such that internally it might be resolved as a logic '1')
tPCH08	All	PROC	PLT	1			ms		SLP_S3# de-assertion to PLT_PWROK assertion. Note: eSPI SLP_S3# virtual wire will assert with or later than the SLP_S3# hard signal. Timing should be measured from version of SLP_S3# used to control the power deliver sub-system for power up
continued...									

Label	Applicable SKU	Required By	Controlled By	Min.	Max.	Typical	Units	Note #	Description
tPCH09	All	PROC	PLT	2, 4, 8, 16			ms		SLP_A# high to Processor assuming ASW rails are stable (95% of full value) Measured from hard SLP_A# signal. Adjustable by OEM via APWROK Timing soft strap (PMC Descriptor Record 0 [22:21]). Default= 2ms Refer to the NVL SPI Programming Guide for details
tPCH12	All	PROC	PLT	400			ns	5, 24	RSMRST# assertion to VCCPRIM reaching -5% of nominal value
tPCH13	All	PROC		0			ms	14	RTCRST# assertion to VCCRSTC_PROG dropping 5% of nominal value (this applies only when RTC battery is removed)
tPCH15	All	PROC	PLT		50, 100		ms		SLP_LAN# (or LANPHYPC) rising to VCCLANPHY high and stable. Default is 100ms. Adjustable by OEM via LAN PHY Power Up Time soft strap (PMC Descriptor Record 0 [19]). Default= 100ms Refer to NVL SPI Programming Guide for details
tPCH18	All		PROC	95			ms		De-assertion of RSMRST# to De-assertion of eSPI_RESET# on exit from G3. Set by Si; given for informational purposes. Not for measurement
tPCH21	All		PROC	Refer note 39			ms	39	Warm Reset PLTRST# assertion duration time
tPCH27	All		PROC	30			us	33	SLP_S4# assertion to SLP_S5# assertion. <i>Note:</i> For eSPI systems the min delay between eSPI SLP_Sx virtual wires may be 0, but is not guaranteed.
tPCH28	All		PROC	30			us	33	SLP_S3# assertion to SLP_S4# assertion. <i>Note:</i> For eSPI systems the min delay between eSPI SLP_Sx# virtual wires may be 0, but is not guaranteed.
tPCH29	All		PROC	0			ms		SLP_S3# assertion to PLT_PWROK deassertion. <i>Note:</i> eSPI SLP_S3# virtual wire will assert with, or later than, the SLP_S3# hard signal. Timing should be measured from version of SLP_S3# used to control the power deliver sub-system for power down.
tPCH30	All		PROC	95			ms		Internal RTC clock stabilization time. Set by Hard Strap. Provided as Information purpose only. Not for measurement. Timed from: RSMRST# L to H
tPCH31	All		PROC		tPC H03 + tPC H30		ms	15, 44	Prim rails ramped and stable until SUSCLK is stable. SUSCLK will start before the max timing, but will not be guaranteed valid until max timing is met. NOTE - While calculating tPCH31, consider measured tPCH03 value.
continued...									

Label	Applicable SKU	Required By	Controlled By	Min.	Max.	Typical	Units	Note #	Description
tPCH33	All		PROC	0, 99			ms	16	PLT_PWROK high to PLTRST# de-assertion. This timing is set by the Processor via Soft strap settings. <i>Note:</i> The eSPI PLTRST# virtual wire may be delayed from the hard PLTRST# wire.
tPCH34	All	PROC	PLT		50		ms		All Primary Rails should ramp up within this window.
tPCH35	All	PROC	PLT		1		ms	20	Measured from PRIMARY VR EN low to last Processor PRIMARY rail reaching Vmin or out of spec. Refer note for restart condition. All IOs will be indeterministic even with weak PU/PD and the functionality will not be guaranteed during power down
tPCH36	All	PROC	PLT		100		mV/us		PRIMARY Power rails instantaneous slew rate
tPCH41	All	PROC	PROC	1			ms		PLT_PWROK high to PCIe clock outputs stable. (set by Si, provided as informational)
tPCH43	All	PROC	PLT	tPCH30			ms	28	RSMRST# de-assertion to PWRBTN# monitored
tPCH45	All		PROC	(1, 5, 50, 100) + tPCH46			ms	41	tPCH09 expiring to CL_RST# high SoC / PCH PCIe ref Clock outputs stable to PLTRST de-assertion. Timing set by PCH via Soft Strap settings Adjustable by OEM via tCPU45 Timing soft strap (PMC Descriptor Record 0 [9:8]). Default=100ms See NVL SPI Programming Guide for details
tPCH46	A		PROC	1, 2, 5, 10			ms	41	Addition delay added at expiration of tPCH45 between tPCH45 and PLTRST# deassertion. This timing is not applicable for eSPI systems. Adjustable by OEM via tCPU46 Timing soft strap (PMC Descriptor Record 0 [11:10]). Default=2ms NOTE: Timing can be adjusted through the FIT tool. See NVL SPI Programming Guide for details
tPCH50	All	PROC	PROC		100		ms		Global reset entry to power button monitored after entry into global reset flow, as might initiated by a power button hard shutdown. To be referenced from the de-assertion of eSPI_RESET# after the global reset entry. <i>Note:</i> A power button wake event detected during the reset flow will not initiate a transition to S0 until after the global reset completes, which could take multiple seconds to complete.
tPLT01	All		PROC	200			ms	2	RSMRST# de-assertion to PRIMPWRDNACK valid. Timing set by Processor.
tPLT02	All			0	90		ms		RSMRST# de-assertion to ACPRESENT valid (not floating).
tPLT04	All	PROC	PLT	1			ms	3, 19	ALL_SYS_PWRGD assertion to PLT_PWROK. This timing must be controlled on the platform.
tPLT14	All		PROC	4			s		Programmable power cycle duration timer. Set by the Power Cycle Duration (PWR_CYC_DUR) field in the register.
continued...									

Label	Applicable SKU	Required By	Controlled By	Min.	Max .	Typical	Units	Note #	Description
									The value in this register determines the minimum time a platform will stay in reset (SLP_S3#, SLP_S4#, SLP_S5# asserted and also SLP_A# and SLP_LAN# asserted if applicable) during a host partition reset with power cycle or a global reset. The duration programmed in this register takes precedence over the applicable SLP_# stretch timers in these reset scenarios. This field is not writable when the SLP_Sx# Stretching Policy Lock-Down bit is set. This field is specified in 20ms increments starting at 20ms. For example : - 0x00 = 20ms - 0x01 = 40ms - 0xC7 = 4 seconds - 0xFF = 5.12 seconds. Note that the duration programmed in this register should never be smaller than the stretch duration programmed in the following registers - - GEN_PMCON_3.SLP_S3_MIN_ASST_WDTH - GEN_PMCON_3.S4MAW - PM_CFG.SLP_A_MIN_ASST_WDTH - PM_CFG.SLP_LAN_MIN_ASST_WDTH
tPLT15	All	PLT (MEM)	PLT		200		us	40	SLP_S4# assertion to VDD2 and other memory VRs disabled. Memory technology (JEDEC) dependent Refer note reference.
tPLT16	All	PLT (MEM)	PLT	Refer Notes			ms	30	Memory rail ramp timings.
continued...									

Label	Applicable SKU	Required By	Controlled By	Min.	Max.	Typical	Units	Note #	Description
tPLT17	All	PROC	PLT		2		ms	10, 35	SLP_S3# low to IMVP_VR_ON de-assertion. Note: eSPI SLP_S3# Virtual Wire will assert with or later than the SLP_S3# hard signal. Timing should be measured from version of SLP_S3# used to control the power delivery sub-system for power up.
tPLT19	All	PLT	PLT		10		us	22	RSMRST_PWRGD (platform generated signal) de-assertion to RSMRST# assertion

- Notes:**
- Processor Primary Rails must never be active while VCCRTC_PROC is OFF
 - RSMRST# should be de-asserted for at least 200ms before PROC will consider PRIMWRDNACK valid.
 - Processor can delay de-assertion of PLTRST# with a soft strap setting. Refer tPCH45 (t573) and tPCH46 (t1001) and the SPI Programming Guide for more details.
 - For catastrophic/surprise power failures only.
 - RTC corruption avoidance on PRIM power down and unexpected / surprise power down. RSMRST# must be low tPCH12 before any of the rails PRIM rails fall out of spec to avoid RTC corruption.
 - PLT_PWROK has no edge rate requirement, but edges must be monotonic. tCPU02 must be measured from VR_READY of IMVP High to PLT_PWROK assertion.
 - NA.
 - NA
 - NA
 - It is strongly recommended that the SLP_S3# be a qualifying input signal to ALL_SYS_PWRGD logic, which drives IMVP VR_ON (Enable Pin) input.
 - NA
 - NA
 - Applies to all Processor power supply rails
 - There are no perceived functional failures if this minimum is violated since all data is lost anyway when VCCRTC_PROC drops. The only real requirement is to avoid driving a high signal level into the powered down device for an extended period of time, assuming an external device(s) on a different power supply than VCCRTC_PROC is driving RTCRST# or any other RTC well input signal.
 - SUSCLK is now powered in PRIM well.
 - Processor can delay de-assertion of PLTRST# with a soft strap setting. If tPCH33 is not enabled, the platform is responsible for controlling the assertion timing of PLT_PWROK in such a way that it satisfies platform component timing requirements of power stable to reset de-assertion. Refer SPI Programming Guide for more details. On a platform level this timing doesn't necessarily need to meet 99ms, PCH soft straps guarantees a min of 99ms.
 - NA
 - NA
 - PLT_PWROK assertion assumes Processor voltage rails are ramped and stable and ready for VID command, for the minimum required time
 - No ramp down order requirement between rails. All VRs should stop regulation based on PRIM_VR_EN signal, or another common PRIMARY rail shutdown signal. Platform must ensure that rails are below 200mV prior to restart. All IOs will be indeterministic even with weak PU/PD and the functionality will not be guaranteed during power down**
 - NA
 - 10us max limit is an estimation and will vary based on platform VR and EC implementation details. Platform designers are responsible for ensuring that tPCH12 is not violated during normal pG3 entries based on EC and VR response times to assertion of RSMRST_PWRGD
 - NA
 - Applicable to all power down sequences where PRIM well rails are turned off and to emergency power loss scenario. Refer RSMRST# Special Requirements section
 - NA
 - The capacitors used in the RTC external RC reset circuits should be evaluated with regards to aging, variation with voltage, and variation with temperature to ensure reliable operation and reset timing in the intended operating application. Refer RTC External RTCRST# Circuit Section.
 - NA
 - Applicable to all G3 exits where GEN_PMCON_B.AG3E=1. The PWRBTN# has a de-bounce of 16ms and must assert for at least 18ms for a guaranteed detection after the minimum tPCH43 timing before Processor will detect PWRBTN# as a wake event
 - Dead coin-cell scenario - For scenarios where the coin-cell is dead or missing VCCRTC must ramp before any PRIM supplies start ramp. There is not special accommodation for dead coin-cell cases; all normal sequencing requirements must be met.**
 - Generally, JEDEC specifications require both power up and power down sequences for a given memory technology. Designers are responsible for making sure their memory rail sequence meet all JEDEC requirements for the memory technology of interest. Designers must account for platform level circuitry (output decoupling, discharge circuits, etc) that may impact memory rail sequencing.
 - NA
 - NA
 - Not applicable to eSPI mode virtual wires, refer to the eSPI Compatibility Specification (#508740) for eSPI implementations
 - NA
 - Timing to VR being disabled, not until the VR is fully ramped down
 - NA
 - Applies to all power state transitions including emergency power loss and G3 entry
 - NA
 - During a warm reset sequence, the Processor sequencing flows will internally re-execute several timing parameters and tPCH45 and tPCH46 are two of the parameters that are re-executed. tPCH45 and tPCH46 are primarily for G3/Sx to S0 flows, but they are executed for cold boot, cold reset, and warm reset. The nominal setting for these timings is partially controlled by the setting of SPI soft straps associated with tPCH45 and tPCH46. The minimum possible PLTRST# assertion time will be equal to the soft strap settings for tPCH45 + tPCH46. The maximum PLTRST# assertion time is not guaranteed across all warm reset cycles, but the typical worst case assertion time is approximately equal to (tPCH45 soft strap setting) + (tPCH46 soft strap setting) + 20ms.

Label	Applicable SKU	Required By	Controlled By	Min.	Max.	Typical	Units	Note #	Description
40. This is a platform timing recommendation to help ensure that memory device power down sequencing requirements are likely to be met. This is not a requirement for the Processor itself. Refer to the appropriate JEDEC memory specification for you memory technology for memory power down sequencing requirements. Recommend not to exceed 200us delay with respect to SLP_S4#. 41. This parameter is programmable, refer to the Flash Descriptor Record in the SPI Programming Guide in the ME firmware package 42. NA 43. For the dead RTC coincell and coincell-less designs with depleted main battery RTC cases, where an RTC well is ramped by an LDO, and the RTC resets are RC based, the board designer should set the PRIM rail stable to RSMRST# assertion delay (tPCH03) for 30ms or more. This is required to ensure that SRTCST# and RTCRST# de-assert after VCCRTC is stable, but before RSMRST# de-assertion. Failure to meet this requirement may result in RSMRST# asserting with, or before, SRTCST# and RTCRST# reach VIH, which is a sequencing violation and can result in a non-booting system scenario. Refer Real Time Clock (RTC) Design Guidelines on page 394 and External RTCRST# Circuit on page 398 and External SRTCST# Circuit on page 398 for SRTCST# and RTCRST# RC timing network details 44. SUSCLK stable means the clock is toggling and is within it is defined parameters. This timing specification is applicable to all G3 exits, excluding the G3 exit immediately following the ramp of VCCRTC_PROC, which may have a longer delay 45. NA 46. NA. 47. For measurement details, reference RTC Reset Timing Technical Advisory - #610459. Additional Notes: Unless otherwise noted, all specifications in this table apply to all processor frequencies.									

10.11.7 Rail-to-Rail Power Sequencing Requirements

Processor Rail Order Requirements

RTC and PRIM rail ramp order

RTC must ramp before start of Processor PRIM rail ramp.

RTC and Primary power up order

RTC (Note: RTC must complete ramp before start of VCCPRIM ramp for dead coincell case).

tPCH01 delay (9ms min). Measured from VCCRTC_PROC=1.08V to the point in time where the voltage on RTCRST# or SRTCST# equals 0.65 time the voltage present on VCCRTC_PROC during its ramp. This measurement should be made from VCCRTC_PROC = 1.08V to the first of RTCRST# or SRTCST# reaching a voltage that equals 0.65 times the voltage present on the VCCRTC_PROC rail.

RTCRST#=H and SRTCST#=H

tPCH04 delay (9ms min for coincell / 30ms min for coinless RTC)

PRIM rails enabled by platform (VCCPRIM_VNNAON --> VCCPRIM_IO --> VCCPRIM_1P8_PROC).

tPCH03 delay (2ms min)

RSMRST#=H

Power Down: No specific order, but RSMRST# must be low before any of the PRIM rails fall out of spec (tPCH12) and the rail decay must meet tPCH35 requirements.

VDD2

Power Up (Processor)

VDD2 rail to Processor must ramp before VDDQ and IMVP Rails VCC_CORE, VCC_LP_ECORE, VCCGT, VCCSA

VDD2 must be stable tCPU01 prior to assertion of PLT_PWROK for Sx to S0 transitions.

Power Down (Processor)

DDR sequencing requirements set by JEDEC spec. See JEDEC spec for memory technology of interest (LPDDR5). No specific rail order requirement for VDD2 with respect to any Processor rails

PLT_PWROK must go low tCPU22 before VDD2 rail falls out of specification.

VDDQ

Power Up (Processor)

VDDQ rail to Processor must ramp after VDD2 ramp and before IMVP Rails VCC_CORE, VCC_LP_ECORE, VCCGT, VCCSA

VDDQ must be stable tCPU01 prior to assertion of PLT_PWROK for Sx to S0 transitions.

Power Down (Processor)

DDR sequencing requirements set by JEDEC spec. See JEDEC spec for memory technology of interest (LPDDR5). No specific rail order requirement for VDDQ with respect to any Processor rails

PLT_PWROK must go low tCPU22 before VDDQ rail falls out of specification.

Memory Rails - VDD1, VDD2H, VDD2L and VDDQ

Power Up

Must ramp before IMVP Rails - VCC_CORE, VCC_LP_ECORE, VCCGT, VCCSA. DDR sequencing requirements set by JEDEC specification. Refer to the JEDEC specification for memory technology of interest.

Power Down

DDR sequencing requirements set by JEDEC specification. Refer to the JEDEC specification for memory technology of interest.

Must ramp according to JEDEC sequencing requirements for specific memory technology of interest.

VDDQ for LPCAMM

In order to support LPCAMM2 E1 modules, Intel recommends using the "VDDQ common Method" as defined in the JEDEC standard. System boards designed with the common method will supply VDDQ to the CAMM2 connector and must also tie the "VDDQ_DISABLE" CAMM connector pin to "VIN_BULK" CAMM connector pin as recommended in JEDEC standard. To ensure the proper power-up sequence for LPCAMM2 memory rails, the "PWR_GOOD" output from the CAMM2 connector should be part of the mother board VDDQ regulator enable logic. The mother board VDDQ regulator must complete its soft-start within the memory power-up sequence specified in the memory JEDEC standard. During the power-down sequence VDDQ must quickly discharge the output voltage to meet the memory power down sequence specified in the memory JEDEC standard.

For systems using a common VDDQ rail for the processor and DRAM, the mother board VDDQ regulator shall consider supporting memory DVFSQ mode. DVFSQ requires supporting inrush current capability calculated with the maximum slew rate chosen for the DVFSQ mode (for example, 15 mV/usec $\pm$ 10%), total capacitance (including memory CAMM2 module/DRAM de-coupling and Processor VDDQ rail de-coupling plus mother board de-coupling capacitance if applicable). Refer to the latest JEDEC standards.

VCC_CORE, VCC_LP_ECORE, VCCGT, VCCSA

Power Up

IMVP is enabled after all other platform rails are stable. VCC_CORE Vboot = 0V, VCC_LP_ECORE Vboot = 0V, VCCGT Vboot=0V and VCCSA Vboot=0V. VIDs set post PLT_PWROK assertion (post EPD_ON).

Power Down

VR should be disabled with SLP_S3# (or a derivative of that signal) to ensure power good goes low as fast as possible.

- **General Rail-to-Rail Sequencing**

All platform rails should follow a sequential order for system power on which are already called out in the [Table 757](#) on page 736.

All Primary rails should ramp within 50ms of each other. For power down, there are no explicit timing requirements/relationships between the various Primary Well voltages on power down, but it is required that all Primary Well voltage regulators stop regulation simultaneously based on PRIM_VR_EN de-assertion. The power down sequencing should not be staggered from one VR to the next. Natural variance between VRs disabling due to differences in voltage regulator disable time is acceptable on the order of tens of microseconds. It is required that all rails ramp down within tPCH35.

For surprise power down requirements, refer to the [Surprise Power Down Sequencing Considerations](#) on page 744 for assumptions.

RSMRST# Special Requirements

1. When the system is powered off (G3), RSMRST# must not glitch from their Low states while the corresponding Processor/PCH PRIM rails are not powered or are below normal operating voltage specifications to ensure RTC corruption does not happen.
2. When a system is in S0-S5 state and entering pG3 or G3 state, the RSMRST# must be driven low (by an external controller such as EC) at least 400ns before any of the PRIM rails go out of specification. Failure to meet this requirement may result in unexpected Processor/PCH behavior, including failure to boot (which may only be recovered through a G3 cycling) and possible RTC corruption. This is true for all power states transitions including emergency power loss.

RSMRST# Min LOW Time during Cycling

RSMRST# assertion min low time is 100ms. The platform must then follow the normal power sequence (normal G3 exit sequence). Platform must ensure that all PRIM rails reach its Vmin or less, within tPCH35 timing of the falling edge of PRIM VR EN. Refer to Platform Sequencing Timing Requirements Table under section Power Sequencing Timing Diagrams Legacy Signals.

RTCRST# Min LOW Time during Cycling

For RTCRST# de-assertion min low time is 100ms, after the min time platform must follow the normal power sequence (normal G3 exit sequence).

Surprise Power Down Sequencing Considerations

Surprise power down events will be treated slightly differently to past generation platforms. The main goal of the various power down timing specification such as tPCH12, etc, is to ensure proper isolation between the associated power well and the RTC well to guarantee that RTC contents are not accidentally corrupted. There are many events that could cause a surprise power down. The following is a short list of some events, but is not exhaustive:

- VR failure (over current, over voltage, IC failure)
- AC removal with no DC Battery present
- Removal of the primary battery

Properly designed platforms generally should not be experiencing VR failures of any kind; therefore the focus of this section is on the unexpected power removal caused by user interaction, which could be an end user, factory technicians, or system level induced power down that removes all power from Processor.

To ensure RTC is not corrupted, and prevent potential Processor damage, the platform must de-assert the appropriate power good signals BEFORE the rails go out of their defined tolerance range. This implies that the platform should monitor the highest voltage available which is usually the main power supply like the battery voltage to determine when it has dropped too low and VR failure/shutdown is eminent. At that point, the power good signals (PLT_PWROK , RSMRST#) should be driven Low before their associated rails turn off and droop below the defined tolerance.

Finally, regardless of the reason, any time that PRIM_VR_EN is taken low, the platform is responsible for ensuring that tPCH12 is met. Additionally, once RSMRST# is taken low, the platform is responsible for taking the system to G3 before attempting a restart. Refer to the RSMRST# Special Requirements for details regarding RSMRST# requirements.

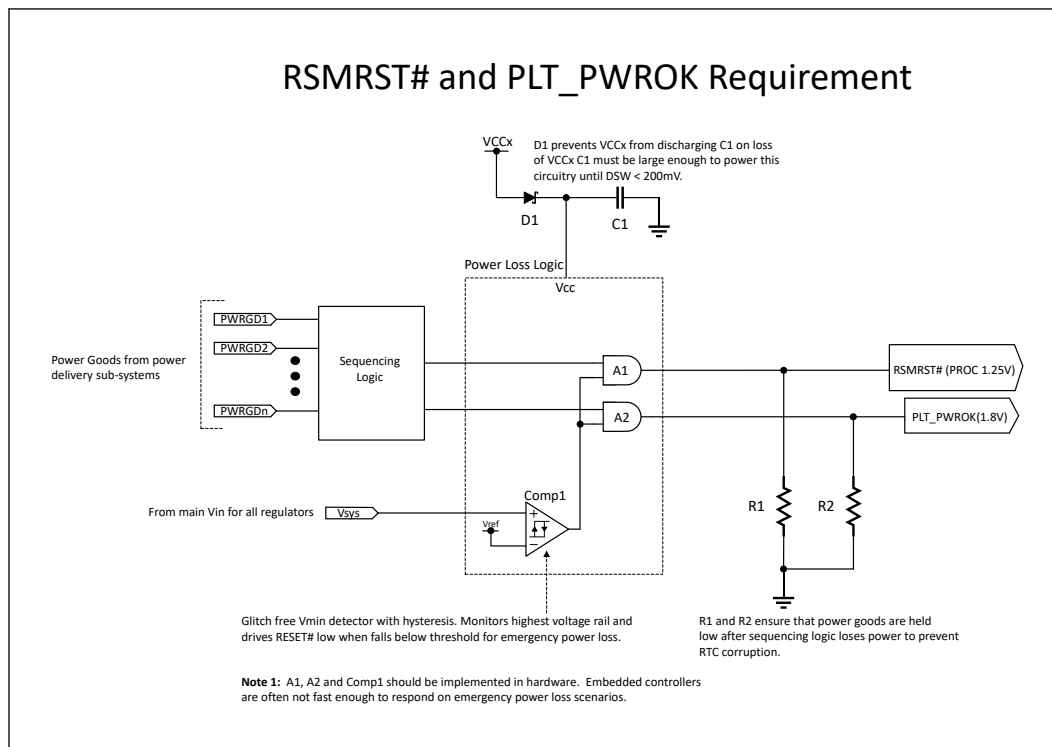
Critical Design Notes:

1. Designers **MUST** meet tPCH12 for RSMRST# low to respective rails out of spec . Failing to meet these requirements could result in RTC corruption and possible permanent damage to the Processor.
2. PCs are expected to operate even after the battery is fully exhausted (at end of life and unable to hold charge). Once the battery is exhausted, the system is only powered from the brick, which can be accidentally removed during use. System Designers should not assume an internal battery will prevent any of the emergency loss scenarios. Systems should be designed to meet tPCH12 as if they

only run only from brick. Failing to design in this manner may expose the Processor to RTC corruption and possible permanent damage to the Processor in systems where the battery is at end of life.

The below figure shows one such possible architectural implementation for ensuring that the PWROK go low and are locked out before the rails go out of spec on any emergency power loss.

Figure 460. PLT_PWROK and RSMRST# Requirement for Power Loss



eSPI Considerations for Sequencing

In general, eSPI support does not have any major impact to power sequencing requirements. However, there are a few behavioral differences worth noting that could have side effects to platform behavior that should be considered.

With eSPI, the EC may or may not take in or drive physical pins that historically would have been supported by the PROC. The values of these signals (ex. SLP_S4#) are tunneled over the eSPI interface between the PROC / EC as virtual wires. If the EC is using a combination of hard signals from the PROC and Virtual Wires over eSPI, there is no guarantee of relative timing between the two signal types, other than the hard signals will change state first.

Only the physical signals directly on the PROC are guaranteed to refer the PROC-defined timing relationships. Example, SLP_S4# de-assertion. SLP_S3# de-assertion relationship is defined as 30us for the physical pins but could be shorter for the virtual wire relationship on the EC.

PRIMPWRDNACK Virtual Wire (VW) is just valid after the rising edge of RSMRST# for eSPI designs and no need to wait for 1 -> 0 transition to signal it is valid. However, in non-eSPI mode, PRIMPWRDNACK status is to be considered valid after its first 1 -> 0 transition.

11.0 Power Integrity Recommendations

This chapter describes the decoupling and layout recommendations for platform power rails with reference stack-up. The decoupling solution recommended in this section is based on a specific VR switching frequency/bandwidth assumption. The decoupling solution needs to be adjusted if a different VR solution or stack-up is used.

If you are unable to follow the design guide recommendation due to capacitor selection availability/routing constraints, you can utilize the "Power Integrity Model Set" provided to perform the analysis to ensure the electrical targets are met.

It is important to design both power and ground carefully to reduce inductance and resistance. Following decoupling capacitor counts/values is insufficient to guarantee the design quality. Connectivity from the loads to the capacitors are crucial to ensure effective decoupling. General design considerations can be found in the "Power Integrity General Guidelines Technical Advisory" which includes detail descriptions for power integrity best design practices on component placement, layer arrangement and via placement. By following these guidelines you can increase the overall quality of the design, including capacitor effectiveness, and when combined with the analysis using the "Power Integrity Model Set" you can potentially meet the electrical targets with minimum capacitor count.

NOTES

1. Customers might find out discrepancies between Intel and VR vendor decoupling capacitor count recommendations, in such cases we recommend to follow the recommendation with the highest capacitor count among the two and to validate the solution using the VRTT tool. The decoupling recommendations in this chapter are designed to meet the electrical targets of the power rail beyond the bandwidth of the voltage regulator, for the reference stackup and layout detailed in this document; these recommendations do not take into account vendor specific requirements for the VR solution.
2. If common motherboard approach is planned to accommodate the SKUs - 8C/ H16C 4Xe/ H16C 12Xe it is recommended to follow the 12Xe BOM for the motherboard design. An extra bulk cap is proposed in the 12Xe PI BOM since GT DC_LL is 0, and 12Xe has a bigger graphics die to meet the overshoot requirements.
3. If common motherboard approach is planned for 8C/H16C 4Xe , it is recommended to choose any of the preferred BOMs based on cost/area trade-offs.
4. It is recommended to ignore the placeholder caps. The PDG will be updated in the next release.
5. For PI models, refer to the Nova Lake H Power Integrity Model ([# 866209](#)).

Some of the power rails specify the maximum inductance/resistance to meet the design target. "Inductance Calculation Method Technical Advisory" illustrates how the inductance value specified in the design guide is calculated.

Table 758. Reference Documents

Document Title	Document Number
Nova Lake H Power Integrity Model	866209
Power Integrity General Guidelines Technical Advisory	727656
Inductance Calculation Method Technical Advisory	642578
Routing Near Switch Mode Power Supply (SMPS) Noise Sources Technical White Paper	726825
Acoustic Noise Mitigation Guidance Technical Advisory	575216

11.1 Processor Decoupling and Layout Recommendations for NVL 8C/ H16C

11.1.1 Processor Power Rails

11.1.1.1 Type-3, 0.8mm, 8 Layer

PCB Type: Type-3

PCB Thickness: 0.8mm

PCB Layer Count: 8

PCB Assembly: Double Sided

11.1.1.1.1 VCC_CORE(8C/T3/0.8mm PCB)

Table 759. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_CORE_8C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 89 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_CORE_8C_T3_0.8mm_4

Figure 461. VCC_CORE_8C_T3_0.8mm_1

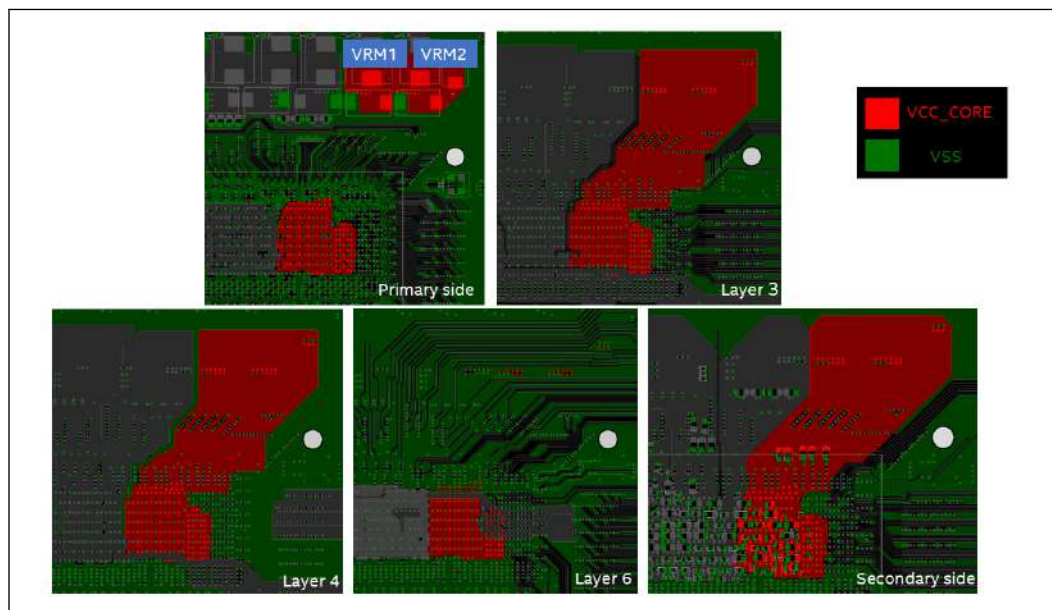


Figure 462. VCC_CORE_8C_T3_0.8mm_4

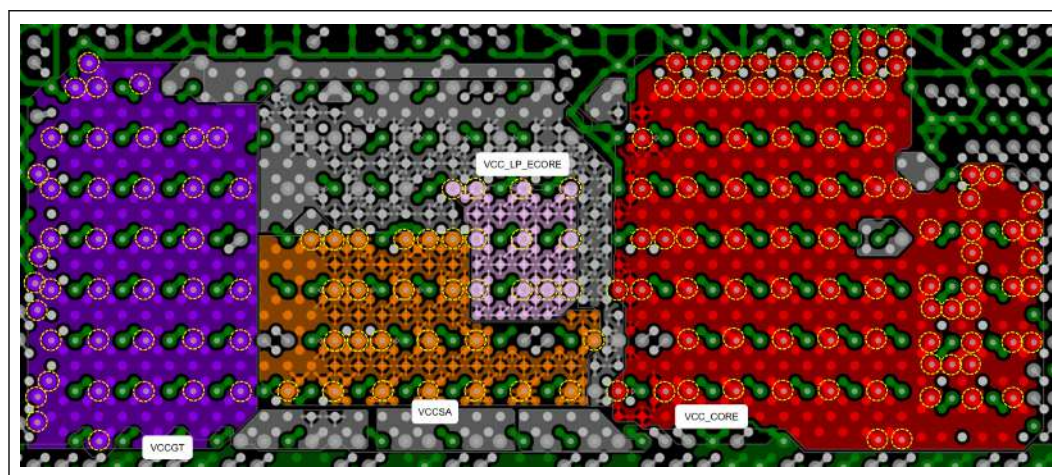


Table 760. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
<i>continued...</i>

Note
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 761. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_CORE_8C_T3_0.8mm_2
Primary side	7343	Placeholder	1		VCC_CORE_8C_T3_0.8mm_2
Secondary side	0201	1 uF	12	Place beneath BGA.	VCC_CORE_8C_T3_0.8mm_3
Secondary side	0402	10 uF	10	Place beneath BGA.	VCC_CORE_8C_T3_0.8mm_3
Secondary side	0603	22 uF	4	Place beneath BGA.	VCC_CORE_8C_T3_0.8mm_3
Secondary side	0603	Placeholder	8		VCC_CORE_8C_T3_0.8mm_3

Figure 463. VCC_CORE_8C_T3_0.8mm_2

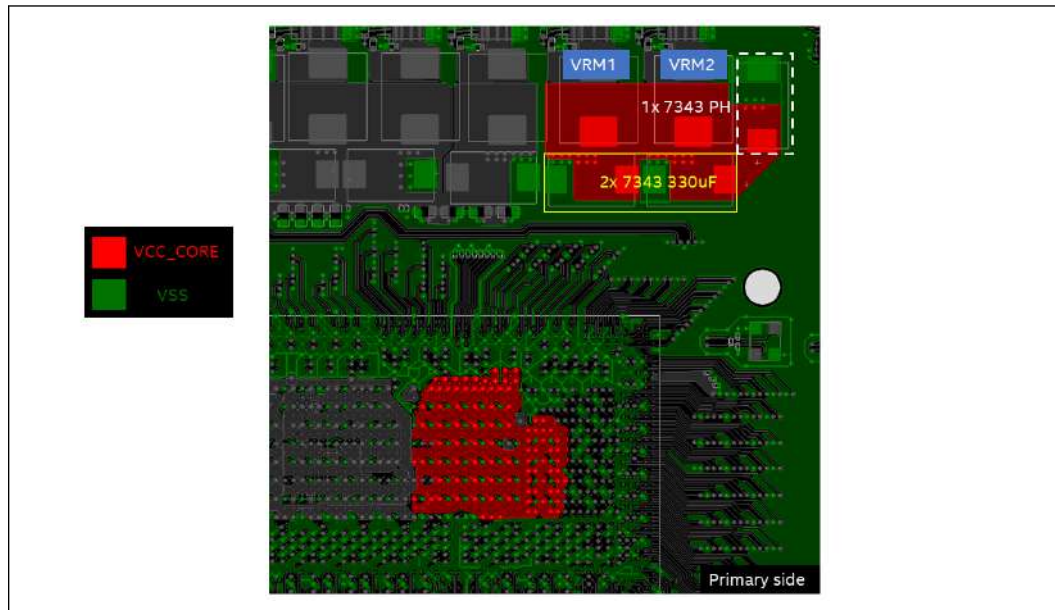


Figure 464. VCC_CORE_8C_T3_0.8mm_3



11.1.1.1.2 VCCGT(8C/T3/0.8mm PCB)

Table 762. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCGT_8C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 40 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCGT_8C_T3_0.8mm_4

Figure 465. VCCGT_8C_T3_0.8mm_1

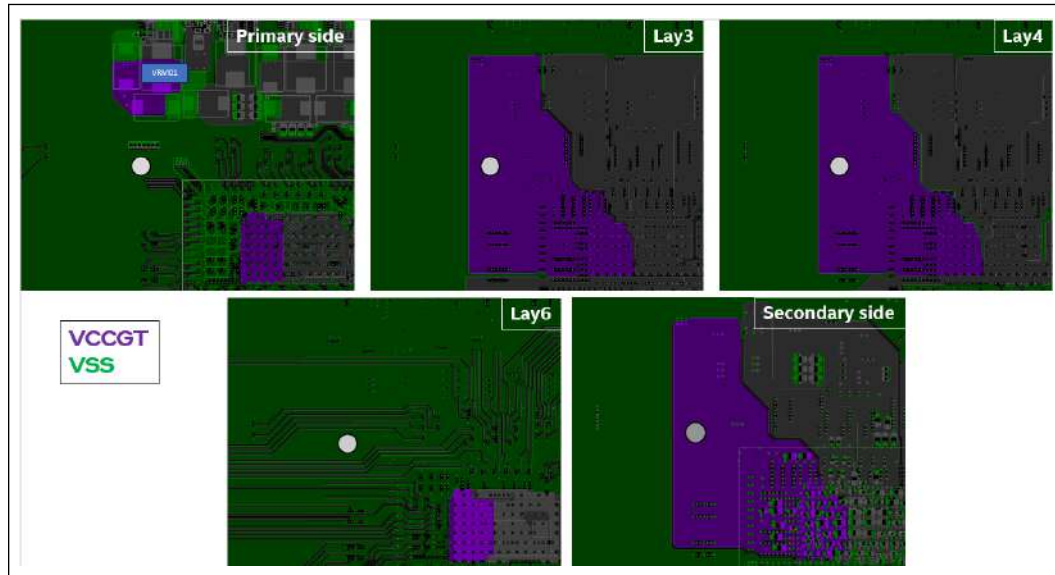


Figure 466. VCCGT_8C_T3_0.8mm_4

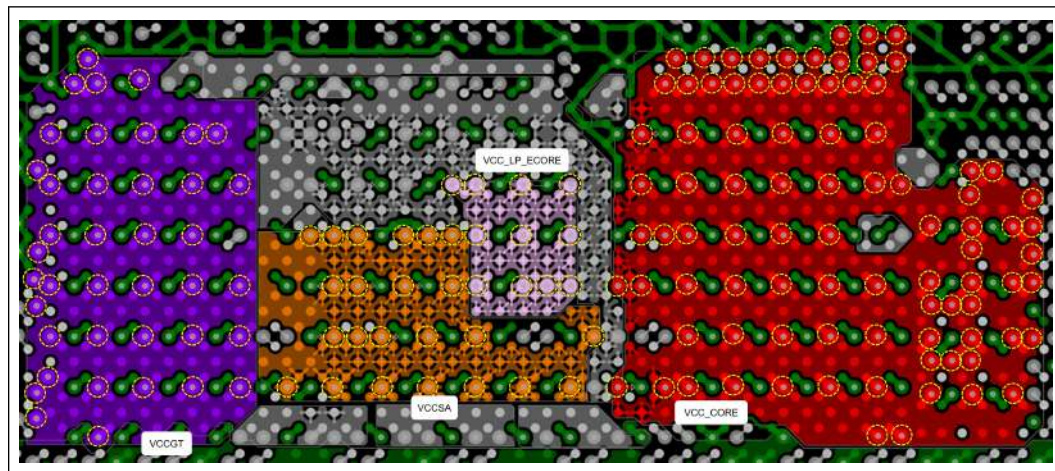


Table 763. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
<i>continued...</i>

Note
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 764. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCGT_8C_T3_0.8mm_2
Primary side	7343	Placeholder	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCGT_8C_T3_0.8mm_2
Secondary side	0201	1uF	2		VCCGT_8C_T3_0.8mm_3
Secondary side	0402	10 uF	15		VCCGT_8C_T3_0.8mm_3
Secondary side	0603	22uF	9		VCCGT_8C_T3_0.8mm_3
Secondary side	0201	Placeholder	3		VCCGT_8C_T3_0.8mm_3
Secondary side	0402	Placeholder	3		VCCGT_8C_T3_0.8mm_3
Secondary side	0603	Placeholder	3		VCCGT_8C_T3_0.8mm_3

Figure 467. VCCGT_8C_T3_0.8mm_2

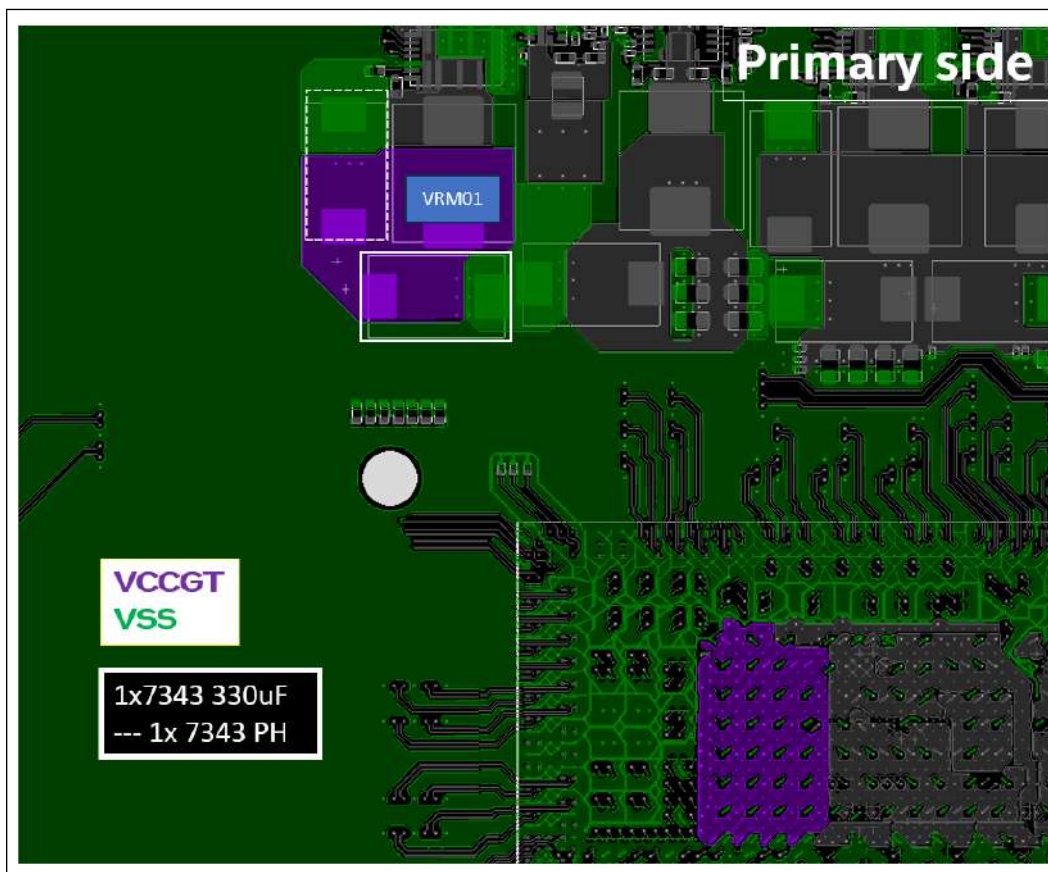
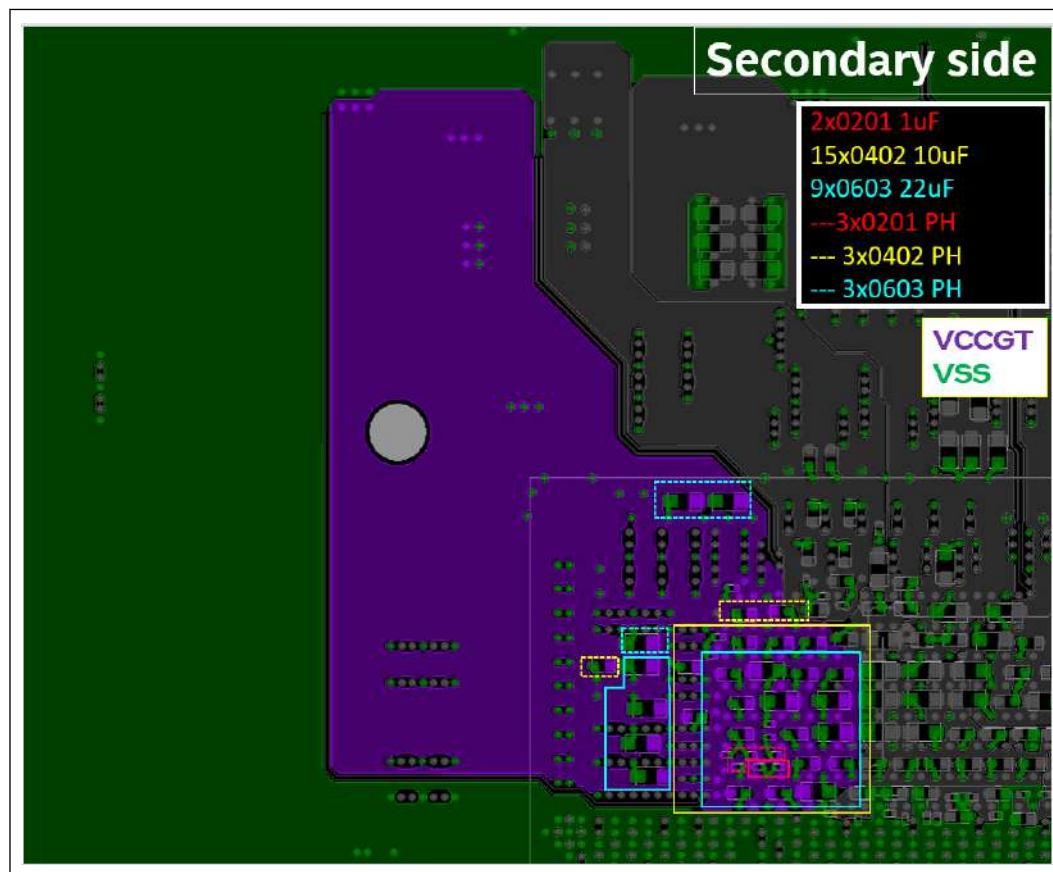


Figure 468. VCCGT_8C_T3_0.8mm_3



11.1.1.1.3 VCCSA(8C/T3/0.8mm PCB)

Table 765. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCSA_8C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
	Populate minimum 23 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCSA_8C_T3_0.8mm_4

Figure 469. VCCSA_8C_T3_0.8mm_1

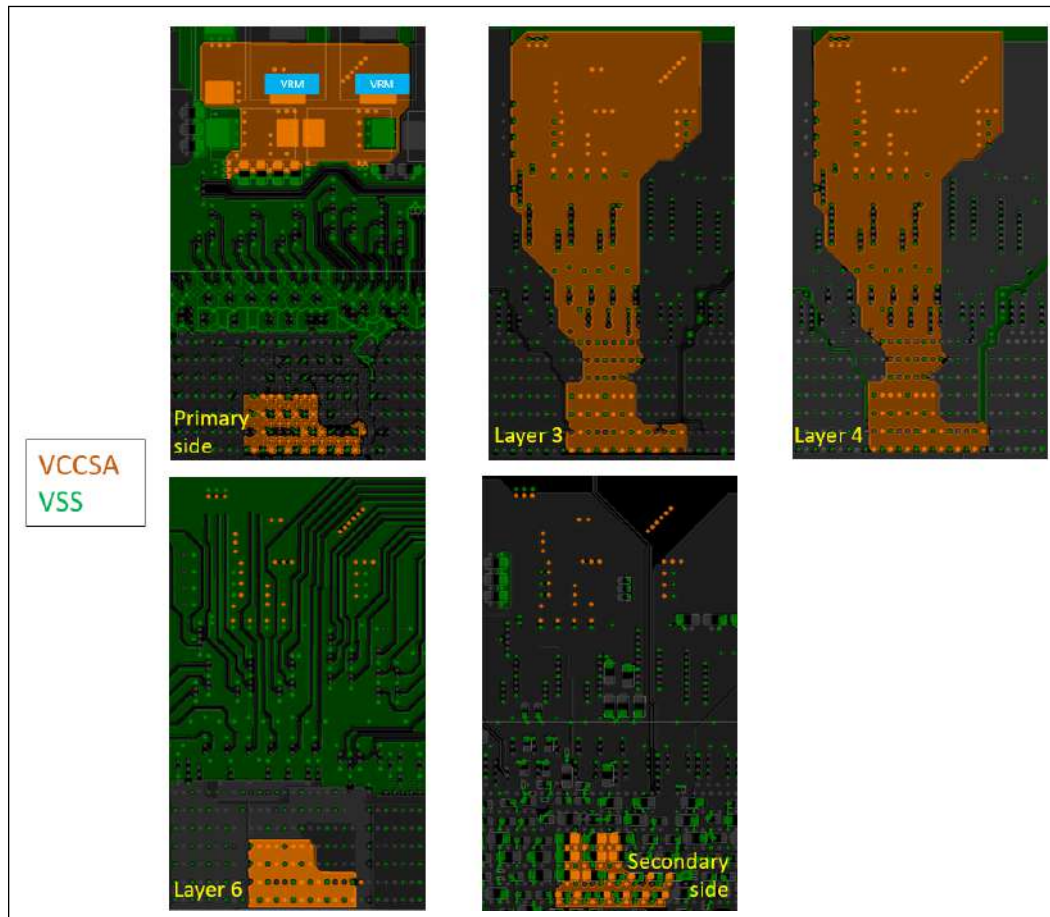


Figure 470. VCCSA_8C_T3_0.8mm_4

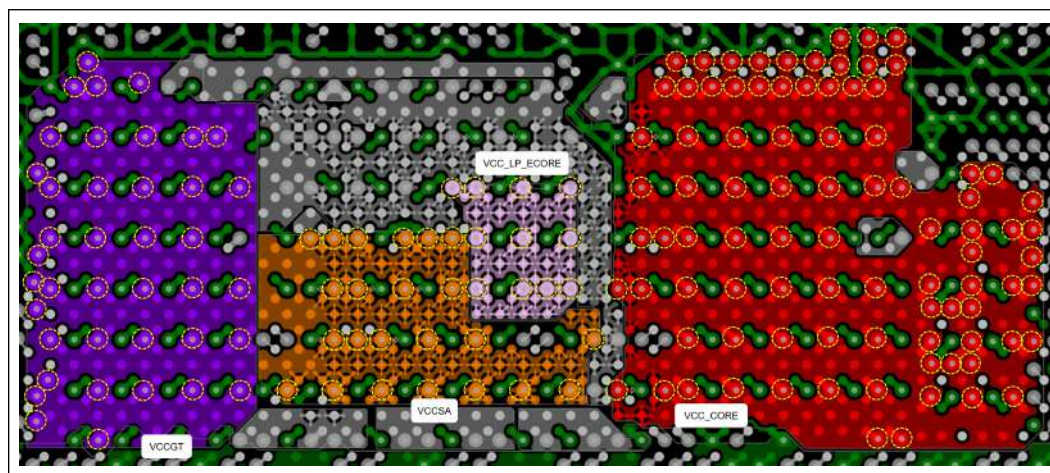


Table 766. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 767. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	0603	22uF	2		VCCSA_8C_T3_0.8mm_2
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCSA_8C_T3_0.8mm_2
Primary side	0603	Placeholder	2		VCCSA_8C_T3_0.8mm_2
Primary side	7343	Placeholder	1		VCCSA_8C_T3_0.8mm_2
Secondary side	0402	10 uF	10	Place beneath BGA.	VCCSA_8C_T3_0.8mm_3
Secondary side	0603	22 uF	8	Place beneath BGA.	VCCSA_8C_T3_0.8mm_3

Figure 471. VCCSA_8C_T3_0.8mm_2

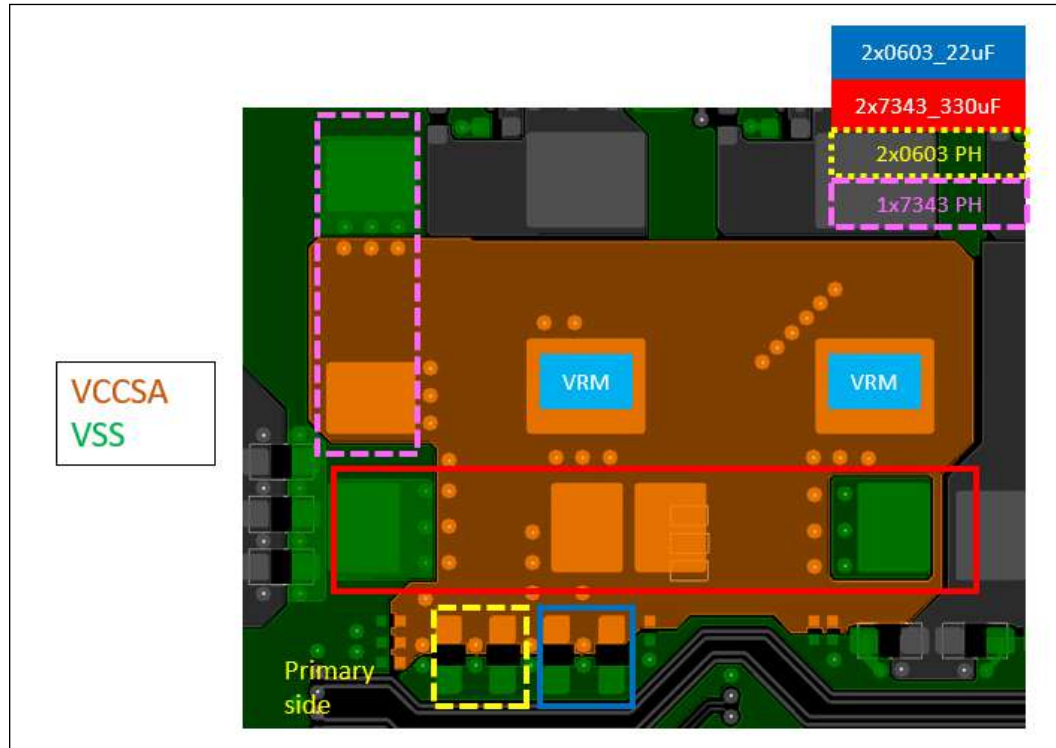
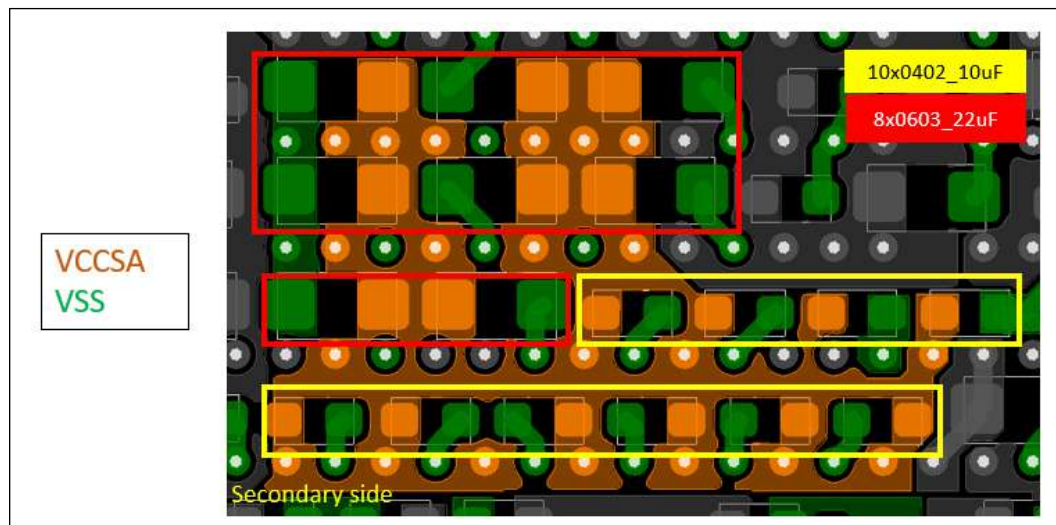


Figure 472. VCCSA_8C_T3_0.8mm_3



11.1.1.1.4 VCC_LP_ECORE(8C/T3/0.8mm PCB)

Table 768. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_LP_ECORE_8C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 11 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_LP_ECORE_8C_T3_0.8mm_4

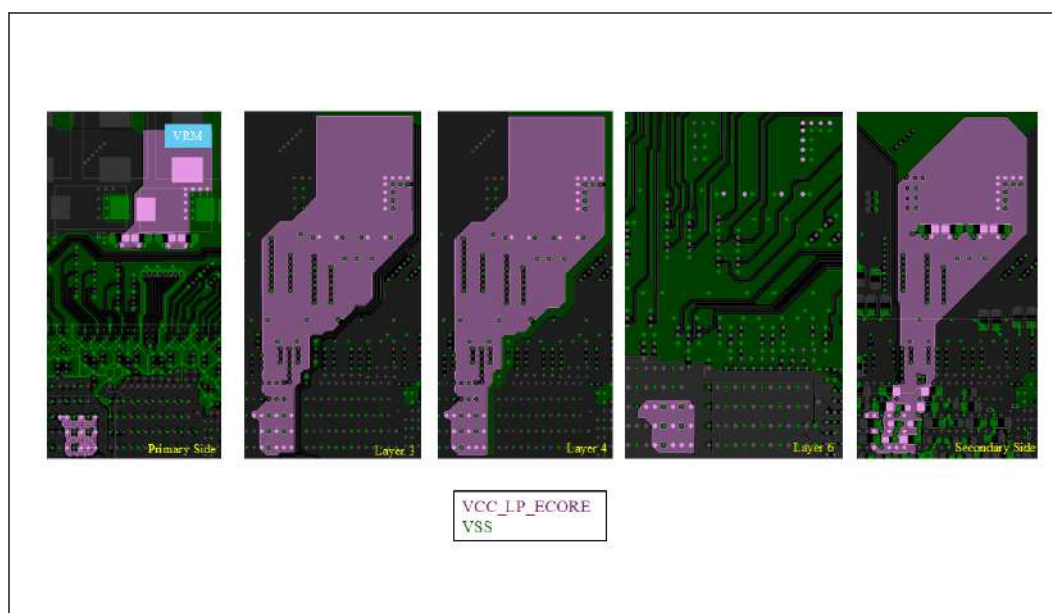
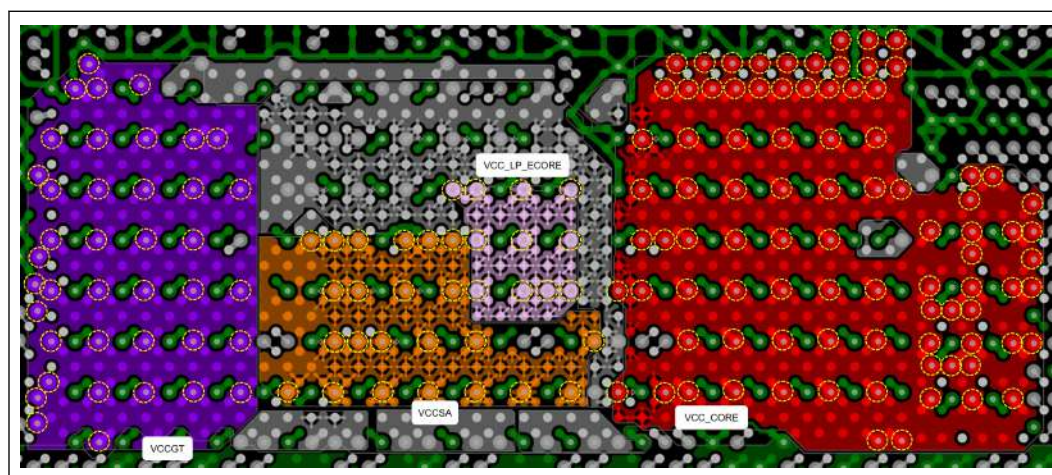
Figure 473. VCC_LP_ECORE_8C_T3_0.8mm_1**Figure 474. VCC_LP_ECORE_8C_T3_0.8mm_4**

Table 769. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0402 ->0603 ->Bulk -> VR.
2. For bottom layer capacitor placement, place 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 770. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_LP_ECORE_8C_T3_0.8mm_2
Primary side	0603	Placeholder	4		VCC_LP_ECORE_8C_T3_0.8mm_2
Secondary side	0402	10uF	4	Place beneath BGA	VCC_LP_ECORE_8C_T3_0.8mm_3
Secondary side	0603	22uF	7	Place beneath BGA	VCC_LP_ECORE_8C_T3_0.8mm_3
Secondary side	0603	Placeholder	2	Place near package edge	VCC_LP_ECORE_8C_T3_0.8mm_3

Figure 475. VCC_LP_ECORE_8C_T3_0.8mm_2

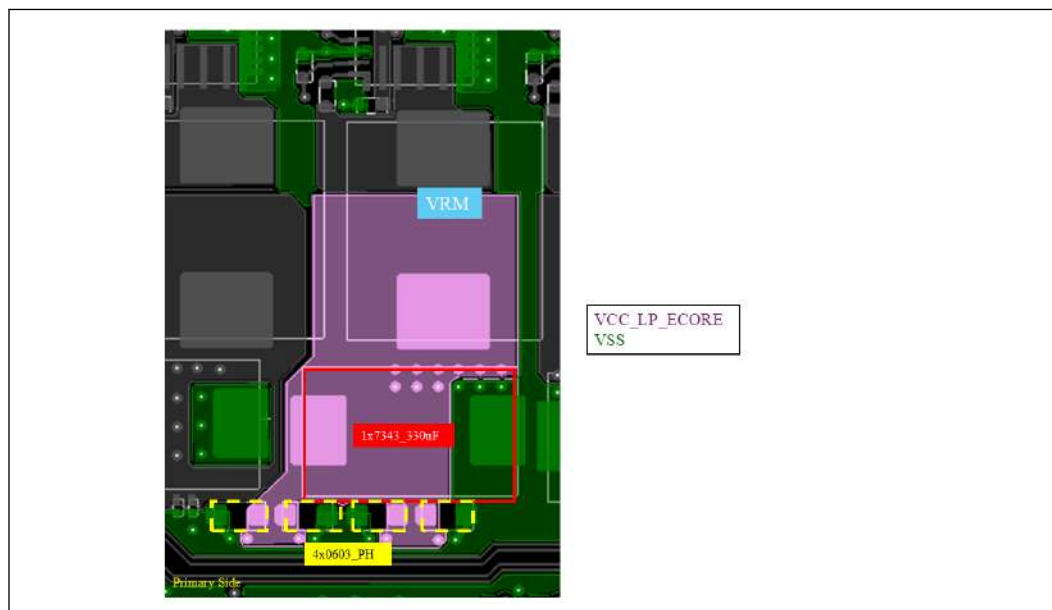
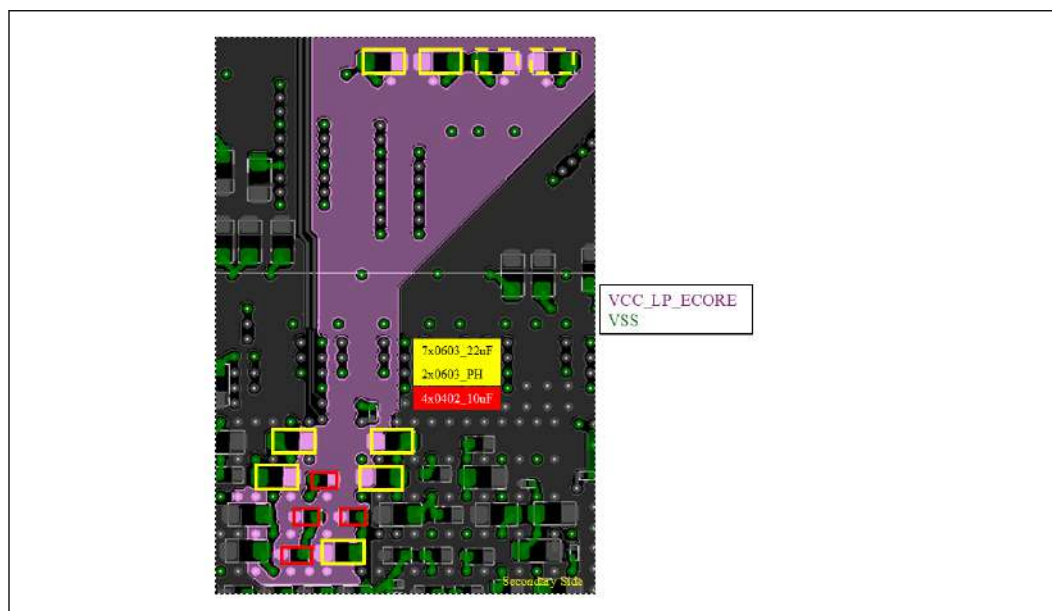


Figure 476. VCC_LP_ECORE_8C_T3_0.8mm_3



11.1.1.1.5 VCCPRIM_VNNAON (T3 PCB)

Table 771. Reference Layout Design

Index	Design Note	Reference Design
1	To meet Rpath of <3.5mΩ from VRM to VCCPRIM_VNNAON BGA groups, route VCCPRIM_VNNAON plane on at least three 30um-thick copper layers from VRM output to processor BGA with the following conditions : ► When routing on the top or bottom layer, keep plane width at least 15mm at VRM region & not lesser than 7mm upon approaching BGA region ► When routing on buildup layers (minimum 2 layers), keep plane width at least 10mm at VRM region & not lesser than 6mm upon approaching BGA region	VCCPRIM_VNNAON_1
2	Short all VCCPRIM_VNNAON BGAs with plane on at least of two 30um-thick copper layers	VCCPRIM_VNNAON_2
3	VCCPRIM_VNNAON VRM uses pseudo differential sensing to compensate for voltage drops along the power delivery path. The PWR & GND sense lines are connected to BP14 & BP12 BGAs respectively	VCCPRIM_VNNAON_3
4	Short both VCCPRIM_VNNAON_FLTR BGAs (CM16 & CM18) with plane	VCCPRIM_VNNAON_4
5	Merge both VCCPRIM_VNNAON_FLTR & VCCPRIM_VNNAON power planes using 1 x 0603 1uH (120mΩ max DCR) inductor & 1 x 0603 22uF cap to form an LC-filter with the following conditions : ► Place the LC-filter <1mm distance from or underneath VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR BGAs ► Ensure the merging point for both VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR planes are <1mm distance from or underneath VCCPRIM_VNNAON_FLTR BGAs Note : Refer diagram for design implementation guidance	VCCPRIM_VNNAON_5
6	Yellow dotted circles highlights minimum count of VCCPRIM_VNNAON PTH recommended in bga region to prevent current overstress in PTHs. Ensure GND PTHs neatly placed close to PWR PTHs as shown in illustration to minimize return current loop	VCCPRIM_VNNAON_6

Figure 477. VCCPRIM_VNNAON_1

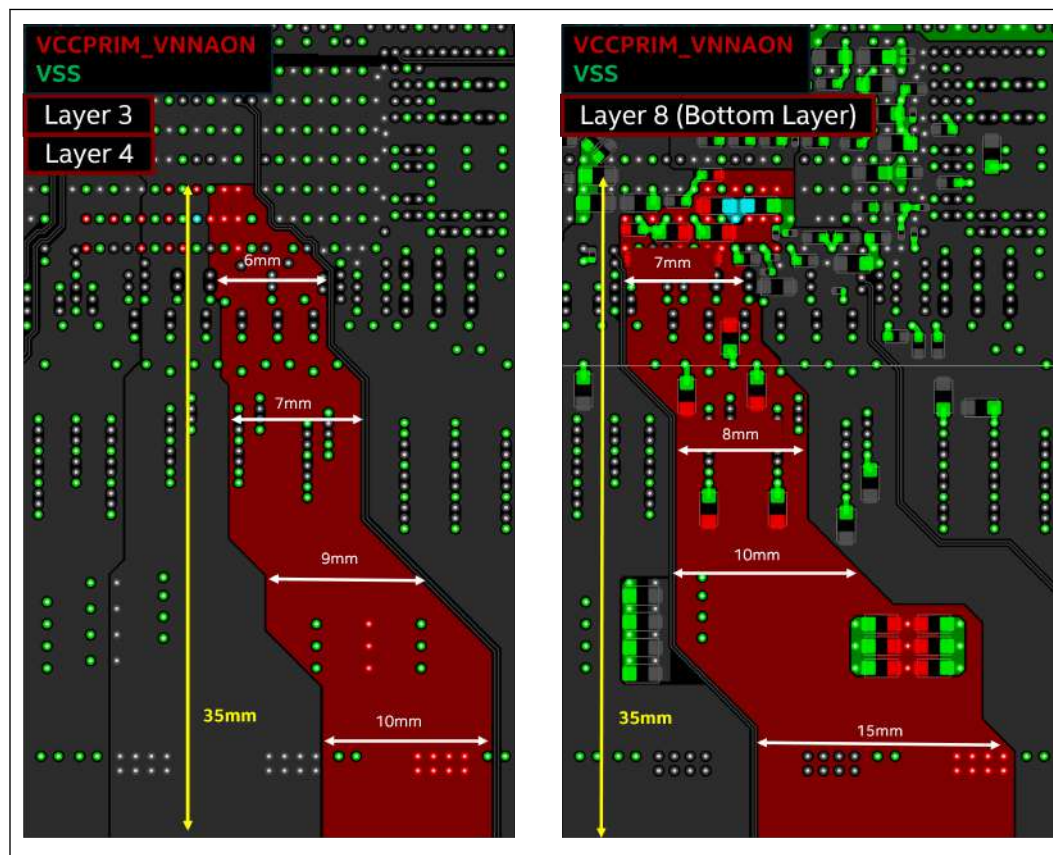


Figure 478. VCCPRIM_VNNAON_2

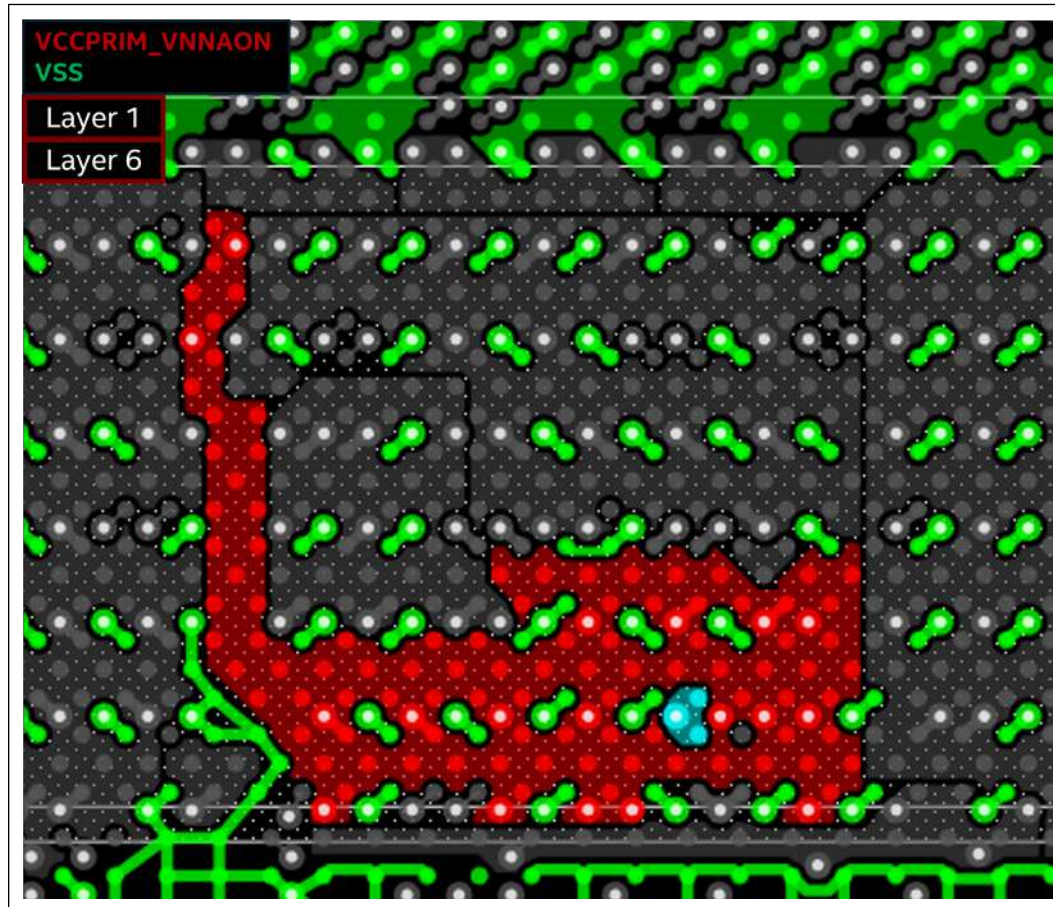


Figure 479. VCCPRIM_VNNAON_3

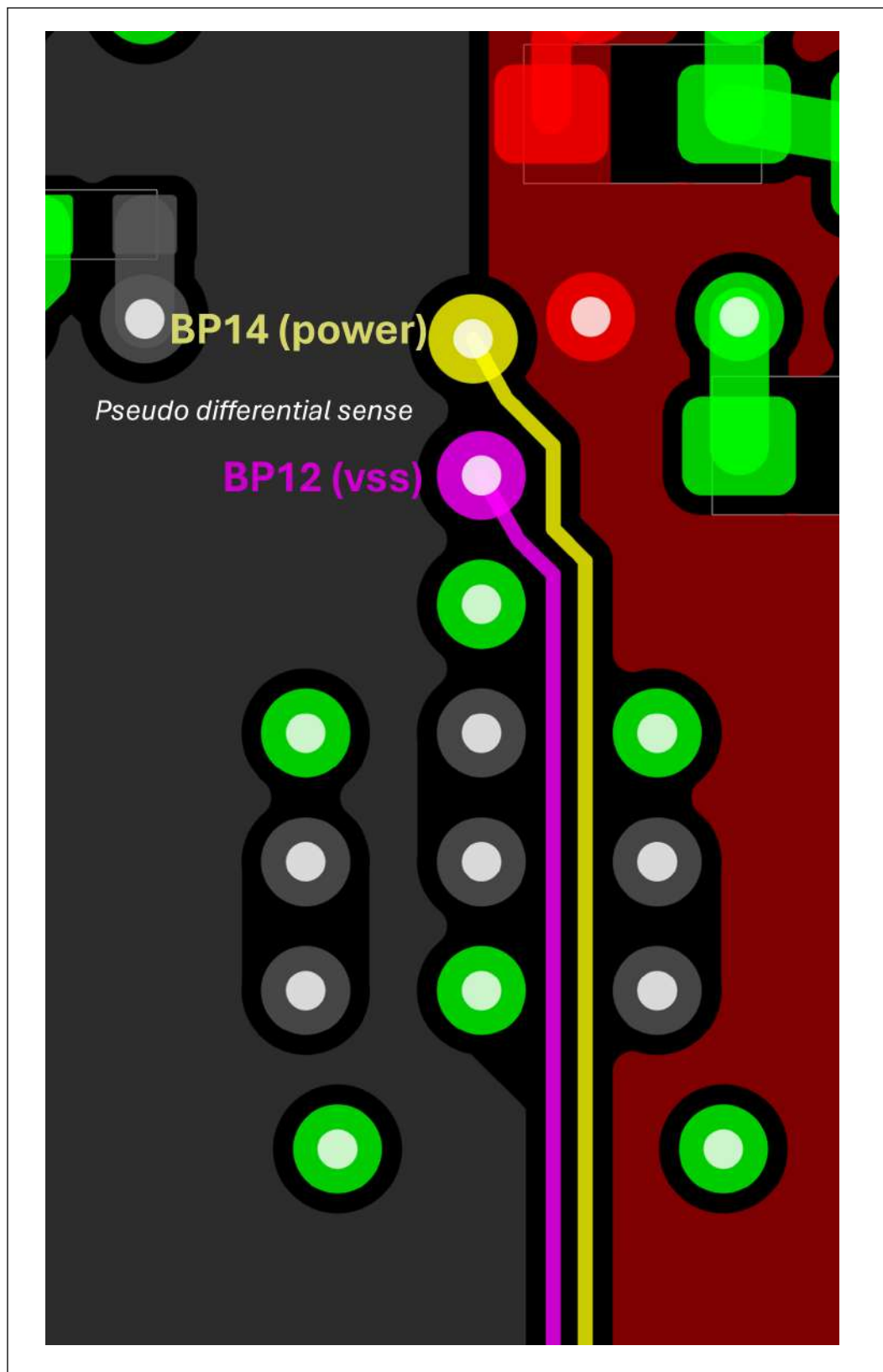


Figure 480. VCCPRIM_VNNAON_4

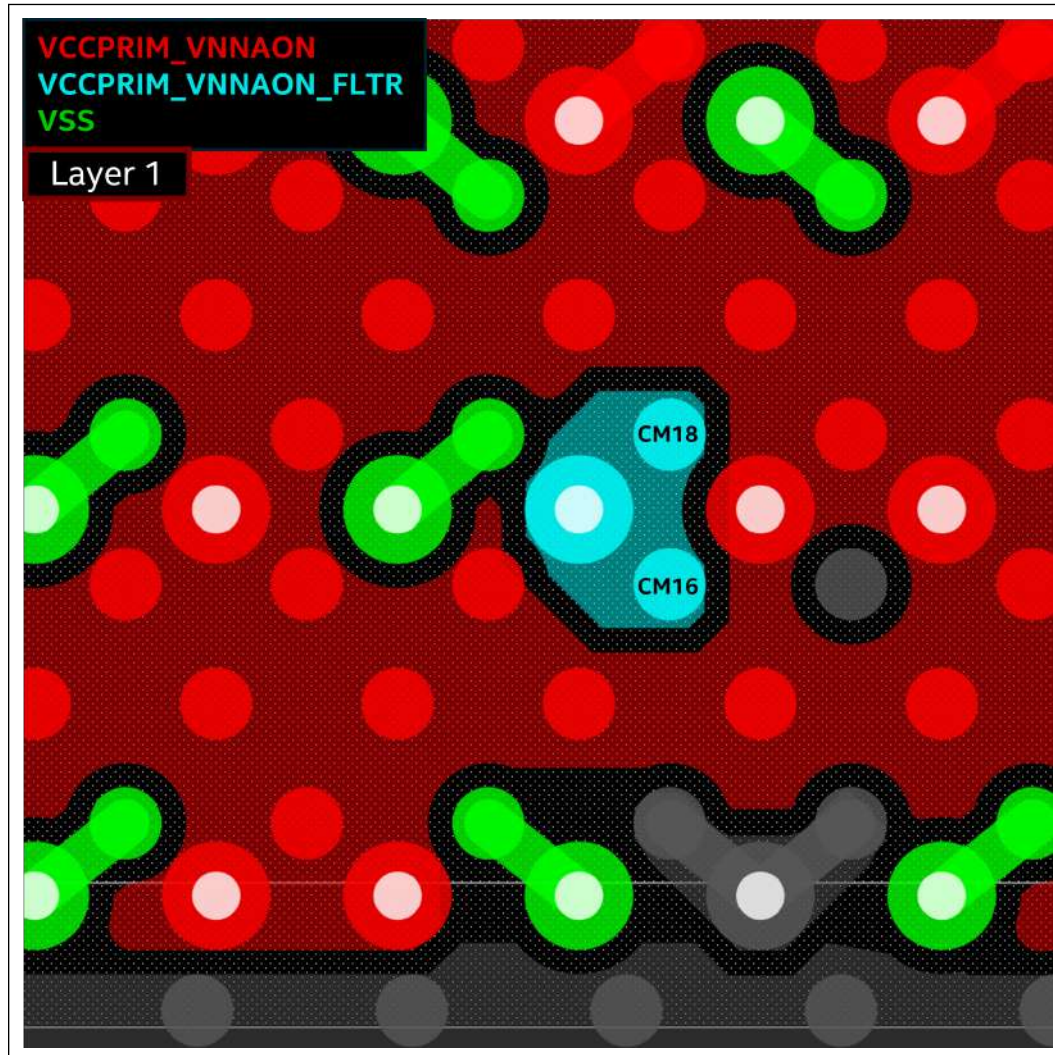


Figure 481. VCCPRIM_VNNAON_5

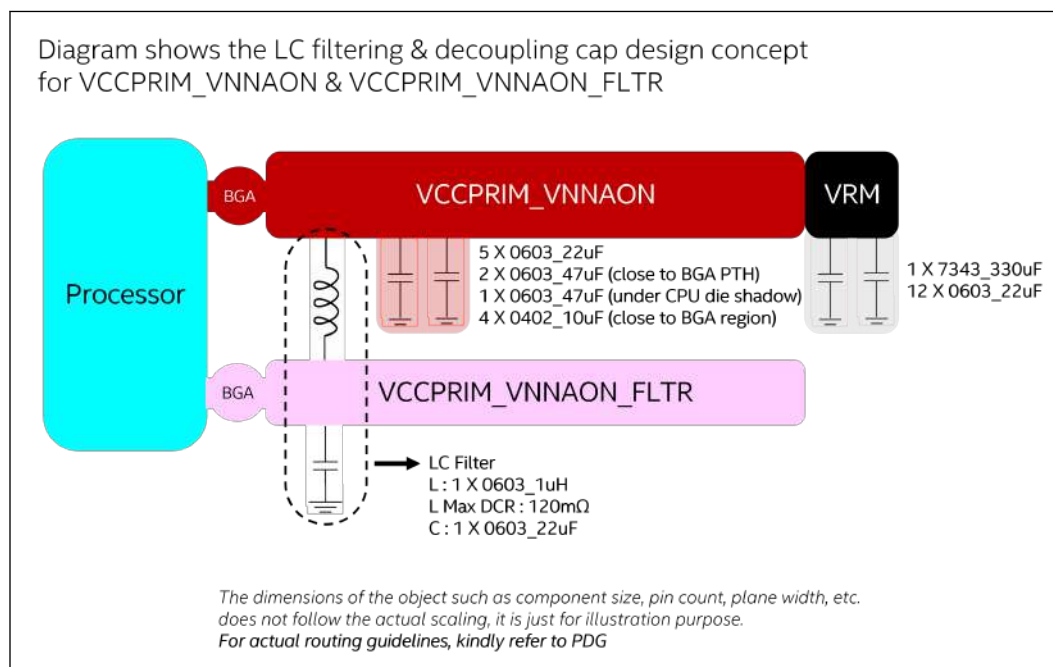


Figure 482. VCCPRIM_VNNAON_6

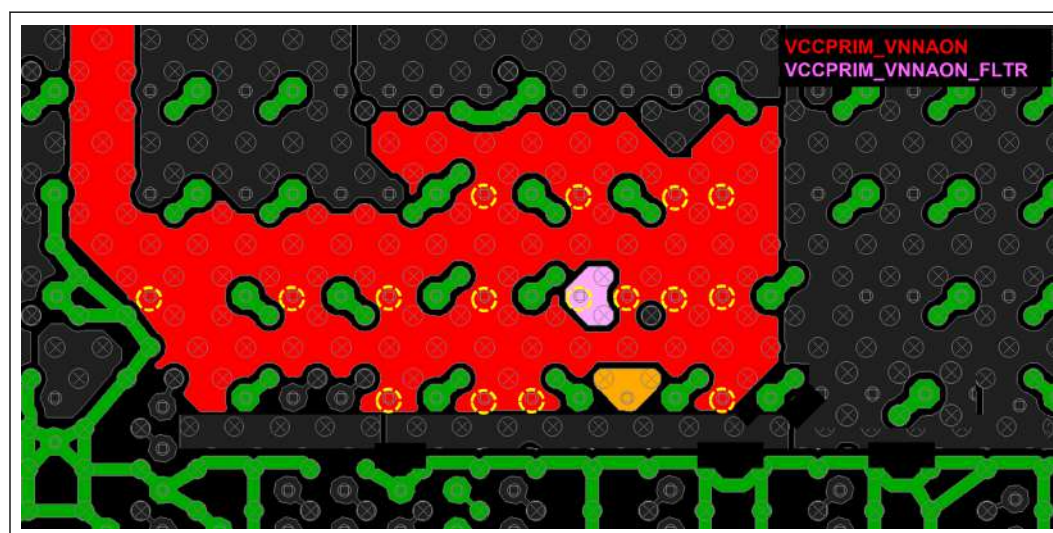


Table 772. Notes

Note
Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
Heighten capacitor effectiveness by placing them from VRM output to package in the following order : 0402 → 0603 → 0805 → Bulk
<i>continued...</i>

Note
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
When placing capacitors, align their pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor provided the caps meet the recommended parasitic capacitance.

Table 773. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	7343	330uF	1	Place bulk cap near to VRM output	VCCPRIM_VNNAON_6
Primary Side	0603	22uF	6	Place caps near to VRM output	VCCPRIM_VNNAON_6
Secondary Side	0603	22uF	6	Place caps near to VRM output	VCCPRIM_VNNAON_7
Secondary Side	0603	47uF	1	Place 1 x 0603 47uF cap under processor die shadow of VCCPRIM_VNNAON BGAs	
Secondary side	0603	47uF	2	Place 2 x 0603 47uF caps <1mm from VCCPRIM_VNNAON PTH located underneath BGA	VCCPRIM_VNNAON_8
Secondary Side	0603	22uF	5	Place 5 x 0603 22uF caps <10mm from package edge. Make sure PWR-GND PTH pair is placed very close to cap terminals	VCCPRIM_VNNAON_8
Secondary Side	0402	10uF	4	Place 4 x 0402 10uF caps underneath BGA or <1mm from package edge. Make sure PWR-GND PTH pair is placed very close to cap terminals	VCCPRIM_VNNAON_8
Secondary Side	0603	22uF	1	Pair 1 x 0603 22uF VCCPRIM_VNNAON_FLTR cap with 1 x 0603 1uH (120mΩ max DCR) inductor <1mm from or underneath VCCPRIM_VNNAON_FLTR BGAs to form an LC-filter	VCCPRIM_VNNAON_8
Secondary Side	0603	1 uH_120mΩ DCR	1	Merge both VCCPRIM_VNNAON_FLTR & VCCPRIM_VNNAON power planes using 1 x 0603 1uH (120mΩ max DCR) inductor & 1 x 0603 22uF cap to form an LC-filter with the following conditions : ► Place the LC-filter <1mm distance from or underneath VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR BGAs ► Ensure the merging point for both VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR planes are <1mm distance from or underneath VCCPRIM_VNNAON_FLTR BGAs	VCCPRIM_VNNAON_8

Figure 483. VCCPRIM_VNNAON_6

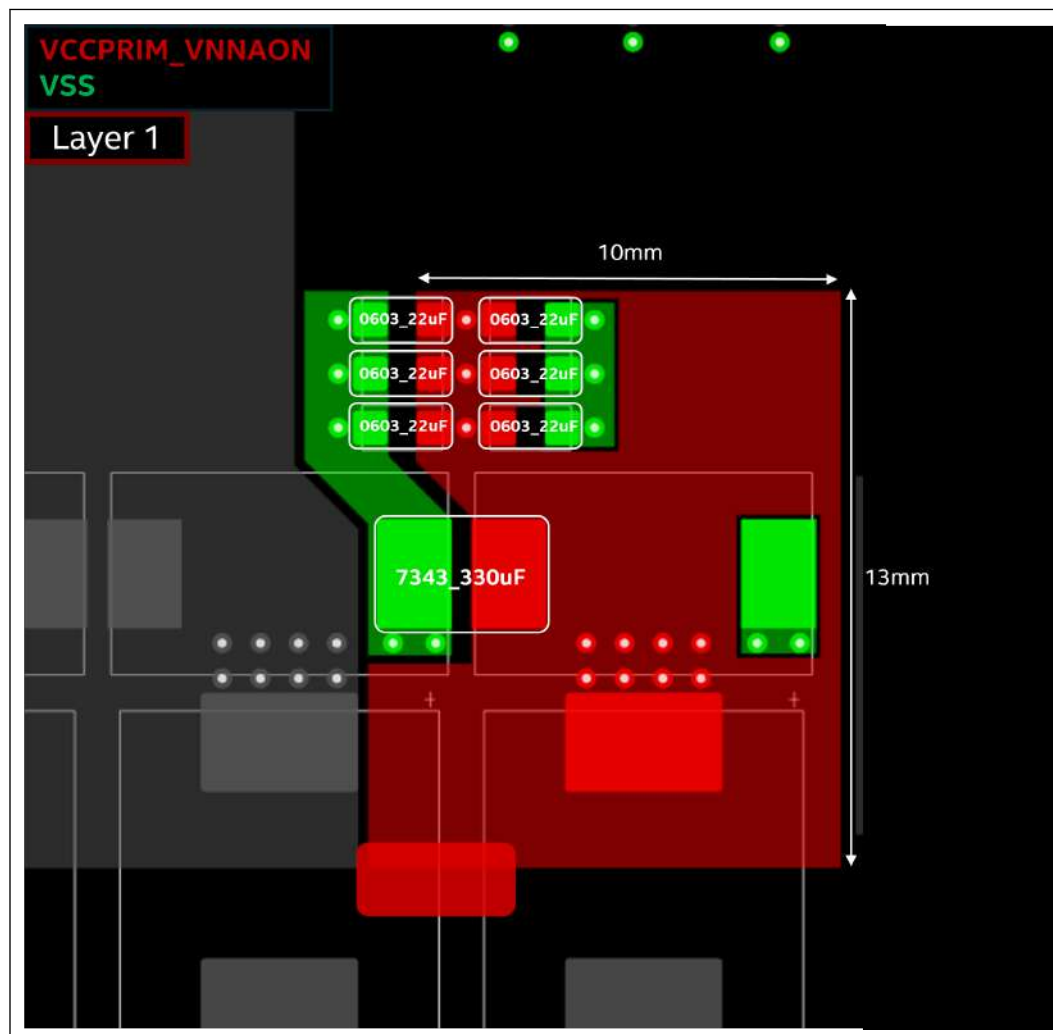


Figure 484. VCCPRIM_VNNAON_7

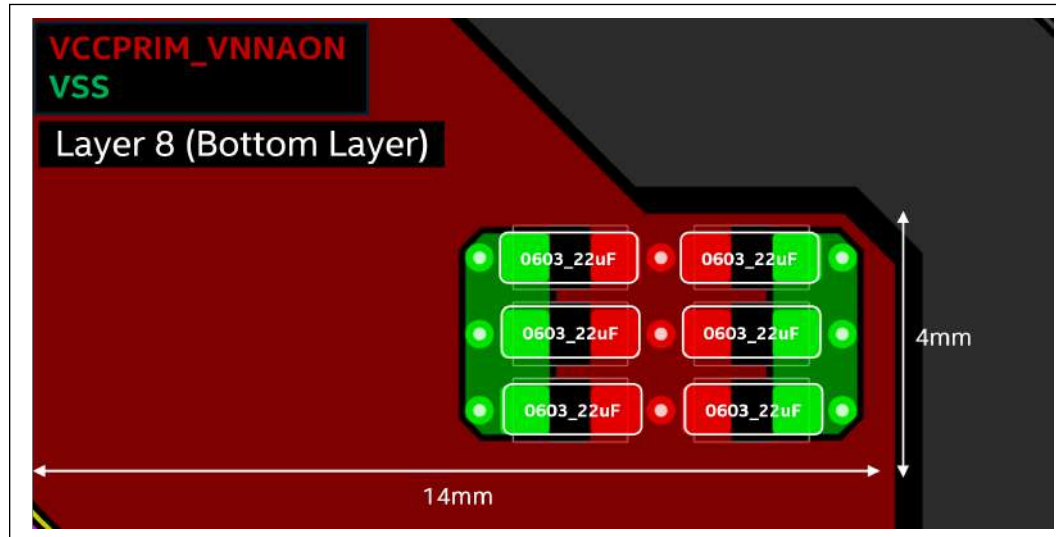
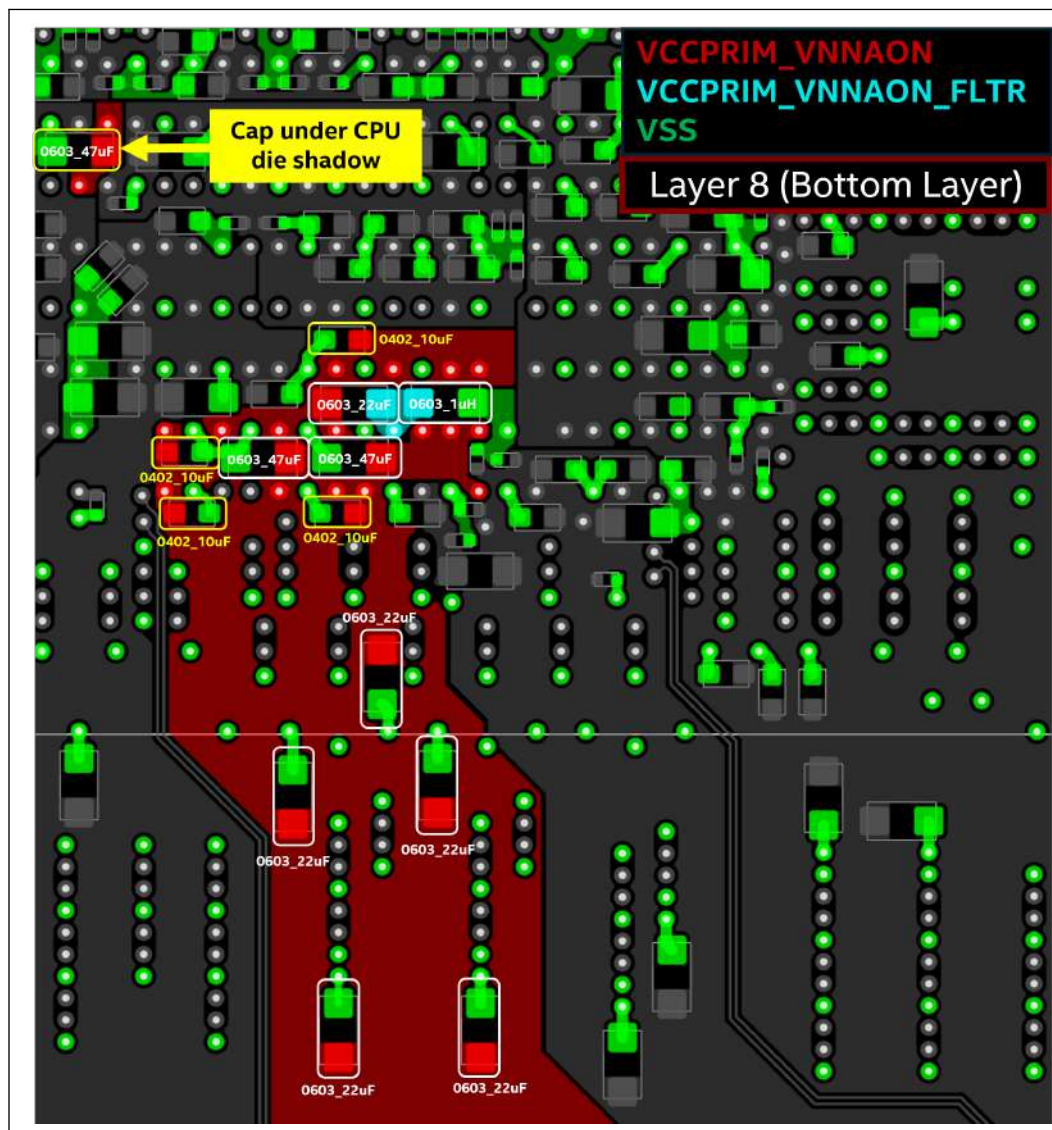


Figure 485. VCCPRIM_VNNAON_8



11.1.1.1.6 VCCPRIM_IO (T3 PCB)

Table 774. Reference Layout Design

Index	Design Note	Reference Design
1	Route VCCPRIM_IO power plane from VR output inductor to the BGA on Layer 1, Layer 6 and Layer 8. Adjacent layer from VCCPRIM_IO power plane should be ground.	VCCPRIM_IO_1
2	VCCPRIM_IO_FLTR should be merged with VCCPRIM_IO after LC filter on Layer 8	VCCPRIM_IO_1
3	VCCPRIM_IO VRM uses differential sensing. The power sense line is connected to SENSE_VCCPRIM_IO BGA(DF8) and ground sense line connected to SENSE_VSS_VCCPRIM_IO(DF9)	VCCPRIM_IO_2
4	The SVID 1.25 V supply for pull up resistors should be sourced from VCCPRIM_IO_OUT_SVID BGA EP5.	

Figure 486. VCCPRIM_IO_1

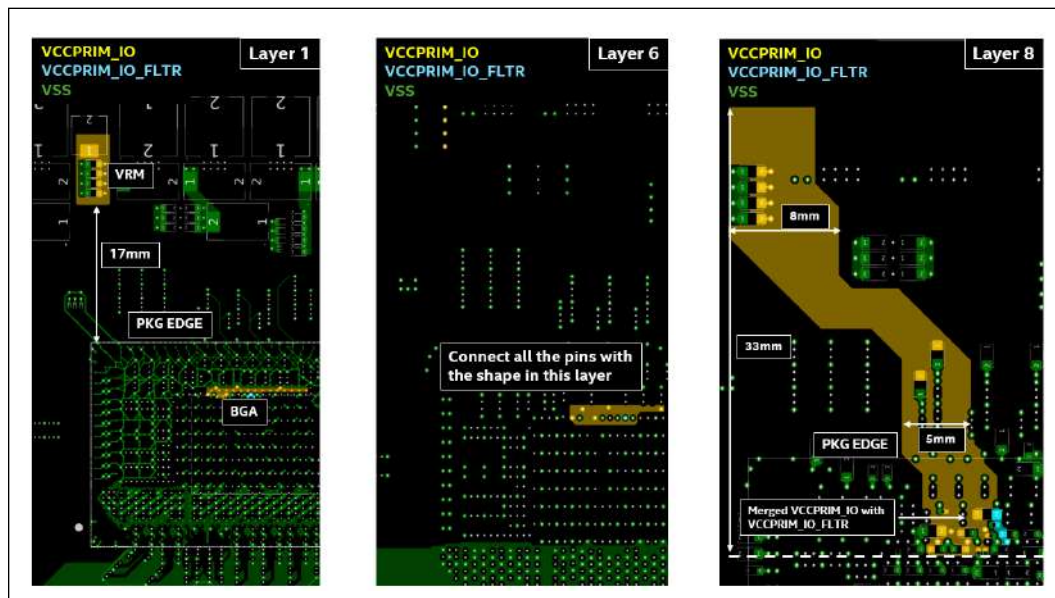


Figure 487. VCCPRIM_IO_2

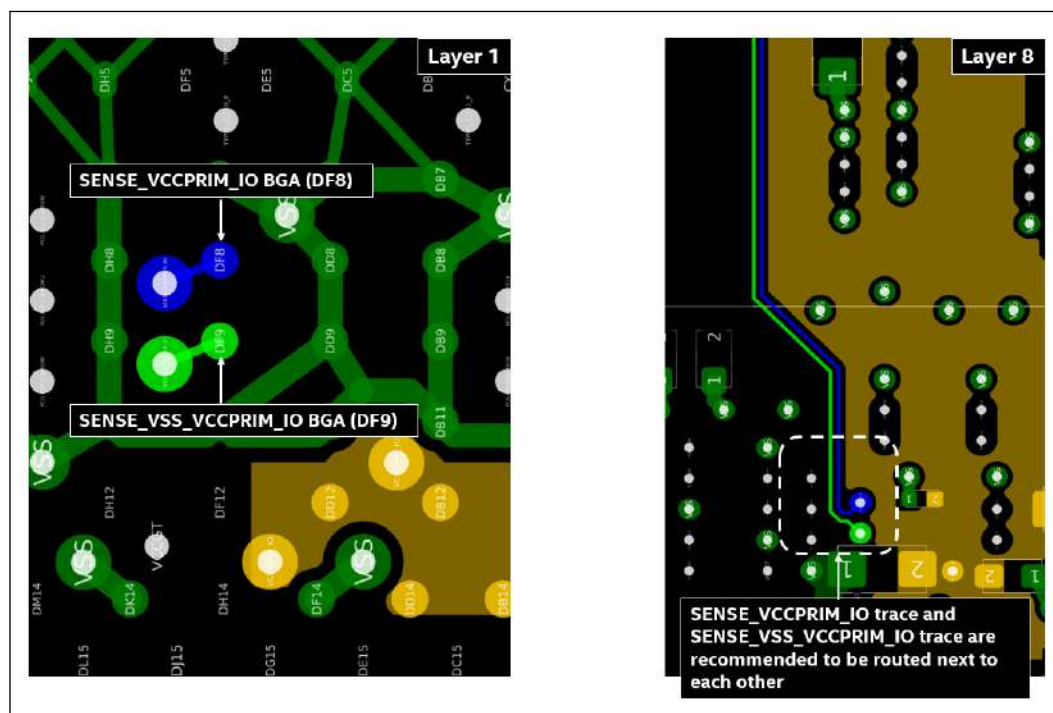


Table 775. Notes

Note
Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
Heighten capacitor effectiveness by placing them from VRM to package in the following order : 0402 → 0603 → 0805 → Bulk
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
Align capacitor pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 776. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	0805	47uF	4	Place near the VCCPRIM_IO VRM output	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0603	22 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402	10 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	4	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4

continued...

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side (Inductor)	0603	1 uH	1	Inductor component of the LC filter from VCCPRIM_IO_FLTR rail with ESR = 350mΩ	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402/0603	22 uF	1	Decoupling capacitor placed on VCCPRIM_IO_FLTR rail for LC filtering purpose	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	1	Place right below the VCCPRIM_IO_FLTR BGA	VCCPRIM_IO_3,VCCPRIM_IO_4

Figure 488. VCCPRIM_IO_3

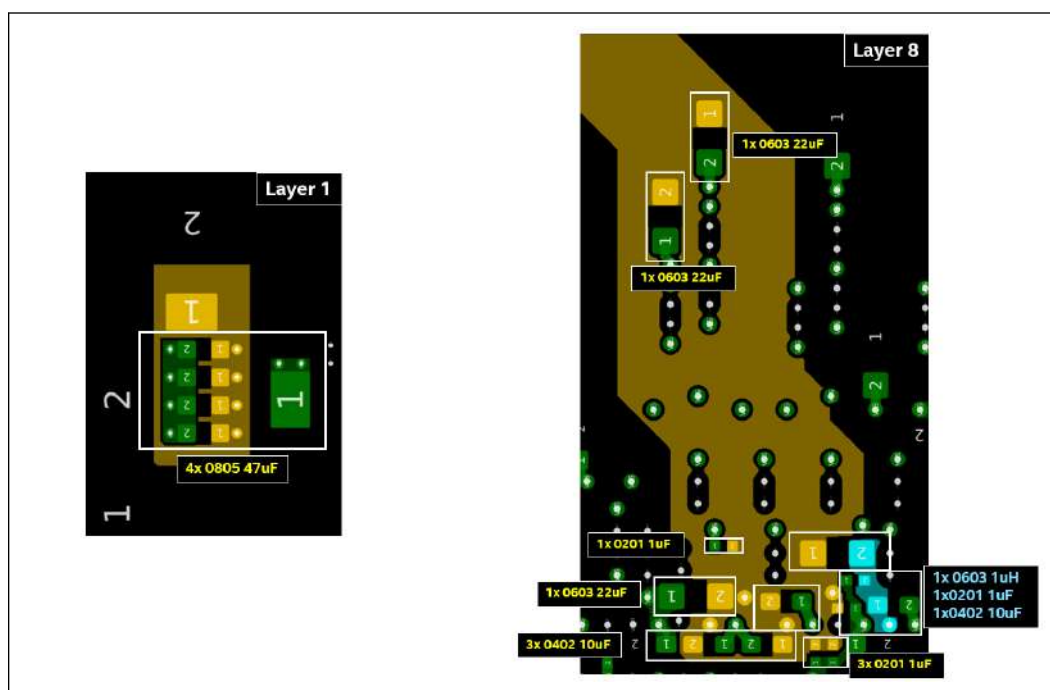
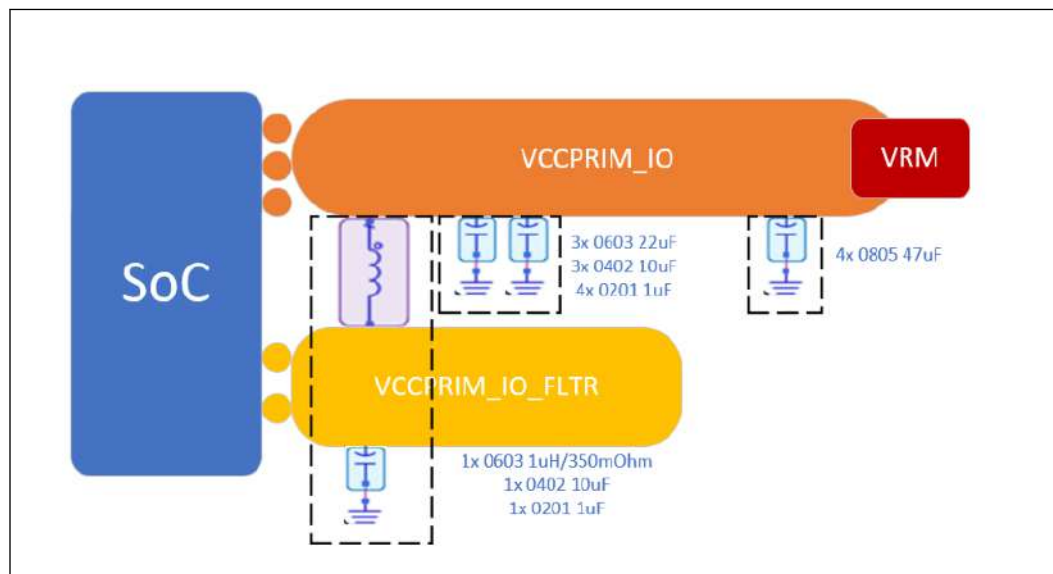


Figure 489. VCCPRIM_IO_4



11.1.1.1.7 VCCPRIM_1P8_PROC

Table 777. Reference Layout Design

Index	Design Note	Reference Design
1	Route the VCCPRIM_1P8_PROC, VCCPRIM_1P8_PROC_FLTRA and VCCPRIM_1P8_PROC_FLTRB on Layer 4 & Layer 6, with immediate ground reference. VCCPRIM_1P8_PROC_FLTRA should be merged with VCCPRIM_1P8_PROC after LC filter on Layer 1.	VCCPRIM_1P8_PROC_1
2	VCCPRIM_1p8_FLTRB should be merged with VCCPRIM_1p8 on Layer4 after ~15mm from VCCPRIM_1p8_BGAs.	VCCPRIM_1P8_PROC_2

Figure 490. VCCPRIM_1P8_PROC_1

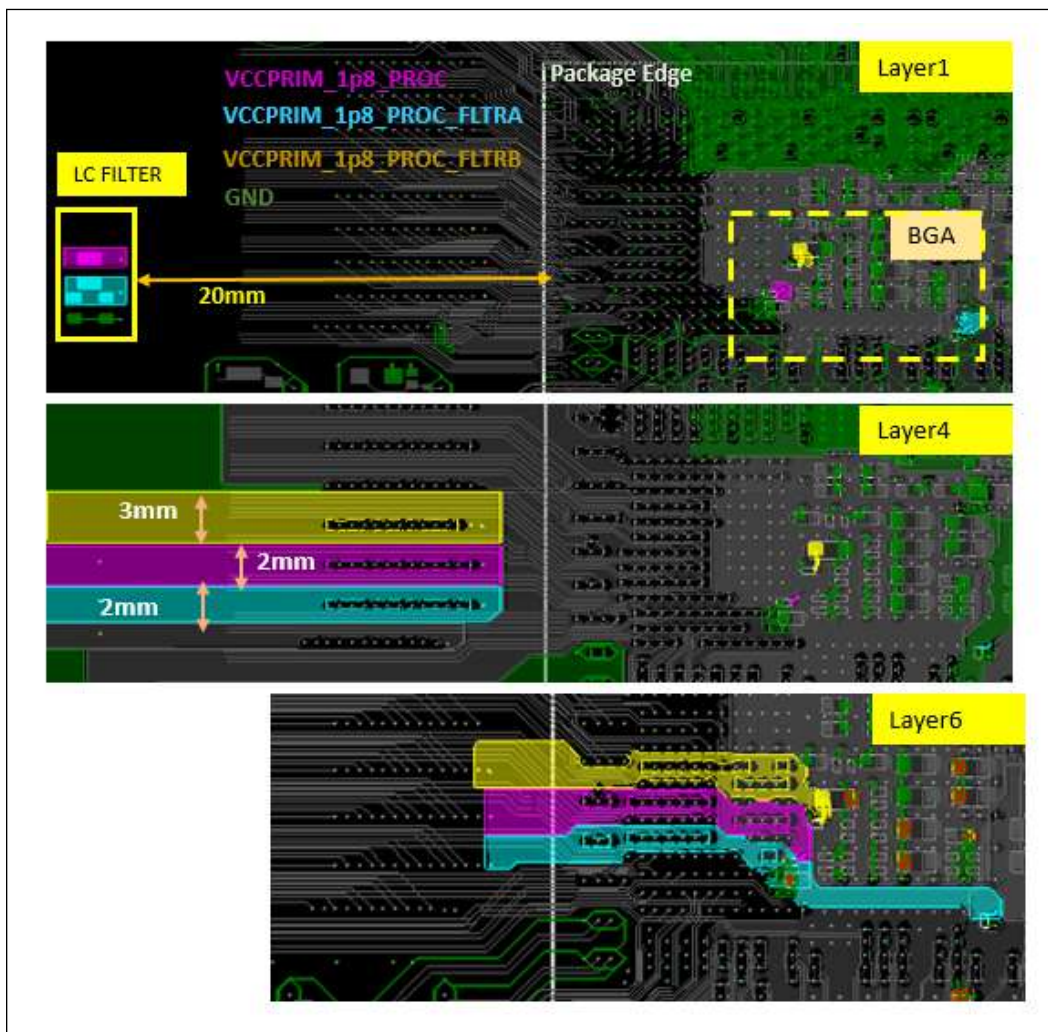


Figure 491. VCCPRIM_1P8_PROC_2



Table 778. Notes

Note
Power and ground via pairs should be placed close to capacitor pads to provide proper vertical connections when placing capacitor on the secondary side.
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 779. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side (Inductor)	0805 / 0603	1uH	1	Inductor component of the LC filter from VCCPRIM_1P8_PROC rail to VCCPRIM_1P8_PROC_FLTRA rail. It must meet electrical requirements: max DCR < 100mOhm and rated current >200 mA.	VCCPRIM_1P8_PROC_2
Primary side	0603	22uF	2	Decoupling capacitor(C1,C2) placed on VCCPRIM_1P8_PROC_FLTRA rail for LC filtering purpose.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C3 under VCCPRIM_1P8_PROC_FLTRA BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0603	Placeholder	1	Place C4 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C5 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0402 / 0201	Placeholder	1	Place C6 under VCCPRIM_1P8_PROC BGAs	VCCPRIM_1P8_PROC_2

Figure 492. VCCPRIM_1P8_PROC_2



11.1.1.1.8 VCCRTC_PROC

Table 780. Reference Layout Design

Index	Design Note	Reference Design
1	Require 1 layer of VCCRTC_PROC plane from vrm to board pin with immediate GND reference on one of the adjacent layers. Refer to reference design image for design implementation.	VCCRTC_PROC_1

Figure 493. VCCRTC_PROC_1

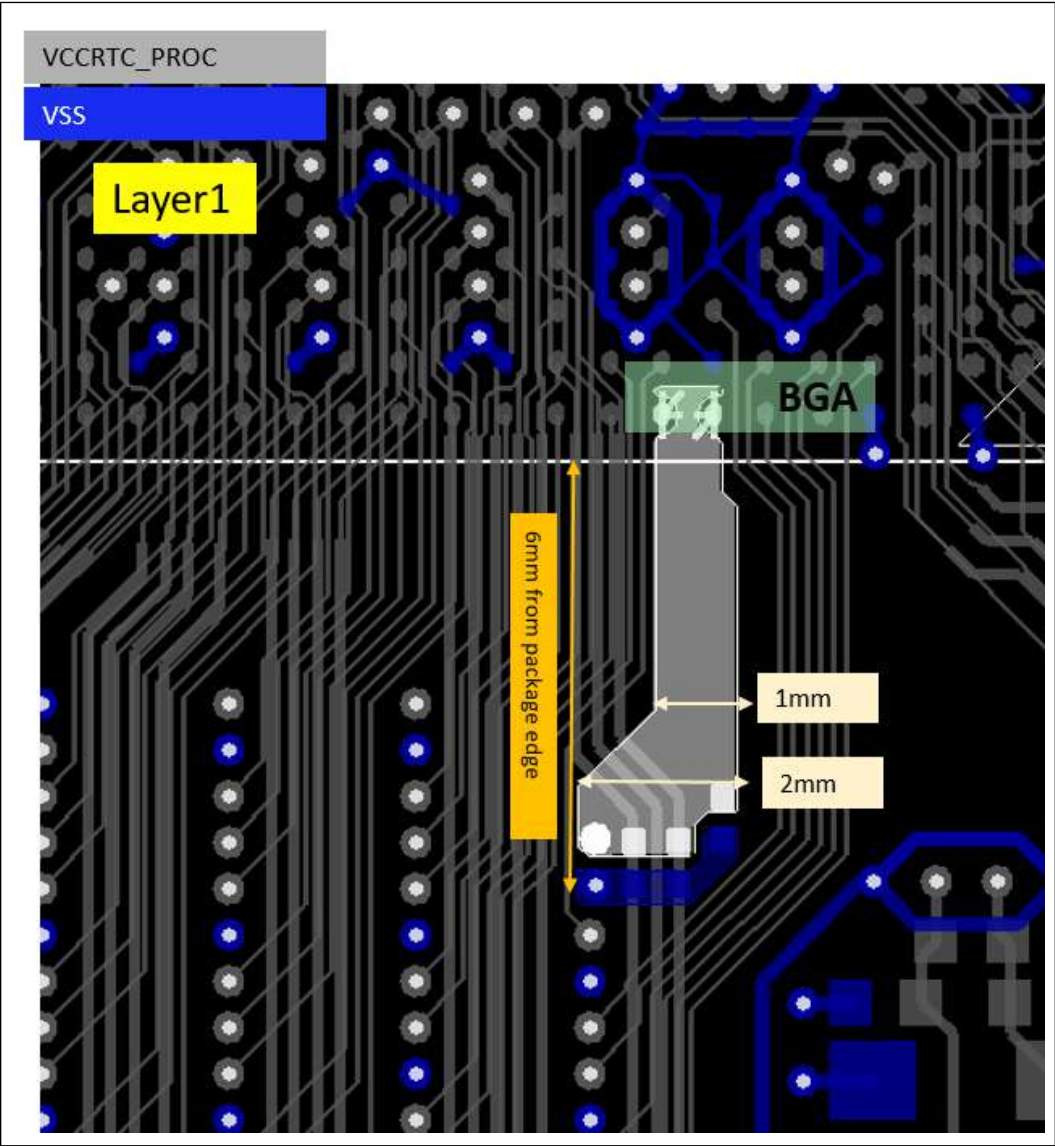


Table 781. Notes

Note
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 782. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	0201	1 uF	1	Place C1 within 6mm from package edge.	VCCRTC_PROC_2
Primary side	0402 / 0201	0.1 uF	1	Place C2 within 6mm from package edge.	VCCRTC_PROC_2

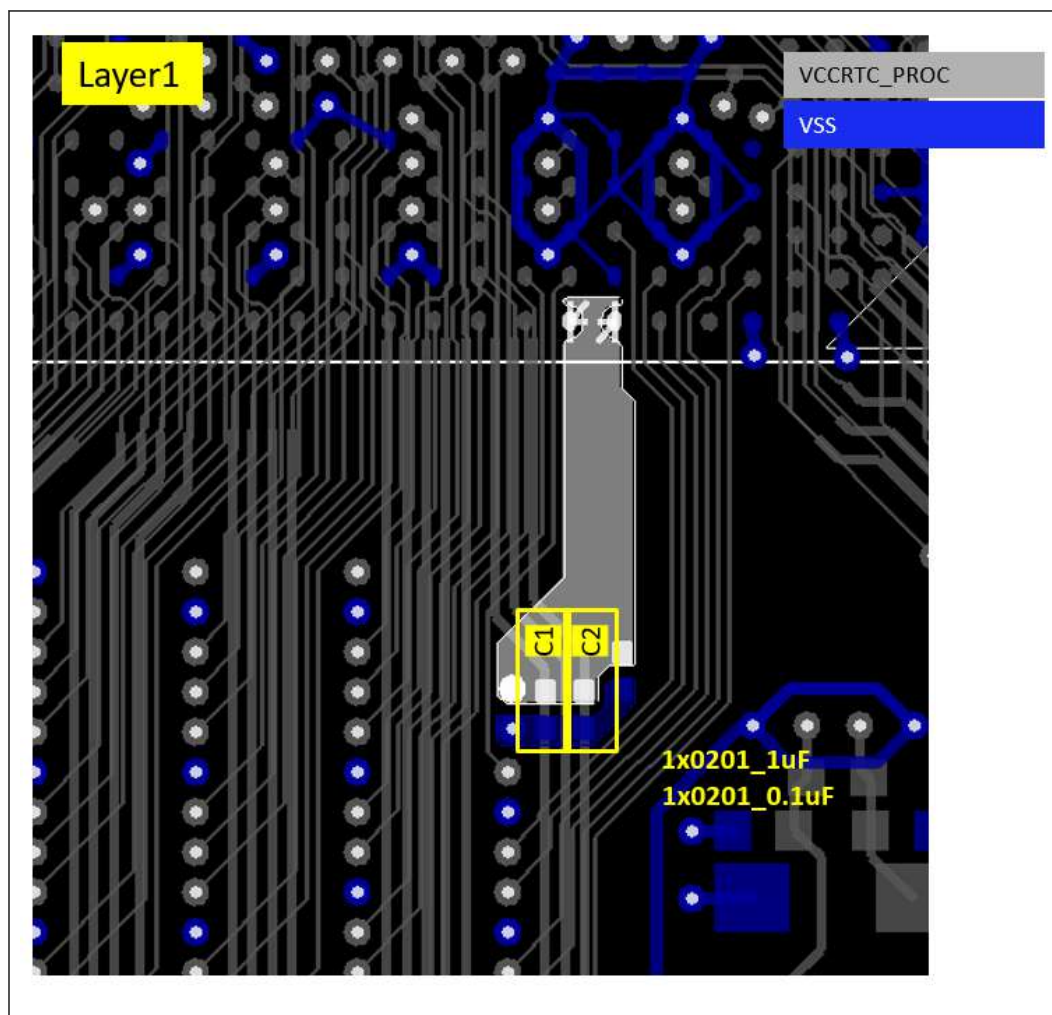
Figure 494. VCCRTC_PROC_2**11.1.1.1.9 VDD2 LPDDR5x LPCAMM2**

Table 783. Reference Layout Design

Index	Design Note	Reference Design
1	VDD2 Type-3 8L CAMM motherboard reference layout showing the plane routing	VDD2_T3_CAMM8L_1
2	Adjacent solid ground to power planes for good return path	

Figure 495. VDD2_T3_CAMM8L_1

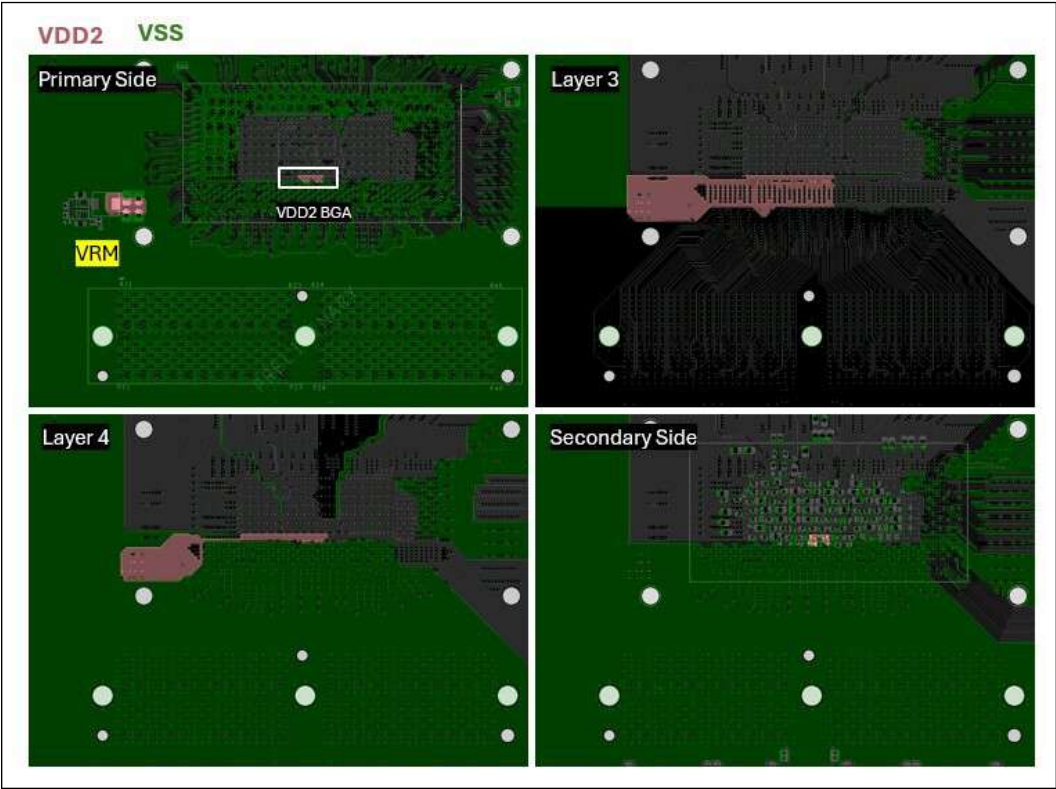


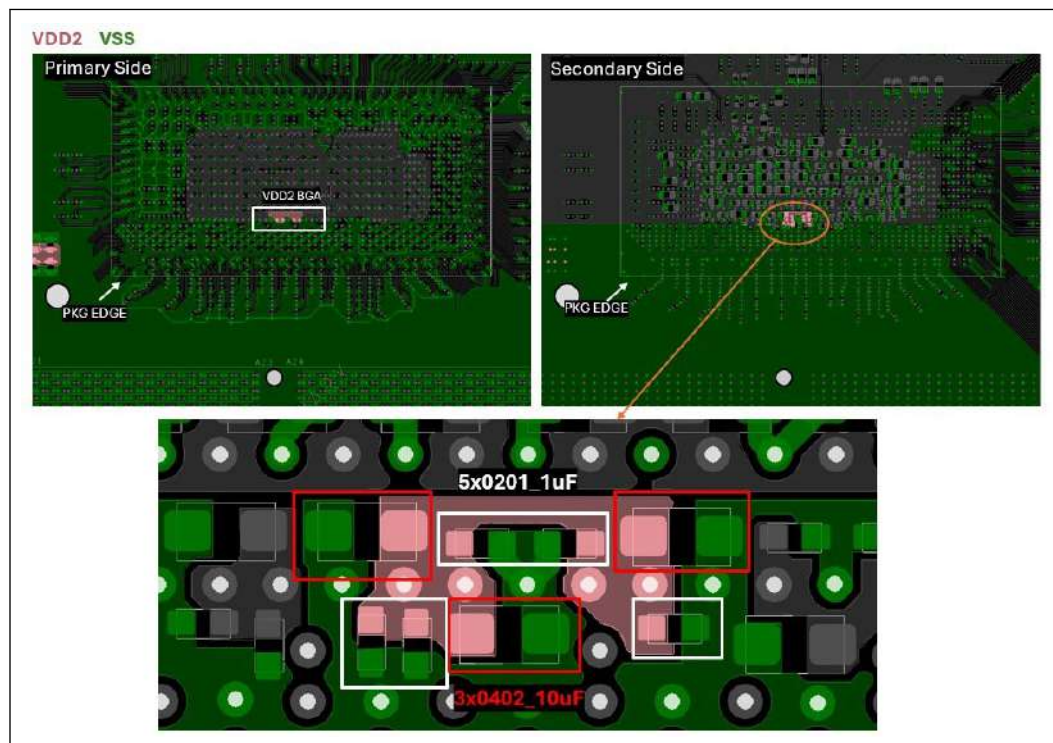
Table 784. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 785. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side	0201	1uF	5	Place beneath BGA	VDD2_T3_CAMM8L_2
Secondary side	0402	10 uF	3	Place beneath BGA	VDD2_T3_CAMM8L_2

Figure 496. VDD2_T3_CAMM8L_2



11.1.1.1.10 VDDQ LPDDR5x LPCAMM2

Table 786. Reference Layout Design

Index	Design Note	Reference Design
1	VDDQ Type-3 8L CAMM motherboard reference layout showing the plane routing	VDDQ_T3_CAMM8L_1
2	Adjacent solid ground to power planes for good return path	

Figure 497. VDDQ_T3_CAMM8L_1

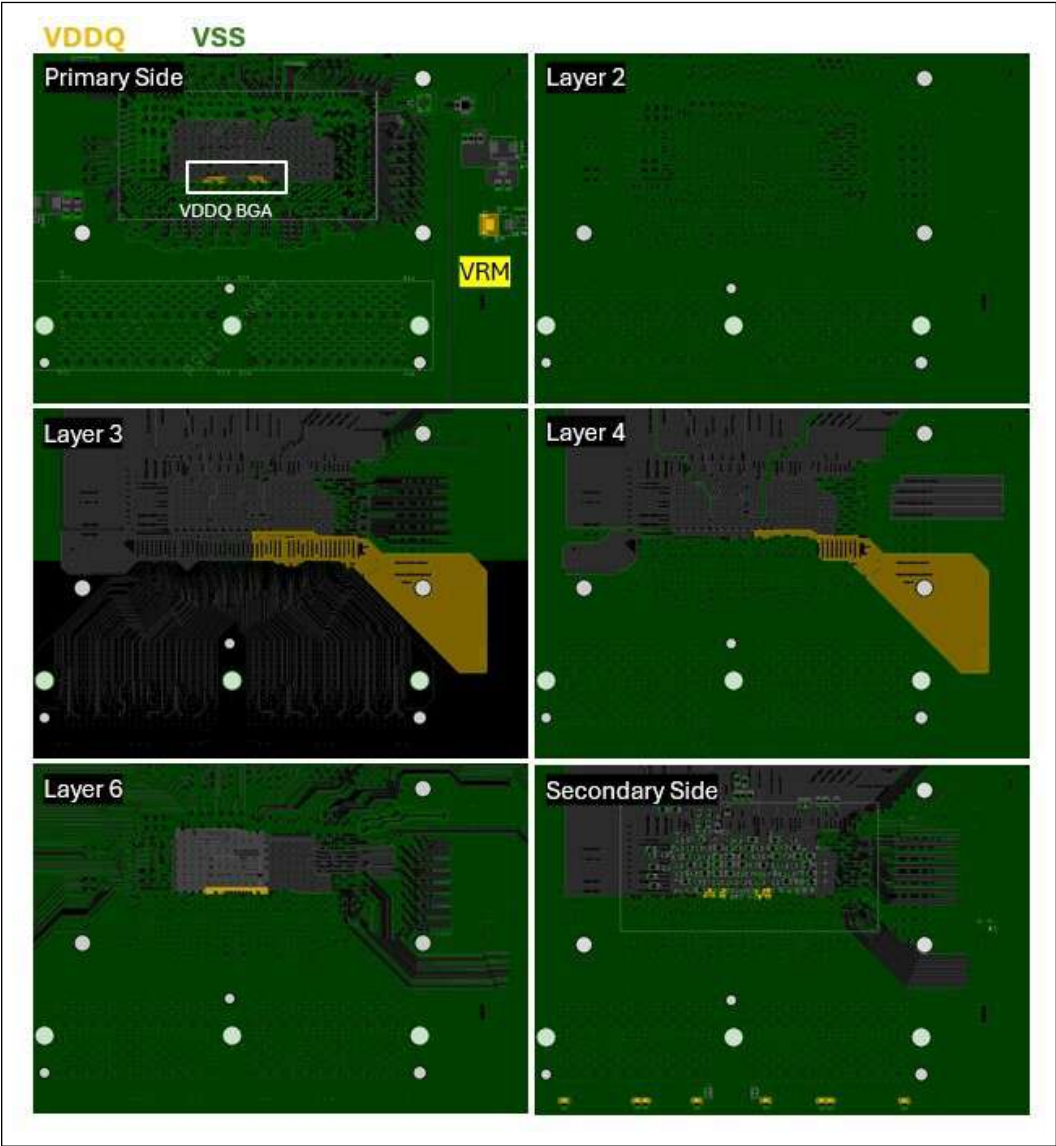
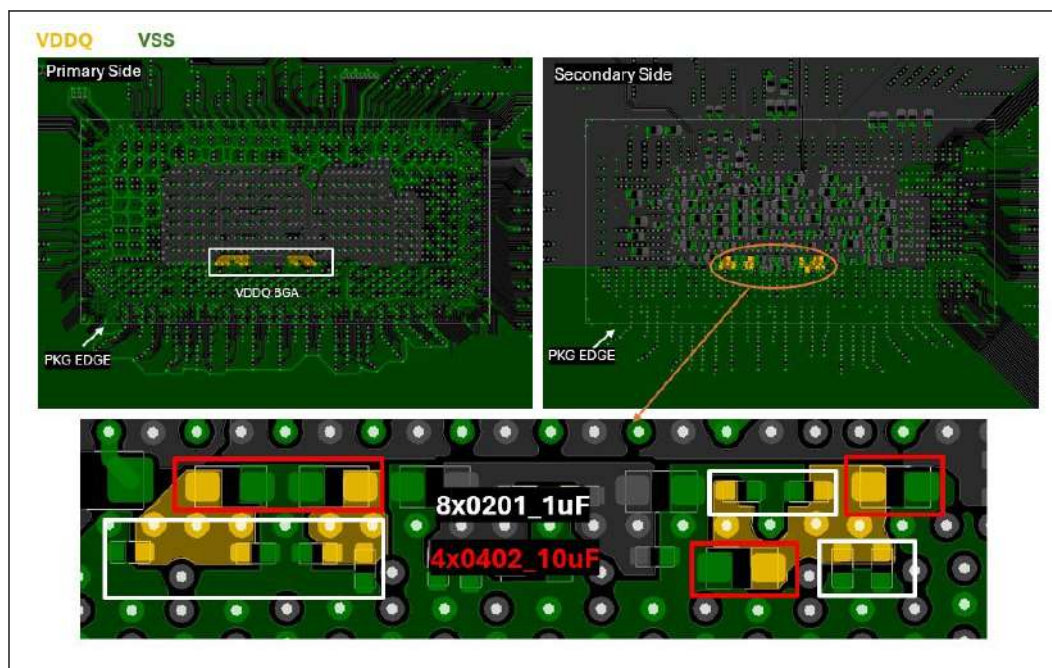


Table 787. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 788. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary Side	0201	1uF	8	Place beneath BGA	VDDQ_T3_CAMM8L_2
Secondary Side	0402	10uF	4	Place beneath BGA	VDDQ_T3_CAMM8L_2

Figure 498. VDDQ_T3_CAMM8L_2**11.1.1.1.11 VDD2 DDR5 (T3 PCB)****Table 789. Reference Layout Design**

Index	Design Note	Reference Design
1	VDD2 Type-3 DDR5 motherboard reference layout showing the plane routing	VDD2_T3_DDR5_8L_1
2	Adjacent solid ground to power planes for good return path	

Figure 499. VDD2_T3_DDR5_8L_1



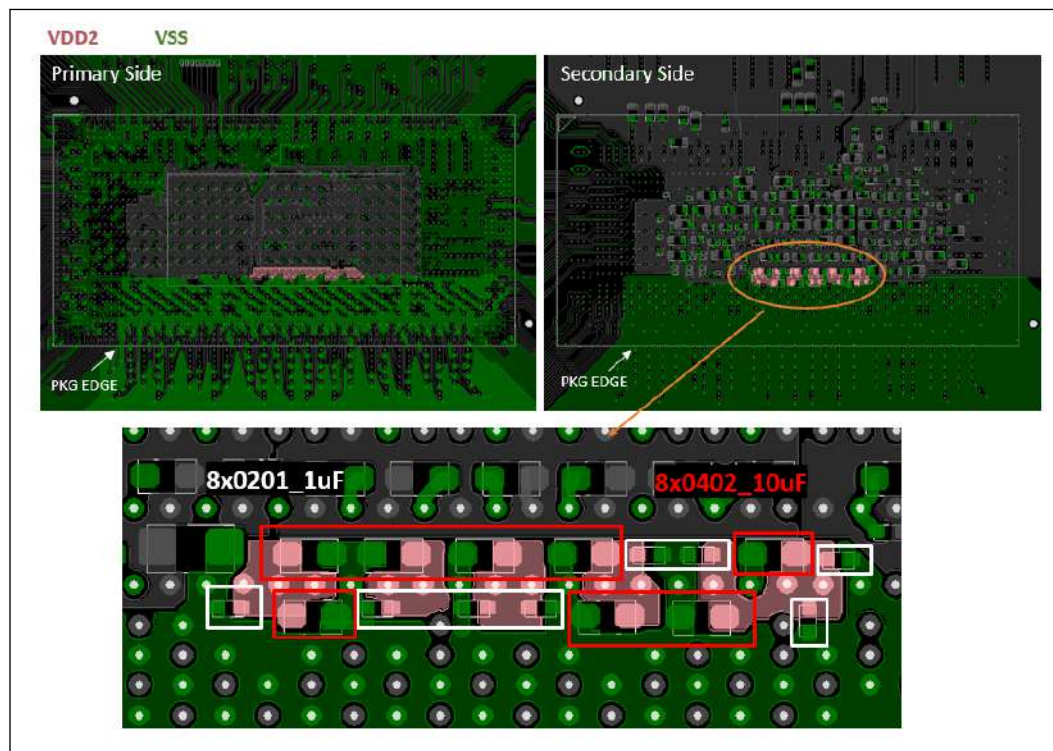
Table 790. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 791. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side	0201	1uF	8	Place beneath BGA	VDD2_T3_DDR5_8L_2
Secondary side	0402	10 uF	8	Place beneath BGA	VDD2_T3_DDR5_8L_2

Figure 500. VDD2_T3_DDR5_8L_2



11.1.1.2 Type-3, 0.9mm, 10 Layer

PCB Type: Type-3

PCB Thickness: 0.9mm

PCB Layer Count: 10

PCB Assembly: Double Sided

11.1.1.2.1 VCC_CORE(H16C_12Xe/T3/0.8mm & 0.9mm PCB)

Table 792. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_1

continued...

Index	Design Note	Reference Design
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 89 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_4

Figure 501. VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_1

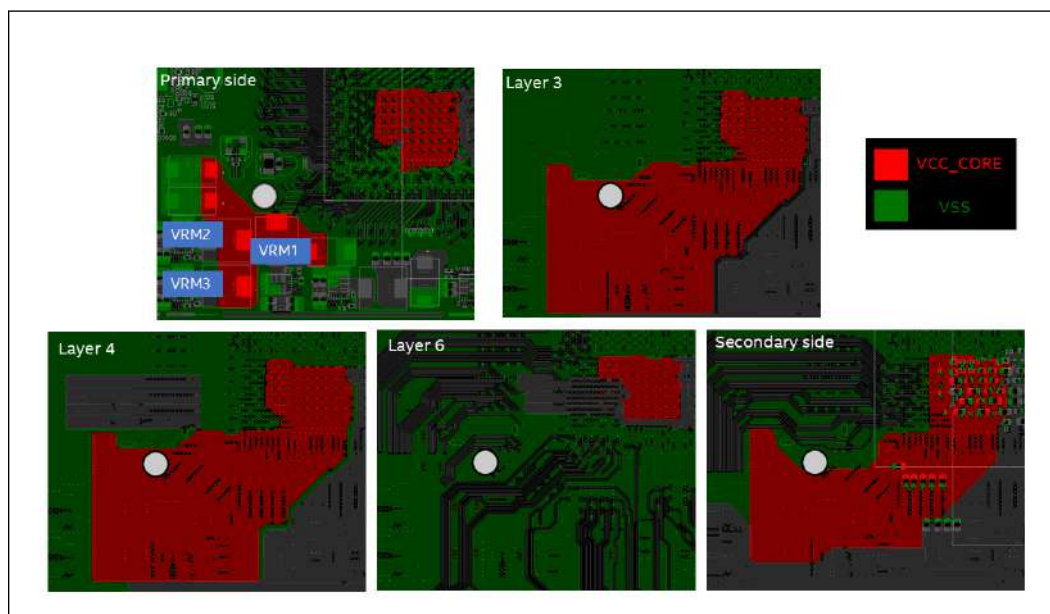


Figure 502. VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_4

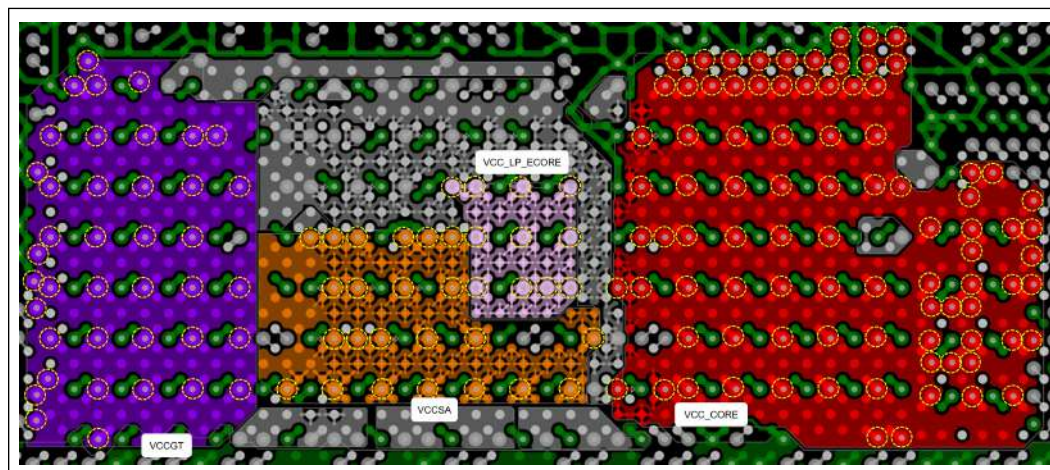


Table 793. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 794. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Primary side	7343	Placeholder	1		VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0201	1 uF	5	Place beneath BGA.	VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0402	10 uF	10	Place beneath BGA.	VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	22 uF	3	Place beneath BGA.	VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0201	Placeholder	7		VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	Placeholder	9		VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3

Figure 503. VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_2

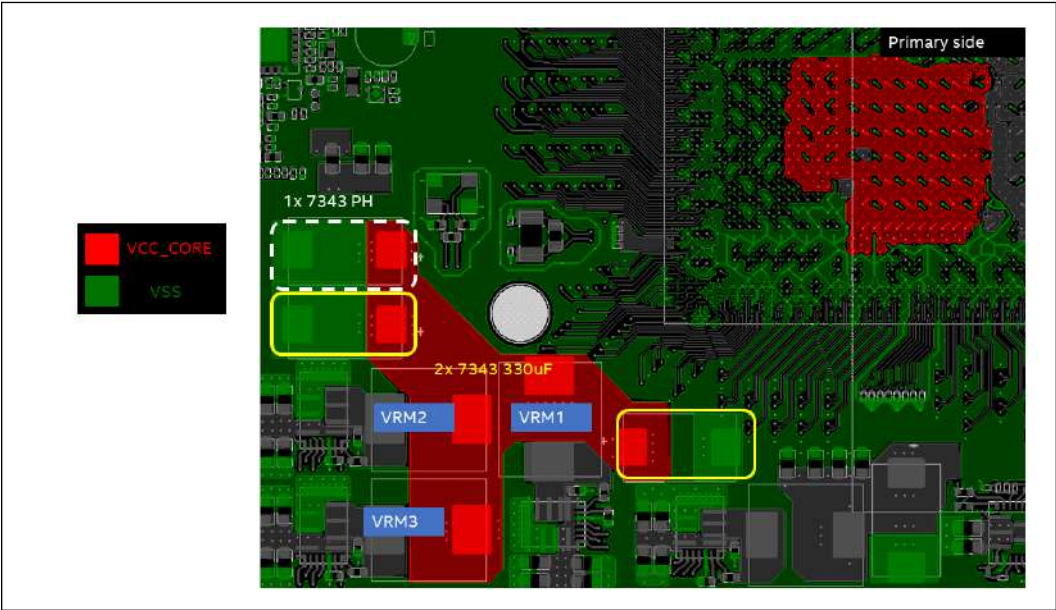
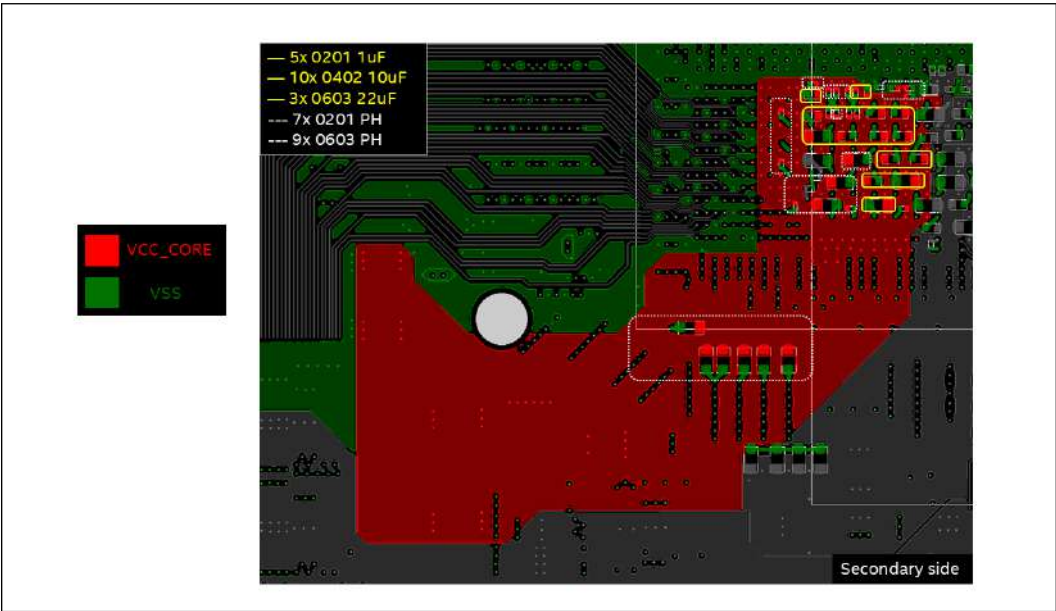


Figure 504. VCC_CORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.2 VCC_CORE(H16C_4Xe/T3/0.8mm & 0.9mm PCB)

Table 795. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA.	VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_1
continued...		

Index	Design Note	Reference Design
	Please see reference image as an example of feasible placement.	
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 89 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_4

Figure 505. VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_1

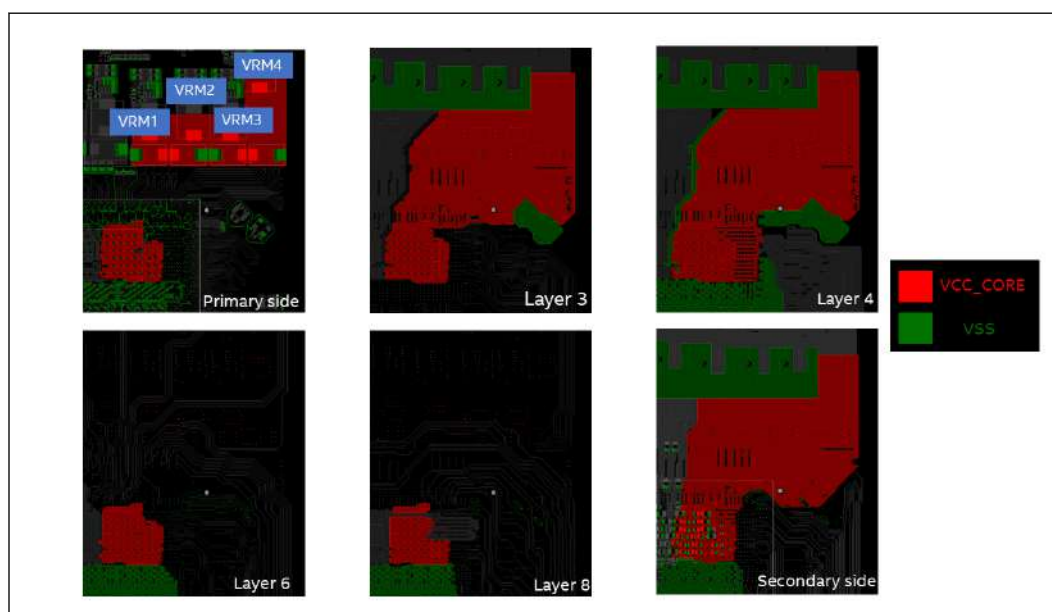


Figure 506. VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_4

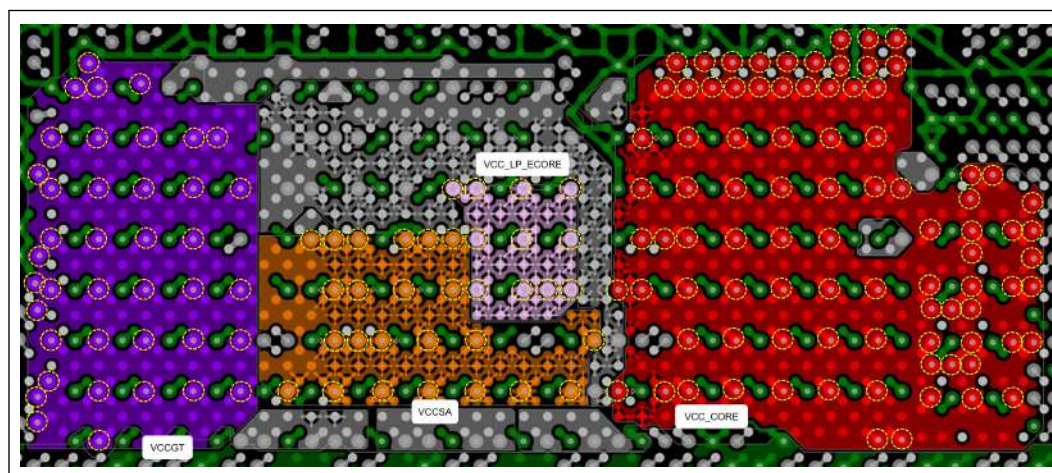


Table 796. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 797. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Primary side	7343	Placeholder	2		VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0201	1 uF	5	Place beneath BGA.	VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0402	10 uF	12	Place beneath BGA.	VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0402	Placeholder	5		VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	Placeholder	5		VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3

Figure 507. VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_2

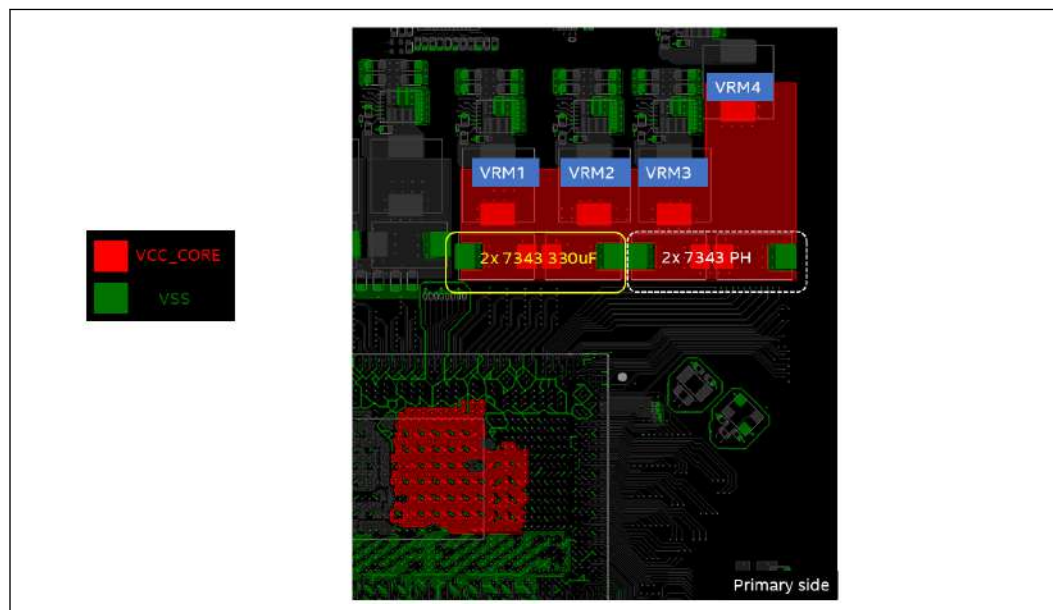


Figure 508. VCC_CORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.3 VCCGT(H16C_12Xe/T3/0.8mm & 0.9mm PCB)

Table 798. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA.	VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_1
<i>continued...</i>		

Index	Design Note	Reference Design
	Please see reference image as an example of feasible placement.	
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 40 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_4

Figure 509. VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_1

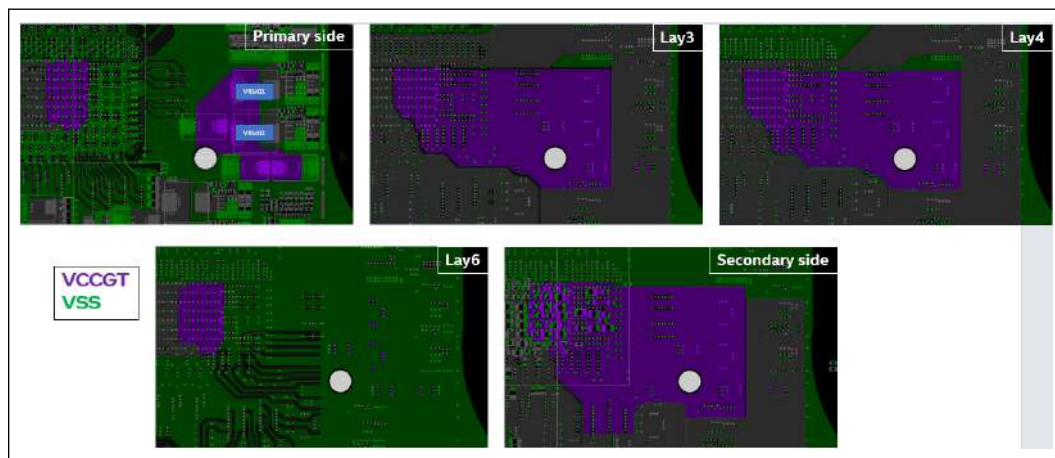


Figure 510. VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_4

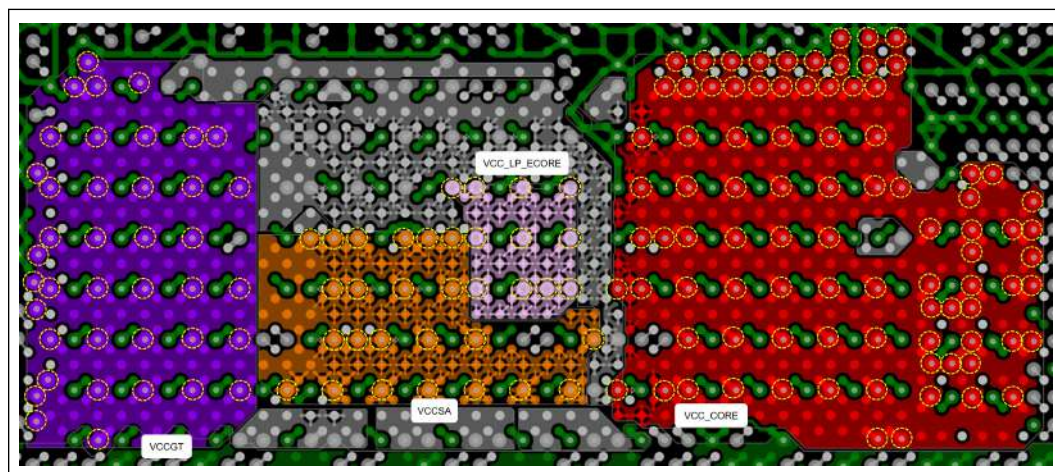


Table 799. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 800. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5 mOhm with 2.5V voltage rating.	VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Primary side	7343	Placeholder	1		VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0402	10 uF	13		VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0402	Placeholder	5		VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	Placeholder	12		VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_3

Figure 511. VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_2

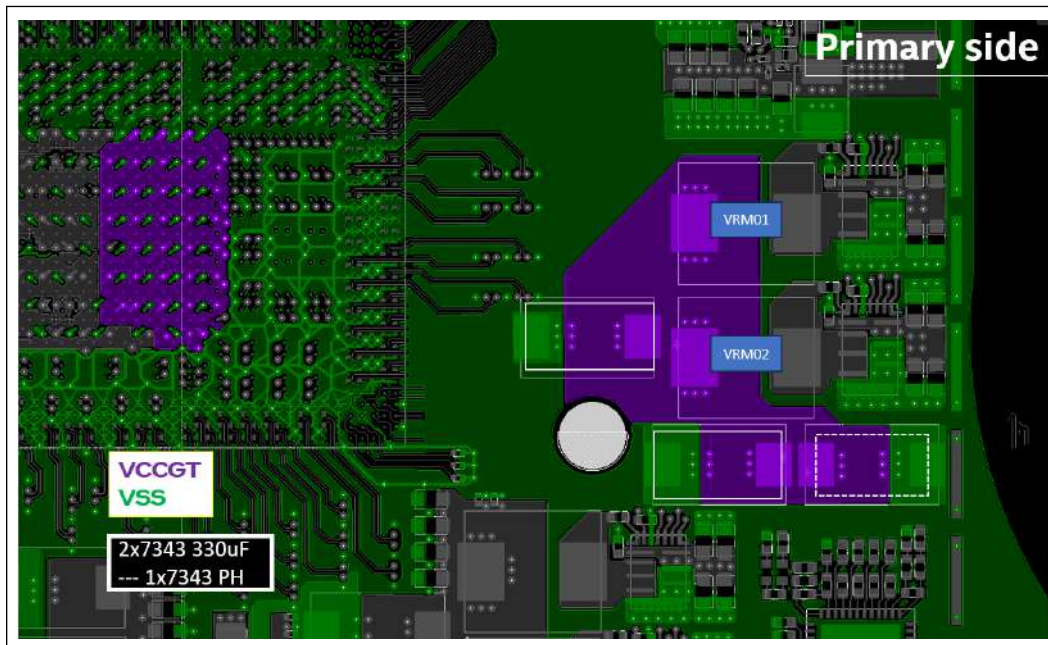
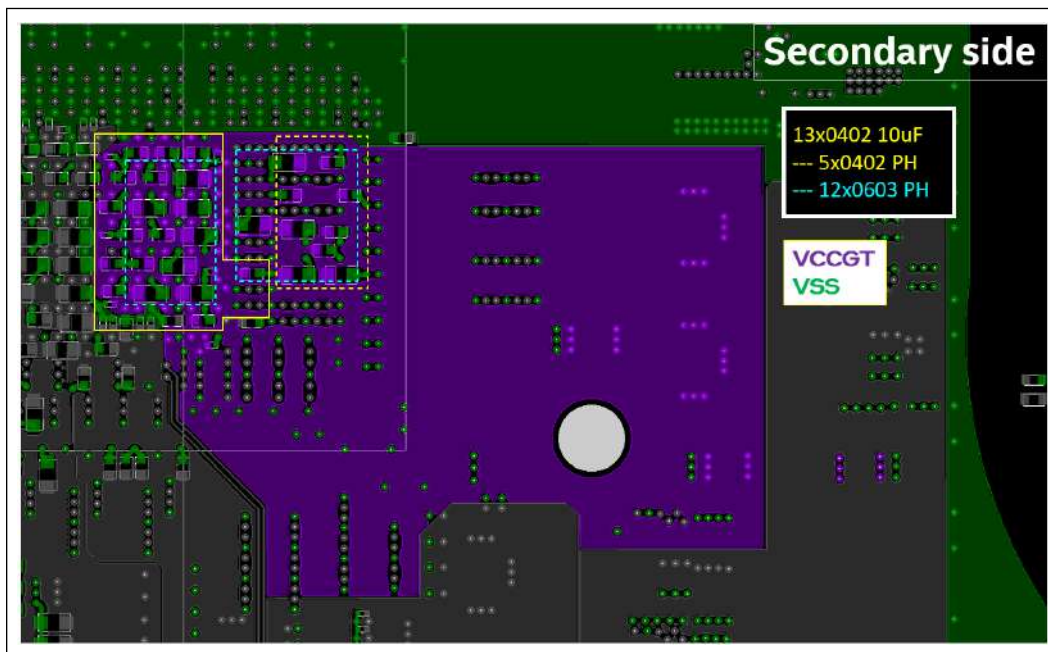


Figure 512. VCCGT_H16C_12Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.4 VCCGT(H16C_4Xe/T3/0.8mm & 0.9mm PCB)

Table 801. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 40 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_4

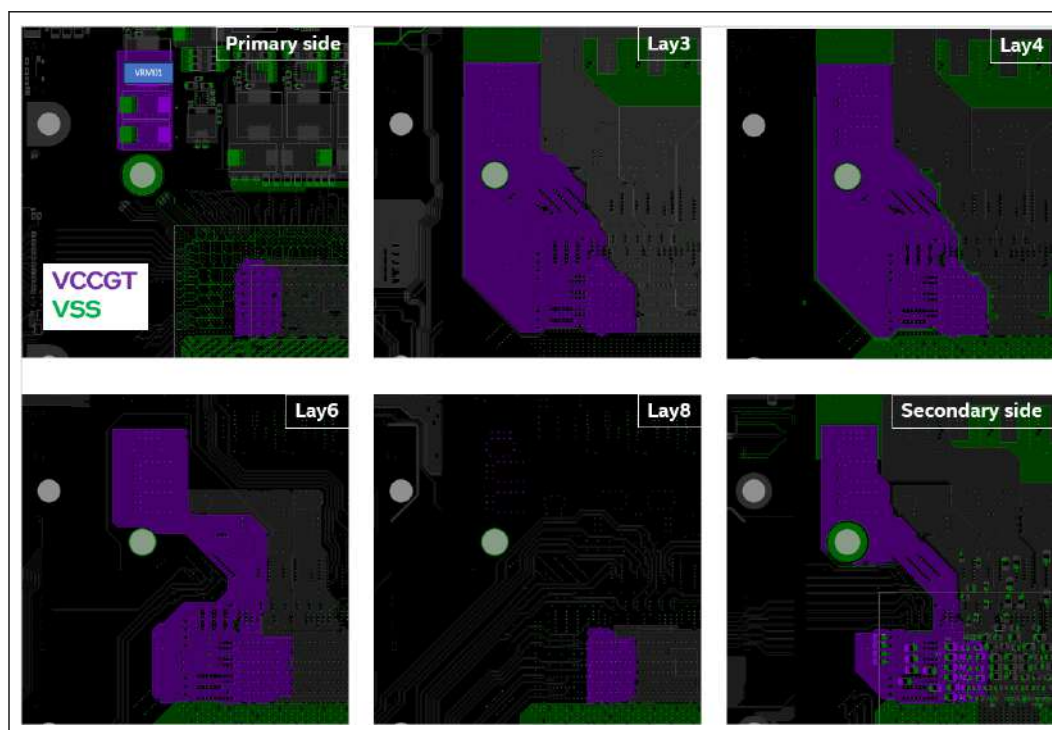
Figure 513. VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_1

Figure 514. VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_4

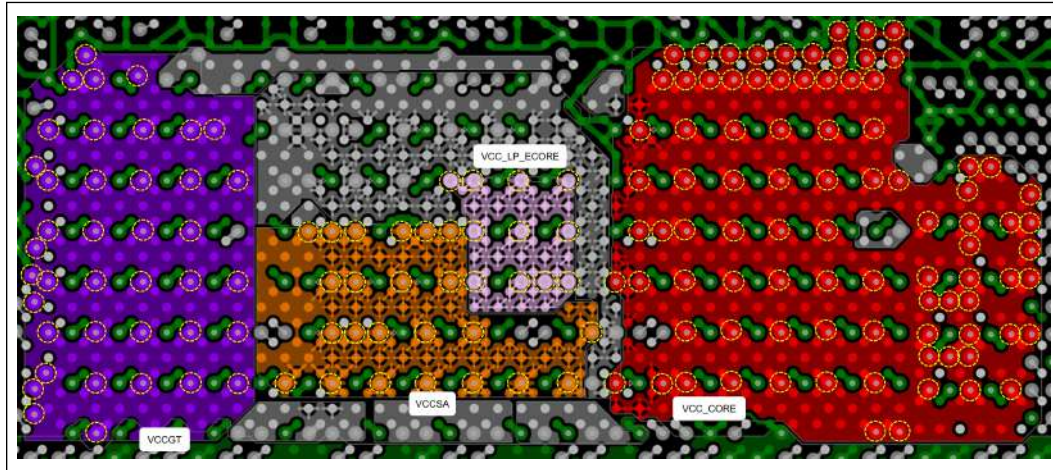


Table 802. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 803. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Primary side	7343	Placeholder	1		VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0402	10 uF	11		VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	22uF	10		VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	Placeholder	6		VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_3

Figure 515. VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_2

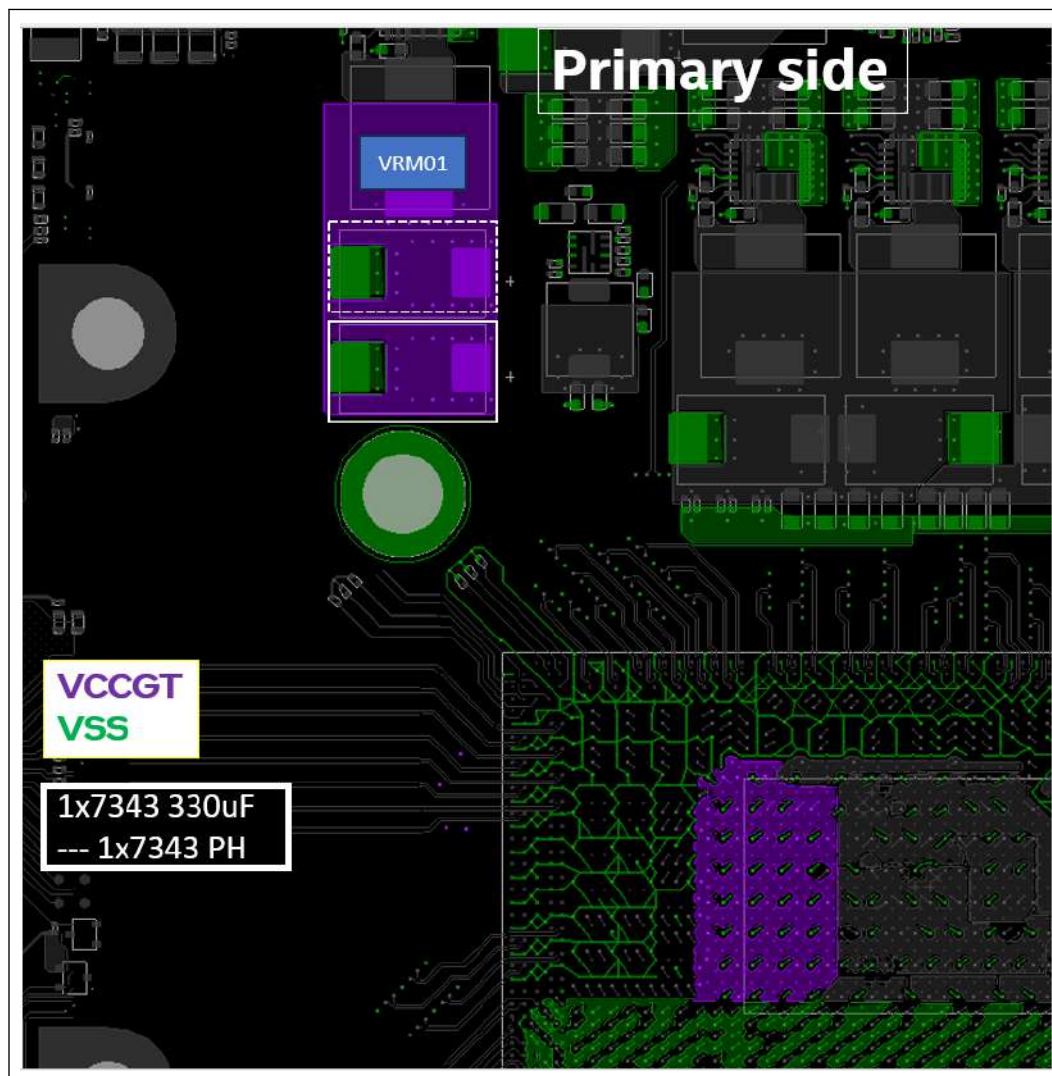
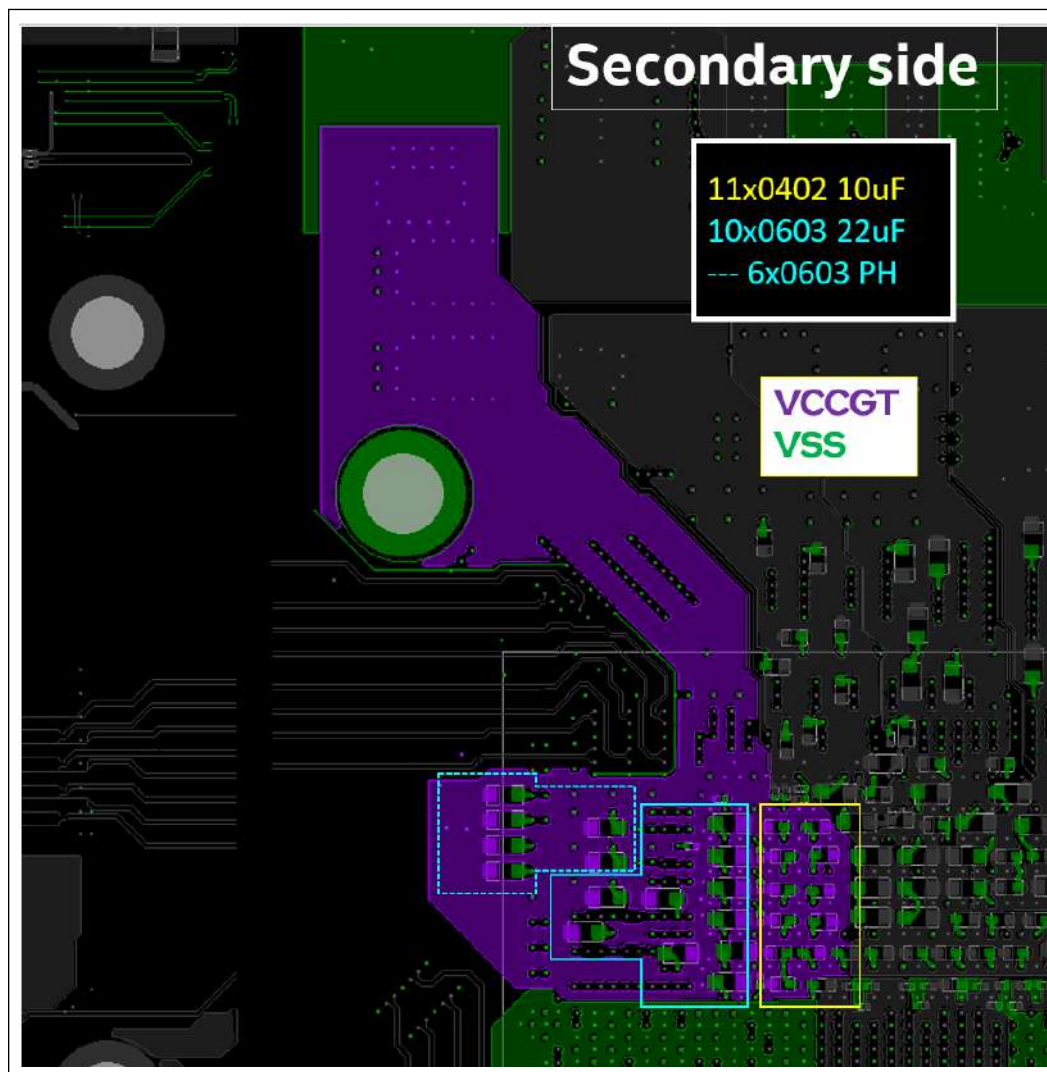


Figure 516. VCCGT_H16C_4Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.5 VCCSA(H16C_12Xe/T3/0.8mm & 0.9mm PCB)

Table 804. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA.	VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_1
continued...		

Index	Design Note	Reference Design
	Please see reference image as an example of feasible placement.	
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 23 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_4

Figure 517. VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_1

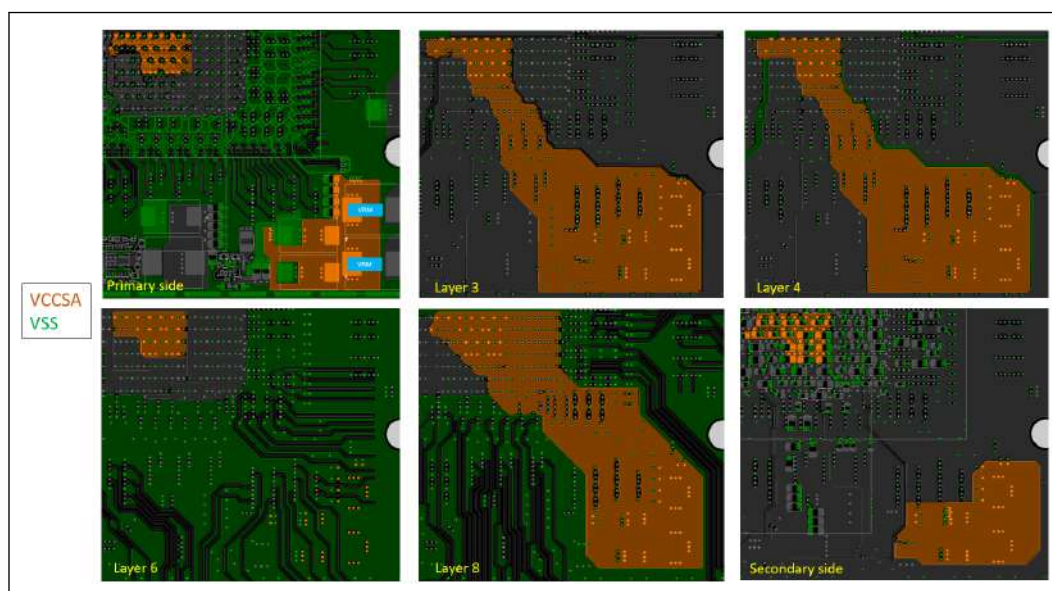


Figure 518. VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_4

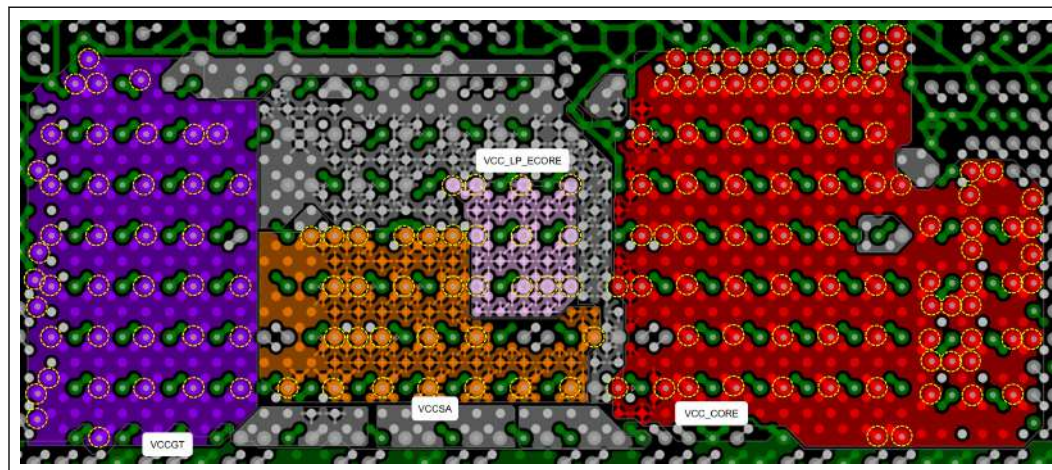


Table 805. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 806. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Primary side	0603	Placeholder	4		VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0402	10 uF	10	Place beneath BGA.	VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	22 uF	6	Place beneath BGA.	VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	Placeholder	2		VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_3

Figure 519. VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_2

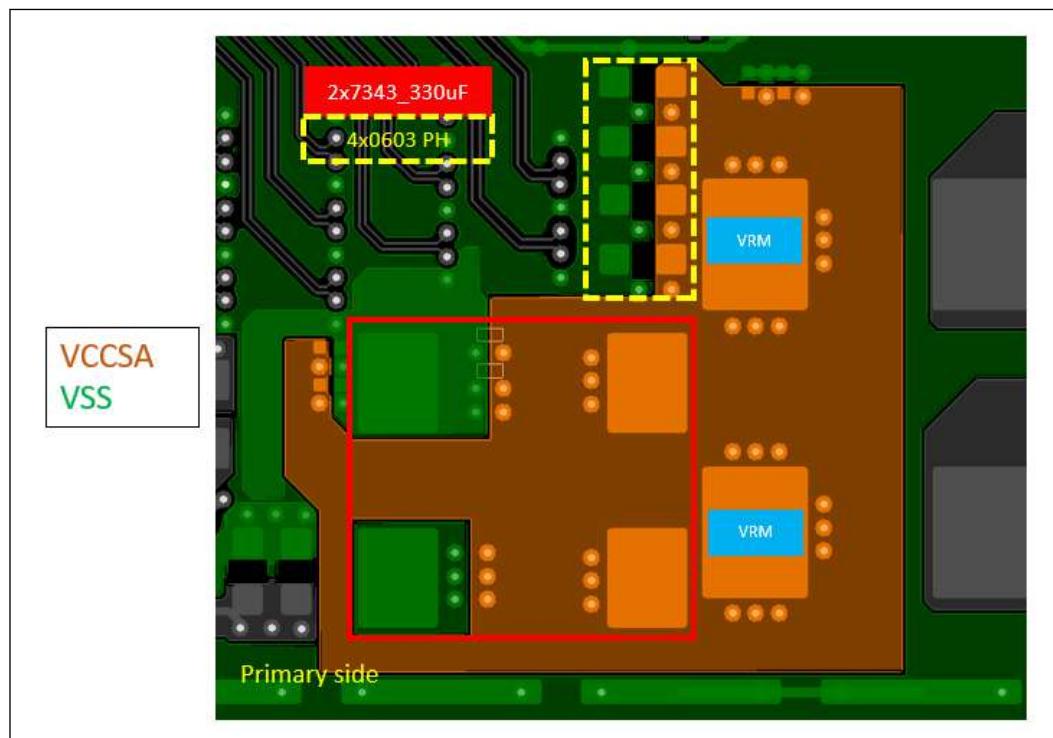


Figure 520. VCCSA_H16C_12Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.6 VCCSA(H16C_4Xe/T3/0.8mm & 0.9mm PCB)

Table 807. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 23 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_4

Figure 521. VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_1

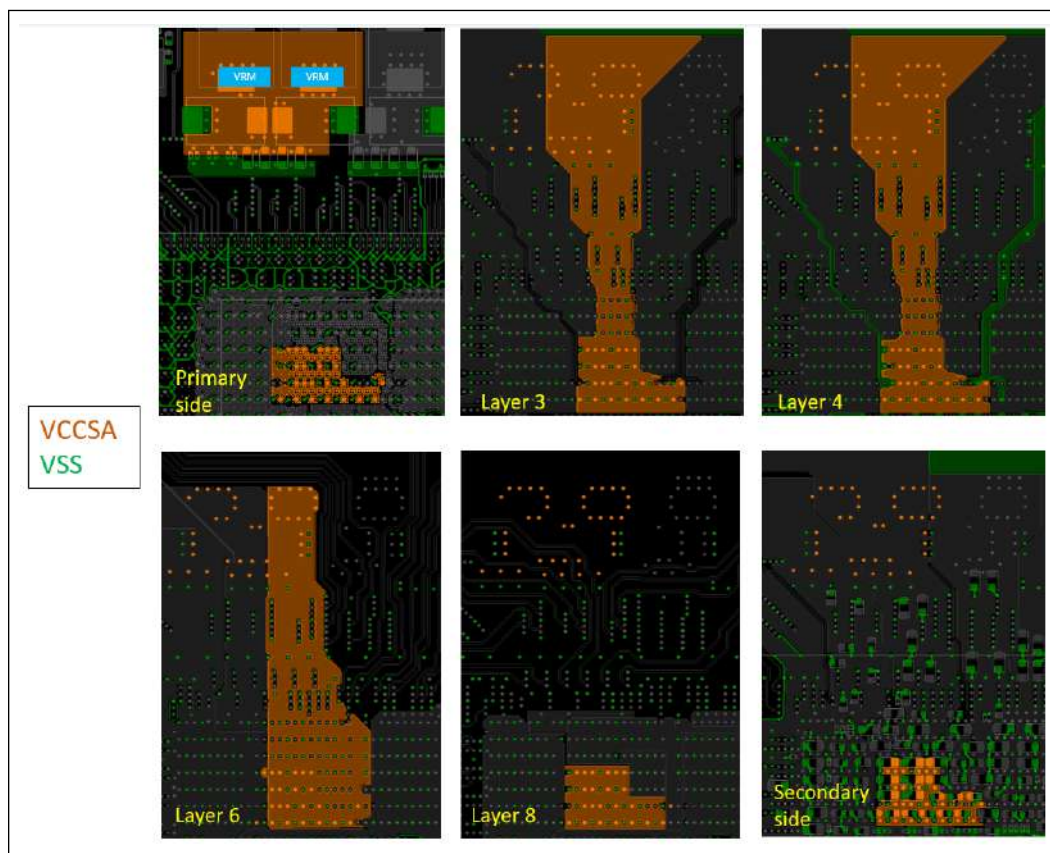


Figure 522. VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_4

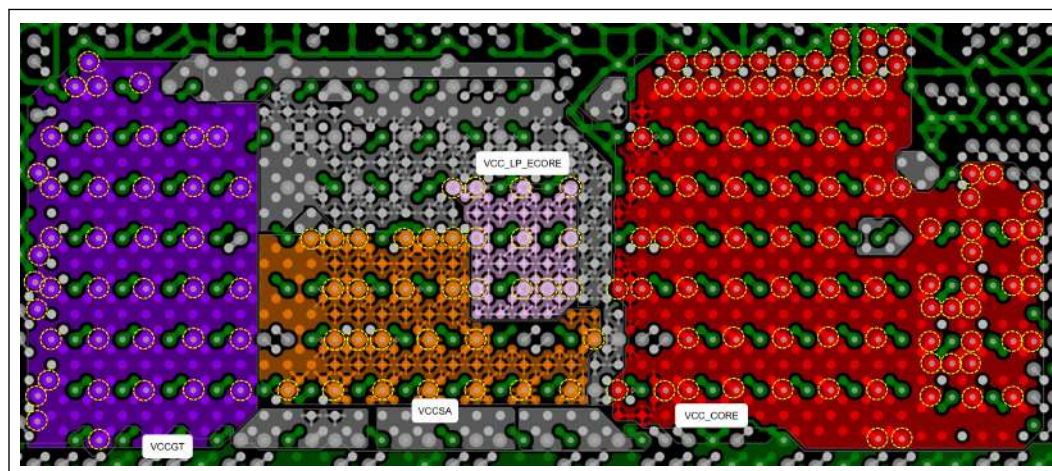


Table 808. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 809. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Primary side	0603	Placeholder	4		VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0402	10 uF	7	Place beneath BGA.	VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	22 uF	8	Place beneath BGA.	VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0402	Placeholder	2		VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_3

Figure 523. VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_2

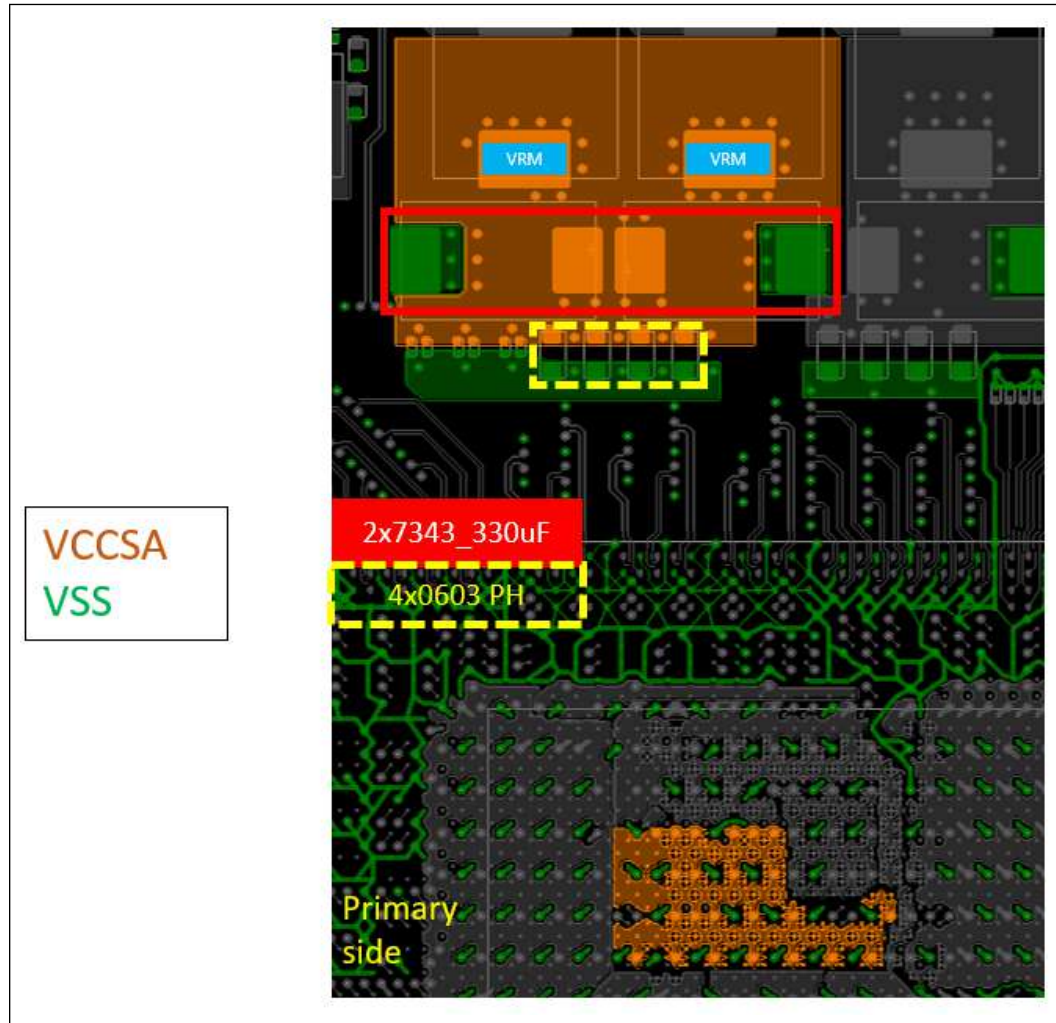
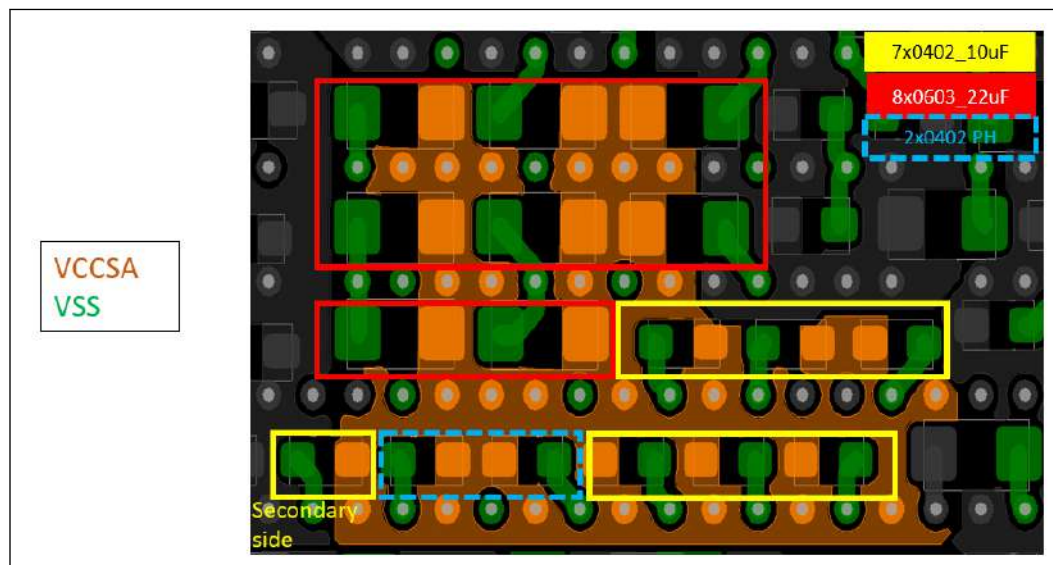


Figure 524. VCCSA_H16C_4Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.7 VCC_LP_ECORE(H16C_12Xe/T3/0.8mm & 0.9mm PCB)

Table 810. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 11 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_4

Figure 525. VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_1



Figure 526. VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_4

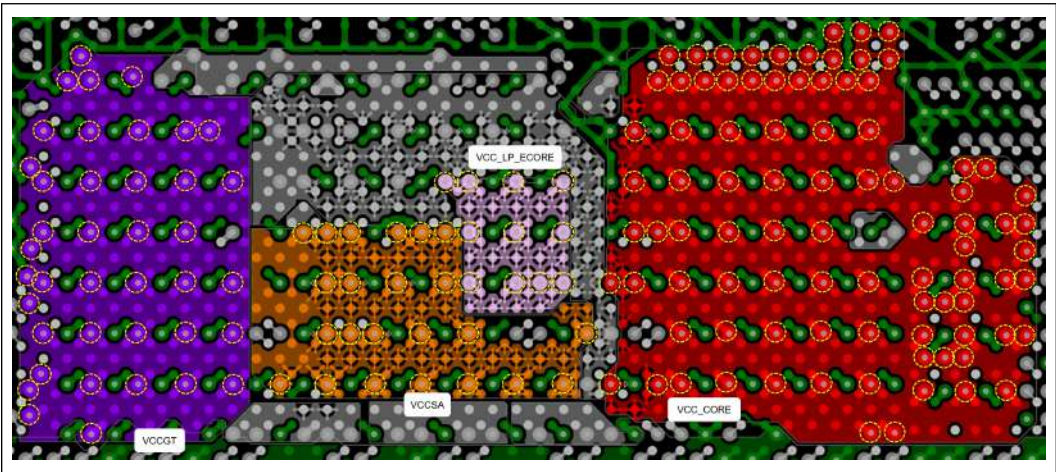


Table 811. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0402 ->0603 ->Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
<i>continued...</i>

Note
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 812. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Primary side	0603	Placeholder	4		VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0402	10uF	4	Place beneath BGA	VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	22uF	9	Place beneath BGA	VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3

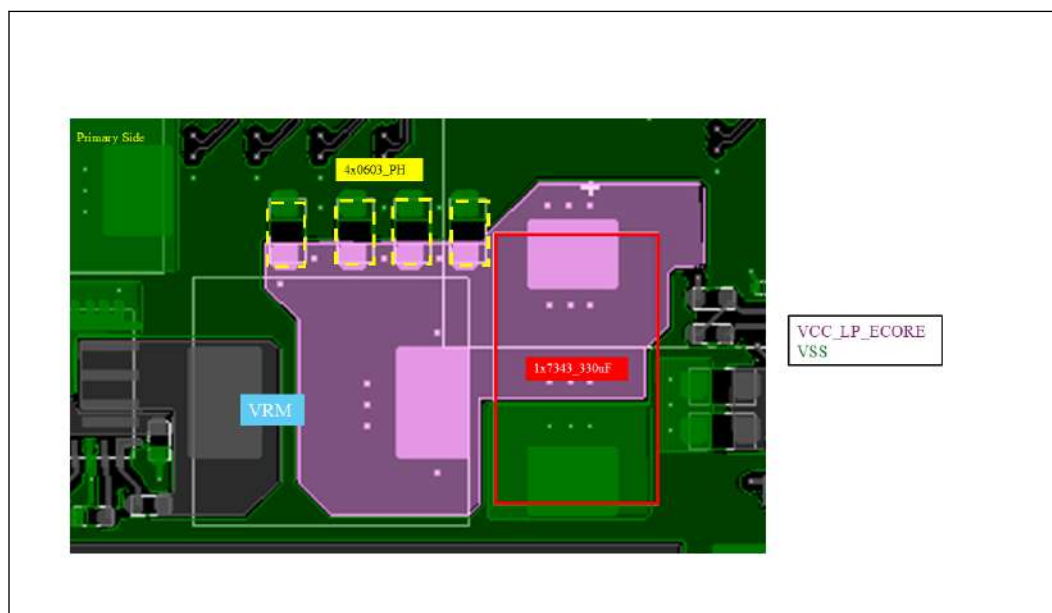
Figure 527. VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_2

Figure 528. VCC_LP_ECORE_H16C_12Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.8 VCC_LP_ECORE(H16C_4Xe/T3/0.8mm & 0.9mm PCB)

Table 813. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 11 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_4

Figure 529. VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_1



Figure 530. VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_4

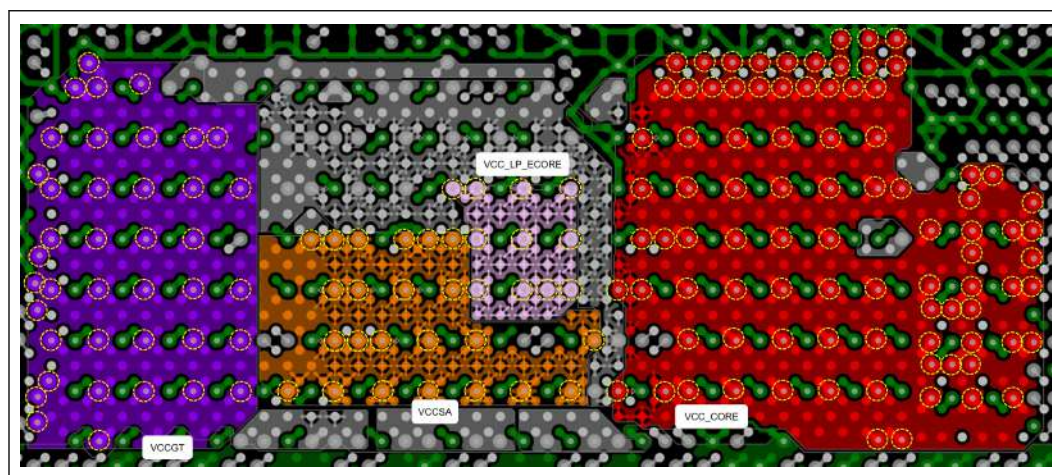


Table 814. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
<i>continued...</i>

Note
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 815. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Primary side	0603	Placeholder	4	Place the 0603 cap near package edge.	VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_2
Secondary side	0402	10uF	4	Place beneath BGA	VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	22uF	8	Place beneath BGA or close to package edge.	VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3
Secondary side	0603	Placeholder	1	Place beneath BGA or close to package edge.	VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3

Figure 531. VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_2

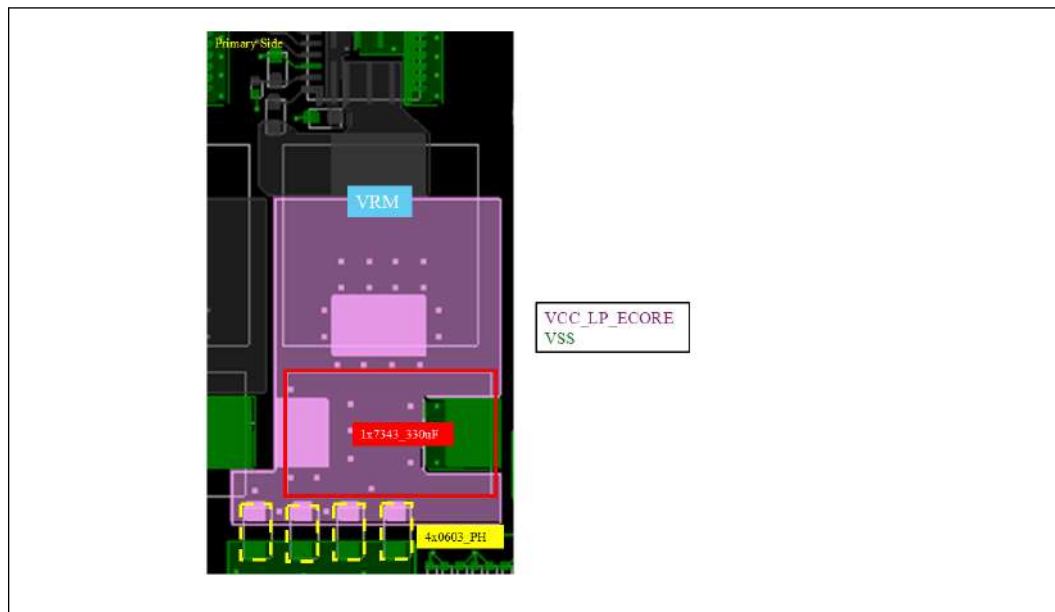
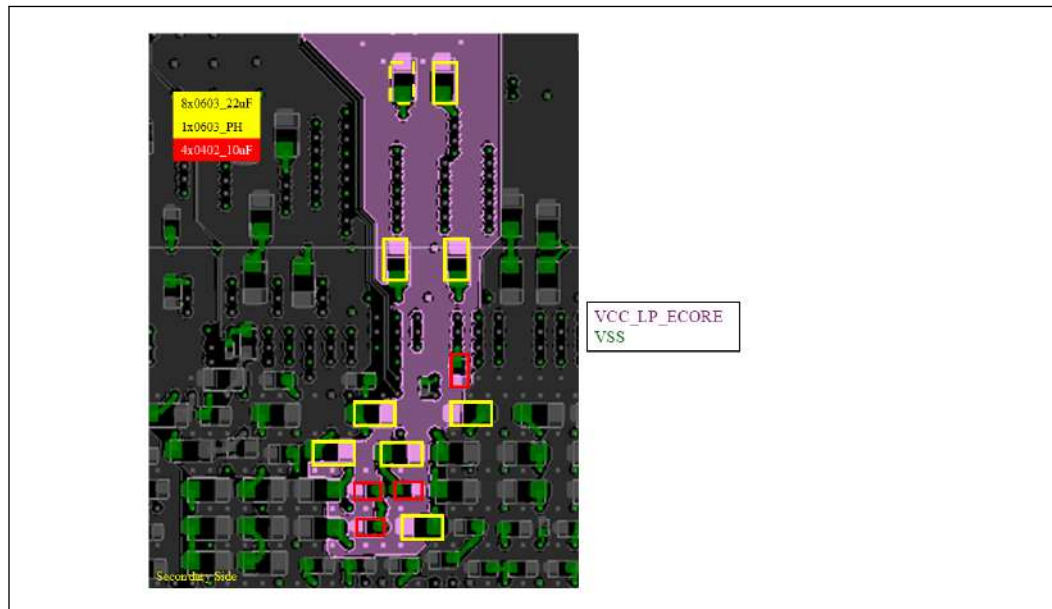


Figure 532. VCC_LP_ECORE_H16C_4Xe_T3_0.8mm_&_0.9mm_3



11.1.1.2.9 VCCPRIM_VNNAON (T3 PCB)

Table 816. Reference Layout Design

Index	Design Note	Reference Design
1	To meet Rpath of <3.5mΩ from VRM to VCCPRIM_VNNAON BGA groups, route VCCPRIM_VNNAON plane on at least three 30um-thick copper layers from VRM output to processor BGA with the following conditions : ► When routing on the top or bottom layer, keep plane width at least 15mm at VRM region & not lesser than 7mm upon approaching BGA region ► When routing on buildup layers (minimum 2 layers), keep plane width at least 10mm at VRM region & not lesser than 6mm upon approaching BGA region	VCCPRIM_VNNAON_1
2	Short all VCCPRIM_VNNAON BGAs with plane on at least of two 30um-thick copper layers	VCCPRIM_VNNAON_2
3	VCCPRIM_VNNAON VRM uses pseudo differential sensing to compensate for voltage drops along the power delivery path. The PWR & GND sense lines are connected to BP14 & BP12 BGAs respectively	VCCPRIM_VNNAON_3
4	Short both VCCPRIM_VNNAON_FLTR BGAs (CM16 & CM18) with plane	VCCPRIM_VNNAON_4
5	Merge both VCCPRIM_VNNAON_FLTR & VCCPRIM_VNNAON power planes using 1 x 0603 1uH (120mΩ max DCR) inductor & 1 x 0603 22uF cap to form an LC-filter with the following conditions : ► Place the LC-filter <1mm distance from or underneath VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR BGAs ► Ensure the merging point for both VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR planes are <1mm distance from or underneath VCCPRIM_VNNAON_FLTR BGAs Note : Refer diagram for design implementation guidance	VCCPRIM_VNNAON_5
6	Yellow dotted circles highlights minimum count of VCCPRIM_VNNAON PTH recommended in bga region to prevent current overstress in PTHs. Ensure GND PTHs neatly placed close to PWR PTHs as shown in illustration to minimize return current loop	VCCPRIM_VNNAON_6

Figure 533. VCCPRIM_VNNAON_1

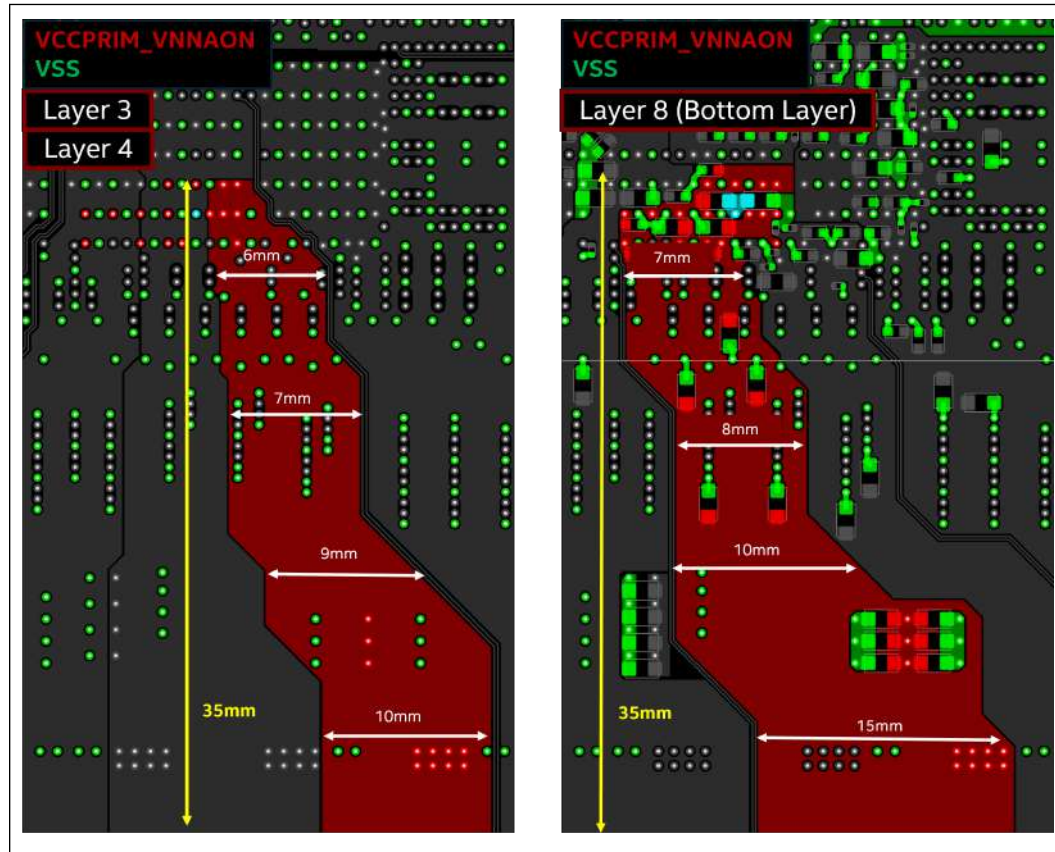


Figure 534. VCCPRIM_VNNAON_2



Figure 535. VCCPRIM_VNNAON_3

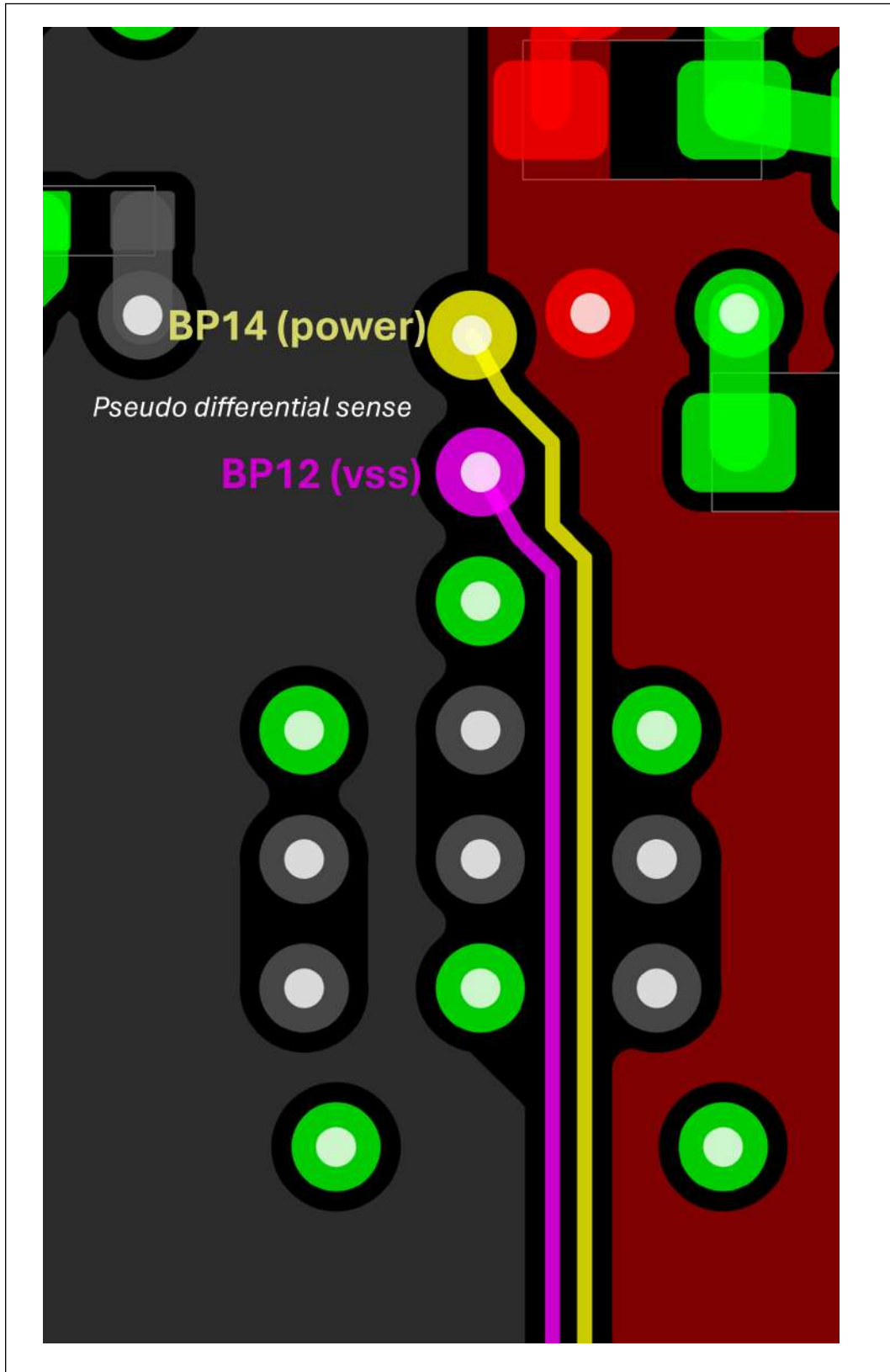


Figure 536. VCCPRIM_VNNAON_4

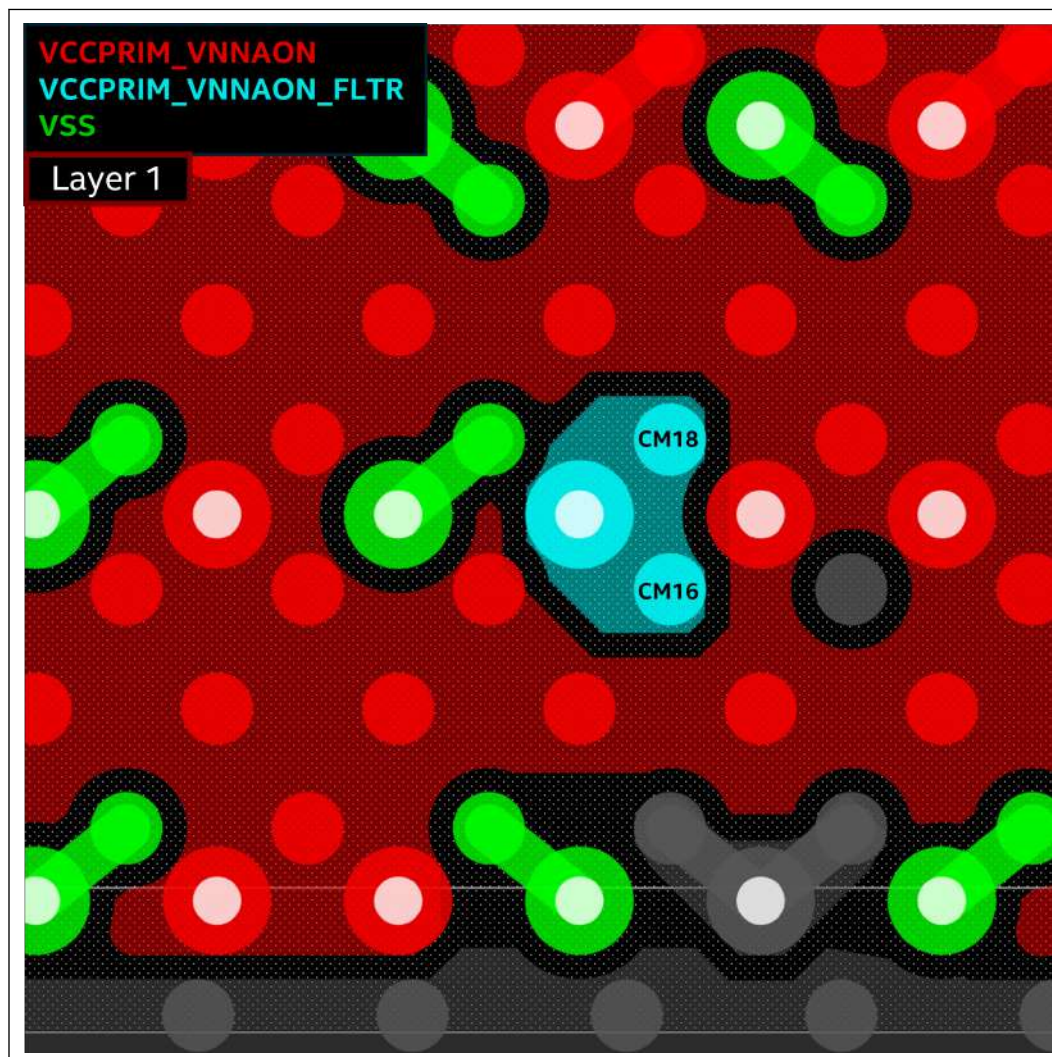


Figure 537. VCCPRIM_VNNAON_5

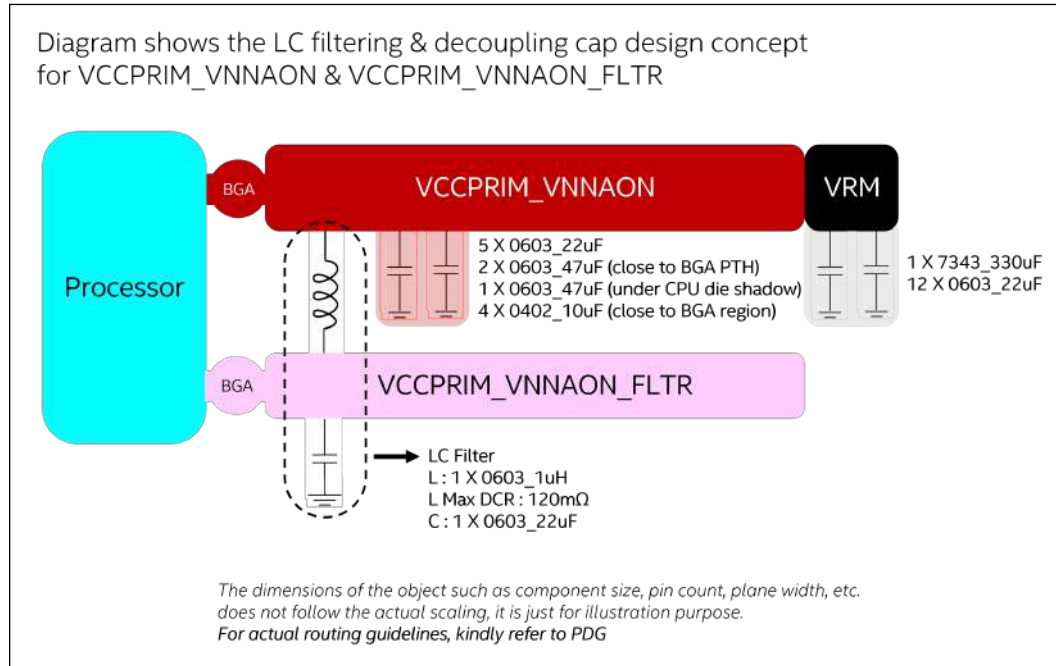


Figure 538. VCCPRIM_VNNAON_6

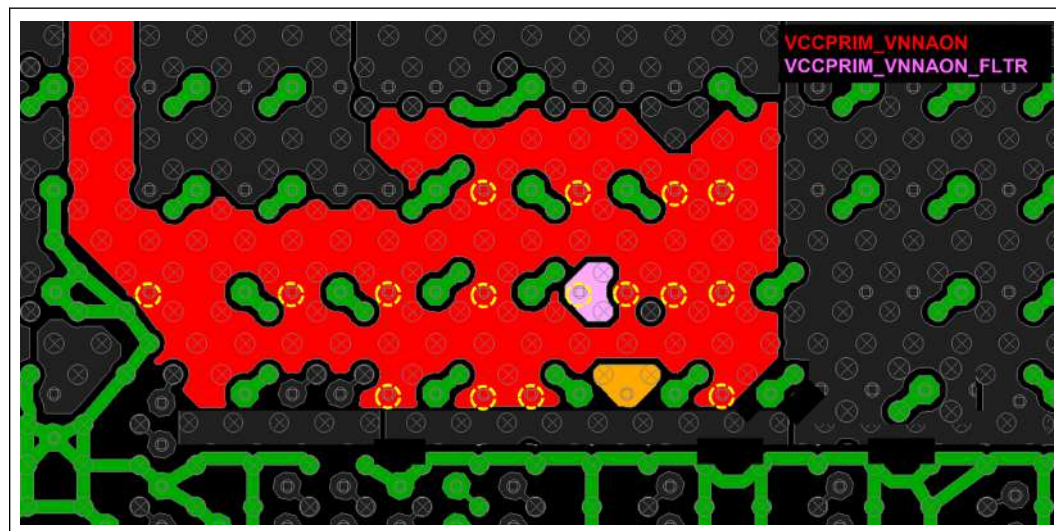


Table 817. Notes

Note
Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
Heighten capacitor effectiveness by placing them from VRM output to package in the following order : 0402 → 0603 → 0805 → Bulk
<i>continued...</i>

Note
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
When placing capacitors, align their pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor provided the caps meet the recommended parasitic capacitance.

Table 818. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	7343	330uF	1	Place bulk cap near to VRM output	VCCPRIM_VNNAON_6
Primary Side	0603	22uF	6	Place caps near to VRM output	VCCPRIM_VNNAON_6
Secondary Side	0603	22uF	6	Place caps near to VRM output	VCCPRIM_VNNAON_7
Secondary Side	0603	47uF	1	Place 1 x 0603 47uF cap under processor die shadow of VCCPRIM_VNNAON BGAs	
Secondary side	0603	47uF	2	Place 2 x 0603 47uF caps <1mm from VCCPRIM_VNNAON PTH located underneath BGA	VCCPRIM_VNNAON_8
Secondary Side	0603	22uF	5	Place 5 x 0603 22uF caps <10mm from package edge. Make sure PWR-GND PTH pair is placed very close to cap terminals	VCCPRIM_VNNAON_8
Secondary Side	0402	10uF	4	Place 4 x 0402 10uF caps underneath BGA or <1mm from package edge. Make sure PWR-GND PTH pair is placed very close to cap terminals	VCCPRIM_VNNAON_8
Secondary Side	0603	22uF	1	Pair 1 x 0603 22uF VCCPRIM_VNNAON_FLTR cap with 1 x 0603 1uH (120mΩ max DCR) inductor <1mm from or underneath VCCPRIM_VNNAON_FLTR BGAs to form an LC-filter	VCCPRIM_VNNAON_8
Secondary Side	0603	1 uH_120mΩ DCR	1	Merge both VCCPRIM_VNNAON_FLTR & VCCPRIM_VNNAON power planes using 1 x 0603 1uH (120mΩ max DCR) inductor & 1 x 0603 22uF cap to form an LC-filter with the following conditions : ► Place the LC-filter <1mm distance from or underneath VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR BGAs ► Ensure the merging point for both VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR planes are <1mm distance from or underneath VCCPRIM_VNNAON_FLTR BGAs	VCCPRIM_VNNAON_8

Figure 539. VCCPRIM_VNNAON_6

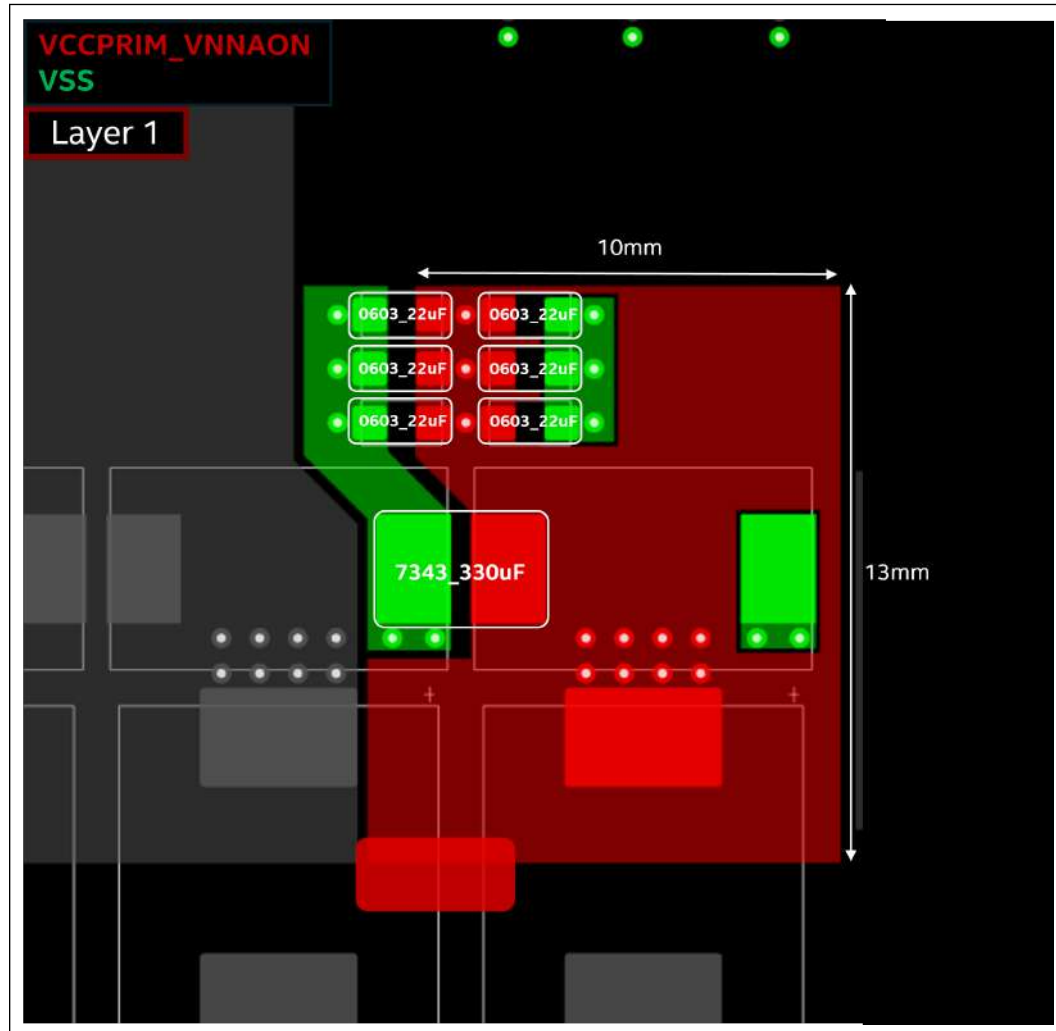


Figure 540. VCCPRIM_VNNAON_7

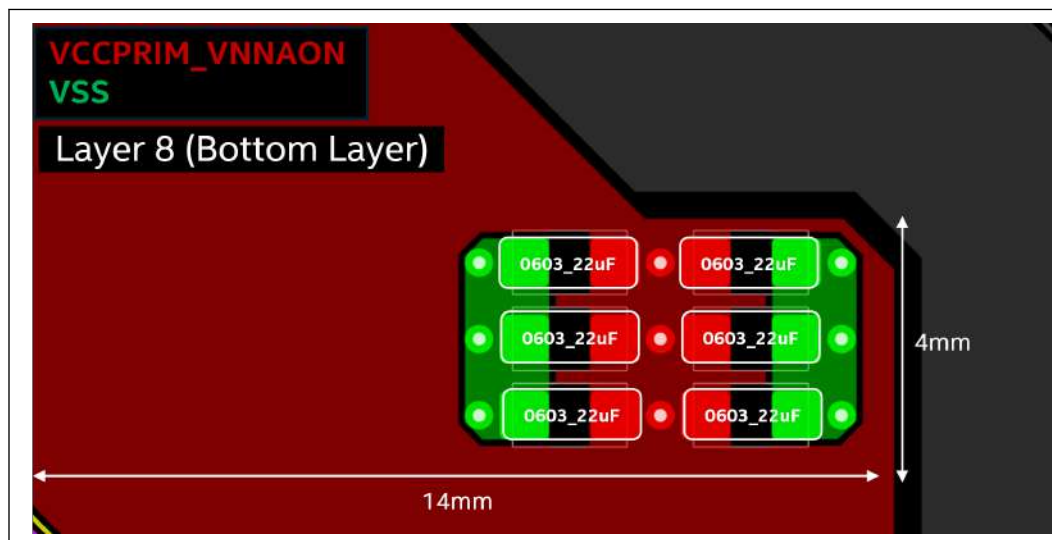
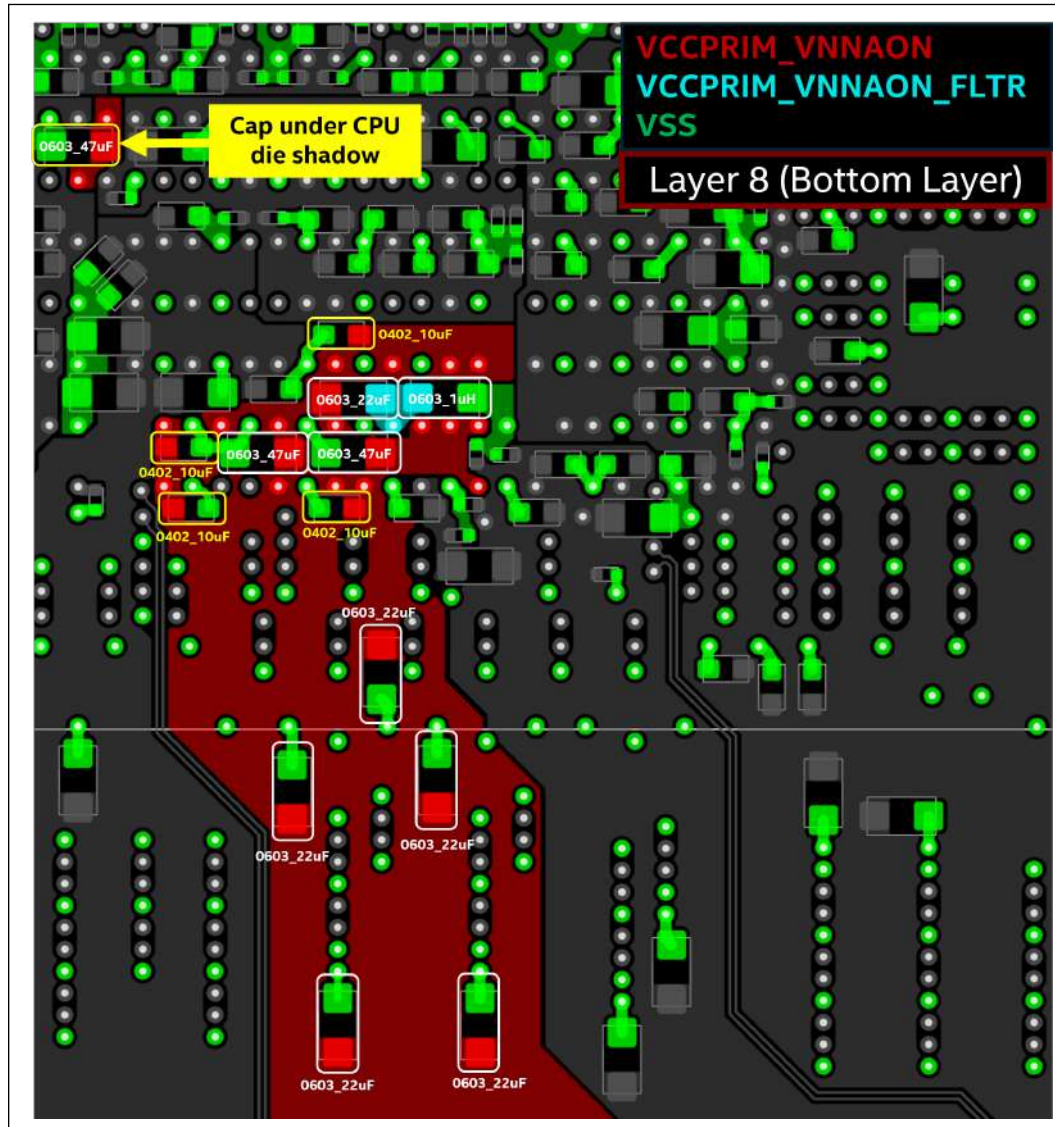


Figure 541. VCCPRIM_VNNAON_8



11.1.1.2.10 VCCPRIM_IO (T3 PCB)

Table 819. Reference Layout Design

Index	Design Note	Reference Design
1	Route VCCPRIM_IO power plane from VR output inductor to the BGA on Layer 1, Layer 6 and Layer 8. Adjacent layer from VCCPRIM_IO power plane should be ground.	VCCPRIM_IO_1
2	VCCPRIM_IO_FLTR should be merged with VCCPRIM_IO after LC filter on Layer 8	VCCPRIM_IO_1
3	VCCPRIM_IO VRM uses differential sensing. The power sense line is connected to SENSE_VCCPRIM_IO BGA(DF8) and ground sense line connected to SENSE_VSS_VCCPRIM_IO(DF9)	VCCPRIM_IO_2
4	The SVID 1.25 V supply for pull up resistors should be sourced from VCCPRIM_IO_OUT_SVID BGA EP5.	

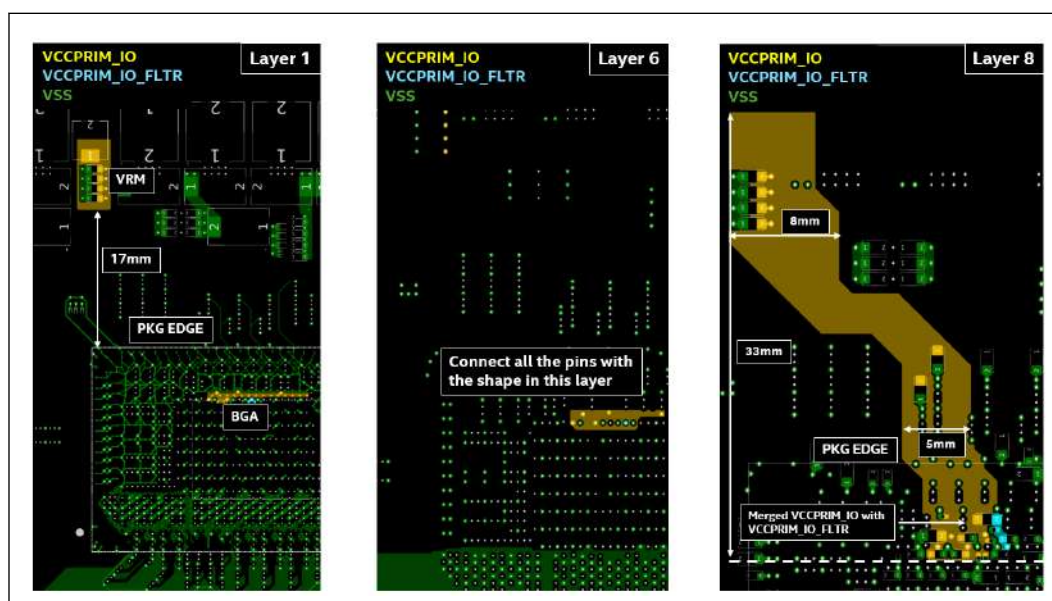
Figure 542. VCCPRIM_IO_1

Figure 543. VCCPRIM_IO_2

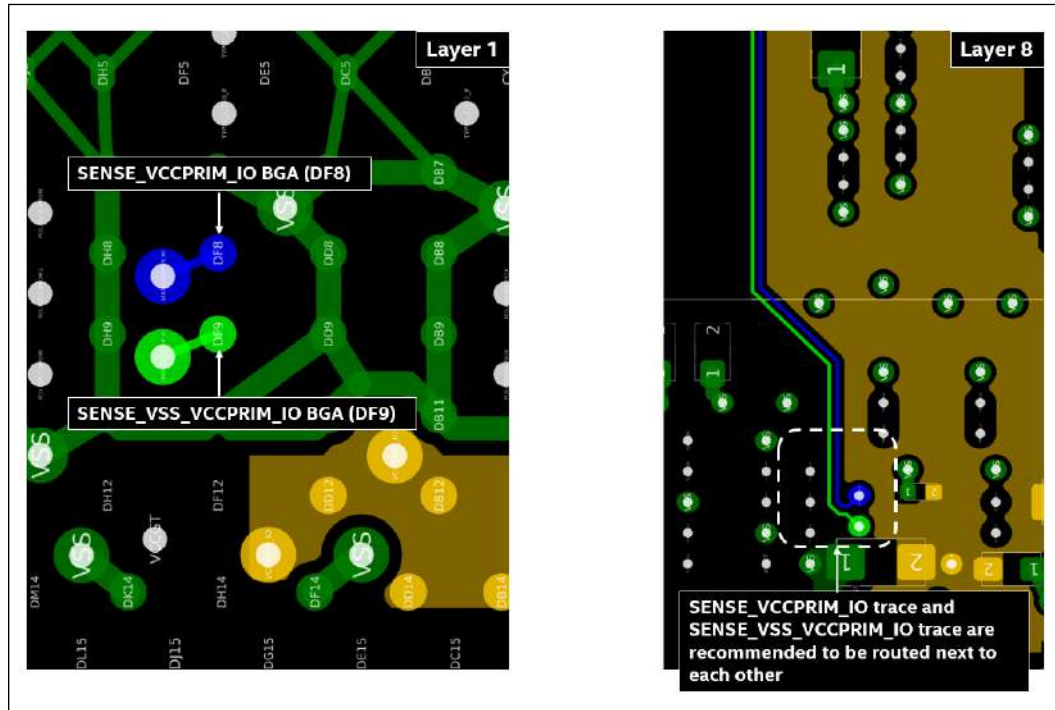


Table 820. Notes

Note
Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
Heighten capacitor effectiveness by placing them from VRM to package in the following order : 0402 → 0603 → 0805 → Bulk
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
Align capacitor pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 821. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	0805	47uF	4	Place near the VCCPRIM_IO VRM output	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0603	22 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402	10 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	4	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4

continued...

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side (Inductor)	0603	1 uH	1	Inductor component of the LC filter from VCCPRIM_IO_FLTR rail with ESR = 350mΩ	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402/0603	22 uF	1	Decoupling capacitor placed on VCCPRIM_IO_FLTR rail for LC filtering purpose	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	1	Place right below the VCCPRIM_IO_FLTR BGA	VCCPRIM_IO_3,VCCPRIM_IO_4

Figure 544. VCCPRIM_IO_3

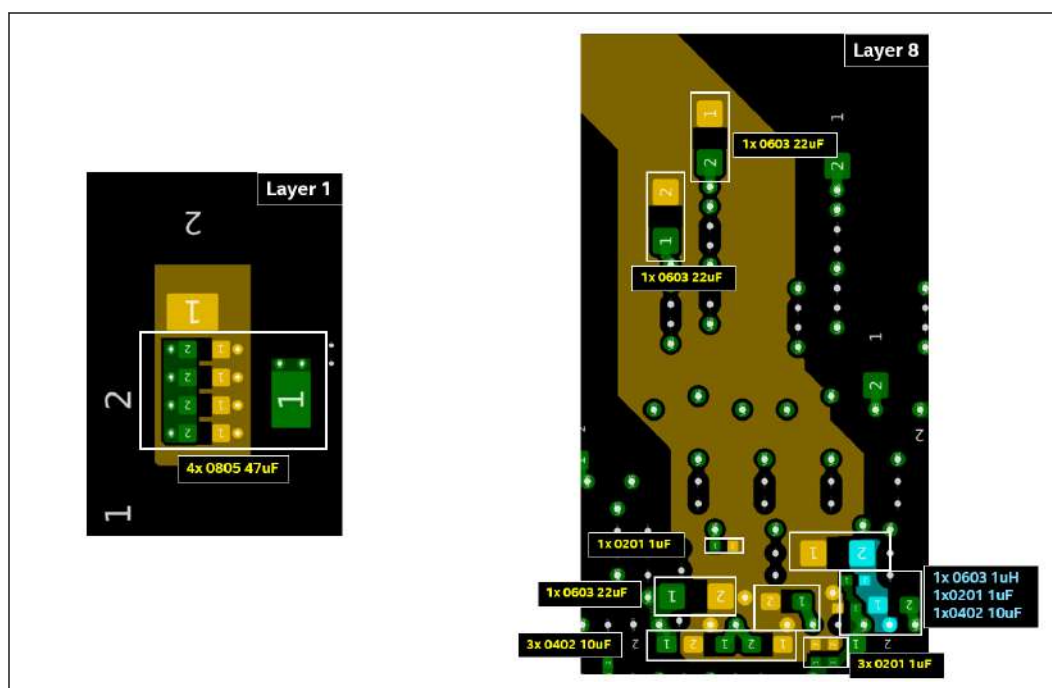
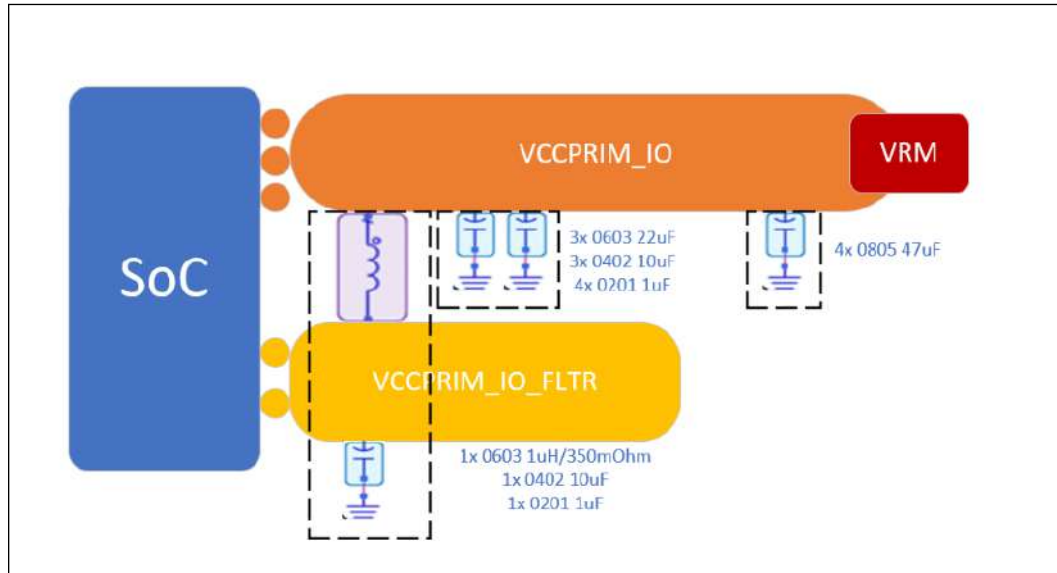


Figure 545. VCCPRIM_IO_4



11.1.1.2.11 VCCPRIM_1P8_PROC

Table 822. Reference Layout Design

Index	Design Note	Reference Design
1	Route the VCCPRIM_1P8_PROC, VCCPRIM_1P8_PROC_FLTRA and VCCPRIM_1P8_PROC_FLTRB on Layer 4 & Layer 6, with immediate ground reference. VCCPRIM_1P8_PROC_FLTRA should be merged with VCCPRIM_1P8_PROC after LC filter on Layer 1.	VCCPRIM_1P8_PROC_1
2	VCCPRIM_1p8_FLTRB should be merged with VCCPRIM_1p8 on Layer4 after ~15mm from VCCPRIM_1p8_BGAs.	VCCPRIM_1P8_PROC_2

Figure 546. VCCPRIM_1P8_PROC_1

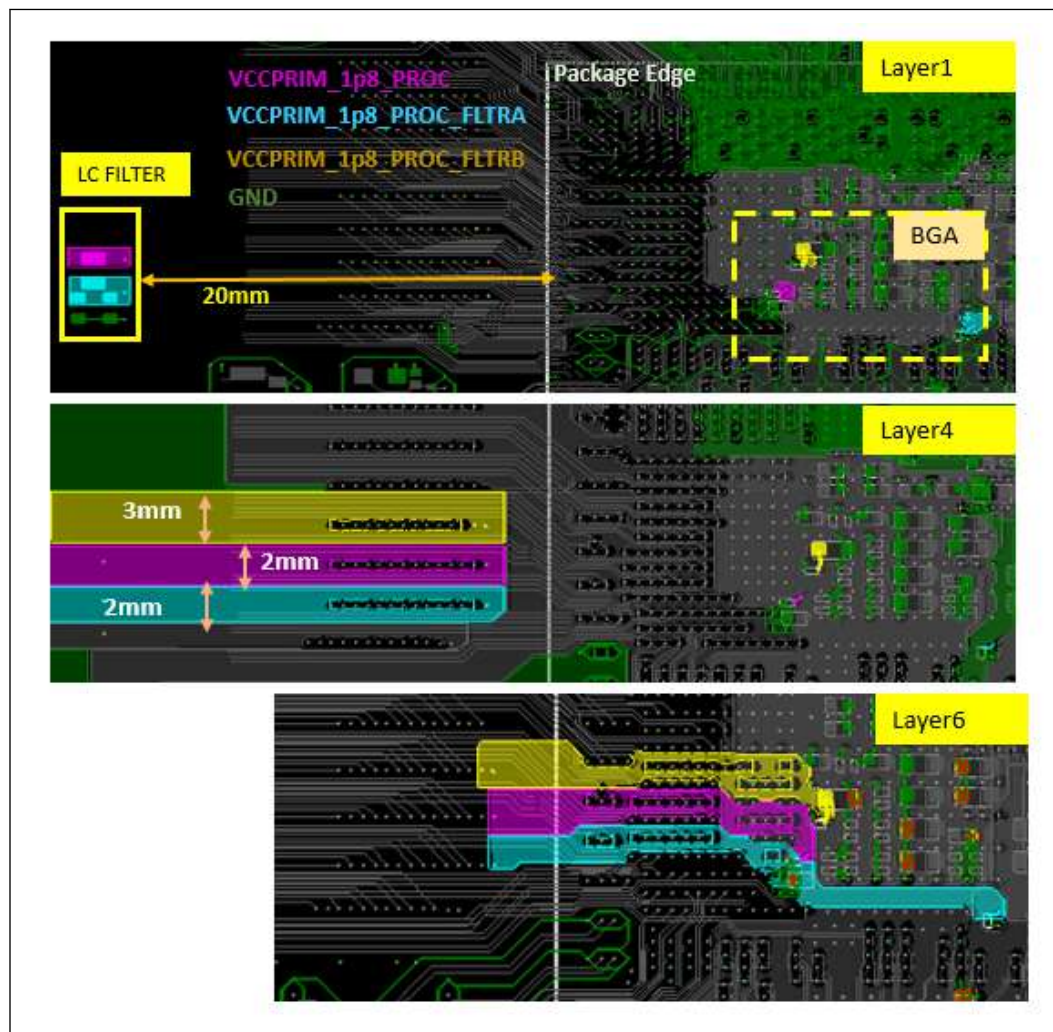


Figure 547. VCCPRIM_1P8_PROC_2



Table 823. Notes

Note
Power and ground via pairs should be placed close to capacitor pads to provide proper vertical connections when placing capacitor on the secondary side.
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 824. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side (Inductor)	0805 / 0603	1uH	1	Inductor component of the LC filter from VCCPRIM_1P8_PROC rail to VCCPRIM_1P8_PROC_FLTRA rail. It must meet electrical requirements: max DCR < 100mOhm and rated current >200 mA.	VCCPRIM_1P8_PROC_2
Primary side	0603	22uF	2	Decoupling capacitor(C1,C2) placed on VCCPRIM_1P8_PROC_FLTRA rail for LC filtering purpose.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C3 under VCCPRIM_1P8_PROC_FLTRA BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0603	Placeholder	1	Place C4 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C5 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0402 / 0201	Placeholder	1	Place C6 under VCCPRIM_1P8_PROC BGAs	VCCPRIM_1P8_PROC_2

Figure 548. VCCPRIM_1P8_PROC_2



11.1.1.2.12 VCCRTC_PROC

Table 825. Reference Layout Design

Index	Design Note	Reference Design
1	Require 1 layer of VCCRTC_PROC plane from vrm to board pin with immediate GND reference on one of the adjacent layers. Refer to reference design image for design implementation.	VCCRTC_PROC_1

Figure 549. VCCRTC_PROC_1

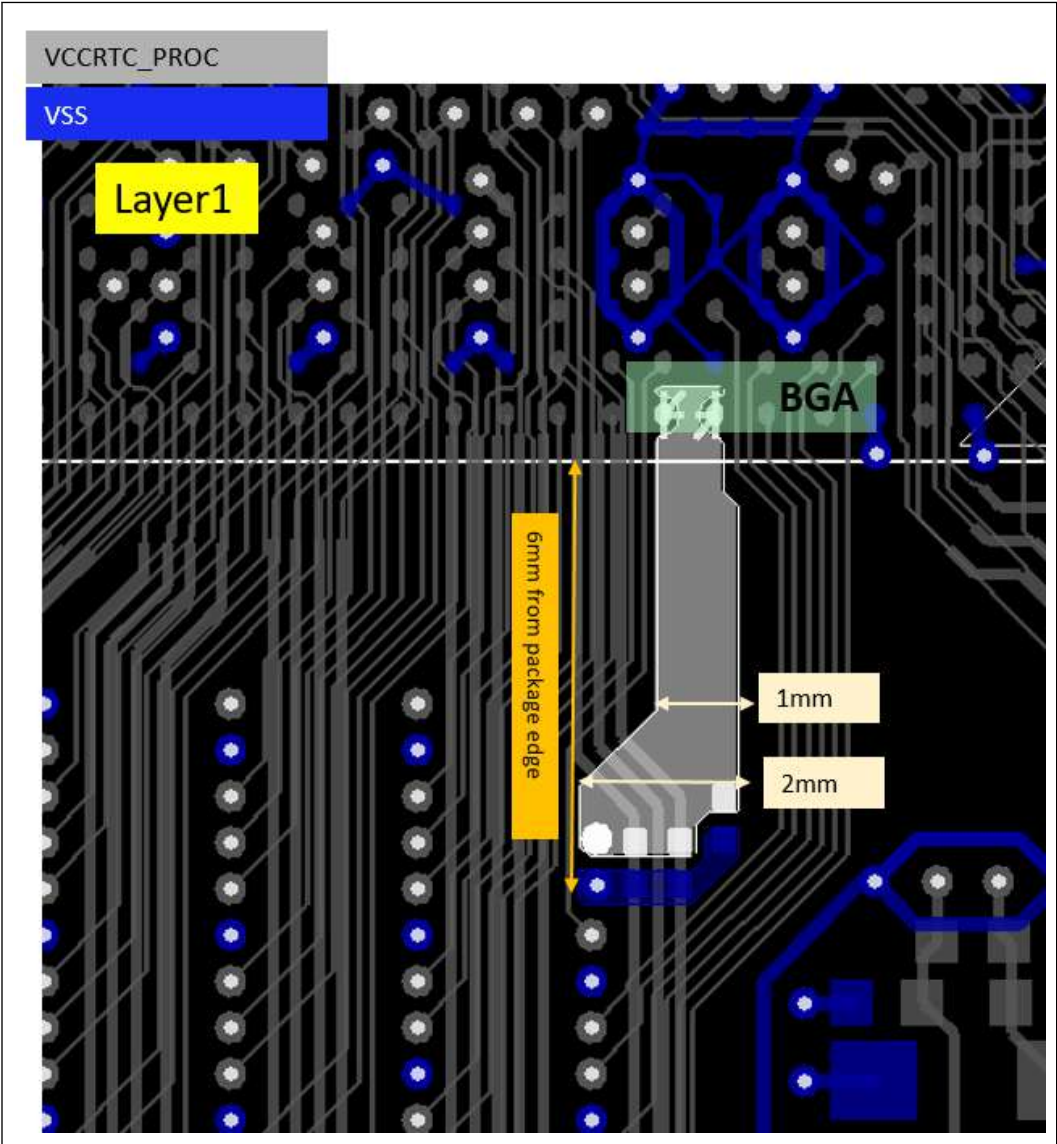


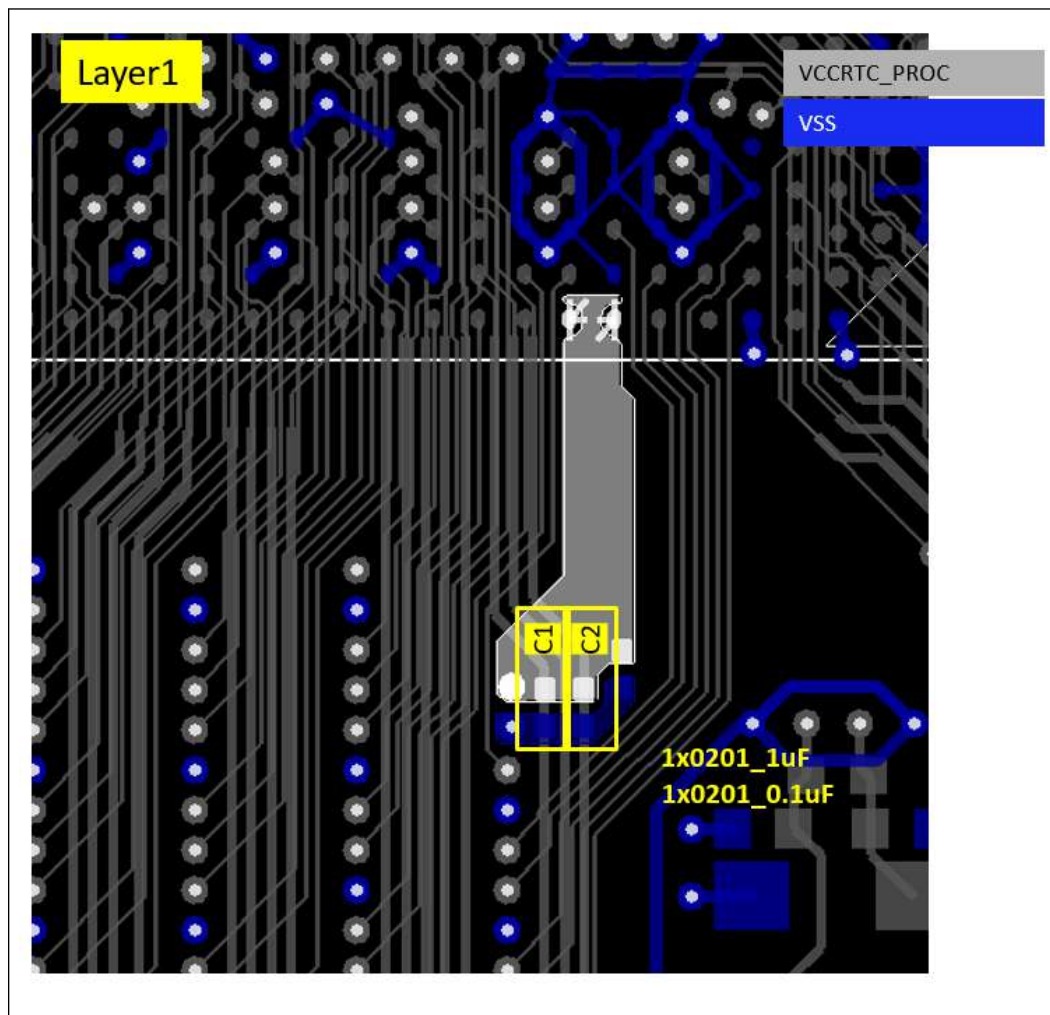
Table 826. Notes

Note
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 827. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	0201	1 uF	1	Place C1 within 6mm from package edge.	VCCRTC_PROC_2
Primary side	0402 / 0201	0.1 uF	1	Place C2 within 6mm from package edge.	VCCRTC_PROC_2

Figure 550. VCCRTC_PROC_2



11.1.1.2.13 VDD2 LPDDR5x MEMORY DOWN (T3 PCB)

Table 828. Reference Layout Design

Index	Design Note	Reference Design
1	VDD2 Type-3 LP5 motherboard reference layout showing the plane routing	VDD2_T3_LP5_1
2	Adjacent solid ground to power planes for good return path	

Figure 551. VDD2_T3_LP5_1



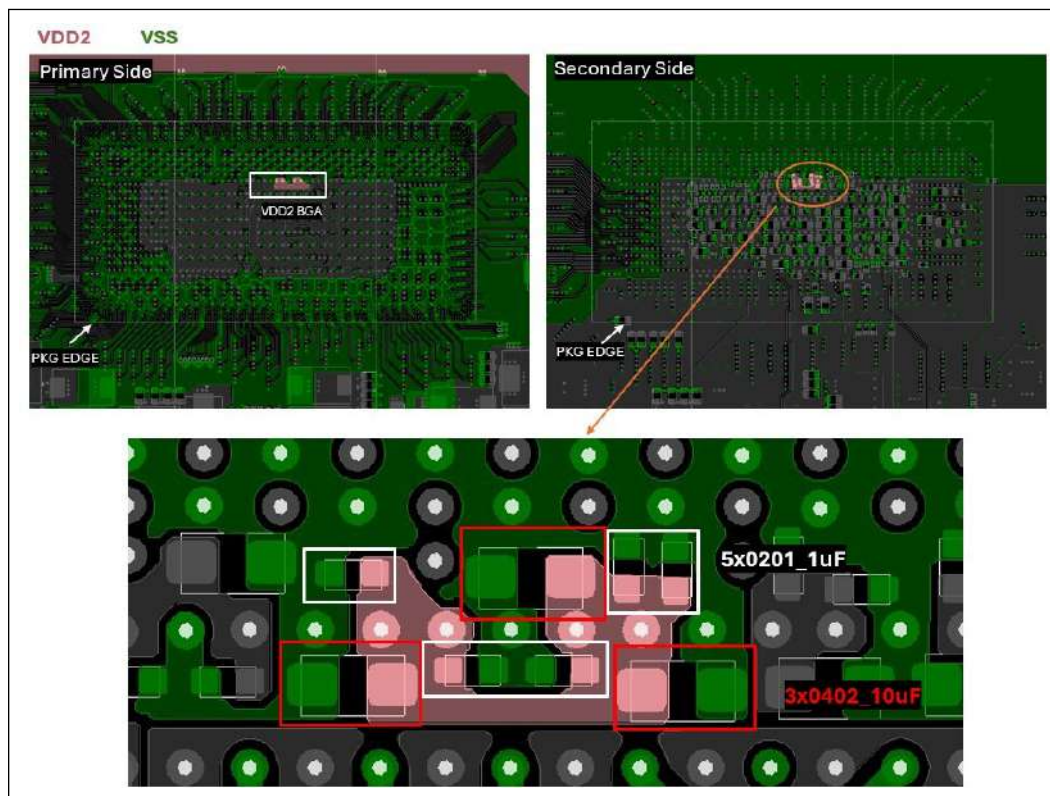
Table 829. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 830. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side	0201	1uF	5	Place beneath BGA	VDD2_T3_LP5_2
Secondary side	0402	10 uF	3	Place beneath BGA	VDD2_T3_LP5_2

Figure 552. VDD2_T3_LP5_2



11.1.1.2.14 VDDQ LPDDR5x MEMORY DOWN (T3 PCB)

Table 831. Reference Layout Design

Index	Design Note	Reference Design
1	VDDQ Type-3 LP5 motherboard reference layout showing the plane routing	VDDQ_T3_LP5_1
2	Adjacent solid ground to power planes for good return path	

Figure 553. VDDQ_T3_LP5_1

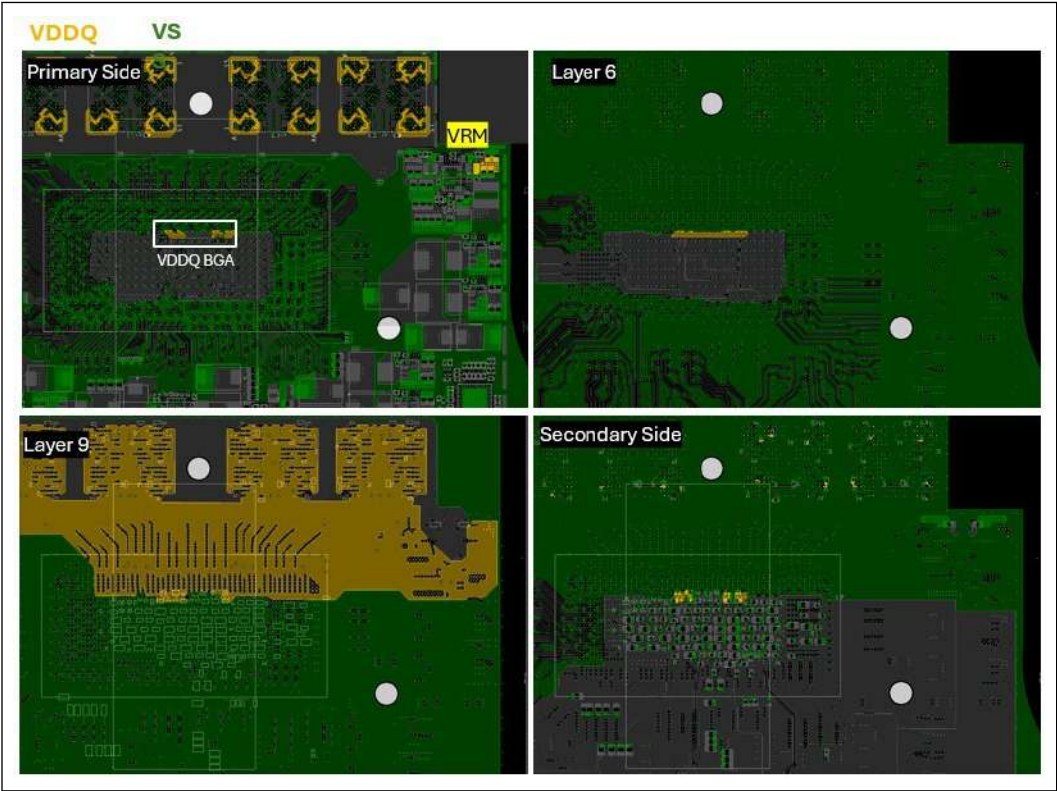


Table 832. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 833. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary Side	0201	1uF	8	Place beneath BGA	VDDQ_T3_LP5_2
Secondary Side	0402	10uF	4	Place beneath BGA	VDDQ_T3_LP5_2

Figure 554. VDDQ_T3_LP5_2



11.1.1.3 Type-4, 0.8mm, 10 Layer

PCB Type: Type-4

PCB Thickness: 0.8mm

PCB Layer Count: 10

PCB Assembly: Double Sided

11.1.1.3.1 VCC_CORE(H16C_12Xe/T4/0.8mm PCB)

Table 834. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_CORE_H16C_12Xe_T4_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	

Figure 555. VCC_CORE_H16C_12Xe_T4_0.8mm_1

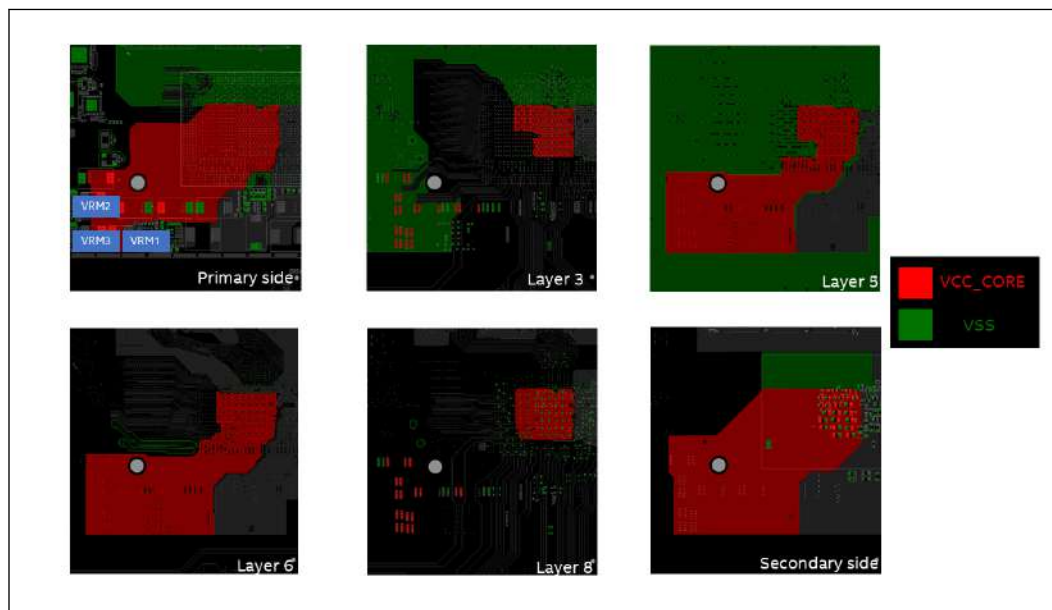


Table 835. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 836. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_CORE_H16C_12Xe_T4_0.8mm_2
Primary side	7343	Placeholder	1		VCC_CORE_H16C_12Xe_T4_0.8mm_2
Secondary side	0201	1 uF	5	Place beneath BGA.	VCC_CORE_H16C_12Xe_T4_0.8mm_3
Secondary side	0402	10 uF	8	Place beneath BGA.	VCC_CORE_H16C_12Xe_T4_0.8mm_3
continued...					

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side	0603	22 uF	1	Place beneath BGA.	VCC_CORE_H16C_12Xe_T4_0.8mm_3
Secondary side	0201	Placeholder	7		VCC_CORE_H16C_12Xe_T4_0.8mm_3
Secondary side	0402	Placeholder	2		VCC_CORE_H16C_12Xe_T4_0.8mm_3
Secondary side	0603	Placeholder	11		VCC_CORE_H16C_12Xe_T4_0.8mm_3

Figure 556. VCC_CORE_H16C_12Xe_T4_0.8mm_2

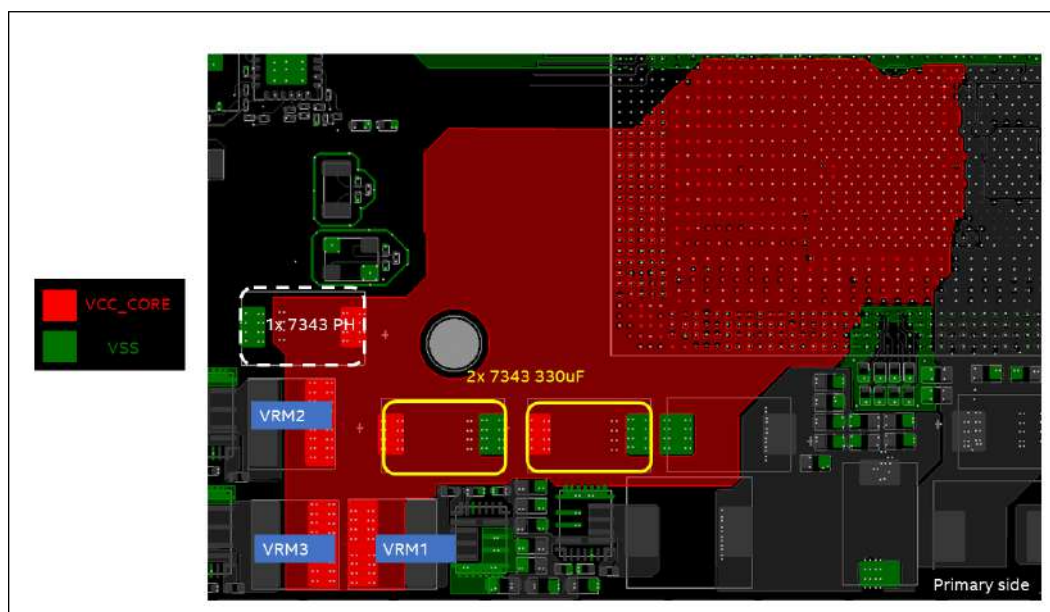
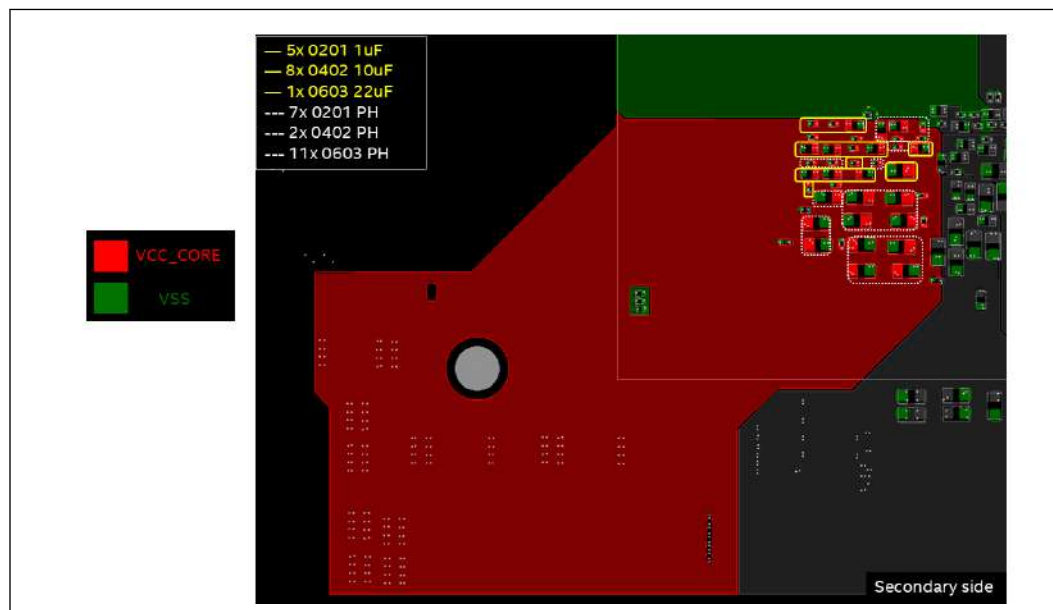


Figure 557. VCC_CORE_H16C_12Xe_T4_0.8mm_3



11.1.1.3.2 VCCGT(H16C_12Xe/T4/0.8mm PCB)

Table 837. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCGT_H16C_12Xe_T4_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	

Figure 558. VCCGT_H16C_12Xe_T4_0.8mm_1

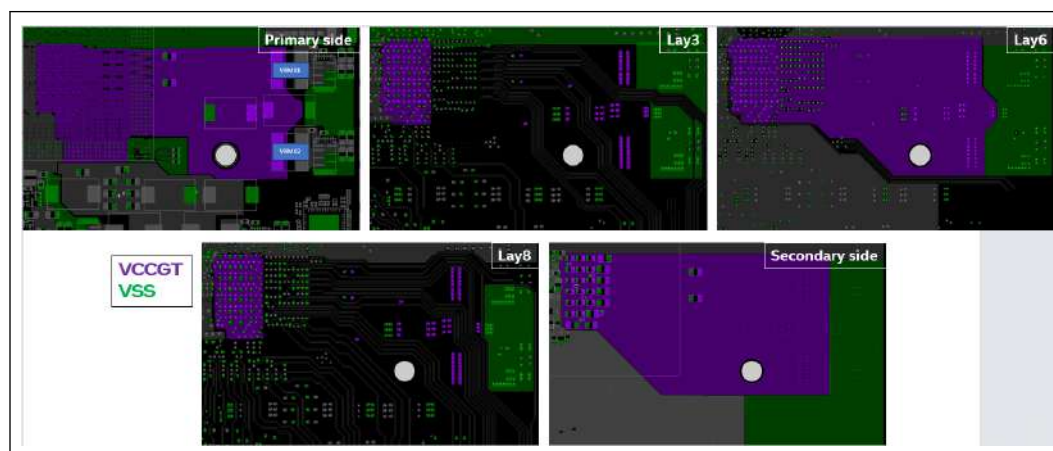


Table 838. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 839. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5 mOhm with 2.5V voltage rating.	VCCGT_H16C_12Xe_T4_0.8mm_2
Primary side	0603	Placeholder	2		VCCGT_H16C_12Xe_T4_0.8mm_2
Secondary side	0402	10 uF	11		VCCGT_H16C_12Xe_T4_0.8mm_3
Secondary side	0402	Placeholder	6		VCCGT_H16C_12Xe_T4_0.8mm_3
Secondary side	0603	Placeholder	10		VCCGT_H16C_12Xe_T4_0.8mm_3

Figure 559. VCCGT_H16C_12Xe_T4_0.8mm_2

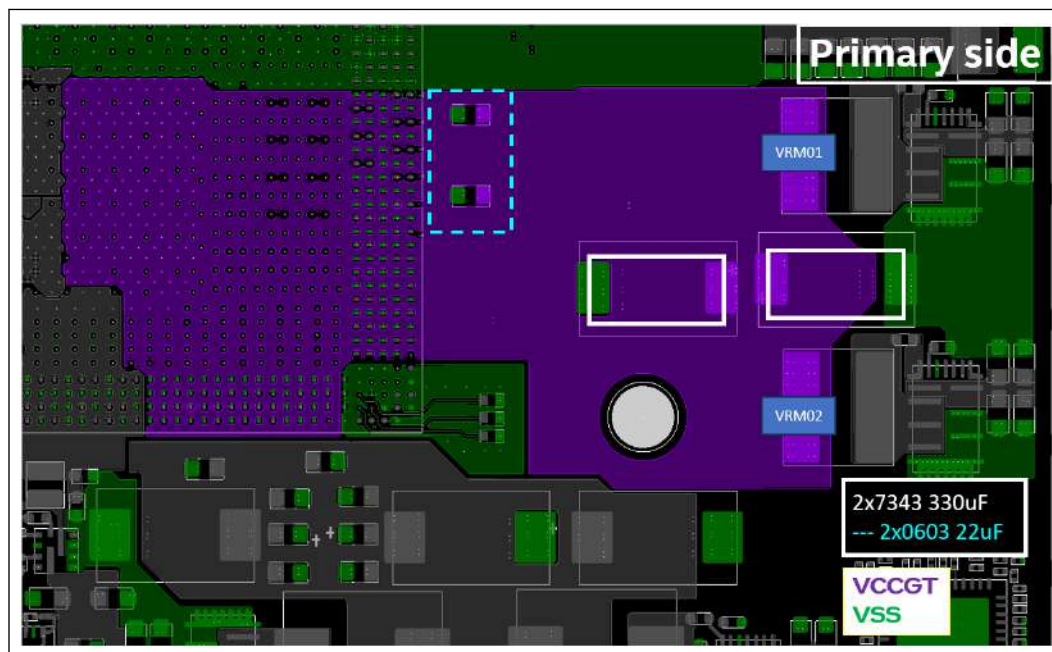
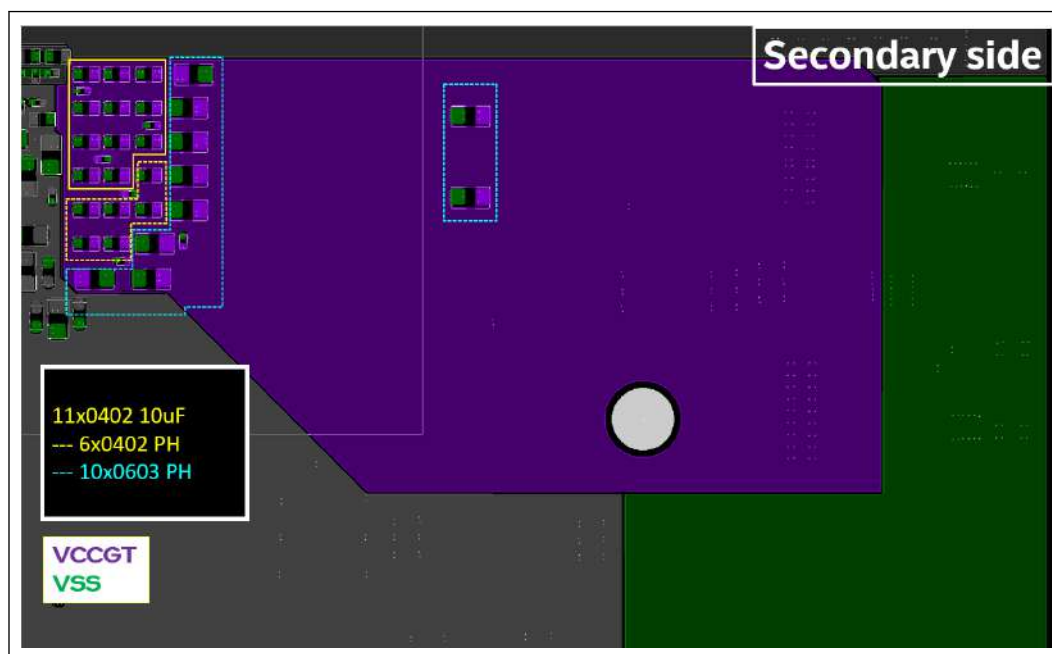


Figure 560. VCCGT_H16C_12Xe_T4_0.8mm_3



11.1.1.3.3 VCCSA(H16C_12Xe/T4/0.8mm PCB)

Table 840. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCSA_H16C_12Xe_T4_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	

Figure 561. VCCSA_H16C_12Xe_T4_0.8mm_1

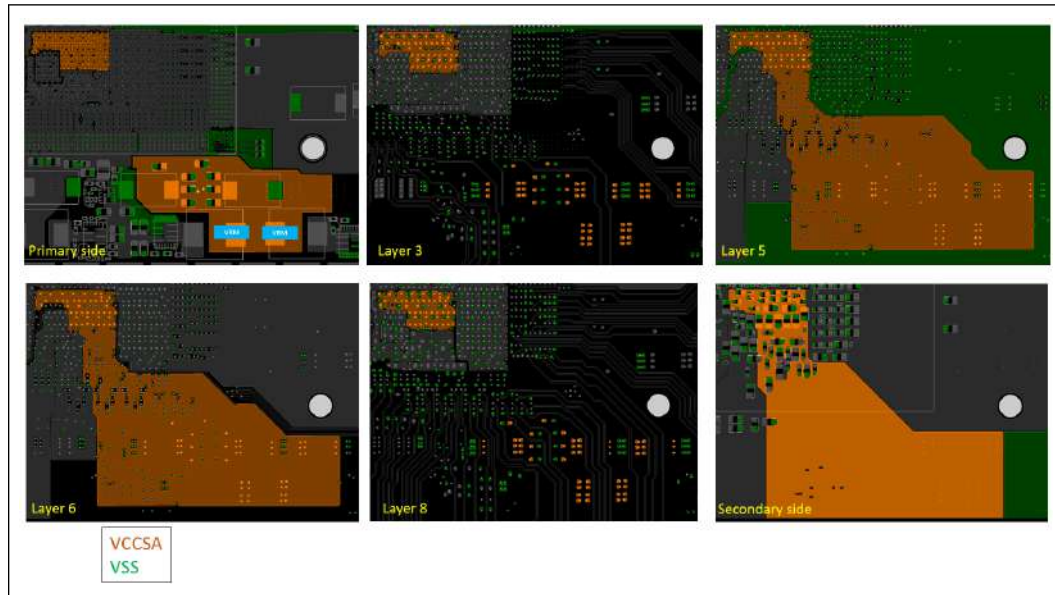


Table 841. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 842. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCSA_H16C_12Xe_T4_0.8mm_2
Primary side	0603	Placeholder	8		VCCSA_H16C_12Xe_T4_0.8mm_2
Secondary side	0402	10 uF	7	Place beneath BGA.	VCCSA_H16C_12Xe_T4_0.8mm_3
Secondary side	0603	22 uF	5	Place beneath BGA.	VCCSA_H16C_12Xe_T4_0.8mm_3
Secondary side	0402	Placeholder	3		VCCSA_H16C_12Xe_T4_0.8mm_3
Secondary side	0603	Placeholder	3		VCCSA_H16C_12Xe_T4_0.8mm_3

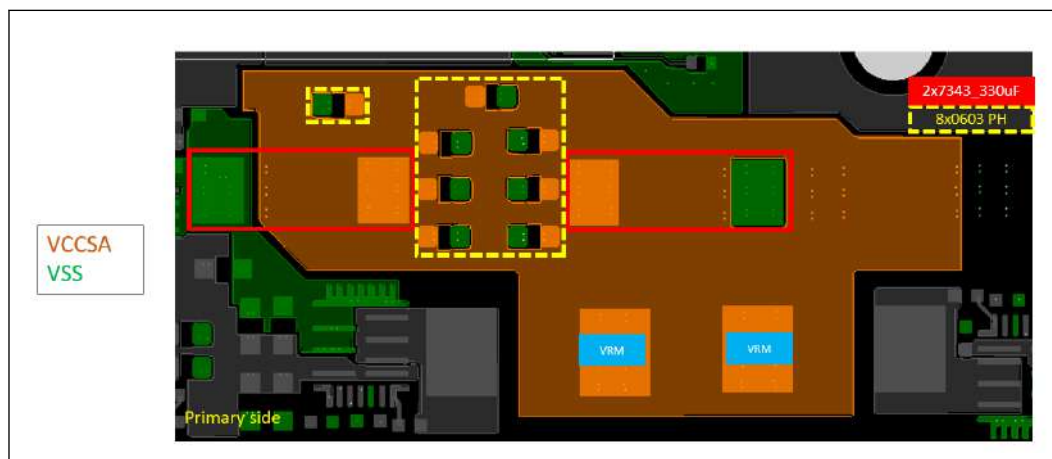
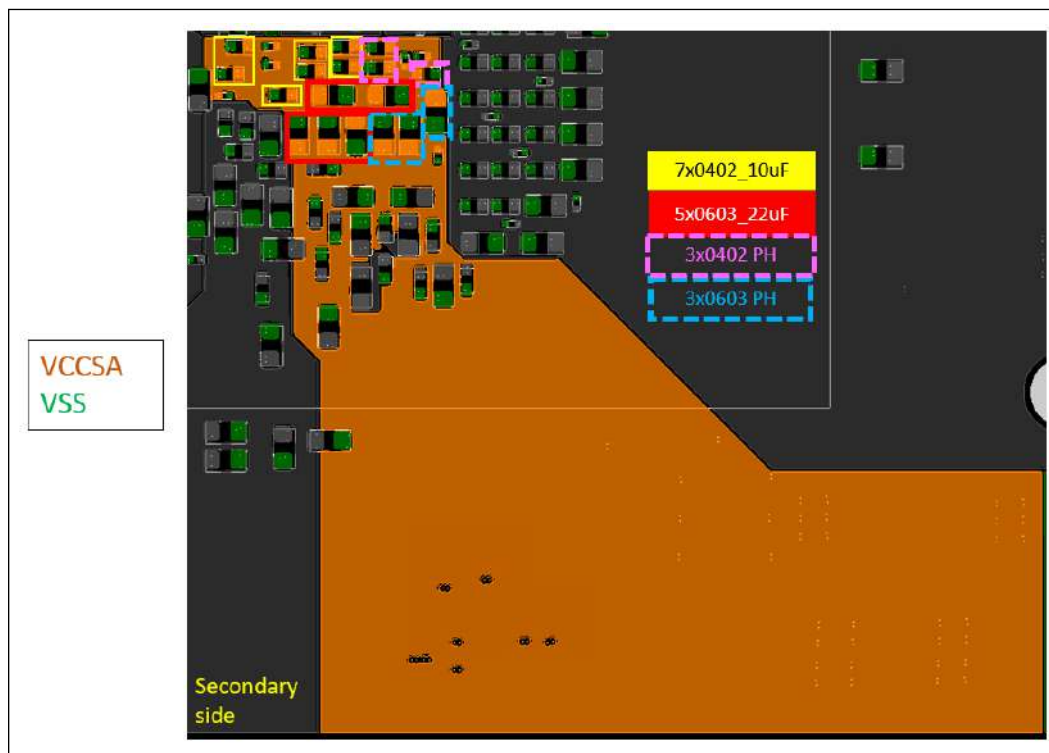
Figure 562. VCCSA_H16C_12Xe_T4_0.8mm_2

Figure 563. VCCSA_H16C_12Xe_T4_0.8mm_3



11.1.1.3.4 VCC_LP_ECORE(H16C_12Xe/T4/0.8mm PCB)

Table 843. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	

Figure 564. VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_1

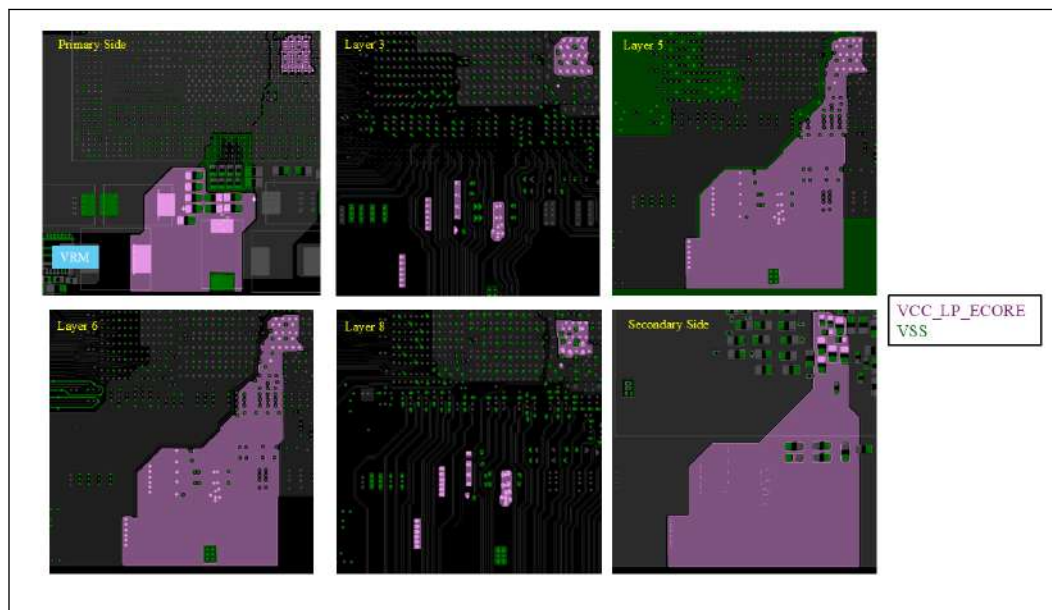


Table 844. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.
7. It is recommended to retain 20% of the existing placeholder capacitors in PDG.

Table 845. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	0603	22uF	1		VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_2
Primary side	0603	Placeholder	8		VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_2
Primary side	7343	330uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_2
continued...					

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	Placeholder	1	Place near VRM	VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_2
Secondary side	0402	10uF	4	Place beneath BGA	VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_3
Secondary side	0603	22uF	5	Place beneath BGA	VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_3

Figure 565. VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_2

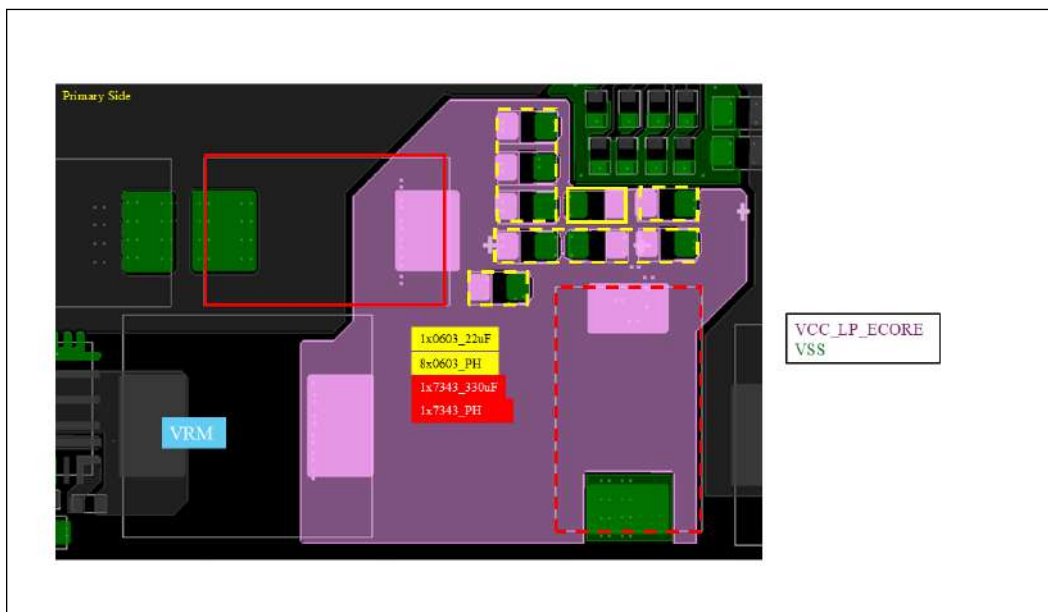
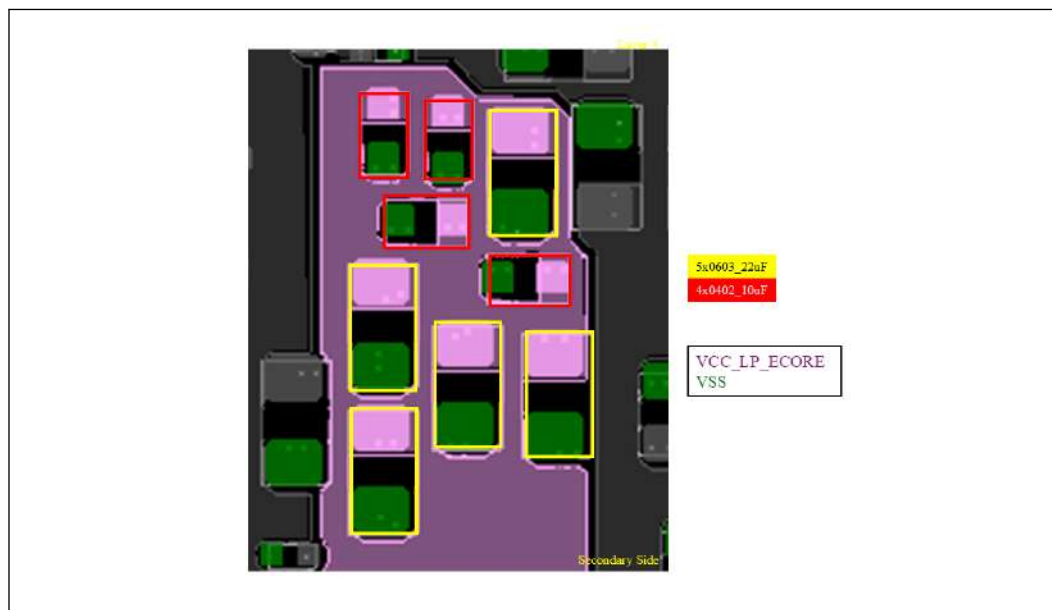


Figure 566. VCC_LP_ECORE_H16C_12Xe_T4_0.8mm_3



11.1.1.3.5 VCCPRIM_VNNAON (T4 PCB)

Table 846. Reference Layout Design

Index	Design Note	Reference Design
1	Design at least two solid and wide power planes as continuous current path from VR output all the way to BGAs. Advise to use 30um copper thickness layer for these two main routing layers. Ensure solid ground reference plane in nearby layers. Connect BGA pins when possible	VCCPRIM_VNNAON_1, VCCPRIM_VNNAON_2
2	The VCCPRIM_VNNAON_FLTR rail require very clean supply, isolated from VCCPRIM_VNNAON in package and advise merge back to VCCPRIM_VNNAON only on board secondary side through an LC filter. VCCPRIM_VNNAON has two power pins CM16 and CM18.	VCCPRIM_VNNAON_3
3	VCCPRIM_VNNAON VRM uses differential sensing BP14 (SENSE_VCCPRIM_VNNAON) and BP12(SENSE_VSS_VCCPRIM_VNNAON).	
4	Maximum Rpath from VRM output inductor location to VCCPRIM_VNNAON BGA groups is 3.5mOhm.	
5	Yellow dotted circles highlights minimum count of VCCPRIM_VNNAON PTH recommended in bga region to prevent current overstress in PTHs. Ensure GND PTHs neatly placed close to PWR PTHs as shown in illustration to minimize return current loop	VCCPRIM_VNNAON_6

Figure 567. VCCPRIM_VNNAON_1

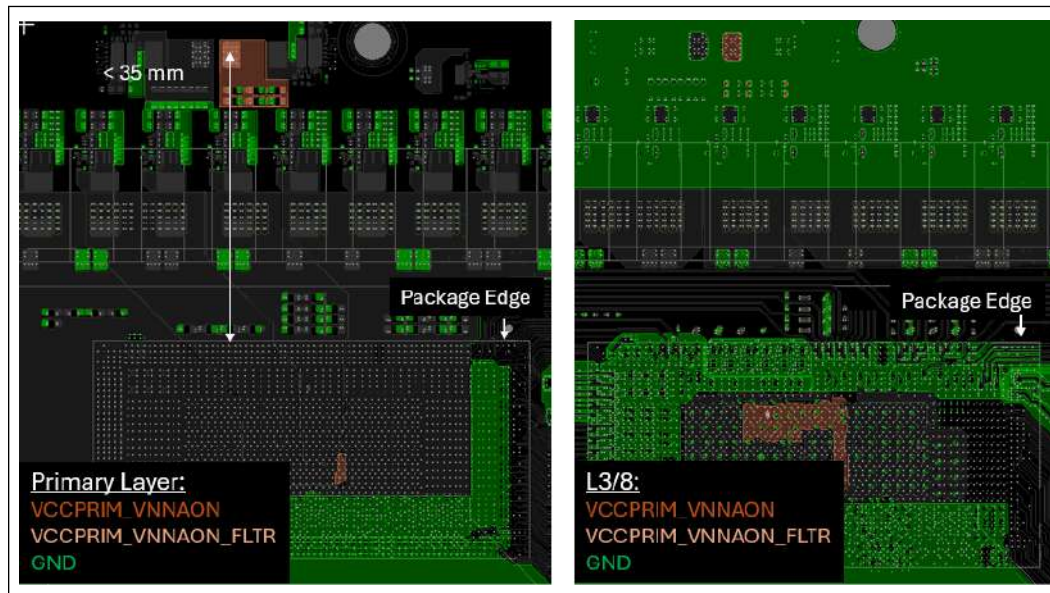


Figure 568. VCCPRIM_VNNAON_2

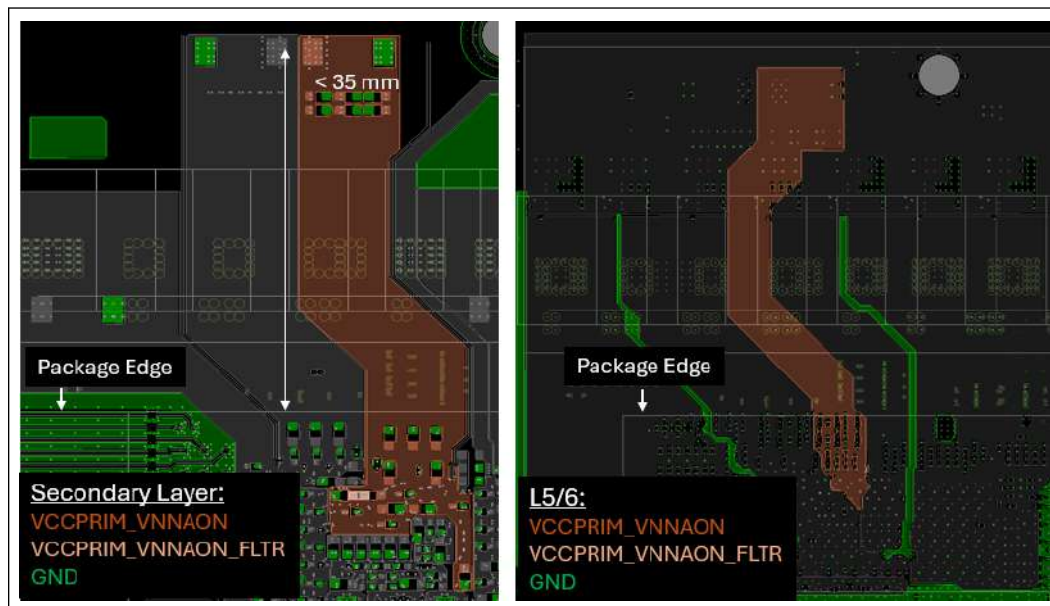


Figure 569. VCCPRIM_VNNAON_3

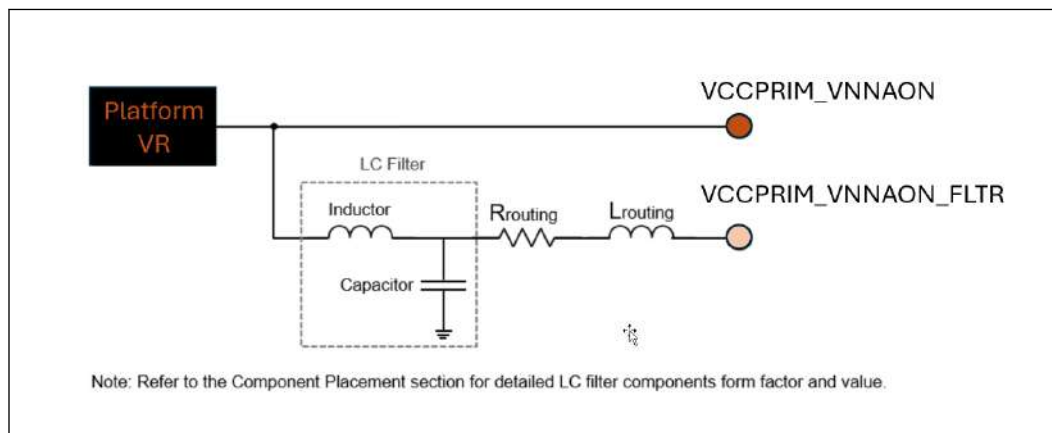


Figure 570. VCCPRIM_VNNAON_6

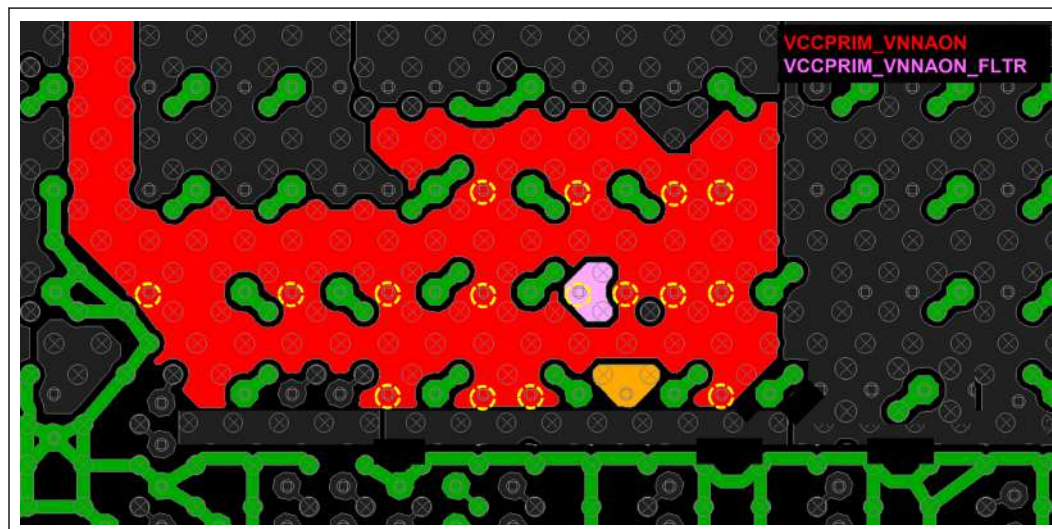


Table 847. Notes

Note
Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
Heighten capacitor effectiveness by placing them from VRM output to package in the following order : 0402 → 0603 → 0805 → Bulk
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
When placing capacitors, align their pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor provided the caps meet the recommended parasitic capacitance.

Table 848. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	7343	330uF	1	Place near VRM. ESR = 4.5 mΩ	VCCPRIM_VNNAON_4
Primary/Secondary Side	0603	22uF	12	Place near VRM	VCCPRIM_VNNAON_4, VCCPRIM_VNNAON_5
Secondary side	0603	47uF	3	Place 3 x 0603 47uF caps under VCCPRIM_VNNAON BGA region. One of the cap place need to place in processor die area.	VCCPRIM_VNNAON_5
Secondary Side	0603	22uF	5	Place 5 x 0603 22uF with distance <10mm from Package edge.	VCCPRIM_VNNAON_5
Secondary Side	0402	10uF	4	Place 4 x 0402 10uF caps under VCCPRIM_VNNAON BGA region.	VCCPRIM_VNNAON_5
Secondary Side	0603	22uF	1	Capacitor of LC filter	VCCPRIM_VNNAON_5
Secondary Side	0402/0603	1 uH_120mΩ DCR	1	Inductor of LC filter. Electrical requirement of the inductor: rated current > 200mA, DCR < 0.12Ω	VCCPRIM_VNNAON_5

Figure 571. VCCPRIM_VNNAON_4

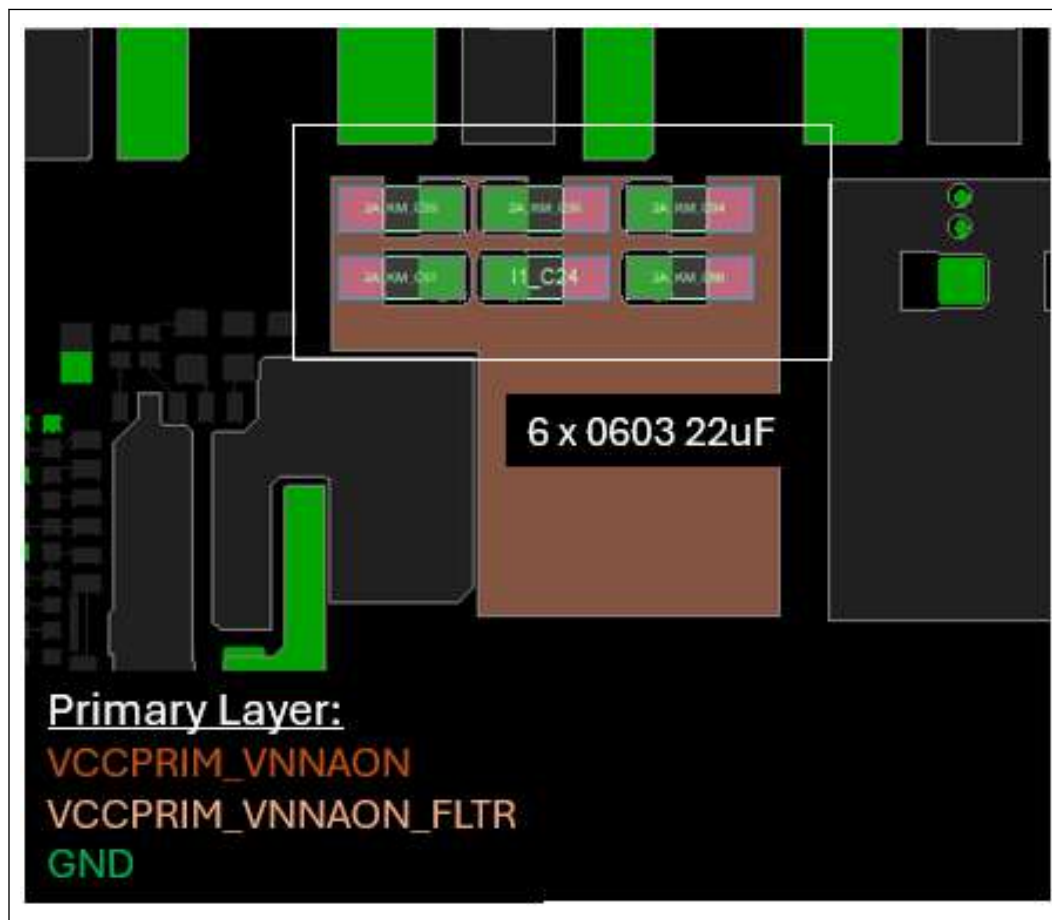
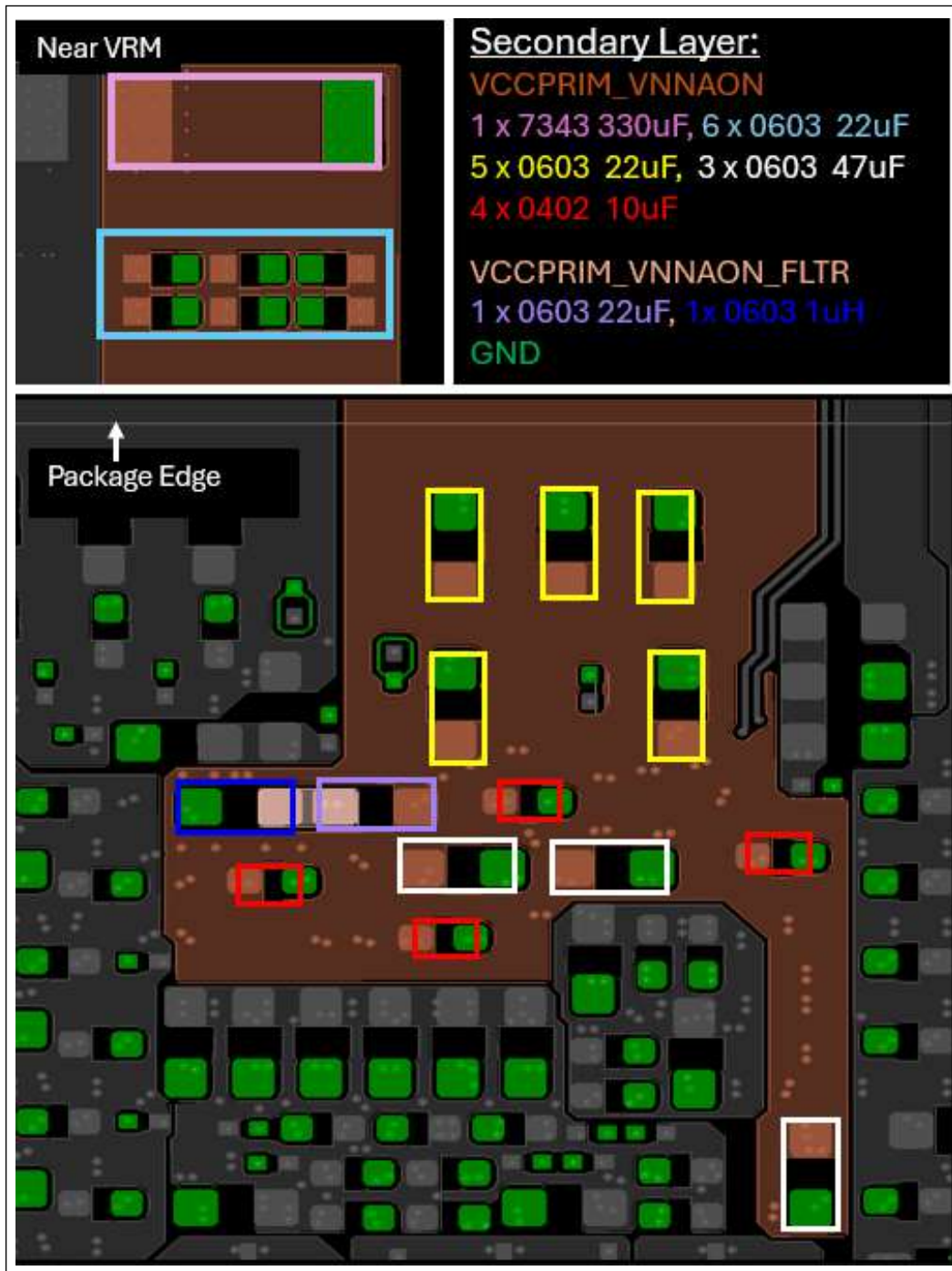


Figure 572. VCCPRIM_VNNAON_5



11.1.1.3.6 VCCPRIM_IO (T4 PCB)

Table 849. Reference Layout Design

Index	Design Note	Reference Design
1	Route VCCPRIM_IO power plane from VR output inductor to the BGA on Layer 1, Layer 3, Layer 8 and Layer 10. Adjacent layer from VCCPRIM_IO power plane should be ground.	VCCPRIM_IO_1
2	VCCPRIM_IO_FLTR should be merged with VCCPRIM_IO after LC filter on Layer 10	VCCPRIM_IO_1
3	VCCPRIM_IO VRM uses differential sensing. The power sense line is connected to SENSE_VCCPRIM_IO BGA(DF8) and ground sense line connected to SENSE_VSS_VCCPRIM_IO(DF9)	VCCPRIM_IO_2
4	The SVID 1.25 V supply for pull up resistors should be sourced from VCCPRIM_IO_OUT_SVID BGA EP5.	

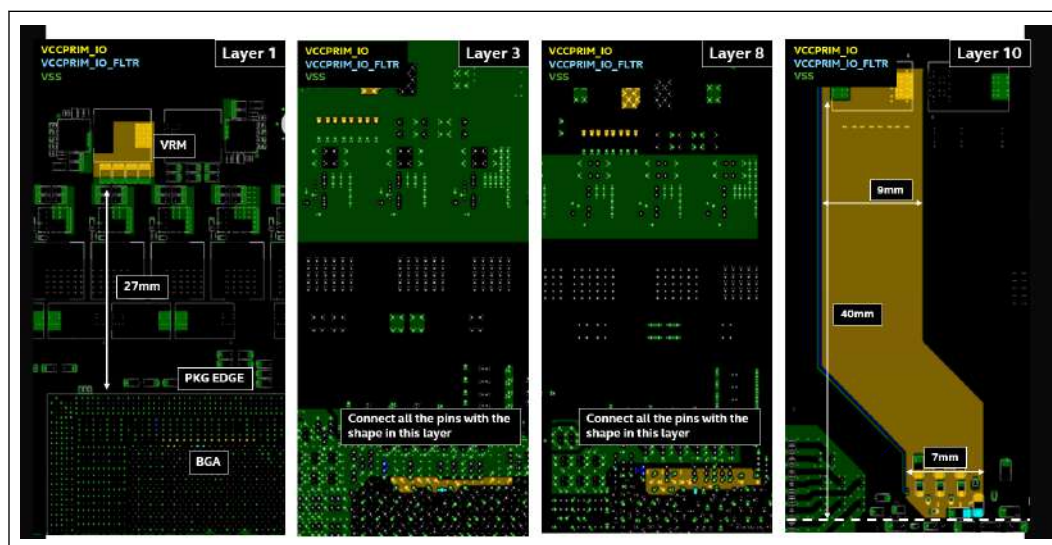
Figure 573. VCCPRIM_IO_1

Figure 574. VCCPRIM_IO_2

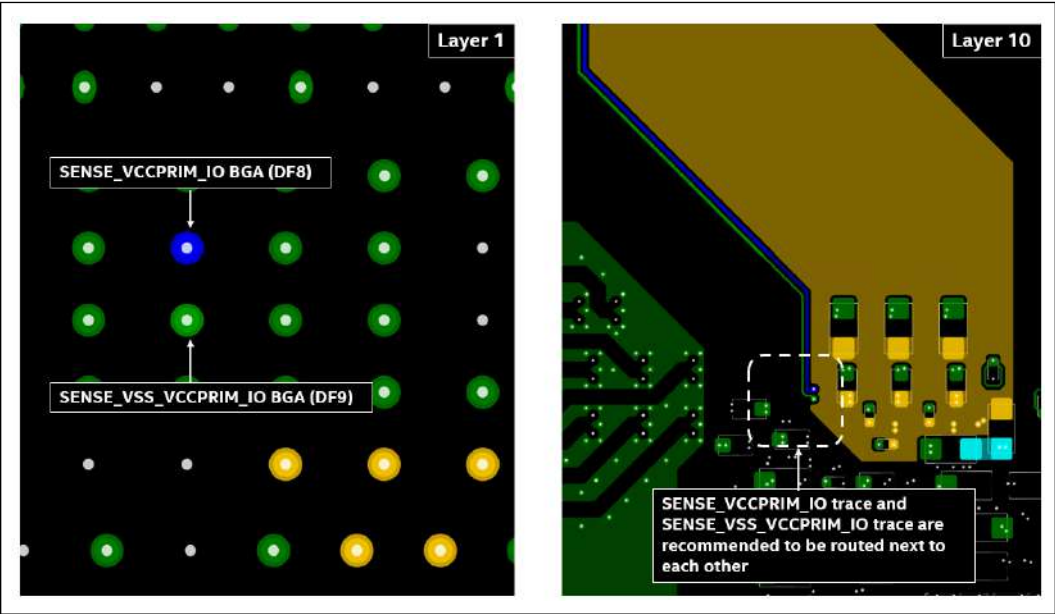


Table 850. Notes

Note
Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
Heighten capacitor effectiveness by placing them from VRM to package in the following order : 0402 → 0603 → 0805 → Bulk
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
Align capacitor pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 851. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	0805	47uF	4	Place near the VCCPRIM_IO VRM output	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0603	22 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402	10 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	4	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
continued...					

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side (Inductor)	0603	1 uH	1	Inductor component of the LC filter from VCCPRIM_IO_FLTR rail with ESR = 350mΩ	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402/0603	22 uF	1	Decoupling capacitor placed on VCCPRIM_IO_FLTR rail for LC filtering purpose	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	1	Place right below the VCCPRIM_IO_FLTR BGA	VCCPRIM_IO_3,VCCPRIM_IO_4

Figure 575. VCCPRIM_IO_3

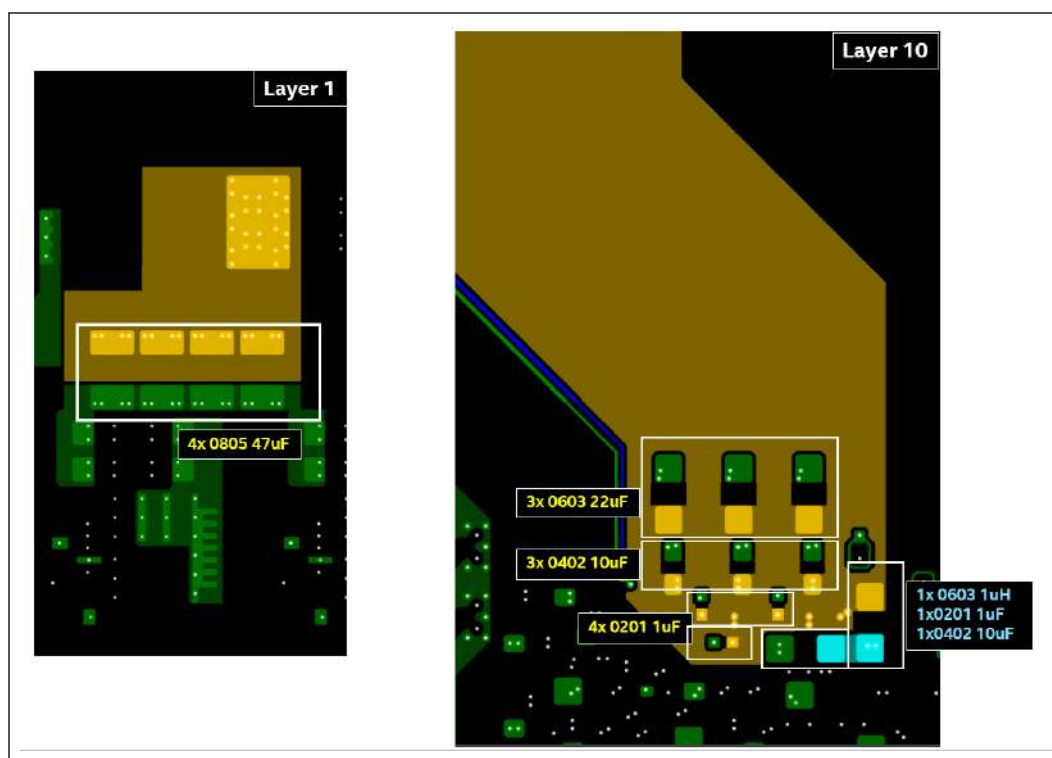
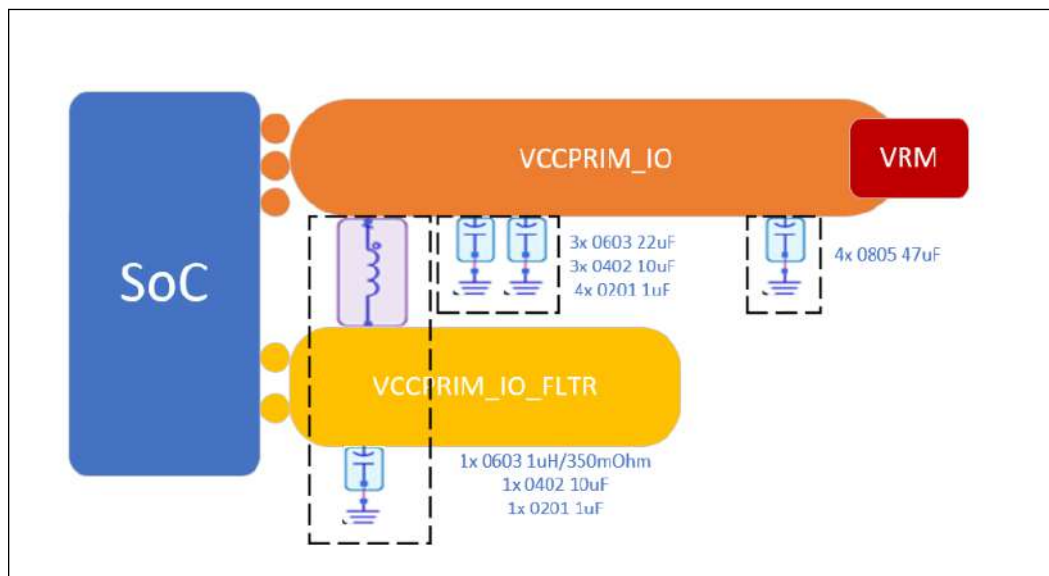


Figure 576. VCCPRIM_IO_4



11.1.1.3.7 VCCPRIM_1P8_PROC

Table 852. Reference Layout Design

Index	Design Note	Reference Design
1	Route the VCCPRIM_1P8_PROC, VCCPRIM_1P8_PROC_FLTRA and VCCPRIM_1P8_PROC_FLTRB on Layer 4 & Layer 6, with immediate ground reference. VCCPRIM_1P8_PROC_FLTRA should be merged with VCCPRIM_1P8_PROC after LC filter on Layer 1.	VCCPRIM_1P8_PROC_1
2	VCCPRIM_1p8_FLTRB should be merged with VCCPRIM_1p8 on Layer4 after ~15mm from VCCPRIM_1p8_BGAs.	VCCPRIM_1P8_PROC_2

Figure 577. VCCPRIM_1P8_PROC_1

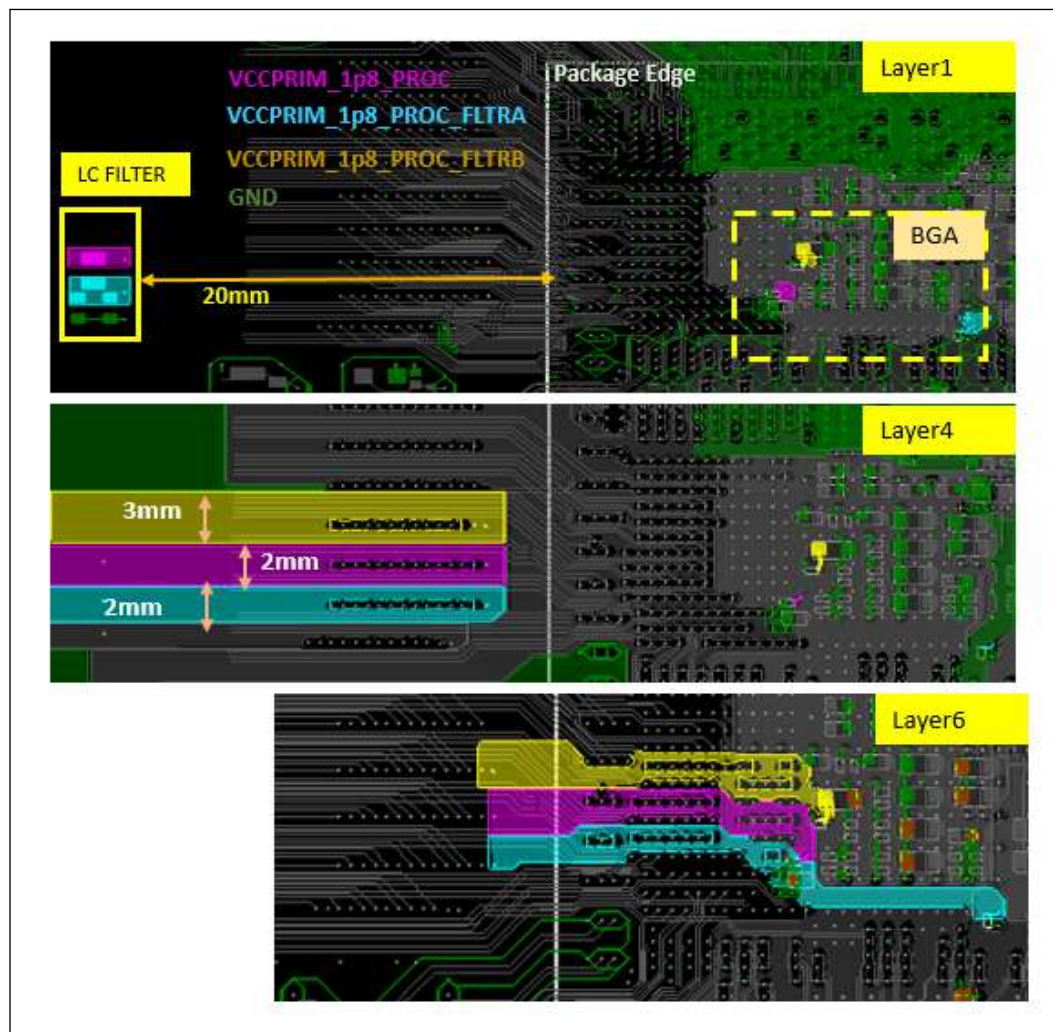


Figure 578. VCCPRIM_1P8_PROC_2



Table 853. Notes

Note
Power and ground via pairs should be placed close to capacitor pads to provide proper vertical connections when placing capacitor on the secondary side.
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 854. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side (Inductor)	0805 / 0603	1uH	1	Inductor component of the LC filter from VCCPRIM_1P8_PROC rail to VCCPRIM_1P8_PROC_FLTRA rail. It must meet electrical requirements: max DCR < 100mOhm and rated current >200 mA.	VCCPRIM_1P8_PROC_2
Primary side	0603	22uF	2	Decoupling capacitor(C1,C2) placed on VCCPRIM_1P8_PROC_FLTRA rail for LC filtering purpose.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C3 under VCCPRIM_1P8_PROC_FLTRA BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0603	Placeholder	1	Place C4 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C5 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0402 / 0201	Placeholder	1	Place C6 under VCCPRIM_1P8_PROC BGAs	VCCPRIM_1P8_PROC_2

Figure 579. VCCPRIM_1P8_PROC_2



11.1.1.3.8 VCCRTC_PROC

Table 855. Reference Layout Design

Index	Design Note	Reference Design
1	Require 1 layer of VCCRTC_PROC plane from vrm to board pin with immediate GND reference on one of the adjacent layers. Refer to reference design image for design implementation.	VCCRTC_PROC_1

Figure 580. VCCRTC_PROC_1

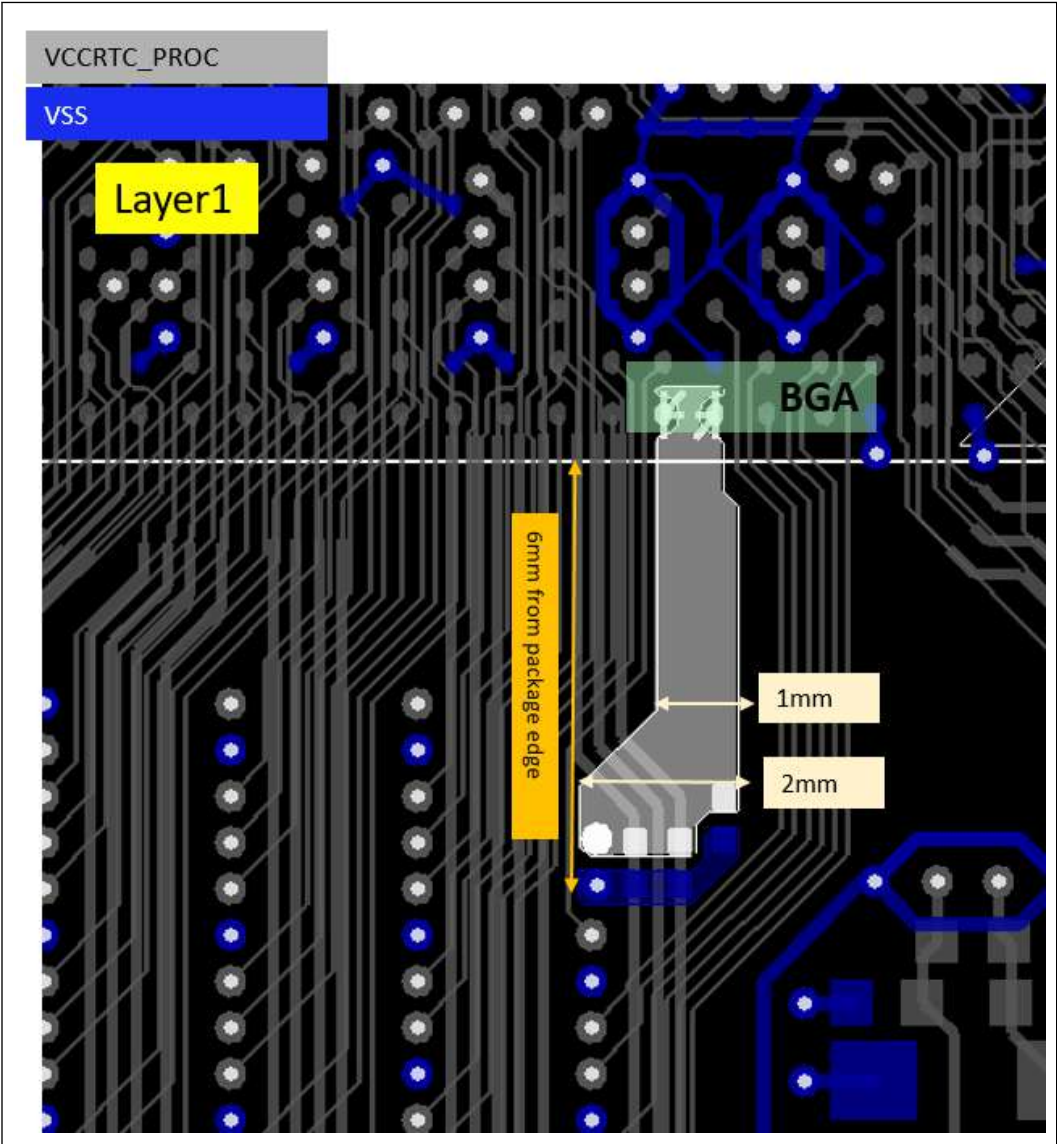


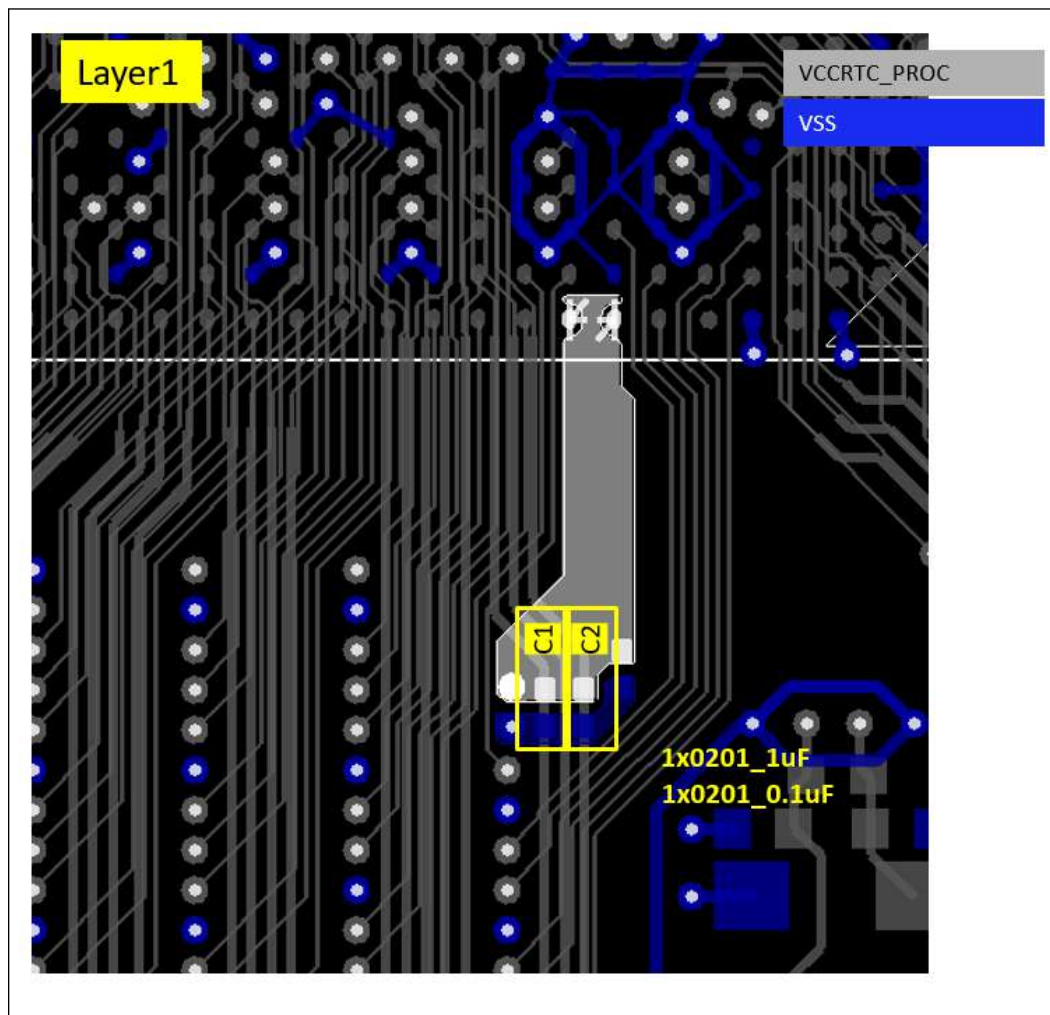
Table 856. Notes

Note
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 857. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	0201	1 uF	1	Place C1 within 6mm from package edge.	VCCRTC_PROC_2
Primary side	0402 / 0201	0.1 uF	1	Place C2 within 6mm from package edge.	VCCRTC_PROC_2

Figure 581. VCCRTC_PROC_2



11.1.1.3.9 VDD2 LPDDR5x MEMORY DOWN (T4 PCB)

Table 858. Reference Layout Design

Index	Design Note	Reference Design
1	VDD2 Type-4 LP5 motherboard reference layout showing the plane routing	VDD2_T4_LP5_1
2	Adjacent solid ground to power planes for good return path	

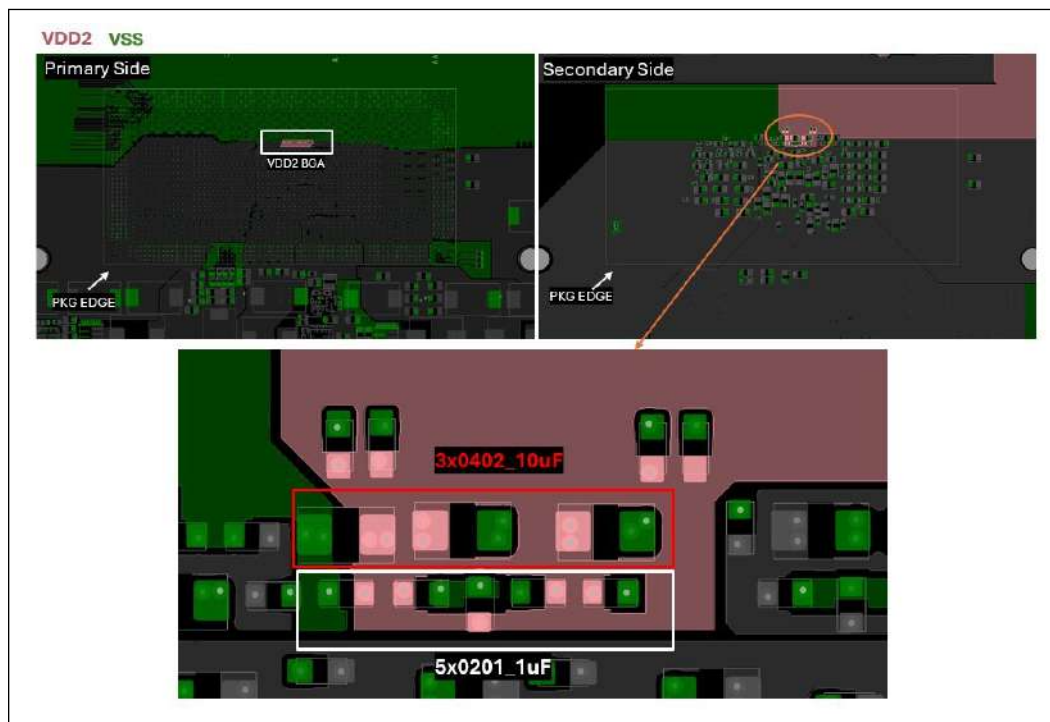
Figure 582. VDD2_T4_LP5_1**Table 859. Notes**

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 860. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side	0201	1uF	5	Place beneath BGA	VDD2_T4_LP5_2
Secondary side	0402	10 uF	3	Place beneath BGA	VDD2_T4_LP5_2

Figure 583. VDD2_T4_LP5_2



11.1.1.3.10 VDDQ LPDDR5x MEMORY DOWN (T4 PCB)

Table 861. Reference Layout Design

Index	Design Note	Reference Design
1	VDDQ Type-4 LP5 motherboard reference layout showing the plane routing	VDDQ_T4_LP5_1
2	Adjacent solid ground to power planes for good return path	

Figure 584. VDDQ_T4_LP5_1

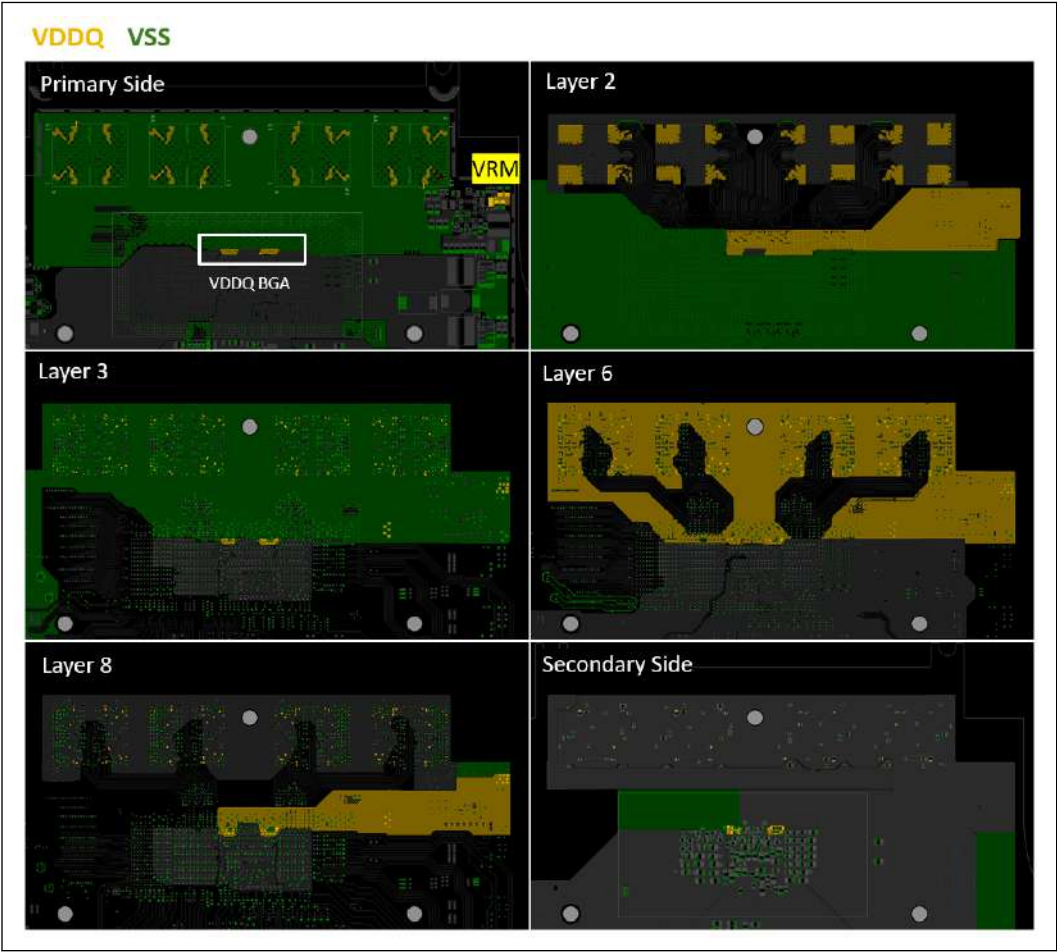


Table 862. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 863. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary Side	0201	1uF	8	Place beneath BGA	VDDQ_T4_LP5_2
Secondary Side	0402	10uF	4	Place beneath BGA	VDDQ_T4_LP5_2

Figure 585. VDDQ_T4_LP5_2



11.2 Processor Decoupling and Layout Recommendations for NVL 6C

11.2.1 Processor Power Rails

11.2.1.1 Type-3, 0.8mm, 8 Layer

PCB Type: Type-3

PCB Thickness: 0.8mm

PCB Layer Count: 8

PCB Assembly: Double Sided

11.2.1.1.1 VCC_CORE(6C/T3/0.8mm PCB)

Table 864. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_CORE_6C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 89 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_CORE_6C_T3_0.8mm_4

Figure 586. VCC_CORE_6C_T3_0.8mm_1

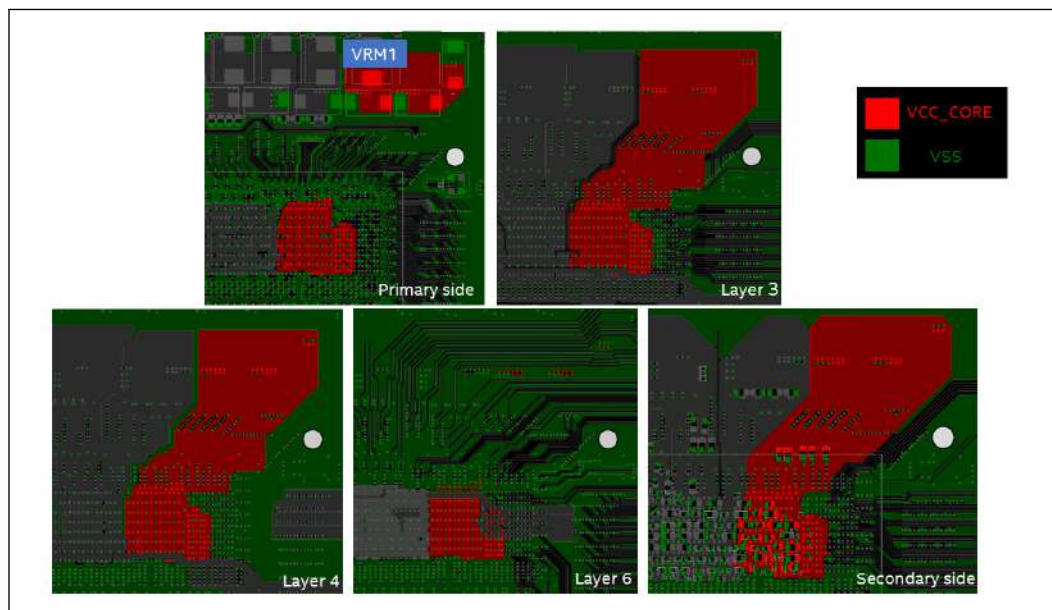


Figure 587. VCC_CORE_6C_T3_0.8mm_4

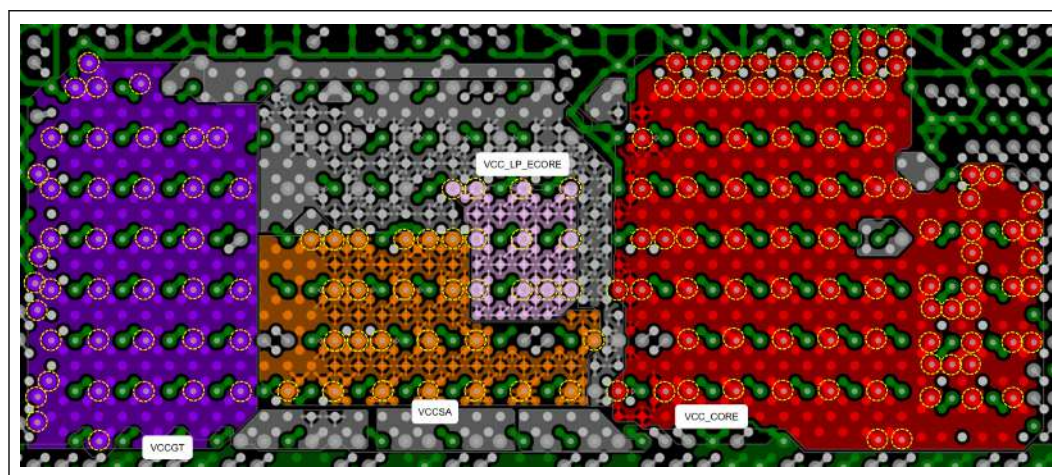


Table 865. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
<i>continued...</i>

Note
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.

Table 866. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_CORE_6C_T3_0.8mm_2
Primary side	7343	Placeholder	2		VCC_CORE_6C_T3_0.8mm_2
Secondary side	0201	1 uF	12	Place beneath BGA.	VCC_CORE_6C_T3_0.8mm_3
Secondary side	0402	10 uF	10	Place beneath BGA.	VCC_CORE_6C_T3_0.8mm_3
Secondary side	0603	22 uF	5	Place beneath BGA.	VCC_CORE_6C_T3_0.8mm_3
Secondary side	0603	Placeholder	7		VCC_CORE_6C_T3_0.8mm_3

Figure 588. VCC_CORE_6C_T3_0.8mm_2

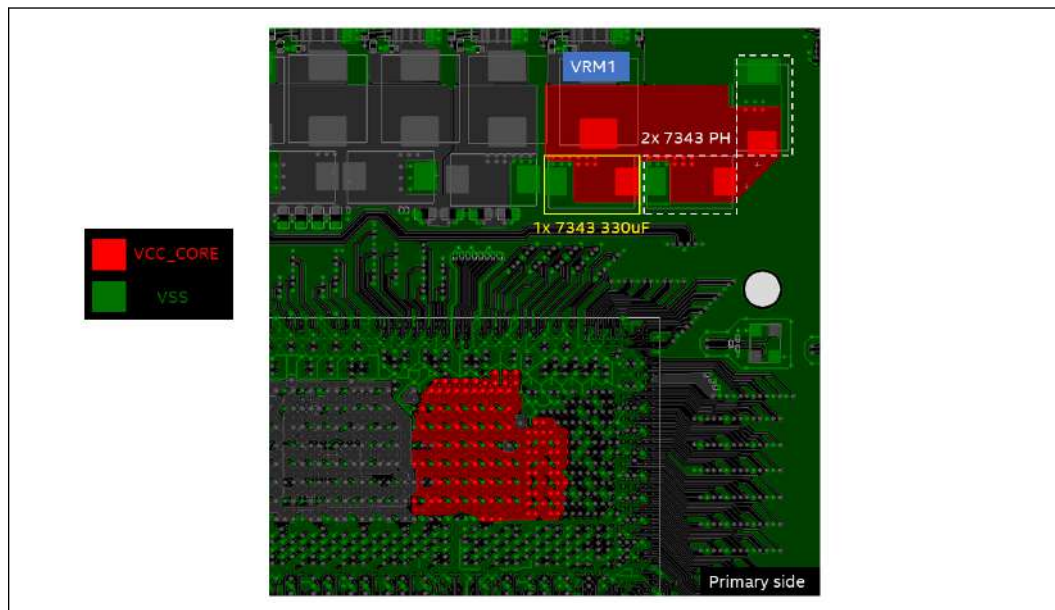


Figure 589. VCC_CORE_6C_T3_0.8mm_3



11.2.1.1.2 VCCGT(6C/T3/0.8mm PCB)

Table 867. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCGT_6C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 40 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCGT_6C_T3_0.8mm_4

Figure 590. VCCGT_6C_T3_0.8mm_1

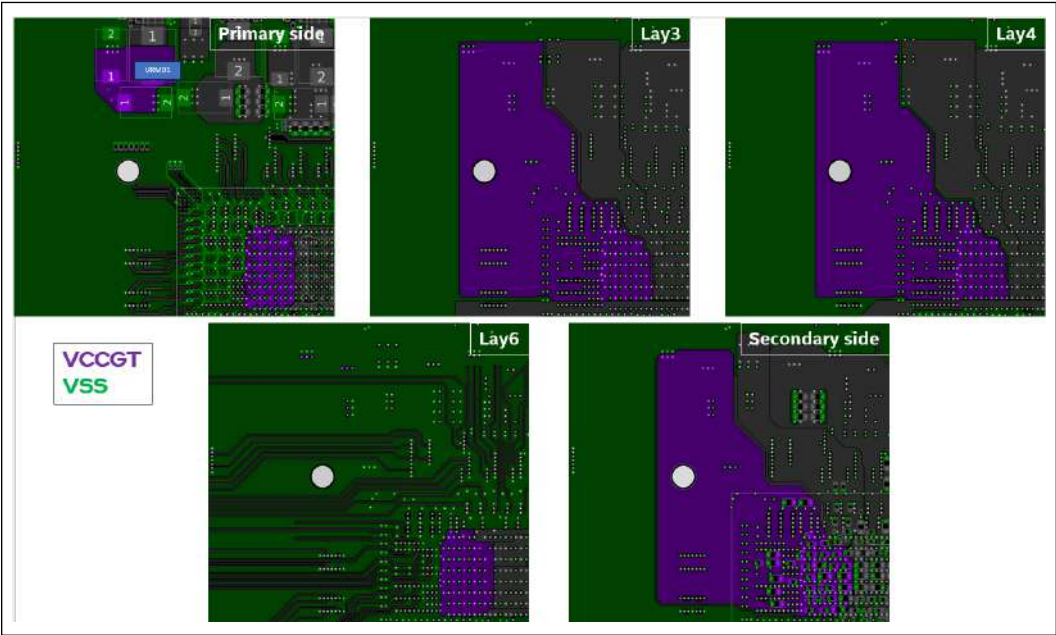


Figure 591. VCCGT_6C_T3_0.8mm_4

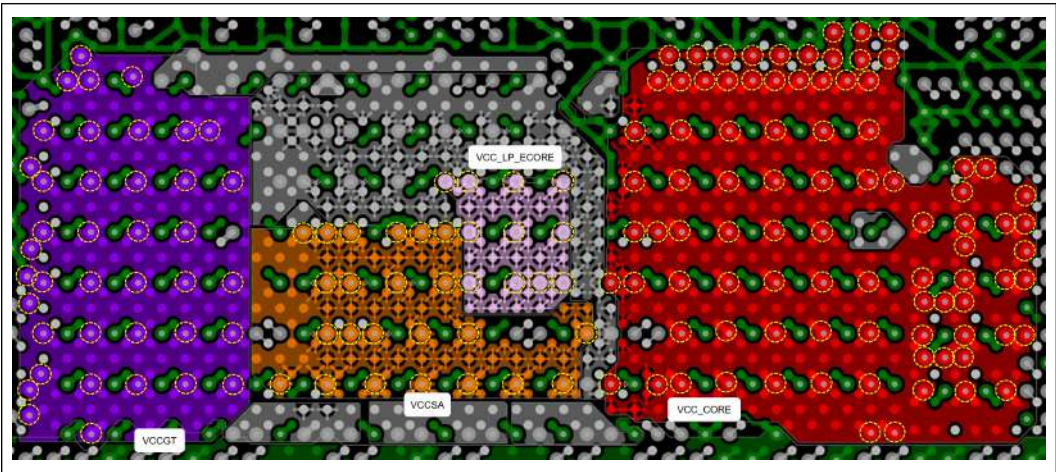


Table 868. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
<i>continued...</i>

Note
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.

Table 869. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330 uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCGT_6C_T3_0.8mm_2
Primary side	7343	Placeholder	1		VCCGT_6C_T3_0.8mm_2
Secondary side	0201	1uF	3		VCCGT_6C_T3_0.8mm_3
Secondary side	0402	10 uF	9		VCCGT_6C_T3_0.8mm_3
Secondary side	0201	Placeholder	7		VCCGT_6C_T3_0.8mm_3
Secondary side	0402	Placeholder	9		VCCGT_6C_T3_0.8mm_3

Figure 592. VCCGT_6C_T3_0.8mm_2

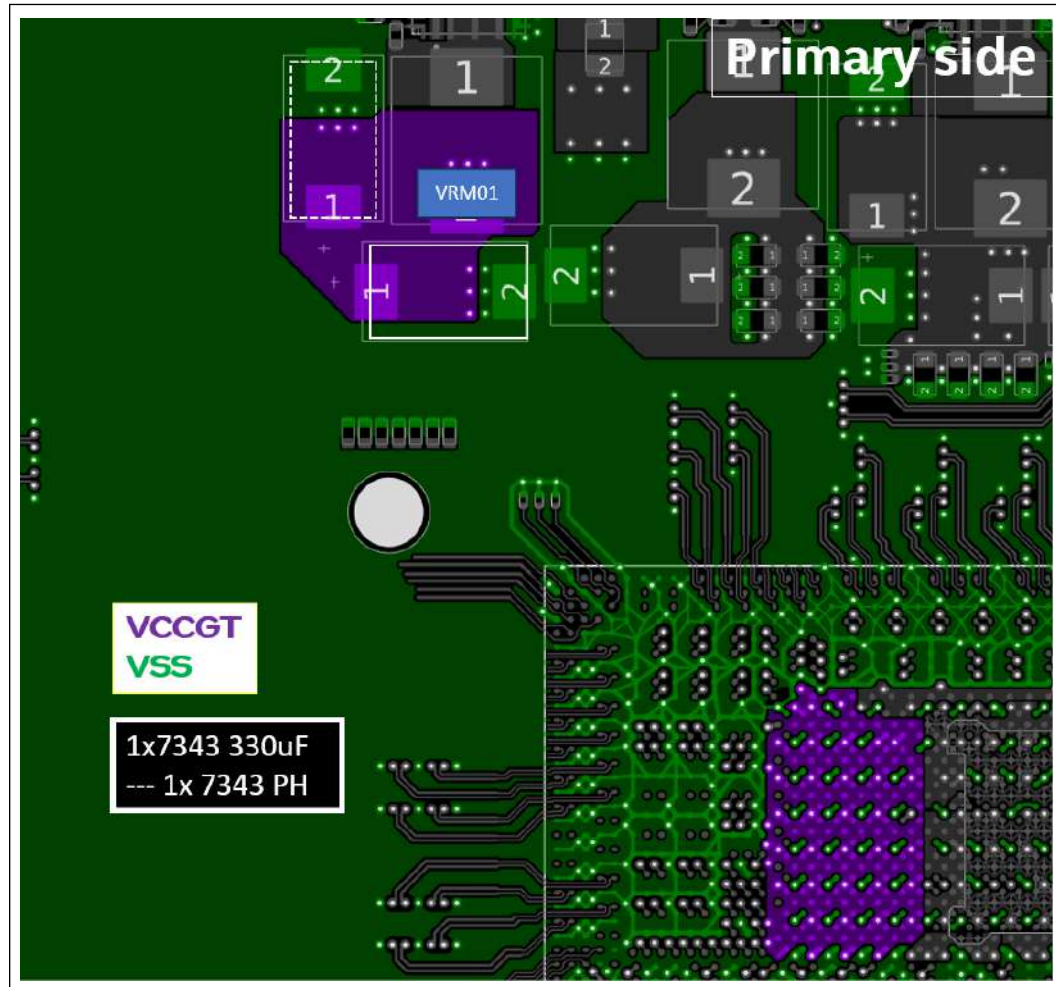
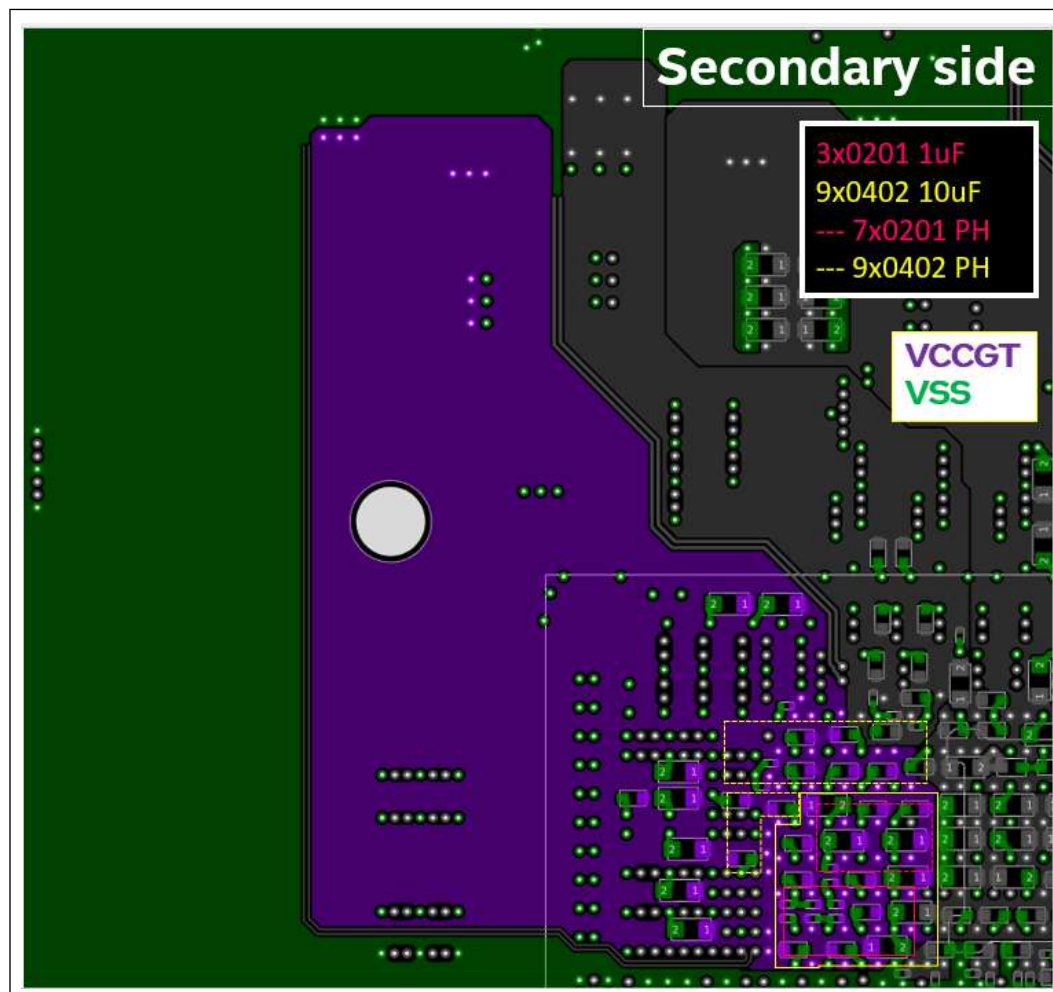


Figure 593. VCCGT_6C_T3_0.8mm_3



11.2.1.1.3 VCCSA(6C/T3/0.8mm PCB)

Table 870. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCCSA_6C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 23 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCCSA_6C_T3_0.8mm_4

Figure 594. VCCSA_6C_T3_0.8mm_1

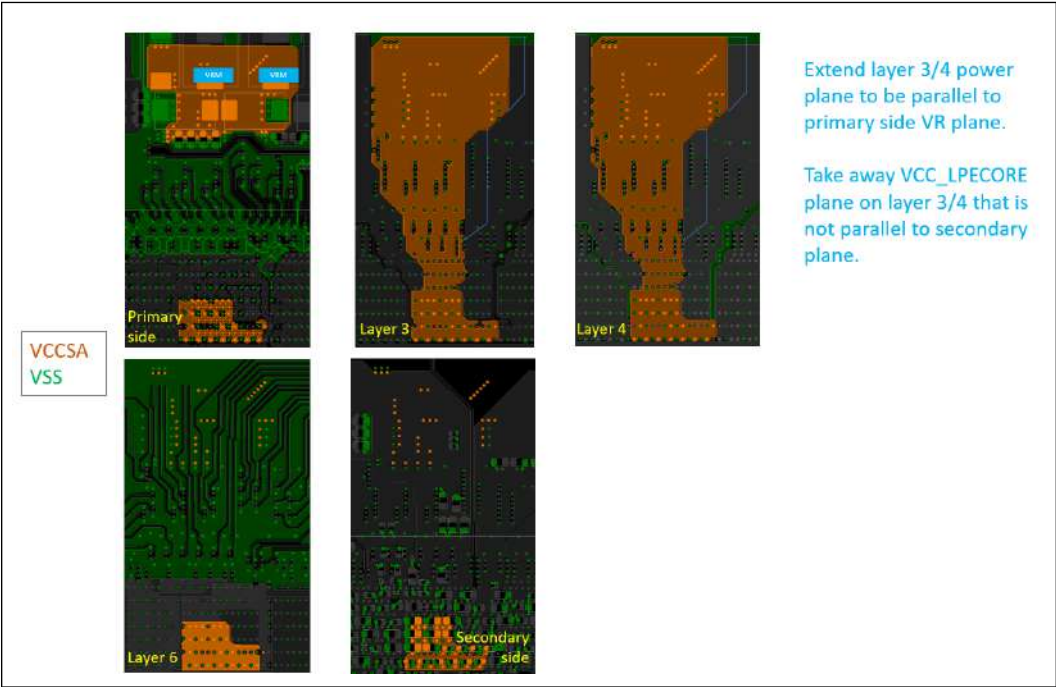


Figure 595. VCCSA_6C_T3_0.8mm_4

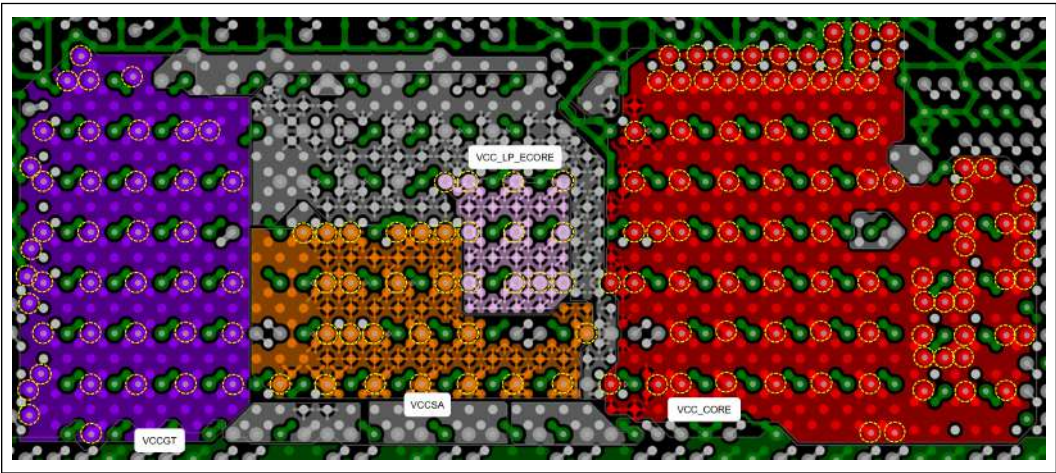


Table 871. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0201 -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0201, 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150kHz VR bandwidth or 600 kHz switching frequency, assuming BW=1/4 of Fsw.
<i>continued...</i>

Note
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.

Table 872. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	0603	22uF	4		VCCSA_6C_T3_0.8mm_2
Primary side	7343	330 uF	2	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCCSA_6C_T3_0.8mm_2
Primary side	7343	Placeholder	1		VCCSA_6C_T3_0.8mm_2
Secondary side	0402	10 uF	10	Place beneath BGA.	VCCSA_6C_T3_0.8mm_3
Secondary side	0603	22 uF	8	Place beneath BGA.	VCCSA_6C_T3_0.8mm_3

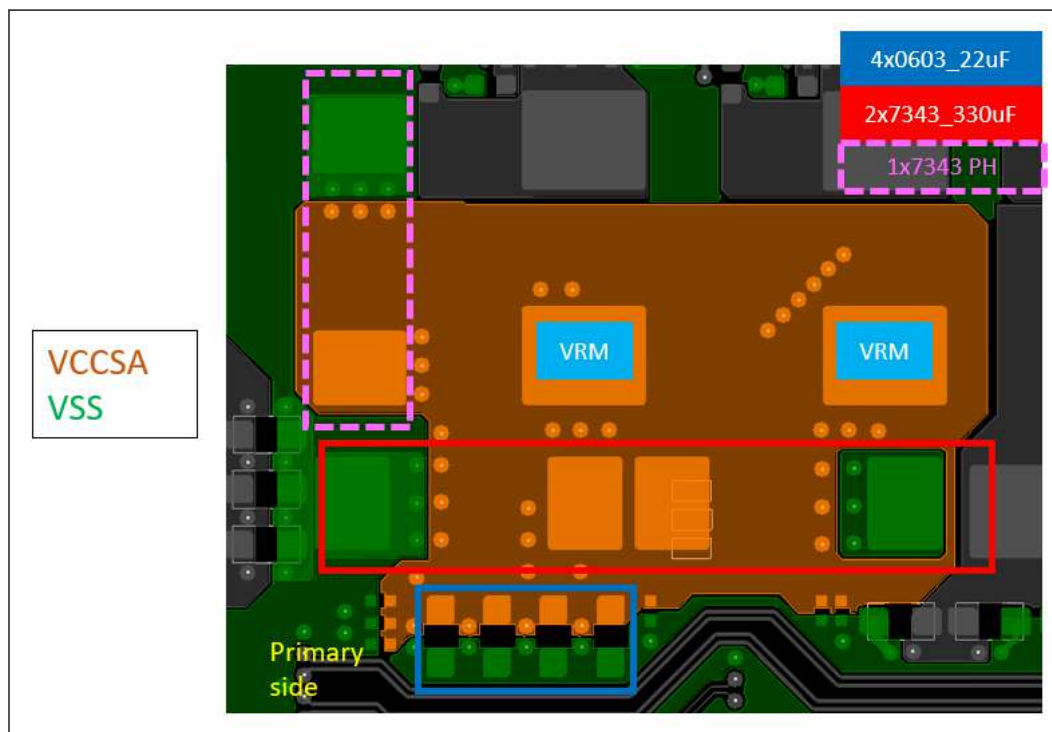
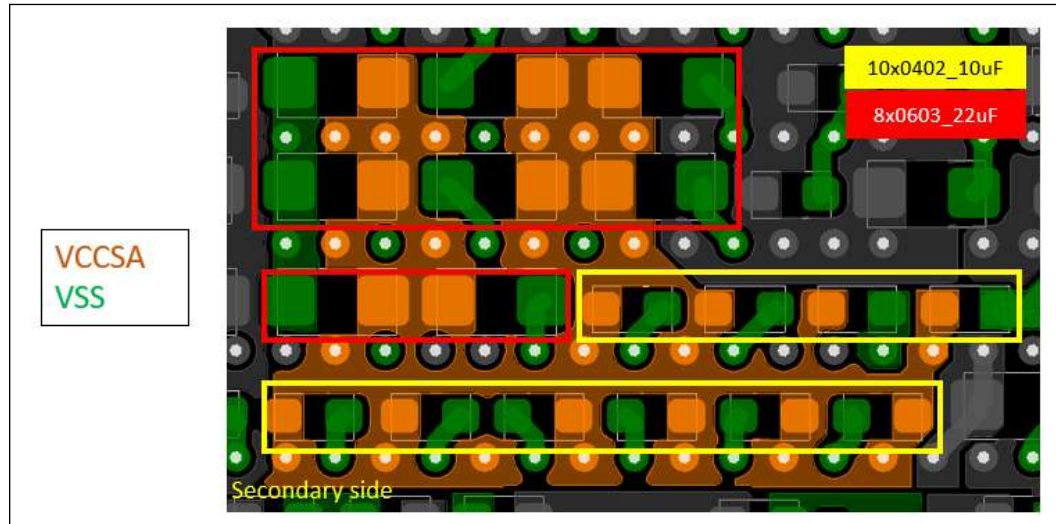
Figure 596. VCCSA_6C_T3_0.8mm_2

Figure 597. VCCSA_6C_T3_0.8mm_3



11.2.1.1.4 VCC_LP_ECORE(6C/T3/0.8mm PCB)

Table 873. Reference Layout Design

Index	Design Note	Reference Design
1	Design solid planes for a continuous current flow path from VR output to BGA. Please see reference image as an example of feasible placement.	VCC_LP_ECORE_6C_T3_0.8mm_1
2	Adjacent solid ground to power planes for good return path.	
3	Sense pins routed as differential pair to VR. Avoid routing adjacent to noisy power plane and have adjacent solid ground plane for good return path.	
4	Populate minimum 11 PTHs under the BGA to achieve optimum performance, see diagram for recommended locations.	VCC_LP_ECORE_6C_T3_0.8mm_4

Figure 598. VCC_LP_ECORE_6C_T3_0.8mm_1

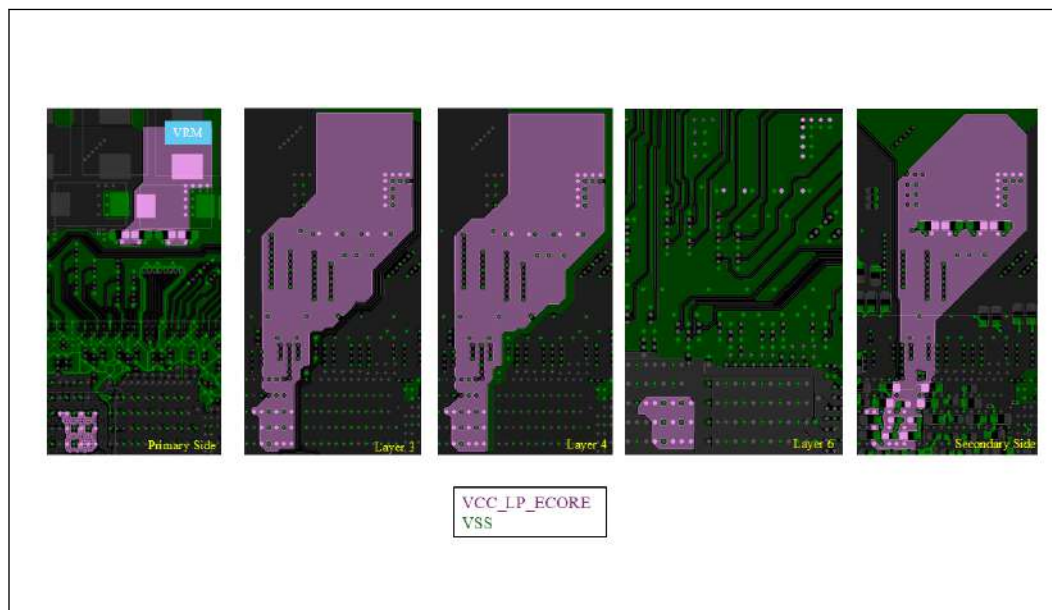


Figure 599. VCC_LP_ECORE_6C_T3_0.8mm_4

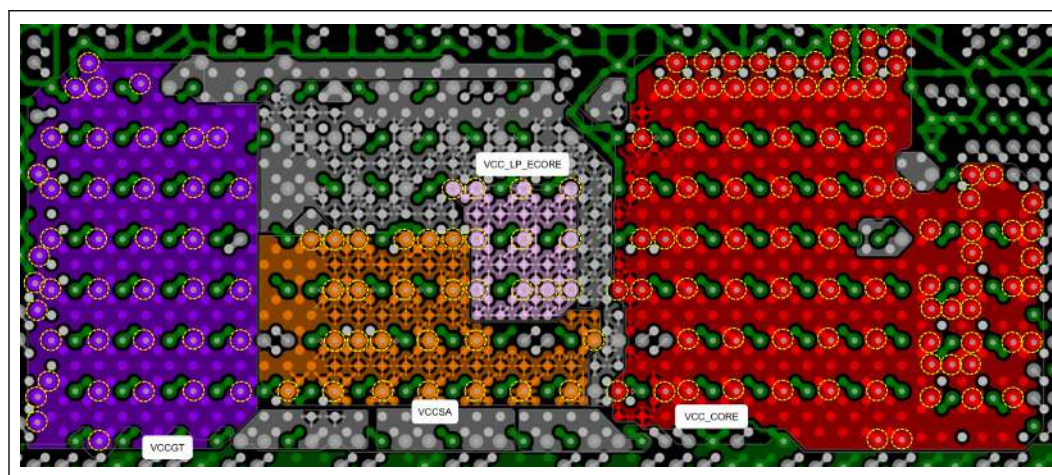


Table 874. Notes

Note
1. Smaller capacitors should be placed closest to the package. For example: Package -> 0402 -> 0603 -> Bulk -> VR.
2. For bottom layer capacitor placement, place 0402, 0603 capacitors directly under the package and cannot be placed at the edge.
3. Capacitor recommendation based on VR solution with 150 kHz VR bandwidth or 600 kHz switching frequency, assuming $BW=1/4$ of F_{sw} .
<i>continued...</i>

Note
4. Align capacitor pads along current flow path from VR to package. Make sure decoupling capacitors have power/ ground via pairs connected as close as possible to the capacitors pads.
5. Use MLCC components with 6.3 V voltage rating.
6. Please refer to acoustic noise mitigation technical advisory document #575216 for more details.

Table 875. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	7343	330uF	1	Recommended capacitor ESR = 4.5mOhm with 2.5V voltage rating.	VCC_LP_ECORE_6C_T3_0.8mm_2
Primary side	0603	Placeholder	4		VCC_LP_ECORE_6C_T3_0.8mm_2
Secondary side	0402	10uF	4	Place beneath BGA	VCC_LP_ECORE_6C_T3_0.8mm_3
Secondary side	0603	22uF	3	Place beneath BGA	VCC_LP_ECORE_6C_T3_0.8mm_3
Secondary side	0603	Placeholder	6	Place near package edge	VCC_LP_ECORE_6C_T3_0.8mm_3

Figure 600. VCC_LP_ECORE_6C_T3_0.8mm_2

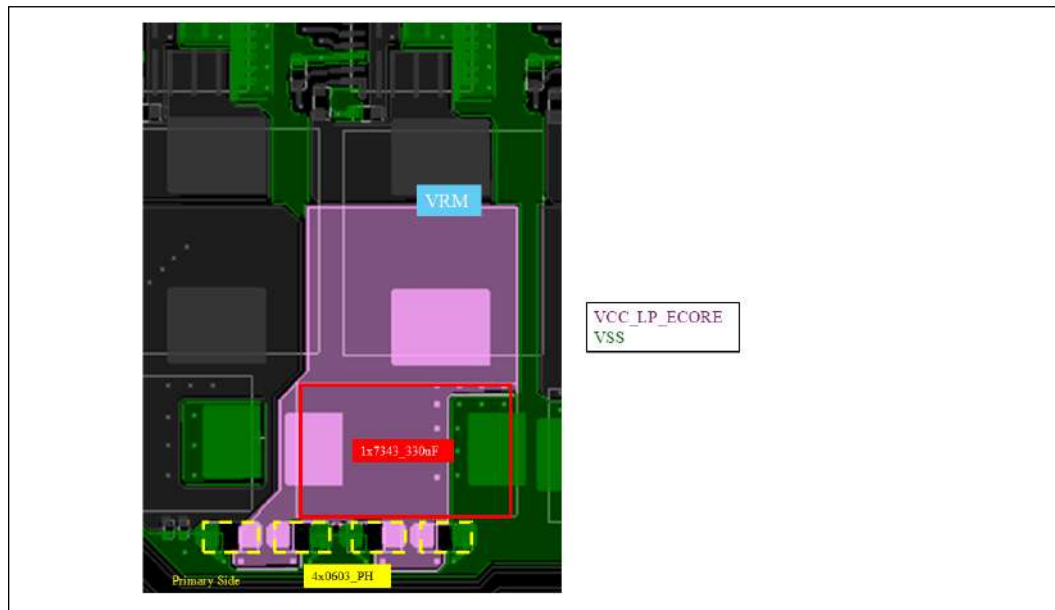
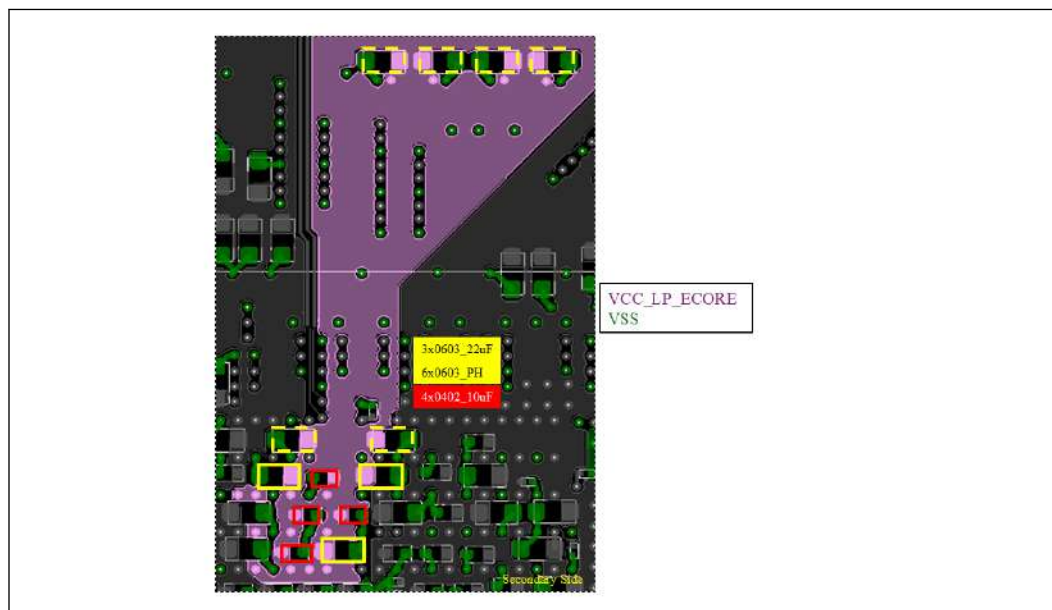


Figure 601. VCC_LP_ECORE_6C_T3_0.8mm_3



11.2.1.1.5 VCCPRIM_VNNAON (T3 PCB)

Table 876. Reference Layout Design

Index	Design Note	Reference Design
1	To meet Rpath of <3.5mΩ from VRM to VCCPRIM_VNNAON BGA groups, route VCCPRIM_VNNAON plane on at least three 30um-thick copper layers from VRM output to processor BGA with the following conditions : ► When routing on the top or bottom layer, keep plane width at least 15mm at VRM region & not lesser than 7mm upon approaching BGA region ► When routing on buildup layers (minimum 2 layers), keep plane width at least 10mm at VRM region & not lesser than 6mm upon approaching BGA region	VCCPRIM_VNNAON_1
2	Short all VCCPRIM_VNNAON BGAs with plane on at least of two 30um-thick copper layers	VCCPRIM_VNNAON_2
3	VCCPRIM_VNNAON VRM uses pseudo differential sensing to compensate for voltage drops along the power delivery path. The PWR & GND sense lines are connected to BP14 & BP12 BGAs respectively	VCCPRIM_VNNAON_3
4	Short both VCCPRIM_VNNAON_FLTR BGAs (CM16 & CM18) with plane	VCCPRIM_VNNAON_4
5	Merge both VCCPRIM_VNNAON_FLTR & VCCPRIM_VNNAON power planes using 1 x 0603 1uH (120mΩ max DCR) inductor & 1 x 0603 22uF cap to form an LC-filter with the following conditions : ► Place the LC-filter <1mm distance from or underneath VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR BGAs ► Ensure the merging point for both VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR planes are <1mm distance from or underneath VCCPRIM_VNNAON_FLTR BGAs	VCCPRIM_VNNAON_5

continued...

Index	Design Note	Reference Design
	Note : Refer diagram for design implementation guidance	
6	Yellow dotted circles highlights minimum count of VCCPRIM_VNNAON PTH recommended in bga region to prevent current overstress in PTHs. Ensure GND PTHs neatly placed close to PWR PTHs as shown in illustration to minimize return current loop	VCCPRIM_VNNAON_PTH

Figure 602. VCCPRIM_VNNAON_1

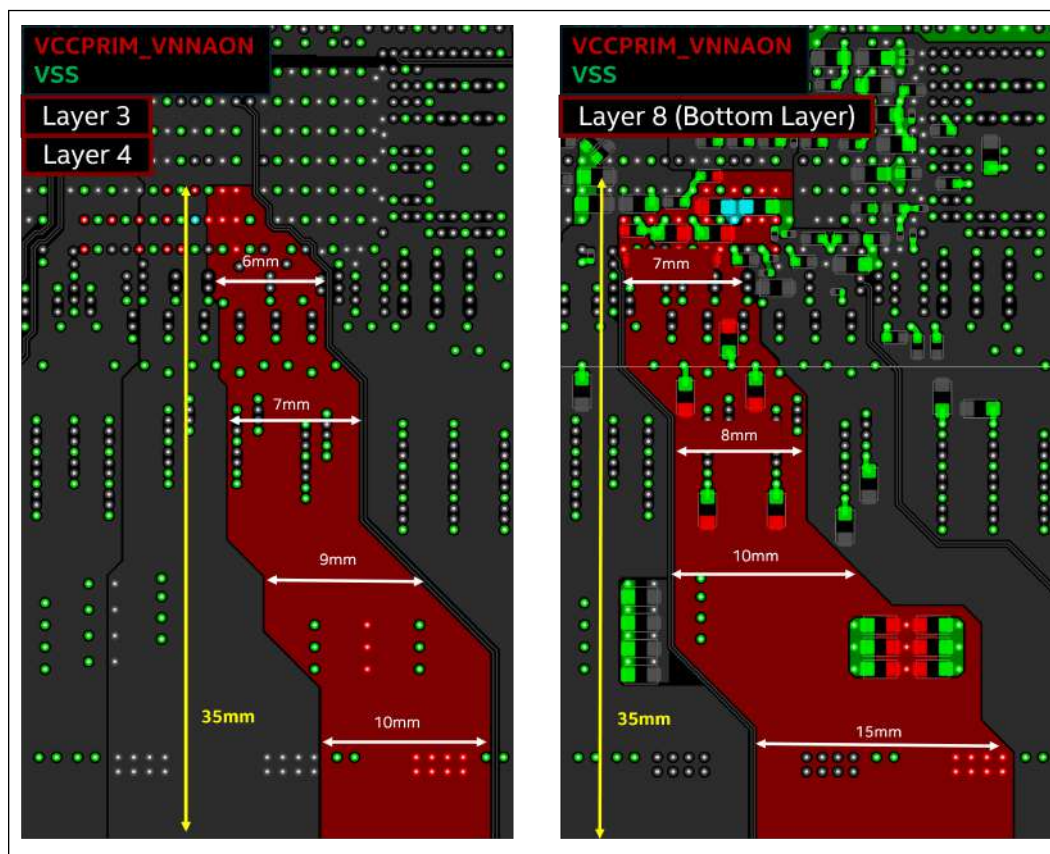


Figure 603. VCCPRIM_VNNAON_2



Figure 604. VCCPRIM_VNNAON_3

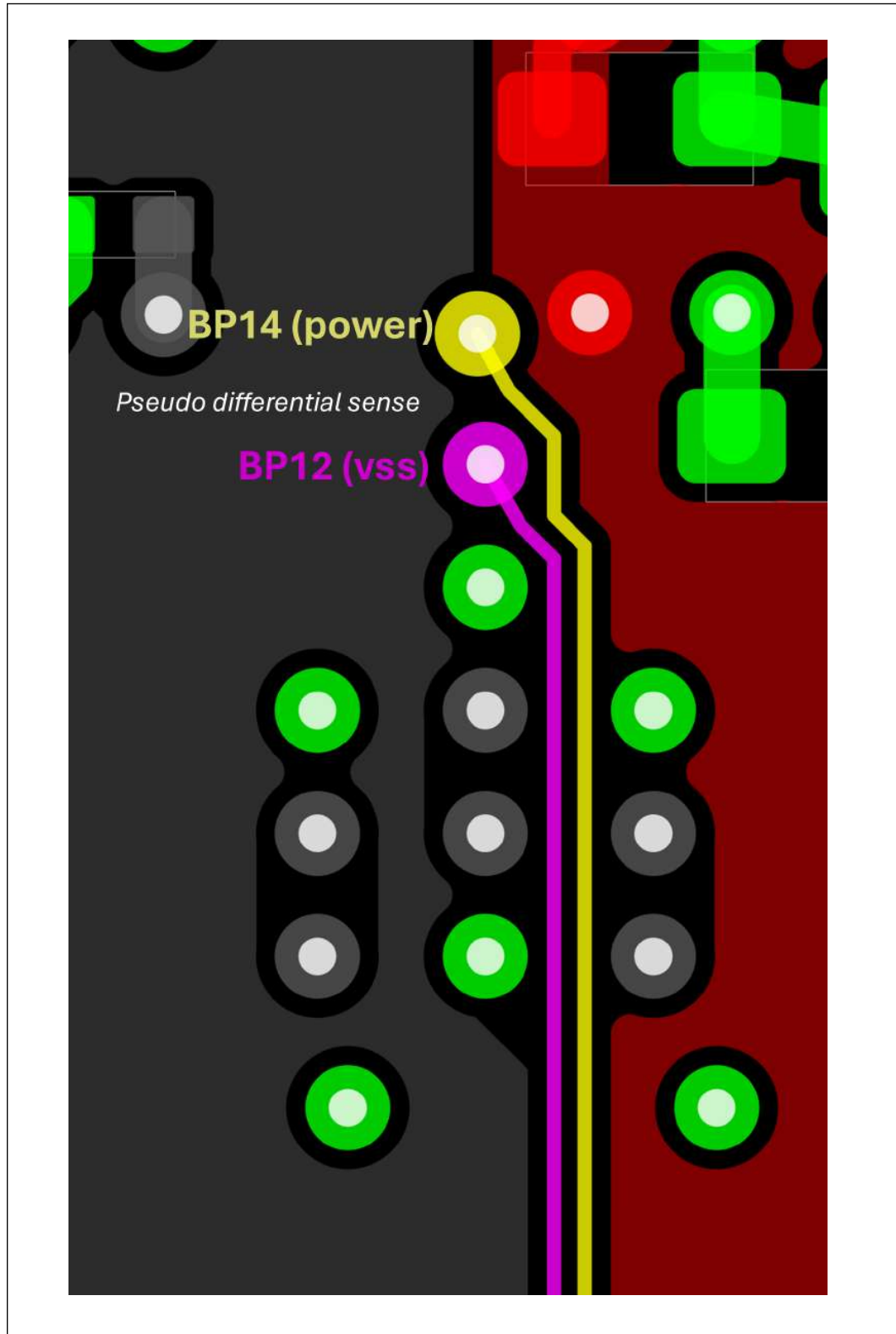


Figure 605. VCCPRIM_VNNAON_4

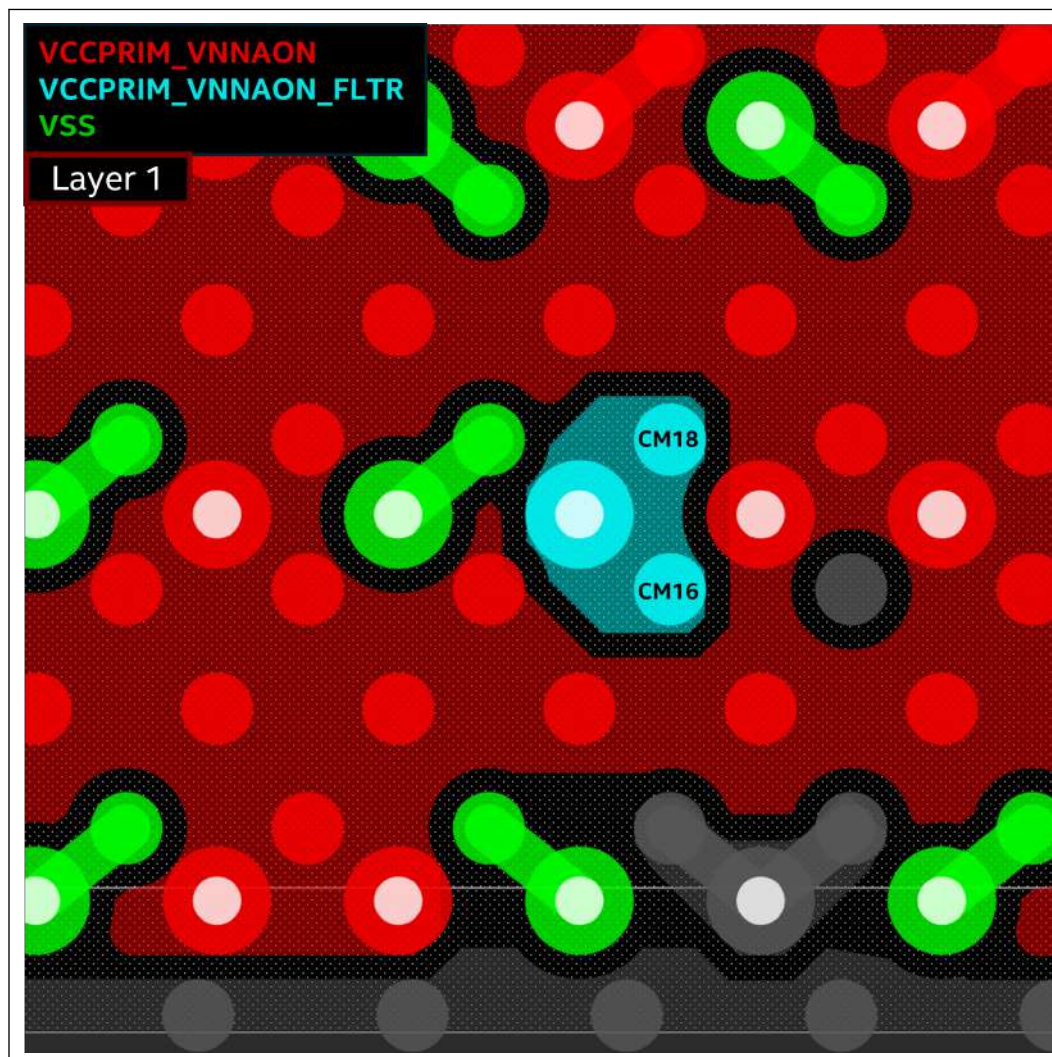


Figure 606. VCCPRIM_VNNAON_5

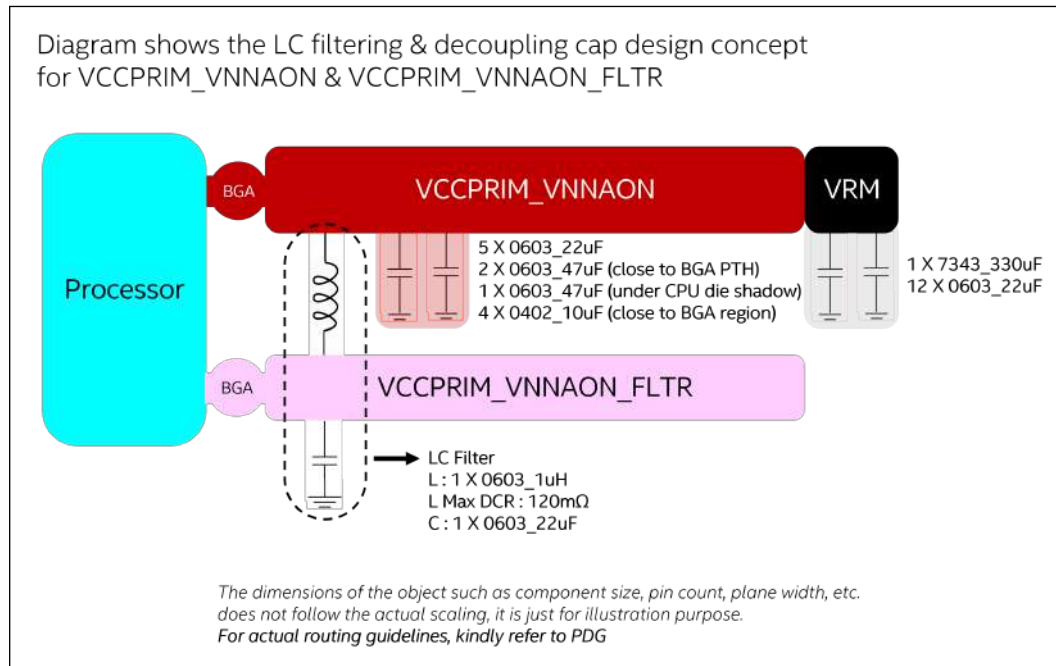


Figure 607. VCCPRIM_VNNAON_PTH

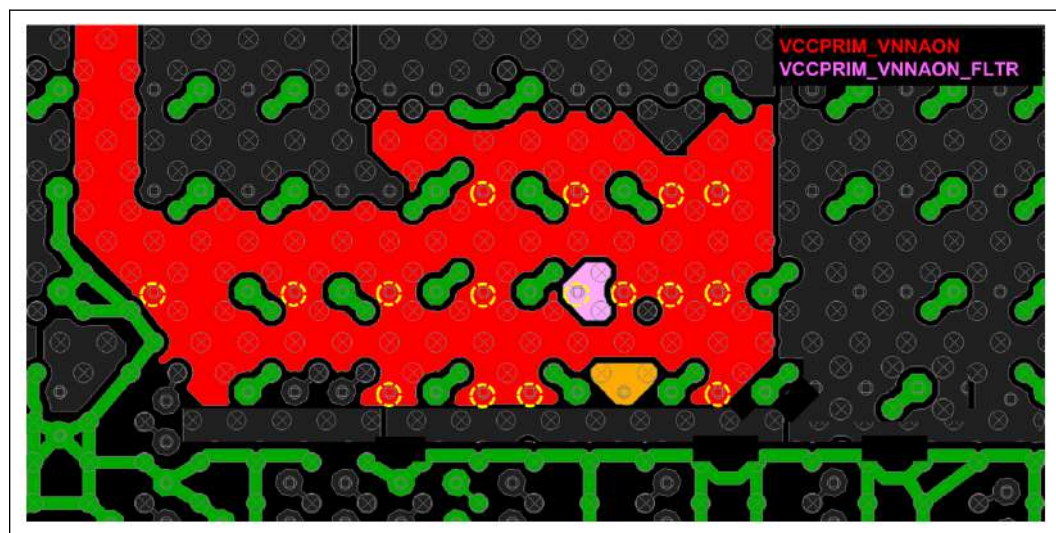


Table 877. Notes

	Note
	Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
	Heighten capacitor effectiveness by placing them from VRM output to package in the following order : 0402 → 0603 → 0805 → Bulk
	<i>continued...</i>

Note
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
When placing capacitors, align their pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor provided the caps meet the recommended parasitic capacitance.

Table 878. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	7343	330uF	1	Place bulk cap near to VRM output	VCCPRIM_VNNAON_6
Primary Side	0603	22uF	6	Place caps near to VRM output	VCCPRIM_VNNAON_6
Secondary Side	0603	22uF	6	Place caps near to VRM output	VCCPRIM_VNNAON_7
Secondary Side	0603	47uF	1	Place 1 x 0603 47uF cap under processor die shadow of VCCPRIM_VNNAON BGAs	
Secondary side	0603	47uF	2	Place 2 x 0603 47uF caps <1mm from VCCPRIM_VNNAON PTH located underneath BGA	VCCPRIM_VNNAON_8
Secondary Side	0603	22uF	5	Place 5 x 0603 22uF caps <10mm from package edge. Make sure PWR-GND PTH pair is placed very close to cap terminals	VCCPRIM_VNNAON_8
Secondary Side	0402	10uF	4	Place 4 x 0402 10uF caps underneath BGA or <1mm from package edge. Make sure PWR-GND PTH pair is placed very close to cap terminals	VCCPRIM_VNNAON_8
Secondary Side	0603	22uF	1	Pair 1 x 0603 22uF VCCPRIM_VNNAON_FLTR cap with 1 x 0603 1uH (120mΩ max DCR) inductor <1mm from or underneath VCCPRIM_VNNAON_FLTR BGAs to form an LC-filter	VCCPRIM_VNNAON_8
Secondary Side	0603	1 uH_120mΩ DCR	1	Merge both VCCPRIM_VNNAON_FLTR & VCCPRIM_VNNAON power planes using 1 x 0603 1uH (120mΩ max DCR) inductor & 1 x 0603 22uF cap to form an LC-filter with the following conditions : ► Place the LC-filter <1mm distance from or underneath VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR BGAs ► Ensure the merging point for both VCCPRIM_VNNAON & VCCPRIM_VNNAON_FLTR planes are <1mm distance from or underneath VCCPRIM_VNNAON_FLTR BGAs	VCCPRIM_VNNAON_8

Figure 608. VCCPRIM_VNNAON_6

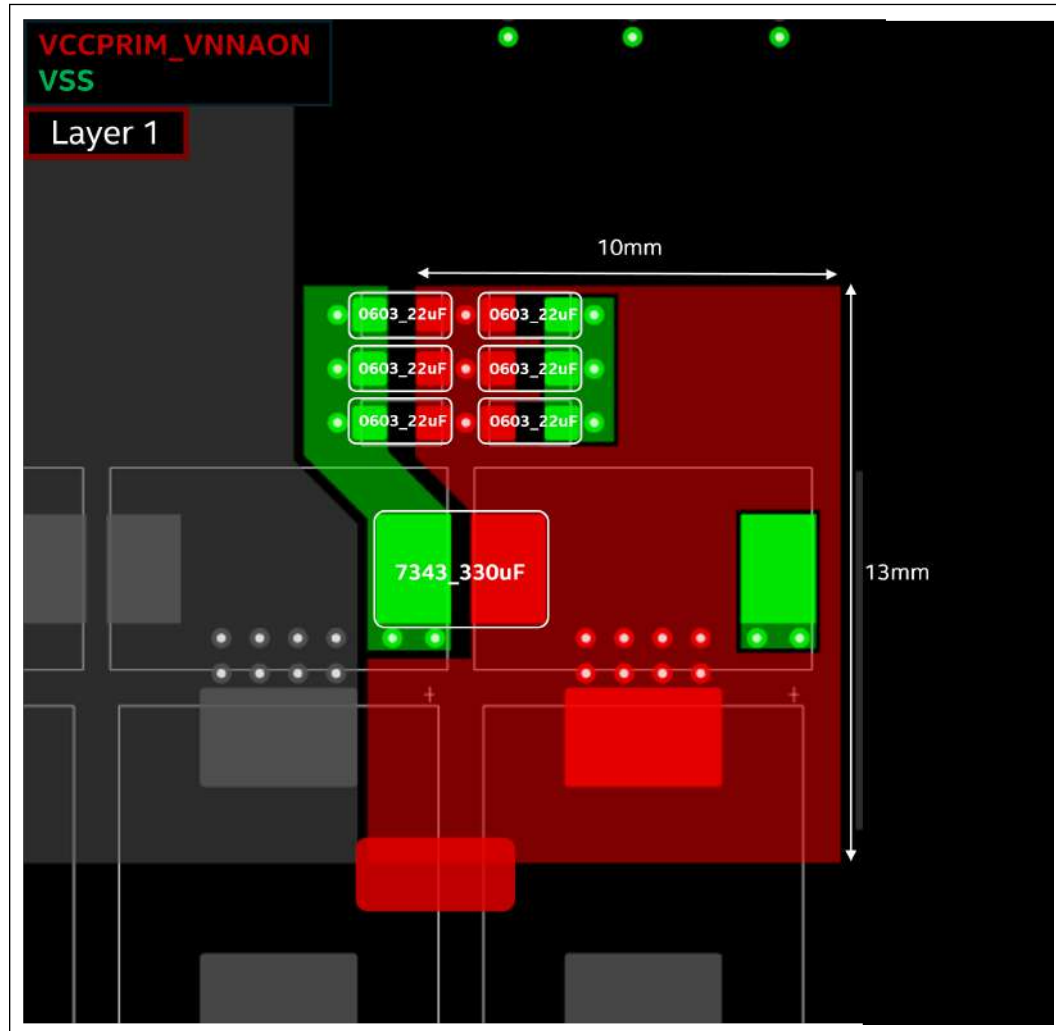


Figure 609. VCCPRIM_VNNAON_7

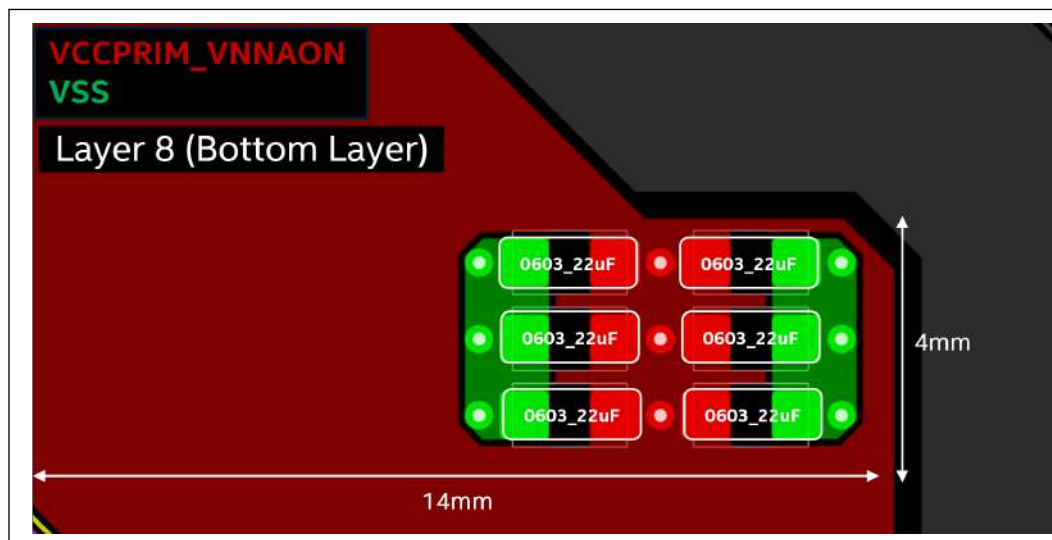
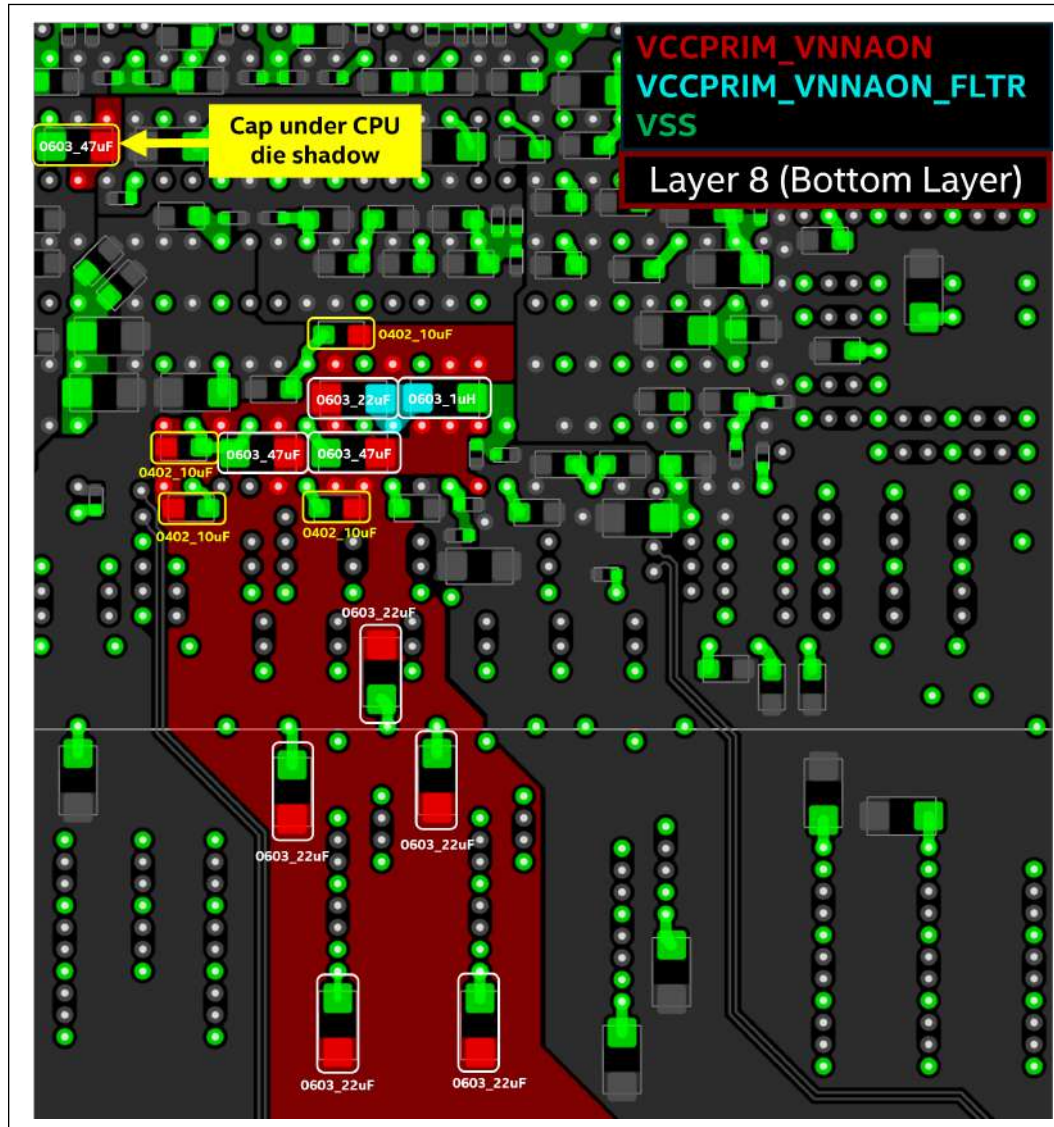


Figure 610. VCCPRIM_VNNAON_8



11.2.1.1.6 VCCPRIM_IO (T3 PCB)

Table 879. Reference Layout Design

Index	Design Note	Reference Design
1	Route VCCPRIM_IO power plane from VR output inductor to the BGA on Layer 1, Layer 6 and Layer 8. Adjacent layer from VCCPRIM_IO power plane should be ground.	VCCPRIM_IO_1
2	VCCPRIM_IO_FLTR should be merged with VCCPRIM_IO after LC filter on Layer 8	VCCPRIM_IO_1
3	VCCPRIM_IO VRM uses differential sensing. The power sense line is connected to SENSE_VCCPRIM_IO BGA(DF8) and ground sense line connected to SENSE_VSS_VCCPRIM_IO(DF9)	VCCPRIM_IO_2
4	The SVID 1.25 V supply for pull up resistors should be sourced from VCCPRIM_IO_OUT_SVID BGA EP5.	

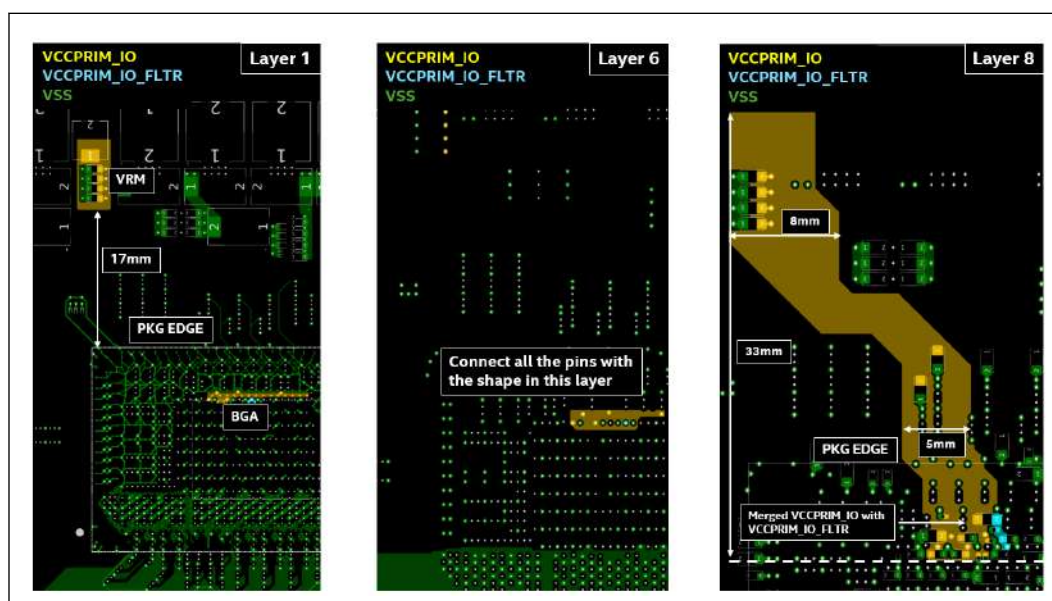
Figure 611. VCCPRIM_IO_1

Figure 612. VCCPRIM_IO_2

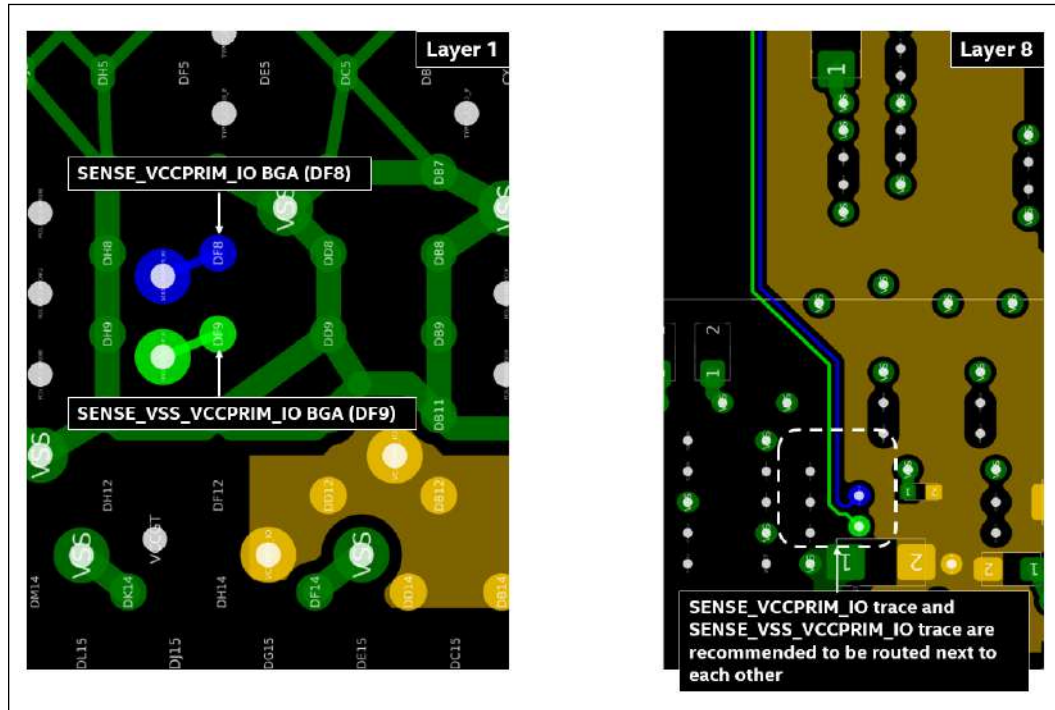


Table 880. Notes

Note
Place VRM bulk cap or additional MLCC caps to keep sense voltage below threshold limit & to fulfill PD requirements
Heighten capacitor effectiveness by placing them from VRM to package in the following order : 0402 → 0603 → 0805 → Bulk
Establish robust return path for decoupling capacitor by minimizing the distance between power & ground via pairs as well as placing the via pairs as close as possible to the capacitor pads
Align capacitor pads to the current path flowing from VR to package
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 881. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary Side	0805	47uF	4	Place near the VCCPRIM_IO VRM output	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0603	22 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402	10 uF	3	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	4	Place right below the VCCPRIM_IO BGA	VCCPRIM_IO_3,VCCPRIM_IO_4
continued...					

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side (Inductor)	0603	1 uH	1	Inductor component of the LC filter from VCCPRIM_IO_FLTR rail with ESR = 350mΩ	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0402/0603	22 uF	1	Decoupling capacitor placed on VCCPRIM_IO_FLTR rail for LC filtering purpose	VCCPRIM_IO_3,VCCPRIM_IO_4
Secondary side	0201	1uF	1	Place right below the VCCPRIM_IO_FLTR BGA	VCCPRIM_IO_3,VCCPRIM_IO_4

Figure 613. VCCPRIM_IO_3

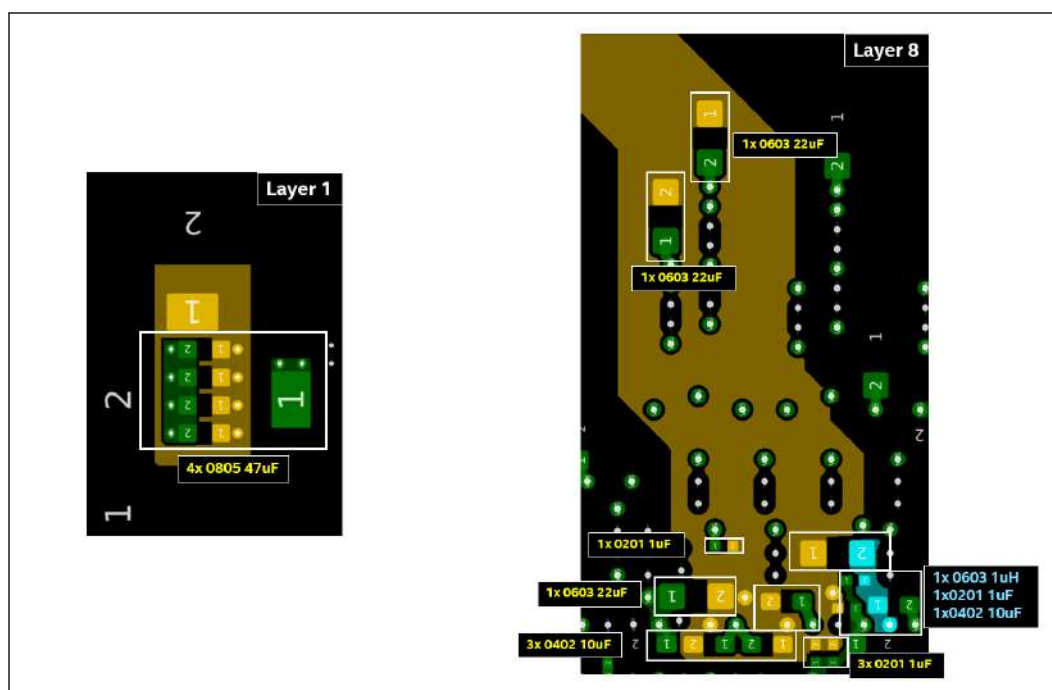
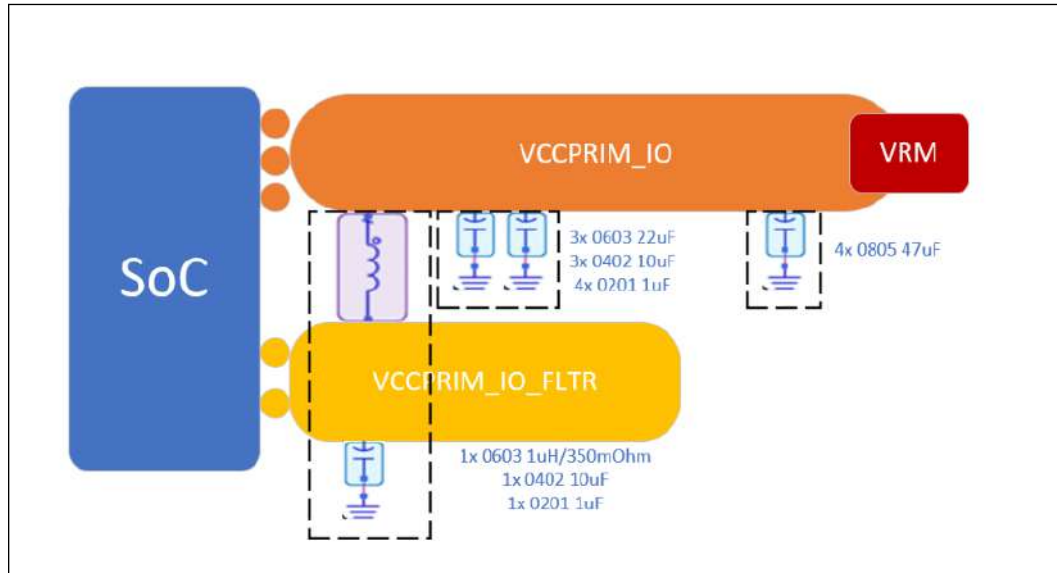


Figure 614. VCCPRIM_IO_4



11.2.1.1.7 VCCPRIM_1P8_PROC

Table 882. Reference Layout Design

Index	Design Note	Reference Design
1	Route the VCCPRIM_1P8_PROC, VCCPRIM_1P8_PROC_FLTRA and VCCPRIM_1P8_PROC_FLTRB on Layer 4 & Layer 6, with immediate ground reference. VCCPRIM_1P8_PROC_FLTRA should be merged with VCCPRIM_1P8_PROC after LC filter on Layer 1.	VCCPRIM_1P8_PROC_1
2	VCCPRIM_1p8_FLTRB should be merged with VCCPRIM_1p8 on Layer4 after ~15mm from VCCPRIM_1p8_BGAs.	VCCPRIM_1P8_PROC_2

Figure 615. VCCPRIM_1P8_PROC_1

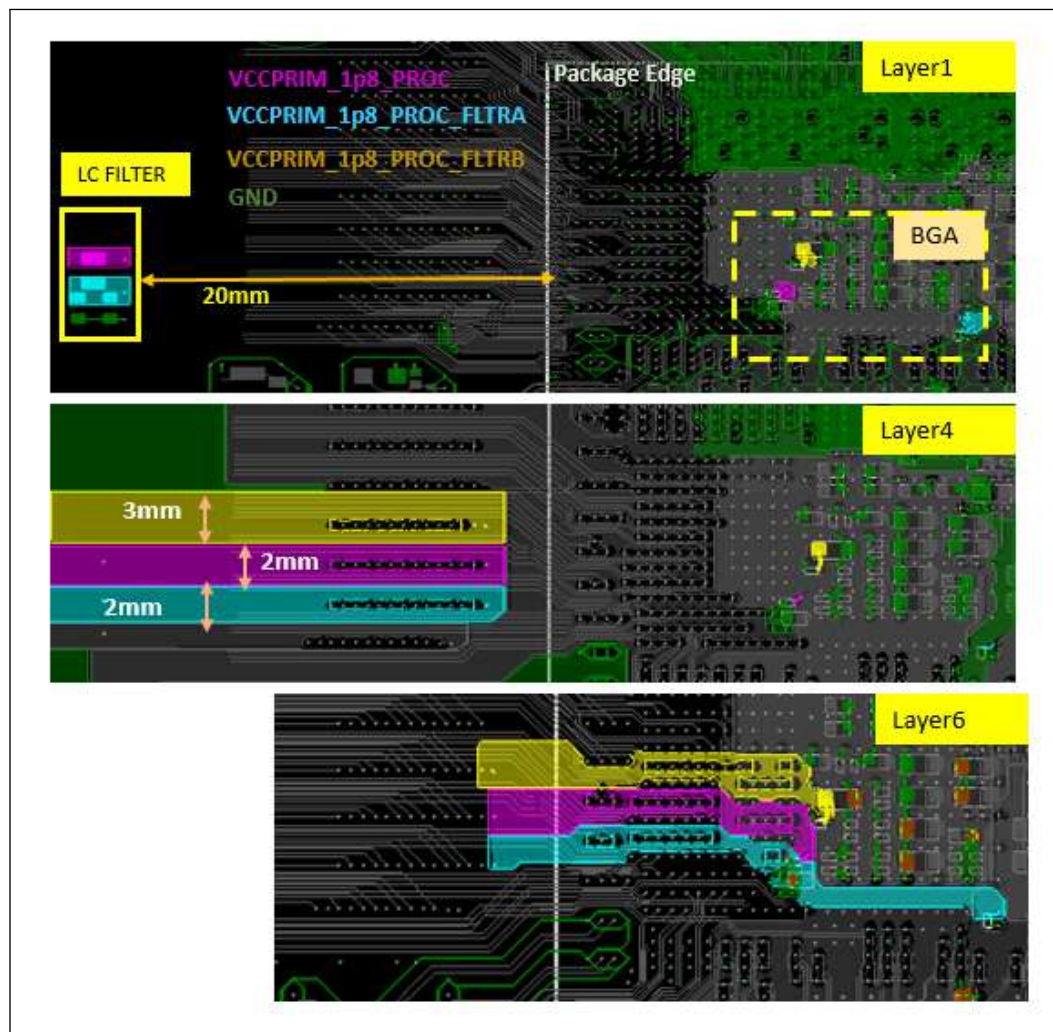


Figure 616. VCCPRIM_1P8_PROC_2



Table 883. Notes

Note
Power and ground via pairs should be placed close to capacitor pads to provide proper vertical connections when placing capacitor on the secondary side.
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 884. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side (Inductor)	0805 / 0603	1uH	1	Inductor component of the LC filter from VCCPRIM_1P8_PROC rail to VCCPRIM_1P8_PROC_FLTRA rail. It must meet electrical requirements: max DCR < 100mOhm and rated current >200 mA.	VCCPRIM_1P8_PROC_2
Primary side	0603	22uF	2	Decoupling capacitor(C1,C2) placed on VCCPRIM_1P8_PROC_FLTRA rail for LC filtering purpose.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C3 under VCCPRIM_1P8_PROC_FLTRA BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0603	Placeholder	1	Place C4 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0201	Placeholder	1	Place C5 under VCCPRIM_1P8_PROC_FLTRB BGAs.	VCCPRIM_1P8_PROC_2
Secondary side	0402 / 0201	Placeholder	1	Place C6 under VCCPRIM_1P8_PROC BGAs	VCCPRIM_1P8_PROC_2

Figure 617. VCCPRIM_1P8_PROC_2


11.2.1.1.8 VCCRTC_PROC

Table 885. Reference Layout Design

Index	Design Note	Reference Design
1	Require 1 layer of VCCRTC_PROC plane from vrm to board pin with immediate GND reference on one of the adjacent layers. Refer to reference design image for design implementation.	VCCRTC_PROC_1

Figure 618. VCCRTC_PROC_1

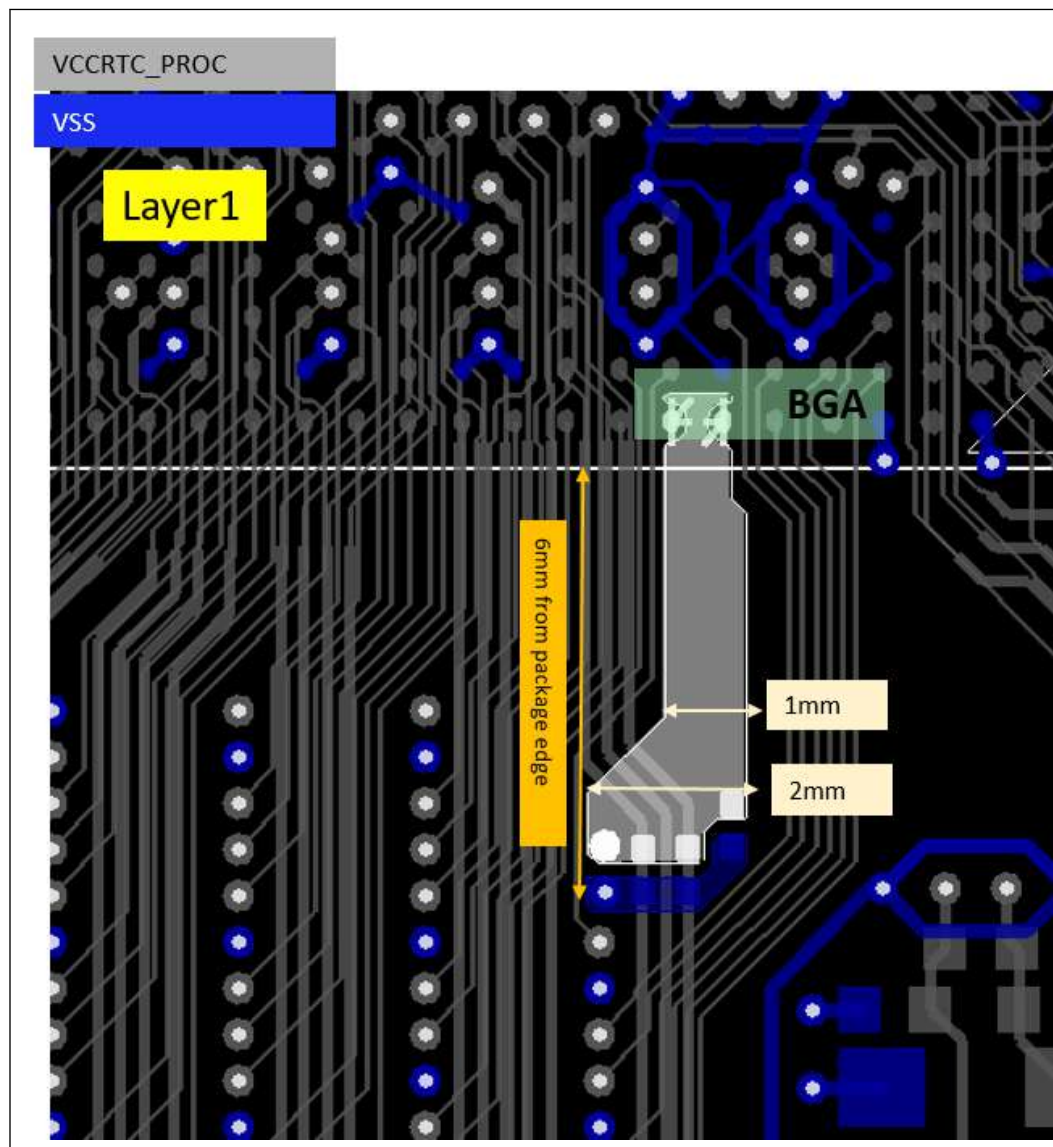


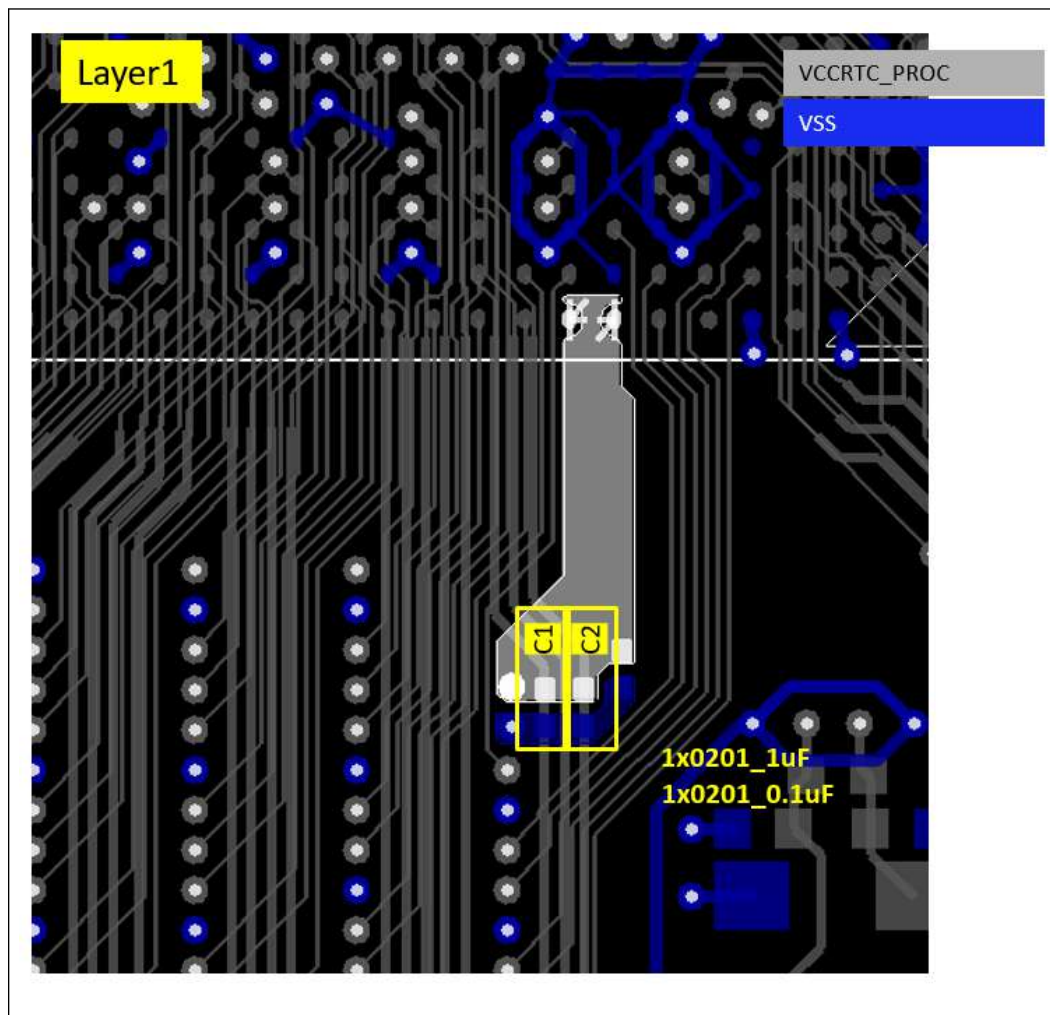
Table 886. Notes

Note
Customers can choose capacitors of any form factor if the capacitors meet the recommended parasitic capacitance.

Table 887. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Primary side	0201	1 uF	1	Place C1 within 6mm from package edge.	VCCRTC_PROC_2
Primary side	0402 / 0201	0.1 uF	1	Place C2 within 6mm from package edge.	VCCRTC_PROC_2

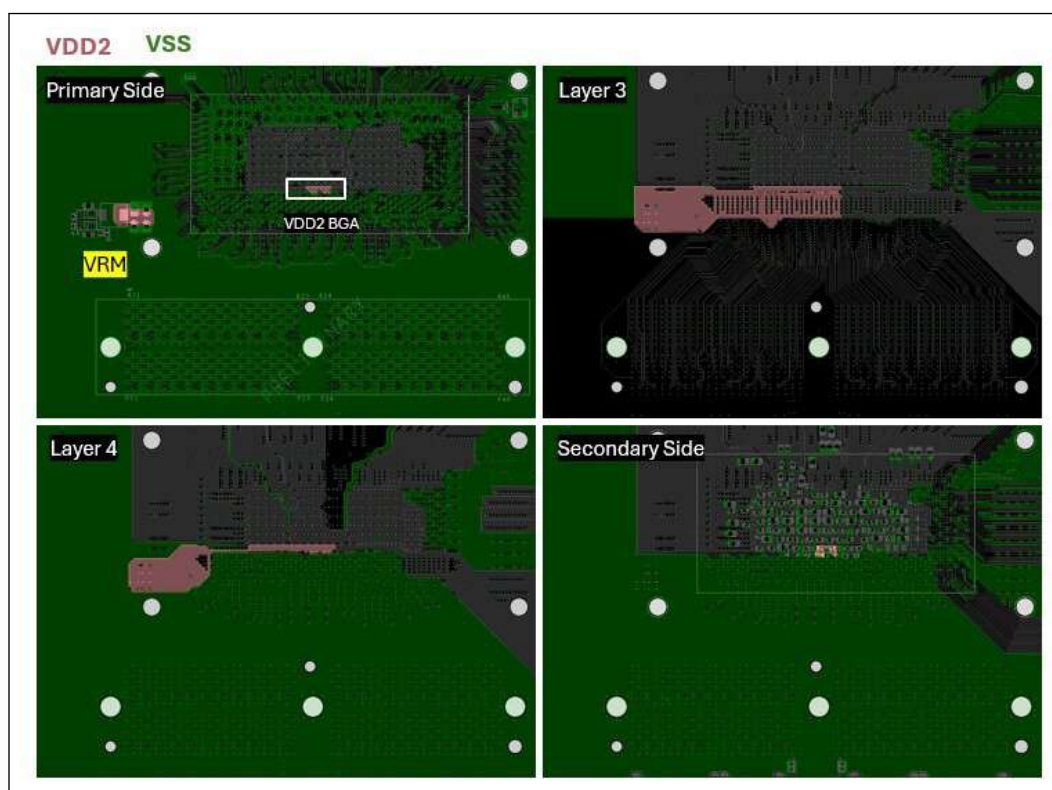
Figure 619. VCCRTC_PROC_2



11.2.1.1.9 VDD2 LPDDR5x LPCAMM2

Table 888. Reference Layout Design

Index	Design Note	Reference Design
1	VDD2 Type-3 8L CAMM motherboard reference layout showing the plane routing	VDD2_T3_CAMM8L_1
2	Adjacent solid ground to power planes for good return path	

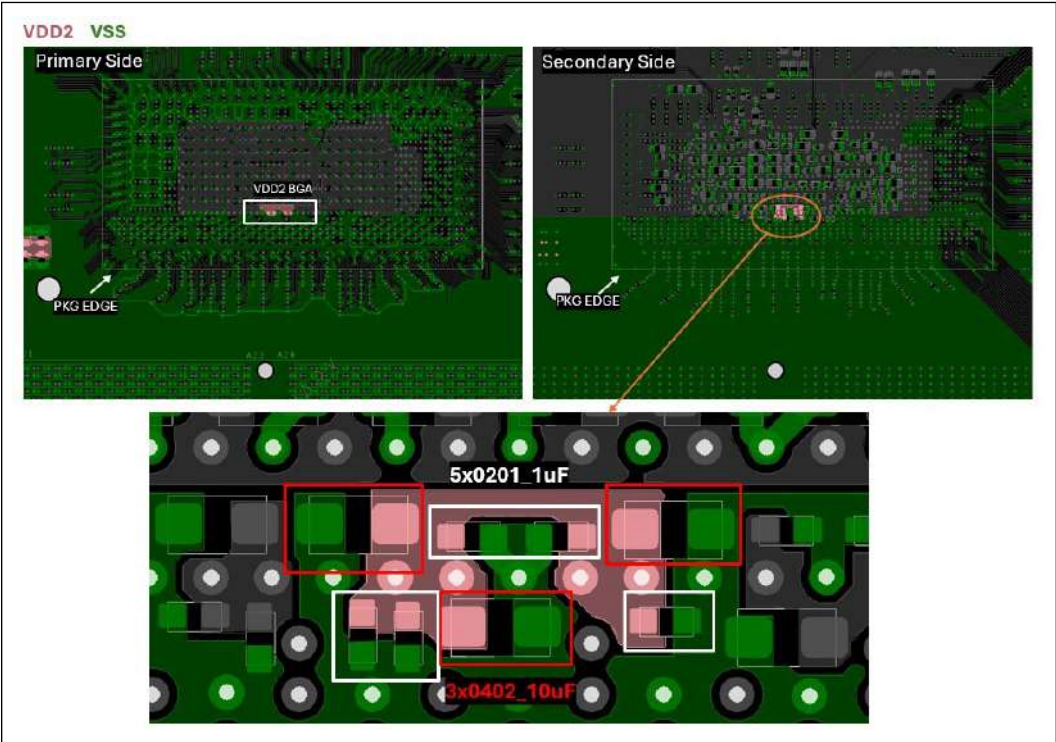
Figure 620. VDD2_T3_CAMM8L_1**Table 889. Notes**

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 890. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side	0201	1uF	5	Place beneath BGA	VDD2_T3_CAMM8L_2
Secondary side	0402	10 uF	3	Place beneath BGA	VDD2_T3_CAMM8L_2

Figure 621. VDD2_T3_CAMM8L_2



11.2.1.1.10 VDDQ LPDDR5x LPCAMM2

Table 891. Reference Layout Design

Index	Design Note	Reference Design
1	VDDQ Type-3 8L CAMM motherboard reference layout showing the plane routing	VDDQ_T3_CAMM8L_1
2	Adjacent solid ground to power planes for good return path	

Figure 622. VDDQ_T3_CAMM8L_1

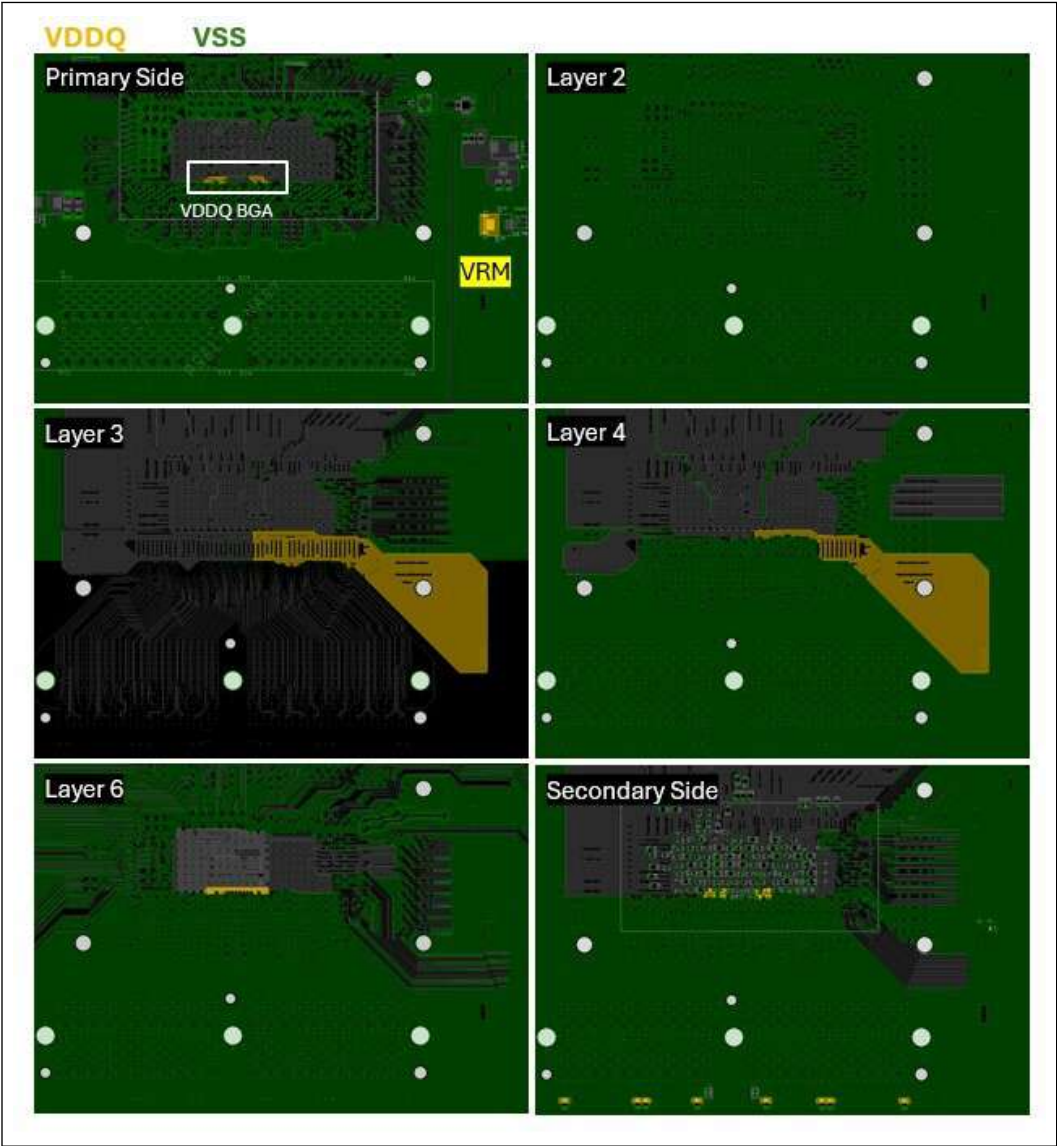


Table 892. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 893. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary Side	0201	1uF	8	Place beneath BGA	VDDQ_T3_CAMM8L_2
Secondary Side	0402	10uF	4	Place beneath BGA	VDDQ_T3_CAMM8L_2

Figure 623. VDDQ_T3_CAMM8L_2



11.2.1.1.11 VDD2 DDR5 (T3 PCB)

Table 894. Reference Layout Design

Index	Design Note	Reference Design
1	VDD2 Type-3 DDR5 motherboard reference layout showing the plane routing	VDD2_T3_DDR5_8L_1
2	Adjacent solid ground to power planes for good return path	

Figure 624. VDD2_T3_DDR5_8L_1



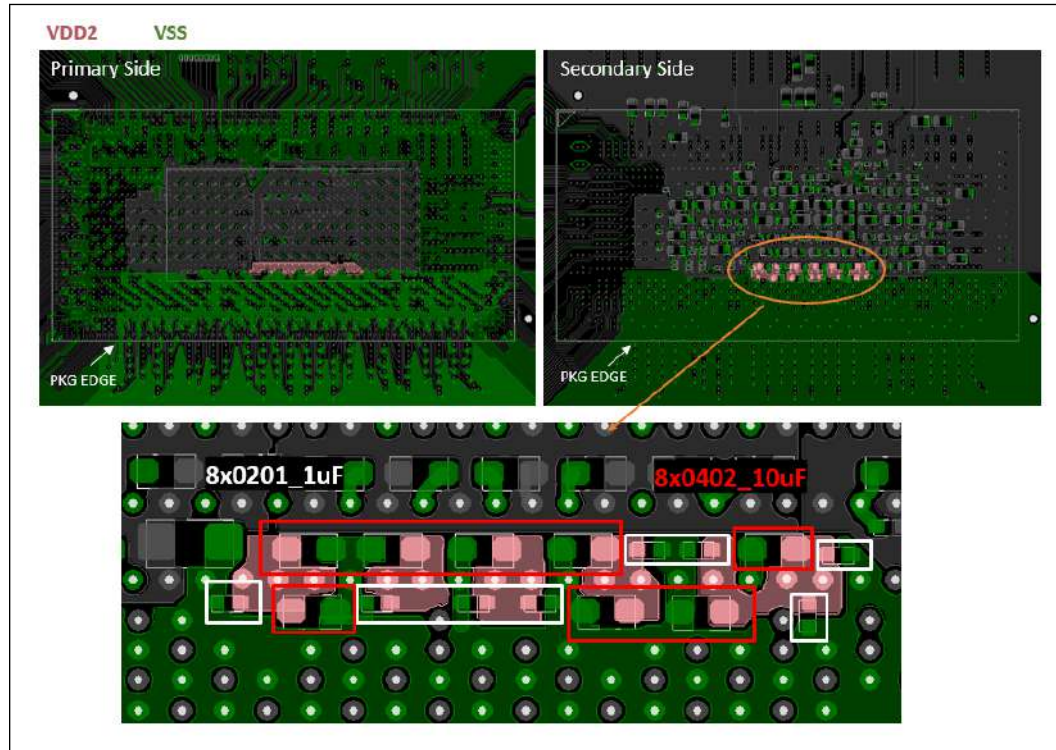
Table 895. Notes

Note
1. Power and ground via pairs should be placed close to capacitor pads to provide proper connection to BGAs

Table 896. Decoupling Solution And Filter Recommendation

Component Placement	Form Factor	Value	Quantity	Note	Reference Design
Secondary side	0201	1uF	8	Place beneath BGA	VDD2_T3_DDR5_8L_2
Secondary side	0402	10 uF	8	Place beneath BGA	VDD2_T3_DDR5_8L_2

Figure 625. VDD2_T3_DDR5_8L_2



11.3 Impedance Spectrum Tool (IST) Testing Requirements and Recommendations

Impedance Spectrum Tool testing requirements and recommendations are shown below.

NOTE

IST is not directly available to customers, Intel will need these trigger points to support debug of customer issues at Intel validation labs. Intel recommends customers implement the IST tool requirements from this section.

Table 897. IST Tests

Signal Names	Feature	As Shown In	Required or Recommended	Notes
VCC_CORE_SENSE / VCC_CORE_VSS_SENSE	Trigger signal	Table 898 on page 897 and Figure 626 on page 898	Required	Route to un-stuffed header as shown in Figure 626 on page 898
	Differential sense nets	NA	Required	VR sense lines to monitor for IFDIM/IST
	Current sense resistors	Figure 627 on page 898	Recommended	One sense resistor (1-2mOhm) per phase. VRTT load line data is required if sense resistors are not provided.
continued...				

Signal Names	Feature	As Shown In	Required or Recommended	Notes
VCC_LP_ECORE_SENSE / VCC_LP_ECORE_VSS_SENSE	Trigger signal	Table 898 on page 897 and Figure 626 on page 898	Required	Route to un-stuffed header as shown in Figure 626 on page 898
	Differential sense nets	NA	Required	VR sense lines to monitor for IFDIM/IST
	Current sense resistors	Figure 627 on page 898	Recommended	One sense resistor (1-2mOhm) per phase. VRTT load line data is required if sense resistors are not provided.
VCCGT_SENSE / VCCGT_VSS_SENSE	Trigger signal	Table 898 on page 897 and Figure 626 on page 898	Required	Route to un-stuffed header as shown in Figure 626 on page 898
	Differential sense nets	NA	Required	VR sense lines to monitor for IFDIM/IST
	Current sense resistors	Figure 627 on page 898	Recommended	One sense resistor (1-2mOhm) per phase. VRTT load line data is required if sense resistors are not provided.
VCCSA_SENSE / VCCSA_VSS_SENSE	Trigger signals	Table 898 on page 897 and Figure 626 on page 898	Required	Route to un-stuffed header as shown in Figure 626 on page 898
	Differential Sense Nets	NA	Required	VR sense lines to monitor for IFDIM/IST
	Current sense resistors	Figure 627 on page 898	Recommended	One sense resistor (1-2mOhm) per phase. VRTT load line data is required if sense resistors are not provided

Table 898. Signals Required for IST/IFDIM

Feature	Required or Recommended	Netname View Pins	Ball/Pin #
IST / IFDIM trigger	Required	BPM[0]	M19
IST / IFDIM trigger	Optional	BPM[1]	L19

Routing Requirements

- ITP/XDP uses same BPM[*] line. Recommend using steering resistor for access to trigger signal
- Termination on board can be a regular Test Point or SMP connector

Figure 626. IST Trigger Point Implementation Requirement

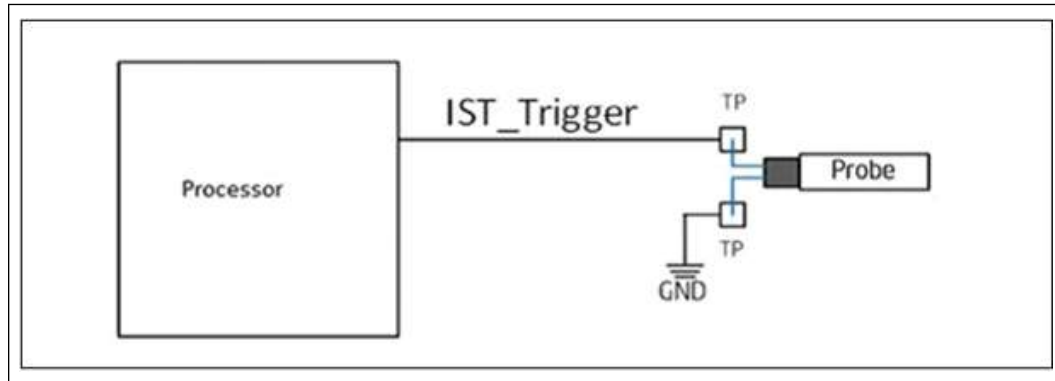
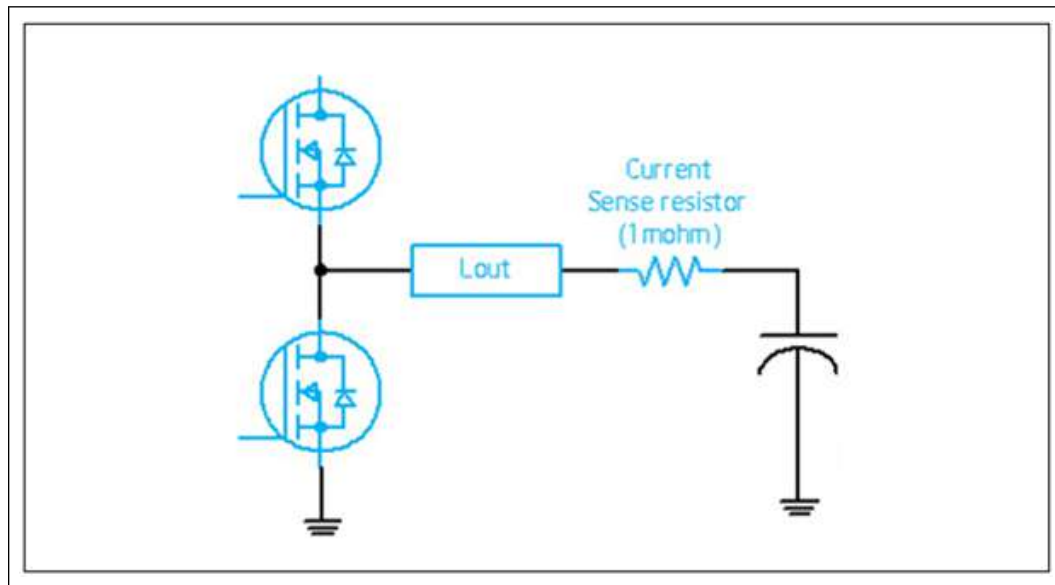


Figure 627. Current Sense Resistor Recommendation



11.4 Processor Decoupling Considerations for Fast V-Mode

When the load exceeds the VR cycle-by-cycle current limiting, Fast V-Mode relies on the power rail decoupling to make up the difference between current demanded by the load and the current provided by the regulator. For more details on Fast V-Mode operation refer to the Power Delivery section.

Customers need to ensure that their design has a minimum amount of decoupling capacitance, to meet the *FVM Max ΔV* specified in the Power Map. All the processor decoupling recommended in this Power Integrity section already meets this requirement, however if customers want to deviate from the recommended decoupling, they need to use the following equation to calculate the minimum required capacitance for their application.

$$C_{decap} = \frac{(I_{ccMAX} - I_{ccMAX.App}) * t_{THROTTLE}}{FVM\ Max\ \Delta V}$$

Where:

C_{decap} (μF) is the minimum derated capacitance required on a given power rail to meet the $FVM\ Max\ \Delta V$

I_{ccMAX} (A) is the Max. Current for Processor Rail (from Processor Electrical Specification or Power Map)

$I_{ccMAX.App}$ (A) is the Max. Application Current for Processor Rail (from Processor Electrical Specification or Power Map)

$FVM\ Max\ \Delta V$ (mV) is the maximum allowable ΔV at VR output during Fast V-mode cycle by cycle current limit over time $t_{THROTTLE}$ (from Power Map)

$t_{THROTTLE}$ (ns) is the SoC throttle time for Fast-Vmode event when voltage falls below SoC internal threshold (from Power Map)

When customers design their power rail decoupling, they need to ensure that the total derated capacitance from their power rail is equal or higher than C_{decap} .

NOTE

The derated capacitance must be used and not nominal capacitance, this because capacitor materials are not ideal, and the nominal capacitance will be derated based on voltage, temperature, and aging conditions.

12.0 Platform Debug and Test Hooks

For information on the debug diagram, guidance and enabling, refer to the Chapter 4 in Open and Closed Chassis Debug Nova Lake Onwards User Guide (# [870129](#)).

NOTE

For more updates, refer to the Nova Lake JTAG On Die Termination Workaround Technical Advisory, (# [862716](#)).

13.0 System Meter Overview

13.1 Acronyms

SN	Acronym	Expansion
1	EMI	Energy Meter Interface
2	PAC	Power Monitor with Accumulation Chip

13.2 Introduction of System Meter

- System Meter is a simple, yet powerful mechanism to measure critical SoC and other platform subsystem power.
- System Meter uses a combination of onboard hardware power accumulators and software to measure power, the accuracy compared to open chassis instrumentation such as National Instruments DAQ is $\leq 5\%$.
- Benefits
 - Suitable for Close Chassis measurement. (Debug Possible)
 - Ultra-Low cost and simple to use
 - Cost-Adder controllable by BOM configs
 - No board rework/blue wire is required for power measurement
 - No additional Host machine is required
 - Telemetry and First Level Triage

Figure 628. System Meter Implementation Steps

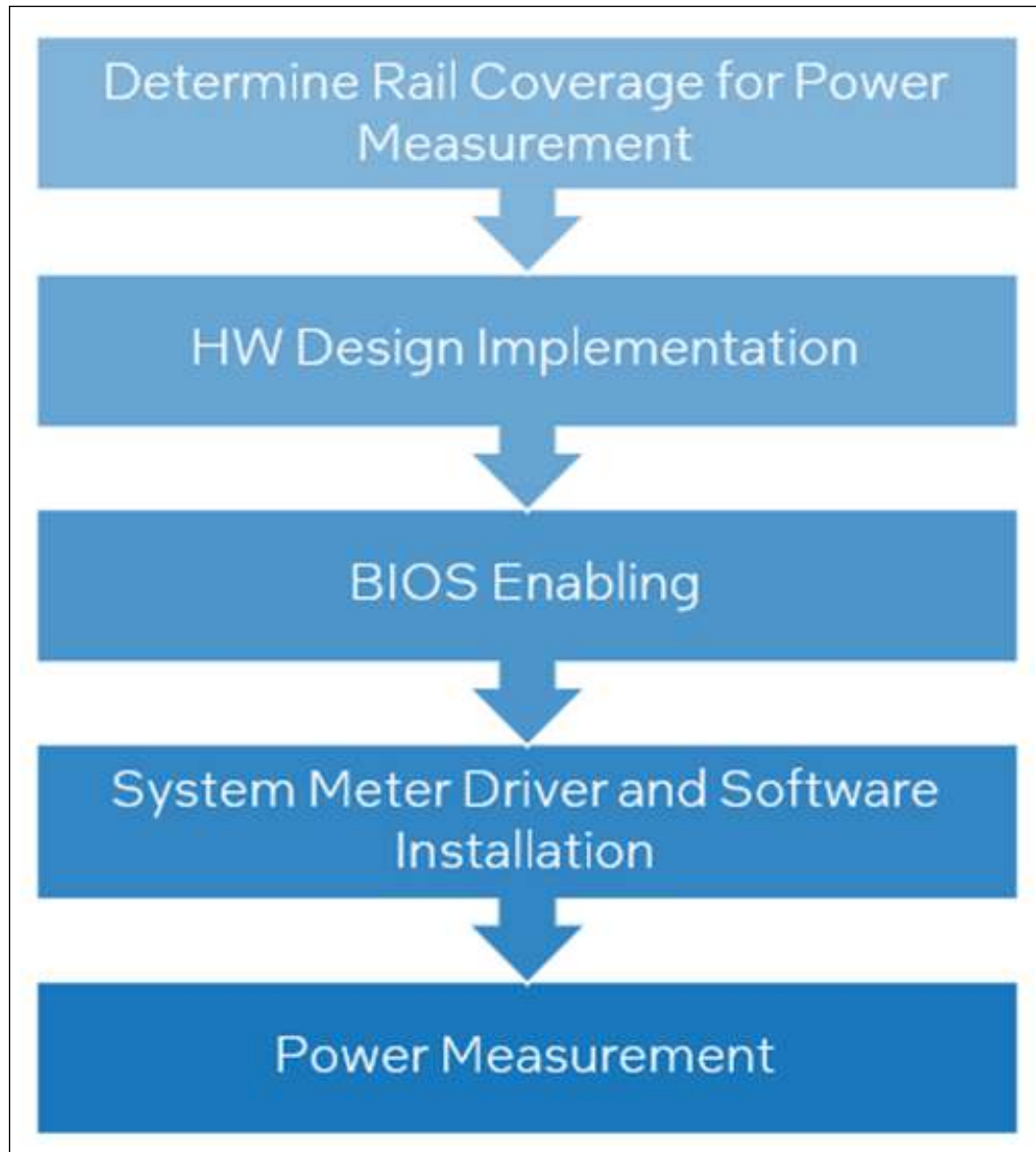


Table 899. System Meter High Level Overview

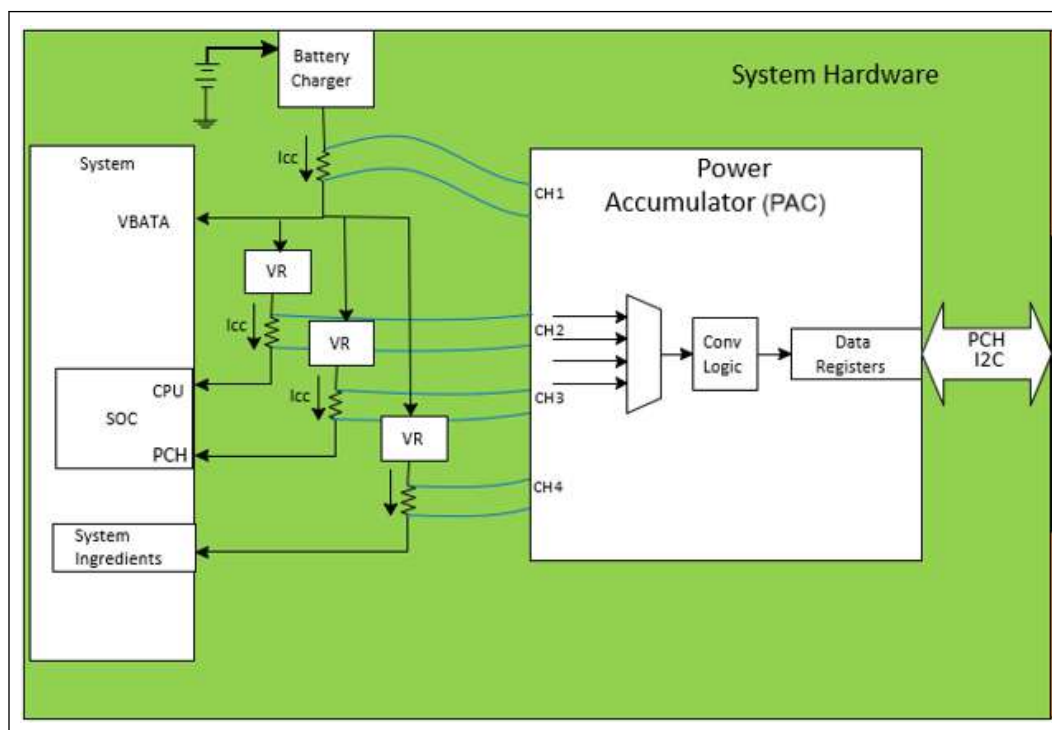
Use Cases		Target Based (on SUT)
Support Solution		Hardware ADC (PAC)
Sampling Rate		$\geq 1\text{mS}$ (BIOS can configure 8-1024 samples per second)
Voltage Measurement Range		0-32V
Power Data Reporting Rate (SW Capability)		$\sim 1\text{S}$ (or 1-60S configurable at SW)
Tolerance Comparing with 3rd Party DAQ	ModS -Idle Scenarios (Power < 150mW)	$\leq 5\%$
continued...		

Use Cases		Target Based (on SUT)
	Active Scenarios (Power > 150mW)	$\leq 2\%$
Self-Power Monitoring	Power Reading Interface	I2C (SoC / PCH)
PCB Space Need (Worst Case)		For each rail: <u>59.98 - 136.12mm²</u>
MSFT Support		System Meter SW and MSFT EMI compatible (System Meter can read power of Devices with Energy Meter Interface (EMI support))
		MSFT SW can read System Meter power numbers (MSFT E3 Driver/API compatible)
Good for		In chassis, Self Power Measurement for Real Life Usage (one BIOS setup / enablement).
		Intel and MSFT aligned Self Power Monitoring solution.
		Wider possibility of Intel System analysis SW integration and one voice (HW / SW) with MSFT SW.

13.3 System Meter Conceptual Block Diagram

- System Meter uses the Power Accumulator (PAC) to perform power calculations and energy accumulation through the Current Senses on each rail.
- PAC has an I²C interface for digital control and reading results.

Figure 629. System Meter Conceptual Block Diagram



13.4 Implementation Flow Considerations

Must

- Intercept Self-Power Monitoring Solution (From circuit design in architectural phase).
- Export schematic netlist with the solution then import to PCB layout design.
- Includes the design on initial PCB placement stage.

Optional

- Refer to the Platform Power and Logic Test Instrumentation Technical Advisory (# 570953) if you would like to design a Flexible Instrumented Platform (FIP) to measure the power of your projects.

13.5 Schematic Design Considerations

13.5.1 Power Rails Coverage

Choice of rails for power is driven by following key factors.

- Key power contributors of the platform.
- Features / Ingredients that need power assessment.
- Below are the key rails a typical platform.
 - Platform Power
 - Panel (Backlight / Electronic Logic)
 - CPU Rail Output Voltage (VCCCORE / VCCGT / VCCSA / VCC_LP_ECORE...etc.)
 - PCH Rails (VCCPRIM_3P3, VCCPRIM_1P8, VNNAON, VCCPRIM_IO)
 - Memory Rail Output Voltage (VDD1, VDD2H, VDD2L, VDDQ)
 - Storage (SSD / HDD)
 - Communications (Wi-Fi)
 - Other Component Power (Camera, Touch)
- Refer to the [PAC IC Sensing Rails Recommendation on Nova Lake](#) on page 904 for the recommended priorities on Nova Lake platform.

13.5.2 PAC IC Sensing Rails Recommendation on Nova Lake

- Below are the Nova Lake sensing rails recommendation depending on how many PAC ICs that could be placed on the board for reference.

Table 900. PAC Device Configuration

Rail#	Power Rails	Domain	Naming Convention in BIOS	Comments
1	VCC_CORE**	CPU	CPU_VCC_CORE	(** Keep PMR at the IMVP VRs Input)
2	VCC_LP_ECORE*		CPU_VCC_LP_ECORE	
continued...				

Rail#	Power Rails	Domain	Naming Convention in BIOS	Comments
3	VCCGT**		CPU_VCCGT	
4	VCCSA**		CPU_VCCSA	
5	VCCPRIM_IO		CPU_VCCPRIM_IO	
6	VCCPRIM_VNNAO N**		CPU_VCCPRIM_VNNA ON	
7	VCCPRIM_1P8_P ROC		CPU_VCCPRIM_1P8_P ROC	
8	VDD2*	MEMORY	MEMORY_VDD2	*Option depends on DDR Types
	VIN_BULK (DDR5)*		MEMORY_VIN_BULK	
9	V3P3A_SSD	STORAGE	STORAGE_SSD	
10	V3P3A_WLAN	CONNECTIVIT Y	CONNECTIVITY_WLAN	
11	V3P3DX_EDP	DISPLAY	DISPLAY_EDP_TCON	
12	VCC_EDP_BKLT	DISPLAY	DISPLAY_EDP_BACKLI GHT	
13	VBATA	SYSTEM	SYSTEM_VBATA	
14	V_CHGR_VIN	ADAPTER	ADAPTER_VAC_IN	
15	PLATFORM V3P3A #	ROP	ROP_V3P3A	#Rest of the platform as per system requirements
16	PLATFORM V1P8A #	ROP	ROP_1P8A	
17	V0P85A_PCH	PCHIOE	PCHIOE_V0P85A	PCH IOE
18	V1P25A_PCH	PCHIOE	PCHIOE_V1P25A	
19	V1P8A_PCH	PCHIOE	PCHIOE_V1P8A	
20	V3P3A_PCH	PCHIOE	PCHIOE_V3P3A	
Notes: 1. Recommended to have minimum 4 PAC devices (16 channels) for NVL H and system with PCH IOE may have 5 PAC (20 channels) 2. Select PMR resistance value and power rating as per the current requirement of each rails 3. Select the PMR values such that, the voltage at the CPU/PCH pins are with in the voltage specification for the maximum current. Recommend to have 10 mΩ at the VR inputs and 5 mΩ at the low voltage rails for better accurate measurement. 4. Make sure the feedback for the VRs are taken after PMR, for the platform rails VBATA, V3P3A, V1P8A				

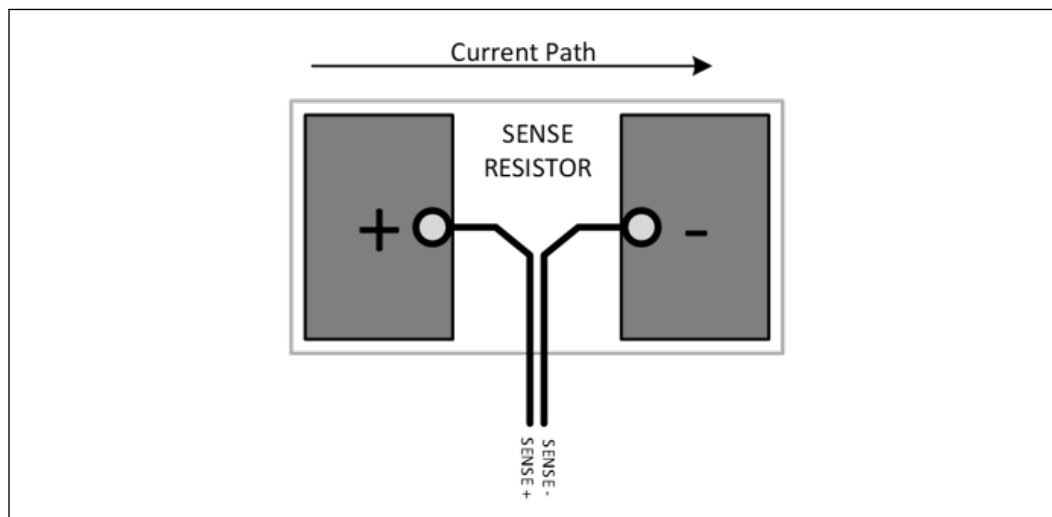
Figure 630. Rails Considered for System Meter in NVL H DDR5 SODIMM RVP

S. No.	Power Meter	CH	Platform Rail	Rails to be referred by BIOS	Domain	Rsense(mΩ)	SMBus Address
1	1	1	+VCCORE_PH1_OUT	SOC_VCCORE_PH1_OUT	SOC	2	0x18
2		2	+VCCORE_PH2_OUT	SOC_VCCORE_PH2_OUT	SOC	2	
3		3	+VCCORE_PH3_OUT	SOC_VCCORE_PH3_OUT	SOC	2	
4		4	+VCCORE_PH4_OUT	SOC_VCCORE_PH4_OUT	SOC	2	
5	2	1	+VCCORE_PH5_OUT	SOC_VCCORE_PH5_OUT	SOC	2	0x1E
6		2	+VCCORE_PH6_OUT	SOC_VCCORE_PH6_OUT	SOC	2	
7		3	+VCCGT_PH1_OUT	SOC_VCCGT_PH1_OUT	SOC	2	
8		4	+VCCGT_PH2_OUT	SOC_VCCGT_PH2_OUT	SOC	2	
9	3	1	+VCCSA_PH1_OUT	SOC_VCCSA_PH1_OUT	SOC	2	0x11
10		2	+VCCSA_PH2_OUT	SOC_VCCSA_PH2_OUT	SOC	2	
11		3	+VCC_LP_CORE	SOC_VCC_LP_CORE	SOC	2	
12		4	+VCCPRIM_VNNAON	SOC_VCCPRIM_VNNAON	SOC	1	
13	4	1	+VCCPRIM_IO	SOC_VCCPRIM_IO	SOC	4	0x15
14		2	+V0P85_PCH	PCH_V0P85_PCH	PCH	2	
15		3	+VCCIO_PCH_VR_OUT	PCH_V1P25_PCH	PCH	10	
16		4	+V3P3A_PCH	PCH_V3P3A_PCH	PCH	22	
17	5	1	+VDD2H_CPU	SOC_VDD2H_CPU	SOC	5	0x19
18		2	+VIN_BULK_MEM	DRAM_VIN_BULK_MEM	DRAM	2	
19		3	+V3P3A_PCH_GEN5_SSD1	STORAGE_V3P3A_PCH_SSD1	STORAGE	5	
20		4	+V3P3A_PCH_GEN5_SSD2	STORAGE_V3P3A_PCH_SSD2	STORAGE	5	
21	6	1	+V125_PCIE_X8_SLOT	PCIE_V125_PCIEx8_SLOT	PCIE	2	0x1C
22		2	+V3P3S_PCIE_X8_SLOT	PCIE_V3P3S_PCIEx8_SLOT	PCIE	5	
23		3	+V125_PCIE_X4_SLOT	PCIE_V125_PCIEx4_SLOT	PCIE	2	
24		4	+V3P3S_PCIE_X4_SLOT	PCIE_V3P3S_PCIEx4_SLOT	PCIE	5	
25	7	1	+V3P3DX_EDP	DISPLAY_V3P3DX_EDP	DISPLAY	10	0x17
26		2	+V3P3A_WLAN	CONNECTIVITY_V3P3A_WLAN	CONNECTIVITY	5	
27		3	+V1P8S_MCSI_CAM1	CAMERA_V1P8S_MCSI_CAM1	CAMERA	10	
28		4	+V3P3S_MCSI_CAM1	CAMERA_V3P3S_MCSI_CAM1	CAMERA	10	
29	10	1	+VJT_B8_TYPEC_OUT	CHGRIN_V_CHGR_VIN	CHGRIN	5	0x1B
30		2	+VSY5	SYSTEM_V3A1A_IMVP	SYSTEM	1	
31		3	+VBATA_RCP	SYSTEM_V3A1A_RCP	SYSTEM	2	
32		4	+VCC_EDP_BKLT	DISPLAY_EDP_BKLT	DISPLAY	10	
33	11	1	+V1P8A_PCH_VR_OUT	PCH_V1P8A_PCH	PCH	5	0x14
34		2	+VCCPRIM_1P8	SOC_VCCPRIM_1P8	SOC	10	
35		3	+VCCRTC	SOC_VCCRTC	SOC	22	
36		4	+VCCPRTC_3P3_PCH	PCH_VCCPRTC_3P3	PCH	22	

13.5.3 Schematic Design Considerations/R-Sense Selection

Schematic

- Ensure source side of Current Sense (PMR or R-Sense) is connected to "SENSE+" and load side connected to "SENSE-".

Figure 631. Kelvin Connection Across PMR

- Connect the Power Accumulator IC (PAC IC) I2C link to the I2C bus on SoC or PCH.
 - .
 - Power the PAC IC with the right voltage corresponding to the I2C bus levels of the SoC / PCH.
 - Device powered with 3.3V.
 - Device I/O powered with appropriate I2C level as supported by SoC or PCH (3.3V or 1.8V).
 - Select right I2C address for each of the PAC ICs implemented and avoid address conflicts on the bus.

R-Sense Selection

- Accuracy of the measurement dependent on the sense resistor (R-Sense) value. Following factors need to be considered when selecting sense resistor.
 - Calculate R-Sense considering the maximum current sensed that includes current peaks well beyond nominal current.
 - Select the R-Sense value to ensure sufficient dV available across the sense resistors for measuring power accurately while maintaining Vmin margins.
 - Select small footprint sense resistor while keeping power dissipation in consideration while Vmin Margin at the max interface current.
 - Ensure to use 1% or lower tolerance R-Sense.
- Use 'Kelvin Sensing' or any VIA in pad type of Current Sense symbols to avoid excessive/unquantifiable IR drops.
- For FIP Board R-Sense Layout Guidance, refer to the Platform Power and Logic Test Instrumentation Technical Advisory (# [570953](#)).

13.5.4 Reference Schematics (RVP)

- Refer to the Nova Lake HX/H DDR5 SODIMM 1DPC T3 RVP Technical Documentation Kit (# [856341](#)) implementation.

13.5.5 BOM List/Supported ICs

Refer to the RVP Technical Documentation Kit [842753](#) for BOM and supported IC details.

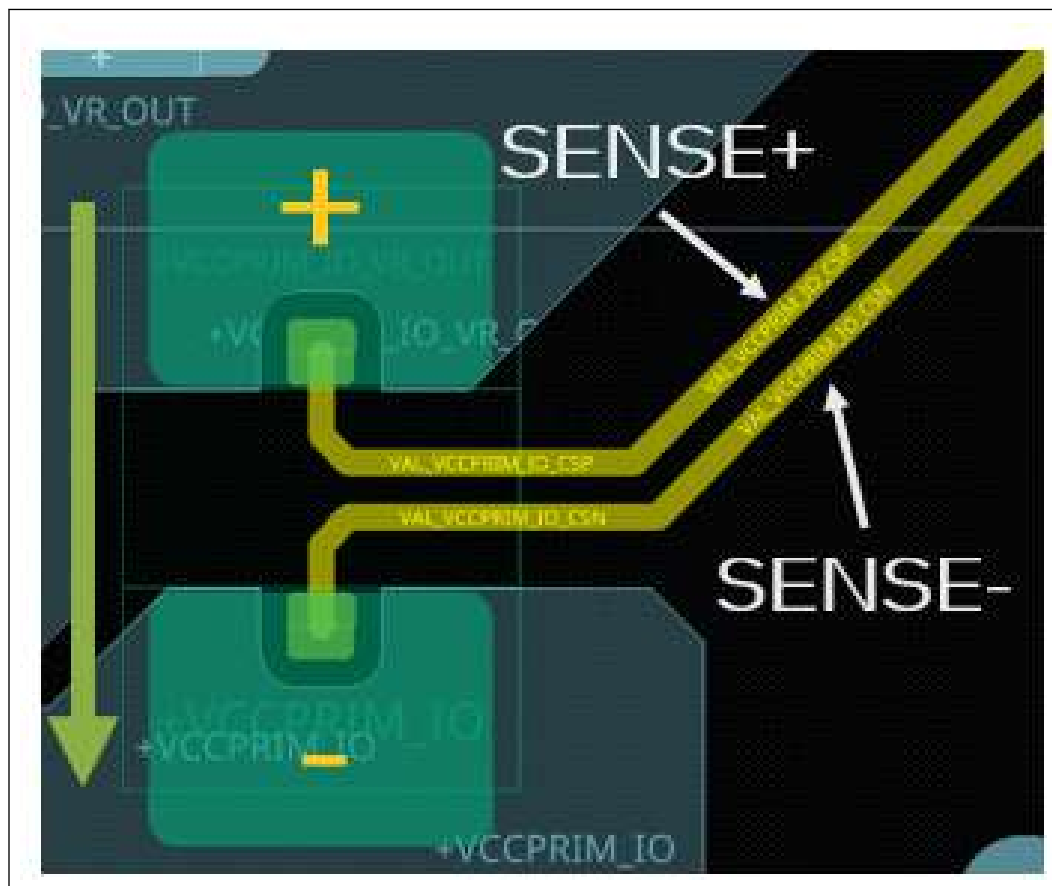
13.6 Layout Design Considerations

13.6.1 R-Sense Layout Considerations

NOTE

For low RSENSE values, resistor connections and solder joints can significantly impact accuracy.

- Ensure sense lines from Current Sense (PMR or R-Sense) are tapped closer to the inner center of pad area.
- PCB Pattern:
 - Use a wide metal trace for the high-current path.
- Connections:
 - **VSource** connects to the + terminal.
 - **Load** connects to the - terminal.
- Kelvin Connection:
 - Route SENSE+ and SENSE- directly to the sense resistor to avoid high-current interference in the VSENSE measurement. Refer to the [Figure 632](#) on page 909.
- Differential Pair Routing:
 - SENSE+ and SENSE- should be routed as a differential pair to the PAC chip's SENSE inputs.
 - Route Kelvin Sensed differential pairs from Current Sense to the PAC IC with solid GND reference.
 - Avoid referencing to noisy planes for better noise margin.
- I2C:
 - Follow the design requirements defined in Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# [844316](#)).
- For detailed layout considerations, refer to the PAC datasheet.

Figure 632. Kelvin Connections Layout Reference for NVL -H RVP

13.6.2 Reference Layout (RVP)

- The below mentioned details are only for reference. Refer to the RVP Technical Documentation Kit (TDK) for implementation.

Figure 633. PAC1954 Layout Example on NVL-H RVP, BOT Layer

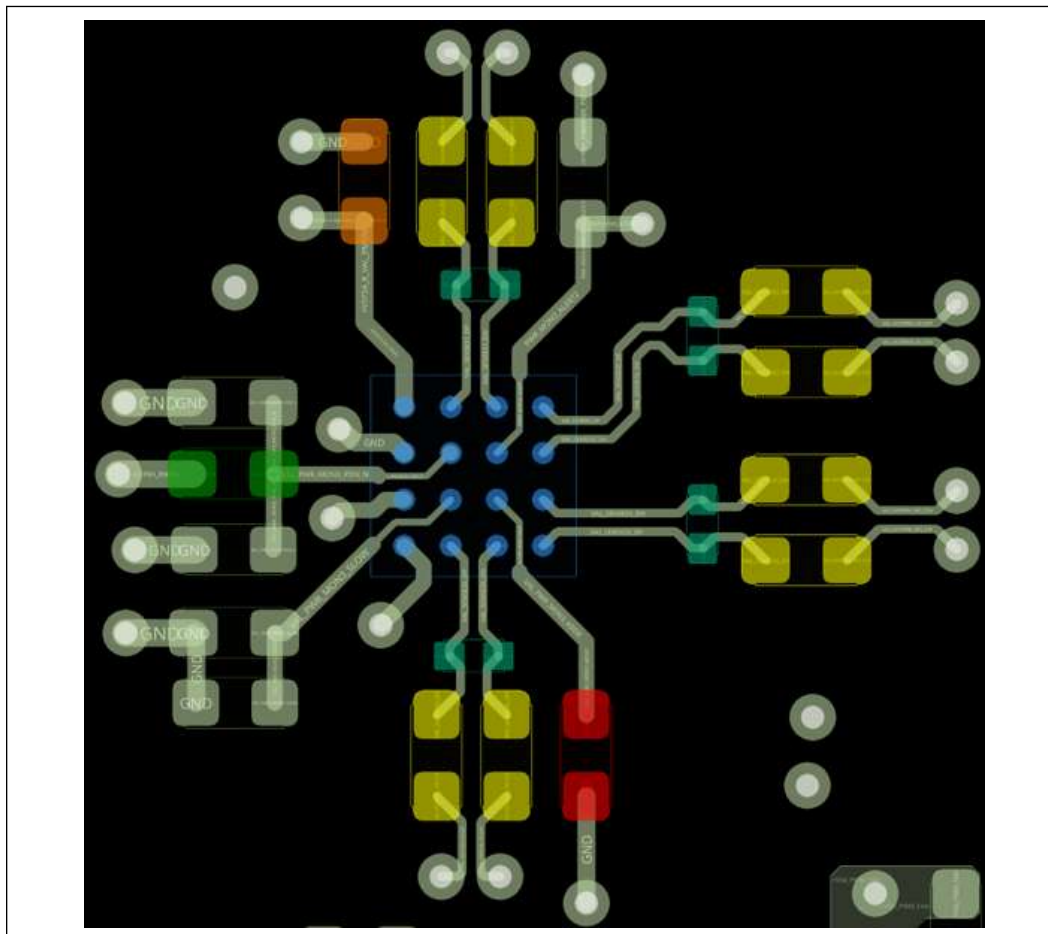
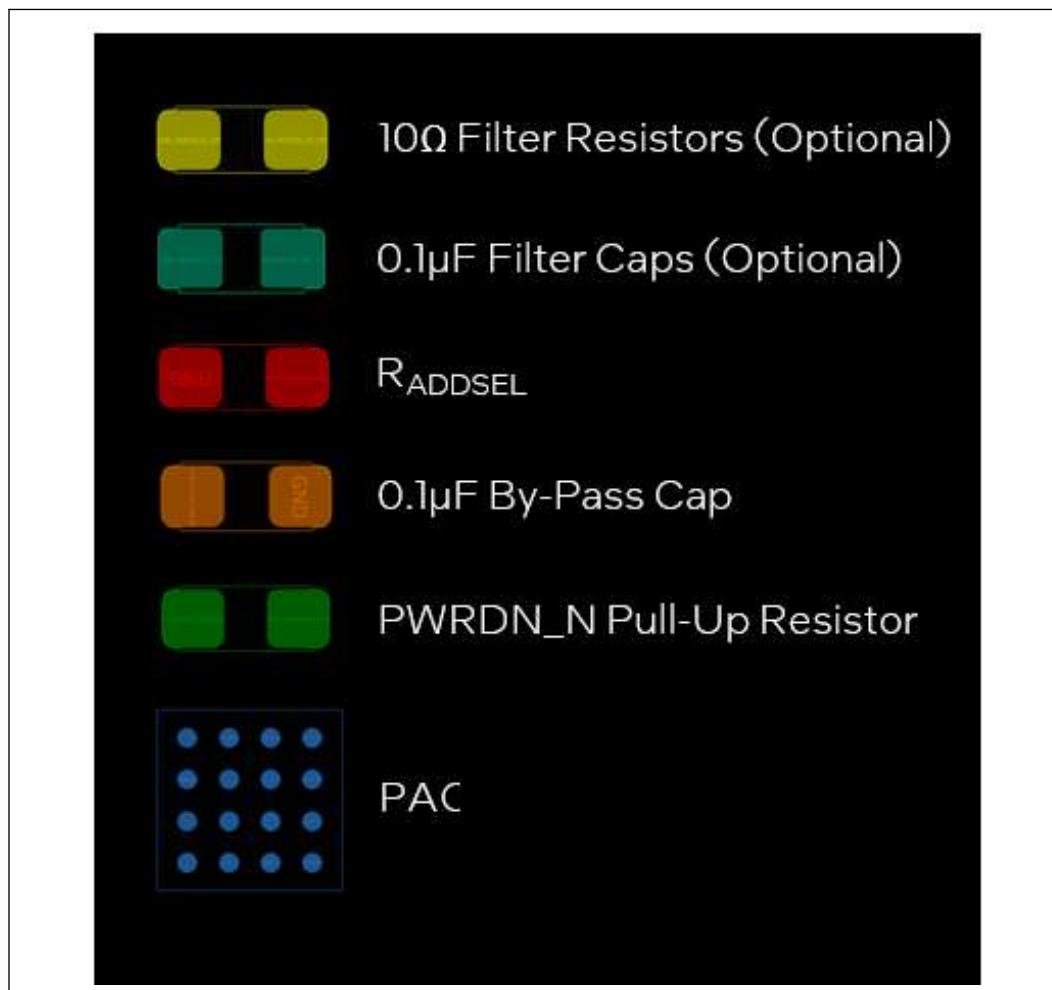


Figure 634. Color Code Explained**NOTE**

For additional examples of system meter implantation and more details, refer to the System Meter Design Guide Reference Guide ([# 838411](#)).